Medical Dictionary by Specialty

A Comprehensive Reference of Medical Terminology

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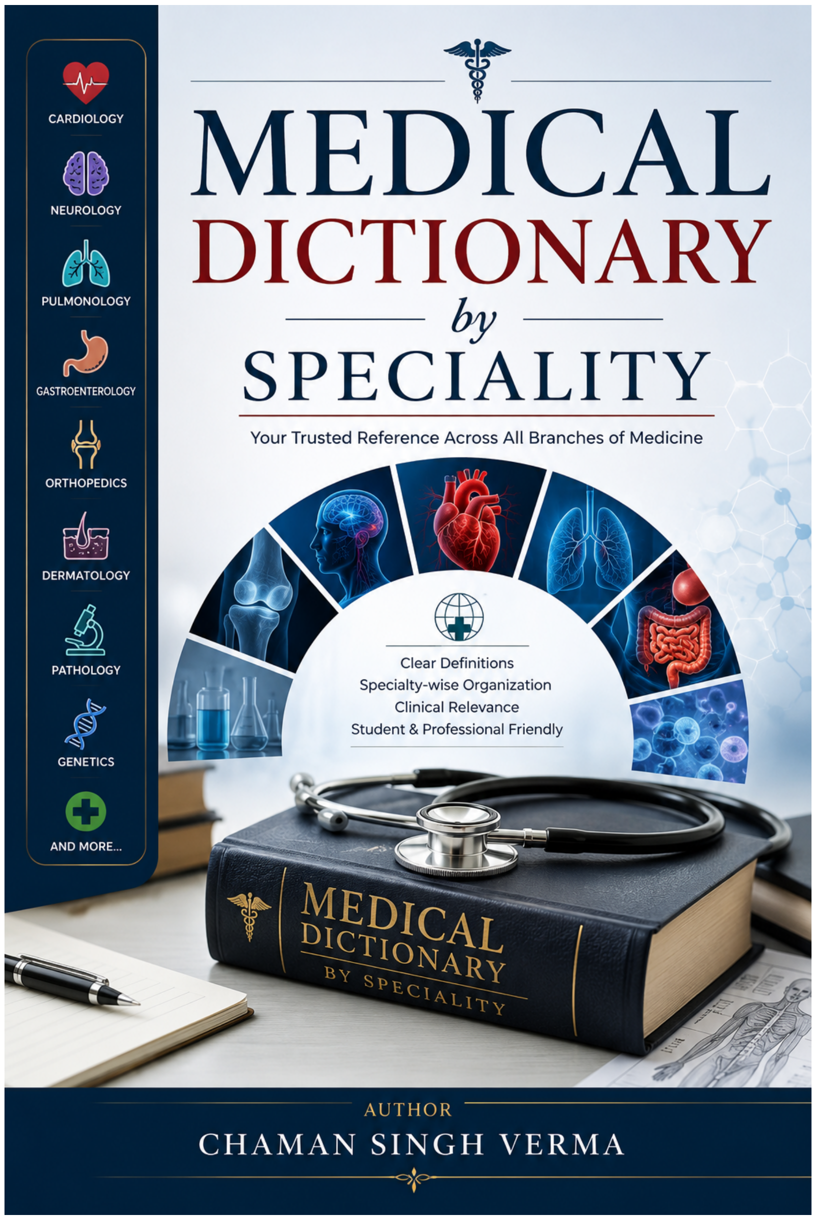
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Preface

Medical dictionaries have traditionally been organised alphabetically, an arrangement well suited for looking up an unfamiliar term when you already know its name. But medicine is not learned alphabetically, nor is it practised that way. A cardiologist thinking through a differential diagnosis reaches for cardiac terms, not a mixture of cardiology wedged between caesarean section and cataract.

This dictionary is organised by medical specialty for precisely that reason. Each of the 35 chapters corresponds to a recognised clinical or preclinical discipline, with terms listed alphabetically within that chapter. The result is a reference that mirrors the structure of medical knowledge itself: renal terms in Nephrology, antiarrhythmics in Cardiology, autoimmune diseases in Rheumatology. Cross-references redirect the reader from variant names and alternative spellings to the main entry, so "Pertussis" still leads to "Whooping Cough" and "Hepatolenticular Degeneration" points to "Wilson Disease."

The entries are drawn primarily from Black's Medical Dictionary (42nd Edition) and supplemented with definitions reflecting contemporary clinical practice. Each entry is written in a concise single-line format, with the term displayed in bold red and automatically indexed. The dictionary runs to over 5,600 entries across 772 pages, covering anatomy, physiology, pathology, pharmacology, clinical medicine, surgery, and the allied health sciences.

This work is a living document. The LaTeX source is designed to be easily extended, corrected, and reorganised. Contributions, corrections, and suggestions are welcome.

Chaman Singh Verma

Contents

Chapter 1

Anatomy

Abdomen

– The abdomen is the region of the trunk between the thorax and pelvis, bounded superiorly by the diaphragm and inferiorly by the pelvic inlet. It contains the stomach, liver, spleen, pancreas, kidneys, intestines, and major blood vessels. Clinically divided into nine regions by two vertical midclavicular and two horizontal planes: right and left hypochondriac, epigastric, umbilical, iliac, and hypogastric. Surface anatomy is critical for physical examination and diagnosis of appendicitis, cholecystitis, and bowel obstruction.

Abdominal Aorta

– From diaphragmatic hiatus at T12 to common iliac bifurcation at L4. Major branches: celiac trunk (T12), SMA (L1), IMA (L3), renal arteries (L1-L2), gonadal arteries, lumbar arteries. Anterior to the vertebral column. Abdominal aortic aneurysm is common in elderly males (13th leading cause of death). Rupture causes hypotension, back pain, and pulsatile mass with over 80 percent mortality.

Acetabulum

– Cup-shaped lateral hip socket from fused ilium, ischium, and pubis. Articulates with the femoral head at the ball-and-socket hip joint. The acetabular labrum deepens the socket. Oriented with approximately 15 degrees anteversion and 45 degrees inclination. Fractures are serious high-energy trauma injuries.

Acromion

– Bony process forming the shoulder's highest point, extending laterally from the scapular spine. Articulates with the clavicle at the acromioclavicular joint and arches over the glenohumeral joint. Attachment for deltoid and trapezius. The subacromial space contains the supraspinatus tendon and subacromial bursa. Its shape influences impingement syndrome susceptibility.

Amygdala

– Almond-shaped nucleus cluster in the anterior medial temporal lobe, part of the limbic system, primarily processing emotions (especially fear/anxiety). Receives thalamic/cortical sensory input; projects to hypothalamus, brainstem, and cortex for autonomic/behavioral fear responses. Essential for emotional memory (fear conditioning) and social signal processing. Lesions (Urbach-Wiethe disease) impair fear recognition. Hyperactive in anxiety disorders and PTSD.

Anatomical Snuffbox

– Triangular depression on the dorsal wrist, bounded by extensor pollicis longus medially and extensor pollicis brevis/abductor pollicis longus laterally. Floor contains the scaphoid, trapezium, and radial artery. Tenderness after a fall on an outstretched hand suggests scaphoid fracture. The radial artery pulse is palpable here.

Ankle Joint

– Uniaxial hinge synovial joint between distal tibia, lateral malleolus, and superior talus, allowing dorsiflexion/plantarflexion. Medial (deltoid) ligament resists eversion; lateral complex resists inversion. The ankle mortise provides bony stability. Ankle sprains are the most common sports injury (anterior talofibular ligament most commonly affected). Fractures classified by the Weber system.

Antecubital Fossa

– Triangular depression on the anterior elbow, bounded laterally by brachioradialis, medially by pronator teres, and superiorly by a line connecting the epicondyles. Floor formed by brachialis and supinator. Contains biceps tendon, brachial artery, and median nerve. Primary site for venipuncture and blood pressure measurement.

Anterior Tibial Artery

– Arises from the popliteal artery, passes through the anterior intermuscular septum to the anterior compartment, continuing to the ankle as the dorsalis pedis artery. Supplies anterior leg and dorsal foot muscles. Palpable anterior to the lateral malleolus as the dorsalis pedis pulse. Important for vascular assessment; vulnerable to injury in tibial fractures and anterior compartment syndrome.

Aorta

– Largest artery, from the left ventricle distributing oxygenated blood systemically. Approximately 30 cm long, 2.5 cm diameter. Four segments: ascending aorta, aortic arch (brachiocephalic trunk, left common carotid, left subclavian), descending thoracic aorta, descending abdominal aorta (bifurcating into common iliacs at L4). Three layers: intima, media (elastic tissue), adventitia. Aneurysm and dissection are life-threatening.

Aortic Arch

– Curves from second right sternocostal joint superiorly and left, then posteriorly and inferiorly to T4. From right to left: brachiocephalic trunk, left common carotid, left subclavian. Related to left recurrent laryngeal nerve (hooking under it), trachea, esophagus, left phrenic nerve. Arch aneurysms may cause hoarseness and SVC syndrome.

Aortic Valve

– Semilunar valve between left ventricle and aorta with right, left, and posterior cusps. Each cusp forms a sinus of Valsalva from which coronary arteries originate. Opens during systole; closes during diastole. Aortic stenosis is the most common elderly valvular disease from calcific degeneration, causing a systolic crescendo-decrescendo murmur. Bicuspid aortic valve is the most common congenital valvular abnormality.

Artery

– A blood vessel that carries oxygenated blood away from the heart to the tissues, except for the pulmonary arteries which carry deoxygenated blood to the lungs. Arteries have thick, elastic, muscular walls that withstand high pressure and distribute blood throughout the body. The aorta is the largest artery. Arterial structure: tunica intima (endothelium), tunica media (smooth muscle and elastic fibers), tunica adventitia (connective tissue). Arteries branch into smaller arterioles and eventually capillaries. Diseases: atherosclerosis, aneurysms, arteritis, emboli.

Ascending Aorta

– From left ventricle at the aortic valve to the sternal angle (approximately 5 cm), enclosed in pericardium. Gives rise to right/left coronary arteries at the aortic sinuses. Most common site of aneurysm and dissection, especially in Marfan syndrome, bicuspid aortic valve, and hypertension. Aneurysms may compress adjacent structures and cause aortic regurgitation.

Ascending Colon

– Extends from cecum to hepatic flexure below the liver, approximately 15-20 cm, retroperitoneal (fixed to posterior abdominal wall). Anteriorly: abdominal wall. Posteriorly: right kidney. Absorbs water/electrolytes and compacts fecal matter. Tumors may present as a palpable right-sided mass. The right colic flexure marks its transition to the transverse colon.

Atlas (C1)

– First cervical vertebra articulating with occipital condyles at the atlanto-occipital joint for nodding movement. Lacks a body and spinous process; has anterior/posterior arches and lateral masses. Transverse foramina transmit vertebral arteries. The anterior arch has a facet for the axis dens. Commonly fractured in axial loading injuries such as diving accidents.

Atrium

– One of the two upper chambers of the heart that receive blood returning to the heart. The right atrium receives deoxygenated blood from the superior and inferior venae cavae and the coronary sinus; the left atrium receives oxygenated blood from the four pulmonary veins. The interatrial septum separates the two atria; the fossa ovalis marks the site of the fetal foramen ovale. Atrial contraction (atrial kick) contributes 15–25% of ventricular filling. The atrial appendages (auricles) increase atrial capacity. Conditions: atrial fibrillation, atrial septal defect, atrial myxoma.

Axilla

– Pyramidal space between the upper lateral thorax and medial arm. Bounded by anterior fold (pectoralis major/minor), posterior fold (latissimus dorsi/teres major), medial wall (ribs/serratus anterior), and lateral wall (humerus). Contains axillary artery/vein, brachial plexus cords, and lymph nodes. Axillary nodes are significant in breast cancer staging.

Axillary Nerve

– From posterior cord (C5-C6), passing through the quadrangular space with the posterior circumflex humeral artery. Innervates deltoid and teres minor; provides sensation to the regimental badge area (lateral deltoid skin). Vulnerable in anterior shoulder dislocations and surgical neck humeral fractures. Injury causes loss of lateral deltoid sensation and weakness of abduction beyond 15 degrees.

Axis (C2)

– Second cervical vertebra with the dens (odontoid process) projecting superiorly, serving as a pivot for atlas rotation, allowing approximately 50 percent of cervical rotation. Has a large spinous process and flat superior articular facets. Dens fractures (Types I-III) may cause atlanto-axial instability and spinal cord compression, often requiring surgical fixation.

Basal Ganglia

– Deep gray matter nuclei in the cerebral hemispheres for motor control, cognition, and emotion. Key structures: caudate nucleus, putamen, globus pallidus (internal/external segments), subthalamic nucleus, substantia nigra. Direct pathway facilitates movement; indirect inhibits it. Receive cortical/thalamic input, project back to cortex via thalamus. Diseases: Parkinson's (substantia nigra dopamine depletion), Huntington's (caudate atrophy), hemiballismus (subthalamic lesion).

Biceps Brachii Muscle

– Two-headed anterior arm muscle. Long head from supraglenoid tubercle, short head from coracoid process, both inserting on the radial tuberosity. Innervated by musculocutaneous nerve (C5-C6). Most powerful elbow flexor in supination; supinates forearm; weakly flexes shoulder. Distal rupture causes Popeye deformity. Long head susceptible to tenosynovitis.

Bone

– A rigid, mineralised connective tissue that forms the skeleton, providing structural support, protection of vital organs, hematopoiesis (bone marrow), and mineral storage (calcium, phosphate). Bone composition: organic matrix (90% type I collagen) and inorganic mineral (hydroxyapatite, calcium phosphate). Types: cortical (compact, 80% of bone mass) and cancellous (trabecular, 20%). Cells: osteoblasts (bone formation), osteocytes (mechanosensation), osteoclasts (bone resorption). Ossification: intramembranous (flat bones) and endochondral (long bones). Remodelling: continuous cycle of resorption and formation.

Brachialis Muscle

– Flat strong muscle deep to the biceps, from distal anterior humerus to ulnar tuberosity and coronoid process. Innervated by musculocutaneous nerve (C5-C6). Primary elbow flexor regardless of forearm position ("workhorse" of elbow flexion). Less visible than biceps but significantly contributes to arm strength.

Brachial Plexus

– Nerve network (C5-T1) formed by roots, trunks, divisions, and cords, giving rise to upper limb nerves. Roots unite into superior/middle/inferior trunks, dividing into anterior/posterior divisions, recombining into lateral/posterior/medial cords. Major branches: musculocutaneous, axillary, radial, median, ulnar. Birth injuries cause Erb's palsy (C5-C6) or Klumpke's palsy (C8-T1). Vulnerable to shoulder trauma and thoracic outlet syndrome.

Brachial Plexus Cords

– Formed by divisions recombining behind the pectoralis minor. Lateral cord (anterior divisions of upper/middle trunks), posterior cord (all three posterior divisions), medial cord (anterior division of lower trunk). Named for relationship to the axillary artery. Lateral cord: lateral pectoral nerve, contributes to median nerve. Posterior cord: subscapular, thoracodorsal, axillary, radial nerves. Medial cord: medial pectoral, cutaneous, and ulnar nerves.

Brachial Plexus Roots

– Ventral rami of C5-T1 emerging from intervertebral foramina, passing between anterior and middle scalene muscles (interscalene triangle). C5-C6 form the upper trunk, C7 continues as middle trunk, C8-T1 form the lower trunk. Root injuries cause specific dermatome/myotome deficits. C5 avulsion causes loss of deltoid/biceps function. Roots are vulnerable in birth trauma, motorcycle accidents, and disc herniation radiculopathy.

Brachial Plexus Trunks

– Formed in the posterior neck triangle. Upper (C5-C6), middle (C7), and lower (C8-T1) trunks each divide into anterior/posterior divisions behind the clavicle. Upper trunk is most commonly injured in birth injuries (Erb-Duchenne palsy), causing waiter's tip position. Lower trunk injury (Klumpke's palsy from upward arm traction) causes claw hand.

Brainstem

– Connects cerebrum to spinal cord, with midbrain (superior), pons (middle), and medulla oblongata (inferior). Contains cranial nerve nuclei (III-XII), ascending/descending tracts, and vital autonomic centers for cardiac, respiratory, and vasomotor function. Reticular formation regulates consciousness and sleep-wake cycles. Contains all brain-spinal cord pathways. Lesions are often devastating and fatal, causing respiratory arrest, cardiac arrhythmias, and bilateral cranial nerve deficits.

Bronchus

– One of the two main branches of the trachea that conduct air into the lungs. The right main bronchus is wider, shorter, and more vertical than the left, making it more susceptible to aspiration. Each main bronchus divides into lobar bronchi (secondary), segmental bronchi (tertiary), and progressively smaller branches (bronchioles). Bronchial walls contain ciliated pseudostratified columnar epithelium, smooth muscle, cartilage plates, and mucous glands. Bronchoconstriction occurs in asthma. Bronchiectasis is irreversible dilatation. Bronchoscopy allows visualization and biopsy.

Cadaver

– A dead human body used for anatomical dissection, surgical training, and medical research. Cadavers are preserved through embalming (arterial injection of formaldehyde-based fixatives) to prevent decomposition and maintain tissue integrity. Anatomical dissection of cadavers remains a cornerstone of medical education, providing three-dimensional understanding of human anatomy, anatomical variation, and surgical approaches. Whole-body donation programs supply cadavers to medical schools; ethical considerations include informed consent, respectful treatment, and compliance with anatomical gift laws.

Calcaneus

– Largest tarsal bone forming the heel. Articulates with talus proximally and cuboid anteriorly. Lever for the calf muscles via the Achilles tendon on its posterior tuberosity. Predominantly trabecular bone, the most commonly fractured tarsal bone from falls from height. Bohler's angle assesses fractures on lateral radiographs.

Cantlie Line

– Functional liver division extending from the gallbladder fossa to the IVC, corresponding to the middle hepatic vein plane. Unlike the falciform ligament (anatomical division), it represents the true functional blood supply/biliary drainage division. Right hepatic artery and portal vein supply the right half; left supply the left. Basis for anatomic hepatic segmentectomy and liver transplantation.

Capitate

– Largest carpal bone, centrally located in the distal row as the carpal keystone. Rounded head articulates with scaphoid and lunate; distal surface with the third metacarpal. Last carpal bone to ossify. Wrist joint axis, bearing significant axial load. Avascular necrosis may occur with fractures similar to scaphoid injuries.

Carpal Tunnel

– Narrow passageway on the palmar wrist, bounded by carpal bones dorsally and the flexor retinaculum as the roof. Transmits the median nerve and flexor tendons. Carpal tunnel syndrome causes numbness in the lateral 3.5 digits and thenar weakness. It is the most common entrapment neuropathy, associated with repetitive motion, diabetes, hypothyroidism, and pregnancy.

Cartilage

– A tough, flexible avascular connective tissue found in joints, the ear, nose, rib cage, and intervertebral discs. Composed of chondrocytes embedded in an extracellular matrix of collagen fibers, proteoglycans, and water, providing resilience and compressive strength. Types: hyaline cartilage (most common, covers articular surfaces of joints, costal cartilages), elastic cartilage (ear, epiglottis), and fibrocartilage (intervertebral discs, menisci, pubic symphysis). Cartilage lacks blood vessels, nerves, and lymphatics; it receives nutrition by diffusion from synovial fluid, limiting healing capacity.

Cecum

– First and widest large intestinal part in the right iliac fossa below the ileocecal valve, a blind-ended pouch approximately 6 cm long. Appendix arises from its posteromedial wall approximately 2 cm below the valve, containing abundant lymphoid tissue. Cecal volvulus causes large bowel obstruction. Absorbs water/electrolytes and compacts fecal matter.

Celiac Trunk

– First major abdominal aortic branch at T12-L1, the foregut artery. Divides into left gastric (lesser curvature), splenic (spleen, pancreas, greater curvature), and common hepatic (proper hepatic/gastroduodenal) arteries. Surrounded by the celiac plexus. Compression by the median arcuate ligament causes celiac trunk compression syndrome with postprandial pain.

Cell

– The fundamental structural and functional unit of all living organisms. Human cells are eukaryotic, characterized by a nucleus containing DNA and membrane-bound organelles. Major components: cell membrane (phospholipid bilayer with embedded proteins, regulates transport and signaling), cytoplasm (cytosol + organelles), nucleus (chromosomes, nucleolus), mitochondria (ATP production through oxidative phosphorylation), endoplasmic reticulum (proteinlipid synthesis), Golgi apparatus (modification and sorting of proteins), lysosomes (intracellular digestion), peroxisomes (fatty acid oxidation), and cytoskeleton (microfilaments, intermediate filaments, microtubules–maintains shape and enables movement). Cells differentiate into 200 specialized types with distinct functions (neurons, myocytes, hepatocytes, etc.). Cell division occurs via mitosis (somatic cells) and meiosis (gametes). Cell death occurs by apoptosis (programmed) or necrosis (pathologic).

Cerebellum

– Largest brain portion (approximately 10 percent of volume but over 50 percent of neurons) in the posterior cranial fossa behind the brainstem. Two hemispheres and a vermis, divided into vestibulocerebellum (balance, eye movements), spinocerebellum (posture, gait, coordination), and cerebrocerebellum (motor planning, cognition). Receives mossy/climbing fiber input from spinal cord, brainstem, and cortex; outputs via deep nuclei to thalamus/motor cortex. Lesions cause ataxia, dysmetria, intention tremor, nystagmus, and scanning speech.

Cerebral Lobes

– Four lobes divided by sulci. Frontal (anterior to central sulcus): executive function, motor control (precentral gyrus), Broca's area. Parietal (between central and parieto-occipital sulci): somatosensory (postcentral gyrus), spatial awareness. Temporal (below lateral sulcus): auditory processing, memory (hippocampus), Wernicke's area. Occipital (posterior): visual processing. Insula (deep within lateral sulcus): taste, emotion, interoception.

Ciliary Body

– Circumferential structure between iris and choroid, with ciliary muscle (accommodation control via zonular fibers) and ciliary processes (aqueous humor production). Part of the uveal tract (iris, ciliary body, choroid). Parasympathetic CN III fibers cause muscle contraction, relaxing zonular fibers and allowing lens convexity for near vision. Ciliary processes secrete aqueous humor into the posterior chamber, flowing through the pupil to the anterior chamber and draining at the trabecular meshwork and Schlemm's canal.

Clavicle

– Only horizontally oriented long bone, serving as a shoulder-trunk strut. Medial end articulates with the manubrium (sternoclavicular joint); lateral end with the acromion (acromioclavicular joint). Most common fracture at the middle-lateral third junction from falls on an outstretched hand. Protects the underlying brachial plexus and subclavian vessels.

Coccyx

– Terminal vertebral column part with 3-5 fused vertebrae. Articulates with the sacrum, allowing limited flexion important during childbirth. Provides attachment for gluteus maximus, coccygeus, and levator ani (pelvic floor). A common site of pain (coccydynia) following falls, direct trauma, and prolonged sitting.

Cochlea

– Snail-shaped hearing organ in the inner ear, coiled approximately 2.5 turns around the modiolus. Three fluid-filled chambers: scala vestibuli (perilymph), scala media (endolymph, organ of Corti), scala tympani (perilymph). The organ of Corti has inner/outer hair cells transducing mechanical vibrations into electrical signals via stereocilia deflection. Basilar membrane vibrates differentially along its length (tonotopic: high frequencies at base, low at apex). Hair cell damage from noise, ototoxicity, and ageing causes irreversible sensorineural hearing loss, most commonly affecting high frequencies.

Colon

– The largest part of the large intestine, extending from the cecum to the rectum. Anatomically divided into ascending colon (right side), transverse colon (across the abdomen), descending colon (left side), and sigmoid colon (S-shaped, leading to the rectum). Functions: absorption of water and electrolytes, fermentation of undigested carbohydrate by gut microbiota, formation and storage of feces. The colon has a distinct appearance with taeniae coli (three longitudinal muscle bands), haustra (sacculations), and appendices epiploicae (fat appendages). Common diseases: diverticulosis, colitis, colorectal cancer, irritable bowel syndrome.

Common Iliac Arteries

– Terminal abdominal aortic branches at L4, each approximately 4 cm long, dividing into internal and external iliacs. Supply the pelvis and lower extremities. Right crosses anterior to the right common iliac vein. Aneurysms may occur with AAA and compress ureters causing hydronephrosis. They unite to form the IVC at L5.

Common Peroneal Nerve

– Sciatic division (L4-S2) winding around the fibular neck, the body's most vulnerable major nerve to external compression. Divides into superficial peroneal (peroneus longus/brevis, dorsal foot sensation) and deep peroneal (tibialis anterior, extensors, first web space sensation). Injury causes foot drop (loss of dorsiflexion/eversion) with dorsal foot sensory loss.

Coracoid Process

– Hook-like scapular projection extending anteriorly and laterally. Attachment for 3 muscles (pectoralis minor, coracobrachialis, short head of biceps) and 3 ligaments. Important surgical landmark. Fractures uncommon but may occur with direct trauma or shoulder dislocations.

Cornea

– Transparent, avascular, dome-shaped anterior eye surface providing approximately two-thirds of total refractive power. Five layers: epithelium (stratified squamous, rapidly regenerating), Bowman's membrane, stroma (thickest, regular collagen), Descemet's membrane, endothelium (single cell layer maintaining transparency). Innervated by CN V1 (ophthalmic trigeminal), extremely sensitive. Conditions: abrasion, ulcers, keratoconus (progressive thinning and cone-shaped protrusion). Contact lens wear is a major risk factor for microbial keratitis, which can rapidly lead to corneal ulceration, perforation, and sight loss if not treated promptly.

Corpus Callosum

– Brain's largest white matter tract with approximately 200 million axons connecting left and right hemispheres. Four parts: rostrum, genu, body, splenium. Facilitates interhemispheric communication for sensory, motor, and cognitive integration. Genu connects prefrontal cortices; body connects motor/sensory cortices; splenium connects temporal/occipital cortices. Callosotomy is a last-resort epilepsy treatment producing split-brain syndromes with disconnection phenomena.

Cuboid

– Cube-shaped lateral foot tarsal bone articulating with calcaneus proximally and 4th/5th metatarsals distally. Attachment for the long plantar ligament and peroneus longus tendon. Keystone of the lateral longitudinal arch. Cuboid syndrome (subluxation) is a common lateral foot pain cause in athletes from excessive pronation.

Deltoid Muscle

– Large triangular muscle forming the shoulder contour, from lateral clavicle, acromion, and scapular spine to the deltoid tuberosity. Innervated by the axillary nerve (C5-C6). Anterior fibers flex/medially rotate, middle fibers abduct (primary abductor beyond 15 degrees), posterior fibers extend/laterally rotate. Essential for abduction; common injection site. Axillary nerve injury causes loss of abduction.

Dermis

– Thick connective tissue layer beneath the epidermis providing structural support, elasticity, and nourishment. Two layers: papillary (superficial, loose connective tissue with dermal papillae interdigitating with epidermal ridges) and reticular (deep, dense irregular connective tissue). Contains collagen (type

I, tensile strength), elastin, fibroblasts, blood vessels, lymphatics, nerves, hair follicles, sebaceous glands, and sweat glands. The dermal-epidermal junction is critical for adhesion; disruption causes blistering disorders such as pemphigus vulgaris and toxic epidermal necrolysis.

Descending Colon

– From splenic flexure to left iliac fossa, approximately 25-30 cm, retroperitoneal and fixed by phrenicocolic ligament. Anteriorly: small intestine. Posteriorly: left kidney. Absorbs water/electrolytes and compacts fecal matter. Left-sided pathology (diverticulitis, cancer) is more common here and in the sigmoid. Narrower than ascending colon, more susceptible to obstruction.

Descending Thoracic Aorta

– From aortic arch at T4 to diaphragm at T12. Gives off bronchial, esophageal, mediastinal, and posterior intercostal arteries. In the posterior mediastinum, initially left of the vertebral column. Related to esophagus, thoracic duct, and azygos vein. Aneurysms are mostly atherosclerotic, treatable with thoracic endovascular aortic repair.

Diaphragm

– Dome-shaped musculotendinous partition separating thoracic/abdominal cavities, primary inspiration muscle. Central tendon with muscular periphery from xiphoid, lower six costal cartilages, and lumbar vertebrae via crura and arcuate ligaments. Innervated by phrenic nerves (C3-C5). Contraction flattens the dome increasing thoracic volume. Three openings: caval (T8), esophageal (T10), aortic (T12). Paralysis causes hemiparesis or paradoxical breathing.

Duodenum

– Shortest first small intestinal segment (25 cm), C-shaped around the pancreatic head. Four parts. Descending part has the major duodenal papilla for bile/pancreatic duct entry via the hepatopancreatic ampulla. Brunner's glands produce alkaline mucus. Most common peptic ulcer site. Primary iron and calcium absorption site.

Dural Venous Sinuses

– Venous channels between dural layers, draining brain blood and CSF from the subarachnoid space into the internal jugular veins. Major sinuses: superior sagittal, transverse, sigmoid, cavernous, straight. Valveless with bidirectional flow possible. Thrombosis (cerebral venous sinus thrombosis) causes increased intracranial pressure, headache, seizures, and focal deficits.

Epidermis

– Outermost skin layer, stratified squamous keratinized epithelium approximately 0.1 mm thick, with five layers (superficial to deep): stratum corneum (dead keratinized cells), lucidum (clear layer, thick skin only), granulosum (keratinization begins), spinosum (spiny layer, Langerhans cells), basale (stem cells, melanocytes, Merkel cells). Keratinocytes (80 percent of cells) produce keratin for waterproofing and protection. Avascular, nourished by dermal diffusion. Melanocytes produce melanin for UV protection and skin pigmentation.

Epididymis

– Coiled duct approximately 6 meters long on the posterolateral aspect of each testis, divided into head (caput, receiving efferent tubules from the testis), body (corpus), and tail (cauda, continuing as the vas deferens). Functions include sperm maturation (gaining motility over approximately 12 days), storage (in the tail), and transport. The epididymis is a common site of infection (epididymitis), typically from sexually transmitted organisms in young men or urinary tract organisms in older men.

Epigastric Region

– Central superior region between the hypochondriac regions and above the umbilical region. Overlies the stomach, duodenum, pancreas, and portions of the liver. The aorta and inferior vena cava pass through it. Epigastric pain associates with gastric ulcers, pancreatitis, GERD, and aortic aneurysms. Critical for detecting pulsatile masses and organomegaly.

Epithelium

– One of the four primary tissue types, composed of cells that line the surfaces of the body (skin, mucous membranes, serous membranes) and form glands. Epithelial tissues are classified by cell shape (squamous–flat, cuboidal–cube-shaped, columnar–tall, transitional–variable) and layering (simple–single layer, stratified–multiple layers, pseudostratified–appears layered but all cells contact the basement membrane). Functions: protection (stratified squamous epithelium of skin), absorption (simple columnar of intestinal lining), secretion (glandular epithelium), filtration (simple squamous of capillary endothelium), and sensory reception (olfactory epithelium). Epithelium rests on a basement membrane, is avascular (receives nutrition by diffusion), and has high regenerative capacity. Epithelial-mesenchymal transition is a key process in development and cancer metastasis.

Ethmoid Bone

– Delicate bone between the orbits forming parts of the nasal septum and orbital/nasal roofs. Has the cribriform plate (olfactory nerve filaments), perpendicular plate (superior nasal septum), and labyrinths (nasal conchae, ethmoid air cells). The crista galli provides falx cerebri attachment. Cribriform fractures cause CSF rhinorrhea and anosmia.

Eustachian Tube

– Auditory/pharyngotympanic tube connecting the middle ear to the nasopharynx, approximately 36 mm long in adults. Cartilaginous medial two-thirds and bony lateral third. Functions: equalizing middle ear/atmospheric pressure, draining middle ear secretions, and protecting from nasopharyngeal pathogens. Opens during swallowing, yawning, sneezing via tensor/levator veli palatini muscles. Children have shorter, wider, more horizontal tubes predisposing to otitis media. Dysfunction causes middle ear effusion (glue ear), the most common cause of conductive hearing loss in children.

Extensor Digitorum

– Posterior forearm muscle from lateral epicondyle via the common extensor tendon, dividing into 4 tendons inserting on extensor expansions of digits 2-5. Innervated by posterior interosseous nerve (C7-C8). Extends MCP joints, assists PIP/DIP extension. Lateral epicondylitis (tennis elbow) is an overuse injury of its common origin. Extensor hood coordinates finger movements.

External Auditory Canal

– Tube from the concha to the tympanic membrane, approximately 2.5 cm long in adults. Lateral third is cartilaginous; medial two-thirds are bony (temporal bone). Skin-lined with ceruminous glands (producing cerumen/earwax) and hair follicles. Cerumen has antimicrobial and insect-repellent properties. Innervated by auriculotemporal nerve (V3) laterally and vagus nerve (Arnold's nerve) medially, explaining why canal touching may trigger coughing. Otitis externa (swimmer's ear) is common.

External Iliac Artery

– Larger common iliac terminal branch, continuing as the femoral artery below the inguinal ligament. Gives off inferior epigastric and deep circumflex iliac arteries. Related to external iliac vein medially and ureter anteriorly. Commonly used for cardiac catheterization and endovascular access. Atherosclerosis causes thigh/calf claudication.

External Jugular Vein

– Formed by posterior auricular and retromandibular vein divisions, crossing the sternocleidomastoid superficially to drain into the subclavian vein. Visible through the skin; used for peripheral IV access when other sites are unavailable. Distended in right heart failure and SVC obstruction. A useful physical examination landmark.

External Oblique Muscle

– Most superficial flat abdominal muscle from outer lower eight ribs to linea alba, pubic tubercle, and iliac crest. Innervated by T7-T12. Contralateral trunk rotation and flexion; compresses abdomen. Forms the inguinal ligament from its free inferior border. Superficial inguinal ring is an opening in its aponeurosis. Strains common in twisting athletes.

Extraocular Muscles

– Six muscles controlling eye movement: four rectus (superior, inferior, medial, lateral) and two oblique (superior, inferior). Medial rectus (CN III) adducts; lateral rectus (CN VI) abducts; superior rectus (CN III) elevates/intorts; inferior rectus (CN III) depresses/extorts; superior oblique (CN IV) depresses/intorts; inferior oblique (CN III) elevates/extorts. Hering's law: yoke muscles receive equal innervation. Sherrington's law: agonist contraction accompanies antagonist relaxation. Paralysis of individual muscles causes characteristic strabismus (squint) and diplopia.

Facet Joints

– Zygapophysial joints connecting adjacent vertebral articular processes. Orientation varies: cervical (45 degrees, allowing rotation), thoracic (coronal, allowing rotation but limiting flexion/extension), lumbar (sagittal, allowing flexion/extension but limiting rotation). Innervated by medial branches of dorsal rami. Common source of low back pain treated with facet injections and medial branch blocks.

Fallopian Tubes

– Paired muscular tubes (approximately 10-12 cm long) extending from the uterus to near the ovaries, serving as the site of fertilization. Four parts: infundibulum (with fimbriae sweeping the oocyte), ampulla (widest part, most common fertilization site), isthmus (narrow medial part), and intramural (passing through the uterine wall). The tubal mucosa has ciliated epithelium propelling the oocyte/embryo toward the uterus. Tubal pathology (infection, adhesions) can cause ectopic pregnancy or infertility.

False Ribs

– Ribs 8-10 not articulating directly with the sternum; their cartilages join rib 7's cartilage forming the costal margin. Greater mobility and flexibility of the lower thoracic wall. More susceptible to fractures at their angles due to greater length and less rigid attachment. The costal margin is an important clinical landmark for abdominal examination.

Femoral Artery

– Continues from external iliac through the femoral triangle and adductor canal to become the popliteal artery. Branches: superficial/deep circumflex femoral, deep femoral (profunda femoris), muscular. Palpable in the femoral triangle; most common site for arterial catheterization. Related to femoral vein medially and nerve laterally. Atherosclerosis causes peripheral arterial disease.

Femoral Nerve

– Largest lumbar plexus nerve (L2-L4), passing through psoas major and entering the thigh under the inguinal ligament lateral to the femoral artery. Innervates quadriceps, sartorius, pectineus; sensation to anterior thigh and medial leg (saphenous nerve). Used for knee surgery nerve block anesthesia. Neuropathy (retroperitoneal hemorrhage, hip surgery) causes quadriceps weakness and loss of knee jerk.

Femoral Triangle

– Upper anterior thigh space bounded superiorly by the inguinal ligament, laterally by sartorius, and medially by adductor longus. Contents from lateral to medial: femoral nerve, artery, vein, and deep inguinal lymph nodes. The femoral artery is used for arterial catheterization. The femoral vein is accessed for central venous lines. Femoral hernias occur through the femoral canal medial to the vein.

Femur

– Longest, heaviest, strongest body bone. Proximal: head, neck, trochanters. Shaft: linea aspera. Distal: condyles, epicondyles. The neck connects head to shaft at approximately 125 degrees. Neck fractures are common in elderly osteoporotic patients with avascular necrosis risk. Bears entire upper body weight during standing.

Fibula

– Slender lateral leg bone, not weight-bearing, serving for muscle attachment and forming the lateral malleolus. Head provides biceps femoris and lateral collateral ligament attachment; the common peroneal nerve wraps around its neck, vulnerable to injury. Lateral malleolus extends more distally and posteriorly than the medial malleolus. Often fractured with tibial fractures.

Flexor Digitorum Profundus

– Deep anterior forearm muscle with ulnar and humeroulnar heads, inserting on distal phalanges bases of digits 2-5. Medial half (digits 4-5) innervated by ulnar nerve; lateral half (digits 2-3) by anterior interosseous nerve (C8-T1). Flexes DIP, PIP, MCP joints, and wrist. Anterior interosseous palsy causes inability to pinch with the index finger.

Flexor Digitorum Superficialis

– Large anterior forearm muscle with humeroulnar, radial, and ulnar heads, inserting on middle phalanges of digits 2-5. Innervated by median nerve (C8-T1). Flexes PIP joints, MCP joints, and wrist. Important for median nerve function assessment. Tendons pass through carpal tunnel, susceptible to stenosing tenosynovitis (trigger finger).

Floating Ribs

– Ribs 11-12 with short costal cartilages terminating in abdominal wall musculature, not articulating with the sternum or other ribs. Most mobile and least protected, susceptible to fracture from direct trauma. Fractures may injure underlying kidneys, requiring hematuria and flank tenderness assessment in trauma.

Fovea Centralis

– Central macular pit (approximately 1.5 mm) with the highest cone density (approximately 200,000 per square millimeter) and no rods, providing the sharpest visual acuity and color vision. Avascular with overlying retinal layers displaced laterally for direct light access to cones. The foveola (central 0.35 mm) has the thinnest retina and highest cone density. Pathology here (macular holes, central serous chorioretinopathy, macular degeneration) severely impairs central vision.

Frontal Bone

– Forms the forehead and superior orbits, with the squama frontalis, orbital plates, and nasal part. Contains frontal sinuses draining into the nasal cavity via the frontonasal duct. Articulates with parietal bones at the coronal suture. The metopic suture fuses by age six. Fractures may cause pneumocephalus or CSF rhinorrhea through the sinuses.

Ganglion

– A collection of neuronal cell bodies located outside the central nervous system (CNS). Ganglia are classified as: (1) Sensory ganglia (dorsal root ganglia of spinal nerves, cranial nerve ganglia–trigeminal, geniculate, vestibular, spiral, petrosal, nodose), containing pseudounipolar neurons whose peripheral processes receive sensory input and central processes enter the CNS. (2) Autonomic ganglia (sympathetic chain ganglia, prevertebral ganglia–celiac, superior mesenteric, inferior mesenteric, and parasympathetic ganglia–ciliary, pterygopalatine, submandibular, otic, enteric–Auerbach and Meissner plexuses), containing multipolar neurons with efferent output to target organs. (3) Basal ganglia (CNS structures involved in motor control: caudate, putamen, globus pallidus, substantia nigra, subthalamic nucleus–nonpathological usage differs from peripheral ganglia). Clinically: dorsal root ganglia are sites of latency for herpes zoster; basal ganglia dysfunction causes Parkinson disease, Huntington disease, and dystonia. The term "ganglion" is also used for benign cystic swellings arising from tendon sheaths or joint capsules (ganglion cyst), which contain mucinous fluid and are not neural in origin.

Gastrocnemius Muscle

– Most superficial posterior leg muscle forming the calf prominence, with medial/lateral femoral condyle heads merging with soleus to form the Achilles tendon on the calcaneus. Innervated by tibial nerve (S1-S2). Plantarflexes ankle and flexes knee (two-joint muscle). Active in walking/running/jumping. Achilles tendon is the body's strongest but vulnerable to rupture during explosive activities.

Gland

– An organized group of epithelial cells specialized for secretion. Glands are classified by: (1) Location: exocrine (secrete onto a surface via ducts–sweat, salivary, mammary, sebaceous, lacrimal, pancreatic exocrine) or endocrine (secrete hormones directly into the bloodstream–thyroid, pituitary, adrenal, pancreatic islets, parathyroid, gonads). (2) Number of cells: unicellular (goblet cells) or multicellular. (3) Duct structure: simple (unbranched duct) or compound (branched duct). (4) Secretory unit shape: tubular, acinar (alveolar), or tubuloacinar. (5) Secretion mechanism: merocrine (exocytosis, e.g., salivary, pancreatic), apocrine (apical cell membrane pinches off, e.g., mammary, axillary sweat), holocrine (entire cell disintegrates, e.g., sebaceous). (6) Secretion type: serous (watery, enzyme-rich), mucous (viscous, mucin-rich), mixed (seromucous).

Glenohumeral Joint

– Ball-and-socket synovial joint between humeral head and glenoid cavity. Most mobile body joint allowing flexion, extension, abduction, adduction, medial/lateral rotation, circumduction. Mobility trades for stability, making it the most commonly dislocated joint. Stabilized by glenoid labrum, glenohemoral ligaments, and rotator cuff. Loose, redundant capsule.

Glenoid Cavity

– Shallow pear-shaped depression on the lateral scapular angle articulating with the humeral head. The glenoid labrum deepens the socket by approximately 50 percent. Due to its shallowness, the glenohumeral joint is the most commonly dislocated joint. Bankart lesions (inferior labral tears) are associated with anterior dislocations and recurrent instability.

Glisson's Capsule

– Thin fibrous connective tissue envelope surrounding the liver, sending septa inward dividing it into lobules. Richly innervated by phrenic and intercostal nerve pain fibers, sensitive to distension and inflammation. Covered by visceral peritoneum except at the bare area. Hepatic injuries may cause subcapsular hematomas that can rupture causing hemoperitoneum.

Globin

– A family of heme-binding proteins involved in oxygen transport and storage. Structure: globin fold (8 alpha-helices arranged in a globular structure), binds one heme molecule (iron-protoporphyrin IX) per subunit. Types: (1) hemoglobin (alpha2-beta2 tetramer in adults, HbA; alpha2-gamma2 in fetal, HbF): transports oxygen in red blood cells. (2) Myoglobin (single subunit): stores oxygen in skeletal and cardiac muscle, facilitates oxygen diffusion. (3) Neuroglobin: expressed in neurons, protects against hypoxia. (4) Cytoglobin: expressed in fibroblasts and other tissues, involved in nitric oxide metabolism and oxidative stress. Mutations in globin genes cause haemoglobinopathies: sickle cell disease (beta-globin Glu6Val), thalassaemias (reduced alpha or beta globin chain production).

Globulin

– A group of serum proteins (globular proteins) that are less soluble in water but soluble in dilute salt solutions. On serum protein electrophoresis, globulins migrate in the alpha-1, alpha-2, beta, and gamma regions. Alpha-1 globulins: alpha-1-antitrypsin, alpha-fetoprotein. Alpha-2 globulins: haptoglobin, alpha-2-macroglobulin, ceruloplasmin. Beta globulins: transferrin, beta-2-microglobulin, complement factors (C3, C4). Gamma globulins: immunoglobulins (antibodies). Abnormal globulin levels: polyclonal increase (chronic infection, autoimmune disease, cirrhosis), monoclonal gammopathy (multiple myeloma producing a monoclonal peak, MGUS, Waldenström macroglobulinaemia), decreased (immunodeficiency, nephrotic syndrome).

Glomerulus

– The tuft of capillaries in the renal corpuscle (Bowman's capsule) where blood is filtered to form urine. The glomerular filtration barrier consists of: (1) fenestrated endothelium (70-100 nm pores), (2) glomerular basement membrane (GBM, 300-350 nm, type IV collagen network), (3) podocyte foot processes (slit diaphragms formed by nephrin, NEPH1, podocin, and CD2AP). The barrier selectively retains cells and proteins >69 kDa while allowing water and small solutes to pass. Glomerular filtration rate (GFR): 125 mL/min (180 L/day). Mesangial cells support the capillary loops and regulate flow by contraction. Glomerular injury causes proteinuria and haematuria (glomerulonephritis).

Glucagon

– A peptide hormone produced by the alpha cells of the pancreatic islets (Langerhans). Primary function: counter-regulatory hormone to insulin, maintaining blood glucose by: (1) stimulating hepatic glycogenolysis (breakdown of glycogen to glucose), (2) promoting gluconeogenesis (synthesis of glucose from lactate, amino acids, glycerol), (3) inhibiting glycolysis and glycogen synthesis in the liver. Secreted in response to: hypoglycaemia, catecholamines, amino acids (arginine), and vagal stimulation. Secretion inhibited by: glucose, insulin, somatostatin. Glucagon also relaxes smooth muscle in the gastrointestinal tract (used therapeutically as IV glucagon for severe hypoglycaemia and for relaxation of the GI tract during endoscopy or imaging). Glucagonoma: a rare pancreatic neuroendocrine tumor producing glucagon, causing diabetes, weight loss, necrolytic migratory erythema (characteristic rash).

Glucose

– A monosaccharide ($C_6H_{12}O_6$) that is the primary energy source for cells. Types: D-glucose (dextrose, naturally occurring enantiomer) and L-glucose (synthetic). Routes of entry: dietary carbohydrates (digested to glucose), gluconeogenesis (synthesis from non-carbohydrate precursors), glycogenolysis (breakdown of stored glycogen). Cellular metabolism: glycolysis (cytosol, 2 ATP per glucose), TCA cycle (mitochondria, 36 ATP total), pentose phosphate pathway (generates NADPH and ribose-5-phosphate). Glucose transport into cells: facilitated diffusion via GLUT transporters (GLUT1-4, 12 isoforms; GLUT4 is insulin-sensitive in muscle and adipose tissue). Blood glucose regulation: insulin (lowers), glucagon, cortisol, catecholamines, growth hormone (raise). Clinical significance: hyperglycaemia (diabetes), hypoglycaemia, glucose tolerance test (OGTT) for diabetes diagnosis. Intravenous dextrose is used for treating hypoglycaemia and as a component of parenteral nutrition.

Gluteus Maximus Muscle

– Largest, most superficial gluteal muscle forming the buttock bulk, from posterior iliac gluteal line, sacrum, coccyx, and sacrotuberous ligament to the iliotibial tract and gluteal tuberosity. Innervated by inferior gluteal nerve (L5-S2). Primary hip extensor and lateral rotator. Important for stair climbing, rising from seated, and posture. Common intramuscular injection site.

Gluteus Medius Muscle

– Thick fan-shaped muscle between maximus and minimus, from external ilium to the greater trochanter. Innervated by superior gluteal nerve (L4-S1). Primary hip abductor essential for pelvic stability during single-leg stance. Superior gluteal nerve injury causes Trendelenburg sign with contralateral pelvic drop. Commonly involved in hip pain and trochanteric bursitis.

Gluteus Minimus Muscle

– Smallest, deepest gluteal muscle from external ilium to the greater trochanter. Innervated by superior gluteal nerve (L4-S1). Hip abductor and medial rotator stabilizing the hip during walking. Works synergistically with medius for pelvic stability. Tendinopathy commonly causes lateral hip pain mimicking trochanteric bursitis.

Glycogen

– A branched polysaccharide composed of glucose units linked by alpha-1,4 glycosidic bonds (linear) and alpha-1,6 bonds (branching points, every 8-10 residues). Glycogen serves as the main storage form of glucose in animals. Stored primarily in: liver (up to 100 g, 3-5% wet weight, released as glucose for systemic use) and skeletal muscle (approximately 400 g, 1-2% wet weight, used locally for exercise). Glycogenesis: synthesis of glycogen from glucose (insulin-stimulated). Glycogenolysis: breakdown of glycogen to glucose-1-phosphate (glucagon-stimulated in liver, epinephrine-stimulated in muscle). Glycogen storage diseases: von Gierke (type I, G6Pase deficiency, severe fasting hypoglycaemia), Pompe (type II, acid alpha-glucosidase deficiency, fatal cardiomyopathy), Cori (type III, debranching enzyme), McArdle (type V, muscle phosphorylase deficiency, exercise intolerance).

Greater Curvature

– Convex lateral stomach border from gastroesophageal junction to pylorus, approximately four times longer than the lesser curvature. Attachment site for the greater omentum. Blood supply from short gastric and left gastroepicolic vessels. Common site for gastric ulcers and cancer. The gastrocolic ligament connects it to the transverse colon.

Greater Trochanter

– Large quadrilateral lateral proximal femoral prominence. Insertion for gluteus medius/minimus and piriformis. Palpable landmark and site of trochanteric bursa. Intertrochanteric fractures between trochanters are extracapsular with lower avascular necrosis risk because periosteal blood supply is preserved.

Hair Follicle

– Complex dermal/hypodermal mini-organ producing hair. Has infundibulum (surface opening), isthmus (from sebaceous duct to arrector pili insertion), and lower segment (bulb with matrix and dermal papilla). Hair cycle: anagen (active growth, 2-7 years), catagen (regression, 2-3 weeks), telogen (resting, 2-3 months). Arrector pili contraction causes hair erection (goosebumps). Androgenetic alopecia results from DHT-induced follicle miniaturization. Alopecia areata is autoimmune follicle destruction.

Hamate

– Wedge-shaped distal row carpal bone on the ulnar side with the hook of the hamate. The hook forms Guyon's canal medial border and serves as flexor retinaculum attachment. Articulates with 4th/5th metacarpals. Hook fractures associated with racquet sports and ulnar nerve compression.

Hamstring Muscles

– Three posterior thigh muscles: biceps femoris (long/short heads), semitendinosus, semimembranosus, mostly from the ischial tuberosity (short head of biceps from linea aspera). Long head/semimembranosus/semitendinosus innervated by tibial sciatic nerve; short head by common peroneal division. Flex knee, extend hip. Essential for running/deceleration. Strains are among the most common sports injuries.

Heart

– Hollow muscular mediastinal organ pumping blood through the circulatory system. Approximately fist-sized (250-350 g), beating 100,000 times daily to pump 7,500 liters. Four chambers: right/left atria and ventricles. Right side handles deoxygenated blood to lungs; left side pumps oxygenated blood systemically. Enclosed in pericardium; wall has epicardium, myocardium, endocardium. Cardiac conduction system coordinates the heartbeat.

Hip Joint

– Multiaxial ball-and-socket synovial joint between femoral head and acetabulum. Designed for stability and mobility, bearing entire upper body weight. Acetabular labrum deepens the socket; strong ligaments (iliofemoral, pubofemoral, ischiofemoral) reinforce it. Ligamentum teres contains the femoral head artery. Allows flexion, extension, abduction, adduction, rotation, circumduction. Total hip arthroplasty is a common treatment for osteoarthritis.

Hippocampus

– Curved medial temporal lobe structure, part of the limbic system, critical for new declarative memory formation (episodic and spatial) and short-to-long-term memory conversion. Has dentate gyrus, hippocampus proper (CA1-CA4), subiculum, and entorhinal cortex. One of few adult neurogenesis sites (dentate gyrus). Bilateral damage (herpes simplex encephalitis, bilateral PCA strokes) causes anterograde amnesia. Atrophy is an early Alzheimer's finding.

Humeroradial Joint

– Between capitulum and radial head, allowing flexion-extension and contributing to pronation-supination via radial head rotation. Radial collateral ligament provides lateral stability. Capitulum has articular cartilage only anteriorly. Less stable than humeroulnar joint; more susceptible to dislocation, particularly nursemaid's elbow in children.

Humeroulnar Joint

– Uniaxial hinge between trochlea and trochlear notch of the ulna, allowing only flexion/extension (0 to approximately 145 degrees). Ulnar collateral ligament is the primary valgus restraint. Trochlear notch provides inherent bony stability, making it the most congruent elbow joint. Commonly affected by osteoarthritis and rheumatoid arthritis.

Humerus

– Single upper arm bone, the longest of the upper limb. Proximal end: head, anatomical/surgical necks, tubercles. Shaft: deltoid tuberosity, radial groove, epicondyles. Distal end: trochlea, capitulum, olecranon fossa. The surgical neck is the common fracture site potentially injuring the axillary nerve. Midshaft fractures may injure the radial nerve in the radial groove.

Hyoid Bone

– U-shaped bone in the anterior neck at C3, unique for not directly articulating with any other bone. Suspended by muscles and the stylohyoid ligament. Provides attachment for suprahyoid (digastric, stylohyoid, mylohyoid, geniohyoid) and infrahyoid (sternohyoid, sternothyroid, thyrohyoid, omohyoid) muscles. Critical for swallowing and speech. Fracture is a classic finding in manual strangulation.

Hypodermis

– Subcutaneous tissue/superficial fascia, the deepest skin layer with loose connective tissue and adipose tissue. Attaches skin to underlying muscles/bones via fibrous septa. Provides thermal insulation, energy storage (fat), mechanical cushioning, and contains blood vessels, lymphatics, and nerves. Subcutaneous injections are administered here. Fat distribution is hormonally influenced (estrogen promotes gynoid; testosterone promotes android). Lipoatrophy may occur from repeated insulin injections, causing subcutaneous fat loss and sometimes requiring rotation of injection sites to prevent dermal scarring.

Hypogastric Region

– Central inferior region below the umbilical region, containing the urinary bladder, sigmoid colon, rectum, and in females the uterus and ovaries. Assessed for bladder distension, uterine enlargement, and pelvic masses. Suprapubic pain may indicate cystitis, calculi, or pelvic pathology.

Hypothalamus

– Small region below the thalamus forming the third ventricle's floor and lower lateral walls, weighing approximately 4 grams, the master homeostasis regulator. Controls autonomic and endocrine systems (via the pituitary through the hypophyseal portal system), temperature, hunger, thirst, sleep-wake cycles, and emotion. Key nuclei: suprachiasmatic (circadian rhythm), arcuate (appetite), supraoptic/paraventricular (ADH/oxytocin), mammillary bodies (memory). Lesions cause diabetes insipidus, hypothalamic dysfunction (temperature dysregulation, appetite disturbance), and endocrine disorders such as hypopituitarism.

Ileum

– Longest distal segment (3.5 m) with thinner walls, shorter villi, and Peyer's patches on the antimesenteric border. Absorbs vitamin B12, bile salts, remaining nutrients. Ileocecal valve prevents colonic backflow. Crohn's disease most commonly affects the terminal ileum; resection may cause B12 deficiency and bile salt malabsorption. Meckel's diverticulum is a vitelline duct remnant here.

Ilium

– Largest hip bone forming the superior portion. Body contributes to acetabulum; wing has a concave iliac fossa. Iliac crest runs ASIS to PSIS, a palpable landmark for lumbar puncture level. Sacroiliac joint articulates with sacrum. Gluteal muscles originate from its external surface; iliacus fills its fossa.

Inferior Mesenteric Artery

– Third major abdominal aortic branch at L3, the hindgut artery. Supplies distal transverse colon through rectum. Branches: left colic, sigmoid, superior rectal. The marginal artery of Drummond provides SMA-IMA collateral circulation. Occlusion may cause ischemic colitis at the splenic flexure watershed (Griffiths' point).

Inferior Vena Cava

– Body's largest vein, formed by common iliac vein union at L5, returning lower body deoxygenated blood to the right atrium. Ascends right of the aorta, pierces the diaphragm at T8, enters the pericardium. Receives hepatic, renal, gonadal, and lumbar veins. IVC thrombosis/compression causes bilateral lower extremity edema and DVT.

Inguinal Region

– Area at the thigh-trunk junction traversed by the inguinal ligament (ASIS to pubic tubercle). Clinically significant for inguinal hernias: indirect (through deep inguinal ring, most common) and direct (through Hesselbach's triangle). Contains the spermatic cord in males and round ligament in females, along with the inferior epigastric vessels.

Intermediate Cuneiform

– Smallest cuneiform, centrally located between medial and lateral cuneiforms. Articulates with navicular, medial/lateral cuneiforms, and second metatarsal. Forms the transverse arch apex. Relatively fixed, contributing to midfoot stability. Injuries are typically associated with high-energy midfoot trauma.

Internal Iliac Artery

– Smaller common iliac terminal branch (hypogastric), supplying the pelvis, perineum, gluteal region, and medial thigh. Posterior division: iliolumbar, lateral sacral, superior gluteal. Anterior division: inferior gluteal, internal pudendal, obturator, vesical, uterine, vaginal, middle rectal arteries. Primary uterine blood supply, ligated during hysterectomy for hemorrhage. Embolization controls pelvic hemorrhage.

Internal Jugular Vein

– Descends in the carotid sheath with the internal carotid artery and vagus nerve, draining brain, face, and neck. Begins at the jugular foramen as a sigmoid sinus continuation, joining the subclavian vein to form the brachiocephalic vein. Contains backflow-preventing valves. Used for central venous catheterization. JVD assessment indicates right heart failure or elevated central venous pressure.

Internal Oblique Muscle

– Middle flat abdominal layer deep to external oblique, from thoracolumbar fascia, iliac crest, and lateral inguinal ligament to lower ribs, linea alba, and pubic crest. Innervated by T7-T12 and L1. Ipsilateral rotation (opposite of external oblique). Contributes to inguinal canal and abdominal wall integrity.

Iris

– Colored circular muscular diaphragm between cornea and lens, controlling pupil size and light entry. Two muscles: sphincter pupillae (parasympathetic CN III, constricts in bright light) and dilator pupillae (sympathetic, dilates in darkness). Iris root attaches to the ciliary body. The anterior chamber angle between iris and cornea is where aqueous humor drains. Narrow angles may cause acute angle-closure glaucoma with sudden IOP elevation.

Ischium

– Inferior-posterior hip bone. Ischial tuberosity bears seated weight and serves as hamstring attachment. Ischial spine is a pudendal nerve block landmark and determines pelvic outlet level. Lesser sciatic notch lies between spine and tuberosity.

Jejunum

– Middle segment (2.5 m) with prominent circular folds and tall villi for a velvety appearance. Richer blood supply and thicker walls than the ileum. Primary nutrient absorption site for carbohydrates, amino acids, fatty acids, vitamins, minerals. Fewer Peyer's patches. More commonly affected by Crohn's disease. Robust blood supply preferred for surgical anastomoses.

Jugular Vein

– One of the major veins draining blood from the head, face, and neck. There are two pairs: internal jugular vein (IJV) and external jugular vein (EJV). Internal jugular vein: the larger vessel, runs within the carotid sheath (medial to the carotid artery), descends from the jugular foramen (at the base of the skull) to the brachiocephalic vein. Receives blood from the brain (via dural venous sinuses, sigmoid sinus), face, and superficial neck. The IJV is the preferred site for central venous access (internal jugular central line) because of its straight course, predictable anatomy, and lower risk of pneumothorax compared to the subclavian approach. It is also used for measuring central venous pressure (CVP) and for venous sampling in endocrine testing (inferior petrosal sinus sampling for ACTH-dependent Cushing syndrome). External jugular vein: superficial vein formed by the retromandibular and posterior auricular veins, descends obliquely across the sternocleidomastoid muscle, pierces the deep fascia, and drains into the subclavian vein. Easily visible when it distends (Valsalva, heart failure, superior vena cava obstruction). The EJV is used for peripheral venous access when other sites are compromised. Jugular venous pressure (JVP): the height of the IJV pulsation in the neck reflects right atrial pressure, assessed with the patient at 45 degrees; the a-wave reflects atrial contraction (absent in atrial fibrillation, cannon a-wave in tricuspid atresia, AV dissociation), and the v-wave reflects atrial filling during ventricular systole (giant v-wave in tricuspid regurgitation).

Kidney

– A paired, bean-shaped retroperitoneal organ responsible for filtering blood, excreting waste products as urine, and regulating fluid and electrolyte balance, acid-base homeostasis, and blood pressure (via the renin-angiotensin-aldosterone system). Each kidney contains approximately 1 million nephrons, the functional filtering units. Structure: outer renal cortex, inner renal medulla (organized into renal pyramids), renal pelvis (collecting urine). The renal artery supplies blood; the renal vein drains it. Kidneys also produce erythropoietin (stimulates red blood cell production) and activate vitamin D. Common diseases: chronic kidney disease, nephrolithiasis, glomerulonephritis, pyelonephritis.

Knee Joint

– Largest, most complex joint with three compartments: medial/lateral tibiofemoral and patellofemoral. Tibiofemoral is a modified hinge allowing flexion-extension with rotation when flexed. Stabilized by cruciate (ACL/PCL) and collateral (MCL/LCL) ligaments; menisci cushion and distribute load. The suprapatellar bursa is the body's largest. Common injuries: ACL tears, meniscal tears, patellar dislocations.

Lacrimal Apparatus

– Produces, distributes, and drains tears for corneal moisture and protection. Components: lacrimal gland (superolateral orbit, parasympathetic CN V1, produces aqueous tear layer), superior/inferior puncta (medial eyelid openings), canaliculi, lacrimal sac (lacrimal groove), and nasolacrimal duct (drains into inferior meatus). Dry eye syndrome (keratoconjunctivitis sicca) from decreased production or excessive evaporation is common. Nasolacrimal duct obstruction causes epiphora and may cause dacryocystitis (lacrimal sac infection) presenting with epiphora and mucopurulent discharge.

Lacrimal Gland

– The paired almond-shaped exocrine glands that secrete the aqueous layer of the tear film, located in the lacrimal fossa of the superior lateral orbit (orbital part) and the upper eyelid (palpebral part). The lacrimal gland is a tubuloacinar serous gland, innervated by parasympathetic fibers from the facial nerve (CN VII) via the pterygopalatine ganglion and the zygomatic and lacrimal nerves. Its secretion is stimulated by corneal irritation, emotional responses, and trigeminal reflexes. Tears drain via the lacrimal puncta (upper and lower) into the lacrimal canaliculi, common canaliculus, lacrimal sac, and nasolacrimal duct, emptying into the inferior meatus of the nasal cavity. The tear film has three layers: lipid (meibomian glands, prevents evaporation), aqueous (lacrimal gland, contains lysozyme, lactoferrin, IgA, electrolytes), and mucin (conjunctival goblet cells, anchors tears to cornea). Diseases: dacryoadenitis (inflammation, usually viral or autoimmune–Sjogren syndrome, sarcoidosis), dacryocystitis (infection of the lacrimal sac, commonly obstructed nasolacrimal duct), and tumors (pleomorphic adenoma most common benign, adenoid cystic carcinoma most common malignant).

Large Intestine

– Final GI segment (approximately 1.5 meters) from ileocecal valve to rectum. Six segments: cecum, ascending, transverse, descending, sigmoid colon, rectum. Absorbs water, electrolytes, and vitamins (K, biotin) from colonic bacteria. Characterized by haustra, taeniae coli, and epiploic appendages. Houses approximately 100 trillion bacteria forming a diverse microbiome essential for metabolism and immunity.

Lateral Cuneiform

– Located between intermediate cuneiform and cuboid, articulating with navicular, intermediate cuneiform, cuboid, and third metatarsal. Contributes to the transverse arch and lateral foot column. Injuries typically from high-energy midfoot trauma.

Latissimus Dorsi Muscle

– Broadest back muscle forming the posterior axillary wall, from T7-L5 spinous processes, thoracolumbar fascia, iliac crest, and lower ribs to the intertubercular groove floor. Innervated by thoracodorsal nerve (C6-C8). Adducts, medially rotates, extends the arm powerfully in swimming/rowing/climbing. Common reconstructive surgery flap. Susceptible to overhead strain.

Left Atrium

– Receives oxygenated blood from 4 pulmonary veins. Most posterior chamber forming the cardiac base on chest radiography. Smooth walls from absorbed pulmonary veins; small auricle with pectinate muscles. Mitral valve separates it from the left ventricle. Enlargement is common in mitral disease and atrial fibrillation, potentially causing esophageal compression or recurrent laryngeal nerve hoarseness.

Left Hypochondriac Region

– Upper left abdomen lateral to the epigastric region, containing the spleen, stomach fundus, splenic flexure, and pancreatic tail. Important for evaluating splenomegaly, gastric pathology, and splenic infarction. The spleen is normally not palpable below the costal margin; enlargement into this region is a significant finding.

Left Iliac Region

– Lower left abdomen containing the descending colon, sigmoid colon, and left ureter. Important for evaluating diverticulosis and diverticulitis, most commonly affecting the sigmoid colon here. Palpation may reveal masses from inflammatory bowel disease or colorectal cancer.

Left Ventricle

– Thickest cardiac chamber (13-15 mm), pumping oxygenated blood into the aorta for systemic distribution. Generates approximately 120 mmHg systolic pressure. Receives blood through the mitral valve; ejects through the aortic valve. Left ventricular hypertrophy from chronic hypertension or aortic stenosis is a risk factor for heart failure, arrhythmias, and sudden death.

Lens

– Transparent biconvex avascular structure behind the iris/pupil, suspended by zonular fibers from the ciliary body. Provides approximately one-third of refractive power and allows accommodation (focusing on near objects by changing shape). Concentric lens fiber layers from surface epithelium, surrounded by a capsule. Absorbs ultraviolet light. Cataracts (opacification) are the most common cause of reversible blindness worldwide. Accommodation decreases with age (presbyopia).

Lesser Curvature

– Concave medial stomach border from gastroesophageal junction to pylorus. Attachment site for the lesser omentum connecting to the liver, containing right/left gastric arteries. Most common gastric ulcer site. The angular incisure marks the body-antrum junction. Cancer here may cause early obstruction from the narrow lumen.

Lesser Trochanter

– Small conical posteromedial proximal femoral prominence. Insertion for iliopsmus (primary hip flexor). Isolated fractures in adults may indicate pathologic fracture from metastatic disease; in adolescents may result from iliopsoas avulsion. Visible on lateral hip radiographs.

Ligament

– A dense, fibrous connective tissue band that connects bone to bone, providing passive joint stability and guiding joint motion. Ligaments are composed predominantly of type I collagen fibers (parallel orientation, crimped pattern) and elastin fibers (10-20% in some ligaments), arranged in bundles. The collagen fibers give ligaments high tensile strength (up to 30-40 N/mm2 for the ACL), while elastin allows elastic recoil. Ligaments have a limited blood supply (attachments are relatively avascular), leading to slow healing after injury. Mechanoreceptors (Ruffini, Pacinian, Golgi-type) within ligaments provide proprioceptive feedback about joint position and tension. The ligament response to injury: acute inflammatory phase (days), proliferative phase (weeks, fibroblast proliferation, collagen deposition, initially type III then type I), and remodelling phase (months-years, orientation of collagen along lines of stress). Classification of ligament injuries (grades): grade 1 (mild stretching, microscopic tears, no instability), grade 2 (partial tear, some laxity), grade 3 (complete rupture, gross instability). Healing is better in intra-articular ligaments (ACL has poor healing potential, often requires surgical reconstruction) vs extra-articular (MCL heals well non-operatively). Ligaments weaken with age and corticosteroid use, and sprains are common ankle sprain (anterior talofibular ligament most common), ACL tear (knee, pivot sport), MCL tear (knee, valgus stress), and ulnar collateral ligament (thumb, skier's thumb/gamekeeper's thumb).

Linea Aspera

– Prominent longitudinal posterior femoral shaft ridge from converging muscle attachment lines. Insertion for adductor magnus, vastus medialis/lateralis, and short head of biceps femoris. Thickest femoral cortex, resisting bending forces. Diverges distally into supracondylar lines. The profunda femoris artery pierces the adductor magnus near it.

Liver

– Largest internal organ and gland (approximately 1.5 kg) in the right upper quadrant protected by the rib cage. Over 500 functions: bile production, protein synthesis, detoxification, metabolism. Four lobes (right, left, caudate, quadrate). Enclosed by Glisson's capsule. Dual blood supply: hepatic artery (25 percent oxygenated) and portal vein (75 percent nutrient-rich). Cantlie line divides it functionally. Remarkable regenerative capacity.

Lumbar Vertebrae (L1-L5)

– Largest and strongest mobile vertebrae for weight-bearing. Large kidney-shaped bodies, short hatchet-shaped spinous processes, sagittally-oriented facets allowing primarily flexion/extension. Lordotic curvature. Most common disc herniation site, particularly L4-L5 and L5-S1 causing sciatica. Lumbar puncture performed between L3-L4 or L4-L5.

Lunate

– Crescent-shaped proximal row carpal bone between scaphoid and triquetrum. Most commonly dislocated carpal bone from falls on extended wrists. Dislocation may compress the median nerve causing acute carpal tunnel syndrome. Vulnerable to avascular necrosis (Kienbock disease) from its precarious blood supply.

Lung

– The paired, cone-shaped respiratory organs located in the thoracic cavity, responsible for gas exchange (oxygen uptake, carbon dioxide elimination). Each lung is divided into lobes: the right lung has three (superior, middle, inferior), the left lung has two (superior, inferior) with a cardiac notch accommodating the heart. Structure: trachea branches into bronchi, bronchioles, alveolar ducts, and alveoli (300–500 million, providing a surface area of approximately 70–100 m^2). The pleura (visceral and parietal) surrounds each lung. Ventilation is driven by the diaphragm and intercostal muscles. Common diseases: pneumonia, COPD, asthma, lung cancer, pulmonary embolism.

Lymph Node Structure

– Small bean-shaped lymphoid organs (1-25 mm) distributed along lymphatic vessels, filtering lymph and mounting immune responses. Each has a cortex (outer, containing B-cell follicles with germinal centers), paracortex (deep cortex, containing T-cells and dendritic cells), and medulla (inner, containing medullary cords and sinuses). Afferent vessels enter the cortex; efferent vessels exit at the hilum (also containing the artery, vein, and efferent lymphatic). Macrophages and dendritic cells within the node process and present antigens to lymphocytes, initiating adaptive immune responses and germinal center formation.

Macula Lutea

– Small oval yellowish central retinal area approximately 5.5 mm in diameter for high-acuity vision. Yellow color from xanthophyll pigments (lutein, zeaxanthin) filtering blue light and protecting photoreceptors. The fovea centralis is the central depression (1.5 mm) with highest cone density and no rods. Fovea is avascular, supplied by underlying choriocapillaris. Age-related macular degeneration is the leading cause of irreversible vision loss in developed countries.

Mandible

– Largest facial bone forming the lower jaw, with horseshoe-shaped body and two rami ending in condylar (TMJ) and coronoid processes. Mental foramen transmits the mental nerve. Mandibular foramen is the entry point for inferior alveolar nerve blocks. Most commonly fractured facial bone, typically at the condyle, angle, or symphysis.

Masseter Muscle

– Thick rectangular muscle of mastication on the mandibular ramus, from the zygomatic arch to mandibular angle/ramus. Innervated by the masseteric nerve (V3). Most powerful masticatory muscle, elevating the mandible with up to 70 kg force. Involved in bruxism, TM disorders, and myofascial pain. Hypertrophy can widen facial appearance.

Maxilla

– Paired bone forming the upper jaw, central face, and parts of orbit, nasal cavity, and hard palate. Body contains the maxillary sinus (largest paranasal sinus). Four processes: frontal, zygomatic, alveolar (upper teeth), and palatine (anterior hard palate). Fractures classified using the Le Fort system. The infraorbital nerve emerges from its foramen.

Medial Cuneiform

– Largest cuneiform, on the medial foot between navicular and first metatarsal. Insertion for tibialis anterior and posterior tendons supporting the medial longitudinal arch. The first tarsometatarsal joint (Lisfranc joint) is an important landmark; Lisfranc injuries involve fracture-dislocation at this level.

Median Nerve

– From lateral and medial cords (C5-T1), passing through the arm without branching, through the cubital fossa, into the forearm innervating most flexor muscles. In the hand innervates thenar muscles (opponens, abductor brevis, flexor brevis pollicis) and first two lumbricals. Sensation to lateral 3.5 digits. Carpal tunnel syndrome is the most common entrapment, causing thenar wasting and lateral palm/finger sensory loss.

Medulla Oblongata

– Most inferior brainstem segment, continuous with the spinal cord at the foramen magnum. Contains vital autonomic centers: cardiovascular center (heart rate, blood pressure), respiratory center (dorsal/ventral respiratory groups), and area postrema (vomiting center). Houses the pyramidal decussation (corticospinal tracts crossing, explaining contralateral motor control), olives (inferior olivary nuclei for cerebellar motor learning), and cranial nerve IX-XII nuclei. Lesions (Chiari malformation, syringobulbia, lateral medullary syndrome) cause ipsilateral cranial nerve deficits and contralateral long tract signs, often with vertigo, dysphagia, and respiratory compromise.

Melanocytes

– Dendritic pigment-producing neural crest-derived cells in the epidermal basal layer and hair follicles, comprising approximately 5-10 percent of basal keratinocytes. Produce melanin (eumelanin, brown-black; pheomelanin, yellow-red) in melanosomes, transferred to keratinocytes via dendritic processes for UV protection. Each melanocyte supplies approximately 30-40 keratinocytes (epidermal melanin unit). UV exposure stimulates melanin production (tanning). Cell of origin for melanoma, a potentially fatal skin cancer that metastasises early and requires prompt surgical excision with clear margins.

Metacarpals

– Five long bones (I-V) forming the palm framework. Each has base, shaft, and head. The first is shortest and most mobile, articulating with the trapezium at a saddle joint for opposition. Second/third are firmly fixed; fourth/fifth more mobile. Bennett's fracture affects the first metacarpal base; boxer's fracture affects the fifth neck.

Metatarsals

– Five long bones (I-V) forming the forefoot. Each has base, shaft, and head. First is shortest/thickest, bearing approximately one-third of forefoot load. Stress fractures commonly affect the second and third (march fractures). Morton's neuroma typically affects the nerve between 3rd/4th metatarsal heads. MTP joints are condyloid allowing flexion, extension, limited abduction/adduction.

Midbrain

– Most superior brainstem segment connecting pons to diencephalon. Contains cerebral peduncles (corticospinal/corticobulbar tracts), substantia nigra (dopaminergic neurons degenerated in Parkinson's), red nucleus (rubrospinal tract for coordination), and periaqueductal gray (pain modulation). Superior/inferior colliculi on the tectum process visual/auditory reflexes. The cerebral aqueduct connects third/fourth ventricles. Syndromes (Weber's, Benedikt's, Claude's, Nothnagel's) result from vascular lesions affecting specific midbrain regions, producing characteristic combinations of ipsilateral cranial nerve palsy and contralateral hemiplegia or ataxia.

Mitral Valve

– Left atrioventricular (bicuspid) valve with anterior and posterior cusps supported by chordae tendineae and papillary muscles. Anterior cusp is larger and proximate to the aortic valve. Mitral valve prolapse is the most common valvular abnormality. Mitral stenosis (most commonly rheumatic) and regurgitation are significant. Most commonly affected valve in rheumatic heart disease.

Muscle

– A specialized contractile tissue responsible for movement, posture, and heat production. Three types: skeletal muscle (voluntary, striated, attached to bones via tendons, responsible for locomotion), cardiac muscle (involuntary, striated, found only in the heart, with intercalated discs for synchronized contraction), and smooth muscle (involuntary, nonstriated, found in walls of blood vessels, gastrointestinal tract, bladder, uterus). Skeletal muscle structure: epimysium (outer sheath), perimysium (surrounds fascicles), endomysium (surrounds individual fibers). Contraction mechanism: sliding filament theory (actin-myosin cross-bridge cycling, ATP-dependent, calcium-regulated via troponin-tropomyosin).

Musculocutaneous Nerve

– From lateral cord (C5-C7), piercing coracobrachialis, running between biceps and brachialis to supply these muscles and lateral forearm skin (lateral antebrachial cutaneous nerve). Sole motor supply to the anterior arm compartment. Injury causes weakness of elbow flexion/forearm supination with lateral forearm sensation loss. Vulnerable in axillary injuries and proximal humeral fractures.

Navicular

– Boat-shaped tarsal bone between talus and three cuneiforms. Navicular tuberosity is a palpable medial prominence serving as posterior tibial tendon insertion, key medial longitudinal arch stabilizer. Central third has a precarious blood supply, susceptible to stress fractures and avascular necrosis (Kohler disease in children). Most commonly fractured midfoot tarsal bone.

Nerve

– A bundle of axons (nerve fibers) surrounded by connective tissue sheaths that transmits electrical impulses between the central nervous system and the periphery. Structure: epineurium (outer layer), perineurium (surrounds fascicles), endoneurium (surrounds individual axons). Types: sensory (afferent, carry impulses toward CNS), motor (efferent, carry impulses away from CNS to muscles/glands), and mixed nerves. Myelinated axons conduct impulses faster (saltatory conduction) than unmyelinated axons. Cranial nerves (12 pairs) arise from the brain; spinal nerves (31 pairs) arise from the spinal cord. Nerve regeneration is limited in the CNS but possible in the PNS.

Obturator

– A structure that occludes an opening. In anatomy, the obturator foramen is a large opening in the pelvic bone, closed by the obturator membrane. The obturator nerve (L2–L4) innervates the adductor muscles of the thigh and provides sensory innervation to the medial thigh. The obturator artery supplies the adductor compartment. The term also applies to surgical obturators—prosthetic devices used to close palatal defects (cleft palate, post-surgical).

Occipital Bone

– Forms the posterior-inferior cranium surrounding the foramen magnum (spinal cord passage). Has basilar, lateral, and squamous parts. Occipital condyles articulate with C1 for nodding movement. The external occipital protuberance provides muscle attachment. Fractures through the foramen magnum may cause brainstem injury and are life-threatening.

Occipital Lobe

– The most posterior of the four cerebral lobes, located behind the parieto-occipital sulcus and above the tentorium cerebelli. The occipital lobe is the primary visual processing center of the brain. The primary visual cortex (V1, Brodmann area 17) receives input from the lateral geniculate nucleus via the optic radiations; damage causes contralateral homonymous hemianopia or quadrantanopia. Higher-order visual areas (V2–V5) process colour, motion, and object recognition. The lingual gyrus and cuneus are key occipital landmarks. Lesions beyond V1 can cause visual agnosia, alexia without agraphia, and visual hallucinations (peduncular hallucinosis).

Odontoid Process

– Also called the dens, a tooth-like projection superiorly from the body of the axis (C2 vertebra). The odontoid process projects into the atlas (C1) and is held in place by the transverse ligament of the atlas, allowing rotation of the head (the atlantoaxial joint accounts for 50% of cervical rotation). Fractures of the odontoid process (Anderson–D'Alonzo classification types I–III) can cause atlantoaxial instability and spinal cord compression. Type II fractures (at the base of the dens) have the highest non-union rate. Os odontoideum is a congenital variant where the dens is not fused to the axis body.

Olfactory Nerve

– Cranial nerve I (CN I), responsible for the sense of smell. Olfactory receptor neurons in the olfactory epithelium of the nasal roof project axons through the cribriform plate of the ethmoid bone to the olfactory bulb, then via the olfactory tract to the primary olfactory cortex (piriform cortex, amygdala, entorhinal cortex). Loss of smell (anosmia) can result from head trauma (shearing of fibers at the cribriform plate), viral infections (COVID-19, common cold), nasal polyps, neurodegenerative diseases (Parkinson, Alzheimer, Lewy body dementia), and meningiomas of the olfactory groove. Testing uses the University of Pennsylvania Smell Identification Test (UPSIT) or sniffin' sticks.

Omentum

– A double layer of peritoneum connecting the stomach to other abdominal organs. The greater omentum (gastrocolic omentum) hangs from the greater curvature of the stomach, covers the transverse colon and small intestine, and is rich in fat and immune cells (milky spots—aggregates of macrophages and lymphocytes). It is mobile and can migrate to sites of inflammation to wall off infection (the 'policeman of the abdomen'). The lesser omentum (gastrohepatic omentum) connects the lesser curvature of the stomach and duodenum to the liver, containing the hepatic artery, portal vein, and common bile duct (within the hepatoduodenal ligament). Omentectomy is performed in gastric and ovarian cancer surgery.

Optic Nerve

– CN II exiting through the optic disc (blind spot), passing through the optic canal, and partially decussating at the optic chiasm. Technically a CNS tract with oligodendrocyte myelination and meningeal coverings, making it susceptible to demyelinating disease (optic neuritis in MS). Papilledema (disc swelling from raised ICP) and optic atrophy are important findings. Carries approximately 1.2 million ganglion cell axons.

Ossicles

– Three smallest body bones forming the ossicular chain in the middle ear: malleus (hammer, attached to tympanic membrane), incus (anvil), stapes (stirrup, footplate in oval window). Amplify and transmit sound vibrations from tympanic membrane to inner ear, providing approximately 22-fold amplification (impedance matching between air and fluid). Tensor tympani (CN V3) attaches to the malleus; stapedius (CN VII) to the stapes, forming the acoustic reflex dampening loud sounds. Ossicular erosion or disruption (from chronic otitis media, cholesteatoma, or trauma) causes conductive hearing loss, often correctable with tympanoplasty or ossicular reconstruction.

Osteoblast

– A bone-forming cell derived from mesenchymal stem cells in the bone marrow. Osteoblasts synthesise and deposit the bone matrix (osteoid), composed primarily of type I collagen and non-collagenous proteins (osteocalcin, osteonectin, alkaline phosphatase). They then regulate matrix mineralisation by releasing matrix vesicles containing calcium and phosphate. Osteoblasts also produce RANKL (receptor activator of nuclear factor kappa-B ligand), which stimulates osteoclast differentiation, and osteoprotegerin (OPG), which inhibits RANKL. Once encased in mineralised matrix, they become osteocytes. Impaired osteoblast function contributes to osteoporosis (uncoupling of bone formation and resorption).

Osteoclast

– A large, multinucleated bone-resorbing cell derived from the monocyte–macrophage lineage (hematopoietic stem cells). Osteoclasts attach to bone surfaces via integrins, form a sealed acidic compartment (ruffled border), and secrete hydrogen ions (dissolve hydroxyapatite) and proteolytic enzymes (cathepsin K, matrix metalloproteinases) to degrade the bone matrix. They are regulated by RANK–RANKL signalling (stimulates differentiation/activity) and calcitonin (inhibits). Overactive osteoclasts cause bone loss in osteoporosis, Paget disease of bone, rheumatoid arthritis, and multiple myeloma. Bisphosphonates, denosumab (RANKL inhibitor), and calcitonin are therapeutic inhibitors of osteoclast activity.

Ovary

– The female gonad, a paired almond-shaped organ located in the pelvic cavity on each side of the uterus. Functions: production of oocytes (ova) and secretion of female sex hormones (oestrogen, progesterone, and small amounts of androgens). Each ovary contains follicles at various stages of development. Monthly ovulation releases a mature oocyte. After ovulation, the follicle becomes the corpus luteum, which secretes progesterone. Ovarian structure: outer cortex (containing follicles) and inner medulla (containing blood vessels and connective tissue). Common conditions: polycystic ovary syndrome, ovarian cysts, ovarian tumors, premature ovarian insufficiency.

Pancreas

– Retroperitoneal organ from the duodenum to the spleen with exocrine (digestive enzymes from acinar cells) and endocrine (hormones from islets of Langerhans) functions. Head (in duodenal C-loop, with uncinate process), body, and tail. Main pancreatic duct joins the common bile duct at the hepatopancreatic ampulla (of Vater) opening at the major duodenal papilla. Pancreatic adenocarcinoma commonly affects the head, causing painless jaundice.

Parietal Bones

– Paired curved bones forming the cranial vault roof and sides. Articulate at the sagittal suture, with frontal at coronal, temporal at squamous, and occipital at lambdoid sutures. The pterion (where four bones meet) overlies the middle meningeal artery; fractures here may cause epidural hematomas.

Parotid Gland

– The largest of the three major salivary glands, located in the preauricular region, extending from the zygomatic arch superiorly to the angle of the mandible inferiorly. The parotid gland is encased in the parotid fascia (investing layer of deep cervical fascia). It produces serous saliva (thin, watery, rich in amylase), secreted via Stensen duct (parotid duct), which pierces the buccinator muscle and opens opposite the second upper molar tooth. Structures passing through the parotid gland (from superficial to deep): facial nerve (CN VII—its five branches: temporal, zygomatic, buccal, marginal mandibular, and cervical), retromandibular vein, and external carotid artery (dividing into maxillary and superficial temporal arteries). Clinical: parotitis (mumps—most common cause), Sjögren syndrome, sialolithiasis (parotid duct stones), and parotid tumors (80% benign—pleomorphic adenoma most common; 20% malignant—mucoepidermoid carcinoma, adenoid cystic carcinoma). Parotidectomy is the surgical treatment; the facial nerve must be identified and preserved.

Patella

– Largest sesamoid bone, embedded in the quadriceps tendon. Articulates with the trochlear groove forming the patellofemoral joint. Increases quadriceps mechanical advantage by approximately 30 percent. Triangular with inferior apex. Covered with the thickest articular cartilage (up to 7 mm). Common conditions: fractures, dislocations, chondromalacia.

Pectoralis Major Muscle

– Thick fan-shaped chest muscle with clavicular and sternocostal heads, inserting on the lateral intertubercular groove lip. Innervated by medial/lateral pectoral nerves (C5-T1). Clavicular head flexes; sternocostal head extends from flexion. Adducts and medially rotates powerfully. Rupture uncommon, may occur during weightlifting at the musculotendinous junction.

Pectoralis Minor Muscle

– Thin triangular muscle deep to pectoralis major, from ribs 3-5 to the coracoid process. Innervated by medial pectoral nerve (C8-T1). Draws scapula anteriorly/inferiorly, stabilizing it against the thorax. Assists forced inspiration. Tightness contributes to rounded shoulder posture and thoracic outlet syndrome.

Pelvis

– Bony ring from 2 hip bones, sacrum, and coccyx. Each hip bone fuses from ilium, ischium, and pubis meeting at the acetabulum. Divided into false (greater, above brim) and true (lesser, below brim) pelvis. Males: narrower, deeper, heart-shaped inlet. Females: wider, shallower, oval inlet for childbirth.

Peripheral Nervous System

– The portion of the nervous system consisting of nerves and ganglia outside the brain and spinal cord. The PNS connects the central nervous system (CNS) to peripheral tissues and organs. Components: (1) Cranial nerves (12 pairs)—emerging from the brain/brainstem, innervating head, neck, and viscera. (2) Spinal nerves (31 pairs)—emerging from the spinal cord via dorsal and ventral roots, each dividing into dorsal and ventral rami. (3) Autonomic nervous system—sympathetic (thoracolumbar, 'fight or flight') and parasympathetic (craniosacral, 'rest and digest'), regulating involuntary functions. Peripheral nerves are composed of axons bundled into fascicles by connective tissue layers: endoneurium (surrounds individual axons), perineurium (surrounds fascicles—blood–nerve barrier), and epineurium (outer layer). Myelination (by Schwann cells in the PNS) enables saltatory conduction. Peripheral nerve injury is classified by Seddon (neuropraxia—conduction block, axonotmesis—axon disrupted but sheath intact, neurotmesis—complete transection) or Sunderland (grades I–V). Peripheral neuropathy (polyneuropathy, mononeuritis multiplex) has many causes.

Phalanges

– 14 finger bones. Thumb: 2 (proximal, distal). Each other finger: 3 (proximal, middle, distal). MCP joints are condyloid allowing flexion, extension, abduction, adduction, circumduction. IP joints are hinge joints allowing only flexion/extension. Mallet finger and jersey finger are common tendon avulsion injuries.

Pharynx

– A muscular tube extending from the base of the skull (clivus) to the level of the cricoid cartilage (C6), where it continues as the esophagus. The pharynx is divided into three parts: nasopharynx (posterior to the nasal cavity, contains the pharyngeal tonsil/adenoids, Eustachian tube openings), oropharynx (posterior to the oral cavity, contains the palatine tonsils, lingual tonsil, base of tongue, soft palate), and laryngopharynx/hypopharynx (posterior to the larynx, extends from the hyoid bone to the cricoid cartilage, leads to esophagus and larynx). The pharynx functions in swallowing (deglutition—voluntary oral phase followed by involuntary pharyngeal phase, coordinated by the swallowing center in the medulla), breathing (conducts air from nasal/oral cavities to the larynx), and speech (resonance). The pharyngeal constrictor muscles (superior, middle, inferior) contract sequentially in peristalsis to propel the bolus. The pharynx is innervated by the pharyngeal plexus (CN IX and X).

Phrenic Nerve

– Primarily from C4 (with C3 and C5: "C3, 4, 5 keeps the diaphragm alive"), traveling between anterior and middle scalene muscles, descending anterior to the anterior scalene across the first rib. Right phrenic passes anterior to the lung root; left passes anterior and lateral. Innervates the ipsilateral diaphragm and provides sensory innervation to central diaphragmatic pleura, pericardium, and peritoneum. Injury causes ipsilateral hemidiaphragm paralysis reducing vital capacity. Phrenic nerve palsy causes ipsilateral hemidiaphragmatic paralysis, detected as an elevated hemidiaphragm on chest radiograph with paradoxical movement on fluoroscopic sniff test.

Pinna

– Visible external ear portion, elastic cartilage covered by skin, shaped to funnel sound into the external auditory canal. Key features: helix (outer rim), antihelix (inner ridge), tragus (canal prominence), antitragus, concha (canal-leading depression), lobule (earlobe, only non-cartilaginous part). Blood supply: superficial temporal and posterior auricular arteries. Innervated by great auricular nerve (C2-C3) and auriculotemporal nerve (V3). Perichondritis can cause permanent deformity.

Pisiform

– Small pea-shaped sesamoid bone in the flexor carpi ulnaris tendon, in the proximal row. Articulates only with the triquetrum. Forms the medial border of Guyon's canal (ulnar nerve/artery passage). Only sesamoid carpal bone, subcutaneous and easily palpable on the ulnar palm.

Pons

– Middle brainstem segment with prominent anterior bulge containing pontine nuclei (relaying cortical input to cerebellum), corticospinal fibers, and cranial nerve V, VI, VII, and VIII nuclei. Fourth ventricle floor is partly formed by the pons. Pneumotaxic center (parabrachial nucleus) regulates breathing pattern. Bilateral ventral pontine infarction causes locked-in syndrome with preserved consciousness and vertical eye movements but complete paralysis.

Popliteal Artery

– Continues from femoral artery at the adductor hiatus, dividing into anterior/posterior tibial arteries at the lower popliteus border. Deepest popliteal fossa structure, related anteriorly to vein and tibial nerve. Palpable at the fossa's inferior margin. Popliteal aneurysms are the most common peripheral arterial aneurysms, frequently bilateral and associated with AAA. Vulnerable to injury in knee dislocations.

Popliteal Fossa

– Diamond-shaped space on the posterior knee, bounded by hamstrings superiorly and gastrocnemius heads inferiorly. Contains popliteal artery/vein, tibial nerve, and common peroneal nerve. The popliteal pulse is palpated at the inferior margin. Significant for evaluating aneurysms, Baker's cysts, and vascular status in peripheral artery disease.

Portal System

– Venous network draining the GI tract, spleen, and pancreas to the liver via the portal vein (formed by superior mesenteric and splenic vein union behind the pancreatic neck). Carries nutrient-rich deoxygenated blood for hepatic processing before systemic entry. Receives left gastric, right gastric, cystic, and paraumbilical veins. Portal hypertension (cirrhosis) causes esophageal varices, caput medusae, and ascites.

Posterior Tibial Artery

– Arises from the popliteal artery, courses deep in the posterior leg, palpable behind the medial malleolus. Divides into medial/lateral plantar arteries for the sole. Supplies posterior leg muscles. Important for peripheral vascular disease assessment. The pedal pulse behind the medial malleolus is a key clinical examination component.

Prostate

– Walnut-sized gland (approximately 20 g) surrounding the urethra just below the bladder, producing approximately 30 percent of seminal fluid. It has peripheral, central, and transition zones. The peripheral zone is the most common site of prostate cancer; the transition zone enlarges in benign prostatic hyperplasia (BPH), compressing the urethra and causing urinary obstruction. The prostate secretes prostate-specific antigen (PSA), citric acid, and zinc. The ejaculatory ducts pass through the prostate and open at the verumontanum in the prostatic urethra.

Proximal Radioulnar Joint

– Pivot synovial joint between radial head circumference and ulnar radial notch, held by the annular ligament. Allows pronation/supination with approximately 150 degrees range. Parallel bones in supination; radius crosses ulna in pronation. Significant in radial head fractures, Monteggia fracture-dislocations, and nursemaid's elbow.

Pubis

– Anterior-inferior hip bone with body, superior ramus, and inferior ramus. Two pubic bones meet at the pubic symphysis (cartilaginous joint widening in childbirth). Pubic tubercle is the lateral inguinal ligament attachment. Direct hernias emerge medial to inferior epigastric vessels near it.

Pulmonary Valve

– Semilunar valve between right ventricle and pulmonary trunk with three cusps. Most superiorly located cardiac valve with thinner cusps than the aortic valve. Pulmonic stenosis is a congenital defect causing right ventricular hypertrophy. Less commonly affected by endocarditis than mitral/aortic valves. A systolic ejection murmur at the left upper sternal border suggests pulmonic stenosis.

Pupil

– Central round iris aperture for light entry, size controlled by sphincter pupillae (CN III parasympathetic, constricts to 2-3 mm) and dilator pupillae (sympathetic, dilates to 8 mm). The pupillary light reflex tests CN II (afferent) and CN III (efferent): light in one eye causes bilateral constrination (direct and consensual). Argyll Robertson pupils (small, irregular, Accommodate but do not React) suggest neurosyphilis. Fixed dilated pupils may indicate CN III compression from uncal herniation.

Quadriceps Femoris Muscle

– Four-muscle anterior thigh group: rectus femoris, vastus lateralis, vastus medialis, vastus intermedius, all inserting on the tibial tuberosity via the patellar ligament. Innervated by femoral nerve (L2-L4). Primary knee extensor for walking, running, rising from seated. VMO is important for patellar tracking. Weakness common after knee injury, surgery, and aging.

Radial Nerve

– From posterior cord (C5-T1), the plexus's largest branch. In the arm runs in the radial groove with the profunda brachii artery, innervating triceps and anconeus. At the elbow divides into superficial (sensory to dorsal hand) and deep (posterior interosseous) branches. Deep branch innervates forearm/hand extensors. Midshaft humeral fracture palsy causes wrist drop with preserved grip strength.

Radius

– Lateral forearm bone in the anatomical position. Proximal end: head, neck, radial tuberosity. Shaft. Distal end: styloid process, articular surfaces. The radial head articulates with the capitulum and ulnar notch, allowing pronation/supination. Bears approximately 80 percent of wrist axial load. Colles' fracture (distal radius with dorsal displacement) is among the most common fractures.

Rectus Abdominis Muscle

– Long flat vertical anterior abdominal wall muscle from pubic symphysis to xiphoid process/costal cartilages 5-7, divided by tendinous intersections ("six-pack"). Innervated by T7-T12. Flexes trunk, compresses abdomen, increases intra-abdominal pressure. Diastasis recti common in pregnancy/obesity. Common TRAM flap donor site for breast reconstruction.

Reflex Arc

– The neural pathway that mediates a reflex action, bypassing conscious brain processing for rapid, automatic responses. A reflex arc consists of five components: (1) Receptor—sensory nerve ending detecting a stimulus (e.g., pain, stretch). (2) Sensory (afferent) neuron—carries the signal from the receptor to the spinal cord (dorsal root ganglion). (3) Integration center—a single synapse (monosynaptic reflex, e.g., patellar reflex) or multiple interneurons (polysynaptic reflex, e.g., withdrawal reflex) within the spinal cord gray matter. (4) Motor (efferent) neuron—carries the signal from the spinal cord to the effector organ. (5) Effector—muscle or gland that executes the response. Examples: monosynaptic—stretch reflex (patellar reflex, testing L3–L4 nerve root). Polysynaptic—flexor withdrawal reflex (pain causes limb withdrawal), crossed extensor reflex (opposite limb extends to maintain posture), and superficial reflexes (abdominal, cremasteric, plantar—Babinski sign). The reflex arc is the functional unit of the nervous system and is assessed in the neurological examination to localise lesions.

Renal Cortex

– Outer kidney portion between capsule and medulla. Granular tissue with glomeruli, convoluted tubules, and cortical collecting ducts. Extends between pyramids as renal columns. Receives approximately 90 percent of renal blood flow; primary filtration site. Cortical nephrons (85 percent) have cortical glomeruli; juxtamedullary (15 percent) are near the corticomedullary junction. Essential for GFR regulation and erythropoietin production.

Renal Medulla

– Inner kidney with 8-18 renal pyramids containing loops of Henle, vasa recta, and medullary collecting ducts in countercurrent arrangement for urine concentration. Pyramid bases face cortex; apices (papillae) project into minor calyces. Lower blood flow (approximately 10 percent) and higher osmolality (up to 1200 mOsm/kg at papillary tip). Corticomedullary gradient is critical for ADH-mediated water reabsorption. More susceptible to ischemic injury than cortex.

Retina

– Innermost eye layer, a thin neural tissue layer lining the posterior two-thirds of the globe for phototransduction (light to neural signals). Ten layers, with photoreceptors (rods for low-light vision, cones for color vision) in the outer nuclear layer. Macula lutea is the central high-acuity area; fovea centralis has the highest cone density and sharpest vision. Supplied by the central retinal artery; drained by the central retinal vein. Conditions: retinal detachment, diabetic retinopathy, age-related macular degeneration (ARMD), retinal vein occlusion, and retinitis pigmentosa.

Retroperitoneum

– Space posterior to the peritoneum. Primary retroperitoneal (never intraperitoneal): kidneys, adrenals, ureters, aorta, IVC. Secondary (originally intraperitoneal): duodenum 2nd-4th parts, pancreas head/body, ascending/descending colon. Divided into anterior pararenal, perirenal, and posterior pararenal spaces. Contains retroperitoneal fibrosis, hematomas, and tumors.

Rib

– One of 12 paired, curved bones forming the thoracic cage, protecting the heart, lungs, and great vessels. True ribs (1–7) attach directly to the sternum via costal cartilage. False ribs (8–10) attach indirectly to the sternum via the costal cartilage of rib 7. Floating ribs (11–12) have no anterior attachment. Each rib has a head, neck, tubercle, shaft, and costal groove (housing the intercostal neurovascular bundle). Intercostal muscles occupy the spaces between ribs and are essential for respiration. Rib fractures are common and can cause pneumothorax or haemothorax.

Right Atrium

– Receives deoxygenated blood from superior vena cava, inferior vena cava, and coronary sinus. Smooth posterior part (sinus venosus) and rough anterior auricle with pectinate muscles. The fossa ovalis is the fetal foramen ovale remnant. The SA node pacemaker is near the superior vena cava opening. Contracts to fill the right ventricle through the tricuspid valve.

Right Hypochondriac Region

– One of nine abdominal regions in the upper right abdomen, bounded medially by the right midclavicular line and superiorly by the right costal margin. Contains the cecum, appendix, ascending colon, and inferior right liver lobe. Pain here may indicate appendicitis (McBurney's point), cecal pathology, or hepatic disease. Important for assessing hepatomegaly and gallbladder referred pain.

Right Iliac Region

– Lower right abdomen containing the cecum, appendix, and terminal ileum. McBurney's point (junction of lateral and middle thirds from umbilicus to ASIS) is critical for appendicitis diagnosis. Also significant for evaluating inguinal and femoral hernias, and the right ureter's course.

Right Lymphatic Duct

– Short lymphatic vessel (approximately 1.5 cm) draining lymph from the right upper quadrant: right arm, right side of the head and neck, and right thorax. Formed by the junction of right jugular, subclavian, and bronchomediastinal trunks, emptying into the right venous angle (junction of right internal jugular and subclavian veins). Blockage may cause right-sided chylothorax, though this is less common than left-sided from thoracic duct injury.

Right Ventricle

– Receives deoxygenated blood from the right atrium through the tricuspid valve, pumping into the pulmonary trunk via the pulmonary valve. Thinner walls (3-5 mm) than the left ventricle due to low pulmonary resistance. Features: trabeculae carneae, 3 papillary muscles with tricuspid chordae tendineae, moderator band with right bundle branch. Pressure overload from pulmonary hypertension causes hypertrophy and failure.

Sacroiliac Joint

– The synovial joint between the sacrum and the ilium (auricular surfaces), forming the connection between the axial skeleton and the pelvis. The sacroiliac joint is reinforced by strong ligaments (anterior, interosseous, and posterior sacroiliac ligaments, sacrotuberous and sacrospinous ligaments). It has limited mobility (2–4 degrees of rotation, 1–2 mm of translation) and functions primarily to transmit weight from the upper body to the lower limbs and provide stability. Sacroiliac joint dysfunction (sacroiliitis) is a common cause of low back pain, often referred to the buttock, groin, and posterior thigh. Causes: inflammatory (ankylosing spondylitis, psoriatic arthritis, reactive arthritis, inflammatory bowel disease-associated arthropathy), degenerative (osteoarthritis), pregnancy and postpartum (hormonally-mediated ligamentous laxity), trauma, and infection (septic arthritis, rare). Diagnosis: clinical examination (FABER test, Gaenslen sign, compression/distraction), imaging (X-ray, CT, MRI with STIR sequences for inflammation), and diagnostic intra-articular injection. Treatment: physiotherapy (core strengthening, pelvic stabilisation), NSAIDs, local corticosteroid injection, and radiofrequency ablation for refractory pain.

Sacrum

– Large triangular bone from 5 fused sacral vertebrae between the hip bones. Articulates with L5 superiorly and coccyx inferiorly. The sacral promontory is an obstetric landmark. Contains 4 pairs of sacral foramina. Forms the posterior pelvic wall. Sacroiliac joints are important for force transmission between spine and lower limbs.

Salivary Glands

– Exocrine glands that produce saliva. Three pairs of major salivary glands: parotid glands (largest, located preauricularly, secrete serous saliva via Stensen duct), submandibular glands (located in the submandibular triangle, secrete mixed serous and mucous saliva via Wharton duct), and sublingual glands (smallest, located in the floor of the mouth, secrete predominantly mucous saliva via multiple ducts of Rivinus). Numerous minor salivary glands (600–1000) are scattered throughout the oral mucosa, palate, pharynx, and lips. Saliva (1–1.5 L/day) contains water (99%), electrolytes, amylase (starch digestion), mucus (lubrication), lysozyme and IgA (antimicrobial), and lingual lipase (fat digestion). Salivary secretion is controlled by the autonomic nervous system: parasympathetic (CN VII and IX—profuse, watery secretion) and sympathetic (scanty, viscous secretion).

Scaphoid

– Largest proximal carpal row bone on the radial wrist side. Most commonly fractured carpal bone from falls on an outstretched hand. Unique distal blood supply makes waist fractures prone to proximal fragment avascular necrosis. Anatomical snuffbox tenderness is characteristic. Initial radiographs may be negative, requiring repeat imaging or MRI.

Scapula

– Flat triangular bone on the posterior thorax between ribs 2-7. Key features: acromion, coracoid process, glenoid cavity, spine, supraspinous/infraspinous fossae. Stabilized by 17 muscles and serves as rotator cuff attachment. Has 3 borders, 3 angles, and 2 surfaces. Fractures are uncommon, usually from high-energy trauma.

Sciatic Nerve

– Body's largest nerve (L4-S3), formed from the lumbosacral plexus, exiting via the greater sciatic foramen below the piriformis. Descends in the posterior thigh, dividing into tibial and common peroneal nerves at the popliteal fossa. Innervates hamstrings and provides lower leg/foot motor/sensory innervation via its branches. Sciatica is pain radiating along its distribution, commonly from L4-L5 or L5-S1 disc herniation.

Sclera

– Dense fibrous opaque outer eye coat providing structural integrity and extraocular muscle attachment. White, covering approximately five-sixths of the globe. Continuous with cornea anteriorly (limbus) and dura mater posteriorly through the optic nerve sheath. May appear yellowish in elderly or with jaundice (icterus). Scleritis (inflammation) is associated with autoimmune diseases like rheumatoid arthritis and may be sight-threatening.

Scrotum

– Pouch of skin and superficial fascia containing the testes, epididymides, and lower spermatic cords. The scrotal wall has multiple layers reflecting the abdominal wall layers. The dartos muscle (smooth muscle in the superficial fascia) and cremaster muscle (skeletal muscle from the internal oblique) regulate testicular temperature by contracting (bringing testes closer to the body in cold) or relaxing (lowering them in warmth). The scrotum is divided by a median septum into two compartments, each containing one testis with its epididymis and the lower portion of the spermatic cord.

Sebaceous Glands

– Holocrine glands associated with hair follicles (pilosebaceous units) secreting sebum (triglycerides, wax esters, squalene, cholesterol). Lubricate and waterproof skin/hair, have antimicrobial properties (fatty acids), and maintain skin barrier function. Most numerous on the face and scalp. Hyperactivity contributes to acne vulgaris; hypoactivity causes dry skin. Stimulated by androgens (explaining puberty acne). Fordyce spots are ectopic glands on oral mucosa and genitalia.

Semicircular Canals

– Three fluid-filled (endolymph) loops in each inner ear, oriented approximately orthogonally to detect angular (rotational) acceleration in three planes: horizontal (lateral), anterior (superior), and posterior. Each has an ampulla with a crista containing hair cells embedded in a gelatinous cupula. Head rotation deflects the cupula, bending stereocilia and stimulating hair cells. Horizontal canal excitatory direction is ampullopetal flow. BPPV results from displaced otoconia entering the canals, causing benign paroxysmal positional vertigo with brief episodes of spinning provoked by head movement.

Seminal Vesicles

– Paired lobulated glands behind the bladder and anterior to the rectum, each approximately 5 cm long, producing approximately 60-70 percent of seminal volume. The secretion is alkaline (neutralizing vaginal acidity), rich in fructose (energy source for sperm), prostaglandins (stimulating female reproductive tract contractions), and clotting proteins (causing semen coagulation after ejaculation). Each seminal vesicle duct joins the vas deferens to form the ejaculatory duct. Seminal vesicle pathology (infection, cysts, or invasion by prostate cancer) may affect ejaculate volume, cause haemospermia, and impair male fertility.

Serratus Anterior Muscle

– Fan-shaped muscle from ribs 1-8/9 to the medial scapular border. Innervated by long thoracic nerve (C5-C7). Protracts scapula, upwardly rotates glenoid for overhead elevation, stabilizes against thorax. Long thoracic nerve injury causes medial scapular winging, especially visible during wall push-ups. Commonly injured in breast surgery.

Sigmoid Colon

– S-shaped segment from descending colon to rectum in the left iliac fossa, approximately 40-50 cm, with complete mesentery (intraperitoneal, mobile). Most common diverticular disease site from narrow lumen and high pressure. Common volvulus site, particularly in elderly/institutionalized patients. Second most common colorectal cancer site. Transition between water-absorbing colon and rectal storage.

Skin

– The largest organ of the body (approximately 1.5–2 m^2), forming a protective barrier against physical injury, pathogens, UV radiation, and water loss. Layers: epidermis (stratified squamous epithelium, keratinocytes, melanocytes, Langerhans cells, Merkel cells), dermis (connective tissue with collagen and elastin fibers, blood vessels, nerves, hair follicles, sweat and sebaceous glands), and hypodermis (subcutaneous fat). Functions: thermoregulation (sweating, vasodilation/vasoconstriction), sensation (touch, pain, temperature), vitamin D synthesis, immune surveillance. Common diseases: dermatitis, psoriasis, skin cancer (basal cell, squamous cell, melanoma), acne, infections.

Skull

– Bony framework of the head with 22 bones: cranium (8: frontal, 2 parietal, 2 temporal, occipital, sphenoid, ethmoid) and facial (14). Joined by sutures: coronal, sagittal, lambdoid, squamous. The skull base has foramina for cranial nerves and vessels. Protects the brain and provides attachment for muscles of mastication and facial expression.

Small Intestine

– Longest GI segment (approximately 6 meters) from pylorus to ileocecal valve. Primary nutrient digestion and absorption site. Three segments: duodenum (25 cm, chyme/bile/pancreatic enzymes), jejunum (2.5 m, prominent folds/villi for absorption), ileum (3.5 m, Peyer's patches for immunity). Specialized for absorption via plicae circulares, villi, and microvilli. Ileocecal valve prevents colonic backflow.

Soleus Muscle

– Broad flat deep posterior leg muscle from posterior tibia, fibula, and tendinous arch to the calcaneus via the Achilles tendon. Innervated by tibial nerve (S1-S2). Powerful ankle plantarflexor, the primary anti-gravity muscle for upright posture. One-joint muscle. High slow-twitch fiber proportion for fatigue-resistant sustained postural activities and walking.

Sphenoid Bone

– Butterfly-shaped bone at the skull base, with body (sphenoid sinuses), greater/lesser wings, and pterygoid processes. The sella turcica houses the pituitary gland. Important foramina include the optic canal (CN II, ophthalmic artery), superior orbital fissure (CN III, IV, V1, VI), foramen rotundum (V2), ovale (V3), and spinosum (middle meningeal artery). Articulates with most cranial bones.

Spinal Cord

– The cylindrical bundle of nerve fibers extending from the medulla oblongata (at the foramen magnum) to the conus medullaris (L1–L2 in adults), surrounded by the vertebral column and meninges (dura mater, arachnoid mater, pia mater). The spinal cord is divided into 31 segments: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal. Each segment gives rise to a pair of spinal nerves (dorsal root—sensory, ventral root—motor). Internal structure: central gray matter (butterfly-shaped—dorsal horn: sensory; ventral horn: motor; lateral horn: autonomic) surrounded by white matter (ascending tracts—spinothalamic, dorsal columns, spinocerebellar; descending tracts—corticospinal, reticulospinal, vestibulospinal). The spinal cord conducts sensory information from the periphery to the brain, carries motor commands from the brain to muscles, and mediates spinal reflexes (stretch, withdrawal). Spinal cord injury causes paralysis, sensory loss, and autonomic dysfunction below the level of the lesion.

Spine

– See Spinal Cord and Vertebral Column.

Spleen

– Largest lymphoid organ in the left upper quadrant under ribs 9-11, approximately 12 cm long and 150-200 g. Red pulp filters blood (removing old/damaged RBCs and platelets); white pulp mounts immune responses (lymphoid follicles and periarteriolar lymphoid sheaths). Enclosed by a fibroelastic capsule with trabeculae. Connected to the stomach by the gastrosplenic ligament. The splenic hilum is where vessels enter and exit. Asplenia (congenital or surgical) increases infection risk from encapsulated bacteria, particularly Streptococcus pneumoniae, Neisseria meningitidis, and Haemophilus influenzae.

Spleen Hilum

– Concave medial surface where the tortuous splenic artery (celiac trunk branch) enters and splenic vein exits, with lymphatics and nerves. Artery divides into 5 segmental branches at the hilum (basis for partial splenectomy). Splenic vein joins the superior mesenteric vein forming the portal vein. Attaches the splenorenal ligament containing vessels and pancreatic tail. Pancreatic pathology may involve splenic vessels here.

Sternocleidomastoid Muscle

– Strap-like anterior neck muscle from manubrium and medial clavicle to the mastoid process. Innervated by CN XI and C2-C3. Divides neck into anterior/posterior triangles. Unilaterally: laterally flexes to opposite side, rotates face to same side. Bilaterally: flexes neck, assists forced inspiration. Key examination landmark; commonly affected by congenital torticollis.

Sternum

– Flat bone in the anterior midline thorax, with manubrium, body, and xiphoid process. The sternal angle (of Louis) at the manubriosternal junction is a key landmark at the 2nd costal cartilage, T4-T5, and tracheal bifurcation. Common site for bone marrow biopsy. Median sternotomy is the standard cardiac surgical approach.

Stomach

– J-shaped muscular organ in the left upper quadrant between esophagus and duodenum. Food reservoir with mechanical and chemical digestion. Five regions: cardia, fundus, body, antrum, pylorus. Lesser and greater curvatures. Mucosa: parietal cells (HCl, intrinsic factor), chief cells (pepsinogen), mucous cells (mucus), G cells (gastrin). Produces approximately 2 liters of secretions daily; holds 1-1.5 liters.

Superior Mesenteric Artery

– Second major abdominal aortic branch at L1, the midgut artery. Supplies distal duodenum through proximal two-thirds of transverse colon. Branches: inferior pancreaticoduodenal, jejunal, ileal, ileocolic, right colic, middle colic. Arises anterior to the pancreatic uncinate process, crosses the third duodenal part. Occlusion causes acute mesenteric ischemia with severe pain disproportionate to examination. SMA syndrome compresses the duodenum.

Superior Vena Cava

– Formed by brachiocephalic vein junction at the first right costal cartilage, returning upper body deoxygenated blood to the right atrium. Approximately 7 cm, valveless. Related to right lung/pleura, right phrenic nerve, and azygos vein. SVC syndrome from compression (commonly lung cancer) causes facial swelling, plethora, and upper extremity edema.

Talus

– Second largest tarsal bone articulating with tibia/fibula proximally at the ankle. Approximately 60 percent of its surface is articular cartilage with no muscular attachments. Transmits entire body weight to the calcaneus. Talar neck fractures may disrupt blood supply causing avascular necrosis in up to 50 percent. Also participates in the subtalar joint for inversion/eversion.

Tarsus

– The collection of seven bones forming the posterior part of the foot, between the tibia/fibula and the metatarsals. The tarsal bones: talus (articulates with tibia and fibula at the ankle joint), calcaneus (heel bone—largest tarsal bone, insertion of Achilles tendon), navicular (medial midfoot), cuboid (lateral midfoot), and three cuneiforms (medial, intermediate, lateral—articulate with the metatarsals). The tarsus transmits body weight from the leg to the foot, provides a flexible arch, and permits inversion/eversion movements at the subtalar and midtarsal joints. Tarsal coalition (congenital fusion of two or more tarsal bones—calcaneonavicular and talocalcaneal most common) causes peroneal spastic flatfoot and limited subtalar motion. Tarsal tunnel syndrome: compression of the posterior tibial nerve beneath the flexor retinaculum (medial ankle), causing burning pain, paraesthesia, and numbness on the plantar surface of the foot.

Temporal Bones

– Paired bones forming the cranial floor and lateral wall, with squamous, petrous, mastoid, tympanic, and styloid parts. Contain the external auditory canal, middle and inner ear, and foramina: carotid canal (internal carotid), jugular foramen (CN IX-XI, IJV), and foramen lacerum. Fractures may cause hearing loss, facial nerve palsy, and CSF otorrhea.

Temporomandibular Joint

– Synovial ginglymoarthrodial joint between mandibular condyle and temporal bone's mandibular fossa, separated by an articular disc. Only bilateral body joint, allowing hinge and sliding movements. Disc divides into superior/inferior compartments. Temporomandibular disorders cause pain, clicking, limited movement, common with bruxism.

Testis (Testicle)

– The male gonad, a paired ovoid organ (4–5 cm × 2–3 cm, weight 15–20 g) located in the scrotum, responsible for sperm production (spermatogenesis) and testosterone secretion. Each testis is covered by the tunica vaginalis (peritoneal remnant) and tunica albuginea (dense fibrous capsule). Internally: 200–300 lobules, each containing 1–4 seminiferous tubules (site of spermatogenesis—60–70 days), lined by Sertoli cells (support, nourishment, and regulation of spermatogenesis; produce inhibin B, anti-Müllerian hormone) and containing germ cells (spermatogonia → primary spermatocytes → secondary spermatocytes → spermatids → spermatozoa). Leydig cells (interstitial cells) in the connective tissue between tubules produce testosterone (regulated by LH). The testis develops in the abdomen and descends into the scrotum at 28–32 weeks gestation under the influence of androgens, guided by the gubernaculum. Undescended testis (cryptorchidism) increases the risk of infertility and testicular cancer. The testis is innervated by the genitofemoral nerve (L1–L2) and supplied by the testicular artery.

Thalamus

– Bilateral ovoid gray matter structure forming the third ventricle's lateral walls, the brain's relay station for nearly all sensory (except olfaction), motor, and limbic information to the cortex. Numerous nuclei: VPL (spinothalamic, dorsal column), VPM (trigeminal), LGN (visual), MGN (auditory), VL/VA (motor). Lesions cause contralateral sensory loss, Dejerine-Roussy thalamic pain syndrome, and cognitive impairment.

Thoracic Duct

– Body's largest lymphatic vessel, approximately 40 cm long, draining lymph from the entire body below the diaphragm and left side above it (approximately 75 percent of total lymph). Begins at the cisterna chyli (abdominal lymphatic sac) at the L1-L2 level, ascends through the aortic hiatus, through the posterior mediastinum between the aorta and azygos vein, and empties into the junction of the left internal jugular and left subclavian veins. Blocked or damaged thoracic duct causes chylothorax (leakage of chyle into the pleural space), typically on the left side, manifesting as a milky pleural effusion rich in triglycerides.

Thoracic Vertebrae (T1-T12)

– Have costal facets on bodies and transverse processes for rib articulation. Heart-shaped bodies, long downward spinous processes, coronally-oriented facets allowing rotation and lateral flexion but limiting flexion/extension. Small circular vertebral foramen. Most rigid vertebral region due to the rib cage, least affected by disc herniations.

Thymus Gland

– A bilobed lymphoid organ located in the superior mediastinum, posterior to the sternum and anterior to the great vessels. The thymus is the primary lymphoid organ for T-cell development and maturation. It is largest relative to body size in infancy (weighs 20–30 g at birth), continues to grow until puberty (peak 30–40 g), then progressively involutes with age, replaced by adipose tissue (involution—60–80% loss by age 50). Structure: each lobe is divided into lobules by fibrous septa; each lobule has an outer cortex (immature T-lymphocytes / thymocytes, epithelial reticular cells, macrophages) and an inner medulla (mature T-cells, Hassall corpuscles—concentric whorls of epithelial cells, site of negative selection). T-cell development: bone marrow-derived lymphoid progenitors enter the thymus at the corticomedullary junction, undergo positive selection (cortex—T-cells with functional T-cell receptors that recognise self-MHC molecules are retained) and negative selection (medulla—T-cells that react strongly with self-antigens are deleted by apoptosis, mediated by AIRE gene expression), producing a self-tolerant, self-MHC-restricted T-cell repertoire. Disorders: thymic hyperplasia (myasthenia gravis—associated with thymoma or hyperplasia), thymoma (benign or malignant epithelial tumor, associated with autoimmune diseases—myasthenia gravis, pure red cell aplasia, hypogammaglobulinaemia), DiGeorge syndrome (22q11.2 deletion—thymic aplasia/hypoplasia, T-cell deficiency, recurrent infections).

Thyroid Gland

– An endocrine gland located in the anterior neck, consisting of two lobes (right and left) connected by an isthmus, lying anterior to the trachea between the cricoid cartilage and the suprasternal notch. The thyroid produces tri-iodothyronine (T_3) and thyroxine (T_4)—iodinated tyrosine derivatives that regulate metabolism, growth, and development—and calcitonin (produced by parafollicular/C-cells, regulates calcium metabolism). Thyroid hormone synthesis: iodide is actively transported into follicular cells (NIS/sodium-iodide symporter), oxidised by thyroid peroxidase (TPO), and iodinated to tyrosine residues on thyroglobulin (Tg) to form MIT and DIT, which couple to form T_3 and T_4. Secretion is regulated by the hypothalamic–pituitary–thyroid axis (TRH $\longrightarrow$ TSH $\longrightarrow$ thyroid hormone $\longrightarrow$ negative feedback). Disorders: hyperthyroidism (Graves disease, toxic nodular goitre, thyroiditis), hypothyroidism (Hashimoto thyroiditis, iodine deficiency, post-surgical/radioiodine), thyroid nodules and thyroid cancer (papillary—most common, follicular, medullary, anaplastic), goitre, and thyroiditis. The thyroid gland develops from the foramen cecum at the base of the tongue (week 4 of gestation) and descends to its final position; thyroglossal duct remnants can form midline neck cysts.

Tibia

– Larger leg bone, the main weight-bearer transmitting body weight from knee to ankle. Proximal: tibial plateau (medial/lateral condyles) and tibial tuberosity (patellar ligament insertion). Shaft: sharp subcutaneous anterior border. Distal: medial malleolus articulating with the talus at the ankle. Tibial plateau fractures are common in high-energy trauma.

Tibialis Anterior Muscle

– Superficial anterior leg muscle from lateral tibial condyle and shaft to medial cuneiform and first metatarsal base. Innervated by deep peroneal nerve (L4-L5). Primary dorsiflexor for foot clearance during gait swing phase; inverts foot. Weakness causes foot drop with steppage gait. Commonly affected by anterior compartment syndrome and shin splints.

Tibialis Posterior Muscle

– Deepest posterior leg muscle from posterior tibia, fibula, and interosseous membrane to navicular tuberosity, medial cuneiform, and sustentaculum tali. Innervated by tibial nerve (L4-L5). Plantarflexes and inverts foot; primary dynamic medial longitudinal arch stabilizer. Dysfunction causes adult-acquired flatfoot with arch flattening, hindfoot valgus, and medial ankle pain.

Tibial Nerve

– Sciatic division (L4-S3) continuing from posterior thigh through the popliteal fossa into the deep posterior leg, innervating gastrocnemius, soleus, tibialis posterior, flexor digitorum longus, and flexor hallucis longus. Passes behind the medial malleolus as the tarsal tunnel, dividing into medial/lateral plantar nerves for the foot. Sensation to the sole. Tarsal tunnel syndrome causes burning pain and paresthesias in the sole.

Tonsils

– Aggregated lymphoid tissue forming Waldeyer's ring at the pharyngeal entrance. Pharyngeal tonsils (adenoids) in the nasopharynx, palatine tonsils at the oropharyngeal pillars, lingual tonsils at the tongue base, and tubal tonsils near the Eustachian tube orifices. They are the first line of defense against inhaled or ingested pathogens. Palatine tonsils are most commonly infected (tonsillitis) and surgically removed (tonsillectomy) for recurrent infections or obstructive sleep apnoea. The tonsils are part of Waldeyer's ring, a protective lymphoid barrier at the pharyngeal inlet that samples inhaled and ingested antigens, with maximal activity in early childhood.

Transverse Colon

– Longest, most mobile colonic segment from hepatic to splenic flexure, approximately 45 cm, with complete mesentery (transverse mesocolon). Crosses anterior to duodenum, pancreas, and major vessels. Splenic flexure is the highest colon point and common gas trapping site. Most common diverticular disease site along with the sigmoid.

Transversus Abdominis Muscle

– Deepest flat abdominal muscle with horizontal fibers, from inguinal ligament, iliac crest, thoracolumbar fascia, and inner lower six costal cartilages. Innervated by T7-T12 and L1. Compresses abdomen without trunk movement, providing core stability and increased intra-abdominal pressure during coughing, lifting, Valsalva. Key for core stability exercises and low back pain rehabilitation.

Trapezium

– Saddle-shaped distal row carpal bone on the radial side, articulating with the first metacarpal at the CMC joint of the thumb. This saddle joint allows thumb opposition essential for grip and fine motor function. Basal joint arthritis is common in postmenopausal women, causing pinch grip pain and weakness.

Trapezius Muscle

– Large triangular muscle from occipital bone and C7-T12 spinous processes to clavicle, acromion, and scapular spine. Innervated by CN XI and C3-C4. Upper fibers elevate scapula, middle fibers retract, lower fibers depress. Essential for shoulder stability and scapulothoracic motion. CN XI injury causes scapular winging and difficulty with abduction above 90 degrees.

Trapezoid

– Smallest distal row carpal bone between trapezium and capitate, articulating with the second metacarpal. Wedge shape provides stability for the firmly fixed second metacarpal. Fractures rare (less than 1 percent of carpal fractures). Contributes to the radial column wrist stability.

Triceps Brachii Muscle

– Three-headed posterior arm muscle. Long head from infraglenoid tubercle, lateral head from posterior humerus above radial groove, medial head below. All insert on the olecranon. Innervated by radial nerve (C7-C8). Primary elbow extensor; assists shoulder extension/adduction. Triceps reflex (C7) tests radial nerve/spinal cord integrity.

Tricuspid Valve

– Right atrioventricular valve with anterior, posterior, and septal cusps supported by chordae tendineae and three papillary muscles. Largest cardiac valve. Most commonly affected in IV drug use-associated endocarditis. Tricuspid regurgitation from right ventricular dilation, endocarditis, or carcinoid. Stenosis is rare, mostly rheumatic. Thinner and more delicate than the mitral valve.

Trigeminal Nerve

– Cranial nerve V (CN V), the largest cranial nerve, providing sensory innervation to the face and anterior scalp (ophthalmic V_1, maxillary V_2, mandibular V_3 divisions) and motor innervation to the muscles of mastication (masseter, temporalis, medial and lateral pterygoids, tensor veli palatini, tensor tympani, mylohyoid, anterior belly of digastric). The trigeminal nerve emerges from the lateral pons as a large sensory root (trigeminal ganglion/ semilunar/Gasserian ganglion in Meckel cave) and a smaller motor root. Sensory distribution: V_1 (ophthalmic)—forehead, scalp, upper eyelid, cornea, nasal tip; V_2 (maxillary)—cheek, lower eyelid, side of nose, upper lip, upper teeth and gums, palate; V_3 (mandibular)—lower lip, chin, lower teeth and gums, anterior two-thirds of tongue (general sensation), floor of mouth, temporal region. Motor: V_3 supplies the muscles of mastication, tensor veli palatini, tensor tympani, mylohyoid, and anterior belly of digastric. Reflexes: corneal reflex (afferent V_1, efferent CN VII), jaw jerk (afferent and efferent V_3—monosynaptic). Lesions: trigeminal neuralgia (V_2/V_3), trigeminal schwannoma, cavernous sinus syndrome, and brainstem stroke.

Triquetrum

– Pyramidal proximal row carpal bone on the ulnar side, articulating with lunate, hamate, pisiform, and TFCC. Second most commonly fractured carpal bone. Contributes to proximal carpal row and ulnar wrist stability. Pisiform articulates with its volar surface.

True Ribs

– Ribs 1-7 articulating directly with the sternum via their own costal cartilages. Each has a head, neck, tubercle, and shaft. The first two are short and strongly curved for neck/shoulder muscle attachment. Costal cartilages provide thoracic cage elasticity for respiratory expansion. Less commonly fractured than false ribs due to shorter length and more protected position.

Tympanic Membrane

– Thin translucent cone-shaped membrane separating external and middle ear, approximately 10 mm diameter. Three layers: outer cutaneous, middle fibrous, inner mucous. Malleus handle is visible through it. Light reflex (cone of light) in the anteroinferior quadrant indicates normal position. Perforation from trauma, infection (otitis media), or barotrauma. Myringotomy and tympanostomy tube placement treat chronic otitis media with effusion. Vibrates in response to sound, transmitting energy to the ossicular chain and ultimately to the cochlear fluid for auditory transduction.

Ulna

– Medial forearm bone, longer than the radius. Proximal end: trochlear notch, olecranon, coronoid process. Shaft. Distal end: head, styloid process. The trochlear notch articulates with the trochlea forming the primary elbow hinge. Bears approximately 20 percent of wrist axial load. The olecranon is the bony elbow prominence and triceps insertion. Serves as the forearm rotation axis.

Ulnar Nerve

– From medial cord (C8-T1), passing behind the medial epicondyle (cubital tunnel), into the forearm innervating flexor carpi ulnaris and medial FDP half. In the hand innervates hypothenar muscles, interossei, medial two lumbricals, and adductor pollicis. Sensation to medial 1.5 digits. Injury (elbow trauma, Guyon's canal compression) causes claw hand (MCP hyperextension with IP flexion of digits 4-5) and loss of grip/pinch strength.

Umbilical Region

– Central region around the umbilicus containing small intestine, transverse colon, and great vessels. The umbilicus is a landmark for assessing ascites, caput medusae in portal hypertension, and umbilical hernias. Percussion and auscultation here are essential components of the abdominal examination.

Umbilicus (Navel)

– The scar on the anterior abdominal wall marking the site of attachment of the umbilical cord in fetal life, located at the level of the L3-L4 intervertebral disc, approximately midway between the xiphoid process and the pubic symphysis. The umbilicus is a remnant of fetal structures: the umbilical vein (ligamentum teres hepatis, running to the liver), the umbilical arteries (medial umbilical ligaments), and the urachus (median umbilical ligament, running to the bladder apex). It is a common site of hernia (umbilical hernia) and is used as a laparoscopic port entry site (Hasson technique, Veress needle insertion for pneumoperitoneum). Dermatomal supply is T10. The aorta is palpable at the level of the umbilicus, and the iliac bifurcation lies at the L4 level, approximately 1 cm below it.

Uterus

– Hollow muscular organ in the pelvis between the bladder and rectum, approximately 7.5 cm long in adult females, responsible for menstruation, implantation, and fetal development during pregnancy. Three layers: endometrium (inner mucosa, cyclically shed during menstruation), myometrium (middle smooth muscle layer, contracts during labor), and perimetrium (outer serous coat). The uterus has a fundus (superior portion above the tube openings), body, and cervix (lower portion protruding into the vagina. The cervix has an internal and external os, with the endocervical canal being a common site for cervical cancer screening via smear tests.

Vagina

– Muscular fibromembranous canal extending from the cervix to the vulva, approximately 7-9 cm long in adult females. Functions include receiving the penis during intercourse, serving as the birth canal, and channeling menstrual flow. The vaginal walls have rugae (transverse folds allowing distension) and are lined with stratified squamous epithelium (non-keratinized, responding to estrogen by producing glycogen). Lactobacilli in the vaginal flora metabolize glycogen to lactic acid, maintaining an acidic environment (pH 4-5) that protects against pathogenic colonisation by opportunistic organisms.

Vas Deferens

– Muscular duct (approximately 45 cm long) transporting sperm from the epididymal tail to the ejaculatory duct. It ascends through the spermatic cord, passes through the inguinal canal, enters the pelvis, crosses over the ureter, and joins the seminal vesicle duct to form the ejaculatory duct (in the prostate). The vas deferens has thick smooth muscle walls providing peristaltic propulsion of sperm. It is the structure cut during vasectomy (male sterilization), typically at the scrotal level where it is easily palpable through the scrotal skin, allowing reliable identification and division with minimal morbidity.

Vein

– A blood vessel that carries deoxygenated blood toward the heart (except pulmonary veins, which carry oxygenated blood). Veins have thinner walls and larger lumens than arteries, with less smooth muscle and elastic tissue. Many veins contain one-way valves (especially in the legs) that prevent backflow and assist venous return against gravity. Venous return is facilitated by the skeletal muscle pump, respiratory pump, and cardiac suction. Venous system: superficial veins (subcutaneous), deep veins (accompany arteries), and perforating veins (connect superficial and deep systems). Conditions: varicose veins, deep vein thrombosis, chronic venous insufficiency.

Ventricle

– One of the two lower, thicker-walled chambers of the heart that pump blood out of the heart. The right ventricle pumps deoxygenated blood into the pulmonary artery (low-pressure system); the left ventricle pumps oxygenated blood into the aorta (high-pressure system, generating systemic blood pressure). The left ventricular wall is 3–4 times thicker than the right. The interventricular septum separates the ventricles. Ventricular contraction (systole) ejects blood; ventricular relaxation (diastole) allows filling. Conditions: ventricular hypertrophy, ventricular failure, ventricular septal defect, ventricular tachycardia/fibrillation. Also refers to the four fluid-filled cavities within the brain (two lateral ventricles, third, and fourth ventricles) that produce and circulate cerebrospinal fluid (CSF).

Vertebral Column

– 33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, 4 fused coccygeal. Protects the spinal cord, supports the head/trunk, provides muscle/rib attachment. Each typical vertebra has a body, vertebral arch, and 7 processes. Intervertebral discs (nucleus pulposus/annulus fibrosus) provide cushioning. Four curvatures: cervical lordosis, thoracic kyphosis, lumbar lordosis, sacral kyphosis.

Vestibule

– Central bony labyrinth between cochlea and semicircular canals, containing utricle (horizontal linear acceleration and head tilt detection) and saccule (vertical linear acceleration detection). Each has a macula with hair cells covered by gelatinous membrane embedded with otoconia (calcium carbonate crystals). Forces deflect otoconia, bending stereocilia and generating nerve impulses. The vestibular system works with visual and proprioceptive systems for balance. Vestibular neuritis and labyrinthitis cause acute vertigo, nystagmus, and balance disturbance, while Mnière's disease involves episodic vertigo with tinnitus and hearing loss.

Viscera

– The internal organs of the body, particularly those within the thoracic and abdominal cavities. The thoracic viscera include the heart, lungs, esophagus, trachea, and thymus. The abdominal viscera are divided into the digestive organs (stomach, small intestine, large intestine, liver, gallbladder, pancreas), the urinary organs (kidneys, ureters, bladder), and the reproductive organs (uterus, ovaries in females;

prostate, seminal vesicles in males). Visceral pain is poorly localised, vague, and referred to dermatomes that share the same spinal cord level (e.g., cardiac ischemic pain referred to the left arm and jaw, gallbladder pain referred to the right shoulder). Visceral peritoneum covers the abdominal organs; the mesentery suspends the intestines and carries blood vessels, lymphatics, and nerves. The viscera receive dual autonomic innervation: sympathetic (fight or flight) and parasympathetic (rest and digest, primarily via the vagus nerve).

Vulva

– External female genitalia, including the mons pubis (fatty mound over the pubic symphysis), labia majora (outer fatty folds covered with hair), labia minora (inner hairless folds), clitoris (erectile tissue homologous to the penis, with the glans being the most sensitive erogenous zone), vestibular glands (Bartholin's glands lubricating the vaginal opening), and the urethral and vaginal openings. The vulva is innervated by the pudendal nerve. Vulvar skin conditions include lichen sclerosus, vulvar intraepithelial neoplasia (VIN), and vulvovaginitis, each presenting with pruritus, irritation, and visible changes to the vulvar skin and mucosa.

Wrist Joint

– Condyloid synovial joint between distal radius and articular disc proximally and scaphoid/lunate/triquetrum distally. Allows flexion, extension, radial/ulnar deviation, circumduction. The triangular fibrocartilage complex absorbs ulnar side load. Commonly affected by rheumatoid arthritis, osteoarthritis, and traumatic injuries including scaphoid fractures and Colles' fractures.

Zygoma (Zygomatic Bone)

– The cheekbone, a paired, quadrangular bone of the facial skeleton (viscerocranium) that forms the prominence of the cheek and the lateral wall and floor of the orbit. The zygomatic bone articulates with four bones: the maxilla (at the zygomaticomaxillary suture, forming the infraorbital rim), the temporal bone (at the zygomaticotemporal suture, forming the zygomatic arch, which is completed posteriorly by the temporal process of the zygomatic bone and anteriorly by the zygomatic process of the temporal bone), the frontal bone (at the zygomaticofrontal suture, forming the lateral orbital margin), and the sphenoid bone (at the zygomaticosphenoid suture). The zygomatic bone has three surfaces: lateral (facial, for attachment of the zygomaticus major and minor muscles), orbital (forming the lateral and inferior orbital walls, containing the zygomaticofacial and zygomaticotemporal foramina for the corresponding nerves), and temporal (forming part of the temporal fossa). The zygomatic nerve (branch of V2, maxillary division of the trigeminal nerve) runs through the zygomatic bone, providing sensory innervation. Zygomatic fractures (tripod/trimalar fracture, zygomaticomaxillary complex fracture) are the second most common facial fracture (after nasal bone fractures) and cause flattening of the malar prominence, periorbital ecchymosis, subconjunctival hemorrhage, trismus (from impingement of the coronoid process of the mandible), and infraorbital nerve paraesthesia (numbness of the cheek, upper lip, and lateral nose).

Chapter 2

Anesthesiology

Acute Pain

– A sudden, short-duration pain signal that alerts the body to injury or illness, typically lasting less than three months. It arises from tissue damage and resolves as healing occurs, serving a protective biological function. Common causes include surgery, fractures, burns, cuts, and infections.

Airway Management

– Airway management encompasses techniques and devices used to maintain ventilation and oxygenation in patients with compromised airways. The modified Mallampati classification predicts difficult laryngoscopy based on oropharyngeal anatomy visible with mouth open: Class I (soft palate, uvula, fauces, pillars visible), Class II (soft palate, uvula, fauces visible), Class III (soft palate and uvula visible), Class IV (only hard palate visible). Other predictors of difficult intubation include reduced mouth opening (less than 3 cm), reduced thyromental distance (less than 6.5 cm), limited neck extension (less than 35 degrees), and obesity; equipment including the difficult airway trolley should be immediately available when difficulty is anticipated.

Anaphylaxis During Anesthesia

– A rare but life-threatening IgE-mediated hypersensitivity reaction, with incidence of 1 in 5,000 to 20,000 anesthetics. Most common triggers are neuromuscular blocking agents (rocuronium, succinylcholine), latex, and antibiotics (vancomycin, piperacillin-tazobactam). Latex allergy is less common with modern latex-free operating rooms. Signs include hypotension, tachycardia, bronchospasm, cutaneous manifestations (urticaria, flushing), and elevated end-tidal CO2 in the immediate diagnosis and management of anaphylaxis; treatment follows the ABCDE approach with removal of the suspected trigger, administration of 100% oxygen, intravenous fluids, and adrenaline (epinephrine) as the first-line drug at a dose of 50-100 mcg IV titrated to effect; second-line agents include antihistamines and corticosteroids, though their role in acute management is limited.

Anesthesia

– The use of medications to prevent sensation, particularly pain, during medical procedures or surgery. Types: general anesthesia (reversible loss of consciousness and sensation, involving hypnotics, analgesics, and muscle relaxants), regional anesthesia (blocking sensation in a specific body region, e.g., epidural, spinal, brachial plexus block), local anesthesia (numbing a small area, e.g., infiltration, topical), and sedation (reduced consciousness while maintaining airway reflexes and spontaneous ventilation). General anesthesia involves three components: hypnosis (loss of consciousness, propofol, volatile anesthetics), analgesia (pain relief, opioids, ketamine, lidocaine), and muscle relaxation (neuromuscular blocking agents). Monitoring: ECG, BP, SpO_2, end-tidal CO_2, anesthetic gas concentration, and temperature.

Anesthesia for Coagulopathy

– Coagulopathy increases bleeding risk during surgery. Assessment includes history, physical examination, and laboratory tests (INR, PT, aPTT, platelet count, fibrinogen). INR greater than 1.5 or aPTT greater than 1.5 times control increases bleeding risk. Platelet count less than 50,000 for major surgery or less than 100,000 for neurosurgery requires platelet transfusion. Fibrinogen less than 150 mg/dL requires cryoprecipitate or fibrinogen concentrate. Fresh frozen plasma (15 to 20 mL/kg) correcrrects coagulopathy effectively. Vitamin K (10 mg IV slow push, 30 minutes onset) is used for warfarin reversal. Prothrombin complex concentrate (PCC) is preferred over FFP for urgent reversal (INR correction in 15 minutes). Antifibrinolytics (tranexamic acid 1 g IV) reduce bleeding in trauma and cardiac surgery.

Anesthesia for Epilepsy

– Patients with epilepsy require continuation of antiepileptic drugs perioperatively. Missed doses increase seizure risk. Propofol and benzodiazepines are anticonvulsant. Sevoflurane in high concentrations may lower seizure threshold. Hyperventilation and hypocapnia provoke seizures. Avoid meperidine (normeperidine is proconvulsant). Tramadol lowers seizure threshold. Postoperative pain and sleep deprivation are seizure triggers. Ensure antiepileptic drugs are given on schedule. Status epilepticuss requires immediate treatment with benzodiazepines (lorazepam 4 mg IV, midazolam, or diazepam 20 mg IV) and escalation of care.

Anesthesia for Hemophilia

– Hemophilia A (factor VIII deficiency) and Hemophilia B (factor IX deficiency) require factor replacement before surgery. Mild cases: desmopressin (DDAVP 0.3 mcg/kg IV) releases stored factor VIII and von Willebrand factor. Moderate to severe cases: factor concentrate (factor VIII or IX) to achieve 80 to 100 percent factor level. Dosing: factor VIII 40 units/kg raises level by 80 percent; factor IX 50 units/kg raises by 100 percent. Repeat dosing every 12 to 24 hours. Avoid intramuscular injectioions and intramuscular injections; avoid neuraxial blockade if factor levels are not normalised. Monitor for bleeding postoperatively.

Anesthesia for Neurosurgery

– Neuroanesthesia aims to maintain cerebral perfusion pressure (CPP = mean arterial pressure minus intracranial pressure), optimize brain relaxation, and minimize secondary brain injury. Induction agents (propofol, etomidate) reduce cerebral metabolic rate and intracranial pressure. Volatile anesthetics increase cerebral blood flow at concentrations greater than 1 MAC. Hyperventilation ($PaCO_2$ 30 to 35 mmHg) causes cerebral vasoconstriction and reduces intracranial pressure acutely. Mannitol (0.5 to 1 g/kg IV) reduces intracranial pressure through osmotic diuresis. Maintain CPP at 60 to 70 mmHg. Avoid hypotension and hypoxia. Emergence should be smooth to prevent coughing and hypertension. Postoperative neurological assessment is essential.

Anesthesia for Pheochromocytoma

– Pheochromocytoma is a catecholamine-secreting tumor causing episodic hypertension, tachycardia, headache, sweating, and pallor. Preoperative alphablockade (phenoxybenzamine 10 to 14 days) is essential before beta-blockade (propranolol) to prevent unopposed alpha stimulation. Blood pressure and heart rate variability requires aggressive hemodynamic monitoring (arterial line, central venous pressure). Induction and intubation cause massive catecholamine release. Tumor manipulation causes hypertennsive episodes intraoperatively; resection causes hypotension from catecholamine withdrawal. Postoperative management requires continued alpha and beta blockade.

Anesthesia for Porphyria

– Acute intermittent porphyria is a metabolic disorder of heme synthesis causing abdominal pain, neuropathy, psychiatric symptoms, and dark urine. Certain anesthetics are porphyrinogenic (triggering attacks): barbiturates, sulfonamides, carbamazepine, and possibly volatile anesthetics (though sevoflurane is considered safe). Safe agents include propofol, opioids, benzodiazepines (low dose), succinylcholine, and non-depolarizing neuromuscular blockers. Porphyria-safe drug databases should be consullted and porphyria-safe drug lists verified before administering any anesthetic agent.

Anesthesia Gas Analyzers

– Anesthesia gas analyzers monitor delivered and exhaled gas concentrations. Oxygen analyzer (paramagnetic or fuel cell) is required on every anesthesia machine and provides continuous SpO_2 and FiO_2 monitoring. Capnography (infrared CO_2 analyzer) measures end-tidal CO_2, confirming endotracheal tube position, ventilation adequacy, and detecting esophageal intubation immediately. End-tidal CO_2 normally equals arterial CO_2 within 5 mmHg. Pulse oximetry (spectrophotometry) measures hemoglobin oxygen ssaturation. Temperature monitoring (nasopharyngeal or oesophageal) detects malignant hyperthermia early. Neuromuscular monitoring (TOF) guides muscle relaxant dosing. Invasive arterial pressure is indicated for major cases, haemodynamic instability, and patients with cardiorespiratory disease. Central venous pressure and cardiac output monitoring (pulmonary artery catheter, TTE/TEE, FloTrac, LiDCO) are used for major surgery and sick patients.

Anesthesia Informatics

– Anesthesia informatics integrates electronic health records, decision support systems, and data analytics into clinical practice. Automated documentation reduces charting time. Clinical decision support alerts for drug interactions and dosing errors. Big data analytics identify patterns in anesthesia outcomes.

Anesthesia Machine

– The anesthesia machine delivers precise mixtures of oxygen, air, and volatile anesthetics to the patient. Components include gas cylinders (E-cylinder oxygen contains 660 liters at 2000 psi), pipeline supplies (hospital oxygen at 50 psi), pressure regulators, fail-safe valve (oxygen pressure below 25 psi shuts off nitrous oxide), vaporizers (variable-bypass sevoflurane, desflurane uses heated pressurized system), scavenging system (removes waste anesthetic gases), and breathing circuit (Circle system) comprises fresh gas inlet, inspiratory and expiratory valves, reservoir bag, CO2 absorber, adjustable pressure limiting valve, and ventilator. The circle system reduces gas consumption and heat loss. Safety features include oxygen failure alarm, hypoxia prevention system, and vapouriser interlock (prevents simultaneous use of two volatile agents).

Anesthesia Monitoring

– Anesthesia monitoring encompasses continuous assessment of patient physiology during surgery. Standard ASA monitoring includes electrocardiography (detecting arrhythmias and ischemia), pulse oximetry (continuous SpO2 measurement), non-invasive blood pressure (oscillometric every 3 to 5 minutes), end-tidal CO2 (confirming tube position and ventilation adequacy), temperature monitoring, and inspired oxygen concentration. Invasive arterial pressure monitoring provides beat-to-beat blood pressure and allows precise vasopressor management. Central venous pressure and cardiac output monitoring are used in high-risk patients. Neuromuscular monitoring (train-of-four) is used for muscle relaxant management. Temperature monitoring is essential for detecting malignant hyperthermia and hypothermia. The goal is to maintain physiological parameters within acceptable ranges and detect abnormalities early.

Anesthesia Pharmacogenomics

– Anesthesia pharmacogenomics studies genetic variations affecting drug response. Pseudocholinesterase deficiency causes prolonged succinylcholine paralysis. CYP2D6 polymorphisms affect codeine metabolism to morphine. Malignant hyperthermia susceptibility is linked to RYR1 gene mutations. Pharmacogenomic testing guides personalized anesthesia.

Aspiration Pneumonitis

– Chemical lung injury caused by inhalation of gastric contents, characterized by bronchospasm, hypoxemia, pulmonary edema, and atelectasis. Risk factors include full stomach, emergency surgery, obesity, pregnancy, gastroesophageal reflux, bowel obstruction, and decreased consciousness. Prevention includes rapid sequence induction with cricoid pressure (Sellick maneuver), endotracheal intubation, and preoperative fasting. Treatment includes suctioningng the airway, bronchoscopy, and supportive care with mechanical ventilation if needed. Most cases resolve within 24-48 hours with appropriate management.

Atropine

– A competitive muscarinic antagonist used to treat symptomatic bradycardia, reduce secretions, and reverse cholinergic side effects of neostigmine. Atropine blocks parasympathetic effects on the heart, increasing heart rate. Dose for bradycardia is 0.5 mg IV, repeat every 3 to 5 minutes, maximum 3 mg. Preoperative dose is 0.4 mg IV or IM. Atropine crosses the blood-brain barrier at high doses, causing central anticholinergic syndrome (confusion, hallucinations, agitation). Side effects include dry mouth, blurred vision, urinary retention, and tachycardia. Atropine is contraindicated in angle-closure glaucoma, obstructive uropathy, and myasthenia gravis. It is a tertiary amine that freely crosses the blood-brain barrier, unlike glycopyrrolate, producing central anticholinergic effects (agitation, confusion, hallucinations) at high doses.

Autologous Blood Donation

– Autologous blood donation involves preoperative collection and storage of the patient's own blood for intraoperative or postoperative transfusion. It eliminates risks of alloimmune reactions, transfusion-transmitted infections, and immune modulation. Indications include rare blood types, multiple alloantibodies, elective orthopedic surgery, and religious objections to allogeneic transfusion. Preoperative donation requires hemoglobin greater than 11 g/dL and at least 3 weeks before surgery. Intraoperative cell salvage and acute normovolaemic haemodilution reduce allogeneic transfusion requirements. Complications include bacterial contamination, clerical error, and fluid overload.

Awake Fiberoptic Intubation

– The gold standard for anticipated difficult airways, performed with the patient conscious and cooperatively maintaining spontaneous ventilation. Topical anesthesia is achieved through lidocaine nebulization, nerve blocks (glossopharyngeal, superior laryngeal, transtracheal), and spray-as-you-go technique. Sedation with dexmedetomidine or remifentanil provides anxiolysis while maintaining respiratory drive. The fiberoptic bronchoscope is passed through the vocal cords, and the endotracheal tube is passed over the bronchoscope into the trachea. Awake fiberoptic intubation is used in patients with known difficult airway, limited neck mobility (cervical spine injury, rheumatoid arthritis), head and neck tumours, and morbid obesity. Contraindications include patient refusal, severe hypoxia, and bleeding diathesis.

Awareness Under Anesthesia

– Unintended recall of events during general anesthesia, most commonly occurring during procedures where neuromuscular blockade prevents movement. Incidence: 0.1-0.2% of all general anesthetics, higher in cardiac, trauma, and cesarean section. Presents with postoperative recall of conversations, surgical sounds, or pain (worst outcomes). Risk factors: inadequate anesthesia (planned light anesthesia, hemodynamic instability), equipment failure, and certain drugs (ketamine, etomidate, nitrous oxide). Prevention: adequate dosing, volatile anesthetic end-tidal monitoring (MAC >0.7), BIS monitoring (target 40-60), and pre-oxygenation with a protocol. Management: documented debriefing, apology, psychological support, and referral to post-traumatic stress disorder services.

Blood Products

– Blood products used in anesthesia include packed red blood cells (increase oxygen-carrying capacity, each unit raises hemoglobin by approximately 1 g/dL), fresh frozen plasma (contains all coagulation factors, used for warfarin reversal, massive transfusion, and coagulopathy), platelets (prevent or treat thrombocytopenia, 1 apheresis unit raises count by 30,000 to 60,000), cryoprecipitate (rich in fibrinogen, factor VIII, von Willebrand factor, used for fibrinogen deficiency), and whole blood (is indicated for massive hemorrhage with specific ratios of PRBC:FFP:platelets (1:1:1). Transfusion reactions include acute haemolytic reaction, febrile non-haemolytic reaction, allergic reaction, transfusion-associated circulatory overload (TACO), transfusion-related acute lung injury (TRALI), and transfusion-transmitted infections.

Brain Monitoring

– Brain monitoring during anesthesia assesses the depth of hypnosis and neurological function. Bispectral index (BIS) processes the frontal EEG into a number from 0 (isoelectric) to 100 (fully awake), with 40 to 60 recommended for general anesthesia. BIS monitoring reduces awareness risk and may reduce anesthetic requirements. Entropy monitoring (state and response entropy) is an alternative. Processed EEG is less reliable in patients with brain pathology, hypothermia, and certain anesthetics (ketetamine and nitrous oxide). Monitoring raw EEG waveforms is valuable for detecting cerebral ischaemia and epileptiform activity in patients undergoing neurosurgery and vascular surgery.

Bronchospasm

– Involuntary constriction of bronchial smooth muscle causing airway obstruction, wheezing, and increased airway pressures. Triggers include airway manipulation, volatile anesthetics (sevoflurane is bronchodilating; desflurane can cause bronchospasm), histamine-releasing drugs (morphine, atracurium), and reactive airway disease. Treatment includes deepening anesthesia with sevoflurane, beta-2 agonists (albuterol nebulized), epinephrine (10 to 100 mcg IV), and corticosteroids (hydrocortisone 100 mg IV). Volatile anesthetics may need to be reduced or switched to intravenous agents (propofol) to permit effective bronchodilator therapy, and severe cases may require intravenous adrenaline infusion and magnesium sulphate (2 g IV).

Bupivacaine

– A long-acting amide local anesthetic with onset of 5 to 15 minutes and duration of 2 to 4 hours for peripheral nerve blocks, 1 to 2 hours for epidural. It is available in 0.25 percent, 0.5 percent, and 0.75 percent concentrations. Bupivacaine provides dense sensory and motor blockade. Maximum dose is 2.25 mg/kg (170 mg for 75 kg adult). Bupivacaine is cardiotoxic in overdose, causing refractory ventricular arrhythmias and cardiac arrest. Lipid emulsion therapy (20 percent Intralippid) 1.5 mL/kg bolus followed by infusion is the specific antidote for bupivacaine cardiotoxicity. Bupivacaine is contraindicated for intravenous regional anesthesia (Bier block) due to risk of systemic toxicity.

Cannot Intubate Cannot Oxygenate

– The most critical airway emergency — inability to intubate the trachea and inability to oxygenate the patient by face mask or supraglottic airway despite optimal efforts. CICO is a life-threatening situation requiring immediate surgical airway access. Signs: SpO2 decreasing despite 100% O2, absent end-tidal CO2, no chest rise. Management (CICO algorithm): declare CICO, call for surgical airway, perform scalpel cricothyroidotomy (vertical skin incision, horizontal membrane incision, tracheal hook, insert endotracheal tube or bougie). Alternative: needle cricothyroidotomy with jet ventilation (temporizing). Prevention: early recognition of difficult airway, awake techniques, and capnography confirmation of every intubation.

Carbon Dioxide Absorber

– The carbon dioxide absorber in the Circle breathing system contains soda lime or baralyme, which chemically reacts with exhaled CO2 to form calcium carbonate and water, allowing gas recycling. Soda lime contains calcium hydroxide with sodium hydroxide and potassium hydroxide as activators, and phenolphthalein as indicator (turns purple with CO2 absorption and desiccation). Desiccated CO2 absorbers react with sevoflurane to produce Compound A, a fluorinated vinyl compound nephrotoxic in rats. Clinical significance in humans is uncertain, but fresh gas flows below 2 L/min should be avoided with sevoflurane. Baralyme is more reactive than soda lime and is no longer available in most regions.

Chronic Pain

– Pain persisting longer than three months or beyond normal tissue healing time, often lacking an ongoing identifiable cause. It involves complex interactions between biological, psychological, and social factors that can amplify pain signaling in the nervous system. Management is multimodal, combining medications, physical therapy, cognitive behavioral therapy, nerve blocks, and complementary approaches.

Chronic Pain Management

– Chronic pain management extends beyond the perioperative period. Opioid therapy for chronic pain requires careful patient selection, informed consent, and regular reassessment. Multimodal approach includes NSAIDs, acetaminophen, gabapentinoids, antidepressants (SNRIs, TCAs), topical agents (lidocaine patches, capsaicin), and interventional procedures (nerve blocks, radiofrequency ablation, spinal cord stimulation). Opioid rotation (switching to a different opioid) is used for tolerance or side effects; methadone is used for both pain and opioid use disorder due to its long half-life (24-36 hours) and NMDA receptor antagonism, which may reduce tolerance. Buprenorphine is a partial mu-opioid agonist with ceiling effect on respiratory depression, making it safer in overdose but with a theoretical risk of precipitating withdrawal. Adjuvant therapies include physical therapy, cognitive behavioural therapy, and neuromodulation (spinal cord stimulation, peripheral nerve stimulation). Long-term opioid use carries risks of hypogonadism, immunosuppression, and opioid-induced hyperalgesia.

Colloids

– Solutions containing large molecules that remain in the intravascular space, providing greater volume expansion per milliliter than crystalloids. Albumin (5 percent, 25 percent) is a natural plasma protein used for volume expansion and specific indications (cirrhosis with paracentesis, nephrotic syndrome). Synthetic colloids include hydroxyethyl starch (HES), which is associated with acute kidney injury, coagulopathy, and increased mortality, leading to restricted use. Dextran (40 and 70 kDa) and dextran 70 (70 kDa) — are associated with anaphylactic reactions, interference with blood cross-matching, and impaired platelet function. Gelatin (succinylated gelatin, polygeline) is a colloid with lower molecular weight and shorter duration of action, rarely used due to risk of anaphylactoid reactions. The SAFE study demonstrated no mortality benefit of albumin over saline in ICU patients, and albumin use is limited to specific indications due to its cost and lack of proven benefit over crystalloids in most clinical settings.

Cricoid Pressure

– Cricoid pressure (Sellick maneuver) applies posterior displacement of the cricoid cartilage against the cervical vertebrae, occluding the esophagus and reducing aspiration risk during rapid sequence intubation. The technique involves applying 30 N pressure (20 N once consciousness is lost) with the thumb and forefinger on the cricoid cartilage. Evidence for effectiveness is mixed, with some studies showing no reduction in aspiration and potential difficulty with laryngoscopy. Recent guidelines ssuggest individualising cricoid pressure and considering alternative techniques (laryngeal mask airway, video laryngoscopy, suction-assisted laryngoscopy) in difficult situations.

Crystalloids

– Aqueous solutions of mineral salts that distribute between intravascular and interstitial compartments. Normal saline (0.9 percent NaCl) has 154 mEq/L each of sodium and chloride, pH 5.4, and is associated with hyperchloremic metabolic acidosis at large volumes. Lactated Ringer's contains sodium 130, potassium 4, calcium 3, chloride 109, and lactate 28 mEq/L, with pH 6.5. Lactate is metabolized to bicarbonate, making it physiologically balanced. Balanced crystalloids are preferred over normal saline for large-volume resuscitation (>2 L) to reduce the risk of hyperchloraemic metabolic acidosis and acute kidney injury, though isotonic saline remains the most widely available crystalloid worldwide and is appropriate for most clinical situations.

Delayed Emergence

– Prolonged time to regain consciousness after discontinuation of anesthetic agents. Failure to regain consciousness within 30-60 minutes of completing surgery. Causes: anesthetic drug overdose, residual volatile anesthetic (closed circuit, long duration), synergistic sedative effects (benzodiazepines, opioids), residual neuromuscular blockade, metabolic (hypoglycemia, hyponatremia, hypothermia, hypoxia, hypercarbia), and CNS injury (stroke, cerebral edema, seizure). Workup: ABCs, check TOF, end-tidal gas, temperature, glucose, electrolytes, and arterial blood gas. CT head and EEG for suspected neurologic injury.

Delirium in Anesthesia

– Delirium is an acute confusional state characterized by fluctuating consciousness, inattention, disorganized thinking, and altered level of awareness. Postoperative delirium occurs in 15 to 50 percent of elderly surgical patients, with higher rates in hip fracture and cardiac surgery. Risk factors include advanced age, preexisting cognitive impairment, sensory deprivation, sleep disruption, pain, opioids, anticholinergic drugs, and metabolic derangements. Delirium increases length of stay, complications, and mortality; prevention focuses on early recognition of at-risk patients, optimisation of pain and sleep, minimising use of anticholinergic medications, providing sensory aids (glasses, hearing aids), early mobilisation, and family involvement in care.

Desflurane

– A volatile inhalation anesthetic with the lowest blood-gas partition coefficient (0.42) among commonly used agents, providing the fastest induction and emergence of all volatile anesthetics. MAC is 6 percent at sea level. Desflurane causes airway irritation and can trigger sympathetic stimulation (tachycardia, hypertension) if inspired concentration exceeds 1.3 MAC abruptly. It is not suitable for inhalation induction in adults due to pungency. Desflurane is metabolized minimally (0.02 percent) and is the most resistant volatile to biodegradation, with negligible metabolism and no known organ toxicity, though it is a potent greenhouse gas.

Dexmedetomidine

– A highly selective alpha-2 adrenergic agonist providing sedation, anxiolysis, and analgesic sparing effects without significant respiratory depression. It produces cooperative sedation (patients can be aroused and followed commands) resembling natural sleep. Dexmedetomidine is used for procedural sedation, ICU sedation, and as an adjunct to general anesthesia. Dose: loading dose 1 mcg/kg over 10 minutes, then maintenance 0.2 to 0.7 mcg/kg/hour. Side effects include bradycardiaa, hypotension, and dry mouth. Bradycardia is the most common adverse effect, typically responding to dose reduction or anticholinergic agents (glycopyrrolate, atropine). Dexmedetomidine is unique among sedatives in providing sedation with minimal respiratory depression while maintaining a patent airway and spontaneous ventilation, making it particularly useful in the intensive care unit for patients requiring sedation during mechanical ventilation.

Difficult Airway

– A clinical situation in which a trained anesthesiologist or emergency physician encounters difficulty with bag-mask ventilation, supraglottic airway placement, laryngoscopy, or endotracheal intubation. Difficult mask ventilation: inability to maintain SpO2 >92% using 100% oxygen and positive pressure with a mask. Difficult laryngoscopy: Cormack-Lehane grade III or IV view (only epiglottis or no glottic structures visible). Difficult intubation: >3 attempts or >10 minutes required. Predictors: LEMON (Look, Evaluate, Mallampati, Obstruction, Neck mobility), 3-3-2 rule, upper lip bite test, obesity, and OSA. Management: ASA difficult airway algorithm; call for help, optimize position, use video laryngoscope, bougie, and consider awake fiberoptic intubation for anticipated difficult airway.

Difficult Intubation

– Inability to perform direct laryngoscopy or endotracheal intubation despite appropriate expertise and equipment. Predictors include Mallampati Class III or IV, limited mouth opening (less than 3 cm), limited neck mobility, short thyromental distance (less than 6.5 cm), obesity, beard, facial trauma, and prior difficult intubation. The difficult airway algorithm emphasizes preoperative assessment, preparation, and sequential approach: preoxygenation, direct laryngoscopy with Macintosh blade, bougie, video laryngoscopy, supraglottic airway (intubating LMA, i-gel), and surgical airway (cricothyroidotomy). The cannot intubate, cannot oxygenate (CICO) situation requires immediate surgical airway access.

Emergence Agitation

– A state of confusion, restlessness, and inconsolable crying during emergence from anesthesia, occurring in 10 to 80 percent of pediatric patients and 5 to 10 percent of adults. Risk factors include sevoflurane induction, short procedures, parental separation, and preexisting behavioral disorders. Prevention includes gentle handling, parental presence during emergence, adequate analgesia, and minimizing stimulation. Treatment involves ensuring adequate oxygenatioation, adequate analgesia, and minimal stimulation. Treatment may require pharmacological intervention with propofol (0.5-1 mg/kg), midazolam (0.05 mg/kg), or dexmedetomidine (0.5 mcg/kg) in severe cases.

Epidural Analgesia

– Epidural analgesia provides segmental pain relief by infusing local anesthetic (bupivacaine 0.1 to 0.2 percent) with or without opioids (fentanyl 2 to 4 mcg/mL, hydromorphone 10 to 20 mcg/mL) through an epidural catheter. Indications include major abdominal, thoracic, and orthopedic surgery, and labor analgesia. Epidural analgesia reduces postoperative pulmonary complications, ileus, and thromboembolic events. Side effects include hypotension (sympathetic blockade), urinary retention, pruritus (uritus (opioid-related), and motor blockade. Epidural haematoma and abscess are rare but serious complications requiring urgent neurosurgical consultation. Contraindications include coagulopathy, local infection at insertion site, increased intracranial pressure, patient refusal, and severe hypovolaemia.

Epidural Anesthesia

– A regional anesthetic technique involving injection of local anesthetic and/or opioid into the epidural space (the potential space between the dura mater and the ligamentum flavum within the spinal canal), producing segmental analgesia and sympathetic blockade. Indications: labor analgesia (most common use), major abdominal, thoracic, and orthopedic surgery (provides intraoperative and postoperative analgesia), and chronic pain management. Common local anesthetics: bupivacaine 0.0625–0.125%, ropivacaine 0.1–0.2%, lidocaine 1–2%. Opioids (fentanyl, morphine) are added to enhance analgesia and reduce local anesthetic dose. Side effects: hypotension (sympathetic blockade), motor blockade, urinary retention, pruritus (opioid-related), nausea. Serious complications: epidural hematoma (risk increased with anticoagulation), epidural abscess, post-dural puncture headache, and nerve injury. Contraindicated in coagulopathy, infection at the insertion site, increased intracranial pressure, and patient refusal.

Esmolol

– An ultra-short-acting cardioselective beta-1 adrenergic blocker with onset of 1 to 2 minutes and half-life of 9 minutes, metabolized by red blood cell esterases. It is administered as IV infusion (loading dose 500 mcg/kg over 1 minute, then 50 to 300 mcg/kg/min) for titratable heart rate and blood pressure control. Esmolol is used for intraoperative tachycardia, hypertension, arrhythmias, and prevention of extubation-induced sympathetic response. It is ideal for patients requiring rappid emergence and recovery, making it suitable for short procedures, day-case surgery, and patients undergoing procedures where rapid neurological assessment is needed.

Etomidate

– An intravenous induction agent with rapid onset (30 seconds) and short duration of action. It is hemodynamically neutral, making it the induction agent of choice in hemodynamically unstable patients, trauma, and sepsis. Etomidate acts as a GABA-A receptor agonist producing unconsciousness without significant cardiovascular effects. It decreases cerebral metabolic rate and intracranial pressure. Etomidate suppresses adrenal steroidogenesis by inhibiting 11-beta-hydroxylase, reducing cortisol and aldosterone levels for 12-24 hours after a single induction dose, which may be clinically significant in septic patients and those with adrenal insufficiency; this has led to controversy about its use in septic shock and increased preference for ketamine as an alternative in haemodynamically unstable patients.

Extubation Criteria

– Extubation requires the patient to meet specific criteria: fully awake and following commands, adequate tidal volume and respiratory rate, strong cough and gag reflex, adequate oxygenation (SpO2 greater than 95 percent on room air or baseline), hemodynamic stability, and return of neuromuscular function (TOF ratio greater than 0.9, sustained head lift for 5 seconds). Extubation should be performed at light anesthesia or fully awake. Awake extubation is preferred in difficult airway, obese, or high risk of aspiration (obstetric patients, obesity, gastro-oesophageal reflux disease, full stomach). Complications of extubation include laryngospasm, bronchospasm, aspiration, airway edema, and hypoxemia. Use of airway exchange catheters is recommended in patients with a known difficult airway to allow re-intubation if needed.

Failed Intubation

– Inability to successfully place an endotracheal tube after multiple attempts. In the operating room, a failed intubation algorithm is followed: call for help, maintain oxygenation (face mask, supraglottic airway), and decide to wake the patient (if non-emergent) or proceed with a supraglottic airway (if surgery is urgent). In the emergency department or ICU: difficult airway cart, video laryngoscopy, bougie, and surgical airway (cricothyroidotomy) if cannot intubate and cannot oxygenate. Failed intubation is more common in obstetric anesthesia (1 in 200-300) due to airway changes in pregnancy.

Fasting Guidelines

– Preoperative fasting guidelines minimize aspiration risk while reducing patient discomfort and catabolism. Clear liquids (water, clear juice without pulp, black coffee, sports drinks) are allowed up to 2 hours before surgery. Breast milk up to 4 hours. Infant formula up to 6 hours. Light meal (toast and clear liquid) up to 6 hours. Heavy/fatty meal up to 8 hours. The American Society of Anesthesiologists (ASA) fasting guidelines are evidence-based. Preoperative carbohydrate loading (clear carbohohydrate drink) 2-3 hours before surgery reduces perioperative insulin resistance, thirst, and hunger without increasing aspiration risk, supporting enhanced recovery after surgery protocols.

Fentanyl

– A synthetic opioid analgesic with potency 50 to 100 times that of morphine, rapid onset (30 seconds IV), and short duration of action (30 to 60 minutes). It is lipid-soluble, allowing rapid brain penetration. Fentanyl is used for intraoperative analgesia, patient-controlled analgesia, and regional analgesia (epidural or intrathecal). It causes dose-dependent respiratory depression, bradycardia, and chest wall rigidity at rapid high-dose administration. Fentanyl does not cause histamiamine release, making it haemodynamically stable. Intraoperative doses are 1-2 mcg/kg IV; patient-controlled analgesia bolus is 10-20 mcg IV with lockout of 6-10 minutes.

Fentanyl Derivatives

– Opioid derivatives used in anesthesia include fentanyl (most common, 50 to 100 mcg IV), sufentanil (5 to 10 times more potent than fentanyl, 5 to 10 mcg IV, used in cardiac anesthesia), remifentanil (ultra-short-acting, metabolized by esterases, 0.1 to 0.5 mcg/kg/min infusion), alfentanil (shorter duration than fentanyl, used for brief procedures), and carfentanil (10,000 times more potent than fentanyl, used for wildlife immobilization). All opioids cause dose-dependent respiratory depression, histamine release, hypotension, and bradycardia at high doses and rapid administration. Fentanyl is lipid-soluble, allowing rapid brain penetration and making it suitable for transdermal delivery (fentanyl patch) for chronic pain. Fentanyl causes chest wall rigidity at rapid high-dose administration (greater than 150 mcg IV bolus), requiring positive pressure ventilation or neuromuscular blockade to overcome.

General Anesthesia

– A controlled, reversible state of unconsciousness, analgesia, amnesia, and muscle relaxation induced by pharmacologic agents, enabling surgical procedures without patient awareness or pain. The components of general anesthesia are induction (rapid onset of unconsciousness), maintenance (sustained anesthetic state), and emergence (recovery of consciousness). Induction agents include propofol (1.5 to 2.5 mg/kg IV, onset 15 to 30 seconds, duration 5 to 10 minutes, the most common induction agent), etomidate (0.2-0.3 mg/kg, haemodynamically neutral), and ketamine (1-2 mg/kg, dissociative anesthetic producing sympathetic stimulation and bronchodilation). Maintenance: volatile anesthetics (sevoflurane, isoflurane, desflurane) or total intravenous anesthesia (TIVA) with propofol and remifentanil infusions. Muscle relaxation can be achieved with depolarizing (succinylcholine) or non-depolarizing (rocuronium, vecuronium, atracurium) neuromuscular blocking agents. Emergence involves discontinuing anesthetics, reversing muscle relaxation, and extubating the patient once criteria are met.

Geriatric Anesthesia

– Geriatric patients have decreased physiologic reserve and increased anesthesia risk. Age-related changes include decreased cardiac output, reduced pulmonary function, decreased renal function, altered pharmacokinetics (reduced volume of distribution, decreased hepatic metabolism), and increased brain sensitivity to anesthetics. Frailty assessment predicts postoperative complications. Delirium is a common postoperative complication, particularly in patients with dementia. Minimally invasive technchniques, regional anesthesia, and multimodal analgesia improve outcomes. Preoperative optimisation of comorbidities reduces perioperative risk. Postoperative cognitive dysfunction and functional decline are concerns requiring appropriate discharge planning and follow-up.

Glycopyrrolate

– An anticholinergic agent used to counteract muscarinic side effects of neostigmine and other cholinesterase inhibitors. It does not cross the blood-brain barrier, unlike atropine, producing fewer central side effects. Glycopyrrolate is administered at 0.2 to 0.4 mg IV for each 1 mg of neostigmine. It causes dry mouth, urinary retention, and tachycardia. Glycopyrrolate is also used preoperatively to reduce secretions and prevent bradycardia during vagal stimulation. It has a slower onset of action compared to atropine (2-3 minutes IV). It does not effectively cross the placenta and has minimal effect on fetal heart rate, making it preferred for obstetric anesthesia.

Hypertension in Anesthesia

– Hypertension during anesthesia causes increased bleeding, myocardial ischemia, and stroke risk. Common causes include light anesthesia, pain, hypercarbia, hypoxia, bladder distension, shivering, and drug withdrawal (beta-blockers, clonidine). Treatment involves deepening anesthesia, providing analgesia (fentanyl, remifentanil), and antihypertensive agents (esmolol for tachycardia-associated hypertension, labetalol, nicardipine, nitroglycerin). In neurosurgery, blood pressure control prevents cerebral perfusion pressure management and avoidance of hypertension-induced bleeding.

Hyperthermia in Anesthesia

– Perioperative hyperthermia (temperature greater than 38 degrees Celsius) has multiple causes. Malignant hyperthermia (discussed separately) is the most urgent. Sepsis, thyroid storm, serotonin syndrome, neuroleptic malignant syndrome, and central fever are other causes. Fever increases metabolic rate by 10 to 13 percent per degree Celsius, increasing oxygen consumption and carbon dioxide production. Treatment involves identifying and addressing the underlying cause. Antipyretics (acetaminophen, and NSAIDs) are used. Cooling measures include forced air warming (paradoxically), cold intravenous fluids, ice packs, and cooling blankets. Avoid shivering if possible (meperidine 12.5-25 mg IV, dexmedetomidine). Identify and treat the underlying cause: dantrolene for malignant hyperthermia, beta-blockers for thyroid storm, cyproheptadine for serotonin syndrome.

Hypotension in Anesthesia

– Mean arterial pressure less than 65 mmHg or a decrease greater than 20 percent from baseline during anesthesia. Causes include anesthetic depth (volatile agents, propofol, opioids), hypovolemia (fasting, blood loss, third spacing), vasodilation (neuraxial blockade, sepsis, anaphylaxis), and cardiac dysfunction (myocardial depression, arrhythmias). Treatment involves identifying and addressing the cause: reducing anesthetic depth, fluid resuscitation, vasopressors (phenylephrine 50-100 mcg IV, ephedrine 5-10 mg IV, norepinephrine infusion), and inotropic support (epinephrine, dobutamine) if cardiac dysfunction is suspected or confirmed.

Hypothermia in Anesthesia

– Hypothermia (core temperature less than 36 degrees Celsius) occurs in 50 to 80 percent of surgical patients due to anesthetic-induced thermoregulatory dysfunction and heat redistribution. Initial phase (first 30 to 60 minutes) involves core-to-peripheral heat redistribution causing 0.5 to 1.5 degree Celsius drop. Prolonged surgery causes heat loss through radiation, convection, evaporation, and conduction. Hypothermia increases surgical site infection risk, prolongs drug metabolism, causes coaguagulopathy, and increases transfusion requirements. Prevention includes forced-air warming blankets, warmed intravenous fluids, and maintaining operating room temperature above 21 degrees Celsius.

Inotropes in Anesthesia

– Inotropes increase myocardial contractility. Dobutamine is a beta-1 and beta-2 agonist increasing cardiac output with minimal effect on heart rate. Dose: 2 to 20 mcg/kg/min infusion. Milrinone is a phosphodiesterase-3 inhibitor increasing contractility and causing vasodilation. Dose: loading dose 50 mcg/kg over 10 minutes, then 0.375 to 0.75 mcg/kg/min. Epinephrine has potent beta-1 inotropic effects at low doses and alpha effects at high doses. Dose: 0.01 to 0.5 mcg/kg/min. Inotropes are used for cardiac dysfunction (myocardial depression, heart failure) and vasodilatory shock (sepsis, anaphylaxis). Adverse effects of inotropes include arrhythmias, increased myocardial oxygen consumption, and tolerance with prolonged use.

Intraoperative Blood Transfusion

– Intraoperative blood transfusion replaces lost blood volume and maintains oxygen-carrying capacity during major surgery. Indications include hemoglobin less than 7 g/dL in stable patients, less than 8 g/dL in cardiac disease, or active hemorrhage with hemodynamic instability. Packed red blood cells are the most commonly transfused component, with each unit increasing hemoglobin by approximately 1 g/dL. Massive transfusion protocol (greater than 10 units in 24 hours or greater than 4 units in 1 hhour) involves predefined ratios of PRBC, FFP, and platelets (1:1:1). Complications of massive transfusion include hypothermia, acidosis, hypocalcaemia (citrate toxicity), hyperkalaemia, and dilutional coagulopathy. Point-of-care testing (TEG, ROTEM) guides component therapy in real time.

Intraoperative Neuromonitoring

– Intraoperative neuromonitoring (IONM) uses electrophysiological techniques to monitor neural pathway integrity during surgery. Somatosensory evoked potentials (SSEP) monitor sensory pathways. Motor evoked potentials (MEP) monitor motor pathways. Electromyography (EMG) monitors cranial nerves and spinal roots. IONM is used in spine surgery (decompressive procedures, scoliosis correction), vascular surgery (carotid endarterectomy, aortic aneurysm repair), and skull base surgery. Changes in IONM signals require prompt communication with the surgical team to guide intervention (e.g., reducing retraction, altering surgical approach, or completing resection). anesthetic management must be tailored to IONM requirements, avoiding neuromuscular blocking agents during MEP monitoring and using total intravenous anesthesia (propofol and remifentanil) to minimise interference with evoked potential monitoring.

Intraoperative Temperature Monitoring

– Core temperature monitoring is essential during prolonged or major surgery. Sites include esophageal (most accurate for cardiac surgery), nasopharyngeal, bladder (with continuous irrigation), tympanic, and pulmonary artery (gold standard, requires PA catheter). Skin temperature (forehead, axilla) reflects peripheral temperature and is inaccurate for core estimation. Temperature probes are placed before induction for baseline measurement. Continuous monitoring allows early detection of hypothermiermia, which is common in prolonged surgery. Forced-air warming blankets are the most effective active warming method. Maintaining normothermia is essential for patient comfort, wound healing, and coagulation.

Isoflurane

– A volatile inhalation anesthetic with blood-gas partition coefficient of 1.4 and MAC of 1.15 percent. It was the most commonly used volatile anesthetic for many years but has been largely replaced by sevoflurane and desflurane. Isoflurane causes dose-dependent vasodilation and myocardial depression, maintaining cardiac output relatively well. It increases cerebral blood flow and intracranial pressure. Isoflurane is metabolized approximately 0.2 percent by CYP2E1 to inorganic fluoride (free fluoride ion) and trifluoroacetic acid, but hepatic metabolism is not clinically significant at usual doses. Isoflurane preserves hepatic and renal blood flow. It is a bronchodilator and is suitable for patients with reactive airway disease.

Ketamine

– A dissociative anesthetic acting as a non-competitive NMDA receptor antagonist, producing profound analgesia and amnesia while maintaining airway reflexes and spontaneous ventilation. It causes sympathetic stimulation, increasing heart rate, blood pressure, and cardiac output, making it ideal for hypotensive or traumatized patients. Ketamine is a bronchodilator and is used in asthmatic patients. It increases cerebral blood flow and intracranial pressure, limiting use in neurosurgery.Ketamine increases cerebral blood flow and intracranial pressure, limiting its use in neurosurgery. It is also used for treatment-resistant depression, acute and chronic pain, and as a bronchodilator in status asthmaticus. Psychotomimetic effects (emergence phenomena, hallucinations) are reduced in children and with benzodiazepine co-administration.

Laryngospasm

– A protective reflex causing violent adduction of the vocal cords, resulting in complete airway obstruction. It is most common during light anesthesia, emergence, and in pediatric patients. Triggers include secretions, blood, airway manipulation, and suctioning. Signs include paradoxical chest-abdominal movement, suprasternal retractions, and absent air entry. Treatment includes positive pressure ventilation with 100 percent oxygen via face mask, jaw thrust, and deepening anesthesia with propofol (0.5-1 mg/kg IV) or succinylcholine (0.1-0.5 mg/kg IV). If oxygenation fails, cricothyroidotomy or tracheostomy is required. Prevention includes maintaining adequate depth of anesthesia before airway manipulation and avoiding airway stimulation during light planes of anesthesia.

Lidocaine

– An amide local anesthetic and Class IB antiarrhythmic drug. As a local anesthetic, it is used for infiltration, nerve blocks, epidural, and spinal anesthesia (onset 5 to 10 minutes, duration 60 to 120 minutes). It is often combined with epinephrine (1:200,000) to prolong duration and reduce systemic absorption. Maximum dose is 4.5 mg/kg without epinephrine and 7 mg/kg with epinephrine. Lidocaine reduces intraocular pressure and is used to blunt airway reflexes during laryngoscopy (11.5 mg/kg IV prior to laryngoscopy). Lidocaine undergoes hepatic metabolism by CYP3A4 and has an elimination half-life of 1.5-2 hours which is prolonged in heart failure and hepatic disease. Toxicity (perioral numbness, metallic taste, tinnitus, seizures, arrhythmias) occurs at plasma concentrations greater than 5 mcg/mL, progressing to CNS depression and cardiovascular collapse at levels greater than 10 mcg/mL.

Lipid Emulsion Therapy

– The antidote for local anesthetic systemic toxicity. The mechanism involves the lipid sink theory (lipid phase sequesters lipophilic local anesthetics from neural and cardiac tissue) and metabolic theory (lipid provides fatty acid substrate to ischemic myocardium). Dosing: bolus 1.5 mL/kg (approximately 100 mL for 70 kg adult) over 1 minute, followed by infusion 0.25 mL/kg/min. If cardiovascular instability persists, repeat bolus every 3 to 5 mininutes, up to maximum 10 mL/kg. Continue infusion until haemodynamic stability is restored. Lipid emulsion is effective for bupivacaine and other lipophilic local anesthetics. Prepare lipid emulsion immediately when performing regional anesthesia with high-risk agents. Propofol (lipid emulsion vehicle) is not a substitute for Intralipid therapy as it delivers insufficient lipid volume.

Local Anesthetic

– A medication that causes reversible loss of sensation in a specific, localized area of the body without affecting consciousness. Local anesthetics block voltage-gated sodium channels in nerve cell membranes, preventing action potential generation and propagation. The non-ionized (lipophilic) form crosses the nerve sheath; the ionized form binds to the sodium channel intracellularly, stabilizing it in the inactivated state. Small nerve fibers (A-delta and C, pain and temperature) are blocked before larger fibers (A-beta, touch and motor). Types: amides (lidocaine, bupivacaine, ropivacaine, prilocaine — metabolized in the liver) and esters (procaine, tetracaine, benzocaine — metabolized by plasma pseudocholinesterases). Maximum safe doses: lidocaine 4.5 mg/kg without epinephrine, 7 mg/kg with epinephrine; bupivacaine 2.25 mg/kg. Contraindications include allergy to same class, severe hepatic impairment (amides), and pseudocholinesterase deficiency (esters). Toxicity presents with CNS symptoms (perioral numbness, metallic taste, tinnitus, seizures, coma) followed by cardiovascular collapse (hypotension, arrhythmias, cardiac arrest). Treatment: stop injection, airway support, benzodiazepines for seizures, and Intralipid 20% for cardiovascular toxicity. Addition of epinephrine prolongs duration and reduces systemic absorption.

Local Anesthetic Systemic Toxicity

– A life-threatening complication caused by elevated plasma concentration of local anesthetic, most commonly from inadvertent intravascular injection or rapid absorption. Presents with progressive CNS symptoms (perioral numbness, metallic taste, tinnitus, agitation, seizures) followed by cardiovascular effects (arrhythmias, bradycardia, hypotension, cardiac arrest). Cardiovascular collapse typically precedes CNS symptoms with bupivacaine (highly cardiotoxic). Prevention: test dose, incremental injection, aspiration, and ultrasound guidance. Treatment: STOP injection, call for help, airway management, lipid emulsion therapy (Intralipid 20%: 1.5 mL/kg bolus, then 0.25 mL/kg/min infusion). Avoid vasopressin and calcium channel blockers. Continue CPR until lipid levels are therapeutic.

Malignant Hyperthermia

– A rare, life-threatening pharmacogenetic disorder of skeletal muscle triggered by volatile inhalation anesthetics and succinylcholine, with incidence of approximately 1 in 15,000 to 50,000 anesthetics. The pathophysiology involves a defect in the ryanodine receptor (RYR1) on the sarcoplasmic reticulum, causing uncontrolled calcium release and sustained muscle contraction. Early signs include rising end-tidal CO2 (the earliest and most sensitive sign), tachycardia, masseseter muscle rigidity (masseter spasm), hyperthermia (late and fulminant sign), and metabolic acidosis. Treatment: discontinue all triggering agents immediately, hyperventilate with 100% oxygen, administer dantrolene (2.5 mg/kg IV bolus, repeated as needed, up to 10 mg/kg), active cooling, correct metabolic acidosis, treat hyperkalaemia with insulin and glucose, and monitor end-tidal CO2 and core temperature. Dantrolene inhibits calcium release from the sarcoplasmic reticulum by blocking the ryanodine receptor. Post-crisis monitoring in intensive care for 24-48 hours is mandatory as recurrence can occur.

Mechanical Ventilation

– Mechanical ventilation provides respiratory support by delivering controlled or spontaneous breaths through an endotracheal tube or tracheostomy. Ventilator modes include volume-controlled ventilation (delivers set tidal volume regardless of airway pressure), pressure-controlled ventilation (delivers breaths at set inspiratory pressure, tidal volume varies based on compliance), pressure support ventilation (spontaneous breaths augmented by set pressure level), and synchronized intermittent mandandatory ventilation (SIMV) which delivers set mandatory breaths between spontaneous breaths. Lung-protective ventilation uses low tidal volumes (6-8 mL/kg ideal body weight), plateau pressure less than 30 cmH2O, positive end-expiratory pressure (PEEP 5-10 cmH2O), and permissive hypercapnia. Driving pressure (plateau pressure minus PEEP) greater than 15 cmH2O is associated with increased mortality in ARDS.

Mendelson Syndrome

– See Aspiration Pneumonitis. Mendelson syndrome is a historical eponym for aspiration pneumonitis, originally described in obstetric patients.

Midazolam

– A water-soluble benzodiazepine with rapid onset and short duration of action, used for anxiolysis, sedation, amnesia, and seizure control. It acts on GABA-A receptors producing dose-dependent sedation. Midazolam is commonly used for procedural sedation, premedication, and continuous sedation in intensive care. It causes dose-dependent respiratory depression, particularly when combined with opioids. Midazolam is metabolized hepatically by CYP3A4 to active metabolite 1-hydroxymidazolaolam, which has sedative activity contributing to its effects, particularly in patients with hepatic impairment and the elderly. Flumazenil is the specific benzodiazepine antagonist.

Monitored Anesthesia Care

– A distinct service where an anesthesia professional provides premedication, administration of analgesics and/or sedatives, continuous monitoring of the patient's cardiorespiratory status, and intervention for any emergent or unavoidable complications that may arise during a diagnostic or therapeutic procedure. MAC is used for endoscopy, colonoscopy, interventional radiology, and minor surgical procedures. The patient may progress from moderate sedation to deep sedation, where airway intervention may be required. MAC requires at least one qualified anesthesia professional continuously present to monitor vital signs, administer sedation and analgesia, and be prepared to manage the airway and haemodynamic emergencies at any time during the procedure.

Neostigmine

– An acetylcholinesterase inhibitor used to reverse non-depolarizing neuromuscular blockade. It prevents breakdown of acetylcholine at the neuromuscular junction, restoring neuromuscular transmission. Neostigmine is administered with an anticholinergic agent (glycopyrrolate or atropine) to prevent muscarinic side effects (bradycardia, salivation, bronchospasm). Dose is 0.04 to 0.07 mg/kg IV, maximum 5 mg. Neostigmine is ineffective against depolarizing blockade (succinylcholine) andshould not be used for routine reversal of rocuronium due to the availability of sugammadex, which provides faster and more predictable reversal without muscarinic side effects.

Neuromuscular Junction

– The neuromuscular junction is the synapse between motor neuron and skeletal muscle fiber, where acetylcholine released from presynaptic terminals binds to nicotinic acetylcholine receptors on the postsynaptic motor end plate, causing depolarization and muscle contraction. Acetylcholinesterase in the synaptic cleft breaks down acetylcholine, terminating the signal. Neuromuscular blocking agents interfere with this process: depolarizing agents (succinylcholine) cause persistent depolarization, whihile non-depolarizing agents (rocuronium, vecuronium, atracurium) competitively block acetylcholine binding, producing flaccid paralysis that can be reversed with anticholinesterases or sugammadex.

Nitrous Oxide

– A volatile inhalation anesthetic with blood-gas partition coefficient of 0.47 and MAC of 105 percent, meaning it cannot produce surgical anesthesia alone. It is typically used as an adjunct to other anesthetics, reducing the requirement for volatile agents by 30 to 40 percent. Nitrous oxide has analgesic properties at subanesthetic concentrations and is widely used in obstetric analgesia and dentistry. It diffuses into air-filled cavities, expanding trapped gas volumes andincreasing the risk of tension pneumothorax, bowel distension, and expansion of air emboli. Nitrous oxide causes diffusion hypoxia upon emergence, where nitrous oxide rapidly exits the blood into the alveoli, diluting alveolar oxygen; this is prevented by administering 100% oxygen during emergence. Chronic exposure causes vitamin B12 oxidation and inactivation of methionine synthase, leading to megaloblastic anaemia and peripheral neuropathy with prolonged use.

Opioid Reversal

– Achieved with naloxone, a competitive mu-opioid receptor antagonist. Dose: 0.04 to 0.1 mg IV, titrated to effect, repeat every 2 to 3 minutes. Maximum typical dose is 2 to 4 mg. Naloxone has shorter half-life (30 to 80 minutes) than most opioids, risking recurrent respiratory depression. Continuous infusion (2 to 4 mg/hour diluted in 500 mL saline) may be needed for long-acting opioids (methadone, sustained-release formulations). Nalmefene and naltrexone are longer-acting oralantagonists with half-lives of 4-10 hours and 4 hours respectively. Long-acting reversal agents should be used postoperatively and in monitored settings. Opioid reversal can precipitate acute withdrawal syndrome in opioid-dependent patients, characterized by agitation, hypertension, tachycardia, vomiting, and diarrhoea. Partial agonists and mixed agonist-antagonists (buprenorphine, nalbuphine) have ceiling effects on respiratory depression but can precipitate withdrawal in opioid-dependent patients.

Other Pain Conditions

– Other pain types encompass pain conditions not primarily classified as nociceptive (somatic or visceral) or classic neuropathic pain, including nociplastic pain (altered nociception without clear tissue or nerve damage, such as fibromyalgia, chronic tension-type headache, temporomandibular disorders, and chronic pelvic pain), complex regional pain syndrome (CRPS Type I and II—persistent regional pain disproportionate to inciting event with autonomic and motor features), central pain syndromes (post-stroke pain, spinal cord injury pain), and chronic visceral pain syndromes (interstitial cystitis/bladder pain syndrome, irritable bowel syndrome, chronic pancreatitis, endometriosis). These conditions share mechanisms of central sensitization, dysfunctional descending modulation, and psychologic contributors. Diagnosis is clinical, often requiring exclusion of alternative pathology. Management is multimodal: patient education, physical therapy, cognitive-behavioral therapy, medications (tricyclic antidepressants, gabapentinoids, SNRIs), and neuromodulation; opioids are generally ineffective and not recommended for chronic non-cancer pain.

Pain

– An unpleasant sensory and emotional experience associated with actual or potential tissue damage, serving as a protective alerting system but becoming maladaptive when chronic (lasting >3 months). The biopsychosocial model recognizes nociception (tissue damage), pain perception (sensory-discriminative, affective-motivational, cognitive-evaluative dimensions), and pain behaviors influenced by genetics, psychology, social context, and comorbidities. Classification: nociceptive (somatic—well-localized, aching; visceral—diffuse, cramping), neuropathic (nerve injury—burning, shooting, allodynia), and nociplastic (altered nociception without clear lesion—fibromyalgia). Acute pain resolves with healing; chronic pain affects 20% of adults and is a leading cause of disability. Assessment: PQRST (provoking/palliating, quality, region/radiation, severity, timing), functional impact, and psychologic screening (depression, catastrophizing, kinesiophobia). Management requires an individualized multimodal approach.

Pain Assessment Scales

– Pain assessment uses standardized scales to quantify pain intensity. Numeric rating scale (NRS) asks patients to rate pain from 0 (no pain) to 10 (worst pain imaginable). Visual analog scale (VAS) is a 10 cm line with anchors. Faces pain scale (Wong-Baker FACES) uses facial expressions for children and cognitively impaired patients. FLACC scale (Face, Legs, Activity, Cry, Consolability) assesses pain in non-verbal children. Critical-care Pain Observation Tool (CPOT) assesses pain in mechanicallyventilated patients in the intensive care unit. PAINAD (Pain Assessment in Advanced Dementia) assesses pain in elderly patients with dementia. Pain should be assessed at rest and with movement. Standardized scales improve pain documentation, treatment consistency, and outcomes. The goal is to maintain pain scores below 3-4 out of 10, allowing the patient to function and participate in recovery.

Pain Relief (Analgesia)

– Achieved through pharmacologic, interventional, physical, and psychologic modalities targeting nociceptive transmission and pain perception. Pharmacologic: World Health Organization (WHO) analgesic ladder recommends non-opioids (acetaminophen, NSAIDs) for mild pain, weak opioids (tramadol, codeine) for moderate pain, and strong opioids (morphine, oxycodone, hydromorphone, fentanyl) for severe pain, with adjuvant medications (antidepressants, anticonvulsants, local anesthetics) for neuropathic pain. Interventional procedures: nerve blocks (facet, epidural, peripheral nerve), radiofrequency ablation, spinal cord stimulation, intrathecal pumps. Physical modalities: ice/heat, transcutaneous electrical nerve stimulation (TENS), physical therapy, massage, acupuncture. Psychologic: cognitive-behavioral therapy, biofeedback, relaxation techniques, mindfulness meditation. Opioid prescribing requires careful risk-benefit assessment given addiction potential (especially with chronic use), respiratory depression, tolerance, and opioid-induced hyperalgesia. Multimodal analgesia—combining agents with different mechanisms—improves efficacy and reduces opioid requirements (opioid-sparing effect).

Patient-Controlled Analgesia

– Patient-controlled analgesia (PCA) allows patients to self-administer small doses of analgesic medication within preset safety limits. IV PCA typically uses morphine (demand 1 mg, lockout 6 to 10 minutes, hourly limit 4 mg), hydromorphone (demand 0.2 mg, lockout 6 to 10 minutes), or fentanyl (demand 10 to 20 mcg, lockout 6 to 10 minutes). PCA provides more consistent pain control than nurse-administered boluses and gives patients a sense of control. Basal infusion is controversial and increases the risk of respiratory depression, especially in opioid-naive patients and those with obstructive sleep apnea. PCA by proxy (family members or nurses activating the button) is dangerous and should be avoided. Contraindications include patient inability to understand the device (cognitive impairment, age <5 years), allergy, and opioid tolerance. Epidural PCA (PCEA) combines a background infusion with patient-controlled boluses and is used for postoperative and labour analgesia.

Pediatric Anesthesia

– Pediatric anesthesia requires age-appropriate dosing, equipment, and techniques. Children have higher metabolic rates, increased oxygen consumption, smaller functional residual capacity, and higher risk of desaturation. Airway anatomy differs (larger tongue, more anterior larynx, narrower subglottic area). Induction is commonly performed via inhalation (sevoflurane in oxygen) or intravenous. Dosing is based on weight (mg/kg) with adjustments for neonates and infants. Temperature management is critical in neonates and infants due to high surface area-to-volume ratio. Emergence delirium is more common in children (sevoflurane) and is managed with propofol, dexmedetomidine, or gentle awakening. Postoperative nausea and vomiting is also common and treated with ondansetron, dexamethasone, and total intravenous anesthesia.

Peripheral Nerve Blocks

– Peripheral nerve blocks provide targeted anesthesia and analgesia for specific surgical sites. Ultrasound-guided blocks improve success rates and reduce complications. Common blocks include interscalene (shoulder surgery), supraclavicular (arm surgery), infraclavicular (arm and hand), axillary (hand surgery), femoral (knee surgery), adductor canal (knee surgery with preserved quadriceps function), sciatic (foot and ankle), popliteal (foot and ankle), and ilioinguinal/iliohypogastric (inguinal hernia repair). Local anesthetics (bupivacaine 0.25-0.5%, ropivacaine 0.2-0.5%, lidocaine 1-2%) are used with or without epinephrine to prolong block duration and reduce systemic absorption. Ultrasound guidance has become the standard of care, reducing the risk of inadvertent vascular puncture, intraneural injection, and pneumothorax while improving block success rates and reducing local anesthetic volume requirements.

Postanesthesia Care Unit

– The postanesthesia care unit (PACU) is a monitored recovery area where patients are observed during the immediate postoperative period. Admission assessment includes vital signs, level of consciousness, pain assessment, and surgical site evaluation. The Aldrete scoring system assesses recovery criteria: activity, respiration, circulation, consciousness, and oxygen saturation. Common PACU problems include postoperative nausea and vomiting (treated with ondansetron), pain (treated with PCA or regional pain management. Modified Aldrete scoring system uses 5 criteria: activity (2: moves all extremities; 1: moves 2 extremities; 0: moves 0), respiration (2: deep breaths/cough; 1: dyspnea; 0: apnea), circulation (2: BP +/- 20 of preop; 1: +/- 20-50; 0: +/- >50), consciousness (2: fully awake; 1: arousable; 0: unresponsive), and oxygen saturation (2: >92% on RA; 1: >92% with O2; 0: <92% with O2). Total score >9 indicates readiness for discharge. Discharge criteria include stable vital signs, adequate oxygenation, controlled pain, minimal nausea/vomiting, and ability to void (for spinal/epidural).

Post-Dural Puncture Headache

– A positional headache occurring after accidental dural puncture during epidural placement or lumbar puncture, caused by cerebrospinal fluid leakage and intracranial hypotension. Symptoms include frontal or occipital headache worse when upright and relieved when supine, neck stiffness, tinnitus, photophobia, and nausea. Incidence is 1 to 2 percent with 25 gauge pencil-point needles, 5 to 10 percent with 22 gauge cutting needles. Treatment includes conservative: bed rest, hydration, caffeine (500 mg IV or oral), and analgesics. Invasive: epidural blood patch (15-20 mL of autologous blood injected into the epidural space at the level of the dural puncture), which is highly effective (80-95% success rate). Prevention: use of pencil-point (Whitacre, Sprotte) instead of cutting needles (Quincke), small gauge (25-27 G), and orientation of the needle bevel parallel to the dural fibers.

Postoperative Cognitive Dysfunction

– Decline in cognitive function (memory, concentration, executive function) after surgery and anesthesia, more common in elderly patients. POCD can be early (weeks to months) or persistent (>1 year). Incidence: 25-40% in elderly patients at hospital discharge, 10-15% at 3 months. Risk factors: age, lower education level, preoperative cognitive impairment, prolonged surgery, respiratory complications, and postoperative delirium. Pathophysiology: neuroinflammation (surgical stress response, cytokines), anesthetics (GABAergic drugs), and microemboli. Prevention: minimize depth and duration of anesthesia, avoid anticholinergic drugs, early mobilization, and cognitive stimulation. Management: neuropsychological evaluation and cognitive rehabilitation.

Postoperative Delirium

– A common complication after major surgery, particularly in elderly patients. It is categorized as hyperactive (agitation, restlessness, hallucinations), hypoactive (lethargy, withdrawal, decreased responsiveness), or mixed. Hypoactive delirium is the most common and most missed form. The Confusion Assessment Method (CAM) is the standard diagnostic tool. Postoperative delirium is associated with increased mortality, longer hospital stay, and accelerated cognitive declinethat may persist for months to years. Prevention includes comprehensive geriatric assessment, preoperative cognitive screening, multidisciplinary team approach, and avoidance of high-risk medications (anticholinergics, benzodiazepines, meperidine). Management includes non-pharmacological measures (reorientation, family presence, early mobilization, sleep hygiene, sensory aids) and, if needed, low-dose antipsychotics (haloperidol 0.5-1 mg oral/IM, olanzapine, quetiapine) for hyperactive symptoms causing safety concerns.

Postoperative Nausea and Vomiting

– Nausea, retching, or vomiting occurring within 24-48 hours after surgery. PONV is one of the most common and distressing complications. Risk factors (Apfel score): female gender, non-smoker, history of motion sickness/PONV, and postoperative opioids. 0-1 risk factors: 10-20% incidence; 2: 40%; 3: 60%; 4: 80%. Prophylaxis: multimodal approach — 5-HT3 antagonists (ondansetron 4 mg), dexamethasone 4-8 mg, droperidol 0.625 mg, scopolamine patch, and total intravenous anesthesia with propofol. Treatment: 5-HT3 antagonist + dexamethasone + droperidol for rescue; consider NK1 antagonists (aprepitant) for high-risk patients.

Postoperative Pain Management

– Postoperative pain management uses multimodal analgesia combining pharmacologic and regional techniques. Opioids (morphine, hydromorphone, fentanyl) provide potent analgesia but cause respiratory depression, nausea, and ileus. Acetaminophen (1000 mg IV every 6 hours) provides baseline analgesia. NSAIDs (ketorolac, celecoxib) reduce opioid requirements. Gabapentinoids (gabapentin, pregabalin) address neuropathic pain. Local anesthetic wound infiltration and regional blocks provide targeted analgegesia. Regional techniques (epidural, spinal, peripheral nerve blocks) provide superior analgesia for specific surgical sites. Multimodal analgesia reduces opioid requirements and opioid-related side effects by targeting different pain pathways. The goal of postoperative pain management is to facilitate early mobilization, reduce pulmonary complications, and improve patient satisfaction.

Post-Surgical Pain

– Acute pain following operative procedures, caused by tissue incision, retraction, dissection, and inflammation, mediated by nociceptors sensitized by prostaglandins, bradykinin, and cytokines. Intensity varies by procedure (thoracotomy, joint replacement, and laparotomy produce more severe pain). Uncontrolled post-surgical pain increases risk of chronic postsurgical pain (CPSP, prevalence 10–50% depending on procedure), promotes complications (atelectasis, immobilization, thromboembolism, ileus, delayed wound healing), and prolongs hospital stay. Assessment: numeric rating scale (0–10) at rest and with movement. Multimodal analgesia (opioid-sparing): regional anesthesia (nerve blocks, neuraxial anesthesia, local infiltration), NSAIDs/acetaminophen (around-the-clock), gabapentinoids, ketamine (for opioid-tolerant patients), and patient-controlled analgesia (PCA) with opioids for moderate-severe pain. Enhanced Recovery After Surgery (ERAS) protocols integrate multimodal analgesia, early mobilization, and patient education.

Preoperative Assessment

– Preoperative assessment evaluates surgical risk and optimizes medical conditions before anesthesia. It includes medical history, physical examination, airway assessment, cardiovascular risk stratification (Revised Cardiac Risk Index), pulmonary function assessment, and laboratory testing based on clinical indication. The ASA Physical Status Classification System grades patient fitness: ASA I (healthy), ASA II (mild systemic disease), ASA III (severe systemic disease), ASA IV (life-threatening disease (ASA IV), moribund patient not expected to survive 24 hours (ASA V), and brain-dead organ donor (ASA VI). The ASA class correlates with perioperative mortality: class I (<0.1%), II (0.2%), III (1.8%), IV (7.8%), and V (9.4%). Preoperative testing should be guided by patient age, comorbidities, surgery type, and anesthesia technique rather than routine batteries; focused testing based on clinical indication is more cost-effective and avoids unnecessary interventions.

Propofol

– The most widely used intravenous induction and maintenance agent, acting as a GABA-A receptor agonist producing rapid onset (30 to 40 seconds) and short duration of action (5 to 10 minutes). It causes dose-dependent hypotension through vasodilation and decreased cardiac output, and is contraindicated in patients with propofol allergy (egg or soybean oil hypersensitivity). Propofol has antiemetic properties and is used for total intravenous anesthesia (TIVA) when combined with remifentanil infusion. Propofol infusion syndrome (rare but fatal) is associated with prolonged high-dose infusions (>4-5 mg/kg/hour for >48 hours) in children and critically ill patients, presenting as metabolic acidosis, rhabdomyolysis, renal failure, and cardiac failure. Propofol is formulated in a lipid emulsion (soybean oil, egg lecithin, glycerol) providing 1.1 kcal/mL and requiring aseptic handling to prevent infection.

Rapid Sequence Induction

– An intubation technique designed to minimize aspiration risk in patients with full stomach. Steps include preoxygenation with 100 percent oxygen for 3 to 5 minutes (8 vital capacity breaths or 3 minutes), application of cricoid pressure, administration of induction agent (propofol 2 mg/kg or etomidate 0.3 mg/kg) followed immediately by succinylcholine (1.5 mg/kg) or rocuronium (1.2 mg/kg), and intubation without bag-mask ventilation. If rocuronium is used, sugammadex (16 mg/kg) is available as a rescue reversal agent, providing rapid and complete reversal of rocuronium-induced neuromuscular blockade. RSI is indicated in full stomach, emergency surgery, obesity, pregnancy, gastrointestinal obstruction, and severe gastro-oesophageal reflux disease.

Regional Anesthesia

– Regional anesthesia involves the administration of local anesthetic agents to specific nerves or nerve plexuses to provide surgical anesthesia and postoperative analgesia for a defined body region. Spinal anesthesia involves injection of local anesthetic (bupivacaine 0.5 percent, 7.5 to 15 mg) into the subarachnoid space at the L3-L4 or L2-L3 interspace, producing rapid onset of dense motor, sensory, and sympathetic blockade below the level of injection. Epidural anesthesia involves placement ofa catheter into the epidural space, allowing continuous infusion of local anesthetic with or without opioids for surgical anesthesia and postoperative analgesia, with the advantage of titratable block level and duration. Contraindications to regional anesthesia include coagulopathy, infection at insertion site, patient refusal, increased intracranial pressure (dural puncture), and severe hypovolaemia.

Remifentanil

– An ultra-short-acting synthetic opioid analgesic metabolized by non-specific plasma and tissue esterases, producing context-sensitive half-time of approximately 3 to 10 minutes regardless of infusion duration. This makes it ideal for total intravenous anesthesia (TIVA) combined with propofol, and for procedures requiring rapid emergence. Remifentanil is 200 to 300 times more potent than morphine. It causes dose-dependent respiratory depression and bradycardia. Remifentanil-induceuced bradycardia and hypotension at bolus doses greater than 1 mcg/kg, and postoperative hyperalgesia at high infusion rates (>0.3 mcg/kg/min), requiring careful dose selection to balance opioid-induced hyperalgesia against inadequate analgesia.

Rocuronium

– A non-depolarizing aminosteroid neuromuscular blocking agent with onset of 60 to 90 seconds and duration of 30 to 45 minutes at intubating dose. It is the primary alternative to succinylcholine for rapid sequence intubation when administered at 1.2 mg/kg, providing comparable intubating conditions. Rocuronium does not cause histamine release, hyperkalemia, or malignant hyperthermia. It is metabolized hepatically and excreted in bile. Reversal is achieved with sugammadex (2 to 4 mg/kg) depending on depth of blockade. Sugammadex has largely replaced neostigmine for rocuronium reversal due to faster onset, predictable reversal regardless of blockade depth, and absence of muscarinic side effects. Rocuronium causes minimal cardiovascular effects and does not cause histamine release.

Ropivacaine

– A long-acting amide local anesthetic with sensory-motor dissociation, providing effective analgesia with less motor blockade than bupivacaine. It is used for epidural analgesia (0.2 percent), peripheral nerve blocks, and postoperative pain management. Ropivacaine is less cardiotoxic than bupivacaine and has a safer toxicity profile. Maximum dose is 3 mg/kg. Onset is 5 to 15 minutes, duration is 2 to 4 hours. Ropivacaine is available in 0.2 percent, 0.5 percent, and 1 percent concentrations. Ropivacaine is the least cardiotoxic amide local anesthetic and is preferred for high-volume peripheral nerve blocks and epidural infusions, particularly in ambulatory surgery where rapid recovery of motor function is desirable.

Scavenging System

– The anesthesia gas scavenging system collects and removes waste anesthetic gases from the operating room environment to protect staff from chronic exposure. Active scavenging uses negative pressure from vacuum source; passive scavenging vents to exhaust through gravity. The interface between the anesthesia machine and scavenging system includes a positive pressure relief valve (to prevent barotrauma) and negative pressure relief valve (to prevent excessive suction). Occupational exposure limits for nitrous oxide are 25 ppm (time-weighted average over 8 hours) and for volatile agents 0.5-2 ppm depending on the agent. Proper scavenging is a critical occupational health and safety requirement in all anaesthetising locations, with periodic environmental monitoring to verify adequate system function and compliance with exposure limits.

Sedation Levels

– Defined by the ASA continuum: minimal sedation (anxiolysis, normal response to verbal commands), moderate sedation (response to verbal or light tactile stimulation, airway patent, spontaneous ventilation adequate), deep sedation (purposeful response after repeated or painful stimulation, airway may require assistance, spontaneous ventilation inadequate), and general anesthesia (unarousable even with painful stimulus, airway intervention often required, spontaneous ventilationmay be inadequate.

The ASA recommends that practitioners be able to rescue patients from the level of sedation immediately above that intended. Sedation is administered by qualified anesthesia professionals or trained non-anesthesia personnel under appropriate supervision with monitoring of oxygen saturation, heart rate, blood pressure, and level of consciousness.

Sevoflurane

– A volatile inhalation anesthetic with low blood-gas partition coefficient (0.65), providing rapid induction and emergence. It is non-pungent and widely used for inhalation induction in pediatric patients. Minimum alveolar concentration (MAC) is 2 percent in adults, 2.5 percent in children. Sevoflurane causes dose-dependent myocardial depression and vasodilation. It is metabolized minimally (2 to 5 percent) by CYP2E1 to inorganic fluoride. Compound A is formed when sevoflurane reaccts with desiccated CO2 absorbents (soda lime, baralyme), producing Compound A, a vinyl ether that is nephrotoxic in rats at high concentrations, though clinical significance in humans at low fresh gas flows (>2 L/min) is debated; this concern has limited the use of low-flow anesthesia with sevoflurane.

Spinal Analgesia

– Spinal analgesia involves single-shot injection of local anesthetic and/or opioid into the subarachnoid space for rapid, dense sensory and motor blockade. Bupivacaine 0.5 percent (10 to 15 mg) provides surgical anesthesia lasting 2 to 4 hours. Intrathecal morphine (0.1 to 0.3 mg) provides postoperative analgesia for 12 to 24 hours but carries risk of delayed respiratory depression, particularly in obese or elderly patients. Spinal analgesia is used for lower extremity surgery, cesarean section, and urological procedures, and provides rapid onset and reliable blockade. Hypotension from sympathetic blockade is the most common complication, treated with intravenous fluids and vasopressors (phenylephrine, ephedrine). Post-dural puncture headache is a risk, particularly with larger-gauge or cutting needles.

Succinylcholine

– A depolarizing neuromuscular blocking agent with rapid onset (30 to 60 seconds) and short duration of action (5 to 10 minutes), making it the drug of choice for rapid sequence intubation. It acts by binding to nicotinic acetylcholine receptors at the neuromuscular junction, causing persistent depolarization and initial fasciculations followed by paralysis. Succinylcholine is metabolized by plasma pseudocholinesterase. Adverse effects include hyperkalemia (contraindicated in buburns, major trauma, neuromuscular disease (muscle atrophy), and malignant hyperthermia susceptibility. Succinylcholine causes increased intraocular, intragastric, and intracranial pressure. Fasciculations can be prevented by a small priming dose of a non-depolarizing muscle relaxant (defasciculating dose). Prolonged paralysis occurs in patients with pseudocholinesterase deficiency (genetic variant or acquired).

Sugammadex

– A selective relaxant binding agent that reverses aminosteroid neuromuscular blockers (rocuronium and vecuronium) by encapsulating them in a cyclodextrin molecule, forming a 1:1 complex that is renally excreted. Onset of action is within 2 to 3 minutes. Dosing: 2 mg/kg for moderate reversal (second dose of rocuronium), 4 mg/kg for rapid reversal (3 minutes after rocuronium), and 16 mg/kg for immediate reversal (within 3 minutes of rocuronium). Sugammadex eliminates the need for anticholinesterase drugs (neostigmine) for aminosteroid reversal, as it encapsulates the steroid molecule directly without affecting acetylcholinesterase, avoiding muscarinic side effects. Anaphylactic reactions to sugammadex have been reported but are rare. Sugammadex is the reversal agent of choice for rapid sequence intubation with rocuronium, providing reliable and complete reversal within 3 minutes at the 16 mg/kg dose.

Supraglottic Airway Devices

– Supraglottic airway devices (SADs) sit above the vocal cords, providing a seal around the laryngeal inlet without tracheal intubation. First-generation SADs (classic LMA, i-gel) provide airway seal but do not protect against aspiration. Second-generation SADs (LMA ProSeal, LMA Supreme, i-gel) include a gastric drainage port and higher sealing pressures, allowing ventilation and aspiration protection. SADs are used for short procedures, as rescue airway devices, and in difficult airway algorithmsalgorithms. SAD complications include airway obstruction, laryngospasm, and aspiration. Correct placement is confirmed by chest rise, capnography, and no leak. Contraindications include full stomach, reduced pulmonary compliance, and pharyngeal pathology.

Target-Controlled Infusion

– A computerized infusion pump system that maintains a predetermined drug concentration in plasma or effect site using pharmacokinetic modeling. TCI systems use three-compartment pharmacokinetic models to calculate infusion rates that achieve and maintain target concentration. Propofol TCI models include Marsh (plasma-targeted) and Schnider (effect-site targeted). Remifentanil TCI uses the Minto model. TCI provides more stable anesthetic depth compared to manualmanual infusion techniques by maintaining a stable predicted concentration, reducing haemodynamic variability and anesthetic waste. Limitations include the accuracy of pharmacokinetic models in individual patients (obesity, elderly, paediatrics, organ dysfunction), variability in drug potency, and the potential for infusion pump programming errors.

Total Intravenous Anesthesia

– Total intravenous anesthesia (TIVA) uses intravenous agents exclusively for induction and maintenance of anesthesia, avoiding volatile inhalation agents. The most common TIVA regimen combines propofol infusion (50 to 200 mcg/kg/min) with remifentanil infusion (0.1 to 0.5 mcg/kg/min). TIVA advantages include rapid emergence, reduced postoperative nausea and vomiting, suitability for patients with malignant hyperthermia risk, and use in procedures requiring volatile anesthetic avoidance. TIVA disadvantages include the need for dedicated infusion pumps, risk of propofol infusion syndrome with prolonged high-dose infusions, and the lack of an end-tidal volatile anesthetic concentration monitor to confirm anesthetic depth; BIS or entropy monitoring is typically used with TIVA to assess depth of anesthesia and reduce the risk of intraoperative awareness.

Train-of-Four Monitoring

– Train-of-four (TOF) monitoring assesses neuromuscular blockade by delivering four supramaximal electrical stimuli at 2 Hz to a peripheral nerve (usually ulnar nerve at the wrist). The ratio of the fourth twitch height to the first twitch height indicates the degree of blockade. TOF ratio of 0 indicates complete blockade, 0.25 indicates profound blockade, 0.5 indicates moderate blockade, and 0.7 to 0.9 indicates partial recovery. A TOF ratio greater than 0.9 is required for adequate neuromuscularrecovery before extubation, and for prevention of residual neuromuscular blockade. Residual blockade (TOF ratio <0.9) occurs in 20-40% of patients receiving non-depolarizing neuromuscular blockers and is associated with increased risk of postoperative pulmonary complications, hypoxemia, and aspiration, making objective TOF monitoring essential before extubation.

Tranexamic Acid

– An antifibrinolytic agent that inhibits plasminogen activation, reducing fibrinolysis and blood loss. TXA is used in trauma (CRASH-2 trial: 1 g IV over 10 minutes, then 1 g infusion over 8 hours), cardiac surgery (reduces re-exploration and transfusion), orthopedic surgery (joint replacement reduces blood loss by 30 to 50 percent), and heavy menstrual bleeding. Dose is 10 to 20 mg/kg IV. TXA is contraindicated in active intravascular clotting, history of thromboembolism,thromboembolism, and subarachnoid hemorrhage. Side effects are rare but include seizures, thrombotic events, and anaphylaxis. Routine use in cardiac surgery is supported by evidence, but its use in trauma beyond the first 3 hours is not recommended.

Vasopressors in Anesthesia

– Vasopressors maintain blood pressure during anesthesia-induced vasodilation. Phenylephrine is a pure alpha-1 agonist causing vasoconstriction without cardiac effects, used for pure vasodilatory hypotension. Dose: bolus 50 to 100 mcg IV or infusion 0.05 to 0.5 mcg/kg/min. Ephedrine stimulates alpha and beta receptors, increasing both heart rate and blood pressure. Dose: 5 to 10 mg IV bolus. Norepinephrine is a potent alpha-1 and beta-1 agonist, used for refractory hypotension and septic shock. Dose: 0.05 to 0.5 mcg/kg/min infusion. Dopamine acts on dopamine, beta, and alpha receptors in a dose-dependent manner: low dose (1-5 mcg/kg/min) mainly acts on dopamine receptors dilating renal and mesenteric vessels; intermediate dose (5-10 mcg/kg/min) acts on beta-1 receptors increasing contractility and heart rate; high dose (>10 mcg/kg/min) acts on alpha-1 receptors causing vasoconstriction.

Video Laryngoscopy

– Video laryngoscopy uses a camera-equipped blade to visualize the glottis on a screen, providing indirect laryngoscopy with improved glottic exposure. Common devices include GlideScope, C-MAC, and McGrath. Video laryngoscopy is particularly useful in Mallampati Class III or IV patients, limited neck mobility, and obesity. It provides a magnified, well-illuminated view of the vocal cords. The line-of-sight principle differs from direct laryngoscopy, requiring different technique (lifting anteriorlly and superior glottic view on screen). Video laryngoscopy is rapidly becoming the standard of care for both anticipated and unexpected difficult airways, and for routine intubation in many institutions due to its improved first-pass success rate and reduced oesophageal intubation rate compared to direct laryngoscopy.

Chapter 3

Biochemistry

Base Excision Repair

– The repair pathway for small, non-helix-distorting base lesions including oxidized bases (8-oxoguanine), deaminated bases (uracil from cytosine deamination), alkylated bases, and abasic (AP) sites. Steps: (1) a specific DNA glycosylase recognizes and removes the damaged base by hydrolyzing the N-glycosidic bond, creating an AP (apurinic/apyrimidinic) site. Different glycosylases recognize different lesions: uracil-DNA glycosylase (UNG) removes uracil from DNA; OGG1 removes 8-oxoguanine; MPG (methylpurine DNA glycosylase) removes alkylated bases. After glycosylase action, AP endonuclease 1 (APE1) nicks the DNA backbone 5' to the AP site, DNA polymerase beta replaces the missing nucleotide, and DNA ligase seals the nick.

Carbohydrate

– A class of organic compounds composed of carbon, hydrogen, and oxygen (general formula $C_m(H_2O)_n$), serving as the primary energy source for the body. Classification: monosaccharides (glucose, fructose, galactose–single sugar units), disaccharides (sucrose, lactose, maltose–two units), oligosaccharides (3–10 units), and polysaccharides (starch, glycogen, cellulose, fiber–many units). Glucose is the essential fuel for the brain and red blood cells. The body stores limited glycogen (liver 100 g, muscle 400 g). Dietary carbohydrates are digested to monosaccharides, absorbed in the small intestine, and metabolized via glycolysis, the TCA cycle, and oxidative phosphorylation to produce ATP. Excess carbohydrate is converted to fat (lipogenesis). The glycemic index ranks foods by their effect on blood glucose.

Cholesterol Synthesis

– Occurs in all nucleated cells (primarily liver and intestine, approximately 1 g/day production), using acetyl-CoA as the precursor through approximately 30 enzymatic steps. Key stages: (1) two acetyl-CoA condense to acetoacetyl-CoA, then to HMG-CoA (3-hydroxy-3-methylglutaryl-CoA); (2) HMG-CoA reductase (ER-membrane enzyme, the rate-limiting step) reduces HMG-CoA to mevalonate using 2 NADPH. This step is inhibited by statins (competitive inhibitors, the most widely prescribed cholesterol-lowering drugs). After mevalonate, isopentenyl pyrophosphate (IPP) is formed, and six IPP molecules condense to form squalene, which is cyclized to lanosterol and then converted through nineteen additional steps to cholesterol.

CRISPR-Cas9 Gene Editing

– A revolutionary genome editing technology adapted from the bacterial adaptive immune system. Bacteria store fragments of invading phage DNA in CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) arrays, which are transcribed and processed into guide RNAs (crRNAs) that direct Cas nucleases to cleave matching foreign DNA. The engineered system uses: a guide RNA (gRNA = crRNA + tracrRNA, or a single guide RNA/sgRNA) complementary to the target DNA sequence (approximately 20 nucleotides). The Cas nuclease (most commonly Cas9 from Streptococcus pyogenes) creates a double-strand break (DSB) at the target site, which is repaired by non-homologous end joining (NHEJ, causing insertions/deletions and gene knockout) or homology-directed repair (HDR, enabling precise edits using a donor template).

Deoxyribonucleic Acid (DNA)

– The molecule that carries the genetic instructions used in the development, functioning, and reproduction of all living organisms. DNA is a double-stranded helix composed of nucleotides, each containing a deoxyribose sugar, a phosphate group, and one of four nitrogenous bases: adenine (A), thymine (T), guanine (G), and cytosine (C). Base pairing is specific: A pairs with T (two hydrogen bonds), G pairs with C (three hydrogen bonds). The sequence of bases encodes genetic information. DNA is organized into chromosomes within the nucleus (nuclear DNA, 3 billion base pairs in humans) and is also present in mitochondria (mtDNA, 16,569 base pairs). DNA replication is semiconservative. The human genome contains 20,000 protein-coding genes. DNA analysis techniques (PCR, sequencing, microarray) are fundamental to modern medicine.

DNA Replication: Semi-Conservative

– The mechanism of DNA duplication producing two identical daughter molecules, each containing one original (parental) strand and one newly synthesized strand. The Meselson-Stahl experiment (1958) demonstrated semi-conservative replication using nitrogen isotopes (15N/14N) and density gradient centrifugation. Replication begins at origins of replication (single oriC in E. coli; multiple origins in eukaryotic chromosomes, allowing timely replication of large genomes). Key enzymes and steps: (1) helicase unwinds the DNA at the origin; (2) single-strand binding proteins stabilise the separated strands; (3) primase synthesises short RNA primers to provide a free 3'-OH group; (4) DNA polymerase III (prokaryotes) or DNA polymerase delta/epsilon (eukaryotes) extends the primer, synthesising the new strand in the 5' to 3' direction; (5) on the lagging strand, discontinuous Okazaki fragments are joined by DNA ligase after primer removal by DNA polymerase I (prokaryotes) or RNase H and FEN1 (eukaryotes).

Enzymes

– Proteins (or occasionally RNA molecules–ribozymes) that act as biological catalysts, accelerating chemical reactions by lowering activation energy without being consumed in the process. Enzymes are highly specific, typically catalyzing a single reaction or class of reactions. Nomenclature: substrate + -ase (e.g., lactase breaks down lactose). Classification (EC system): oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases. Factors affecting enzyme activity: temperature, pH, substrate concentration, and inhibitors. Clinically important enzymes: cardiac troponin (myocardial infarction), creatine kinase (muscle injury), alanine transaminase (liver injury), lipase (pancreatitis), alkaline phosphatase (bone/liver disease), and lactate dehydrogenase (tissue damage). Therapeutic enzymes: tissue plasminogen activator (thrombolysis), asparaginase (leukemia), and pancreatic enzyme replacement.

Fatty Acid Synthesis

– Occurs in the cytoplasm of hepatocytes, adipocytes, and lactating mammary glands, converting acetyl-CoA to long-chain fatty acids (primarily palmitate, C16:0). The process occurs on fatty acid synthase (FAS), a large homodimeric multifunctional enzyme. Malonyl-CoA is the two-carbon donor, synthesized from acetyl-CoA by acetyl-CoA carboxylase (ACC, the rate-limiting enzyme, requiring biotin, activated by insulin/citrate, inhibited by palmitoyl-CoA and AMPK-mediated phosphorylation). The cycle uses malonyl-CoA for each two-carbon addition, with seven complete cycles producing palmitate (C16:0) from acetyl-CoA. NADPH (from the pentose phosphate pathway) provides reducing equivalents for each cycle.

Galactose

– A monosaccharide (C6H12O6) that is an epimer of glucose (differing at the C4 position). Galactose is produced from the hydrolysis of lactose (milk sugar) in the small intestine by lactase. It is converted to glucose via the Leloir pathway: galactose to galactose-1-phosphate to UDP-galactose to UDP-glucose to glucose-1-phosphate. Deficiency of enzymes in this pathway causes galactosemia (GALT, GALK, or GALE deficiency). Galactose is also a component of glycoproteins, proteoglycans, and glycolipids (including blood group antigens).

Galactosemia

– Inborn errors of galactose metabolism, with three main types. Classic galactosemia (GALT deficiency, most severe, autosomal recessive): galactose-1-phosphate uridylyltransferase converts galactose-1-phosphate to glucose-1-phosphate. Deficiency causes accumulation of galactose-1-phosphate (toxic to liver, kidney, brain, and lens) and galactitol (from reduction of galactose by aldose reductase, accumulating in the lens causing osmotic cataracts). Newborns present after milk feeding (lactose = glucose + galactose) with vomiting, diarrhea, failure to thrive, hypoglycemia, jaundice, and hepatomegaly. Long-term complications include cataracts, intellectual disability, and premature ovarian failure.

Gaucher Disease

– The most common lysosomal storage disease (incidence approximately 1 in 40,000-60,000; 1 in 800 in Ashkenazi Jews), caused by deficiency of glucocerebrosidase (glucosylceramidase, GBA, chromosome 1q22, autosomal recessive), which cleaves the terminal glucose from glucocerebroside (glucosylceramide). Accumulated glucocerebroside is phagocytosed by macrophages, creating pathognomonic "Gaucher cells" with a crinkled tissue paper cytoplasmic appearance. Type 1 (non-neuronopathic, approximately 90 percent of cases): hepatosplenomegaly, anemia, thrombocytopenia, bone pain crises, and Erlenmeyer flask deformity of the femurs. Type 2 (acute neuronopathic, infantile): severe neurodegeneration with early death. Type 3 (chronic neuronopathic): later onset with milder neurological involvement.

Gene Therapy Vectors

– Delivery vehicles for introducing therapeutic nucleic acids into target cells. Viral vectors exploit the natural ability of viruses to infect cells and deliver genetic material: adeno-associated virus (AAV, non-enveloped, single-stranded DNA, non-integrating, low immunogenicity, tropism for multiple tissues, limited packaging capacity approximately 4.7 kb; serotypes AAV2, AAV8, AAV9, AAVrh10, etc., with different tissue tropisms; used for Luxturna/RPE65, Zolgensma/SMN1, hemophilia gene therapy); lentivirus (based on HIV, single-stranded RNA, integrates into host genome, packaging capacity approximately 8 kb, pseudotyped with VSV-G for broad tropism; used for CAR-T cell therapy, hemoglobinopathies, and SCID-X1); adenovirus (double-stranded DNA, episomal, highly immunogenic, large capacity approximately 37 kb; used for vaccines, oncolytic therapy, and cancer gene therapy).

Gluconeogenesis

– The synthesis of glucose from non-carbohydrate precursors (lactate, glycerol, glucogenic amino acids, propionate) primarily in the liver (90 percent) and renal cortex (10 percent). It reverses most glycolytic reactions but bypasses the three irreversible steps using four unique enzymes: (1) pyruvate carboxylase (pyruvate to oxaloacetate, requiring biotin, in the mitochondria) and PEPCK (oxaloacetate to phosphoenolpyruvate, requiring GTP); (2) fructose-1,6-bisphosphatase converts fructose-1,6-bisphosphate to fructose-6-phosphate; (3) glucose-6-phosphatase converts glucose-6-phosphate to free glucose, completing gluconeogenesis. The process is energy-intensive, requiring 6 ATP equivalents per glucose synthesized, and is regulated by glucagon (stimulatory) and insulin (inhibitory).

Glycogenesis

– The synthesis of glycogen from glucose for energy storage in the liver (approximately 100 g, maintaining blood glucose) and skeletal muscle (approximately 400 g, local energy). Two key enzymes: glycogen synthase adds glucose residues from UDP-glucose to the non-reducing ends of glycogen via alpha-1,4-glycosidic bonds (the rate-limiting step, activated by dephosphorylation and allosteric activation by glucose-6-phosphate); branching enzyme (amylo-1,4 to 1,6 transglucosidase) creates alpha-1,6 branch points, allowing glycogen to be more soluble and providing multiple non-reducing ends for rapid glucose release.

Glycogenolysis

– The breakdown of glycogen to glucose, occurring in the liver (to maintain blood glucose) and muscle (for local ATP generation). Key enzyme: glycogen phosphorylase, which cleaves alpha-1,4-glycosidic bonds by phosphorolysis (using inorganic phosphate rather than water), producing glucose-1-phosphate. The debranching enzyme has two activities: transferase (moves a trisaccharide segment from a branch to the main chain) and alpha-1,6-glucosidase (cleaves the remaining branch-point glucose, releasing free glucose). Glycogen phosphorylase is activated by phosphorylation (cascade involving glucagon and epinephrine) and allosterically by AMP; inactivated by insulin and glucose.

Glycolysis

– The metabolic pathway converting one molecule of glucose (6-carbon) to two molecules of pyruvate (3-carbon) in the cytoplasm, occurring in all cells.

Ten enzymatic steps in two phases: the energy investment phase (steps 1-5, consuming 2 ATP) and the energy payoff phase (steps 6-10, producing 4 ATP and 2 NADH). Step 1: hexokinase (or glucokinase in the liver) phosphorylates glucose to glucose-6-phosphate (G6P), trapping it in the cell. Step 3: phosphofructokinase-1 (PFK-1) converts fructose-6-phosphate to fructose-1,6-bisphosphate, the committed step and primary regulatory point, allosterically activated by AMP and fructose-2,6-bisphosphate, inhibited by ATP and citrate. (4) Aldolase cleaves fructose-1,6-bisphosphate into dihydroxyacetone phosphate (DHAP) and glyceraldehyde-3-phosphate. (5) Triose phosphate isomerase interconverts DHAP and G3P. The energy payoff phase follows with substrate-level phosphorylation producing ATP and NADH.

JAK-STAT Signaling

– The Janus kinase-signal transducer and activator of transcription pathway mediates signaling for cytokines, interferons, and interleukins whose receptors lack intrinsic enzymatic activity. Mechanism: (1) cytokine binding induces receptor dimerization; (2) associated JAKs (JAK1, JAK2, JAK3, TYK2) are activated by trans-phosphorylation; (3) activated JAKs phosphorylate tyrosine residues on the receptor cytoplasmic domain; (4) STAT proteins (STAT1-6) are recruited to the phosphorylated receptor via their SH2 domains; (5) recruited STATs are phosphorylated by JAKs; (6) phosphorylated STATs form homo- or heterodimers and translocate to the nucleus; (7) STAT dimers bind gamma-interferon activation site (GAS) or interferon-stimulated response element (ISRE) DNA sequences, regulating gene transcription.

Ketogenesis

– Hepatic synthesis of ketone bodies (acetoacetate, beta-hydroxybutyrate, acetone) during prolonged fasting, starvation, uncontrolled diabetes, or very low carbohydrate diets. When oxaloacetate is depleted for gluconeogenesis, accumulated acetyl-CoA cannot enter the TCA cycle and is diverted to ketogenesis in the mitochondrial matrix. Key steps: (1) two acetyl-CoA condense to acetoacetyl-CoA by thiolase; (2) HMG-CoA synthase (the rate-limiting enzyme for ketogenesis, activated by fasting, inhibited by insulin and succinyl-CoA) converts acetoacetyl-CoA to HMG-CoA; (3) HMG-CoA lyase cleaves HMG-CoA to acetoacetate and acetyl-CoA. Acetoacetate is reduced to beta-hydroxybutyrate by beta-hydroxybutyrate dehydrogenase (using NADH) or spontaneously decarboxylated to acetone.

Lipid

– A diverse group of hydrophobic organic compounds that are essential components of cell membranes and serve as energy storage, signalling molecules, and structural components. Lipids are broadly classified into: (1) Fatty acids: saturated (no double bonds, e.g., palmitic, stearic–solid at room temperature), unsaturated (monounsaturated–oleic; polyunsaturated–linoleic acid, omega-3 fatty acids such as eicosapentaenoic EPA, docosahexaenoic DHA; essential fatty acids that must be obtained from diet: linoleic acid and alpha-linolenic acid). (2) Triglycerides (triacylglycerols): esters of glycerol with three fatty acids, stored in adipose tissue, the main form of energy storage. (3) Phospholipids: glycerol backbone with two fatty acids and a phosphate group linked to a polar head group (choline, ethanolamine, serine, inositol); the major component of cell membranes (phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, sphingomyelin). (4) Sterols: cholesterol (cell membrane stability, precursor of steroid hormones, bile acids, vitamin D). (5) Sphingolipids: ceramide, sphingomyelin, glycosphingolipids (cerebrosides, gangliosides–abundant in the nervous system). (6) Eicosanoids: prostaglandins, thromboxanes, leukotrienes, prostacyclin–short-lived signalling molecules derived from arachidonic acid (omega-6) via cyclooxygenase (COX) and lipoxygenase pathways. Clinical significance: hyperlipidemia (elevated cholesterol and/or triglycerides, major risk factor for atherosclerosis, coronary artery disease, stroke), lipid-lowering therapy (statins, ezetimibe, PCSK9 inhibitors, fibrates).

Lipoprotein

– A complex of lipids (cholesterol, triglycerides, phospholipids) and apolipoproteins that transports hydrophobic lipids through the aqueous bloodstream. Lipoproteins are classified into five major classes based on density: chylomicrons, VLDL, IDL, LDL, and HDL.

Lipoprotein Metabolism

– Lipoproteins are spherical particles with a hydrophobic core (triglycerides and cholesterol esters) and amphipathic surface (phospholipids, free cholesterol, apolipoproteins) for transporting lipids in aqueous plasma. Five classes: chylomicrons (largest, least dense; ApoB-48; from intestine; transport dietary triglycerides; acted on by lipoprotein lipase on capillary endothelium). VLDL (from liver; ApoB-100; transport endogenous triglycerides; converted to IDL by lipoprotein lipase). LDL (from VLDL metabolism by hepatic lipase, ApoB-100, the major cholesterol carrier, taken up by LDL receptors on hepatocytes and peripheral cells; elevated LDL is the primary atherogenic lipoprotein, and LDL receptor deficiency causes familial hypercholesterolaemia). HDL (from liver and intestine, ApoA-I, reverse cholesterol transport, the "good" cholesterol; protective against atherosclerosis).

Methylmalonic Acidemia

– An inborn error of organic acid metabolism caused by deficiency of methylmalonyl-CoA mutase (MUT, requiring adenosylcobalamin/B12 as cofactor) or defects in adenosylcobalamin synthesis. The enzyme converts methylmalonyl-CoA to succinyl-CoA (linking odd-chain fatty acid and branched-chain amino acid catabolism to the TCA cycle). Deficiency causes accumulation of methylmalonic acid and other organic acids, leading to metabolic acidosis with elevated anion gap, ketonuria, hyperammonemia (from secondary inhibition of the urea cycle by accumulated organic acids), and neurological deterioration. Treatment includes protein restriction, carnitine supplementation, and vitamin B12 therapy for responsive subtypes.

Nucleotide Excision Repair

– The primary repair pathway for bulky, helix-distorting DNA lesions such as UV-induced pyrimidine dimers (cyclobutane pyrimidine dimers and 6-4 photoproducts), bulky chemical adducts (benzo[a]pyrene-guanine from cigarette smoke), and intrastrand crosslinks. In eukaryotes: (1) the XPC-RAD23B complex recognizes the distortion; (2) TFIIH (with XPD helicase) opens the DNA around the lesion; (3) XPA verifies the damage; (4) RPA stabilizes the single-stranded DNA; (5) XPF-ERCC1 endonuclease makes a 5' incision; (6) XPG endonuclease makes a 3' incision; (7) the oligonucleotide fragment (approximately 24-32 nucleotides) containing the lesion is excised; (8) DNA polymerase delta/epsilon resynthesises the gap using the complementary strand as a template; (9) DNA ligase I seals the nick. Deficiencies cause xeroderma pigmentosum (extreme UV sensitivity, skin cancer predisposition) and Cockayne syndrome.

Oxidative Deamination

– The removal of the amino group from glutamate, catalyzed by glutamate dehydrogenase (GDH) in the mitochondrial matrix, producing free ammonia (NH4+) and alpha-ketoglutarate. This reaction is the primary route for amino group disposal, collecting amino groups from all transamination reactions (since most aminotransferases funnel amino groups to glutamate). GDH is a versatile enzyme that can use either NAD+ or NADP+ as a cofactor and is allosterically activated by ADP (indicating energy need) and inhibited by GTP and ATP (indicating energy sufficiency), linking amino acid catabolism to cellular energy status.

Pentose Phosphate Pathway

– A cytoplasmic glucose oxidation pathway running parallel to glycolysis, especially active in the liver, adipose tissue, adrenal cortex, and red blood cells. The oxidative phase (irreversible): glucose-6-phosphate dehydrogenase (G6PD, the rate-limiting enzyme, producing NADPH) and 6-phosphogluconate dehydrogenase (producing a second NADPH and ribulose-5-phosphate). The non-oxidative phase (reversible): transketolase (requires TPP, B1) and transaldolase interconvert sugar phosphates (ribose-5-phosphate and xylulose-5-phosphate) to produce fructose-6-phosphate and glyceraldehyde-3-phosphate, which can re-enter glycolysis. The pathway provides NADPH for reductive biosynthesis and antioxidant defence, and ribose-5-phosphate for nucleotide synthesis.

Phenylketonuria (PKU)

– The most common and best-studied inborn error of amino acid metabolism, caused by deficiency of phenylalanine hydroxylase (PAH, chromosome 12q, autosomal recessive, approximately 1000 mutations identified). PAH converts phenylalanine to tyrosine using tetrahydrobiopterin (BH4) as a cofactor and molecular oxygen. Deficiency causes accumulation of phenylalanine (greater than 1200 micromol/L in untreated classic PKU) and its transamination products: phenylpyruvate (a phenylketone detected by ferric chloride testing, giving a green color), phenyllactate, and phenylacetate, which cause a characteristic musty odour in urine and neurotoxic effects if untreated.

Proteins

– Large, complex macromolecules composed of amino acids linked by peptide bonds, essential for virtually all biological processes. Proteins are polymers of 20 standard L-α-amino acids, each with a central carbon bonded to an amino group, carboxyl group, hydrogen atom, and a unique side chain (R group). Amino acids are classified as essential (must be obtained from diet—histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine) and non-essential (synthesized in the body). Protein structure has four levels: primary (linear sequence of amino acids), secondary (α-helices and β-pleated sheets stabilized by hydrogen bonds), tertiary (three-dimensional folding stabilized by disulfide bonds, hydrophobic interactions, ionic bonds, hydrogen bonds), and quaternary (assembly of multiple polypeptide subunits—e.g., hemoglobin is a tetramer of two α and two β globin chains). Protein functions: enzymes (catalyze biochemical reactions), structural (collagen, keratin, actin, tubulin), transport (hemoglobin, albumin, transferrin), signalling (hormones, receptors, cytokines), immune defence (antibodies, complement), and gene regulation (transcription factors). Protein synthesis: transcription (DNA $\rightarrow$ mRNA in the nucleus), translation (mRNA $\rightarrow$ protein on ribosomes in the cytoplasm), and post-translational modification (folding, glycosylation, phosphorylation, ubiquitination). Recommended dietary allowance (RDA): 0.8 g/kg body weight/day.

Ribonucleic Acid (RNA)

– A nucleic acid composed of ribonucleotides (adenine, guanine, cytosine, uracil) linked by phosphodiester bonds, with a ribose sugar backbone. RNA is transcribed from DNA by RNA polymerase and serves multiple functions in gene expression and regulation. Main types: (1) Messenger RNA (mRNA)—carries the genetic code from DNA to ribosomes for protein synthesis (translation). Pre-mRNA undergoes splicing (removal of introns, joining of exons), 5' capping, and 3' polyadenylation. (2) Transfer RNA (tRNA)—adaptor molecules that bring specific amino acids to the ribosome during translation; each tRNA has an anticodon complementary to the mRNA codon. (3) Ribosomal RNA (rRNA)—structural and catalytic component of ribosomes (28S, 18S, 5.8S, 5S in eukaryotes), undergoes site-specific methylation and pseudouridylation. (4) Small nuclear RNA (snRNA)—components of the spliceosome (U1, U2, U4, U5, U6), catalyzes pre-mRNA splicing. (5) MicroRNA (miRNA, 22 nucleotides)—regulates gene expression post-transcriptionally by binding complementary mRNA sequences, leading to translational repression or mRNA degradation. (6) Small interfering RNA (siRNA)—mediates RNA interference (RNAi), gene silencing. (7) Long non-coding RNA (lncRNA, >200 nucleotides)—regulates gene expression, chromatin remodeling, and X-inactivation (XIST). RNA is also the genetic material of RNA viruses (retroviruses—HIV, influenza, SARS-CoV-2).

Tay-Sachs Disease

– A fatal lysosomal storage disease caused by deficiency of hexosaminidase A (Hex A, composed of alpha and beta subunits encoded by HEXA and HEXB genes, respectively), leading to accumulation of GM2 ganglioside in neurons of the central nervous system. Classic infantile Tay-Sachs (HEXA mutations, autosomal recessive): normal at birth, progressive neurodegeneration begins at 3-6 months with hypotonia, followed by an exaggerated startle response (to loud noises), loss of acquired motor skills, progressive loss of vision (cherry-red macula on fundoscopic examination due to lipid accumulation in retinal ganglion cells), macrocephaly from neuronal lipid storage, seizures, and death by age 4. Carrier frequency is approximately 1 in 30 in Ashkenazi Jewish populations, where HEXA mutations include the common 4-base-pair insertion in exon 11.

Transamination

– The transfer of an amino group from an amino acid to an alpha-keto acid, catalyzed by aminotransferases (transaminases) requiring pyridoxal phosphate (PLP, vitamin B6) as a cofactor. The reaction proceeds through a ping-pong mechanism: PLP accepts the amino group from the donor amino acid, becoming pyridoxamine phosphate (PMP), then transfers it to the acceptor alpha-keto acid, regenerating PLP. ALT (alanine aminotransferase, formerly SGPT): transfers the amino group from alanine to alpha-ketoglutarate, forming glutamate and pyruvate. AST (aspartate aminotransferase, formerly SGOT): transfers the amino group from aspartate to alpha-ketoglutarate, forming glutamate and oxaloacetate. These enzymes are clinically important markers of hepatocellular injury.

Triglycerides

– The main form of stored energy in the body, composed of a glycerol backbone esterified with three fatty acid chains. Triglycerides are synthesized in the liver (from excess carbohydrates and fatty acids) and stored in adipose tissue. They are transported in the blood as part of very-low-density lipoproteins (VLDL) and chylomicrons. Normal fasting serum triglyceride level: <1.7 mmol/L (<150 mg/dL). hypertriglyceridemia: mild (1.7–5.6 mmol/L)—associated with increased cardiovascular risk; severe (>5.6 mmol/L)—risk of acute pancreatitis (risk increases exponentially with levels >10 mmol/L). Causes: obesity, metabolic syndrome, type 2 diabetes, excessive alcohol intake, hypothyroidism, chronic kidney disease, and genetic disorders (familial hypertriglyceridemia, familial combined hyperlipidemia, lipoprotein lipase deficiency). Management: lifestyle modification (weight loss, reduced carbohydrate and alcohol intake, increased exercise), statins (if LDL also elevated), fibrates (fenofibrate, bezafibrate—first-line for severe hypertriglyceridemia), omega-3 fatty acids (fish oil—4 g/day), and icosapent ethyl (EPA—REDUCE-IT trial: reduced cardiovascular events). Niacin is rarely used due to side effects.

Vitamin B2: Riboflavin

– Riboflavin (vitamin B2) is a water-soluble vitamin that serves as the precursor for two essential coenzymes: flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD). FMN and FAD are prosthetic groups in flavoproteins (flavoenzymes), accepting and donating electrons (as reduced FADH2 and FMNH2) in oxidation-reduction reactions. Key flavoenzymes include: succinate dehydrogenase (Complex II of the ETC, converting succinate to fumarate), NADH dehydrogenase (Complex I), acyl-CoA dehydrogenase (fatty acid oxidation), and xanthine oxidase (purine metabolism). Riboflavin deficiency (ariboflavinosis) causes angular stomatitis, cheilitis, glossitis, seborrheic dermatitis, and corneal vascularization.

Vitamin B7: Biotin

– Biotin (vitamin B7) is a water-soluble vitamin that serves as a prosthetic group (covalently bound via an amide bond to lysine residues) for five carboxylases in humans: (1) pyruvate carboxylase (pyruvate to oxaloacetate, anaplerosis of the TCA cycle and gluconeogenesis); (2) acetyl-CoA carboxylase (acetyl-CoA to malonyl-CoA, the rate-limiting step of fatty acid synthesis); (3) propionyl-CoA carboxylase (propionyl-CoA to methylmalonyl-CoA, in odd-chain fatty acid and branched-chain amino acid catabolism); and (5) 3-methylcrotonyl-CoA carboxylase (leucine catabolism). Biotin deficiency is rare but can occur from excessive raw egg consumption (avidin in egg white binds biotin, preventing absorption), causing dermatitis, alopecia, neurological symptoms, and metabolic acidosis.

Western Blot Analysis

– A immunodetecting technique for identifying and quantifying specific proteins in complex biological samples (cell lysates, tissue homogenates, serum). Steps: (1) sample preparation: cells/tissues are lysed in buffer (containing protease inhibitors, phosphatase inhibitors, detergents) to extract proteins; protein concentration is measured (Bradford, BCA, or Lowry assay); (2) SDS-PAGE: proteins are denatured by heating with SDS (sodium dodecyl sulfate, an anionic detergent that coats proteins uniformly with a negative charge proportional to their mass, allowing separation by size through gel electrophoresis); (3) transfer (blotting): separated proteins are transferred from the gel to a membrane (nitrocellulose or PVDF) by electroblotting; (4) blocking: nonspecific binding sites on the membrane are blocked with a protein solution (BSA, milk, or serum); (5) primary antibody incubation: a specific antibody against the target protein is applied; (6) secondary antibody incubation: an enzyme-conjugated antibody targeting the primary antibody is applied; (7) detection: a chemiluminescent or chromogenic substrate is added, producing a signal proportional to protein quantity that is visualized by film exposure or digital imaging.

Chapter 4

Cardiology

Acute Coronary Syndrome

– ACS encompasses unstable angina, NSTEMI, and STEMI. It results from acute myocardial ischemia due to coronary artery plaque rupture, erosion, or thrombosis. Presentation includes chest pain, ECG changes, and elevated cardiac biomarkers. Management involves antiplatelet therapy, anticoagulation, revascularization (PCI or CABG), and risk factor modification. Early recognition and treatment improve outcomes.

Afterload

– The resistance against which the ventricles must eject blood during systole. Left ventricular afterload is primarily determined by systemic vascular resistance (SVR); right ventricular afterload is determined by pulmonary vascular resistance (PVR). Afterload is estimated by blood pressure or calculated as wall stress. Increased afterload (as in hypertension or aortic stenosis) reduces stroke volume and increases myocardial oxygen demand.

Alcoholic Cardiomyopathy

– Dilated cardiomyopathy caused by chronic excessive alcohol consumption, one of the most common reversible causes of dilated cardiomyopathy. Direct toxic effect of ethanol and its metabolite acetaldehyde on cardiomyocytes. Presentation is identical to other dilated cardiomyopathies. Abstinence is the cornerstone of treatment and can lead to significant recovery of LV function. Other treatments follow standard heart failure guidelines.

Angina Pectoris

– Chest discomfort caused by myocardial ischemia without infarction. Stable angina occurs predictably with exertion and resolves with rest. Unstable angina occurs at rest or with minimal exertion and is an acute coronary syndrome. Prinzmetal (variant) angina occurs at rest due to coronary vasospasm. Diagnosis involves ECG (may show ST depression during episodes) and stress testing. Treatment includes nitrates, beta-blockers, and revascularization when indicated.

Anticoagulant Monitoring

– Unfractionated heparin is monitored with aPTT (target 1.5-2.5 x control). Low molecular weight heparins generally do not require monitoring. Warfarin is monitored with INR (target 2-3 for most indications, 2.5-3.5 for mechanical mitral valve). DOACs (rivaroxaban, apixaban, dabigatran) have predictable pharmacokinetics and do not require routine monitoring, though anti-Xa levels can be measured for rivaroxaban/apixaban and dilute thrombin time for dabigatran.

Aortic Aneurysm

– Permanent dilation of the aorta exceeding 50% of normal diameter. Abdominal aortic aneurysm (AAA) is more common than thoracic aortic aneurysm and is usually caused by atherosclerosis. AAA >5.5 cm or rapid expansion (>0.5 cm/year) warrants surgical repair (open or endovascular). Thoracic aortic aneurysms may be caused by Marfan syndrome, bicuspid aortic valve, or hypertension. Screening ultrasound is recommended for men aged 65-75 who have ever smoked.

Aortic Dissection

– A tear in the intima of the aorta allowing blood to dissect through the media, creating a false lumen. It is classified as Stanford Type A (ascending aorta involvement, requiring emergency surgery) or Type B (descending aorta only, managed medically unless complicated). Risk factors include hypertension, Marfan syndrome, Ehlers-Danlos syndrome, bicuspid aortic valve, and aortic instrumentation.

Symptoms include sudden, severe, tearing chest pain radiating to the back.

Aortic Regurgitation

– Incomplete closure of the aortic valve allowing blood to flow backward from the aorta to the left ventricle during diastole. Causes include bicuspid aortic valve, endocarditis, aortic root dilation, hypertension, and connective tissue disorders. Murmur: early diastolic decrescendo murmur at the left sternal border. Signs include wide pulse pressure, bounding pulses, and de Musset sign. Severe AR requires aortic valve replacement.

Aortic Regurgitation Murmur

– Early diastolic decrescendo murmur heard best at the left sternal border with the patient sitting up and leaning forward in end-expiration. Wide pulse pressure, water-hammer pulses (Corrigan pulse), de Musset sign (head bobbing with each pulse), and Duroziez sign (femoral bruit with compression). Severity correlates with murmur duration and pulse pressure width. Severe AR requires aortic valve replacement before irreversible LV dysfunction develops.

Aortic Stenosis

– Narrowing of the aortic valve opening, obstructing left ventricular outflow. It is the most common valvular disease in developed countries, usually caused by calcific degeneration. Symptoms include angina, syncope, and heart failure (SAD triad). Murmur: crescendo-decrescendo systolic murmur at the right upper sternal border radiating to the carotids. Severe AS (valve area <1.0 cm2) requires aortic valve replacement (surgical or TAVR).

Aortic Stenosis Grading

– Severity grading: mild (valve area >1.5 cm2, mean gradient <20 mmHg), moderate (valve area 1.0-1.5 cm2, mean gradient 20-40 mmHg), severe (valve area <1.0 cm2, mean gradient >40 mmHg). Very severe (valve area <0.6 cm2, mean gradient >60 mmHg) indicates critical stenosis. Grading guides timing of intervention. Echocardiography is the primary modality for assessment, with hemodynamic data from cardiac catheterization when echocardiographic data is inconclusive.

ARB-Neprilysin Inhibitor (ARNI)

– Sacubitril/valsartan combines an ARB with a neprilysin inhibitor, increasing natriuretic peptide levels while blocking the renin-angiotensin system. The PARADIGM-HF trial demonstrated superior reduction in cardiovascular death and heart failure hospitalization compared to enalapril. ARNI is now a cornerstone of HFrEF therapy, recommended as replacement for ACE inhibitors in patients who remain symptomatic on optimal medical therapy.

Arteriosclerosis

– See Atherosclerosis.

Artificial Pacemaker

– See Cardiac Pacemaker.

Aspirin

– An antiplatelet agent that irreversibly inhibits cyclooxygenase-1 (COX-1), reducing thromboxane A2 production and platelet aggregation. Low-dose aspirin (81 mg daily) is used for secondary prevention of cardiovascular events. In acute MI, a loading dose of 162-325 mg is chewed for rapid absorption. Aspirin is also used for pain, inflammation, and fever. Side effects include GI bleeding, tinnitus, and aspirin-exacerbated respiratory disease.

Atherosclerosis

– Chronic inflammatory disease of medium and large arteries characterized by accumulation of lipids, inflammatory cells, and fibrous tissue within the arterial wall, forming atherosclerotic plaques. Initiating event is endothelial dysfunction from risk factors (hypertension, smoking, diabetes, dyslipidemia). Progression involves LDL oxidation, monocyte recruitment, foam cell formation, and smooth muscle proliferation. Complicated plaques may rupture, causing acute thrombosis and ischemic events (myocardial infarction, stroke, or peripheral ischemia).

Atrial Fibrillation

– The most common sustained cardiac arrhythmia, characterized by rapid, irregular atrial activity (350-600 bpm) with irregular ventricular response. ECG shows absent P waves, irregularly irregular RR intervals, and narrow QRS complexes (unless aberrant conduction). AFib increases stroke risk 5-fold due to atrial stasis and thrombus formation, particularly in the left atrial appendage. Treatment targets rate control, rhythm control, and anticoagulation.

Atrial Fibrillation Ablation

– Catheter-based procedure to eliminate atrial fibrillation by creating pulmonary vein isolation (PVI), electrically disconnecting the pulmonary veins from the left atrium. Indicated for symptomatic AFib refractory to or intolerant of antiarrhythmic drugs. Success rates: 70-80% for paroxysmal AFib, 50-60% for persistent AFib. Complications include pulmonary vein stenosis, pericardial tamponade, atrio-esophageal fistula (rare but life-threatening), and stroke. Anticoagulation is continued post-procedure.

Atrial Flutter

– A rapid, regular atrial rhythm (typically 250-350 bpm) caused by a macro-reentrant circuit, most commonly in the right atrium. ECG shows sawtooth flutter waves best seen in leads II, III, and aVF, with regular or irregular ventricular response depending on AV conduction ratio (commonly 2:1 or 4:1). Atrial flutter carries similar stroke risk to AFib and requires anticoagulation. Treatment includes cardioversion and catheter ablation.

Atrial Septal Defect

– Congenital opening in the interatrial secundum allowing left-to-right shunting of blood. Small defects may close spontaneously. Larger defects cause right heart volume overload, right atrial and ventricular dilation, pulmonary hypertension, and eventually right heart failure. Paradoxical embolism may occur with right-to-left shunting. Systolic crescendo-decrescendo murmur at left upper sternal border with fixed split S2. Echocardiography confirms diagnosis and assesses shunt magnitude. Transcatheter closure is indicated for moderate-to-large defects.

Atrioventricular Node

– Located in the posterior-inferior region of the interatrial septum near the coronary sinus. It receives impulses from the SA node via internodal pathways and delays them by approximately 120-200 milliseconds before transmitting them to the ventricles. This delay allows atrial contraction to complete before ventricular contraction. The AV node is the backup pacemaker with an intrinsic rate of 40-60 beats per minute.

Austin Flint Murmur

– A mid-diastolic, low-pitched rumbling murmur heard at the apex, caused by the regurgitant jet from severe aortic regurgitation striking the anterior mitral leaflet. This functional murmur can mimic mitral stenosis but is distinguished by the absence of an opening snap, presence of wide pulse pressure, and early diastolic decrescendo murmur of aortic regurgitation. It resolves after aortic valve replacement.

Balloon Aortic Valvuloplasty

– Catheter-based procedure using balloon inflation to dilate a stenotic aortic valve. Provides temporary hemodynamic improvement in severe AS but restenosis is common within 6-12 months. Used as a bridge to TAVR or surgical AVR in hemodynamically unstable patients, or as palliation in patients not eligible for definitive valve replacement. Complications include aortic regurgitation, stroke, and vascular access complications.

Bicuspid Aortic Valve

– The most common congenital cardiac anomaly, present in 1-2% of the population. The aortic valve has two leaflets instead of three, predisposing to aortic stenosis (usually by age 50-60), aortic regurgitation, and infective endocarditis. Associated with aortopathy (aortic root dilation and dissection risk) and coarctation of the aorta. Regular echocardiographic surveillance is recommended. Aortic valve replacement and/or aortic root replacement may be required.

Bioprosthetic Valve Degeneration

– Structural valve deterioration occurs in 20-30% of bioprosthetic valves by 10-15 years, more rapidly in younger patients. Manifests as stenosis or regurgitation from calcification and leaflet tearing. Risk factors: younger age, renal failure, hyperparathyroidism. TAVR-in-SAVR (valve-in-valve) is an option for high-risk surgical patients. Transcatheter bioprosthetic valve fracture may improve hemodynamics. Anticoagulation is not required long-term.

Bradycardia

– A heart rate <60 beats per minute. May be physiologic (athletes) or pathologic due to sinus node dysfunction, AV block, or medication effects. Symptomatic bradycardia causes fatigue, dizziness, syncope, or heart failure. Evaluation includes ECG, Holter monitoring, and assessment of reversible causes. Treatment: pacing (temporary or permanent) for symptomatic or high-grade AV block.

Brugada Syndrome

– A genetic channelopathy causing abnormal sodium channel function (SCN5A mutation). ECG pattern:

coved ST elevation >2mm in V1-V3 (type 1) followed by negative T wave. Patients are at risk for ventricular fibrillation and sudden cardiac death, often at rest or during sleep. More common in young Southeast Asian men. Diagnosis: type 1 pattern on ECG (spontaneous or drug-induced with ajmaline/flecainide). Treatment: ICD is the only proven preventive therapy; avoid fever and sodium channel blockers.

Bundle Branch Block

– Delay or block in the electrical conduction through the bundle branches. Left Bundle Branch Block (LBBB): QRS >120 ms, broad notched R waves in lateral leads (I, V5-V6), deep S waves in V1-V2. Right Bundle Branch Block (RBBB): QRS >120 ms, rSR' pattern in V1-V2, wide S waves in I, V5-V6. RBBB can occur in healthy individuals or with RV strain. LBBB often indicates structural heart disease, cardiomyopathy, or anterior MI. Management: evaluate underlying cause; consider cardiac resynchronization therapy for LBBB with heart failure.

Bundle of His

– A narrow band of specialized cardiac tissue that conducts electrical impulses from the AV node through the fibrous cardiac skeleton to the interventricular septum. It divides into the right and left bundle branches. The bundle of His is the only electrical connection between atria and ventricles through the fibrous skeleton. Damage here causes complete (third-degree) heart block.

Calcium Channel Blockers

– Agents that block L-type calcium channels, reducing calcium influx into cardiac and smooth muscle cells. Dihydropyridines (amlodipine, nifedipine) cause vasodilation and are used for hypertension and angina. Non-dihydropyridines (verapamil, diltiazem) reduce heart rate and contractility, used for rate control in atrial fibrillation and angina. Side effects include peripheral edema, constipation (verapamil), and bradycardia. Non-dihydropyridines are contraindicated in HFrEF.

Caput Medusae

– Dilated periumbilical veins radiating outward from the umbilicus, seen in portal hypertension with recanalization of the umbilical vein. Not strictly a cardiac finding but relevant in right heart failure causing hepatic congestion and portal hypertension. It must be distinguished from caval collateral circulation. The sign is named after the Gorgon Medusa of Greek mythology.

Cardiac Amyloidosis

– Deposition of amyloid fibrils in the myocardium, causing restrictive cardiomyopathy with thickened, stiff ventricular walls. The most common cardiac amyloidosis is due to transthyretin (ATTR) amyloidosis, which can be wild-type (age-related) or hereditary. Diagnosis may require endomyocardial biopsy, though noninvasive diagnosis with bone scintigraphy is increasingly used. Treatment includes TTR stabilizers (tafamidis) and diuretics for symptom management.

Cardiac Amyloidosis Diagnosis

– Diagnosis involves a combination of clinical suspicion, imaging, and tissue biopsy. Non-invasive diagnosis using Tc-99m pyrophosphate (PYP) or DPD scintigraphy for ATTR cardiac amyloidosis (with negative serum/urine free light chains to exclude AL amyloidosis). Cardiac MRI shows characteristic late gadolinium enhancement pattern (subendocardial, circumferential). Endomyocardial biopsy remains the gold standard for confirming amyloid deposition.

Cardiac Amyloidosis Staging

– Mayo Clinic staging system for ATTR cardiac amyloidosis: Stage I (NT-proBNP <3000 pg/mL), Stage II (NT-proBNP 3000-8499), Stage III (NT-proBNP >=8500). Median survival: Stage I: 69 months, Stage II: 47 months, Stage III: 14 months. Treatment with tafamidis (TTR stabilizer) improves survival and functional capacity. ATTR-CM is increasingly recognized and may be underdiagnosed in elderly patients with unexplained LV hypertrophy.

Cardiac Arrest

– Sudden cessation of effective cardiac function, resulting in loss of pulse and consciousness. Most common rhythm in adults is ventricular fibrillation or pulseless ventricular tachycardia. Treatment follows ACLS protocol: high-quality CPR, defibrillation for shockable rhythms, and ACLS medications (epinephrine, amiodarone). Targeted temperature management improves neurological outcomes. Survival rates vary by rhythm and location: VF/VT has better prognosis than asystole/PEA.

Cardiac Arrest Survival

– Overall survival to discharge for out-of-hospital cardiac arrest is approximately 10%. For witnessed VF/pVT with early defibrillation, survival can exceed 50%. Chain of survival: early recognition and activation of EMS, high-quality CPR, early defibrillation, advanced life support, and post-arrest care. Targeted temperature management (32-36 degrees C) for 24 hours improves neurological outcomes. Coronary angiography should be considered for all comatose survivors.

Cardiac Catheterization

– Invasive procedure involving insertion of catheters into the heart via peripheral arteries or veins. Diagnostic catheterization measures intracardiac pressures, saturations, and performs coronary angiography. Indications include evaluation of CAD, valvular disease, cardiomyopathy, and congenital heart disease. Therapeutic catheterization includes PCI, valvuloplasty, and device closure. Complications include bleeding, vessel injury, arrhythmias, and contrast-induced nephropathy.

Cardiac Conduction System

– The specialized tissue responsible for generating and conducting electrical impulses through the heart. Components include the sinoatrial (SA) node, internodal pathways, atrioventricular (AV) node, bundle of His, right and left bundle branches, and Purkinje fibers. The system ensures coordinated contraction: atria contract first, followed by ventricles, with appropriate delay at the AV node.

Cardiac CT Angiography

– Noninvasive imaging using CT to visualize coronary arteries and cardiac structures. It has high negative predictive value for excluding coronary artery disease, making it useful for low-to-intermediate risk patients with chest pain. Calcium scoring quantifies coronary artery calcification as a marker of atherosclerosis. Limitations include heart rate control requirements, radiation exposure, and inability to assess hemodynamic significance of stenosis. It also evaluates aortic and congenital heart disease.

Cardiac Cycle

– The sequence of events occurring during one complete heartbeat, consisting of systole (contraction) and diastole (relaxation). During diastole, the ventricles fill with blood; during systole, they contract and eject blood. The cardiac cycle includes isovolumetric relaxation, ventricular filling, isovolumetric contraction, and ventricular ejection phases. The cycle takes approximately 0.8 seconds at a heart rate of 75 bpm.

Cardiac Index

– Cardiac output normalized to body surface area (BSA), calculated as CO / BSA. Normal cardiac index is 2.5-4.2 L/min/m2. It provides a more accurate assessment of cardiac function than cardiac output alone, accounting for body size. A cardiac index below 2.2 L/min/m2 indicates cardiogenic shock. Cardiac index is used to guide therapy in critically ill patients.

Cardiac Metastases

– More common than primary cardiac tumors. Most frequent sources: lung cancer, breast cancer, melanoma, lymphoma, and renal cell carcinoma. Most commonly involve the pericardium (effusion, tamponade), but can affect myocardium and endocardium. Presentation depends on location: pericardial effusion, heart failure, arrhythmias, or embolic events. Diagnosis by echocardiography, cardiac MRI, or CT. Treatment focuses on systemic cancer therapy; cardiac surgery rarely indicated.

Cardiac Output Measurement

– Methods include: thermodilution via pulmonary artery catheter (bolus of cold saline and temperature change measurement), Fick principle (O2 consumption / arteriovenous O2 difference), echocardiographic Doppler (flow velocity time integral x valve area x heart rate), and non-invasive cardiac output monitors (bioimpedance, bioreactance). CO measurement guides management of heart failure, shock, and perioperative hemodynamic optimization.

Cardiac Pacemaker Types

– Single-chamber: pacing one chamber (atrium or ventricle). Dual-chamber (DDD): paces and senses both atrium and ventricle, maintaining AV synchrony. Biventricular (CRT): paces both ventricles for resynchronization in heart failure. Rate-responsive (RR): automatically adjusts heart rate based on activity level using accelerometers or minute ventilation sensing. MRI-conditional pacemakers are now standard, allowing safe MRI scanning under specific conditions.

Cardiac PET Imaging

– Positron emission tomography using radiotracers (Rb-82, N-13 ammonia, O-15 water) for quantitative myocardial perfusion assessment. Advantages over SPECT: superior resolution, shorter scan time, attenuation correction, and absolute myocardial blood flow quantification (mL/g/min). FDG PET identifies hibernating myocardium (viable but ischemic tissue that may benefit from revascularization). Used for viability assessment, sarcoidosis evaluation, and cardiac infection imaging.

Cardiac Radiation Injury

– Late complication of mediastinal radiation therapy (Hodgkin lymphoma, breast cancer, lung cancer). Manifestations include pericardial disease (constrictive pericarditis, effusion), coronary artery disease (accelerated atherosclerosis), valvular disease (fibrosis), myocardial fibrosis, and conduction abnormalities. Risk increases with higher radiation doses and larger treatment fields. Long-term surveillance with echocardiography is recommended for survivors.

Cardiac Rehabilitation

– A supervised program of exercise, education, and counseling for patients with cardiovascular disease. It includes exercise training, risk factor modification, nutrition counseling, and psychosocial support. Cardiac rehab reduces cardiovascular mortality by 20-25% and improves functional capacity, quality of life, and adherence to medical therapy. It is recommended for patients after MI, PCI, CABG, heart failure, and valve surgery.

Cardiac Rehabilitation Phases

– Phase I: in-hospital mobilization and education. Phase II: outpatient supervised exercise and risk factor modification (6-12 weeks post-discharge, 36 sessions). Phase III: long-term maintenance with unsupervised or community-based exercise. Cardiac rehab improves survival, functional capacity, and quality of life while reducing recurrent cardiovascular events and rehospitalization. Despite proven benefits, participation rates remain suboptimal.

Cardiac Rehab Outcomes

– Cardiac rehabilitation reduces all-cause mortality by 13% and cardiovascular mortality by 26% post-MI. It improves exercise capacity, lipid profiles, blood pressure control, and glycemic control. Psychosocial benefits include reduced depression and anxiety. Despite Class I recommendation, only 20-30% of eligible patients participate. Barriers include access, cost, referral processes, and patient motivation. Home-based programs may improve participation rates.

Cardiac Resynchronization Therapy

– Biventricular pacing that synchronizes contraction of the left and right ventricles in patients with heart failure, reduced EF, and prolonged QRS duration (especially LBBB). CRT improves symptoms, exercise capacity, and quality of life, and reduces heart failure hospitalizations and mortality. Response may be incomplete in 30% of patients. Selection criteria include EF 35% or less, NYHA II-IV despite optimal medical therapy, and QRS 150 ms or greater.

Cardiac Sarcoidosis

– Infiltrative disease characterized by non-caseating granulomas in the myocardium, affecting 25-30% of patients with systemic sarcoidosis. Presentation includes conduction abnormalities (AV block), ventricular arrhythmias, and heart failure. Diagnosis requires cardiac MRI showing late gadolinium enhancement (typically basal septal and epicardial), PET showing active inflammation, or endomyocardial biopsy. Treatment: corticosteroids for active inflammation, ICD for arrhythmia prevention.

Cardiac Skeleton

– A dense connective tissue framework composed of four fibrous rings that surround the heart valves and provide structural support. The cardiac skeleton includes the annuli fibrosi (valve rings), the membranous septum, and the left and right fibrous trigones. It serves as electrical insulation between atria and ventricles, forcing impulses through the AV node, and provides attachment points for valve leaflets and cardiac muscle.

Cardiac Stress Testing

– Assessment of cardiac function under physiological or pharmacological stress to detect ischemia. Exercise stress testing involves treadmill or bicycle exercise with continuous ECG monitoring. Pharmacological stress (dobutamine, adenosine, regadenoson) is used when patients cannot exercise. Stress testing combined with imaging (nuclear perfusion, echocardiography, cardiac MRI) improves sensitivity and specificity. Indications include evaluation of chest pain, preoperative risk assessment, and prognosis.

Cardiac Tamponade Physiology

– Accumulation of pericardial fluid causing compression of cardiac chambers, impairing diastolic filling and reducing cardiac output. Hemodynamic consequences include equalization of diastolic pressures (RA = RV diastolic = PCWP), pulsus paradoxus, and decreased cardiac output. Beck's triad (hypotension, muffled heart sounds, JVD) is classic but not always present. Emergent pericardiocentesis is life-saving. Echocardiography shows RA/RV diastolic collapse.

Cardiac Tumors

– Primary cardiac tumors are rare; most common benign is myxoma (75%), typically in the left atrium presenting with embolism, obstruction, or constitutional symptoms. Malignant primary cardiac tumors are rare (angiosarcoma most common). Metastatic tumors to the heart are more common than primary tumors, with lung, breast, melanoma, and lymphoma being common sources. Diagnosis is by echocardiography, cardiac MRI, or CT. Treatment depends on tumor type.

Cardiac Tumors Management

– Myxoma: surgical excision is curative, with low recurrence rate. Benign tumors (rhabdomyoma, fibroma) may be observed if asymptomatic but require surgery if causing obstruction or arrhythmia. Malignant tumors: prognosis is poor; treatment may include surgery, chemotherapy, and radiation depending on tumor type. Metastatic disease: treatment of the primary malignancy; cardiac involvement indicates advanced disease. Pericardial effusion may require drainage.

Cardiogenic Shock Management

– Low cardiac output state with end-organ hyporrfusion, typically from acute MI. Inotropes: dobutamine (increases cardiac output, decreases SVR), milrinone (PDE3 inhibitor, vasodilator). Vasopressors: norepinephrine (first-line for hypotension), dopamine. Mechanical support: IABP (modest support, SHOCK II trial no mortality benefit), Impella (percutaneous LVAD, higher support), VA-ECMO (biventricular support). Emergent revascularization is the definitive treatment.

Cardiology

– The branch of medicine devoted to the study, diagnosis, and treatment of disorders of the heart and circulatory system. Cardiologists manage conditions including coronary artery disease, heart failure, valvular heart disease, arrhythmias, cardiomyopathy, pericardial disease, congenital heart disease, and hypertension. Diagnostic tools: electrocardiography, echocardiography, cardiac catheterization, stress testing, cardiac MRI, CT angiography, and nuclear cardiac imaging. Interventional cardiology performs percutaneous coronary intervention and structural heart procedures (TAVR, MitraClip). Electrophysiology manages rhythm disorders with ablation and device therapy (pacemakers, ICDs). Preventive cardiology focuses on risk factor modification for cardiovascular disease.

Cardiomyopathy Classification

– The MOGE(S) classification system provides phenotypic and genotypic description: M (morphofunctional phenotype), O (organ/system involvement), G (genetic inheritance), E (etiology), S (functional status/NYHA class). This replaces older descriptive classifications and provides more precise characterization for prognosis and family counseling. Each cardiomyopathy can be classified as primary (genetic or acquired) or secondary (due to systemic disease).

Cardio-Oncology

– The subspecialty focusing on cardiovascular care of cancer patients, addressing cancer therapy-related cardiac dysfunction, prevention, and management. Key concerns include anthracycline cardiotoxicity, trastuzumab-related cardiomyopathy, immune checkpoint inhibitor myocarditis, and radiation-induced heart disease. Monitoring includes baseline and serial echocardiography during and after cancer treatment. Early detection and intervention can prevent irreversible cardiac damage and allow continuation of cancer therapy.

Cardiopulmonary Exercise Testing

– Comprehensive assessment of cardiovascular and pulmonary function during graded exercise on a treadmill or cycle ergometer. Measures: VO2 max, anaerobic threshold, ventilatory efficiency (VE/VCO2 slope), and hemodynamic responses. Used to evaluate dyspnea on exertion, determine heart failure prognosis (VO2 max <14 mL/kg/min indicates need for transplant evaluation), assess exercise capacity for disability evaluation, and guide cardiac rehabilitation programs.

Cardiorenal Syndrome

– Dysfunction of the heart and kidneys where acute or chronic failure of one organ contributes to dysfunction of the other. Type 1: acute cardiac dysfunction causing acute kidney injury (e.g., cardiogenic shock). Type 2: chronic cardiac dysfunction causing progressive CKD. Type 3: acute kidney injury causing cardiac dysfunction. Type 4: chronic kidney disease causing cardiac dysfunction. Type 5: systemic conditions affecting both organs. Management is challenging and requires coordinated care.

Cardiotoxicity from Chemotherapy

– Chemotherapy agents that damage the myocardium, most commonly anthracyclines (doxorubicin, daunorubicin). Cardiotoxicity presents as dilated cardiomyopathy with reduced ejection fraction. Risk is dose-dependent (doxorubicin cumulative dose >550 mg/m2). Other cardiotoxic agents include trastuzumab (reversible cardiomyopathy), cyclophosphamide, and 5-fluorouracil. Monitoring with serial echocardiography during treatment is essential. Treatment includes ACE inhibitors and beta-blockers.

Cardiovascular Health

– Cardio is shorthand for cardiovascular health or exercise that elevates heart rate and respiratory effort to strengthen the heart, lungs, and circulatory system. Aerobic activities such as walking, running, cycling, and swimming improve cardiac output, lower blood pressure, and enhance metabolic efficiency. Regular cardiovascular exercise reduces the risk of heart disease, stroke, and diabetes while improving overall longevity.

Cardiovascular System

– The organ system comprising the heart and blood vessels (arteries, arterioles, capillaries, venules, veins) responsible for circulating blood throughout the body. Functions: delivery of oxygen and nutrients to tissues, removal of carbon dioxide and metabolic wastes, transport of hormones and immune cells, thermoregulation, and maintenance of fluid and electrolyte homeostasis. The heart is a four-chambered muscular pump (two atria, two ventricles) that generates the pressure gradient for blood flow. The vascular system forms a closed circuit: systemic circulation (oxygenated blood from left heart to body, deoxygenated blood returns to right heart) and pulmonary circulation (deoxygenated blood from right heart to lungs, oxygenated blood returns to left heart).

Central Venous Pressure

– Pressure in the superior vena cava or right atrium, measured via central venous catheter. Normal: 2-8 mmHg. Elevated CVP indicates right heart failure, fluid overload, cardiac tamponade, or constrictive pericarditis. Low CVP indicates hypovolemia. CVP monitoring guides fluid resuscitation and diuretic therapy in heart failure management. Less invasive alternatives include bedside ultrasound assessment of IVC diameter and collapsibility.

Chagas Disease

– A parasitic infection caused by Trypanosoma cruzi, transmitted by reduviid bugs (kissing bugs), endemic in Latin America. Acute phase: fever, swelling at bite site (chagoma), Romana sign (periorbital edema). Chronic phase (years later): Chagas cardiomyopathy (most common and serious), megaesophagus, and megacolon. Cardiomyopathy: progressive heart failure, apical aneurysm, thrombus formation, heart block, and sudden cardiac death. Diagnosis: serology (ELISA, IFA), PCR in acute phase. Treatment: benznidazole or nifurtimox for acute and early chronic phase. Late chronic cardiomyopathy: symptomatic heart failure management, anticoagulation, and pacemaker/ICD as indicated.

Cholesterol

– A waxy, fat-like substance produced by the liver and obtained from food that is essential for cell membrane structure, hormone synthesis, and vitamin D production. It circulates in the blood bound to lipoproteins: low-density lipoprotein LDL carries cholesterol to tissues and can contribute to artery plaque, while high-density lipoprotein HDL helps remove excess cholesterol. Elevated LDL cholesterol is a major modifiable risk factor for cardiovascular disease.

Cholesterol and Lipids

– Fatty substances in the blood, including LDL cholesterol, HDL cholesterol, and triglycerides, that play critical roles in cellular function and energy metabolism. Abnormal lipid levels, particularly high LDL and triglycerides with low HDL, promote atherosclerosis and increase the risk of heart attack and stroke. Management includes dietary modification, exercise, and statins or other lipid-lowering medications.

Chronic Venous Insufficiency

– Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. Leads to venous hypertension, edema, skin changes (lipodermatosclerosis, hyperpigmentation), and venous stasis ulcers (medial malleolus). Risk factors: varicose veins, DVT, obesity, prolonged standing. Diagnosis: clinical, venous duplex ultrasound. Treatment: compression therapy (graded stockings), leg elevation, exercise, and wound care. Severe cases: endovenous ablation, venous stenting, or surgical reconstruction.

Circumflex Artery

– A branch of the left coronary artery that travels in the left atrioventricular groove. It supplies the lateral wall of the left ventricle, the left atrium, and in left-dominant systems, the inferior wall and posteromedial papillary muscle. Circumflex artery occlusion causes lateral wall myocardial infarction, which may be missed on standard ECG leads.

Class IA Antiarrhythmics

– Sodium channel blockers that slow phase 0 depolarization and prolong the action potential duration and refractory period. Examples include quinidine, procainamide, and disopyramide. Used for atrial and ventricular arrhythmias. Quinidine can cause QT prolongation and torsades de pointes. Procainamide may cause lupus-like syndrome with long-term use. These drugs are generally reserved for cases not controlled by first-line agents due to proarrhythmic potential.

Class IB Antiarrhythmics

– Sodium channel blockers that shorten the action potential duration and are preferentially taken up by ischemic tissue. Examples include lidocaine and mexiletine. Lidocaine is used for ventricular tachycardia and ventricular fibrillation, particularly in the setting of acute myocardial infarction. It has minimal effect on normal myocardium. Side effects include CNS toxicity (seizures, confusion) and cardiac depression.

Class IC Antiarrhythmics

– Sodium channel blockers that significantly slow phase 0 depolarization without affecting action potential duration. Examples include flecainide and propafenone. Used for supraventricular arrhythmias in patients without structural heart disease. The CAST trial showed increased mortality in post-MI patients, so they are contraindicated in that setting. Flecainide is effective for atrial fibrillation and atrial flutter in structurally normal hearts.

Class III Antiarrhythmics

– Potassium channel blockers that prolong the action potential duration and refractory period. Examples include amiodarone, sotalol, dofetilide, and ibutilide. Amiodarone is the most commonly used but has significant toxicity. Sotalol has additional beta-blocking properties and can cause torsades de pointes. Dofetilide requires inpatient initiation with QT monitoring. These drugs are used for atrial fibrillation and ventricular arrhythmias.

Clopidogrel

– An antiplatelet agent that irreversibly inhibits the P2Y12 ADP receptor on platelets, preventing platelet aggregation. It is a prodrug requiring CYP2C19 activation. Used in acute coronary syndromes, post-PCI, and stroke prevention. Clopidogrel resistance is associated with CYP2C19 poor metabolizer status. Dual antiplatelet therapy (aspirin + clopidogrel) is standard after ACS and PCI. Ticagrelor and prasugrel are alternatives with more rapid onset.

Coarctation of the Aorta

– A congenital narrowing of the aorta, typically near the ductus arteriosus. Presents with upper extremity hypertension, diminished femoral pulses, and a blood pressure gradient between arms and legs. Associated with bicuspid aortic valve and other congenital heart defects. Repair is via surgical resection or balloon angioplasty with stent placement.

Cocaine Cardiotoxicity

– Cocaine blocks norepinephrine and dopamine reuptake, causing sympathomimetic effects: tachycardia, hypertension, and coronary vasospasm. Acute myocardial infarction can occur even with normal coronary arteries due to vasospasm, accelerated atherosclerosis, and prothrombotic effects. Treatment: benzodiazepines for sympatholysis. Beta-blockers are contraindicated (unopposed alpha stimulation worsens vasospasm). Calcium channel blockers and nitroglycerin for vasospasm.

Collateral Circulation

– Alternative vascular pathways that develop when coronary arteries become chronically narrowed. Collaterals connect different coronary artery territories and can partially compensate for stenosis. They develop gradually in response to ischemia and are more prominent in patients with chronic coronary artery disease. Well-developed collaterals can reduce infarct size during acute occlusion.

Continuous Murmurs

– Murmurs that persist throughout systole and diastole without interruption, produced by blood flowing continuously through an abnormal connection. The most common cause is patent ductus arteriosus (PDA), where blood flows from the aorta to pulmonary artery throughout the cardiac cycle. Other causes include arteriovenous fistulas and rupture of sinus of Valsalva. The murmur is characteristically machine-like in quality.

Contrast Echocardiography

– Use of microbubble contrast agents to enhance endocardial border definition and assess myocardial perfusion. Indicated for suboptimal non-contrast echo images and assessment of myocardial viability. Microbubbles remain entirely within the intravascular space, making them ideal for perfusion assessment. First-pass perfusion imaging detects areas of reduced myocardial blood flow. Contraindicated with intracardiac shunts (risk of paradoxical embolism).

Coronary Artery Disease

– Narrowing or occlusion of coronary arteries, usually due to atherosclerosis, reducing blood flow to the myocardium. Risk factors include hypertension, diabetes, hyperlipidemia, smoking, family history, and obesity. CAD can present as stable angina, acute coronary syndromes (unstable angina, NSTEMI, STEMI), or sudden cardiac death. Diagnosis includes ECG, stress testing, and coronary angiography. Treatment involves lifestyle modification, medications, and revascularization.

Coronary Calcium Score

– A CT-based measurement of coronary artery calcification, quantified as the Agatston score. A score of 0 indicates very low risk of cardiovascular events; >400 indicates high burden and significant atherosclerosis. It is used for cardiovascular risk stratification in asymptomatic individuals and may guide statin therapy decisions. It does not visualize coronary stenosis directly but reflects total atherosclerotic burden.

Coronary Circulation

– The vascular system supplying blood to the myocardium. The left coronary artery (LCA) divides into the left anterior descending (LAD) and circumflex (LCx) arteries. The right coronary artery (RCA) supplies the right ventricle, SA node (in 60% of people), and inferior left ventricular wall. Coronary arteries branch into arterioles, capillaries, and drain into coronary veins and the coronary sinus.

Coronary Dominance

– Determined by which artery gives rise to the posterior descending artery (PDA). Right coronary dominance (85% of people): RCA gives rise to PDA. Left coronary dominance (8%): LCx gives rise to PDA. Codominance (7%): both RCA and LCx contribute. Dominance affects which territories are supplied by each artery and influences the pattern of myocardial infarction.

Coronary Microvascular Dysfunction

– Impairment of the small coronary arteries and arterioles causing impaired coronary flow reserve despite normal epicardial coronary arteries. Common in women, patients with diabetes, and hypertension. Causes chest pain (angina equivalent) and may contribute to heart failure with preserved ejection fraction. Diagnosis: coronary flow reserve measurement during invasive angiography or by PET/MRI. Treatment: statins, ACE inhibitors, ranolazine, and risk factor modification.

Coronary Thrombosis

– See Myocardial Infarction (Heart Attack).

Corticosteroids

– Synthetic glucocorticoids with potent anti-inflammatory and immunosuppressive effects. Examples include prednisone, prednisolone, methylprednisolone, dexamethasone, and hydrocortisone. They suppress inflammatory gene expression and inhibit immune cell function. Used in autoimmune diseases, organ transplant rejection, severe allergies, and adrenal insufficiency. Side effects include osteoporosis, hyperglycemia, weight gain, avascular necrosis, cataracts, adrenal suppression, and increased infection risk.

Cor Triatriatum

– A rare congenital heart defect where the left atrium is divided into two chambers by a fibromuscular membrane, separating the pulmonary venous chamber (posterior) from the true left atrium (anterior) containing the left atrial appendage and mitral valve. The membrane obstructs pulmonary venous return, mimicking mitral stenosis. Presents in infancy with dyspnea, failure to thrive, recurrent respiratory infections, and pulmonary edema (if severe obstruction). Milder cases may present in adulthood with exertional dyspnea and pulmonary hypertension. Diagnosis: echocardiography shows the characteristic membrane in the left atrium with color Doppler demonstrating flow obstruction. Treatment: surgical excision of the membrane. Prognosis is excellent after repair.

CYP450 System

– A superfamily of heme-containing enzymes primarily in the liver that metabolize drugs and xenobiotics. Major isoforms include CYP3A4 (metabolizes 50% of drugs), CYP2D6, CYP2C9, CYP2C19, CYP1A2, and CYP2E1. CYP inducers (rifampin, carbamazepine, phenytoin) decrease drug levels; CYP inhibitors (ketoconazole, erythromycin, grapefruit juice) increase drug levels. Genetic polymorphisms in CYP enzymes affect drug metabolism, forming the basis of pharmacogenomics.

Deep Vein Thrombosis

– Formation of a blood clot in a deep vein, most commonly in the lower extremities. Risk factors include immobility, surgery, cancer, pregnancy, and thrombophilia. Presents with unilateral leg swelling, pain, and erythema. Diagnosis by venous ultrasound and D-dimer testing. Treatment: anticoagulation (heparin, DOACs, warfarin). Complications: pulmonary embolism and post-thrombotic syndrome.

Dextrocardia

– A congenital cardiac positional anomaly where the heart is located in the right hemithorax with the apex pointing to the right. Most cases are due to situs inversus totalis (complete mirror-image reversal of all thoracic and abdominal organs, affecting 1 in 10,000 births), which carries a normal life expectancy if cardiac anatomy is otherwise normal. Dextrocardia with situs solitus (heart on right but abdominal organs normally positioned) is strongly associated with congenital heart disease, particularly corrected transposition of the great arteries.

Diastole

– The phase of the cardiac cycle when the heart muscle relaxes, allowing the chambers to fill with blood. Ventricular diastole begins with isovolumetric relaxation (all valves closed, pressure falls), followed by rapid ventricular filling, slow filling (diastasis), and atrial contraction (atrial kick, contributing 20% of ventricular filling in normal hearts). During diastole, coronary blood flow to the left ventricular myocardium occurs (systolic compression of coronary arteries prevents flow during contraction). Diastolic dysfunction (impaired relaxation or reduced compliance) is a hallmark of heart failure with preserved ejection fraction (HFpEF) and restrictive cardiomyopathy.

Diastolic Murmurs

– Murmurs occurring between S2 and the next S1. Early diastolic decrescendo murmurs indicate semilunar valve regurgitation (aortic regurgitation, pulmonary regurgitation). Mid-diastolic low-pitched rumbles indicate AV valve stenosis (mitral stenosis, tricuspid stenosis). Late diastolic murmurs may indicate mitral stenosis with sinus rhythm. Diastolic murmurs are almost always pathological and require echocardiographic evaluation.

Digitalis toxicity

– A potentially life-threatening adverse effect of cardiac glycoside therapy (digoxin, digitoxin) characterized by a narrow therapeutic index (0.5-2.0 ng/mL for digoxin). Toxicity occurs through Na+/K+-ATPase inhibition, increasing intracellular sodium, which slows calcium efflux via the Na+/Ca2+ exchanger, causing increased intracellular calcium and enhanced cardiac contractility, while also increasing vagal tone and slowing AV conduction. Risk factors include advanced age, renal impairment (digoxin is renally cleared),.

Dilated Cardiomyopathy

– A cardiomyopathy characterized by ventricular dilation and impaired systolic function, resulting in reduced ejection fraction. It is the most common cause of heart failure and the leading indication for heart transplantation. Causes include idiopathic

(30-50%), familial/genetic (20-35%), alcohol, viral myocarditis, peripartum, chemotherapy (doxorubicin), and tachycardia-mediated. Treatment follows standard heart failure guidelines with ACE inhibitors, beta-blockers, diuretics, and device therapy (ICD, CRT).

Dilated Cardiomyopathy Prognosis

– 5-year survival approximately 50%. Risk factors for poor prognosis: low EF, advanced NYHA class, elevated BNP, ventricular arrhythmias, family history of sudden death, and failure to respond to medical therapy. Sudden cardiac death accounts for 30-50% of mortality. ICD placement for primary prevention in patients with EF <35% despite optimal medical therapy. Heart transplant is definitive therapy for end-stage disease.

Direct Oral Anticoagulants

– DOACs directly inhibit specific coagulation factors. Direct thrombin inhibitors: dabigatran. Factor Xa inhibitors: rivaroxaban, apixaban, edoxaban, betrixaban. They have predictable pharmacokinetics, fixed dosing, and routine monitoring is not required. They have fewer drug interactions than warfarin. Dabigatran and rivaroxaban are reversed by specific agents (idarucizumab, andexanet alfa). DOACs are preferred over warfarin for non-valvular atrial fibrillation and VTE.

Dobutamine Stress Echocardiography

– Pharmacological stress test using dobutamine (beta-1 agonist) to increase heart rate, contractility, and myocardial oxygen demand. New or worsening wall motion abnormalities indicate ischemia. Preferred when patients cannot exercise. Beta-blockers should be held before testing. Sensitivity 80-85% and specificity 80-88% for detecting significant coronary artery disease. Additional atropine may be administered if target heart rate not achieved.

Double Outlet Right Ventricle

– A congenital heart defect where more than 50% of both great arteries arise from the right ventricle, with a VSD as the only outlet from the left ventricle. Classified by the position of the VSD relative to the great arteries: subaortic VSD (TOF-type physiology), subpulmonic VSD (Taussig-Bing malformation, TGA-type physiology), doubly committed VSD, and non-committed VSD (remote from both arteries). Presents with variable physiology: cyanosis or heart failure depending on the degree of pulmonary blood flow. The Taussig-Bing malformation presents with cyanosis and early heart failure. Diagnosis: echocardiography showing both great arteries arising from the RV. Treatment: surgical repair tailored to VSD position — arterial switch operation for Taussig-Bing, intracardiac tunnel repair for subaortic VSD.

Dressler Syndrome

– A delayed pericarditis occurring 2-10 weeks after myocardial infarction or cardiac surgery. It is an autoimmune-mediated inflammation caused by antibodies against myocardial antigens released during tissue injury. Symptoms include fever, pleuritic chest pain, and pericardial friction rub. ECG may show diffuse ST elevation. Treatment includes NSAIDs, colchicine, and avoidance of anticoagulants if possible. It is a type of post-cardiac injury syndrome.

Drug Bioavailability

– The fraction of an administered drug that reaches systemic circulation unchanged. Oral bioavailability is affected by first-pass metabolism, absorption, and drug formulation. IV administration has 100% bioavailability. Bioavailability influences drug dosing and route selection. Drugs with low oral bioavailability (e.g., morphine 25%) require higher oral doses compared to IV doses. Food can increase or decrease bioavailability depending on the drug.

Drug Clearance

– The volume of plasma completely cleared of drug per unit time. Hepatic clearance depends on hepatic blood flow and enzyme activity; renal clearance depends on glomerular filtration, tubular secretion, and reabsorption. Total clearance determines maintenance dose. Clearance decreases in hepatic or renal impairment, requiring dose adjustment. Drug clearance is independent of concentration for first-order kinetics but constant for zero-order (saturable) kinetics.

Drug Hypersensitivity Reactions

– Immune-mediated adverse drug reactions classified by Gell and Coombs. Type I (immediate, IgE-mediated: anaphylaxis, urticaria); Type II (cytotoxic, IgG/IgM-mediated: hemolytic anemia, thrombocytopenia); Type III (immune complex-mediated:

serum sickness, vasculitis); Type IV (delayed, T-cell mediated: contact dermatitis, Stevens-Johnson syndrome). Management includes drug discontinuation, supportive care, and in severe cases, immunosuppression. Cross-reactivity between related drugs must be considered.

Drug Overdose Management

– Treatment of drug overdose involves supportive care (airway, breathing, circulation), decontamination (gastric lavage, activated charcoal within 1 hour), enhanced elimination (urinary alkalinization, hemodialysis), and specific antidotes when available. Key antidotes include naloxone (opioids), flumazenil (benzodiazepines), N-acetylcysteine (acetaminophen), digoxin immune Fab (digoxin), and physostigmine (anticholinergic toxicity). Treatment should be tailored to the specific drug involved.

Dual Antiplatelet Therapy

– The combination of aspirin and a P2Y12 inhibitor (clopidogrel, prasugrel, or ticagrelor) for prevention of thrombotic events after acute coronary syndrome and percutaneous coronary intervention. Duration varies: 1 month after bare-metal stent, 6-12 months after drug-eluting stent, and 12 months after ACS. Prasugrel is more potent but increases bleeding risk; ticagrelor is reversible but requires twice-daily dosing. DAPT score guides duration of therapy.

Ductus Arteriosus

– A fetal blood vessel connecting the pulmonary artery to the descending aorta, allowing blood to bypass the non-functioning fetal lungs. In utero, the ductus arteriosus is kept patent by prostaglandin E2 and low oxygen tension. After birth, the ductus normally closes within 24–48 hours (functional closure) and becomes the ligamentum arteriosum by 2–3 weeks (anatomic closure). Failure of closure results in patent ductus arteriosus (PDA), a left-to-right shunt causing continuous machinery murmur, bounding pulses, and volume overload of the left heart. Prostaglandin E1 (alprostadil) is used to maintain ductal patency in ductal-dependent congenital heart disease (e.g., pulmonary atresia, coarctation of the aorta).

DVT

– See Deep Vein Thrombosis.

Ebstein Anomaly

– A rare congenital heart defect accounting for less than 1 percent of all congenital heart disease, characterized by apical displacement of the septal and posterior leaflets of the tricuspid valve into the right ventricle, resulting in atrialization of the proximal right ventricle and a small functional right ventricular chamber. The anterior leaflet is usually large and sail-like. The tricuspid valve is often regurgitant, causing right atrial enlargement and right-to-left shunting across an associated atrial septal defect or patent foramen ovale.

Echocardiography

– Ultrasound imaging of the heart, the primary modality for assessing cardiac structure and function. Transthoracic echocardiography (TTE) is the most common approach; transesophageal echocardiography (TEE) provides better views of posterior structures (left atrium, mitral valve, aorta). Doppler echocardiography assesses blood flow velocities, valve function, and pressure gradients. Echocardiography is used to evaluate ejection fraction, valve disease, pericardial effusion, and structural abnormalities.

Eisenmenger Syndrome

– Reversal of a left-to-right cardiac shunt due to irreversible pulmonary hypertension, resulting in right-to-left shunting and cyanosis. It occurs in untreated congenital heart defects with large left-to-right shunts (VSD, ASD, PDA) that cause progressive pulmonary vascular disease. Features include cyanosis, clubbing, polycythemia, and paradoxical embolism. Management is supportive; pregnancy is contraindicated. Lung transplantation may be considered in severe cases.

Ejection Click

– A high-pitched sound occurring at the beginning of systole, caused by opening of a stenotic semilunar valve. Aortic ejection click is heard in bicuspid aortic valve and aortic stenosis with pliable leaflets. Pulmonary ejection click is heard in pulmonary stenosis and disappears as the valve becomes more calcified and immobile. An ejection click helps distinguish valvular stenosis from subvalvular or supravascular obstruction.

Ejection Fraction

– The percentage of blood ejected from the left ventricle during systole, calculated as (stroke volume / end-diastolic volume) x 100. Normal EF is 55-

70%. Heart failure with reduced ejection fraction (HFrEF): EF 40% or less; heart failure with preserved ejection fraction (HFpEF): EF 50% or greater. EF is measured by echocardiography, cardiac MRI, or ventriculography.

Electrocardiogram (ECG / EKG)

– A recording of the heart's electrical activity using electrodes placed on the body surface. A standard 12-lead ECG provides views of the heart from different angles: limb leads (I, II, III, aVR, aVL, aVF) and precordial leads (V1-V6). ECG is used to diagnose arrhythmias, conduction disturbances, myocardial ischemia/infarction, chamber hypertrophy, electrolyte abnormalities, and drug effects.

Endocarditis

– Infection of the endocardial surface of the heart, most commonly involving the valves. Acute bacterial endocarditis: Staph aureus (most common in IV drug users and prosthetic valves), rapid destruction. Subacute: viridans group streptococci (pre-existing valve disease). Symptoms: fever, heart murmur, petechiae, splinter hemorrhages, Janeway lesions, Osler nodes, Roth spots. Diagnosis: modified Duke criteria, blood cultures (2-3 sets), echocardiography (transthoracic and transesophageal). Complications: valvular regurgitation, heart failure, embolic phenomena (stroke, septic emboli), perivalvular abscess. Treatment: prolonged IV antibiotics based on organism and sensitivity; surgical valve replacement for severe regurgitation, persistent infection, or large vegetations.

Endocarditis Surgical Indications

– Surgery indicated for: heart failure from valve dysfunction, uncontrolled infection (abscess, fistula, vegetation >10 mm with persistent emboli), fungal endocarditis, prosthetic valve dehiscence, and large vegetations (>15 mm) with embolic events. Timing: urgent (within days) for heart failure or uncontrolled infection; emergent (immediate) for acute aortic regurgitation with heart failure or ruptured mycotic aneurysm.

Endocardium

– The innermost layer of the heart wall, a thin membrane of endothelial cells and underlying connective tissue that lines the cardiac chambers and covers the heart valves. The endocardium is continuous with the endothelium of blood vessels and provides a smooth, non-thrombogenic surface. Endocardial damage can predispose to infective endocarditis, and endocardial fibrosis may occur in certain conditions.

Epicardium

– The outermost layer of the heart wall, which is the visceral layer of the serous pericardium. It consists of a single layer of mesothelial cells and underlying connective tissue containing blood vessels, nerves, and adipose tissue. The epicardium produces pericardial fluid that lubricates the heart surface. It can be involved in inflammatory processes such as pericarditis and may contain epicardial fat pads.

Ewart Sign

– Dullness to percussion, decreased breath sounds, and bronchial breathing at the left infrascapular region caused by compressive atelectasis from a large pericardial effusion. It is a physical examination finding of pericardial effusion. The effusion compresses the adjacent left lower lobe, creating a zone of consolidation. Ewart sign is best detected with the patient in the left lateral decubitus position.

Exercise-Induced Arrhythmias

– Catecholamine-sensitive ventricular tachycardia and right ventricular outflow tract VT can be triggered by exercise. HCM-related sudden death risk increases during vigorous exercise. Exercise-induced myocardial ischemia may trigger ventricular arrhythmias in CAD. Arrhythmogenic right ventricular cardiomyopathy frequently presents with exercise-related VT. Athletes with structural heart disease or inherited arrhythmia syndromes may require activity restriction and ICD placement.

Exercise Stress Test

– A noninvasive cardiac diagnostic test that evaluates the cardiovascular response to graded physical exertion, primarily used to diagnose coronary artery disease, assess functional capacity, and guide prognosis. The test is most commonly performed using a treadmill (Bruce protocol: graded increase in speed and incline every 3 minutes) or stationary bicycle ergometer, following the standard Bruce or modified Bruce protocol. During the test, continuous 12-lead ECG monitoring records heart rate, rhythm, and ST-segment changes.

Extracorporeal Membrane Oxygenation

– A temporary mechanical circulatory support system that provides cardiac and/or respiratory support. VA-ECMO (veno-arterial) provides both cardiac and respiratory support; VV-ECMO (venovenous) provides respiratory support only. Used as bridge to recovery or decision in severe cardiogenic shock, refractory cardiac arrest, or acute respiratory failure. Complications include bleeding, thromboembolism, limb ischemia, and neurological injury.

Fainting

– See Syncope (Fainting).

Familial Hypercholesterolemia

– An autosomal dominant genetic disorder causing severely elevated LDL cholesterol from birth, most commonly due to LDL receptor mutations. Heterozygous FH affects 1 in 250 people, causing premature coronary artery disease (MI in 40s-50s). Diagnosis: LDL >190 mg/dL in adults, or >160 mg/dL in children. Treatment: high-intensity statins, ezetimibe, PCSK9 inhibitors. Cascade screening of family members is essential. Lipoprotein apheresis may be needed for severe cases.

First-Degree AV Block

– A conduction delay in which every atrial impulse reaches the ventricles but with a prolonged PR interval (>0.20 seconds). It is caused by delayed conduction through the AV node or His-Purkinje system. First-degree block is usually benign and asymptomatic. Causes include increased vagal tone, drugs (beta-blockers, calcium channel blockers, digoxin), inferior myocardial infarction, and fibrotic degeneration. Treatment of the underlying cause is usually sufficient.

First-Pass Metabolism

– The metabolism of a drug by the liver before it reaches systemic circulation after oral administration. Drugs with high first-pass metabolism (e.g., morphine, nitroglycerin, lidocaine) have low oral bioavailability and may require alternative routes (sublingual, transdermal, IV). First-pass metabolism can be affected by liver disease, enzyme induction, and enzyme inhibition. Prodrugs (e.g., enalapril) require hepatic activation during first-pass metabolism.

Fractional Flow Reserve

– An invasive physiological measurement during coronary catheterization using a pressure wire to assess the hemodynamic significance of a coronary stenosis. FFR = Pa/Pd (aortic pressure / distal coronary pressure) during maximal hyperemia (adenosine). Normal FFR is 1.0; FFR <=0.80 indicates ischemia-producing stenosis warranting revascularization. The FAME and FAME 2 trials demonstrated that FFR-guided PCI improves outcomes compared to angiography-guided PCI.

Frank-Starling Mechanism

– The physiological principle stating that stroke volume increases as end-diastolic volume (preload) increases, up to an optimal point. Within physiological limits, greater ventricular stretch leads to stronger contraction due to increased calcium sensitivity of myofilaments and optimal overlap of actin and myosin. This mechanism allows the heart to automatically match output to venous return. In heart failure, the curve shifts downward and rightward.

Gallavardin Phenomenon

– The dissociation of the murmur of aortic stenosis into two components: a musical quality heard at the apex and a harsh, rough quality heard at the right upper sternal border. This occurs because high-frequency components are transmitted to the apex while low-frequency components are heard at the base. It helps distinguish aortic stenosis from mitral regurgitation, which also presents with an apical murmur.

Half-Life (t1/2)

– The time required for plasma drug concentration to decrease by 50%. It determines dosing frequency and time to steady state (4-5 half-lives). Drugs with long half-lives (e.g., amiodarone: 40-55 days, warfarin: 40 hours) can be given once daily but take longer to reach steady state and to be eliminated. Short half-lives (e.g., nitroglycerin: 3 minutes) require continuous infusion or frequent dosing. Half-life is affected by hepatic and renal function.

Heart

– The muscular organ responsible for pumping blood through the circulatory system. The human heart is a four-chambered, two-sided pump: the right heart (right atrium receiving deoxygenated blood from the body via superior and inferior vena cava, right ventricle pumping to the pulmonary circulation via the pulmonary artery) and the left heart (left atrium receiving oxygenated blood from the lungs via four pulmonary veins, left ventricle pumping to the systemic circulation via the aorta). The heart wall consists of three layers: epicardium (visceral pericardium), myocardium (cardiac muscle with intercalated discs, gap junctions for electrical coupling, and desmosomes for mechanical coupling), and endocardium (lining the chambers and valves). The heart is enclosed in the pericardium (fibrous outer sac with serous inner layer; 10-50 mL of pericardial fluid lubricates). The cardiac skeleton (fibrous rings around valves and membranous septum) provides structural support and electrical insulation between atria and ventricles. Blood supply: left main coronary artery (LCA, divides into left anterior descending–LAD–and circumflex–LCx) and right coronary artery (RCA, posterior descending artery–PDA–in right-dominant circulation). Venous drainage: coronary sinus (receives the great, middle, and small cardiac veins) emptying into the right atrium. Dimensions: approximately 12 x 9 x 6 cm, weight 250-350 grams. The heart beats approximately 100,000 times per day (2.5 billion times in a 70-year lifetime) and pumps about 5 litres/minute at rest (up to 25 litres/minute during exercise).

Heart Attack (Myocardial Infarction)

– Also known as myocardial infarction, the death of heart muscle tissue due to sudden blockage of a coronary artery, typically by a blood clot. It presents with chest pain or pressure, shortness of breath, nausea, and diaphoresis, though symptoms may be atypical in women and diabetics. Emergency treatment includes reperfusion therapy via angioplasty or thrombolytics, followed by long-term medications and lifestyle changes.

Heart Block

– A delay or interruption in the conduction of cardiac impulses from the atria to the ventricles through the atrioventricular (AV) node and His-Purkinje system. Classification: first-degree AV block: PR interval >200 ms, each atrial impulse conducts to the ventricles, typically asymptomatic. Second-degree AV block: Mobitz type I (Wenckebach): progressive PR prolongation then a dropped QRS, usually benign, localised to AV node. Mobitz type II: constant PR interval with intermittent dropped QRS, indicates disease below the AV node (His-Purkinje), risk of progression to complete heart block. Third-degree (complete) AV block: no atrial impulses conducted to ventricles; atria and ventricles beat independently (AV dissociation). Ventricular escape rhythm (wide QRS, rate 30-50 bpm) or junctional escape (narrow QRS, rate 40-60 bpm). Causes: idiopathic fibrosis (Lenegre-Lev disease, most common), ischemic heart disease, myocardial infarction (inferior MI–AV nodal block, usually transient; anterior MI–infranodal block, often permanent, poor prognosis), cardiomyopathy, drugs (beta-blockers, calcium channel blockers, digoxin), electrolyte abnormalities (hyperkalaemia), Lyme disease, Chagas disease, endocarditis (aortic valve abscess), post-cardiac surgery, and infiltrative diseases (sarcoidosis, amyloidosis). Symptoms: syncope, presyncope, dizziness, fatigue, dyspnea, angina (see Stokes-Adams attacks). Treatment: urgent pacing for symptomatic or high-grade AV block (type II second-degree or third-degree). Temporary pacing: transcutaneous (external) or transvenous. Permanent pacemaker: dual-chamber (DDD) preferred for sinus node function, single-chamber (VVI) for atrial fibrillation with AV block. Dual-chamber pacing maintains AV synchrony and reduces atrial fibrillation and heart failure risk.

Heartburn

– A burning sensation, typically retrosternal (behind the sternum), that may radiate to the throat and is often associated with regurgitation of sour or bitter-tasting fluid into the mouth. Despite its name, heartburn is a symptom of gastro-oesophageal reflux disease (GORD), not a cardiac symptom. It is caused by reflux of acidic gastric contents into the lower esophagus, stimulating chemoreceptors in the oesophageal mucosa. Exacerbating factors: large meals, fatty or spicy foods, chocolate, caffeine, alcohol, smoking, bending forward, lying supine after eating, pregnancy, obesity, and hiatus hernia. Patients often self-medicate with antacids (aluminium/magnesium hydroxide), alginates (Gaviscon), H2 antagonists (ranitidine, famotidine), or proton pump inhibitors (omeprazole). Heartburn can be confused with: cardiac chest pain (myocardial ischemia/infarction–pressure, tightness, radiation to left arm/jaw, shortness of breath, diaphoresis), oesophageal spasm (painful, may mimic angina), peptic ulcer disease, and biliary colic. Red flags requiring investigation: dysphagia, odynophagia, weight loss, GI bleeding, anaemia, age >55, family history of oesophageal or gastric cancer. ECG is indicated when cardiac ischemia cannot be distinguished on history alone.

Heart Disease

– A broad term encompassing various conditions affecting the heart and blood vessels, most commonly coronary artery disease, heart failure, arrhythmias, and valvular heart disease. Atherosclerosis, the buildup of plaque in the arteries, is the underlying cause of most cardiovascular events. Risk factors include hypertension, hyperlipidemia, smoking, diabetes, obesity, and physical inactivity.

Heart Failure

– A clinical syndrome resulting from the heart's inability to pump sufficient blood to meet metabolic demands. It is classified by ejection fraction (HFrEF: EF 40% or less; HFpEF: EF 50% or greater) and by symptoms (NYHA class I-IV). Heart failure may be right-sided (causing peripheral edema, JVD, hepatomegaly) or left-sided (causing pulmonary congestion, dyspnea, orthopnea). Diagnosis is supported by elevated BNP or NT-proBNP levels.

Heart Failure Stages

– ACC/AHA staging: Stage A (at risk for HF but no structural disease or symptoms); Stage B (structural heart disease but no symptoms); Stage C (structural disease with prior or current symptoms); Stage D (advanced HF requiring specialized interventions). This staging system emphasizes prevention and progression. Treatment intensifies with each stage: risk factor modification (A), ACE inhibitors/beta-blockers (B), diuretics added (C), and advanced therapies like transplant or LVAD (D).

Heart Health

– The state of cardiovascular well-being, characterized by efficient pumping function, clear arteries, normal blood pressure, and regular heart rhythm. Maintaining heart health involves a heart-healthy diet, regular aerobic and resistance exercise, weight management, stress reduction, and avoidance of tobacco. Preventive cardiology emphasizes early detection and management of risk factors to reduce the burden of cardiovascular disease.

Heart-Healthy Eating

– A dietary approach emphasizing foods that support cardiovascular function and reduce the risk of heart disease. Key principles include limiting saturated fats, trans fats, sodium, and added sugars while prioritizing fruits, vegetables, whole grains, lean proteins, and healthy fats like those from olive oil and fatty fish. This pattern aligns closely with the Mediterranean and DASH dietary guidelines.

Heart Murmur

– An audible extra sound produced by turbulent blood flow within the heart or great vessels, heard on auscultation. Murmurs are described by: timing (systolic, diastolic, continuous), intensity (Levine grading 1-6), location (apex, left sternal border, right upper sternal border), radiation (axilla, carotids, back), pitch (high or low frequency), quality (blowing, rumbling, harsh, musical), and response to maneuvers (inspiration, Valsalva, squatting, standing). Systolic murmurs: (1) Aortic stenosis (ejection systolic, crescendo-decrescendo, right upper sternal border, radiating to carotids); (2) Mitral regurgitation (holosystolic/pansystolic, blowing, apex, radiating to axilla); (3) Mitral valve prolapse (mid-systolic click + late systolic murmur); (4) Hypertrophic obstructive cardiomyopathy (ejection systolic, left sternal border, increases with Valsalva); (5) Ventricular septal defect (holosystolic, pansystolic, left lower sternal border). Diastolic murmurs: (1) Aortic regurgitation (early diastolic, decrescendo, high-pitched blowing, left sternal border); (2) Mitral stenosis (mid-diastolic, low-pitched rumbling, apex, accentuated by left lateral decubitus, presystolic accentuation). Continuous murmurs: patent ductus arteriosus (machinery murmur, left infraclavicular area). Innocent murmurs: Still murmur (vibratory, left lower sternal border, common in children), pulmonary flow murmur, venous hum. Workup: echocardiogram (transthoracic echo, bubble study, transoesophageal echo if needed) to identify valvular abnormality, assess severity, and determine need for intervention.

Heart Rate

– The number of cardiac contractions (heartbeats) per minute. Normal resting heart rate (sinus rhythm): 60-100 bpm in adults; higher in children (70-120 at age 1-2 years, 80-130 in neonates). Heart rate is determined by the intrinsic firing rate of the sinoatrial (SA) node (60-100 bpm), modulated by the autonomic nervous system: sympathetic activation (noradrenaline increases heart rate through beta-1 adrenergic receptors, positive chronotropy) and parasympathetic activation (vagus nerve, acetylcholine acts on M2 receptors, negative chronotropy). Tachycardia: heart rate >100 bpm at rest. Causes: sinus tachycardia (physiological response to exercise, stress, fever, pain, hypovolaemia, anaemia, hyperthyroidism, PE, MI, heart failure), re-entrant tachycardias (AVNRT, AVRT, atrial flutter, atrial fibrillation), ventricular tachycardia. Bradycardia: heart rate <60 bpm. Causes: sinus bradycardia (physiological in athletes, sleep), sick sinus syndrome, heart block (first, second, third degree), drug effect (beta-blockers, calcium channel blockers, digoxin), hypothyroidism, increased intracranial pressure, hypothermia. Assessment: pulse palpation (radial, carotid, apical), ECG (lead II rhythm strip for most accurate rate calculation: 300 / number of large squares between R waves; or number of R waves x 6 for a 10-second strip).

Heart Sounds

– Audible sounds produced by the heart during the cardiac cycle, normally heard as two sounds (lub-dub). S1 (lub) is caused by closure of the mitral and tricuspid valves at the beginning of systole. S2 (dub) is caused by closure of the aortic and pulmonary valves at the end of systole. Additional sounds (S3, S4) may indicate pathology. Heart sounds are auscultated at specific chest wall locations.

Heart Surgery

– Surgical procedures performed on the heart or great vessels to treat structural or functional cardiac conditions. Common operations include coronary artery bypass grafting (CABG), valve repair or replacement, and repair of congenital heart defects. Surgical approaches range from traditional open-heart surgery with sternotomy to minimally invasive and robot-assisted techniques.

Heart Valve Disorders

– Disorders affecting any of the four heart valves, including stenosis (narrowing) and regurgitation (leaking), which impair blood flow through the heart. Causes include congenital malformations, age-related degeneration, rheumatic fever, and endocarditis. Treatment ranges from medication monitoring to valve repair or replacement via surgery or transcatheter interventions.

Heart Valves

– The four cardiac valves ensure unidirectional blood flow through the heart. Atrioventricular valves: mitral valve (bicuspid, between left atrium and left ventricle, two leaflets — anterior and posterior, attached via chordae tendineae to papillary muscles preventing prolapse) and tricuspid valve (between right atrium and right ventricle, three leaflets). Semilunar valves: aortic valve (between left ventricle and aorta, three cusps — left coronary, right coronary, non-coronary) and pulmonary valve (between right ventricle and pulmonary artery, three cusps). Valve dysfunction: stenosis (failure to open fully, causing pressure overload) and regurgitation (failure to close fully, causing volume overload). Common pathologies: aortic stenosis (calcific, bicuspid valve, rheumatic), mitral regurgitation (mitral valve prolapse, ischemic, rheumatic), and infective endocarditis (vegetations on valve leaflets, typically Streptococcus viridans, Staphylococcus aureus). Clinical assessment: auscultation (heart sounds and murmurs), echocardiography (transthoracic and transoesophageal) for anatomy and hemodynamics.

Heart Ventricles

– See Ventricle.

Hemochromatosis Cardiomyopathy

– Iron overload cardiomyopathy from hereditary (HFE gene mutations, most common) or secondary causes. Iron deposition in the myocardium causes restrictive then dilated cardiomyopathy. Diagnosis: elevated transferrin saturation (>45%), ferritin, and genetic testing. Cardiac MRI T2* <20 ms indicates myocardial iron overload. Treatment: phlebotomy for primary hemochromatosis, iron chelation (deferoxamine) for secondary. Early treatment can reverse cardiac dysfunction.

Hepatic Drug Metabolism

– Phase I reactions (oxidation, reduction, hydrolysis) primarily by CYP450 enzymes introduce or expose functional groups. Phase II reactions (conjugation) with glucuronide, sulfate, or acetyl groups increase water solubility for renal excretion. Hepatic metabolism can activate prodrugs, inactivate active drugs, or produce toxic metabolites. Liver disease reduces metabolism, requiring dose adjustment. Enzyme induction increases metabolism (decreasing drug levels); enzyme inhibition decreases metabolism (increasing drug levels).

Hepatojugular Reflux

– Sustained pressure applied over the liver for 15-30 seconds causes a sustained rise in JVP of more than 3 cm above baseline. It indicates right heart failure or fluid overload, reflecting the inability of the right ventricle to accommodate increased venous return. Also seen in cardiac tamponade, constrictive pericarditis, and tricuspid regurgitation. It is a useful bedside test for volume status assessment.

Holosystolic Murmur Differential

– Pansystolic (holosystolic) murmurs are plateau-shaped and extend from S1 to S2. Differential includes: mitral regurgitation (apex, radiates to axilla, increases with handgrip), tricuspid regurgitation (left lower sternal border, increases with inspiration), ventricular septal defect (left lower sternal border, may have thrill), and functional murmurs from increased flow (anemia, thyrotoxicosis, pregnancy).

Hypertension

– Persistently elevated blood pressure, defined as systolic BP >130 mmHg or diastolic BP >80 mmHg (ACC/AHA guidelines). It is the most common modifiable risk factor for cardiovascular disease, stroke, heart failure, and chronic kidney disease. Most cases (90-95%) are essential (primary) hypertension; 5-10% are secondary (renal, endocrine, renovascular). Treatment includes lifestyle modifications (diet, exercise, weight loss) and pharmacotherapy (ACE inhibitors, ARBs, CCBs, thiazide diuretics).

Hypertensive Crisis

– Severe elevation in blood pressure (SBP >180 or DBP >120) requiring urgent medical attention. Two types: hypertensive emergency (with end-organ damage) and hypertensive urgency (no acute end-organ damage). Hypertensive urgency: can be managed with oral antihypertensives (captopril, clonidine, labetalol) and careful outpatient follow-up. Distinction is critical for appropriate management.

Hypertensive Emergency

– Severe hypertension (SBP >180 or DBP >120) with acute target organ damage. Organs involved: brain (hypertensive encephalopathy, stroke), heart (acute coronary syndrome, acute pulmonary edema, aortic dissection), kidneys (acute kidney injury), and eyes (papilledema). Requires immediate BP reduction with IV antihypertensives (nicardipine, labetalol, nitroprusside) and ICU monitoring. Goal: reduce MAP by 10-20% in the first hour, then cautious further reduction.

Hypertensive Heart Disease

– Left ventricular hypertrophy and diastolic/systolic dysfunction resulting from chronic hypertension. LVH is an independent cardiovascular risk factor and precursor to heart failure. Diagnosis by ECG (voltage criteria) or echocardiography (wall thickness >1.1 cm). Treatment of hypertension can regress LVH. Complications include heart failure (HFpEF initially, then HFrEF), atrial fibrillation, and sudden cardiac death. Beta-blockers, ACE inhibitors, and ARBs can regress LVH.

Hypertrophic Cardiomyopathy

– An autosomal dominant genetic disorder characterized by asymmetric left ventricular hypertrophy (typically septal), myocyte disarray, and fibrosis. It is the most common cause of sudden cardiac death in young athletes. Many patients are asymptomatic; symptoms include dyspnea, chest pain, syncope, and palpitations. Dynamic left ventricular outflow tract obstruction occurs in 70% of patients. Treatment includes beta-blockers, calcium channel blockers, myectomy, and ICD for high-risk patients.

Hypertrophic Cardiomyopathy Genetics

– HCM is caused by mutations in genes encoding sarcomeric proteins, most commonly beta-myosin heavy chain (MYH7) and myosin-binding protein C (MYBPC3). Inheritance is autosomal dominant with variable penetrance. Genetic testing identifies causative mutations in 40-60% of cases. First-degree relatives should be screened with echocardiography and ECG. Genotype-negative family members have lower risk but still require periodic surveillance.

Hypertrophic Cardiomyopathy Treatment

– Symptomatic management: beta-blockers (first-line for exertional symptoms), non-dihydropyridine calcium channel blockers (alternative or additive), disopyramide (reduces LVOT gradient). Septal reduction therapies for refractory symptoms: surgical myectomy (gold standard, excellent long-term outcomes) or alcohol septal ablation (less invasive, higher pacemaker requirement). ICD for sudden death prevention. Lifestyle counseling: moderate exercise, avoid competitive sports, avoid dehydration and vasodilators.

Hypoplastic Left Heart Syndrome

– A severe congenital heart defect where the left heart structures (mitral valve, left ventricle, aortic valve, ascending aorta) are underdeveloped and unable to support systemic circulation. The right ventricle pumps blood to both the pulmonary and systemic circulations via a patent ductus arteriosus. Presents in the first days of life with cyanosis, tachypnea, and cardiovascular collapse when the ductus closes. Diagnosis: echocardiography shows a tiny left ventricle, hypoplastic ascending aorta, and ductal-dependent systemic circulation. Staged surgical palliation: Norwood procedure (stage 1, neonatal), Glenn shunt (stage 2, 4-6 months), and Fontan completion (stage 3, 2-4 years). Prognosis has improved significantly but remains guarded. Without surgery, it is uniformly fatal.

Hypotension

– Abnormally low blood pressure, generally systolic <90 mmHg or mean arterial pressure <65 mmHg. Causes include hypovolemia, cardiac failure, sepsis, anaphylaxis, and medications. Orthostatic hypotension is a drop upon standing. Severe hypotension (shock) requires urgent fluid resuscitation and vasopressors. Chronic asymptomatic hypotension is usually benign.

ICD Shock Therapy

– Anti-tachycardia pacing (ATP) for slow VT delivers rapid pacing pulses to terminate the arrhythmia painlessly. Cardioversion shock synchronously terminates VT. Defibrillation shock treats VFib asynchronously. The MADIT-RIT trial showed that higher-rate programming (detection >180-200 bpm) reduced inappropriate shocks and mortality. S-ICD (subcutaneous ICD) avoids transvenous leads but lacks ATP and pacing capability. ICD shocks are associated with psychological morbidity.

Impella Device

– Percutaneous microaxial left ventricular assist device that actively pumps blood from the LV into the aorta, reducing LV workload and increasing cardiac output. Types: Impella 2.5 (up to 2.5 L/min), Impella CP (up to 3.7 L/min), Impella 5.0 (up to 5 L/min, surgically placed). Used in cardiogenic shock, high-risk PCI, and as bridge to decision. Complications: hemolysis, limb ischemia, device malposition, bleeding.

Implantable Cardioverter-Defibrillator

– An implanted device that monitors heart rhythm and delivers electrical therapy (antitachycardia pacing or defibrillation) for life-threatening ventricular arrhythmias. Indications include secondary prevention (survived cardiac arrest, sustained VT) and primary prevention (EF 35% or less with NYHA II-III symptoms despite optimal medical therapy, or post-MI with EF 35% or less after 40 days). ICDs reduce sudden cardiac death and overall mortality in selected patients.

Implantable Loop Recorder

– Subcutaneous device providing continuous ECG monitoring for up to 3 years. Used for evaluation of unexplained syncope, recurrent palpitations, cryptogenic stroke, and occult atrial fibrillation detection. Small size (coin-sized), inserted under local anesthesia in the chest. Automatically records and stores rhythm data during symptoms or detected arrhythmias. Increasingly important in the workup of cryptogenic stroke for detecting silent AFib.

Infective Endocarditis

– Infection of the endocardial surface of the heart, most commonly affecting the heart valves. It is diagnosed using the Duke criteria, which include major criteria (positive blood cultures, echocardiographic evidence of vegetations) and minor criteria (predisposing heart condition, fever, vascular phenomena, immunologic phenomena). Treatment requires prolonged IV antibiotic therapy (4-6 weeks). Surgery is indicated for complications including heart failure, persistent infection, and embolic events.

Interatrial Septum

– The wall separating the right and left atria. It contains the fossa ovalis, a remnant of the foramen ovale from fetal circulation. The interatrial septum has a thinner area where transseptal puncture can be performed during cardiac catheterization. Atrial septal defects (ASD) are congenital openings that allow left-to-right shunting, potentially causing right heart failure.

Interventricular Septum

– The thick muscular wall separating the left and right ventricles. It has two portions: the muscular (inferior, thicker) and membranous (superior, thinner) portions. The septum contains parts of the cardiac conduction system, including the bundle of His. Ventricular septal defects (VSDs) are congenital openings in the septum that allow left-to-right shunting of blood.

Intra-Aortic Balloon Pump

– A mechanical circulatory support device that increases myocardial perfusion and reduces afterload. The balloon inflates during diastole (augmenting coronary perfusion) and deflates during systole (reducing afterload). Inserted percutaneously into the femoral artery. Indications include cardiogenic shock, mechanical complications of MI (papillary muscle rupture, VSD), and high-risk PCI. IABP provides modest hemodynamic support and is being replaced by more effective devices.

Intravascular Ultrasound

– Catheter-based imaging using a miniature ultrasound probe inside coronary arteries to visualize vessel wall layers, plaque composition, and stent deployment. Provides more detailed information than angiography alone, including vessel remodeling, plaque burden, and true vessel diameter. Used to optimize stent deployment (apposition, expansion) and assess intermediate angiographic lesions. OCT (optical coherence tomography) provides higher resolution imaging.

Ischemic Cardiomyopathy

– Dilated cardiomyopathy resulting from extensive myocardial infarction with significant scar burden. EF is typically severely reduced (<35%). The infarcted myocardium is replaced by fibrotic scar tissue, impairing contractile function and promoting ventricular arrhythmias. Viability testing (PET, dobutamine echo, cardiac MRI) may identify hibernating myocardium that could benefit from revascularization. Treatment includes standard heart failure therapy, ICD for arrhythmia prevention, and revascularisation when viable myocardium is present.

Janeway Lesions

– Painless, erythematous or hemorrhagic macules on the palms and soles, caused by septic microemboli in infective endocarditis. They are a minor criterion in the Duke criteria. Unlike Osler nodes, Janeway lesions are painless and represent microabscesses in the dermis. They are transient and may disappear quickly. Both Osler nodes and Janeway lesions indicate active endocarditis and guide diagnosis.

Killip Classification

– A clinical classification system for assessing the severity of myocardial infarction based on physical examination findings. Class I: no heart failure signs. Class II: mild heart failure (crackles, S3 gallop, elevated JVD). Class III: pulmonary edema. Class IV: cardiogenic shock. The Killip class correlates with in-hospital mortality: Class I (6%), Class II (17%), Class III (38%), Class IV (81%). It guides initial management intensity.

Kussmaul Sign

– Paradoxical increase in jugular venous pressure with inspiration, normally the JVP decreases with inspiration. It indicates impaired right ventricular filling during inspiration. Seen in constrictive pericarditis, restrictive cardiomyopathy, severe right heart failure, cardiac tamponade, and massive pulmonary embolism. It is a reflection of ventricular interdependence and loss of normal pericardial compliance.

Left Atrial Myxoma

– Most common primary cardiac tumor (75%), typically originating from the interatrial septum in the left atrium. Presents with embolic events (25%), intracardiac obstruction (mitral valve obstruction mimicking mitral stenosis), and constitutional symptoms (fever, weight loss, elevated ESR mimicking infection or malignancy). Echocardiography shows a mobile, pedunculated mass. Surgical excision is curative. Recurrence is rare but possible, especially in familial syndromes (Carney complex).

Left Atrium

– The upper left chamber that receives oxygenated blood from four pulmonary veins. It has thicker walls than the right atrium (3 mm) and contracts to fill the left ventricle through the mitral valve. The left atrial appendage is a common site for thrombus formation in atrial fibrillation. Left atrial enlargement is associated with mitral valve disease and atrial arrhythmias.

Left Bundle Branch

– A continuation of the bundle of His that divides into anterior and posterior fascicles. The anterior fascicle supplies the anterior and superior left ventricular wall; the posterior fascicle supplies the inferior and posterior wall. Left bundle branch block (LBBB) causes delayed left ventricular activation and is commonly associated with hypertension, coronary artery disease, and cardiomyopathy. LBBB on ECG shows widened QRS with notched R waves in lateral leads.

Left Ventricle

– The lower left chamber and most muscular chamber of the heart, responsible for pumping oxygenated blood into the aorta for systemic distribution. Its walls are 8-15 mm thick and generate systolic pressures of 100-140 mmHg. The left ventricle is the primary chamber affected in systolic heart failure, hypertension, and coronary artery disease. Its function is assessed by ejection fraction (normally 55-70%).

Left Ventricular Noncompaction

– A rare cardiomyopathy characterized by prominent myocardial trabeculations with deep intertrabecular recesses, thought to result from arrested myocardial compaction during embryogenesis. It may present with heart failure, arrhythmias, or thromboembolism. Diagnosis by echocardiography or cardiac MRI showing a noncompacted to compacted myocardial ratio >2.3 (echocardiography) or >2.2 (cardiac MRI). Treatment includes heart failure therapy, anticoagulation for thromboembolism prevention, and arrhythmia management.

Levetiracetam

– An antiepileptic drug with a unique mechanism involving binding to synaptic vesicle protein 2A (SV2A), modulating neurotransmitter release. It is effective for focal, generalized, and myoclonic seizures. It has few drug interactions and does not require therapeutic drug monitoring. Side effects include irritability, behavioral changes, somnolence, and rarely suicidal ideation. It is available in oral and IV formulations, making it useful for acute seizure treatment.

Long QT Syndrome

– A channelopathy characterized by prolonged QT interval (QTc >450 ms in men, >460 ms in women) with risk of torsades de pointes and sudden cardiac death. Acquired: drugs (antiarrhythmics, antipsychotics, antibiotics), electrolyte abnormalities (hypokalemia, hypomagnesemia), bradycardia. Congenital: LQT1 (exercise-triggered, KCNQ1), LQT2 (emotional/auditory, KCNH2), LQT3 (sleep, SCN5A). Diagnosis: ECG, stress test, genetic testing. Treatment: avoid QT-prolonging drugs, beta-blockers for LQT1/2, left cardiac sympathetic denervation, and ICD for high-risk patients.

Marfan Syndrome

– An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene (fibrillin-1). Cardiovascular manifestations include aortic root dilation and dissection, mitral valve prolapse, and aortic regurgitation. Skeletal features include tall stature, arachnodactyly, pectus deformity, and joint hypermobility. Aortic root >5.0 cm or rapid dilation warrants prophylactic aortic root replacement. Beta-blockers or ARBs reduce aortic dilation rate. Regular echocardiographic surveillance is recommended.

Mechanical Circulatory Support

– Devices that provide temporary or permanent cardiac support. Options include intra-aortic balloon pump (IABP), Impella (percutaneous ventricular assist device), TandemHeart, extracorporeal membrane oxygenation (ECMO), and durable left ventricular assist devices (LVADs). Indications include cardiogenic shock, bridge to transplant, and bridge to recovery. LVADs (HeartMate 3) are increasingly used as destination therapy in advanced heart failure ineligible for transplant.

Mid-Systolic Click

– A sharp, high-pitched sound occurring in mid-to-late systole, associated with mitral valve prolapse. The click represents sudden tensing of the elongated chordae tendineae as the prolapsing leaflet billows into the left atrium. It may be followed by a late systolic murmur of mitral regurgitation. The click moves earlier with decreased preload (standing, Valsalva) and later with increased preload (squatting, leg elevation).

MitraClip

– A transcatheter device that creates a double orifice in the mitral valve by clipping the anterior and posterior leaflets together, reducing mitral regurgitation. It is indicated for patients with significant

(3+ or 4+) mitral regurgitation who are at high risk for surgical mitral valve repair. The COAPT trial demonstrated benefit in selected patients with secondary mitral regurgitation. The procedure is performed under transesophageal echocardiographic guidance.

Mitral Regurgitation

– Incomplete closure of the mitral valve allowing blood to flow backward from the left ventricle to the left atrium during systole. Causes include mitral valve prolapse, papillary muscle dysfunction, rheumatic disease, and ischemic heart disease. Murmur: pansystolic (holosystolic) blowing murmur at the apex radiating to the axilla. Chronic MR leads to left atrial and ventricular dilation. Severe MR requires surgical repair or replacement.

Mitral Stenosis

– Narrowing of the mitral valve opening, obstructing left atrial to left ventricular blood flow. Most commonly caused by rheumatic heart disease. Symptoms include dyspnea, fatigue, and atrial fibrillation. Murmur: low-pitched, rumbling mid-diastolic murmur at the apex with opening snap. Severe MS (valve area <1.5 cm2) may require percutaneous balloon mitral valvotomy or surgical replacement. Anticoagulation is indicated for atrial fibrillation.

Mitral Stenosis Murmur

– Low-pitched, rumbling mid-diastolic murmur at the apex, best heard with the bell of the stethoscope in the left lateral decubitus position. Accompanied by an opening snap (early diastolic high-pitched click) and loud S1 in pliable valves. Severe MS may show a diastolic murmur without a preceding S2 gap. Preserved sinus rhythm shows presystolic accentuation. Atrial fibrillation is common and eliminates presystolic component.

Mitral Valve Prolapse

– A valvular heart disease where the mitral valve leaflets prolapse into the left atrium during systole. Most patients are asymptomatic; some develop palpitations, atypical chest pain, or dyspnea. Diagnosis by echocardiography. Complications include mitral regurgitation, infective endocarditis, and arrhythmias. Most cases require only monitoring; severe regurgitation may need surgical repair.

Murmur Classification

– Heart murmurs are classified by timing, grade, shape, location, radiation, and pitch. Timing: systolic (between S1 and S2), diastolic (between S2 and S1), or continuous. Grade: I (barely audible) to VI (heard without stethoscope). Shape: crescendo, decrescendo, or plateau. The grading system helps distinguish pathological from physiological murmurs and guides further diagnostic workup.

Myocardial Perfusion Imaging

– Nuclear cardiology technique using radiotracers (Tc-99m sestamibi, Tl-201) to assess blood flow to the myocardium. Performed at rest and during stress (exercise or pharmacological) to detect reversible ischemia (stress defect with rest redistribution) or fixed defects (scar from prior MI). SPECT (single-photon emission CT) and PET (positron emission tomography) are the modalities used. PET also quantifies myocardial blood flow and can detect balanced ischemia.

Myocarditis

– Inflammation of the myocardium, most commonly caused by viral infections (coxsackievirus, adenovirus, SARS-CoV-2). Presents with chest pain, dyspnea, arrhythmias, and elevated troponin. Diagnosis: ECG, echocardiography, cardiac MRI with Lake Louise criteria, and endomyocardial biopsy. Treatment: supportive care, heart failure management, and immunosuppression for specific causes.

Myocarditis Diagnosis

– Cardiac MRI (Lake Louise criteria) shows myocardial edema (T2-weighted imaging), late gadolinium enhancement (typically epicardial/mid-wall in non-ischemic distribution), and increased myocardial T1/T2 mapping values. Endomyocardial biopsy (Dallas criteria: inflammatory infiltrate + myocyte necrosis) remains the gold standard but is reserved for unclear cases. Blood tests: elevated troponin, ESR, CRP. Viral serology may identify causative agent.

Myocarditis Treatment

– No specific antiviral therapy for most cases. Treatment is supportive: heart failure therapy (diuretics, ACE inhibitors, beta-blockers) for reduced EF, arrhythmia management, and avoidance of exercise during acute phase (risk of sudden death). Immunosuppression may be considered in autoimmune myocarditis or cardiac sarcoidosis. Mechanical circulatory support or transplantation may be required for fulminant myocarditis. Endomyocardial biopsy may guide therapy in selected cases.

Myocardium

– The thick muscular middle layer of the heart wall composed of cardiac muscle cells (cardiomyocytes). The myocardium is responsible for the contractile function of the heart. It contains intercalated discs with gap junctions that allow electrical coupling between cells. The left ventricular myocardium is significantly thicker (8-15 mm) than the right ventricle (3-5 mm) due to higher pressure demands of systemic circulation.

NYHA Functional Classification

– A four-class system grading heart failure severity based on symptoms and exercise tolerance. Class I: no limitation of physical activity. Class II: slight limitation; comfortable at rest but ordinary activity causes symptoms. Class III: marked limitation; comfortable at rest but less than ordinary activity causes symptoms. Class IV: symptoms at rest; unable to carry out any physical activity without discomfort.

Orthostatic Hypotension

– Decrease in SBP >=20 mmHg or DBP >=10 mmHg within 3 minutes of standing. Causes: dehydration, medications (antihypertensives, diuretics, alpha-blockers, antidepressants), autonomic dysfunction (diabetes, Parkinson disease), and prolonged bed rest. Symptoms: dizziness, lightheadedness, syncope, falls. Evaluation: orthostatic vital signs, autonomic reflex testing. Treatment: address reversible causes, increase fluid/salt intake, compression stockings, midodrine or fludrocortisone.

Osler Nodes

– Painful, erythematous nodules on the pads of fingers and toes, caused by immune complex deposition in infective endocarditis. They are a minor criterion in the Duke criteria. Osler nodes are tender and may last hours to days. They must be differentiated from Janeway lesions (painless, erythematous/purpuric macules on palms and soles caused by septic microemboli). Both are classic peripheral stigmata of infective endocarditis.

Pacemaker

– An implanted electronic device that delivers electrical impulses to the heart to regulate heart rate and rhythm. Indications include symptomatic bradycardia, high-degree AV block, and sick sinus syndrome. Single-chamber pacemakers pace the atrium or ventricle; dual-chamber pacemakers pace both. Biventricular pacemakers (cardiac resynchronization therapy) improve symptoms and survival in heart failure with LBBB and prolonged QRS. Modern pacemakers are MRI-compatible.

Pacemaker Complications

– Early: pneumothorax, hematoma, cardiac perforation/tamponade, lead dislodgement. Late: lead fracture, battery depletion, infection, tricuspid regurgitation from lead interference, venous thrombosis, and pacemaker syndrome (AV dyssynchrony in VVI pacing). Electromagnetic interference from devices (cautery, TENS, certain industrial equipment). MRI-conditional pacemakers reduce restrictions but require specific programming. Regular interrogation and follow-up essential.

Palpitation

– The subjective sensation of a rapid, forceful, or irregular heartbeat. Palpitations are a common symptom with a broad differential. Causes: cardiac (arrhythmias—atrial fibrillation, supraventricular tachycardia, ventricular ectopy, atrial flutter; structural heart disease—valvular disease, cardiomyopathy, congenital heart disease), psychiatric (anxiety, panic disorder, stress), physiological (exercise, pregnancy, fever, anaemia, dehydration, caffeine, alcohol, nicotine, illicit drugs—cocaine, amphetamines), endocrine (hyperthyroidism, phaeochromocytoma, hypoglycaemia), medications (sympathomimetics, bronchodilators, antihistamines, antidepressants, thyroid hormone), and electrolyte disturbances (hypokalaemia, hypomagnesaemia). Palpitations described as 'flip-flopping' suggest ectopic beats; 'rapid and regular' suggests SVT; 'rapid and irregular' suggests atrial fibrillation. Evaluation: history (onset, offset, duration, triggers, associated symptoms—chest pain, syncope, dyspnea), ECG (12-lead; if symptoms are intermittent, 24-hour Holter, event monitor, or loop recorder), echocardiogram if structural disease suspected, and blood tests (FBC, TSH, electrolytes). Management: reassurance and lifestyle modification for benign causes; treat underlying arrhythmia or condition. Prognosis is usually excellent if no underlying structural heart disease or significant arrhythmia.

Patent Foramen Ovale

– A normal fetal communication between the right and left atria (the foramen ovale) that fails to close after birth. Present in 25-30% of the general population. Typically small and hemodynamically insignificant, with left-to-right shunting that is usually not clinically important. Most PFOs are incidental findings on echocardiography. However, PFO is associated with cryptogenic stroke (paradoxical embolism — thrombus traverses from the right to left atrium, bypassing the pulmonary circulation), migraine with aura, and decompression sickness in divers. Diagnosis: transthoracic or transesophageal echocardiography with bubble study (agitated saline contrast shows bubbles crossing from right to left atrium, especially with Valsalva maneuver). Treatment: antiplatelet therapy for cryptogenic stroke; percutaneous PFO closure for selected high-risk patients (large shunt, atrial septal aneurysm, recurrent stroke despite medical therapy).

Percutaneous Coronary Intervention

– A catheter-based procedure to open blocked coronary arteries, typically involving balloon angioplasty and stent placement. Indications include acute STEMI (primary PCI), NSTEMI (early invasive strategy), and stable CAD with significant stenosis. Drug-eluting stents have reduced restenosis rates compared to bare-metal stents. Dual antiplatelet therapy is required after stent placement. PCI can be performed via radial or femoral artery access.

Pericardial Effusion Etiology

– Causes: viral pericarditis (most common in developed countries), malignancy (most common in developing countries), uremia, hypothyroidism, autoimmune diseases (SLE, RA), tuberculosis (leading cause worldwide), post-MI (Dressler syndrome), post-cardiac surgery, radiation, and idiopathic. Large effusions (>200 mL) or rapidly accumulating small effusions risk tamponade. Echocardiography is the initial diagnostic test. Pericardiocentesis provides both diagnostic fluid analysis and therapeutic drainage.

Pericardial Knock

– A high-pitched, early diastolic sound caused by sudden cessation of ventricular filling in constrictive pericarditis. It is heard at the left sternal border and may be confused with an S3 or opening snap. The knock is typically louder with inspiration and is a characteristic finding in constrictive pericarditis. It reflects the abrupt restriction of diastolic filling by the rigid pericardium.

Pericardiocentesis

– A procedure to aspirate fluid from the pericardial space, performed emergently for cardiac tamponade or diagnostically for pericardial effusion. The subxiphoid approach is most commonly used, with the needle directed toward the left shoulder under echocardiographic guidance. Complications include myocardial puncture, coronary artery injury, and pneumothorax. Pericardial fluid can be analyzed for cytology, culture, and chemistry to determine the cause of effusion.

Pericarditis

– Inflammation of the pericardium, the fibrous sac surrounding the heart. Acute pericarditis presents with sharp, pleuritic chest pain that improves with leaning forward, pericardial friction rub, and diffuse ST elevation on ECG. Causes include viral infection (most common), bacterial, uremia, MI (Dressler syndrome), radiation, and autoimmune diseases. Treatment includes NSAIDs and colchicine. Complications include pericardial effusion and cardiac tamponade.

Peripartum Cardiomyopathy

– A form of dilated cardiomyopathy occurring in the last month of pregnancy or within 5 months postpartum, with no other identifiable cause. Risk factors include advanced maternal age, multiple gestations, preeclampsia, and African American race. Presents with heart failure symptoms. Treatment includes standard heart failure therapy (ACE inhibitors are safe postpartum but contraindicated during pregnancy). Many patients recover, but some progress to severe heart failure requiring transplant.

Peripartum Cardiomyopathy Prognosis

– Approximately 50% of patients recover normal ventricular function within 6 months. Predictors of poor recovery include: persistent LV dilation, severe LV dysfunction at diagnosis, African American race, and delayed diagnosis. Patients who recover should be counseled regarding risk of recurrence in subsequent pregnancies (30-50% risk of deterioration). ACE inhibitors and beta-blockers should be continued for at least 1 year after diagnosis, even if EF recovers.

Peripheral Artery Disease

– Atherosclerotic narrowing of arteries supplying the lower extremities. Presents with claudication (calf pain with walking, relieved by rest), rest pain, and tissue loss in severe cases. Risk factors: smoking, diabetes, hypertension, hyperlipidemia. Diagnosis: ankle-brachial index <0.90. Treatment: risk factor modification, exercise therapy, antiplatelet agents, and revascularization when indicated.

Peripheral Vascular Disease

– Narrowing of peripheral arteries, most commonly due to atherosclerosis, causing reduced blood flow to extremities. Risk factors include smoking, diabetes, hypertension, and hyperlipidemia. Symptoms include claudication (exercise-induced leg pain), rest pain, and non-healing ulcers. Ankle-brachial index (ABI) <0.90 indicates PVD. Severe PVD (critical limb ischemia) may require revascularization (angioplasty/stenting or bypass surgery) to prevent amputation.

Phenytoin

– An antiepileptic drug that blocks voltage-gated sodium channels, stabilizing neuronal membranes and preventing seizure spread. It is effective for focal and generalized tonic-clonic seizures. Therapeutic drug monitoring is important due to nonlinear (saturable) pharmacokinetics. Side effects include gingival hyperplasia, hirsutism, megaloblastic anemia, osteoporosis, and teratogenicity (fetal hydantoin syndrome). It induces CYP450 enzymes, affecting metabolism of many drugs.

Phlebitis

– Inflammation of a vein, often affecting the lower extremities. Superficial thrombophlebitis: inflammation and thrombosis of a superficial vein (great saphenous vein most common), presenting with erythema, tenderness, warmth, and a palpable cord along the vein. Causes: intravenous cannulation (most common—chemical phlebitis from irritant drugs, mechanical irritation, bacterial infection), varicose veins, thrombophilia, pregnancy, prolonged travel, and Behçet disease. Diagnosis is clinical; doppler ultrasound should be performed to exclude concomitant deep vein thrombosis (15–20% of cases). Treatment: warm compresses, NSAIDs, compression stockings, elevation, and ambulation. Anticoagulation is not required for isolated superficial thrombophlebitis unless it extends to the saphenofemoral junction (then treat as DVT). Septic thrombophlebitis requires IV antibiotics and occasionally surgical excision. Deep vein thrombophlebitis (DVT) is a distinct entity requiring anticoagulation.

Pneumopericardium

– Air in the pericardial space, usually from trauma (penetrating chest injury), esophageal perforation, or iatrogenic causes (central line placement, cardiac surgery). Can cause cardiac tamponade if under tension. Diagnosis by chest X-ray showing air surrounding the heart (air-fluid level if combined with effusion). Treatment: observation if small and asymptomatic; pericardial drainage if causing tamponade.

Post-Cardiac Arrest Management

– Comprehensive care after ROSC: hemodynamic optimization (MAP >65 mmHg, avoid hypotension), ventilation management (target PaCO2 35-45 mmHg), targeted temperature management, coronary angiography for STEMI or high suspicion for ischemia, seizure monitoring and treatment, and neurological prognostication (clinical exam, SSEPs, EEG, MRI at 3-5 days). Withdrawal of care based on multimodal prognostication when no evidence of meaningful recovery.

Preload

– The end-diastolic volume or pressure in the ventricles, representing the degree of ventricular stretch before contraction. Preload is determined by venous return and is estimated by the pulmonary capillary wedge pressure (PCWP) or left ventricular end-diastolic pressure (LVEDP). According to the Frank-Starling mechanism, increased preload (within physiological limits) increases stroke volume. Pathologically elevated preload occurs in heart failure and fluid overload.

Premature Ventricular Contractions

– Ectopic beats originating from the ventricles, appearing as wide, bizarre QRS complexes that occur earlier than expected. PVCs are common in the general population and are usually benign. Frequent PVCs (>10% of total beats) may cause left ventricular dysfunction. Causes include electrolyte imbalances, stimulants, myocardial ischemia, and cardiomyopathy. Treatment is usually unnecessary unless symptomatic or causing cardiomyopathy.

PR Interval

– The time from the onset of the P wave to the onset of the QRS complex, representing the time for electrical conduction through the atria, AV node, and His-Purkinje system. Normal duration is 0.12-0.20 seconds (3-5 small boxes). Prolonged PR interval (>0.20 s) indicates first-degree AV block. Shortened PR interval (<0.12 s) may indicate Wolff-Parkinson-White syndrome or junctional rhythm.

Prosthetic Valve Endocarditis

– Infection of prosthetic (mechanical or bioprosthetic) heart valves, classified as early (<60 days post-surgery, typically nosocomial) or late (>60 days, typically community-acquired). Causative organisms differ from native valve endocarditis: early PVE often due to S. aureus, gram-negative bacilli; late PVE similar to native valve organisms. Higher morbidity and mortality than native valve endocarditis. Often requires surgical intervention in addition to prolonged antibiotic therapy.

Pulmonary Atresia

– A congenital heart defect where the pulmonary valve is completely absent (atretic), obstructing right ventricular outflow. Two main types: pulmonary atresia with intact ventricular septum (PA-IVS) and pulmonary atresia with ventricular septal defect (PA-VSD, also called tetralogy of Fallot with pulmonary atresia). PA-IVS: the right ventricle is small and hypertensive, and pulmonary blood flow depends on a patent ductus arteriosus. PA-VSD: the right ventricle is normal-sized, and pulmonary blood flow is ductal-dependent. Both present with severe cyanosis in the first days of life as the ductus closes. Diagnosis: echocardiography shows absent pulmonary valve tissue, ductal-dependent pulmonary circulation, and RV size. Treatment: PGE1 to maintain ductal patency, followed by staged surgical palliation (BT shunt, RVOT reconstruction, or unifocalization for PA-VSD; RV decompression or Fontan for PA-IVS).

Pulmonary Embolism Treatment

– Anticoagulation: DOACs preferred (rivaroxaban, apixaban) or LMWH bridging to warfarin. Duration: 3 months for provoked (reversible risk factor), indefinite for unprovoked or recurrent. Massive PE: systemic thrombolysis (alteplase 100 mg IV over 2 hours) or surgical catheter-directed embolectomy. Submassive PE: catheter-directed thrombolysis may reduce RV strain. IVC filter: for contraindication to anticoagulation or recurrent PE despite therapy.

Pulmonary Hypertension

– Elevated mean pulmonary artery pressure (>20 mmHg at rest) due to various etiologies. WHO Group 1: pulmonary arterial hypertension (idiopathic, heritable, drug-induced). Group 2: left heart disease. Group 3: lung disease. Group 4: chronic thromboembolic. Group 5: multifactorial. Diagnosis requires right heart catheterization. Treatment depends on group: vasodilators (endothelin receptor antagonists, PDE5 inhibitors, prostacyclins) for Group 1; treat underlying cause for Groups 2-4.

Pulmonary Hypertension Classification

– WHO Classification: Group 1 (PAH: idiopathic, heritable, drug-induced, connective tissue disease-associated). Group 2 (left heart disease: systolic/diastolic dysfunction, valvular). Group 3 (lung disease/hypoxia: COPD, ILD, sleep apnea). Group 4 (chronic thromboembolic). Group 5 (multifactorial: sarcoidosis, hematologic disorders). Right heart catheterization is the gold standard for diagnosis (mean PAP >20 mmHg at rest).

Pulmonary Stenosis

– A right ventricular outflow tract obstruction at the valvular, subvalvular, or supravalvular level. Valvular pulmonary stenosis (most common) is often congenital, frequently associated with Noonan syndrome. Symptoms depend on severity: mild is asymptomatic; moderate presents with dyspnea, fatigue, and syncope; severe presents with cyanosis and right heart failure. Physical exam: harsh systolic ejection murmur at the left upper sternal border, radiating to the back, with a prominent RV heave and a widely split S2 with a diminished P2. Diagnosis: echocardiography with Doppler measures the peak gradient (mild <30 mmHg, moderate 30-60 mmHg, severe >60 mmHg). Treatment: balloon pulmonary valvuloplasty is the procedure of choice for severe valvular stenosis (gradient >40-50 mmHg).

Surgical valvotomy or valve replacement if balloon fails or for dysplastic valves (Noonan syndrome).

Pulmonic Stenosis

– Narrowing of the right ventricular outflow tract at the pulmonic valve. Causes: congenital (most common, often associated with Noonan syndrome or tetralogy of Fallot), rheumatic, carcinoid syndrome. Presents with dyspnea, fatigue, syncope. Signs: harsh systolic ejection murmur at left upper sternal border, RV hypertrophy. Diagnosis: echocardiography (pressure gradient). Treatment: balloon valvuloplasty for moderate-severe congenital PS; surgical valvotomy or valve replacement for failed valvuloplasty.

Pulseless Electrical Activity

– A cardiac arrest rhythm organized electrical activity without a detectable pulse. PEA has many reversible causes (5 H's and 5 T's): hypovolemia, hypoxia, hydrogen ion (acidosis), hypo/hyperkalemia, hypothermia; tension pneumothorax, tamponade (cardiac), toxins, thrombosis (pulmonary), thrombosis (coronary). Treatment focuses on identifying and reversing the underlying cause while performing CPR and administering epinephrine.

Pulsus Paradoxus

– A decrease in systolic blood pressure greater than 10 mmHg during inspiration, normally less than 10 mmHg. Exaggerated in cardiac tamponade (due to ventricular interdependence in the confined pericardial space), severe asthma, COPD exacerbation, and constrictive pericarditis. It is measured by noting the Korotkoff sounds that appear only during expiration on sphygmomanometry. Kussmaul sign (paradoxical JVD increase with inspiration) is a related finding.

Purkinje Fibers

– The terminal branches of the cardiac conduction system, distributed throughout the ventricular myocardium. They conduct impulses rapidly (up to 4 m/s) to ensure coordinated ventricular contraction from apex to base. Purkinje fibers have an intrinsic firing rate of 20-40 beats per minute. They are resistant to ischemia but can become triggered pacemakers in disease states.

Purulent Pericarditis

– Bacterial infection of the pericardial space, most commonly caused by S. aureus, S. pneumoniae, or gram-negative organisms. Often occurs as a complication of pneumonia, endocarditis, chest surgery, or immunosuppression. Presents with fever, pleuritic chest pain, and rapidly accumulating effusion with tamponade physiology. High mortality without treatment. Requires emergent pericardiocentesis or surgical drainage plus prolonged IV antibiotics.

P Wave

– The first deflection on the ECG, representing atrial depolarization. Normal P waves are upright in leads I and II, inverted in aVR, and less than 0.12 seconds (3 small boxes) in duration and 2.5 mm in amplitude. Abnormal P waves indicate atrial enlargement: peaked P waves (right atrial enlargement), notched P waves (left atrial enlargement). Absent P waves suggest atrial fibrillation or junctional rhythm.

QRS Complex

– The deflection representing ventricular depolarization, consisting of a Q wave (initial downward deflection), R wave (upward deflection), and S wave (downward deflection after R). Normal duration is 0.06-0.10 seconds (1.5-2.5 small boxes). Widened QRS (>0.12 s) indicates bundle branch block, ventricular rhythm, or hyperkalemia. Abnormal Q waves indicate prior myocardial infarction.

QT Interval

– The time from the onset of the QRS complex to the end of the T wave, representing total ventricular depolarization and repolarization. Normal QT is 0.35-0.44 seconds. The corrected QT (QTc) adjusts for heart rate using Bazett's formula: QTc = QT / square root of RR interval. QTc prolongation (>450 ms in men, >470 ms in women) increases the risk of torsades de pointes. QTc is shortened in hyperkalemia and digoxin effect.

Raynaud Phenomenon

– Exaggerated vasospastic response of digital arteries to cold or stress, causing well-demarcated color changes (white-blue-red) in fingers or toes. Primary Raynaud: idiopathic, symmetric, no tissue damage, normal capillaries. Secondary Raynaud: associated with connective tissue diseases (scleroderma, SLE), vascular disease, medications (beta-blockers), or occupational vibration. Diagnosis: clinical, nailfold capillaroscopy, ANA testing. Treatment: avoid cold triggers, calcium channel blockers (nifedipine), and prostacyclin analogs for severe cases.

Regadenoson Pharmacological Stress

– Selective A2A adenosine receptor agonist used as a pharmacological vasodilator for nuclear myocardial perfusion imaging. Single IV bolus injection produces coronary vasodilation with fewer side effects than adenosine (less bronchospasm, AV block). Contraindicated in severe asthma or COPD requiring oxygen. Advantages: selective coronary vasodilation, no need for dose adjustment, and simpler administration than adenosine infusion.

Regurgitation

– The backward flow of blood through a heart valve that fails to close properly (valvular incompetence or insufficiency). Regurgitation can affect any of the four heart valves: mitral regurgitation (most common—systolic murmur at apex radiating to axilla, caused by mitral valve prolapse, ischemic heart disease, rheumatic heart disease, dilated cardiomyopathy), aortic regurgitation (diastolic murmur at left sternal edge—chronic causes: bicuspid aortic valve, aortic root dilation, rheumatic, syphilis; acute causes: infective endocarditis, aortic dissection), tricuspid regurgitation (systolic murmur at left lower sternal edge—functional from right ventricular dilation, pulmonary hypertension, or primary from endocarditis, carcinoid), and pulmonic regurgitation (diastolic murmur at left upper sternal edge—usually benign from pulmonary hypertension). Chronic regurgitation allows compensatory ventricular dilation and eccentric hypertrophy, maintaining cardiac output for years before symptoms develop (exertional dyspnea, fatigue, palpitations). Acute severe regurgitation (e.g., flail mitral leaflet, chordal rupture) causes sudden volume overload, pulmonary edema, and cardiogenic shock. Diagnosis: echocardiography (colour Doppler—quantifies severity by jet area, vena contracta, PISA radius, and regurgitant volume). Treatment: medical therapy for chronic disease (vasodilators, diuretics), surgical repair or replacement for severe symptomatic regurgitation or when ventricular dysfunction develops.

Renal Drug Excretion

– The elimination of drugs by the kidney through glomerular filtration, active tubular secretion, and passive tubular reabsorption. Drugs eliminated renally (e.g., aminoglycosides, vancomycin, lithium, metformin) require dose adjustment in renal impairment. Renal clearance is affected by urine pH, tubular flow rate, and drug protein binding. Renal function is estimated by creatinine clearance or eGFR, guiding dosing decisions.

Restrictive Cardiomyopathy

– A cardiomyopathy characterized by impaired ventricular relaxation and filling due to stiff, noncompliant myocardium, with preserved systolic function and normal or reduced ventricular volumes. Causes include amyloidosis, sarcoidosis, hemochromatosis, endomyocardial fibrosis, and radiation. Patients present with signs of heart failure despite normal EF. Prognosis is often poor. Treatment is directed at underlying cause and managing volume overload with diuretics.

Rheumatic Heart Disease

– Valvular damage resulting from acute rheumatic fever, an autoimmune response to Group A streptococcal pharyngitis. Most commonly affects the mitral valve (stenosis/regurgitation), followed by aortic. Prevention through prompt treatment of streptococcal pharyngitis. Management: valve repair/replacement, anticoagulation for atrial fibrillation, and secondary prophylaxis with penicillin.

Right Atrium

– The upper right chamber that receives deoxygenated blood from the superior and inferior vena cava and the coronary sinus. It contains the sinoatrial (SA) node in its wall near the junction with the superior vena cava. The right atrium has thin walls (2 mm) and contracts to fill the right ventricle through the tricuspid valve. Pressure in the right atrium is normally 2-8 mmHg.

Right Bundle Branch

– A continuation of the bundle of His that travels along the right side of the interventricular septum to the right ventricle. It supplies the right ventricular myocardium and Purkinje fibers. Right bundle branch block (RBBB) occurs when this branch is damaged, causing delayed right ventricular activation. RBBB is commonly associated with pulmonary embolism, right ventricular hypertrophy, and aging.

Right Coronary Artery

– Arises from the right aortic sinus and travels in the right atrioventricular groove. It supplies the right atrium, right ventricle, SA node (60% of individuals), AV node (85% of individuals), and the inferior wall of the left ventricle. RCA occlusion causes inferior myocardial infarction, which may be associated with bradycardia and heart block due to AV nodal involvement.

Right Heart Failure

– Failure of the right ventricle to adequately pump blood to the lungs, causing systemic venous congestion. Causes include left heart failure (most common), pulmonary hypertension, right ventricular infarction, pulmonary embolism, and constrictive pericarditis. Signs include elevated JVD, peripheral edema, hepatomegaly, and ascites. Treatment focuses on addressing the underlying cause and managing volume overload with diuretics. Inotropes (dobutamine, milrinone) may be needed for RV failure.

Right Ventricle

– The lower right chamber that receives blood from the right atrium through the tricuspid valve and pumps it through the pulmonary valve into the pulmonary artery. It has crescent-shaped walls (3-5 mm thick) and generates lower pressures than the left ventricle (systolic pressure 25-30 mmHg). The right ventricle is volume-overloaded in conditions like pulmonary hypertension and atrial septal defect.

Roth Spots

– Retinal hemorrhages with pale centers, seen on fundoscopic examination in infective endocarditis. They are a minor criterion in the Duke criteria. Roth spots are caused by immune complex-mediated vasculitis of retinal vessels. They can also be seen in other conditions: severe anemia, leukemia, collagen vascular diseases, and diabetic retinopathy. Their presence in the appropriate clinical context supports endocarditis diagnosis.

S3 Gallop Pathology

– A third heart sound in adults over 40 indicates ventricular dysfunction. It represents rapid ventricular filling against a dilated, noncompliant ventricle. S3 is a hallmark of systolic heart failure (HFrEF) and is associated with increased mortality. It is best heard at the apex with the bell of the stethoscope in the left lateral decubitus position. Common causes: dilated cardiomyopathy, acute MI with volume overload, severe mitral regurgitation.

S4 Gallop Pathology

– A fourth heart sound indicates atrial contraction against a stiff ventricle (decreased compliance). Associated with left ventricular hypertrophy from hypertension, aortic stenosis, and hypertrophic cardiomyopathy. S4 is best heard at the apex with the bell, and is accentuated by exertion, expiration, and isorhythmic AV dissociation. It is a sign of diastolic dysfunction and may precede symptomatic heart failure.

Second-Degree AV Block

– Intermittent failure of AV conduction. Mobitz Type I (Wenckebach): progressive PR prolongation then dropped QRS; usually benign (AV nodal, narrow QRS). Mobitz Type II: constant PR interval with sudden dropped QRS; more serious (infranodal, wide QRS), risk of progression to complete heart block and Stokes-Adams attacks. Treatment: Type I often no treatment needed; Type II requires pacemaker implantation due to risk of high-grade block.

Sick Sinus Syndrome

– Sinus node dysfunction causing a spectrum of arrhythmias: sinus bradycardia, sinus arrest, sinoatrial block, and tachycardia-bradycardia syndrome. Most common in elderly patients. Presents with syncope, fatigue, palpitations, and exercise intolerance. Diagnosis: ECG, Holter monitor (pauses >3 seconds). Treatment: permanent pacemaker implantation for symptomatic bradycardia. Anticoagulation needed for tachy-brady syndrome (atrial fibrillation component).

Single Ventricle

– A group of congenital heart defects where one functional ventricle supports both the systemic and pulmonary circulations. Includes tricuspid atresia, mitral atresia, double inlet left ventricle, hypoplastic left heart syndrome, and unbalanced AV canal. Pulmonary and systemic blood flow must be balanced by restrictive or non-restrictive shunts. Presents with cyanosis, heart failure, or a combination depending on pulmonary blood flow. All single ventricle patients undergo staged Fontan palliation: stage 1 (neonatal: BT shunt, PA band, or Norwood), stage 2 (4-6 months: Glenn shunt — SVC to PA), and stage 3 (2-4 years: Fontan completion — IVC to PA). Long-term complications:

Fontan failure, protein-losing enteropathy, plastic bronchitis, thromboembolism, and arrhythmias.

Sinoatrial Node

– The heart's natural pacemaker, located at the junction of the superior vena cava and right atrium. It consists of specialized cells that spontaneously depolarize at a rate of 60-100 beats per minute, setting the normal sinus rhythm. The SA node receives autonomic innervation: sympathetic stimulation increases heart rate, while parasympathetic (vagal) stimulation decreases it. SA node dysfunction causes sick sinus syndrome.

Splinter Hemorrhages

– Linear hemorrhages under the nail bed, oriented longitudinally. In infective endocarditis, they are caused by septic microemboli to the nail bed capillaries. They are a minor Duke criterion. Splinter hemorrhages can also result from trauma, psoriasis, and systemic vasculitis. Multiple splinter hemorrhages in several nails are more suggestive of endocarditis than a single lesion from trauma.

Stevens-Johnson Syndrome

– A severe, life-threatening mucocutaneous reaction characterized by widespread skin erythema, necrosis, and bullae formation with mucosal involvement (oral, ocular, genital). It is most commonly caused by drug reactions (sulfonamides, anticonvulsants, allopurinol, NSAIDs) and infections (Mycoplasma pneumoniae). It is a medical emergency requiring ICU care, wound management, infection prevention, and often burn center transfer. Mortality is 5-10%.

Strain Imaging

– Echocardiographic or MRI technique measuring myocardial deformation (shortening and thickening). Global longitudinal strain (GLS) by speckle tracking is more sensitive than ejection fraction for detecting subclinical myocardial dysfunction. GLS $<$ -18% is normal. Used for early detection of chemotherapy cardiotoxicity, differentiating cardiomyopathies, and prognosis in heart failure. GLS may detect dysfunction before EF declines.

Stroke Volume

– The volume of blood ejected by the left ventricle with each heartbeat, normally 60-100 mL. It is determined by three factors: preload, afterload, and myocardial contractility. Stroke volume equals end-diastolic volume minus end-systolic volume. Reduced stroke volume occurs in heart failure, hypovolemia, and cardiogenic shock. Stroke volume can be increased by exercise, positive inotropes, or fluid administration.

ST Segment

– The isoelectric period between ventricular depolarization (QRS) and repolarization (T wave). It corresponds to the plateau phase of ventricular action potentials. ST segment elevation ($>$1 mm in limb leads or $>$2 mm in precordial leads) suggests acute myocardial infarction, pericarditis, or early repolarization. ST depression indicates ischemia, digoxin effect, or hypokalemia. ST segment changes are critical for diagnosing acute coronary syndromes.

Supraventricular Tachycardia

– A rapid rhythm (150-250 bpm) originating above the ventricles, characterized by narrow QRS complexes. Most common forms are AV nodal reentrant tachycardia (AVNRT) and AV reentrant tachycardia (AVRT). Episodes are typically paroxysmal with sudden onset and offset. Vagal maneuvers and adenosine are diagnostic and therapeutic. Recurrent SVT may require catheter ablation of the reentrant circuit.

Syncope (Fainting)

– A transient loss of consciousness (TLOC) due to global cerebral hypoperfusion, characterized by rapid onset, short duration, and spontaneous complete recovery. Syncope is common (lifetime prevalence 40%). Three main categories: (1) Reflex (neurally mediated) syncope—vasovagal syncope (most common, triggered by emotion, pain, prolonged standing, fear, venepuncture, heat; prodrome: nausea, sweating, pallor, lightheadedness), situational syncope (micturition, defecation, cough, swallowing, post-exercise), and carotid sinus syncope (head turning, tight collars). (2) Orthostatic hypotension—syncope upon standing due to autonomic failure (Parkinson, diabetes, pure autonomic failure), volume depletion, or medications (α-blockers, diuretics, vasodilators). (3) Cardiac syncope—arrhythmias (bradyarrhythmias—sick sinus syndrome, AV block; tachyarrhythmias—VT, SVT), structural heart disease (aortic stenosis, hypertrophic cardiomyopathy, atrial myxoma, pulmonary embolism, myocardial infarction). Red flags for cardiac syncope: syncope during exertion, supine syncope, associated chest pain or palpitations, family history of sudden cardiac death, abnormal ECG. Evaluation: history (witness account key), ECG, orthostatic BP measurement, echocardiogram, and Holter/event monitor if arrhythmia suspected. Tilt-table testing for recurrent vasovagal syncope. Management: reassurance and trigger avoidance for vasovagal syncope; volume expansion, compression stockings, and fludrocortisone/midodrine for orthostatic hypotension; pacemaker for symptomatic bradycardia; ICD for VT/VF.

Systemic Vascular Resistance

– The resistance against which the left ventricle must pump, calculated as (MAP - CVP) / CO x 80. Normal: 800-1200 dynes/sec/cm5. Elevated SVR occurs in cardiogenic shock (compensatory vasoconstriction), aortic stenosis, and systemic hypertension. Decreased SVR occurs in septic shock, anaphylaxis, and cirrhosis. SVR guides vasopressor and vasodilator therapy in critical care settings.

Systolic Murmurs

– Murmurs occurring between S1 and S2. Ejection systolic murmurs (crescendo-decrescendo) are caused by flow across semilunar valves (aortic stenosis, pulmonary stenosis) or increased flow across normal valves. Pansystolic (holosystolic) murmurs (plateau) are caused by regurgitation across AV valves (mitral regurgitation, tricuspid regurgitation) or left-to-right shunts (VSD). Timing with carotid pulsation helps distinguish aortic from mitral murmurs.

Tachycardia

– A heart rate exceeding 100 beats per minute in adults at rest. Tachycardia is a sign, not a diagnosis, and requires investigation to determine the underlying rhythm and cause. Sinus tachycardia: a physiological response to increased demand—exercise, fever, anxiety, pain, dehydration, anaemia, infection, hyperthyroidism, pulmonary embolism, heart failure, or drugs (β-agonists, caffeine, cocaine). The P wave is present and normal, rate 100–150 (up to 200 in children/young adults). Treatment: address the underlying cause. Narrow QRS tachycardia (QRS $<$0.12 s): supraventricular tachycardia—AV nodal reentrant tachycardia, AV reentrant tachycardia, atrial fibrillation, atrial flutter, atrial tachycardia. Wide QRS tachycardia (QRS $>$0.12 s): ventricular tachycardia (most common), SVT with aberrancy, pre-excited atrial fibrillation (Wolf–Parkinson–White). Stable tachycardia: 12-lead ECG, vagal maneuvers, adenosine (for SVT), amiodarone/procainamide (for VT). Unstable tachycardia (hypotension, altered consciousness, chest pain, acute heart failure, shock): synchronized cardioversion. Palpitations are the most common symptom of tachycardia; syncope suggests hemodynamic compromise.

Takotsubo Cardiomyopathy

– Also known as stress cardiomyopathy or broken heart syndrome, characterized by transient left ventricular apical ballooning resembling a Japanese octopus trap (takotsubo). It is typically triggered by emotional or physical stress. ECG may show ST elevation mimicking STEMI. Coronary angiography shows no obstructive disease. Most patients recover normal ventricular function within weeks. Treatment is supportive with beta-blockers and ACE inhibitors.

Targeted Temperature Management

– Controlled therapeutic hypothermia for comatose survivors of cardiac arrest. Target: 32-36 degrees C for 24-72 hours followed by gradual rewarming. Mechanism: reduces metabolic demand, limits reperfusion injury, and reduces free radical generation. TTM2 trial showed 33 degrees C was not superior to 36 degrees C. Active cooling methods: cold IV saline, surface cooling devices, endovascular cooling catheters. Complications: infection, electrolyte disturbances, coagulopathy.

TCAs (Tricyclic Antidepressants)

– Antidepressants that block reuptake of serotonin and norepinephrine, with additional anticholinergic, antihistaminic, and alpha-adrenergic blocking effects. Examples include amitriptyline, nortriptyline, imipramine, and desipramine. They are effective for depression, neuropathic pain, and migraine prophylaxis. Side effects include sedation, weight gain, dry mouth, urinary retention, constipation, orthostatic hypotension, and cardiac conduction abnormalities. Lethal in overdose due to cardiac sodium channel blockade and QT prolongation.

Third-Degree AV Block

– Complete heart block with no conduction between atria and ventricles. Atria and ventricles beat independently (AV dissociation). Ventricular escape rhythm: junctional (40-60, narrow QRS) or ventricular (20-40, wide QRS). Causes: idiopathic fibrosis, MI (inferior: transient AV nodal; anterior: infranodal), drugs, cardiomyopathy, Lyme disease, post-surgical. Symptoms: syncope, dyspnea, fatigue, heart failure. Treatment: immediate temporary pacing (transcutaneous or transvenous) and definitive permanent pacemaker implantation.

Torsades de Pointes

– A specific form of polymorphic ventricular tachycardia with QRS complexes that twist around the baseline, occurring in the setting of QT prolongation. It may degenerate to ventricular fibrillation. Causes include congenital long QT syndrome, drugs that prolong QT interval (antiarrhythmics, antibiotics, antipsychotics), electrolyte disturbances (hypokalemia, hypomagnesemia), and bradycardia. Treatment includes IV magnesium, temporary pacing, and isoproterenol.

Total Anomalous Pulmonary Venous Return

– A cyanotic congenital heart defect where all four pulmonary veins drain into the systemic venous circulation (right atrium or its tributaries) rather than the left atrium. Supracardiac (most common, 45%): pulmonary veins drain via a vertical vein into the left innominate vein or SVC. Cardiac (25%): drain directly into the coronary sinus or right atrium. Infracardiac (20%): drain via a descending vein into the portal vein or IVC (most obstructed type). Mixed (10%). Obstruction presents with severe cyanosis and respiratory distress in the newborn period. Non-obstructed TAPVR presents with mild cyanosis and right heart volume overload. An obligatory right-to-left shunt via an ASD is required for survival. Diagnosis: echocardiography shows pulmonary veins not connecting to the left atrium. Treatment: surgical reimplantation of the pulmonary veins to the left atrium, closure of the ASD. Emergency surgery for obstructed TAPVR.

Toxic Epidermal Necrolysis

– A severe, life-threatening drug reaction characterized by widespread full-thickness epidermal necrosis and detachment, affecting >30% of body surface area. It represents a more severe form of Stevens-Johnson syndrome. Common causative drugs include anticonvulsants (carbamazepine, phenytoin), sulfonamides, allopurinol, and NSAIDs. Treatment is supportive, similar to burn care, with ICU admission and multidisciplinary care. Mortality is 30-50%.

Tricuspid Regurgitation

– Incomplete closure of the tricuspid valve allowing blood to flow backward from the right ventricle to the right atrium during systole. Causes include right ventricular dilation (from pulmonary hypertension or left heart failure), endocarditis, rheumatic disease, and carcinoid syndrome. Murmur: pansystolic murmur at the left lower sternal border increasing with inspiration. Severe TR may require surgical repair or replacement.

Tricuspid Stenosis

– Narrowing of the tricuspid valve opening, most commonly caused by rheumatic heart disease. It is often associated with mitral stenosis. Symptoms include fatigue, edema, and hepatic congestion. Murmur: low-pitched mid-diastolic rumble at the left lower sternal border that increases with inspiration. Diagnosis is confirmed by echocardiography. Severe TS may require surgical commissurotomy or valve replacement.

Troponins

– Cardiac-specific proteins (troponin I and T) released into the bloodstream during myocardial cell injury. They are the most sensitive and specific biomarkers for myocardial infarction. High-sensitivity troponin assays can detect very low levels, allowing early diagnosis. Troponins rise within 3-6 hours of injury, peak at 12-24 hours, and remain elevated for 7-14 days. Serial measurements are used to diagnose acute MI.

T Wave

– The deflection representing ventricular repolarization. Normal T waves are upright in leads I, II, and V3-V6, and inverted in aVR. They should be at least 10% of the QRS amplitude in limb leads and 25% in precordial leads. Abnormal T waves may indicate ischemia (inversion), hyperkalemia (peaked), or pericarditis (diffuse elevation). T wave changes are nonspecific and require clinical correlation.

U Wave

– A small deflection following the T wave, best seen in precordial leads V2-V3. U waves are thought to represent repolarization of the Purkinje fibers or papillary muscles. Prominent U waves are associated with hypokalemia, bradycardia, and certain drugs (digitalis, quinidine). Inverted U waves may indicate ischemia or left ventricular hypertrophy. U waves are normally small and may be absent.

Valproic Acid

– An antiepileptic drug that enhances GABA-mediated inhibition, blocks sodium channels, and has multiple mechanisms of action. It is effective for generalized seizures, absence seizures, and myoclonic seizures. Side effects include weight gain, tremor, hair loss, hepatotoxicity (rare in adults, higher risk in children under 2), teratogenicity (neural tube defects), and pancreatitis. It inhibits CYP enzymes and is highly protein-bound, with numerous drug interactions.

Varicose Veins

– Dilated, tortuous superficial veins, most commonly in the lower extremities. Caused by venous valve incompetence leading to venous reflux. Risk factors: family history, prolonged standing, obesity, pregnancy, and older age. Symptoms: leg heaviness, swelling, aching. Complications: skin changes, ulceration, thrombophlebitis. Treatment: compression stockings, sclerotherapy, endovenous ablation, and surgical ligation.

Vasovagal Syncope

– Reflex syncope mediated by increased vagal tone and withdrawal of sympathetic activity, causing bradycardia and/or vasodilation resulting in transient cerebral hypoperfusion. Triggers: prolonged standing, heat, pain, emotional stress. Preceded by prodrome (lightheadedness, nausea, diaphoresis, visual tunneling). Characterized by brief loss of consciousness with spontaneous recovery. Diagnosis: clinical history, tilt-table testing. Treatment: education, lifestyle modifications, and rarely, pacemaker implantation.

Venous Thrombosis

– The formation of a blood clot (thrombus) within a vein, most commonly in the deep veins of the lower extremities (deep vein thrombosis, DVT). Pathophysiology (Virchow triad): venous stasis (immobility, surgery, long-haul travel, obesity), hypercoagulability (pregnancy, malignancy, thrombophilia — factor V Leiden, prothrombin gene mutation, antithrombin deficiency, protein C/S deficiency, antiphospholipid syndrome), and endothelial injury (trauma, surgery, central venous catheters). DVT presents with unilateral leg pain, swelling, warmth, erythema, and Homan sign (calf pain on dorsiflexion, unreliable). Complications: pulmonary embolism (PE, the most dangerous complication, causing dyspnea, chest pain, haemoptysis, and sudden death) and post-thrombotic syndrome (chronic venous insufficiency from venous valve damage). Diagnosis: D-dimer (elevated, sensitive but not specific) and compression ultrasound (non-compressible vein is diagnostic). Ventilation-perfusion (V/Q) scan or CT pulmonary angiography for PE. Treatment: anticoagulation (heparin/LMWH bridging to warfarin or DOAC — rivaroxaban, apixaban, edoxaban, dabigatran) for minimum 3-6 months; compression stockings to prevent post-thrombotic syndrome. Duration: unprovoked DVT may require indefinite anticoagulation. Prevention (prophylaxis): early mobilisation, compression devices, and prophylactic anticoagulation in hospitalised at-risk patients.

Ventricular Assist Device

– A mechanical pump that supports heart function in patients with advanced heart failure. LVADs (left ventricular assist devices) are most common, supporting the failing left ventricle. The HeartMate 3 is the current standard with fully magnetically levitated rotors. Indications include bridge to transplant (BTT), destination therapy (DT), and bridge to decision. Complications include bleeding, infection, thromboembolism, and device malfunction.

Ventricular Fibrillation

– Rapid, chaotic ventricular electrical activity resulting in no effective cardiac output, leading to cardiac arrest. ECG shows irregular, disorganized waveforms with no identifiable QRS complexes. VFib is a medical emergency requiring immediate defibrillation. Causes include acute myocardial infarction, electrolyte imbalances, drug toxicity, and electrical injury. Survival decreases approximately 10% for each minute without defibrillation.

Ventricular Septal Defect

– The most common congenital heart defect (30% of all CHD), characterized by an abnormal opening in the interventricular septum allowing left-to-right shunting. Types: perimembranous (most common, 70%, near the membranous septum), muscular (5-20%, completely surrounded by muscle), supracristal (5-7%, above the crista supraventricularis, associated with aortic regurgitation), and inlet (5-8%, near the tricuspid valve, common in Down syndrome). Small VSDs may close spontaneously (especially muscular). Large VSDs present with heart failure, failure to thrive, and frequent respiratory infections. On examination: holosystolic murmur at the left lower sternal border, hyperdynamic precordium, and widely split S2. Diagnosis: echocardiography shows shunt and estimates RV pressure. Treatment: small/asymptomatic — observation; large/symptomatic — surgical patch closure or transcatheter device closure. Pulmonary vascular disease develops if unrepaired (Eisenmenger syndrome).

Ventricular Tachycardia

– A wide-complex tachycardia (QRS >0.12 s) originating from the ventricles, with a rate of 100-250 bpm. Monomorphic VT has uniform QRS complexes; polymorphic VT has varying QRS morphology. VT is life-threatening and may degenerate to ventricular fibrillation. Causes include coronary artery disease, cardiomyopathy, electrolyte abnormalities, and drugs. Treatment includes cardioversion for unstable VT, amiodarone for stable VT, and implantable cardioverter-defibrillator for prevention.

Wolff-Parkinson-White Syndrome

– A pre-excitation syndrome caused by an accessory pathway (bundle of Kent) connecting atria and ventricles, bypassing the AV node. ECG shows short PR interval (<0.12 s), delta wave (slurred upstroke of QRS), and widened QRS. Patients are at risk for supraventricular tachycardia and, if atrial fibrillation occurs, rapid ventricular response potentially degenerating to ventricular fibrillation. Treatment includes catheter ablation of the accessory pathway.

Chapter 5

Dentistry

Amalgam

– A dental restorative material composed of elemental mercury (50%) combined with a metallic alloy powder containing silver, tin, copper, and zinc (most modern: high-copper amalgam). Advantages: durable (lasts 10-15+ years), low cost, moisture-tolerant (can be placed in areas not perfectly dry), longest track record of any restorative material. Disadvantages: esthetic (silver color, not tooth-colored), requires mechanical retention (undercuts, slots), tooth structure removal for retention, and concern over mercury content (FDA: safe for most patients; avoids placement/removal in pregnant women and those with mercury allergy). Placement: cavity preparation with undercuts, mixing amalgam in a capsule (trituration), condensation into the cavity, and carving to contour. Mercury exposure: dental personnel may have higher urine mercury but below toxic thresholds. Amalgam separators (capture >95% of amalgam waste) are required by the EPA. Alternatives: composite resin (tooth-colored), glass ionomer, and ceramic. Use has declined in many countries (Sweden, Norway phased out for environmental reasons).

Apical Periodontitis

– Inflammation and destruction of the periradicular tissues (periodontal ligament and alveolar bone at the apex) caused by bacterial infection of the root canal system (necrotic pulp). Presents with pain on biting/percussion (symptomatic) or asymptomatic with radiographic periapical radiolucency (asymptomatic). Acute apical abscess: severe pain, swelling (facial cellulitis, space infection), fever, and malaise. Chronic apical periodontitis: asymptomatic, well-defined periapical radiolucency. Phoenix abscess: acute exacerbation of chronic lesion. Treatment: root canal therapy (remove infected pulp, clean, shape, disinfect, and obturate the root canal system). Incision and drainage for fluctuant swelling (through the gingiva or mucosa). Antibiotics: for systemic signs, facial swelling, or immunocompromised patients (amoxicillin, clindamycin if penicillin-allergic).

Apicoectomy (Root-End Resection)

– A surgical endodontic procedure performed when conventional root canal therapy or retreatment is unsuccessful due to persistent apical disease. Indications: persistent apical periodontitis after RCT, obstructed canal (calcification, separated instrument, broken file), apical perforation, root-end cyst, and biopsy of periapical lesion. Procedure: local anesthesia, full-thickness mucoperiosteal flap elevation, osteotomy (bone window) to access the root apex, apical root-end resection (3 mm of apex removed), ultrasonic root-end preparation (3 mm deep), root-end filling (MTA mineral trioxide aggregate or Super-EBA), and flap repositioning/suturing. Success rate: 75-90% with MTA apical seal. Post-operative: mild to moderate pain, swelling, avoid heavy chewing for 1 week.

Bridge (Fixed Partial Denture)

– A fixed dental prosthesis that replaces one or more missing teeth by attaching to adjacent natural teeth (abutments) or implants. Typical: a three-unit bridge (two abutments, one pontic — artificial tooth replacing the missing tooth). Types: traditional (full-coverage crowns on abutments, pontic in between), cantilever (pontic supported on only one side — more conservative but higher failure rate), resin-bonded/Maryland bridge (wings bonded to the lingual surfaces of adjacent teeth — minimal tooth reduction, for single anterior tooth replacement), and implant-supported bridge (most conservative of adjacent teeth — no preparation of healthy adjacent teeth). Pontic design: modified ridge-lap or ovate pontic (hygienic, esthetic). Materials: PFM, full ceramic (lithium disilicate, zirconia), and metal. Relative contraindications: long span, heavy occlusal forces (bruxism), short clinical crowns (insufficient retention). Longevity: 10-15 years. Failure: abutment tooth fracture/decay (most common, 10-20% at 10 years), debonding, and pontic fracture.

Bruxism

– A repetitive jaw-muscle activity characterized by clenching or grinding of the teeth, occurring during sleep (sleep bruxism) or wakefulness (awake bruxism). Prevalence: 8-15% of adults, up to 30% of children. Etiology: multifactorial — psychosocial stress, anxiety, sleep disorders (OSA, snoring), occlusal interference, genetics, caffeine, alcohol, smoking, and certain medications (SSRIs have been implicated — may cause or worsen). Clinical consequences: tooth wear/attrition (incisal and occlusal surfaces — flattened cusps, exposed dentin), tooth sensitivity, fractures of restorations/cusps, masticatory muscle hypertrophy (masseter hypertrophy visible), TMJ pain, morning headache, and tongue/cheek scalloping. Diagnosis: clinical exam (wear facets), history (reports of grinding from bed partner), and polysomnography (for sleep bruxism, definitive, shows rhythmic masticatory muscle activity RMMA). Treatment: occlusal splint/night guard (hard acrylic or soft, reduces wear and muscle activity), stress management, biofeedback, sleep hygiene, and in severe cases, botulinum toxin injection into masseter and temporalis muscles. Pharmacotherapy: clonidine, clonazepam (low-dose, short-term). No cure; management is symptomatic.

Composite Resin (Tooth-Colored Filling)

– An esthetic dental restorative material composed of a resin matrix (Bis-GMA, UDMA, TEGDMA) filled with inorganic particles (silica, barium glass, zirconia). Categories by filler size: microfilled (0.04 µm, polishable but weak, for anterior), hybrid (0.5-1 µm + microfillers, universal), nanofilled/nanohybrid (5-100 nm nanoparticles, best properties). Advantages: esthetic (color matched to the tooth), minimal tooth preparation (bonded restoration, preservation of sound tooth structure), and repairable. Application: acid-etch (37% phosphoric acid for 15-30 seconds — removes smear layer, creates microporosity for micromechanical bond), apply bonding agent (primer + adhesive), place composite in increments (<2 mm each to reduce polymerization shrinkage), light-cure (LED light 400-500 nm for 10-40 seconds), finish and polish. Polymerization shrinkage: 1.5-3%, can cause marginal gaps, post-operative sensitivity, microleakage, and secondary caries. Bulk-fill composites (can be placed in 4-5 mm increments) reduce technique sensitivity. C-factor: ratio of bonded to unbonded surfaces; higher C-factor = more shrinkage stress. Longevity: 5-8 years in posterior teeth, longer in anterior.

Crown (Dental Crown)

– A full-coverage dental restoration that encases the entire visible portion of a tooth above the gum line, restoring its shape, size, strength, and appearance. Indications: large caries/cracks (insufficient tooth structure for a filling), crown fracture, after root canal therapy (posterior teeth — endodontically treated teeth are brittle), cosmetic enhancement, and as a bridge abutment. Crown materials: full metal (gold alloy — most durable, wear-resistant, minimal tooth reduction — 1-1.5 mm), PFM (porcelain fused to metal — good esthetics and strength, metal margin may show with gingival recession), all-ceramic (lithium disilicate — e.max, most esthetic, high strength, 1-1.5 mm reduction), and zirconia (yttria-stabilized zirconia — highest strength, used for posterior crowns and bridges, CAD/CAM fabricated, opaque). Preparation: 1-2 mm tooth reduction circumferentially with a shoulder or chamfer margin. Impression: digital (intraoral scanner — iTero, Trios, PrimeScan) or conventional (polyvinyl siloxane or alginate). Temporary crown: acrylic or bis-acryl composite temporary for 1-2 weeks while the permanent crown is fabricated. Cementation: resin cement (for ceramic and zirconia), glass ionomer, or zinc phosphate. Longevity: 10-15+ years, with proper hygiene and regular recall.

Deep Sedation and General Anesthesia in Dentistry

– Pharmacologically induced states of consciousness suppression for dental treatment, used for patients with severe dental phobia, special needs (developmental delay, autism, cerebral palsy), young children requiring extensive treatment, and lengthy/complex surgical procedures (impacted third molars, dental implants, orthognathic surgery). Practitioner: DDS/DMD with a dental anesthesia residency permit or certified registered nurse anesthetist (CRNA). Levels: moderate conscious sedation (IV: midazolam, fentanyl; oral: triazolam, hydroxyzine — patient can maintain airway and respond to verbal commands), deep sedation (propofol, ketamine — purposeful response to painful stimulation, may require airway support), and general anesthesia (endotracheal intubation, controlled ventilation). Monitoring: pulse oximetry, capnography, ECG, NIBP. Patient selection: ASA I-II generally safe, ASA III-IV require medical consultation. Risks: respiratory depression (opioids, propofol), laryngospasm, aspiration, hypotension, cardiac arrhythmia. Recovery: monitored PACU with Aldrete scoring.

Dental Abscess

– A localized collection of pus in the oral cavity, classified by origin: periapical (from necrotic pulp — most common), periodontal (from periodontal pocket), and pericoronal (pericoronitis around partially erupted tooth, most commonly lower third molar). Signs: severe, throbbing pain, tooth tender to percussion (periapical), swelling (intraoral, extraoral, facial), cellulitis (diffuse swelling, systemic involvement — Ludwig angina for submandibular space infection, life-threatening airway compromise), lymphadenopathy, fever, and malaise. Radiographic: periapical radiolucency (periapical abscess may not be visible early — bone demineralization >30% needed for radiographic visibility). Treatment: establish drainage (root canal therapy — access the pulp chamber and allow drainage through the canal, or incision and drainage for fluctuant swelling), removal of the cause (RCT, extraction), and antibiotics for systemic involvement (amoxicillin 500 mg TID x 5-7 days, clindamycin 300 mg QID if penicillin-allergic). Severe infections: IV antibiotics, hospital admission, and surgical drainage.

Dental Calculus

– See Tartar (Dental Calculus).

Dental Caries (Tooth Decay)

– A chronic, progressive, multifactorial disease characterized by demineralization of tooth enamel and dentin by organic acids produced by bacterial fermentation of dietary carbohydrates. Etiology: cariogenic bacteria (Streptococcus mutans, Lactobacillus, Actinomyces), fermentable carbohydrates (sucrose most cariogenic), susceptible tooth surface, and time. Caries classification: pit and fissure (occlusal surfaces), smooth surface (interproximal, buccal/lingual), root caries (exposed root surfaces in recession), and recurrent caries (around existing restorations). Depth: incipient/reversible (enamel only, white spot lesion), moderate (dentin involvement, may be reversible if <1/3 into dentin), deep (approaching pulp, risk of pulpitis), and pulpal involvement (pulpitis, necrosis). Diagnosis: visual-tactile exam (ICDAS criteria), bitewing radiographs (detect interproximal caries not visible clinically), transillumination (DiFOTI), and laser fluorescence (DIAGNOdent). Prevention: fluoride (toothpaste, varnish, water fluoridation), dietary counseling (limit sugar frequency), pit and fissure sealants, and regular recall visits. Treatment: minimally invasive (resin infiltration for incipient lesions, GI/temporary), direct restoration (composite resin, amalgam, glass ionomer), indirect restoration (inlay/onlay/crown), and root canal therapy for pulpal involvement. ICDAS (International Caries Detection and Assessment System): codes 0-6 based on visual appearance from sound to extensive cavitation.

Dental Emergencies

– severe toothache, knocked-out teeth, cracked or fractured teeth, abscesses, and uncontrolled bleeding from the mouth. A knocked-out permanent tooth should be placed back in the socket or in milk and seen by a dentist within 30 minutes for the best chance of reimplantation. Dental abscesses require prompt treatment to prevent spread of infection to adjacent tissues or the bloodstream.

Dental Health

– Dental health involves the care of teeth, gums, and supporting structures to prevent common problems like cavities, gum disease, and tooth loss. Regular brushing with fluoride toothpaste, flossing, and professional cleanings control plaque buildup and detect issues early. Poor dental health is linked to systemic conditions including cardiovascular disease, diabetes complications, and adverse pregnancy outcomes.

Dental Public Health

– The non-clinical dental specialty focused on preventing oral disease and promoting oral health at the population level through community-based programs, policy development, and research. Activities: community water fluoridation (optimal 0.7 mg/L, reduces caries by 25-40%), school-based sealant programs, fluoride varnish programs, oral health education, epidemiology of oral disease (NHANES data), and access-to-care initiatives. Key metrics: DMFT index (Decayed, Missing, Filled permanent Teeth), DMFS (surfaces), dmft for primary dentition. Public health challenges: disparities in oral health access (low-income, rural, racial/ethnic minorities), dental caries prevalence (most common chronic disease of childhood — 5x more common than asthma), and oral cancer screening.

Dentin

– The calcified tissue of the tooth that lies beneath the enamel (crown) and cementum (root), forming the bulk of the tooth. Dentin is composed of approximately 70% hydroxyapatite (mineral), 20% organic matrix (primarily type I collagen), and 10% water. It contains microscopic tubules (dentinal tubules) that radiate from the pulp chamber to the enamel–dentin junction, housing odontoblast processes and allowing transmission of sensory stimuli (heat, cold, pressure) to the pulp as pain. Dentin is produced by odontoblasts throughout life (secondary and tertiary dentin). When exposed by caries or attrition, dentin hypersensitivity may occur. Dentinogenesis imperfecta is a genetic disorder causing opalescent, weakened dentin.

Dentition

– The arrangement, number, and type of teeth in the dental arch at a given stage of life. Primary dentition (deciduous teeth, baby teeth): 20 teeth, erupting from 6 months to 2.5 years. Mixed dentition: phase when both primary and permanent teeth are present (6–12 years). Permanent dentition (adult): 32 teeth including 8 incisors, 4 canines, 8 premolars, and 12 molars (including 4 third molars/wisdom teeth). Eruption timing varies but follows a predictable sequence. Dental formula records the number and type of teeth in each quadrant. Abnormalities: anodontia (congenital absence), supernumerary teeth (extra teeth), malocclusion, and impacted teeth.

Denture

– A removable dental prosthesis that replaces missing teeth and adjacent tissues. Complete denture: replaces all teeth in an arch (maxillary or mandibular). Immediate denture: placed immediately after extraction of remaining teeth (allows patient to never be without teeth; requires relining after bone resorption). Partial denture (RPD — removable partial denture): replaces some but not all teeth in an arch; held in place by clasps (cast metal — cobalt-chromium, or flexible nylon Valplast) and rests on abutment teeth. Implant-retained overdenture: denture that snaps onto 2-4 dental implants (greatly improved retention and stability, especially for the mandibular complete denture — 95% of lower denture problems solved by 2 implants). Implant-supported fixed complete denture: fixed prosthesis on 4-6 implants (All-on-4 concept). Adaptation period: 2-4 weeks for speech, eating, and saliva control. Denture stomatitis (denture sore mouth): Candida albicans infection under the denture — improved by removing the denture at night, proper cleaning, and antifungal (nystatin, clotrimazole). Complications: poor fit due to bone resorption (requires reline — hard or soft tissue conditioner), fracture (may require repair), and sore spots (adjustment with acrylic bur).

Dry Socket (Alveolar Osteitis)

– A common post-extraction complication characterized by severe, throbbing pain starting 2-3 days after tooth extraction, caused by premature loss of the blood clot from the extraction socket, exposing underlying bone. Risk factors: tobacco use (nicotine reduces blood supply, clot dissolution from sucking), traumatic extraction, oral contraceptives (estrogen increases fibrinolysis), poor oral hygiene, and mandibular teeth (especially third molars — higher density bone, less blood supply). Treatment: gentle socket irrigation with warm saline to clear debris, placement of a medicated dressing (Alvogyl — butamben, eugenol, iodoform; or eugenol-soaked ribbon gauze — zinc oxide-eugenol) for immediate pain relief, and NSAIDs. The dressing may need to be changed every 1-3 days for 5-10 days until granulation tissue covers the bone. Prevention:

post-operative instructions (no smoking, no spitting, no drinking through straws for 48-72 hours).

Endodontics

– The dental specialty concerned with the morphology, physiology, and pathology of the dental pulp and periradicular tissues. Endodontists perform root canal therapy (pulpectomy, pulpotomy), root canal retreatment (for failed prior RCT), endodontic surgery (apicoectomy — root-end resection with retrograde filling), and management of traumatic dental injuries (crown fractures, root fractures, luxation, avulsion). Common diagnoses: irreversible pulpitis (severe, lingering pain to thermal stimuli), pulp necrosis (no response to cold, periapical radiolucency), and acute apical abscess (pain, swelling, drainage). Treatment involves cleaning and shaping the root canal system (hand files and rotary NiTi files), disinfection (sodium hypochlorite irrigation 1-6%, chlorhexidine, calcium hydroxide), and obturation (gutta-percha with sealer). Cone-beam CT (CBCT) is used for complex anatomy (extra canals, root resorption, fractures). Success rate: >90% for initial root canal treatment.

General Dentistry

– The primary dental care provider responsible for the diagnosis, prevention, and treatment of common oral diseases including dental caries (cavities), periodontal disease, and oral pathology. General dentists perform restorative procedures (fillings, crowns, bridges), simple extractions, root canal therapy, and preventive care (cleanings, fluoride, sealants). They also coordinate comprehensive treatment plans and refer to specialists when needed. Patients are typically seen every 6 months for routine recall examinations and prophylaxis.

Gingival Hyperplasia (Overgrowth)

– Abnormal enlargement of the gingiva. Drug-induced: phenytoin (antiepileptic — most severe, 50% of patients), cyclosporine (immunosuppressant, 30%), and calcium channel blockers (nifedipine, amlodipine — 20%). Other causes: hereditary gingival fibromatosis (rare, autosomal dominant), hormonal (pregnancy, puberty), inflammatory (chronic plaque-induced gingivitis), and leukemic infiltration (acute monocytic/myelomonocytic leukemia). Presents: firm, pink, lobulated enlargement of the interdental papillae and marginal gingiva, may cover the clinical crowns. Bleeding and secondary inflammation common when plaque control is poor. Treatment: improved oral hygiene (reduces inflammatory component, drug-induced overgrowth persists), chlorhexidine mouthwash, surgical excision (gingivectomy with scalpel, electrosurgery, or laser — CO2, diode). Discontinuation or substitution of the causative drug if possible (consult with prescribing physician). Phenytoin substitute: levetiracetam, lamotrigine. Recurrence is common if the drug continues.

Gingivitis

– Reversible inflammation of the gingiva (gums) caused by bacterial plaque biofilm accumulation at the gingival margin. Clinical signs: erythema, edema (blunted papilla), bleeding on probing (BOP — first sign, before visible redness), and loss of gingival stippling. No bone loss or clinical attachment loss (true gingivitis). Pregnancy gingivitis: hormone-mediated exaggerated response to plaque in the second/third trimester. Gingival hyperplasia/enlargement: phenytoin, cyclosporine, calcium channel blockers, and hormonal changes. Treatment: professional prophylaxis (scaling and polishing), improved oral hygiene (brushing twice daily, flossing, antigingivitis toothpaste — stannous fluoride, triclosan), and chlorhexidine mouthwash for short-term use. Resolution: 10-14 days with plaque control.

Impacted Tooth

– A tooth that fails to fully erupt into the normal functional position within the expected time frame, most commonly the third molars (wisdom teeth — 90% of impactions), followed by maxillary canines (2-3%), mandibular premolars, and maxillary central incisors (mesiodens, supernumerary). Third molar impaction classification: relationship to the ramus and second molar (Classes I, II, III), depth (Level A, B, C), and angulation (mesioangular most common, vertical, horizontal, distoangular). Indications for extraction: pericoronitis (recurrent infection of the operculum over partially erupted tooth), caries on the distal of the second molar, non-restorable caries, periodontal disease, root resorption, orthodontic reasons, cyst formation (dentigerous cyst), and prophylactic removal (controversial, NICE guidelines recommend against routine). Complications of extraction: dry socket (alveolar osteitis — most common, 1-5%, severe pain 2-3 days post-op, treat with irrigation and medicated dressing — Alvogyl, eugenol paste), inferior alveolar nerve injury (temporary 5-10%, permanent <1%), lingual nerve injury, maxillary sinus communication, mandible fracture (rare), and infection.

Ludwig Angina

– A severe, rapidly spreading cellulitis of the sublingual, submandibular, and submental spaces, most commonly from an odontogenic source (lower second/third molar infection). Life-threatening due to airway compromise from massive bilateral swelling elevating and posteriorly displacing the tongue (airway obstruction). Signs: bilateral submandibular swelling (bull neck appearance), elevation of the tongue, trismus, dysphagia, drooling, fever, and impending airway obstruction. Pathogens: Streptococcus viridans, anaerobes (Peptostreptococcus, Bacteroides, Fusobacterium), and mixed oral flora. Airway management is the priority: awake fiberoptic intubation or cricothyrotomy/tracheostomy (may be extremely difficult — early intervention before deterioration). Empiric IV antibiotics: high-dose penicillin + metronidazole (or ampicillin-sulbactam, clindamycin). Surgical drainage: bilateral submandibular and submental incisions with placement of Penrose drains. Extraction of the causative tooth. ICU monitoring.

Malocclusion

– Misalignment or incorrect positioning of the teeth when the jaws are closed, affecting function (chewing, speech) and/or aesthetics. Angle classification: Class I (neutrocclusion — normal molar relationship, but teeth may be crowded, rotated, spaced), Class II (retrognathism — mandible posterior to maxilla, overjet — typically >3-4 mm, Division 1: protruding maxillary incisors, Division 2: retroclined central incisors), Class III (prognathism — mandible anterior to maxilla, anterior crossbite, underbite). Other descriptors: overjet (horizontal overlap of maxillary incisors over mandibular, normal 2-3 mm), overbite (vertical overlap, normal 2-3 mm, deep bite >5 mm or covering >50% of lower incisors), open bite (anterior or posterior, no vertical overlap), crossbite (anterior or posterior, buccal or lingual), crowding, spacing, midline deviation, and impacted teeth. Etiology: genetic (skeletal discrepancies), environmental (thumb sucking, tongue thrust, mouth breathing, early loss of primary teeth), and acquired (periodontal disease, trauma). Treatment: orthodontic (braces, aligners, expanders, headgear), orthognathic surgery (for severe skeletal discrepancy), and restorative (crowns, veneers to correct minor malocclusion).

Mesiodens

– The most common supernumerary tooth (extra tooth), located in the midline of the maxilla between the central incisors, shaped like a peg or cone. Prevalence: 0.15-1.9%, more common in males (2:1). May be single or multiple, erupted or impacted. Complications: delayed eruption or impaction of permanent central incisors, midline diastema, root resorption of adjacent teeth, and cyst formation (dentigerous cyst). Most are asymptomatic and found incidentally on radiographs (occlusal or panoramic). Management: early removal (when roots of central incisors are 2/3 formed, around age 8-9) prevents orthodontic complications. Orthodontic closure of space after extraction. Associated with cleidocranial dysplasia and Gardner syndrome.

Mucogingival Deformity

– Condition involving the relationship between the gingiva and oral mucosa. Gingival recession (most common): apical migration of the gingival margin exposing the root surface, classified by Miller classification (Class I: not extending to mucogingival junction; Class II: extending to/beyond MGJ but no interproximal loss; Class III: interproximal loss; Class IV: severe interproximal loss). Causes: traumatic brushing, periodontal disease, tooth position (prominent root), thin biotype/fenestration, orthodontic movement, and frenum pull. Clinical: root exposure, dentin hypersensitivity (cold, air, touch), esthetic concern, and root caries risk. Treatment: non-surgical (desensitizing toothpaste, bonding/glass ionomer for root coverage), surgical root coverage (coronally advanced flap, subepithelial connective tissue graft — SCTG from the palate — gold standard for Class I and II, acellular dermal matrix allograft ADM, tunneling technique, and pinhole surgical technique). Free gingival graft (FGG): increases the zone of attached gingiva, not primarily for root coverage.

Odontology

– The scientific study of the structure, development, and diseases of the teeth and associated oral structures. Odontology encompasses dental anatomy, oral pathology, forensic odontology (identification of human remains through dental records, bite mark analysis), and clinical dentistry. Forensic odontology is particularly valuable when other identification methods fail, as teeth are the most durable structures in the human body and survive fire, decomposition, and trauma. Dental age estimation (through eruption patterns and cemental annulation) aids in both forensic identification and legal age determination.

Oral and Maxillofacial Surgery (OMFS)

– The dental specialty that diagnoses and surgically treats diseases, injuries, and defects of the mouth, jaws, face, and neck. Scope: dentoalveolar surgery (impacted third molar extraction — wisdom teeth, surgical extractions, pre-prosthetic surgery), dental implant placement, orthognathic surgery (corrective jaw surgery for malocclusion, skeletal discrepancies — Le Fort I maxillary osteotomy, bilateral sagittal split osteotomy BSSO for mandible), temporomandibular joint surgery (arthrocentesis, arthroscopy, discectomy, total joint replacement), facial trauma (mandible fractures, Le Fort II/III midface fractures, zygomaticomaxillary complex ZMC fractures, orbital floor blowout fractures, nasal fractures), oral pathology (biopsy, excision of lesions, management of oral cancer — composite resection, neck dissection, microvascular reconstruction), cleft lip and palate repair, and cosmetic facial surgery (rhinoplasty, facelift, blepharoplasty). Training: 4-6 year residency after dental school, often with an MD degree.

Oral Medicine

– The dental specialty focused on the oral health care of medically complex patients and the diagnosis and non-surgical management of oral mucosal and orofacial diseases. Areas: oral mucosal diseases (lichen planus, pemphigus vulgaris, mucous membrane pemphigoid, erythema multiforme, Sjogren syndrome — autoimmune exocrinopathy causing xerostomia and dry eyes), orofacial pain (temporomandibular disorders TMD, trigeminal neuralgia, burning mouth syndrome BMS, atypical odontalgia), salivary gland disorders (xerostomia, sialolithiasis, sialadenitis, Sjogren syndrome), oral complications of cancer therapy (mucositis, osteoradionecrosis, xerostomia, infection), and management of patients with systemic disease (diabetes, cardiovascular disease, bleeding disorders, bisphosphonate-related osteonecrosis of the jaw MRONJ).

Oral Pathology

– The dental specialty concerned with the diagnosis and management of diseases of the oral and maxillofacial region. Oral pathologists examine biopsies and provide histopathologic diagnosis. Common oral lesions: aphthous ulcers (canker sores — minor, major, herpetiform, recurrent aphthous stomatitis), herpes labialis (cold sores, HSV-1 reactivation), oral candidiasis (pseudomembranous thrush, erythematous, angular cheilitis), leukoplakia (white patch, cannot be scraped off, premalignant — 10-15% dysplastic), erythroplakia (red patch, higher malignant potential 50-70%), lichen planus (reticular white striae Wickham striae, erosive), oral lichenoid reaction, mucocele (mucus extravasation phenomenon), pyogenic granuloma (pregnancy tumor), fibroma, papilloma, and squamous cell carcinoma (most common oral malignancy, risk: tobacco, alcohol, HPV-16, betel nut).

Oral Squamous Cell Carcinoma (OSCC)

– The most common malignancy of the oral cavity, accounting for >90% of oral cancers. Risk factors: tobacco (smoking and smokeless — 75% of cases), alcohol (synergistic with tobacco), HPV-16 (oropharynx, base of tongue, tonsil — better prognosis), betel nut (Southeast Asia), immunosuppression (HIV, transplant), and sun exposure (lip). Premalignant lesions: leukoplakia (10-15% dysplastic or malignant), erythroplakia (50-70% high-risk), oral submucous fibrosis. Common sites: lateral tongue (most common), floor of mouth, buccal mucosa, gingiva, palate, and lip. Clinical: non-healing ulcer (>2 weeks), red/white patch, exophytic mass, pain, dysphagia, otalgia, loose teeth, cervical lymphadenopathy. Diagnosis: biopsy with histopathology, CT/MRI/PET-CT for staging. Treatment: surgical resection with negative margins (1 cm), neck dissection (therapeutic or elective for N0 with >15-20% occult metastasis risk), primary radiation (for early stage), adjuvant chemoradiation for high-risk features (positive margins, extracapsular nodal extension, perineural invasion). Prognosis: 60-70% 5-year survival for early stage, 30-50% for advanced.

Orthodontics and Dentofacial Orthopedics

– The dental specialty concerned with the diagnosis, prevention, and correction of malocclusion and dentofacial abnormalities. Orthodontists use fixed appliances (braces — metal, ceramic, lingual), clear aligners (Invisalign, Spark, ClearCorrect), and removable appliances to move teeth through bone by applying controlled forces. Key concepts: Angle classification of malocclusion — Class I (neutrocclusion, normal molar relationship, may have crowding/spacing), Class II (distocclusion, mandibular retrusion, overjet — Division 1 protruding maxillary incisors, Division 2 retroclined central incisors), Class III (mesiocclusion, mandibular prognathism, anterior crossbite). Cephalometric analysis: assesses skeletal and dental relationships on lateral cephalogram (SNA, SNB, ANB angles). Treatment phases: Phase I (interceptive, mixed dentition ages 7-10 — expander, partial braces, habit appliances), Phase II (comprehensive, full permanent dentition — braces/aligners, retention). Retention: fixed lingual retainers and/or clear vacuum-formed retainers to prevent relapse. Duration: 12-30 months. Complications: root resorption, white spot lesions (decalcification), gingival inflammation.

Pediatric Dentistry (Pedodontics)

– The dental specialty focusing on oral health care from infancy through adolescence, including children with special health care needs. Services: infant oral health exams (first visit by age 1), caries risk assessment, preventive care (fluoride varnish, sealants), restorative treatment for primary and young permanent teeth (composite fillings, stainless steel crowns for primary molars, pulp therapy), behavior management (tell-show-do, positive reinforcement, nitrous oxide sedation, protective stabilization, general anesthesia for extensive treatment), management of dental trauma (crown fractures, luxation, avulsion — reimplantation within 30 minutes best outcomes), space maintenance after early loss of primary teeth (band-and-loop, lingual arch, Nance appliance), and interceptive orthodontics (habit appliances — thumb/finger habit, tongue thrust). Early childhood caries (ECC): caries in primary teeth before age 6, severe if smooth-surface caries in children <3 years. Nursing/bottle caries: upper incisors affected by night-time bottle feeding.

Periodontal Abscess

– A localized purulent infection in the periodontal pocket wall, usually an acute exacerbation of chronic periodontitis. Presents with ovoid, erythematous, fluctuant swelling on the gingiva along the lateral root surface, severe throbbing pain, tenderness to palpation/probing, tooth mobility, and purulent exudate on probing. The tooth is vital (unlike apical abscess where pulp is necrotic). Radiographic: advanced bone loss, often vertical or crater-shaped defects. Treatment: incision and drainage through the pocket, scaling and root planing (SRP), irrigation with chlorhexidine, occlusal adjustment if traumatic occlusion, and bite-block/tooth-stabilizing splint. Antibiotics if systemic involvement (amoxicillin with or without metronidazole, or clindamycin). Periodontal abscess must be distinguished from periapical abscess (pulp vitality testing, radiographic periapical vs. lateral lesion).

Periodontal Disease

– See Periodontics.

Periodontal (Gum) Disease

– Also known as periodontal disease, an inflammatory condition affecting the gums and supporting structures of the teeth. It begins as gingivitis with redness, swelling, and bleeding of the gums and can progress to periodontitis, causing gum recession, bone loss, and tooth loss. Prevention and treatment involve good oral hygiene, professional dental cleanings, and in advanced cases, scaling, root planing, or surgery.

Periodontics

– The dental specialty concerned with the supporting structures of the teeth (periodontium: gingiva, periodontal ligament, cementum, alveolar bone), including diagnosis and treatment of periodontal disease. Gingivitis: reversible inflammation of the gingiva without bone loss, caused by bacterial plaque biofilm. Periodontitis: destructive inflammatory disease with progressive loss of alveolar bone and connective tissue attachment, classified by stage (I-IV, based on severity and complexity) and grade (A-C, based on rate of progression). Chronic periodontitis (most common): slow progression, adults >30. Aggressive periodontitis: rapid bone loss, younger patients, familial pattern. Necrotizing periodontal diseases: NUG (necrotizing ulcerative gingivitis — pain, fetor oris, pseudomembranes, fever) and NUP (necrotizing ulcerative periodontitis — associated with HIV, severe immunosuppression). Periodontal probing: 6-point pocket depth measurement (normal <3 mm), bleeding on probing (active inflammation), clinical attachment level (CAL). Non-surgical treatment: scaling and root planing (SRP, deep cleaning under local anesthesia). Surgical: flap surgery (open flap debridement), osseous surgery (recontour bone), guided tissue regeneration (GTR with barrier membrane), bone grafting, and platelet-rich plasma (PRP). Maintenance: periodontal maintenance q3-4 months in treated patients.

Prosthodontics

– The dental specialty concerned with the restoration and replacement of teeth using fixed and removable prostheses. Fixed prosthodontics: crowns (full coverage restoration for damaged teeth — metal, PFM, all-ceramic/e.max, zirconia), bridges (fixed partial denture replacing 1-2 missing teeth — traditional cantilever, resin-bonded Maryland bridge, implant-supported bridge), inlays and onlays (partial coverage restorations). Removable prosthodontics: complete dentures (full set for edentulous patients — conventional immediate, implant-retained overdenture), partial dentures (removable partial denture RPD with clasps, precision attachments). Implant prosthodontics: single crowns, implant-retained overdentures, implant-supported fixed complete dentures (All-on-4, All-on-6 concept using 4-6 implants with a full-arch fixed prosthesis). Maxillofacial prosthetics: obturator (for palatal defects after cancer surgery), speech aid prosthesis, nasal/auricular/ocular prosthesis, and radiation stent. Materials: polymethyl methacrylate (PMMA) denture base, cobalt-chromium alloy for RPD frameworks, ceramic (feldspathic, lithium disilicate, zirconia), and composite resin.

Space Maintainer

– A passive appliance used in pediatric dentistry to maintain the space created by the premature loss of a primary tooth, preventing adjacent teeth from drifting into the space and preserving sufficient space for the permanent successor to erupt. Types: band-and-loop (most common, for single tooth loss, unilateral — a band is cemented on the tooth adjacent to the space, with a loop extending across the edentulous space to contact the tooth on the other side), lingual arch (mandibular, maintains space when multiple primary molars are lost), Nance holding appliance (maxillary, similar to lingual arch with an acrylic button on the palate), and distal shoe (extends mesially into the extraction site to guide eruption of the first permanent molar). Crown and loop: for a missing primary molar when the adjacent tooth requires a full crown. Failure of space maintenance: space closure, crowding, impaction of the permanent tooth, and asymmetric eruption.

Submandibular Space Infection

– An infection of the submandibular space, usually from an odontogenic source (mandibular molar periapical infection spreading below the mylohyoid muscle). Presents with unilateral submandibular swelling, tenderness, and trismus. May progress to Ludwig angina (bilateral involvement with airway compromise). Imaging: CT with contrast (identifies involved spaces and abscess formation). Treatment: surgical drainage through the submandibular approach (incision parallel to the inferior border of the mandible, through platysma into the submandibular space, blunt dissection, Penrose drain), extraction of the causative tooth, and antibiotics (amoxicillin-clavulanate, clindamycin, or penicillin + metronidazole).

Tartar (Dental Calculus)

– Hardened, calcified dental plaque that forms on the teeth, both above the gumline (supragingival calculus) and below the gumline (subgingival calculus). Calculus is primarily composed of calcium phosphate (hydroxyapatite, brushite, whitlockite) and dead bacteria, mineralised from saliva (supragingival) or gingival crevicular fluid (subgingival). It is rough and porous, providing a surface for further plaque accumulation, and is a major risk factor for periodontal disease (gingivitis → periodontitis). Calculus cannot be removed by brushing; it requires professional scaling and root planing (dental cleaning) using ultrasonic scalers and hand instruments (curettes, scalers). Subgingival calculus appears brownish-black (stained by blood products). Tartar control toothpastes (pyrophosphates, zinc citrate) can reduce new calculus formation. Systemic risk factors: smoking (more calculus), xerostomia (less protective saliva), and diet (soft, processed foods).

Teeth

– See Tooth / Teeth and Dentition.

Temporomandibular Joint Disorders (TMJ)

– Temporomandibular joint disorders encompass a group of conditions affecting the jaw joint and the surrounding muscles that control jaw movement. Symptoms include jaw pain, clicking or popping sounds with movement, difficulty chewing, and locking of the jaw, often resulting from bruxism, arthritis, trauma, or malocclusion. Diagnosis is based on clinical examination and imaging when indicated; treatment ranges from conservative measures such as soft foods, stress reduction, and oral splints to physical therapy, medications, and in severe cases, arthrocentesis or surgery.

Temporomandibular Joint Disorder (TMD)

– A group of conditions causing pain and dysfunction in the temporomandibular joint, masticatory muscles, and associated structures. Etiology: multifactorial — malocclusion, bruxism (teeth grinding/clenching), trauma (whiplash, microtrauma), arthritis (osteoarthritis, rheumatoid, psoriatic joint disease), connective tissue disease, stress/anxiety (muscle hyperactivity), and parafunctional habits. Clinical: TMJ pain (preauricular, worse with jaw function), limited mouth opening (<35-40 mm), clicking/popping/crepitus (disc displacement with reduction = click on opening, without reduction = closed lock), locking (open or closed), muscle tenderness (masseter, temporalis, pterygoid), headaches, ear pain. Imaging: panoramic X-ray, CBCT for bone assessment, MRI for disc position and morphology. Treatment: conservative first-line (80% respond) — soft diet, NSAIDs, muscle relaxants, night guard/splint (flat-plane occlusal splint), physical therapy (jaw exercises, ultrasound, TENS), moist heat/ice. Second-line: arthrocentesis (joint lavage), arthroscopy (lysis and lavage, disc repositioning), and corticosteroid injection. Surgical: discectomy, total joint replacement (for severe degenerative disease). Botulinum toxin (Botox) for bruxism and masseter hypertrophy.

TMJ Dislocation

– Anterior displacement of the mandibular condyle from the glenoid fossa, usually occurring during wide mouth opening (yawning, dental treatment, vomiting, intubation). Presents with the mouth open, inability to close, preauricular depression (empty glenoid fossa), deviation of the mandible to the contralateral side, and pain. Diagnosis: clinical, confirmed by panoramic radiograph or CT. Reduction: the operator stands in front of the patient, places thumbs on the posterior mandibular molars (intraoral technique) or on the external oblique ridge, and applies downward and backward pressure to guide the condyle back into the fossa. Gauze between the thumbs to prevent being bitten. Muscle relaxants, sedation, or benzodiazepines may facilitate reduction. Post-reduction: soft diet, avoid yawning widely, NSAIDs. Recurrent dislocation: treated with autologous blood injection, capsular tightening, eminectomy, or zygomatic arch augmentation.

Toothache (Odontalgia)

– Pain originating from the tooth or its supporting structures. This is the most common reason for emergency dental visits. Common causes: dental caries (most common — deep caries extending to dentin/pulp), pulpitis (reversible: sharp, brief pain to cold/sweet, resolves quickly; irreversible: spontaneous, lingering, throbbing pain to cold, may last minutes), abscess (periapical or periodontal — severe pain, tenderness to biting/percussion), cracked tooth syndrome (sharp, fleeting pain on biting/release — may be intermittent, difficult to diagnose), dry socket (alveolar osteitis — severe post-extraction pain with referral to ear, 2-3 days post-op), pericoronitis (pain/swelling around erupting third molar), sinusitis (maxillary — referred pain to upper teeth, pain on bending forward), and trigeminal neuralgia (electric-shock unilateral pain triggered by light touch). Diagnosis: history (character of pain, duration, triggers), clinical exam (caries probing, percussion, palpation, mobility, periodontal probing), vitality testing (cold test — refrigerant spray, electric pulp tester — or heat test for diagnosis of irreversible pulpitis), and radiographs (bitewing, periapical, panoramic, CBCT). Treatment: depends on diagnosis — restoration (filling/crown), pulp capping, root canal therapy, extraction, periodontal treatment, or anti-inflammatory/muscle relaxant for TMD.

Tooth Decay and Cavities

– Tooth decay, also known as dental caries, is the progressive destruction of tooth enamel and dentin caused by acid produced by bacteria metabolizing sugars in the mouth. The process begins with demineralization of enamel and can progress through the dentin to the pulp, potentially causing pain, infection, and tooth loss. Prevention relies on regular brushing with fluoride toothpaste, flossing, limiting sugary foods and drinks, and professional applications of fluoride or dental sealants, while established cavities are treated with fillings, root canals, or extraction.

Tooth Eruption

– The process by which a tooth moves from its developmental position within the alveolar bone through the oral mucosa into functional occlusion. Primary (baby/deciduous) teeth: 20 teeth, eruption begins at 6 months (lower central incisors first), complete by 30 months. Order: central incisors (6-10 months), lateral incisors (8-12 months), first molars (12-16 months), canines (16-20 months), second molars (20-30 months). Permanent teeth: 32 teeth, begins at 6 years (first molar, lower central incisor), third molars (wisdom teeth ages 17-25). Timing variability is normal. Delayed eruption: caused by systemic conditions (hypothyroidism, hypoparathyroidism, rickets, cleidocranial dysplasia, Down syndrome) or local factors (over-retained primary tooth, supernumerary tooth, cyst, scar tissue from trauma). Eruption cyst: bluish soft tissue swelling over an erupting tooth in children; resolves spontaneously without treatment. Natal teeth: teeth present at birth (most common lower central incisors, may be mobile — extraction if risk of aspiration or injury during breastfeeding).

Tooth Loss and Dental Implants

– The absence of one or more teeth, most commonly resulting from advanced periodontal disease, severe tooth decay, or trauma, and can impair chewing, speech, and facial aesthetics. Dental implants are titanium posts surgically placed into the jawbone that serve as artificial tooth roots, providing a stable foundation for crowns, bridges, or dentures. Implant-supported restorations offer superior function and longevity compared to traditional removable prosthetics, though they require adequate bone density, good oral hygiene, and a healing period of several months.

Wisdom Tooth (Third Molar)

– The third molar, the last tooth in the dental arch, typically erupting between ages 17 and 25 (hence wisdom tooth). Humans have four third molars (maxillary right and left, mandibular right and left). Wisdom teeth are the most commonly impacted (unerupted or partially erupted) teeth due to insufficient space in the jaw, with an impaction rate of 25-75%. Impaction types: mesioangular (most common, angled forward), distoangular (angled backward), vertical (upright but blocked), horizontal (lying on side, highest risk of mandibular nerve injury during extraction), and buccal/lingual. Complications: pericoronitis (inflammation and infection of the soft tissue around a partially erupted wisdom tooth, presenting with pain, trismus, swelling, halitosis, and purulent discharge), caries (distal surface of second molar), periodontal disease, cyst formation (dentigerous cyst around the impacted crown), root resorption of adjacent teeth, and rarely tumor (ameloblastoma, odontogenic keratocyst). Management: prophylactic extraction of asymptomatic, disease-free third molars is controversial; NICE guidelines recommend against prophylactic extraction of healthy, non-pathological wisdom teeth. Indications for extraction: recurrent pericoronitis (2+ episodes), caries or periodontal disease not amenable to restoration, pathology (cyst, tumor), external root resorption of the second molar, and for orthodontic reasons (lack of space, relapse prevention). Surgical extraction: performed under local anesthesia, intravenous sedation, or general anesthesia. Assessment: panoramic radiograph (orthopantomogram, OPG) to evaluate root morphology, number of roots (1-4), root curvature, proximity to the inferior alveolar nerve canal (mandibular canal). Complications of extraction: dry socket (alveolar osteitis, 5-10% of mandibular extractions), pain, swelling, trismus, bleeding, infection, and nerve injury (inferior alveolar nerve — lip paraesthesia/anesthesia, incidence 0.5-5%; lingual nerve — tongue paraesthesia, incidence 0.1-2%). Risk of nerve injury is increased with horizontal and distoangular impactions, deep impaction, and when the nerve canal is superimposed on the root tips on OPG.

Chapter 6

Dermatology

Acne Rosacea Subtypes

– Rosacea subtypes include erythematotelangiectatic (persistent facial erythema and flushing), papulopustular (red bumps and pus-filled pimples), phymatous (skin thickening, rhinophyma), and ocular (eye irritation, blepharitis). Demodex mite overgrowth contributes to pathogenesis. Triggers include sun, heat, spicy foods, alcohol, and stress. Treatment includes metronidazole cream, ivermectin cream, oral doxycycline, and laser therapy for telangiectasias.

Acne Vulgaris

– A chronic inflammatory disease of the pilosebaceous unit (hair follicle and associated sebaceous gland), affecting approximately 85 percent of adolescents and young adults between ages 12 and 24, with a prevalence of 12 to 22 percent in young adult women and 3 percent in men, making it the most common skin condition worldwide. Pathogenesis is multifactorial: excess sebum production (driven by androgens, particularly dihydrotestosterone [DHT] converted by 5-alpha-reductase from ttestosterone by 5-alpha-reductase type 1 in the sebaceous gland), follicular hyperkeratinisation (abnormal desquamation of follicular epithelial cells leading to microcomedone formation), Cutibacterium acnes colonisation and proliferation within the follicle, and inflammation mediated by both innate and adaptive immune responses targeting C. acnes antigens. Comedones (open blackheads and closed whiteheads) are the primary non-inflammatory lesions, while papules, pustules, and nodules represent inflammatory lesions.

Actinic Keratosis

– A premalignant lesion caused by chronic UV exposure, appearing as rough, scaly, erythematous papules on sun-exposed skin. AK is the precursor to squamous cell carcinoma, with approximately 10 percent progressing to invasive cancer. Risk factors include fair skin, outdoor occupation, immunosuppression, and human papillomavirus coinfection. Field cancerization describes widespread actinic damage. Treatment includes cryotherapy, topical 5-fluorouracil, imiquimod, diclofendiclofenac gel, and photodynamic therapy. Regular surveillance for progression to squamous cell carcinoma is necessary.

Actinic Keratosis Management

– Actinic keratosis management requires field-directed and lesion-directed therapy. Lesion-directed: cryotherapy (liquid nitrogen, 5 to 10 second freeze-thaw cycle), curettage, electrodessication. Field-directed: 5-fluorouracil cream (1 or 5 percent daily for 2 to 4 weeks), imiquimod cream (3 or 5 percent), diclofenac gel (twice daily for 60 to 90 days), photodynamic therapy (ALA or MAL followed by blue or red light), and topical tirbanibulin. Regular skin cancer screening is essential for patientpatients with regular follow-up and lesion mapping.

Alopecia

– Hair loss from any cause. Androgenetic alopecia (male/female pattern hair loss): most common, related to DHT sensitivity. Alopecia areata: autoimmune, patchy hair loss with potential progression to totalis (scalp) or universalis (entire body). Telogen effluvium: diffuse shedding 2-3 months after a stressor (illness, childbirth, weight loss, medication). Traction alopecia: from tight hairstyles. Scarring alopecias: lichen planopilaris, discoid lupus, frontal fibrosing alopecia. Treatment: minoxidil, finasteride, corticosteroids, JAK inhibitors (for alopecia areata), and hair transplantation.

Alopecia Areata

– A common, autoimmune, non-scarring hair loss disorder affecting approximately 1 to 2 percent of the population worldwide, with a lifetime risk of 2 percent, occurring equally across sexes and ethnicities, with peak onset between 15 and 29 years. Pathogenesis involves loss of immune privilege of the hair follicle: in normal physiology, the hair follicle bulb maintains an immune-privileged microenvironment (low MHC class I expression, production of TGF-beta, alpha-MSH, and and IL-15), leading to collapse of the immune-privileged state, with CD8+ cytotoxic T cells attacking the hair follicle bulb (the anagen-stage hair follicle), causing premature catagen entry and cessation of hair growth. The condition presents as well-circumscribed, round to oval, smooth patches of non-scarring hair loss on the scalp or any body hair, and severity ranges from a single patch (patchy AA) to complete scalp hair loss (alopecia totalis) or total body hair loss (alopecia universalis).

Androgenetic Alopecia

– The most common cause of hair loss, affecting 50 percent of men and 25 percent of women by age 50. It is caused by miniaturization of hair follicles driven by dihydrotestosterone (DHT). Male pattern: receding hairline and vertex thinning. Female pattern: diffuse crown thinning with preserved frontal hairline. Treatment: minoxidil 5 percent topical (both sexes), finasteride 1 mg daily (men only), dutasteride, spironolactone (women), low-level laser therapy, and hair transhair transplantation. Finasteride and dutasteride are not recommended in women of childbearing potential due to teratogenicity.

Atopic Dermatitis

– A chronic, relapsing inflammatory skin disease characterized by intense pruritus and xerosis, with a prevalence of 10 to 20 percent in children and 3 to 7 percent in adults worldwide, with peak onset in infancy (60 percent develop in the first year of life, 85 percent by age 5) and an increasing prevalence in industrialized nations, suggesting an environmental contribution to disease expression. Pathogenesis involves a complex interplay of epidermal barrier dysfunctdysfunction (filaggrin loss-of-function mutations, the strongest known genetic risk factor, present in 20 to 40 percent of patients), immune dysregulation (TH2/TH22 polarisation with IL-4, IL-13, IL-31, and thymic stromal lymphopoietin overexpression, leading to IgE production, eosinophilia, and pruritus), and altered skin microbiome (Staphylococcus aureus colonisation, reduced microbial diversity). The disease follows a relapsing-remitting course with flares triggered by irritants, allergens, stress, infection, and climatic factors, and follows the atopic march with subsequent development of food allergy, asthma, and allergic rhinitis.

Balanitis

– Inflammation of the glans penis, often involving the prepuce (balanoposthitis). Causes include poor hygiene, irritants (soaps, detergents), infections (Candida albicans most common; also Streptococcus, Gardnerella, Trichomonas), and dermatologic conditions (lichen planus, psoriasis, cicatricial lichen sclerosuslichen sclerosus et atrophicus). Presents with erythema, swelling, discharge, pruritus, and discomfort. Treatment depends on etiology: antifungals for Candida, antibiotics for bacterial infection, topical corticosteroids for inflammatory causes.

Baldness

– The common term for hair loss or alopecia, most frequently referring to androgenetic alopecia (male or female pattern hair loss). Affects up to 50% of men by age 50 and 25% of women by age 60. Driven by dihydrotestosterone (DHT)-mediated miniaturization of hair follicles in susceptible individuals. Other causes include alopecia areata, telogen effluvium, traction alopecia, and scarring alopecias. Treatments: topical minoxidil, oral finasteride (men), low-level laser therapy, platelet-rich plasma injections, and hair transplantation.

Barrier Creams

– Topical preparations applied to the skin to create a physical protective layer against irritants, moisture, and pathogens. Common formulations include zinc oxide, petrolatum, dimethicone, and titanium dioxide. Indicated for preventing irritant contact dermatitis (occupational exposure to water, detergents, chemicals), managing incontinence-associated dermatitis, protecting periwound skin from maceration, and preventing moisture-associated skin damage. Barrier creams complement but do not replace regular cleansing and moisturizing.

Basal Cell Carcinoma

– The most common malignant skin tumor worldwide, accounting for approximately 80 percent of non-melanoma skin cancers, with an incidence of over 4 million cases per year in the United States and a lifetime risk of approximately 30 percent in fair-skinned populations. BCC arises from the pluripotent cells of the basal layer of the epidermis and hair follicle epithelium, driven by ultraviolet radiation-induced mutations (pyrimidine dimers, particularly in tumor suppressuppressor genes TP53 and PTCH1), with additional involvement of the Hedgehog signalling pathway (loss of PTCH1 function or, less commonly, activating SMO mutations, leading to uncontrolled proliferation of basal cells). BCC presents as pearly, translucent papules or nodules with telangiectatic vessels, with several clinical subtypes: nodular (most common, pearly nodule with rolled border and central ulceration, so-called rodent ulcer, on the face), superficial (scaly, erythematous patch or thin plaque with thread-like border, most common on the trunk), morphoeic (sclerosing, scar-like plaque with ill-defined borders, high-risk for subclinical extension and recurrence), and pigmented (dark brown or black nodule, more common in darker skin types).

Bedbug Bites

– Cutaneous reactions to bites of Cimex lectularius (bedbugs), nocturnal blood-feeding insects. Bites are painless but cause erythematous, pruritic papules, often in a linear arrangement (breakfast, lunch, dinner sign) or clusters on exposed skin (face, neck, arms, hands). Delayed hypersensitivity develops over days. Some have bullous or urticarial reactions. Diagnosis: clinical; finding bedbugs or blood spots on sheets. Treatment: symptomatic (topical corticosteroids, oral antihistamines). Secondary infection may require antibiotics. Management: professional pest control, wash bedding in hot water, encase mattresses.

Boil (Furuncle)

– A painful, deep-seated infection of a hair follicle caused most commonly by Staphylococcus aureus, extending into the surrounding subcutaneous tissue. Presents as a tender, erythematous, fluctuant nodule that eventually discharges pus. When multiple adjacent follicles coalesce, it forms a carbuncle, often with systemic symptoms. Common sites: neck, face, armpits, and buttocks. Treatment: warm compresses, incision and drainage for fluctuant lesions, and antibiotics for surrounding cellulitis, large lesions, or immunocompromised patients. Recurrent furunculosis may require decolonization (mupirocin nasal ointment, chlorhexidine washes).

Bullous Pemphigoid

– An autoimmune subepidermal blistering disease affecting elderly patients. It presents with tense blisters on erythematous or normal-appearing skin, often preceded by pruritic urticarial plaques. Autoantibodies target BP180 and BP230 (hemidesmosome proteins). Diagnosis is by direct immunofluorescence showing linear IgG and C3 at the basement membrane. Treatment includes high-potency topical or systemic corticosteroids and steroid-sparing agents (doxycycline, azathioprine, mymycophenolate mofetil). Rituximab is used for refractory disease. Prognosis is generally good but disease can be chronic with relapses.

Candidal Intertrigo

– A Candida infection of skin folds (axillae, inframammary, inguinal, intergluteal, pannus) characterized by erythematous, macerated plaques with satellite pustules at the periphery. Most common in obese, diabetic, and immunocompromised patients, and with prolonged antibiotic use. Presents with itching, burning, and soreness. Diagnosis: clinical; KOH prep shows pseudohyphae and budding yeasts. Treatment: topical antifungals (clotrimazole, miconazole, nystatin); drying powders; keep skin folds dry. Severe or refractory cases: oral fluconazole. Prevention: weight loss, moisture management, glycemic control.

Chemical Peels

– Chemical peels use acids to exfoliate skin layers. Superficial peels (glycolic acid 20 to 70 percent, salicylic acid 20 to 30 percent) affect the epidermis. Medium peels (trichloroacetic acid 35 to 50 percent) reach the papillary dermis. Deep peels (phenol 88 percent) reach the reticular dermis. Indications include acne, photoaging, hyperpigmentation, and fine wrinkles. Recovery time increases with peel depth. Post-peel care includes sun protection and emollients. Complications include infectioninfection, post-inflammatory hyperpigmentation, and scarring. Proper patient selection and pre-treatment preparation minimise adverse effects.

Contact Allergen Testing

– Patch testing identifies allergens causing allergic contact dermatitis. Standard panel includes 35 common allergens (nickel, fragrance mix, balsam of Peru, formaldehyde, rubber accelerators, topical antibiotics, preservatives). Patches are applied to the back for 48 hours and read at 48, 72, and 96 hours. Positive reactions indicate allergens to avoid. Contact allergen databases and patient education are essential for management. False positives and negatives can occur.

Contact Dermatitis

– An inflammatory skin reaction caused by direct contact with an external agent,divided into two main two main types: allergic contact dermatitis (ACD, type IV delayed hypersensitivity reaction, mediated by sensitized T lymphocytes) and irritant contact dermatitis (ICD, non-immunologic, dose-dependent direct chemical or physical damage to the epidermis, accounting for 80 percent of cases). Allergic contact dermatitis requires prior sensitization to the allergen (which can occcur over hours to days, while irritant contact dermatitis occurs immediately after exposure to the inciting agent). Clinical presentation varies by duration and severity, with acute ACD presenting with erythema, vesiculation, oozing, and crusting, and chronic ACD presenting with lichenification, fissuring, scaling, and hyperpigmentation. The distribution of the rash provides important clues to the causative agent. Diagnosis is primarily clinical, supported by the history of exposure and confirmed by patch testing, which is the gold standard for identifying the specific allergen.

Corns and Calluses

– Corns (heloma) are focal areas of hyperkeratosis caused by pressure and friction, typically on toes. Hard corns (heloma durum) occur on dorsal toes; soft corns (heloma molle) occur between toes. Calluses (tyloma) are diffuse hyperkeratosis on weight-bearing surfaces. Treatment includes padding, shoe modification, keratolytic agents (salicylic acid), and debridement. Diabetics should avoid self-treatment due to ulcer risk.

Cosmetic Dermatology

– Cosmetic dermatology encompasses aesthetic procedures. Botulinum toxin injections (Botox, Dysport, Xeomin) relax facial muscles reducing wrinkles. Dermal fillers (hyaluronic acid, calcium hydroxylapatite, poly-L-lactic acid, polymethylmethacrylate) restore volume. Chemical peels (glycolic, salicylic, TCA) improve skin texture and pigmentation. Laser therapy (ablative CO2, Er:YAG, non-ablative fractional) for wrinkles, scars, and pigmentation. Intense pulsed light (IPL) for photorejuvenation.

Cutaneous Drug Eruptions

– a spectrum of reactions. Exanthematous (maculopapular) is the most common, appearing 4 to 14 days after drug initiation. Urticarial appears within hours. Photoallergic occurs at sun-exposed sites. Lichenoid resembles lichen planus. Pigmented purpuric dermatosis presents as cayenne pepper spots. Erythema nodosum presents as tender nodules on shins. Each pattern suggests different drug classes and mechanisms.

Cutaneous Lymphoma

– Cutaneous lymphomas are heterogeneous group of T-cell and B-cell malignancies primarily involving the skin. Primary cutaneous B-cell lymphomas include marginal zone lymphoma, follicle center lymphoma, and diffuse large B-cell lymphoma leg type. Primary cutaneous T-cell lymphomas include mycosis fungoides and Sézary syndrome. Diagnosis requires multiple biopsies and immunohistochemical staining. Treatment is stage-dependent, ranging from topical therapy to systemic chemotherapy and radiation.

Cutaneous T-Cell Lymphoma

– A malignancy of skin-homing T lymphocytes. Mycosis fungoides is the most common subtype, presenting as patches, plaques, and tumors in a clothing distribution. Sézary syndrome is the leukemic variant with erythroderma, lymphadenopathy, and circulating malignant cells. Diagnosis requires serial skin biopsies and T-cell receptor gene rearrangement studies. Treatment ranges from topical steroids and phototherapy to systemic therapies and extracorporeal photopherephotopheresis. Prognosis is variable; early-stage mycosis fungoides has a favourable prognosis, while tumour-stage and Sezary syndrome have poorer outcomes.

Dandruff and Scalp Conditions

– Dandruff is a common scalp condition presenting as flaking, itching, and mild inflammation caused by overgrowth of the yeast Malassezia, dry skin, or seborrheic dermatitis. Other scalp conditions include psoriasis, contact dermatitis, ringworm tinea capitis, and head lice. Treatment typically involves medicated anti-dandruff shampoos containing zinc pyrithione, coal tar, salicylic acid, ketoconazole, or selenium sulfide.

Dermal Structure

– The dermis has two layers: papillary dermis (superficial, thin, containing capillaries, lymphatics, and Meissner corpuscles) and reticular dermis (deep, thick, containing dense irregular connective tissue, collagen, elastin, hair follicles, sweat glands, and sebaceous glands). The dermo-epidermal junction contains type IV collagen and laminin, anchoring the epidermis to the dermis. Dermis provides tensile strength, elasticity, and nourishment to the avascular epidermis.

Dermatitis

– Inflammation of the skin, often used interchangeably with eczema. Atopic dermatitis: chronic pruritic inflammation in flexural areas, associated with asthma and allergies, driven by skin barrier dysfunction and type 2 inflammation. Contact dermatitis: allergic (type IV hypersensitivity) or irritant (direct toxicity). Seborrheic dermatitis: erythematous patches with greasy yellow scale on scalp, face, and chest, associated with Malassezia yeast. Stasis dermatitis: due to chronic venous insufficiency, lower legs. Treatment: emollients, topical corticosteroids, calcineurin inhibitors, and avoidance of triggers.

Dermatofibroma

– A common benign dermal nodule, typically on the lower extremities of young adults. It presents as a firm, hyperpigmented papule that dimples when pinched (Fitzpatrick sign). It is thought to result from reactive fibrosis following minor trauma or insect bites. Dermatofibromas are asymptomatic and require no treatment. Excision is performed for diagnostic uncertainty or cosmetic reasons. They may recur after excision.

Dermatofibrosarcoma

– Dermatofibrosarcoma protuberans is a locally aggressive fibrohistiocytic tumor of the skin. It typically presents as a slowly enlarging, indurated plaque on the trunk, which may develop nodules over time. It has a high local recurrence rate (up to 50 percent with simple excision) but low metastatic potential. Wide local excision with 2 to 3 cm margins or Mohs surgery is treatment. Imatinib is effective for unresectable or metastatic disease.

Dermatofibrosarcoma Protuberans

– A rare locally aggressive cutaneous sarcoma arising from dermal fibroblasts. It presents as a slow-growing, firm, reddish-purple plaque or nodule, typically on the trunk. It has a high local recurrence rate but low metastatic potential. Treatment is wide local excision with Mohs micrographic surgery or imatinib for unresectable disease. It is characterized by the COL1A1-PDGFB fusion gene.

Dermatopathology

– The microscopic examination of skin specimens. Histopathologic features include acanthosis (epidermal thickening), spongiosis (intercellular edema), parakeratosis (retained nuclei in stratum corneum), hyperkeratosis (thickened stratum corneum), dyskeratosis (individual cell keratinization), and Civatte bodies (apoptotic keratinocytes). Special stains include PAS (fungi), Fontana-Masson (melanin), trichrome (collagen), and Congo red (amyloid). Direct immunofluorescence diagnosdiagnoses autoimmune blistering diseases (pemphigus, bullous pemphigoid, lupus erythematosus) by detecting in vivo bound immunoglobulins and complement at specific skin locations.

Digital Mucous Cyst

– A benign ganglion cyst arising from the distal interphalangeal joint, presenting as a translucent, dome-shaped nodule on the dorsal aspect of the finger near the nail fold. It is associated with osteoarthritis of the DIP joint. Treatment includes observation, intralesional corticosteroid injection, cryotherapy, and surgical excision with joint capsule removal to prevent recurrence.

DRESS Syndrome

– Drug Reaction with Eosinophilia and Systemic Symptoms, a severe, potentially life-threatening adverse drug reaction. Typically occurs 2-8 weeks after drug initiation (latency longer than SJS/TEN). Causative drugs: anticonvulsants (phenytoin, carbamazepine, lamotrigine), allopurinol, sulfonamides, dapsone, minocycline. Presents with fever, facial edema, generalized rash (morbilliform, exfoliative dermatitis), lymphadenopathy, eosinophilia, and atypical lymphocytosis. Internal organ involvement: hepatitis (most common), nephritis, pneumonitis, myocarditis, thyroiditis. Diagnosis: RegiSCAR criteria. Treatment: immediate drug withdrawal, systemic corticosteroids, supportive care. Mortality 5-10%.

Drug Hypersensitivity Syndrome

– A severe delayed drug reaction occurring 2 to 8 weeks after drug initiation. It presents with fever, rash (morbilliform, facial edema, exfoliative dermatitis), lymphadenopathy, eosinophilia, and internal organ involvement (hepatitis most common, also nephritis, pneumonitis, myocarditis). Culprit drugs include anticonvulsants (carbamazepine, phenytoin, lamotrigine), allopurinol, sulfonamides, and minominocycline, typically associated with a latency period of 2 to 8 weeks. Diagnosis is based on the RegiSCAR criteria. Treatment consists of immediate withdrawal of the culprit drug, systemic corticosteroids (prednisolone 0.5 to 2 mg/kg/day) with slow tapering over weeks to months, and supportive care including wound care, hydration, and treatment of organ-specific complications. The mortality rate is approximately 10 percent.

Eczema

– A group of inflammatory skin conditions characterized by pruritic, erythematous, scaly patches. Atopic dermatitis is the most common form. Nummular eczema presents as coin-shaped lesions. Dyshidrotic eczema affects palms and soles with vesicles. Eczema herpeticum is a disseminated HSV infection in eczema patients, a medical emergency. Stasis dermatitis occurs in lower extremities with venous insufficiency. Treatment includes emollients, topical corticosteroids, calcineurininhibitors (tacrolimus, pimecrolimus), and systemic immunosuppressants for severe cases.

Eczema (Atopic Dermatitis)

– A chronic inflammatory skin condition characterized by dry, itchy, and inflamed patches of skin. It is driven by a combination of genetic predisposition, immune system dysfunction, and environmental triggers. Management involves moisturizers, topical corticosteroids, avoidance of irritants, and sometimes systemic medications for moderate to severe cases.

Epidermal Inclusion Cyst

– A common benign cyst lined by stratified squamous epithelium with keratin debris. It presents as a firm, mobile, subcutaneous nodule, typically on the face, neck, and back. A central punctum is characteristic. It may become inflamed or infected. Treatment includes complete excision (including the cyst wall) to prevent recurrence. Incision and drainage alone leads to recurrence.

Epidermal Layers

– The epidermis has five layers from deep to superficial: stratum basale (germinative layer with keratinocyte stem cells and melanocytes), stratum spinosum (prickle cell layer with desmosomes), stratum granulosum (keratohyalin granules), stratum lucidum (transparent layer, only in thick skin), and stratum corneum (dead, flattened keratinocytes providing barrier function). The epidermis renews every 28 to 40 days. Keratinocytes, melanocytes, Langerhans cells, and Merkel cells are the four main epidepidermal cell types in the epidermis. The basement membrane zone anchors the epidermis to the dermis and contains type IV collagen, laminin, and integrins.

Epidermolysis Bullosa

– A group of genetic disorders causing skin fragility and blistering with minimal trauma. Simplex (intraepidermal blistering), junctional (blistering at lamina lucida), dystrophic (sublamina densa blistering), and Kindler syndrome (mixed). Dystrophic EB can cause scarring, contractures, and squamous cell carcinoma. Management includes wound care, pain control, nutritional support, and genetic counseling. New treatments include gene therapy and protein replacement.

Erysipelas

– A superficial form of cellulitis involving the upper dermis and superficial lymphatics, most commonly caused by Streptococcus pyogenes. It presents as a bright red, well-demarcated, raised plaque with sharp borders, typically on the face or lower extremities. Fever and chills precede the rash. Complications include abscess, bacteremia, and recurrent episodes. Treatment is penicillin or amoxicillin. Recurrent erysipelas can cause lymphedema (elephantiasis nostras).

Erythema

– Redness of the skin or mucous membranes caused by increased blood flow to dilated superficial capillaries, often a sign of inflammation, infection, or other underlying conditions. Erythema blanch on pressure (distinguishes it from purpura, which does not blanch). Causes: localized inflammation (cellulitis, dermatitis, insect bite, sunburn), systemic conditions (fever, polycythemia, carcinoid syndrome, rosacea), drug reactions, and autoimmune disease (lupus, dermatomyositis). Specific patterns include erythema multiforme (target lesions, HSV-related), erythema nodosum (painful red nodules on shins, panniculitis), erythema migrans (Lyme disease–expanding annular rash with central clearing), and erythema ab igne (reticulated hyperpigmentation from chronic heat exposure).

Erythema Multiforme

– An acute, self-limited immune-mediated reaction, most commonly triggered by herpes simplex virus (90 percent of cases) and Mycoplasma pneumoniae. It presents as target lesions (iris lesions with three zones: central duskiness, middle pale edematous ring, outer erythematous halo) on extremities. Minor EM has few mucosal lesions; major EM has extensive mucosal involvement. Treatment is supportive, with acyclovir prophylaxis for HSV-triggered recurrent EM.

Erythema Nodosum

– The most common form of panniculitis, characterized by painful, erythematous, tender nodules and plaques on the bilateral shins, representing an inflammatory reaction of the subcutaneous fat. The condition is a hypersensitivity reaction to a variety of inciting antigens, most commonly infections (Streptococcal pharyngitis, the most frequent cause in children; tuberculosis; coccidioidomycosis; histoplasmosis; blastomycosis; Yersinia; Salmonella; Mycoplasma; and viral infections including Epstein-Epstein-Barr virus and hepatitis B and C), medications (oral contraceptives, sulfonamides, penicillins, and NSAIDs), inflammatory bowel disease (Crohn disease and ulcerative colitis), sarcoidosis, Behcet disease, pregnancy, and malignancies (lymphoma and leukaemia). The onset is acute, with the appearance of exquisitely tender, warm, erythematous nodules, approximately 1 to 10 cm in diameter, typically on the pretibial surfaces. Individual lesions undergo an evolution of color changes resembling a bruise, resolving over 2 to 6 weeks without ulceration or scarring. Recurrence is common, particularly with streptococcal infections.

Erythrasma

– A superficial bacterial infection of the skin caused by Corynebacterium minutissimum, a gram-positive, aerobic bacillus that proliferates in moist, occluded environments. The infection is most common in warm, humid climates and affects the intertriginous areas, particularly the groin (most common site), axillae, inframammary folds, perineum, and interdigital web spaces of the feet (toe web spaces). Risk factors include obesity, diabetes mellitus, hyperhidrosis, advanced age, poor hygiene, and imimmunosuppression (HIV, organ transplant, corticosteroid use). The infection presents as well-demarcated, reddish-brown to coral-red patches or plaques with a fine scale in the intertriginous areas, often asymptomatic with mild pruritus. Diagnosis is clinical, confirmed by Wood lamp examination demonstrating characteristic coral-red fluorescence. Treatment includes topical antibacterial agents (erythromycin gel, clindamycin lotion) or topical antifungals (clotrimazole, miconazole) for mild disease, and oral erythromycin, clarithromycin, or tetracycline for extensive or refractory cases.

Fixed Drug Eruption

– A recurrent drug reaction appearing at the same site with each exposure to the offending drug. It presents as well-demarcated, round to oval, erythematous to violaceous plaques that may blister. Common culprits include sulfonamides, tetracyclines, NSAIDs, and barbiturates. After resolution, lesions leave post-inflammatory hyperpigmentation. Genitalia, lips, and hands are common sites. Treatment is identification and avoidance of the causative drug.

Folliculitis

– Inflammation of hair follicles, presenting as pustules centered on follicles. Bacterial folliculitis (Staphylococcus aureus) is most common. Hot tub folliculitis (Pseudomonas) occurs after hot tub exposure. Gram-negative folliculitis complicates long-term antibiotic use. Pseudofolliculitis barbae (razor bumps) results from ingrown hairs. Treatment includes topical antibiotics (mupirocin), benzoyl peroxide, and avoiding shaving. Recurrent staphylococcal folliculitis may require dede-colonisation with mupirocin nasal ointment and chlorhexidine washes.

Fungal Infection (Tinea/Dermatophytosis)

– Superficial fungal infections of skin, hair, or nails caused by dermatophytes (Trichophyton, Microsporum, Epidermophyton). Tinea corporis (ringworm): annular scaling plaque with central clearing. Tinea pedis (athlete's foot): interdigital scaling, maceration. Tinea cruris (jock itch): groin rash sparing scrotum. Tinea capitis: scaly patches with broken hairs. Onychomycosis: thickened, discolored, dystrophic nails. Diagnosis: KOH prep, culture. Treatment: topical antifungals (terbinafine, azoles); systemic (terbinafine, itraconazole, griseofulvin) for tinea capitis and onychomycosis.

Fungal Skin Infections

– Skin infections caused by dermatophytes or yeasts, including ringworm (tinea), athlete's foot, and candidiasis. These infections thrive in warm, moist environments and present with red, itchy, scaly, or ring-shaped lesions. Treatment typically involves topical or oral antifungal medications and measures to keep the skin clean and dry.

Furuncle and Carbuncle

– Furuncle (boil) is a deep infection of the hair follicle caused by Staphylococcus aureus, presenting as a painful, erythematous nodule that may drain pus. Carbuncle is a cluster of interconnected furuncles causing a larger, more deeply invasive infection. Treatment includes warm compresses, incision and drainage, and antibiotics (dicloxacillin, cephalexin, or TMP-SMX for MRSA). Recurrent furunculosis suggests carrier state or immunodeficiency.

Genodermatoses

– Genetic skin disorders. Inherited ichthyoses (lamellar, X-linked, congenital ichthyosiform erythroderma). Epidermolysis bullosa (simplex, junctional, dystrophic). Neurocutaneous syndromes (neurofibromatosis, tuberous sclerosis, Sturge-Weber). Ectodermal dysplasias (hypohidrotic, hypotrichosis, nail dystrophy). Incontinentia pigmenti (X-linked dominant, lethal in males). Bloom syndrome, Rothmund-Thomson syndrome, and Werner syndrome are DNA repair disorders with photosensitivitphotosensitivity and increased risk of skin cancer. The RASopathies (Costello, Noonan, cardiofaciocutaneous, LEOPARD syndromes) present with characteristic facies, cardiac defects, and papillomas. Multidisciplinary management and genetic counselling are essential.

Geriatric Dermatology

– Geriatric dermatology addresses skin changes and diseases of aging. Senile purpura (easy bruising from fragile vessels). Actinic keratoses (premalignant from sun exposure). Seborrheic keratoses (benign proliferations). Pruritus senilis (generalized itching from xerosis). Skin cancer incidence increases with age. Impaired wound healing and increased infection risk. Polypharmacy increases adverse drug reactions. Careful medication review and skin cancer screening are essential.

Glomus Tumor

– A rare benign neoplasm arising from glomus bodies (thermoregulatory shunts), presenting as a small, blue-red, exquisitely tender subungual or digital nodule. The classic triad is severe pain, point tenderness, and cold sensitivity. Diagnosis is clinical with positive Love pin test and Hildreth sign (tumor disappears with ischemia). Treatment is surgical excision. Recurrence is uncommon if completely removed.

Granuloma Annulare

– A benign inflammatory condition characterized by annular, erythematous to skin-colored plaques with raised borders and central clearing. It most commonly affects the hands, feet, and elbows. Localized form is most common; generalized form is associated with diabetes mellitus. Diagnosis is clinical with biopsy showing palisading granulomas around necrobiotic collagen. Self-resolving within 2 years. Treatment of persistent cases includes topical or intralesional corticosteroisteroids, phototherapy, and oral corticosteroids for severe disseminated disease.

Hair Loss (Alopecia)

– The partial or complete loss of hair from areas where it normally grows, most commonly the scalp. Causes include genetic predisposition (androgenetic alopecia), autoimmune conditions (alopecia areata), hormonal changes, medications, stress, and nutritional deficiencies. Treatment depends on the underlying cause and may include topical minoxidil, oral finasteride, corticosteroid injections, or hair transplantation.

Hair Loss Classification

– Hair loss (alopecia) is classified as scarring (cicatricial) or non-scarring. Non-scarring alopecias are potentially reversible: androgenetic alopecia (male and female pattern), alopecia areata, telogen effluvium, anagen effluvium, and tinea capitis. Scarring alopecias cause permanent hair loss: lichen planopilaris, discoid lupus, frontal fibrosing alopecia, and folliculitis decalvans. Diagnosis requires clinical examination, trichoscopy, and sometimes biopsy.

Herpes Zoster

– A painful, vesicular eruption resulting from reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia or cranial nerve ganglia, where the virus has remained dormant since the primary varicella (chickenpox) infection, with an incidence of approximately 1 to 3 per 1,000 person-years overall and up to 10 per 1,000 person-years in adults over 60. Risk factors include advancing age (over 50 years, 50 percent of cases), immunosupprimmunosuppression (HIV, organ transplantation, haematological malignancies, chemotherapy, and long-term corticosteroid or anti-TNF therapy). The prodromal phase (2 to 3 days) consists of headache, photophobia, and malaise, followed by the eruption phase, where painful, grouped, vesicular lesions on an erythematous base appear in a unilateral, single dermatomal distribution, most commonly the thoracic dermatomes (50 to 60 percent), followed by the trigeminal nerve distribution, particularly the ophthalmic division (V1, 12 to 15 percent, which requires urgent ophthalmology referral due to the risk of herpes zoster ophthalmicus). The acute neuritis and the zoster rash typically resolve over 2 to 4 weeks.

Hidradenitis

– Hidradenitis suppurativa is a chronic inflammatory follicular occlusion disease affecting apocrine gland-bearing areas (axillae, groin, inframammary). It presents with recurrent painful nodules, abscesses, sinus tracts, and scarring. Hurley staging: Stage I (single or multiple abscesses without sinus tracts), Stage II (recurrent abscesses with sinus tracts and bridging scars), Stage III (diffuse or multi-tract involvement). Smoking and obesity are risk factors. Treatment includes lifestyle modiflifestyle modification (weight loss, smoking cessation), topical clindamycin, oral antibiotics (tetracyclines, rifampicin-clindamycin combination), biologic therapy (adalimumab, the only FDA-approved biologic for HS), and surgical deroofing or excision for advanced disease.

Hidradenitis Suppurativa

– A chronic, inflammatory, follicular occlusion disease characterized by recurrent, painful nodules, abscesses, sinus tracts, and scarring in apocrine gland-bearing areas (axillae [most common, 70 to 80 percent], groin and inguinal folds [30 to 40 percent], perineum and perianal region [20 to 35 percent], buttocks [20 to 30 percent], and inframammary folds [10 to 20 percent]), with a prevalence of 1 to 4 percent worldwide (likely underdiagnosed, with average diagnoaverage diagnosis delay of 7-10 years). The pathogenesis begins with follicular occlusion (hyperkeratinisation of the terminal hair follicle infundibulum leading to dilation and rupture of the follicle), which triggers an intense inflammatory response with release of keratin and cell debris into the dermis, activating neutrophils, macrophages, and TH1/TH17 lymphocytes with elevated levels of TNF-alpha, IL-17, and IL-1 beta. Risk factors include smoking (the strongest environmental risk factor, present in 70 to 90 percent of patients), obesity, hormonal influences, and genetic factors (mutations in NCSTN, PSENEN, and PSEN1 encoding components of the gamma-secretase complex).

Hives (Urticaria)

– A common skin condition characterized by transient, well-circumscribed, intensely pruritic, erythematous wheals (hives) of variable size and shape that blanch with pressure and resolve within 24 hours without scarring, resulting from mast cell degranulation and histamine release in the dermis. Urticaria is classified by duration: acute urticaria (less than 6 weeks, most common, accounting for 70 percent of cases, typically caused by IgE-mediated hypersensitivity reactions to foods [nuts, shellfishellfish, eggs, milk) and medications (NSAIDs, ACE inhibitors, antibiotics, opiates) in acute urticaria, with physical triggers (dermatographism, cholinergic, cold, solar, aquagenic, delayed pressure, vibratory) accounting for 20 to 30 percent of cases; and chronic urticaria (lasting 6 weeks or longer, affecting 1 to 2 percent of the population, more common in middle-aged women, with a mean duration of 1 to 5 years, and spontaneous onset in 70 to 80 percent of cases, of which approximately 40 percent have an autoimmune basis with IgG autoantibodies targeting the high-affinity IgE receptor Fc-epsilon-RI or IgE itself). Treatment follows a stepwise approach: first-line, second-generation H1-antihistamines (cetirizine, levocetirizine, fexofenadine, loratadine, desloratadine, bilastine) up to fourfold the standard dose; second-line, add H2-antagonist or leukotriene receptor antagonist (montelukast); third-line, add omalizumab (monoclonal anti-IgE antibody, FDA-approved for chronic spontaneous urticaria) or cyclosporine for severe refractory cases.

Hyperhidrosis

– Excessive sweating beyond thermoregulatory needs. Primary focal hyperhidrosis affects palms, soles, axillae, and face without underlying cause. Secondary generalized hyperhidrosis is caused by medications, endocrine disorders, infections, or malignancies. Treatment includes topical aluminum chloride (first-line), oral anticholinergics (glycopyrrolate, oxybutynin), iontophoresis for palms and soles, botulinum toxin injection, and endoscopic thoracic sympathectomy for severe refrasevere refractory cases. Secondary hyperhidrosis requires treatment of the underlying cause.

Hypertrophic Scar

– Raised scar tissue confined to the original wound margins, caused by excessive collagen production during healing. Unlike keloids, hypertrophic scars regress over time and do not extend beyond wound boundaries. They are common after burns, surgery, and trauma. Treatment includes pressure garments, silicone sheeting, intralesional corticosteroids, and laser therapy. Early intervention prevents progression.

Hypohidrosis

– Reduced or absent sweating, leading to heat intolerance and risk of hyperthermia. Causes include autonomic neuropathy (diabetes, Parkinson disease), skin diseases (scleroderma, burns), medications (anticholinergics), and genetic disorders (hypohidrotic ectodermal dysplasia). Diagnosis is confirmed by sweat test (starch-iodine test, thermoregulatory sweat test). Treatment includes cooling measures, hydration, and avoiding heat exposure.

Ichthyosis

– A group of disorders characterized by dry, scaly skin resembling fish scales. Ichthyosis vulgaris is the most common, autosomal dominant, associated with filaggrin mutation and atopy. X-linked ichthyosis is caused by steroid sulfatase deficiency. Lamellar ichthyosis is a severe congenital form. Treatment includes keratolytics (urea, lactic acid, ammonium lactate), emollients, and retinoids for severe cases.

Immunomodulators in Dermatology

– Topical immunomodulators treat inflammatory skin diseases without corticosteroid side effects. Pimecrolimus 1 percent cream and tacrolimus 0.03 and 0.1 percent ointments are calcineurin inhibitors used for atopic dermatitis, facial eczema, and intertriginous areas. They are steroid-sparing agents. Off-label uses include vitiligo, lichen planus, and psoriasis. JAK inhibitors (ruxolitinib cream for vitiligo and atopic dermatitis) represent a new class of topical immunomodulators.

Immunotherapy in Dermatology

– Cancer immunotherapy has revolutionized skin cancer treatment. Checkpoint inhibitors (pembrolizumab, nivolumab, ipilimumab) block PD-1 or CTLA-4, enhancing T-cell mediated tumor killing. They are first-line for advanced melanoma and metastatic SCC. Cemiplimab is approved for advanced BCC and SCC. Adverse events include dermatitis, colitis, hepatitis, pneumonitis, and endocrinopathies. Skin immune-related adverse events occur in 30 to 40 percent of patients and may be the first manifestation of ssystemic immunotherapy toxicity, particularly when the skin is involved as a target organ of the immune response.

Ingrown Toenail

– Ingrown toenail (onychocryptosis) occurs when the nail edge grows into the surrounding skin, causing pain, redness, and swelling. It is common in adolescents and adults, exacerbated by improper nail trimming, tight shoes, and trauma. Treatment includes partial nail avulsion with matrixectomy (phenol application) for recurrent cases. Conservative management includes cotton wedge under the nail edge and proper nail trimming.

Intertrigo

– An inflammatory rash of opposing skin surfaces (skin folds) caused by friction, moisture, heat, and lack of air circulation. Affects intertriginous areas: axillae, inframammary, inguinal, abdominal folds, and intergluteal cleft. Presents with erythema, maceration, fissuring, and erosions. Often secondarily infected with Candida albicans (satellite pustules), bacteria (erythrasma, Corynebacterium), or dermatophytes. Risk factors: obesity, diabetes, hyperhidrosis, incontinence, and poor hygiene. Diagnosis: clinical; KOH prep and Wood lamp exam for secondary infections. Treatment: keep area dry (drying powders, barrier creams), topical antifungals for Candida, low-potency corticosteroids for inflammation. Prevention: weight loss, moisture-wicking clothing.

Isotretinoin

– A systemic retinoid used for severe, recalcitrant nodular acne. It reduces sebum production, normalizes follicular keratinization, decreases Propionibacterium acnes, and has anti-inflammatory effects. Dose is 0.5 to 1 mg/kg/day for 15 to 20 weeks, targeting cumulative dose of 120 to 150 mg/kg. Side effects include cheilitis (almost universal), dry skin, photosensitivity, elevated triglycerides and liver enzymes, and teratogenicity (absolute contraindication in pregnancy). Two forTwo forms of contraception are required for women of childbearing potential one month before, during, and one month after therapy due to teratogenicity (iPLEDGE risk management programme in the US). Monitoring includes monthly pregnancy tests, lipid profile, and liver function tests. Other side effects include arthralgias, myalgias, headache, pseudotumor cerebri (rare), and decreased night vision. A second course may be considered after a washout period of at least 8 to 16 weeks.

Kaposi Sarcoma

– A vascular tumor caused by human herpesvirus 8 (HHV-8), presenting as violaceous plaques and nodules on skin and mucous membranes. Four types: classic (elderly Mediterranean men), endemic (African), iatrogenic (immunosuppression post-transplant), and AIDS-associated. Cutaneous lesions are the most common presentation in AIDS. Treatment includes antiretroviral therapy, local therapy (radiation, cryotherapy), and chemotherapy for widespread disease.

Keloid

– An overgrowth of scar tissue extending beyond the original wound margins, caused by excessive collagen deposition during wound healing. Keloids are more common in darker skin types (Fitzpatrick IV to VI) and occur at sites of trauma, surgery, piercings, and burns. They are pruritic and painful. Treatment includes intralesional corticosteroid injections, silicone sheeting, cryotherapy, laser therapy, and surgical excision with adjuvant radiation. Recurrence is common.

Keratosis Pilaris

– A common, benign genetic follicular condition characterized by keratotic follicular papules. Small, rough, flesh-colored or red bumps on the extensor surfaces of proximal arms, thighs, and sometimes buttocks and face. Caused by abnormal keratinization and plugging of hair follicles with retained keratin. Associated with ichthyosis vulgaris and atopic dermatitis. Onset in childhood, may improve with age. Diagnosis: clinical. Treatment: emollients with keratolytics (lactic acid, urea, salicylic acid, ammonium lactate), topical retinoids (tretinoin, adapalene), and gentle exfoliation. Laser therapy for cosmetically bothersome erythema.

Leukoplakia

– A clinical term for a white patch or plaque on the oral mucosa that cannot be rubbed off and cannot be characterized clinically as any other definable lesion (WHO). Leukoplakia is a potentially malignant disorder of the oral cavity, with a risk of malignant transformation to squamous cell carcinoma of approximately 1-5% per year. Risk factors: tobacco smoking, alcohol consumption, areca nut/betel quid chewing, and chronic friction (e.g., from sharp teeth or dentures). Clinical subtypes: homogeneous (flat, thin, uniform white, low risk) and non-homogeneous (speckled–red and white, nodular, verrucous, higher risk). Sites: buccal mucosa, alveolar ridge, floor of mouth (highest risk), tongue, palate. The term may also rarely be used for vulvar, penile, or laryngeal white patches. Diagnosis: biopsy (optional for homogeneous lesions, always indicated for non-homogeneous or suspicious lesions) to confirm histology (hyperkeratosis, acanthosis, dysplasia). The presence of epithelial dysplasia (mild, moderate, severe) predicts malignant potential. Management: elimination of risk factors (smoking cessation, alcohol reduction), observation for homogeneous lesions, surgical excision for dysplastic or non-homogeneous lesions, CO2 laser ablation, cryotherapy. Photodynamic therapy and topical retinoids have limited evidence. Long-term follow-up is essential.

Lice Infestations

– Pediculosis is infestation by head lice (Pediculus humanus capitis), body lice (Pediculus humanus corporis), or pubic lice (Pthirus pubis). Head lice cause pruritus of the scalp and posterior neck; nits (eggs) are attached to hair shafts. Body lice live in clothing and cause dermatitis. Pubic lice (crabs) affect the pubic region and cause pruritic maculae ceruleae. Treatment includes permethrin 1 percent shampoo, oral ivermectin, malathion 0.5 percent lotion, and mechanical removal of nits. ClotClothing and bedding should be washed in hot water (60 degrees C) and dried on high heat to prevent reinfestation. Household contacts should be examined and treated if infested.

Lichen

– A descriptive dermatological term referring to a flat-topped, firm, papular skin lesion resembling the lichen plant (a symbiotic organism of algae and fungi growing on rocks and trees). Lichenoid describes lesions that resemble lichen planus or lichenoid drug eruptions. Specific lichenoid conditions include: lichen planus, lichen sclerosus, lichen simplex chronicus (localised neurodermatitis from chronic scratching), lichen striatus, lichen nitidus, lichen myxoedematosus (papular mucinosis), and lichenoid drug eruptions (e.g., ACE inhibitors, antimalarials, gold, quinidine, NSAIDs, thiazides). See individual entries.

Lichen Planus

– A T-cell mediated inflammatory condition affecting skin, mucous membranes, nails, and hair. Classic presentation: the 6 Ps (pruritic, purple, polygonal, planar, papules, plaques), often with Wickham striae (white lacy network on surface). Oral lichen planus affects buccal mucosa and may be erosive. Hypertrophic lichen planus affects shins. Nail lichen planus causes longitudinal ridging and pterygium. Treatment includes topical corticosteroids, systemic corticosteroids for severesevere disease, and acitretin. Oral and genital lichen planus may require topical calcineurin inhibitors (tacrolimus, pimecrolimus) as first-line therapy due to the risk of corticosteroid-induced atrophy of mucosal surfaces.

Lichen Sclerosus

– A chronic inflammatory skin disease most commonly affecting anogenital skin. It presents as ivory-white, atrophic plaques with wrinkled paper-like appearance. In women, it predominantly affects the vulva and perianal region (figure-eight distribution). In men, it affects the glans penis and foreskin (balanitis xerotica obliterans). Lichen sclerosus is associated with increased risk of squamous cell carcinoma. Treatment includes ultra-potent topical corticosteroids (clobetasolclobetasol propionate 0.05 percent ointment) as first-line therapy, applied once or twice daily for 12 to 24 weeks, with gradual tapering to maintenance. Topical calcineurin inhibitors (tacrolimus, pimecrolimus) are used for corticosteroid-sparing effect. Systemic therapy with methotrexate, mycophenolate mofetil, or phototherapy is reserved for refractory, extensive, or extragenital disease. Long-term monitoring is required due to the risk of malignant transformation (squamous cell carcinoma develops in 3 to 5 percent of patients with vulval lichen sclerosus).

Lichen Simplex Chronicus

– A thickened, leathery plaque caused by chronic scratching and rubbing (itch-scratch cycle). It commonly affects the neck, ankles, lower legs, and genital area. It is a form of neurodermatitis associated with stress and anxiety. Treatment includes breaking the itch-scratch cycle with topical corticosteroids, antihistamines, and behavioral modification. Underlying atopic dermatitis or contact dermatitis should be treated.

Lipoma

– The most common benign soft tissue tumor, composed of mature adipocytes. It presents as a soft, mobile, painless subcutaneous nodule, commonly on the trunk, neck, and proximal extremities. Lipomas grow slowly and may be multiple (familial multiple lipomatosis). Diagnosis is clinical; ultrasound or MRI for deep or atypical lesions. Treatment is observation or surgical excision for symptomatic or cosmetically concerning lipomas. Angiolipoma is a painful variant with vascular proliferatiovascular proliferation that is more symptomatic than typical lipomas.

Medical Tattooing

– Medical tattooing (permanent makeup, micropigmentation) covers depigmented skin (vitiligo, vitiligo camouflage), restores areolas after mastectomy, and masks scars. Trichopigmentation simulates hair on bald scalp. Medical tattooing uses specialized pigments and techniques. Contraindications include keloid propensity, active skin disease, and pregnancy. Complications include infection, allergic reaction, pigment migration, and dissatisfaction. Removal requires laser therapy.

Melanoma

– A malignant neoplasm arising from melanocytes (pigment-producing cells derived from the neural crest, located in the basal layer of the epidermis, hair follicles, and mucous membranes), representing the most lethal form of skin cancer, with an incidence of approximately 20 to 30 per 100,000 persons per year in fair-skinned populations and a lifetime risk of approximately 2 to 3 percent. Risk factors include ultraviolet radiation exposure (intermittent intense sun exposure with blisteblistering sunburns during childhood or adolescence (the single most important modifiable risk factor), high density of melanocytic nevi (greater than 50 nevi or the presence of atypical or dysplastic nevi, which confer a 5- to 20-fold increased risk), red or blonde hair, fair skin (Fitzpatrick I and II), positive family history of melanoma, personal history of melanoma (5 to 10 percent risk of developing a second primary melanoma), and immunosuppression. The four main histopathological subtypes are superficial spreading melanoma (the most common, accounting for 70 percent of all melanomas), nodular melanoma (15 percent, presenting as a rapidly growing, elevated nodule without a clinically appreciable radial growth phase, often diagnosed late with a worse prognosis), lentigo maligna melanoma (10 percent, arising on sun-damaged facial skin of elderly patients), and acral lentiginous melanoma (5 percent, the most common melanoma in darker skin types, Fitzpatrick IV to VI, arising on the palms, soles, and subungual sites). The diagnosis is confirmed by excisional biopsy with a 1 to 2 mm clinical margin for histopathological evaluation.

Melanoma Surveillance

– regular skin examination, patient education on self-examination, and dermoscopic monitoring of atypical nevi. High-risk patients (personal or family history, numerous atypical nevi, immunosuppression) require 6-month follow-up. Standard patients require annual examination. Total body photography establishes baseline for change detection. Sentinel lymph node biopsy is recommended for melanoma 1 mm or greater or with ulceration.

Melanoma Treatment

– Stage-dependent. Stage 0 to IA: wide local excision. Stage IB to IIA: wide excision with sentinel lymph node biopsy. Stage IIB to IIC: wide excision, SLNB, and adjuvant therapy. Stage III: wide excision, SLNB or lymph node dissection, and adjuvant immunotherapy (pembrolizumab, nivolumab) or targeted therapy (dabrafenib plus trametinib for BRAF-mutant). Stage IV: systemic therapy (immunotherapy, targeted therapy, chemotherapy). BRAF and c-KIT mutations guide targeted therapy-targeted therapy). Palliative radiation for brain or bone metastases. Regular surveillance with imaging and skin examinations is required.

Merkel Cell Carcinoma

– An aggressive neuroendocrine skin cancer arising from Merkel cells. It presents as a rapidly growing, painless, red-violet nodule on sun-exposed skin of elderly patients. Approximately 80 percent are associated with Merkel cell polyomavirus. Risk factors include immunosuppression, UV exposure, and fair skin. Treatment includes wide local excision, sentinel lymph node biopsy, and radiation therapy. Prognosis is poor with high rates of local recurrence and metastasis.

Milia

– Small, white, keratin-filled cysts appearing on the face, particularly around the eyes. They are common in newborns (neonatal milia) and adults (primary milia). Secondary milia occur after blistering injuries, dermabrasion, or laser therapy. Milia are asymptomatic and may resolve spontaneously. Treatment includes extraction with a sterile needle, topical retinoids, and chemical peels.

Molluscum Contagiosum

– A viral skin infection caused by a poxvirus, presenting as firm, dome-shaped, umbilicated papules with central dimpling, 2 to 5 mm in diameter. It is transmitted by direct skin contact, autoinoculation, and sexual contact in adults. Immunocompetent hosts clear lesions spontaneously within 6 to 12 months. Immunocompromised patients may have hundreds of lesions. Treatment includes cryotherapy, curettage, topical cantharidin, potassium hydroxide, and imiquimod. Lesions resoLesions resolve within weeks to months after treatment. Spontaneous immunity is not lifelong.

Morton Neuroma

– A perineural fibrosis of the plantar digital nerve, most commonly between the third and fourth metatarsal heads. It presents as burning pain, numbness, and a feeling of walking on a pebble. It is not a true neuroma but a mechanical entrapment neuropathy. Treatment includes wide shoes, metatarsal pads, corticosteroid injection, and surgical excision for refractory cases. Alcohol injection sclerotherapy is an alternative.

Mpox (monkeypox)

– A zoonotic viral disease caused by the monkeypox virus, an orthopoxvirus in the same genus as smallpox (variola) and vaccinia. The virus has two clades: clade I (formerly Congo Basin, Central African) with higher virulence and mortality (up to 10 percent), and clade II (formerly West African) with lower mortality (less than 1 percent). The 2022 global outbreak involved clade IIb, which spread predominantly through intimate contact, including sexual contact, among men who have sex with men, with with cases concentrated in men who have sex with men and their close contacts. The incubation period is typically 5 to 21 days. The prodrome consists of fever, headache, myalgia, and lymphadenopathy (the most distinguishing feature from smallpox), followed by the eruption of a deep-seated, well-circumscribed, umbilicated vesiculopustular rash that evolves synchronously through macular, papular, vesicular, pustular, and crusting stages over 2 to 4 weeks, with centrifugal distribution. Oral antiviral therapy with tecovirimat (TPOXX) is used for severe disease, immunocompromised patients, and those at high risk of complications.

Nail Psoriasis

– Nail psoriasis affects 50 percent of patients with psoriasis and 80 percent with nail involvement having joint disease. Features include pitting (most common), onycholysis (nail separation), oil drop sign (salmon patch), subungual hyperkeratosis, and nail plate crumbling. Nail psoriasis is associated with psoriatic arthritis. Treatment is challenging: topical corticosteroids, intralesional corticosteroids, systemic therapy for psoriasis, and apremilast. Nail avulsion is rarely needed.

Neonatal Skin Conditions

– Common and usually benign. Erythema toxicum neonatorum (red papules and pustules on erythematous base, appears day 2 to 3, resolves by 2 weeks). Milia (tiny white cysts on nose and cheeks). Miliaria crystallina (clear vesicles from sweat duct obstruction). Sebaceous hyperplasia (yellow papules on nose and cheeks). Transient neonatal pustular melanosis (vesiculopustular rash with collarette of scale, resolves spontaneously). Acne neonatorum (comedones and papules, matmatures with sebaceous gland development and resolves without scarring). Treatment is reassurance and education for parents. Topical treatments are rarely needed.

Nettle Rash

– See Urticaria (Hives / Nettle Rash).

Neurofibroma

– A benign peripheral nerve sheath tumor. Solitary neurofibromas are common and present as soft, flesh-colored papules or nodules. Multiple neurofibromas are characteristic of neurofibromatosis type 1 (NF1). Cutaneous neurofibromas are diagnostically significant in NF1 (greater than or equal to 2 neurofibromas is one of the diagnostic criteria). Plexiform neurofibromas are pathognomonic for NF1 and carry risk of malignant transformation to malignant peripheral nerve sheath tumor (MMPNST), occurring in 5 to 10 percent of patients with NF1, presenting as rapid growth, new or increasing pain, or a change in consistency of a previously stable plexiform neurofibroma.

Nutritional Dermatoses

– Nutritional deficiencies cause distinctive skin findings. Vitamin C deficiency (scurvy): perifollicular hemorrhage, corkscrew hairs, gingival bleeding. Niacin deficiency (pellagra): photosensitive dermatitis in sun-exposed areas, diarrhea, dementia. Riboflavin deficiency (ariboflavinosis): cheilosis, angular stomatitis, glossitis, seborrheic dermatitis-like eruption. Zinc deficiency (acrodermatitis enteropathica): periorificial and acral dermatitis, alopecia, diarrhea. Vitamin A deficiency: follfollicular hyperkeratosis, night blindness, and xerophthalmia (dryness of the conjunctiva and cornea, Bitot spots, keratomalacia). Vitamin E deficiency: acanthocytosis, peripheral neuropathy. Vitamin K deficiency: ecchymoses, hemorrhage. Iron deficiency: kollonychia (spoon nails), angular stomatitis, glossitis, pallor. Zinc deficiency causes acral dermatitis, periorificial dermatitis, alopecia, diarrhea, and growth retardation in children (acrodermatitis enteropathica). Early recognition and appropriate replacement therapy are essential.

Occupational Dermatoses

– Work-related skin diseases. Occupational contact dermatitis accounts for 90 percent of occupational skin diseases. Irritant contact dermatitis is more common (hands, wet work) than allergic contact dermatitis. Occupations at risk include healthcare workers, hairdressers, construction workers, food handlers, and metalworkers. Common occupational allergens include rubber accelerators, fragrances, preservatives, and metals. Prevention includes hazard identification, protprotective equipment, barrier creams, and education. Management includes identification and avoidance of causative agents, topical corticosteroids, and emollients. Workers with occupational dermatitis may require job modification or retraining.

Onychomycosis

– Fungal nail infection, most commonly caused by dermatophytes (Trichophyton rubrum), affecting toenails more than fingernails. It presents as nail thickening, discoloration (yellow-brown), subungual debris, and nail separation. Diagnosis is confirmed by KOH preparation or fungal culture. Treatment includes oral terbinafine (first-line for dermatophyte), itraconazole, or topical ciclopirox for mild disease. Cure rates are 50 to 80 percent with oral therapy.

Paronychia

– Inflammation of the nail fold. Acute paronychia is bacterial (Staphylococcus aureus, Streptococcus), presenting as painful, erythematous, swollen nail fold with possible abscess. Treatment includes warm compresses, antibiotics, and incision and drainage if abscessed. Chronic paronychia is fungal (Candida) or irritant-induced, lasting greater than 6 weeks, common in wet workers. Treatment includes topical antifungals, corticosteroids, and avoiding moisture.

Patch Testing

– Patch testing identifies allergens causing allergic contact dermatitis. The standard series includes 35 allergens. Additional custom trays include occupational and personal allergens. Patches are applied to the upper back for 48 hours and removed. Reading at 48 hours (initial) and 72 to 96 hours (delayed) identifies positive reactions. Positive reactions are graded (1+ to 4+) based on erythema, edema, and vesiculation. Results guide allergen avoidance and patient education.

Pediatric Dermatology

– Pediatric dermatology encompasses conditions unique to children. Congenital lesions (birthmarks, hemangiomas, port-wine stains, café-au-lait macules). Infantile hemangiomas (strawberry marks) proliferate in first months of life and involute over years. Propranolol is first-line treatment for problematic hemangiomas. Mongolian spots (congenital dermal melanocytosis) are benign and resolve spontaneously. Neonatal acne, erythema toxicum, and milia are self-limited.

Pediatric Dermatology Conditions

– Distinct from adult dermatology. Birthmarks include vascular (hemangiomas, port-wine stains) and pigmented (café-au-lait, congenital nevi, Mongolian spots). Infantile hemangiomas proliferate in first months then involute. Congenital melanocytic nevi carry melanoma risk proportional to size. Juvenile xanthogranuloma is a benign histiocytosis. Neonatal acne resolves spontaneously. Diaper dermatitis is common and treated with barrier creams.

Pediatric Eczema

– Pediatric eczema (atopic dermatitis in children) typically begins in infancy (2 to 6 months) with facial and extensor surface involvement. In older children, flexural involvement (antecubital and popliteal fossae) predominates. Pruritus is the hallmark, causing sleep disturbance and scratching. Associated conditions include asthma, allergic rhinitis, and food allergies (atopic march). Management includes emollients, topical corticosteroids, topical calcineurin inhibitors, and systemic therapy fotherapy for severe cases. Barrier repair and proper bathing technique (lukewarm water, gentle cleanser, immediate emollient application) are essential maintenance strategies.

Pediatric Skin Tumors

– Pediatric skin tumors differ from adult tumors. Infantile hemangioma is the most common benign tumor of infancy. Congenital melanocytic nevus carries risk of melanoma (larger nevi have higher risk). Juvenile xanthogranuloma is a benign histiocytic lesion. Pyogenic granuloma is a common vascular lesion. Dermoid cysts are congenital. Malignant melanoma is rare but increasing in children. Regular monitoring of congenital nevi is recommended.

Pediculosis

– Infestation with lice. Pediculus humanus capitis (head lice): most common in school-age children, spread by direct head-to-head contact; presents with scalp pruritus, nits (eggs) attached to hair shafts. Pediculus humanus corporis (body lice): associated with poor hygiene, found in clothing seams; vector for trench fever, relapsing fever, and typhus. Phthirus pubis (pubic lice/crabs): sexually transmitted, infests pubic area. Treatment: permethrin, pyrethrin, malathion, ivermectin, and wet combing for head lice; hygiene, washing clothing for body lice.

Pemphigus Vulgaris

– An autoimmune intraepidermal blistering disease caused by IgG antibodies against desmoglein 1 and 3. It presents with flaccid blisters that rupture easily, leaving painful erosions of skin and mucous membranes. Nikolsky sign is positive (lateral pressure causes epidermal sloughing). Diagnosis is by biopsy showing acantholysis and direct immunofluorescence showing intercellular IgG deposition. Treatment includes systemic corticosteroids and rituximab (first-line steroid-sparsteroid-sparing agent). Prognosis is poor without treatment; immunosuppression has dramatically improved outcomes, with mortality reduced to less than 10 percent.

Perioral Dermatitis

– A papulopustular eruption around the mouth, sparing the vermilion border. It is more common in young women and associated with topical corticosteroid use. Other triggers include fluoridated toothpaste, cosmetics, and inhaler steroids. Treatment includes discontinuation of topical steroids, topical metronidazole, erythromycin, or oral tetracyclines. It may take weeks to months to resolve.

Pilar Cyst

– A benign cyst arising from the outer root sheath of hair follicles, most commonly on the scalp. It presents as a firm, mobile, subcutaneous nodule without a central punctum. Unlike epidermoid cysts, pilar cysts have smooth walls and contain dense keratin. They may be multiple in familial cases. Treatment is simple excision. Malignant transformation is extremely rare.

Pityriasis Rosea

– An acute, self-limited exanthematous skin condition of unknown cause (possibly viral, HHV-6/7 reactivation). Begins with a herald patch (solitary, 2-10 cm, annular, salmon-colored, scaly plaque on trunk), followed 1-2 weeks later by diffuse eruption of smaller, oval, erythematous papules and plaques with fine central scale following skin cleavage lines (Christmas tree pattern on back). Trunk and proximal extremities involved; face spared. Pruritus mild to moderate. Typically resolves in 6-8 weeks without treatment. Diagnosis: clinical. Treatment: symptomatic (emollients, antihistamines, topical steroids). UVB phototherapy for severe pruritus.

Plantar Wart (Verruca Plantaris)

– A wart (verruca) located on the sole of the foot, caused by human papillomavirus (HPV, typically types 1, 2, 4, 63). Plantar warts are endophytic (grow inward due to pressure) rather than exophytic like common warts, and are often painful with direct pressure (walking, standing). They appear as well-circumscribed, hyperkeratotic papules or plaques on the sole, with a rough surface and loss of normal skin lines over the lesion. Paring reveals punctate black dots (thrombosed capillaries, distinguishing them from corns). Multiple warts may coalesce into a mosaic wart (mosaic verruca). Diagnosis: clinical, confirmed by paring. Plantar warts are frequently confused with corns (heloma) but are distinguished by the presence of thrombosed capillaries (black dots) and disruption of dermatoglyphics. Treatment: spontaneous resolution occurs in 30-50% within 1-2 years. First-line: salicylic acid (40% topical, daily with paring) and cryotherapy (liquid nitrogen, 10-15 second freeze, repeated every 2-4 weeks). Second-line: laser therapy (pulsed dye, CO2), intralesional immunotherapy (Candida or mumps antigen injection), and topical immunotherapy (imiquimod, 5-fluorouracil). Refractory or mosaic warts may require surgical excision or photodynamic therapy. Immunocompromised patients (transplant, HIV) may have more extensive and resistant disease.

Porphyrias

– Metabolic disorders of heme synthesis causing cutaneous and neurological symptoms. Cutaneous porphyrias include porphyria cutanea tarda (blisters on sun-exposed skin, hypertrichosis, sclerodermoid changes), erythropoietic protoporphyria (immediate pain and burning on sun exposure without visible changes), and variegate porphyria. PCT is associated with hepatitis C, HIV, iron overload, and estrogen use. Treatment of PCT includes phlebotomy and hydroxychloroquine. Acute porphyrias (porphyrias (acute intermittent porphyria, variegate porphyria, hereditary coproporphyria) present with acute attacks of abdominal pain, neuropsychiatric symptoms, and autonomic instability. Diagnosis is made by porphyrin analysis in urine, plasma, and faeces. Treatment of acute attacks includes intravenous haem arginate, glucose loading, and avoidance of precipitating factors (drugs, alcohol, fasting, hormones). Long-term management includes sun avoidance, protective clothing, and regular monitoring for complications such as liver disease and hepatocellular carcinoma in PCT.

Pressure Sore (Decubitus Ulcer)

– Decubitus ulcer (bedsores, pressure ulcers) are localised areas of tissue necrosis caused by prolonged pressure over bony prominences, most commonly in immobile patients. Sites: sacrum (most common), heels, elbows, hips, and occiput. Pathophysiology: pressure exceeding capillary closing pressure (32 mmHg) causes ischaemia, followed by reperfusion injury, inflammation, and necrosis. Risk factors: immobility, sensory loss, malnutrition, moisture (incontinence), advanced age, and poor perfusion. NPUAP Staging: Stage 1 (non-blanchable erythema), Stage 2 (partial-thickness skin loss), Stage 3 (full-thickness loss without bone exposure), Stage 4 (full-thickness loss with bone/tendon/muscle involvement), Unstageable (eschar or slough covering base), Deep Tissue Injury (purple/maroon intact skin). Management: pressure redistribution (turn q2h, specialized mattresses), wound debridement, moist wound care, infection control, nutritional support, and surgical repair for advanced stages. Prevention is paramount: risk assessment (Braden scale), regular repositioning, pressure-relieving surfaces, skin care, and nutrition.

Pressure Ulcer

– Localized damage to skin and underlying tissue caused by unrelieved pressure, typically over bony prominences. Staging: Stage I (non-blanchable erythema), Stage II (partial-thickness loss), Stage III (full-thickness loss exposing fat), Stage IV (exposing bone, tendon, or muscle). Unstageable (covered by slough or eschar). Sacrum, heels, and trochanters are most common sites. Prevention includes pressure relief, turning schedules, specialty mattresses,specialty mattresses, and nutritional support. Treatment includes pressure offloading, wound debridement, infection control, moisture management, and advanced wound dressings. Surgical closure may be required for stage III and IV ulcers.

Psoriasis

– A chronic, immune-mediated inflammatory skin disease characterized by well-demarcated, erythematous plaques with silvery-white scale, affecting approximately 2 to 3 percent of the global population, with a bimodal age of onset (peak at 15 to 25 years and 50 to 60 years), a roughly equal male-to-female ratio, and strong genetic predisposition (concordance rate of 70 percent in monozygotic twins; HLA-Cw6 associated with early-onset psoriasis, present in 60 percent of psoriasis patientpatients), and environmental triggers (trauma, infection, stress, medications including lithium, beta-blockers, chloroquine, interferon, and ACE inhibitors, and smoking) acting on a background of genetic susceptibility. The pathophysiology is driven by the IL-23/TH17 axis (dendritic cell-derived IL-23 activates CD4+ and CD8+ T cells to produce IL-17 and IL-22, which stimulate keratinocyte proliferation and the production of antimicrobial peptides, chemokines, and proinflammatory cytokines, with tumour necrosis factor alpha playing a key amplifying role), leading to the characteristic histological triad of acanthosis (epidermal thickening), hyperkeratosis (thickened stratum corneum with retained nuclei, parakeratosis), and a dense dermal inflammatory infiltrate of

T cells, dendritic cells, and neutrophils. The clinical presentation is typically of well-demarcated, erythematous, indurated plaques with a silvery-white micaceous scale, located symmetrically on the extensor surfaces (elbows, knees), scalp, lumbosacral area, and nails.

Psychodermatology

– Psychodermatology addresses the mind-skin connection. Stress exacerbates psoriasis, eczema, acne, and urticaria. Psychiatric disorders (depression, anxiety, body dysmorphic disorder) are associated with skin conditions. Dermatologic conditions cause psychiatric morbidity: psoriasis and depression, acne and anxiety, vitiligo and social phobia. Integrated care addressing both skin and psychological aspects improves outcomes. Cognitive behavioral therapy, mindfulness, and antidepressants may benefimay benefit patients with psychodermatological conditions.

Pyogenic Granuloma

– A benign vascular proliferation presenting as a rapidly growing, friable, red papule or nodule that bleeds easily. It commonly occurs on fingers, face, and lips, and may be triggered by trauma or pregnancy (epulis gravidarum). Diagnosis is clinical. Treatment includes shave excision with electrocautery, laser ablation, or topical timolol for small lesions. Recurrence rate is 15 percent.

Rash

– A general term for any visible skin eruption, change in skin color, texture, or appearance. Rashes can be localised or generalised, acute or chronic, and caused by a wide variety of aetiologies: infectious (viral—measles, rubella, varicella, roseola, erythema infectiosum, hand-foot-and-mouth disease, infectious mononucleosis, HIV seroconversion; bacterial—scarlet fever, impetigo, meningococcaemia, Lyme disease, syphilis; fungal—dermatophytosis, candidiasis), allergic/drug-induced (urticaria, maculopapular drug eruption, Stevens–Johnson syndrome/toxic epidermal necrolysis, fixed drug eruption, contact dermatitis), inflammatory (eczema/dermatitis, psoriasis, lichen planus, seborrhoeic dermatitis, pityriasis rosea), autoimmune (lupus erythematosus, dermatomyositis, scleroderma, vasculitis), and neoplastic (cutaneous T-cell lymphoma, Kaposi sarcoma, basal cell carcinoma). Morphological classification is essential for diagnosis: macule (flat, <1 cm), papule (raised, <1 cm), plaque (raised, >1 cm), patch (>1 cm flat), nodule (>1 cm deep), vesicle (<1 cm fluid-filled), bulla (>1 cm fluid-filled), pustule (pus-filled), wheal (transient, oedematous), and petechiae/purpura (non-blanching hemorrhagic). Distribution (central vs peripheral, flexural vs extensor, sun-exposed vs intertriginous) and configuration (annular, linear, reticulate, dermatomal) further narrow the differential. Diagnostic tools: history (onset, duration, fever, sick contacts, medications, travel), dermatoscopy, Wood lamp, skin scraping (KOH prep for fungus, scabies), Tzanck smear (herpes), patch testing, and skin biopsy. Treatment: directed at the underlying cause.

Rashes

– See Rash.

Ringworm

– See Tinea.

Rosacea

– A chronic inflammatory skin disease characterized by facial erythema, telangiectasia, papules, pustules, and in severe cases, rhinophyma, affecting approximately 5 to 10 percent of fair-skinned adults worldwide, with a female-to-male ratio of 3 to 1 (though severe phenotypes are more common in males), with peak onset between 30 and 50 years of age. The 2002 National Rosacea Society (NRS) expert committee classification and the 2017 updated standard classification system divide rosaceadivide rosacea into four subtypes based on the predominant clinical features: erythematotelangiectatic rosacea (ETR, characterized by persistent centrofacial erythema with telangiectasias, frequent flushing, and a tendency towards burning, stinging, and dryness), papulopustular rosacea (PPR, presenting with persistent erythema with transient, dome-shaped papules and pustules without comedones, distinguishing it from acne vulgaris, which requires treatment with metronidazole, ivermectin, azelaic acid, or oral doxycycline), phymatous rosacea (most commonly rhinophyma, with thickened, irregular, bulbous enlargement of the nose due to sebaceous gland hyperplasia and fibrosis, with a male predominance), and ocular rosacea (manifesting as blepharitis, conjunctivitis, keratitis, iritis, and chalazia, requiring ophthalmology referral and treatment with topical cyclosporine, oral doxycycline, and lid hygiene).

Sarcoidosis Skin Manifestations

– Cutaneous sarcoidosis affects 25 percent of patients with systemic sarcoidosis. Lesions include erythema nodosum (tender shin nodules, Löfgren syndrome with bilateral hilar lymphadenitis and fever), lupus pernio (violaceous plaques on nose, cheeks, ears), maculopapular eruptions, and scar sarcoidosis (granulomatous infiltration of old scars). Diagnosis requires biopsy showing non-caseating granulomas. Systemic evaluation is essential.

Scar Management

– Scar management aims to minimize hypertrophic and keloid formation. Early intervention includes silicone gel or sheeting (first-line evidence-based treatment), pressure therapy (especially for burn scars), and massage. Intralesional corticosteroids (triamcinolone) for hypertrophic scars and keloids. 5-fluorouracil injection for resistant keloids. Laser therapy (pulsed dye laser for erythema, fractional CO2 for texture). Surgical revision for mature, problematic scars. Sun protection prevents hypprevents hyperpigmentation of scars and improves overall appearance.

Schwannoma

– A benign peripheral nerve sheath tumor arising from Schwann cells. It presents as a solitary, slow-growing, painless nodule along a peripheral nerve course. It can be moved laterally but not longitudinally along the nerve. Vestibular schwannoma (acoustic neuroma) arises from CN VIII. Diagnosis is by MRI. Treatment is surgical enucleation preserving nerve function. Schwannomas are encapsulated and separable from the nerve.

Seborrheic Dermatitis

– A chronic inflammatory skin condition affecting sebum-rich areas: scalp, face (nasolabial folds, eyebrows), chest, and intertriginous areas. Presents with erythematous patches with greasy, yellowish scales. Dandruff is a mild form. More common and severe in Parkinson disease, HIV/AIDS, and Down syndrome. Related to Malassezia yeast colonization and immune response. Treatment: ketoconazole shampoo/cream, selenium sulfide, zinc pyrithione, topical corticosteroids, and calcineurin inhibitors for facial involvement.

Seborrheic Keratosis

– A benign epidermal tumor appearing as stuck-on, waxy, hyperpigmented plaques with a verrucous surface. They are extremely common in older adults and increase in number with age. Seborrheic keratoses are not premalignant. The sign of Leser-Tralat indicates sudden eruption of multiple seborrheic keratoses, which may signal internal malignancy (gastric adenocarcinoma). Treatment is cosmetic removal by cryotherapy, curettage, or shave excision.

Seborrhoea

– See Seborrheic Dermatitis.

Skin Aging

– intrinsic (chronological) and extrinsic (photoaging) components. Photoaging causes wrinkle formation, telangiectasias, lentigines, leather-like texture, and actinic keratoses. Intrinsic aging causes thinning, wrinkling, and delayed wound healing. Prevention includes sun protection (broad-spectrum SPF 30 or greater daily). Treatment includes topical retinoids (tretinoin), vitamin C, alpha hydroxy acids, chemical peels, laser therapy, and fillers. Tretinoin is the most evidencemost evidence-based topical treatment for photoaging, with demonstrated improvement in fine wrinkles, roughness, and mottled hyperpigmentation after 3 to 6 months of regular use.

Skin Aging and Appearance

– Skin aging is a complex biological process influenced by both intrinsic factors, such as genetics and natural aging, and extrinsic factors, particularly ultraviolet radiation exposure. Chronological aging leads to thinning, dryness, and loss of elasticity, while photoaging from sun exposure causes wrinkles, pigmentation changes, and textural irregularities. Preventive measures include consistent sun protection, moisturizing, and avoiding smoking, while treatments such as topical retinoids, antioxidants, laser therapy, and injectable fillers can help improve the appearance of aging skin.

Skin Allergies and Contact Dermatitis

– Inflammatory skin reactions triggered by exposure to allergens or irritants that come into direct contact with the skin. Allergic contact dermatitis results from a delayed immune response to substances such as nickel, poison ivy, or fragrances, while irritant contact dermatitis occurs when chemicals or physical agents damage the skin barrier directly. Symptoms include redness, itching, swelling, and blistering; management involves identifying and avoiding triggers, using topical corticosteroids, and maintaining skin barrier function with moisturizers.

Skin Allergy Testing

– prick testing (immediate hypersensitivity, IgE-mediated) and intradermal testing (more sensitive, used for drug and venom allergy). Prick testing introduces allergen extract into the epidermis; wheal and flare reaction at 15 minutes indicates sensitization. Intradermal testing injects dilute allergen into the dermis. Both tests require stopping antihistamines 3 to 7 days before testing. Negative predictive value is high; positive results require clinical correlationclinical correlation to confirm relevance.

Skin and Aging

– Skin aging involves intrinsic and extrinsic changes. Intrinsic aging causes thinning, wrinkling, and fragility. Extrinsic (photo)aging causes wrinkles, telangiectasias, lentigines, and actinic keratoses. Collagen loss, elastin degradation, and decreased cell turnover contribute to aging skin. Prevention includes sun protection, smoking cessation, and antioxidants. Treatment includes retinoids, vitamin C, alpha hydroxy acids, chemical peels, laser therapy, and fillers. Skin cancer screening is esis essential as part of the regular check-up for elderly patients.

Skin and Environment

– Environmental factors significantly affect skin health. Pollution causes oxidative stress, inflammation, and accelerated aging. Particulate matter penetrates skin, generating free radicals. Blue light from screens causes hyperpigmentation, particularly in darker skin tones. Indoor heating and air conditioning cause dryness. Water quality (hard water) affects eczema. Environmental protection includes antioxidant skincare, cleansing, and sun protection.

Skin Appendages

– hair follicles, sebaceous glands, eccrine sweat glands, and apocrine sweat glands. Hair follicles cycle through anagen (growth), catagen (transition), and telogen (resting) phases. Sebaceous glands produce sebum, an oily substance lubricating skin and hair. Eccrine glands (distributed over entire body) produce watery sweat for thermoregulation. Apocrine glands (axillae, groin) produce viscous secretion active at puberty, responsible for body odor when metabolized by bacteby bacteria that break down fatty acids and proteins in the secretion.

Skin Barrier Function

– The skin barrier prevents water loss and protects against environmental insults. The stratum corneum acts as a physical barrier, while the acid mantle (pH 4 to 6) provides chemical defense. Lipids (ceramides, cholesterol, free fatty acids) in the intercellular space form a waterproof barrier. Tight junctions in the stratum granulosum prevent paracellular transport. Barrier dysfunction is central to atopic dermatitis, psoriasis, and ichthyosis. Emollients restore barrier function by filling interintercellular lipids, while moisturisers reduce water loss and improve barrier function.

Skin Biopsy Methods

– Skin biopsy methods vary by lesion characteristics. Punch biopsy (2 to 6 mm circular blade) provides full-thickness specimen for most diagnoses. Shave biopsy (razor or scalpel) removes superficial lesions; deep shave (saucerization) includes subcutaneous fat for suspected melanoma. Excisional biopsy removes entire lesion with narrow margins, preferred for suspected melanoma. Incisional biopsy removes representative portion of large lesions. Site and technique should be chosen to optimize diagnosdiagnostic yield. Proper anesthesia and haemostasis are important for optimal specimen quality.

Skin Biopsy Techniques

– Skin biopsy is essential for diagnosing skin conditions. Punch biopsy (2 to 6 mm) is most common, providing full-thickness skin sample for histopathology and direct immunofluorescence. Shave biopsy (razor or scalpel) is used for superficial lesions but may not include deep dermis. Excisional biopsy removes entire lesion with margins, used for suspected melanoma. Incisional biopsy removes representative portion of large lesions. Site selection depends on suspected diagnosis: perilesional for immuimmunofluorescence (perilesional normal skin) or center of lesion for routine histology.

Skin Cancer Epidemiology

– Skin cancer epidemiology reveals increasing incidence worldwide. Basal cell carcinoma is the most common cancer (1 million cases annually in the US).

Squamous cell carcinoma is second (approximately 200,000 cases annually). Melanoma is less common but most deadly (about 100,000 new cases annually in the US). Incidence of melanoma has increased 200 percent over the past 30 years. UV radiation is the primary risk factor. Skin cancer screening and sun protection are public health priorities.

Skin Cancer Prevention

– Skin cancer prevention focuses on UV radiation protection. Broad-spectrum sunscreen (SPF 30 or greater) applied daily and reapplied every 2 hours during sun exposure. Protective clothing, hats, and sunglasses. Seeking shade during peak UV hours (10 AM to 4 PM). Avoiding tanning beds. Vitamin D supplementation for those with strict sun avoidance. Sunscreen reduces melanoma risk by 50 percent with intermittent use. Sunscreen use in children is particularly important.

Skin Cancer Screening

– Skin cancer screening uses the ABCDE criteria for melanoma, the ugly duckling sign, and total body skin examination. Dermoscopy (dermatoscopy) magnifies skin lesions revealing structures invisible to the naked eye, improving diagnostic accuracy. Skin self-examination teaches patients to monitor moles for changes. Annual screening is recommended for high-risk patients (personal or family history of melanoma, numerous atypical nevi, immunosuppression).

Skin Cancer Staging

– Skin cancer staging guides treatment and prognosis. Melanoma: Breslow thickness (most important prognostic factor), ulceration, mitotic rate, sentinel lymph node status (AJCC 8th edition). Basal cell carcinoma: rarely metastasizes; high-risk features include size greater than 2 cm, location on face, perineural invasion, and recurrent tumors. Squamous cell carcinoma: AJCC staging based on size, depth, perineural invasion, and nodal metastasis. High-risk features include thickness greater than 2 mgreater than 2 mm for SCC and greater than 2 cm for BCC. Sentinel lymph node biopsy is indicated for high-risk SCC and melanoma.

Skin Cancer Surgery

– Skin cancer surgical techniques maximize cure while minimizing morbidity. Wide local excision with appropriate margins (BCC: 4 mm, SCC: 4 to 6 mm, melanoma: based on Breslow thickness). Mohs micrographic surgery provides 99 percent cure rate for BCC and 97 percent for SCC with tissue preservation. Sentinal lymph node biopsy for melanoma greater than 1 mm or with ulceration. Reconstruction ranges from primary closure to flaps and grafts.

Skin Flaps

– Skin flaps transfer tissue with its blood supply. Rotation flaps rotate adjacent tissue into defect. Advancement flaps slide tissue forward. Transposition flaps move tissue across intervening skin. Random pattern flaps rely on dermal plexus; axial pattern flaps have named vascular pedicle. Flaps provide better color and texture match than grafts for facial reconstruction. Design follows relaxed skin tension lines and aesthetic subunits.

Skin Grafts

– Skin grafts replace lost skin. Split-thickness grafts (epidermis and part of dermis) heal by re-epithelialization of donor site, allowing reuse. Full-thickness grafts (entire dermis) provide better cosmetic results but require primary closure of donor site. Sheet grafts provide smooth surface; mesh grafts cover larger areas but have quilted appearance. Graft take requires vascular ingrowth from wound bed (plasmatic imbibition day 1 to 2, inosculation day 2 to 3, revascularization day 3 to 5).

Skin Immune System

– The skin immune system (SALT: skin-associated lymphoid tissue) provides immunological surveillance. Langerhans cells (dendritic cells in epidermis) capture antigens and present them to T cells. Dermal dendritic cells, mast cells, and resident T cells form an innate and adaptive immune network. Cytokines (IL-1, IL-6, TNF-alpha, IL-17, IL-23) mediate inflammatory responses. The skin immune system is involved in contact dermatitis, psoriasis, atopic dermatitis, and skin cancer immunosurveillance.

Skin Microbiome

– The skin microbiome is the community of microorganisms (bacteria, fungi, viruses) residing on the skin. Staphylococcus aureus colonization is increased in atopic dermatitis and psoriasis. Cutibacterium acnes plays a role in acne. Diversity is decreased in diseased skin. Probiotics and prebiotics are being studied for dermatologic conditions. Topical antimicrobials and emollients modulate the microbiome. Disruption of the microbiome contributes to skin disease pathogenesis.

Skin of Color

– Skin of color (Fitzpatrick IV to VI) has unique dermatologic considerations. It is more prone to hyperpigmentation disorders (post-inflammatory hyperpigmentation, melasma, acanthosis nigricans) and keloid formation. It is less prone to skin cancer but presents with more advanced disease when it occurs. Dermatoscopic features differ in pigmented skin. Misdiagnosis is common: vitiligo appears as ash-gray macules, lupus may present with hyperpigmentation rather than erythema. Culturally competent dculturally competent dermatological care acknowledges these differences and ensures accurate diagnosis and appropriate management for patients with skin of color.

Skin Pigmentation

– Determined by melanin produced by melanocytes in the stratum basale. Eumelanin (brown-black) and pheomelanin (red-yellow) are the two types. Melanin production is stimulated by UV radiation through melanocyte-stimulating hormone (MSH). Fitzpatrick skin type classification (I to VI) describes pigmentation and sun response. Hyperpigmentation disorders include melasma, post-inflammatory hyperpigmentation, and lentigines. Hypopigmentation disorders include vitiligo, pityriasis apityriasis alba (post-inflammatory hypopigmentation) and albinism (congenital hypopigmentation, oculocutaneous or ocular). Treatment of hyperpigmentation includes sun protection, topical hydroquinone, azelaic acid, kojic acid, vitamin C, retinoids, chemical peels, and laser therapy.

Skin Tags

– Benign fibroepithelial polyps occurring in intertriginous areas (neck, axillae, groin, under breasts). They are associated with obesity, insulin resistance, and pregnancy. Skin tags are asymptomatic but may become irritated or bleed. Treatment includes snipping, cryotherapy, electrocautery, and shave excision. They are harmless and require no biopsy unless atypical.

Squamous Cell Carcinoma

– Cutaneous squamous cell carcinoma (cSCC) is the second most common non-melanoma skin cancer, accounting for approximately 20 percent of non-melanoma skin cancers, with an incidence of over 1 million cases per year in the United States and a lifetime risk of approximately 7 to 11 percent in fair-skinned populations, with increasing incidence. cSCC arises from keratinocytes of the epidermis (squamous cells), driven by ultraviolet radiation-induced mutations (TP53 tumor suppressor mutations in greagreat part due to TP53 hotspot mutations in sun-exposed skin), ultraviolet B (UVB) signature mutations (C to T and CC to TT transitions at dipyrimidine sites), and chronic immunosuppression (organ transplant recipients have a 65 to 250-fold increased risk of cSCC). cSCC typically presents as an erythematous, hyperkeratotic, indurated papule or plaque on sun-exposed areas (face, ears, scalp, dorsal hands, forearms, lower legs) and may arise de novo, from actinic keratosis (the precursor lesion), from Bowen disease (squamous cell carcinoma in situ), or from erythroplasia of Queyrat (penile carcinoma in situ). High-risk features for local recurrence and metastasis include tumour thickness greater than 2 mm, Clark level IV or V, perineural invasion, lymphovascular invasion, poor differentiation, location on the ear or lip, diameter greater than 2 cm, and recurrence after prior treatment.

Subcutaneous Tissue

– Subcutaneous tissue (hypodermis, superficial fascia) lies beneath the dermis and connects skin to underlying structures. It contains adipose tissue (fat), blood vessels, nerves, and the superficial muscular aponeurotic system (SMAS). Subcutaneous fat provides insulation, cushioning, and energy storage. The thickness varies by body site, age, sex, and nutritional status. Liposuction removes subcutaneous fat. Subcutaneous injections (insulin, heparin, vaccines) deposit medication in this layer.

Sun Protection (Photoprotection)

– Sun protection involves measures taken to reduce exposure to ultraviolet radiation from the sun, which is the primary environmental risk factor for skin cancer and photoaging. Recommended strategies include applying broad-spectrum sunscreen with a sun protection factor of at least 30, wearing protective clothing and wide-brimmed hats, seeking shade during peak ultraviolet hours between 10 a.m. and 4 p.m., and avoiding tanning beds. Consistent sun protection throughout life significantly reduces the risk of melanoma, squamous cell carcinoma, and basal cell carcinoma, as well as preventing premature skin aging.

Systemic Antifungals

– Systemic antifungals treat extensive or refractory fungal infections. Terbinafine is first-line for onychomycosis (6 weeks for fingernails, 12 weeks for toenails) and tinea capitis. Itraconazole pulse therapy (200 mg twice daily for 1 week per month) treats onychomycosis. Fluconazole treats tinea corporis, cruris, and versicolor. Griseofulvin treats tinea capitis (requires 12 to 16 weeks). Voriconazole and posaconazole treat invasive fungal infections. Hepatic function monitoring is required.

Telogen Effluvium

– Diffuse hair shedding caused by premature entry of hair follicles into the telogen (resting) phase. Triggers include severe illness, surgery, childbirth, rapid weight loss, emotional stress, medications (beta-blockers, retinoids, anticoagulants), and nutritional deficiencies (iron, zinc, vitamin D). Acute TE resolves within 6 months; chronic TE lasts longer than 6 months. Diagnosis is clinical with positive hair pull test and normal scalp examination. Treatment addresses theaddresses the underlying trigger; recovery of normal hair growth occurs over 3 to 6 months, though the condition may become chronic if the triggering factor persists.

Tinea Capitis

– A dermatophyte infection of the scalp and hair shafts, most common in children. It presents as scaly patches, alopecia, black dots (broken hairs), kerion (inflamed boggy plaque), and impetigo of Hilgart (pyogenic reaction). Trichophyton tonsurans and Microsporum canis are the most common organisms. Diagnosis is by KOH preparation and fungal culture. Treatment requires oral antifungals (griseofulvin, terbinafine, itraconazole) for 4 to 8 weeks. Topical agents alone are ineffectivineffective. Selenium sulfide shampoo can reduce spore shedding. Treatment of close contacts and fomite disinfection prevent spread.

Tinea Corporis

– A dermatophyte infection of glabrous skin, presenting as annular, erythematous, scaly plaques with raised, active borders and central clearing. It is contagious through direct contact with infected persons, animals, or contaminated objects. Diagnosis is by KOH preparation showing branching septate hyphae. Treatment includes topical antifungals (terbinafine, clotrimazole, miconazole) for 2 to 4 weeks. Oral terbinafine or itraconazole for extensive or refractory disease.

Tinea (Dermatophytosis)

– Superficial fungal infection by dermatophytes. Tinea corporis (ringworm): annular, scaly, erythematous plaque with central clearing. Tinea pedis (athlete's foot): interdigital scaling, maceration, and fissuring. Tinea cruris (jock itch): erythematous, pruritic rash in groin with raised border. Tinea capitis: scaly scalp patches with broken hairs, more common in children; may cause kerion (inflammatory mass). Diagnosis: KOH preparation and culture. Treatment: topical terbinafine or azoles for mild to moderate; systemic terbinafine, itraconazole for extensive, scalp, or nail involvement.

Tinea Pedis

– A dermatophyte infection of the feet. Moccasin type (diffuse scaling of sole and sides), interdigital type (most common, between toes), vesiculobullous type (acute inflammation), and ulcerative type. Associated with tinea unguium (nail infection). Treatment includes topical antifungals (terbinafine, butenafine, clotrimazole). Oral terbinafine for refractory cases. Preventive measures include keeping feet dry, wearing sandals in public showers, and treating nail infenail infection (onychomycosis) concurrently.

Tinea Versicolor

– A superficial fungal infection caused by Malassezia species, presenting as hypo- or hyperpigmented macules with fine scale on the trunk and proximal extremities. It is more noticeable after sun exposure due to unaffected tanning of surrounding skin. Diagnosis is clinical with KOH showing spaghetti and meatballs appearance (hyphae and spores). Treatment includes selenium sulfide shampoo, topical antifungals (ketoconazole, ciclopirox), and oral fluconazooral fluconazole (200 to 400 mg single dose or 300 mg weekly for 2 weeks) for extensive or recurrent disease, with a 50 to 80 percent response rate but high relapse rates (more than 50 percent within 2 years, requiring maintenance therapy).

Topical Antibiotics

– Topical antibiotics treat superficial bacterial skin infections. Mupirocin 2 percent ointment is first-line for impetigo and folliculitis (inhibits bacterial protein synthesis). Retapamulin 1 percent ointment is an alternative. Fusidic acid is used outside the US. Topical gentamicin and bacitracin are less effective and may cause contact dermatitis. Neomycin has high allergy rates and is no longer recommended. Topical antibiotics should be used for limited areas and short durations to prevent reprevent resistance. Widespread or deep infections require systemic antibiotics.

Topical Antifungals

– Topical antifungals treat superficial fungal infections. Allylamines (terbinafine, naftifine) inhibit squalene epoxidase, fungicidal. Azoles (clotrimazole, miconazole, ketoconazole, econazole) inhibit ergosterol synthesis, fungistatic. Ciclopirox has broad-spectrum activity. Tolnaftate is effective against dermatophytes. Application frequency: allylamines once or twice daily for 1 to 2 weeks; azoles once or twice daily for 2 to 4 weeks. Treatment should continue 1 to 2 weeks after clinical resolclinical resolution to prevent recurrence.

Topical Calcineurin Inhibitors

– Non-steroidal anti-inflammatory agents used for atopic dermatitis and facial eczema. They inhibit T-cell activation by blocking calcineurin, preventing cytokine production. Advantages over corticosteroids: no skin atrophy, safe for face and intertriginous areas. Side effects include transient burning and itching at application site. Black box warning for malignancy risk has not been substasubstantiated by real-world use, though FDA requires it. Intermittent therapy is safe for long-term use.

Topical Corticosteroid Potency

– Topical corticosteroids are classified into seven potency groups. Group 1 (ultra-high): clobetasol propionate 0.05 percent. Group 2 (high): betamethasone dipropionate 0.05 percent. Group 3 to 4 (medium): triamcinolone acetonide 0.1 percent. Group 5 to 6 (low): hydrocortisone 1 percent. Group 7 (lowest): hydrocortisone 0.5 percent. Face, groin, and intertriginous areas require low-potency steroids. Thick plaques on palms and soles may require ultra-high potency. Prolonged use causes skin atrophy,skin atrophy, telangiectasias, striae, and perioral dermatitis. Occlusion increases potency. Short-term use of appropriate potency minimises adverse effects while maximising efficacy.

Topical Retinoids

– Vitamin A derivatives used for acne, photoaging, and hyperpigmentation. Tretinoin (all-trans retinoic acid) is the most studied. Adapalene is more stable and less irritating. Tazarotene is the most potent. Mechanism: normalizes follicular keratinization, increases cell turnover, and stimulates collagen production. Side effects include irritation, erythema, peeling, and photosensitivity. Start with low concentration, alternate nights, and use moisturizer. Sunscreen is mandatmandatory during retinoid therapy. Results are seen over 4 to 6 months. Maintenance therapy with lower frequency is needed.

Urticaria

– A common skin condition characterized by the transient appearance of wheals (raised, erythematous or white plaques with surrounding flare), angioedema (swelling of the deeper dermis and subcutaneous tissue, often involving the lips, eyelids, tongue, and extremities), or both, with a prevalence of 15 to 20 percent of the population experiencing at least one episode during their lifetime, and chronic urticaria (episodes occurring on most days for greater than 6 weeks) affectinaffecting 1 to 2 percent of the population, more common in middle-aged women. The diagnosis is clinical. Treatment follows a stepwise approach: second-generation antihistamines first-line, with updosing as needed; add H2-antagonist or montelukast second-line; add omalizumab or cyclosporine third-line.

Verruca (Wart)

– Benign epidermal hyperplasia caused by human papillomavirus (HPV). Common warts (verruca vulgaris): rough papules on hands/fingers. Plantar warts: painful lesions on soles, grow inward due to pressure. Flat warts (verruca plana): smooth, flat-topped papules on face, hands, legs. Filiform warts: finger-like projections, often on face. Genital warts (condyloma acuminata): sexually transmitted. Diagnosis: clinical. Treatment: cryotherapy, salicylic acid, cantharidin, imiquimod, podophyllin, and laser therapy. Most warts resolve spontaneously over months to years.

Vitiligo

– An acquired, autoimmune pigmentary disorder characterized by well-demarcated, white macules and patches resulting from the destruction of epidermal melanocytes, affecting approximately 0.5 to 2 percent of the global population, with equal distribution across sexes and races, and an age of onset typically before 30 years (50 percent before age 20). Pathogenesis involves autoimmune destruction of melanocytes: CD8+ cytotoxic T cells infiltrate the lesional and perilesional epidermis, tatargeting melanocyte-specific antigens such as tyrosinase, Melan-A/MART-1, and gp100, with the destruction mediated by the release of cytotoxic granules containing perforin and granzyme B. The exact trigger for the loss of self-tolerance is unknown but is thought to involve a combination of genetic susceptibility (more than 30 susceptibility loci identified, most importantly the MHC class II region on chromosome 6p21.3, including HLA-DRB1*04 and HLA-DQB1*03), environmental triggers (oxidative stress, mechanical trauma, Koebner phenomenon, and chemical exposure to monobenzone and other phenolic compounds), and autoimmune diathesis (association with other autoimmune diseases, particularly thyroid disease, found in 15 to 25 percent of patients, most commonly Hashimoto thyroiditis).

Vitiligo and Pigmentation Disorders

– Vitiligo is a chronic skin condition characterized by the loss of melanocytes, resulting in well-defined white patches on the skin and sometimes mucous membranes. The exact cause is not fully understood but involves an autoimmune process in which the body attacks its own pigment-producing cells, and it may be associated with other autoimmune diseases. Pigmentation disorders also include melasma, characterized by brown patches on the face often triggered by hormonal changes, and post-inflammatory hyperpigmentation, which follows skin injury or inflammation; treatments range from topical corticosteroids and phototherapy to depigmentation and cosmetic camouflage.

Vitiligo Treatment

– Vitiligo treatment options include topical corticosteroids (first-line for limited disease), topical calcineurin inhibitors (tacrolimus, pimecrolimus for face and intertriginous areas), narrowband UVB phototherapy (first-line for widespread disease), excimer laser for localized lesions, and systemic immunosuppressants for rapidly progressive disease. JAK inhibitors (ruxolitinib cream) are FDA-approved for vitiligo. Surgical options include autologous melanocyte transplantation for stable diseasestable disease (lesions stable for at least 6 to 12 months), including suction blister grafting, punch grafting, and cultured melanocyte transplantation, with repigmentation rates of 50 to 90 percent in properly selected patients.

Warts

– See Verruca (Wart).

Wen (Epidermoid Cyst)

– See Epidermal Inclusion Cyst.

Wound Healing Phases

– Wound healing occurs in overlapping phases: hemostasis (immediate, platelet aggregation and clot formation), inflammation (days 1 to 4, neutrophils and macrophages clear debris and bacteria), proliferation (days 4 to 21, granulation tissue, angiogenesis, epithelialization, collagen deposition), and remodeling (day 21 to 2 years, collagen maturation, scar strengthening to 80 percent of original strength). Factors impairing healing include diabetes, infection, ischemia, malnutrition, and steroids.

Xanthelasma

– Xanthelasma palpebrarum are soft, yellow, lipid-filled papules or plaques on the medial canthus of the eyelids (upper more commonly than lower), the most common form of cutaneous xanthoma. They are composed of foam cells (lipid-laden macrophages) in the superficial dermis. Xanthelasma is associated with hyperlipidemia (elevated LDL, low HDL) in approximately 50% of patients; the other 50% are normolipidaemic. It may also be associated with diabetes, obesity, and cardiovascular disease (xanthelasma is an independent risk factor for atherosclerotic cardiovascular disease). Diagnosis: clinical. Treatment: options for cosmetic removal include surgical excision (elliptical excision with primary closure, risks of ectropion or scar), laser therapy (CO2, argon, or pulsed dye laser), chemical cautery (trichloroacetic acid, which should be used sparingly to avoid eyelid scarring), and radiofrequency ablation. Underlying hyperlipidemia should be screened and managed.

Xanthoma

– Cutaneous deposits of lipids (cholesterol, triglycerides) in the dermis or tendons, presenting as yellow-orange papules, nodules, or plaques. Xanthomas are classified by clinical morphology and location: eruptive xanthomas (sudden eruption of multiple small, red-yellow papules on buttocks, extensor surfaces — pathognomonic for hypertriglyceridaemia, chylomicronaemia syndrome, lipoprotein lipase deficiency), tuberous xanthomas (firm, painless, yellow-red nodules on elbows, knees, buttocks — seen in familial hypercholesterolaemia, Sitosterolaemia), tendinous xanthomas (subcutaneous nodules within tendons, most commonly the Achilles tendon and extensor tendons of the hands — pathognomonic for familial hypercholesterolaemia, cerebrotendinous xanthomatosis), and planar xanthomas (flat or slightly raised yellow plaques on palms, flexural creases, or intertriginous areas — associated with type III hyperlipoproteinaemia/dysbetalipoproteinaemia). Diagnosis: clinical, serum lipid panel, and biopsy (foam cells, Touton giant cells). Xanthomas are not harmful themselves but are markers of underlying lipid disorders that require treatment to reduce cardiovascular risk.

Xeroderma

– See Xerosis.

Xerosis

– Abnormally dry skin, common in elderly patients and winter months. It results from decreased sebaceous and sweat gland activity, impaired skin barrier function, and environmental factors. Treatment includes regular emollient application (cream or ointment rather than lotion), gentle cleansing, and humidification. Severe xerosis may progress to asteatotic eczema (eczema craquelé) with cracked, porcelain-like appearance.

Zinc Ointment (Zinc Oxide)

– Zinc oxide is a topical agent with mild antiseptic, astringent, and barrier-protective properties, used in a wide range of dermatological preparations. Mechanism: zinc is an essential trace element and cofactor for numerous metalloenzymes involved in wound healing (matrix metalloproteinases, DNA polymerases, superoxide dismutase). Zinc oxide has anti-inflammatory effects (inhibits neutrophil degranulation and mast cell degranulation), provides UV protection (physical sunscreen, blocks UVA and UVB), and forms a protective barrier over irritated or denuded skin that reduces moisture loss and protects from friction and irritants. Indications: nappy/diaper rash (the most common use — as a barrier cream that protects the perineal skin from moisture and faecal enzymes), minor wounds, abrasions, and burns, sun protection (physical sunscreen, often combined with titanium dioxide), and as a topical treatment for acne vulgaris and seborrhoeic dermatitis. Zinc sulphate is used in oral zinc supplementation for zinc deficiency and for treatment of Wilson disease (blocks intestinal copper absorption). Adverse effects: topical — skin irritation, contact dermatitis (rare). Oral zinc: nausea, gastric irritation, copper deficiency with long-term high doses (zinc inhibits copper absorption via metallothionein induction).

Chapter 7

Emergency Medicine

Abscess

– A localized collection of pus within a tissue, organ, or cavity, typically caused by bacterial infection. Abscesses form when bacteria (most commonly Staphylococcus aureus including MRSA) invade tissue, triggering an intense inflammatory response with neutrophil accumulation and tissue necrosis. The body attempts to wall off the infection, creating a walled-off cavity filled with purulent material (dead neutrophils, bacteria, necrotic tissue). Clinical presentation includes a painful, swollen, erythematous, warm mass that may fluctuate (indicating fluid collection). Systemic symptoms (fever, leukocytosis) may be present. Diagnosis is primarily clinical; ultrasound confirms the presence and size of a fluid collection. Treatment requires incision and drainage (I and D) for Source control, as antibiotics alone cannot penetrate the abscess wall effectively. Small abscesses (< 2 cm) may respond to warm compresses and antibiotics. Culture should be obtained to guide antibiotic therapy, particularly for recurrent abscesses or MRSA risk factors. Complications include cellulitis, bacteremia, and sepsis if untreated. Common locations include skin and soft tissue (furuncles, carbuncles), perianal (perianal abscess), intra-abdominal (from perforated viscus), dental (dental abscess), and brain (cerebral abscess).

Acute Aortic Syndromes

– A spectrum of life-threatening conditions involving the thoracic aorta, including aortic dissection, intramural hematoma (IMH), and penetrating aortic ulcer (PAU). Aortic dissection involves an intimal tear creating a false lumen. Classification (Stanford): Type A involves the ascending aorta (surgical emergency); Type B involves only the descending aorta (medical management unless complicated). Clinical presentation: sudden onset of severe, tearing or ripping chest or back pain radiating to the interscapular region, hypertension or hypotension, pulse deficits, aortic regurgitation murmur, or neurological deficits. Diagnosis: CT angiography (gold standard), transesophageal echocardiography (TEE), or MRI. Management: anti-impulse therapy (IV beta-blockers like esmolol/labetalol targeting HR < 60 bpm and SBP 100–120 mmHg), followed by emergency surgery for Type A or TEVAR/medical management for Type B.

Acute Coronary Syndromes

– A spectrum of conditions comprising unstable angina (UA), non-ST-elevation myocardial infarction (NSTEMI), and ST-elevation myocardial infarction (STEMI). Pathophysiology: atherosclerotic plaque rupture or erosion leading to thrombus formation and acute coronary artery occlusion. STEMI: complete occlusion presenting with ST-segment elevation in contiguous leads (≥ 1 mm in limb leads, ≥ 2 mm in precordial leads) or new LBBB. NSTEMI: partial occlusion causing cardiac troponin elevation without ST elevation. Unstable angina: ischemic symptoms at rest or increasing frequency without troponin elevation. Management: STEMI requires immediate reperfusion via primary percutaneous coronary intervention (PCI) within 90 minutes (door-to-balloon) or fibrinolysis within 30 minutes if PCI is unavailable within 120 minutes. Medical therapy: dual antiplatelet therapy (aspirin + P2Y12 inhibitor), anticoagulation (heparin), high-intensity statin, beta-blocker, and ACE inhibitor.

Adrenal Crisis

– An acute, life-threatening manifestation of severe adrenal insufficiency. Causes: primary (Addison disease), secondary (pituitary failure), or tertiary (exogenous corticosteroid withdrawal). Precipitants: systemic infection, trauma, surgery, or abrupt steroid cessation. Clinical presentation: severe hypotension refractory to vasopressors and fluid resuscitation, fever, abdominal pain, vomiting, hyponatremia, hyperkalemia, hypoglycemia, and altered mental status. Emergency management: immediate administration of IV hydrocortisone 100 mg bolus, followed by 200 mg/24 hours continuous infusion; aggressive isotonic saline fluid resuscitation; dextrose for hypoglycemia; and identification and treatment of the underlying precipitating cause.

Altitude Illness

– Acute mountain sickness (AMS): headache + 1 symptom (nausea, fatigue, dizziness, insomnia) at >2500m. HACE: AMS + ataxia/altered mental status (cerebral edema). HAPE: dyspnea at rest, cough, frothy sputum, crackles, hypoxemia (noncardiogenic pulmonary edema). Prevention: gradual ascent (300-500m/day above 3000m), acetazolamide 125 mg q12h prophylaxis. Treatment: descent (immediate for HACE/HAPE), oxygen, acetazolamide, dexamethasone 8 mg load then 4 mg q6h (HACE), nifedipine 30 mg ER q12h (HAPE), portable hyperbaric chamber (Gamow bag).

Anaphylaxis

– A severe, potentially life-threatening systemic hypersensitivity reaction characterized by rapid onset of multisystem involvement, primarily affecting the skin, respiratory tract, cardiovascular system, and gastrointestinal tract. It is mediated by IgE-dependent mast cell and basophil degranulation (Type I hypersensitivity) or by direct mast cell activation (non-IgE mediated, e.g., radiocontrast, opioids, vancomycin). Common triggers: foods (peanuts, tree nuts, shellfish, fish, eggs, milk, soy, wheat), medications (penicillins, NSAIDs, neuromuscular blocking agents, radiocontrast), insect stings (Hymenoptera: bees, wasps, fire ants), and exercise (exercise-induced anaphylaxis, often food-dependent). Clinical criteria: acute onset (minutes to hours) with involvement of skin/mucosa PLUS respiratory compromise (wheeze, stridor, dyspnea, hypoxia) OR reduced blood pressure (systolic less than 90 mmHg in adults, or greater than 30 percent decrease from baseline) OR gastrointestinal symptoms (vomiting, crampy abdominal pain) with skin involvement. Management: immediate intramuscular epinephrine (0.3-0.5 mg of 1:1,000 solution in the mid-outer thigh, repeat every 5-15 minutes as needed) is the first-line and most critical intervention. Adjunctive treatments: high-flow oxygen, IV fluids (normal saline or lactated Ringer's for hypotension), H1 antihistamine (diphenhydramine 25-50 mg IV), H2 antihistamine (famotidine 20 mg IV), corticosteroids (methylprednisolone 125 mg IV or prednisone 40-60 mg PO), and albuterol nebulizer for bronchospasm. Patients should be observed for at least 4-6 hours (biphasic reaction occurs in 1-20 percent of cases). All patients discharged after anaphylaxis should receive an epinephrine auto-injector and referral to an allergist.

Asphyxia

– A life-threatening lack of oxygen due to drowning, choking, or an obstruction of the airways.

Aspiration

– The inhalation of foreign material (oropharyngeal secretions, gastric contents, food, liquid, or blood) into the larynx and lower respiratory tract. Aspiration pneumonitis (Mendelson syndrome) is a chemical lung injury caused by acidic gastric fluid. Aspiration pneumonia is an infectious lung process developing after inhalation of colonized oral secretions. Risk factors: impaired consciousness (alcohol intoxication, stroke, seizure, general anesthesia), dysphagia, enteral tube feeding, and vomiting. Clinical features: sudden cough, dyspnea, wheezing, hypoxemia, fever, and infiltrate on chest radiograph (commonly right lower lobe due to steeper right mainstem bronchus angle). Management: airway suctioning, supplemental oxygen, bronchodilators, and empiric antibiotics for aspiration pneumonia.

Bends (Decompression Sickness)

– A condition resulting from rapid ascent from deep water or pressurized environment, causing dissolved inert gases (primarily nitrogen) to form bubbles in tissues and blood. Symptoms range from joint pain (the bends), skin rash (cutis marmorata), and neurologic deficits (paresthesias, weakness, confusion) to respiratory distress (the chokes) and cardiovascular collapse. Treatment: 100% oxygen and recompression in a hyperbaric chamber. Prevention: staged ascent with decompression stops and adherence to dive tables or computers.

Burns and Scalds

– Tissue damage caused by thermal, chemical, electrical, or radiation exposure. Burns from dry heat (flame, hot surfaces) and scalds from moist heat (hot liquids, steam) are classified by depth: superficial (first-degree: epidermal involvement, painful erythema without blisters), partial-thickness (second-degree: epidermal and dermal involvement, painful blisters), and full-thickness (third-degree: destruction of all skin layers and subcutaneous tissue, painless leathery or charred appearance). Total Body Surface Area (TBSA) is estimated using the Rule of Nines or Lund-Browder chart. Initial management: airway assessment for inhalation injury, fluid resuscitation using the Parkland formula (4 mL x kg x %TBSA over 24 hours, half given in the first 8 hours), pain management, sterile dressings, and referral to a burn center.

Carbon Monoxide Poisoning

– Occurs when inhaled carbon monoxide (CO) binds to hemoglobin with 200–250 times greater affinity than oxygen, forming carboxyhemoglobin (COHb), shifting the oxyhemoglobin dissociation curve to the left and severely impairing tissue oxygen delivery. Sources: faulty gas heaters, engine exhaust in enclosed areas, house fires, and charcoal grills. Clinical presentation: headache, nausea, dizziness, confusion, syncope, chest pain, seizures, and coma. Classic 'cherry-red' skin color is rare. Diagnosis: co-oximetry blood gas measuring COHb levels (standard pulse oximetry cannot distinguish COHb from oxyhemoglobin). Treatment: 100% high-flow normobaric oxygen via non-rebreather mask (shortens COHb half-life from 320 to 80 minutes); hyperbaric oxygen (HBO) therapy for severe cases (COHb $>$ 25%, syncope, seizures, coma, or pregnancy).

Cardiac Tamponade

– A life-threatening condition caused by fluid accumulation within the pericardial space under increased pressure, compromising cardiac diastolic filling and severely reducing cardiac output. Causes: trauma, pericarditis, post-cardiac surgery, aortic dissection, or malignancy. Clinical features: Beck triad (hypotension, muffled heart sounds, jugular venous distension), pulsus paradoxus (systolic BP drop $>$ 10 mmHg during inspiration), tachycardia, and tachypnea. Diagnosis: bedside echocardiography demonstrating pericardial effusion with right ventricular diastolic collapse and IVC plethora. Treatment: emergency pericardiocentesis (ultrasound-guided) or surgical pericardial window, alongside IV fluid boluses to maintain preload while preparing for drainage.

Cardiopulmonary Resuscitation (CPR)

– An emergency life-saving procedure performed to restore circulation and breathing in a person in cardiac arrest. Basic life support (BLS) sequence: check responsiveness, activate emergency medical services, and begin chest compressions (rate 100–120/min, depth 5–6 cm in adults, allowing full chest recoil). Give rescue breaths at a 30:2 ratio (compressions to breaths). Compression-only CPR is recommended for untrained bystanders. Advanced cardiovascular life support (ACLS) includes defibrillation for shockable rhythms (VF/pulseless VT), advanced airway management, IV/IO vascular access, and pharmacotherapy (epinephrine 1 mg every 3–5 min, amiodarone 300 mg bolus). High-quality CPR with minimal interruptions is the single most important determinant of cardiac arrest survival.

Coma

– A state of profound unconsciousness characterized by the inability to wake up, lack of awareness, and absence of voluntary responses to external stimuli. Causes are categorized as structural CNS lesions (stroke, hemorrhage, trauma, brain tumor, herniation) or metabolic/toxic encephalopathies (hypoxia, hypoglycemia, diabetic ketoacidosis, hepatic encephalopathy, drug overdose, severe hyponatremia, sepsis). Assessment: Glasgow Coma Scale (GCS $<$ 8 indicates severe impairment requiring endotracheal intubation for airway protection), pupillary light reflex, corneal reflex, and doll's eye (oculocephalic) reflex. Emergency management: immediate ABCs, administration of the 'coma cocktail' if indicated (dextrose for hypoglycemia, thiamine, naloxone for opioid toxicity), non-contrast head CT, and treatment of the underlying cause.

Defibrillation

– The delivery of a controlled electrical shock to the heart to terminate life-threatening arrhythmias, specifically ventricular fibrillation (VF) and pulseless ventricular tachycardia (pVT), allowing the sinoatrial node to resume normal rhythm. Defibrillation is time-critical: survival decreases by 7–10% per minute without it. Automated external defibrillators (AEDs) analyze rhythm and deliver shocks with voice prompts for bystander use. Biphasic waveforms (150–200 J) are more effective than monophasic (360 J) with less myocardial damage. For shockable rhythms: defibrillate immediately, then resume CPR for 2 minutes before rhythm check. Energy selection: adult 120–200 J biphasic; pediatric 2–4 J/kg.

Dehydration

– A state of total body water depletion resulting from excessive fluid loss, inadequate fluid intake, or both. Causes: gastrointestinal losses (vomiting, diarrhea), renal losses (diabetic ketoacidosis, diabetes insipidus, diuretic overuse), skin losses (burns, excessive sweating), or inadequate intake. Classified by severity: mild (3–5% body weight loss), moderate (6–9%), and severe ($\geq$ 10%, leading to hypovolemic shock). Clinical signs: dry mucous membranes, decreased skin turgor, sunken eyes, tachycardia, hypotension, oliguria, and altered mental status. Management: oral rehydration therapy with glucose-electrolyte solutions for mild-to-moderate dehydration; rapid IV fluid resuscitation (isotonic 0.9% normal saline or lactated Ringer's) for severe dehydration or shock.

Heat-Related Illnesses (Heat Stroke and Heat Exhaustion)

– Heat-related illnesses range from mild (heat cramps, heat edema, heat syncope) to severe (heat exhaustion, heat stroke). Heat exhaustion: hypovolemia and electrolyte depletion from excessive sweating (headache, dizziness, nausea, fatigue, core temp 37–40 °C).Heatstroke : life − threateningemergencydefinedbycoretemperature > 40 °C(104 °F)accompaniedbycentralnervoussystemdysfunction(alteredmentalstatus, delirium, seizures, coma).Cla thermoregulatoryfailureleadingtodirectcellulartoxicity, SIRS, rhabdomyolysis, AKI, ARDS, DIC, andmulti− organfailure.Treatment : immediateaggressivecooling(coldwaterimmersion, evaporativemistingandfan, icepackstogroin/axillae/neck, coldIV 39 °Cwithin30minutes.

Hyperkalemic Emergency

– Severe hyperkalemia (serum potassium > 6.5 mEq/L) or any hyperkalemia accompanied by ECG changes represents a life-threatening cardiac emergency. ECG progression: peaked T waves, prolonged PR interval, flattened P waves, QRS widening, sine-wave pattern, and ventricular fibrillation/asystole. Staged emergency treatment: (1) Cardiac membrane stabilization: IV calcium gluconate 1–2 g or calcium chloride 1 g over 2–5 minutes; (2) Intracellular potassium shift: IV regular insulin 10 units with 50% dextrose (D50W), inhaled albuterol 10–20 mg, and IV sodium bicarbonate (if metabolic acidosis present); (3) Potassium elimination: loop diuretics (furosemide), gastrointestinal cation exchangers (sodium zirconium cyclosilicate, patiromer), or emergency hemodialysis.

Hypothermia

– A medical emergency defined by a core body temperature below 35 °C(95 °F).Classifiedasmild(32 − −35 °C : shivering, dysarthria, apathy), moderate(28 − −32 °C : cessationofshivering, delirium, OsbornJwavesonEC 28 °C : coma, fixedpupils, severebradycardia, ventricularfibr gentlehandlingtoavoidtriggeringventricularfibrillation; pass −42 °C, heatedhumidifiedoxygen, cavitylavage, ECMO/card to−severehypothermia.Patientsshouldnotbepronounceddeadu 32 °C('notdeaduntilwarmanddead').

Nucleus pulposus

– The gel-like shock-absorbing central portion of each spinal disc.

Obstetric Emergencies

– Life-threatening conditions occurring during pregnancy, labor, or the postpartum period. Includes ectopic pregnancy (ruptured: surgical emergency; stable: methotrexate), preeclampsia/eclampsia (hypertension, proteinuria, seizures; treated with IV magnesium sulfate for seizure prophylaxis and IV labetalol/hydralazine for BP control), placental abruption (painful vaginal bleeding, uterine tenderness), placenta previa (painless vaginal bleeding), and postpartum hemorrhage (PPH: blood loss > 1000 mL, managed with uterine massage, oxytocin, methylergonovine, misoprostol, and tranexamic acid).

Pediatric Emergencies

– Urgent medical conditions in pediatric patients evaluated using the Pediatric Assessment Triangle (PAT: Appearance, Work of Breathing, Circulation to Skin). Common conditions include severe croup (stridor at rest, treated with nebulized racemic epinephrine and dexamethasone 0.6 mg/kg), bronchiolitis (wheezing/crackles in infants, supportive care), pediatric anaphylaxis (epinephrine 0.01 mg/kg IM), febrile seizures, pediatric dehydration, and septic shock (treated with 20 mL/kg isotonic fluid boluses and early broad-spectrum antibiotics).

Post-traumatic headache

– A persistent headache resulting from a head or neck injury, sometimes lasting for a year or more.

Procedural Sedation

– Levels: minimal (anxiolysis), moderate (conscious sedation), deep, general anesthesia. Pre-procedure: fasting (2h clear liquids, 6h light meal), ASA class, airway assessment (Mallampati), consent, monitoring (BP, HR, SpO2, ETCO2, ECG), rescue equipment. Agents: propofol (0.5-1 mg/kg bolus, 2575 mcg/kg/min infusion; hypotension, apnea), ketamine (1-2 mg/kg IV, 4-5 mg/kg IM; preserves airway/respiration, emergence reaction, hypertension), midazolam (0.02-0.04 mg/kg; amnesia, respiratory depression with opioids), fentanyl (1-2 mcg/kg; respiratory depression), etomidate (0.2-0.3 mg/kg; hemodynamic stability, myoclonus, adrenal suppression). Reversal: flumazenil (BZD), naloxone (opioids). Discharge: Aldrete score >=9, responsible adult, no driving.

Psychiatric Emergencies

– Acute behavioral disturbances, severe agitation, suicidal ideation, or psychosis posing immediate danger to the patient or others. Management of the acutely agitated patient: verbal de-escalation techniques first-line; pharmacologic restraint if de-escalation fails (IM olanzapine 10 mg, ziprasidone 20 mg, or haloperidol 5 mg plus lorazepam 2 mg). Suicide risk assessment: evaluation of ideation, plan, intent, and means, requiring safety precautions and involuntary psychiatric hold if imminent risk is identified.

Pulmonary Embolism

– Obstruction of pulmonary arteries by thrombus (usually from DVT). Risk factors: Virchow triad (stasis, hypercoagulability, endothelial injury). Wells score (clinical probability), PERC rule (low risk exclusion). D-dimer (age-adjusted: age x10 ug/L if >50) for low/intermediate probability. CTPA (gold standard); VQ scan (pregnant, contrast allergy); echocardiography (RV strain: McConnell sign, 60/60 sign, TAPSE). Risk stratification: massive (hemodynamic instability, SBP <90 or vasopressors), submassive (normotensive + RV dysfunction + biomarker elevation), low-risk (normotensive, no RV dysfunction). Management: anticoagulation (LMWH, DOAC, UFH, warfarin). Massive PE: systemic thrombolysis (alteplase 100 mg over 2h, tenecteplase 0.5 mg/kg single bolus) if no contraindication; catheter-directed thrombolysis/mechanical thrombectomy if contraindication; surgical embolectomy if failed. Submassive: consider catheter-directed therapy if clinical deterioration. IVC filter if anticoagulation contraindicated.

Snakebite Envenomation

– Envenomation resulting from venomous snakebites (pit vipers: rattlesnakes, copperheads, cottonmouths; elapids: coral snakes). Pit viper envenomation causes localized tissue destruction (pain, progressive edema, bullae, ecchymosis) and systemic manifestations (coagulopathy, thrombocytopenia, hypotension). Management: limb immobilization at heart level, avoidance of harmful remedies (no tourniquets, ice, or incision/suction), and prompt administration of antivenom (Crotalidae polyvalent immune Fab / CroFab or Anavip) for progressive edema, coagulopathy, or systemic toxicity.

Status Epilepticus

– A continuous seizure lasting $\geq$ 5 minutes, or two or more discrete seizures without complete recovery of consciousness between episodes. Staged treatment protocol: Phase 1 (0–5 min): IV lorazepam 4 mg or IM midazolam 10 mg first-line; Phase 2 (5–20 min): IV non-sedating antiseizure medication (fosphenytoin 20 mg PE/kg, levetiracetam 60 mg/kg, or valproate 40 mg/kg); Phase 3 (20–40 min, refractory status epilepticus): continuous IV anesthetic infusion (propofol, midazolam, or pentobarbital) with endotracheal intubation and continuous EEG monitoring.

Tension Pneumothorax

– A life-threatening emergency occurring when air enters the pleural space via a one-way valve mechanism, continuously increasing intrapleural pressure, causing complete collapse of the ipsilateral lung, mediastinal shift to the contralateral side, compression of the vena cava, severely compromised venous return, and obstructive shock. Clinical presentation: severe dyspnea, cyanosis, hypotension, tachycardia, absent breath sounds on the affected side, hyperresonance to percussion, tracheal deviation away from the affected side, and distended neck veins. Immediate intervention: emergency needle decompression (14-gauge needle placed in the 2nd intercostal space midclavicular line or 5th intercostal space anterior axillary line) before chest radiograph, followed by tube thoracostomy (chest tube).

Troponins

– Proteins found in heart muscle that leak into the circulation during a heart attack or other heart injury.

Chapter 8

Endocrinology

Acromegaly

– Chronic disorder of excess growth hormone secretion after epiphyseal closure in adults causing progressive enlargement of acral structures including hands, feet, jaw, and soft tissues. Most commonly caused by a pituitary somatotroph adenoma accounting for approximately ninety-five percent of cases. Clinical features include coarsened facial features, prognathism, macroglossia, widening of the hands and feet, soft tissue swelling causing carpal tunnel syndrome, excessive sweating, hypertension, diabetes mellitus, and cardiovascular disease. Diagnosis is confirmed by elevated IGF-1 levels and failure of growth hormone suppression during oral glucose tolerance testing. Treatment options include transsphenoidal surgery for adenoma resection, medical therapy with somatostatin analogues, dopamine agonists, or growth hormone receptor antagonists, and radiation therapy for residual or refractory disease.

Acromegaly Complications

– Systemic complications of chronic growth hormone excess include diabetes mellitus from insulin resistance, cardiovascular disease with cardiomyopathy and arrhythmias, sleep apnea from macroglossia and upper airway soft tissue hypertrophy, hypertension, colon polyps and increased colorectal cancer risk, osteoarthritis, carpal tunnel syndrome, and increased mortality. Screening colonoscopy is recommended at diagnosis and at regular intervals. Cardiac assessment with echocardiography is important. Regular dermatological surveillance for skin lesions and colonoscopic screening according to standard guidelines are recommended.

Addison Disease

– Primary adrenal insufficiency resulting from destruction or dysfunction of the adrenal cortex, leading to deficiency of cortisol, aldosterone, and adrenal androgens. Most common cause in developed countries is autoimmune adrenalitis (autoimmune polyglandular syndrome type 2), with anti-21-hydroxylase antibodies present in 85-90 percent of cases. Other causes include infections (tuberculosis, histoplasmosis, CMV in HIV/AIDS), hemorrhage, metastatic infiltration, and genetic disorders (congenital adrenal hyperplasia, adrenoleukodystrophy, and autoimmune polyglandular syndrome type 1). Presents with fatigue, hyperpigmentation of the skin and oral mucosa, weight loss, hypotension, and salt craving. Diagnosis is made by elevated ACTH with a low cortisol level and a poor response to cosyntropin stimulation testing. Treatment requires lifelong glucocorticoid and mineralocorticoid replacement therapy with stress dose adjustments during illness or surgery.

Addisonian Crisis

– Life-threatening emergency resulting from acute cortisol deficiency in patients with adrenal insufficiency. Precipitated by infection, trauma, surgery, or abrupt discontinuation of chronic glucocorticoid therapy. Presents with severe hypotension or shock unresponsive to fluids and vasopressors, profound weakness, nausea, vomiting, abdominal pain, fever, confusion, and hypoglycemia. Laboratory findings include hyponatremia, hyperkalemia, hypoglycemia, and eosinophilia. Treatment is immediate IV hydrocortisone, aggressive fluid resuscitation with normal saline and dextrose to correct hypoglycemia, vasopressor support if needed, and treatment of the underlying precipitating cause.

Adrenal Cortex Physiology

– The adrenal cortex produces three classes of steroid hormones from the three zones. The zona glomerulosa produces mineralocorticoids primarily aldosterone regulating sodium and potassium balance under the influence of angiotensin II and potassium. The zona fasciculata produces glucocorticoids primarily cortisol regulating glucose metabolism protein catabolism and the stress response under the influence of ACTH. The zona reticularis produces adrenal androgens including DHEA and DHEA-S under the influence of ACTH.

Adrenal Hemorrhage

– Bleeding into the adrenal gland most commonly occurring in critically ill patients with sepsis, coagulopathy, or anticoagulation therapy. Bilateral adrenal hemorrhage may cause acute adrenal insufficiency with circulatory collapse. Waterhouse-Friderichsen syndrome: adrenal hemorrhage associated with meningococcal sepsis. Diagnosis is by CT showing enlarged adrenal glands with hemorrhage. Treatment includes stress-dose corticosteroids for bilateral involvement.

Adrenal Insufficiency

– Deficient production of cortisol by the adrenal cortex classified as primary adrenal insufficiency or Addison disease from adrenal gland destruction, secondary from pituitary ACTH deficiency, and tertiary from chronic exogenous glucocorticoid suppression of the hypothalamic-pituitary-adrenal axis. Primary causes include autoimmune adrenalitis accounting for eighty percent in developed countries, tuberculosis, fungal infections, metastatic disease, and hemorrhage. Presents with fatigue, weakness, weight loss, hyperpigmentation, hypotension, and salt craving in primary disease, whereas secondary deficiency presents without hyperpigmentation or hyperkalaemia.

Adrenal Medulla Physiology

– The adrenal medulla produces catecholamines primarily epinephrine and norepinephrine in response to sympathetic nervous system activation. Epinephrine accounts for approximately eighty percent and norepinephrine approximately twenty percent of catecholamine output. These hormones mediate the fight-or-flight response through alpha and beta adrenergic receptors causing tachycardia hypertension bronchodilation and metabolic mobilization including glycogenolysis and lipolysis. Pheochromocytomas arise from the adrenal medulla and secrete excess catecholamines causing episodic hypertension, headache, sweating, and palpitations.

Adrenal Myelolipoma

– Benign adrenal tumor composed of mature adipose tissue and hematopoietic elements. Usually asymptomatic and discovered incidentally. No malignant potential. May cause adrenal hemorrhage if large. Diagnosis is by imaging showing fat attenuation on CT. No treatment required unless symptomatic from mass effect or hemorrhage.

Adrenal Venous Sampling

– Interventional radiology procedure for lateralizing aldosterone secretion in primary hyperaldosteronism. Catheterization of both adrenal veins allows measurement of aldosterone and cortisol levels to calculate selectivity index confirming successful cannulation and lateralization ratio. Lateralization to one gland suggests aldosterone-producing adenoma amenable to adrenalectomy. Bilateral secretion suggests bilateral adrenal hyperplasia managed medically. The procedure is technically challenging and requires an experienced interventional radiologist, with success rates of approximately seventy to eighty percent for right adrenal vein cannulation and higher for the left.

Aicardi-Goutieres Syndrome

– Rare autosomal recessive inflammatory disorder mimicking congenital viral infection. Characterized by cerebral calcifications, white matter disease, cerebral atrophy, and cerebrospinal fluid lymphocytosis. Presents with severe neurodevelopmental impairment, seizures, and spasticity. Some forms involve interferon pathway dysregulation with elevated interferon-alpha in cerebrospinal fluid. Diagnosis is by clinical features, brain imaging showing calcifications, and genetic testing. No specific treatment is available; management is supportive and includes anticonvulsants, physiotherapy, and management of feeding difficulties. Prognosis is poor with many children not surviving beyond early childhood.

Alpha-Glucosidase Inhibitors

– Oral glucose-lowering medications including acarbose, miglitol, and voglibose that delay carbohydrate digestion and glucose absorption in the small intestine by inhibiting brush border alpha-glucoside enzymes. Reduce postprandial glucose excursions without affecting fasting glucose significantly. Side effects include flatulence, bloating, and diarrhea from undigested carbohydrates reaching the colon. Contraindicated in inflammatory bowel disease and hepatorenal impairment. Effective A1c reduction of approximately 0.5 to 0.8 percent is modest compared with other oral agents.

Amiodarone-Induced Thyroid Dysfunction

– Thyroid abnormalities occurring in approximately fifteen to twenty percent of patients on amiodarone due to the high iodine content of the molecule. Type one results from iodine-induced increased thyroid hormone synthesis in patients with underlying thyroid disease. Type two results from destructive thyroiditis. Diagnosis is by thyroid function tests, thyroid uptake, and color flow Doppler. Type one is treated with thionamides. Type two is treated with corticosteroids. Amiodarone-induced hypothyroidism is managed with levothyroxine replacement without discontinuing amiodarone if it is clinically indicated. Amiodarone should be discontinued if possible when hyperthyroidism develops.

Amylin Analogues

– Pramlintide is a synthetic analogue of amylin a co-secreted hormone from pancreatic beta cells that slows gastric emptying suppresses postprandial glucagon secretion and promotes satiety. Used as adjunctive therapy with insulin in type one and type two diabetes. Reduces postprandial glucose excursions and A1c. Side effects include nausea and injection site reactions. Risk of hypoglycemia when combined with insulin requiring dose reduction.

Antidiuretic Hormone Physiology

– Antidiuretic hormone also known as vasopressin is synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and stored in the posterior pituitary. Released in response to increased serum osmolality detected by hypothalamic osmoreceptors and decreased blood volume detected by baroreceptors. Acts on the collecting duct through V2 receptors inserting aquaporin-2 water channels causing water reabsorption and urine concentration. Also causes vasoconstriction through V1 receptors at higher concentrations causing vasoconstriction. Deficiency leads to diabetes insipidus, while excess causes syndrome of inappropriate antidiuretic hormone secretion.

Aromatase Excess Syndrome

– Rare autosomal dominant condition caused by mutations in the CYP19A1 gene leading to overexpression of aromatase enzyme and excessive estrogen production. Presents in males with gynecomastia, advanced bone age, and accelerated linear growth with early epiphyseal closure. In females with precocious puberty, macroclitoris, and virilization. Diagnosis is by elevated estradiol and suppressed gonadotropins. Treatment is with aromatase inhibitors anastrozole or letrozole to reduce estrogen synthesis and manage symptoms.

Blood Glucose Monitoring

– Blood sugar monitoring involves measuring glucose levels in the blood to manage diabetes, typically using a glucometer with finger-stick testing or continuous glucose monitors that track interstitial glucose in real time. Regular monitoring helps patients and clinicians adjust insulin doses, diet, and physical activity to maintain target glycemic ranges. Advances include flash glucose monitoring systems and smartphone-integrated sensors that improve convenience and data trends.

Bone Densitometry

– Dual-energy X-ray absorptiometry measures bone mineral density at the lumbar spine and hip. Results are expressed as T-score comparing to young adult mean and Z-score comparing to age-matched controls. T-score of minus 1.0 to minus 2.5 indicates osteopenia. T-score of minus 2.5 or less indicates osteoporosis. T-score of minus 2.5 with fragility fracture indicates severe osteoporosis. Fracture risk assessment using FRAX incorporates clinical risk factors and BMD to estimate ten-year probability of hip fracture over ten years.

Bone Turnover Markers

– Biochemical markers reflecting the rate of bone remodeling. Formation markers include bone-specific alkaline phosphatase, osteocalcin, and N-terminal and C-terminal propeptides of type I procollagen. Resorption markers include C-terminal and N-terminal telopeptides of type I collagen and deoxypyridinoline. Elevated markers indicate increased bone turnover as in primary hyperparathyroidism Paget disease and osteomalacia. Suppressed markers indicate low bone turnover as in adynamic bone disease. Useful for monitoring response to osteoporosis therapy and assessing metabolic bone disease.

Calcitonin

– A 32-amino-acid peptide hormone secreted by the parafollicular C-cells of the thyroid gland in response to elevated serum calcium. It lowers blood calcium by inhibiting osteoclast-mediated bone resorption and increasing renal calcium excretion. Its physiologic role in humans is minor compared to PTH and vitamin D; however, calcitonin is a sensitive tumor marker for medullary thyroid carcinoma (MTC). Synthetic calcitonin (salmon calcitonin, more potent than human) was historically used for Paget disease of bone, osteoporosis, and hypercalcemia but has been largely replaced by bisphosphonates, denosumab, and other agents due to modest efficacy and association with malignancy risk with long-term use.

Calcium Sensing Receptor

– G-protein coupled receptor expressed primarily on parathyroid glands and renal tubular cells that senses extracellular ionized calcium levels. On parathyroid glands activation suppresses PTH secretion. On renal tubular cells activation increases calcium excretion and decreases phosphate reabsorption. Inactivating mutations cause familial hypocalciuric hypercalcemia and neonatal severe hyperparathyroidism. Activating mutations cause autosomal dominant hypocalcemia. The calcium sensing receptor is the target for cinacalcet, a calcimimetic used in secondary hyperparathyroidism of chronic kidney disease and parathyroid carcinoma.

Carcinoid Heart Disease

– Cardiac complication of carcinoid syndrome occurring in approximately fifty percent of patients with hepatic metastases. Serotonin and other vasoactive substances bypass hepatic metabolism and cause endocardial fibrosis affecting the right-sided heart valves particularly the tricuspid and pulmonary valves. Tricuspid regurgitation is the most common finding followed by pulmonary stenosis. Presents with right heart failure including edema ascites and fatigue. Diagnosis is by echocardiography. Treatment includes diuretics for heart failure and valve replacement surgery for severe valvular disease. Somatostatin analogues may slow disease progression.

Catecholamine Crisis

– Acute life-threatening elevation of catecholamines causing severe hypertension, tachycardia, and multi-organ dysfunction. Occurs in pheochromocytoma and paraganglioma precipitated by anesthesia, surgery, trauma, or medications that stimulate catecholamine release. Treatment includes immediate IV phentolamine or nitroprusside for blood pressure control followed by beta-blockade once adequate alpha-blockade is established. Beta-blockers should never be given before alpha-blockade due to the risk of unopposed alpha-adrenergic stimulation causing hypertensive crisis.

Closed-Loop Insulin Delivery

– Automated insulin delivery systems that combine continuous glucose monitoring with an insulin pump and control algorithm. The algorithm adjusts basal insulin delivery based on sensor glucose values in real time. Hybrid closed-loop systems require user-initiated meal boluses but automatically manage basal rates. Fully automated systems are in development. Current systems include Medtronic 780G, Omnipod 5, Tandem Control-IQ, and CamAPS FX. Improve time in range and reduce hypoglycemia compared to conventional insulin pump therapy.

Congenital Adrenal Hyperplasia

– Group of autosomal recessive disorders caused by enzyme deficiencies in the adrenal steroidogenesis pathway. Twenty-one hydroxylase deficiency is the most common accounting for approximately ninety-five percent of cases. Impaired cortisol synthesis leads to elevated ACTH driving adrenal androgen excess causing virilization in females and precocious puberty in males. Severe forms present with salt-wasting crisis in neonates from aldosterone deficiency. Diagnosis is by elevated 17-hydroxyprogesterone levels and genetic testing. Treatment includes glucocorticoid replacement to suppress ACTH and androgen excess, mineralocorticoid replacement for salt-wasting forms, and corrective surgery for virilised genitalia in females.

Congenital Hyperinsulinism

– Condition of inappropriate insulin secretion in neonates and infants causing severe hypoglycemia. Most commonly caused by mutations in the KATP channel genes ABCC8 and KCNJ11. Histologically classified into focal and diffuse forms. Presents with severe symptomatic hypoglycemia in the first days of life. Diagnosis requires demonstration of inappropriately elevated insulin and C-peptide during hypoglycemia. Treatment includes continuous glucose infusion, diazoxide as first-line medical therapy, and octreotide for diazoxide-unresponsive cases. Surgical resection of focal lesions or near-total pancreatectomy for diffuse disease may be necessary.

Congenital Hyperthyroidism

– Thyrotoxicosis present at birth most commonly caused by maternal Graves disease with transplacental passage of thyroid-stimulating immunoglobulins. Presents with tachycardia, irritability, poor feeding, weight loss, and goiter. Transient neonatal hyperthyroidism resolves as maternal antibodies are cleared over weeks to months. Neonatal Graves disease requires treatment with methimazole and beta-blockers. Rare causes include TSH receptor mutations causing constitutive activation and McCune-Albright syndrome.

Congenital Hypothyroidism

– Thyroid hormone deficiency present at birth occurring in approximately one in two thousand to four thousand live births. Most common cause is thyroid dysgenesis including aplasia, ectopia, and hypoplasia. Universal newborn screening with TSH or T4 allows early detection and treatment preventing intellectual disability. Presentation in unscreened infants includes prolonged jaundice, constipation, poor feeding, lethargy, hypotonia, macroglossia, and umbilical hernia. Treatment is levothyroxine replacement starting at ten to fifteen micrograms per kilogram daily with dose adjustment based on thyroid function tests.

Continuous Glucose Monitoring

– Technology measuring interstitial glucose levels continuously providing real-time glucose data, trends, and alerts. Sensors are inserted subcutaneously typically in the abdomen or arm and measure glucose every five minutes. Devices include Dexcom, FreeStyle Libre, and Medtronic Guardian systems. Provides time in range the percentage of time glucose remains between seventy and one hundred eighty milligrams per deciliter, glucose variability metrics, and trend arrows. Alerts notify patients of impending or current hypoglycemia and hyperglycemia. Data can be shared remotely with caregivers and healthcare providers, improving diabetes management.

Cushing Disease

– Hypercortisolism caused by a pituitary ACTH-secreting adenoma, the most common cause of endogenous Cushing syndrome (70%). Presents with central obesity, moon face, buffalo hump, violaceous striae, proximal myopathy, easy bruising, osteoporosis, hypertension, hyperglycemia, depression, and menstrual irregularities. Diagnosis: elevated 24-hour urinary free cortisol, abnormal dexamethasone suppression test, late-night salivary cortisol; ACTH level differentiates ACTH-dependent from independent causes; pituitary MRI localizes adenoma. Treatment: transsphenoidal adenomectomy is first-line; bilateral adrenalectomy for refractory cases.

Cushing Syndrome

– Clinical state resulting from chronic exposure to excess glucocorticoids regardless of cause. Most common endogenous cause is Cushing disease from pituitary ACTH-secreting adenoma accounting for approximately seventy percent. Ectopic ACTH syndrome from non-pituitary tumors including small cell lung cancer and carcinoid tumors accounts for approximately fifteen percent. Adrenal adenomas and carcinomas account for approximately fifteen percent. Exogenous glucocorticoid administration is the most common cause of exogenous Cushing syndrome from prescribed corticosteroids including prednisolone and dexamethasone for inflammatory conditions.

Cushing Syndrome Complications

– Untreated Cushing syndrome carries significant morbidity and mortality from cardiovascular disease, infections, thromboembolic events, and metabolic derangements. Cardiovascular complications include accelerated atherosclerosis, hypertension, and dyslipidemia. Infection risk is increased from immunosuppression. Venous thromboembolism occurs at high rates. Metabolic complications include diabetes, osteoporosis, and myopathy. Psychiatric manifestations include depression, cognitive impairment, and psychosis. Cardiovascular risk remains elevated even after biochemical cure, necessitating long-term surveillance.

Dawn Phenomenon

– Physiological increase in blood glucose occurring in the early morning hours between approximately three and eight AM driven by counter-regulatory hormone surges including growth hormone, cortisol, and catecholamines. More pronounced in patients with diabetes who lack the compensatory insulin response. Can be identified by comparing overnight and fasting glucose values. Differentiated from the Somogyi effect which is rebound hyperglycemia after nocturnal hypoglycemia. Management involves adjustment of basal insulin rates and consideration of continuous glucose monitoring to detect and address early morning hyperglycemia.

Delayed Puberty

– Failure to develop secondary sexual characteristics by age thirteen in girls and age fourteen in boys. Causes include constitutional delay which is the most common, chronic illness, malnutrition, hypogonadotropic hypogonadism from Kallmann syndrome or pituitary disease, and hypergonadotropic hypogonadism from gonadal failure. Diagnosis includes bone age assessment, hormone levels including LH, FSH, testosterone or estradiol, and imaging. Treatment is sex steroid replacement to induce puberty with monitoring of growth and bone age progression. Underlying causes should be addressed where possible.

Denosumab for Osteoporosis

– Fully human monoclonal antibody targeting RANK ligand preventing osteoclast formation activation and survival. Reduces fracture risk at the spine hip and non-vertebral sites. Administered as subcutaneous injection every six months. Advantages include no renal dose adjustment and effectiveness even with oral bisphosphonate intolerance. Disadvantages include risk of vertebral fractures after discontinuation due to rebound bone loss requiring bisphosphonate transition therapy or extended denosumab therapy indefinitely.

Diabetes and Metabolic Health

– Closely linked through insulin resistance, obesity, dyslipidemia, and hypertension, collectively known as the metabolic syndrome. Poor metabolic health increases the risk of developing type 2 diabetes and worsens glycemic control in existing diabetes. Improving metabolic health through weight loss, dietary changes, and physical activity can prevent or reverse type 2 diabetes in some individuals.

Diabetes Complications

– Diabetes complications arise from chronic hyperglycemia damaging blood vessels and nerves over time. Microvascular complications include diabetic retinopathy leading to vision loss, nephropathy progressing to kidney failure, and neuropathy causing pain and loss of sensation. Macrovascular complications include accelerated atherosclerosis manifesting as heart attack, stroke, and peripheral artery disease.

Diabetes insipidus

– A disorder characterized by polyuria (excessive urine output exceeding 40 mL/kg/24 hours) and polydipsia due to deficient secretion of antidiuretic hormone (ADH, vasopressin) from the posterior pituitary (central diabetes insipidus) or renal resistance to ADH action (nephrogenic diabetes insipidus). Central DI is caused by destruction of the hypothalamic-pituitary axis from head trauma, neurosurgery, tumors (craniopharyngioma, germinoma), infiltrative diseases (Langerhans cell histiocytosis, sarcoidosis), and idiopathic causes. Nephrogenic DI is caused by genetic mutations (AVPR2 or AQP2), drugs (lithium, demeclocycline), hypercalcaemia, hypokalaemia, and chronic kidney disease. Diagnosis involves water deprivation test measuring urine osmolality and response to desmopressin. Central DI shows marked improvement with desmopressin, whereas nephrogenic DI shows minimal response. Treatment includes desmopressin for central DI, correction of underlying cause for nephrogenic DI, and thiazide diuretics with a low-sodium diet for partial nephrogenic DI.

Diabetes Medications

– Diabetes medications lower blood glucose through various mechanisms: metformin reduces hepatic glucose production; sulfonylureas stimulate insulin secretion; SGLT2 inhibitors increase urinary glucose excretion; GLP-1 receptor agonists enhance incretin effects; and insulin replaces deficient hormone. Choice of therapy depends on diabetes type, glycemic targets, patient preferences, cardiovascular and renal status, and cost.

Diabetes Mellitus

– A metabolic disorder characterized by chronic hyperglycemia resulting from insufficient insulin production, insulin resistance, or both. Type 1 diabetes results from autoimmune destruction of pancreatic beta cells, while type 2 is driven by genetic and lifestyle factors leading to insulin resistance. Long-term complications affect the eyes, kidneys, nerves, and cardiovascular system; management focuses on glycemic control, diet, exercise, and medications.

Diabetes Mellitus Type 1

– An autoimmune disease characterized by destruction of pancreatic beta cells leading to absolute insulin deficiency. Usually presents in childhood or young adulthood with acute onset of polyuria, polydipsia, weight loss, and diabetic ketoacidosis (DKA) at diagnosis in many. Laboratory: hyperglycemia, positive autoantibodies (GAD, IA-2, ZnT8), low C-peptide. Lifelong insulin therapy required. Intensive glycemic control reduces microvascular and macrovascular complications.

Diabetes Mellitus Type 2

– A progressive metabolic disorder characterized by insulin resistance and relative insulin deficiency, strongly associated with obesity and physical inactivity. Typically presents in adults but increasingly seen in younger populations. Diagnosis: fasting glucose $\geq$126 mg/dL, HbA1c $\geq$6.5%, or OGTT $\geq$200 mg/dL. Management: lifestyle modification, metformin first-line, followed by additional oral agents, GLP-1 receptor agonists, SGLT2 inhibitors, and ultimately insulin as beta-cell function declines.

Diabetes Prevention

– Preventing diabetes focuses on reducing the risk of developing type 2 diabetes through lifestyle interventions and risk factor management. Maintaining a healthy body weight, engaging in regular physical activity, and following a balanced diet low in refined sugars and saturated fats are cornerstone strategies. For individuals with prediabetes, structured programs involving dietary changes, exercise, and sometimes metformin can significantly lower the likelihood of progression to overt diabetes.

Diabetic Foot Disease

– Complex multifactorial condition resulting from the combination of peripheral neuropathy, peripheral arterial disease, and impaired immune function in diabetes. Neuropathy leads to loss of protective sensation and foot deformities. Peripheral arterial disease impairs healing. Minor trauma progresses to ulceration and infection. The Wagner classification grades ulcers from zero to five based on depth and presence of gangrene. Treatment includes offloading, wound care, infection management, revascularisation for peripheral arterial disease, and amputation for non-salvageable tissue loss. Multidisciplinary foot care teams are essential for optimal outcomes.

Diabetic Ketoacidosis

– An acute, life-threatening complication of diabetes mellitus, more common in type 1. Characterized by hyperglycemia (>250 mg/dL), metabolic acidosis (pH <7.3, bicarbonate <15 mEq/L), and ketonemia. Precipitating factors: infection, missed insulin doses, MI, pancreatitis, and stress. Presents with polyuria, polydipsia, nausea, vomiting, abdominal pain, Kussmaul respirations, and altered mental status. Treatment: fluid resuscitation, insulin infusion, potassium replacement, and correction of precipitating cause.

Diabetic Ketoacidosis Monitoring

– Monitoring during DKA management includes hourly blood glucose measurement, serum electrolytes every two to four hours, venous blood gas assessment, beta-hydroxybutyrate levels, and fluid balance. Resolution criteria include blood glucose less than two hundred milligrams per deciliter, bicarbonate greater than fifteen milliequivalents per liter, venous pH greater than seven point three, and anion gap less than twelve. Transition to subcutaneous insulin occurs when resolution criteria are met and the patient is eating and drinking normally. Frequent monitoring of potassium is essential because levels fall during insulin therapy and require aggressive replacement to prevent hypokalaemia.

Diabetic Neuropathy

– Most common complication of diabetes affecting approximately fifty percent of patients. Distal symmetric polyneuropathy presents with stocking-glove sensory loss, burning pain, and loss of proprioception. Autonomic neuropathy causes gastroparesis, orthostatic hypotension, erectile dysfunction, and sudomotor dysfunction. Mononeuropathies include cranial nerve palsies particularly third nerve palsy and carpal tunnel syndrome. Diagnosis is by clinical examination and nerve conduction studies. Treatment includes optimised glycaemic control, symptomatic management of neuropathic pain with agents such as gabapentin, pregabalin, amitriptyline, or duloxetine, and management of autonomic complications.

Diabetic Retinopathy

– A microvascular complication of diabetes mellitus and a leading cause of blindness in working-age adults. Non-proliferative DR (NPDR): microaneurysms, dot-blot hemorrhages, exudates (hard and cotton-wool spots), and venous beading. Proliferative DR (PDR): retinal neovascularization, vitreous hemorrhage, and tractional retinal detachment. Macular edema (DME) can occur at any stage and is the most common cause of vision loss. Risk factors: poor glycemic control, hypertension, duration of diabetes. Management: strict glucose and blood pressure control; anti-VEGF injections (ranibizumab, aflibercept, bevacizumab) and focal laser for DME; panretinal photocoagulation for PDR.

Dipeptidyl Peptidase-4 Inhibitors

– Oral glucose-lowering medications including sitagliptin, saxagliptin, linagliptin, and alogliptin that inhibit the enzyme degrading incretin hormones GLP-1 and GIP. Resulting increase in active incretin levels enhances glucose-dependent insulin secretion and suppresses glucagon. Neutral effect on weight and low hypoglycemia risk. Cardiovascular safety has been demonstrated though some agents showed increased heart failure hospitalizations. Dose adjustment required in renal impairment for all except linagliptin which is hepatobiliary excreted.

Dwarfism

– Short stature resulting from a medical condition, most commonly achondroplasia (the most common form of skeletal dysplasia, caused by gain-of-function mutation in FGFR3, presenting with rhizomelic limb shortening, macrocephaly, frontal bossing, and trident hands). Other causes: growth hormone deficiency, hypothyroidism, Turner syndrome, Noonan syndrome, skeletal dysplasias (osteogenesis imperfecta, spondyloepiphyseal dysplasia), and intrauterine growth restriction. Diagnosis: history, physical exam, growth chart analysis, skeletal survey, genetic testing, and endocrine evaluation. Management: growth hormone therapy for GH deficiency; limb lengthening procedures in selected cases; management of complications (spinal stenosis, obesity, obstructive sleep apnea, otitis media).

Ectopic ACTH Syndrome

– Adrenocorticotropic hormone secretion by non-pituitary tumors causing Cushing syndrome. Most commonly caused by small cell lung cancer and bronchial carcinoid tumors. Other sources include thymic carcinoid, pancreatic neuroendocrine tumors, and medullary thyroid carcinoma. Ectopic ACTH typically causes more severe hypercortisolism than Cushing disease with pronounced hypokalemia and metabolic alkalosis. Diagnosis requires demonstration of non-suppressible cortisol after high-dose dexamethasone suppression test, inferior petrosal sinus sampling for localisation of ACTH source, and imaging of the pituitary and adrenal glands.

Ectopic Thyroid Tissue

– Thyroid tissue located outside the normal pretracheal position resulting from failure of normal thyroid descent during embryological development. Most commonly found at the base of the tongue termed lingual thyroid. May be the only functioning thyroid tissue. Symptoms include dysphagia, dysphonia, and upper airway obstruction. Diagnosis is by imaging with thyroid scan confirming uptake in the ectopic location. Treatment includes levothyroxine supplementation to suppress TSH and reduce ectopic tissue size. Surgical excision is reserved for symptomatic lesions.

Empty Sella Syndrome

– Herniation of subarachnoid space into the sella turcica, compressing the pituitary gland and flattening it against the sellar floor. Primary: due to congenital weakness of the diaphragma sella. Secondary: post-surgery, post-radiation, or pituitary apoplexy (Sheehan syndrome). Most are asymptomatic. Symptoms when present: headache, visual field defects, and endocrine dysfunction (hypopituitarism, growth hormone deficiency, hyperprolactinemia). Diagnosis: MRI shows CSF filling the sella with compressed pituitary. Treatment: endocrine replacement as needed; no treatment for the empty sella itself.

Endocrine Disrupting Chemicals

– Environmental chemicals that interfere with endocrine function including bisphenol A, phthalates, pesticides, and polychlorinated biphenyls. Potential associations with diabetes, obesity, thyroid dysfunction, reproductive disorders, and hormone-sensitive cancers. Evidence is strongest for associations with type two diabetes and obesity. Avoidance strategies include limiting plastic food containers choosing organic produce and avoiding personal care products with parabens and phthalates. Regulatory bodies continue to evaluate the health impact of these chemicals and develop guidelines for safe exposure limits.

Endocrine Glands

– Ductless glands that secrete hormones directly into the bloodstream to regulate distant target organs. Major endocrine glands: hypothalamus (releasing/inhibiting hormones regulating the pituitary), pituitary gland (anterior: GH, TSH, ACTH, FSH, LH, prolactin; posterior: ADH, oxytocin), thyroid gland (T4, T3, calcitonin), parathyroid glands (PTH), adrenal glands (cortex: cortisol, aldosterone, androgens; medulla: epinephrine, norepinephrine), pancreas (islets of Langerhans: insulin, glucagon, somatostatin), ovaries (estrogen, progesterone), testes (testosterone), pineal gland (melatonin), and kidneys (erythropoietin, renin). The endocrine system maintains homeostasis through feedback loops (primarily negative feedback), with the hypothalamic-pituitary axis serving as the master regulator.

Endocrinology

– The branch of medicine concerned with the study, diagnosis, and treatment of disorders of the endocrine system–the network of glands that secrete hormones regulating metabolism, growth and development, tissue function, sexual function, reproduction, sleep, and mood. Endocrinologists manage conditions including diabetes mellitus, thyroid disorders (hyperthyroidism, hypothyroidism, nodules, cancer), adrenal disorders (Cushing syndrome, Addison disease, pheochromocytoma), pituitary disorders (acromegaly, prolactinoma, hypopituitarism), parathyroid and calcium disorders (hyperparathyroidism, osteoporosis), and reproductive endocrinology (infertility, PCOS, menopause). Diagnostic tools: hormone assays, stimulation/suppression tests, and imaging (ultrasound, CT, MRI, nuclear medicine).

Euthyroid Sick Syndrome

– Abnormal thyroid function tests in patients with non-thyroidal illness (critical illness, starvation, major surgery) without intrinsic thyroid disease. Pattern: low T3 (most common), low T4, reverse T3 elevated, normal or low TSH. The severity correlates with illness severity. In critical illness: low T3 and T4 are associated with worse prognosis. T4 and TSH may rise during recovery. No treatment indicated; thyroid hormone replacement does not improve outcomes. Correct the underlying illness.

Familial Hypocalciuric Hypercalcemia

– Autosomal dominant condition caused by inactivating mutations in the calcium-sensing receptor gene leading to a higher set point for calcium-mediated PTH suppression. Presents with mild hypercalcemia, inappropriately normal or mildly elevated PTH, and low urinary calcium excretion with calcium-to-creatinine clearance ratio less than 0.01. Benign condition that does not require parathyroidectomy. Distinguished from primary hyperparathyroidism by urinary calcium excretion to prevent unnecessary surgery. Genetic testing confirms the diagnosis and identifies the specific mutation in the calcium-sensing receptor gene for definitive classification.

Follicular Thyroid Carcinoma

– Second most common thyroid carcinoma accounting for approximately ten percent of cases. Distinguished from follicular adenoma by capsular or vascular invasion on histological examination which cannot be determined by fine-needle aspiration. Metastasizes hematogenously to bone, lung, and liver. Treatment includes total thyroidectomy with radioactive iodine ablation. Prognosis is generally favorable though worse than papillary carcinoma particularly with distant metastases.

Galactina

– A hormone produced by the placenta that stimulates prolactin secretion from the anterior pituitary. Involved in mammary gland development during pregnancy and the onset of lactation postpartum. Elevated levels are associated with pathological galactorrhoea and certain pituitary tumors. Measurement is a component of the assessment of hyperprolactinaemia.

Galactorrhoea

– The spontaneous flow of milk from the breast not associated with childbirth or breastfeeding. Galactorrhoea can be unilateral or bilateral and is caused by hyperprolactinaemia (elevated prolactin). Aetiologies: pituitary prolactinoma (most common), hypothalamic-pituitary stalk compression, medications (antipsychotics, metoclopramide, verapamil, oral contraceptives, opioids), hypothyroidism (TRH stimulates prolactin release), chronic renal failure, chest wall irritation (herpes zoster, surgery, trauma). Galactorrhoea may also be idiopathic. Diagnostic workup: serum prolactin, TSH, pregnancy test (beta-hCG), MRI pituitary with contrast. Management: treat underlying cause; dopamine agonists (cabergoline, bromocriptine) for prolactinoma.

Galactosaemia

– A rare autosomal recessive disorder of galactose metabolism caused by deficiency of one of three enzymes: galactose-1-phosphate uridyltransferase (GALT, classic galactosaemia, type I), galactokinase (GALK, type II), or UDP-galactose epimerase (GALE, type III). Classic galactosaemia (GALT deficiency, incidence 1:30,000-60,000) presents in the neonatal period within days of milk feeding: vomiting, diarrhea, failure to thrive, jaundice, hepatomegaly, cataracts (due to galactitol accumulation), hypoglycemia, and E. coli sepsis. Diagnosis: newborn screening (GALT enzyme activity in dried blood spot), urine reducing substances. Long-term complications despite dietary restriction: speech and learning difficulties, ataxia, ovarian failure (in females), decreased bone density. Treatment: lifelong lactose/galactose-free diet.

Gastrinoma

– Gastrin-secreting neuroendocrine tumor causing Zollinger-Ellison syndrome characterized by gastric acid hypersecretion leading to severe peptic ulcer disease, diarrhea, and esophagitis. Approximately sixty percent are malignant with liver metastases. Multiple endocrine neoplasia type 1 association in approximately twenty percent. Diagnosis is by elevated fasting gastrin level and positive secretin stimulation test. Treatment includes high-dose proton pump inhibitors for acid control and surgical resection of localised disease. Somatostatin analogues control symptoms in functioning tumors. Chemotherapy and peptide receptor radionuclide therapy are options for metastatic disease.

Gestational Diabetes Mellitus

– Glucose intolerance first recognized during pregnancy affecting approximately six to nine percent of pregnancies. Results from insulin resistance induced by placental hormones including human placental lactogen, cortisol, and progesterone combined with insufficient beta cell compensation. Risk factors include advanced maternal age, obesity, family history of diabetes, previous gestational diabetes, polycystic ovary syndrome, and certain ethnicities. Universal screening is performed at twenty-four to twenty-eight weeks of gestation with a seventy-five gram oral glucose tolerance test. Treatment includes lifestyle modification and insulin or metformin if glycaemic targets are not met. Glucose levels normalise after delivery but affected women have increased long-term risk of type two diabetes.

Gigantism

– A rare condition of excessive growth and tall stature caused by growth hormone (GH) hypersecretion that begins before epiphyseal plate closure in childhood, before the onset of puberty. The most common cause is a GH-secreting pituitary adenoma (usually a benign somatotroph adenoma, often macroadenoma greater than 1 cm), which may be sporadic or associated with genetic syndromes (McCune-Albright syndrome, multiple endocrine neoplasia type 1, Carney complex, familial isolated pituitary adenoma from germline mutations in the AIP gene). Presents with accelerated linear growth, tall stature, large hands and feet, coarse facial features, and delayed skeletal maturation. Diagnosis is confirmed by elevated IGF-1 and GH levels that fail to suppress on glucose tolerance testing. Pituitary MRI identifies the responsible adenoma. Treatment includes transsphenoidal adenomectomy, medical therapy with somatostatin analogues, and rarely radiotherapy. Early diagnosis and intervention improve final height outcomes.

Glucagonoma

– A rare pancreatic neuroendocrine tumor secreting glucagon. Classic presentation: 4 Ds — diabetes mellitus (mild, new onset), dermatitis (necrolytic migratory erythema, characteristic painful, erythematous, blistering rash on groin, perineum, extremities), diarrhea, and deep vein thrombosis. Presents with weight loss, glossitis, and hypoaminoacidemia. Most are malignant and metastatic at diagnosis. Diagnosis: elevated glucagon level (>500 pg/mL) and pancreatic tumor on imaging. Treatment: surgical resection; somatostatin analogs for symptom control; chemotherapy and PRRT for advanced disease.

Glucocorticoid Replacement Therapy

– Replacement of cortisol deficiency in adrenal insufficiency using hydrocortisone in divided doses. Typical regimen is ten to fifteen milligrams in the morning and five to ten milligrams in the early afternoon mimicking the physiological cortisol diurnal rhythm. Alternative preparations include prednisolone and dexamethasone though these have longer half-lives and are less physiological. Fludrocortisone is added for mineralocorticoid replacement in primary adrenal insufficiency. Stress dosing requires doubling or tripling of the daily hydrocortisone dose during intercurrent illness, fever, or minor surgery, with parenteral administration for major surgery or when oral intake is not possible. Patient education about sick-day rules and provision of an emergency injectable glucocorticoid kit are essential to prevent adrenal crisis.

Glucocorticoids

– A class of steroid hormones produced by the zona fasciculata of the adrenal cortex, regulated by the hypothalamic-pituitary-adrenal (HPA) axis. Cortisol is the primary endogenous glucocorticoid, essential for: (1) carbohydrate metabolism (gluconeogenesis, glycogenolysis, peripheral insulin resistance); (2) protein metabolism (catabolism in muscle, connective tissue, bone); (3) fat metabolism (lipolysis, fat redistribution to face and trunk); (4) anti-inflammatory and immunosuppressive effects (inhibit phospholipase A2, COX-2, pro-inflammatory cytokines); (5) stress response; (6) maintenance of blood pressure (permissive effect on catecholamines). Synthetic glucocorticoids (prednisolone, dexamethasone, betamethasone, hydrocortisone) are used therapeutically for: autoimmune and inflammatory diseases (rheumatoid arthritis, asthma, inflammatory bowel disease, SLE), allergic reactions, organ transplant rejection, adrenal insufficiency (replacement therapy), cerebral edema (dexamethasone), and fetal lung maturation (betamethasone). Side effects of chronic use: Cushing syndrome (central obesity, moon face, buffalo hump, striae), osteoporosis, diabetes, hypertension, immunodeficiency, cataract, growth suppression in children.

Glucose Management Indicator

– Derived metric from continuous glucose monitoring that estimates the hemoglobin A1c equivalent based on mean sensor glucose values. Formula converts mean glucose to an A1c-equivalent value allowing comparison between CGM data and traditional A1c. Provides a more comprehensive picture of glycemic control than A1c alone because it reflects glucose variability and the complete glucose profile rather than a weighted average.

Glycaemia

– The concentration of glucose in the blood, normally maintained within a narrow range (fasting: 3.9-5.5 mmol/L or 70-100 mg/dL) through the coordinated action of insulin (lowers glucose) and counter-regulatory hormones (glucagon, cortisol, catecholamines, growth hormone). Disorders of glycaemia: hyperglycemia (elevated glucose, diagnostic of diabetes mellitus when fasting >7.0 mmol/L or HbA1c >48 mmol/mol), hypoglycemia (low glucose, <3.0 mmol/L clinically significant, causes: insulin overdose, sulfonylurea excess, insulinoma, liver failure, alcohol excess). Glycaemic control is assessed by: self-monitored blood glucose (SMBG), continuous glucose monitoring (CGM), HbA1c (glycated haemoglobin, reflects average glucose over 2-3 months).

Glycosuria

– The presence of glucose in the urine. Normally, the renal tubules reabsorb filtered glucose via sodium-glucose co-transporters (SGLT2 in proximal tubule). Glycosuria occurs when blood glucose exceeds the renal threshold (approximately 10 mmol/L or 180 mg/dL), overwhelming reabsorptive capacity. Causes: (1) hyperglycemia (diabetes mellitus, most common); (2) renal glycosuria (benign condition with low renal threshold, due to SGLT2 mutation); (3) pregnancy (decreased renal threshold due to increased GFR). Diagnostic significance: glycosuria is a screening finding for diabetes; its absence does not exclude diabetes. SGLT2 inhibitors (dapagliflozin, empagliflozin) are a class of antidiabetic drugs that intentionally induce glycosuria by blocking renal glucose reabsorption.

Goiter

– Enlargement of the thyroid gland, which can be diffuse or nodular. Causes: iodine deficiency (most common worldwide), Hashimoto thyroiditis, Graves disease, multinodular goiter, and thyroid cancer. May be associated with normal, hyperthyroid, or hypothyroid function. Large goiters can cause compressive symptoms: dysphagia, dysphonia, and stridor. Diagnosis: thyroid ultrasound, TSH, and thyroid scan. Treatment depends on etiology and symptoms.

Gonadotrophins

– Hormones that stimulate the gonads. The anterior pituitary secretes follicle-stimulating hormone (FSH) and luteinising hormone (LH), both glycoprotein hormones sharing a common alpha subunit (identical to TSH and hCG alpha) with a unique beta subunit. FSH: females–stimulates ovarian follicle growth and estrogen production; males–stimulates Sertoli cells for spermatogenesis. LH: females–triggers ovulation and corpus luteum formation (progesterone); males–stimulates Leydig cells to produce testosterone. Human chorionic gonadotrophin (hCG) is produced by the placenta and maintains the corpus luteum in early pregnancy. Therapeutically, gonadotrophins are used in assisted reproductive technology (IVF, ovulation induction, spermatogenesis induction).

Gonads

– The primary reproductive organs: testes in males and ovaries in females. Functions: (1) gamete production (spermatogenesis in testes, oogenesis in ovaries); (2) sex steroid hormone secretion (testosterone from testicular Leydig cells; estrogen and progesterone from ovarian follicular cells and corpus luteum). Gonadal function is regulated by the hypothalamic-pituitary-gonadal axis: GnRH from hypothalamus stimulates gonadotrophins (LH and FSH) from anterior pituitary, which act on the gonads. Disorders: hypogonadism (primary: gonadal failure; secondary: pituitary/hypothalamic deficiency), hypergonadism (precocious puberty), gonadal dysgenesis (Turner syndrome, Klinefelter syndrome).

Graves Disease

– Autoimmune disorder and most common cause of hyperthyroidism characterized by thyroid-stimulating immunoglobulins that bind and activate the thyroid-stimulating hormone receptor causing unregulated thyroid hormone synthesis and secretion. The diffuse goiter reflects TSH receptor stimulation. Ophthalmopathy occurs in approximately twenty-five percent with proptosis, periorbital edema, diplopia, and in severe cases optic nerve compression. Pretibial myxedema is a localized dermopathy. Diagnosis is confirmed by elevated free T4 and suppressed TSH, positive thyroid-stimulating immunoglobulins, and increased radioactive iodine uptake on thyroid scintigraphy.

Growth Hormone Deficiency

– Insufficient secretion of growth hormone from the anterior pituitary, causing short stature in children and metabolic effects in adults. In children: delayed growth, short stature, delayed bone age, and increased fat mass. In adults: decreased muscle mass, increased fat mass, reduced bone density, fatigue, and dyslipidemia. Diagnosis: low IGF-1, GH stimulation testing. Treatment: recombinant human GH (somatropin) replacement.

Growth Hormone Physiology

– Growth hormone is secreted in a pulsatile pattern by pituitary somatotroph cells regulated by hypothalamic GHRH which stimulates release and somatostatin which inhibits release. GH acts directly on tissues and indirectly through IGF-1 produced primarily by the liver. GH promotes linear growth in children and in adults regulates body composition with anabolic effects on muscle catabolic effects on adipose tissue effects on bone metabolism and glucose homeostasis. GH secretion decreases with aging, contributing to sarcopenia and increased adiposity in the elderly. Testing for GH deficiency requires provocative testing with insulin tolerance test or glucagon stimulation test, as random GH levels are unreliable due to pulsatile secretion.

Gynaecomastia

– The benign enlargement of breast tissue in males. Pathophysiology: an imbalance between estrogen (stimulates breast tissue growth) and androgen (suppresses) activity, resulting in increased estrogen-to-testosterone ratio. Gynaecomastia can be: (1) Physiologic: neonatal (transient, maternal estrogen), pubertal (60-70% of boys, peak age 13-14, typically resolves within 2 years), senescent (ageing-related increase in aromatase activity and decrease in testosterone). (2) Pathologic: hypogonadism (primary–

Klinefelter syndrome, secondary–pituitary failure), hyperthyroidism (increased aromatisation), liver cirrhosis (impaired estrogen metabolism), testicular tumors (Leydig cell, Sertoli cell, choriocarcinoma–hCG stimulates estrogen), renal failure, refeeding after starvation. (3) Drug-induced: anti-androgens (spironolactone, cimetidine, finasteride, bicalutamide), estrogens, anabolic steroids, GnRH agonists, calcium channel blockers, digoxin, isoniazid, HIV drugs, tricyclic antidepressants, methadone, marijuana. Diagnosis: true gynaecomastia (palpable concentric disc of firm tissue >0.5 cm) vs pseudogynaecomastia (adipomastia, fatty enlargement without breast tissue). Workup: breast examination, testicular exam, LFT, TFT, serum hCG, LH, FSH, testosterone (total and free), prolactin, estradiol. Imaging: mammography if asymmetric or suspicious. Management: reassurance for physiologic gynaecomastia (spontaneous regression); discontinuation of offending drugs; medical therapy: tamoxifen (SERM, reduces estrogen effect on breast) or raloxifene; severe/long-standing cases (fibrotic stage >12 months): subcutaneous mastectomy (liposuction or surgical excision).

Hashimoto Thyroiditis

– Autoimmune thyroiditis characterized by lymphocytic infiltration and progressive destruction of thyroid follicles leading to hypothyroidism. Most common cause of hypothyroidism in iodine-sufficient areas. Anti-thyroid peroxidase and anti-thyroglobulin antibodies are present. The thyroid may be diffusely enlarged and firm or atrophic. Patients present with fatigue, weight gain, cold intolerance, constipation, dry skin, and depression. Subclinical hypothyroidism with elevated TSH and normal free T4 levels is common and may progress to overt hypothyroidism. Treatment is levothyroxine replacement titrated to achieve normal TSH levels.

Healthy Eating for Diabetes

– A dietary approach designed to maintain stable blood glucose levels in individuals with diabetes mellitus. It emphasizes carbohydrate consistency, high fiber intake, lean proteins, and healthy fats while minimizing refined sugars and simple carbohydrates. Meal planning often involves carbohydrate counting or the plate method, and coordination with medication and physical activity is essential.

Hemoglobin A1c

– Glycated hemoglobin reflecting average blood glucose concentration over the preceding two to three months corresponding to the red blood cell lifespan. Formed by non-enzymatic glycation of hemoglobin at the N-terminal valine of the beta chain. The percent of glycated hemoglobin is directly proportional to mean glucose levels. Used for diagnosis of diabetes with greater than or equal to 6.5 percent indicating diabetes, 5.7 to 6.4 percent indicating prediabetes, and less than 5.7 percent indicating normoglycaemia. Limitations include interference from haemoglobinopathies, anaemia, and conditions affecting red blood cell lifespan, which may give falsely low or high values independent of glycaemic control.

Hemoglobin A1c Variability

– Factors that affect hemoglobin A1c accuracy include hemoglobinopathies such as sickle cell disease and thalassemias which alter red blood cell lifespan and glycation rates, iron deficiency anemia which shortens red blood cell survival lowering A1c, vitamin B12 and folate deficiency, pregnancy, chronic kidney disease, erythropoietin therapy, and recent blood transfusion. Conditions that shorten red blood cell lifespan lower A1c while those that prolong lifespan raise A1c independent of glycemic control. When A1c and glucose data are discordant, alternative methods such as fructosamine or glycated albumin may provide useful complementary information, and a careful review for conditions affecting red blood cell turnover is warranted.

Hirsutism

– Excess terminal hair growth in women in a male-pattern distribution (face–upper lip, chin, sideburns; chest, back, lower abdomen, inner thighs). Hirsutism affects 5-10% of women of reproductive age. Pathophysiology: increased androgen production or increased end-organ sensitivity. The modified Ferriman-Gallwey (mFG) score quantifies hirsutism (score >8 indicates hirsutism). Causes: (1) Polycystic ovary syndrome (PCOS, 70-80% of cases): elevated LH:FSH ratio, hyperandrogenism, anovulation. (2) Idiopathic hirsutism (15-20%): normal serum androgens, increased 5-alpha-reductase activity (converts testosterone to DHT in the hair follicle). (3) Non-classic congenital adrenal hyperplasia (NCCAH, 5-10%): 21-hydroxylase deficiency (elevated 17-hydroxyprogesterone). (4) Androgen-secreting tumors (ovarian or adrenal, rare, rapidly progressive): testosterone >200 ng/dL, DHEAS >700 mcg/dL. (5) Cushing syndrome: central obesity, striae, hypertension, hyperglycemia. (6) Medications: danazol, testosterone, anabolic steroids, ciclosporin, valproate, phenytoin. (7) Other: HAIR-AN syndrome (hyperandrogenism, insulin resistance, acanthosis nigricans), hyperprolactinaemia. Diagnosis: serum total and free testosterone, SHBG, DHEAS, 17-OHP, LH, FSH, prolactin, TSH, free T4; pelvic ultrasound for PCOS. Management: (1) Mechanical: hair removal by shaving, waxing, electrolysis, laser hair removal (most effective, targets melanin in hair bulb). (2) Pharmacological: combined oral contraceptive pill (first line for PCOS, reduces ovarian androgen production, increases SHBG), anti-androgens (spironolactone 100-200 mg/day, finasteride, cyproterone acetate, flutamide–teratogenic, requires contraception), metformin (insulin sensitizer, modest benefit), eflornithine cream (topical, inhibits ornithine decarboxylase, slows hair growth). (3) Treatment of underlying cause (CAH: glucocorticoids; tumor: surgical removal).

Hormone

– A chemical messenger produced by endocrine glands, secreted into the bloodstream, and transported to distant target cells where it binds to specific receptors to exert a physiological effect. Hormones regulate: growth and development (growth hormone, thyroid hormone, sex steroids), metabolism (insulin, glucagon, cortisol, thyroid hormone, adrenaline), reproduction (LH, FSH, estrogen, progesterone, testosterone, prolactin), fluid and electrolyte balance (ADH, aldosterone), calcium homeostasis (PTH, calcitonin, vitamin D), and stress response (cortisol, adrenaline, noradrenaline). Hormone classification: (1) Peptide/protein hormones (water-soluble): hypothalamic releasing factors, pituitary hormones, insulin, glucagon, PTH, calcitonin; bind to cell surface receptors (GPCRs, receptor tyrosine kinases), act via second messengers (cAMP, IP3/DAG, Ca2+). (2) Steroid hormones (lipid-soluble): cortisol, aldosterone, estrogen, progesterone, testosterone, vitamin D; synthesised from cholesterol, bind to intracellular receptors (cytoplasmic/nuclear receptors), act as transcription factors (slow genomic effects, hours-days). (3) Amino acid-derived: thyroid hormones (T3/T4, iodothyronines, nuclear receptors), catecholamines (adrenaline/noradrenaline, GPCRs), melatonin. (4) Prostaglandins/leukotrienes (eicosanoids): paracrine/autocrine. Hormone secretion is regulated by feedback loops (most commonly negative feedback: e.g., T4 inhibits TRH and TSH; insulin lowers glucose which reduces further insulin secretion) and by the hypothalamic-pituitary axis. Disorders: overproduction (hyperfunction) or underproduction (hypofunction) of hormones cause endocrine diseases.

Hormone Replacement Therapy (HRT)

– The administration of exogenous estrogen (with or without progestogen) to relieve symptoms of estrogen deficiency and prevent long-term health consequences in postmenopausal women. Indications: moderate-severe vasomotor symptoms (hot flushes, night sweats, 80% of women experience at some point), urogenital atrophy (vaginal dryness, dyspareunia, recurrent UTIs), prevention of osteoporosis (in women at high fracture risk who cannot tolerate bisphosphonates), and premature ovarian insufficiency (POI, <40 years, HRT needed at least until average age of menopause). Estrogen-only: for women post-hysterectomy; reduces risk of endometrial cancer. Combined estrogen-progestogen: for women with intact uterus (unopposed estrogen increases endometrial cancer risk; progestogen protects by inducing endometrial shedding). Administration routes: oral (conjugated equine estrogens 0.3-0.625 mg, estradiol 1-2 mg; medroxyprogesterone acetate 2.5-5 mg), transdermal (patches/gel, avoids first-pass hepatic metabolism, lower VTE risk), vaginal (cream, ring, tablets for urogenital symptoms only, minimal systemic absorption). Risks: VTE (2-4 fold increased, highest in first year), breast cancer (5-6 extra cases per 10,000 women-years with combination HRT >5 years, lower with estrogen-only), stroke, gallbladder disease, endometrial cancer (if unopposed estrogen). The Women's Health Initiative (WHI) 2002 found overall risks outweighed benefits for chronic disease prevention in older women (mean age 63), leading to reduced HRT use. Current guidelines (NICE, IMS): HRT is safe and effective for symptom management in women aged <60 or <10 years from menopause onset; lowest effective dose for shortest duration (no arbitrary upper limit for symptom management); tibolone (synthetic steroid) is an alternative. Contraindications: breast cancer (current or prior), estrogen-dependent tumors, unexplained vaginal bleeding, active VTE, history of VTE if not anticoagulated, arterial disease, pregnancy, porphyria.

Hungry Bone Syndrome

– Severe hypocalcemia occurring after parathyroidectomy particularly in patients with severe preoperative hyperparathyroidism and high bone turnover. Rapid bone remineralization after removal of excess PTH sequesters calcium and phosphorus from the serum causing symptomatic hypocalcemia. Risk factors include severe preoperative hyperparathyroidism, high alkaline phosphatosis, and osteitis fibrosa cystica. Prevention includes preoperative calcium and vitamin D supplementation. Treatment is aggressive intravenous and oral calcium and calcitriol supplementation, often requiring high doses that are gradually tapered as bone turnover normalises over weeks to months.

Hyperaldosteronism Complications

– Beyond hypertension and hypokalemia, primary hyperaldosteronism causes target organ damage including left ventricular hypertrophy, fibrosis, stroke, coronary artery disease, atrial fibrillation, and chronic kidney disease. These complications occur at higher rates than in essential hypertension of similar severity. Cardiovascular risk persists even after treatment of the hyperaldosteronism supporting early diagnosis and intervention. Screening is recommended for all patients with resistant hypertension, spontaneous or diuretic-induced hypokalaemia, or adrenal incidentaloma, and is cost-effective given the high prevalence and cardiovascular morbidity.

Hypercalcemia

– Serum calcium greater than 10.5 milligrams per deciliter. Causes summarized by mnemonic mnemonic MALIGNANT: Myeloma, Metastases, Metabolic bone disease, Hyperparathyroidism, Adrenal insufficiency, Paget disease, Immobilization, Thiazide diuretics, Addison disease, Neonatal, Tumor-derived PTHrP. Symptoms include stones bones groans and psychic moans referring to nephrolithiasis, bone pain, constipation, and depression. Treatment includes saline hydration, loop diuretics after hydration, bisphosphonates for sustained calcium lowering, and denosumab for refractory hypercalcaemia. Underlying cause should be addressed.

Hyperglycemia

– Elevated blood glucose levels above normal range (>140 mg/dL postprandial or >126 mg/dL fasting). Acute hyperglycemia can cause osmotic diuresis, dehydration, and electrolyte disturbances. Chronic hyperglycemia leads to microvascular (retinopathy, nephropathy, neuropathy) and macrovascular complications. Management: treat underlying diabetes, insulin or oral hypoglycemics as indicated, and address precipitating factors.

Hypernatremia

– Serum sodium greater than 145 mEq/L. Always indicates free water deficit relative to sodium. Causes include inadequate water intake particularly in elderly and cognitively impaired patients, diabetes insipidus, osmotic diuresis from hyperglycemia, and GI or renal water losses. Treatment is free water replacement with rate of correction dependent on chronicity. Overly rapid correction risks cerebral edema particularly in chronic hypernatremia. D5W or hypotonic saline is used for correction.

Hyperosmolar Hyperglycemic State

– Hyperglycemic emergency characterized by severe hyperglycemia greater than 600 milligrams per deciliter, hyperosmolality greater than 320 milliosmoles per kilogram, and profound dehydration without significant ketosis. Primarily occurs in type two diabetes where enough endogenous insulin is present to prevent ketogenesis but insufficient to control glucose. Presents with extreme dehydration, altered mental status, seizures, and focal neurological deficits. Mortality rates range from five to twenty percent. Treatment is aggressive IV fluid resuscitation, insulin infusion to lower glucose and shift fluids intracellularly, and careful electrolyte monitoring and replacement.

Hyperparathyroidism

– A disorder of excessive parathyroid hormone (PTH) secretion from one or more of the four parathyroid glands, classified as primary, secondary, or tertiary. Primary hyperparathyroidism is caused by a single parathyroid adenoma (80-85 percent), multigland hyperplasia (10-15 percent, associated with MEN1 and MEN2A), or parathyroid carcinoma (less than 1 percent). It is the most common cause of hypercalcemia in the outpatient setting, with a prevalence of 1-4 per 1000 adults, most common in postmenopausal women with a prevalence of approximately two percent in those over fifty-five years. Symptoms of hypercalcaemia include polyuria, polydipsia, nephrolithiasis, bone pain, constipation, fatigue, and cognitive dysfunction. Severe hypercalcaemia above fourteen milligrams per decilitre may cause altered mental status, cardiac arrhythmias, and acute kidney injury. Diagnosis requires elevated serum calcium with inappropriately normal or elevated PTH. Localisation is performed with sestamibi scan, ultrasound, or four-dimensional CT to identify the abnormal parathyroid gland. Parathyroidectomy is curative for symptomatic or complicated disease. Asymptomatic patients may be monitored if they meet specific criteria including age, calcium level, bone density, and renal function parameters.

Hyperprolactinemia

– Elevated serum prolactin levels most commonly caused by prolactin-secreting pituitary adenomas termed prolactinomas. Other causes include medications particularly antipsychotics and metoclopramide, hypothyroidism, stalk effect from nonfunctioning pituitary adenomas, renal failure, and chest wall lesions. Women present with galactorrhea, amenorrhea, and infertility. Men present with decreased libido, erectile dysfunction, and rarely galactorrhea. Macroprolactinomas may present with mass effects including headache, visual field defects, and hypopituitarism. Diagnosis includes serum prolactin measurement to exclude stalk effect, MRI of the pituitary, and full pituitary hormone assessment to evaluate for hypopituitarism. Treatment of prolactinomas is with dopamine agonists, while non-functioning adenomas causing mass effects are managed with transsphenoidal surgery.

Hyperthyroidism

– A condition of excess thyroid hormone production and secretion from the thyroid gland, resulting in thyrotoxicosis (elevated serum T4 and T3 concentrations). The most common cause is Graves disease (60-80 percent of cases), an autoimmune disorder caused by thyroid-stimulating immunoglobulins (TSI) that bind to and activate the TSH receptor on thyroid follicular cells, leading to unregulated thyroid hormone production and thyroid gland enlargement. Other causes include toxic multinodular goiter (toxic nodular goiter) caused by autonomously functioning thyroid nodules in older adults with long-standing multinodular goiter, often presenting with subclinical hyperthyroidism, atrial fibrillation, and cardiovascular complications. Other causes include thyroiditis (subacute, silent, postpartum), exogenous thyroid hormone excess, and rare causes such as struma ovarii, TSH-secreting pituitary adenoma, and hCG-mediated hyperthyroidism in trophoblastic disease. Clinical features reflect increased metabolic rate: weight loss despite increased appetite, heat intolerance, diaphoresis, tremor, palpitations, tachycardia, atrial fibrillation, anxiety, irritability, insomnia, frequent bowel movements, oligomenorrhoea, and proximal myopathy. In older patients, apathetic hyperthyroidism may present with lethargy, depression, weight loss, and atrial fibrillation without classic signs of adrenergic excess. Laboratory findings include suppressed TSH with elevated free T4 and T3. T3 toxicosis, where only T3 is elevated with normal T4, may occur early in Graves disease or in toxic nodular goiter. Thyroid scintigraphy shows diffuse increased uptake in Graves disease, heterogeneous uptake in toxic multinodular goiter, and a focal hot nodule in toxic adenoma, whereas uptake is low in thyroiditis.

Hyperthyroidism (Thyrotoxicosis)

– See Hyperthyroidism.

Hypocalcemia

– Serum calcium less than 8.5 milligrams per deciliter or ionized calcium less than 4.5 milligrams per deciliter. Causes include hypoparathyroidism, vitamin D deficiency, hypomagnesemia, hyperphosphatemia, and critical illness. Symptoms include perioral numbness, paresthesias, muscle cramps, carpopedal spasm, Chvostek sign, Trousseau sign, seizures, and QTc prolongation. Acute symptomatic hypocalcemia is treated with IV calcium gluconate. Chronic management includes oral calcium and calcitriol supplementation, along with magnesium repletion if hypomagnesaemia is present to correct the impaired PTH secretion.

Hypoglycemia

– Low blood glucose (<70 mg/dL) with autonomic (sweating, palpitations, tremor, anxiety) and neuroglycopenic (confusion, seizures, coma) symptoms. In diabetic patients, common causes: excessive insulin or sulfonylurea doses, missed meals, exercise, and alcohol intake. Severe hypoglycemia requires glucagon or IV dextrose. Whipple triad: symptoms consistent with hypoglycemia, low plasma glucose, and symptom resolution with glucose administration.

Hypoglycemia in Diabetes

– Blood glucose less than 70 milligrams per deciliter in patients with diabetes. Classification by level: level one less than 70, level two less than 54 which is clinically significant hypoglycemia, and level three severe hypoglycemia requiring assistance. Causes include insulin or sulfonylurea overdose, missed meals, excessive exercise, and alcohol consumption. Counter-regulatory hormone responses including glucagon and epinephrine cause tremor, sweating, palpitations, and anxiety. Neuroglycopenic symptoms including confusion, drowsiness, visual disturbance, slurred speech, seizures, and loss of consciousness occur with progressive cerebral glucose deprivation. Treatment is immediate ingestion of fast-acting carbohydrate for conscious patients, or intramuscular glucagon or intravenous dextrose for severe hypoglycemia with altered consciousness.

Hypoglycemia Unawareness

– Loss of the warning symptoms of hypoglycemia including tremor, sweating, and palpitations occurring in patients with recurrent hypoglycemia particularly type one diabetes. Results from attenuated counter-regulatory hormone responses particularly impaired glucagon and epinephrine responses. Significantly increases the risk of severe hypoglycemia. Associated with long diabetes duration, recurrent hypoglycemic episodes, and autonomic neuropathy. Management includes liberalization of glycemic targets, structured education in hypoglycemia recognition, avoidance of hypoglycemia to restore symptomatic awareness, and use of continuous glucose monitoring with alerts.

Hypogonadism

– Decreased functional activity of the gonads (testes or ovaries) with reduced sex hormone production. Male hypogonadism: low testosterone causing decreased libido, erectile dysfunction, infertility, reduced muscle mass, and osteoporosis. Female: estrogen deficiency causing amenorrhea, hot flashes, and osteoporosis. Primary: gonadal failure (e.g., Klinefelter, Turner). Secondary: pituitary/hypothalamic dysfunction (e.g., prolactinoma, hypothalamic amenorrhea). Diagnosis: sex hormone levels, LH/FSH. Treatment: hormone replacement therapy.

Hypogonadism in Men

– Testosterone deficiency resulting from inadequate testicular function classified as primary testicular failure or secondary hypothalamic-pituitary dysfunction. Primary causes include Klinefelter syndrome, cryptorchidism, orchitis, trauma, and chemotherapy. Secondary causes include Kallmann syndrome, pituitary tumors, hyperprolactinemia, and chronic illness. Symptoms include decreased libido, erectile dysfunction, fatigue, decreased muscle mass, increased body fat, osteoporosis, and depression. Diagnosis includes morning total testosterone on at least two occasions, LH and FSH to distinguish primary from secondary hypogonadism, prolactin, and iron studies to exclude haemochromatosis. Testosterone replacement therapy is indicated for symptomatic men with confirmed deficiency after excluding contraindications including prostate cancer and untreated sleep apnoea.

Hypogonadism in Women

– Testosterone deficiency in women causing decreased libido, fatigue, depressed mood, and reduced sense of well-being. Causes include oophorectomy, premature ovarian insufficiency, adrenal insufficiency, hypopituitarism, and medications. Diagnosis is challenging due to the lack of well-established reference ranges for female testosterone levels. Treatment with transdermal testosterone at physiological doses may improve symptoms. Benefits and risks of long-term therapy are still under investigation, and long-term safety data are limited. Individualised treatment decisions should weigh potential benefits against possible adverse effects on cardiovascular health and breast cancer risk.

Hypomagnesemia

– Serum magnesium less than 1.7 milligrams per deciliter. Causes include gastrointestinal losses from diarrhea, renal losses from diuretics and alcohol, malnutrition, and medications including proton pump inhibitors. Manifestations include muscle cramps, tremor, seizures, cardiac arrhythmias including torsades de pointes, and refractory hypokalemia and hypocalcemia. Treatment is magnesium replacement orally for mild cases and intravenously for severe or symptomatic hypomagnesemia. Always consider checking serum magnesium in any patient with unexplained hypokalaemia or hypocalcaemia, as hypomagnesaemia impairs renal potassium conservation and PTH secretion, making electrolyte repletion difficult until magnesium is normalised.

Hyponatremia

– Serum sodium less than 135 mEq/L. Most common electrolyte disorder in hospitalized patients. Causes are classified by volume status: hypovolemic from GI losses and diuretics, euvolemic from SIADH and hypothyroidism, and hypervolemic from heart failure, cirrhosis, and nephrotic syndrome. Acute hyponatremia with neurological symptoms requires rapid correction with hypertonic saline. Chronic hyponatremia should be corrected slowly at no more than eight milliequivalents per liter per twenty-four hours to prevent osmotic demyelination syndrome. Use of vasopressin receptor antagonists (vaptans) may be considered in euvolemic and hypervolemic hyponatremia when conventional therapy is insufficient.

Hypoparathyroidism

– Deficiency or absence of parathyroid hormone causing hypocalcemia and hyperphosphatemia. Most commonly occurs after thyroid or parathyroid surgery from inadvertent removal of parathyroid glands. Autoimmune hypoparathyroidism may occur alone or as part of autoimmune polyendocrine syndrome type one. Symptoms of hypocalcemia include perioral tingling, muscle cramps, carpopedal spasm, Chvostek sign, Trousseau sign, seizures, and cardiac arrhythmias. Chronic hypocalcemia causes basal ganglia calcifications. Treatment requires oral calcium and active vitamin D (calcitriol) supplementation to maintain serum calcium in the low-normal range and avoid hypercalciuria.

Hypoparathyroidism Management

– Treatment of chronic hypocalcemia from hypoparathyroidism includes oral calcium supplementation in divided doses and active vitamin D calcitriol to enhance calcium absorption. Goal is to maintain serum calcium in the low-normal range to prevent hypercalciuria and renal complications. Thiazide diuretics reduce urinary calcium excretion. Magnesium supplementation may be needed if hypomagnesemia is present. PTH replacement therapy with recombinant human PTH is approved for refractory cases not controlled with conventional therapy, but its use is limited by cost and the need for twice-daily injections.

Hypopituitarism

– Deficiency of one or more pituitary hormones most commonly from pituitary adenomas, craniopharyngiomas, pituitary surgery or radiation, traumatic brain injury, or infiltrative diseases. Growth hormone deficiency is typically the first hormone lost followed by gonadotropins then TSH and ACTH. Diagnosis requires dynamic testing of pituitary reserve. Treatment is replacement of deficient hormones with levothyroxine, hydrocortisone, sex steroids, desmopressin for DI, and growth hormone.

Hypothalamic-Pituitary-Adrenal Axis Physiology

– The hypothalamus releases CRH in a diurnal rhythm peaking in the early morning stimulating pituitary corticotrophs to secrete ACTH which stimulates the adrenal cortex to produce cortisol. Cortisol exerts negative feedback on both the hypothalamus and pituitary suppressing CRH and ACTH release. The system is activated during physiological stress overriding normal feedback mechanisms. Assessment of axis integrity uses the overnight dexamethasone suppression test for Cushing screening and the ACTH stimulation test for adrenal insufficiency assessment.

Hypothalamic-Pituitary-Gonadal Axis Physiology

– The hypothalamus releases GnRH in a pulsatile fashion stimulating pituitary gonadotrophs to secrete LH and FSH. In males LH stimulates Leydig cells to produce testosterone while FSH stimulates Sertoli cells to support spermatogenesis.

In females FSH stimulates follicular development and estrogen production while the LH surge triggers ovulation. Sex steroids exert negative feedback on the hypothalamus and pituitary. Disruption at any level causes reproductive dysfunction.

Hypothalamic-Pituitary-Thyroid Axis Physiology

– The hypothalamus secretes TRH stimulating pituitary thyrotrophs to release TSH which stimulates the thyroid gland to produce T4 and T3. T3 the more biologically active form is produced by peripheral conversion of T4 by deiodinase enzymes. T3 and T4 exert negative feedback on both the hypothalamus and pituitary suppressing TRH and TSH release. The system maintains thyroid hormone levels within a narrow range with TSH being the most sensitive indicator of axis dysfunction.

Hypothyroidism

– A common condition of inadequate thyroid hormone production, with a prevalence of 4-10 percent in the general population and up to 20 percent in women over 60 years. Primary hypothyroidism accounts for 99 percent of cases, most commonly caused by chronic autoimmune (Hashimoto) thyroiditis (lymphocytic infiltration leading to gradual follicular destruction, associated with elevated anti-thyroid peroxidase [TPO] and anti-thyroglobulin antibodies), followed by iatrogenic causes (radioactive iodine therapy for Graves disease, radiation therapy for head and neck cancers, and medications (amiodarone, lithium, tyrosine kinase inhibitors). Clinical features reflect reduced metabolic rate including fatigue, lethargy, cold intolerance, weight gain, constipation, dry skin, hair loss, hoarseness, bradycardia, and delayed tendon reflexes. Severe hypothyroidism may cause myxedema, pericardial effusion, and cognitive impairment. Myxedema coma is a life-threatening complication with hypothermia, altered mental status, and respiratory depression. Diagnosis is confirmed by elevated TSH with low free T4. Positive TPO antibodies indicate autoimmune etiology. Treatment is levothyroxine replacement with a starting dose based on patient age, weight, and cardiovascular risk, titrated to achieve normal TSH. In elderly patients or those with ischemic heart disease, a low starting dose of twenty-five to fifty micrograms daily with gradual titration is recommended to avoid precipitating myocardial ischemia or arrhythmia.

Inhaled Insulin

– Rapid-acting insulin delivered via inhalation as a powder using the Afrezza device. Approved for mealtime insulin coverage in adults with diabetes. Onset of action within twelve minutes and duration of approximately three hours providing a more physiological postprandial insulin profile than subcutaneous rapid-acting analogs. Contraindicated in smokers and patients with chronic lung disease. Spirometry monitoring is required before initiation and periodically during therapy. Reduces postprandial glucose excursions and may reduce hypoglycemia compared with injectable mealtime insulin in selected patients. Absorption variability and limited dosing options restrict widespread use.

Insulin

– A hormone produced by pancreatic beta cells that promotes glucose uptake by cells, reduces hepatic glucose production, and inhibits lipolysis. Types include rapid-acting (lispro, aspart), short-acting (regular), intermediate-acting (NPH), long-acting (glargine, detemir), and ultra-long-acting (degludec). Used in type 1 diabetes and type 2 diabetes when oral agents fail. Side effects include hypoglycemia and weight gain. Insulin is administered by injection or insulin pump.

Insulin Injection Technique

– Proper technique for subcutaneous insulin injection includes selecting appropriate injection sites rotating between abdomen thighs buttocks and upper arms using appropriate needle length four to eight millimeters for most adults pinching skin for thin patients to avoid intramuscular injection rotating injection sites within an area to prevent lipohypertrophy and using a new pen needle or syringe for each injection. Site selection affects absorption rate with abdomen being fastest followed by arm and thigh being slower. Proper disposal of sharps using a sharps container is essential for safety.

Insulin Pen Devices

– Pre-filled or reusable insulin pen devices providing convenient and accurate insulin delivery. Benefits include portability precision dosing discrete injection and reduced needle phobia. Modern pen devices provide half-unit dosing capabilities memory features and Bluetooth connectivity for integration with continuous glucose monitoring systems and insulin dose tracking applications. Insulin pen needles are typically four to eight millimeters in length with thirty-one gauge diameter for minimal discomfort. Smart insulin pens integrate with digital health platforms for automated dose logging and decision support, enhancing diabetes self-management.

Insulin Preparations

– Hormone synthesized by pancreatic beta cells that regulates glucose homeostasis by promoting glucose uptake in skeletal muscle and adipose tissue, suppressing hepatic glucose production, and inhibiting lipolysis. Available preparations include rapid-acting insulin analogs including lispro, aspart, and glulisine with onset fifteen to thirty minutes and duration three to five hours, short-acting regular insulin with onset thirty to sixty minutes and duration six to eight hours, intermediate-acting NPH insulin with onset one to two hours and peak four to twelve hours, long-acting insulin analogs including glargine, detemir, and degludec with relatively flat pharmacokinetic profiles and duration up to twenty-four to forty-two hours respectively, and pre-mixed preparations containing fixed ratios of rapid-acting and intermediate-acting insulin.

Insulin Pump Therapy

– Continuous subcutaneous insulin infusion delivering rapid-acting insulin analog through a catheter placed subcutaneously typically in the abdomen. Provides basal insulin at programmed rates and bolus doses at mealtimes. Advantages include reduced glycemic variability, improved time in range, flexibility in meal timing, and reduced hypoglycemia compared to multiple daily injections. Advanced systems integrate with continuous glucose monitors as hybrid closed-loop or automated insulin delivery systems. Disadvantages include risk of infusion site infection, catheter occlusion, and diabetic ketoacidosis from infusion set failure. Candidates require appropriate training and commitment to pump management.

Insulin Resistance

– Diminished biological response to physiological insulin concentrations resulting in impaired glucose disposal in peripheral tissues and failure to suppress hepatic glucose production. Pathophysiological hallmark of type two diabetes and metabolic syndrome. Develops from complex interactions between genetic susceptibility, obesity particularly visceral adiposity, physical inactivity, and inflammatory mediators. Mechanisms include post-receptor signaling defects in the insulin signaling cascade including IRS-1 phosphorylation, PI3K activation, and GLUT4 translocation to the cell membrane. Management primarily involves lifestyle modification with weight loss, physical activity, and dietary changes, along with pharmacotherapy including metformin, thiazolidinediones, and other insulin sensitizers.

Insulin Resistance Syndromes

– Rare inherited severe insulin resistance syndromes include Donohue syndrome also known as leprechaunism, Rabson-Mendenhall syndrome, and type A insulin resistance syndrome caused by mutations in the insulin receptor gene. Donohue syndrome is the most severe with death in infancy. Rabson-Mendenhall syndrome presents with dental and nail abnormalities and pineal hyperplasia. Type A insulin resistance presents with acanthosis nigricans, hyperandrogenism in females, and severe insulin resistance. Treatment includes high-dose insulin therapy, metformin, and consideration of recombinant human IGF-1 in selected cases. Genetic counselling is recommended for affected families.

Kallmann Syndrome

– Congenital form of hypogonadotropic hypogonadism caused by failure of GnRH neuron migration from the olfactory placode to the hypothalamus during embryonic development. Characterized by delayed puberty with low LH and FSH and anosmia or hyposmia from olfactory bulb hypoplasia. Associated features include renal agenesis, cleft palate, and synkinesia. Treatment is pulsatile GnRH therapy for fertility or sex steroid replacement for virilization or feminization.

Klinefelter Syndrome

– Most common sex chromosome aneuploidy in males with a karyotype of 47,XXY occurring in approximately one in six hundred live male births. Characterized by small firm testes, gynecomastia, tall stature with long extremities, and infertility from azoospermia. Testosterone levels are low with elevated gonadotropins reflecting primary testicular failure. Learning disabilities and behavioral issues may occur. Testosterone replacement therapy beginning at puberty improves virilization, bone density, and quality of life. Fertility options including assisted reproductive techniques should be discussed as spermatogenesis is not restored by testosterone therapy alone.

Liddle Syndrome Treatment

– Liddle syndrome is an autosomal dominant cause of early-onset hypertension with hypokalemia and metabolic alkalosis caused by gain-of-function mutations in ENaC sodium channel subunits. Aldosterone levels are suppressed. Unlike primary hyperaldosteronism the hypertension responds to ENaC blockers amiloride or triamterene but not to spironolactone because the channel is constitutively active independent of aldosterone. Genetic testing confirms the diagnosis. Dietary sodium restriction is important for management as it distinguishes the condition from primary hyperaldosteronism and directs appropriate therapy.

Lipodystrophy Syndromes

– Rare conditions characterized by selective loss of adipose tissue leading to severe insulin resistance, diabetes, and hypertriglyceridemia. Congenital generalized lipodystrophy involves loss of all body fat from birth. Familial partial lipodystrophy involves loss of limb and truncal fat with fat accumulation in face and neck. Acquired forms occur with antiretroviral therapy particularly protease inhibitors. Diagnosis is by clinical examination and imaging showing absent or abnormal fat distribution on dual-energy X-ray absorptiometry. Management focuses on metabolic complications including metformin for diabetes, fibrates and omega-3 fatty acids for hypertriglyceridaemia, and leptin replacement therapy with metreleptin for congenital generalized lipodystrophy.

Low Testosterone (Hypogonadism)

– Low testosterone (male hypogonadism) is a clinical syndrome of deficient testosterone production by the testes (primary) or inadequate stimulation by the pituitary/hypothalamus (secondary), defined as total testosterone <300 ng/dL on two morning measurements with consistent symptoms. Symptoms include low libido, erectile dysfunction, fatigue, depressed mood, decreased muscle mass and bone density, increased body fat, hot flashes, and reduced energy. Causes: primary (Klinefelter syndrome, testicular trauma/orchitis, chemotherapy, aging), secondary (pituitary tumors, hemochromatosis, obesity, sleep apnea, opioids, glucocorticoids). Diagnosis requires confirmatory labs: total/free testosterone, LH, FSH, prolactin, iron studies; pituitary MRI if secondary. Testosterone replacement therapy (gels, injections, pellets, buccal tablets) improves symptoms but requires monitoring of hematocrit, PSA, and fertility impact; contraindicated in untreated prostate or breast cancer.

Maturity Onset Diabetes of the Young

– A monogenic form of diabetes with autosomal dominant inheritance, typically presenting before age 25 with non-insulin-dependent hyperglycemia. Caused by mutations in transcription factors or enzymes of beta-cell function. MODY subtypes: GCK-MODY (MODY 2, mild fasting hyperglycemia, no treatment needed), HNF1A-MODY (MODY 3, most common, sulfonylurea-responsive), HNF4A-MODY (MODY 1, sulfonylurea-responsive). Distinguished from type 1 diabetes by negative autoantibodies and detectable C-peptide. Distinguished from type 2 diabetes by young age, lack of obesity, and strong family history. Diagnosis: genetic testing. Treatment: varies by subtype; diet, sulfonylureas, or insulin.

McCune-Albright Syndrome

– Sporadic condition caused by activating mutations in the GNAS1 gene encoding the Gs alpha subunit leading to constitutive activation of adenylyl cyclase. Classic triad includes polyostotic fibrous dysplasia of bone, cafe-au-lait skin spots with irregular coast-of-Maine borders, and autonomous endocrine hyperfunction including precocious puberty, hyperthyroidism, growth hormone excess, and Cushing syndrome. Diagnosis is clinical and confirmed by genetic testing of affected tissue. Treatment targets the specific endocrine abnormality with surgery, medical therapy, or radiation as appropriate for the individual endocrine tumor type. Orthopaedic management of fibrous dysplasia may include bisphosphonates and surgical correction of deformities.

Medullary Thyroid Carcinoma

– Neuroendocrine tumor arising from parafollicular C cells that produce calcitonin. Accounts for approximately five percent of thyroid carcinomas. May be sporadic or hereditary as part of multiple endocrine neoplasia type two A or two B. Serum calcitonin serves as a tumor marker. Diagnosis requires FNA with calcitonin immunohistochemistry. Total thyroidectomy with central lymph node dissection is the standard treatment. RET proto-oncogene mutation testing is mandatory in all patients to identify hereditary medullary thyroid carcinoma and guide prophylactic thyroidectomy in affected family members. Tyrosine kinase inhibitors such as vandetanib and cabozantinib are used for advanced or metastatic disease.

Menopausal Hormone Therapy (MHT)

– Menopausal hormone therapy (MHT), formerly called hormone replacement therapy (HRT), is the use of estrogen with or without progestogen to alleviate moderate-to-severe menopausal vasomotor symptoms (hot flashes, night sweats), genitourinary syndrome of menopause (vaginal dryness, dyspareunia, urinary urgency), and prevent osteoporosis. Estrogen-alone therapy is used in post-hysterectomy women; combined therapy (estrogen + progestogen) is required for those with an intact uterus to prevent endometrial hyperplasia/cancer. Regimens: systemic (oral, transdermal patch/gel, spray) for vasomotor symptoms; low-dose vaginal (cream, tablet, ring) for genitourinary symptoms with minimal systemic absorption. The Women's Health Initiative (WHI) established that MHT increases risk of breast cancer, venous thromboembolism, and stroke, though absolute risk is low in women aged <60 or within 10 years of menopause onset. Treatment should use the lowest effective dose for the shortest duration needed, individualizing based on risk profile (cardiovascular, thrombotic, breast cancer).

Menopause and Perimenopause

– Menopause is the permanent cessation of menstruation for 12 consecutive months, reflecting ovarian follicle depletion and estrogen decline, occurring at a median age of 51–52 years in North America. Perimenopause (menopausal transition) begins 4–8 years before the final menstrual period, characterized by variable cycle length, vasomotor symptoms (hot flashes, night sweats—affecting 80% of women), sleep disturbance, mood changes, vaginal dryness, and declining fertility. Diagnosis is clinical based on age and menstrual pattern changes; FSH elevation (>25 IU/L) and AMH decline confirm ovarian aging but are not required. Health implications include accelerated bone loss (osteoporosis risk), increased cardiovascular risk, weight gain (central adiposity), and genitourinary syndrome. Management: lifestyle modification (cooling layers, exercise, calcium/vitamin D), MHT for moderate–severe symptoms (most effective), non-hormonal options (SSRIs/SNRIs, gabapentin, oxybutynin) for vasomotor symptoms, and vaginal estrogen for urogenital atrophy.

Metabolic Acidosis

– A primary acid-base disorder characterized by decreased serum bicarbonate (<22 mEq/L) and pH <7.35. Causes are classified by anion gap (AG). High AG: diabetic ketoacidosis, lactic acidosis, renal failure, starvation, toxins (methanol, ethylene glycol, salicylates). Normal AG (hyperchloremic): diarrhea, renal tubular acidosis, acetazolamide. Compensation: respiratory alkalosis (Kussmaul respirations). Treatment: address the underlying cause; sodium bicarbonate reserved for severe acidosis (pH <7.0).

Metabolic Syndrome

– Cluster of interrelated metabolic risk factors increasing cardiovascular disease and type two diabetes risk. Defined by the presence of at least three of five criteria: waist circumference greater than 40 inches in men or greater than 35 inches in women, triglycerides greater than 150 milligrams per deciliter, HDL cholesterol less than 40 milligrams per deciliter in men or less than 50 in women, blood pressure greater than 130/85 millimeters of mercury, and fasting glucose greater than 100 milligrams per deciliter, with treatment thresholds and targets individualised based on overall cardiovascular risk profile and presence of other metabolic risk factors.

Metformin in Prediabetes

– Metformin reduces progression from prediabetes to type two diabetes by approximately thirty-one percent in high-risk individuals. Most effective in younger patients under sixty with BMI greater than 35 and women with prior gestational diabetes. The Diabetes Prevention Program demonstrated sustained benefit over ten years of follow-up. Recommended by many guidelines as an adjunct to lifestyle modification in high-risk prediabetic individuals.

Mitochondrial Diabetes

– Diabetes caused by mutations in mitochondrial DNA particularly the m.3243A>G mutation in the MT-TL1 gene. Maternally inherited. Often associated with sensorineural hearing loss termed maternally inherited diabetes and deafness. Patients are typically lean with impaired insulin secretion. Diagnosis requires genetic testing. Insulin therapy is usually necessary though some patients respond to sulfonylureas. Screening for associated conditions including cardiac disease, seizures, and gastrointestinal dysmotility, and psychiatric disturbance. Genetic counselling is important given the maternal inheritance pattern.

Morbid Obesity

– Severe obesity defined clinically as BMI ≥40 kg/m^2 or BMI ≥35 kg/m^2 with obesity-related comorbidities (type 2 diabetes, hypertension, obstructive sleep apnoea, non-alcoholic fatty liver disease, osteoarthritis, cardiovascular disease). Morbid obesity significantly increases all-cause mortality (hazard ratio 2.5–3.0 compared with normal weight). Management: intensive lifestyle intervention, pharmacotherapy (GLP-1 receptor agonists—semaglutide 2.4 mg weekly, tirzepatide; phentermine–topiramate; naltrexone–bupropion), and bariatric/metabolic surgery (Roux-en-Y gastric bypass, sleeve gastrectomy, adjustable gastric band, biliopancreatic diversion with duodenal switch). Bariatric surgery achieves 25–35% total body weight loss and induces remission of type 2 diabetes in 60–80% of patients (the SOS study and STAMPEDE trial). Eligibility per NICE guidelines: BMI ≥40, or BMI ≥35 with recent-onset type 2 diabetes; consideration of psychological suitability, commitment to lifelong follow-up, and nutritional supplementation.

Multiple Endocrine Neoplasia Type 1

– An autosomal dominant tumor syndrome caused by MEN1 gene mutation (menin). Characterized by tumors of the parathyroid (hyperparathyroidism in 95% of patients by age 50, most common), pancreatic islet cells (gastrinoma causing Zollinger-Ellison syndrome, insulinoma, glucagonoma, VIPoma), and anterior pituitary (prolactinoma most common, GH, ACTH). Other tumors: carcinoid, adrenal adenoma, lipomas. Diagnosis: clinical criteria, genetic testing for MEN1 mutation. Management: surgical resection of tumors; peptide receptor radionuclide therapy for advanced neuroendocrine tumors.

Multiple Endocrine Neoplasia Type 2

– An autosomal dominant tumor syndrome caused by RET proto-oncogene mutations. Type 2A (95%): medullary thyroid carcinoma (MTC, nearly 100% penetrance), pheochromocytoma (50%), and hyperparathyroidism (20-30%). Type 2B: MTC, pheochromocytoma, mucosal neuromas, marfanoid habitus; more aggressive MTC. Management: prophylactic thyroidectomy based on RET mutation risk stratification; screen for pheochromocytoma before thyroid surgery; annual lab and imaging surveillance.

Myxedema Coma

– Severe decompensated hypothyroidism characterized by altered mental status, hypothermia, bradycardia, hypoventilation, hyponatremia, and hypoglycemia. Most commonly occurs in elderly patients with long-standing untreated hypothyroidism precipitated by infection, surgery, or sedative medications. Mortality rates exceed fifty percent. Diagnosis is clinical with low free T4 and elevated TSH. Treatment includes IV levothyroxine and IV hydrocortisone to prevent adrenal crisis, passive rewarming, and supportive care in the intensive care unit setting including respiratory and haemodynamic support.

Neonatal Diabetes Mellitus

– Diabetes diagnosed within the first six months of life. Can be transient or permanent. Transient neonatal diabetes mellitus is caused by abnormalities at chromosome 6q24 and usually resolves but may relapse in adolescence. Permanent neonatal diabetes mellitus is most commonly caused by mutations in the KCNJ11 or ABCC8 genes encoding the KATP channel subunits. Patients with KCNJ11 mutations can often transition from insulin to sulfonylureas with excellent glycemic control due to the specific mechanism of action targeting the KATP channel defect.

Neuroendocrine Tumors Grading

– Classification of neuroendocrine tumors by proliferative rate using Ki-67 index and mitotic count. Well-differentiated low-grade tumors have Ki-67 less than three percent and fewer than two mitoses per ten high-power fields. Intermediate-grade tumors have Ki-67 of three to twenty percent. Poorly differentiated high-grade tumors have Ki-67 greater than twenty percent. Grading guides prognosis and treatment intensity with low-grade tumors treated with somatostatin analogs and high-grade tumors require chemotherapy including everolimus and temozolomide and peptide receptor radionuclide therapy.

Neurofibromatosis Type 1

– Autosomal dominant disorder caused by mutations in the NF1 gene encoding neurofibromin on chromosome seventeen. Diagnostic criteria include two or more of the following: six or more cafe-au-lait spots, two or more neurofibromas or one plexiform neurofibroma, axillary or inguinal freckling, optic pathway glioma, two or more Lisch nodules, bony abnormalities, and a first-degree relative with NF1. Endocrine manifestations include pheochromocytoma and growth hormone excess.

Non-Functioning Pituitary Adenoma

– Pituitary adenomas that do not secrete clinically significant hormones. The most common pituitary adenomas to present with mass effects. Typically macroadenomas at diagnosis causing headaches, visual field defects particularly bitemporal hemianopia from optic chiasm compression, and hypopituitarism from compression of normal pituitary tissue. Diagnosis is by MRI showing a pituitary mass and hormone testing to exclude functioning adenomas. Management includes transsphenoidal surgery for symptomatic macroadenomas causing visual compromise or hypopituitarism. Post-operative surveillance is required with MRI and hormone monitoring to detect recurrence.

Normocalcemic Hyperparathyroidism

– Condition of elevated parathyroid hormone with persistently normal serum calcium and normal vitamin D levels. Exclusion of secondary causes of elevated PTH including vitamin D deficiency, renal insufficiency, and medications is required. Represents an early or mild form of primary hyperparathyroidism. May progress to overt hypercalcemia. Monitoring includes annual calcium PTH levels and bone density assessment. Parathyroidectomy is considered for patients with complications or progressive disease. Management includes monitoring calcium, PTH, and bone density annually, with parathyroidectomy considered for patients who develop complications such as worsening bone density, nephrolithiasis, or progression to hypercalcaemia.

Obesity

– Abnormal or excessive fat accumulation that presents a health risk. Defined by BMI: $\geq$30 kg/m^2 (class I: 30-34.9, class II: 35-39.9, class III: $\geq$40). Major risk factor for type 2 diabetes, hypertension, cardiovascular disease, sleep apnea, osteoarthritis, certain cancers, and reduced life expectancy. Management: lifestyle interventions, pharmacotherapy (GLP-1 agonists, phentermine-topiramate, naltrexone-bupropion), and bariatric surgery for severe obesity.

Oestrogen Replacement Therapy

– See Hormone Replacement Therapy (HRT).

Osteomalacia

– Impaired mineralization of newly formed osteoid in adults resulting from vitamin D deficiency, phosphate depletion, or defective mineralization. Causes include vitamin D deficiency from inadequate intake, malabsorption, and impaired synthesis, phosphate depletion from tumor-induced osteomalacia with fibroblast growth factor twenty-three excess, and medications including bisphosphonates and aluminum. Presents with diffuse bone pain, muscle weakness particularly proximal, and increased fracture risk. Diagnosis is confirmed by low 25-hydroxyvitamin D levels, elevated alkaline phosphatase, and characteristic radiographic findings including Looser zones (pseudofractures) on X-ray. Treatment is high-dose vitamin D supplementation with calcium, and treatment of the underlying cause where possible.

Papillary Thyroid Carcinoma

– Most common histological subtype of thyroid carcinoma accounting for approximately eighty percent of cases. Characterized by nuclear features including nuclear grooves, intranuclear cytoplasmic inclusions, and Orphan Annie eye nuclei. Often multifocal and bilateral with lymph node metastases. Excellent prognosis with greater than ninety-five percent ten-year survival. Treatment includes total thyroidectomy with radioactive iodine ablation for intermediate and high-risk disease. TSH suppression with levothyroxine for high-risk patients to reduce recurrence risk.

Parathyroid Crisis

– Severe hypercalcemia with calcium greater than fourteen milligrams per deciliter causing altered mental status, cardiac arrhythmias, and acute kidney injury. Most commonly occurs in patients with primary hyperparathyroidism precipitated by dehydration, immobility, or thiazide diuretics. Treatment requires aggressive IV normal saline hydration, loop diuretics after adequate hydration, calcitonin for rapid but transient calcium reduction, bisphosphonates for sustained calcium lowering, and denosumab for refractory cases. Urgent parathyroidectomy is the definitive treatment once the patient is stabilised.

Parathyroid Hormone Physiology

– Parathyroid hormone is the primary regulator of calcium homeostasis secreted by the four parathyroid glands in response to low ionized calcium levels detected by the calcium-sensing receptor. PTH increases serum calcium through three mechanisms: stimulating osteoclastic bone resorption to release calcium and phosphorus, increasing renal calcium reabsorption in the distal tubule, and stimulating renal production of active vitamin D which enhances intestinal calcium absorption. PTH also increases phosphate excretion in the proximal tubule, thereby lowering serum phosphate. PTH secretion is tightly regulated by ionised calcium through negative feedback on the calcium-sensing receptor.

PCOS

– See Polycystic Ovary Syndrome.

Pheochromocytoma

– Catecholamine-secreting tumor arising from chromaffin cells of the adrenal medulla or extra-adrenal paraganglia. Approximately ten percent are bilateral, ten percent are extra-adrenal, and ten percent are malignant. Classic presentation includes episodic headache, sweating, and palpitations with sustained or paroxysmal hypertension. Associated with hereditary syndromes including von Hippel-Lindau, multiple endocrine neoplasia type two, and neurofibromatosis type one. Diagnosis is by plasma free metanephrines or 24-hour urinary fractionated metanephrines, which have high sensitivity for detection. Imaging with CT or MRI of the abdomen and pelvis localises the tumor. MIBG scintigraphy or PET-CT with FDG or gallium-DOTATATE may be used for staging and detection of metastatic disease. Definitive treatment is laparoscopic adrenalectomy following adequate preoperative alpha-blockade with phenoxybenzamine or doxazosin for at least ten to fourteen days, combined with beta-blockade after alpha-blockade is established. Metastatic disease requires debulking surgery, radionuclide therapy, or tyrosine kinase inhibitors.

Phosphate Homeostasis

– Serum phosphate is regulated by the interplay of PTH vitamin D and fibroblast growth factor 23. PTH promotes renal phosphate excretion. Active vitamin D stimulates intestinal phosphate absorption. FGF23 produced by osteocytes inhibits renal phosphate reabsorption and suppresses 1-alpha-hydroxylase reducing active vitamin D production. Hyperphosphatemia occurs in advanced CKD and is associated with cardiovascular calcification and increased mortality. Management includes dietary phosphate restriction and phosphate binders in chronic kidney disease.

Pituitary Apoplexy

– Acute hemorrhage or infarction within a pituitary adenoma causing sudden headache, visual field defects, ophthalmoplegia, and altered consciousness. Medical emergency requiring urgent glucocorticoid administration to prevent adrenal crisis from acute ACTH deficiency. Severe cases with visual compromise require emergent transsphenoidal surgery. Mild cases may be managed conservatively with close monitoring and hormone replacement. Risk factors include pregnancy, dopamine agonist therapy, and anticoagulation as risk factors. Immediate management focuses on haemodynamic stabilisation, glucocorticoid administration, and urgent neurosurgical consultation.

Polycystic Ovary Syndrome

– Common endocrine disorder affecting six to twelve percent of reproductive-aged women characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound. Diagnostic criteria include two of three features: oligo-anovulation, clinical or biochemical hyperandrogenism, and polycystic ovaries after exclusion of other causes. Associated with insulin resistance, obesity, type two diabetes, and cardiovascular risk. Treatment includes lifestyle modification for weight management, combined oral contraceptive or progestin therapy for menstrual regulation and endometrial protection, antiandrogens like spironolactone for hirsutism, metformin for insulin resistance and glucose intolerance, and ovulation induction with letrozole or clomiphene citrate for infertility. Long-term monitoring for cardiovascular risk factors and glucose intolerance is essential.

Precocious Puberty

– Onset of secondary sexual characteristics before age eight in girls and age nine in boys. Central precocious puberty results from early activation of the hypothalamic-pituitary-gonadal axis. Causes include CNS tumors, hydrocephalus, and idiopathic. Peripheral precocious puberty results from autonomous sex steroid production from adrenal tumors, gonadal tumors, or congenital adrenal hyperplasia. Diagnosis includes bone age X-ray, GnRH stimulation test, and imaging of the brain and adrenal glands. Treatment is with GnRH agonists for central precocious puberty to suppress the axis and slow pubertal progression, while peripheral causes are managed by addressing the underlying source of sex steroid production.

Prediabetes

– Intermediate state of hyperglycemia between normal glucose homeostasis and diabetes defined by fasting glucose 100 to 125 milligrams per deciliter, two-hour glucose during oral glucose tolerance test 140 to 199 milligrams per deciliter, or A1c 5.7 to 6.4 percent. Individuals are at increased risk for progression to type two diabetes with annual progression rates of approximately five to ten percent. Cardiovascular disease risk is already elevated. Lifestyle intervention with weight loss of five to seven percent and increased physical activity is the cornerstone of prevention and is more effective than metformin. For individuals with additional risk factors, metformin may be considered.

Premature Ovarian Insufficiency

– Loss of ovarian function before age forty characterized by amenorrhea, hypergonadotropic hypogonadism, and estrogen deficiency. Causes include genetic disorders particularly Turner syndrome and fragile X premutations, autoimmune oophoritis, iatrogenic from chemotherapy or radiation, and idiopathic. Associated with decreased bone density, cardiovascular risk, and infertility. Diagnosis requires four months of amenorrhea with elevated FSH greater than twenty-five milli-international units per millilitre on two occasions four weeks apart. Treatment includes hormone replacement therapy with estrogen and progesterone, and fertility options including oocyte donation for those desiring pregnancy.

Pretibial Myxedema

– Localized dermopathy occurring in approximately five percent of patients with Graves disease caused by glycosaminoglycan deposition in the dermis particularly over the pretibial area. Presents as non-pitting indurated plaques with a peau dorange appearance. Pathogenesis involves orbital fibroblasts stimulated by thyroid autoantibodies producing excessive hyaluronic acid. Usually occurs in patients with severe ophthalmopathy. Treatment includes topical corticosteroids and compression bandaging through a multidisciplinary approach with dermatology and endocrinology input.

Primary Aldosteronism

– Overproduction of aldosterone by the adrenal glands (Conn syndrome: adrenal adenoma; or bilateral adrenal hyperplasia). Causes hypertension with hypokalemia (40% normokalemic). Presents with treatment-resistant hypertension with or without hypokalemia, metabolic alkalosis, low plasma renin, and high aldosterone-to-renin ratio (ARR >30). Diagnosis: elevated ARR, confirmatory testing (saline suppression test, captopril challenge, oral sodium loading), adrenal CT and adrenal vein sampling to distinguish unilateral from bilateral. Treatment: unilateral adrenalectomy for adenoma; mineralocorticoid receptor antagonists (spironolactone, eplerenone) for bilateral hyperplasia.

Primary Hyperaldosteronism

– Autonomous aldosterone secretion most commonly caused by an aldosterone-producing adrenal adenoma termed Conn syndrome. Leads to hypertension, hypokalemia, metabolic alkalosis, and suppressed renin. Accounts for approximately five to ten percent of hypertensive patients. Screening is by elevated aldosterone-to-renin ratio. Diagnosis requires confirmatory salt loading test demonstrating failure to suppress aldosterone. Localization is by adrenal CT and adrenal vein sampling to distinguish unilateral from bilateral disease. Unilateral aldosterone-producing adenoma is treated with laparoscopic adrenalectomy. Bilateral hyperplasia is managed medically with mineralocorticoid receptor antagonists such as spironolactone or eplerenone.

Primary Hyperparathyroidism

– Autonomous overproduction of parathyroid hormone from one or more abnormal parathyroid glands causing hypercalcemia. Most commonly caused by a single parathyroid adenoma accounting for approximately eighty-five percent. Multigland hyperplasia accounts for approximately ten to fifteen percent and parathyroid carcinoma is rare. Clinical manifestations include nephrolithiasis, osteoporosis, neuropsychiatric symptoms, gastrointestinal symptoms, and cardiovascular disease. Many patients are asymptomatic at diagnosis, identified incidentally through routine biochemical testing. Surgery is recommended for all symptomatic patients and for asymptomatic patients who meet criteria including age under fifty, serum calcium greater than one milligram per decilitre above the upper limit of normal, osteoporosis, nephrolithiasis, or renal impairment.

Prolactinoma Treatment

– Dopamine agonists are first-line medical therapy for prolactinomas. Cabergoline is preferred over bromocriptine due to higher efficacy, less side effects, and twice-weekly dosing improving compliance. Normalize prolactin levels in approximately eighty to ninety percent of patients and significantly reduce tumor size in most. Surgery is reserved for dopamine agonist-resistant tumors, patient intolerance, or cerebrospinal fluid leak from macroadenomas. Radiation therapy is reserved for refractory cases. Long-term monitoring with serial prolactin measurements and MRI is required to detect recurrence.

Prolactin Physiology

– Prolactin is the only pituitary hormone primarily under tonic inhibitory control by the hypothalamus through dopamine acting on D2 receptors. Prolactin stimulates lactation suppresses GnRH and is important for reproductive function. Physiological elevations occur during pregnancy lactation breast stimulation sleep and stress. Hyperprolactinemia suppresses the hypothalamic-pituitary-gonadal axis causing amenorrhea infertility and galactorrhea in women and erectile dysfunction and hypogonadism in men. Management of hyperprolactinemia involves treatment of the underlying cause and dopamine agonist therapy when indicated.

Radioactive Iodine Ablation

– Treatment of differentiated thyroid carcinoma using iodine-131 to destroy residual thyroid tissue after total thyroidectomy. Indications include intermediate and high-risk disease for remnant ablation and treatment of residual or recurrent disease. Patients must be hypothyroid or receive recombinant TSH to maximize iodine uptake. Preparation includes low-iodine diet for one to two weeks and discontinuation of thyroid hormone or levothyroxine withdrawal. Side effects include sialadenitis, taste changes, xerostomia, and rarely radiation thyroiditis. Long-term risks include secondary malignancies such as leukaemia and salivary gland tumors. Gonadal dysfunction may occur, and sperm or oocyte cryopreservation should be considered before treatment. Pregnancy is contraindicated for six to twelve months after ablation.

Renin-Angiotensin-Aldosterone System Physiology

– The RAAS regulates blood pressure and fluid electrolyte balance. Renin released from juxtaglomerular cells cleaves angiotensinogen to angiotensin I. Angiotensin converting enzyme converts angiotensin I to angiotensin II which causes vasoconstriction and stimulates aldosterone secretion from the adrenal cortex. Aldosterone acts on the collecting duct to promote sodium reabsorption and potassium secretion. The system is activated by decreased renal perfusion decreased sodium delivery to the macula densa, and decreased sodium chloride delivery. Dysregulation of this system contributes to hypertension, heart failure, and chronic kidney disease. Pharmacological inhibitors of the RAAS including ACE inhibitors, angiotensin receptor blockers, and mineralocorticoid receptor antagonists are cornerstones of cardiovascular and renal protective therapy.

Rickets

– A childhood disorder of impaired mineralization of growing bone due to vitamin D deficiency, calcium deficiency, or phosphate deficiency. Presents with bone pain, deformities (bowed legs, wrist enlargement, rachitic rosary), growth retardation, and hypocalcemic seizures. Predisposing factors: dark skin, limited sun exposure, exclusive breastfeeding without supplementation. Diagnosis: low 25-hydroxyvitamin D, elevated ALP, characteristic X-ray findings. Treatment: high-dose vitamin D and calcium supplementation.

Sheehan Syndrome

– Postpartum pituitary necrosis resulting from severe hemorrhage and hypotension during delivery. The pituitary gland is enlarged during pregnancy and particularly vulnerable to ischemic injury from blood loss. Presentation varies from fatigue and failure to lactate to panhypopituitarism. Growth hormone and prolactin deficiency occur first. Diagnosis is by hormone testing. Treatment is lifelong hormone replacement of all deficient axes.

Statins in Diabetes

– HMG-CoA reductase inhibitors are first-line therapy for cardiovascular risk reduction in diabetic patients. Most adults with diabetes aged forty to seventy-five benefit from moderate-intensity statin therapy regardless of baseline LDL cholesterol. High-intensity statin therapy with atorvastatin forty to eighty milligrams or rosuvastatin twenty to forty milligrams is indicated for patients with established atherosclerotic cardiovascular disease or multiple risk factors. Benefits include reduction in major adverse cardiovascular events and all-cause mortality. Statin therapy should be continued regardless of baseline lipid levels, and adherence is critical for optimal cardiovascular prevention.

Subacute Thyroiditis

– Inflammatory condition of the thyroid most likely viral in etiology presenting with painful tender thyroid enlargement, fever, and malaise. Associated with elevated erythrocyte sedimentation rate and C-reactive protein. Disease course includes thyrotoxicosis from hormone release during gland destruction, followed by hypothyroidism, and then recovery in most patients. Diagnosis is clinical with the typical presentation and characteristic elevated inflammatory markers. Treatment includes NSAIDs for pain and inflammation, and beta-blockers for symptomatic thyrotoxicosis during the initial phase. Glucocorticoids may be used in severe cases. Thyroid function should be monitored as the patient may require levothyroxine replacement during the subsequent hypothyroid phase.

Teriparatide for Osteoporosis

– Recombinant human parathyroid hormone fragments one through thirty-four that stimulate osteoblast-mediated bone formation. Indicated for severe osteoporosis with fractures or very low T-scores and for patients who have failed or are intolerant to bisphosphonates. Daily subcutaneous injection for up to two years. Increases bone density more rapidly than anti-resorptive agents. Must be followed by anti-resorptive therapy with bisphosphonates or denosumab to maintain gains. Romosozumab an anti-sclerostin antibody, builds bone rapidly and is given as monthly subcutaneous injections for one year, after which antiresorptive therapy is required to maintain bone density gains.

Testosterone

– The primary male sex hormone, an androgen produced mainly in the testicles that is essential for the development of male reproductive tissues, secondary sexual characteristics, and overall health. In addition to its role in sexual function and sperm production, testosterone influences muscle mass, bone density, fat distribution, red blood cell production, and mood. Testosterone levels naturally decline with age, and abnormally low levels can cause reduced libido, erectile dysfunction, fatigue, loss of muscle mass, and osteoporosis; testosterone replacement therapy is available but requires careful consideration of potential risks.

Tetany

– A clinical syndrome of neuromuscular hyperexcitability caused by hypocalcaemia (most common—total calcium <2.1 mmol/L, ionised calcium <1.1 mmol/L), but also caused by hypomagnesaemia, hypokalaemia, and alkalosis (respiratory or metabolic). Symptoms: perioral paresthesias (tingling around the mouth), numbness and tingling in the fingers and toes, muscle cramps, carpopedal spasm (flexion of the wrist and metacarpophalangeal joints, extension of the interphalangeal joints, adduction of the thumb—'obstetrician's hand'), and laryngospasm (stridor, respiratory distress—life-threatening). Latent tetany: elicited by Trousseau sign (carpopedal spasm induced by inflation of a BP cuff above systolic pressure for 3 minutes) and Chvostek sign (tapping the facial nerve anterior to the ear causes ipsilateral facial muscle twitching). Causes of hypocalcaemia: hypoparathyroidism (post-surgical—most common, autoimmune, DiGeorge syndrome), vitamin D deficiency/insufficiency, chronic kidney disease, acute pancreatitis, pseudohypoparathyroidism, and drugs (bisphosphonates, denosumab, furosemide, phenytoin). Treatment: IV calcium gluconate 10–20 mL of 10% solution (for acute symptomatic tetany) given slowly with cardiac monitoring, followed by oral calcium and vitamin D/calcitriol. Treat underlying cause.

Thiazolidinediones

– Insulin sensitizers activating peroxisome proliferator-activated receptors primarily in adipose tissue and skeletal muscle. Agents include pioglitazone and rosiglitazone. Mechanisms include enhanced insulin-mediated glucose uptake, altered adipokine secretion, and reduced hepatic glucose output. Effective A1c reduction of one to one point five percent. Side effects include weight gain, fluid retention leading to heart failure exacerbation, increased fracture risk particularly in postmenopausal women and in premenopausal women with polycystic ovary syndrome where they may promote ovulation. Due to safety concerns, pioglitazone is more commonly used than rosiglitazone, and both require careful monitoring for adverse effects.

Thyroglobulin Surveillance

– Thyroglobulin produced only by thyroid follicular cells serves as a tumor marker after total thyroidectomy for differentiated thyroid carcinoma. In patients who have undergone complete thyroidectomy and radioactive iodine ablation thyroglobulin should be undetectable. Rising thyroglobulin levels indicate residual or recurrent thyroid tissue. Stimulated thyroglobulin after recombinant human TSH or thyroid hormone withdrawal provides greater sensitivity for detecting recurrence. The thyroglobulin antibody assay is essential as autoantibodies can interfere with the measurement causing falsely low or undetectable levels, limiting the utility of the test in antibody-positive patients.

Thyroglossal Duct Cyst

– Congenital midline neck mass resulting from persistence of the thyroglossal duct the embryological tract of thyroid descent. Most common congenital neck mass presenting in childhood. Located at or near the hyoid bone and moves with swallowing and tongue protrusion. Diagnosis is by ultrasound demonstrating a cystic midline neck mass. Thyroid tissue may be present within the cyst. Treatment is surgical excision with Sistrunk procedure which includes removal of the central portion of the hyoid bone to prevent recurrence related to the embryological tract. Recurrence is rare after complete excision.

Thyroid Cancer Staging

– American Joint Committee on Cancer TNM staging for differentiated thyroid carcinoma incorporates tumor size, extent of invasion, lymph node metastases, and distant metastases. Age is a critical prognostic factor with patients under fifty having more favorable outcomes. Stage I is any tumor without metastases in patients under fifty. Stage II includes distant metastases or extrathyroidal extension in patients under fifty and larger tumors in patients over fifty. Risk stratification systems include the American Thyroid Association risk stratification, which categorises patients into low, intermediate, and high-risk groups to guide post-operative management including the need for radioactive iodine ablation and the intensity of TSH suppression therapy.

Thyroid Disorders

– Conditions affecting the thyroid gland, a butterfly-shaped endocrine organ in the neck that produces hormones regulating metabolism, growth, and development. Hyperthyroidism involves excessive thyroid hormone production, causing symptoms such as weight loss, tachycardia, heat intolerance, and anxiety, while hypothyroidism results from insufficient hormone production and leads to fatigue, weight gain, cold intolerance, and depression. Autoimmune diseases including Graves disease and Hashimoto thyroiditis are common causes; management includes medications, radioactive iodine therapy, or thyroid surgery depending on the specific disorder.

Thyroid Eye Disease

– Autoimmune inflammatory disorder of the orbital tissues associated with Graves disease characterized by expansion of extraocular muscles and orbital fat. Pathogenesis involves orbital fibroblasts expressing the TSH receptor targeted by the same thyroid-stimulating immunoglobulins affecting the thyroid. Clinical features include proptosis, periorbital edema, chemosis, lid retraction, and diplopia. Severe disease can cause optic nerve compression. Treatment includes smoking cessation, selenium supplementation which has been shown to slow progression of mild to moderate thyroid eye disease, glucocorticoids for active inflammatory disease, and orbital radiotherapy or surgical decompression for sight-threatening proptosis or optic neuropathy. Teprotumumab, an IGF-1 receptor inhibitor, is a newer targeted therapy that reduces proptosis and diplopia in active disease.

Thyroid Function Tests

– Laboratory assessment of the hypothalamic-pituitary-thyroid axis. TSH is the most sensitive screening test with elevated levels indicating primary hypothyroidism and suppressed levels indicating hyperthyroidism. Free T4 measures the biologically active unbound thyroxine. Total T4 and T3 reflect bound and unbound hormone and are affected by binding protein abnormalities. T3 thyrotoxicosis occurs when T3 is preferentially elevated in early Graves disease or toxic adenoma. TSH receptor antibodies can also be measured to confirm autoimmune etiology. Interpretation of thyroid function tests must account for non-thyroidal illness, medication effects, and binding protein abnormalities that can alter total hormone levels without affecting free hormone concentrations.

Thyroiditis

– Inflammation of the thyroid gland with various etiologies. Hashimoto thyroiditis: autoimmune, most common cause of hypothyroidism in iodine-sufficient areas. Subacute (De Quervain) thyroiditis: post-viral, presents with painful thyroid and transient hyperthyroidism followed by hypothyroidism. Postpartum thyroiditis: autoimmune, occurs within 1 year of delivery. Silent thyroiditis: painless lymphocytic thyroiditis. Diagnosis: TSH, T4, TPO antibodies, thyroid ultrasound, and radioactive iodine uptake.

Thyroiditis Classification

– Inflammatory conditions of the thyroid classified by clinical presentation and etiology. Hashimoto thyroiditis is chronic autoimmune with hypothyroidism. Graves disease is autoimmune with hyperthyroidism. Subacute thyroiditis is painful and likely viral. Silent thyroiditis and postpartum thyroiditis are painless autoimmune causing transient thyrotoxicosis followed by hypothyroidism. Riedel thyroiditis is rare idiopathic fibrosis causing a stony-hard goiter that may compress adjacent structures. Diagnosis is by clinical presentation, thyroid function tests, ultrasound, and radioactive iodine uptake which is low in inflammatory thyroiditis and high in Graves disease. Management is directed at the underlying etiology.

Thyroid Nodule Evaluation

– Systematic approach to evaluation of thyroid nodules detected by physical examination or imaging. Thyroid ultrasound characterizes nodule features including size, composition, echogenicity, margins, calcifications, and shape. TI-RADS classification stratifies malignancy risk and guides biopsy decisions. Fine-needle aspiration biopsy is indicated for nodules meeting size and suspicious feature criteria. Bethesda cytology classification categorizes results from non-diagnostic through suspicious for malignancy to malignant. Management of benign nodules is observation with serial ultrasound, while malignant or suspicious nodules require surgical resection. Molecular testing of biopsy specimens may help refine risk stratification and guide surgical decision-making.

Thyroid Storm

– Life-threatening exacerbation of thyrotoxicosis presenting with hyperpyrexia, tachycardia, heart failure, altered mental status, and multi-organ dysfunction. Precipitated by infection, surgery, trauma, or iodine load in patients with untreated hyperthyroidism. Mortality rates range from twenty to thirty percent. Diagnosis is clinical based on Burch-Wartofsky scoring system. Treatment includes beta-blockers to control heart rate, thionamides to block new thyroid hormone synthesis, iodine to block thyroid hormone release (the Plummer effect), corticosteroids to reduce peripheral conversion of T4 to T3 and support haemodynamics, and aggressive supportive care including cooling, hydration, and management of heart failure. Definitive treatment of hyperthyroidism with radioactive iodine or thyroidectomy should follow after stabilisation.

Time in Range

– Percentage of time that continuous glucose monitoring readings remain within the target glucose range of seventy to one hundred eighty milligrams per deciliter for most adults with diabetes. General target is greater than seventy percent corresponding to approximately seventeen hours per day. Time below range less than seventy is less than four percent. Time below fifty-four is less than one percent. Time above range greater than one hundred eighty is less than twenty-five percent. Time in range correlates well with A1c and is a more dynamic measure of glycaemic control. Each ten percent increase in time in range corresponds to an approximate 0.5 percent reduction in A1c. Time in range targets may be individualised for older adults, pregnant women, and those with frequent hypoglycemia.

Turner Syndrome

– Chromosomal abnormality in females with complete or partial absence of one X chromosome most commonly with a karyotype of 45,X occurring in one in two thousand five hundred live female births. Features include short stature, webbed neck, shield chest, lymphedema, streak gonads causing primary amenorrhea and infertility, congenital heart defects particularly bicuspid aortic valve and coarctation of the aorta, and renal anomalies. Ovarian failure leads to estrogen deficiency requiring hormone replacement therapy for breast development and maintenance of secondary sexual characteristics, and estrogen replacement for bone health and cardiovascular protection. Growth hormone therapy may be considered during childhood to improve adult height. Fertility options including oocyte donation should be discussed.

Type 1 Diabetes Mellitus

– Type 1 diabetes is an autoimmune disease in which the immune system attacks and destroys the insulin-producing beta cells of the pancreas, resulting in absolute insulin deficiency. It typically presents in childhood or adolescence with symptoms of polyuria, polydipsia, weight loss, and fatigue, and requires lifelong insulin therapy through multiple daily injections or an insulin pump. Management involves careful monitoring of blood glucose levels, carbohydrate counting, and balancing insulin doses with physical activity to maintain near-normal glycemia and prevent acute complications like diabetic ketoacidosis and long-term complications affecting the eyes, kidneys, nerves, and cardiovascular system.

Type 2 Diabetes Mellitus

– Type 2 diabetes is a chronic metabolic disorder characterized by insulin resistance and progressive pancreatic beta-cell dysfunction, leading to hyperglycemia. It is strongly associated with obesity, physical inactivity, and genetic predisposition, and typically develops in adulthood, though increasing rates are seen in younger populations. Management begins with lifestyle modifications including weight loss, dietary changes, and exercise, then progresses to oral medications such as metformin, and may eventually require insulin as beta-cell function declines over time.

Vasopressin (Antidiuretic Hormone, ADH)

– See Antidiuretic Hormone (ADH).

VIPoma

– A rare pancreatic neuroendocrine tumor secreting vasoactive intestinal peptide (VIP), causing WDHA syndrome: watery diarrhea (large volume, 3-5 L/day), hypokalemia, and achlorhydria/acidosis. Most are malignant and located in the pancreatic tail. Diagnosis: elevated VIP levels, pancreatic tumor on imaging. Treatment: surgical resection for localized disease; somatostatin analogs (octreotide) control symptoms; chemotherapy or peptide receptor radionuclide therapy for metastatic disease.

Virilisation

– The development of male secondary sexual characteristics in females, resulting from excessive androgen production or exposure. Causes: congenital adrenal hyperplasia (21-hydroxylase deficiency, the most common), androgen-secreting tumors (ovarian or adrenal), Cushing syndrome, exogenous androgen use (anabolic steroids, testosterone therapy), and polycystic ovary syndrome (mild hyperandrogenism). Clinical features: hirsutism (male-pattern hair growth on face, chest, back, and abdomen), temporal hair recession (male-pattern baldness), deepening of the voice, clitoromegaly, increased muscle mass, acne, and amenorrhoea/oligomenorrhoea. The severity and age of onset depend on the cause: prenatal virilisation (ambiguous genitalia at birth, classically in congenital adrenal hyperplasia) vs. postpubertal virilisation (gradual onset in PCOS, rapid onset suggests tumor). Evaluation: serum testosterone (total and free), DHEAS, 17-hydroxyprogesterone, and imaging (ovarian/adrenal ultrasound, CT/MRI). Treatment: directed at the underlying cause — glucocorticoid replacement for CAH, surgical resection of tumors, antiandrogens (spironolactone, cyproterone acetate) and cosmetic measures (electrolysis, laser hair removal) for PCOS.

Von Hippel-Lindau Disease

– Autosomal dominant syndrome caused by mutations in the VHL tumor suppressor gene on chromosome three predisposing to hemangioblastomas of the central nervous system and retina, clear cell renal cell carcinoma, pheochromocytoma, pancreatic cysts and neuroendocrine tumors, and endolymphatic sac tumors. Retinal hemangioblastomas are the most common initial manifestation. CNS hemangioblastomas occur most commonly in the cerebellum and spinal cord. Screening includes annual ophthalmological examination, imaging of the brain and spine every two to three years, and abdominal imaging for renal cell carcinoma and pancreatic lesions. Treatment is surgical resection of symptomatic tumors, with targeted therapies such as sunitinib and belzutifan for advanced renal cell carcinoma.

Wolfram Syndrome

– Autosomal recessive disorder caused by mutations in the WFS1 gene encoding wolframin. Classic triad of diabetes mellitus, optic atrophy, and diabetes insipidus. Additional features include sensorineural deafness, neurodegeneration, and psychiatric illness. Onset of diabetes typically in the first decade. Progressive optic atrophy leads to blindness. Diagnosis is by clinical criteria and genetic testing. Treatment is supportive with insulin for diabetes and desmopressin for diabetes insipidus.

Chapter 9

Ear, Nose, and Throat (ENT)

Acoustic Neuroma

– Acoustic neuroma, also known as vestibular schwannoma, is a benign, slow-growing tumor arising from the Schwann cells of the vestibular portion of the eighth cranial nerve (vestibulocochlear nerve), typically within the internal auditory canal or the cerebellopontine angle, and it accounts for approximately eight to ten percent of all intracranial tumors. The tumor usually presents with unilateral sensorineural hearing loss, tinnitus, and disequilibrium, and less commonly with vertigo, facial numbness (from trigeminal nerve compression), or facial weakness (from facial nerve compression).

Adenoidectomy

– The surgical removal of the pharyngeal tonsils (adenoids), which are lymphoid tissue structures located in the nasopharynx, and it is commonly performed in children for obstructive sleep apnea, chronic nasal obstruction, recurrent otitis media, and chronic rhinosinusitis that is refractory to medical management. The adenoids are typically most prominent between the ages of three and seven years and undergo spontaneous involution during childhood, and adenoid hypertrophy can.

Allergic Rhinitis

– Allergic rhinitis, commonly known as hay fever, is a type I IgE-mediated hypersensitivity reaction of the nasal mucosa to inhaled allergens, characterized by sneezing, nasal congestion, rhinorrhea (clear, watery nasal discharge), nasal pruritus, and conjunctival symptoms including itching, tearing, and redness.

Anosmia

– The complete loss of the sense of smell, while hyposmia: a reduced sense of smell, and these conditions can significantly impair quality of life by reducing the ability to appreciate food flavors, detect environmental hazards such as gas leaks and smoke, and enjoy activities. The causes of anosmia are numerous and include post-viral olfactory loss (the most common cause, often following upper respiratory viral infections including SARSCoV-2), chronic rhinosinusitis with nasal polyps, head trauma (particularly frontal lobe injuries that disrupt the olfactory nerves), neurodegenerative diseases (Alzheimer disease, Parkinson disease), congenital conditions (Kallmann syndrome), endocrine disorders (hypothyroidism, diabetes), medication side effects, and toxic exposures (smoking, industrial chemicals). The diagnosis involves olfactory testing using standardized smell identification and threshold tests, nasal endoscopy to evaluate for anatomical causes, and MRI to assess the olfactory bulbs and frontal lobes. Treatment depends on the underlying cause and may include intranasal corticosteroids for inflammatory causes, olfactory training for postviral anosmia, and management of underlying medical conditions.

Aphthous Ulcer

– Recurrent, painful, round or oval ulcers of the oral mucosa, commonly called canker sores. Three types: minor (most common, <1 cm, heal in 7-14 days without scarring), major (>1 cm, deeper, heal over weeks with scarring), and herpetiform (multiple pinhead-sized ulcers that fuse). Etiology unclear: immune-mediated, genetic predisposition, stress, trauma, nutritional deficiencies (B12, folate, iron), and food sensitivities. Diagnosis: clinical. Treatment: symptomatic (topical corticosteroids, amlexanox, lidocaine, oral hygiene). Severe cases: systemic corticosteroids, colchicine, or dapsone.

Arytenoid Cartilage

– The arytenoid cartilages are a pair of small, pyramid-shaped cartilages located on the posterior aspect of the larynx, articulating with the superior border of the cricoid cartilage lamina, and they serve as the attachment points for the vocal cords and the intrinsic laryngeal muscles that control vocal cord movement. Each arytenoid cartilage has a vocal process (anteriorly), to which the vocal cord ligament attaches, and a muscular process (laterally), to which the posterior cricoarytenoid and lateral cricoarytenoid muscles attach. The movement of the arytenoid cartilages on the cricoid cartilage during phonation, breathing, and swallowing causes the vocal cords to adduct (close), abduct (open), and tense, and the posterior cricoarytenoid muscle is the sole abductor of the vocal cords, making it critical for maintaining airway patency. The arytenoid cartilages may become displaced or subluxed by trauma, intubation injury, or surgery, and arytenoid dislocation can cause hoarseness and vocal cord paresis.

Audiogram

– A chart that shows a person's ability to hear at different pitches or frequencies.

Audiologist

– A health professional who assesses hearing and fits hearing aids.

Audiometry

– A complete hearing test that involves listening to sounds of different frequencies and volume.

Cerumen

– A substance that helps keep dirt out of the ear and lubricates the skin in the ear. More commonly known as earwax.

Cerumen Impaction

– The accumulation of cerumen (earwax) in the external auditory canal to the point where it causes symptoms or prevents adequate visualization of the tympanic membrane, and it is a common condition affecting approximately five to ten percent of children and up to thirty percent of elderly individuals and cognitively impaired adults. Cerumen is a normal, protective secretion of the ceruminous glands in the cartilaginous portion of the external auditory canal, composed of secretions from modified apocrine glands mixed with dead skin cells and hair, and it serves to moisturize the canal skin, trap debris, and provide antimicrobial protection. Symptoms of impaction include aural fullness, decreased hearing, tinnitus, pruritus, pain, and a feeling of obstruction, and impacted cerumen may be visible as a yellow, brown, or dark mass obscuring the tympanic membrane. Treatment includes softening agents such as mineral oil, docusate sodium, or hydrogen peroxide, followed by manual removal using curettes, irrigation, or suction, and patients should be advised against inserting cotton swabs or other objects into the ear canal.

Cholesteatoma

– A pathological accumulation of keratinizing squamous epithelium within the middle ear, mastoid, or temporal bone that behaves like a locally invasive, destructive lesion, causing progressive conductive hearing loss, chronic otorrhea, and potential complications including ossicular erosion, labyrinthine fistula, facial nerve paralysis, and intracranial infections. The condition may be congenital (arising from embryonic epithelial remnants within the temporal bone) or acquired (typically developing from retraction pockets of the pars flaccida or perforations of the tympanic membrane), and it grows by the accumulation and desquamation of keratin debris within the squamous epithelial-lined sac. The cholesteatoma matrix produces proteolytic enzymes and inflammatory mediators that erode adjacent bony structures, including the ossicles, the bony labyrinth, and the mastoid cortex. Diagnosis is confirmed by otoscopy showing a retraction pocket or pearly white mass behind the tympanic.

Chronic Rhinosinusitis

– A heterogeneous group of disorders characterized by persistent inflammation of the paranasal sinuses lasting more than twelve weeks, with symptoms including nasal obstruction, purulent rhinorrhea (anterior or posterior), facial pain or pressure, and olfactory dysfunction, and it is classified into chronic rhinosinusitis without nasal polyps and chronic rhinosinusitis with nasal polyps based on the presence or absence of nasal polyps on endoscopic examination. The pathogenesis is multifactorial, involving bacterial biofilm formation, mucociliary dysfunction, immunological dysregulation (type 2 inflammation in polyp disease), and anatomical factors that impair sinus drainage. The diagnosis is confirmed by nasal endoscopy showing mucosal edema, polyps, or purulent drainage from the sinus ostia, and CT imaging of the sinuses showing mucosal thickening and sinus opacification. Treatment includes intranasal corticosteroids, saline irrigation, antibiotics for acute exacerbations, and surgical intervention with functional endoscopic sinus surgery for patients who fail medical therapy.

Cochlear Implant

– A cochlear implant is an electronic medical device that bypasses damaged or absent inner ear hair cells to provide direct electrical stimulation of the auditory nerve, restoring a perception of sound in individuals with severe to profound sensorineural hearing loss who derive limited benefit from hearing aids. The device consists of an external component (microphone, speech processor, and transmitter coil) and an internal component (receiver-stimulator and electrode array inserted into the cochlea), and it works by converting sound into electrical signals that are transmitted to the electrode array, which stimulates the spiral ganglion neurons of the auditory nerve. The implant does not restore normal hearing but provides a representation of environmental sounds and speech that the brain learns to interpret through auditory rehabilitation. Indications include bilateral severe to profound sensorineural hearing loss in adults who receive limited benefit from appropriately fitted hearing aids, and bilateral severe to profound sensorineural hearing loss in children (typically implanted at twelve to eighteen months of age) to support speech and language development. The surgery is performed under general anesthesia and is generally safe, with risks including facial nerve injury, meningitis, and device failure.

Conductive Hearing Loss

– A type of hearing loss caused by impairment of the transmission of sound through the outer ear, middle ear, or both, preventing sound vibrations from reaching the cochlea efficiently, and it is typically characterized by a conductive gap (an air-bone gap on audiometry indicating a difference between air conduction and bone conduction thresholds). Common causes in-.

Cricoid Cartilage

– The cricoid cartilage is a complete, ring-shaped cartilage located at the base of the larynx, directly above the trachea and below the thyroid cartilage, and it is the only complete cartilaginous ring in the airway, providing structural support and maintaining airway patency. The cricoid cartilage is shaped like a signet ring, with a narrow anterior arch and a broad posterior lamina that articulates with the arytenoid cartilages superiorly. The cricothyroid membrane, connecting the thyroid and cricoid cartilages anteriorly, is the site of emergency cricothyrotomy (cricothyroidotomy) for securing the airway when intubation is not possible. The cricothyroid muscle, which attaches to the cricoid and thyroid cartilages, is the only intrinsic laryngeal muscle that is not innervated by the recurrent laryngeal nerve, and it functions to tense the vocal cords. The cricoid cartilage is prone to fracture in blunt neck trauma, and cricoid fracture can cause airway compromise, subglottic stenosis, and recurrent laryngeal nerve injury.

CSF Rhinorrhea

– Cerebrospinal fluid rhinorrhea is the leakage of cerebrospinal fluid from the subarachnoid space through a defect in the skull base into the nasal cavity or paranasal sinuses, resulting in a clear, watery nasal discharge that may be mistaken for allergic rhinitis or sinusitis. The most common causes include trauma (skull base fractures), iatrogenic injury (following sinus or skull base surgery), spontaneous.

Deafness

– Complete or partial loss of the ability to hear, classified by type: conductive deafness (impaired sound transmission through the external or middle ear, causes include cerumen impaction, otitis media, otosclerosis, tympanic membrane perforation), sensorineural deafness (damage to the cochlea or auditory nerve, causes include presbycusis, noise exposure, ototoxic drugs, Meniere disease, acoustic neuroma), and mixed hearing loss. Diagnosis: audiometry (pure-tone, speech), tympanometry, and otoacoustic emissions. Management: hearing aids, cochlear implants (for severe-to-profound sensorineural loss), bone-anchored hearing aids (for conductive or single-sided deafness), and surgical correction of underlying causes.

Decibel

– A logarithmic unit (dB) used to measure the intensity of sound. The decibel scale compares a given sound pressure level to a reference level (0 dB =

20 micropascals, the threshold of normal hearing at 1 kHz). The scale is logarithmic: an increase of 10 dB represents a tenfold increase in sound intensity. Normal conversation: 60 dB. Threshold of pain: 120 dB. Prolonged exposure to sounds >85 dB causes noise-induced hearing loss. Audiometry measures hearing thresholds in dB hearing level (dB HL) across frequencies 250–8000 Hz.

Dix-Hallpike Maneuver

– The Dix-Hallpike maneuver is a diagnostic test used to identify benign paroxysmal positional vertigo by detecting the characteristic nystagmus and vertigo that occur when displaced otoconia within the posterior semicircular canal move in response to gravity during specific head positioning. The patient is seated on the examination table, and the examiner rotates the patient's head forty-five degrees toward the side being tested, then quickly lays the patient back with the head hanging below the level of the shoulders, maintaining the head rotation.

Earache (Otalgia)

– Pain originating from the ear, a common symptom with diverse causes. Otalgia is classified as primary (originating within the ear itself) or secondary/referred (pain perceived in the ear but arising from another site). Primary causes: otitis media, otitis externa, Eustachian tube dysfunction, cholesteatoma, foreign body, cerumen impaction, barotrauma, mastoiditis. Referred causes: temporomandibular joint dysfunction, dental pathology (molar caries, abscess, impacted wisdom teeth), tonsillitis, pharyngitis, sinusitis, cervical spine disorders, trigeminal neuralgia, and malignancies of the pharynx or larynx. Evaluation: otoscopy, pneumatic otoscopy, tuning fork tests (Rinne, Weber), and assessment of surrounding structures. Management depends on etiology.

Ear canal

– A tube leading from the eardrum to the outer ear.

Ear Diseases

– Any disorder affecting the structures of the ear–external ear (pinna, ear canal), middle ear (tympanic membrane, ossicles, Eustachian tube), or inner ear (cochlea, vestibular apparatus). Broad categories: congenital (microtia, atresia, preauricular sinus), infectious (otitis externa, otitis media, mastoiditis), inflammatory (Eustachian tube dysfunction, cholesteatoma), traumatic (tympanic membrane perforation, temporal bone fracture, barotrauma), neoplastic (acoustic neuroma, glomus tumor, squamous cell carcinoma), and degenerative (presbycusis, Meniere disease, otosclerosis). Diagnosis: otoscopy, audiometry, tympanometry, and imaging (CT temporal bone, MRI internal auditory meatus). Management is etiology-specific.

Eardrum

– A thin membrane separating the ear canal and middle ear.

Ear Infections

– Infections of the ear, most commonly affecting the middle ear (otitis media) or the outer ear canal (otitis externa). They are often caused by bacteria or viruses and present with ear pain, drainage, hearing loss, and sometimes fever. Recurrent ear infections in children may require tympanostomy tubes or other interventions to prevent complications.

Ear, Nose, and Throat (ENT)

– A medical specialty (otolaryngology) focused on the diagnosis and treatment of disorders affecting the ears, nose, throat, head, and neck. Conditions managed include hearing loss, sinusitis, tonsillitis, voice disorders, and head and neck cancers. Otolaryngologists also perform surgical procedures such as tonsillectomy, sinus surgery, and cochlear implantation.

Earwax

– Cerumen, a waxy secretion produced by the ceruminous glands in the outer third of the external auditory canal. Earwax lubricates and protects the skin of the ear canal, trapping dust, debris, and microorganisms to prevent them from reaching the tympanic membrane. Normal self-cleaning mechanism migrates wax laterally through epithelial migration. Impaction occurs when cerumen accumulates excessively, causing conductive hearing loss, itching, tinnitus, a feeling of fullness, or discomfort. Excessive use of cotton buds can cause impaction. Treatment: cerumenolytic drops (sodium bicarbonate, hydrogen peroxide, olive oil) or microsuction/ear irrigation. Irrigation is contraindicated with known tympanic membrane perforation, previous ear surgery, or active otitis media.

Epistaxis

– Epistaxis, commonly known as a nosebleed, is hemorrhage from the nasal mucosa that can range from minor, self-limited bleeding to severe, lifethreatening hemorrhage requiring emergency intervention, and it is a common presenting complaint in both primary care and emergency departments.

Epley Maneuver

– The Epley maneuver, also known as the canalith repositioning maneuver, is a therapeutic procedure used to treat posterior canal benign paroxysmal positional vertigo by repositioning displaced calcium carbonate crystals (otoconia) from the posterior semicircular canal back into the utricle through a series of sequential head and body position changes guided by the physician. The procedure begins with the patient seated, the head rotated forty-five degrees toward the affected ear, and the patient is rapidly laid back with the head hanging (the Dix-Hallpike position), maintaining the head rotation for thirty to sixty seconds until the nystagmus resolves. The head is then rotated ninety degrees to the opposite side while maintaining the neck extension, held for thirty to sixty seconds, and the patient is then rolled onto their side with the nose pointing toward the floor, held for thirty to sixty seconds, before sitting upright. The maneuver relies on gravity to guide the otoconia through the semicircular canal and back into the utricle, with a success rate of approximately eighty percent after one to two treatments, and patients are advised to avoid lying flat on the affected side for several days afterward.

Esophageal Foreign Body

– Ingestion of a foreign object lodged in the esophagus. Most common in children (coins, batteries, toys) and adults with dentures (food bolus), psychiatric illness, or prisoners. Common sites: cricopharyngeus (C6, 60%), aortic arch (T4), and gastroesophageal junction. Symptoms: dysphagia, odynophagia, drooling, and foreign body sensation. Sharp objects: risk of perforation. Button batteries: require emergency removal due to rapid tissue necrosis and perforation. Diagnosis: X-ray neck/chest/abdomen (AP and lateral). Treatment: endoscopic removal (urgent for sharp, batteries, large objects, or complete obstruction). Glucagon IV for distal esophageal food impaction may facilitate passage. Bougienage for smooth coins in children.

Ethmoidal Sinus

– The ethmoidal sinuses, also known as ethmoidal air cells, are a collection of multiple small, thin-walled, air-containing cavities within the ethmoid bone between the orbits and above the nasal cavity, and they are divided into anterior and posterior groups based on their drainage sites. The anterior ethmoidal air cells, typically numbering six to twelve, drain through the ethmoidal infundibulum into the middle meatus, while the posterior ethmoidal air cells, typically numbering three to eight, drain into the superior meatus. The ethmoidal sinuses are separated from the orbits by the paper-thin lamina papyracea, and from the anterior cranial fossa by the fovea ethmoidalis (the roof of the ethmoid labyrinth), making complications of ethmoidal sinusitis potentially dan-.

External otitis

– An infection or irritation of the outer ear, ear canal, or both. Also called swimmer's ear.

FESS

– Functional endoscopic sinus surgery is a minimally invasive surgical technique that uses nasal endoscopes and specialized instruments to restore normal sinus drainage and ventilation by removing obstructive tissue and opening blocked sinus ostia, while preserving as much normal sinus mucosa and anatomy as possible. The procedure is indicated for chronic rhinosinusitis with or without nasal polyps that has failed medical therapy, recurrent acute sinusitis, and complications of sinusitis including mucocele and orbital complications. FESS involves the uncinectomy (removal of the uncinate process to access the maxillary sinus ostium), ethmoidectomy (removal of ethmoidal air cells), maxillary antrostomy (enlargement of the maxillary sinus ostium), frontal sinusotomy (opening of the frontal sinus drainage pathway), and sphenoidotomy (opening of the sphenoid sinus), depending on the extent of disease. The procedure is performed under general anesthesia using rigid endoscopes with angles of 0, 30, 45, and 70 degrees, and it offers the advantages of direct visualization, targeted tissue removal, preservation of normal anatomy, and faster recovery compared to traditional open sinus surgery.

Frontal Sinus

– The frontal sinuses are paired, air-filled cavities located within the frontal bone above the orbits and the root of the nose, separated by a bony septum that may be deviated to one side, and they are the last of the paranasal sinuses to develop, typically appearing around age seven and reaching full size by late adolescence. The frontal sinuses are irregular in.

Glossitis

– Inflammation of the tongue. Causes: (1) Infection: bacterial (Streptococcus, Actinomyces), viral (HSV, Coxsackie–herpangina, HIV), fungal (Candida–oral thrush). (2) Nutritional deficiencies: iron (atrophic glossitis, smooth red tongue), B12 (magenta tongue), riboflavin (purple-red tongue with fissures), niacin (pellagra, beefy red tongue), folate. (3) Autoimmune: geographic tongue (benign migratory glossitis), lichen planus, pemphigus, Behçet disease. (4) Trauma: burns (hot food/drink), bite injury, ill-fitting dentures or orthodontic appliances. (5) Allergic: contact stomatitis (toothpaste, mouthwash, food). (6) Systemic: amyloidosis, sarcoidosis, Crohn disease. Symptoms: pain, burning, swelling, redness, altered taste, difficulty eating. Diagnosis: clinical examination, nutritional bloods, swab for culture, biopsy if persistent. Treatment: directed at cause (antifungals for Candida, B12 injections for pernicious anaemia, avoidance of irritants).

Glottis

– The opening between the vocal folds (true vocal cords) in the larynx, through which air passes during respiration and phonation. Boundaries: anterior (anterior commissure, attachment of vocal folds to thyroid cartilage), posterior (arytenoid cartilages and posterior commissure). The glottis has three functional states: (1) Open (respiration): vocal folds abducted by posterior cricoarytenoid muscles, creating a triangular opening. (2) Closed (swallowing, valsalva): vocal folds adducted, protecting the lower airway. (3) Oscillating (phonation): vocal folds vibrate as air passes through, producing sound. Glottic aperture size directly affects airway resistance and voice production. Glottic edema (angioedema, anaphylaxis) causes stridor and airway obstruction.

Hair Cells

– The sensory receptor cells of the inner ear that convert mechanical sound vibrations into electrical neural signals, and they are named for the bundle of stereocilia projecting from their apical surface. The inner hair cells, arranged in a single row along the length of the cochlea, are the primary auditory receptors, with approximately 95 percent of the auditory nerve fibers innervating them, and they are responsible for transmitting sound information to the brain. The outer hair cells, arranged in three rows, function as cochlear amplifiers through a unique property called electromotility, in which the cells change length in response to electrical stimulation, actively amplifying the motion of the basilar membrane and enhancing the ear's sensitivity to quiet sounds and its ability to distinguish between different frequencies. Hair cells are highly susceptible to damage from loud noise exposure, ototoxic medications (including aminoglycoside antibiotics, cisplatin, and loop diuretics), aging, and genetic mutations, and once damaged in mammals, they do not regenerate, leading to permanent sensorineural hearing loss.

Hay Fever (Allergic Rhinitis)

– An IgE-mediated, type I hypersensitivity reaction to airborne allergens (pollen, dust mites, mould spores, animal dander). Hay fever: seasonal allergic rhinitis (caused by grass, tree, and weed pollens), while perennial allergic rhinitis occurs year-round (house dust mites, pet dander, mould). Pathophysiology: allergen cross-links IgE on sensitised mast cells in nasal mucosa, triggering degranulation (histamine, leukotrienes, prostaglandins) within minutes (early phase: sneezing, itching, rhinorrhea, nasal congestion). Late phase (4-6 hours): eosinophil and basophil recruitment, causing persistent congestion and hyperreactivity. Symptoms: sneezing paroxysms, anterior rhinorrhea (clear watery discharge), nasal congestion, post-nasal drip, nasal pruritus, palatal/ear itching, conjunctival symptoms (itchy, red, watery eyes, periorbital edema). Associated conditions: asthma (80% of asthmatics have rhinitis, unified airway concept), eczema, sinusitis, nasal polyps, otitis media with effusion. Diagnosis: clinical (symptom pattern, seasonal/perennial triggers), allergen-specific IgE testing (skin prick test or serum specific IgE, RAST). Management: (1) Allergen avoidance (encasing mattresses and pillows, HEPA filters, avoiding outdoor activity during high pollen counts, saline nasal irrigation). (2) Pharmacological: oral/nasal antihistamines (cetirizine, loratadine, fexofenadine), intranasal corticosteroids (beclomethasone, fluticasone, mometasone–first-line for moderate-severe intermittent or persistent disease), leukotriene receptor antagonists (montelukast–adjunctive with asthma), intranasal mast cell stabilisers (cromoglycate), oral decongestants (pseudoephedrine–short-term only). (3) Immunotherapy: allergen-specific subcutaneous (SCIT) or sublingual (SLIT) immunotherapy, modifies the immune response (induces IgG-blocking antibodies, regulatory T cells, shifts from Th2 to Th1), highly effective for pollen and dust mite allergies.

Hearing aid

– An electronic device worn in or behind the ear by people with hearing problems that makes sounds louder.

Hearing Aids and Devices

– Electronic devices worn in or behind the ear that amplify sound for individuals with hearing loss. Modern hearing aids include digital signal processing, directional microphones, and connectivity to smartphones and other audio sources. Cochlear implants are surgically implanted devices that directly stimulate the auditory nerve for those with severe-to-profound hearing loss who do not benefit from conventional hearing aids.

Hearing Loss

– Reduced ability to hear, classified as conductive (outer/middle ear pathology: cerumen impaction, otitis media, otosclerosis, tympanic membrane perforation), sensorineural (inner ear or auditory nerve: presbycusis, noise exposure, ototoxic drugs, Meniere disease, acoustic neuroma), or mixed. Presents with difficulty understanding speech, especially in background noise. Diagnosis: audiometry (pure tone average, speech discrimination), tympanometry. Treatment: hearing aids, cochlear implants for severe-to-profound SNHL, surgical correction for conductive causes. Prevention: noise protection, ototoxic medication monitoring.

Impedance hearing testing

– A test that sends sound waves to the eardrum to determine if a problem in the middle ear is causing hearing loss.

Incus

– The incus, commonly known as the anvil, is the middle ossicle of the three bones in the middle ear, articulating with the malleus at the incudomalleolar joint and with the stapes at the incudostapedial joint. It is the largest of the three ossicles and is shaped like a premolar tooth, with a body and two processes: the long process, which descends parallel to the manubrium of the malleus and articulates with the stapes via the lenticular process, and the short process, which projects posteriorly and is attached to the posterior wall of the middle ear by the posterior incudal ligament. The incus functions as a lever in the ossicular chain, amplifying sound vibrations from the tympanic membrane to the oval window with an amplification factor of approximately 1.3, and combined with the area ratio between the tympanic membrane and oval window, provides a total sound pressure amplification of approximately 22-fold. The long process of the incus is particularly vulnerable to ischemic necrosis due to its tenuous blood supply, and erosion of this structure is a common complication of chronic otitis media and cholesteatoma.

Inner ear

– The deepest part of the ear, consisting of the cochlea and the labyrinth.

Kiesselbach Plexus

– Kiesselbach plexus, also known as Little area, is a highly vascularized region located on the anterior nasal septum where five arterial anastomoses converge, making it the most common site of epistaxis (nosebleed), accounting for approximately ninety.

Labyrinth

– The inner ear. It contains the cochlea, which is responsible for hearing, as well as structures that are needed for balance.

Labyrinthitis

– Inflammation of the labyrinth, the inner ear structure containing the cochlea and vestibular apparatus, resulting in both auditory and vestibular symptoms including sudden, severe vertigo, sensorineural hearing loss, tinnitus, nausea, and vomiting. The condition is distinguished from vestibular neuritis, which involves inflammation of the vestibular nerve without cochlear involvement, by the presence of hearing loss. Labyrinthitis may be caused by viral infections (the most common cause, often following upper respiratory infections), bacterial infections (typically spreading from middle ear infections or meningitis), autoimmune diseases, or ototoxic medications. The viral form is generally self-limited, with gradual improvement over days to weeks, though some patients experience persistent vestibular symptoms and residual hearing loss. Bacterial labyrinthitis is more severe and can result in permanent hearing loss and vestibular dysfunction, requiring aggressive antibiotic treatment and potentially surgical drainage of the middle ear. Treatment includes vestibular suppressants for acute vertigo, antiemetics, and vestibular rehabilitation therapy for persistent symptoms.

Laryngitis

– Inflammation of the larynx and vocal cords that presents with hoarseness, voice loss, sore throat, dry cough, and a sensation of a lump in the throat (globus sensation), and it may be acute or chronic. Acute laryngitis is most commonly caused by viral upper respiratory infections, and it is usually self-limited, resolving within one to two weeks, though it may be accompanied by fever, malaise, and cervical lymphadenopathy. Chronic laryngitis lasting more than three weeks may be caused by voice abuse, laryngopharyngeal reflux, chronic sinusitis with postnasal drip, smoking, environmental irritants, allergic rhinitis, and certain medications, and it can lead to chronic hoarseness and permanent vocal cord changes if not addressed. The diagnosis is typically clinical, based on history and laryngoscopy findings of vocal cord erythema, edema, and sometimes hemorrhagic spots. Treatment of acute laryngitis includes voice rest, hydration, humidification, and avoidance of irritants, while chronic laryngitis requires identification and treatment of the underlying cause.

Laryngoscopy

– A medical procedure used to visualise the larynx (voice box) and adjacent structures of the throat. Types: (1) Indirect laryngoscopy: the simplest technique, using a laryngeal mirror angled at 120 degrees to reflect the laryngeal image, with a headlight or external light source. (2) Flexible nasolaryngoscopy (or transnasal flexible laryngoscopy): a thin, flexible fibreoptic endoscope passed through the nostril to visualise the nasal cavity, pharynx, and larynx. Performed under topical anesthesia (lidocaine spray, decongestant). The patient is awake and phonates to assess vocal fold movement. Used for evaluating hoarseness, stridor, dysphagia, laryngeal masses, and vocal cord paralysis. (3) Direct laryngoscopy: performed under general anesthesia using a rigid laryngoscope (e.g., Macintosh or Miller blade for intubation; suspension laryngoscope for surgery), providing a direct view of the larynx and subglottis. Used for biopsy, removal of laryngeal lesions (polyps, nodules, papillomas), laser surgery, and foreign body removal. (4) Video laryngoscopy: uses a camera at the tip of a rigid laryngoscope blade, providing an indirect view of the glottis on a screen. Improves glottic visualisation in difficult airways (e.g., Cormack-Lehane grade 3-4, obesity, limited neck mobility). Devices: GlideScope, McGrath, C-MAC, Airtraq.

Larynx

– The larynx, commonly known as the voice box, is a complex organ located in the anterior neck between the pharynx and the trachea that serves three primary functions: phonation (voice production), protection of the lower airway from aspiration during swallowing, and regulation of airflow during respiration. The laryngeal skeleton consists of several cartilages including the thyroid cartilage (the largest, commonly known as the Adam's apple), the cricoid cartilage (the only complete cartilaginous ring in the airway), the paired arytenoid cartilages (which articulate with the vocal cords), and the epiglottis (which covers the laryngeal inlet during swallowing). The larynx contains two pairs of folds: the true vocal cords (vocal folds), which are responsible for phonation and are covered by stratified squamous epithelium, and the false vocal cords (vestibular folds), which are located superior to the true cords and play a role in airway protection. The larynx is innervated by the superior laryngeal nerve (sensory to the supraglottic larynx) and the recurrent laryngeal nerve (motor to all intrinsic laryngeal muscles except the cricothyroid), and injury to these nerves,.

Malleus

– The malleus, commonly known as the hammer, is the largest of the three ossicles in the middle ear and is the most laterally positioned, with its handle (manubrium) embedded in the fibrous layer of the tympanic membrane and its head articulating with the incus at the incudomalleolar joint. The malleus is an irregularly shaped bone approximately 8 to 9 millimeters in length, consisting of a head, neck, anterior process, lateral process, and manubrium, and it transmits vibrations from the tympanic membrane to the incus and subsequently to the stapes and oval window. The tensor tympani muscle attaches to the handle of the malleus and contracts reflexively in response to loud sounds, increasing the stiffness of the ossicular chain and reducing the transmission of potentially damaging low-frequency sounds to the inner ear. During middle ear infections, the malleus may become eroded by cholesteatoma or chronic inflammation, leading to conductive hearing loss, and its assessment during otoscopy is important for evaluating middle ear pathology.

Mastoidectomy

– A surgical procedure that involves the removal of diseased mastoid air cells and bone from the mastoid portion of the temporal bone, performed for chronic otitis media with cholesteatoma, mastoiditis, or as a component of surgery for acoustic neuroma and other skull base tumors. The canal wall up (closed) mastoidectomy technique preserves the bony wall between the external auditory canal and the mastoid cavity, maintaining the normal anatomy and eliminating the need for ongoing mastoid cavity care, while the canal wall down (open) mastoidectomy removes the bony wall, creating a common cavity between the external auditory canal and the mastoid, which is indicated for extensive cholesteatoma that cannot be completely removed through a canal wall up approach. The canal wall down technique results in a larger mastoid cavity that requires periodic cleaning and is prone to accumulation of debris and recurrent infection, but it provides better access and visualization for complete removal of cholesteatoma. The procedure is performed under general anesthesia using a postauricular.

Mastoiditis

– Bacterial infection of the mastoid air cells, typically a complication of acute otitis media. Presents with postauricular pain, swelling, erythema, and tenderness, fever, and otorrhea. Diagnosis: CT temporal bone shows mastoid air cell opacification with bony destruction. Complications: subperiosteal abscess, Bezold abscess (pus tracking into sternocleidomastoid), facial nerve palsy, sigmoid sinus thrombosis, meningitis, and intracranial abscess. Treatment: IV antibiotics (ceftriaxone, vancomycin), myringotomy with tube placement, and mastoidectomy for refractory cases.

Meatuses

– The nasal meatuses are the air passages located beneath each corresponding turbinate on the lateral wall of the nasal cavity, and they serve as the drainage pathways for the paranasal sinuses and the nasolacrimal duct. The inferior meatus, located beneath the inferior turbinate, receives drainage from the nasolacrimal duct, which carries tears from the eye to the nasal cavity. The middle meatus, located beneath the middle turbinate, receives drainage from the maxillary sinus through the hiatus semilunaris, the frontal sinus through the frontonasal duct, and the anterior ethmoidal air cells through the ethmoidal infundibulum. The superior meatus, located beneath the superior turbinate, receives drainage from the posterior ethmoidal air cells. The sphenoethmoidal recess, located above and behind the superior turbinate, receives drainage from the sphenoid sinus. Understanding the drainage pathways of the paranasal sinuses through the meatuses is essential for diagnosing and treating sinusitis, as obstruction of these narrow drainage pathways can.

Meniere Disease

– An inner ear disorder characterized by recurrent episodes of vertigo lasting 20 minutes to 12 hours, with fluctuating sensorineural hearing loss, tinnitus, and aural fullness. Due to endolymphatic hydrops. Diagnosis: clinical criteria; audiometry shows low-frequency SNHL during attacks. Treatment: low-sodium diet, diuretics (hydrochlorothiazide), intratympanic steroids, intratympanic gentamicin for vertigo control, and surgical interventions (endolymphatic sac decompression, vestibular nerve section) for refractory cases.

Middle Ear

– The middle ear, also known as the tympanic cavity, is an air-filled, mucosa-lined space within the temporal bone that lies between the tympanic membrane laterally and the medial wall of the middle ear (abyrinthine wall) medially, and it contains the three ossicles (malleus, incus, and stapes) that transmit sound vibrations from the tympanic membrane to the oval window of the inner ear. The middle ear is divided into three compartments: the mesotympanum (directly behind the tympanic membrane), the epitympanum or attic (above the level of the tympanic membrane), and the hypotympanum (below the level of the tympanic membrane). The middle ear is connected to the nasopharynx by the Eustachian tube, which equalizes air pressure on both sides of the tympanic membrane, and it is lined by a mucous membrane that is continuous with the lining of the Eustachian tube and the pharynx. The.

Mixed Hearing Loss

– A combination of both conductive and sensorineural hearing loss occurring in the same ear, resulting in a hearing deficit that involves impaired sound transmission through the outer or middle ear in addition to cochlear or neural dysfunction. The audiometric pattern shows both an air-bone gap indicating the conductive component and elevated bone conduction thresholds indicating the sensorineural component, and the total hearing loss is the sum of both components. Common causes include chronic otitis media or cholesteatoma with superimposed presbycusis, otosclerosis with cochlear involvement, and noise exposure with concurrent middle ear pathology. The management of mixed hearing loss requires addressing both components: treating the conductive component medically or surgically when possible (such as repairing a tympanic membrane perforation or reconstructing the ossicular chain) and managing the sensorineural component with hearing amplification or cochlear implantation as indicated.

Nasal Cavity

– The nasal cavity is the paired, air-filled space located between the nasal septum medially and the lateral nasal wall, extending from the nares (nostrils) anteriorly to the choanae (posterior openings into the nasopharynx) posteriorly, and it serves three primary functions: warming, humidifying, and filtering inspired air; olfaction (smell); and resonance of the voice. The lateral nasal wall contains three scroll-like bony projections called turbinates (superior, middle, and inferior), which increase the surface area of the nasal mucosa and create turbulent airflow to optimize air conditioning. Beneath each turbinates lies a corresponding meatus: the superior, middle, and inferior meatuses, which receive drainage from the paranasal sinuses. The nasal septum, composed of both cartilage and bone, divides the nasal cavity into two halves and is lined by a.

Nasal Polyps

– Benign, pedunculated, edematous outgrowths of the sinus and nasal mucosa that arise most commonly from the ethmoidal sinuses and protrude into the nasal cavity through the middle meatus, causing nasal obstruction, anosmia (loss of smell), and rhinorrhea. The polyps are composed of edematous stroma with a sparse inflammatory infiltrate including eosinophils, lymphocytes, and plasma cells, and they are associated with type 2 inflammation, elevated tissue and serum IgE levels, and aspirin sensitivity in approximately thirty to forty percent of patients (Samter triad: aspirin sensitivity, asthma, and nasal polyposis). Nasal polyps are more common in adults, particularly males over the age of forty, and they are associated with chronic rhinosinusitis, asthma, aspirin sensitivity, cystic fibrosis, and primary ciliary dyskinesia.

Nasal Septum

– The nasal septum is the midline structure that divides the nasal cavity into two halves, consisting of an anterior cartilaginous portion (the septal cartilage or quadrilateral cartilage), a posterior bony portion (the perpendicular plate of the ethmoid bone and the vomer), and the membranous portion (the columella) at the anterior nasal opening. The septum provides structural support for the nose, divides the nasal airflow, and supports the nasal mucosa, which is richly vascularized by branches of the internal and external carotid arteries. Deviation of the nasal septum is extremely common, with virtually all individuals having some degree of asymmetry, and significant deviation can cause nasal obstruction, impaired sinus drainage, recurrent.

Nasopharyngeal Carcinoma

– A malignancy arising from the epithelial lining of the nasopharynx, and it has a unique geographical and ethnic distribution, with highest incidence in southern China, Southeast Asia, and North Africa.

The tumor is strongly associated with Epstein-Barr virus infection, which is present in virtually all cases of undifferentiated nasopharyngeal carcinoma, and dietary factors including salt-preserved fish and other nitrosaminecontaining foods. The tumor typically arises in the lateral wall of the nasopharynx (fossa of Rosenmuller) and often presents with a painless neck mass (cervical lymphadenopathy), nasal obstruction, epistaxis, hearing loss from Eustachian tube obstruction, and cranial nerve palsies. Diagnosis is confirmed by endoscopic examination and biopsy, and staging involves MRI of the nasopharynx and neck and PET-CT. Nasopharyngeal carcinoma is highly radiosensitive, and the primary treatment is radiation therapy combined with chemotherapy for locally advanced disease, with chemotherapy regimens typically including cisplatin and 5-fluorouracil.

Neck Cancer

– Human papillomavirus, particularly HPV-16, is an increasingly important etiological factor in head and neck squamous cell carcinoma, particularly oropharyngeal cancer, and it is responsible for the rising incidence of oropharyngeal cancer in younger patients without traditional risk factors of tobacco and alcohol use. HPV-positive oropharyngeal cancer has distinct epidemiological, clinical, and molecular characteristics compared to HPV-negative disease, including a younger age of presentation, better performance status, smaller primary tumors with larger cervical lymph node metastases, and significantly improved prognosis, with five-year survival rates of approximately eighty to ninety percent compared.

Non-Allergic Rhinitis

– Non-allergic rhinitis, also known as vasomotor rhinitis, is a chronic nasal condition characterized by nasal symptoms including congestion, rhinorrhea, and sneezing that are not caused by an IgE-mediated allergic response, and it is distinguished from allergic rhinitis by the absence of allergen-specific IgE, negative allergy testing, and the absence of the systemic symptoms typically associated with allergic rhinitis. The pathogenesis involves dysfunction of the autonomic nervous system regulation of the nasal mucosal blood vessels and glands, leading to exaggerated vasodilation and increased mucus production in response to environmental triggers.

Oral Leukoplakia

– A white patch or plaque on the oral mucosa that cannot be scraped off and cannot be characterized as any other diagnosable disease, considered a potentially malignant disorder. Risk factors: tobacco use (most common), alcohol, and betel nut chewing. Most common sites: buccal mucosa, alveolar ridge, and tongue. Dysplasia (mild, moderate, severe) found on biopsy in 10-40% of cases. Progression to invasive squamous cell carcinoma: 1% per year for non-dysplastic, higher with dysplasia. Diagnosis: biopsy for histology. Treatment: excision or laser ablation for dysplastic lesions; observation for non-dysplastic. Risk factor modification essential. Long-term surveillance required.

Ossicles

– Three bones in the middle ear that move in response to sound vibrations.

Otic capsule

– The otic capsule is the dense bony shell (otic capsule bone) that surrounds and protects the inner ear structures, including the cochlea, vestibule, and semicircular canals, forming the hardest bone in the human body. It develops from the otic placode and undergoes endochondral ossification, and unlike other temporal bone components, it undergoes very little remodeling after development, making it resistant to otosclerosis but susceptible to otic capsule dehiscence (superior semicircular canal dehiscence syndrome), in which thinning or absence of the bone over the superior semicircular canal creates a third window phenomenon, causing sound- or pressure-induced vertigo (Tullio phenomenon), conductive hearing loss, and autophony.

Otitis Externa

– Otitis externa, commonly known as swimmer's ear, is an infection or inflammation of the external.

Otitis media

– A group of inflammatory conditions of the middle ear, most commonly caused by viral or bacterial infection, with Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis being the most frequent bacterial pathogens. It is classified into acute otitis media (AOM), characterized by rapid-onset otalgia, fever, and bulging erythematous tympanic membrane with middle ear effusion; otitis media with effusion (OME), in which non-purulent fluid persists in the middle ear without signs of acute infection, often causing conductive hearing loss and speech delay in children; and chronic suppurative otitis media (CSOM), defined by persistent otorrhea through a tympanic membrane perforation for more than 6 weeks. Eustachian tube dysfunction is the primary predisposing factor, as impaired drainage and ventilation of the middle ear allow pathogens to ascend from the nasopharynx. Treatment of AOM includes observation or antibiotics (amoxicillin 80-90 mg/kg/day for 10 days in children) depending on age and severity, while OME may require tympanostomy tube placement if effusion persists beyond 3 months with hearing loss. Complications include tympanic membrane perforation, mastoiditis, cholesteatoma, and intracranial spread.

Otolaryngology (ENT)

– The surgical specialty concerned with the diagnosis and treatment of disorders of the ear, nose, throat, head, and neck. Otolaryngologists (ENT surgeons) manage conditions ranging from otitis media and sinusitis to head and neck cancer, thyroid disease, and voice disorders. Subspecialties: otology/neurotology (ear disorders, hearing loss, balance), rhinology (sinus disease, anterior skull base), laryngology (voice, swallowing), head and neck oncology (cancer surgery including neck dissection, laryngectomy, thyroidectomy), paediatric ENT (cleft palate, cochlear implants), and facial plastic and reconstructive surgery. Training in the UK requires 2 years of core surgical training followed by 6 years of specialty training (ST3–ST8).

Otology

– The branch of medicine and surgery concerned with the study and treatment of ear disorders. Otology encompasses hearing loss (conductive, sensorineural, mixed), tinnitus, vertigo and balance disorders (Meniere disease, benign paroxysmal positional vertigo, vestibular neuritis), otitis media, otosclerosis, cholesteatoma, and ear trauma. Diagnostic tools: pure-tone audiometry, tympanometry, otoacoustic emissions, auditory brainstem response (ABR), and CT/MRI of the temporal bone. Surgical procedures include myringotomy, tympanoplasty, mastoidectomy, stapedectomy, cochlear implantation, and ossiculoplasty. Neurotology is a subspecialty addressing the interface between the ear and brain (acoustic neuroma, CSF leak repair, skull base surgery).

Otorrhea

– Discharge from the ear. Classification: acute otorrhea (<6 weeks) and chronic otorrhea (>6 weeks). Causes: acute otitis media with tympanic membrane perforation (purulent discharge, often in children), otitis externa (watery or mucoid discharge, ear canal inflammation), chronic suppurative otitis media (persistent purulent discharge through a non-healing perforation, associated with cholesteatoma), cerebrospinal fluid otorrhea (clear, watery, β-2 transferrin positive, indicates skull base fracture or surgical complication), and foreign body. otorrhea with pain, fever, and mastoid tenderness raises concern for acute mastoiditis. Management: topical antibiotic drops (ciprofloxacin—first-line, avoids ototoxicity of aminoglycosides), aural toilet (microsuction), treatment of underlying cause; for CSF otorrhea, neurosurgical repair is required.

Otosclerosis

– An abnormal bony growth in the middle ear that causes hearing loss.

Otoscopy

– A clinical examination technique used to visualize the external auditory canal and tympanic membrane, and it is an essential component of the evaluation of ear complaints including hearing loss, ear pain, otorrhea, and tinnitus. Pneumatic otoscopy involves the use of a specialized otoscope with a sealed speculum that allows insufflation of air into the ear canal while observing the tympanic membrane, assessing the mobility of the membrane in response to pressure changes, which is reduced in otitis media with effusion and absent in middle ear fluid. The normal tympanic membrane appears as a translucent, pearly gray, cone-shaped membrane with a visible light reflex in the anteroinferior quadrant, and the handle of the malleus and the umbo (the central point of the tympanic membrane) are visible landmarks. Abnormal findings include bulging, erythema, effusion, perforation, retraction, cholesteatoma, and tympanostomy tubes, and proper otoscopic technique requires pulling the pinna superiorly and posteri-.

Outer ear

– The external part of the ear, as well as the external ear canal and the eardrum.

Ozaena

– A chronic condition characterized by atrophic rhinitis with crusting, foul-smelling nasal discharge (fetor), and a sensation of nasal obstruction despite a patent airway. Ozaena is caused by progressive atrophy of the nasal mucosa and underlying bone, often associated with Klebsiella ozaenae infection. It is more common in women, typically presents in middle age, and is more prevalent in tropical and subtropical regions. The nasal mucosa is pale, dry, and crusted; removal of crusts reveals a widely patent nasal cavity. Atrophy may extend to the olfactory epithelium, causing anosmia. Treatment: nasal saline irrigation, topical antibiotics (neomycin, gentamicin), oral antibiotics (ciprofloxacin, rifampicin for Klebsiella), oestrogen nasal spray (improves mucosal blood flow), and surgical closure of the nasal cavity (Young procedure) in severe refractory cases.

Paranasal Sinuses

– The paranasal sinuses are four pairs of air-filled, mucosa-lined cavities within the facial bones that communicate with the nasal cavity through narrow drainage pathways called ostia, and they serve multiple functions including lightening the weight of the skull, humidifying and warming inspired air, producing mucus for nasal clearance, contributing to vocal resonance, and providing structural support for the face. The maxillary sinuses, the largest of the paranasal sinuses, are located in the maxillary bones beneath the orbits and lateral to the nasal cavity, with their ostia opening into the middle meatus. The frontal sinuses, located in the frontal bone above the orbits, drain through the frontonasal duct into the middle meatus. The ethmoidal sinuses, consisting of multiple small air cells within the ethmoid bone between the orbits, are divided into anterior and posterior groups, with the anterior cells draining into the middle meatus and the posterior cells draining into the superior meatus. The sphenoid sinuses, located in the sphenoid bone posterior to the ethmoidal sinuses, drain into the sphenoethmoidal recess. Sinusitis occurs when the ostia become obstructed, preventing mucus drainage and creating a favorable environment for bacterial infection.

Parotitis

– Inflammation of the parotid gland. Acute parotitis: most commonly caused by mumps virus (paramyxovirus—epidemic parotitis, typically in unvaccinated children, bilateral in 70%, resolves spontaneously over 7–10 days). Suppurative (bacterial) parotitis: Staphylococcus aureus (most common), streptococci, anaerobes—occurs in dehydrated, post-surgical, or immunocompromised patients; presents with painful, erythematous swelling, purulent discharge from Stensen duct, fever, and leukocytosis; treated with IV antibiotics (flucloxacillin), hydration, sialagogues, and surgical drainage if abscess forms. Chronic parotitis: recurrent episodes due to duct obstruction (sialolithiasis), Sjögren syndrome, sarcoidosis, or IgG4-related disease. Diagnosis: clinical, ultrasound, sialography, MRI. Complications of bacterial parotitis: parotid abscess, facial nerve palsy, extension to the parapharyngeal space. Mumps parotitis is prevented by MMR vaccination (two doses).

Peritonsillar Abscess

– Peritonsillar abscess, also known as quinsy, is a collection of pus in the peritonsillar space between the tonsillar capsule and the superior pharyngeal constrictor muscle, typically developing as a complication of acute tonsillitis, and it is the most common deep neck space infection. The condition presents with severe, unilateral sore throat, odynophagia, dysphagia, trismus (difficulty opening the mouth due to spasm of the pterygoid muscles), hot potato voice (muffled speech), and deviation of the uvula toward the contralateral side, with erythematous, bulging of the soft palate and tonsillar pillar on the affected side. The most common causative organisms.

Pharyngitis

– Inflammation of the pharynx, the most common presentation of which is sore throat (odynophagia) with or without fever, and it is classified as bacterial or viral based on the causative organism, with viral pharyngitis accounting for approximately eighty to ninety percent of cases and bacterial pharyngitis (primarily Group A Streptococcus) accounting for approximately ten to twenty percent. Viral pharyngitis typically presents with a gradual onset of sore throat, cough, rhinorrhea, conjunctivitis, and hoarseness, and is caused by adenoviruses, rhinoviruses, coronaviruses, influenza, and.

Pleomorphic Adenoma

– Pleomorphic adenoma, also known as benign mixed tumor, is the most common benign salivary gland tumor, accounting for approximately sixty to seventy percent of all parotid gland neoplasms and a significant proportion of minor salivary gland tumors, and it is characterized by a diverse histological appearance with a mixture of epithelial, myoepithelial, and mesenchymal-like components in a variable stroma that may be mucoid, chondroid, or osseous. The tumor typically presents as a slow-growing, painless, firm, mobile mass in the parotid gland, most commonly in the superficial lobe, and it is more common in women (female-to-male ratio of approximately 2:1), with a peak incidence in the fourth to.

Pneumatic Otoscopy

– A diagnostic technique that combines visualization of the tympanic membrane with assessment of its mobility in response to positive and negative air pressure changes, providing information about the presence of middle ear fluid and the condition of the Eustachian tube. The procedure uses a pneumatic otoscope with a soft rubber bulb that generates positive and negative pressure, and the examiner observes the tympanic membrane for movement in response to these pressure changes. Normal tympanic membranes move freely with pressure changes, while middle ear effusion reduces or eliminates mobility, and the degree of reduced mobility correlates with the presence and volume of middle ear fluid. Pneumatic otoscopy is recommended as the primary diagnostic tool for acute otitis media in children, as it provides both structural and functional information about the middle ear, and it can detect middle ear effusion with a sensitivity of approximately ninety percent when performed by experienced examiners. The technique is inexpensive, non-invasive, and does not require sedation, making it ideal for use in young children.

Presbycusis

– Age-related hearing loss caused by the death of hair cells in the inner ear.

Prosthesis

– An artificial device such as a hearing aid, artificial joint, or dentures that substitutes for a missing body part.

Pure Tone Audiometry

– The standard behavioral test used to measure hearing sensitivity across the frequency range important for speech perception, typically from 250 to 8000 Hertz, by determining the lowest intensity level at which a pure tone can be detected approximately fifty percent of the time (the hearing threshold). Air conduction thresholds are measured using headphones or insert earphones, and bone conduction thresholds are measured using a bone vibrator placed on the mastoid process or forehead, and the comparison between air and bone conduction thresholds distinguishes between conductive, sensorineural, and mixed hearing loss. The audiogram plots the hearing thresholds in decibels hearing level (dB HL) against frequency in Hertz (Hz), with normal hearing defined as thresholds of 25 dB HL or better. The degree of hearing loss is classified as normal (0-25 dB HL), mild (26-40 dB HL), moderate (41-55 dB HL), moderately severe (56-70 dB HL), severe (71-90 dB HL), and profound (greater than 90 dB HL). Pure tone audiometry is the foundation of audiometric assessment and is essential for diagnosing hearing loss, determining its type and degree, and guiding treatment decisions.

Quinsy

– See Peritonsillar Abscess.

Retropharyngeal Abscess

– A collection of pus in the retropharyngeal space, located between the posterior pharyngeal wall and the prevertebral fascia. Most common in children aged 2-4 years, arising from suppurative lymphadenitis of the retropharyngeal lymph nodes that drain the nasopharynx, adenoids, and sinuses. In adults, it typically results from direct extension of pharyngeal infection (tonsillitis, dental infection) or from penetrating trauma (foreign body, instrumentation). Presents with fever, sore throat, neck pain, dysphagia, drooling, cervical lymphadenopathy, and neck stiffness. In children, the classic presentation includes refusal to feed, stridor, or respiratory distress. Diagnosis is confirmed by contrast-enhanced CT of the neck showing a hypodense collection with rim enhancement in the retropharyngeal space. Lateral neck X-ray may show prevertebral soft tissue swelling (greater than 7 mm at C2 or greater than 14 mm at C6). Treatment: airway management, IV antibiotics (e.g., clindamycin or ampicillin-sulbactam), and surgical drainage via transoral incision or transcervical approach for larger abscesses or those not responding to antibiotics. Complications include airway obstruction, mediastinitis, jugular vein thrombosis, and sepsis.

Rhinitis

– Inflammation of the nasal mucosa. Allergic rhinitis: IgE-mediated response to environmental allergens (pollen, dust mites, pet dander), presenting with sneezing, rhinorrhea, nasal congestion, and pruritus. Non-allergic rhinitis: vasomotor rhinitis (triggers: irritants, temperature changes, strong odors), infectious (viral URI), medication-induced (rhinitis medicamentosa from topical decongestant overuse), hormonal (pregnancy, hypothyroidism), and gustatory rhinitis (food-induced). Diagnosis: clinical, allergy testing. Treatment: intranasal corticosteroids (first-line), oral antihistamines, decongestants, saline irrigation, allergen immunotherapy for allergic rhinitis.

Rhinoplasty

– A surgical procedure to alter the shape, size, or function of the nose, performed for cosmetic reasons (aesthetic rhinoplasty—improving nasal appearance) or functional reasons (functional rhinoplasty—improving nasal airflow by correcting internal/external nasal valve collapse, deviated septum, or turbinate hypertrophy). Approaches: open rhinoplasty (transcolumellar incision—better exposure, preferred for complex cases) and closed (endonasal) rhinoplasty (incisions within the nostril—no external scar, less swelling). Components: hump reduction (dorsal bony and cartilaginous hump), osteotomies (narrowing the nasal bones), tip rhinoplasty (refinement of the nasal tip—cephalic trim, tip sutures, columellar strut graft), alar base reduction, and septoplasty (correction of septal deviation). Grafts: autologous cartilage (septal, auricular, costal) for structural support or contour augmentation. Post-operative: swelling and bruising (peaks at 48 hours, resolves over 1–2 weeks), nasal splint for 1 week, avoidance of glasses and strenuous activity for 4–6 weeks. Complications: bleeding, infection, nasal obstruction, asymmetry, saddle nose deformity (over-resection of dorsum), and need for revision (5–15%).

Rinne Test

– The Rinne test is a screening test that compares air conduction and bone conduction hearing using a vibrating tuning fork (typically 512 Hz) to help differentiate between conductive and sensorineural hearing loss. The test is performed by first placing the vibrating tuning fork on the mastoid process to assess bone conduction, asking the patient to indicate when the sound is no longer heard, and then immediately holding the tuning fork adjacent to the external auditory canal to assess air conduction, asking the patient to indicate whether the sound is still audible. A positive Rinne test (normal result) occurs when air conduction is heard longer than bone conduction, indicating normal middle ear function or sensorineural hearing loss. A negative.

Salivary Gland Tumors

– Most salivary tumors are benign (80% in parotid, 50% in submandibular, 50% in minor glands). Benign: pleomorphic adenoma (most common overall), Warthin tumor (bilateral in 10%, smokers). Malignant: mucoepidermoid carcinoma (most common malignant), adenoid cystic carcinoma (perineural invasion), acinic cell carcinoma. Presents with painless, slow-growing mass. Malignant features: rapid growth, pain, facial nerve paralysis, skin fixation, and cervical lymphadenopathy. Diagnosis: fine needle aspiration (FNA), CT/MRI for staging. Treatment: superficial parotidectomy for benign; total parotidectomy with neck dissection for malignant, with postoperative radiation as needed.

Semicircular Canals

– The semicircular canals are three bony, fluid-filled, tube-like structures within the vestibular portion of the inner ear that detect rotational (angular) acceleration.

Septal Hematoma

– Accumulation of blood between the perichondrium and the septal cartilage of the nose, typically following nasal trauma (nasal fracture). Presents with nasal obstruction, pain, and a bluish, boggy swelling of the nasal septum. An untreated septal hematoma causes avascular necrosis of the nasal cartilage (cartilage receives nutrition from perichondrium), leading to saddle nose deformity (septal perforation and collapse) within days. Diagnosis: nasal speculum exam. Treatment: immediate incision and drainage (small incision at the inferior aspect), packing to prevent re-accumulation, and antibiotics to prevent infection (septal abscess). Follow-up for re-accumulation.

Sialolithiasis

– Sialolithiasis, commonly known as salivary stones, is the formation of calcified concretions within the salivary gland ducts or parenchyma, most commonly affecting the submandibular gland (approximately eighty percent of cases) due to the long, dependent course of the Wharton duct, the alkaline pH of submandibular saliva, and its higher mucin and calcium content, followed by the parotid gland (approximately ten percent) and the sublingual gland (approximately five percent). The stones are composed primarily of calcium phosphate and calcium carbonate, with an organic matrix of glycoproteins and bacteria, and they range in size from a few millimeters to several centimeters. Patients typically present with intermittent, recurrent episodes of glandular swelling and pain that worsen during meals (when salivary flow is stimulated), and the stone may be palpable in the floor of the mouth (submandibular stones) or intraorally. Diagnosis is confirmed by plain radiography (submandibular stones are radiopaque in approximately eighty percent), ultrasound, or CT imaging. Treatment depends on stone size and location, with small stones managed with sialagogues and hydration, and larger or impacted stones managed with sialendoscopy, lithotripsy, or surgical removal.

Sialorrhoea

– See Drooling / Excessive Salivation.

Sinusitis

– Inflammation of the paranasal sinus mucosa. Acute sinusitis: viral (most common, resolves in <10 days) or bacterial (symptoms >10 days, double worsening, or severe: Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis). Chronic sinusitis: symptoms >12 weeks with radiographic evidence. Presents with nasal congestion, purulent rhinorrhea, facial pain/pressure, headache, and hyposmia. Diagnosis: clinical, CT sinus for chronic/recurrent. Treatment: analgesics, saline irrigation, intranasal corticosteroids; antibiotics for acute bacterial. Endoscopic sinus surgery for refractory chronic sinusitis.

Sinusitis and Sinus Problems

– Sinus problems refer to conditions affecting the paranasal sinuses, which are air-filled cavities in the skull that can become inflamed or infected. Acute sinusitis typically follows a viral upper respiratory infection and causes facial pain, nasal congestion, and headache, while chronic sinusitis persists for weeks or longer and may involve nasal polyps or allergic components. Treatment ranges from saline rinses and decongestants to intranasal corticosteroids, antibiotics for bacterial infection, and in refractory cases, endoscopic sinus surgery.

Speech Discrimination

– Speech discrimination testing, also known as word recognition scoring, measures the ability to understand speech at a comfortable listening level, and it is an important component of the audiometric evaluation that provides information about cochlear and neural function beyond what pure tone audiometry alone can reveal. The test typically uses a standardized list of phonetically balanced monosyllabic words presented at a suprathreshold level (usually 40 dB above the speech recognition threshold), and the patient repeats the words, with the percentage of words correctly identified representing the speech discrimination score. Normal speech discrimination scores range from ninety to one hundred percent, while reduced scores indicate cochlear pathology (particularly in patients with high-tone sensorineural hearing loss) or retrocochlear pathology (as in acoustic neuroma, where scores may be disproportionately.

Sphenoidal Sinus

– The sphenoid sinuses are paired, air-filled cavities located within the body of the sphenoid bone, the most posterior of the paranasal sinuses, situated deep within the skull at the center of the cranial base, and they are in close proximity to critical neurovascular structures including the optic nerves, the internal carotid arteries, the cavernous sinuses, the pituitary gland, and the brainstem. The sphenoid sinuses vary considerably in size and pneumatization among individuals, and they drain through the sphenoethmoidal recess located above and posterior to the superior turbinate into the superior meatus.

Stapedectomy

– A microsurgical procedure performed to treat otosclerosis, a condition characterized by abnormal bone remodeling in the otic capsule that fixes the stapes footplate in the oval window, preventing its normal vibratory movement and causing progressive conductive hearing loss. The procedure involves creating a small opening in the tympanic membrane, elevating the tympanomeatal flap to visualize the middle ear, identifying the fixed stapes, and either removing the entire stapes footplate (stapedectomy) or creating a small fenestration in the footplate (stapedotomy) and inserting a prosthetic device (typically a titanium or fluoroplastic piston) that connects the incus to the oval window, bypassing the fixed stapes and restoring the ossicular chain's ability to transmit sound vibrations. The procedure is performed under local or general anesthesia, and the success rate for hearing improvement is approximately ninety to ninety-five percent, with closure of the air-bone gap to within 10 decibels in most patients. Complications include sensorineural hearing loss (from perilymph leak or inner ear trauma), vertigo, taste disturbance, and tympanic membrane perforation.

Stapes

– The stapes, commonly known as the stirrup, is the smallest bone in the human body and the most medially positioned ossicle in the middle ear, articulating with the incus at the incudostapedial joint and transmitting sound vibrations to the inner ear through its footplate, which is attached to the oval window by the annular ligament. The stapes consists of a head, two crura (anterior and posterior), and a footplate approximately 3.2 millimeters by 1.4 millimeters, and it acts as a piston, converting the lever action of the malleus and incus into a piston-like movement that transmits sound energy into the fluid-filled inner ear. The stapedius.

Stomatitis

– Inflammation of the oral mucosa, presenting as erythema, ulceration, pain, and difficulty eating. Causes: infections (viral—herpes simplex (herpetic gingivostomatitis), Coxsackie (hand-foot-and-mouth disease), herpes zoster; bacterial—acute necrotising ulcerative gingivitis (Vincent angina); fungal—Candida albicans (oral thrush, common in immunocompromised, infants, denture wearers, after antibiotics/inhaled corticosteroids)), physical trauma (biting, burns, ill-fitting dentures), chemical irritation (aspirin burn, alcohol, smoking), aphthous ulcers (recurrent aphthous stomatitis—minor, major, herpetiform; associated with stress, trauma, nutritional deficiencies, Crohn disease, Behçet disease), autoimmune (pemphigus vulgaris, mucous membrane pemphigoid, lichen planus), drug-induced (methotrexate, NSAIDs, chemotherapy—mucositis), and nutritional deficiencies (B vitamins, iron, folate, zinc). Diagnosis: clinical, swab for culture/PCR, patch testing, and biopsy. Treatment: treat underlying cause; symptomatic—topical analgesics (lidocaine), antiseptic mouthwashes (chlorhexidine), corticosteroids (topical or systemic), antifungals for Candida, antivirals for herpes.

Sublingual Gland

– The sublingual glands are the smallest of the major salivary glands, located in the floor of the mouth beneath the tongue, and they produce primarily mucous saliva that is delivered into the oral cavity through multiple small ducts (ducts of Rivinus) that open along the sublingual fold, and through the major sublingual duct (Bartholin duct), which joins the Wharton duct of the submandibular gland. The sublingual glands are rarely affected by sialolithiasis due to the short, wide ducts, and they are more commonly involved in autoimmune conditions including.

Thyroid Cartilage

– The thyroid cartilage is the largest of the laryngeal cartilages, consisting of two laminae that fuse anteriorly at an angle of approximately ninety degrees in males and one hundred twenty degrees in females, creating the laryngeal prominence (commonly known as the Adam's apple), which is more prominent in males due to the acute angle and the effects of testosterone during puberty. The thyroid cartilage forms the anterior and lateral walls of the larynx and provides structural support and protection for the vocal cords and the intrinsic laryngeal muscles. The superior and inferior horns of the thyroid cartilage articulate with the hyoid bone and the cricoid cartilage, respectively, and the cricothyroid muscle, the only intrinsic laryngeal muscle innervated by the external branch of the superior laryngeal nerve, attaches to the thyroid cartilage and acts to tense the vocal cords, raising the pitch of the voice. The thyroid cartilage is prone to fracture in blunt neck trauma, and thyroid cartilage fractures can cause airway compromise, subcutaneous emphysema, and hoarseness.

Thyroid Nodule

– A thyroid nodule is a discrete lesion within the thyroid gland that is palpably or radiographically distinct from the surrounding thyroid parenchyma, and it is a common clinical finding, present in approximately five percent of palpable neck examinations and up to fifty percent of individuals on ultrasound imaging, with the vast majority being benign. The evaluation of thyroid nodules focuses on identifying the small minority that represent thyroid carcinoma, and it involves ultrasound assessment of nodule characteristics (size, composition, echogenicity, margins, shape, and calcifications), fine-needle aspiration biopsy of suspicious nodules, and thyroid function testing. The Thyroid Imaging Reporting.

Tinnitus

– The perception of sound in the absence of an external acoustic stimulus, commonly described as ringing, buzzing, hissing, or roaring in the ears, and it affects approximately fifteen to twenty percent of the global population, with a significant impact on quality of life in severe cases. Subjective tinnitus, the most common form, is perceived only by the patient and is most commonly associated with sensorineural hearing loss, noise ex-.

TI-RADS

– The Thyroid Imaging Reporting and Data System is a standardized ultrasound-based risk stratification system developed by the American College of Radiology to evaluate thyroid nodules and guide recommendations for fine-needle aspiration biopsy, reducing unnecessary biopsies of benignappearing nodules while ensuring adequate sampling of suspicious nodules. The TI-RADS scoring system assigns points based on five ultrasound characteristics: composition (solid, mixed cystic-solid, predominantly cystic, or spongiform), echogenicity (anechoic, hypoechoic, isoechoic, or markedly hypoechoic), shape (wider-than-tall or taller-thanwide), margin (smooth, ill-defined, lobulated, or extra-thyroidal extension), and echogenic foci (none, macrocalcifications, rim calcifications, or microcalcifications). The total TI-RADS score determines the category from TR1 (benign, no biopsy needed) to TR5 (highly suspicious, biopsy recommended for nodules one centimeter or larger), and the system provides evidence-based size thresholds for biopsy recommendations at each TI-RADS level, optimizing the balance between cancer detection and unnecessary procedures.

Tonsillectomy

– The surgical removal of the palatine tonsils, and it is one of the most commonly performed surgical procedures in children, traditionally indicated for recurrent acute tonsillitis (seven episodes in one year, five per year for two years, or three per year for three years), chronic tonsillitis, and obstructive sleep apnea caused by tonsillar hypertrophy. The procedure is performed under general anesthesia, and the most commonly used techniques include cold dissection with electrocautery, bipolar diathermy, intracapsular tonsillectomy (partial tonsillectomy using a microdebrider or coblation), and total tonsillectomy. The most common complications include postoperative hemorrhage (the most significant complication, occurring in approximately two to four percent of cases), dehydration from pain-related reduced oral intake, and velopharyngeal insufficiency. Postoperative pain is typically moderate to severe and lasts seven to ten days, and patients should avoid aspirin and non-steroidal anti-inflammatory drugs due to the increased risk of bleeding. The intracapsular technique preserves a thin rim of tonsillar tissue, reducing postoperative pain and hemorrhage rates, though with a small risk of tonsillar regrowth.

Tonsillitis

– Inflammation of the tonsils, most commonly the palatine tonsils, which are lymphoid tissue structures located in the oropharynx between the palatoglossal and palatopharyngeal.

Turbinates

– The nasal turbinates, also known as conchae, are three pairs of scroll-like bony projections on the lateral wall of the nasal cavity that are covered by a highly vascularized respiratory mucosa, and they function to increase the surface area available for air conditioning, create turbulent airflow for optimal filtration and warming of inspired air, and regulate nasal airflow through alternating congestion and decongestion of the venous sinusoids within the mucosa (nasal cycle). The inferior turbinate is the largest and is independently mobile, while the middle and superior turbinates are parts of the ethmoid bone. The inferior turbinate mucosa contains large venous sinusoids that can rapidly engorge with blood, causing nasal congestion, and its size fluctuates throughout the day in response to autonomic nervous system input, environmental temperature, allergens, and infection. Chronic enlargement of the inferior turbinates, called turbinate hypertrophy, is a common cause of nasal obstruction and may require medical treatment with topical corticosteroids and antihistamines, or surgical reduction procedures including turbinate cauterization, submucosal resection, or outfracture.

Turbinates Hypertrophy

– Chronic enlargement of the nasal turbinates, most commonly the inferior turbinates, resulting in nasal airway obstruction, which can be caused by chronic inflammation from allergic rhinitis, non-allergic rhinitis, chronic sinusitis, hormonal changes, or vasomotor dysfunction. The inferior turbinates contain large venous sinusoids that can rapidly engorge with blood, and chronic stimulation from allergens, irritants, or infection leads to sustained engorgement and eventual hypertrophy of both the mucosal and bony components of the turbinate. The condition contributes to nasal obstruction, mouth breathing, snoring, and impaired quality of life, and it may be difficult to distinguish from the reversible congestion of acute rhinitis. Medical management includes intranasal corticosteroids, intranasal antihistamines, and avoidance of triggers, while surgical reduction may be considered for refractory cases and includes turbinate outfracture, submucosal resection (using radiofrequency, electrocautery, or microdebrider), and partial turbinectomy, with care taken to avoid excessive turbinate resection which can cause empty nose syndrome.

Tympanic membrane

– The tympanic membrane (eardrum) is a thin, cone-shaped, translucent membrane at the medial end of the external auditory canal that separates the external ear from the middle ear and serves as the interface for sound transmission by converting acoustic energy into mechanical vibrations of the ossicular chain. It is composed of three layers: an outer squamous epithelial layer continuous with the skin of the ear canal, a middle fibrous layer (lamina propria) containing radial and circular collagen fibers that provide structural integrity, and an inner mucosal layer continuous with the lining of the middle ear cavity. The pars tensa constitutes the majority of the membrane and contains a fibrous layer, while the smaller pars flaccida (Shrapnell membrane) lacks a fibrous layer and is more susceptible to retraction pocket formation and cholesteatoma development. Landmarks visible on otoscopy include the handle of the malleus (manubrium), the umbo at its tip, the cone of light reflex in the anteroinferior quadrant, and the pars flaccida superiorly. Perforation of the tympanic membrane may result from trauma, infection, or barotrauma and typically heals spontaneously within weeks, though persistent perforations may require tympanoplasty.

Tympanometry

– An objective, quantitative test that measures the compliance (mobility) of the tympanic membrane and the pressure within the middle ear as a function of air pressure in the external auditory canal, providing information about middle ear function that supplements otoscopic examination. The test uses a tympanometer that delivers a probe tone (typically 226 Hz) into the sealed ear canal while varying the air pressure from positive to negative, measuring the reflected sound energy at each pressure level to generate a tympanogram.

Uvula

– A small, fleshy flap of tissue that hangs from the back of the throat over the root of the tongue.

Vertigo

– A false sensation of rotational or linear movement of the self or the environment, and it is a symptom rather than a disease, requiring differentiation from dizziness, disequilibrium, and presyncope. Peripheral vertigo, caused by dysfunction of the vestibular apparatus or vestibular nerve, is typically characterized by episodic, rotational vertigo that is severe, lasting seconds to hours, often with nausea and vomiting, and associated with nystagmus that is suppressed by visual fixation, hearing loss, and tinnitus. Central vertigo, caused by dysfunction.

Videostroboscopy

– A diagnostic technique used to evaluate the structure and function of the vocal cords by producing a slow-motion, composite image of vocal cord vibration using stroboscopic light flashes synchronized to the frequency of vocal cord vibration, allowing the assessment of vocal cord mucosal wave, amplitude, symmetry, phase closure, and.

Vocal Cord Nodules

– Vocal cord nodules, also known as singer's nodules or contact nodules, are bilateral, symmetric, benign epithelial thickenings that develop on the anterior third of the vocal cords at the junction of the membranous and cartilaginous portions, caused by chronic vocal abuse, misuse, or overuse that results in repeated mechanical trauma to the vocal cord mucosa. The nodules are composed of fibrosis, edema, and thickened epithelium in the superficial lamina propria, and they occur at the point of maximum vibration and contact stress on the vocal cords. Patients typically present with hoarseness, breathiness, and vocal fatigue, and the diagnosis is confirmed by laryngoscopy, which reveals bilateral, symmetric, sessile or pedunculated masses on the vocal cords that may touch when the cords are adducted (kissing nodules). Treatment involves voice therapy with modification of vocal habits, avoidance of whispering and shouting, and hydration, and surgical removal is reserved for nodules that are refractory to voice therapy, are large enough to obstruct the glottis, or are suspected of harboring dysplasia.

Vocal Cord Paralysis

– Immobility of one or both vocal cords due to disruption of the recurrent laryngeal nerve (RLN). Unilateral vocal cord paralysis: hoarseness, breathy voice, aspiration, and ineffective cough. Causes: iatrogenic (thyroid, cardiac, spinal, thoracic surgery — most common), malignancy (lung, thyroid, esophageal, mediastinal), trauma, and idiopathic (10-20%). Bilateral vocal cord paralysis: stridor, dyspnea, airway obstruction, but the voice may be normal. Diagnosis: laryngoscopy, CT neck/chest to identify underlying cause, laryngeal EMG. Treatment: unilateral: voice therapy, vocal cord injection (temporary), medialization thyroplasty (permanent). Bilateral: tracheostomy, partial arytenoidectomy, or posterior cordotomy for airway.

Vocal Cord Paresis

– Partial paralysis or weakness of one or both vocal cords due to impaired function of the recurrent laryngeal nerve (RLN) or the vagus nerve. Unilateral paresis is more common and presents with hoarseness, breathy dysphonia, vocal fatigue, and sometimes aspiration. Causes include iatrogenic injury (thyroid, parathyroid, carotid, or thoracic surgery), neoplasms (lung, thyroid, oesophageal), and idiopathic origin. Bilateral paresis presents with inspiratory stridor and dyspnea due to paramedian cord position, often requiring tracheostomy or cord lateralization. Diagnosis: laryngoscopy shows reduced or absent vocal cord movement on the affected side. Laryngeal electromyography can help differentiate neurogenic from mechanical causes. Management depends on severity: voice therapy for mild cases, medialization thyroplasty or injection laryngoplasty for glottic insufficiency, and arytenoid adduction for symptomatic unilateral paresis. Bilateral cases may require posterior cordotomy, arytenoidectomy, or tracheostomy.

Vocal Cord Polyp

– Vocal cord polyps are benign, unilateral, sessile or pedunculated lesions that arise from the vocal cord mucosa, typically as a result of a single traumatic vocal event such as shouting, screaming, or coughing, causing hemorrhage and edema in the superficial lamina propria (Reinke space). Unlike vocal cord nodules, which are bilateral and symmetric, polyps are typically unilateral and may be located anywhere along the vocal cord, though they are most common on the anterior third. The polyps may be translucent (edematous), hemorrhagic, or fibrotic, and they interfere with normal vocal cord vibration, causing hoarseness, breathiness, and a rough voice quality. Diagnosis is confirmed by laryngoscopy, which reveals a smooth, pedunculated or sessile mass protruding from the vocal cord. Treatment involves voice therapy for small polyps, surgical removal for larger polyps or those that are refractory to voice therapy, and avoidance of the vocal abuse that caused the initial trauma to prevent recurrence.

Vocal Cords

– The vocal cords, also known as vocal folds, are two pairs of horizontal folds of tissue within the.

Voice and Speech

– The products of complex coordinated activity of the respiratory, laryngeal, and articulatory systems. Voice (phonation) is produced by airflow from the lungs passing through the larynx causing vibration of the vocal cords (vocal folds); fundamental frequency (pitch) is determined by vocal cord length and tension, intensity (loudness) by subglottic air pressure, and quality by the pattern of vocal cord vibration. Speech (articulation) involves modification of the laryngeal sound by the pharynx, oral cavity, tongue, lips, teeth, and palate (resonance). Voice disorders (dysphonia): organic (vocal cord nodules, polyps, Reinke edema, laryngitis, laryngeal cancer, vocal cord paralysis/paresis) or functional (muscle tension dysphonia, psychogenic dysphonia, puberphonia). Speech disorders: dysarthria (neurological weakness/incoordination of articulators — stroke, Parkinson disease, ALS, cerebellar disease), apraxia of speech (motor planning deficit), and articulation disorders. Voice/speech assessment: laryngoscopy (direct or videostroboscopy) and acoustic analysis.

Weber Test

– The Weber test is a screening test for lateralization of hearing that uses a vibrating tuning fork (typically 512 Hz) placed on the midline of the skull (vertex or forehead) to determine whether the sound is heard equally in both ears or louder in one ear, providing information about the type and side of hearing loss. In a patient with normal hearing, the sound is perceived equally in both ears or slightly louder in one ear due to normal asymmetry. In conductive hearing loss, the sound lateralizes to the affected ear because ambient noise is reduced on that side, enhancing the perception of bone-conducted sound. In sensorineural hearing loss, the sound lateralizes to the unaffected ear because the damaged cochlea is less sensitive to the bone-conducted sound.

Xerostomia (Dry Mouth)

– A subjective sensation of oral dryness resulting from reduced or absent saliva production (salivary gland hypofunction). Xerostomia affects approximately 20% of the general population, increasing with age and with the number of medications taken. Causes: medications (the most common cause, over 500 drugs cause xerostomia — anticholinergics, tricyclic antidepressants, SSRIs, antipsychotics, antihistamines, decongestants, antihypertensives, diuretics, opioids, and proton pump inhibitors), radiation therapy (head and neck cancer radiotherapy causes irreversible damage to salivary glands, dose-dependent, >30 Gy causes permanent hypofunction), systemic diseases (Sjogren syndrome, an autoimmune exocrinopathy affecting lacrimal and salivary glands; diabetes mellitus; HIV/AIDS; sarcoidosis; amyloidosis), and dehydration (poor fluid intake, vomiting, diarrhea, polyuria). Consequences: increased risk of dental caries (root caries, smooth surface caries), oral candidiasis (thrush), dysphagia (difficulty swallowing dry foods), dysgeusia (taste disturbance), oral mucosal discomfort and burning, halitosis, and difficulty wearing dentures. Diagnosis: sialometry (unstimulated whole salivary flow rate <0.1 mL/min is hyposalivation), minor salivary gland biopsy for suspected Sjogren syndrome. Treatment: symptomatic — frequent water sips, sugar-free chewing gum (stimulates residual salivary flow), artificial saliva substitutes (carboxymethylcellulose, mucin-based sprays, gels, and lozenges), and salivary stimulants (pilocarpine 5 mg tid, cevimeline 30 mg tid — muscarinic agonists that stimulate residual salivary tissue, contraindicated in uncontrolled asthma, narrow-angle glaucoma, and iritis). Preventive: meticulous oral hygiene, regular fluoride application (high-fluoride toothpaste, fluoride varnish), chlorhexidine mouthwash for caries prevention, and antifungal therapy for oral candidiasis..

Chapter 10

Gastroenterology

Abdominal Compartment Syndrome

– Condition defined by sustained intra-abdominal pressure greater than twenty millimeters of mercury associated with new organ dysfunction. Causes include massive ascites, bowel distension from obstruction or ileus, hemorrhage, and post-surgical conditions particularly after large abdominal wall hernia repairs. Elevated intra-abdominal pressure impairs venous return reducing cardiac output, restricts diaphragmatic excursion causing respiratory failure, compresses the renal veins causing oliguria, and can progress to abdominal compartment syndrome with multi-organ failure.

Acute Liver Failure

– Rapid deterioration of liver function in a previously healthy individual without prior liver disease. Causes include acetaminophen toxicity, viral hepatitis, drug reactions, autoimmune hepatitis, and Wilson disease. Features include coagulopathy, encephalopathy, and multiorgan failure. Presents with jaundice, confusion, and bleeding. Management in intensive care includes monitoring, treatment of reversible causes, and consideration for liver transplantation which is the definitive treatment for irreversible acute liver failure.

Acute Pancreatitis

– Acute inflammatory process of the pancreas most commonly caused by gallstones and alcohol accounting for approximately eighty percent of cases. The pathogenesis involves premature trypsin activation within pancreatic acinar cells triggering autodigestion. Clinical presentation includes severe epigastric pain radiating to the back, nausea, and vomiting. Diagnosis requires two of three criteria: characteristic pain, serum lipase at least three times upper limit of normal, and characteristic imaging findings of pancreatic inflammation on CT.

Adenoma-Carcinoma Sequence

– Established model of stepwise colorectal carcinogenesis progressing from normal epithelium through adenomas to carcinoma over ten to fifteen years. The sequence begins with APC tumor suppressor gene inactivation, followed by KRAS oncogene activation promoting adenoma growth, and TP53 inactivation associated with progression to invasive carcinoma. This model explains why polyp removal during colonoscopy prevents colorectal cancer. It accounts for seventy to eighty percent of colorectal cancers. The pathway underscores the importance of screening colonoscopy and polypectomy for cancer prevention.

Anal Abscess

– Collection of pus in the perianal or ischiorectal space usually from infected anal glands. Presents with throbbing anal pain, swelling, fever, and difficulty sitting. Diagnosed clinically with tenderness and fluctuance. Treatment is incision and drainage under local anesthesia. Antibiotics are not routinely required for simple abscesses. Anoscopy should be performed to evaluate for an underlying fistula. Recurrence is common if an associated fistula is not identified and treated.

Anal Fissure

– Linear tear in the distal anal mucosa usually posterior in the midline. Presents with painful defecation and bright red blood on wiping or in the toilet. Associated with constipation, hard stools, and chronic diarrhea. The internal anal sphincter spasm perpetuates the fissure by reducing blood flow to the posterior midline. Managed with stool softeners, dietary fiber, topical nitrates or calcium channel blockers, botulinum toxin injection, and lateral internal sphincterotomy for refractory cases.

Anal Fistula

– Abnormal tract connecting the anal canal to the perianal skin usually following a drained abscess. Presents with recurrent drainage, pain, and pruritus. Parks classification categorizes fistulae by relationship to the sphincter muscles. Diagnosed by examination under anesthesia and MRI. Treated surgically with fistulotomy for low fistulae, seton placement for high fistulae, advancement flap repair, or the LIFT procedure. Crohn-related fistulae require medical optimization with biologic agents before surgical intervention to optimise wound healing and reduce recurrence.

Anastomotic Leak

– Breakdown of a surgical anastomosis in the gastrointestinal tract causing spillage of luminal contents into the peritoneal cavity. Presents with fever, tachycardia, abdominal pain, leukocytosis, and peritonitis. Diagnosed by CT with oral or rectal contrast showing extraluminal contrast or fluid collection. Risk factors include malnutrition, steroid use, obesity, and distal anastomosis. Managed with broad-spectrum antibiotics, percutaneous drainage of abscesses, and possible reoperation for diffuse peritonitis.

Angiodysplasia

– Vascular ectasia consisting of dilated fragile blood vessels in the gastrointestinal mucosa and submucosa, most commonly found in the cecum and ascending colon. They are the most common cause of acute lower gastrointestinal bleeding in elderly patients over sixty years of age. Patients typically present with intermittent painless hematochezia or melena and chronic iron deficiency anemia. Diagnosis is best achieved by colonoscopy revealing cherry-red spots. Management of actively bleeding lesions includes endoscopic therapy with cautery, argon plasma coagulation, or clipping.

Anorectal Manometry

– Measurement of pressures in the anal canal and rectum to assess sphincter function and rectal sensation. A pressure-sensing catheter is inserted into the rectum and slowly withdrawn through the anal canal. Measures resting and squeeze pressures reflecting internal and external sphincter function respectively. Rectal sensation is assessed by balloon inflation. Used to evaluate fecal incontinence, chronic constipation, and to screen for Hirschsprung disease in children. The study helps guide treatment decisions for these conditions.

APACHE II Score

– Severity scoring system for critically ill patients incorporating physiological variables, age, and chronic health status to generate scores from zero to seventy-one. Higher scores indicate greater severity and higher mortality. In acute pancreatitis, scores of eight or greater indicate severe disease. The score is calculated within twenty-four hours of ICU admission. It is validated across multiple ICU populations and used for severity assessment, quality improvement, and research. Limitations include the complexity of calculation and the need for accurate physiological data.

Appendicitis

– Inflammation of the vermiform appendix, most commonly due to luminal obstruction by fecalith, lymphoid hyperplasia, or foreign body. Presents with periumbilical pain migrating to the right lower quadrant, nausea, anorexia, and low-grade fever. Diagnosis: clinical, elevated WBC, CT scan. Treatment: laparoscopic appendectomy. Complications: perforation, peritonitis, abscess formation if untreated.

Ascites

– Pathological accumulation of fluid within the peritoneal cavity representing the most common decompensating event in cirrhosis. The pathogenesis involves splanchnic arterial vasodilation reducing effective arterial blood volume and activating neurohormonal mechanisms leading to sodium and water retention. Diagnosis is established by clinical examination and confirmed by ultrasonography. Diagnostic paracentesis determines the serum-ascites albumin gradient, with a gradient of one point one grams per decilitre or greater indicates portal hypertension.

Autoimmune Gastritis

– Chronic atrophic gastritis caused by antibodies against parietal cells leading to achlorhydria and vitamin B12 deficiency. Associated with pernicious anemia and increased risk of gastric carcinoid tumors and type 1 gastric neuroendocrine tumors.

Anti-parietal cell antibodies are present in approximately ninety percent of patients. Diagnosis is by serology showing elevated gastrin levels and decreased pepsinogen I to II ratio, confirmed by endoscopic biopsy showing oxyntic gland atrophy. Management involves iron supplementation and endoscopic surveillance for gastric neuroendocrine tumors.

Autoimmune Hepatitis

– Chronic hepatitis caused by immune-mediated destruction of hepatocytes characterized by interface hepatitis with lymphoplasmacytic infiltrate, elevated transaminases, hypergammaglobulinemia, and autoantibodies. Type 1 autoimmune hepatitis is associated with anti-nuclear and anti-smooth muscle antibodies while type 2 is associated with anti-liver kidney microsomal antibodies. Diagnosis is based on the International Autoimmune Hepatitis Group scoring system. Treatment involves corticosteroids with azathioprine as steroid-sparing agents for maintenance therapy.

Autoimmune Pancreatitis

– Chronic inflammatory condition characterized by dense lymphoplasmacytic infiltrate with IgG4-positive plasma cells representing the pancreatic manifestation of IgG4-related disease. Type 1 is more common affecting older men with elevated serum IgG4 and extrapancreatic involvement. Type 2 affects younger patients with higher prevalence of inflammatory bowel disease. Clinical presentation includes painless obstructive jaundice and new-onset diabetes. Imaging shows a sausage-shaped pancreas with peripancreatic fat stranding and capsule-like rim enhancement.

Barium Swallow

– Fluoroscopic examination of the esophagus after ingestion of barium contrast. Used to evaluate esophageal anatomy, motility, strictures, and structural abnormalities. The study demonstrates mucosal detail, filling defects, strictures, and motility disorders. A barium follow-through extends the examination to the small bowel. Double-contrast technique using barium and air provides superior mucosal detail. The study has been largely replaced by upper endoscopy for direct visualization but remains useful in selected patients who cannot tolerate endoscopy.

Barrett Esophagus

– Metaplastic transformation of the normal stratified squamous epithelium of the distal esophagus to specialized intestinal-type columnar epithelium containing goblet cells, occurring as a consequence of chronic gastroesophageal reflux disease. This condition represents an important premalignant lesion because it confers a significantly increased risk of esophageal adenocarcinoma, estimated at approximately 0.5 percent per year in patients without dysplasia. The pathogenesis involves chronic acid exposure leading to inflammation, ulceration, and metaplastic change, with progression driven by bile reflux and inflammatory mediators.

Barrett Esophagus Surveillance

– Systematic endoscopic monitoring of patients with Barrett esophagus to detect dysplasia and early esophageal adenocarcinoma. Patients without dysplasia undergo surveillance every three to five years with four-quadrant biopsies every two centimeters throughout the Barrett segment. Patients with low-grade dysplasia should be offered endoscopic eradication therapy with radiofrequency ablation. Patients with high-grade dysplasia or intramucosal carcinoma should undergo endoscopic mucosal resection or oesophagectomy, depending on depth of invasion and patient fitness.

Bile Salt Malabsorption

– Condition characterized by impaired reabsorption of bile acids in the terminal ileum leading to excessive bile acids entering the colon where they stimulate water and electrolyte secretion causing chronic watery diarrhea. Normally approximately ninety-five percent of bile acids are reabsorbed in the terminal ileum through the apical sodium-dependent bile acid transporter. When this mechanism is disrupted by surgical resection, inflammatory bowel disease, or idiopathic causes, excess bile acids in the colon stimulate colonic secretion causing watery diarrhea, which responds to bile acid sequestrants such as colestyramine.

Biliary Atresia

– A progressive, obliterative cholangiopathy of the extrahepatic bile ducts in neonates, leading to biliary obstruction, cholestasis, and ultimately biliary cirrhosis if untreated. It is the most common cause of neonatal jaundice requiring surgery and the leading indication for pediatric liver transplantation, with an incidence of approximately 1 in 10,000 to 15,000 live births. The etiology is unknown but likely involves an inflammatory insult (possible viral trigger: rotavirus, reovirus) superimposed on a developmental susceptibility of the neonatal bile duct epithelium.

Biliary Dyskinesia

– Functional gallbladder disorder with impaired gallbladder ejection fraction in the absence of gallstones. Presents with biliary colic-like right upper quadrant pain. Diagnosed by CCK-HIDA scan showing ejection fraction less than thirty-five percent. The diagnosis requires exclusion of other causes of biliary-type pain. Treatment is cholecystectomy, which provides symptom relief in approximately eighty percent of patients with confirmed biliary dyskinesia.

Bleeding Esophageal Varices

– Life-threatening hemorrhage from dilated submucosal veins in the lower esophagus, occurring as a complication of portal hypertension, most commonly from cirrhosis. Varices form when the portal venous pressure gradient exceeds 10 mmHg (the threshold for variceal development). Risk of first variceal hemorrhage is approximately 15-25 percent per year, and each episode carries a mortality of 15-30 percent. Clinical presentation includes hematemesis (bright red blood or coffee-ground emesis), melena, and signs of hypovolaemic shock.

Boerhaave Syndrome

– Transmural spontaneous rupture of the esophagus typically occurring after forceful vomiting in the setting of a full stomach. Most commonly involves the left posterolateral wall of the distal esophagus. Presents with severe chest pain, subcutaneous emphysema, mediastinal emphysema, and rapidly progressive sepsis. This is a surgical emergency with mortality exceeding fifty percent when diagnosis and treatment are delayed. Diagnosis is suggested by chest radiography showing pneumomediastinum or pleural effusion on chest radiograph, and confirmed by CT showing esophageal wall thickening with extraluminal air.

Bouveret Syndrome

– Rare variant of gallstone ileus where a large gallstone erodes through a cholecystoduodenal fistula and impacts in the duodenum or pylorus causing gastric outlet obstruction. Represents approximately two to three percent of gallstone ileus cases. Presents with nausea, vomiting, epigastric pain, and anorexia. Diagnosis is by CT or upper gastrointestinal series showing gastric dilation with a filling defect in the duodenum and pneumobilia. Treatment is endoscopic or surgical removal of the impacted stone.

Bowel Incontinence

– The involuntary loss of solid or liquid stool through the anus, also known as fecal incontinence, affecting approximately 2-15 percent of the general population and increasing in prevalence with age. Causes are classified by mechanism: structural sphincter damage (obstetric anal sphincter injury from vaginal delivery, the most common cause in women; anorectal surgery; trauma), neurogenic dysfunction (diabetic neuropathy, spinal cord injury, stroke, multiple sclerosis, cauda equina syndrome), and cognitive impairment.

Budd-Chiari Syndrome

– Obstruction of hepatic venous outflow at any level from small hepatic venules to the inferior vena cava junction with the right atrium. The most common predisposing condition is myeloproliferative disorders accounting for forty to fifty percent of cases. Other risk factors include thrombophilia and oral contraceptive use. Clinical presentation varies from acute right upper quadrant pain with ascites to chronic presentation mimicking cirrhosis. Diagnosis is established by Doppler ultrasonography showing absent or reversed hepatic venous flow, and confirmed by venography.

C282Y Mutation

– Most common genetic cause of hereditary hemochromatosis, a C to T transition in the HFE gene resulting in cysteine to tyrosine substitution at position 282. Homozygosity occurs in approximately one in two hundred individuals of Northern European descent. The mutation disrupts HFE protein interaction with transferrin receptor leading to inappropriately low hepcidin production and unregulated iron absorption. Penetrance is incomplete with only approximately twenty-eight percent of male and one per cent of female C282Y homozygotes develop clinical iron overload.

Cambridge Classification

– ERCP-based grading system for chronic pancreatitis severity evaluating the main pancreatic duct, branches, and common bile duct. Normal pancreas shows smooth duct without abnormalities. Equivocal shows fewer than three abnormal side branches. Mild shows three or more abnormal side branches. Moderate shows abnormal main duct. Severe shows calculi, cavities, or strictures. The classification has been largely superseded by MRCP and endoscopic ultrasound which provide both ductal and parenchymal assessment.

Cameron Lesions

– Linear gastric erosions or ulcers found on the mucosal folds at the diaphragmatic hiatus in patients with large hiatal hernias. They result from mechanical trauma as the gastric mucosa prolapses through the diaphragmatic crura during respiration and movement. May be a source of occult or overt gastrointestinal bleeding presenting with iron deficiency anemia or melena. Diagnosis requires careful endoscopic examination with particular attention to the hiatal hernia sac. Treatment involves proton pump inhibitors and treatment of the underlying mechanical cause.

Capsule Endoscopy

– Non-invasive diagnostic technique using a swallowable wireless endoscopic device to capture images of the small intestinal mucosa. The capsule contains a miniature video camera and transmitter that sends images to a receiver worn on the patient's belt. The procedure is primarily used to evaluate the small bowel for obscure gastrointestinal bleeding, suspected Crohn disease, small bowel tumors, and celiac disease. The patient ingests the capsule after an overnight fast and it is passed naturally in the stool within 24 to 48 hours.

Carcinoid Syndrome

– Serotonin-secreting neuroendocrine tumor causing flushing, diarrhea, bronchospasm, and right-sided heart disease. Occurs when vasoactive substances bypass hepatic metabolism through venous drainage or liver metastases. Diagnosis is with twenty-four hour urine 5-hydroxyindoleacetic acid or serum chromogranin A. Octreotide controls symptoms by suppressing hormone release. Surgical debulking reduces tumor burden. Telotristat ethyl inhibits serotonin synthesis for refractory diarrhea. Right-sided valvular heart disease, particularly tricuspid regurgitation and pulmonary stenosis, is managed with valve replacement in severe cases.

Celiac Crisis

– Acute life-threatening presentation of celiac disease characterized by severe dehydration, electrolyte imbalances, metabolic acidosis, hypocalcemic tetany, and hypovolemic shock. Most commonly occurs in infants and young children after a viral illness or introduction of gluten-containing foods. Presents with profuse watery diarrhea, vomiting, abdominal distension, and signs of severe dehydration. Treatment requires aggressive fluid and electrolyte resuscitation, correction of hypocalcemia and hypoglycaemia, and administration of intravenous corticosteroids.

Celiac Disease

– An immune-mediated enteropathy triggered by gluten ingestion in genetically predisposed individuals (HLA-DQ2/DQ8). Gliadin peptides from wheat, barley, and rye activate an autoimmune response against tissue transglutaminase (tTG), causing villous atrophy, crypt hyperplasia, and intraepithelial lymphocytosis in the small intestine. Presents with diarrhea, steatorrhea, weight loss, bloating, iron deficiency anemia, and failure to thrive in children. Extraintestinal: dermatitis herpetiformis, osteoporosis, infertility, and neuropathy. Diagnosis: positive tTG-IgA, EMA, and duodenal biopsy showing villous atrophy. Management: strict lifelong gluten-free diet. Complications: refractory celiac disease, enteropathy-associated T-cell lymphoma (rare).

Celiac Trunk Stenosis

– Narrowing of the celiac trunk typically caused by extrinsic compression from the median arcuate ligament of the diaphragm, known as median arcuate ligament syndrome or celiac artery compression syndrome. Causes chronic abdominal pain particularly after eating and weight loss due to postprandial mesenteric ischemia. Diagnosis is by CT angiography or conventional angiography showing characteristic hook-shaped narrowing of the celiac trunk and collateral vessel formation. Treatment is surgical division of the median arcuate ligament with or without celiac artery reconstruction.

Charcot Triad

– Clinical constellation of fever, right upper quadrant pain, and jaundice representing the classic presentation of ascending cholangitis, infection of the biliary tree from common bile duct obstruction by a gallstone. The full triad is found in approximately fifty to seventy percent of patients. When hypotension and altered mental status are added, it constitutes Reynolds pentad indicating suppurative cholangitis. Initial management involves intravenous fluids, broad-spectrum antibiotics, and urgent biliary drainage by ERCP with endoscopic sphincterotomy.

Child-Pugh Classification

– Clinical scoring system assessing prognosis of chronic liver disease by evaluating five parameters: total bilirubin, serum albumin, international normalized ratio, ascites, and hepatic encephalopathy. Each parameter is scored one to three points. Child-Pugh A with five to six points represents well-compensated cirrhosis with one-year survival exceeding ninety-five percent. Child-Pugh B with seven to nine points represents significant functional compromise with one-year survival approximately eighty percent. Child-Pugh C with ten to fifteen points represents decompensated cirrhosis with one-year survival of approximately fifty percent.

Cholangitis

– Bacterial infection of the biliary tree, most commonly caused by choledocholithiasis (common bile duct stones) obstructing bile flow, but also from biliary strictures, stent occlusion, and malignancy. The classic clinical presentation is Charcot triad: fever, right upper quadrant pain, and jaundice (present in 50-70 percent of cases). Suppurative cholangitis with Reynolds pentad (Charcot triad plus hypotension and altered mental status) indicates severe disease requiring urgent decompression. Complications include pancreatitis, perforation, bleeding, and cholangitis.

Choledochal Cyst

– Congenital dilatation of the biliary tree classified into five types according to the Todani classification. Type I involving dilation of the common bile duct is the most common. Presents with the classic triad of abdominal pain, jaundice, and palpable mass though all three are present in less than twenty percent of cases. Risk of cholangiocarcinoma is significantly increased. Diagnosed by MRCP or CT showing biliary dilation. Treated with complete surgical excision of the cyst and biliary reconstruction with Roux-en-Y hepaticojejunostomy.

Choledocholithiasis

– The presence of gallstones within the common bile duct, occurring in approximately 10-20 percent of patients with cholelithiasis. Stones may form primarily in the common bile duct (primary stones, brown pigment stones, associated with biliary stasis and infection) or migrate from the gallbladder (secondary stones, most common, identical to gallbladder stones). Clinical presentation ranges from asymptomatic (incidental finding on imaging) to classic biliary colic, obstructive jaundice, and cholangitis. Endoscopic retrograde cholangiopancreatography is both diagnostic and therapeutic, allowing stone extraction after sphincterotomy.

Cholelithiasis

– Presence of gallstones in the gallbladder. Types: cholesterol stones (80%, radiolucent, due to supersaturated bile), pigment stones (black: hemolysis, cirrhosis; brown: infection, bile stasis). Risk factors: female, fertile, fat, forty (5 Fs), obesity, rapid weight loss, hemolytic disorders, cirrhosis. Most are asymptomatic (incidental finding). Symptomatic cholelithiasis: episodic postprandial RUQ/epigastric pain (biliary colic). Complications: acute cholecystitis, choledocholithiasis, cholangitis, gallstone pancreatitis. Diagnosis: right upper quadrant ultrasound (gold standard). Treatment: laparoscopic cholecystectomy for symptomatic stones; observation for asymptomatic.

Chronic Intestinal Pseudo-Obstruction

– Chronic condition of recurrent episodes of intestinal obstruction without mechanical blockage due to neuromuscular dysfunction of the gastrointestinal tract. May involve the small bowel, colon, or both. Causes include systemic sclerosis, diabetes, Parkinson disease, amyloidosis, and paraneoplastic syndromes. Presents with abdominal distension, pain, nausea, vomiting, constipation, and malabsorption. Diagnosed by imaging showing dilated bowel loops without transition point, and and antroduodenal manometry showing abnormal motility patterns in the small bowel.

Chronic Mesenteric Ischemia

– Recurrent abdominal pain occurring after eating resulting from insufficient blood flow to meet the metabolic demands of the gastrointestinal tract. Most commonly caused by atherosclerotic stenosis of at least two of the three major mesenteric arteries. Classic presentation is sitophobia, fear of eating due to postprandial abdominal pain typically beginning thirty minutes to two hours after meals and lasting one to two hours. Patients develop food avoidance and significant weight loss. Diagnosis is confirmed by CT angiography or conventional angiography demonstrating stenosis of at least two mesenteric vessels.

Chronic Pancreatitis

– Progressive irreversible inflammatory condition characterized by permanent structural damage causing exocrine and endocrine impairment. The most common cause is chronic alcohol consumption. Other causes include hereditary pancreatitis, autoimmune pancreatitis, and metabolic factors. Clinical manifestations include chronic abdominal pain, steatorrhea, and diabetes mellitus. Diagnosis is established by imaging showing calcifications, ductal dilatation, and parenchymal atrophy. The Cambridge classification is used for staging severity.

Clostridioides difficile Infection

– Bacterial infection of the colon caused by C. difficile toxin, usually following antibiotic use that disrupts normal colonic flora. Presents with watery diarrhea, fever, and abdominal pain. Diagnosed by stool toxin assay or nucleic acid amplification test. Treatment involves discontinuation of the offending antibiotic and initiation of oral vancomycin or fidaxomicin. Fecal microbiota transplantation is highly effective for recurrent infection. Complications include toxic megacolon, perforation, and death.

CMV Colitis

– Colonic inflammation caused by cytomegalovirus reactivation in immunocompromised patients including those with HIV and organ transplant recipients. Presents with bloody diarrhea and abdominal pain. Diagnosed by colonoscopy with biopsy showing characteristic owl-eye intranuclear inclusions in enlarged cells. Treated with intravenous ganciclovir or oral valganciclovir. CMV colitis should be considered in any immunocompromised patient with colitis unresponsive to standard therapy.

Colitis

– Inflammation of the colon, with a broad differential diagnosis and diverse clinical presentations. Acute colitis presents with diarrhea (often bloody), abdominal pain, tenesmus, and urgency. The most common causes include infectious colitis (bacterial: Shigella, Salmonella, Campylobacter, E. coli O157:H7, C. difficile; viral: CMV; parasitic: amebiasis), ischemic colitis (from hypoperfusion, typically at watershed areas of splenic flexure and rectosigmoid), and idiopathic inflammatory bowel disease (ulcerative colitis and Crohn disease).

Collagenous Colitis

– Chronic inflammatory bowel disease characterized by thickening of the subepithelial collagen band in the colonic mucosa to greater than ten micrometers normally less than five micrometers with preservation of the epithelial surface. Presents with chronic watery non-bloody diarrhea primarily in middle-aged women. The collagen band disrupts water and electrolyte absorption leading to secretory diarrhea. Diagnosis requires colonoscopy with random colonic biopsies demonstrating the characteristic subepithelial collagen deposition.

Colonic Ischemia

– Decreased blood flow to the colon causing mucosal or transmural injury. Most commonly affects watershed areas particularly the splenic flexure and rectosigmoid junction. Causes include non-occlusive hypoperfusion from cardiogenic shock, vasoconstrictive medications, and vasculitis. Presents with sudden onset crampy left lower quadrant pain followed by bloody diarrhea within twenty-four hours. Most cases are self-limited with mucosal ischemia resolving within one to two weeks. Severe transmural infarction requires surgical resection.

Colonoscopy

– Visualization of the entire colon and terminal ileum using a flexible colonoscope. It is the gold standard for colorectal cancer screening and the preferred method for evaluating colonic pathology. The procedure requires bowel preparation with purgative solution to clear the colon. Therapeutic capabilities include polypectomy, hemostasis, dilation, and stent placement. Complications include perforation, bleeding, and post-polypectomy syndrome. Adequate adenoma detection rate is a key quality metric in colonoscopy.

Colorectal Cancer Screening

– Public health strategy to detect colorectal cancer and precancerous polyps in asymptomatic individuals. Current guidelines recommend screening beginning at age forty-five. Options include fecal immunochemical test annually, multi-target stool DNA test every three years, colonoscopy every ten years, CT colonography every five years, and flexible sigmoidoscopy every five years. The choice depends on patient preference, access, risk assessment, and shared decision-making. Screening reduces cancer incidence and mortality.

Constipation

– A common symptom-based disorder defined by the Rome IV criteria as having two or more of the following for at least 3 months: straining during more than 25 percent of bowel movements, lumpy or hard stools, sensation of incomplete evacuation, sensation of anorectal obstruction, manual maneuvers to facilitate defecation, or fewer than 3 bowel movements per week. Constipation is classified as primary (normal transit, slow transit, or defecatory dysfunction from pelvic floor dyssynergia or structural abnormalities of the anorectum.

Courvoisier Sign

– Clinical finding of a palpable non-tender gallbladder in obstructive jaundice suggesting malignant biliary obstruction. In gallstone disease, the gallbladder is typically fibrotic and not palpable, whereas in malignant obstruction it is distended because the wall is initially normal and compliant. The most common cause is pancreatic head carcinoma. The sign has a positive predictive value of approximately eighty-five to ninety percent. Physical examination requires careful right upper quadrant palpation.

Crohn Disease

– Chronic transmural inflammatory condition that can affect any segment of the gastrointestinal tract from the oral cavity to the perianal region, though the terminal ileum and ileocolonic region are most commonly involved. Unlike ulcerative colitis, Crohn disease is characterized by transmural inflammation that extends through all layers of the bowel wall, segmental or skip lesions with intervening normal bowel, and a cobblestone mucosal appearance resulting from intersecting lines of ulceration. The transmural nature of inflammation predisposes to fistula formation, abscesses, and strictures.

Crohn Disease Cobblestoning

– Endoscopic appearance of the colonic or small bowel mucosa in Crohn disease characterized by a bumpy, irregular surface resembling cobblestones. This distinctive appearance results from intersecting lines of deep linear and serpiginous ulceration that surround areas of relatively preserved or edematous but intact mucosa. The combination of raised mucosal islands between ulcer tracks creates the characteristic cobblestone pattern. This finding is highly suggestive of Crohn disease and reflects the underlying pathology of the disease.

CT Colonography

– CT imaging of the colon after bowel preparation and insufflation with air or carbon dioxide. Used for colorectal cancer screening in patients unable to undergo optical colonoscopy and for evaluating complete colonoscopy when the cecum was not reached. Detects polyps greater than six millimeters with high sensitivity. Advantages include noninvasiveness and examination of extracolonic structures. Disadvantages include radiation exposure, need for adequate bowel preparation, and inability to perform polypectomy or biopsy at the time of imaging.

CT Enterography

– CT imaging of the small bowel after administration of large volumes of neutral oral contrast distending the small bowel with IV contrast. Excellent for evaluating Crohn disease, small bowel tumors, and obscure gastrointestinal bleeding. Demonstrates bowel wall thickening, hyperenhancement, mesenteric fat stranding, lymphadenopathy, and complications including strictures, fistulaes, and abscesses. The technique provides high spatial resolution and rapid acquisition. It is limited by ionizing radiation exposure.

Curling Ulcer

– Stress ulceration of the stomach or duodenum associated with severe burns. Part of the Curling triad with hypovolemia and sepsis. The pathogenesis involves reduced mucosal blood flow during the systemic inflammatory response to burn injury. Prevention with stress ulcer prophylaxis using proton pump inhibitors or H2 blockers is standard in burn unit patients. Treatment of established Curling ulcers involves acid suppression and surgical intervention for complications including hemorrhage and perforation.

Cushing Ulcer

– Stress ulceration associated with central nervous system disease, trauma, or surgery. Caused by vagally mediated gastric acid hypersecretion, distinguishing it from Curling ulcer which results from mucosal ischemia. The ulcers are typically solitary, located in the stomach or duodenum, and present with gastrointestinal bleeding. Management involves acid suppression with proton pump inhibitors and treatment of the underlying neurological condition.

Defecography

– Dynamic imaging of the rectum and pelvic floor during defecation using fluoroscopy or MRI. The rectum is filled with contrast paste and the patient performs defecation while images are captured. Evaluates rectal prolapse, intussusception, rectocele, enterocele, and pelvic floor dyssynergia. MR defecography provides superior soft tissue detail without radiation and is the preferred modality when available. The study is particularly useful in evaluating patients with obstructed defecation syndrome, particularly when standard therapies have failed.

Diarrhea

– Frequent passage of loose, watery stools, typically defined as more than 3 bowel movements per day with stool weight exceeding 200 grams. Acute diarrhea (less than 14 days) is most commonly infectious in etiology: viral (norovirus, rotavirus, adenovirus), bacterial (Campylobacter, Salmonella, Shigella, E. coli, C. difficile), and parasitic (Giardia, Cryptosporidium, Entamoeba). Chronic diarrhea (more than 4 weeks) requires evaluation for malabsorption syndromes (celiac disease, pancreatic insufficiency, bile acid malabsorption, and microscopic colitis).

Diarrhea, Constipation, Gas, and Bloating

– These common gastrointestinal symptoms arise from disruptions in normal digestive function caused by diet, infections, medications, or underlying conditions like irritable bowel syndrome. Diarrhea involves increased stool frequency and water content, while constipation involves infrequent or difficult passage of stool. Gas and bloating result from swallowed air or colonic fermentation of undigested food and may cause abdominal distension and discomfort.

Dieulafoy Lesion

– Large caliber persistent submucosal artery in the stomach wall that can cause massive gastrointestinal bleeding. The artery is abnormally large and superficial, lacking the normal tapering into smaller arterioles. Presents with hematemesis or melena without a visible ulcer. The lesion is typically found within six centimeters of the gastroesophageal junction. Diagnosed and treated by endoscopic therapy with clips, thermal coagulation, or band ligation. Recurrent bleeding may occur and repeat endoscopy with hemostatic therapy.

Digestion

– The process by which the body breaks down ingested food into smaller molecules that can be absorbed and utilized. Digestion begins in the mouth (mechanical–mastication; enzymatic–salivary amylase for starch, lingual lipase). In the stomach: gastric acid (HCl) denatures proteins, pepsin digests proteins, and mechanical churning produces chyme. In the small intestine: pancreatic enzymes (trypsin, chymotrypsin, amylase, lipase, nucleases) break down macromolecules; bile salts emulsify fats. Brush border enzymes (lactase, sucrase, maltase, peptidases) complete digestion. Absorption occurs primarily in the small intestine. The colon absorbs water and electrolytes and ferments undigested fiber.

Digestive Health

– The efficient functioning of the gastrointestinal tract in breaking down food, absorbing nutrients, and eliminating waste. Digestive health depends on factors such as diet, hydration, gut microbiome balance, and the integrity of the digestive mucosa. Common digestive disorders include irritable bowel syndrome, gastroesophageal reflux disease, and inflammatory bowel disease.

Diverticular Disease

– Condition characterized by the presence of diverticula, which are outpouchings of the colonic mucosa and submucosa through defects in the muscularis propria at sites where vasa recta penetrate the bowel wall. Diverticulosis, the presence of diverticula without inflammation, is extremely common in Western populations, affecting more than half of individuals over age sixty, and is thought to result from a combination of low dietary fiber intake, increased intraluminal pressure from straining, and age-related weakening of the colonic wall.

Diverticulitis

– Inflammation and infection of colonic diverticula, a common complication of diverticulosis affecting 10-25 percent of individuals with diverticula. The pathophysiology involves microperforation of a diverticulum with fecal contents entering the pericolic tissues, inciting inflammation and potential abscess formation. Clinical presentation includes acute onset of left lower quadrant abdominal pain (typically constant, localized, worsened by movement), fever, nausea, vomiting, and altered bowel habit, and anorexia.

Diverticulosis

– The presence of diverticula, small mucosal outpouchings through the colonic wall at sites of vascular penetration, most commonly in the sigmoid colon (90 percent of cases). Prevalence increases with age, affecting over 50 percent of individuals older than 60 years and nearly 70 percent by age 85. The pathogenesis involves a combination of low dietary fiber intake leading to increased intraluminal pressure, segmental colonic muscle hypertrophy, and weakening of the colonic wall at points of nutrient arteries at the point of vasa recta penetration through the circular muscle layer.

Diverticulosis and Diverticulitis

– Diverticulosis is a condition in which small, bulging pouches (diverticula) form in the lining of the digestive tract, most commonly in the colon. Diverticulitis occurs when these pouches become inflamed or infected, causing abdominal pain, fever, and changes in bowel habits. Treatment ranges from dietary modification and antibiotics for mild cases to surgical intervention for severe or recurrent episodes.

Double-Balloon Enteroscopy

– Advanced endoscopic technique using an enteroscope equipped with balloons at the tip of both the endoscope and an overtube allowing systematic advancement and visualization of the entire small bowel. The balloons are alternately inflated and deflated to anchor the bowel wall and advance the endoscope. This technique allows deep intubation of the small bowel from both oral and anal approaches. Double-balloon enteroscopy is the most comprehensive method for evaluating small bowel pathology, with the ability to perform biopsies and therapeutic interventions throughout the entire small bowel.

Dubin-Johnson syndrome

– A benign, autosomal recessive disorder of conjugated hyperbilirubinemia caused by mutations in the ABCC2 gene encoding the multidrug resistance-associated protein 2 (MRP2), which transports conjugated bilirubin from hepatocytes into bile. This results in impaired excretion of conjugated bilirubin and other organic anions, leading to elevated serum conjugated bilirubin levels (typically 2-5 mg/dL). The liver has a characteristic gross black appearance due to accumulation of dark pigment (epinephrine metabolites) deposited in the hepatocytes.

Dumping Syndrome

– Rapid gastric emptying of hypertonic contents into the small bowel after gastrectomy or gastric bypass surgery. Early dumping occurs within thirty minutes with vasomotor symptoms including flushing, palpitations, diarrhea, and dizziness from fluid shifts into the intestinal lumen. Late dumping occurs one to three hours later with reactive hypoglycemia from exaggerated insulin secretion. Managed with dietary modifications including small frequent meals, avoidance of simple sugars, and lying down after meals. Pharmacological therapy includes octreotide and acarbose to slow gastric emptying and blunt the insulin response.

Duodenal Ulcer

– Peptic ulcer in the duodenal mucosa strongly associated with H. pylori infection. Presents with epigastric pain relieved by food typically occurring several hours after meals and at night. Complications include hemorrhage, perforation, and gastric outlet obstruction. Diagnosis is by upper endoscopy. Treatment involves proton pump inhibitor therapy for four to eight weeks and H. pylori eradication. Duodenal ulcers have a lower risk of malignancy compared to gastric ulcers but require follow-up endoscopy to document healing.

Dysentery

– Severe diarrheal illness with blood and mucus in stool, typically caused by Shigella, Salmonella, Campylobacter, or Entamoeba histolytica. Presents with frequent bloody stools, abdominal cramps, tenesmus, and fever. Diagnosis: stool culture, microscopy, PCR. Treatment: rehydration and targeted antibiotics based on pathogen and sensitivity. Complications: toxic megacolon, hemolytic uremic syndrome (Shiga toxin-producing E. coli).

Dyspepsia

– Chronic or recurrent pain or discomfort centered in the upper abdomen. Organic causes: peptic ulcer disease, GERD, gastritis, gastric cancer. Functional dyspepsia is diagnosed after exclusion of organic pathology with normal upper endoscopy. Rome IV criteria define subtypes: epigastric pain syndrome and postprandial distress syndrome. Management: H. pylori test and treat, PPI trial, prokinetics, and low-dose antidepressants.

Dysphagia

– Difficulty swallowing, classified as oropharyngeal (difficulty initiating a swallow, choking, coughing, nasal regurgitation) or esophageal (sensation of food sticking retrosternally). Oropharyngeal causes: neurological (stroke, Parkinson's disease, motor neurone disease, myasthenia gravis), structural (pharyngeal pouch, tumors), and muscular (dermatomyositis). esophageal causes: mechanical obstruction (stricture, Schatzki ring, esophageal cancer, eosinophilic esophagitis) and motility disorders (achalasia, diffuse esophageal spasm, scleroderma). Investigations: videofluoroscopic swallow study for oropharyngeal; upper endoscopy, barium swallow, and esophageal manometry for esophageal. Complications: aspiration pneumonia, malnutrition, and dehydration.

Emesis (Vomiting)

– The forceful expulsion of gastric contents through the mouth, a complex reflex coordinated by the vomiting center in the medulla oblongata. The reflex involves: prodromal phase (nausea, salivation, autonomic activation–tachycardia, sweating), retching (rhythmic contractions of diaphragm and abdominal muscles against a closed glottis), and expulsion (forceful contraction of abdominal muscles with relaxation of the lower esophageal sphincter and diaphragm). Causes: gastrointestinal (gastroenteritis, gastroparesis, obstruction), vestibular (motion sickness, labyrinthitis), CNS (increased intracranial pressure, migraine), metabolic (pregnancy, uremia, DKA, hypercalcemia), medications (chemotherapy, opioids, antibiotics), toxins (food poisoning, alcohol), and psychogenic. Complications: aspiration, dehydration, electrolyte disturbances, Mallory-Weiss tear. Antiemetics target different receptors: 5-HT3, D2, H1, M1, NK1.

Endoscopic Mucosal Resection

– Advanced endoscopic technique for removal of superficial gastrointestinal neoplasms by lifting the lesion through submucosal injection and resecting with an electrosurgical snare. The technique involves injection of saline with vasoconstrictive agent into the submucosa, capturing the elevated lesion with a snare, and resecting with electrocautery. Indications include sessile colorectal polyps, early gastric cancer, Barrett esophagus with high-grade dysplasia, and large duodenal lesions. Complications include bleeding, perforation, and stricture formation, but these are uncommon in experienced hands.

Endoscopic Retrograde Cholangiopancreatography

– Combined endoscopic and fluoroscopic procedure used to visualize and treat disorders of the biliary and pancreatic ducts. A duodenoscope is advanced to the second portion of the duodenum and the ampulla of Vater is cannulated. Contrast dye is injected under fluoroscopic guidance creating cholangiograms and pancreatograms. Therapeutic interventions include endoscopic sphincterotomy, stone extraction using baskets and balloons, stent placement for strictures, and nasobiliary drainage. The most common complication is post-ERCP pancreatitis, occurring in 3-10 percent of cases.

Endoscopic Submucosal Dissection

– Advanced endoscopic technique for en bloc resection of large gastrointestinal neoplasms using specialized electrosurgical knives to dissect through the submucosa. The technique allows complete histological assessment and is particularly valuable for early gastric and colorectal cancers. Unlike piecemeal EMR, ESD achieves higher en bloc resection rates, lower recurrence rates, and accurate staging. Complications include bleeding in five to ten percent and perforation in three to five percent, though the risk of perforation is higher than with EMR and requires careful patient selection and operator expertise.

Endoscopic Ultrasound

– Diagnostic and therapeutic endoscopic procedure combining endoscopy with high-frequency ultrasound to provide detailed imaging of the gastrointestinal wall layers and adjacent organs. EUS is particularly valuable for evaluating pancreatic masses, staging gastrointestinal malignancies, evaluating submucosal lesions, and assessing the biliary tree. EUS-guided fine-needle aspiration allows tissue sampling of lesions not accessible by conventional biopsy. The staging accuracy for esophageal and rectal cancers, and for evaluating pancreatic lesions and mediastinal lymphadenopathy.

Endoscopy

– Visualization of the gastrointestinal tract using flexible fiber-optic or video endoscopes. Upper endoscopy examines the esophagus, stomach, and duodenum. Colonoscopy examines the colon and terminal ileum. Both allow direct visualization, biopsy, and therapeutic interventions including polypectomy, hemostasis, and dilation. Endoscopic procedures require patient preparation, sedation, and monitoring. Complications are uncommon but include perforation, bleeding, and adverse reactions to sedation. Most procedures are safe when performed by trained endoscopists with appropriate monitoring.

Enema

– The introduction of fluid into the rectum and colon through the anus for diagnostic or therapeutic purposes. Types: cleansing enema (warm water, saline, or soap solution to relieve constipation or prepare for colonoscopy/surgery), retention enema (mineral oil for fecal impaction, corticosteroid or mesalamine for inflammatory bowel disease), barium enema (radiographic contrast study of the colon–now largely replaced by colonoscopy and CT colonography), and medicinal enema (administration of drugs such as laxatives, anti-inflammatories, or anesthetics). Potential complications: electrolyte disturbance (with excessive tap water enemas), bowel perforation (rare, with improper technique), and vagal reaction.

Enteritis

– Inflammation of the small intestine, most commonly caused by infectious agents (viral, bacterial, or parasitic). Viral enteritis (norovirus, rotavirus, adenovirus, astrovirus) presents with acute onset of watery diarrhea, nausea, vomiting, and abdominal cramps, typically self-limited over 2–7 days. Bacterial enteritis (Campylobacter, Salmonella, Shigella, E. coli, Yersinia) more often presents with fever, bloody diarrhea, and tenesmus. Chronic enteritis occurs in Crohn disease, celiac disease, radiation enteritis, and eosinophilic gastroenteritis. Diagnosis: clinical history, stool studies (culture, PCR, ova and parasites), and imaging. Management: rehydration (oral or IV), antiemetics, and antibiotics only for specific bacterial infections.

Eosinophilic Esophagitis

– Chronic immune-mediated condition characterized by dense eosinophilic infiltration of the esophageal mucosa defined as fifteen or more eosinophils per high-power field, in the absence of other causes of esophageal eosinophilia. The condition is driven by a T helper type two inflammatory response and is strongly associated with atopic conditions. The clinical presentation varies by age: infants present with feeding difficulties; school-age children with dysphagia; and adults with progressive dysphagia, and food bolus impaction.

Eosinophilic Gastroenteritis

– Rare chronic inflammatory condition characterized by eosinophilic infiltration of the gastrointestinal tract beyond the esophagus involving any segment from stomach to rectum and extending beyond the mucosa to involve the muscularis propria or serosa. The condition is classified based on depth of infiltration: mucosal disease presents with pain and diarrhea; muscular disease causes obstruction; serosal disease causes eosinophilic ascites. Diagnosis is established by tissue biopsy demonstrating eosinophilic infiltration in the affected tissue.

Erosive Gastritis

– Inflammation of the stomach lining with superficial erosions but no full-thickness ulceration. Commonly caused by NSAIDs, alcohol, or critical illness. May present with epigastric pain or gastrointestinal bleeding. Diagnosis is by upper endoscopy showing erosions and erythema. Management involves proton pump inhibitors and removal of offending agents. Stress ulcer prophylaxis with proton pump inhibitors or H2 blockers is indicated in critically ill patients at risk for stress-related mucosal disease.

Esophageal Achalasia

– Primary esophageal motility disorder characterized by failure of the lower esophageal sphincter to relax during swallowing and absent peristalsis in the distal esophagus, resulting from selective loss of inhibitory neurons in the myenteric plexus. The condition typically presents with progressive dysphagia to both solids and liquids, regurgitation, chest pain, and weight loss. Complications include aspiration pneumonia and increased risk of esophageal squamous cell carcinoma. Diagnosis is confirmed by high-resolution manometry showing incomplete lower esophageal sphincter relaxation and absent peristalsis.

Esophageal Candidiasis

– Fungal infection of the esophagus caused by Candida species common in immunocompromised patients including those with HIV, organ transplant recipients, and patients on inhaled corticosteroids. Presents with odynophagia and dysphagia. Diagnosed by endoscopy showing white adherent plaques and biopsy confirming fungal hyphae. Treated with oral fluconazole as first-line therapy. Esophageal candidiasis is an AIDS-defining illness when the CD4 count is below two hundred cells per microliter.

Esophageal Dilation

– Endoscopic passage of bougies or balloons to widen esophageal strictures. Used for symptomatic strictures from peptic disease, radiation, caustic injury, or eosinophilic esophagitis. Savary bougies are passed over a guidewire with progressive dilation. Through-the-scope balloons allow controlled radial dilation under direct visualization. Complications include perforation, which occurs in approximately one percent, and bleeding. Serial dilations are often required for refractory strictures. Dilation should be performed gradually over multiple sessions to reduce perforation risk.

Esophageal Diverticulum

– Outpouching of the esophageal wall classified by location. Zenker diverticulum occurs in the hypopharynx and is the most common esophageal diverticulum. Mid-esophageal diverticula are traction diverticula caused by extrinsic inflammatory disease including tuberculosis and histoplasmosis. Epiphrenic diverticula occur in the distal esophagus above the lower esophageal sphincter and are associated with esophageal motility disorders including achalasia and diffuse esophageal spasm. Most diverticula are asymptomatic and discovered incidentally; symptomatic cases may require surgical diverticulectomy and myotomy.

Esophageal Intramural Pseudodiverticulosis

– Condition characterized by multiple small outpouchings of the esophageal submucosal glands seen on barium swallow as flask-shaped contrast collections oriented longitudinally along the esophagus. The pseudodiverticula are not true diverticula but rather dilated excretory ducts of submucosal glands. Associated with gastroesophageal reflux disease, diabetes mellitus, and chronic alcoholism. Most patients present with progressive dysphagia due to associated esophageal stricture formation. Diagnosis is confirmed by barium swallow showing characteristic flask-shaped outpouchings, and management focuses on treating the underlying stricture with dilation.

Esophageal Leukoplakia

– White plaques on the esophageal mucosa representing squamous cell hyperplasia with hyperkeratosis. Similar to oral leukoplakia it is considered a potentially premalignant condition. Associated with chronic irritation from smoking, alcohol, and gastroesophageal reflux. Diagnosis is by endoscopy with biopsy to exclude dysplasia or carcinoma. Management includes surveillance with endoscopy and biopsy, smoking and alcohol cessation, and treatment of reflux disease. The risk of malignant transformation is low, but regular surveillance with endoscopic biopsy is recommended.

Esophageal Manometry

– Diagnostic test measuring esophageal motor function by recording intraluminal pressures during swallowing. A catheter with pressure sensors is passed through the nose into the stomach and withdrawn to record pressures at multiple levels. High-resolution manometry using closely spaced sensors provides a spatiotemporal color plot. The Chicago classification categorizes esophageal motility disorders into major disorders including achalasia and distal esophageal spasm, and minor disorders including ineffective esophageal motility (IEM).

Esophageal Perforation

– Full-thickness rupture of the esophageal wall, most commonly Boerhaave syndrome from forceful vomiting occurring on the left posterolateral distal esophagus. Iatrogenic perforation during endoscopy is another common cause. Presents with chest pain, subcutaneous emphysema, and mediastinitis. A surgical emergency requiring emergent repair and drainage. Delayed diagnosis significantly increases mortality. Management includes nil per os, intravenous antibiotics, and surgical repair with tissue flap reinforcement for large defects.

Esophageal Spasm

– Esophageal motility disorder characterized by uncoordinated or excessive contractions impairing bolus transport. Diffuse esophageal spasm shows simultaneous contractions on manometry. Jackhammer esophagus features extremely high-amplitude peristaltic contractions. Patients present with episodic dysphagia and chest pain that may mimic cardiac pain. Diagnosis is by high-resolution manometry demonstrating characteristic pressure patterns. Barium swallow may show the corkscrew esophagus pattern. Treatment includes calcium channel blockers, nitrates, phosphodiesterase inhibitors, and endoscopic botulinum toxin injection for symptomatic relief.

Esophageal Stricture

– Narrowing of the esophageal lumen from chronic gastroesophageal reflux disease, caustic ingestion, or eosinophilic esophagitis. Presents with progressive dysphagia initially to solids then liquids. Diagnosed by upper endoscopy and barium swallow. Peptic strictures are the most common type and are treated with endoscopic balloon dilation combined with aggressive acid suppression. Recurrent strictures may require repeated dilations. Malignant strictures require stent placement or surgical resection with reconstruction.

Esophageal Varices

– Dilated, tortuous submucosal veins in the lower esophagus, most commonly caused by portal hypertension (cirrhosis—hepatic fibrosis with increased portal venous pressure >10 mmHg). They develop as portosystemic collaterals connecting the portal venous system (left gastric vein) to the systemic venous system (azygos vein). Prevalence: 50% of patients with cirrhosis, 30% of whom will bleed within 2 years. Ruptured esophageal varices cause acute upper GI hemorrhage (hematemesis, melena, hemodynamic instability), with mortality of 15–20% per episode. Diagnosis: upper endoscopy—emergency for suspected variceal bleeding (red wale signs, cherry-red spots, fibrin plug). Management of acute bleeding: resuscitation, vasoactive drugs (terlipressin or octreotide), prophylactic antibiotics (ceftriaxone 1 g IV, reduces bacterial peritonitis and mortality), endoscopic band ligation (first-line, 90% hemostasis), and covered transjugular intrahepatic portosystemic shunt (TIPS) for refractory bleeding. Secondary prophylaxis: nonselective beta-blockers (propranolol, carvedilol) plus repeat band ligation every 2–4 weeks until varices obliterated. Primary prophylaxis for medium/large varices: beta-blockers or band ligation.

Esophageal Web

– Thin mucosal fold protruding into the esophageal lumen most commonly in the postcricoid region or upper esophagus. Webs may be asymptomatic or cause dysphraga particularly to solids. Multiple webs can occur in association with iron deficiency anemia as part of Plummer-Vinson syndrome or Paterson-Kelly syndrome. Diagnosis is by barium swallow showing thin filling defects and upper endoscopy which may inadvertently displace the web. Symptomatic webs are treated with endoscopic dilation. The significance of an esophageal web lies primarily in its association with Plummer-Vinson syndrome and the increased risk of postcricoid squamous cell carcinoma.

Esophagitis

– Inflammation of the esophageal mucosa. Causes: gastro-esophageal reflux disease (GERD—most common, 60–70% of cases), infectious (Candida albicans—most common in immunocompromised, HIV, diabetes, post-antibiotics; herpes simplex virus—painful ulcers; CMV—large ulcerations in immunocompromised), eosinophilic esophagitis (EoE—allergic/atopic, characteristic rings, furrows, and white plaques on endoscopy), pill-induced (doxycycline, bisphosphonates, NSAIDs, potassium chloride, quinidine—typically in the mid-esophagus at areas of extrinsic compression), and radiation esophagitis (during thoracic radiotherapy). Symptoms: heartburn, odynophagia (painful swallowing), dysphagia (especially for solids in EoE and strictures), and retrosternal chest discomfort. Diagnosis: upper endoscopy with biopsy is the gold standard; biopsies from the distal and mid-esophagus help distinguish reflux esophagitis (basal hyperplasia, eosinophils >15/hpf in EoE, neutrophilic infiltration in severe reflux). GERD esophagitis is graded by the Los Angeles classification (A–D). Management: lifestyle measures plus proton pump inhibitors (omeprazole 20 mg, lansoprazole 30 mg daily) for 4–8 weeks; EoE requires topical swallowed corticosteroids (fluticasone, budesonide) and dietary elimination; infectious esophagitis is treated with antifungals or antivirals as appropriate.

Esophagus (esophagus)

– A muscular tube approximately 25 cm long connecting the pharynx to the stomach, extending from the cricopharyngeus muscle (at the level of C6) to the gastroesophageal junction (T11). The esophagus has three anatomic constrictions: cervical (upper esophageal sphincter–cricopharyngeus at C6), thoracic (crossing of the aortic arch and left main bronchus), and diaphragmatic (hiatus). The wall consists of mucosa (non-keratinized stratified squamous epithelium), submucosa, muscularis propria (upper third: skeletal muscle; middle third: mixed; lower third: smooth muscle), and adventitia (no serosa). Functions: transport of swallowed food and liquid from pharynx to stomach via coordinated peristaltic contractions. The lower esophageal sphincter prevents gastric reflux. Common disorders: GERD, esophagitis, Barrett esophagus, esophageal cancer, achalasia, and strictures.

Familial Adenomatous Polyposis

– Hereditary condition with hundreds to thousands of adenomas throughout the colon caused by germline APC gene mutations with autosomal dominant inheritance. Polyps develop during the second decade and colorectal cancer occurs by age forty if untreated. Diagnosis is by colonoscopy showing numerous polyps and genetic testing. Management requires prophylactic colectomy typically in the second decade. Upper gastrointestinal endoscopy surveillance is needed for duodenal and periampullary polyps. Genetic counselling and testing of at-risk family members are essential components of management.

Fatty Liver Disease (MASLD)

– A condition characterized by the accumulation of excess fat in liver cells, most commonly associated with obesity, insulin resistance, and metabolic syndrome. Non-alcoholic fatty liver disease (NAFLD) is the most common form and ranges from simple steatosis to non-alcoholic steatohepatitis (NASH), which can progress to fibrosis and cirrhosis. Management focuses on weight loss, dietary modification, exercise, and control of metabolic risk factors.

Fecal Calprotectin

– Neutrophil-derived protein measured in stool as a marker of intestinal inflammation. It is stable in stool at room temperature for up to seven days. Useful for differentiating inflammatory from functional bowel disease and monitoring inflammatory bowel disease activity. Levels above fifty micrograms per gram suggest intestinal inflammation. Fecal calprotectin has high sensitivity and specificity for detecting endoscopically active inflammatory bowel disease and can reduce the need for endoscopic surveillance when used as a triage tool.

Fecal Elastase

– Pancreatic enzyme measured in stool as a marker of exocrine pancreatic function. Low levels below two hundred micrograms per gram indicate pancreatic insufficiency, and levels below one hundred are consistent with severe exocrine insufficiency. Useful for diagnosing chronic pancreatitis and cystic fibrosis-related pancreatic insufficiency. The test is not affected by enzyme supplementation. Fecal elastase has good sensitivity and specificity for detecting moderate to severe pancreatic exocrine insufficiency.

Fecal Incontinence

– Inability to control bowel movements ranging from minor soiling to complete loss of continence. Causes include sphincter injury from obstetric trauma or surgery, neuropathy from diabetes or spinal cord injury, diarrhea, rectal prolapse, and cognitive impairment. Diagnosed by anorectal manometry, endoanal ultrasound, and pudendal nerve terminal motor latency testing. Managed with pelvic floor exercises and biofeedback as first-line therapy, dietary modification to optimize stool consistency, bulk-forming laxatives as initial management, with prescription medications such as loperamide for diarrhea-predominant symptoms and linaclotide for constipation-predominant symptoms.

Fecal Microbiota Transplantation

– Transfer of stool from a healthy donor into the gastrointestinal tract of a recipient to restore microbiome diversity. Highly effective for recurrent Clostridioides difficile infection with cure rates exceeding eighty-five percent. Administered via colonoscopy, enema, nasoduodenal tube, or oral capsules. The procedure is generally safe with rare complications including aspiration during administration. Donor screening is essential to prevent transmission of infectious diseases. Research is ongoing into the role of FMT in other conditions such as inflammatory bowel disease and metabolic syndrome.

Feeding Jejunostomy

– Surgical or endoscopic placement of a feeding tube into the jejunum for enteral nutrition when upper gastrointestinal access is not feasible. Used in patients with head and neck cancers, esophageal obstruction, or severe dysphagia. Provides long-term nutritional support and avoids the complications of parenteral nutrition. The tube can be placed percutaneously using endoscopic or radiological guidance. Complications include tube displacement, occlusion, leakage, and local infection.

FibroScan

– Non-invasive transient elastography technique measuring liver stiffness to assess hepatic fibrosis. Values correlate with fibrosis stage with higher values indicating more advanced fibrosis. The technique avoids the need for liver biopsy in many patients. Results may be unreliable in obesity, ascites, and narrow intercostal spaces. Controlled attenuation parameter measures hepatic steatosis simultaneously. FibroScan is useful for screening, diagnosis, and monitoring of liver fibrosis in chronic liver disease, particularly in patients with chronic hepatitis B, non-alcoholic fatty liver disease, and recurrent alcohol-related liver disease.

Fistula-in-Ano

– Abnormal epithelialized tract connecting the anal canal to the perianal skin, representing a common and distressing complication of Crohn disease and also occurring following perianal abscess drainage or from cryptoglandular infection. In Crohn disease, fistulae result from transmural inflammation that penetrates through the bowel wall and tracks to the skin surface. Parks classification categorizes fistulae based on their relationship to the anal sphincter muscles: intersphincteric fistulae course between the internal and external anal sphincters, trans-sphincteric fistulae pass through the external sphincter, supra-sphincteric fistulae loop over the puborectalis, and extrasphincteric fistulae track outside the sphincter complex entirely.

Fitz-Hugh-Curtis Syndrome

– Perihepatitis associated with pelvic inflammatory disease caused by ascension of Neisseria gonorrhoeae or Chlamydia trachomatis from the pelvis to the liver capsule along the right paracolic gutter. Presents with right upper quadrant pain, fever, and tender hepatomegaly often in young sexually active women. May mimic cholecystitis or hepatitis. Diagnosis requires demonstration of pelvic inflammatory disease with cervical cultures or nucleic acid amplification testing and imaging showing perihepatic enhancement on CT or MRI.

Focal Nodular Hyperplasia

– Benign liver lesion with a characteristic central stellate scar visible on imaging. The lesion represents a hyperplastic response to a pre-existing vascular malformation rather than a true neoplasm. Associated with oral contraceptive use though it also occurs in men and children. No risk of hemorrhage or malignant transformation. Diagnosed by imaging showing a well-circumscribed enhancing mass with central scar. Managed expectantly with no treatment required. Differentiation from hepatic adenoma, which carries risk of hemorrhage and malignant transformation and is managed differently.

Foodborne Illness

– Any illness resulting from the ingestion of contaminated food, encompassing infections and intoxications caused by pathogens, toxins, or chemical contaminants. It affects millions annually and poses particular risks to pregnant women, young children, older adults, and immunocompromised individuals. Prevention relies on proper food handling, cooking temperatures, hand hygiene, and safe storage practices.

Food Poisoning

– Illness caused by consuming food contaminated with bacteria, viruses, parasites, or their toxins. Common pathogens include Salmonella, Escherichia coli, Listeria, and norovirus, with symptoms such as nausea, vomiting, diarrhea, abdominal cramps, and fever. Most cases resolve without treatment, but severe infections may require hospitalization and supportive care.

Functional Dyspepsia

– Chronic epigastric pain or discomfort without identifiable structural cause. The Rome IV criteria include postprandial distress syndrome with early satiety and postprandial fullness, and epigastric pain syndrome with epigastric pain or burning unrelated to meals. Pathophysiology involves visceral hypersensitivity, altered gastric accommodation, and duodenal eosinophilia. Management includes dietary modifications, proton pump inhibitors, histamine H2 receptor antagonists, prokinetics, and neuromodulators such as tricyclic antidepressants and serotonin-norepinephrine reuptake inhibitors.

Gall-Bladder

– A pear-shaped sac located beneath the liver on the visceral surface, in the gallbladder fossa between the right and quadrate lobes of the liver. Capacity: approximately 30-50 mL. Functions: (1) concentrates and stores bile (10-fold concentration by absorption of water and electrolytes), (2) releases bile into the duodenum via the cystic duct and common bile duct in response to cholecystokinin (CCK) released after a fatty meal, (3) secretes mucus (protects gallbladder mucosa from bile acids). The gallbladder is absent in some species (rat, horse) and can be removed (cholecystectomy) without significant digestive impairment. Bile composition: bile salts, bilirubin, cholesterol, phospholipids.

Gall-Bladder, Diseases of

– Pathological conditions affecting the gallbladder. Types: (1) Cholelithiasis (gallstones)–most common, affects 10-15% of adults in Western countries. (2) Acute cholecystitis–inflammation of the gallbladder, usually due to stone obstruction of the cystic duct (90%). (3) Chronic cholecystitis–persistent inflammation with fibrosis, almost always associated with stones. (4) Acalculous cholecystitis–inflammation without stones, seen in critically ill patients, burns, trauma, total parenteral nutrition. (5) Gallbladder polyps–most are benign cholesterol polyps; adenomatous polyps have malignant potential. (6) Gallbladder carcinoma–rare (1-2% of gallstone patients), more common with large stones (>3 cm), porcelain gallbladder, and chronic Salmonella typhi carriage. (7) Choledocholithiasis–stones in the common bile duct. (8) Mirizzi syndrome–stone impacted in the cystic duct causing extrinsic compression of the common hepatic duct.

Gallstone Ileus

– Large gallstone eroding through the gallbladder wall into the duodenum causing mechanical small bowel obstruction. Diagnosed by CT showing Rigler triad of small bowel obstruction, pneumobilia, and ectopic gallstone. The most common site of obstruction is the ileum where the bowel caliber narrows. Treatment is surgical with enterolithotomy to relieve the obstruction. Cholecystectomy and fistula repair may be performed simultaneously or as a staged procedure depending on the patient's condition.

Gallstones

– Also known as cholelithiasis, crystalline deposits that form in the gallbladder composed predominantly of cholesterol (80 percent of stones in Western populations, yellow-green, radiolucent on plain x-ray, develop when bile is supersaturated with cholesterol) or bilirubin pigment (black pigment stones: sterile bile with high unconjugated bilirubin from hemolysis, cirrhosis, age; brown pigment stones: infected bile from bacterial beta-glucuronidase activity, associated with biliary stasis and parasitic infection in East Asian populations).

Gallstones (Cholelithiasis)

– See Gallstones.

Gastralgia

– Pain in the stomach or epigastric region. The term is non-specific and encompasses various causes of epigastric discomfort: functional dyspepsia (most common), peptic ulcer disease (gastric or duodenal), gastritis, gastro-esophageal reflux disease (GERD), gastric cancer, pancreatitis, cholecystitis, and referred pain (myocardial ischemia, pleurisy). Clinical assessment: quality (burning, gnawing, cramping), timing (postprandial, nocturnal, fasting), aggravating/relieving factors, associated symptoms (nausea, vomiting, weight loss, hematemesis, melena). Management: identify and treat underlying cause; H. pylori testing and eradication if present; trial of PPI if GERD or peptic ulcer suspected.

Gastrectomy

– Surgical removal of all (total gastrectomy) or part (partial/subtotal gastrectomy) of the stomach. Indications: gastric cancer (most common), severe peptic ulcer disease (rarely needed now due to PPIs and H. pylori treatment), gastric lymphoma, gastrointestinal stromal tumor (GIST), perforation, bleeding. Types: Billroth I (partial gastrectomy with gastroduodenostomy), Billroth II (partial gastrectomy with gastrojejunostomy), Roux-en-Y (total gastrectomy with oesophagojejunostomy). Complications: anastomotic leak (highest risk), dumping syndrome, bile reflux gastritis (post Billroth II), malabsorption (iron, B12, calcium, vitamin D, fat-soluble vitamins), bacterial overgrowth, weight loss, afferent/efferent loop syndrome. After total gastrectomy, intramuscular vitamin B12 supplementation is required lifelong due to loss of intrinsic factor.

Gastric Bezoar

– Accumulation of indigestible material in the stomach. Phytobezoars consist of vegetable fiber, trichobezoars of hair, and pharmacobezoars of medications. Associated with gastroparesis, prior surgery, and impaired gastric motility. Diagnosed by endoscopy. Treated with enzymatic dissolution using cellulase, papain, or cola, endoscopic fragmentation, or surgical removal for refractory cases. Bezoars are an important cause of gastric outlet obstruction and ulceration.

Gastric Carcinoid Tumor

– Neuroendocrine tumor arising from enterochromaffin-like cells in the gastric fundus. Type I gastric carcinoids are associated with autoimmune atrophic gastritis and hypergastrinemia, are typically small and multifocal, and carry a good prognosis. Type II gastric carcinoids are associated with Zollinger-Ellison syndrome and multiple endocrine neoplasia type 1. Type III gastric carcinoids are sporadic, solitary, larger, and more aggressive. Diagnosed by endoscopy with biopsy. Treatment depends on tumor size, type, and stage: small type I tumors can be managed by endoscopic resection and surveillance, while types II and III often require surgical resection with lymphadenectomy.

Gastric Emptying Study

– Nuclear medicine diagnostic test measuring the rate of gastric emptying. The protocol involves ingestion of a radiolabeled meal followed by serial gamma camera images obtained at one, two, and four hours. Gastric emptying is considered delayed if more than ten percent of the meal remains at four hours. The test is primarily used to diagnose gastroparesis, a condition of delayed gastric emptying without mechanical obstruction most commonly occurring as a complication of diabetes mellitus. The study may also help differentiate gastroparesis from functional dyspepsia.

Gastric Lymphoma

– Malignant proliferation of lymphoid tissue in the stomach most commonly mucosa-associated lymphoid tissue lymphoma. Strongly associated with H. pylori infection. Early disease may regress completely with H. pylori eradication therapy alone. More advanced disease may require chemotherapy with rituximab and cyclophosphamide, doxorubicin, vincristine, and prednisone. Surgery is rarely needed. Diffuse large B-cell lymphoma of the stomach is the other common type and requires combination chemotherapy with R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone).

Gastric Outlet Obstruction

– Blockage of gastric outflow at the pylorus or duodenum. Causes include peptic ulcer scarring, gastric cancer, and pyloric stenosis in infants. Presents with projectile vomiting, dehydration, and metabolic alkalosis. Diagnosis is by upper endoscopy and CT scan. Management requires decompression with nasogastric tube, fluid resuscitation, correction of electrolyte abnormalities, and definitive treatment of the underlying cause. Pyloric stenosis in infants is treated surgically with pyloromyotomy.

Gastric Polyps

– Protruding lesions of the gastric mucosa classified by histological type. Fundic gland polyps are the most common type associated with long-term proton pump inhibitor use and familial adenomatous polyposis. Hyperplastic polyps are associated with chronic gastritis and H. pylori infection and carry a small risk of dysplasia. Adenomatous polyps are true neoplasms with malignant potential requiring complete removal and surveillance. Inflammatory polyps occur in the setting of inflammatory bowel disease and may regress with treatment of the underlying colitis.

Gastric Ulcer

– Peptic ulcer in the gastric mucosa associated with

H. pylori, NSAIDs, and smoking. Pain is typically worsened by food. There is a higher risk of malignancy than duodenal ulcers. Diagnosis is by upper endoscopy with biopsy to exclude malignancy. Treatment involves proton pump inhibitors and H. pylori eradication. Follow-up endoscopy at eight weeks is recommended to confirm healing and exclude malignancy in gastric ulcers, particularly in patients over forty-five years.

Gastric Ulcer (see Peptic Ulcer)

– See Peptic Ulcer.

Gastric Volvulus

– Rotation of the stomach along its axis causing obstruction. It may present with Borchardt triad of epigastric pain, retching, and inability to pass a nasogastric tube. It is a surgical emergency requiring decompression and gastropexy. Organoaxial volvulus is the most common type where the stomach rotates along its longitudinal axis. Mesenteroaxial volvulus involves rotation along an axis perpendicular to the long axis. Diagnosis is by CT scan or upper endoscopy.

Gastrin

– A peptide hormone produced by G cells in the gastric antrum and duodenum. Secreted in response to: (1) gastric distension, (2) amino acids and peptides (especially from protein-rich meals), (3) vagal stimulation (via gastrin-releasing peptide, GRP), (4) hypercalcaemia. Actions: stimulates gastric acid secretion by parietal cells (via histamine release from enterochromaffin-like cells and direct effect on CCK-B receptors), stimulates gastric motility, promotes growth of gastric mucosa (trophic effect). Inhibited by somatostatin (from D cells in response to low antral pH). Hypergastrinaemia: Zollinger-Ellison syndrome (gastrinoma, usually pancreatic), pernicious anemia (achlorhydria, loss of feedback inhibition), chronic PPI use. Fasting serum gastrin is measured in suspected gastrinoma.

Gastritis

– Inflammation of the gastric mucosa, classified by duration (acute or chronic), etiology, and histologic pattern. Acute gastritis is most commonly caused by Helicobacter pylori infection (acute H. pylori infection causes transient hypochlorhydria with epigastric pain, nausea, and vomiting), NSAIDs (inhibiting mucosal prostaglandin synthesis, especially in the gastric antrum, leading to erosions and bleeding), excessive alcohol consumption, severe physiologic stress (stress-related gastritis in critically ill patients). Chronic gastritis is most commonly caused by persistent H. pylori infection, which can progress to atrophic gastritis, intestinal metaplasia, and gastric adenocarcinoma.

Gastrocnemius

– The large superficial muscle of the posterior calf, forming the bulk of the calf contour. Origin: medial and lateral femoral condyles. Insertion: calcaneus via the Achilles tendon. Nerve: tibial nerve (S1-S2). Action: plantar flexion of the ankle (standing, walking, running, jumping); also flexes the knee. The gastrocnemius works with the soleus as the triceps surae. Clinical relevance: medial gastrocnemius strain (tennis leg), gastrocnemius rupture, deep vein thrombosis (DVT) may cause calf swelling and must be distinguished from muscle injury. Power: can generate up to 3-4 times body weight during running.

Gastroduodenal Crohn Disease

– Involvement of the stomach and duodenum by Crohn disease occurring in approximately five percent of patients with Crohn. Gastric Crohn typically involves the antrum and pylorus and can cause gastric outlet obstruction from edema, fibrosis, or stricture. Duodenal Crohn presents with epigastric pain, nausea, vomiting, and weight loss. Diagnosis is by upper endoscopy with biopsy showing non-caseating granulomas and chronic inflammation. Treatment involves medical therapy with corticosteroids for induction of remission, followed by immunomodulators or biologic therapy for maintenance.

Gastroenteritis

– Inflammation of the stomach and intestinal mucosa, most commonly caused by viral infections (norovirus, rotavirus, adenovirus), bacterial pathogens (E. coli, Salmonella, Campylobacter), or parasites. Presents with diarrhea, vomiting, abdominal cramps, and dehydration. Self-limited in most cases. Management: oral rehydration, antiemetics, and symptomatic care. Antibiotics reserved for specific bacterial infections.

Gastroenterology

– The medical specialty focused on the diagnosis and management of diseases of the gastrointestinal tract (esophagus, stomach, small intestine, colon, rectum, pancreas, gallbladder, bile ducts, liver). Subspecialties: hepatology (liver), pancreatology, inflammatory bowel disease, gastrointestinal oncology, neurogastroenterology. Diagnostic tools: upper endoscopy (esophagogastroduodenoscopy, OGD), colonoscopy, capsule endoscopy, endoscopic retrograde cholangiopancreatography (ERCP), endoscopic ultrasound (EUS), breath tests (H. pylori, lactose intolerance, SIBO), manometry, pH metry. Therapeutic procedures: polypectomy, mucosal resection, hemostasis (injection, cautery, clips, banding), dilation, stent placement, percutaneous endoscopic gastrostomy (PEG).

Gastroesophageal Reflux Disease (GERD)

– A chronic condition in which gastric contents reflux backward into the esophagus, causing troublesome symptoms and/or complications. Epidemiology: affects 10–20% of adults in Western countries. Pathophysiology: transient lower esophageal sphincter relaxations (TLESRs) are the primary mechanism; contributing factors include a hypotensive lower esophageal sphincter, hiatal hernia, impaired esophageal clearance, delayed gastric emptying, and obesity. Symptoms: classic symptoms include heartburn (retrosternal burning) and acid regurgitation; atypical symptoms include non-cardiac chest pain, chronic cough, laryngitis, asthma exacerbation, and globus sensation. Complications: erosive esophagitis, esophageal stricture, Barrett esophagus (metaplasia from squamous to columnar epithelium, increasing risk of esophageal adenocarcinoma). Diagnosis: clinical evaluation, PPI trial, upper endoscopy (for alarm symptoms such as dysphagia, weight loss, GI bleeding, anemia, or age > 55), and 24-hour ambulatory pH/impedance monitoring for refractory cases. Management: lifestyle modifications (weight loss, elevation of head of bed, avoiding trigger foods and late meals), proton pump inhibitors (PPIs like omeprazole 20 mg daily), H2 receptor antagonists, and laparoscopic Nissen fundoplication or magnetic sphincter augmentation for refractory cases.

Gastrointestinal Bleeding

– Hemorrhage from any part of the gastrointestinal tract classified as upper or lower based on location relative to the ligament of Treitz. Upper gastrointestinal bleeding originating proximal to this landmark typically presents with hematemesis or melena. Common causes include peptic ulcer disease, esophageal and gastric varices from portal hypertension, Mallory-Weiss tears, and erosive gastritis. Lower gastrointestinal bleeding originating distal to the ligament of Treitz usually presents with haematochezia or bright red blood per rectum, along with abdominal pain.

Gastrointestinal Stromal Tumor

– Mesenchymal tumor arising from interstitial cells of Cajal in the gastrointestinal wall most commonly in the stomach. Often driven by gain-of-function mutations in the KIT proto-oncogene or PDGFRA gene. The majority arebut approximately twenty to thirty percent are malignant. Diagnosed by endoscopic ultrasound with fine-needle aspiration and immunohistochemistry showing CD117 or DOG1 positivity. Treated with imatinib a tyrosine kinase inhibitor for unresectable or metastatic disease and surgical resection for localized disease, with imatinib as first-line therapy for advanced or metastatic GIST.

Gastroparesis

– Delayed gastric emptying without mechanical obstruction most commonly caused by diabetes mellitus, post-surgical changes, or idiopathic causes. Presents with nausea, vomiting of undigested food, early satiety, bloating, and abdominal pain. Diagnosed by gastric emptying scintigraphy showing more than ten percent retention at four hours. Complications include poor glycemic control, bezoar formation, and malnutrition. Managed with dietary modifications including small frequent meals and low-fat low-fibre diet, prokinetic agents such as metoclopramide and domperidone, and antiemetics. In severe cases, gastric electrical stimulation or jejunostomy tube placement may be considered.

Gastrostomy

– Surgical creation of an opening (stoma) through the abdominal wall into the stomach for enteral feeding or decompression. Methods: percutaneous endoscopic gastrostomy (PEG, most common–pull or introducer technique), radiologically inserted gastrostomy (RIG, under fluoroscopic guidance), surgical gastrostomy (open or laparoscopic; Stamm or Witzel technique). Indications: neurological dysphagia (stroke, motor neuron disease, dementia), head and neck cancer, esophageal obstruction, major trauma/burns, prolonged mechanical ventilation. Contraindications: gastric outlet obstruction, severe ascites, gastric varices, coagulopathy, previous abdominal surgery (relative). Complications: peristomal infection (most common, 5-30%), tube displacement, buried bumper syndrome, gastrocolic fistula, peritonitis, pneumoperitoneum, bleeding.

Gavage

– Feeding through a tube passed into the stomach (nasogastric, orogastric, or gastrostomy tube). Used when oral feeding is impossible or inadequate. In neonates: gavage feeding is used for premature infants unable to coordinate suck-swallow-breathe. Types: bolus (intermittent, 100-400 mL over 10-30 minutes) or continuous (drip, 50-150 mL/hour via pump). Complications: tube misplacement (aspiration), tube blockage, nasal erosion, sinusitis, gastro-esophageal reflux, diarrhea (hyperosmolar feeds).

Giardiasis

– Infection with Giardia lamblia (G. intestinalis), a protozoan parasite causing diarrheal illness. Transmission via contaminated water (fecal-oral). Presents with foul-smelling, greasy diarrhea, bloating, abdominal cramps, weight loss, and steatorrhea. Diagnosis: stool antigen testing or microscopy. Treatment: metronidazole or tinidazole. Prevention: water purification, hand hygiene.

Gluten

– A complex of storage proteins (prolamins and glutelins) found in wheat (gliadin), barley (hordein), and rye (secalin). Gluten provides elasticity to dough. In susceptible individuals, gluten triggers an autoimmune response (celiac disease) or non-autoimmune sensitivity (non-celiac gluten sensitivity, NCGS). Gluten-free diet is the treatment for celiac disease. Oats contain avenin (structurally different) and are tolerated by most celiac patients if certified gluten-free (avoid cross-contamination). Gluten is also implicated in: dermatitis herpetiformis (cutaneous manifestation of celiac disease), gluten ataxia, and wheat allergy (IgE-mediated, different pathogenesis).

Gluten-Sensitive Enteropathy (Celiac Disease)

– An autoimmune enteropathy triggered by ingestion of gluten (wheat, barley, rye) in genetically predisposed individuals. Incidence: 1% worldwide, higher in HLA-DQ2 (90%) and HLA-DQ8 carriers. Pathophysiology: gluten peptides (especially 33-mer alpha-gliadin) are deamidated by tissue transglutaminase (tTG), presented by HLA-DQ2/DQ8, activating T-cell-mediated damage to small intestinal villi (villous atrophy, crypt hyperplasia, increased intraepithelial lymphocytes). Symptoms: (1) Classical: chronic diarrhea, steatorrhea, weight loss, bloating, abdominal pain, failure to thrive (children). (2) Non-classical: iron-deficiency anemia, fatigue, osteoporosis, elevated transaminases, infertility, neuropathy, dermatitis herpetiformis. (3) Asymptomatic: detected by screening in at-risk groups (family history, type 1 diabetes, Down syndrome, autoimmune thyroiditis). Diagnosis: positive serology (IgA anti-tTG, confirm with anti-EMA or anti-DGP), duodenal biopsy (Marsh grade 0-IIIc). Treatment: lifelong strict gluten-free diet. Complications: refractory celiac disease (unresponsive to >12 months of diet), enteropathy-associated T-cell lymphoma (EATL), ulcerative jejunoileitis, increased mortality (hazard ratio 1.2-1.4).

Graft-Versus-Host Disease

– Immunological reaction where donor T cells attack recipient tissues after allogeneic bone marrow transplant. Affects the skin, liver, and gastrointestinal tract. GI manifestations include diarrhea, abdominal pain, nausea, anorexia, and gastrointestinal bleeding. Acute graft-versus-host disease occurs within the first hundred days and chronic occurs after day hundred. Diagnosis is by endoscopy with biopsy showing characteristic apoptosis and crypt abscesses. Treated with corticosteroids, calcineurin inhibitors (tacrolimus), mycophenolate mofetil, and anti-TNF agents.

Griping (see Colic)

– Severe abdominal pain, typically spasmodic in nature. See Colic.

Gut Microbiome

– The complex community of trillions of microorganisms, including bacteria, viruses, and fungi, residing in the gastrointestinal tract. The gut microbiome plays a crucial role in digestion, immune function, vitamin synthesis, and protection against pathogens. Its composition is influenced by diet, medications, and lifestyle, and imbalances have been linked to conditions such as obesity, inflammatory bowel disease, and metabolic disorders.

Halitosis

– An unpleasant odour of the breath that may be socially distressing. The most common cause (85-90%) is poor oral hygiene– accumulation of bacteria on the tongue, dental plaque, periodontitis, and gingivitis. The odour arises from volatile sulphur compounds (VSCs–hydrogen sulphide, methyl mercaptan, dimethyl sulphide) produced by Gram-negative anaerobic bacteria (especially Porphyromonas gingivalis, Treponema denticola, Tannerella forsythia) that break down sulphur-containing amino acids (cysteine, methionine). Other oral causes: dry mouth (xerostomia, reduced salivary cleansing), dental caries, oral infections, tongue coating, poorly fitting dentures. Non-oral causes: sinusitis (post-nasal drip), tonsillitis/tonsilloliths, gastro-esophageal reflux disease (GERD), Helicobacter pylori infection, liver failure (fetor hepaticus–a sweet, musty odour), renal failure (uraemic fetor–ammoniacal/urine-like), diabetic ketoacidosis (fruity/acetone), lung abscess, bronchiectasis (putrid odour), and trimethylaminuria (fish malodour syndrome). Diagnosis: clinical assessment (oral examination), organoleptic scoring (sensory assessment by clinician), gas chromatography, BANA test (benzoyl-DL-arginine-naphthylamide for bacterial enzymes). Management: good oral hygiene (tongue scraping, toothbrushing, flossing), treatment of underlying dental/medical conditions, mouthwash (chlorhexidine, zinc-containing, chlorine dioxide), avoidance of tobacco and alcohol.

Hartmann Procedure

– Surgical resection of the rectosigmoid colon with creation of an end colostomy and rectal stump. Used for emergency resection of complicated diverticulitis with peritonitis and colorectal cancer with obstruction or perforation. The procedure avoids primary anastomosis in a contaminated field. Reversal with restoration of intestinal continuity is performed electively when the patient recovers, though not all patients are candidates for reversal. Laparoscopic reversal may be technically demanding due to adhesions and fibrosis and carries significant morbidity.

Helicobacter pylori

– Gram-negative, microaerophilic, spiral-shaped bacterium specifically adapted to colonize the human gastric mucosa. It is estimated that approximately half the world's population is infected, making it one of the most common chronic bacterial infections globally. H. pylori colonizes the gastric antrum and body, surviving the extremely acidic environment by producing large quantities of the enzyme urease, which converts urea to ammonia and carbon dioxide, creating a protective alkaline microenvironment around the organism that neutralizes gastric acid and allows colonization.

Hematemesis

– Vomiting of blood, indicating upper gastrointestinal bleeding proximal to the ligament of Treitz. Blood may be bright red (acute, brisk bleeding) or resemble coffee grounds (altered blood that has been in contact with gastric acid). Common causes: peptic ulcer disease (most common, 50%), esophageal varices (15%), Mallory-Weiss tears (10%), erosive gastritis, esophagitis, and gastric cancer. Initial management: ABC approach, large-bore IV access, fluid resuscitation, blood transfusion (target Hb >80 g/L in hemodynamically stable patients), and proton pump inhibitor infusion. Prognostic scoring: Rockall score and Glasgow-Blatchford score predict need for intervention and mortality. Upper endoscopy within 24 hours is both diagnostic and therapeutic (hemostatic clips, adrenaline injection, thermal coagulation, or band ligation for varices).

Hemochromatosis

– Iron overload disorder characterized by excessive intestinal iron absorption leading to progressive iron deposition in parenchymal organs. Hereditary hemochromatosis is most commonly caused by homozygous C282Y mutations in the HFE gene affecting approximately one in two hundred individuals of Northern European descent. Clinical manifestations include hepatic cirrhosis, cardiomyopathy, diabetes mellitus, arthropathy, and hypogonadism. Diagnosis is established by elevated transferrin saturation, elevated serum ferritin, and confirmatory genetic testing for HFE mutations.

Hemorrhoids

– Dilated veins in the rectal and anal canal. Internal hemorrhoids are above the dentate line, innervated by visceral nerves, and present with painless bright red bleeding. External hemorrhoids are below the dentate line, innervated by somatic nerves, and may thrombose causing severe pain. Grading: first degree bleed but do not prolapse; second degree prolapse with straining but reduce spontaneously; third degree require manual reduction; fourth degree are irreducible. Managed with dietary fiber, increased fluid intake, stool softeners, and topical treatments. Surgical intervention including haemorrhoidectomy, stapled haemorrhoidopexy, or rubber band ligation is reserved for advanced grades and refractory symptoms.

Hepatic Cyst

– Fluid-filled cavity in the liver most commonly congenital and asymptomatic. Simple hepatic cysts are lined by biliary epithelium and contain clear fluid. May cause pain or compressive symptoms if large or complicated by hemorrhage, infection, or rupture. Simple cysts require no treatment. Polycystic liver disease is an autosomal dominant condition associated with autosomal dominant polycystic kidney disease and may cause significant hepatomegaly and symptoms requiring surgical intervention including fenestration, aspiration sclerotherapy, or surgical resection for symptomatic giant cysts.

Hepatic Encephalopathy

– Neuropsychiatric syndrome occurring in advanced liver disease resulting from accumulation of neurotoxic substances, particularly ammonia. Ammonia crosses the blood-brain barrier and is metabolized by astrocytes into glutamine causing osmotic swelling and impaired neurotransmission. The West Haven grading classifies severity from grade one with mild confusion to grade four with coma. Precipitating factors include gastrointestinal bleeding, infection, constipation, and dehydration. Diagnosis is primarily clinical, supported by elevated serum ammonia, and exclusion of other causes of altered mental status. Management includes lactulose, rifaximin, dietary protein restriction, and treatment of precipitating factors.

Hepatic Hemangioma

– Most common benign liver tumor composed of dilated blood-filled vascular spaces lined by endothelial cells. Usually asymptomatic and found incidentally on imaging. Most are small and require no treatment. Giant hemangiomas exceeding five centimeters may cause pain, compression of adjacent structures, or rarely consumptive coagulopathy with Kasabach-Merritt syndrome. Diagnosed characteristically on imaging showing peripheral nodular enhancement with centripetal fill-in on contrast-enhanced CT or MRI, with a characteristic discontinuous centripetal filling pattern.

Hepatic Hydrothorax

– Large pleural effusion occurring in patients with cirrhosis and portal hypertension in the absence of primary cardiac or pulmonary disease. Most commonly right-sided resulting from transit of ascitic fluid through small defects in the diaphragm into the pleural space driven by the pressure gradient between the abdomen and thorax. Presents with dyspnea, cough, and hypoxemia. Diagnosis requires thoracentesis demonstrating transudative fluid with a high ascitic fluid to pleural fluid protein gradient, along with treatment of the underlying liver disease and portal hypertension.

Hepatitis A Virus

– Picornavirus transmitted through the fecal-oral route most commonly via contaminated food or water. The incubation period is fifteen to fifty days. Clinical presentation ranges from asymptomatic infection in young children to symptomatic acute hepatitis with jaundice, fatigue, nausea, elevated transaminases, dark urine, and pale stools. Unlike hepatitis B and C, hepatitis A does not cause chronic infection. Fulminant hepatitis occurs in approximately zero point one to zero point three percent with overall mortality increasing with age and underlying liver disease.

Hepatitis B Virus

– Hepadnavirus transmitted through percutaneous exposure, sexual contact, and perinatally causing acute and potentially chronic hepatitis. The virus has a partially double-stranded circular DNA genome. Chronic hepatitis B affects approximately two hundred forty million people worldwide, increasing the risk of cirrhosis and hepatocellular carcinoma. Diagnosis is made by serological testing: surface antigen indicates infection, surface antibody indicates immunity, core antibody indicates past or current infection. Vaccination is recommended for all susceptible individuals and is highly effective at preventing infection and its complications.

Hepatitis C Virus

– Flavivirus transmitted primarily through percutaneous exposure with injection drug use being the most common route. Approximately seventy to eighty percent of acute infections progress to chronic hepatitis C, the leading cause of cirrhosis, hepatocellular carcinoma, and liver transplantation in many Western countries. Diagnosis involves anti-hepatitis C antibody screening followed by confirmatory RNA measurement. Treatment with direct-acting antivirals achieves sustained virological response rates exceeding 95 percent, with minimal side effects.

Hepatitis D Virus

– Defective RNA virus requiring co-infection with hepatitis B to replicate, using hepatitis B surface antigen for its envelope. Hepatitis D causes more severe liver disease than hepatitis B alone with accelerated progression to cirrhosis and higher risk of hepatocellular carcinoma. Co-infection with hepatitis D and B during acute hepatitis B is generally self-limited, but superinfection of chronic hepatitis B carriers leads to chronic hepatitis D in seventy to ninety percent of cases. Diagnosis is by serological testing for anti-hepatitis D antibodies and confirmed by HDV RNA detection.

Hepatitis E Virus

– Non-enveloped RNA virus transmitted through the fecal-oral route most commonly via contaminated drinking water. It is the most common cause of acute viral hepatitis worldwide. Clinical presentation is similar to hepatitis A with jaundice and elevated transaminases. The infection is usually self-limited in immunocompetent individuals, but fulminant hepatic failure occurs in approximately one to two percent overall with significantly higher mortality in pregnant women reaching twenty-five percent in the third trimester.

Hepatocellular Jaundice

– Jaundice caused by dysfunction of hepatocytes, impairing the liver's ability to take up, conjugate, or excrete bilirubin. See Jaundice for complete overview. Causes: viral hepatitis (A, B, C, D, E), alcoholic hepatitis, drug-induced liver injury (paracetamol, isoniazid), cirrhosis, autoimmune hepatitis, Wilson disease, haemochromatosis. Laboratory features: mixed elevation of unconjugated and conjugated bilirubin, elevated transaminases (ALT/AST), and impaired synthetic function (low albumin, prolonged INR). Differentiated from obstructive jaundice by normal bile ducts on imaging.

Hepatoma

– See Hepatocellular Carcinoma.

Hepatopulmonary Syndrome

– Triad of liver disease, intrapulmonary vascular dilatation, and impaired oxygenation causing hypoxemia. Presents with platypnea which is dyspnea worsening in the upright position and orthodeoxia which is oxygen desaturation in the upright position. Diagnosed by contrast-enhanced echocardiography showing right-to-left shunt or macroaggregated albumin lung perfusion scan. The condition is a indication for liver transplantation as it may improve or resolve after transplant. There is no effective medical therapy beyond supportive care and oxygen supplementation.

Hepatorenal Syndrome

– Functional renal failure in advanced liver disease characterized by intense renal vasoconstriction in the setting of splanchnic vasodilation. Type 1 hepatorenal syndrome is rapidly progressive with median survival of approximately two weeks without treatment. Type 2 is slowly progressive associated with refractory ascites. Diagnosis requires exclusion of other causes of renal failure. Treatment options include vasoconstrictor therapy with terlipressin and albumin, transjugular intrahepatic portosystemic shunt (TIPS) for refractory cases, and liver transplantation as definitive treatment.

Hiatal Hernia

– Protrusion of part of the stomach through the esophageal hiatus into the thoracic cavity. Type I (sliding, 95%): gastroesophageal junction above the diaphragm. Types II-IV (paraesophageal): fundus or other portions herniate alongside the esophagus. Sliding hernias are associated with GERD. Paraesophageal hernias can cause obstruction, volvulus, and ischemia. Treatment: PPIs for GERD; surgical repair for symptomatic paraesophageal hernias.

Hiccups (Singultus)

– Involuntary, spasmodic contractions of the diaphragm and intercostal muscles followed by sudden glottic closure, producing the characteristic "hic" sound. The reflex arc involves the phrenic nerve (C3-C5), vagus nerve, and brainstem centers. Acute hiccups (<48 hours) are common and benign, triggered by gastric distension (overeating, carbonated beverages, aerophagia), temperature changes (hot/cold food or drink), alcohol, excitement, or stress. Persistent hiccups (>48 hours) and intractable hiccups (>1 month) require evaluation for underlying pathology: gastrointestinal (GERD, hiatal hernia, gastritis), neurologic (stroke, brainstem tumor, multiple sclerosis, meningitis), thoracic (pneumonia, pleurisy, pericarditis, myocardial infarction), metabolic (uremia, hyponatremia, diabetes), or drug-induced (dexamethasone, benzodiazepines, opioids, barbiturates, chemotherapy). Imaging: chest X-ray, CT head/chest/abdomen, upper endoscopy, and brain MRI when neurologic signs are present. Treatment for acute hiccups: vagal maneuvers (breath-holding, Valsalva, drinking water rapidly, swallowing dry sugar, pulling knees to chest, carotid sinus massage), and rebreathing into a bag (raises CO2, may suppress the reflex arc). Pharmacotherapy for persistent hiccups: chlorpromazine (first-line, FDA-approved), metoclopramide, baclofen, gabapentin, or haloperidol; severe refractory cases may require phrenic nerve block or vagus nerve stimulation.

Hirschsprung Disease

– Congenital condition characterized by the absence of enteric ganglion cells in a segment of the distal colon, resulting from failure of neural crest cell migration during embryonic development. The aganglionic segment, which most commonly involves the rectosigmoid colon, remains tonically contracted because it lacks the inhibitory neural input required for relaxation, causing functional bowel obstruction. Affected infants typically present with failure to pass meconium within forty-eight hours of birth, along with abdominal distension, bilious vomiting, and delayed passage of meconium. Diagnosis is confirmed by barium enema showing a transition zone between the dilated proximal ganglionic segment and the narrow distal aganglionic segment, and definitive treatment is surgical resection of the aganglionic segment with pull-through anastomosis.

Hirschsprung's Disease

– A congenital disorder characterized by the absence of parasympathetic ganglion cells (aganglionosis) in the distal colon, resulting in failure of relaxation and functional obstruction. Incidence: 1 in 5,000 live births, male predominance (4:1). The aganglionic segment extends proximally from the internal anal sphincter for a variable distance: short-segment (most common, 80%, rectosigmoid), long-segment (15%, extends beyond the sigmoid), total colonic aganglionosis (5%). Pathophysiology: during embryonic development (weeks 5-12), neural crest cells fail to migrate completely along the hindgut. The aganglionic segment has absent submucosal (Meissner) and myenteric (Auerbach) plexuses, causing tonic contraction and functional obstruction. Mutation in RET proto-oncogene is the most common genetic cause (50% of familial, 15% of sporadic cases). Associated with Down syndrome (5-10%) and other genetic syndromes (Waardenburg, MEN IIA). Clinical presentation: neonates–failure to pass meconium in the first 24-48 hours (99% of normal infants pass meconium within 24 hours), abdominal distension, bilious vomiting, enterocolitis (fever, explosive diarrhea, sepsis). Older infants/children–chronic constipation, abdominal distension, failure to thrive, enterocolitis. Diagnosis: barium enema (transition zone–narrow aganglionic segment proximal to dilated ganglionic colon), anorectal manometry (absent rectoanal inhibitory reflex), full-thickness rectal biopsy (golden standard–absent ganglion cells, hypertrophied nerve trunks on acetylcholinesterase staining). Treatment: surgical resection of the aganglionic segment with pull-through procedure (Soave, Swenson, Duhamel). Laparoscopic-assisted and transanal endorectal pull-through are preferred. Prognosis: excellent for short-segment disease; long-term issues with constipation, soiling, and enterocolitis in some patients.

Ileal Pouch-Anal Anastomosis

– Surgical creation of a J-pouch from the terminal ileum after proctocolectomy to preserve intestinal continuity for ulcerative colitis. Avoids permanent ileostomy by allowing defecation through the anal route. The procedure involves removal of the entire colon and rectum with construction of a reservoir from distal ileum. Complications include pouchitis, which is the most common complication, cuffitis, anastomotic stricture, and pouch failure. Pouchitis typically responds to antibiotic therapy with metronidazole or ciprofloxacin, and chronic pouchitis may require probiotic therapy, topical steroids, or biologic agents.

Inflammatory Bowel Disease

– Chronic relapsing inflammatory conditions of the gastrointestinal tract encompassing Crohn disease and ulcerative colitis, resulting from a dysregulated immune response in genetically susceptible individuals triggered by environmental factors and alterations in the gut microbiome. The incidence and prevalence of IBD have increased substantially over recent decades, particularly in newly industrialized countries, suggesting important roles for urbanization, dietary changes, and reduced microbial diversity, which collectively drive the chronic inflammatory response.

Inflammatory Bowel Disease (IBD)

– A group of chronic inflammatory conditions affecting the gastrointestinal tract, primarily Crohn's disease and ulcerative colitis. Crohn's disease can involve any part of the digestive tract with transmural inflammation, while ulcerative colitis is confined to the colonic mucosa. Symptoms include abdominal pain, diarrhea, rectal bleeding, and weight loss, managed with anti-inflammatory medications, immunosuppressants, and sometimes surgery.

Internal Hernia

– Herniation of bowel through a mesenteric defect common after Roux-en-Y gastric bypass creating a defect at the jejunojejunostomy. May cause intermittent or acute small bowel obstruction. Presents with episodic abdominal pain, nausea, and vomiting. Diagnosed by CT showing clustered dilated small bowel loops and the swirling pattern of mesenteric vessels. Treated surgically with reduction of the hernia and closure of the mesenteric defect. Internal hernias after bariatric surgery are a surgical emergency requiring prompt operative intervention to prevent bowel ischemia and infarction.

Intestinal Ganglioneuromatosis

– Benign proliferation of ganglion cells and nerve fibers in the gastrointestinal tract wall. May be diffuse involving the entire gastrointestinal tract or localized as ganglioneuromas. Associated with multiple endocrine neoplasia type 2B where intestinal ganglioneuromatosis may precede the development of medullary thyroid carcinoma. Can cause constipation, diarrhea, abdominal pain, or pseudo-obstruction. Diagnosis is by endoscopic biopsy showing ganglion cells within the lamina propria and muscularis mucosae, often associated with nerve fibre hypertrophy.

Intestinal Obstruction

– A mechanical or functional blockage of intestinal contents. Small bowel obstruction (SBO): most common cause is adhesions (60%), followed by hernias, tumors, and Crohn disease. Large bowel obstruction (LBO): most common is colorectal cancer, volvulus (sigmoid most common), and diverticulitis. Symptoms: abdominal pain (crampy, colicky), vomiting, abdominal distension, and obstipation. In SBO, vomiting is early and prominent. In LBO, distension is prominent and vomiting is late. Diagnosis: plain abdominal X-ray (air-fluid levels, dilated loops), CT scan (gold standard, identifies etiology and strangulation). Treatment: NPO, NG decompression, IV fluids, and electrolyte correction. Complete SBO often requires surgery (laparoscopy or laparotomy with adhesiolysis). LBO: resection, Hartmann procedure, or colonic stenting as bridge to surgery.

Intestinal Perforation

– Full-thickness breach in the bowel wall allowing luminal contents to enter the peritoneal cavity. Causes include diverticulitis, appendicitis, peptic ulcer disease, and trauma. Presents with acute abdomen, peritonitis, and free air on imaging. A surgical emergency requiring emergent laparotomy with primary repair or resection depending on location and degree of contamination. Postoperative management includes broad-spectrum antibiotics and intensive care support.

Intestinal Pseudo-Obstruction

– Acute or chronic dilation of the bowel without mechanical obstruction due to neuromuscular dysfunction. Acute colonic pseudo-obstruction also called Ogilvie syndrome occurs in hospitalized or post-surgical patients. Chronic intestinal pseudo-obstruction may be idiopathic or secondary to systemic diseases including scleroderma, diabetes, and Parkinson disease. Managed with decompression, prokinetic agents including metoclopramide and erythromycin, and treatment of underlying conditions. Neostigmine is the first-line pharmacological treatment for acute colonic pseudo-obstruction, with resolution rates exceeding 80 percent, but requires cardiac monitoring due to risk of bradycardia.

Intra-Abdominal Abscess

– Localized collection of pus within the abdomen commonly in the pelvis, between bowel loops, or beneath the diaphragm. Results from intra-abdominal infection or surgery. Presents with fever, abdominal pain, and leukocytosis. Diagnosed by CT with contrast showing rim-enhancing fluid collection. Treated with percutaneous CT-guided drainage and broad-spectrum antibiotics. Surgery is reserved for multiloculated abscesses not amenable to percutaneous drainage or failure of percutaneous management.

Intra-Abdominal Hypertension

– Sustained elevation of intra-abdominal pressure above twelve millimeters of mercury measured through the urinary bladder. Graded as grade one twelve to fifteen, grade two sixteen to twenty, grade three twenty-one to twenty-five, and grade four greater than twenty-five millimeters of mercury. Causes include fluid resuscitation, abdominal surgery, ileus, ascites, and abdominal hemorrhage. Intra-abdominal hypertension can progress to abdominal compartment syndrome with organ dysfunction. Monitoring of intra-abdominal pressure is essential in critically ill and post-surgical patients to guide interventions and prevent progression to abdominal compartment syndrome.

Intramural Hematoma

– Collection of blood within the wall of the gastrointestinal tract most commonly the duodenum. Caused by trauma, anticoagulation, or coagulopathy. Presents with abdominal pain, vomiting, and sometimes obstructive symptoms. Diagnosis is by CT showing circumferential wall thickening with high-attenuation hematoma. Managed conservatively with bowel rest, nasogastric decompression, and correction of coagulopathy. Most resolve spontaneously without surgical intervention.

Intussusception

– Condition in which a segment of intestine telescopes into the adjacent distal segment causing bowel obstruction and compromise of the mesenteric blood supply. It is the most common cause of intestinal obstruction in infants and children between six months and three years of age. Classic presentation includes episodic colicky abdominal pain, vomiting, a sausage-shaped abdominal mass, and current jelly stool. Diagnosis is confirmed by ultrasound demonstrating a target sign or donut sign. In stable patients with non-ischemic intussusception, air or contrast enema reduction is both diagnostic and therapeutic, with success rates exceeding 80 percent in children.

Irritable Bowel Syndrome

– Functional gastrointestinal disorder characterized by recurrent abdominal pain associated with altered bowel habits without identifiable structural abnormalities. The Rome IV criteria define IBS as recurrent abdominal pain occurring at least one day per week associated with changes in defecation or stool form. Subtypes include IBS with constipation, IBS with diarrhea, and IBS with mixed bowel habits. Pathophysiology involves visceral hypersensitivity, altered gut motility, gut-brain axis dysfunction, and psychosocial factors. Management includes dietary modifications, stress reduction, fibre supplementation, antispasmodics, and neuromodulators.

Irritable Bowel Syndrome (IBS)

– A chronic functional gastrointestinal disorder characterized by recurrent abdominal pain and altered bowel habits (diarrhea, constipation, or mixed) without identifiable structural pathology. Pathophysiology involves dysregulation of the gut–brain axis, visceral hypersensitivity, altered motility, and imbalances in the gut microbiome. Diagnosis is clinical based on Rome IV criteria: recurrent abdominal pain at least 1 day/week for 3 months, associated with defecation or change in stool frequency/form. Management includes dietary modifications (low-FODMAP diet), fiber supplementation, antispasmodics, probiotics, and neuromodulators (tricyclic antidepressants, SSRIs) for refractory symptoms.

Ischemic Colitis

– Colonic mucosal ischemia resulting from reduced blood flow through the marginal artery and end-arteries, most commonly occurring at watershed areas between vascular territories, particularly the splenic flexure. Unlike acute mesenteric ischemia affecting the small bowel, ischemic colitis most often occurs with non-occlusive hypoperfusion from cardiac failure or dehydration. The clinical presentation includes sudden onset crampy left-sided abdominal pain followed by bloody diarrhea. Most cases are self-limiting and resolve with supportive care and bowel rest, with full recovery expected within one to two weeks without specific intervention.

Jaundice

– Yellowish discoloration of the skin, sclerae, and mucous membranes caused by hyperbilirubinemia (serum bilirubin greater than 2-3 mg/dL), resulting from an imbalance between bilirubin production and clearance. Bilirubin metabolism: heme from senescent red blood cells is converted by heme oxygenase to biliverdin, then to unconjugated (indirect) bilirubin, which is transported to the liver bound to albumin. Unconjugated bilirubin is taken up by hepatocytes and conjugated with glucuronic acid by uridine diphosphate glucuronosyltransferase (UGT) in the hepatocyte. Conjugated (direct) bilirubin is water-soluble and excreted into bile. Hyperbilirubinaemia is classified as predominantly unconjugated (pre-hepatic or hepatic) or conjugated (hepatic or post-hepatic/obstructive).

Jaundice (Icterus)

– See Jaundice.

Kayser-Fleischer Rings

– Golden-brown copper deposition rings in Descemet membrane of the cornea representing a pathognomonic finding of Wilson disease. These rings are visible on slit lamp examination and present in virtually all patients with neurological manifestations of Wilson disease though they may be absent in isolated hepatic disease. The rings typically begin at the superior and inferior poles and progress circumferentially. They may gradually diminish after initiation of chelation therapy. Detection requires slit-lamp examination by an experienced ophthalmologist.

Lactose

– The primary disaccharide found in milk and dairy products (glucose + galactose, beta-1,4 glycosidic bond). See Lactose Intolerance.

Lactose intolerance

– A common digestive disorder caused by lactase deficiency leading to malabsorption of lactose, the primary disaccharide in milk and dairy products. The lactase enzyme is produced by enterocytes of the small intestinal brush border and hydrolyzes lactose into glucose and galactose for absorption. Three types exist: primary lactase deficiency (most common, involves the genetically programmed decline in lactase activity after weaning, affecting 70-75 percent of the world's adult population, with highest prevalence in individuals of East Asian, African, and Middle Eastern descent where lactase persistence is less common.

Large Bowel Obstruction

– Mechanical blockage of the colon most commonly caused by colorectal cancer, volvulus, or diverticular stricture. Clinical presentation includes progressive abdominal distension, constipation, and obstipation. The cecum is at highest risk of perforation with distal obstruction. Diagnosis is by CT scan identifying the site and cause. Management depends on cause: sigmoid volvulus may be managed by endoscopic detorsion, cecal volvulus requires emergent surgery, and colorectal cancer obstruction may require a diverting colostomy or colonic stent as a bridge to elective resection.

Liver Abscess

– Focal collection of pus within the liver parenchyma. Pyogenic abscesses are most commonly from biliary tract infection or hematogenous spread from portal or systemic infection. Amoebic abscesses are caused by Entamoeba histolytica predominantly in tropical regions. Presents with fever, right upper quadrant pain, and hepatomegaly. Diagnosed by CT showing rim-enhancing fluid collection. Treatment involves percutaneous CT-guided drainage and broad-spectrum antibiotics for pyogenic abscesses. Metronidazole is the treatment of choice for amoebic liver abscesses, with aspiration reserved for large abscesses at risk of rupture.

Liver Biopsy

– Percutaneous, transjugular, or laparoscopic sampling of liver tissue for histological evaluation. Indicated for staging fibrosis, diagnosing specific liver diseases, and assessing treatment response. The procedure is guided by ultrasound for accuracy. Complications include bleeding, pain, and pneumothorax. Contraindications include uncorrected coagulopathy and thrombocytopenia. Transjugular biopsy is performed when percutaneous biopsy is contraindicated due to coagulopathy or ascites. Histologic evaluation remains the gold standard for assessing fibrosis stage and grade of inflammatory activity in chronic liver disease.

Liver Cirrhosis

– Advanced hepatic fibrosis with regenerative nodules surrounded by dense fibrous septa representing the end result of chronic liver injury. Common etiologies include chronic viral hepatitis, alcoholic liver disease, and nonalcoholic steatohepatitis. The pathogenesis involves activation of hepatic stellate cells producing excessive extracellular matrix. Cirrhosis progresses through compensated and decompensated phases. Decompensation manifests with ascites, variceal hemorrhage, encephalopathy, and ultimately hepatic failure.

Liver Diseases

– A broad category of pathological conditions affecting the structure and function of the liver. See individual entries: Hepatitis, Liver Cirrhosis, Fatty Liver, Hepatocellular Carcinoma, etc.

Liver Health

– Essential for metabolism, detoxification, protein synthesis, bile production, and immune function. Common liver disorders include nonalcoholic fatty liver disease (NAFLD, now termed metabolic dysfunction–associated steatotic liver disease or MASLD) affecting 30% of adults, alcoholic liver disease (steatosis, hepatitis, cirrhosis), viral hepatitis (HAV, HBV, HCV), drug-induced liver injury (acetaminophen toxicity most common), and autoimmune hepatitis. MASLD progresses from steatosis to steatohepatitis (NASH/MASH) with fibrosis and cirrhosis, driven by insulin resistance, obesity, and metabolic syndrome. Screening: liver enzymes (ALT, AST), viral serologies, abdominal ultrasound, FibroScan for fibrosis, and liver biopsy when indicated. Management: weight loss ≥7% improves steatohepatitis in MASLD, alcohol cessation, antiviral therapy for HBV/HCV (direct-acting antivirals cure >95% of HCV), and vaccination for HAV/HBV.

Liver Transplantation

– Surgical replacement of a diseased liver with a healthy donor organ indicated for end-stage liver disease, acute liver failure, and select hepatocellular carcinomas within Milan criteria. Requires lifelong immunosuppression with calcineurin inhibitors, antimetabolites, and corticosteroids. Living donor liver transplantation uses a portion of a living donor's liver which regenerates in both donor and recipient. Complications include rejection, infection, vascular complications, biliary complications (anastomotic strictures, bile leaks), and hepatic artery thrombosis.

Low FODMAP Diet

– Dietary intervention for irritable bowel syndrome that restricts fermentable oligosaccharides, disaccharides, monosaccharides, and polyols, which are short-chain carbohydrates poorly absorbed in the small intestine and rapidly fermented by colonic bacteria producing gas and drawing water into the intestinal lumen. The approach involves an elimination phase of two to six weeks, a reintroduction phase identifying personal triggers, and a long-term personalization phase. Studies demonstrate symptom improvement in 50-80 percent of patients with irritable bowel syndrome.

Lymphocytic Colitis

– Chronic inflammatory bowel disease characterized by increased intraepithelial lymphocytes exceeding twenty per one hundred surface epithelial cells in the colonic mucosa without significant subepithelial collagen band thickening. Shares clinical features with collagenous colitis including chronic watery nonbloody diarrhea primarily in middle-aged women. The pathogenesis may involve an immune response to luminal antigens. Diagnosis requires colonoscopy with random colonic biopsies showing increased intraepithelial lymphocytes on histology.

Lynch Syndrome

– Hereditary nonpolyposis colorectal cancer syndrome caused by mismatch repair gene mutations including MLH1, MSH2, MSH6, and PMS2 with autosomal dominant inheritance. It increases the risk of colorectal, endometrial, ovarian, gastric, and urinary tract cancers. Diagnosis is suggested by microsatellite instability testing and immunohistochemistry showing loss of mismatch repair proteins, confirmed by genetic testing. Management involves intensive colonoscopic surveillance every one to two years starting at age 20-25, and prophylactic hysterectomy and salpingo-oophorectomy after completion of childbearing for women with Lynch syndrome.

Malabsorption

– Defective absorption of nutrients from the small intestine. Causes: celiac disease, pancreatic insufficiency, bacterial overgrowth, Crohn disease, short bowel syndrome, and infections. Presents with diarrhea, steatorrhea, weight loss, and deficiency of specific nutrients (iron, B12, vitamin D, calcium). Diagnosis: stool studies, D-xylose test, breath tests, small bowel biopsy. Treatment: address underlying cause and replace deficient nutrients.

Mallory-Weiss Tear

– Longitudinal mucosal laceration at the gastroesophageal junction caused by forceful vomiting or retching. It typically presents with hematemesis. Most bleeds spontaneously within twenty-four to forty-eight hours. Diagnosis is made by upper endoscopy showing a linear mucosal tear. Management includes observation, proton pump inhibitors, and endoscopic hemostasis with clips or thermal therapy if active bleeding is identified. Recurrence is uncommon and the prognosis is generally excellent.

Meckel Diverticulum

– Congenital remnant of the omphalomesenteric duct located on the antimesenteric border of the distal ileum approximately sixty centimeters from the ileocecal valve. It follows the rule of twos: occurs in approximately two percent of the population, located approximately two feet from the ileocecal valve, approximately two inches in length, and may contain two types of ectopic tissue most commonly gastric and pancreatic mucosa. The most common presentation in children is painless lower gastrointestinal bleeding from ulceration of ectopic gastric mucosa.

Meconium Ileus

– Intestinal obstruction in neonates caused by inspissated meconium most commonly associated with cystic fibrosis. Presents with abdominal distension, bilious vomiting, and failure to pass meconium within forty-eight hours. Diagnosis is suggested by abdominal radiography showing a ground-glass appearance and may be confirmed by CT showing calcified meconium. Treatment includes water-soluble contrast enema which can be both diagnostic and therapeutic by loosening the inspissated meconium. Surgical intervention is required when non-operative management fails, with laparotomy and enterotomy for meconium removal and possible bowel resection for complicated cases.

Meconium Peritonitis

– Sterile chemical peritonitis resulting from in utero intestinal perforation allowing meconium to spill into the peritoneal cavity. Most commonly associated with cystic fibrosis causing inspissated meconium obstruction, intestinal atresia, and volvulus. May be detected prenatally as fetal ascites with echogenic peritoneal calcifications. After birth presents with abdominal distension, vomiting, and signs of peritonitis. Diagnosis is by abdominal radiography showing intraperitoneal calcifications. Treatment is supportive with bowel rest, nasogastric decompression, and parenteral nutrition; surgery is reserved for ongoing perforation or obstruction.

MELD Score

– Model for End-Stage Liver Disease predicting three-month mortality in advanced liver disease used for liver transplant allocation. The score is calculated using serum bilirubin, creatinine, and international normalized ratio generating a continuous score from approximately six to forty points. Higher scores indicate more severe disease and higher mortality risk. The updated MELD score includes serum sodium. MELD exception points are granted for conditions including hepatocellular carcinoma within the Milan criteria (single tumor up to 5 cm, or up to 3 tumors each up to 3 cm, without extrahepatic spread or major vascular invasion).

melena

– Passage of black, tarry, foul-smelling stools due to the presence of digested blood, indicating upper gastrointestinal bleeding (usually proximal to the ligament of Treitz). The black color results from oxidation of haemoglobin to haematin by intestinal enzymes and bacteria. At least 200 mL of blood is typically required in the upper GI tract to produce melena. Common causes: peptic ulcer disease (most common), esophageal varices, Mallory-Weiss tear, erosive gastritis, Dieulafoy lesion, and gastric cancer. melena may persist for several days after a single bleeding episode. Differentiation from black stools caused by iron supplements, bismuth, or liquorice is important (these are not tarry and test negative for occult blood). Management: same as for hematemesis (resuscitation, PPI, urgent upper endoscopy).

Microscopic Colitis

– Umbrella term encompassing both collagenous colitis and lymphocytic colitis which share the clinical presentation of chronic watery non-bloody diarrhea and the pathophysiological feature of colonic mucosal inflammation. Endoscopic appearance is typically normal necessitating random biopsies for diagnosis. More common in women and associated with autoimmune diseases, celiac disease, and certain medications including proton pump inhibitors, selective serotonin reuptake inhibitors, and nonsteroidal anti-inflammatory drugs (NSAIDs), which can cause mucosal injury through inhibition of cyclooxygenase and prostaglandin synthesis.

Mirizzi Syndrome

– Condition where a gallstone impacted in the cystic duct or Hartmann pouch extrinsic compression of the common hepatic duct causing obstructive jaundice. Classified into four types based on the degree of erosion and fistula formation. Type one is extrinsic compression without fistula. Types two through four involve cholecystobiliary fistula of increasing size. Presents with obstructive jaundice, cholangitis, and abdominal pain. Diagnosis is by MRCP or endoscopic ultrasound demonstrating the impacted gallstone on the bile duct, and cholecystectomy with fistula repair is the definitive management.

MR Enterography

– MRI of the small bowel after oral contrast administration. Superior soft tissue contrast for evaluating Crohn disease activity, fistulas, and abscesses without ionizing radiation. Demonstrates mural thickening, edema, enhancement patterns, and extraintestinal complications. Diffusion-weighted imaging helps assess disease activity. Preferred imaging modality for young patients requiring serial follow-up due to absence of radiation. The technique is limited by longer examination time, availability, and susceptibility to motion artefact.

Neuroendocrine Tumors

– Heterogeneous group of neoplasms arising from neuroendocrine cells throughout the gastrointestinal tract and pancreas. May be functional producing hormones or non-functional. Graded by Ki-67 index and mitotic count into well-differentiated low-grade, intermediate-grade, and poorly differentiated high-grade tumors. Functional tumors include gastrinomas, insulinomas, glucagonomas, VIPomas, and carcinoid tumors secreting serotonin. Diagnosis is by hormone levels, imaging with CT, MRI, or octreotide scintigraphy (OctreoScan), and tissue biopsy for histological grading. Management includes somatostatin analogues for symptom control, peptide receptor radionuclide therapy (PRRT), and surgical resection for localised disease.

Non-Alcoholic Fatty Liver Disease

– Hepatic steatosis in the absence of significant alcohol consumption, strongly associated with metabolic syndrome, obesity, and type 2 diabetes. NAFLD includes simple steatosis and non-alcoholic steatohepatitis (NASH) with inflammation and fibrosis. NASH can progress to cirrhosis and hepatocellular carcinoma. Diagnosis: abdominal ultrasound, elevated liver enzymes, transient elastography, and liver biopsy for staging. Management: weight loss, exercise, vitamin E, and pioglitazone for NASH.

Nonalcoholic Steatohepatitis

– Progressive form of NAFLD characterized by hepatic steatosis with lobular inflammation and hepatocyte ballooning with significant risk of progression to fibrosis, cirrhosis, and hepatocellular carcinoma. NASH affects approximately three to five percent of the general population. The histological features include macrovesicular steatosis, inflammation, hepatocyte ballooning, and perisinusoidal fibrosis. Progressive fibrosis is the most important predictor of liver-related mortality. Management involves lifestyle modification with weight loss, dietary changes, exercise, vitamin E therapy in non-diabetic patients with biopsy-proven NASH, and consideration of obeticholic acid or GLP-1 receptor agonists in selected patients.

Obstructive Jaundice

– Jaundice caused by obstruction of bile flow from the liver to the duodenum, resulting in accumulation of conjugated bilirubin in the blood. See Jaundice for complete overview. Causes: (1) Extrahepatic obstruction: choledocholithiasis (common bile duct stones, most common), pancreatic head carcinoma (painless jaundice, Courvoisier law), cholangiocarcinoma, ampullary tumor, primary sclerosing cholangitis, biliary stricture (post-surgical, post-ERCP), Mirizzi syndrome. (2) Intrahepatic obstruction: primary biliary cholangitis (now PBC), drug-induced cholestasis, infiltrative diseases (sarcoidosis, amyloidosis), sepsis, total parenteral nutrition. Laboratory features: elevated conjugated (direct) bilirubin, raised ALP and GGT (cholestatic enzymes), transaminases may be normal or mildly elevated. Imaging: ultrasound (dilated bile ducts, stone, mass), MRCP, ERCP (therapeutic). Complications: cholangitis, secondary biliary cirrhosis, fat-soluble vitamin deficiency (A, D, E, K), pruritus.

Occult Blood

– Blood present in feces that is not visible to the naked eye, detectable only by chemical or immunochemical testing. fecal occult blood testing (FOBT) is used for colorectal cancer screening. Guaiac-based FOBT (gFOBT) detects haem peroxidase activity (requires dietary restriction—avoid red meat, horseradish, vitamin C). fecal immunochemical test (FIT) uses antibodies specific for human haemoglobin (no dietary restriction, more sensitive, quantitative; threshold typically 10–20 μg Hb/g feces). Positive FIT requires follow-up colonoscopy. Screening with FIT every 1–2 years reduces colorectal cancer mortality by 15–33%. False positives can occur with upper GI bleeding, NSAIDs, anticoagulants, and menstrual blood. Other causes of occult GI bleeding include peptic ulcer disease, angiodysplasia, and inflammatory bowel disease.

Ogilvie Syndrome

– Acute colonic pseudo-obstruction characterized by massive colonic dilation in the absence of mechanical obstruction typically occurring in hospitalized patients with serious comorbidities including post-surgical states, severe infections, and medication effects. Pathophysiology involves autonomic imbalance with sympathetic overactivity causing inhibition of colonic motility. The cecum is disproportionately dilated carrying risk of perforation which occurs when the cecal diameter exceeds twelve centimetres, with management including nasogastric decompression, correction of electrolyte abnormalities, discontinuation of exacerbating medications, neostigmine administration, and colonoscopic decompression for refractory cases.

Pancreatic Cancer

– Aggressive malignancy with pancreatic ductal adenocarcinoma accounting for approximately eighty-five percent of cases. It is the fourth leading cause of cancer-related death with a five-year survival of approximately ten percent. Risk factors include smoking, chronic pancreatitis, diabetes, and genetic syndromes. The most common presentation is painless obstructive jaundice from head of pancreas tumors. The Courvoisier sign suggests malignant biliary obstruction. Diagnosis involves CT, endoscopic ultrasound with fine-needle aspiration for tissue diagnosis, and tumor marker CA 19-9. Staging uses CT and endoscopic ultrasound.

Paracentesis

– Percutaneous needle aspiration of ascitic fluid from the peritoneal cavity. Used for diagnostic analysis to determine the cause of ascites and for therapeutic removal of large volumes for symptom relief. The serum-ascites albumin gradient is calculated to differentiate portal hypertensive from non-portal hypertensive causes. A gradient of one point one grams per deciliter or greater indicates portal hypertension. Diagnostic paracentesis with cell count, culture, and cytology should be performed before initiating treatment in all patients with new-onset ascites.

Peak Flow Meter

– See Peak Expiratory Flow Rate.

Peptic Ulcer Disease

– Condition characterized by ulceration of the gastrointestinal mucosa extending through the muscularis mucosae into the submucosa or deeper layers, occurring most commonly in the duodenal bulb and gastric antrum. The two predominant etiologies are Helicobacter pylori infection and nonsteroidal anti-inflammatory drug use, which together account for the vast majority of peptic ulcers. Duodenal ulcers are strongly associated with H. pylori and present with epigastric pain that is classically relieved by food and antacids, whereas gastric ulcers cause pain that is worsened by eating.

Peritoneal Carcinomatosis

– Diffuse metastatic seeding of the peritoneal cavity from gastrointestinal or ovarian malignancies. Presents with ascites, bowel obstruction, and cachexia. Diagnosed by CT and paracentesis showing malignant cells. Prognosis is generally poor. Management includes systemic chemotherapy, palliative care, and in selected cases, cytoreductive surgery with hyperthermic intraperitoneal chemotherapy. The peritoneal cancer index assesses disease distribution for surgical planning.

Peritoneal Dialysis Catheter

– Flexible tube surgically placed into the peritoneal cavity for peritoneal dialysis. The Tenckhoff catheter is the most commonly used type with one or two cuffs that promote tissue ingrowth and anchor the catheter. Placement can be done surgically, laparoscopically, or percutaneously under radiological guidance. Complications include peritonitis, catheter-related infection, tunnel infection, outflow failure, and hernia formation. Proper catheter placement technique and exit site care are essential for preventing infectious complications and ensuring catheter longevity.

Pilonidal Disease

– A chronic inflammatory condition of the sacrococcygeal region involving hair follicles. Pilonidal sinus: epithelialized tract containing hair and debris. Pilonidal abscess: acute infection of the sinus. Most common in young, hirsute males, exacerbated by prolonged sitting, obesity, and local trauma. Presents with pain, swelling, and purulent drainage in the natal cleft. Treatment: incision and drainage for acute abscess; definitive surgical treatment (excision with secondary healing, flap closure, or pit-picking) for chronic disease. Recurrence is common without good hygiene and hair removal.

Plummer-Vinson Syndrome

– Triad of iron deficiency anemia, esophageal web, and dysphagia occurring predominantly in middle-aged women. The esophageal webs are thin mucosal folds typically located in the postcricoid region of the hypopharynx and proximal esophagus. The pathogenesis is thought to involve epithelial atrophy from chronic iron deficiency affecting rapidly dividing mucosal cells. The syndrome is considered a premalignant condition with increased risk of postcricoid squamous cell carcinoma. Diagnosis is established by barium swallow demonstrating the characteristic thin mucosal web in the proximal esophagus, and treatment consists of iron replacement therapy and endoscopic dilation of the web.

Pneumatosis Cystoides Intestinalis

– Gas-filled cysts within the bowel wall that may be primary or secondary to bowel ischemia, infection, chronic lung disease, or medications. Often an incidental finding on imaging. Primary pneumatosis is benign and asymptomatic. Secondary pneumatosis may indicate serious underlying disease. Managed by treating the underlying condition. Surgery is reserved for complications including pneumoperitoneum, obstruction, or hemorrhage.

Pneumatosis Intestinalis

– Presence of gas within the bowel wall appearing as linear or cystic radiolucencies on imaging. Primary pneumatosis is benign and idiopathic accounting for approximately fifteen percent of cases. Secondary pneumatosis is associated with intestinal ischemia, necrotizing enterocolitis in neonates, obstructive pulmonary disease, and connective tissue diseases. The cystic form appears as clusters of gas-filled cysts in the submucosa or subserosa. The linear form suggests intramural gas tracking along the mesenteric vessels, which may indicate bowel necrosis and requires urgent surgical intervention.

Polyp Surveillance Colonoscopy

– Interval-based endoscopic follow-up of patients with previously identified colorectal polyps. Intervals are determined by polyp number, size, and histological features. One to two small adenomas require surveillance at seven to ten years. Three to four adenomas at three to five years. Five to ten adenomas or adenomas with villous features or high-grade dysplasia at three years. More than ten adenomas at one year. Sessile serrated polyps greater than ten millimeters require three to five year surveillance intervals.

Porcelain Gallbladder

– Calcification of the gallbladder wall visible on imaging representing chronic inflammatory condition associated with cholelithiasis and chronic cholecystitis. Historically considered a strong risk factor for gallbladder carcinoma, recent studies suggest the cancer risk may be lower than previously estimated. Most patients are asymptomatic and the condition is discovered incidentally. When cholecystectomy is performed, the gallbladder should be carefully examined for occult malignancy. Management is cholecystectomy, preferably by the laparoscopic approach.

Portal Hypertension

– Elevated pressure in the portal venous system defined as hepatic venous pressure gradient exceeding five millimeters of mercury with clinically significant portal hypertension occurring when the gradient exceeds ten. The most common cause is cirrhosis where increased intrahepatic resistance combined with splanchnic vasodilation creates a hyperdynamic circulatory state. Consequences include esophageal variceal hemorrhage, ascites, splenomegaly with hypersplenism, and hepatic encephalopathy. Diagnosis is established by measurement of the hepatic venous pressure gradient, upper gastrointestinal endoscopy to evaluate for varices, and imaging to assess for features of portal hypertension such as splenomegaly and portosystemic collaterals.

Portal Hypertensive Gastropathy

– Diffuse mucosal changes in the stomach due to portal hypertension causing chronic blood loss. Characterized by mosaic pattern and cherry red spots on endoscopy representing dilated mucosal vessels. Most common in patients with advanced cirrhosis and portal hypertension. Managed with beta-blockers to reduce portal pressure and iron supplementation for anemia. Argon plasma coagulation may be used for severe bleeding. The condition is distinct from gastritis and does not respond to acid suppression therapy.

Portopulmonary Syndrome

– Development of pulmonary arterial hypertension in the setting of portal hypertension or liver disease. Results from increased cardiac output and splanchnic vasodilation leading to pulmonary vascular remodeling. Classified as mild, moderate, or based on hemodynamic parameters with mean pulmonary artery pressure greater than twenty millimeters of mercury. Presents with exertional dysphagia, fatigue, and exercise intolerance. Diagnosis requires right heart catheterization confirming elevated pulmonary artery pressure on right heart catheterisation, with treatment including pulmonary vasodilator therapy and liver transplantation in selected cases.

Postcholecystectomy Syndrome

– Persistence or recurrence of symptoms after cholecystectomy. Causes include retained common bile duct stones, sphincter of Oddi dysfunction, bile salt malabsorption, irritable bowel syndrome, and other gastrointestinal conditions. Evaluation includes liver function tests, MRCP, and endoscopic ultrasound. Treatment depends on the identified cause. Bile salt malabsorption responds to bile acid sequestrants. Sphincter of Oddi dysfunction may require endoscopic sphincterotomy.

Prebiotics and Probiotics

– Prebiotics are nondigestible food ingredients (fiber and oligosaccharides—inulin, fructooligosaccharides, galactooligosaccharides) that selectively stimulate the growth and activity of beneficial gut bacteria (Bifidobacterium, Lactobacillus), promoting host health. Probiotics are live microorganisms that confer a health benefit when administered in adequate amounts—commonly *Lactobacillus* species, *Bifidobacterium*, *Saccharomyces boulardii* (yeast), and *Streptococcus thermophilus*. Evidence supports probiotics for prevention of antibiotic-associated diarrhea (50% risk reduction), reduction of *C. difficile* infection recurrence (alongside standard antibiotics), management of irritable bowel syndrome (bloating, pain), treatment of acute infectious gastroenteritis (reduces duration by 1 day), and prevention of necrotizing enterocolitis in preterm infants. Synbiotics combine prebiotics and probiotics. The microbiome modulates immune function, metabolism, and gut–brain signaling. Quality control varies widely; products should specify genus, species, strain, and colony-forming units (CFU) per dose. Fermented foods (yogurt, kefir, kimchi, sauerkraut) naturally provide beneficial microbes.

Primary Biliary Cholangitis

– Chronic autoimmune cholestatic liver disease characterized by progressive destruction of small intrahepatic bile ducts. The disease predominantly affects middle-aged women. Anti-mitochondrial antibodies are found in approximately ninety-five percent of patients. The hallmark histological finding is florid duct lesion with granulomatous destruction of interlobular bile ducts. Clinical presentation includes fatigue and pruritus. Diagnosis is established by elevated alkaline phosphatase, positive anti-mitochondrial antibodies, elevated serum IgM, and characteristic histology on liver biopsy. First-line treatment is ursodeoxycholic acid, which improves liver biochemistry and delays disease progression.

Primary Sclerosing Cholangitis

– Chronic cholestatic liver disease characterized by progressive fibrosis and strictures of intrahepatic and extrahepatic bile ducts. The disease has a strong association with ulcerative colitis occurring in seventy to eighty percent of patients. Diagnosis is established by MRCP showing multifocal strictures and dilatations giving a beaded appearance. Patients have a significantly increased risk of cholangiocarcinoma developing in ten to fifteen percent. There is no proven effective medical therapy, and liver transplantation is the definitive treatment for end-stage disease.

Proctitis

– Inflammation of the rectal mucosa causing tenesmus, rectal pain, and bloody discharge. Causes include inflammatory bowel disease, sexually transmitted infections including gonorrhea, chlamydia, and herpes, radiation therapy, and ischemia. Diagnosed by proctoscopy with biopsy. Treatment is directed at the underlying etiology. Empiric treatment for sexually transmitted infections should be initiated while awaiting culture results. Inflammatory proctitis responds to topical mesalamine or corticosteroid enemas.

Pruritus Ani

– Persistent itching around the anus with multiple potential causes including hemorrhoids, fissures, infections such as pinworms and candidiasis, dermatologic conditions including psoriasis and lichen sclerosus, and idiopathic causes. Managed by addressing the underlying cause, strict hygiene measures with gentle cleansing after bowel movements, avoiding scratching, and topical treatments including hydrocortisone cream and barrier preparations. Cotton underwear and avoidance of irritating foods such as coffee, spicy foods, and citrus fruits should also be advised.

Pseudomembranous Colitis

– Colonic inflammation caused by C. difficile infection showing yellow-white plaques on colonoscopy. The plaques consist of fibrin, inflammatory cells, and necrotic epithelial debris overlying mucosal ulceration. Presents with profuse watery diarrhea and abdominal pain. Diagnosis is confirmed by endoscopy with biopsy showing characteristic pseudomembranes. Treated with oral vancomycin or fidaxomicin. Severe cases may require intravenous metronidazole and surgical intervention for fulminant colitis unresponsive to medical therapy.

Psoas Abscess

– Collection of pus in the iliopsoas muscle compartment. Primary psoas abscess results from hematogenous spread of Staphylococcus aureus typically in immunocompromised patients. Secondary psoas abscess results from contiguous spread from adjacent structures including Crohn disease, diverticulitis, appendicitis, and vertebral osteomyelitis. Presents with fever, flank or abdominal pain, and characteristic hip flexion posture. Diagnosis is by CT of the abdomen and pelvis showing the psoas collection. Treatment is CT-guided percutaneous drainage and intravenous antibiotics directed at the causative organism.

Push Enteroscopy

– Endoscopic technique using a longer endoscope to visualize and intervene in the proximal small bowel beyond the reach of standard upper endoscopy. Unlike capsule endoscopy, push enteroscopy allows direct visualization with the ability to obtain biopsies and perform hemostasis. The procedure is performed with the endoscope advanced through the mouth into the proximal and mid-jejunum. The primary indication is evaluation of obscure gastrointestinal bleeding identified on capsule endoscopy. The diagnostic yield of push enteroscopy is highest when performed within 24-48 hours of active bleeding, and it enables targeted therapeutic intervention.

Pyloric Stenosis

– Hypertrophy of the pyloric sphincter muscle causing gastric outlet obstruction most common in young infants. Presents with non-bilious projectile vomiting typically beginning at two to eight weeks of age. Dehydration and metabolic alkalosis with hypochloremia and hypokalemia develop as the condition progresses. Diagnosis is by ultrasound showing pyloric muscle thickness greater than three millimeters and length greater than fifteen millimeters. Treated surgically with Ramstedt pyloromyotomy, which typically resolves with full recovery.

Ranson Criteria

– Clinical scoring system for acute pancreatitis severity incorporating eleven parameters at two time points. Admission parameters include age, white blood cell count, glucose, LDH, and AST. Forty-eight hour parameters include hematocrit drop, BUN increase, calcium, oxygen tension, base deficit, and fluid sequestration. A score of three or more indicates severe pancreatitis with mortality increasing with score. The Ranson criteria require forty-eight hours to complete, making them less useful for early severity assessment, and the Bedside Index for Severity in Acute Pancreatitis (BISAP) score is increasingly preferred for early risk stratification.

Rectal Prolapse

– Protrusion of rectal tissue through the anal opening. Complete prolapse involves all layers of the rectal wall and presents as a concentric cylindrical mass with visible mucosal folds. Partial prolapse involves only the mucosa. Associated with chronic straining, pelvic floor dysfunction, and neurological conditions. Treatment is surgical with rectopexy being the preferred approach. Perineal approaches including Delorme procedure and Altemeier procedure are options for elderly or high-risk patients.

Rectovaginal Fistula

– Abnormal connection between the rectum and vagina causing passage of stool or gas through the vagina. Causes include obstetric injury particularly third and fourth degree perineal tears, Crohn disease, radiation therapy, and surgical complications. Presents with vaginal discharge of stool, flatus, or recurrent vaginal infections. Classified by location as low involving the anal sphincter or high above the sphincter. Treated surgically with approaches including transanal advancement flap, transperineal repair, or with gracilis muscle flap interposition for complex recurrent fistulae.

Reynolds Pentad

– Extension of the Charcot triad to include hypotension and altered mental status representing suppurative cholangitis, a severe life-threatening form of ascending cholangitis. Patients present with high fever, rigors, right upper quadrant pain, jaundice, hypotension, and confusion. Management involves aggressive fluid resuscitation, vasopressor support, broad-spectrum antibiotics, and emergent biliary decompression by ERCP. Without prompt treatment, suppurative cholangitis carries mortality exceeding 50 percent without prompt intervention.

Salmonellosis

– Infection with Salmonella bacteria (non-typhoidal), typically acquired from contaminated food (eggs, poultry, meat). Presents with acute gastroenteritis: diarrhea, fever, and abdominal cramps within 6-72 hours of ingestion. Self-limited in immunocompetent patients. Bacteremia can occur, especially in immunocompromised. Treatment: supportive care; antibiotics reserved for severe disease, bacteremia, or immunocompromised hosts.

Schatzki Ring

– Thin circumferential mucosal ring located at the squamocolumnar junction of the distal esophagus at the superior margin of a hiatal hernia. The ring is composed of mucosa and submucosa. May be asymptomatic or cause episodic dysphagia particularly to solid food and acute food bolus impaction. Diagnosis is by barium swallow showing a thin circumferential narrowing at the gastroesophageal junction. Upper endoscopy may miss the ring if the esophagus is overdistended with air. Treatment of symptomatic Schatzki rings is endoscopic dilation or disruption of the ring with biopsy forceps or electrocautery.

Serrated Pathway

– Alternative molecular pathway to colorectal carcinogenesis accounting for fifteen to thirty percent of cases. It involves serrated polyps as precursors including sessile serrated lesions with dysplasia. Molecular features include CpG island methylator phenotype, BRAF mutations, and microsatellite instability from MLH1 silencing. Sessile serrated lesions tend to be flat, sessile, mucus-covered, and located in the proximal colon making them difficult to detect. Recognition has led to updated colonoscopic screening and surveillance guidelines that emphasize careful inspection of the proximal colon.

Shigellosis

– Acute infectious colitis caused by Shigella species (S. sonnei most common in developed countries). Presents with fever, abdominal cramps, tenesmus, and frequent bloody, mucoid stools. Highly contagious with low infectious dose. Diagnosis: stool culture. Treatment: azithromycin, ciprofloxacin, or ceftriaxone (resistance patterns vary). Complications: reactive arthritis, hemolytic uremic syndrome (S. dysenteriae type 1).

Short Bowel Syndrome

– Clinical condition resulting from surgical resection or functional loss of a significant portion of the small intestine leading to inadequate absorption of nutrients. Clinical manifestations depend on the length and location of the remaining bowel, the presence or absence of the ileocecal valve, and the adaptability of the remaining intestine. Patients typically experience profuse diarrhea, dehydration, electrolyte imbalances, and progressive malnutrition. Those losing the terminal ileum lack vitamin B12 absorption, requiring lifelong parenteral supplementation, while those with an intact colon can absorb short-chain fatty acids and salvage some calories and fluid.

Small Bowel Obstruction

– Mechanical blockage of the small intestine most commonly caused by adhesions from prior surgery accounting for sixty to seventy-five percent, followed by hernias and malignancy. Clinical presentation includes colicky abdominal pain, bilious vomiting, distension, and obstipation. Diagnosis is established by abdominal CT demonstrating the transition point. Management involves nasogastric decompression, intravenous fluids, and monitoring for ischemia. Complete obstruction with strangulation require emergency laparotomy with resection of non-viable bowel.

Somatostatinoma

– Rare pancreatic or duodenal tumor producing somatostatin causing diabetes, gallstones, steatorrhea, and hypochlorhydria from suppression of insulin, gallbladder contraction, pancreatic enzymes, and gastric acid respectively. Diagnosed by elevated somatostatin levels. Treatment is surgical resection for localized disease. Somatostatin analogs may control symptoms but do not treat the underlying tumor. The prognosis depends on tumor stage and completeness of surgical resection.

Spontaneous Bacterial Peritonitis

– Infection of ascitic fluid without an identifiable intra-abdominal source representing a serious complication of cirrhosis with ascites. The pathogenesis involves impaired immune function, bacterial translocation from the intestinal lumen, and seeding of ascitic fluid which provides a favorable medium for bacterial growth. The most common causative organism is Escherichia coli. Patients present with fever, abdominal pain, and altered mental status, though symptoms may be absent in thirty percent of cases, necessitating a high index of suspicion and a low threshold for diagnostic paracentesis.

Stomach Ulcer

– See Gastric Ulcer.

Tenesmus

– A sensation of incomplete defecation, characterized by a constant feeling of needing to pass stool accompanied by straining, pain, and cramping, despite little or no stool being passed. Tenesmus is a symptom, not a diagnosis. Causes: inflammatory bowel disease (ulcerative colitis—most common, Crohn disease), infectious colitis (Shigella, Salmonella, Campylobacter, Clostridioides difficile, amoebic dysentery), radiation proctitis (post-pelvic radiotherapy), colorectal cancer (especially rectal cancer—should be excluded), anorectal conditions (hemorrhoids, anal fissure, solitary rectal ulcer syndrome, proctalgia fugax), and irritable bowel syndrome. Pathophysiology: inflammation, infiltration, or irritation of the rectal mucosa and nerve endings (S2–S4, pelvic splanchnic nerves) triggers the urge to defecate. Assessment: proctoscopy/sigmoidoscopy or colonoscopy with biopsy, stool culture, and imaging (CT, MRI) if tumor suspected. Treatment: directed at the underlying cause—mesalazine, steroids, and biologic therapy for IBD; antibiotics for infectious colitis; surgery for malignancy; topical anaesthetics, biofeedback, and amitriptyline for functional disorders.

Toxic Megacolon

– Acute life-threatening dilation of the colon associated with systemic toxicity, representing one of the most serious complications of inflammatory bowel disease, particularly ulcerative colitis. It can also occur in infectious colitis caused by Clostridioides difficile and cytomegalovirus. The pathophysiology involves transmural inflammation extending beyond the mucosa, causing destruction of the myenteric plexus and smooth muscle dysfunction leading to loss of colonic tone. The diagnosis requires radiologic evidence of colonic dilation exceeding 6 cm in diameter, with loss of haustrations, accompanied by clinical signs of toxicity including fever, tachycardia, leukocytosis, and systemic inflammatory response.

Tropical Sprue

– Chronic diarrhea and malabsorption syndrome of uncertain etiology occurring predominantly in tropical and subtropical regions. The condition is characterized by villous atrophy of the small intestinal mucosa leading to malabsorption of nutrients including folate, iron, and vitamin B12. The exact pathogenesis remains unclear but may involve an infectious etiology, altered gut microbiota, or nutritional deficiencies. The clinical presentation includes chronic diarrhea, steatorrhea, weight loss, and anemia. Treatment includes a prolonged course of tetracycline or doxycycline with folic acid supplementation, and the condition responds well to antibiotic therapy if diagnosed early.

Typhlitis

– Inflammation of the cecum in immunocompromised patients particularly those with neutropenia from chemotherapy. Also known as neutropenic enterocolitis. Presents with right lower quadrant pain, fever, and diarrhea. Diagnosis is by CT showing cecal wall thickening. Managed with bowel rest, broad-spectrum antibiotics covering gram-negative and anaerobic organisms, and resolution of neutropenia. Surgery is reserved for complications including perforation and hemorrhage.

Typhoid Fever

– A systemic infection caused by Salmonella typhi, acquired through contaminated food or water. Incubation 7-14 days. Presents with prolonged fever, headache, abdominal pain, rose spots, and relative bradycardia. Diagnosis: blood culture (first week), stool culture (second week), Widal test. Treatment: azithromycin, ceftriaxone, or fluoroquinolones. Complications: intestinal perforation, hemorrhage. Prevention: vaccination and water sanitation.

Ulcerative Colitis

– Chronic inflammatory disease limited to the colon and rectum, characterized by continuous mucosal and submucosal inflammation that invariably begins in the rectum and extends proximally in an uninterrupted pattern. The distribution follows a predictable anatomical pattern classified as proctitis when confined to the rectum, left-sided colitis extending to the splenic flexure, or extensive colitis extending beyond the splenic flexure. Unlike Crohn disease, the inflammation in ulcerative colitis is confined to the mucosa and submucosa, without transmural involvement.

Variceal Hemorrhage

– Life-threatening complication of portal hypertension involving rupture of dilated esophageal or gastric varices. The risk is determined by variceal size, red color signs, liver disease severity, and portal hypertension degree. Acute management involves hemodynamic resuscitation, vasoactive drugs, prophylactic antibiotics, and emergent endoscopic band ligation. Transjugular intrahepatic portosystemic shunt is indicated for refractory bleeding. Prevention involves screening endoscopy and prophylactic non-selective beta-blockers or endoscopic variceal ligation for primary prophylaxis, depending on variceal size and patient tolerance.

Volvulus

– Twisting of a loop of intestine around its mesenteric attachment point causing luminal obstruction and compromise of the mesenteric blood supply which can rapidly progress to bowel ischemia. The two most common sites are the sigmoid colon accounting for approximately seventy-five percent of colonic volvulus in elderly patients with chronic constipation and redundant sigmoid, and the cecum accounting for approximately twenty-five percent in younger patients with a mobile cecum. Sigmoid volvulus presents with abdominal distension, pain, and obstipation, and is initially managed with endoscopic detorsion using a sigmoidoscope, followed by elective sigmoid resection to prevent recurrence.

Whipple Disease

– A rare systemic bacterial infection caused by Tropheryma whipplei, affecting the small intestine and multiple organ systems. Presents with arthralgias (prodromal, years before GI symptoms), weight loss, chronic diarrhea, steatorrhea, and abdominal pain. Extraintestinal: endocarditis, neurologic symptoms (dementia, ophthalmoplegia, myoclonus), and lymphadenopathy. Diagnosis: small bowel biopsy showing PAS-positive macrophages containing rod-shaped bacteria, PCR of tissue or CSF. Treatment: prolonged antibiotic therapy (ceftriaxone IV induction, followed by TMP-SMX or doxycycline + hydroxychloroquine for 1-2 years). Untreated is fatal.

Whipple Procedure

– Pancreaticoduodenectomy for cancers of the pancreatic head, distal bile duct, or ampulla. Involves removal of the pancreatic head, duodenum, distal bile duct, and gallbladder with three anastomoses: hepaticojejunostomy, pancreaticojejunostomy, and gastrojejunostomy. The pylorus-preserving variant maintains the entire stomach. Mortality at high-volume centers is approximately two to five percent. Complications include pancreatic fistula, delayed gastric emptying, and hemorrhage. Adjuvant chemotherapy with gemcitabine or 5-fluorouracil-based regimens is given after resection to improve survival, but the prognosis remains poor.

Wireless Motility Capsule

– Ingestible capsule that measures pH, pressure, and temperature throughout the gastrointestinal tract. Provides regional transit times through the stomach, small bowel, and colon in a single test. The pH sensor identifies transition points between segments based on characteristic pH changes. Useful for evaluating gastric emptying, small bowel transit, and colonic transit in patients with suspected motility disorders. Non-invasive alternative to scintigraphy with the advantage of providing whole gut transit assessment in a single non-invasive study.

Zenker Diverticulum

– Pulsion diverticulum at the junction of the thyropharyngeus and cricopharyngeus muscles in the proximal esophagus. It presents with dysphagia, regurgitation of undigested food, halitosis, and aspiration. Diagnosis is by barium swallow showing the characteristic outpouching. Treatment is surgical with diverticulectomy or cricopharyngeal myotomy. The endoscopic approach with cricopharyngeal myotomy is increasingly used. Complications include aspiration pneumonia and, rarely, perforation.

Zieve's Syndrome

– A triad of haemolytic anemia, hyperlipidaemia, and jaundice occurring in the setting of chronic alcohol use disorder and alcoholic liver disease (acute alcoholic steatohepatitis or alcoholic cirrhosis). First described by Leslie Zieve in 1958 in patients with alcoholic liver disease presenting with transient haemolytic anemia (Coombs-negative, likely due to oxidative damage to erythrocyte membranes from acetaldehyde and lipid peroxidation products), markedly elevated serum triglycerides (hyperlipidaemia, often severe, >1000 mg/dL, due to alcohol-induced increase in hepatic VLDL secretion and reduced lipoprotein lipase activity), and jaundice (from alcoholic hepatitis, intrahepatic cholestasis, and haemolysis). The syndrome is typically self-limited, resolving with alcohol abstinence over 2-4 weeks, but recurrence is common if alcohol consumption resumes. Management: supportive care, alcohol cessation, and management of hypertriglyceridaemia (dietary fat restriction, fibrates) to prevent acute pancreatitis.

Zollinger-Ellison Syndrome

– Gastrin-secreting tumor called gastrinoma causing gastric acid hypersecretion, severe peptic ulcers, and diarrhea. Most tumors are located in the pancreas or duodenum with approximately sixty percent being malignant. Presents with multiple or refractory peptic ulcers, diarrhea, and esophageal reflux. Diagnosed by elevated fasting gastrin levels greater than one thousand picograms per milliliter and positive secretin stimulation test. Treatment involves high-dose proton pump inhibitors and surgical resection for localized gastrinoma, with a cure rate of approximately 30 percent.

Chapter 11

Genetics

Allelic Heterogeneity

– Phenomenon where different mutations at the same gene locus cause the same disease, often with variable expressivity or severity. For example, over 2,000 CFTR mutations cause cystic fibrosis with phenotypes ranging from mild (isolated congenital absence of the vas deferens) to severe (classic multisystem CF).

Chromosomal Abnormality

– Any deviation from the normal number (46 chromosomes, 23 pairs) or structure of chromosomes. Numerical abnormalities: aneuploidy (trisomy 21/Down syndrome, trisomy 18/Edwards syndrome, trisomy 13/Patau syndrome, monosomy X/Turner syndrome), polyploidy (triploidy, tetraploidy). Structural abnormalities: deletions (Cri-du-chat, 22q11.2 deletion), duplications, translocations (Robertsonian, reciprocal), inversions, and ring chromosomes. Mosaicism: presence of two or more genetically different cell lines in the same individual. Screening: combined test (nuchal translucency, hCG, PAPP-A), NIPT (cell-free fetal DNA), quadruple test. Diagnosis: chorionic villus sampling (11-14 weeks), amniocentesis (15+ weeks). Karyotype, FISH, or chromosomal microarray confirms the diagnosis.

Chromosomes

– The thread-like structures within the cell nucleus that carry genetic information in the form of DNA. Humans have 46 chromosomes arranged in 23 pairs: 22 pairs of autosomes and 1 pair of sex chromosomes (XX in females, XY in males). Each parent contributes one chromosome to each pair. Chromosomes are visible under a light microscope during cell division and are arranged in a standardized format called a karyotype (ordered by size, centromere position, and banding pattern, using G-banding or spectral karyotyping). The short arm is designated p (petit) and the long arm is q. Chromosomal abnormalities can involve number (aneuploidy) or structure (deletions, duplications, translocations, inversions). Telomeres: protective caps at chromosome ends. Centromere: region where sister chromatids are joined and spindle fibers attach during cell division.

Cystic Fibrosis

– An autosomal recessive multisystem disorder caused by mutations in the CFTR gene on chromosome 7q31.2, encoding the cystic fibrosis transmembrane conductance regulator, a cAMP-regulated chloride channel expressed on epithelial surfaces. Over 2,000 CFTR mutations have been identified; the most common is F508del (deletion of phenylalanine at position 508), accounting for approximately 70 percent of alleles worldwide. CFTR mutations are classified by defect type: Class I (no protein production – nonsense, frameshift); Class II (defective trafficking and degradation – F508del); Class III (defective gating – G551D); Class IV (defective chloride conductance); Class V (reduced synthesis); Class VI (accelerated turnover). Clinical features: chronic suppurative lung disease (bronchiectasis, Pseudomonas aeruginosa, progressive obstructive airways disease), exocrine pancreatic insufficiency (steatorrhea, fat-soluble vitamin deficiency), meconium ileus in neonates, congenital bilateral absence of the vas deferens (male infertility), and elevated sweat chloride (>60 mmol/L, measured by pilocarpine iontophoresis). Management: chest physiotherapy, mucolytics (dornase alfa), inhaled antibiotics (tobramycin), pancreatic enzyme replacement, fat-soluble vitamin supplementation, and CFTR modulator therapy (ivacaftor for Class III; lumacaftor/ivacaftor and tezacaftor/ivacaftor for

Class II; elexacaftor/tezacaftor/ivacaftor triple therapy). Lung transplantation in end-stage disease.

Friedreich Ataxia

– The most common inherited ataxia, an autosomal recessive disorder caused by GAA trinucleotide repeat expansion in the FXN gene on chromosome 9q13, encoding frataxin, a mitochondrial protein essential for iron-sulfur cluster biogenesis. The repeat expansion (typically 66-1,700 repeats; normal is 5-30) causes gene silencing through heterochromatin formation, leading to frataxin deficiency (20-30 percent of normal levels). Frataxin deficiency impairs mitochondrial iron metabolism, causing iron accumulation within mitochondria, defective iron-sulfur cluster assembly, impaired oxidative phosphorylation, and increased oxidative stress. Clinical triad: progressive ataxia (cerebellar and sensory), hypertrophic cardiomyopathy (most common cause of premature death), and diabetes mellitus or glucose intolerance. Additional features: dysarthria, lower limb areflexia, loss of vibration and proprioception (dorsal columns), pyramidal weakness, pes cavus, scoliosis, and optic atrophy. Age of onset typically 5-25 years; earlier onset correlates with larger repeat expansions. Diagnosis: genetic testing for GAA expansion in the FXN gene. Management is supportive: physical therapy, speech therapy, cardiac monitoring (annual echocardiogram, ECG), diabetes screening. No disease-modifying therapy available; omaveloxolone approved in some populations. Prognosis: loss of ambulation 10-20 years after onset; death usually from cardiomyopathy.

Gamete

– A mature haploid reproductive cell (23 chromosomes) that fuses with another gamete during fertilization to form a diploid zygote. Male gamete: spermatozoon (flagellated, motile, produced by spermatogenesis in testes, approximately 100 million per day). Female gamete: oocyte (ovum, non-motile, larger, produced by oogenesis in ovaries; approximately 1 million present at birth, reduced to 400-500 ovulated in a lifetime). Gametogenesis involves meiosis (reduction division) to achieve haploidy. Abnormal gametes may occur from meiotic non-disjunction, leading to aneuploidy in offspring (Down syndrome, Turner syndrome, Klinefelter syndrome).

Gaucher Disease

– Most common lysosomal storage disorder caused by glucocerebrosidase deficiency. Results in accumulation of glucocerebroside in macrophages. Type 1 is the most common and affects the liver, spleen, and bones.

Gene

– The fundamental physical and functional unit of heredity. A gene is a segment of DNA (or RNA in some viruses) that encodes a functional product (typically a protein or, in the case of non-coding RNA genes, a functional RNA molecule). The human genome contains 20,000-25,000 protein-coding genes. Structure: promoter region (regulatory sequence upstream, binds RNA polymerase and transcription factors), exons (coding sequences), introns (non-coding intervening sequences, spliced out), and untranslated regions (UTRs, 5 and 3 prime). Gene expression is regulated at multiple levels: transcriptional (transcription factors, enhancers, silencers, epigenetic modifications), post-transcriptional (splicing, RNA editing, mRNA stability), translational (initiation factors, miRNA, ribosome binding), and post-translational (phosphorylation, ubiquitination, glycosylation). Gene nomenclature: official symbols assigned by HGNC (HUGO Gene Nomenclature Committee), e.g., CFTR, BRCA1, TP53.

Genetic Code

– The set of rules by which information encoded in nucleotide sequences (DNA and RNA) is translated into proteins. The code is: (1) Triplet: three consecutive nucleotides (codon) specify one amino acid. (2) Degenerate: 64 possible codons encode 20 amino acids and 3 stop codons (UAA, UAG, UGA). (3) Universal: nearly identical across all organisms. (4) Non-overlapping: each nucleotide belongs to only one codon. (5) Commaless: codons are read sequentially without gaps. Key features: start codon (AUG, methionine, also sets reading frame), wobble hypothesis (third base pairing is flexible, allowing fewer tRNAs to decode all codons).

Genetic Engineering

– The direct manipulation of an organism's genes using biotechnology. Techniques include: (1) Recombinant DNA technology: insertion of foreign DNA into a vector (plasmid, virus) for expression in a host organism (bacteria, yeast, mammalian cells). (2) CRISPR-Cas9: RNA-guided nuclease that cuts DNA at a specific sequence, enabling gene knockout, knock-in, and precise editing. (3) Gene silencing: RNA interference (siRNA, shRNA) to reduce gene expression. (4) Gene therapy: delivery of functional genes to treat genetic disorders. Applications: production of therapeutic proteins (insulin, growth hormone, clotting factors, monoclonal antibodies), genetically modified organisms (GMOs, crop improvement), gene therapy (SCID, hemophilia, sickle cell disease, Leber congenital amaurosis), synthetic biology, and xenotransplantation. Ethical and safety concerns: off-target effects, germline editing consequences, ecological impact of GMOs.

Genetics

– The branch of biology concerned with the study of genes, genetic variation, and heredity in living organisms. Major subdisciplines: (1) Mendelian genetics: patterns of inheritance of single-gene traits (autosomal dominant, autosomal recessive, X-linked, Y-linked, mitochondrial). (2) Molecular genetics: structure and function of DNA, RNA, and proteins at the molecular level. (3) Population genetics: allele frequency distribution and change under the influence of natural selection, genetic drift, mutation, and gene flow (Hardy-Weinberg equilibrium). (4) Quantitative genetics: inheritance of complex traits influenced by multiple genes and environment (heritability estimates). (5) Genomics: the comprehensive study of entire genomes (structural, functional, comparative, epigenomics). (6) Clinical genetics: diagnosis and management of genetic disorders, genetic counseling, prenatal diagnosis, and carrier screening.

Genome

– The complete set of genetic material (DNA) of an organism. Human genome: approximately 3.2 billion base pairs, 20,000-25,000 protein-coding genes (only 1-2% of total DNA), organized into 23 pairs of chromosomes (22 autosomes + sex chromosomes). Non-coding DNA (98%) includes: regulatory sequences (promoters, enhancers), transposable elements, introns, telomeres, centromeres, pseudogenes, microRNAs, long non-coding RNAs, and repetitive sequences. The Human Genome Project (1990-2003) sequenced the human genome. Genomic medicine uses genome sequencing for diagnosis, pharmacogenomics, and prediction of disease risk.

Genotype

– The genetic constitution of an organism, determined by the specific alleles present at a given genetic locus. The genotype is the inherited genetic information carried by an individual, as distinct from the phenotype (observable physical or biochemical characteristics). Genotype is influenced by genotype-environment interactions. In Mendelian genetics, a homozygous genotype has two identical alleles at a locus (e.g., AA or aa), while a heterozygous genotype has two different alleles (Aa). Genotype frequency in a population is determined by Hardy-Weinberg equilibrium. Genotyping methods: PCR, DNA sequencing, microarray, SNP analysis.

Hemochromatosis

– Hereditary hemochromatosis (HH) is an autosomal recessive disorder of iron metabolism caused most commonly by homozygosity for the C282Y mutation in the HFE gene on chromosome 6p21.3, leading to inappropriately low hepcidin production and excessive intestinal iron absorption (1.5-2 mg/day normally, increasing to 3-4 mg/day in HH). Iron accumulates progressively in parenchymal organs (liver, pancreas, heart, pituitary, joints, skin), causing tissue damage through oxidative stress (Fenton reaction generating hydroxyl radicals that damage lipids, proteins, and DNA). Clinical features: bronze or gray skin pigmentation (bronze diabetes), diabetes mellitus, cirrhosis (increased risk of hepatocellular carcinoma), dilated or restrictive cardiomyopathy, hypogonadotropic hypogonadism, arthritis (typically second and third metacarpophalangeal joints), and osteoporosis. Penetrance is incomplete; many homozygotes show only biochemical abnormalities without clinical disease. Diagnosis: elevated transferrin saturation (>45 percent) and serum ferritin; HFE genetic testing confirms; liver MRI or biopsy quantifies iron burden. Treatment: graded phlebotomy (500 mL weekly until ferritin <50 mcg/L, then maintenance every 2-4 months). Iron chelation (deferasirox, deferoxamine) for patients who cannot tolerate phlebotomy. First-degree relatives should be screened with iron studies and HFE genotyping.

Hemophilia A

– X-linked recessive bleeding disorder caused by deficiency of coagulation factor VIII. Severity correlates with factor activity level. Characterized by hemarthrosis, soft tissue bleeding, and prolonged aPTT.

Hemophilia B

– X-linked recessive bleeding disorder caused by deficiency of coagulation factor IX. Also known as Christmas disease. Clinically indistinguishable from hemophilia A. Diagnosed by factor IX activity assay.

Huntington Disease

– An autosomal dominant neurodegenerative disorder caused by an expanded CAG trinucleotide repeat in the HTT gene on chromosome 4p16.3, encoding huntingtin protein. Normal alleles have fewer than 26 CAG repeats; 27-35 repeats are intermediate (not pathogenic but may expand in offspring); 36-39 repeats show reduced penetrance; 40 or more repeats cause full penetrance disease. The expanded polyglutamine tract causes mutant huntingtin to misfold, aggregate, and disrupt cellular processes (transcription, mitochondrial function, axonal transport, synaptic transmission), leading to neuronal death (most prominently in the caudate nucleus, putamen, and cerebral cortex). Clinical triad: chorea (involuntary, dance-like movements; the most characteristic motor feature), progressive dementia (executive dysfunction, memory loss, visuospatial deficits), and psychiatric disturbance (depression (most common), anxiety, irritability, obsessive-compulsive symptoms, psychosis). Mean age of onset 35-44 years; juvenile-onset HD (CAG >60 repeats) presents with rigidity, bradykinesia, and seizures rather than chorea. Anticipation occurs – the CAG repeat expands during male meiosis (paternal transmission), resulting in earlier onset in successive generations. Diagnosis: genetic testing for CAG repeat expansion in the HTT gene (with pre-test and post-test genetic counseling). Treatment is symptomatic: tetrabenazine or deutetrabenazine for chorea; antidepressants; antipsychotics. No disease-modifying therapy exists. Prognosis: mean survival 15-20 years from onset.

Li-Fraumeni Syndrome

– Autosomal dominant cancer predisposition syndrome most commonly caused by TP53 mutations. Characterized by early-onset cancers including sarcomas, breast cancer, brain tumors, and adrenocortical carcinoma. Very high lifetime cancer risk.

Marfan Syndrome

– An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene on chromosome 15q21.1, encoding fibrillin-1, a major component of microfibrils that form the scaffold for elastin in connective tissue. Fibrillin-1 mutations also lead to increased transforming growth factor-beta (TGF-beta) signaling, contributing to the pathogenesis. The Ghent nosology (2010 revision) diagnoses Marfan syndrome when aortic root dilation (Z-score >=2) AND ectopia lentis are present; or aortic root dilation AND a pathogenic FBN1 mutation; or aortic root dilation AND a sufficient systemic score (points for wrist and thumb signs, pectus carinatum, hindfoot deformity, pneumothorax, dural ectasia, protrusio acetabuli, facial features, skin striae, myopia, mitral valve prolapse). Skeletal features: tall stature, arachnodactyly, pectus deformities, scoliosis, joint hypermobility. Cardiovascular: aortic root aneurysm (leading cause of mortality), aortic regurgitation, mitral valve prolapse, aortic dissection (type A). Ocular: ectopia lentis (upward lens dislocation), myopia, retinal detachment. Diagnosis: revised Ghent criteria; FBN1 genetic testing; serial echocardiography for aortic root measurement; slit-lamp ophthalmology. Management: beta-blockers (atenolol) and losartan to slow aortic root growth; serial echocardiography; annual ophthalmology review; avoidance of contact sports and heavy exertion. Elective prophylactic aortic root replacement (Bentall procedure) when aortic root diameter reaches 50 mm (or 45 mm with rapid progression or family history of dissection).

Phenylketonuria

– An autosomal recessive aminoacidopathy caused by deficiency of phenylalanine hydroxylase (PAH), the enzyme that converts phenylalanine to tyrosine. PAH deficiency leads to toxic accumulation of phenylalanine and its metabolites (phenylketones: phenylacetate, phenyllactate, phenylpyruvate) in blood and brain, causing irreversible intellectual disability if untreated. PKU is caused by mutations in the PAH gene on chromosome 12q23.2, with over 1,000 mutations identified. Classic PKU: blood phenylalanine >1,200 micromol/L (20 mg/dL) on normal diet; mild PKU: 600-1,200 micromol/L; mild hyperphenylalaninaemia: 120-600 micromol/L (no treatment required). Newborn screening by tandem mass spectrometry (Guthrie test) detects elevated phenylalanine levels in all developed countries; prompt initiation of dietary therapy prevents neurological damage. Treatment: lifelong low-phenylalanine diet (phenylalanine-free medical formula and restricted natural protein); tetrahydrobiopterin (BH4, sapropterin) response in some mutation types; pegvaliase (PEGylated recombinant phenylalanine ammonia lyase) for adults with poorly controlled PKU. Maternal PKU: women with PKU must maintain strict phenylalanine control before conception and throughout pregnancy to prevent fetal damage (microcephaly, intellectual disability, congenital heart disease, low birth weight).

Sickle Cell Disease

– A group of inherited hemoglobinopathies caused by a point mutation (glutamic acid to valine at position 6) in the beta-globin gene (HBB) on chromosome 11, producing hemoglobin S (HbS). Homozygous HbSS is the most common and severe form. Under conditions of deoxygenation, HbS polymerizes, causing red blood cells to assume a rigid sickle shape. Sickled cells are less deformable, leading to vaso-occlusion, microvascular ischemia, and hemolytic anemia. SCD manifests through acute and chronic complications: vaso-occlusive crises (bone pain, dactylitis, priapism), acute chest syndrome (fever, chest pain, pulmonary infiltrates, hypoxia), stroke (overt or silent cerebral infarction), bacterial infection risk (functional asplenia, encapsulated organisms – Streptococcus pneumoniae, Haemophilus influenzae), aplastic crisis (parvovirus B19-induced temporary arrest of erythropoiesis), splenic sequestration, pulmonary hypertension, leg ulcers, avascular necrosis of femoral head, and chronic kidney disease. Diagnosis: hemoglobin electrophoresis or high-performance liquid chromatography (HPLC) showing HbS (typically >80 percent in HbSS). Newborn screening (isoelectric focusing) universal in high-prevalence regions. Treatment: hydroxyurea (increases fetal hemoglobin HbF, reducing sickling); regular blood transfusion for stroke prevention; penicillin prophylaxis and pneumococcal vaccination (functional asplenia); pain management; acute chest syndrome management (antibiotics, transfusion, bronchodilators). hematopoietic stem cell transplantation is curative; gene therapy (LentiGlobin, CRISPR-based approaches) is emerging. Prognosis: mean survival 40-50 years in high-income countries.

Sporadic

– Occurring in isolated, random, or scattered instances, without a predictable pattern or clear cause. In medical genetics, sporadic cases of a disease occur in individuals with no family history of the condition, resulting from de novo (new) mutations rather than inherited germline mutations. Sporadic cancers arise from acquired (somatic) mutations. Examples: most cases of breast, ovarian, and colorectal cancers are sporadic. Many genetic syndromes can occur sporadically through de novo mutations (e.g., neurofibromatosis type 1, CHARGE syndrome, Cornelia de Lange syndrome). Sporadic cases have a low recurrence risk in siblings unless gonadal mosaicism is present.

Tay-Sachs Disease

– Autosomal recessive lysosomal storage disorder caused by hexosaminidase A deficiency. Results in accumulation of GM2 ganglioside in neurons causing progressive neurodegeneration. Most common in Ashkenazi Jewish populations.

Teratogenesis

– The process by which congenital malformations (birth defects) are produced in a developing fetus by exposure to teratogens—agents that interfere with normal embryonic or fetal development. Teratogenesis is most sensitive during the period of organogenesis (weeks 3–8 of gestation), when major organs are forming. Principles: (1) Susceptibility depends on the genotype of the embryo and the timing, dose, and duration of exposure. (2) Different teratogens produce characteristic patterns of malformations. (3) The critical period for each organ system is when it is undergoing rapid differentiation. Major teratogens: (1) Medications—thalidomide (limb defects—phocomelia, amelia), isotretinoin (CNS, cardiac, craniofacial defects), valproic acid (neural tube defects, learning difficulties), warfarin (nasal hypoplasia, stippled epiphyses), methotrexate (CNS, skeletal), ACE inhibitors (renal dysplasia, oligohydramnios), tetracycline (bone and tooth discoloration, bone growth suppression), and androgens/lithium/carbamazepine/phenytoin. (2) Maternal infections—rubella (deafness, cataracts, congenital heart disease), cytomegalovirus, toxoplasmosis, varicella, Zika virus. (3) Maternal conditions—diabetes (neural tube defects, congenital heart disease, macrosomia), phenylketonuria (microcephaly, intellectual disability). (4) Environmental exposures—alcohol (fetal alcohol syndrome—facial dysmorphism, growth restriction, neurodevelopmental impairment), ionizing radiation (microcephaly, intellectual disability), methylmercury (cerebral palsy, blindness), tobacco smoke (IUGR, cleft palate, preterm birth). Prevention: preconception counseling, folic acid supplementation (4–5 mg/day in high-risk women—reduces neural tube defects by 70%), and avoidance of known teratogens during pregnancy.

Thalassemia

– Group of autosomal recessive disorders characterized by reduced or absent globin chain synthesis. Alpha thalassemia results from mutations in the HBA genes. Beta thalassemia results from mutations in the HBB gene.

Trisomy

– The presence of an extra chromosome (three copies instead of the normal pair), resulting in a total of 47 chromosomes. Trisomy is the most common type of aneuploidy and usually arises from non-disjunction during meiosis (failure of homologous chromosomes or sister chromatids to separate). Trisomies involving autosomes (non-sex chromosomes) are generally incompatible with life, except for trisomy 21, 18, and 13. Trisomy 21 (Down syndrome): most common viable autosomal trisomy (incidence 1 in 700 live births), associated with intellectual disability, congenital heart disease (AVSD, VSD, ASD), hypotonia, characteristic facial features (flat facies, upslanting palpebral fissures, epicanthal folds, protruding tongue), increased risk of leukemia (ALL), Alzheimer disease (by age 40), and atlantoaxial instability. Trisomy 18 (Edwards syndrome): incidence 1 in 5,000, severe intellectual disability, rocker-bottom feet, clenched hands with overlapping fingers, congenital heart disease, and omphalocele; median survival <1 year. Trisomy 13 (Patau syndrome): incidence 1 in 15,000, severe intellectual disability, cleft lip/palate, polydactyly, holoprosencephaly, and congenital heart disease; median survival 7–10 days. Sex chromosome trisomies: XXY (Klinefelter syndrome), XXX (triple X syndrome), XYY (Jacob syndrome)—generally milder phenotypes. Prenatal screening: combined test (nuchal translucency, β-hCG, PAPP-A) at 11–13 weeks, NIPT (cell-free fetal DNA), and diagnostic testing (CVS or amniocentesis).

Wilson Disease

– An autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene on chromosome 13q14.3, encoding a copper-transporting ATPase. The defective ATP7B protein cannot incorporate copper into ceruloplasmin or excrete excess copper into bile, leading to toxic copper accumulation primarily in the liver and brain (basal ganglia, particularly the putamen and globus pallidus), with secondary deposition in the cornea (Kayser-Fleischer rings – golden-brown copper deposits in Descemet's membrane at the limbus, visible on slit-lamp examination, pathognomonic for Wilson disease). Hepatic presentation: acute hepatitis, fulminant hepatic failure (with Coombs-negative hemolytic anemia), chronic hepatitis, cirrhosis, and portal hypertension. Neuropsychiatric presentation: movement disorders (tremor, dysarthria, dystonia, chorea, parkinsonism), dysphagia, personality changes, depression, psychosis, and cognitive decline. Other features: renal tubular dysfunction (Fanconi syndrome), arthropathy, and cardiomyopathy. Diagnosis: reduced serum ceruloplasmin (<0.2 g/L), elevated 24-hour urinary copper excretion (>1.6 micromol/24h), elevated hepatic copper on liver biopsy (>250 mcg/g dry weight); Kayser-Fleischer rings on slit-lamp examination; ATP7B genetic testing. The Leipzig diagnostic scoring system aids diagnosis. Treatment: lifelong copper chelation (D-penicillamine as first-line, trientine as second-line) and zinc acetate (blocks intestinal copper absorption, used as maintenance or for pre-symptomatic patients). Avoid copper-rich foods (shellfish, liver, nuts, chocolate). Liver transplantation for fulminant hepatic failure or decompensated cirrhosis. Prognosis: excellent with lifelong treatment; progressive and fatal without treatment.

Chapter 12

Hematology

ABO Incompatibility

– A condition occurring when ABO blood group antigens on red blood cells are mismatched between a mother and fetus, leading to hemolytic disease of the fetus and newborn (HDFN). ABO incompatibility is the most common cause of HDFN, occurring more frequently than Rh incompatibility but generally causing milder disease. It most commonly affects neonates of blood group A or B born to group O mothers, because group O individuals produce IgG anti-A and anti-B antibodies that can cross the placenta. Unlike Rh disease, ABO incompatibility frequently affects first pregnancies and rarely causes severe hemolysis requiring exchange transfusion.

Acute leukemia

– leukemia characterized by a rapid onset, clonal proliferation of immature blast cells (>20% of bone marrow cells), and rapid progression without treatment. Subtypes: acute lymphoblastic leukemia (ALL, B-cell or T-cell, most common childhood cancer, peak age 2-5) and acute myeloid leukemia (AML, more common in adults, median age 65-70). Clinical features: bone marrow failure (anemia–fatigue, pallor; thrombocytopenia–bruising, petechiae, bleeding; neutropenia–fever, recurrent infections), organ infiltration (lymphadenopathy, hepatosplenomegaly, bone pain, gum hypertrophy in AML M4/M5, skin infiltration). Emergency complications: hyperleukocytosis (blast count >100,000, risk of leukostasis, requiring urgent leukapheresis), tumor lysis syndrome, disseminated intravascular coagulation (especially in AML M3/promyelocytic). Diagnosis: full blood count and blood film, bone marrow aspirate and trephine biopsy, flow cytometry (immunophenotyping), cytogenetics (karyotype, FISH), and molecular genetics (NPM1, FLT3-ITD, CEBPA, RUNX1, BCR-ABL for ALL, PML-RARA for APL).

Acute Lymphoblastic Leukemia

– A clonal malignancy of lymphoid progenitor cells, most common in children (peak age 2-5 years) but also occurring in adults with a worse prognosis with increasing age. B-cell ALL (85 percent of childhood, 75 percent of adult ALL) is classified by immunophenotype: early pre-B (CD19+, CD10-, TdT+), common ALL (CD19+, CD10+, TdT+), pre-B (CD19+, cytoplasmic mu heavy chain+), and mature B-cell (Burkitt-type, CD19+, surface immunoglobulin+, CD10+). T-cell ALL (15% of childhood, 25% of adult ALL) presents with mediastinal mass and has a poorer prognosis.

Acute Megakaryoblastic Leukemia

– A rare subtype of acute myeloid leukemia characterized by the proliferation of megakaryoblasts in the bone marrow and peripheral blood. It accounts for approximately 1-2 percent of AML in adults but is the most common AML subtype in children with Down syndrome (trisomy 21), where it occurs in 1-5 percent of cases (called transient abnormal myelopoiesis in neonates, which often resolves spontaneously). AMKL is associated with extensive bone marrow fibrosis and a poor prognosis.

Acute Promyelocytic Leukemia

– A distinct subtype of acute myeloid leukemia (AML) characterized by the t(15;17)(q24;q21) translocation, which fuses the promyelocytic leukemia (PML) gene on chromosome 15 with the retinoic acid receptor alpha (RARA) gene on chromosome 17. This PML-RARA fusion protein blocks myeloid differentiation at the promyelocyte stage, leading to accumulation of abnormal promyelocytes in the bone marrow and peripheral blood. APL accounts for approximately 5-10 percent of AML cases.

Anaemia

– A condition in which the number of red blood cells or the hemoglobin concentration is below the normal range, reducing the oxygen-carrying capacity of the blood. Classified by etiology: blood loss (acute or chronic), decreased production (iron deficiency, vitamin B12/folate deficiency, bone marrow failure, anemia of chronic disease), and increased destruction (hemolytic anemias). Classified by morphology: microcytic (low MCV — iron deficiency, thalassemia, anemia of chronic disease), normocytic (normal MCV — acute blood loss, hemolysis, renal failure), macrocytic (high MCV — B12/folate deficiency, alcohol, MDS, hypothyroidism). Symptoms: fatigue, pallor, dyspnea, dizziness, tachycardia. Severe anemia can cause heart failure. Diagnosis: CBC, reticulocyte count, iron studies, vitamin B12/folate, and peripheral smear. Treatment: address underlying cause, iron supplementation, B12/folate replacement, erythropoietin, and blood transfusion for severe symptomatic anemia.

Anemia

– A reduction in the oxygen-carrying capacity of the blood, defined by a hemoglobin concentration below the normal range for age and sex: less than 13.5 g/dL in adult males, less than 12.0 g/dL in adult females. Anemia is classified by mechanism into three categories: decreased production (iron deficiency, vitamin B12 deficiency, folate deficiency, chronic disease, aplastic anemia, myelodysplastic syndrome), increased destruction (hemolytic anemias including hereditary spherocytosis, sickle cell disease, paroxysmal nocturnal haemoglobinuria), or blood loss.

Antiphospholipid Syndrome

– A systemic autoimmune thrombophilia characterized by arterial and venous thrombosis, pregnancy morbidity, and persistent antiphospholipid antibody positivity. Antiphospholipid antibodies are a heterogeneous group of autoantibodies targeting phospholipid-binding proteins: lupus anticoagulant (LA, detected by prolongation of phospholipid-dependent coagulation assays that fails to correct with mixing study, paradoxically associated with thrombosis), anticardiolipin antibodies (aCL, IgG or IgM, detected by ELISA), and anti-beta-2 glycoprotein I antibodies.

Antithrombin Deficiency

– A rare (1 in 2,000-5,000), strongly thrombophilic inherited disorder (autosomal dominant). Antithrombin (previously antithrombin III) is a serine protease inhibitor that inactivates thrombin and factors Xa, IXa, XIa, and XIIa. Heterozygous: 10-50 fold increased VTE risk, more severe than factor V Leiden. Presents with DVT and PE at a young age (<30), recurrent thrombosis, and unusual sites (mesenteric, cerebral veins). Diagnosis: low antithrombin activity (type I: reduced antigen and activity; type II: reduced activity). Heparin resistance is a clue. Management: anticoagulation for acute VTE; may require high-dose heparin; lifelong anticoagulation. Prophylaxis in high-risk situations is essential.

Aplastic Anemia

– A bone marrow failure syndrome characterized by pancytopenia (anemia, leukopenia, thrombocytopenia) and a hypocellular bone marrow with replacement by fat cells, resulting from immune-mediated destruction of hematopoietic stem cells. Acquired aplastic anemia accounts for 80-90 percent of cases, with autoimmune T-cell-mediated destruction of CD34+ hematopoietic progenitors as the primary mechanism, supported by response to immunosuppressive therapy. Causes include idiopathic (60-70%), drugs, toxins, viruses, and pregnancy.

Bleeding

– The escape of blood from the vascular system, which may be external (visible hemorrhage from a wound, orifice, or natural opening) or internal (bleeding into a body cavity, organ, or tissue). Bleeding is classified by source: arterial bleeding (bright red, pulsatile, and difficult to control due to high pressure), venous bleeding (dark red, steady flow), and capillary bleeding (slow oozing from superficial wounds). The clinical significance of bleeding depends on the rate and volume of blood loss, as well as the location of bleeding, with hemorrhage in critical areas such as the intracranial cavity, pericardium, or airway being immediately life-threatening even with relatively small volumes of blood loss.

Bleeding Disorders

– A heterogeneous group of conditions characterized by abnormal hemostasis, resulting from defects in blood vessels, platelets, or coagulation factors. They are classified into three categories: vascular disorders (Ehlers-Danlos syndrome, Henoch-Schonlein purpura, hereditary hemorrhagic telangiectasia), platelet disorders (thrombocytopenia from immune destruction [immune thrombocytopenia], decreased production [aplastic anemia, chemotherapy], or sequestration [hypersplenism]; and platelet function disorders (von Willebrand disease, uraemia, aspirin, NSAIDs)), and coagulation disorders (hemophilia A and B, von Willebrand disease, factor deficiencies, vitamin K deficiency, liver disease, anticoagulant therapy, and DIC).

Blood Group

– The classification of blood based on the presence or absence of specific antigens (agglutinogens) on the surface of red blood cells. The ABO system has four groups: A (A antigen), B (B antigen), AB (both A and B), and O (neither A nor B). The Rh system: RhD-positive (85% of population) or RhD-negative (15%). Blood type compatibility is critical for transfusion: group O RhD-negative is the universal donor (no ABO/RhD antigens); group AB RhD-positive is the universal recipient (no ABO/RhD antibodies). In pregnancy, RhD-negative women carrying RhD-positive fetuses may develop anti-D antibodies, causing hemolytic disease of the newborn in subsequent pregnancies — prevented by antenatal and postnatal anti-D immunoglobulin (RhoGAM).

Blood Transfusion

– The administration of whole blood or blood components (packed red blood cells, platelets, fresh frozen plasma, cryoprecipitate) to restore or maintain adequate circulating blood volume and oxygen-carrying capacity. Indications: packed red blood cells for symptomatic anemia (hemoglobin less than 7-8 g/dL in stable patients; higher thresholds for patients with acute coronary syndrome, hemoglobin less than 10 g/dL, or ongoing hemorrhage); platelets for active bleeding with thrombocytopenia less than 50×10^9/L; FFP for coagulopathy with INR >1.5; cryoprecipitate for fibrinogen <100 mg/dL.

Burkitt Lymphoma

– A highly aggressive B-cell lymphoma characterized by MYC translocation (t(8;14) IGH-MYC, t(8;22) IGL-MYC, t(2;8) IGK-MYC) and a "starry sky" morphology (tingible body macrophages). Three variants: endemic (African, EBV+, jaw/facial bones), sporadic (abdominal, ileocecal), immunodeficiency-associated (HIV). Clinical: rapidly growing mass, LDH extremely high, TLS risk. Treatment: intensive short-course chemotherapy (CODOX-M/IVAC, hyper-CVAD, LMB) + rituximab + CNS prophylaxis with intrathecal methotrexate.

Chronic leukemia

– leukemia characterized by an insidious onset, clonal proliferation of relatively mature blood cells, and a more indolent clinical course. Subtypes: chronic lymphocytic leukemia (CLL, most common adult leukemia, median age 72, monoclonal B-cell proliferation CD5+, CD23+, often asymptomatic, managed with watchful waiting in early stage) and chronic myeloid leukemia (CML, characterized by Philadelphia chromosome t(9;22) BCR-ABL fusion, trilineage myeloproliferation, progresses through chronic phase to accelerated phase to blast crisis).

Chronic Lymphocytic Leukemia

– The most common leukemia in Western countries, characterized by the clonal proliferation of mature-appearing but functionally incompetent B lymphocytes that accumulate in blood, bone marrow, and lymphoid tissues. Diagnosis requires a monoclonal B-lymphocyte count of 5,000 per microliter or greater with a characteristic immunophenotype: CD5+, CD19+, CD23+, CD200+, with weak surface immunoglobulin expression and weak CD20. CLL has a highly variable clinical course, with some patients having indolent disease and others progressing rapidly.

Chronic Myeloid Leukemia

– A myeloproliferative neoplasm characterized by the Philadelphia chromosome [t(9;22)(q34;q11.2)], which creates the BCR-ABL1 fusion gene encoding a constitutively active tyrosine kinase. CML has three phases: chronic phase (most common at diagnosis, 85-90 percent, characterized by low blast count less than 10 percent, thrombocytosis, and basophilia), accelerated phase (increasing blasts 10-19 percent, basophilia greater than 20 percent, or additional cytogenetic abnormalities), and blast phase (>20% blasts, acute leukemia-like, poor prognosis).

Chronic Myelomonocytic Leukemia

– A rare myelodysplastic/myeloproliferative neoplasm characterized by persistent monocytosis (absolute monocyte count greater than 1,000 per microliter and greater than 10 percent of white blood cells) in the peripheral blood and bone marrow, with dysplastic features and variable blast count. CMML bridges the categories of MDS and myeloproliferative neoplasm, with clinical features of both: cytopenias and dysplasia (MDS features) plus splenomegaly and monocytosis, and increased risk of transformation to AML (15-30%).

Coagulation Factor Deficiencies

– Rare factor deficiencies: Factor VII (most common rare bleeding disorder, prolonged PT, normal aPTT, mucosal bleeding, hemarthrosis; recombinant FVIIa for treatment), Factor X (prolonged PT/aPTT), Factor XI (Ashkenazi Jews, prolonged aPTT, bleeding with surgery/trauma, FFP for replacement), Factor XIII (normal PT/aPTT, clot solubility test, umbilical stump bleeding, intracranial hemorrhage; FXIII concentrate), Fibrinogen (afibrinogenemia, hypofibrinogenemia, dysfibrinogenemia; cryoprecipitate, fibrinogen concentrate.

Direct Oral Anticoagulants

– Direct oral anticoagulants (DOACs) target specific coagulation factors: dabigatran (direct thrombin inhibitor, factor IIa), rivaroxaban, apixaban, edoxaban (direct factor Xa inhibitors). Indications: VTE treatment/prevention, stroke prevention in atrial fibrillation (non-valvular), post-orthopedic surgery prophylaxis. Advantages: fixed dosing, no routine monitoring, rapid onset/offset, fewer drug/food interactions (vs warfarin). Reversal: idarucizumab (anti-dabigatran monoclonal antibody), andexanet alfa (anti-Xa reversal). DOACs are preferred over warfarin for most indications.

Disseminated Intravascular Coagulation

– A pathological state of simultaneous systemic activation of coagulation and fibrinolysis, leading to widespread microvascular thrombosis that causes organ ischemia and dysfunction, paradoxically accompanied by consumptive coagulopathy and increased bleeding risk. DIC is triggered by conditions that expose tissue factor to the circulation: sepsis (most common trigger, especially gram-negative and fungal), trauma (tissue factor release), malignancy (particularly adenocarcinomas and acute promyelocytic leukemia),.

Dyscrasia

– An historic term broadly referring to any abnormal condition of the blood or bone marrow, now used primarily in the context of plasma cell dyscrasias (also called monoclonal gammopathies). These include multiple myeloma, monoclonal gammopathy of undetermined significance (MGUS), Waldenstrom macroglobulinemia, AL amyloidosis, and POEMS syndrome. The term indicates a disorder of blood cell production or function, often involving the overproduction of a monoclonal immunoglobulin (M protein) by a clonal plasma cell population. Diagnosis: serum protein electrophoresis, immunofixation, serum free light chain assay, and bone marrow biopsy.

Erythropoietin

– A glycoprotein hormone (EPO) that regulates erythropoiesis–the production of red blood cells in the bone marrow. Produced primarily by interstitial peritubular cells in the kidney (90%) and hepatocytes (10%) in response to tissue hypoxia. EPO binds to EPOR receptors on erythroid progenitor cells in the bone marrow, promoting their survival, proliferation, and differentiation into mature erythrocytes. Deficiency causes anemia of chronic kidney disease (the most clinically significant EPO deficiency). Recombinant human EPO (epoetin alfa, darbepoetin alfa) is used to treat anemia in CKD, chemotherapy-induced anemia, and before elective surgery to reduce transfusion needs. Adverse effects: hypertension, thrombosis (especially when hemoglobin targets are too high), and pure red cell aplasia (rare, antibody-mediated, associated with subcutaneous administration of certain epoetin formulations).

Essential Thrombocythemia

– A chronic myeloproliferative neoplasm characterized by sustained platelet count elevation (greater than 450,000 per microliter) due to clonal megakaryocytic proliferation, with a risk of both thrombotic and hemorrhagic complications. The JAK2 V617F mutation is present in 50-60 percent of cases, CALR mutations in 20-30 percent, and MPL mutations in 3-5 percent; approximately 10-15 percent are triple-negative. Clinical presentation ranges from asymptomatic (incidentally found on routine blood count) to thrombotic events (DVT, PE, microvascular occlusions including erythromelalgia and TIAs) and hemorrhagic complications (GI bleeding, epistaxis).

Factor V Leiden

– The most common inherited thrombophilia, caused by a point mutation in the factor V gene (G1691A, R506Q) resulting in resistance to activated protein C (APC resistance). Heterozygous: 5-fold increased VTE risk; homozygous: 50-80 fold increased risk. Most common in Caucasians (3-7% prevalence). Clinical significance: increased risk of DVT, PE, and recurrent pregnancy loss. Less clear association with arterial thrombosis. Management: anticoagulation for VTE (consider DOACs); asymptomatic individuals generally do not require prophylaxis except in high-risk situations (surgery, pregnancy, prolonged immobilization).

Folate Deficiency

– Folate (vitamin B9) deficiency causes megaloblastic anemia identical to B12 deficiency but without neurologic manifestations (except possible cognitive changes). Dietary sources: leafy greens, legumes, liver; daily requirement 400 mcg. Absorption in jejunum. Causes: inadequate intake (alcoholism, malnutrition, elderly), malabsorption (celiac, short bowel), increased demand (pregnancy, hemolysis, dialysis), drugs (methotrexate, trimethoprim, phenytoin, sulfasalazine). Lab: macrocytic anemia, hypersegmented neutrophils on peripheral smear, low folate levels.

Folate-deficiency anemia

– A type of megaloblastic anemia caused by insufficient folic acid, a B vitamin essential for DNA synthesis, cell division, and red blood cell production. Folate deficiency impairs thymidylate synthesis, causing defective DNA replication in rapidly dividing cells of the bone marrow, resulting in megaloblastic erythropoiesis with large, abnormally shaped erythrocyte precursors. Causes include inadequate dietary intake (most common in alcohol use disorder, older adults, fad diets, and anorexia), increased requirements (pregnancy, hemolysis, malignancy, chronic inflammatory disorders), and impaired absorption (short bowel, celiac disease, gastric bypass). Diagnosis confirmed by low serum folate and elevated homocysteine; methylmalonic acid is normal (distinguishes from B12 deficiency).

Follicular Lymphoma

– The most common indolent non-Hodgkin lymphoma, accounting for 20-25 percent of NHL cases, characterized by t(14;18) translocation leading to BCL2 overexpression and follicular growth pattern. Clinical presentation: painless lymphadenopathy, often widespread, with bone marrow involvement in 60-70 percent. Grading: 1-2 (follicular, indolent), 3A (centrocytes + centroblasts), 3B (centroblasts only, aggressive). FLIPI (Follicular Lymphoma International Prognostic Index) incorporates five adverse prognostic factors: age >60, serum LDH > normal, number of nodal sites >4, hemoglobin <12 g/dL, and Ann Arbor stage III-IV.

G6PD Deficiency

– Glucose-6-phosphate dehydrogenase (G6PD) deficiency is an X-linked enzymopathy affecting the pentose phosphate pathway, leading to impaired NADPH production and reduced glutathione regeneration, rendering erythrocytes susceptible to oxidative hemolysis. Common variants: African A- (mild), Mediterranean (severe), Mahidol (SE Asia). Triggers: fava beans, oxidant drugs (primaquine, dapsone, sulfonamides, rasburicase, methylene blue), infections. Clinical: acute hemolytic anemia (back pain, dark urine (haemoglobinuria), jaundice, anemia); self-limiting within 1-2 weeks of trigger removal.

Granulocyte

– A type of white blood cell (leukocyte) containing cytoplasmic granules that are visible on light microscopy. Granulocytes are produced in the bone marrow (myelopoiesis, 3-4 days maturation) and are part of the innate immune system. Three types: (1) Neutrophils (40-70% of WBCs): bilobed nucleus, pale pink granules, primary defence against bacteria and fungi, short half-life (6-8 hours). (2) Eosinophils (1-4%): bilobed nucleus, orange-red granules, defence against parasites and involvement in allergic inflammation. (3) Basophils (<1%): bilobed nucleus, large dark purple granules, release histamine and heparin in allergic reactions. Granulocyte count (total): 2.0–7.5 $\times$ 10^9/L. Neutropenia (ANC <1.5) increases infection risk; agranulocytosis (ANC <0.5) is a medical emergency.

Haemangioma

– A benign vascular tumor composed of proliferating endothelial cells, the most common soft tissue tumor of infancy. Haemangiomas present shortly after birth (not usually visible at birth), undergo a proliferative phase (first 6-12 months, rapid growth), followed by involution (slow regression over 5-10 years with replacement by fibrofatty tissue). 60% occur in the head and neck region. Most are superficial (strawberry naevus), but can be deep (blue-hued) or mixed. Complications: ulceration (10-20%, most common), bleeding, infection, visual axis obstruction (periocular), airway obstruction (subglottic), Kasabach-Merritt phenomenon (platelet trapping, consumptive coagulopathy). Treatment options: observation (majority involute spontaneously), topical timolol for superficial lesions, oral propranolol (2-3 mg/kg/day, first-line for problematic lesions), laser therapy for residual telangiectasia after involution, surgical excision for complications or residual deformity.

Haemarthrosis

– Bleeding into a joint cavity. Causes: trauma (most common), hemophilia (spontaneous haemarthrosis is characteristic, especially in severe deficiency of factor VIII or IX, common in knees, elbows, ankles), anticoagulant therapy, pigmented villonodular synovitis (PVNS, recurrent haemarthrosis), joint surgery, and vascular malformations. In hemophilia, recurrent haemarthrosis leads to synovitis, cartilage damage, joint space narrowing, and ultimately haemophilic arthropathy (chronic pain, stiffness, deformity, ankylosis). Management: factor replacement (aim for 50-100% activity), joint aspiration only if tense and painful (risk of infection), rest, ice, splinting, physiotherapy to prevent contractures.

Haematocrit

– The proportion of blood volume occupied by red blood cells (RBCs), expressed as a percentage or decimal fraction. Normal reference range: men 42-50%, women 37-47% (varies by altitude, age, and laboratory). Haematocrit is measured by centrifugation of whole blood in a microhaematocrit tube (PCV–packed cell volume) or calculated from mean corpuscular volume (MCV) and RBC count (Hct = MCV x RBC count / 10). Clinical interpretation: low haematocrit (anemia–decreased RBC production, increased destruction, or blood loss), high haematocrit (polycythemia–primary: polycythemia vera; secondary: chronic hypoxia, high altitude, COPD, cyanotic heart disease, EPO-secreting tumors, smoking). The haematocrit is used to assess the severity of anemia and to guide transfusion decisions (trigger typically <21-24% or <7-8 g/dL hemoglobin depending on clinical context).

Haematologist

– A medical doctor who specializes in the diagnosis, treatment, and prevention of diseases of the blood and blood-forming organs. Conditions managed: anemia, bleeding disorders (hemophilia, von Willebrand disease), thrombophilia, hematologic malignancies (leukemia, lymphoma, myeloma, MDS, MPN), and hemoglobinopathies (sickle cell disease, thalassemia). Hematologists interpret laboratory tests (CBC, coagulation studies, bone marrow biopsy, flow cytometry, cytogenetics, molecular testing) and administer treatments including chemotherapy, targeted therapy, immunotherapy, anticoagulation, clotting factor replacement, and stem cell transplantation.

Haematoma

– A localised collection of blood outside blood vessels, usually clotted, within an organ or tissue. Haematomas form when trauma or disease causes a blood vessel to rupture. They are classified by location: subcutaneous (ecchymosis/bruise), intramuscular, subdural, epidural, intracerebral, subungual (under nail), perianal, rectus sheath, psoas, septal (nasal), subchorionic/retroplacental (pregnancy), and retroperitoneal. A haematoma may expand (especially in anticoagulated patients) and can cause complications: compression of adjacent structures (nerve compression, compartment syndrome), infection if it becomes secondarily infected, organisation and fibrosis, or calcification. Management depends on size and location: most small haematomas resorb spontaneously over weeks; large or expanding haematomas may require drainage (surgical or percutaneous) and reversal of any coagulopathy.

Haematuria

– The presence of blood in the urine. Classification: gross (visible to the naked eye, indicates bleeding beyond the nephron) vs microscopic (detected only on urinalysis or microscopy, >3 RBCs per high-power field). Aetiologies: (1) Glomerular:

IgA nephropathy (most common cause of macroscopic haematuria), post-streptococcal GN, membranoproliferative GN, Alport syndrome, thin basement membrane nephropathy. Glomerular haematuria often presents with dysmorphic RBCs (acanthocytes) and RBC casts. (2) Non-glomerular: urinary tract infection (most common cause), urolithiasis (renal calculi), malignancy (bladder cancer, RCC, prostate cancer), benign prostatic hyperplasia (BPH), trauma, exercise-induced haematuria, anticoagulation. (3) Factitious: menstrual contamination, factitious haematuria. Evaluation: history (timing, associated symptoms, smoking history, anticoagulant use), urinalysis and culture, urine cytology, imaging (CT urogram or ultrasound), cystoscopy (for persistent or unexplained haematuria, especially in older adults).

Haemodialysis

– A medical procedure that filters waste products, toxins, and excess fluid from the blood when the kidneys are unable to do so (end-stage renal disease, ESRD). Mechanism: blood is pumped through an extracorporeal circuit and passes through a dialyser (artificial kidney) containing thousands of hollow fibers with a semipermeable membrane. Dialysis accomplishes: diffusion (removal of small solutes–urea, creatinine, potassium, phosphate down a concentration gradient), ultrafiltration (removal of excess fluid by applying hydrostatic pressure), and convection (removal of middle molecules by solvent drag). Access: arteriovenous fistula (AVF, gold standard), arteriovenous graft (AVG), or central venous catheter (tunnelled, non-tunnelled). Typical schedule: 3-4 hours, 3 times per week. Complications: hypotension (most common, due to ultrafiltration), muscle cramps, disequilibrium syndrome, air embolism, access complications (stenosis, thrombosis, aneurysm, infection), dialyser reactions (hypersensitivity to ethylene oxide or heparin), amyloidosis (beta-2 microglobulin accumulation).

Haemophilia

– An X-linked recessive bleeding disorder caused by deficiency of coagulation factor VIII (hemophilia A, 80-85%) or factor IX (hemophilia B, Christmas disease, 15-20%). Incidence: 1 in 5,000 male births (hemophilia A), 1 in 30,000 (hemophilia B). Severity correlates with residual factor activity: severe (<1% activity, spontaneous bleeding into joints and muscles), moderate (1-5%, bleeding after minor trauma), mild (6-40%, bleeding after major trauma or surgery). Clinical features: spontaneous haemarthrosis (knees, elbows, ankles–most characteristic), intramuscular haematomas (iliopsoas, calf, forearm), easy bruising, prolonged bleeding after dental extractions or surgery, haematuria, intracranial hemorrhage (most common cause of death). Diagnosis: prolonged aPTT with normal PT and platelet count, factor VIII and IX assays (low). Treatment: factor replacement (recombinant or plasma-derived, on-demand or prophylaxis), desmopressin (DDAVP, releases von Willebrand factor, effective in mild hemophilia A), emicizumab (bispecific monoclonal antibody, mimics factor VIII function, subcutaneous prophylaxis for hemophilia A). Complications: development of inhibitors (alloantibodies against factor VIII/IX, 20-30% of severe hemophilia A, renders standard replacement ineffective), haemophilic arthropathy (chronic joint damage), transfusion-transmitted infections (HIV, hepatitis B/C–now rare due to viral inactivation and recombinant products).

Haemoptysis

– Coughing up blood or blood-stained sputum from the lower respiratory tract. Aetiologies: bronchitis (most common, 25-50%, acute or chronic), bronchiectasis (dilated airways, chronic inflammation), lung cancer (squamous cell most commonly, 20% present with haemoptysis), tuberculosis (especially cavitary or endobronchial), pulmonary embolism (haemoptysis with pleuritic chest pain and dyspnoea), pneumonia (especially Klebsiella–currant jelly sputum), lung abscess, bronchiectasis, aspergilloma (fungus ball in a cavity), mitral stenosis (pulmonary venous hypertension causing rupture of bronchial varices), iatrogenic (bronchoscopy, lung biopsy, anticoagulation), and vasculitides (Wegener granulomatosis, Goodpasture syndrome). Massive haemoptysis (>200 mL/24 hours or >50 mL/hour) is life-threatening due to asphyxiation rather than exsanguination. Diagnostic workup: chest X-ray, CT chest, bronchoscopy, sputum culture and cytology, complete blood count, coagulation profile. Management: ABC, position patient with bleeding side down, bronchial artery embolisation (BAE) for massive haemoptysis, surgery (lobectomy, pneumonectomy) for refractory cases.

Hairy Cell Leukemia

– A rare, indolent B-cell chronic lymphoproliferative disorder characterized by the accumulation of mature B lymphocytes with characteristic cytoplasmic projections ("hairy cells") in the bone marrow, peripheral blood, and spleen. It accounts for less than 2 percent of adult leukemias and predominantly affects middle-aged men (male-to-female ratio approximately 4-5:1). Classic HCL is driven by the BRAF V600E mutation, present in virtually all cases. Clinical presentation includes pancytopenia, splenomegaly (often massive), recurrent infections, and fatigue.

Hematocrit

– The percentage of total blood volume composed of packed red blood cells, one of the most commonly ordered laboratory tests in clinical medicine. The hematocrit is measured by centrifugation of anticoagulated whole blood in a capillary tube or automated analyzer, separating the cellular elements from plasma; the normal range is 41-53 percent in adult men and 36-46 percent in adult women, with lower values in children and during pregnancy (physiologic hemodilution). The hematocrit reflects both the packed cell volume and the plasma volume, so it is influenced by both the red blood cell mass and the plasma volume, with a low haematocrit indicating anemia and a high haematocrit suggesting polycythemia.

hematology

– The branch of medicine concerned with the study, diagnosis, treatment, and prevention of disorders of the blood and blood-forming organs (bone marrow, lymph nodes, spleen). Subspecialties: benign hematology (anemia, haemoglobinopathies–sickle cell disease, thalassemia, hemostatic disorders–hemophilia, von Willebrand disease, immune thrombocytopenia), malignant hematology (acute and chronic leukemias, myelodysplastic syndromes, myeloproliferative neoplasms, lymphoma, multiple myeloma, amyloidosis), transfusion medicine (blood banking, donor management, component therapy, apheresis, stem cell transplantation), haemato-oncology (chemotherapy, targeted therapy, immunotherapy, CAR-T cells, HSCT). Diagnostic tools: full blood count (FBC), blood film, bone marrow aspiration and trephine biopsy, flow cytometry, cytogenetics, molecular genetics (PCR, NGS), coagulation studies, hemoglobin electrophoresis, serum protein electrophoresis.

Hemoglobin

– The oxygen-carrying metalloprotein in red blood cells, composed of four globin subunits each containing an iron-porphyrin (heme) group. Adult hemoglobin (HbA): alpha2-beta2 tetramer (two alpha chains, two beta chains). Fetal hemoglobin (HbF): alpha2-gamma2, higher oxygen affinity for oxygen transfer across the placenta. hemoglobin reversibly binds oxygen (4 O2 molecules per tetramer) through cooperative binding (sigmoidal oxygen dissociation curve, p50 = 26 mmHg). The Bohr effect: decreased pH and increased CO2 shift the curve rightward, increasing oxygen unloading in tissues. Normal: men 13.5-17.5 g/dL, women 12.0-16.0 g/dL. Abnormal haemoglobins: sickle cell disease (HbS, Glu6Val in beta-globin), HbC, HbE, metHb, carboxyHb (CO poisoning, cherry-red appearance). HbA1c: glycated hemoglobin, measures average blood glucose over 2-3 months, used for diabetes monitoring. Buffy coat: anticoagulated blood centrifuged shows three layers: plasma (top), buffy coat (middle, WBCs and platelets), packed RBCs (bottom).

Hemoglobinopathies: Thalassemias

– Thalassemias are inherited disorders of globin chain synthesis imbalance: alpha-thalassemia (HBA1/HBA2 deletions) and beta-thalassemia (HBB mutations). Beta-thalassemia major (transfusion-dependent): severe anemia presenting in infancy, ineffective erythropoiesis, marrow expansion, extramedullary hematopoiesis, iron overload from transfusions. Management: chronic transfusions (maintain Hb >9-10 g/dL), iron chelation (deferasirox, deferoxamine, deferiprone), splenectomy if hypersplenism, luspatercept (activin receptor ligand trap for beta-thalassemia), and allogeneic HSCT as definitive therapy.

hemolysis

– The premature destruction of red blood cells, releasing hemoglobin into the plasma. Extravascular hemolysis (most common): RBCs are destroyed by macrophages in the spleen and liver — causes unconjugated hyperbilirubinemia, increased LDH, and decreased haptoglobin. Intravascular hemolysis: RBCs rupture within the circulation — causes hemoglobinemia, hemoglobinuria, and decreased haptoglobin. Causes: inherited (hereditary spherocytosis, G6PD deficiency, sickle cell disease, thalassemia) and acquired (autoimmune hemolytic anemia, microangiopathic hemolytic anemia — DIC, TTP, HUS; mechanical heart valves, infection, drugs, transfusion reaction). Lab findings: reticulocytosis, elevated LDH, unconjugated bilirubin, decreased haptoglobin, and abnormal peripheral smear (spherocytes, schistocytes, sickle cells). Treatment: address underlying cause, folic acid supplementation, corticosteroids for AIHA, and splenectomy for selected cases.

Hemolytic Anemia

– Anemia due to accelerated destruction of red blood cells (RBCs), classified as intrinsic (RBC defect) or extrinsic (external factors). Intrinsic: hereditary spherocytosis, G6PD deficiency, sickle cell disease, thalassemia. Extrinsic: autoimmune hemolytic anemia (warm or cold agglutinin), microangiopathic hemolytic anemia (TMA, DIC, TTP/HUS), drug-induced, and mechanical (prosthetic valves). Laboratory: elevated LDH, indirect bilirubin, reticulocyte count; decreased haptoglobin; positive direct Coombs test in immune-mediated. Treatment depends on cause: corticosteroids for warm AIHA, splenectomy, and immunosuppressants.

Hemolytic Anemias: Autoimmune

– Autoimmune hemolytic anemia (AIHA): warm (IgG, 70-80%, extravascular hemolysis in spleen, DAT positive for IgG +/- C3, associated with CLL, SLE, drugs, idiopathic), cold agglutinin disease (IgM, complement-mediated, intravascular/extravascular, DAT positive for C3 only, associated with lymphoproliferative, Mycoplasma), mixed (both warm and cold), paroxysmal cold hemoglobinuria (Donath-Landsteiner antibody, biphasic hemolysin, post-viral in children). Drug-induced (penicillins, cephalosporins, methyldopa). Treatment: corticosteroids (prednisone 1 mg/kg/day) for warm AIHA; rituximab, splenectomy, IVIG, and immunosuppressants for refractory cases; avoid transfusions if possible; keep patient warm for cold agglutinin disease.

Hemolytic Anemias: Hereditary

– Hereditary spherocytosis (HS): cytoskeletal protein defects (ankyrin, band 3, protein 4.2, alpha/beta spectrin) -> spherical RBCs, splenic trapping, hemolysis. AD (75%) or AR. Clinical: anemia, jaundice, splenomegaly, gallstones. Spherocytes on smear, increased MCHC, osmotic fragility test, eosin-5-maleimide (EMA) binding (flow cytometry). Treatment: folic acid, splenectomy (if moderate/severe, after age 5-6, pneumococcal/meningococcal/Hib vaccines pre-op), partial splenectomy. Hereditary elliptocytosis: milder, usually asymptomatic; hereditary pyropoikilocytosis: severe hemolysis.

hemolytic Disease of the Newborn

– A condition in which maternal antibodies destroy fetal/neonatal red blood cells (RBCs), causing hemolytic anemia, hyperbilirubinaemia, and in severe cases hydrops fetalis. Most commonly due to Rh incompatibility: an RhD-negative mother carries an RhD-positive fetus; fetomaternal hemorrhage at delivery (or during pregnancy) sensitises the mother to produce anti-D IgG antibodies; subsequent pregnancies with RhD-positive fetuses are vulnerable. ABO incompatibility (mother O, baby A or B) is more common but typically milder. Pathophysiology: maternal IgG crosses the placenta, coats fetal RBCs, and causes extravascular hemolysis in the fetal spleen. Severity ranges from mild jaundice to severe anemia, hepatosplenomegaly, hydrops (anasarca, ascites, pleural effusion, heart failure), and intrauterine death (hydropic). Diagnosis: maternal antibody screening (indirect Coombs test), paternal zygosity testing, fetal genotyping (cell-free fetal DNA), serial MCA-PSV Doppler ultrasound to detect fetal anemia. Management: intrauterine transfusion (IUT) for severe anemia, early delivery, phototherapy for neonatal jaundice, exchange transfusion for severe hyperbilirubinaemia (neurotoxicity–kernicterus). Prevention: routine antenatal anti-D prophylaxis (RAADP) at 28 weeks and postpartum anti-D IgG within 72 hours of delivery for all non-sensitised RhD-negative women. Screening and prophylaxis have reduced HDN incidence by 90%.

Hemolytic Jaundice

– Jaundice resulting from excessive hemolysis (breakdown of red blood cells) causing overproduction of unconjugated bilirubin that exceeds the liver's capacity for conjugation and excretion. See Jaundice for complete overview. Causes: hereditary spherocytosis, G6PD deficiency, sickle cell disease, thalassemia, autoimmune hemolytic anemia, hemolytic transfusion reaction, malaria, and microangiopathic hemolytic anemia (MAHA). Laboratory features: elevated unconjugated (indirect) bilirubin, increased reticulocyte count, low haptoglobin, elevated LDH, raised urobilinogen in urine, and evidence of hemolysis on blood film (spherocytes, schistocytes, bite cells).

Hemolytic Uremic Syndrome

– A thrombotic microangiopathy characterized by microangiopathic hemolytic anemia, thrombocytopenia, and acute kidney injury. Typical (Shiga toxin-producing E. coli, STEC-HUS): O157:H7 (most common), non-O157; follows diarrheal prodrome; children; supportive care (no antibiotics, may increase toxin release); eculizumab not routinely recommended. Atypical (aHUS): complement dysregulation (mutations in CFH, CFI, MCP/CD46, C3, CFB, THBD, or autoantibodies to CFH); requires eculizumab (anti-C5 monoclonal antibody) and plasma exchange. Management of both types includes supportive care with transfusion and plasma exchange.

hemorrhoids (Piles)

– Vascular cushions in the anal canal composed of arteriovenous anastomoses (normal structures present from birth), which become symptomatic when they bleed, prolapse, or thrombose. Anatomy: internal hemorrhoids (above the dentate line, 3 primary positions–left lateral, right anterior, right posterior, covered by columnar epithelium, visceral innervation–painless), external hemorrhoids (below the dentate line, covered by squamous epithelium, somatic innervation–painful if thrombosed). Grading of internal hemorrhoids: grade 1 (confined to anal canal, no prolapse), grade 2 (prolapses on defecation but reduces spontaneously), grade 3 (prolapses and requires manual reduction), grade 4 (irreducible, permanently prolapsed). Risk factors: chronic constipation/straining, prolonged sitting, pregnancy, obesity, low-fibre diet, cirrhosis (portal hypertension). Symptoms: painless rectal bleeding (bright red blood on toilet paper or dripping into the bowl, typically coating stools), prolapse, pruritus ani, mucoid discharge, thrombosis (acute severe pain, perianal lump). Diagnosis: clinical (inspection of external hemorrhoids, digital rectal examination, anoscopy, proctoscopy). Management: lifestyle (high-fibre diet, adequate hydration increased fibre, avoid straining), topical treatments (anaesthetics, steroids, astringents), rubber band ligation (grade 1-2), sclerotherapy, infrared coagulation, haemorrhoidectomy (Milligan-Morgan open, Ferguson closed, stapled, THD–transanal haemorrhoidal dearterialisation).

hemostasis

– The physiological process that stops bleeding at the site of vascular injury. Three phases: (1) Primary hemostasis: vasoconstriction (reflex, endothelin-mediated), platelet adhesion (to exposed subendothelial von Willebrand factor via platelet glycoprotein Ib/IX/V, to collagen via GPVI), platelet activation (shape change, granule release–ADP, thromboxane A2, serotonin), and platelet aggregation (fibrinogen bridges between platelet GPIIb/IIIa receptors), forming a platelet plug. (2) Secondary hemostasis: coagulation cascade, culminating in cross-linked fibrin stabilising the platelet plug. (3) Fibrinolysis: tissue plasminogen activator (tPA) converts plasminogen to plasmin, which degrades fibrin into D-dimers and FDPs. The coagulation cascade is classically described as intrinsic (surface contact–factors XII, XI, IX, VIII) and extrinsic (tissue factor–factor VII) pathways converging on the common pathway (factor X, V, II, I). In vivo, the cell-based model of hemostasis more accurately describes initiation on tissue factor-bearing cells, amplification on platelets, and propagation on platelet surfaces. Disorders: hemophilia (deficiency of VIII or IX), von Willebrand disease (deficiency or dysfunction of vWF), thrombocytopenia, platelet function disorders, anticoagulation therapy.

Heparin-Induced Thrombocytopenia

– An immune-mediated adverse drug reaction caused by IgG antibodies against platelet factor 4 (PF4)-heparin complexes, leading to platelet activation, thrombocytopenia, and a prothrombotic state. HIT typically occurs 5-14 days after heparin exposure (unfractionated heparin > low molecular weight heparin). Clinical: thrombocytopenia (fall >50% from baseline, nadir rarely <20k), thrombosis (venous: DVT/PE; arterial: limb ischemia, stroke, MI) in 20-50%, skin necrosis at injection sites (50%), and adrenal hemorrhage. Diagnosis: 4Ts score, HIT ELISA (PF4-heparin ELISA), serotonin release assay (SRA). Treatment: stop heparin, start non-heparin anticoagulant (argatroban, bivalirudin, fondaparinux, danaparoid). Avoid warfarin until platelets >150 (risk of venous limb gangrene).

Hereditary Spherocytosis

– The most common inherited hemolytic anemia in Northern Europeans (autosomal dominant 75%, recessive 25%). Caused by defects in RBC membrane proteins (spectrin, ankyrin, band 3, protein 4.2), leading to loss of membrane surface area, reduced deformability, and premature destruction in the spleen. Presents with hemolytic anemia, jaundice, splenomegaly, and reticulocytosis. Lab findings: spherocytes on peripheral smear, increased MCHC, negative direct Coombs test, elevated osmotic fragility. Complications: aplastic crisis (parvovirus B19), hemolytic crisis, gallstones (pigment stones), and leg ulcers. Treatment: folic acid supplementation; splenectomy for severe anemia.

Hodgkin Lymphoma

– A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells. Classical HL (cHL) accounts for 95 percent of cases and is subdivided into nodular sclerosis (most common, mediastinal predominance, young adults), mixed cellularity, lymphocyte-rich, and lymphocyte-depleted subtypes. Nodular lymphocyte-predominant HL (NLPHL, 5 percent) features lymphocyte-predominant (LP) cells (CD20+, CD15-, CD30-) instead of Reed-Sternberg cells. Staging with PET-CT. Treatment: ABVD (doxorubicin, bleomycin, vinblastine, dacarbazine) or escalated BEACOPP for advanced disease; involved-field radiation for bulky disease; brentuximab vedotin for relapsed/refractory CD30+ disease.

Hodgkin's Lymphoma

– See Hodgkin Lymphoma.

Immune Thrombocytopenia

– An autoimmune disorder characterized by isolated thrombocytopenia (platelet count <100 x10^9/L) due to autoantibody-mediated platelet destruction and impaired platelet production. Primary ITP (no identifiable cause) vs secondary (SLE, CLL, HIV, HCV, drugs). Clinical: petechiae, purpura, epistaxis, menorrhagia; intracranial hemorrhage rare (<1%). Diagnosis: exclusion of other causes (CBC, blood smear, HIV/HCV, TSH, ANA, bone marrow if >60 or atypical features. First-line: corticosteroids (prednisone). Second-line: TPO receptor agonists (eltrombopag, romiplostim), rituximab, mycophenolate mofetil. Splenectomy reserved for refractory cases. Platelet transfusion for life-threatening bleeding.

Iron Deficiency Anemia

– The most common anemia worldwide, affecting over 1 billion people, resulting from insufficient iron for hemoglobin synthesis. Iron balance requires adequate dietary intake (heme iron from animal sources better absorbed than non-heme iron from plant sources), efficient absorption (duodenal and proximal jejunal enterocytes, regulated by hepcidin), and minimal losses (menstruation, GI bleeding, pregnancy). Causes include chronic blood loss (menorrhagia, GI bleeding from GI lesions), malabsorption, and increased requirements (pregnancy, growth). Lab: microcytic hypochromic anemia, low ferritin, low serum iron, low transferrin saturation, increased TIBC, increased RDW. Diagnosis: iron studies with ferritin <30 ng/mL. Treatment: oral iron (ferrous sulfate 325 mg daily); IV iron for intolerance, malabsorption, severe deficiency, or chronic blood loss. Response measured by reticulocyte count increase within 7-10 days and Hb rise of 1-2 g/dL over 3-4 weeks.

Leucopenia

– A decreased total white blood cell count below the normal reference range (<4.0 × 10^9/L). Leucopenia is often due to a decrease in neutrophils (neutropenia) or lymphocytes (lymphopenia). Causes: infection (viral–HIV, EBV, influenza, hepatitis; overwhelming bacterial sepsis–endotoxin-induced bone marrow suppression; typhoid), bone marrow failure (aplastic anemia, myelodysplasia, leukemia, myelofibrosis), chemotherapy/radiotherapy (most common cause of moderate-severe neutropenia), drugs (carbimazole, methimazole, clozapine, sulfonamides, immunosuppressants, antipsychotics, anticonvulsants), autoimmune (SLE, Felty syndrome–neutropenia with RA and splenomegaly), hypersplenism (cirrhosis, portal hypertension), nutritional deficiency (vitamin B12, folate, copper), and ethnic/benign variants (pseudoneutropenia, Afro-Caribbean and African descent, stable neutrophil count 1.0–2.5 $\times 10^9$/L with no increased infection risk). Clinical significance: increased risk of infection, especially when absolute neutrophil count (ANC) falls <1.0 $\times 10^9$/L. Severe neutropenia (ANC <0.5) requires urgent evaluation. Management: treat underlying cause, stop offending drug, G-CSF (filgrastim) for chemotherapy-induced neutropenia, antibiotic prophylaxis for severe chronic neutropenia.

Leukemia

– A group of hematological malignancies characterized by the clonal proliferation of abnormal white blood cells (blasts) in the bone marrow and peripheral blood. leukemia is classified by: (1) clinical course: acute (rapid onset, >20% blasts in bone marrow) vs chronic (insidious onset, more differentiated cells). (2) cell lineage: lymphoblastic (T-cell or B-cell precursor) vs myeloid (granulocytic, monocytic, erythroid, megakaryocytic). The four main subtypes are: acute lymphoblastic leukemia (ALL, most common in children), acute myeloid leukemia (AML, more common in adults), chronic lymphocytic leukemia (CLL, most common adult leukemia in Western countries), and chronic myeloid leukemia (CML, characterized by Philadelphia chromosome t(9;22) BCR-ABL fusion).

Leukocytosis

– An increased total white blood cell count above the normal reference range (>11 $\times$ 10^9/L in adults). Causes: physiological (exercise, stress, pregnancy, labor), infectious (most common, especially bacterial–neutrophilia; viral–lymphocytosis; parasitic–eosinophilia), inflammatory (autoimmune disease, vasculitis, gout, pancreatitis), tissue necrosis (myocardial infarction, burns, trauma, surgery), malignancy (myeloproliferative neoplasms–CML, polycythemia vera; solid tumors with bone marrow involvement), medications (corticosteroids, lithium, G-CSF), hematologic (hemolytic anemia, hemorrhage), and post-splenectomy. leukocytosis is further characterized by the specific cell line increased: neutrophilia (neutrophils >7.5 $\times$ 10^9/L, bacterial infection, stress, steroids, CML), lymphocytosis (lymphocytes >4.0 $\times$ 10^9/L, viral infection, CLL, pertussis), monocytosis (monocytes >0.8 $\times$ 10^9/L, TB, endocarditis, CMML), eosinophilia (eosinophils >0.5 $\times$ 10^9/L, parasitic infection, asthma, allergy, eosinophilic granulomatosis with polyangiitis, drug reaction), and basophilia (basophils >0.1 $\times$ 10^9/L, CML, polycythemia vera). A normal peripheral blood smear and lack of abnormal cells distinguishes reactive leukocytosis from leukemia.

Lymphoma (Hodgkin's and Non-Hodgkin's)

– See Hodgkin Lymphoma and Non-Hodgkin Lymphoma.

Mantle Cell Lymphoma

– An aggressive B-cell lymphoma characterized by t(11;14) translocation leading to cyclin D1 overexpression and CCND1-IGH fusion. Clinical: older males, widespread lymphadenopathy, splenomegaly, GI involvement (lymphomatous polyposis), bone marrow, peripheral blood (leukemic phase). Ki-67 index and MIPI (MCL International Prognostic Index) stratify risk. Treatment: young/fit: intensive chemoimmunotherapy (R-DHAP, R-HyperCVAD, Nordic MCL2/3) + autologous stem cell transplant. Older/unfit: less intensive regimens (R-CHOP, R-bendamustine, VR-CAP) plus rituximab maintenance. No cure; median survival 5-7 years.

Multiple Myeloma

– A plasma cell neoplasm characterized by clonal proliferation of malignant plasma cells in the bone marrow, producing monoclonal immunoglobulin (M-protein) and causing end-organ damage defined by the CRAB criteria: hypercalcemia (calcium greater than 11 mg/dL), renal insufficiency (creatinine greater than 2 mg/dL), anemia (hemoglobin less than 10 g/dL), and bone lesions (lytic lesions on imaging). The International Myeloma Working Group (IMWG) diagnostic criteria also include clonal bone marrow plasma cells ≥10% or biopsy-proven bony or extramedullary plasmacytoma, and any one or more of the CRAB features or one or more biomarker of malignancy (clonal plasma cell proliferation ≥60%, involved:uninvolved serum free light chain ratio ≥100, or >1 focal lesion on MRI). Treatment: triplet regimens (VRd: bortezomib, lenalidomide, dexamethasone; or Dara-Rd for transplant-ineligible), followed by autologous stem cell transplant in eligible patients.

Myelodysplastic Syndromes

– A heterogeneous group of clonal hematopoietic stem cell disorders characterized by ineffective hematopoiesis, dysplastic morphology in one or more myeloid lineages, cytopenias, and risk of transformation to acute myeloid leukemia. MDS predominantly affects older adults (median age 70-75 years) and may be primary (de novo) or secondary (therapy-related from prior chemotherapy or radiation). Clinical presentation includes fatigue (anemia), infections (neutropenia), and bleeding (thrombocytopenia). IPSS-R (Revised International Prognostic Scoring System) stratifies risk based on cytopenias, blast percentage, and cytogenetics. Management: supportive care (transfusions, growth factors), lenalidomide for del(5q), luspatercept for ring sideroblasts, hypomethylating agents (azacitidine, decitabine), and allogeneic HSCT for high-risk disease.

Myelofibrosis

– A myeloproliferative neoplasm characterized by bone marrow fibrosis, extramedullary hematopoiesis, and JAK2, CALR, or MPL mutations. Presents with anemia, massive splenomegaly, constitutional symptoms (fever, night sweats, weight loss), and leukoerythroblastic blood smear. Diagnosis: bone marrow biopsy (fibrosis, megakaryocytic hyperplasia), JAK2 V617F mutation (50-60%), CALR (20-25%). Prognosis: dynamic IPSS. Treatment: JAK1/2 inhibitors (ruxolitinib, fedratinib), splenectomy for symptomatic splenomegaly, and allogeneic stem cell transplant as only curative option.

Non-Hodgkin Lymphoma

– Hematologic malignancy of B or T/NK cells. B-cell NHL (85%) includes diffuse large B-cell lymphoma (DLBCL, most common aggressive), follicular lymphoma (most common indolent), mantle cell lymphoma, marginal zone lymphoma, and Burkitt lymphoma. T-cell NHL: peripheral T-cell lymphoma, anaplastic large cell lymphoma. Presents with painless lymphadenopathy, B symptoms (fever, night sweats, weight loss), and cytopenias. Diagnosis: excisional biopsy with flow cytometry, cytogenetics, FISH. Treatment: R-CHOP for DLBCL, rituximab-based therapy for follicular, and stem cell transplant for relapsed/refractory disease.

Paroxysmal Nocturnal Hemoglobinuria

– A rare, acquired clonal disorder of hematopoietic stem cells characterized by complement-mediated intravascular hemolysis, thrombosis (particularly in unusual sites such as hepatic, portal, and cerebral veins), and bone marrow failure. The underlying defect is a somatic mutation in the PIGA gene, which encodes a subunit of the GPI-anchor synthesis enzyme complex, resulting in deficient expression of GPI-anchored proteins (CD55 and CD59) on the surface of blood cells. Diagnosis: flow cytometry showing deficient CD55/CD59 on red blood cells. Treatment: eculizumab or ravulizumab (anti-C5 monoclonal antibodies) for hemolysis and thrombosis; anticoagulation for thrombotic events; supportive care with folic acid, transfusion, and iron chelation for transfusion burden.

Pernicious Anemia

– A type of vitamin B12 deficiency anemia caused by autoimmune destruction of gastric parietal cells, leading to lack of intrinsic factor and impaired B12 absorption in the terminal ileum. More common in older adults, those with autoimmune disorders, and pernicious anemia is associated with atrophic gastritis and increased gastric cancer risk. Presents with megaloblastic anemia (macrocytic RBCs, hypersegmented neutrophils), neuropathy (paresthesias, ataxia, cognitive impairment), and glossitis. Diagnosis: low B12, elevated methylmalonic acid and homocysteine, positive anti-intrinsic factor antibodies. Treatment: lifelong parenteral B12 (cyanocobalamin) replacement.

Platelet Function Disorders

– Inherited: Glanzmann thrombasthenia (GPIIb/IIIa deficiency, absent platelet aggregation, mucocutaneous bleeding, recombinant FVIIa, platelet transfusions), Bernard-Soulier syndrome (GPIb/IX/V deficiency, giant platelets, thrombocytopenia, prolonged bleeding time, platelet transfusions), Storage pool deficiency (delta/dense granules, mild bleeding). Acquired: uremia (platelet dysfunction, desmopressin, dialysis), myeloproliferative neoplasms (acquired von Willebrand syndrome in ET/PV, aspirin), myelodysplasia. Diagnosis: bleeding time, PFA-100, platelet aggregation studies, flow cytometry for glycoprotein expression. Treatment: desmopressin for uraemia and mild type 1 VWD; platelet transfusions for Glanzmann and Bernard-Soulier; recombinant FVIIa for Glanzmann with anti-GPIIb/IIIa antibodies.

Platelets (Thrombocytes)

– Small, anucleate, disc-shaped cell fragments (2–4 μm diameter) derived from megakaryocytes in the bone marrow, circulating at 150–400 $\times$ 10^9/L. Platelets are essential for primary hemostasis: they adhere to exposed subendothelial collagen via von Willebrand factor (GPIb receptor), become activated (shape change, degranulation—release ADP, serotonin, thromboxane A_2), and aggregate via fibrinogen bridges (GPIIb/IIIa receptor) to form a platelet plug. They also provide a phospholipid surface for coagulation factor assembly (tenase and prothrombinase complexes). Platelet lifespan: 7–10 days; old platelets are removed by the spleen. Thrombocytopenia (<150 $\times$ 10^9/L) increases bleeding risk; thrombocytosis (>450 $\times$ 10^9/L) is reactive (infection, inflammation, iron deficiency) or clonal (essential thrombocythaemia). Platelet function disorders: Glanzmann thrombasthenia (GPIIb/IIIa deficiency), Bernard–Soulier syndrome (GPIb deficiency), storage pool disease. Drugs affecting platelets: aspirin (irreversible COX-1 inhibition), clopidogrel ($P2Y_{12}$ receptor antagonist), NSAIDs, and GPIIb/IIIa inhibitors.

Polycythemia Vera

– A chronic myeloproliferative neoplasm characterized by increased red blood cell mass, often with increased white blood cell and platelet counts, driven by constitutive activation of the JAK-STAT signaling pathway. The JAK2 V617F mutation is present in approximately 95 percent of PV patients, with an additional 3 percent harboring JAK2 exon 12 mutations. Clinical presentation includes elevated hemoglobin and hematocrit (per WHO criteria: male hemoglobin greater than 16.5 g/dL or haematocrit greater than 49% in men, greater than 16.0 g/dL or haematocrit greater than 48% in women), thrombocytosis, leukocytosis, and splenomegaly in 70% of patients. Treatment: phlebotomy (target Hct <45%), low-dose aspirin for all patients, cytoreduction (hydroxyurea, interferon-alpha, ruxolitinib) for high-risk patients (age >60, prior thrombosis). Risk of thrombosis, hemorrhage, and transformation to myelofibrosis or AML.

Primary Myelofibrosis

– A chronic myeloproliferative neoplasm characterized by bone marrow fibrosis (reticulin and/or collagen fibrosis), extramedullary hematopoiesis (primarily splenic and hepatic), and a leukoerythroblastic peripheral blood picture. It results from clonal proliferation of hematopoietic stem cells carrying JAK2 V617F (50-60 percent), CALR (25-30 percent), or MPL (3-5 percent) mutations, leading to cytokine release that stimulates fibroblast proliferation and collagen deposition of extracellular matrix proteins. Clinical presentation includes progressive anemia, massive splenomegaly with early satiety and abdominal discomfort, constitutional symptoms (fever, night sweats, weight loss, fatigue), and extramedullary hematopoiesis. Peripheral blood shows leukoerythroblastosis with teardrop cells. Treatment: ruxolitinib (JAK1/2 inhibitor) for symptomatic splenomegaly and constitutional symptoms; allogeneic HSCT is the only potentially curative option; supportive care with transfusions, erythropoietin, and targeted therapy for iron overload.

Protein C Deficiency

– An inherited (autosomal dominant) thrombophilia causing increased VTE risk. Protein C is a vitamin K-dependent natural anticoagulant that inactivates factors Va and VIIIa and stimulates fibrinolysis. Heterozygous: 6-10 fold increased VTE risk. Homozygous: severe neonatal purpura fulminans (widespread skin necrosis and thrombosis) requiring protein C replacement. Diagnosis: functional and antigenic protein C levels (low). Caution: levels also low in warfarin therapy, liver disease, DIC, and vitamin K deficiency. Management: anticoagulation for VTE; warfarin initiation requires heparin bridging to avoid warfarin-induced skin necrosis.

Protein S Deficiency

– An inherited (autosomal dominant) thrombophilia. Protein S is a vitamin K-dependent natural anticoagulant that acts as a cofactor for activated protein C in inactivating factors Va and VIIIa. Heterozygous: 2-10 fold increased VTE risk. Homozygous: rare, purpura fulminans. Total, free, and functional protein S levels must be interpreted carefully as levels are affected by pregnancy, oral contraceptives, liver disease, DIC, and acute thrombosis. Diagnosis: confirmed low free protein S on repeat testing. Management: anticoagulation for VTE; lifelong anticoagulation for recurrent or severe thrombosis.

Rhesus (Rh) Factor

– A system of blood group antigens (the Rh system) on the surface of red blood cells, the most important of which is the D antigen (RhD). Individuals are Rh-positive (RhD+) if the D antigen is present (85% of Caucasians, 92% of Africans, 99% of Asians) or Rh-negative (RhD-) if absent. The Rh factor is determined by the RHD gene on chromosome 1. Rh incompatibility occurs when an Rh-negative mother carries an Rh-positive fetus: fetal red cells entering the maternal circulation (fetomaternal hemorrhage—typically at delivery, but also during pregnancy, trauma, or abortion) stimulate maternal anti-D antibody production (IgG). In a subsequent Rh-positive pregnancy, maternal anti-D IgG crosses the placenta and causes hemolytic disease of the fetus and newborn (HDFN—erythroblastosis fetalis), leading to fetal anemia, hydrops fetalis, kernicterus, and neonatal jaundice. Prevention: prophylactic anti-D immunoglobulin (RhoGAM) is given to Rh-negative women at 28 weeks gestation and within 72 hours of delivery (or any sensitising event—amniocentesis, miscarriage, abortion, ectopic pregnancy), which coats fetal Rh-positive red cells and prevents maternal immune sensitisation. Other Rh antigens: C, c, E, e (Kell, Duffy, Kidd, and MNS are other clinically important blood group systems).

Rh Factor

– See Rhesus (Rh) Factor.

Sickle Cell Trait

– Heterozygous carrier state for the HbS mutation (HbAS), present in 8% of African Americans. Red cells contain both HbA and HbS (typically 30-40% HbS). Usually asymptomatic with normal hemoglobin and RBC indices. Complications (rare but recognized): painless hematuria (renal papillary necrosis), urinary concentrating defect, splenic infarction at high altitude, exercise-related sudden death (during extreme exertion), and hyposthenuria. Diagnosis: hemoglobin electrophoresis showing HbA and HbS. Genetic counseling. No treatment needed; avoid extreme dehydration and high-altitude exposure.

Splenomegaly

– Enlargement of the spleen beyond its normal size (12 cm, weight 150–200 g). Splenomegaly can be detected by percussion (Castell sign—dullness in the lowest intercostal space on inspiration) and palpation (normal spleen is not palpable; a palpable spleen indicates at least 2–3 times enlargement). Causes: congestive (portal hypertension—cirrhosis, hepatic vein thrombosis/Budd–Chiari syndrome, splenic vein thrombosis, heart failure), infectious (EBV, CMV, malaria, leishmaniasis, schistosomiasis, tuberculosis, endocarditis, brucellosis, HIV), hematological (hemolytic anemias—hereditary spherocytosis, thalassemia, sickle cell disease; myeloproliferative neoplasms—myelofibrosis, polycythemia vera, CML; leukemia—CLL, ALL, AML; lymphoma—Hodgkin, non-Hodgkin), inflammatory/autoimmune (sarcoidosis, SLE, rheumatoid arthritis—Felty syndrome, Still disease), infiltrative (Gaucher disease, Niemann–Pick, amyloidosis, glycogen storage diseases), and neoplastic (primary splenic tumors—rare). Massive splenomegaly (spleen >20 cm, >1000 g): most commonly myelofibrosis, CML, malaria, kala-azar, Gaucher disease, and schistosomiasis. Complications: hypersplenism (pancytopenia from increased sequestration/destruction of blood cells), splenic infarction (pain, left upper quadrant), and splenic rupture (trauma). Diagnosis: ultrasound, CT, MRI; further workup directed at underlying cause. Treatment: directed at the underlying cause; splenectomy for symptomatic massive splenomegaly, hypersplenism with severe cytopenias, splenic abscess, or splenic rupture.

Stem Cell Transplantation

– Hematopoietic stem cell transplantation (HSCT) replaces diseased or ablated bone marrow with healthy donor or autologous stem cells. Autologous HSCT: patient's own stem cells collected after mobilization (G-CSF with or without plerixafor), providing hematopoietic rescue after high-dose chemotherapy without graft-versus-host disease risk; indications include relapsed lymphoma and multiple myeloma. Allogeneic HSCT: donor stem cells (HLA-matched sibling, haploidentical family member, or matched unrelated donor). Conditioning: myeloablative (high-dose chemo + TBI) or reduced-intensity (older, comorbidities). GVHD prophylaxis: calcineurin inhibitor plus methotrexate or mycophenolate. Complications: GVHD (acute: skin, liver, GI; chronic: skin, mouth, eyes, liver, lungs), infection (CMV, EBV, fungal), VOD/SOS (veno-occlusive disease), graft failure, and relapse.

T-Cell Lymphomas

– Peripheral T-cell lymphomas (PTCL) are a heterogeneous group of aggressive lymphomas including PTCL-NOS (not otherwise specified, most common), angioimmunoblastic T-cell lymphoma (AITL, T-follicular helper origin, RHOA mutations, dysimmune features), anaplastic large cell lymphoma (ALCL, ALK+ vs ALK-), enteropathy-associated T-cell lymphoma (EATL, celiac association), hepatosplenic T-cell lymphoma (gamma-delta, isochromosome 7q), and cutaneous T-cell lymphomas (mycosis fungoides, Sezary syndrome (erythroderma, generalised lymphadenopathy, circulating Sezary cells). Diagnosis: excisional lymph node biopsy with histology, IHC, flow cytometry, cytogenetics, and FISH. Treatment: CHOP-like chemotherapy for aggressive subtypes; novel agents (brentuximab vedotin for ALCL; romidepsin/belinostat for relapsed PTCL); allogeneic HSCT for eligible patients.

thalassemia

– A group of inherited hemoglobin disorders caused by reduced or absent synthesis of one or more globin chains, leading to microcytic hypochromic anemia. Types: α-thalassemia (deletion of α-globin genes on chromosome 16; severity depends on number of deleted genes: 1 deletion—silent carrier, 2—α-thalassemia trait/minor, 3—hemoglobin H disease (HbH—moderate hemolytic anemia), 4—Hb Bart hydrops fetalis (lethal in utero)), β-thalassemia (point mutations in β-globin gene on chromosome 11; β-thalassemia minor/trait—mild anemia, β-thalassemia intermedia—moderate anemia requiring intermittent transfusions, β-thalassemia major (Cooley anemia)—severe anemia presenting at 6–24 months, requiring lifelong transfusions, leads to iron overload, growth retardation, hepatosplenomegaly, and skeletal changes (bossing, chipmunk facies, osteopenia)). Epidemiology: high prevalence in the Mediterranean, Middle East, South Asia, and Southeast Asia (malaria protection). Diagnosis: full blood count (microcytosis, hypochromia), hemoglobin electrophoresis (HbA_2 elevated in β-thalassemia trait, HbF elevated in β-thalassemia major, HbH in α-thalassemia intermedia), and genetic testing. Management: mild forms require no treatment, or folate supplementation. Transfusion-dependent patients: regular packed red cell transfusions (maintain Hb 9–10 g/dL), iron chelation therapy (deferasirox, desferrioxamine, or deferiprone) to prevent iron overload—the main cause of morbidity (cardiac, hepatic, endocrine). Allogeneic hematopoietic stem cell transplantation is the only curative option (best outcomes in children <16 years without significant iron overload). Luspatercept (erythroid maturation agent) reduces transfusion burden in β-thalassemia. Gene therapy (betibeglogene autotemcel—Zynteglo) is approved for transfusion-dependent β-thalassemia.

Thrombectomy

– A surgical or interventional radiology procedure to remove a blood clot (thrombus) from a blood vessel, restoring blood flow. Types: (1) Endovascular thrombectomy (mechanical thrombectomy)—for acute ischemic stroke due to large vessel occlusion (LVO—internal carotid artery, middle cerebral artery M1 segment). Performed via femoral artery access, a stent retriever or aspiration catheter is deployed to remove the clot. Window: up to 6–24 hours from symptom onset (selected by perfusion imaging). Number needed to treat (NNT) of 3 for functional independence. (2) Surgical thrombectomy—open arteriotomy and clot removal, rarely used now (reserved for failed endovascular therapy, massive PE with hemodynamic instability, or peripheral arterial occlusion). (3) Venous thrombectomy—for extensive iliofemoral DVT (phlegmasia cerulea dolens) or superior vena cava syndrome. (4) Vacuum-assisted thrombectomy (AngioJet, Penumbra)—for peripheral arterial or venous thrombosis. Complications: vessel perforation, dissection, distal embolisation, hemorrhage (especially intracranial hemorrhage post-thrombectomy for stroke).

Thrombocytes

– See Platelets (Thrombocytes).

Thrombophilia

– A tendency to develop venous thromboembolism (VTE) due to inherited or acquired hypercoagulable states. Inherited: factor V Leiden, prothrombin G20210A mutation, antithrombin deficiency, protein C deficiency, protein S deficiency. Acquired: antiphospholipid syndrome (lupus anticoagulant, anticardiolipin antibodies, beta-2 glycoprotein I antibodies), cancer, pregnancy, oral contraceptives, and obesity. Presents with DVT and PE, often recurrent or at unusual sites. Management: anticoagulation (DOACs, warfarin, heparin), with duration depending on provoking factors and risk of recurrence.

Thrombophilia Testing

– Thrombophilia testing evaluates for inherited and acquired hypercoagulable states in patients with venous thromboembolism, recurrent pregnancy loss, or strong family history of VTE. Inherited thrombophilias include Factor V Leiden (most common, 5 percent of Caucasians, activated protein C resistance, detected by clot-based or genetic assay), prothrombin G20210A mutation (2 percent of Caucasians, elevated prothrombin levels), antithrombin deficiency (1 percent, most thrombogenic, measured by activity (functional assay), protein C deficiency (activity assay), protein S deficiency (free protein S antigen, activity). Acquired thrombophilias: antiphospholipid syndrome (lupus anticoagulant, anticardiolipin, anti-beta-2 glycoprotein I), malignancy-associated thrombosis. Testing should be performed at least 3 months after a thrombotic event and after discontinuation of anticoagulation to avoid false-positive or false-negative results.

Thrombotic Thrombocytopenic Purpura

– A life-threatening thrombotic microangiopathy caused by severe deficiency of ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin type 1 motif member 13), a von Willebrand factor-cleaving protease. ADAMTS13 deficiency (activity less than 10 percent) leads to accumulation of ultra-large von Willebrand factor multimers that cause spontaneous platelet adhesion and aggregation in the microcirculation, forming platelet-rich thrombi that consume platelets and coagulation factors, causing thrombocytopenia and microangiopathic hemolytic anemia (schistocytes on peripheral smear). Clinical: fever, thrombocytopenia, microangiopathic hemolytic anemia, renal impairment, neurologic symptoms (headache, confusion, seizure, stroke), and fever. Diagnosis: ADAMTS13 activity <10% and detectable anti-ADAMTS13 antibodies. Treatment: daily plasma exchange (PLEX) until platelet count normalizes and hemolysis resolves; corticosteroids for immune suppression; rituximab for refractory/relapsed TTP; caplacizumab (anti-vWF nanobody) for acute TTP.

Transfusion Medicine

– Blood components: PRBCs (leukoreduced, irradiated for at-risk), FFP (coagulation factors), platelets (apheresis pooled, leukoreduced), cryoprecipitate (fibrinogen, vWF, factor VIII, XIII). ABO/Rh compatibility. Transfusion thresholds: restrictive (Hb 7-8 g/dL) for most stable patients (TRICC, FOCUS, TRISS trials); liberal (Hb 9-10) for acute coronary syndrome, severe TBI, acute bleeding. Massive transfusion protocol (MTP): 1:1:1 (PRBC:FFP:platelets) or 1:1:2; TXA within 3h (CRASH-2). Adverse reactions: acute hemolytic (ABO incompatibility, fever, hypotension, DIC, renal failure), febrile non-hemolytic (most common, cytokines in stored blood), allergic (urticaria, anaphylaxis in IgA deficiency), transfusion-related acute lung injury (TRALI, bilateral pulmonary infiltrates, hypoxia), transfusion-associated circulatory overload (TACO, pulmonary edema from volume overload), and transfusion-transmitted infections (HIV, HBV, HCV, West Nile virus, bacterial contamination).

Vitamin B12 Deficiency

– Vitamin B12 (cobalamin) deficiency causes megaloblastic anemia and neurologic dysfunction through impaired DNA synthesis and abnormal fatty acid metabolism in myelin. Dietary sources include animal products (meat, fish, dairy), with daily requirement of 2-3 mcg. Absorption requires adequate gastric acid and pepsin to release B12 from food proteins, binding to intrinsic factor (IF) produced by gastric parietal cells, and absorption in the terminal ileum via the cubilin-amnionless receptor complex. Causes: pernicious anemia (autoimmune destruction of parietal cells, most common in adults), dietary deficiency (vegans, malnutrition, alcoholism), malabsorption (gastric bypass, Crohn disease, celiac disease, pancreatic insufficiency), terminal ileal disease or resection, and drugs (metformin, PPI). Lab: macrocytic anemia (MCV >100), hypersegmented neutrophils, low B12, elevated methylmalonic acid and homocysteine. Treatment: intramuscular cyanocobalamin 1000 mcg daily for one week, then weekly for one month, then monthly for life; high-dose oral B12 (1000-2000 mcg/day) may be adequate for dietary deficiency.

von Willebrand Disease

– The most common inherited bleeding disorder, caused by quantitative or qualitative deficiency of von Willebrand factor (VWF), a multimeric glycoprotein that mediates platelet adhesion to subendothelium and carries factor VIII. Type 1 (partial quantitative deficiency, 70-80%): mild mucocutaneous bleeding, easy bruising, menorrhagia. Type 2 (qualitative defects): 2A (loss of high-molecular-weight multimers, decreased platelet adhesion), 2B (increased affinity for platelet GPIb (thrombocytopenia), 2M (normal multimers, decreased platelet adhesion), 2N (decreased factor VIII binding, mimics hemophilia A). Type 3 (severe quantitative deficiency). Treatment: desmopressin (DDAVP) for mild type 1; von Willebrand factor-containing concentrates (Humate-P) for more severe bleeding or surgical prophylaxis; tranexamic acid for mucocutaneous bleeding; avoid aspirin and NSAIDs.

Waldenstrom Macroglobulinemia

– A rare B-cell lymphoplasmacytic lymphoma characterized by bone marrow infiltration by lymphoplasmacytic cells and production of monoclonal IgM paraprotein. It accounts for less than 2 percent of hematologic malignancies and typically affects older adults (median age 63-65 years). Clinical manifestations result from both marrow infiltration (anemia, cytopenias) and IgM-related effects: hyperviscosity syndrome (visual disturbances, neurologic symptoms, bleeding, and cryoglobulinaemia. Diagnosis: serum protein electrophoresis (M-spike), immunofixation (IgM), bone marrow biopsy (lymphoplasmacytic infiltrate >10%). Treatment: symptomatic or progressive disease: rituximab alone or in combination with bendamustine, cyclophosphamide/dexamethasone, or bortezomib; plasmapheresis for hyperviscosity syndrome; ibrutinib for MYD88-mutated disease; venetoclax for relapsed/refractory.

Warfarin Management

– Warfarin (vitamin K antagonist) inhibits gamma-carboxylation of factors II, VII, IX, X, proteins C and S. Target INR: 2-3 (most indications), 2.5-3.5 (mechanical mitral valve, recurrent VTE on therapeutic INR). Initiation: 5 mg daily x2-3 days, then adjust per INR; genotype-guided (CYP2C9, VKORC1) algorithms. Interactions: numerous (antibiotics, amiodarone, antifungals, NSAIDs, herbal). Monitoring: INR frequency per stability. Bridging: therapeutic LMWH when INR subtherapeutic for high thrombotic risk. Management: VTE treatment with DOACs or LMWH bridging to warfarin. Long-term anticoagulation based on recurrence risk and bleeding risk. Regular INR monitoring, dietary consistency (vitamin K intake), and medication review to avoid interactions.

White Cell

– Also known as leukocytes, the cells of the immune system that defend the body against infection and foreign substances. Produced in the bone marrow from hematopoietic stem cells. Types: neutrophils (50-70%, first responders to bacterial infection, phagocytosis), lymphocytes (20-40%, adaptive immunity — T-cells, B-cells, NK cells), monocytes (2-8%, differentiate into macrophages and dendritic cells), eosinophils (1-4%, allergic reactions, parasitic infections), and basophils (<1%, allergic responses, release histamine). Normal WBC count: 4,000-11,000/μL. Leukocytosis (elevated WBC): infection, inflammation, stress, leukemia. Leukopenia (low WBC): bone marrow suppression, infection, autoimmune disorders.

White Cell Count

– A laboratory measurement of the number of white blood cells (leukocytes) in a given volume of blood, part of the complete blood count (CBC). Normal range: 4,000-11,000 cells/μL (4-11 $\times$ 10^9/L). Differential white cell count measures the proportion of each leukocyte type: neutrophils, lymphocytes, monocytes, eosinophils, basophils. An elevated WBC (leukocytosis) suggests infection, inflammation, tissue damage, or hematologic malignancy. A decreased WBC (leukopenia) suggests bone marrow suppression from chemotherapy, infection (HIV, sepsis), autoimmune destruction, or drug toxicity. Marked leukocytosis (>100,000/μL) is a defining feature of chronic leukemias (CLL, CML). Left shift: increased band neutrophils, indicating acute bacterial infection.

Chapter 13

Immunology

Allergen

– A substance capable of inducing an allergic (IgE-mediated) immune response in susceptible individuals. Allergens are typically proteins or glycoproteins with molecular weights between 10 and 70 kilodaltons, possessing structural features that promote IgE binding including protease resistance and the ability to crosslink IgE on mast cells. Major allergen families include profilins (pan-allergens in pollen, latex, fruits), lipid transfer proteins (LTPs, important in food allergy, particularly to peaches, apples, and nuts), and tropomyosins (shellfish allergens).

Allergic Reactions

– Hypersensitivity reactions mediated by IgE antibodies in response to otherwise harmless environmental substances (allergens). Type I immediate hypersensitivity reactions involve initial sensitization where dendritic cells present allergen to T helper 2 cells, stimulating B cell class switching to IgE production. IgE binds to high-affinity Fc-epsilon-RI receptors on mast cells and basophils, priming these cells. Upon re-exposure, allergen crosslinks surface-bound IgE, triggering degranulation and release of histamine, leukotrienes, and prostaglandins, causing vasodilation, bronchoconstriction, and increased vascular permeability.

Allergic Rhinitis (Hay Fever)

– Allergies occur when the immune system overreacts to harmless environmental substances such as pollen, dust mites, mold, or pet dander. Hay fever (allergic rhinitis) specifically involves nasal symptoms like sneezing, congestion, and itchy eyes triggered by airborne allergens. Management includes avoidance, antihistamines, nasal corticosteroids, and immunotherapy.

Anaphylactic Shock

– See Anaphylaxis.

Ataxia-Telangiectasia

– Autosomal recessive (ATM gene mutation, DNA damage response kinase). Progressive cerebellar ataxia (gait, truncal, speech), oculocutaneous telangiectasias (conjunctiva, ears, face, 3-6 years), sinopulmonary infections (IgA deficiency in 70%, IgG2 subclass deficiency), increased radiosensitivity, high malignancy risk (lymphoma, leukemia, 10-15%), endocrinopathies (diabetes, gonadal failure). Elevated AFP in 95%. Diagnosis: ATM sequencing, radiosensitivity assay. Treatment: IVIG for recurrent infections; avoid live vaccines and unnecessary radiation.

Autoimmune Diseases

– Autoimmune diseases occur when the immune system mistakenly attacks the body's own healthy tissues, producing chronic inflammation and organ damage. There are over 80 types, including rheumatoid arthritis, lupus, type 1 diabetes, multiple sclerosis, and Hashimoto's thyroiditis. Treatment typically focuses on suppressing immune activity with corticosteroids, immunomodulators, or biologics.

Autoimmune Polyendocrine Syndromes

– APS-1 (APECED, AIRE mutation): autosomal recessive, chronic mucocutaneous candidiasis, hypoparathyroidism, adrenal insufficiency (Addison's), plus type 1 diabetes, autoimmune thyroiditis, vitiligo, alopecia, pernicious anemia, hepatitis, asplenia. APS-2 (Schmidt syndrome): autoimmune adrenal insufficiency + autoimmune thyroid disease (Hashimoto's or Graves) + type 1 diabetes; associated with HLA-DR3/DR4. APS-3: autoimmune thyroid disease + type 1 diabetes (no adrenal). APS-4: other combinations of autoimmune endocrinopathies not fitting APS-1, 2, or 3.

Autoimmunity: Mechanisms

– Loss of tolerance: central (thymic negative selection defects, AIRE mutation -> APS-1) and peripheral (Treg dysfunction, anergy failure, ignorance loss). Molecular mimicry: cross-reactivity between pathogen and self (rheumatic fever: Strep M protein + cardiac myosin; Guillain-Barre: Campylobacter + GM1 ganglioside; reactive arthritis). Epitope spreading: immune response expands to new epitopes on same or different antigens (intramolecular/intermolecular). Bystander activation: inflammation activates autoreactive T cells non-specifically in the local milieu (e.g., in type 1 diabetes, myocarditis).

Chediak-Higashi Syndrome

– Autosomal recessive (LYST mutation, lysosomal trafficking regulator). Giant lysosomal granules in neutrophils, melanocytes, neurons. Partial oculocutaneous albinism, silvery hair, photophobia, nystagmus. Recurrent pyogenic infections (Staph, Strep), accelerated phase (HLH-like, 85%): fever, hepatosplenomegaly, pancytopenia, neurologic decline. Peripheral smear: giant azurophilic granules in neutrophils. Treatment: HSCT before accelerated phase; ascorbic acid improves chemotaxis.

Chronic Granulomatous Disease

– A primary immunodeficiency caused by defects in the NADPH oxidase complex in phagocytes (neutrophils, monocytes, macrophages), impairing the respiratory burst and killing of catalase-positive organisms. The most common form is X-linked (CYBB gene mutation, encoding gp91phox, accounting for 65-70 percent of cases); autosomal recessive forms involve mutations in NCF1 (p47phox, 30 percent), NCF2 (p67phox), NCF4 (p40phox), or CYBA (p22phox). The defective phagocyte respiratory burst leads to recurrent infections with catalase-positive organisms including S. aureus, Serratia marcescens, Burkholderia cepacia, Nocardia, and Aspergillus.

Chronic Mucocutaneous Candidiasis

– Persistent/recurrent Candida infections of skin, nails, mucosa (oral, esophageal, vaginal) without systemic dissemination. Autosomal dominant (STAT1 gain-of-function) or recessive (IL17F, IL17RA, ACT1, RORC). Associated with autoimmunity (hypothyroidism, AIHA, ITP, alopecia), endocrinopathies, Staph infections. STAT1 GOF: multi-organ autoimmunity, cerebral aneurysms. Treatment: azole antifungals (fluconazole, itraconazole), JAK inhibitors (ruxolitinib) for STAT1 GOF; IL-17 pathway defects respond to recombinant IL-17 therapy.

Common Variable Immunodeficiency

– The most common symptomatic primary immunodeficiency in adults, affecting approximately 1 in 25,000 individuals. It is characterized by hypogammaglobulinemia (low IgG with low IgA and/or IgM), impaired antibody response to vaccines, and increased susceptibility to infections. The pathogenesis is heterogeneous: approximately 50 percent of cases have identifiable genetic mutations (TNFSF13B/BAFF-R, TACI, MS4A1/CD20, IKZF1, CD19), while the remainder are idiopathic.

Complement C1q Deficiency

– Autosomal recessive (C1QA, C1QB, C1QC). C1q essential for classical pathway activation and immune complex clearance. 85-90% develop SLE-like disease (malar rash, photosensitivity, nephritis, antinuclear antibodies). Recurrent infections with encapsulated bacteria. Low CH50, low C1q, normal C3/C4. Treatment: SLE management (hydroxychloroquine, immunosuppressives); prophylactic antibiotics; IVIG for infections.

Complement C2/C4 Deficiency

– C2 deficiency: most common early complement deficiency (1:20,000 Caucasians). SLE-like disease (10-30%), recurrent sinopulmonary infections. C4 deficiency: similar, strong SLE association (C4A/C4B null alleles). Low CH50, low C2 or C4, normal C3. Treatment: SLE management, prophylactic antibiotics, IVIG for recurrent infections.

Complement Deficiencies

– Rare primary immunodeficiencies caused by genetic defects in complement pathway proteins, increasing susceptibility to specific infections and autoimmune diseases. Terminal complement deficiencies (C5-C9) impair membrane attack complex (MAC) formation, leading to recurrent Neisseria infections (meningococcal disease, gonococcemia) with a 10,000-fold increased risk compared to the general population.

Vaccination against Neisseria meningitidis (quadrivalent ACWY and serogroup B vaccines), prophylactic antibiotics (penicillin), and prompt treatment of infections.

Cytokine Release Syndrome

– CRS: systemic inflammatory response from massive cytokine release (IL-6, IFN-gamma, TNF, IL-10) after T cell engaging therapies (CAR-T, bispecific antibodies, TGN1412). Grading (ASTCT): Grade 1 (fever), Grade 2 (hypoxia/hypotension responsive to O2/fluids), Grade 3 (hypoxia/hypotension requiring O2/vasopressors), Grade 4 (life-threatening). Management: tocilizumab (anti-IL-6R) first-line; corticosteroids for neurotoxicity (ICANS) or refractory CRS; anakinra for steroid-refractory; siltuximab (anti-IL-6) for refractory CRS.

DiGeorge Syndrome

– 22q11.2 deletion syndrome (most common microdeletion syndrome, 1:4000). Thymic hypoplasia/aplasia -> T-cell deficiency (variable, partial in 75%, complete in <1%), cardiac defects (conotruncal: TOF, truncus arteriosus, interrupted aortic arch), hypocalcemia (parathyroid hypoplasia), facial features, palatal abnormalities, learning disabilities. Immunodeficiency: variable CD4+ T-cell lymphopenia, impaired vaccine responses, autoimmunity risk. Treatment: calcium/vitamin D for hypocalcemia; IVIG if antibody deficiency is present; thymus transplant for complete DiGeorge.

Factor H Deficiency

– Autosomal recessive (CFH mutation). Factor H regulates alternative pathway (cofactor for factor I, accelerates decay). Uncontrolled alternative pathway activation -> C3 consumption -> low C3, normal C4. Atypical HUS (aHUS): microangiopathic hemolytic anemia, thrombocytopenia, acute kidney injury. C3 glomerulopathy (C3GN/DDD): C3-dominant deposits on IF. Treatment: eculizumab (anti-C5) for aHUS and C3G; plasma exchange for acute aHUS.

Factor I Deficiency

– Autosomal recessive (CFI mutation). Factor I cleaves C3b/C4b (cofactor: factor H, MCP, C4BP). Uncontrolled complement -> low C3, low C4, low factor H. Recurrent pyogenic infections, aHUS, C3 glomerulopathy, SLE-like disease. Treatment: eculizumab for aHUS/C3G; plasma infusion (contains factor I); IVIG.

Granuloma

– A distinctive pattern of chronic inflammation characterized by the aggregation of modified macrophages (epithelioid histiocytes), often with multinucleated giant cells (Langhans or foreign-body type), surrounded by lymphocytes. Granulomas form when macrophages cannot eliminate a persistent antigen or pathogen. Types: (1) Caseating (central necrosis): tuberculosis, fungal infections (histoplasmosis, coccidioidomycosis), syphilis. (2) Non-caseating: sarcoidosis, Crohn's disease, berylliosis, foreign body (talc, suture, silicone), granulomatosis with polyangiitis (GPA). (3) Suppurative (with neutrophils): actinomycosis, cat-scratch disease. Diagnosis: tissue biopsy (histology), special stains (AFB for TB, GMS for fungi), culture, PCR for specific pathogens. Treatment: depends on etiology (anti-TB therapy, corticosteroids for sarcoidosis, surgical excision for foreign body).

Griscelli Syndrome

– Three types: Type 1 (MYO5A mutation): primary neurologic impairment, no immunodeficiency; Type 2 (RAB27A mutation): immunodeficiency + partial albinism, HLH risk; Type 3 (MLPH mutation): only partial albinism. Type 2: silvery hair, large melanosomes in melanocytes, defective cytotoxic granule release -> HLH. Treatment: HSCT for type 2; supportive for types 1 and 3.

Hemophagocytic Lymphohistiocytosis

– HLH: uncontrolled immune activation with cytokine storm (IFN-gamma, IL-6, IL-10, TNF). Familial HLH (FHL): PRF1 (FHL2), UNC13D (FHL3), STX11 (FHL4), STXBP2 (FHL5), RAB27A (FHL6/Griscelli type 2). Secondary HLH: infections (EBV, CMV), malignancy (lymphoma), autoimmune (SJIA/MAS, SLE), immunodeficiency. HLH-2004 criteria: 5/8 (fever, splenomegaly, cytopenias >=2 lineages, hypertriglyceridemia/hypofibrinogenemia, hemophagocytosis, low NK activity, ferritin >500, sCD25 >2400). Treatment: dexamethasone and etoposide (HLH-2004 protocol); emapalumab (anti-IFN-gamma) for refractory HLH; HSCT for familial HLH.

Hermansky-Pudlak Syndrome

– Autosomal recessive (10+ genes: HPS1-10, BLOC/AP-3 complexes). Oculocutaneous albinism, bleeding disorder (absent dense bodies in platelets -> prolonged bleeding time), pulmonary fibrosis (HPS-1, HPS-4), granulomatous colitis, neutropenia. Diagnosis: electron microscopy of platelets (absent dense bodies), genetic testing. Treatment: pulmonary fibrosis management (pirfenidone, nintedanib), avoid NSAIDs, HSCT not standard.

HIV/AIDS Immunopathogenesis

– HIV-1 infects CD4+ T cells, macrophages, dendritic cells via gp120 binding to CD4 + CCR5 (R5-tropic, early) or CXCR4 (X4-tropic, late). Reverse transcription -> integration -> transcription/translation -> budding. Acute retroviral syndrome (2-4 weeks): fever, lymphadenopathy, pharyngitis, rash, high viral load. Clinical latency (8-10 years): CD4 declines from 500-1500 to <200. AIDS: CD4 <200 or AIDS-defining OIs (PJP <200, toxoplasma encephalitis <100, CMV retinitis <50, MAC <50, cryptococcal meningitis <50).

Hyper-IgM Syndromes

– A group of disorders characterized by normal or elevated IgM with low IgG, IgA, and IgE, due to defective class switch recombination (CSR) and somatic hypermutation (SHM). X-linked (CD40LG mutation, CD40 ligand deficiency, 70% of cases): T cells cannot provide CD40 signals to B cells; presents with recurrent pyogenic infections, Pneumocystis jirovecii pneumonia (PJP), cryptosporidiosis (sclerosing cholangitis), neutropenia, increased malignancy risk. Autosomal recessive: AID (activation-induced cytidine deaminase, AICDA), UNG (uracil N-glycosylase), or other CSR/SHM defects.

Immune and Infectious Diseases

– A medical field encompassing disorders of the immune system and illnesses caused by pathogenic microorganisms. Immune system disorders include autoimmune diseases (e.g., lupus, rheumatoid arthritis), immunodeficiencies, and hypersensitivity reactions. Infectious diseases range from viral respiratory infections to bacterial, fungal, and parasitic illnesses, requiring prevention through vaccination and treatment with antimicrobial agents.

Immunoglobulin Replacement Therapy

– Indications: primary antibody deficiencies (XLA, CVID, hyper-IgM, specific antibody deficiency), secondary hypogammaglobulinemia (CLL, myeloma, post-rituximab, post-transplant) with recurrent infections. IVIG: 400-600 mg/kg every 3-4 weeks (trough IgG >700-800 mg/dL); SCIG: 100-200 mg/kg/week (more stable levels, fewer systemic reactions). Products: varying IgA content (low IgA for anti-IgA antibodies), sucrose-free (renal safety), glycine, stabilizing agents. Adverse: headache, aseptic meningitis, infusion reactions, thrombosis; anaphylaxis in IgA deficiency (use low-IgA products).

Immunotherapy and Biologic Treatments

– Advanced therapies that harness or modify the immune system to treat diseases, particularly cancer, autoimmune disorders, and inflammatory conditions. Biologics are large-molecule drugs derived from living organisms that target specific immune pathways, such as TNF inhibitors, monoclonal antibodies, and checkpoint inhibitors. These treatments offer precision targeting but require careful monitoring for immune-related adverse effects.

Inflammasomes

– Multiprotein complexes activating caspase-1 -> cleavage of pro-IL-1beta/pro-IL-18 -> mature IL-1beta/IL-18 + pyroptosis (gasdermin D pores). Canonical: NLRP3 (crystals, ATP, K+ efflux, ROS, lysosomal rupture), NLRC4 (flagellin, T3SS), AIM2 (cytosolic DNA), Pyrin (RhoA inactivation). Non-canonical: caspase-11 (human caspase-4/5) activated by cytosolic LPS -> GSDMD cleavage -> pyroptosis + NLRP3 activation. Inflammasomopathies: CAPS (NLRP3 GOF), FMF (MEFV/pyrin LOF -> pyrin inflammasome overactivation), NLRC4 GOF (macrophage activation syndrome).

Inflammation

– The body's biological response to harmful stimuli such as pathogens, damaged cells, or irritants, characterized by redness, heat, swelling, pain, and loss of function. Acute inflammation is a protective process that eliminates the initial cause of injury and initiates tissue repair. Chronic inflammation, however, contributes to the pathogenesis of numerous diseases including atherosclerosis, diabetes, arthritis, and cancer.

Innate Lymphoid Cells

– ILCs: lymphoid lineage, no rearranged antigen receptors, mirror Th subsets. ILC1 (T-bet, IFN-gamma): intracellular pathogens, IBD. ILC2 (GATA3, IL-5/13): helminths, allergy, asthma, tissue repair. ILC3 (ROR-gammat, IL-17/22): extracellular bacteria/fungi, lymphoid organogenesis. LTi cells (ILC3 subset): lymph node/Peyer's patch development. NK cells (Eomes, perforin/granzyme): cytotoxic, missing-self recognition. ILCs in disease: ILC2 in asthma/CRSwNP (target: anti-IL-33, anti-TSLP, anti-IL-5); ILC3 in IBD and psoriasis; ILC1 in chronic inflammation.

Interferon Signaling

– Type I IFN (IFN-alpha/beta): all nucleated cells produce; receptor IFNAR (ubiquitous); JAK1/TYK2 -> STAT1/2/IRF9 (ISGF3) -> ISGs (hundreds: antiviral, antiproliferative, immunomodulatory). Type II IFN (IFN-gamma): T/NK cells; receptor IFNGR; JAK1/JAK2 -> STAT1 -> GAS elements. Type III IFN (IFN-lambda): mucosal epithelium; receptor IFNLR. IFN signatures: high in SLE (IFN-alpha), Aicardi-Goutieres (TREX1, RNASEH2, SAMHD1, ADAR1 mutations -> cytosolic nucleic acid accumulation -> IFN), SAVI (STING GOF), and COPA syndrome.

Leukocyte Adhesion Deficiency

– LAD-I (ITGB2 mutation, CD18 deficiency): absent CD11/CD18 integrins -> impaired neutrophil adhesion/transmigration -> delayed umbilical cord separation (>30 days), recurrent soft tissue infections without pus formation (leukocytosis >20,000), poor wound healing, periodontitis. Severe: death by age 2 without HSCT. LAD-II (SLC35C1 mutation, GDP-fucose transporter): absent sialyl Lewis X -> impaired rolling; mental retardation, short stature, Bombay blood group. LAD-III (FERMT3 mutation, kindlin-3): impaired integrin activation leading to a similar phenotype to LAD-I plus Glanzmann-like bleeding tendency.

Lymph

– A clear, watery fluid that circulates through the lymphatic system, derived from interstitial fluid that has entered the lymphatic capillaries. Lymph composition: water, electrolytes, small amounts of protein (especially in liver and intestine), white blood cells (predominantly lymphocytes, 95%, the remainder are monocytes, macrophages, and occasional granulocytes), cellular debris, microorganisms, and absorbed dietary lipids (chylomicrons, giving lymph a milky appearance–chyle). Lymph is formed when hydrostatic and osmotic pressures across the capillary wall cause net filtration of fluid into the interstitial space. Approximately 2-4 L of lymph per day drains through the thoracic duct into the left subclavian vein. Lymph flow is propelled by: contraction of smooth muscle in the wall of collecting lymphatics (intrinsic contractility), compression by skeletal muscle contraction, respiratory motion, and pulsation from adjacent arteries. One-way valves prevent backflow. Lymph nodes filter lymph, removing pathogens and antigens, and serving as sites for immune surveillance and activation. The lymphatic system has crucial roles in: fluid homeostasis (returns excess interstitial fluid to the bloodstream), immune surveillance (lymphocytes recirculate through lymph nodes, detecting and responding to antigens), and fat absorption (intestinal lacteals absorb dietary lipids).

Lymphadenopathy

– Enlargement of lymph nodes (greater than 1 cm in diameter, or >1.5 cm for inguinal nodes). Lymphadenopathy can be localised (single or contiguous region) or generalised (three or more non-contiguous regions). Localised causes: infection (most common–pharyngitis/tonsillitis causing cervical adenopathy, wound infection causing axillary/inguinal adenopathy, TB causing supraclavicular/scalene adenopathy, cat-scratch disease causing axillary/epitrochlear adenopathy), malignancy (lymphoma, metastatic cancer, leukemia), autoimmune (sarcoidosis, SLE, rheumatoid arthritis). Generalised causes: infectious (EBV, CMV, HIV, TB, toxoplasmosis, brucellosis), malignant (lymphoma, CLL, ALL), autoimmune (SLE, sarcoidosis, Still disease), drug-induced (phenytoin, allopurinol), and storage diseases (Gaucher, Niemann-Pick). Diagnostic evaluation: history (duration, size, tenderness, fever, night sweats, weight loss, exposure, travel), physical exam (location, size, consistency–hard/fixed suggests malignancy, soft/mobile suggests infection, matted suggests TB/sarcoidosis, fluctuant suggests abscess), and workup: full blood count, CRP/ESR, serology (EBV, CMV, HIV, toxoplasma), chest X-ray, ultrasound of nodes, and excisional biopsy (indications: node >2 cm, supraclavicular location, firm/fixed, no regression after 4-6 weeks, systemic symptoms, or abnormal CXR).

Lymphangioma

– A benign congenital malformation of the lymphatic system, composed of dilated lymphatic channels. Lymphangiomas are typically present at birth (90% diagnosed by age 2) and result from failure of the lymphatic system to develop normal connections with the venous system. Types: (1) Cystic hygroma: most common (50-60%), large multiloculated cysts, typically in the neck (posterior triangle) or axilla. Antenatal ultrasound may detect as a nuchal translucency (associated with Turner syndrome, Noonan syndrome, trisomy 21). (2) Cavernous lymphangioma: deep, involves neck, tongue, lip (macrodontia, macroglossia). (3) Lymphangioma circumscriptum: cutaneous vesicles on skin surface. Symptoms: soft, fluctuant, transilluminating mass, often painless. May enlarge rapidly with upper respiratory infection or trauma due to increased lymph production. Complications: infection (lymphangitis), intralesional bleeding, compression of airways, dysphagia, cosmetic deformity. Diagnosis: clinical, ultrasound (cystic, anechoic, septated), MRI (T2 hyperintense, fluid-fluid levels, no enhancement). Management: observation (30% may regress spontaneously, especially small ones), surgical excision (complete excision is curative, but recurrence risk of 10-40% if incomplete), sclerotherapy (OK-432, bleomycin, doxycycline, alcoholic solution of zein–Ethibloc, especially for cystic hygroma), sirolimus (mTOR inhibitor, for complex/refractory cases).

Lymphedema (Lymphoedema)

– Chronic swelling of a body part due to impaired lymphatic drainage, leading to accumulation of protein-rich interstitial fluid. Primary lymphedema: caused by congenital or hereditary lymphatic maldevelopment. Milroy disease (autosomal dominant, VEGFR3 mutation, congenital, present at birth, 2:1 female:male, lower limbs, non-pitting), Meige disease (lymphedema praecox, onset at puberty, 2:1 female:male, lower limbs up to knee, mild), and late-onset lymphedema (tarda, >35 years). Secondary lymphedema: more common, acquired damage to lymphatic vessels or nodes. Causes: cancer-related (axillary lymph node dissection for breast cancer–most common cause of upper limb lymphedema, pelvic node dissection for gynaecological/ urological cancers, melanoma), radiation therapy (lymphatic fibrosis), filariasis (Wuchereria bancrofti, Brugia malayi–most common cause worldwide, mosquito-borne nematode infection, tropical regions, causes massive leg and genital swelling, elephantiasis), trauma/surgery (lymph node excision), recurrent cellulitis, liposuction (may worsen), and venous disease (phlebolymphedema). Pathophysiology: lymphatic obstruction leads to accumulation of protein-rich fluid, which triggers chronic inflammation, adipose deposition, and fibrosis. Over time, the skin becomes thickened, hyperkeratotic, and papillomatous. Diagnosis: clinical (non-pitting edema Stemmer sign–inability to pinch the skin of the dorsum of the second toe, specific for lymphedema), lymphoscintigraphy (radionuclide imaging of lymphatic transport, gold standard), MRI (honeycomb pattern, dermal thickening, edema fluid), ICG fluorescence lymphangiography. Management: complex decongestive therapy (CDT, gold standard)–phase 1 (intensive, 2-4 weeks, manual lymphatic drainage MLD, compression bandaging, exercise, skincare) and phase 2 (maintenance, daytime compression garments, nighttime bandaging, self-MLD). Additional: pneumatic compression devices, laser therapy, liposuction (for non-pitting, fibrotic, late-stage lymphedema), surgical (lymphaticovenous anastomosis LVA, vascularised lymph node transfer VLNT, Charles procedure for severe elephantiasis). Antibiotic prophylaxis (penicillin V, 250 mg BD) for recurrent cellulitis. Long-term management with patient education, weight control, and skincare.

Lymph Node / Gland

– Small, bean-shaped organs of the lymphatic system that filter lymph and serve as sites for immune cell activation. Lymph nodes are distributed throughout the body (approximately 500-600 in total) and are concentrated in regions: cervical, axillary, inguinal, mediastinal, mesenteric, para-aortic, and iliac chains. Structure: fibrous capsule, cortex (B-cell follicles–primary and secondary with germinal centers), paracortex (T-cell zone, high endothelial venules), and medulla (medullary cords with plasma cells and macrophages, medullary sinuses). The subcapsular sinus receives afferent lymphatic vessels. The hilum contains the efferent lymphatic vessel and blood vessels. Functions: filtration–macrophages in the sinuses phagocytose particulate matter and microorganisms; immune surveillance–antigens are presented to lymphocytes by dendritic cells, leading to B-cell and T-cell activation, antibody production, and generation of memory cells. Lymph nodes enlarge (lymphadenopathy) in response to infection, inflammation, or malignancy. Palpable lymph nodes in adults: >1 cm in most regions is considered abnormal, except inguinal nodes (1.5 cm). Sentinel lymph node: the first lymph node draining a tumor, biopsied to stage cancer spread (e.g., melanoma, breast cancer).

Lymphocyte

– A type of white blood cell (leukocyte) that is the primary cellular component of the adaptive immune system. Lymphocytes account for 20-40% of circulating white blood cells (1.0–4.8 $\times$ 10^9/L). Three main types: (1) T lymphocytes (T cells): develop in the thymus, responsible for cell-mediated immunity. Subsets: CD4+ helper T cells (Th1, Th2, Th17, Tfh, Treg) orchestrate immune responses by secreting cytokines; CD8+ cytotoxic T cells kill virus-infected cells, tumor cells, and allograft cells. (2) B lymphocytes (B cells): develop in the bone marrow, responsible for humoral immunity. Upon activation, B cells differentiate into plasma cells (secrete antibodies–immunoglobulins IgG, IgA, IgM, IgE, IgD) and memory B cells (long-lived, rapid response upon re-exposure). (3) Natural killer (NK) cells: part of the innate immune system, kill virus-infected cells and tumor cells without prior sensitisation (no antigen receptor rearrangement). Lymphocyte development: hematopoietic stem cells in bone marrow differentiate into common lymphoid progenitors. B cells mature in the bone marrow, T cells migrate to the thymus for maturation (positive selection for self-MHC recognition, negative selection against self-reactivity). After encountering antigen, lymphocytes undergo clonal expansion and differentiate into effector and memory cells. Lymphocyte recirculation: naive lymphocytes travel from blood to lymph nodes via high endothelial venules (HEVs), then to lymph, and back to blood. Lymphocyte disorders: lymphocytosis (viral infection, CLL), lymphopenia (HIV, chemotherapy, immunodeficiency), lymphoma (malignant proliferation of lymphocytes), and autoimmune diseases (dysregulated lymphocyte reactivity).

Macrophage Activation Syndrome

– MAS is a form of secondary HLH complicating rheumatic diseases (SJIA, Kawasaki, SLE, AOSD). MAS in SJIA: cytopenias, hyperferritinemia, elevated LDH, elevated triglycerides, low fibrinogen, transaminitis. High mortality (20-50%). IL-1/IL-6 driven. Treatment: high-dose corticosteroids, anakinra (IL-1 blockade), tocilizumab (IL-6 blockade), cyclosporine A, etoposide for severe.

Mannose-Binding Lectin Deficiency

– MBL2 polymorphisms (common, 10-30% heterozygotes, 1-5% homozygotes). MBL activates lectin pathway. Low MBL associated with increased infections in immunocompromised (chemotherapy, transplant), recurrent infections in children, autoimmune disease susceptibility. Diagnosis: serum MBL level, MBL2 genotyping. Treatment: usually asymptomatic; IVIG for recurrent infections in immunocompromised.

Neutrophil Extracellular Traps

– NETs: web-like structures of chromatin + granular proteins (MPO, NE, histone) released by neutrophils (NETosis: suicidal vs vital). Functions: trap/kill pathogens (bacteria, fungi, viruses); pathological in thrombosis (NETs provide scaffold for platelets, activate coagulation), autoimmunity (SLE: NETs expose LL-37/DNA -> pDC activation -> IFN-alpha; RA: citrullinated antigens in NETs -> ACPA), cancer (NETs trap circulating tumor cells -> metastasis), COVID-19 (immunothrombosis). DNase I degrades NETs; therapeutic DNase (dornase alfa) used in NET-related pathology.

Pattern Recognition Receptors

– PRRs recognize PAMPs (pathogen) and DAMPs (damage). TLRs: TLR1/2 (lipoproteins), TLR2/6 (lipoteichoic acid), TLR3 (dsRNA), TLR4 (LPS, MD-2, CD14), TLR5 (flagellin), TLR7/8 (ssRNA), TLR9 (CpG DNA). Cytosolic: RIG-I/MDA5 (RNA), cGAS (DNA -> cGAMP -> STING -> IFN), NOD1/2 (peptidoglycan). CLRs: Dectin-1 (beta-glucan), Mincle (mycobacterial TDM), DC-SIGN (mannose). NLRs: NOD1/2 (cytosolic bacteria), NLRP3 (inflammasome). Signaling: MyD88 (all TLRs except TLR3) -> NF-kB/IRF; TRIF (TLR3/4) -> IRF3/NF-kB activation.

Phagocytosis

– The process by which specialized cells (phagocytes) engulf and destroy foreign particles, microorganisms, and cellular debris. Professional phagocytes: neutrophils (most abundant, first responders, short-lived), macrophages (tissue-resident—Kupffer cells in liver, alveolar macrophages in lung, microglia in brain, osteoclasts in bone), dendritic cells (antigen presentation to T cells), and eosinophils (parasite defence). Phagocytosis steps: (1) Chemotaxis—phagocytes migrate to the site of infection guided by chemoattractants (IL-8, C5a, bacterial peptides). (2) Recognition and opsonisation—pathogens are coated with opsonins (IgG antibodies, complement C3b, collectins) that bind phagocyte receptors (Fcγ receptors, complement receptors, mannose receptors). (3) Engulfment—the phagocyte extends pseudopods around the particle, forming a phagosome. (4) Phagosome–lysosome fusion—the phagosome fuses with lysosomes to form a phagolysosome; the pathogen is killed by reactive oxygen species (respiratory burst—NADPH oxidase produces superoxide), nitric oxide, antimicrobial peptides (defensins, cathelicidins), and hydrolytic enzymes (lysozyme, proteases). (5) Exocytosis—degraded material is expelled. Defects in phagocytosis cause recurrent infections: chronic granulomatous disease (NADPH oxidase deficiency—inability to produce reactive oxygen species), Chediak–Higashi syndrome (defective phagolysosome fusion), and leukocyte adhesion deficiency (impaired migration).

Properdin Deficiency

– X-linked recessive (CFP gene). Properdin stabilizes alternative pathway C3 convertase. Recurrent Neisseria infections (meningococcemia, gonococcemia). Low AH50, normal CH50, low properdin. Treatment: meningococcal vaccination (ACWY + B), prophylactic penicillin, IVIG for severe infections.

Regulatory T Cells

– Tregs (CD4+ CD25+ FOXP3+): essential for peripheral tolerance. Thymic (tTreg) vs peripheral (pTreg) origin. FOXP3: master transcription factor; mutations cause IPEX. Mechanisms: IL-2 consumption (CD25 high), CTLA-4 (transendocytosis of CD80/86), TGF-beta/IL-10 secretion, cAMP transfer, granzyme/perforin killing of APCs, metabolic disruption (CD39/CD73 -> adenosine). Treg defects: IPEX (FOXP3), autoimmune polyendocrinopathy. Therapeutic: low-dose IL-2 expands Tregs (trials in SLE, GVHD); Treg adoptive transfer (trials in GVHD, type 1 diabetes, transplant rejection).

Secondary Immunodeficiencies

– Acquired immune defects from: HIV/AIDS (CD4+ T-cell depletion), malignancies (CLL: hypogammaglobulinemia; myeloma: hypogammaglobulinemia + impaired specific antibody; lymphoma), immunosuppressive medications (corticosteroids, chemotherapy, biologics: rituximab -> prolonged B-cell depletion and hypogammaglobulinemia; TNF inhibitors -> TB reactivation; alemtuzumab -> profound T-cell depletion), protein-losing states (nephrotic syndrome, enteropathy, burns), splenectomy (encapsulated bacteria risk, asplenia), and advanced age (immunosenescence).

Severe Combined Immunodeficiency

– Severe combined immunodeficiency (SCID) represents a group of genetic disorders characterized by profoundly defective T cell development and function, with variable B and NK cell involvement, resulting in severe susceptibility to life-threatening infections in infancy. It is the most serious primary immunodeficiency, with a combined incidence of approximately 1 in 50,000-100,000 live births. The classic form (X-linked SCID, accounting for approximately 45-50 percent of cases) is caused by mutations in the IL2RG gene (common gamma chain), resulting in absent T cells and NK cells with normal or elevated B cells (T-B+NK- SCID).

T-Cells (T-Lymphocytes)

– A type of lymphocyte that matures in the thymus and is the central effector of cell-mediated immunity. T-cells express the T-cell receptor (TCR) on their surface, which recognises peptide antigens presented by major histocompatibility complex (MHC) molecules on antigen-presenting cells. T-cells are divided into subsets by function and surface markers: (1) CD4+ helper T cells (Th)—recognise MHC class II, orchestrate immune responses. Subsets:

Th1 (IFN-γ, cell-mediated immunity against intracellular pathogens), Th2 (IL-4, IL-5, IL-13, humoral immunity, allergic responses), Th17 (IL-17, neutrophil recruitment, antifungal/extracellular bacterial immunity), Tfh (follicular helper T-cells—help B cells in germinal centers, antibody production), and Treg (regulatory T-cells, FOXP3+, suppress immune responses, maintain self-tolerance). (2) CD8+ cytotoxic T cells (CTL)—recognise MHC class I, kill virus-infected cells, tumor cells, and allograft cells via perforin/granzyme and Fas/FasL pathways. (3) $\gamma\delta$ T-cells—innate-like, recognise non-peptide antigens, bridge innate and adaptive immunity. T-cell development: bone marrow progenitor → thymus (positive selection in the cortex, negative selection in the medulla) → naive T-cell → circulation. Activation: requires TCR binding (signal 1) and co-stimulation (CD28–CD80/86, signal 2). Deficient T-cell function causes severe opportunistic infections (HIV/AIDS, severe combined immunodeficiency—SCID). CD4 count is the key marker of immune function in HIV (normal 500–1500 cells/μL; <200 defines AIDS).

Th1/Th2/Th17/Tfh/Treg Balance

– CD4+ T helper subsets defined by master transcription factors and signature cytokines: Th1 (T-bet, IFN-gamma, IL-12, STAT4): intracellular pathogens, autoimmunity (RA, MS, type 1 diabetes). Th2 (GATA3, IL-4/5/13, IL-4/STAT6): helminths, allergy, asthma. Th17 (ROR-gammat, IL-17A/F, IL-21, IL-22, IL-6+TGF-beta/IL-23/STAT3): extracellular bacteria/fungi, autoimmunity (PSA, AS, MS, IBD). Tfh (Bcl-6, IL-21, CXCR5): germinal center help, antibody affinity maturation, memory B cells. Treg (FOXP3, TGF-beta, IL-10, IL-35): suppression, immune regulation, tolerance.

Trained Immunity

– Innate immune memory: epigenetic/metabolic reprogramming of monocytes/macrophages/NK cells after initial stimulus (BCG, beta-glucan, Candida, oxidized LDL) -> enhanced response to secondary challenge (heterologous protection). Mechanisms: histone modifications (H3K4me3, H3K27ac), metabolic shifts (aerobic glycolysis, glutaminolysis, cholesterol synthesis), mTOR-HIF1alpha pathway. BCG vaccination protects against unrelated infections (neonatal sepsis, respiratory infections). Relevance: sepsis-induced immunoparalysis (trained immunity reversal).

Transplant Immunology

– Hyperacute rejection: preformed anti-donor antibodies (ABO, HLA) -> immediate graft failure (minutes-hours); prevented by crossmatch. Acute cellular rejection: recipient T cells recognize donor alloantigens (direct: donor APC -> recipient T cell; indirect: recipient APC presents donor peptides); days-months; treated with steroids, T cell depletion (thymoglobulin, alemtuzumab). Acute antibody-mediated rejection: de novo DSA -> C4d deposition, capillaritis, glomerulitis; treated with plasmapheresis, IVIG, rituximab, and bortezomib. Chronic rejection: interstitial fibrosis and tubular atrophy (IFTA) with DSA; poorly responsive to therapy.

Vaccine Immunology

– Live attenuated (MMR, varicella, rotavirus, yellow fever, oral typhoid, BCG, intranasal flu): replicate in host -> strong humoral + cellular immunity; contraindicated in immunocompromised. Inactivated/killed (polio IPV, hepatitis A, rabies, injectable flu): safer, weaker immunity, need boosters. Subunit/recombinant (hepatitis B, HPV, pertussis acellular, MenB, RSV): specific antigens, safe, need adjuvants. Toxoid (diphtheria, tetanus): inactivated toxin. mRNA (COVID-19 Pfizer/Moderna): lipid nanoparticles, highly effective, rapid development, require cold chain storage.

Vaccines and Immunizations

– Biological preparations that stimulate the immune system to develop protection against specific infectious diseases. By exposing the body to harmless components of a pathogen, vaccines enable the immune system to generate memory B and T cells that can mount a rapid response upon future exposure. Widespread immunization has dramatically reduced the burden of diseases such as polio, measles, influenza, and COVID-19, and remains one of the most cost-effective public health interventions for preventing morbidity and mortality.

Wiskott-Aldrich Syndrome

– X-linked recessive disorder (WAS gene mutation, encodes WASP protein, regulates actin polymerization in hematopoietic cells). Triad: microthrombocytopenia (small platelets, <70 fl), eczema, recurrent infections (encapsulated bacteria, viruses). Combined immunodeficiency with progressive T-cell lymphopenia. Increased risk of autoimmune disease (AIHA, ITP, vasculitis, IBD) and malignancy (lymphoma, 10-15%). Diagnosis: low WASP expression, WAS gene sequencing. Treatment: IVIG for infections, platelet transfusions for bleeding; splenectomy for refractory thrombocytopenia; HSCT as definitive cure.

X-Linked Agammaglobulinemia

– Caused by mutations in the BTK gene (Bruton tyrosine kinase), essential for B-cell development and maturation. Results in absent or very low mature B cells (<1% CD19+ cells), profoundly low all immunoglobulin isotypes, and absent antibody responses. Presents in male infants at 4-8 months (after maternal IgG wanes) with recurrent sinopulmonary infections (S. pneumoniae, H. influenzae, M. catarrhalis), enteroviral infections (chronic enteroviral meningoencephalitis), and lack of lymph nodes and tonsils. Diagnosis: low immunoglobulins, absent B cells, BTK sequencing. Treatment: lifelong IVIG or SCIG replacement therapy; avoid live vaccines; gene therapy is experimental.

Chapter 14

Infectious Disease

Actinomycosis

– A chronic, suppurative, and granulomatous bacterial infection caused by Actinomyces species, most commonly Actinomyces israelii, which are anaerobic or microaerophilic grampositive branching filaments that are part of the normal oral, gastrointestinal, and female genital flora. Infection typically follows disruption of mucosal barriers (dental procedures, trauma, surgery) and is characterized by slow-progressing, indolent infections that form abscesses, sinus tracts, and sulfur granules (yellow colonies of bacteria embedded in tissue). Cervicofacial actinomycosis ("lumpy jaw") is the most common form (50-60 percent), presenting as a slowly enlarging, indurated mass in the jaw or neck, often following dental extraction or trauma, with draining sinus tracts. Thoracic actinomycosis (20-25 percent) follows aspiration, presenting as a lung mass that may mimic malignancy, with chest pain, cough, fever, and weight loss. Abdominal actinomycosis (15-20 percent) typically follows appendicitis or bowel perforation, presenting as an ileocecal mass with abdominal pain, fever, and draining sinus tracts through the abdominal wall. Pelvic actinomycosis is associated with long-term intrauterine device (IUD) use, presenting with pelvic mass, pain, and abnormal vaginal discharge. Diagnosis requires tissue biopsy showing sulfur granules, gram-positive branching filaments, and culture (difficult, anaerobic). Treatment: high-dose penicillin G IV for 2-6 weeks followed by oral penicillin or amoxicillin for 6-12 months. Iodine-sensitive (hence the name "actino-myces"). Surgical drainage and debridement are often required for large abscesses.

Acute HIV Syndrome

– The acute retroviral syndrome occurring 2-6 weeks after primary HIV infection, corresponding to high-level viremia and immune activation. Presents with fever (most common), lymphadenopathy, pharyngitis, rash (maculopapular, trunk/face), myalgia, arthralgia, headache, nausea, diarrhea, and oral ulcers. High viral load, low CD4 count transiently. Highly infectious period. Diagnosis: HIV RNA (viral load) is positive; HIV antibody may be negative (window period). p24 antigen is detectable. Treatment: immediate ART regardless of CD4 count; pre-exposure prophylaxis for partners. Early diagnosis and treatment improves outcomes and reduces transmission.

Amoebiasis

– Infection caused by the protozoan parasite Entamoeba histolytica, transmitted via the fecal-oral route (contaminated food or water). Most infections are asymptomatic; symptomatic disease ranges from amoebic dysentery (bloody diarrhea, abdominal pain, tenesmus) to extra-intestinal disease, most commonly amoebic liver abscess (right upper quadrant pain, fever, weight loss). Diagnosis: microscopy of stool samples (identification of trophozoites containing ingested red blood cells), antigen detection, PCR, and serology for extra-intestinal disease. Treatment: tinidazole or metronidazole for invasive disease, followed by a luminal agent (paromomycin or diloxanide furoate) to eradicate cyst carriage. Prevention: safe water, hand hygiene, sanitation.

Anthrax

– A serious infectious disease caused by the gram-positive bacterium Bacillus anthracis, which forms highly resistant spores that can persist in the environment for decades, and it is primarily a zoonotic disease of herbivores (cattle, sheep, goats) that can be transmitted to humans through contact with infected animals or their products. In humans, anthrax occurs in three main forms depending on the route of exposure: cutaneous (most common, 95 percent, presenting as a painless pruritic papule that progresses to a vesicle and then to a characteristic black eschar, with surrounding edema), inhalational (the most lethal form, causing hemorrhagic mediastinitis and mediastinal widening on chest radiography, with a biphasic illness of initial nonspecific symptoms followed by sudden respiratory deterioration), and gastrointestinal (following ingestion of contaminated meat, causing abdominal pain, nausea, vomiting, and bloody diarrhea). Diagnosis is confirmed by culture, Gram stain, or PCR from blood, skin lesions, or tissue samples, and serology may be used retrospectively. Treatment involves high-dose IV antibiotics: ciprofloxacin or doxycycline in combination with one or two additional agents (clindamycin, rifampin, meropenem), and antitoxin (raxibacumab or obiltoxaximab) may be considered for inhalational anthrax. Post-exposure prophylaxis consists of ciprofloxacin or doxycycline for 60 days combined with anthrax vaccine adsorbed (AVA, BioThrax).

Antibiotic

– A substance that kills or slows the growth of bacteria.

Antimicrobial

– A general term for antibiotics and other drugs that fight microscopic organisms in the body, such as bacteria, viruses, fungi, and parasites.

Antimicrobial Resistance Mechanisms

– Beta-lactamases: Ambler class A (TEM, SHV, CTX-M ESBLs, KPC carbapenemases), class B (metallo-beta-lactamases: NDM, VIM, IMP zinc-dependent, not inhibited by avibactam), class C (AmpC, chromosomal/inducible, hydrolyzes cephalosporins), class D (OXA carbapenemases,.

Antimicrobial Stewardship

– Core elements (CDC): leadership commitment, accountability, drug expertise, action (prospective audit/feedback, preauthorization), tracking (DOT,.

Antiseptic

– Substances used on wounds to prevent or treat infection; they kill or slow the growth of disease-causing organisms, such as bacteria, on the surface of the body.

Arbovirus

– A virus transmitted by mosquitoes or other member of the arthropod phylum.

Ascariasis

– Infection with Ascaris lumbricoides, the most common helminth infection worldwide (soil-transmitted). Lifecycle: ingestion of embryonated eggs → larvae hatch in small intestine → penetrate intestinal wall → migrate through liver and lungs (Loeffler syndrome: cough, eosinophilia, transient pulmonary infiltrates) → return to small intestine as adults. Most infections are asymptomatic. Heavy burden: intestinal obstruction (especially children), biliary obstruction, pancreatitis, malnutrition in children. Diagnosis: stool microscopy (characteristic eggs). Treatment: albendazole or mebendazole (single dose).

Aspergillosis

– A spectrum of diseases caused by the fungus Aspergillus (most commonly A. fumigatus). Allergic forms: allergic bronchopulmonary aspergillosis (ABPA, in asthma or cystic fibrosis – wheezing, eosinophilia, pulmonary infiltrates), allergic Aspergillus sinusitis. Saprophytic: aspergilloma (fungus ball in pre-existing lung cavity – haemoptysis). Invasive aspergillosis: occurs in immunocompromised patients (neutropenia, transplant, steroid therapy) – hyphal invasion of blood vessels causes tissue infarction, pneumonia, and dissemination to brain, heart, and kidneys (high mortality). Diagnosis: serum galactomannan, beta-D-glucan, CT chest (halo sign, air crescent sign), culture, biopsy. Treatment: voriconazole (first-line), amphotericin B, isavuconazole.

Bacillary Angiomatosis

– A vascular proliferative disease caused by Bartonella henselae or B. quintana, occurring in immunocompromised patients (HIV/AIDS, transplant). Presents with erythematous papules, nodules, or pedunculated lesions resembling Kaposi sarcoma or pyogenic granuloma. Can involve skin, liver (peliosis hepatis), spleen, bone, and lymph nodes. Fever, night sweats, and weight loss may accompany. Diagnosis: biopsy shows lobular proliferations of blood vessels with epithelioid endothelial cells and Warthin-Starry silver stain positive bacilli; PCR confirmation. Treatment: erythromycin or doxycycline for 3 months. Lesions resolve with treatment.

Bacteremia

– The presence of viable bacteria in the bloodstream, a serious condition that can be transient (after dental procedures, toothbrushing), intermittent (undrained abscess), or continuous (endovascular infection like endocarditis). Bacteremia triggers systemic inflammatory responses ranging from fever and leukocytosis to septic shock. Diagnosis requires positive blood cultures; treatment involves IV antibiotics targeting the identified organism and source control of the primary infection site.

Bacterial Infections

– Illnesses caused by pathogenic bacteria proliferating in the body, ranging from mild sore throats and ear infections to life-threatening sepsis and meningitis. They are diagnosed through cultures, PCR testing, or other microbiological methods. Treatment typically involves antibiotics, though rising antimicrobial resistance demands careful stewardship.

Bartonellosis

– A group of infections caused by Bartonella species. Cat scratch disease (B. henselae): regional lymphadenopathy after cat scratch/bite, fever, inoculation papule. Bacillary angiomatosis (B. henselae, B. quintana): vascular proliferative lesions in immunocompromised (HIV), red papules/nodules on skin, bone, and viscera. Oroya fever (B. bacilliformis, endemic in Peru): acute hemolytic anemia, fever, severe myalgia (Carrion disease). Verruga peruana: chronic cutaneous nodular stage. Trench fever (B. quintana): relapsing fever, bone pain, louse-borne. Diagnosis: culture, PCR, serology. Treatment: azithromycin for CSD, doxycycline + rifampin for bacillary angiomatosis.

Bioterrorism Agents

– Category A: anthrax (Bacillus anthracis), smallpox (variola), botulism (Clostridium botulinum), plague (Yersinia pestis), tularemia (Francisella tularensis), viral hemorrhagic fevers (Ebola, Marburg, Lassa, Machupo). Anthrax: inhalational (mediastinal widening, hemorrhagic mediastinitis), cutaneous (eschar), GI. Treatment: ciprofloxacin/doxycycline x60d + vaccine (BioThrax). Smallpox: eradicated 1980, vaccine (ACAM2000, Jynneos), tecovirimat, brincidofovir. Botulism: descending flaccid paralysis, antitoxin (heptavalent). Plague: bubonic (lymphadenopathy/bubo), pneumonic (person-to-person), septicemic. Doxycycline/ciprofloxacin/gentamicin. Tularemia: ulceroglandular, oculoglandular, pneumonic, typhoidal. Streptomycin/gentamicin/doxycycline. VHF: supportive, specific antivirals where available. Public health response: isolation, contact tracing, prophylaxis, risk communication.

Blastomycosis

– A fungal infection caused by Blastomyces dermatitidis, endemic to the Ohio and Mississippi River valleys and Great Lakes region. Inhalation of conidia causes pulmonary infection. Pulmonary: acute pneumonia (often misdiagnosed as bacterial), chronic pneumonia with nodules and cavitation. Disseminated: skin (verrucous or ulcerative lesions, most common extrapulmonary site), bone (osteomyelitis), and GU (prostatitis, epididymitis). Diagnosis: sputum culture, histopathology (broad-based budding yeasts), serum antigen testing. Treatment: itraconazole for mild-moderate; amphotericin B for severe or CNS disease.

Borrelia

– A genus of spiral-shaped bacteria (spirochetes) transmitted by arthropod vectors, most notably ticks and lice. The major human pathogen is Borrelia burgdorferi sensu lato, the causative agent of Lyme disease, transmitted by Ixodes ticks. Borrelia recurrentis causes louse-borne relapsing fever, and Borrelia duttonii causes tick-borne relapsing fever. Borrelia species have a complex genome with multiple linear and circular plasmids. Diagnosis relies on serology (ELISA with Western blot confirmation), PCR, or direct microscopy. Treatment: doxycycline, amoxicillin, or ceftriaxone depending on disease stage.

Botulism

– A rare, potentially fatal neuroparalytic disease caused by botulinum toxin, a potent presynaptic neurotoxin produced by Clostridium botulinum, an anaerobic, spore-forming, gram-positive bacillus. Botulinum toxin blocks the release of acetylcholine at the neuromuscular junction by cleaving SNARE proteins (SNAP-25, VAMP/synaptobrevin, syntaxin), preventing vesicle fusion and causing flaccid paralysis. Four clinical forms exist: foodborne botulism (ingestion of preformed toxin in improperly canned or preserved foods, most commonly home-canned vegetables, honey, and fermented foods), infant botulism (the most common form in the United States, caused by intestinal colonization and in vivo toxin production after ingestion of C. botulinum spores, most commonly from honey), wound bo-.

Brucellosis

– A zoonotic infectious disease caused by gram-negative coccobacilli of the genus Brucella, transmitted to humans through contact with infected animals (cattle, goats, pigs, sheep) or consumption of unpasteurized dairy products. Brucella melitensis (goats and sheep, most severe), B. abortus (cattle), B. suis (pigs), and B. canis (dogs) are the principal species. It is endemic in the Mediterranean basin, Middle East, Central Asia, Mexico, and South America. Clinical presentation is highly variable: acute brucellosis presents with undulant fever (fever that rises and falls in waves), sweats (often malodorous), arthralgia, fatigue, headache, and hepatosplenomegaly. Chronic brucellosis may persist for months to years with relapsing fever, chronic fatigue, depression, arthritic manifestations.

Candidiasis

– Fungal infection caused by Candida species, most commonly C. albicans. Mucocutaneous: oral thrush (white plaques on oral mucosa), esophagitis, vulvovaginitis (itching, discharge), and intertrigo. Invasive candidiasis: candidemia and deep organ infection, typically in immunocompromised or ICU patients. Diagnosis: culture, histopathology. Treatment: topical or oral azoles for mucocutaneous; echinocandins for invasive disease. Risk factors: antibiotics, immunosuppression, diabetes, indwelling catheters.

Cardiac Device Infections

– CIED (pacemaker, ICD) infection: pocket infection (erythema, warmth, purulence, erosion), lead infection (bacteremia, endocarditis). Diagnosis: blood cultures, TEE (lead vegetations), pocket culture. Treatment: complete system removal (leads + generator) - CLASS I. Extraction: transvenous (laser, mechanical sheaths) vs surgical (thoracotomy). Antibiotics: pocket only - 10-14d; bacteremia/endocarditis 4-6w. Relapsing bacteremia if hardware retained. Bridge therapy: temporary transvenous pacing, wearable defibrillator (LifeVest). Reimplantation: contralateral side after clearance.

Cat Scratch Disease

– A regional lymphadenitis caused by Bartonella henselae, a gramnegative bacillus transmitted to humans through cat scratches, bites, or licks on broken skin, particularly from kittens (which carry higher bacterial loads). Clinical presentation: 1-2 weeks after exposure, a papule or pustule develops at the inoculation site, followed by regional lymphadenopathy (axillary, epitrochlear, cervical) that is tender and may suppurate (10-15 percent). Systemic symptoms include low-grade fever, fatigue, headache, anorexia. Atypical presentations: Parinaud oculoglandular syndrome (conjunctivitis + preauricular lymphadenopathy), encephalitis, neuroretinitis, osteomyelitis, hepatosplenic granulomas, endocarditis (culture-negative). Diagnosis: Warthin-Starry silver.

Cat-Scratch Fever

– An infectious disease caused by the bacterium Bartonella henselae, transmitted through a scratch, bite, or lick from an infected cat (typically kittens). After an incubation period of 3–14 days, a papule or pustule forms at the inoculation site, followed by painful regional lymphadenopathy (axillary, epitrochlear, cervical, or submandibular) within 1–3 weeks. Most cases are self-limited, resolving over 2–4 months. Atypical presentations: Parinaud oculoglandular syndrome (conjunctivitis with preauricular lymphadenopathy), hepatosplenic disease (fever, abdominal pain), neuroretinitis, and encephalopathy. Diagnosis: serology (IgG by IFA or ELISA), PCR of lymph node aspirate, or histology (Warthin-Starry silver stain shows pleomorphic bacilli). Treatment: typically supportive; azithromycin for moderate-to-severe lymphadenopathy or immunocompromised patients.

Cell-mediated immunity

– A type of immune response mounted against viruses, certain types of parasites, and perhaps cancer cells.

Cellulitis

– A bacterial infection of the deep dermis and subcutaneous tissue, most commonly caused by Streptococcus pyogenes (Group A Strep) and Staphylococcus aureus. Presents with erythema, warmth, edema, and tenderness, with poorly defined margins. Often on the lower extremities. Risk factors: breaks in skin (trauma, ulcers, tinea pedis), lymphedema, venous insufficiency, and immunosuppression. Diagnosis: clinical; blood cultures if systemic signs. Treatment: antibiotics targeting Strep and Staph (cephalexin, dicloxacillin, clindamycin). Severe cases or facial involvement require IV antibiotics.

Chancroid

– A sexually transmitted infection caused by Haemophilus ducreyi, characterized by painful genital ulcers and tender inguinal lymphadenopathy (bubo). Ulcers are ragged, non-indurated, with purulent exudate and surrounding erythema. More common in men and in resource-limited settings. Associated with increased HIV transmission. Diagnosis: clinical (painful ulcer + tender lymphadenopathy), Gram stain (school of fish appearance), culture (chocolate agar with vancomycin). Treatment: azithromycin (single dose), ceftriaxone (single dose IM), ciprofloxacin, or erythromycin. Fluctuant buboes require aspiration.

Cholera

– An acute diarrheal illness caused by Vibrio cholerae, a gram-negative, comma-shaped, flagellated bacterium, serogroups O1 and O139, that produces cholera toxin, an enterotoxin that irreversibly activates adenylate cyclase in intestinal epithelial cells, causing massive secretion of isotonic fluid into the intestinal lumen. Transmission is fecal-oral through contaminated water and food, with humans as the only natural host. Seven pandemics have occurred since 1817, and cholera remains endemic in over 50 countries, particularly in regions with inadequate water and sanitation. The hallmark presentation is acute onset of profuse, painless, watery diarrhea described as rice-water stool (cloudy white fluid with flecks of mucus) that can exceed one liter per hour, leading to severe dehydration, metabolic acidosis, and hypovolemic shock within hours of onset. Vomiting may precede diarrhea. Clinical dehydration is classified as severe (two or more signs: delayed capillary refill, weak or absent radial pulse, sunken eyes, decreased skin turgor, altered consciousness) or some dehydration (less severe signs). Diagnosis is clinical; stool culture on thiosulfate-citrate-bile salts-sucrose (TCBS) agar grows yellow colonies, and rapid dipstick tests detect V. cholerae O1 and O139 antigen. Treatment is primarily fluid replacement: WHO low-osmolality oral rehydration solution (ORS: 75 mmol/L glucose, 75 mEq/L sodium) for some or no dehydration; aggressive IV Ringer's lactate for severe dehydration (100 mL/kg over 3 hours for adults, 100 mL/kg over 6 hours for children). Antibiotics reduce the duration and volume of diarrhea: azithromycin 1 gram orally as a single dose (preferred for all ages, including pregnant.

Clostridioides difficile Infection

– Clostridioides difficile infection (CDI): toxinmediated colitis following antibiotic disruption of normal flora. Risk factors: antibiotics (clindamycin, fluoroquinolones, cephalosporins, penicillins), age >65, hospitalization, PPIs, immunosuppression, IBD. Clinical: watery diarrhea (3+ unformed stools/24h), abdominal pain, fever, leukocytosis. Severe: WBC >15k, creatinine >1.5x baseline. Fulminant: hypotension, ileus, toxic megacolon. Diagnosis: nucleic acid amplification test (NAAT, sensitive), glutamate dehydrogenase (GDH) + toxin EIA (specific), PCR. Treatment: non-severe: vancomycin 125 mg PO q6h x10d or fidaxomicin 200 mg PO q12h x10d. Severe: vancomycin 125 mg PO q6h (or 500 mg q6h) + metronidazole 500 mg IV q8h. Fulminant: vancomycin 500 mg PO/NG q6h + metronidazole 500 mg IV q8h + consider rectal vancomycin enema; surgery (subtotal colectomy) if refractory. Recurrence (20-30.

CMV

– See Cytomegalovirus Infection.

Coccidioidomycosis

– A fungal infection caused by Coccidioides immitis or C. posadasii, endemic to the southwestern US (San Joaquin Valley fever). Inhalation of arthroconidia causes primary pulmonary infection. Most cases are asymptomatic or mild (flu-like illness with cough, fever, erythema nodosum). Disseminated disease occurs in immunocompromised patients, affecting skin, bone/joints, and meninges. Diagnosis: serology (IgM, IgG), culture, and histopathology (spherules). Treatment: fluconazole or itraconazole; amphotericin B for severe disseminated disease.

Common Cold

– An acute, self-limited viral infection of the upper respiratory tract, one of the most frequent illnesses in humans, with adults experiencing 2-4 episodes annually and children 6-8 episodes. Over 200 viral serotypes cause the common cold, with rhinoviruses (100 serotypes, most common, accounting for 3050 percent of cases) as the leading cause, followed by coronaviruses (15-30 percent, including seasonal HCoV-OC43, -229E, -NL63, -HKU1), respiratory syncytial virus (RSV, 5-15 percent), parainfluenza virus (5-10 percent), adenovirus (5 percent), and enterovirus. Transmission occurs through direct contact, fomites, and respiratory droplets, with peak infectivity in the first 2-3 days of symptoms. The incubation period is 1-3 days. Pathophysiology: viral entry through the nasal epithelium triggers an innate immune response with release.

Communicable disease

– Any disease caused by bacteria, viruses, or other pathogens that is spread from person to person.

Community-Acquired Pneumonia

– CAP: acquired outside hospital. CURB-65 (confusion, uremia >7, respiratory rate >=30, BP <90/60, age >=65) and PSI/PORT score for severity/site of care. Outpatient: amoxicillin 1g TID, doxycycline 100 mg BID, or macrolide (if resistance <25.

COVID-19

– Coronavirus disease 2019 caused by SARS-CoV-2, first identified in December 2019. Ranges from asymptomatic to severe respiratory failure and death. Common symptoms: fever, cough, dyspnea, loss of taste/smell (anosmia), myalgia, and fatigue. Severe disease: pneumonia, ARDS, hyperinflammatory state, and thromboembolic complications. Diagnosis: RT-PCR, rapid antigen testing. Prevention: vaccination (mRNA, adenoviral vector). Treatment: supportive care, remdesivir, dexamethasone for severe disease, and anticoagulation.

Cryptococcosis

– A fungal infection caused by Cryptococcus neoformans or C. gattii, primarily affecting immunocompromised patients (HIV/AIDS, organ transplant, hematologic malignancy). Pulmonary cryptococcosis: cough, dyspnea, pulmonary nodules. Meningoencephalitis: fever, headache, altered mental status, and elevated intracranial pressure. Diagnosis: CSF India ink stain, cryptococcal antigen (CrAg), culture. Treatment: amphotericin B + flucytosine for induction, fluconazole for maintenance. Elevated ICP requires therapeutic lumbar punctures.

Cryptosporidiosis

– A diarrhoeal disease caused by the protozoan parasite Cryptosporidium parvum (zoonotic) or C. hominis (anthroponotic), transmitted via contaminated water (chlorine-resistant oocysts), food, or direct contact. Presents with watery diarrhea, abdominal cramps, nausea, and low-grade fever, lasting 1-2 weeks. In immunocompromised patients (especially HIV with low CD4), it causes chronic, profuse, life-threatening diarrhea and wasting. Diagnosis: modified acid-fast stain of stool (oocysts appear red), antigen detection, PCR. Treatment: supportive (rehydration, electrolyte replacement). Nitazoxanide is effective in immunocompetent patients.

Cystitis

– Inflammation of the urinary bladder, typically from bacterial infection, one of the most common bacterial infections in women. Uncomplicated cystitis occurs in otherwise healthy, non-pregnant women with normal urinary tract anatomy; complicated cystitis occurs in men, pregnant women, children, patients with urinary tract abnormalities, indwelling catheters, immunosuppression, or recent instrumentation. The most common causative organism is Escherichia coli (75-95 percent of uncomplicated cases), followed by Staphylococcus saprophyticus (5-15 percent, particularly in young sexually active women), Klebsiella pneumoniae, and Proteus mirabilis. Clinical presentation includes urinary frequency, urgency, dysuria (painful urination), suprapubic pain, and occasionally hematuria. Fever and flank pain suggest upper tract involvement (pyelonephritis). Diagnosis: urinalysis shows pyuria (leukocyte esterase positive, greater than 5 WBCs per high-power field), nitrite positivity (gram-negative bacteria, not S. saprophyticus or enterococci), and bacteriuria. Urine culture (greater than 103 CFU/mL of a single uropathogen for symptomatic women) confirms the diagnosis and provides antibiotic sensitivity. Treatment: uncomplicated cystitis in women is treated with a short course (3 days) of nitrofurantoin (100 mg twice daily), trimethoprim-sulfamethoxazole (160/800 mg twice daily, if local resistance less than 20 percent), or fosfomycin (3 gram single dose). Beta-lactams (cephalexin, amoxicillin-clavulanate) are less effective and require longer duration (5-7 days). Complicated cystitis requires 7-14 days of culture-directed antibiotics. Men with cystitis require longer treatment (7-14 days) and evaluation for underlying prostatitis or obstruction. Recurrent cystitis (3 or more episodes per year) may be managed with prophylactic antibiotics (postcoital or continuous daily), behavioral modifications (hydration, postcoital voiding), and vaginal estrogen.

Cytomegalovirus Infection

– A widespread herpesvirus infection. Primary infection in immunocompetent individuals is usually asymptomatic or causes mononucleosis-like syndrome. Congenital CMV: most common congenital infection, causes sensorineural hearing loss, microcephaly, and developmental delay. In immunocompromised (HIV/AIDS, transplant): retinitis, colitis, esophagitis, pneumonitis, and encephalitis. Diagnosis: CMV PCR, pp65 antigen, serology. Treatment: ganciclovir, valganciclovir, foscarnet for resistant cases. Prevention: valganciclovir prophylaxis in high-risk transplant recipients.

Dengue fever

– An acute mosquito-borne viral illness caused by dengue virus (DENV-1, -2, -3, -4), a flavivirus transmitted primarily by Aedes aegypti mosquitoes. Dengue is the most rapidly spreading mosquito-borne viral disease worldwide, with an estimated 390 million infections annually and 100 million symptomatic cases, endemic in over 100 countries, particularly in Southeast Asia, the Western Pacific, the Americas, and Africa. The clinical course has three phases: febrile phase (abrupt onset of high fever lasting 2-7 days, severe headache, retro-orbital pain, myalgia, arthralgia, nausea, vomiting, and a characteristic maculopapular rash with islands of sparing); critical phase (occurs around defervescence at days 3-7, lasting 24-48 hours, marked by increased capillary permeability, plasma leakage, and risk of progression to dengue hemorrhagic fever and dengue shock syndrome, characterized by rising hematocrit with falling platelets, serous effusions, and hypoproteinemia); and recovery phase (reabsorption of extravasated fluid, diuresis, and clinical improvement). The World Health Organization classification categorizes severity as dengue without warning signs, dengue with warning signs (abdominal pain, persistent vomiting, fluid accumulation, mucosal bleeding, lethargy, liver enlargement greater than 2 cm, rising hematocrit with rapid platelet drop), and severe dengue (severe plasma leakage leading to shock, severe bleeding, or severe organ involvement). Laboratory findings include leukopenia, thrombocytopenia, elevated liver enzymes, and prolonged aPTT. Diagnosis is confirmed by NS1 antigen detection (positive in the first 5 days), RT-PCR (days 1-5), or serology (IgM positive after days 5-7, IgG rises after 14 days). Treatment is supportive: fever control with acetaminophen (avoid NSAIDs and aspirin due to bleeding risk), aggressive IV fluid resuscitation for severe dengue with plasma leakage (crystalloids, colloids for hypotensive shock), blood transfusion for significant hemorrhage, and close monitoring of vital signs, hematocrit, and platelet counts during the critical phase. No specific an334.

Diphtheria

– An acute, toxin-mediated infectious disease caused by Corynebacterium diphtheriae, a gram-positive aerobic bacillus that produces a potent exotoxin that inhibits protein synthesis via ADP-ribosylation of elongation factor 2, causing local tissue necrosis and systemic toxicity. The bacteria are transmitted through respiratory droplets or direct contact with cutaneous lesions. The classic clinical presentation is respiratory diphtheria: after a 2-5 day incubation period, the patient develops low-grade fever, sore throat, malaise, and a characteristic grayish-white pseudomembrane that forms on the tonsils, pharynx, or larynx composed of fibrin, necrotic epithelial cells, and inflammatory cells. The membrane is firmly adherent, bleeds upon attempted removal, and may extend to obstruct the airway causing stridor, respiratory distress, and asphyxiation. Cervical lymphadenopathy and severe neck edema produce the classic bull neck appearance. Systemic effects of the toxin include myocarditis (occurring in 10-25 percent of cases, presenting with ECG changes, arrhythmias, and heart failure typically 1-2 weeks after onset) and neurologic involvement (palatal paralysis, cranial neuropathies, and peripheral neuritis developing weeks later). Cutaneous diphtheria presents as non-healing ulcers with a gray membrane, more common in tropical settings and individuals with poor hygiene. Diagnosis is clinical confirmed by culture on tellurite or Loeffler medium and the Elek test for toxigenicity. Treatment: equine diphtheria antitoxin (DAT) is the mainstay, neutralizing unbound toxin; it must be given early, intravenously for respiratory cases, with monitoring for serum sickness. Antibiotics eradicate the bacteria and stop toxin production: penicillin G (procaine or IV) or erythromycin for 14 days. Close contacts require throat cultures, prophylaxis with penicillin or erythromycin, and booster immunization. Prevention is through routine childhood immunization with the DTaP vaccine, with boosters (Td or Tdap) every 10.

Donovanosis

– Also known as granuloma inguinale, a chronic, progressive, ulcerative genital infection caused by the intracellular gram-negative bacterium Klebsiella granulomatis (formerly Calymmatobacterium granulomatis). The disease is rare in developed countries but endemic in tropical and subtropical regions, including Papua New Guinea, parts of India, southern Africa, the Caribbean, and South America. Transmission occurs through direct sexual contact with an active ulcer, and the incubation period ranges from 1 week to 3 months after exposure. Clinical presentation begins as a painless, erythematous papule or nodule in the genital region (glans penis, prepuce, labia, vaginal mucosa, cervix, or perianal area) that progressively ulcerates into a painless, beefy-red, friable, granulating ulcer with rolled edges that bleeds easily on contact without significant lymphadenopathy. The ulcers are highly vascular and may have a characteristic beefy red appearance. Without treatment, the ulcer spreads slowly and may cause extensive destruction of genital tissue, fibrosis, lymphedema, and elephantiasis (pseudo-elephantiasis from lymphatic obstruction). Extragenital involvement may occur through autoinoculation to the mouth, lips, and anus. Diagnosis is clinical and confirmed by demonstrating.

Dysentery

– Inflammation of the colon causing frequent, small-volume, bloody, mucoid stools with abdominal cramps, tenesmus, and fever. Two main types: amoebic dysentery (caused by Entamoeba histolytica) and bacillary dysentery (caused by Shigella species). Bacillary dysentery is more common and typically more severe, with higher fever and systemic symptoms. Diagnosis: stool microscopy, culture, PCR. Treatment: rehydration (oral or intravenous), antibiotics for moderate-severe bacillary dysentery (ciprofloxacin, azithromycin, or ceftriaxone – guided by sensitivity), tinidazole/metronidazole for amoebic dysentery. Complications: intestinal perforation, toxic megacolon, hemolytic uraemic syndrome (Shigella dysenteriae type 1).

Ebola Virus Disease

– A severe, often fatal hemorrhagic fever caused by the Ebola virus (Filoviridae family). Transmitted through direct contact with blood or body fluids of infected humans or animals. Incubation 2-21 days. Presents with acute onset of fever, myalgia, headache, followed by vomiting, diarrhea, and hemorrhagic manifestations (petechiae, ecchymosis, bleeding). High case fatality rate. Diagnosis: RT-PCR. Treatment: supportive care, monoclonal antibodies. Outbreak control: isolation, contact tracing, safe burials.

Ehrlichiosis

– A tick-borne bacterial infection caused by Ehrlichia chaffeensis (human monocytic ehrlichiosis) and E. ewingii (granulocytic). Endemic in the southeastern and south-central US. Presents with fever, headache, myalgia, malaise, and thrombocytopenia/leukopenia. Rash less common than RMSF. Diagnosis: clinical suspicion in tick-endemic area, PCR of blood, serology (IFA), intracytoplasmic morulae on peripheral smear. Treatment: doxycycline (first-line). Untreated can be severe with respiratory failure, meningoencephalitis, and DIC.

Elephantiasis

– Massive enlargement of a body part, typically a limb or the scrotum, caused by chronic lymphedema from obstruction of lymphatic vessels. The most common cause worldwide is lymphatic filariasis, a mosquito-borne parasitic infection by Wuchereria bancrofti, Brugia malayi, or Brugia timori. Adult filarial worms inhabit the lymphatic vessels, causing inflammation, dilation, and eventual fibrosis and obstruction of lymph flow. Recurrent secondary bacterial infections (cellulitis, lymphangitis) worsen the edema. The affected limb becomes grossly swollen with thickened, hyperkeratotic, verrucous skin. Other causes: non-filarial elephantiasis (podoconiosis) from chronic exposure to irritant volcanic soil in barefoot agricultural workers, post-surgical lymphedema (axillary node dissection, inguinal node dissection), and congenital lymphatic malformations. Treatment: elevation, compression, meticulous skin hygiene, antibiotics for superinfection, and surgical debulking in severe cases.

Endemic

– Continually present among people in a geographic region.

Endocarditis Prophylaxis

– Infective endocarditis prophylaxis with antibiotics is recommended for patients at highest risk of adverse outcomes from endocarditis who are undergoing dental procedures that involve manipulation of gingival tissue, the periapical region of teeth, or perforation of the oral mucosa. The American Heart.

Febrile

– Feverish; having a high body temperature.

Fungal Infections

– Candidiasis: mucosal (thrush, esophagitis, vulvovaginal) - fluconazole; invasive (candidemia, intraabdominal, UTI) - echinocandin (caspofungin, micafungin, anidulafungin) first-line, step-down to fluconazole if susceptible. Aspergillosis: allergic bronchopulmonary (ABPA, asthma/CF, corticosteroids + itraconazole), chronic pulmonary (CPA, cavitary), invasive (IA, neutropenia, transplant, steroids) voriconazole first-line, isavuconazole alternative. Mucormycosis: rhinocerebral (DKA, neutropenia), pulmonary, cutaneous. Amphotericin B lipid formulation (liposomal 5 mg/kg) + surgical debridement; posaconazole/isavuconazole step-down. Cryptococcosis: meningoencephalitis (HIV CD4 <100), pulmonary. Induction: amphotericin B + flucytosine x2w, consolidation: fluconazole x8w, maintenance: fluconazole x1yr. Endemic mycoses: Histoplasma (Ohio/Mississippi valleys), Coccidioides (Southwest US, Valley fever), Blastomyces (Great Lakes, Southeast), Paracoccidioides (Latin America). Treatment: mild-moderate - itraconazole; severe - amphotericin B then itraconazole. Pneumocystis jirovecii pneumonia (PJP): HIV CD4 <200, immunosuppressed. TMP-SMX (first-line), alternatives: dapsone+TMP, atovaquone, clindamycin+primaquine, pentamidine. Prophylaxis: TMP-SMX if CD4 <200.

Genital Warts (Condylomata Acuminata)

– Anogenital warts caused by human papillomavirus (HPV), most commonly types 6 and 11 (low-risk, >90% of genital warts, not associated with malignancy). Transmission: direct skin-to-skin contact during vaginal, anal, or oral sex; highly infectious (65% transmission rate per sexual encounter). Incubation period: 2 weeks to 8 months (average 3 months). Clinical presentation: flesh-coloured, pink, or hyperpigmented papules or plaques on the genital, perianal, or oral mucosa. Morphologies: exophytic/cauliflower-like (condylomata acuminata), flat papules (smooth, slightly raised, more common on the cervix or penile shaft), keratotic (thick, wart-like, on dry skin of penile shaft or scrotum). Sites: males (foreskin, frenulum, glans, corona, shaft, scrotum, perianal), females (vulva, vaginal introitus, labia, perineum, perianal, cervix). Subclinical infection: detected only by HPV DNA testing or colposcopy. Natural history: 30% regress spontaneously within 4 months, 70% within 2 years. Malignant potential: HPV 6 and 11 rarely cause malignant transformation; HPV 16 and 18 (high-risk types) cause cervical, anal, and oropharyngeal cancer. Diagnosis: clinical (visual inspection, colposcopy, anoscopy). Biopsy if atypical, pigmented, ulcerated, or treatment-resistant. Treatment: patient-applied (imiquimod 5% cream, podophyllotoxin 0.5% solution/gel, sinceatechins 15% ointment); provider-administered (cryotherapy, trichloroacetic acid 80-90%, surgical excision, electrosurgery, laser ablation). No single treatment is universally effective; recurrence rates are 20-50% regardless of modality. Prevention: HPV vaccine (quadrivalent Gardasil targets HPV 6, 11, 16, 18; nonavalent Gardasil 9 targets 9 types) prevents >90% of genital warts when given before sexual debut.

German Measles (Rubella)

– See Rubella.

Giardiasis

– Infection of the small intestine caused by the protozoan parasite Giardia intestinalis (G. lamblia), transmitted via contaminated water (including untreated surface water), food, or person-to-person spread. Many infections are asymptomatic. Symptomatic disease presents 1-2 weeks after exposure with watery, foul-smelling diarrhea, bloating, flatulence, abdominal cramps, nausea, and steatorrhea. Chronic infection can cause malabsorption, weight loss, and failure to thrive in children. Diagnosis: stool microscopy (trophozoites or cysts), antigen detection (ELISA), PCR. Treatment: tinidazole (single dose) or metronidazole (5-7 days). Prevention: water filtration or boiling, hand hygiene.

Glandular Fever (Infectious Mononucleosis)

– See Mononucleosis.

Gonorrhea

– A sexually transmitted infection caused by the Gram-negative diplococcus Neisseria gonorrhoeae. Epidemiology: second most common bacterial STI (after Chlamydia), highest incidence in 15-24 year olds. Clinical features: (1) Men: acute urethritis (purulent discharge, dysuria, 2-7 day incubation); (2) Women: often asymptomatic (50-70%), may present with cervicitis (mucopurulent discharge, intermenstrual bleeding, dyspareunia), urethritis, PID (salpingitis, tubo-ovarian abscess); (3) Extragenital: pharyngeal (oropharyngeal swab–often asymptomatic), rectal (proctitis, discharge, tenesmus); (4) Disseminated: arthritis-dermatitis syndrome (tenosynovitis, migratory polyarthritis, pustular rash), perihepatitis (Fitz-Hugh-Curtis syndrome), meningitis, endocarditis (rare). Neonates: ophthalmia neonatorum (purulent conjunctivitis, can cause blindness without prophylaxis). Diagnosis: nucleic acid amplification test (NAAT) on urine or swab, culture for antibiotic sensitivity. Complications: PID, ectopic pregnancy, infertility (tubal scarring), epididymitis. Treatment: ceftriaxone 500 mg IM (single dose) plus azithromycin 1 g orally (single dose). Antimicrobial resistance: resistance to penicillin, tetracyclines, fluoroquinolones now widespread. Requires partner notification and treatment.

Hansen's Disease (Leprosy)

– A chronic infectious disease caused by Mycobacterium leprae, an acid-fast, obligate intracellular bacillus. Transmission: respiratory droplets (prolonged close contact). Incubation: 2-10 years (average 5). Epidemiology: 200,000 new cases per year globally (mostly India, Brazil, Indonesia). Pathophysiology: M. leprae has a predilection for skin and Schwann cells of peripheral nerves; the clinical spectrum depends on host cell-mediated immunity. Ridley-Jopling classification: (1) Tuberculoid leprosy (TT): strong cell-mediated immunity, few well-defined anesthetic skin lesions (hypopigmented macules with raised edges), thickened nerves, few bacilli. (2) Lepromatous leprosy (LL): poor cell-mediated immunity, numerous symmetric skin lesions (nodules, plaques, diffuse infiltration), extensive nerve involvement (symmetrical, stocking-glove anesthesia), many bacilli (positive slit-skin smear). (3) Borderline forms (BT, BB, BL): unstable, may shift towards TT or LL. Reactions: type 1 (reversal reaction, cell-mediated, worsening skin and nerve lesions), type 2 (erythema nodosum leprosum, immune complex-mediated, fever, painful nodules, neuritis, iritis). Complications: peripheral neuropathy (anesthesia leading to unrecognised injuries, neuropathic ulcers, Charcot joints), claw hand, foot drop, lagophthalmos (facial nerve), nasal deformity, blindness. Diagnosis: clinical (anesthetic skin lesions, thickened nerves), slit-skin smear for acid-fast bacilli, skin or nerve biopsy. Treatment: multidrug therapy (MDT): dapsone + rifampicin +/- clozimine, duration 6-12 months for paucibacillary, 12-24 months for multibacillary. Free MDT from WHO.

Hantavirus

– A genus of enveloped negative-sense RNA viruses of the Bunyaviridae family, transmitted to humans primarily through inhalation of aerosolized excreta.

Healthcare-Associated Pneumonia

– HCAP concept retired (2016 IDSA/ATS); risk factors for MDR now assessed individually. Risk fac-.

Hepatitis

– Inflammation of the liver caused by hepatotropic viruses. See Hepatitis (Viral Hepatitis) for complete definition.

Hepatitis A

– An acute viral infection of the liver caused by hepatitis A virus (HAV), transmitted via fecal-oral route (contaminated food or water). Incubation 15-50 days. Presents with fever, malaise, nausea, jaundice, and elevated transaminases. Self-limited;

does not cause chronic infection. Fulminant hepatitis is rare. Diagnosis: IgM anti-HAV. Prevention: HAV vaccine (recommended for travelers, children) and hygiene. Treatment: supportive care.

Hepatitis A, B, C, D, E

– The five hepatotropic viruses that cause inflammation of the liver. See individual entries: Hepatitis A, Hepatitis B, Hepatitis C, Hepatitis D, Hepatitis E.

Hepatitis B

– A viral infection caused by hepatitis B virus (HBV), transmitted via blood, sexual contact, and perinatally. Can cause acute (self-limited or fulminant) and chronic infection (defined by HBsAg positivity >6 months). Chronic HBV leads to cirrhosis and hepatocellular carcinoma. Diagnosis: HBsAg, anti-HBs, HBeAg, HBV DNA, and liver function tests. Treatment: tenofovir, entecavir, or peg-interferon for chronic infection. Prevention: HBV vaccine (universal infant immunization).

Hepatitis C

– A viral infection caused by hepatitis C virus (HCV), primarily transmitted through blood exposure (IV drug use, blood transfusion before 1992). Acute infection is often asymptomatic; 50-80% progress to chronic hepatitis. Chronic HCV can lead to cirrhosis and hepatocellular carcinoma over decades. Diagnosis: anti-HCV antibody, HCV RNA. Treatment: direct-acting antivirals (DAAs) achieve cure rates >95%. No vaccine available. Screening recommended for at-risk populations.

Hepatitis (Viral Hepatitis)

– Inflammation of the liver caused by hepatotropic viruses. Hepatitis A (HAV): fecal-oral, acute self-limited, no chronic carrier state; vaccine-preventable. Hepatitis B (HBV): blood/sexual/perinatal transmission; acute and chronic infection; leading cause of cirrhosis and HCC. Hepatitis C (HCV): blood-borne; chronic in 50-80%, leading cause of liver transplantation. Hepatitis D (HDV): requires HBV co-infection. Hepatitis E (HEV): fecal-oral, self-limited but severe in pregnancy. Diagnosis: viral serology and PCR. Treatment: supportive for HAV/HEV; antivirals for HBV (tenofovir, entecavir) and HCV (DAAs).

Hepatitis Viruses

– Hepatitis A: fecal-oral, acute self-limited, vaccine (inactivated). Hepatitis B: blood/sexual/perinatal. Acute: HBsAg, IgM anti-HBc. Chronic: HBsAg >6 months, HBV DNA, HBeAg. Treatment: entecavir, tenofovir (TAF/TDF) - suppress DNA, prevent cirrhosis/HCC. HDV: requires HBV, pegylated interferon. Hepatitis C: blood-borne. Acute: HCV RNA. Chronic: genotype (1-6), fibrosis (F0-F4, elastography/biopsy). DAA: pangenotypic (glecaprevir/pibrentasvir 8w, sofosbuvir/velpatasvir 12w, sofosbuvir/velpatasvir/voxilaprevir for retreatment). SVR12 = cure. Hepatitis E: fecal-oral (genotype 1/2), zoonotic (3/4, pork), severe in pregnancy. Hepatitis G/GBV-C: non-pathogenic.

Herpes

– A common viral infection caused by herpes simplex virus (HSV, types 1 and 2) and varicella-zoster virus (VZV, herpes zoster/shingles). See Herpes Simplex and Herpes Zoster.

Herpes Simplex

– A viral infection caused by herpes simplex virus type 1 (HSV-1, primarily orolabial) and type 2 (HSV-2, primarily genital). Primary infection followed by lifelong latency in sensory ganglia with periodic reactivation. Oral herpes: cold sores/fever blisters. Genital herpes: painful vesicular lesions, recurrent outbreaks. Complications: herpes keratitis, encephalitis, and neonatal herpes. Treatment: acyclovir, valacyclovir, famciclovir for acute episodes and suppressive therapy.

Herpes Simplex (Cold Sores / Genital Herpes)

– A viral infection caused by herpes simplex virus type 1 (HSV-1, typically orofacial, though increasingly anogenital) and type 2 (HSV-2, predominantly anogenital). HSV is a double-stranded DNA virus of the Herpesviridae family with latency in sensory ganglia and periodic reactivation. Primary infection: often asymptomatic; when symptomatic, causes painful vesicular lesions on an erythematous base. Orofacial herpes (cold sores): HSV-1, vesicular lesions on the vermilion border of the lips (herpes labialis), triggered by sun exposure, fever, stress, immunosuppression. Genital herpes: HSV-2 (80%) and HSV-1 (20%), painful genital/perianal vesicles that ulcerate and crust over; primary episode lasts 2-4 weeks; recurrent episodes are milder (viral shedding for 3-5 days). Other manifestations: herpetic whitlow (finger), herpes gladiatorum (skin-to-skin contact sports), herpes keratitis (corneal infection, leading cause of infectious blindness in developed countries), eczema herpeticum (Kaposi varicelliform eruption in atopic dermatitis), herpes encephalitis (temporal lobe, most common cause of sporadic viral encephalitis in adults). Neonatal herpes: transmitted during vaginal delivery if active genital lesions, severe with high mortality. Diagnosis: clinical, viral PCR (golden standard), Tzanck smear (multinucleated giant cells). Treatment: acyclovir, valacyclovir, famciclovir. Antiviral therapy can reduce duration and severity of primary and recurrent episodes. Suppressive therapy (daily valacyclovir 500 mg) reduces recurrences by 70-80% and reduces transmission risk. No cure exists.

Herpes Zoster (Shingles)

– A painful, vesicular rash caused by reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia. VZV causes varicella (chickenpox) as a primary infection, then remains latent in sensory ganglia; reactivation occurs when cell-mediated immunity declines (age >50, immunosuppression, HIV, malignancy, chemotherapy, radiotherapy, stress). Incidence: 1 in 3 people develop shingles in their lifetime; incidence rises sharply after age 50. Prodrome: 2-3 days of burning, stabbing pain, paraesthesia in a dermatomal distribution before the rash appears. Rash: unilateral, vesicular, dermatomal (thoracic most common 50%, trigeminal–V1 ophthalmic branch 15%), clusters of clear vesicles on erythematous base, pustules, crusting over 7-10 days, typically resolves within 2-4 weeks. Complications: Postherpetic neuralgia (PHN, most common, 10-20% of patients, defined as pain persisting >90 days after rash onset, incidence increases with age; treatment: gabapentin, pregabalin, tricyclic antidepressants, lidocaine patch, capsaicin cream), herpes zoster ophthalmicus (Hutchinson sign: rash on tip of nose indicates nasociliary nerve involvement, increased risk of keratitis, uveitis, glaucoma, blindness; needs urgent ophthalmology referral), Ramsay Hunt syndrome (VZV in geniculate ganglion: ear pain, vesicles in auditory canal and pinna, facial palsy, tinnitus, hearing loss), disseminated zoster (visceral involvement in severely immunocompromised patients), motor neuropathy (rare, segmental weakness). Diagnosis: clinical, PCR or DFA of vesicle fluid. Treatment: antiviral therapy (acyclovir 800 mg 5x/day, valacyclovir 1 g TID, famciclovir 500 mg TID for 7 days) best within 72 hours of rash onset, reduces acute pain and PHN risk; analgesics (NSAIDs, opioids, gabapentinoids). Prevention: recombinant zoster vaccine (Shingrix, adjuvanted subunit glycoprotein E vaccine, 97% efficacy in adults >50, 2 doses 2-6 months apart, recommended for all immunocompetent adults >50).

Histoplasmosis

– A systemic fungal infection caused by Histoplasma capsulatum, a dimorphic fungus found in nitrogen-rich soil enriched with bat or bird guano, most commonly in the Ohio and Mississippi River valleys of the United States, Central and South America, and parts of Africa and Asia. The organism exists as a mold in the environment (producing infectious microconidia that are aerosolized when soil is disturbed) and converts to a yeast form at body temperature (37 degrees C). Infection occurs through inhalation of conidia, which reach the alveoli and are ingested by alveolar macrophages, where they convert to yeasts and multiply intracellularly. Most immunocompetent individuals (90 percent) develop asymptomatic or self-limited, mild influenza-like illness. The remaining 10 percent develop progressive forms depending on the intensity of exposure and host immune status. Acute pulmonary histo-.

HIV/AIDS

– HIV-1/2: retrovirus, CD4+ T cell depletion. Transmission: sexual, blood, perinatal, breast milk. Acute retroviral syndrome: fever, lymphadenopathy, pharyngitis, rash, high viral load. Clinical latency: CD4 decline over years. AIDS: CD4 <200 or AIDS-defining illness (PJP, toxo, CMV, MAC, crypto, KS, lymphoma, TB). ART: 3+ drugs from 2 classes (2 NRTI + INSTI/NNRTI/PI). Preferred: bictegravir/TAF/FTC, dolutegravir/abacavir/lamivudine (HLA-B*5701 negative), dolutegravir + TAF/FTC. Monitoring: viral load (target <50 copies/mL), CD4, resistance if failure. OI prophylaxis: PJP (TMPSMX if CD4<200), Toxo (if CD4<100), MAC (azithromycin if CD4<50), fungal (fluconazole if high risk). IRIS: paradoxical worsening after ART initiation. PrEP: TDF/FTC or TAF/FTC daily, cabotegravir LA bimonthly. PEP: 3-drug regimen within 72h x28d.

HIV (Human Immunodeficiency Virus)

– A retrovirus (family Retroviridae, genus Lentivirus) that causes acquired immunodeficiency syndrome (AIDS) by infecting and destroying CD4+ T lymphocytes (helper T cells). Two types: HIV-1 (pandemic, most common worldwide) and HIV-2 (less virulent, West Africa). HIV is an enveloped RNA virus; its genome encodes structural proteins (Gag, Env), enzymes (Pol–reverse transcriptase, integrase, protease), and regulatory proteins (Tat, Rev, Nef, Vif, Vpr, Vpu). Transmission: sexual contact (most common worldwide), blood-borne (IV drug use, blood transfusion), mother-to-child (vertical, during pregnancy, labour, or breastfeeding). Pathophysiology: HIV enters CD4+ cells via binding of gp120 to CD4 receptor and co-receptors (CCR5 or CXCR4); reverse transcriptase converts RNA to DNA; integrase incorporates viral DNA into host genome; protease cleaves viral polyproteins for assembly. Without treatment: acute HIV syndrome (2-4 weeks after infection: fever, rash, lymphadenopathy, pharyngitis, myalgia), clinical latency (8-12 years, viral replication continues, CD4 declines), AIDS (CD4 <200 cells/microL, opportunistic infections–PCP, cerebral toxoplasmosis, CMV retinitis, MAC, cryptococcal meningitis, and AIDS-defining cancers–Kaposi sarcoma, non-Hodgkin lymphoma, cervical cancer). Diagnosis: 4th generation antigen/antibody combo test (detects p24 antigen and HIV antibodies), confirm with Western blot or HIV RNA PCR. Management: antiretroviral therapy (ART) for life–combination therapy with 2 NRTIs (tenofovir disoproxil fumarate/emtricitabine) plus a third agent (integrase inhibitor–dolutegravir, bictegravir; NNRTI; or boosted PI). ART reduces viral load to undetectable (<20 copies/mL), restores immune function, prevents AIDS progression, and eliminates sexual transmission risk (U=U–undetectable equals untransmittable). Pre-exposure prophylaxis (PrEP: tenofovir/emtricitabine daily or on-demand for HIV-negative individuals at high risk). Post-exposure prophylaxis (PEP: 28 days of ART, started within 72 hours of exposure). Prevention: condoms, needle exchange, circumcision (reduces female-to-male transmission by 60%), prevention of mother-to-child transmission (ART during pregnancy + formula feeding). Global: 39 million people living with HIV; 1.3 million new infections/year; 630,000 deaths/year.

Host

– A person or other living organism that can be infected by a virus or other pathogen under natural.

Hydrophobia

– See Rabies.

Immunization

– Routine childhood (CDC schedule): HepB, RV, DTaP, Hib, PCV, IPV, Influenza, MMR, VAR, HepA, HPV, MenACWY, MenB, COVID19. Adult: Tdap/Td q10y, influenza annually, zoster (RZV 2-dose), pneumococcal (PCV20 or PCV15+PPSV23), HPV (through 26, shared decision 27-45), HepA/B, MenACWY/MenB (riskbased), COVID-19. Pregnancy: Tdap 27-36w, influenza, COVID-19, RSV (32-36w). Contraindications: anaphylaxis to component, live vaccines in immunocompromised/pregnancy (MMR, VAR, LAIV, yellow fever, oral typhoid). Precautions: moderate/severe illness, GBS <6w post-vaccine. Vaccine safety: VAERS, VSD, CISA. Hesitancy: communication (presumptive recommendation, motivational interviewing).

Incubation period

– The time between when a person is exposed to an infection and when symptoms appear.

Infection

– The growth of harmful organisms that can cause disease, such as bacteria, in the body.

Infections in Pregnancy

– Listeriosis: Listeria monocytogenes (deli meats, soft cheese), meningoencephalitis, fetal loss, neonatal sepsis. Ampicillin + gentamicin. GBS: vaginal colonization 20-30.

Infective Endocarditis

– Infection of the endocardial surface of the heart, most commonly involving valves (native or prosthetic). Etiology: Staphylococcus aureus (most common overall, 30.

Intravascular Catheter Infections

– CLABSI: central line-associated bloodstream infection. Diagnosis: quantitative blood cultures (catheter vs peripheral >5:1), differential time to positivity (>2h), catheter tip culture (>15 CFU).

Catheter-related BSIs: coagulase-negative staphylococci (most common), S. aureus, enterococci, gram-negatives, Candida. Management: remove catheter if S. aureus, Pseudomonas, Candida, mycobacteria, tunnel infection, port abscess, severe sepsis, persistent bacteremia >72h. Exchange over guidewire if no signs of exit site/tunnel infection and non-severe pathogens (CoNS, Bacillus, Corynebacterium). Antibiotics: CoNS 5-7d, S. aureus 2w (4-6w if endocarditis), gram-negative 7-14d, Candida 14d post-clearance. Lock therapy: antibiotic/ethanol/heparin locks for salvage. Prevention: maximal sterile barrier, chlorhexidine prep, daily necessity review, chlorhexidine dressings.

Keshan disease

– Heart disease caused by a lack of selenium, an element that the body needs to function properly.

Lassa Fever

– An acute viral hemorrhagic illness caused by Lassa virus, an arenavirus, endemic in West Africa (Nigeria, Sierra Leone, Guinea, Liberia). Transmission: exposure to excreta (urine, feces) of the multimammate rat (Mastomys natalensis), the natural reservoir. Incubation: 6-21 days. Clinical features: 80% of infections are asymptomatic or mild. Symptomatic illness: gradual onset of fever, malaise, headache, sore throat, myalgia, retrosternal pain, cough, and gastrointestinal symptoms (vomiting, diarrhea). Severe disease (20%): hemorrhage (gingival, nasal, gastrointestinal, vaginal), facial/neck edema, pleural effusion, hypotension/shock, encephalopathy, seizures, multi-organ failure, and sensorineural hearing loss (a unique complication, occurring in 25-30% of survivors, may be permanent). Case fatality rate: 1-2% overall, 15-20% in hospitalised patients, >50% in third trimester of pregnancy (high maternal and fetal mortality). Diagnosis: RT-PCR (most sensitive, first week), Lassa virus-specific IgM/IgG (seroconversion by week 3), virus isolation (BSL-4). Treatment: ribavirin (IV or oral, most effective if given early, within 6 days of onset). Supportive care: fluid resuscitation, blood products, oxygen, vasopressors. Prevention: rodent control, safe food storage, avoidance of rodent contact. No licensed vaccine. Outbreak management: strict infection control (contact and droplet precautions), case surveillance, contact tracing.

Legionnaires' Disease

– A severe form of pneumonia caused by Legionella pneumophila, a Gram-negative aerobic bacillus that is found naturally in freshwater environments (lakes, streams) and artificially in water systems (air conditioning cooling towers, hot tubs, decorative fountains, plumbing systems, showers, humidifiers). First identified after an outbreak at the 1976 American Legion convention in Philadelphia. Transmission: inhalation of aerosolised contaminated water (not person-to-person). Incubation: 2-14 days. Risk factors: age >50, smoking, chronic lung disease, immunocompromised (especially organ transplant recipients), diabetes, and male sex. Clinical features: atypical pneumonia–fever (>39 C), cough (initially non-productive), myalgia, headache, confusion, diarrhea, relative bradycardia, and hyponatraemia. Pontiac fever: a milder, self-limiting flu-like illness without pneumonia caused by the same bacterium. Diagnosis: urinary antigen test (detects L. pneumophila serogroup 1, 70-90% sensitive), sputum culture (requires selective media, BCYE agar), PCR on respiratory samples, direct fluorescent antibody (DFA) staining. Treatment: macrolides (azithromycin 500 mg daily, 7-14 days) or fluoroquinolones (levofloxacin 750 mg daily, 7-14 days) as first-line. Doxycycline is an alternative. Severe cases require IV therapy. Complications: respiratory failure, septic shock, empyema, lung abscess, acute kidney injury, pericarditis. Mortality: 10-15% (higher in immunocompromised). Reportable disease in most jurisdictions.

Leishmaniasis

– A parasitic disease caused by protozoa of the genus Leishmania, transmitted by the bite of infected female phlebotomine sandflies. Visceral leishmaniasis (kala-azar): caused by L. donovani/L. infantum – fever, hepatosplenomegaly, pancytopenia, weight loss, hypergammaglobulinaemia; fatal if untreated. Cutaneous leishmaniasis: skin ulcers at bite sites that heal slowly with scarring (L. major, L. tropica, L. mexicana). Mucocutaneous leishmaniasis: destructive lesions of the nasopharynx (L. braziliensis). Diagnosis: microscopy of tissue smears (amastigotes in macrophages), culture, PCR. Treatment: liposomal amphotericin B (visceral), miltefosine, sodium stibogluconate.

Leprosy

– A chronic bacterial infection caused by Mycobacterium leprae, affecting skin and peripheral nerves. Transmission via nasal droplets. Spectrum of disease determined by immune response: tuberculoid (paucibacillary): one or few hypopigmented, anesthetic skin lesions, thickened nerves. Lepromatous (multibacillary): numerous symmetric nodules, plaques, and nerve involvement with sensory loss and deformities (leonine facies, claw hand, foot drop). Diagnosis: skin biopsy with acid-fast bacilli. Treatment: multi-drug therapy (dapsone + rifampin for paucibacillary; add clofazimine for multibacillary) for 6-12 months. Reactions require corticosteroids.

Leprosy (Hansen's Disease)

– See Hansen's Disease (Leprosy).

Leptospirosis

– A zoonotic bacterial infection caused by spirochaetes of the genus Leptospira, transmitted through contact with water or soil contaminated with the urine of infected animals (rodents, cattle, dogs). Incubation period 5-14 days. Mild form: fever, headache, myalgia (especially calves), conjunctival suffusion. Severe form (Weil disease): jaundice, renal failure, hemorrhage, myocarditis, and pulmonary hemorrhage with respiratory failure. Diagnosis: culture (blood/urine), PCR, serology (microscopic agglutination test). Treatment: doxycycline or penicillin for mild disease; intravenous penicillin or ceftriaxone for severe disease. Prevention: avoiding contaminated water, rodent control, doxycycline prophylaxis for high-risk exposure.

Listeriosis

– A bacterial infection caused by Listeria monocytogenes, transmitted through contaminated food (deli meats, soft cheeses, unpasteurized dairy, produce). Risk groups: pregnant women (fever, flu-like illness), neonates (sepsis, meningitis), immunocompromised, and elderly. Manifestations: febrile gastroenteritis in immunocompetent; bacteremia and meningitis in at-risk. Neonatal: early-onset (sepsis, granulomatosis infantiseptica) and late-onset (meningitis) listeriosis. Diagnosis: blood and CSF culture. Treatment: ampicillin + gentamicin (synergistic); TMP-SMX for PCN-allergic.

Lupus Vulgaris

– A chronic, progressive form of cutaneous tuberculosis (TB of the skin) caused by Mycobacterium tuberculosis. Once common in Europe, now rare in developed countries but still seen in developing regions. Presents as red-brown, gelatinous plaques (apple-jelly nodules on diascopy) typically on the face (nose, cheeks), which slowly enlarge and ulcerate over years without treatment. Can cause extensive tissue destruction and disfigurement. Diagnosis: skin biopsy (epithelioid granulomas with caseous necrosis), acid-fast stain, mycobacterial culture, PCR. Treatment: standard anti-TB therapy (isoniazid, rifampicin, ethambutol, pyrazinamide for 2 months, then isoniazid + rifampicin for 4 months).

Lyme Disease

– A tick-borne infection caused by Borrelia burgdorferi (Ixodes scapularis, deer tick), endemic to the northeastern and upper midwestern US. Early localized: erythema migrans (bullseye rash, >5cm) at tick bite site, fatigue, fever, myalgia. Early disseminated: multiple EM lesions, cranial nerve VII palsy (facial nerve), meningitis, carditis (AV block). Late: arthritis (monoarticular, knee), acrodermatitis chronica atrophicans, encephalopathy. Diagnosis: clinical (EM rash is diagnostic), two-tier serology (ELISA + Western blot). Treatment: doxycycline for early; ceftriaxone IV for neurologic or cardiac involvement. Prevention: tick avoidance, prophylactic doxycycline after high-risk tick bite.

Lymphangitis

– Inflammation of the lymphatic channels, most commonly due to acute bacterial infection spreading from a peripheral wound or cellulitis. The most common causative organism is group A Streptococcus pyogenes (90%), followed by Staphylococcus aureus. Lymphangitis is a sign that infection is spreading and requires urgent medical attention. Clinical features: visible red, warm, tender linear streaks (lymphangitic streaking) extending proximally from the infected wound towards the draining lymph nodes (e.g., from the hand to the axilla, from the foot to the groin). Associated with: cellulitis, ascending lymphadenitis, fever, chills, malaise, and leukocytosis. The draining lymph nodes may be enlarged and tender (lymphadenitis). Diagnosis: clinical (pathognomonic appearance). Blood cultures may be positive. Complications: sepsis, bacteraemia, necrotising fasciitis (if anaerobes or mixed infection), and abscess formation. Treatment: antibiotics active against Streptococcus and Staphylococcus (e.g., cephalexin, clindamycin, flucloxacillin, or IV ceftriaxone/vancomycin for severe cases), elevation of the affected limb, warm compresses, analgesics. Surgical drainage if abscess forms. Prevention: prompt wound cleaning and antisepsis.

Malaria

– A life-threatening mosquito-borne disease caused by Plasmodium parasites (P. falciparum, P. vivax, P. ovale, P. malariae, P. knowlesi), transmitted by Anopheles mosquitoes. Presents with cyclical fevers, chills, headache, myalgia, and anemia. Severe P. falciparum malaria: cerebral malaria, severe anemia, respiratory distress, and multi-organ failure. Diagnosis: blood smear (thick and thin), rapid diagnostic test. Treatment: artemisinin-based combination therapy (ACT); severe malaria requires IV artesunate. Prevention: insecticide-treated nets, chemoprophylaxis.

Meningococcal Disease

– A bacterial infection caused by Neisseria meningitidis (serogroups A, B, C, W, Y), transmitted via respiratory droplets. Meningococcemia: sudden onset fever, petechial/purpuric rash, hypotension, and DIC. Meningococcal meningitis: fever, nuchal rigidity, headache, photophobia. Waterhouse-Friderichsen syndrome: fulminant meningococcemia with bilateral adrenal hemorrhage and septic shock. Diagnosis: blood and CSF culture, Gram stain, PCR. Treatment: IV ceftriaxone urgently; droplet precautions. Chemoprophylaxis for close contacts (rifampin, ciprofloxacin, ceftriaxone). Vaccination: MenACWY and MenB vaccines are available.

Mononucleosis

– An acute viral syndrome caused by Epstein-Barr virus (EBV), primarily affecting adolescents and young adults. Classic triad: fever, pharyngitis/exudative tonsillitis, and cervical lymphadenopathy. Associated with splenomegaly, fatigue, and atypical lymphocytosis. Diagnosis: positive heterophile antibody (Monospot) and EBV serology. Self-limited, supportive care. Avoid contact sports for 3-4 weeks due to splenic rupture risk.

Neonatal Infections

– Early-onset sepsis (EOS, <72h): GBS, E. coli, Listeria. Risk factors: maternal GBS+, chorioamnionitis, PROM >18h. Kaiser sepsis calculator for management. Late-onset sepsis (LOS, >72h): CoNS, S. aureus, gram-negatives, Candida. NEC: prematurity, formula feeding, ischemic gut. Pneumatosis intestinalis, portal venous gas. Treatment: NPO, antibiotics (ampicillin + gentamicin/clindamycin + metronidazole), surgery for perforation. HSV: SEM (skin/eye/mouth), CNS, disseminated. Acyclovir 20mg/kg q8h IV x14d (CNS/disseminated)

Nocardiosis

– A bacterial infection caused by Nocardia species (N. asteroides, N. brasiliensis), found in soil. Most common in immunocompromised patients (transplant, HIV, chronic steroid use). Pulmonary nocardiosis (most common): cough, fever, necrotizing pneumonia, cavitary lesions. Disseminated: brain abscesses (most common site), skin and soft tissue abscesses, bacteremia. Cutaneous nocardiosis: cellulitis, abscess, myectoma after trauma in immunocompetent. Diagnosis: modified acid-fast stain (partially acid-fast), culture (aerobic, grows slowly). Treatment: TMP-SMX (first-line); alternatives: carbapenems, linezolid, amikacin for severe or resistant cases. Prolonged therapy (6-12 months).

Non-Tuberculous Mycobacteria

– NTM: environmental mycobacteria (M. avium complex - MAC, M. kansasii, M. abscessus). Pulmonary NTM: ATS/IDSA criteria (clinical + radiographic + microbiologic: 2+ sputum or 1 BAL + nodules/cavities/bronchiectasis on CT). MAC: nodular bronchiectatic (middle-aged women, "Lady Windermere syndrome") or fibrocavitary (older men, COPD). Treatment: MAC macrolide (azithromycin/clarithromycin) + ethambutol + rifampin x12 months after culture conversion. M. kansasii - INH + RIF + EMB x12m post-conversion. M. abscessus - macrolide + IV amikacin + imipenem/cefoxitin + oral (clarithromycin) - difficult, often requires surgery. Disseminated MAC: AIDS (CD4 <50), MAC bacteremia, fever, weight loss, anemia. Treatment: clarithromycin/azithromycin + ethambutol + rifabutin; prophylaxis if CD4 <50 (azithromycin 1200mg weekly).

Onchocerciasis

– Also known as river blindness, a parasitic disease caused by the filarial nematode Onchocerca volvulus, transmitted by repeated bites of infected blackflies (Simulium species) breeding near fast-flowing rivers. Adult worms form subcutaneous nodules (onchocercomata). Microfilariae migrate through skin and eyes, causing intense pruritus, dermatitis, depigmentation (leopard skin), lichenification, and, most importantly, ocular damage leading to blindness (keratitis, iridocyclitis, chorioretinitis, optic atrophy). Diagnosis: skin snip biopsy (microfilariae), slit-lamp examination of the cornea, PCR. Treatment: ivermectin (annual or semi-annual mass drug administration, kills microfilariae but not adults). No safe, effective macrofilaricide exists.

Opportunistic Infection

– An infection caused by organisms that ordinarily do not cause disease in immunocompetent hosts but cause significant illness when host defences are impaired. Common causative organisms: Pneumocystis jirovecii (PCP pneumonia in HIV with CD4 $<200/\mu$L), Toxoplasma gondii (cerebral toxoplasmosis—HIV CD4 $<100/\mu$L), CMV (retinitis, colitis—HIV CD4 $<50/\mu$L, post-transplant), Mycobacterium avium complex (disseminated—HIV CD4 $<50/\mu$L), Cryptococcus neoformans (meningitis—HIV CD4 $<100/\mu$L), Aspergillus fumigatus (invasive aspergillosis—neutropenia, post-transplant), and Candida species (mucocutaneous and invasive—HIV, ICU, neutropenia). Prophylaxis is given based on CD4 thresholds in HIV (cotrimoxazole for PCP/Toxoplasma, azithromycin for MAC, fluconazole for Cryptococcus in endemic areas). Immune reconstitution inflammatory syndrome (IRIS) can occur when antiretroviral therapy is started in severely immunocompromised HIV patients.

Orchitis

– Inflammation of the testis, often associated with epididymitis (epididymo-orchitis). Causes: infectious—viral (mumps—most common cause of viral orchitis in postpubertal males, typically 4–6 days after parotitis, bilateral in 30%), bacterial (Escherichia coli, Pseudomonas, sexually transmitted—Chlamydia trachomatis, Neisseria gonorrhoeae—typically ascending from urethra/epididymis), Coxsackie virus, and rarely tuberculosis. Non-infectious: trauma, autoimmune, and idiopathic. Presentation: acute scrotal pain and swelling, erythema, fever, and nausea. Diagnosis: clinical plus urine dipstick, urine culture, and Doppler ultrasound (increased testicular blood flow distinguishes orchitis from testicular torsion—torsion shows decreased flow). Treatment: antibiotics (ceftriaxone 500 mg IM single dose plus doxycycline 100 mg BD for 14 days for STI; levofloxacin for enteric organisms), NSAIDs, scrotal elevation, and cold packs; mumps orchitis is supportive (analgesia, steroids for severe inflammation). Complications: abscess, testicular infarction, atrophy, and infertility.

Orf

– A zoonotic viral skin infection caused by the parapoxvirus, transmitted to humans from infected sheep and goats (direct contact with lesions, fomites—virus survives in scabs for months). Orf is an occupational disease of farmers, butchers, and veterinarians. Incubation: 3–7 days. Lesions typically appear on the hands, fingers, or forearms: a single papule at the inoculation site progresses through stages—macule, target (bullseye) lesion, nodule (exudative, weeping), regenerative (dry crust), papillomatous, and regression (resolves over 4–6 weeks without scarring). Lesions may be large (up to 3–5 cm), tender, and occasionally complicated by secondary bacterial infection, lymphangitis, or erythema multiforme. Diagnosis is clinical; electron microscopy of vesicle fluid shows characteristic ovoid virions. Treatment: supportive (hygiene, dressings), no specific antiviral; lesions heal spontaneously. Prevention: gloves during handling of animals with orf lesions, quarantine of infected animals.

Osteomyelitis

– An infection of bone and bone marrow, most commonly caused by bacterial pathogens, with Staphylococcus aureus responsible for the majority of cases. It is classified by etiology (hematogenous, contiguous focus, or direct inoculation), duration (acute less than 2 weeks, subacute 2 weeks to 3 months, chronic greater than 3 months), and route. Hematogenous osteomyelitis typically affects the metaphysis of long bones in children (most commonly the femur, tibia, and humerus) and the vertebral bodies in adults (vertebral osteomyelitis).

Outbreak

– Synonymous with epidemic. Sometimes used to refer to a local epidemic as compared to a larger, general epidemic.

Pandemic

– A disease outbreak affecting large populations or a whole region, country, or continent.

Parasite

– An organism that lives on or within a host organism, deriving nutrients at the host's expense. Medical parasites are classified into three main groups: (1) Protozoa—single-celled organisms (Plasmodium—malaria, Toxoplasma—toxoplasmosis, Giardia—giardiasis, Entamoeba histolytica—amoebic dysentery, Leishmania—leishmaniasis, Trypanosoma—African sleeping sickness, Chagas disease). (2) Helminths—multicellular worms (nematodes/roundworms—Ascaris, hookworm, filariae; cestodes/tapeworms—Taenia solium, Echinococcus; trematodes/flukes—Schistosoma, liver flukes). (3) Ectoparasites—arthropods that infest the skin (scabies mites, lice, fleas, ticks, bed bugs). Parasitic infections disproportionately affect tropical and subtropical regions, causing significant morbidity and mortality (malaria alone causes >600,000 deaths annually). Transmission: vector-borne (mosquitoes, sandflies, ticks), fecal–oral (contaminated water/food), skin penetration (hookworm, Schistosoma), and vertical (Toxoplasma, malaria). Diagnosis: microscopy (blood films, stool examination, tissue biopsy), serology, antigen detection, and PCR. Treatment: antiparasitic drugs (antimalarials, antihelminthics—albendazole, praziquantel, ivermectin, metronidazole). Prevention: vector control, sanitation, hygiene, and health education.

Parasitic Infections

– Malaria: Plasmodium (falciparum, vivax, ovale, malariae, knowlesi). Falciparum: cerebral malaria, severe anemia, ARDS, AKI, hypoglycemia. Diagnosis: thick/thin smear, RDT (HRP2, pLDH),.

Pathogen

– A tiny organism such as a virus, bacterium, or parasite that can invade the body and produce disease.

Pediculus humanus capitis

– A blood-sucking parasite commonly known as the louse (plural, lice) that can cause an itchy scalp; infestations are highly contagious and especially common in school-age children.

Pinworm

– Infection with Enterobius vermicularis, the most common helminth infection in developed countries, primarily in school-age children. Lifecycle: ingestion of eggs → larvae hatch in small intestine → adults colonize the cecum/appendix; female migrates to perianal area at night to deposit eggs. Presents with intense perianal itching (especially at night), restless sleep, and secondary bacterial infection from scratching. Diagnosis: scotch tape test (adhesive tape applied to perianal area in the morning, examined microscopically for eggs). Treatment: albendazole or mebendazole (single dose, repeated after 2 weeks). Treat all household contacts.

Plague

– A bacterial infection caused by Yersinia pestis, transmitted by fleas from rodents. Three forms: bubonic (most common, 80-90%): painful, swollen lymph nodes (buboes) after flea bite, fever, chills. Septicemic: bacteremia without bubo, endotoxic shock, DIC, purpura. Pneumonic: primary (inhalation) or secondary (from bubonic spread); fulminant pneumonia with bloody sputum; highly contagious via respiratory droplets. Untreated bubonic mortality 50-60%; pneumonic nearly 100% untreated. Diagnosis: culture, PCR, Gram stain of bubo aspirate. Treatment: streptomycin, gentamicin, doxycycline, or fluoroquinolones. Isolation for pneumonic plague.

Polio

– Abbreviation for poliomyelitis, an acute viral infection caused by poliovirus, an enterovirus of the Picornaviridae family transmitted via the fecal-oral route, primarily affecting children under 5 years of age. The virus replicates in the oropharynx and intestinal mucosa, then invades regional lymph nodes and the bloodstream; in a small proportion of cases (less than 1 percent), it invades the central nervous system and destroys motor neurons in the anterior horn of the spinal cord and brainstem. Most infections (approximately 95 percent) are asymptomatic or cause only mild nonspecific illness (abortive polio), while non-paralytic polio presents with fever, headache, vomiting, and meningismus. Paralytic polio presents with asymmetric acute flaccid paralysis, most commonly affecting the lower extremities, with loss of deep tendon reflexes but preserved sensation; bulbar polio affects the cranial nerves and respiratory center, potentially causing respiratory failure and death. There is no specific antiviral treatment; management is supportive, including physical therapy and respiratory support. Prevention is highly effective through vaccination with inactivated polio vaccine (IPV, Salk vaccine, injected) or oral polio vaccine (OPV, Sabin vaccine, oral). Global polio eradication efforts have reduced cases by over 99 percent since 1988, with wild poliovirus remaining endemic only in Afghanistan and Pakistan.

Poliomyelitis

– An acute viral infection caused by poliovirus (enterovirus), transmitted fecal-orally. Most infections are asymptomatic (95%). Non-paralytic polio: fever, headache, vomiting, and meningismus. Paralytic polio: asymmetric acute flaccid paralysis due to destruction of anterior horn cells; most commonly affects lower extremities. Respiratory paralysis can be fatal. Post-polio syndrome: progressive muscle weakness decades after initial infection. Diagnosis: stool and throat culture, CSF PCR. No antiviral treatment; supportive care. Prevention: IPV (inactivated polio vaccine). Global eradication is near-complete.

Prophylaxis

– Steps taken to prevent a particular disease or condition, such as taking nitroglycerin to prevent angina.

Prostatitis

– An inflammation of the prostate gland, sometimes caused by a bacterial infection, which may result in painful or difficult urination.

Prosthetic Joint Infection

– PJI definition (MSIS): major (sinus tract, 2+ positive cultures), minor (elevated serum CRP/ESR, synovial WBC >3000 or PMN >80.

Pus

– A thick, yellow or green liquid that is composed of dead cells and bacteria, most often found at the site of a bacterial infection.

Q Fever

– A zoonotic disease caused by Coxiella burnetii, an obligate intracellular Gram-negative bacterium. Q fever is transmitted to humans primarily by inhalation of contaminated aerosols from infected livestock (cattle, sheep, goats) and their products (placenta, birth fluids, urine, feces, milk). Incubation: 2–3 weeks. Acute Q fever: high fever (up to 40°C, may last 1–3 weeks), severe headache, myalgia, arthralgia, fatigue, and atypical pneumonia (cough, chest pain, abnormal chest X-ray). Hepatitis (granulomatous—doughnut granulomas on liver biopsy) is common. Chronic Q fever (<5% of cases): endocarditis (culture-negative, affecting prosthetic and native valves), vascular graft infection, and osteomyelitis. Diagnosis: serology (phase II IgG and IgM—antibodies rise 2–4 weeks after infection; phase I IgG ≥1:800 suggests chronic infection), PCR on blood or tissue during acute illness. Treatment: acute—doxycycline 100 mg BD for 14 days (first-line, shortens fever duration). Chronic—doxycycline plus hydroxychloroquine for ≥18 months. Prevention: pasteurisation of milk, biosecurity measures on farms, and veterinary vaccination (phase I vaccine—Coxevac). Q fever is a notifiable disease in many countries. The Netherlands experienced the largest outbreak (2007–2010, >4,000 cases) linked to intensive goat farming.

Quarantine

– A period of time in which a sick person is kept away from others to prevent the spread of disease.

Relapsing Fever

– An acute febrile illness caused by Borrelia species, characterized by recurrent episodes of fever separated by afebrile intervals. Two forms: (1) Louse-borne relapsing fever (Borrelia recurrentis)—transmitted by the human body louse (Pediculus humanus corporis), associated with overcrowding, poverty, and refugee settings; epidemic potential. (2) Tick-borne relapsing fever (Borrelia duttonii, B. hermsii, B. parkeri)—transmitted by soft ticks (Ornithodoros), endemic in Africa, North America, and the Middle East; tends to have more relapses. Pathophysiology: Borrelia spirochetes multiply in the blood and cause high fever; the immune system clears the organisms by antibody production, but antigenic variation (variable major proteins—Vmp) allows a new variant to emerge and cause a relapse. Clinical: sudden onset of high fever (39–40°C), rigors, severe headache, myalgia, arthralgia, photophobia, and hepatosplenomegaly. The initial febrile episode lasts 3–6 days, followed by an afebrile period of 7–9 days, then 1–3 relapses (tick-borne) or a single relapse (louse-borne). A petechial rash may be present. Diagnosis: darkfield microscopy of blood smears (spirochetes visible during febrile episodes); PCR. Treatment: doxycycline 100 mg BD for 7–10 days (or tetracycline, erythromycin). Jarisch–Herxheimer reaction (fever, rigors, hypotension) occurs within 2–4 hours of starting antibiotics in 50–90% of patients (especially louse-borne)—pre-treated with IV fluids and antipyretics; severe cases may require hydrocortisone.

River Blindness

– See Onchocerciasis.

Salmonella

– A genus of Gram-negative, facultatively anaerobic, rod-shaped bacteria in the Enterobacteriaceae family, a major cause of foodborne illness worldwide. Classification: Salmonella enterica subsp. enterica serovars (>2,500 serovars). Clinically important serovars: S. Typhi and S. Paratyphi A, B, C (cause enteric fever—typhoid and paratyphoid fever—systemic illness, humans are the only reservoir, transmitted by fecal–oral route through contaminated food/water), and non-typhoidal Salmonella (NTS): S. Enteritidis and S. Typhimurium (most common—cause self-limited gastroenteritis, transmitted from animals—poultry, eggs, meat, reptiles). Pathogenesis: bacteria invade intestinal epithelial cells (via type III secretion system), survive within macrophages, and cause inflammation. Clinical: NTS gastroenteritis—nausea, vomiting, non-bloody diarrhea (may become bloody), abdominal cramps, fever (self-limiting, 4–7 days). Enteric fever (typhoid)—incubation 7–14 days, prolonged high fever (stepwise rise), headache, malaise, rose spots on trunk, relative bradycardia, hepatosplenomegaly, constipation initially then diarrhea. Complications of typhoid: intestinal perforation and hemorrhage (most feared). Chronic carriers: S. Typhi persists in the gallbladder (gallstones) and is shed in stool (Typhoid Mary). Diagnosis: stool culture (gastroenteritis), blood culture (enteric fever—most sensitive in first week), bone marrow culture (gold standard), and Widal test (serology, limited utility). Treatment: NTS gastroenteritis—supportive care (antibiotics not recommended for mild cases; prolongs carriage); severe or immunocompromised—ceftriaxone or azithromycin. Enteric fever—ceftriaxone 2 g IV daily for 10–14 days, azithromycin, or ciprofloxacin (if sensitive). Multidrug-resistant (MDR) and extensively drug-resistant (XDR) S. Typhi are emerging (azithromycin and carbapenems for XDR). Prevention: hand hygiene, safe food/water, typhoid vaccines (Vi polysaccharide injectable, Ty21a oral live attenuated).

Scabies

– A contagious skin infestation caused by the mite Sarcoptes scabiei var. hominis, transmitted by prolonged skin-to-skin contact. The female mite burrows into the stratum corneum, laying eggs, causing intense pruritus (worse at night) and a papular erythematous rash. Typical distribution: interdigital finger webs, wrists, elbows, axillae, waist, genitalia (spares the face in adults). Burrows appear as greyish, serpiginous lines. Crusted (Norwegian) scabies occurs in immunocompromised patients. Diagnosis: clinical, confirmed by microscopy of scrapings from burrows (mite, eggs, feces). Treatment: permethrin 5% cream (two applications, 1 week apart) or oral ivermectin. Treat all close contacts.

Sepsis

– The destruction or infection of tissues by disease-causing organisms, usually accompanied by a fever.

Septicaemia

– See Sepsis.

Septicemia

– A condition in which disease-causing organisms have spread to the bloodstream from an infection elsewhere in the body. Also known as blood poisoning.

Sexually Transmitted Infections (STIs)

– Sexually transmitted infections are infections passed from one person to another through sexual contact, including vaginal, anal, and oral sex. Common STIs include chlamydia, gonorrhea, syphilis, herpes simplex virus, human papillomavirus, and HIV. Many STIs can be asymptomatic, making regular screening essential for sexually active individuals; treatment varies by infection, with bacterial STIs being curable with antibiotics, while viral STIs are managed with antiviral medications to reduce symptoms and transmission risk.

Smallpox

– A highly contagious and deadly disease caused by variola virus, eradicated globally by 1980 through vaccination. Characterized by febrile prodrome followed by vesicular/pustular rash (centrifugal distribution, all lesions at same stage). Last known natural case: 1977. Bioterrorism concern due to stored virus in WHO-approved laboratories. Diagnosis: PCR, electron microscopy. No proven treatment; supportive care. Vaccination within 3-4 days of exposure can prevent disease.

Staphylococcal Infection

– Infections caused by Staphylococcus aureus and other staphylococci. Manifestations range from superficial skin infections (folliculitis, furuncles, carbuncles, impetigo) to deep-seated infections (osteomyelitis, septic arthritis, endocarditis, pneumonia, bacteremia, sepsis). Methicillin-resistant S. aureus (MRSA) is a major healthcare and community-associated pathogen. Treatment: beta-lactams for MSSA, vancomycin for MRSA; incision and drainage for abscesses.

Streptococcal Infection

– Infections caused by Streptococcus pyogenes (Group A Strep, GAS). Common presentations: pharyngitis (strep throat), scarlet fever (pharyngitis with erythematous rash), impetigo, cellulitis, and necrotizing fasciitis. Immune-mediated complications: acute rheumatic fever (can cause rheumatic heart disease) and post-streptococcal glomerulonephritis. Diagnosis: rapid antigen test, throat culture. Treatment: penicillin or amoxicillin. Prevention of rheumatic fever requires adequate treatment of GAS pharyngitis.

Strongyloidiasis

– Infection with Strongyloides stercoralis, a unique helminth capable of autoinfection. Most infections asymptomatic or mild (abdominal pain, diarrhea, urticaria). Larvae migration causes larva currens (rapidly migrating serpiginous rash, perianal). Hyperinfection syndrome: life-threatening in immunocompromised (HIV/HTLV-1, corticosteroids, transplant), leading to massive larval dissemination with Gram-negative bacteremia, meningitis, and respiratory failure. Diagnosis: stool microscopy (Baermann technique, agar plate culture), serology. Treatment: ivermectin (first-line); albendazole as alternative. Screening and treatment before immunosuppression is critical.

Swine Flu

– See Influenza.

Taeniasis (Tapeworm Infection)

– Intestinal infection caused by adult tapeworms of the genus Taenia (cestodes). Two important species: Taenia solium (pork tapeworm) and Taenia saginata (beef tapeworm). Transmission: ingestion of undercooked meat containing cysticerci (larval cysts). Adult tapeworms attach to the small intestinal wall via scolex (armed with hooks in T. solium, suckers only in T. saginata) and produce proglottids (segments) containing eggs. Clinical: often asymptomatic; may cause mild abdominal discomfort, nausea, weight loss, increased appetite, and passage of proglottids in stool. Diagnosis: stool microscopy (eggs, proglottids—T. solium eggs are indistinguishable from T. saginata; species identified by uterine branching in gravid proglottids), perianal tape test. Treatment: praziquantel (5–10 mg/kg single dose) or niclosamide. Complications: T. solium can cause neurocysticercosis (ingestion of eggs from autoinfection or contaminated food/water)—larval cysts in the brain cause seizures, hydrocephalus, and focal neurological deficits. Prevention: thorough cooking of meat, improved sanitation, and meat inspection.

Tapeworm Infection

– See Taeniasis (Tapeworm Infection).

Tetanus

– An acute neurological disease caused by tetanospasmin toxin from Clostridium tetani, introduced through wounds. Characterized by generalized rigidity and painful muscle spasms (trismus/lockjaw, risus sardonicus, opisthotonos). Autonomic instability can occur in severe cases. Diagnosis: clinical. Treatment: human tetanus immune globulin (HTIG), wound debridement, metronidazole, and supportive care (benzodiazepines, ICU). Prevention: Tdap/Td vaccination at recommended intervals.

Threadworm (Enterobiasis)

– A common intestinal parasitic infection caused by Enterobius vermicularis (pinworm). Prevalence: most common helminth infection in developed countries, primarily affecting children aged 5–10 years. Transmission: fecal–oral (ingestion of eggs from contaminated hands, food, bedding, or fomites). Eggs are deposited in the perianal region by the female worm at night, causing intense perianal itching (pruritus ani), especially at night. Scratching contaminates fingers and nails, perpetuating the cycle. Diagnosis: perianal tape test (apply transparent adhesive tape to the perianal area in the morning before bathing, examined under microscope for eggs). Stool microscopy has low sensitivity. Treatment: mebendazole 100 mg single dose (or albendazole 400 mg single dose), repeated after 2 weeks (to kill newly hatched worms). All household members should be treated simultaneously. Good hand hygiene, short nails, frequent washing of bedding and pyjamas, and avoidance of scratching. Reinfection is common. Complications: vulvovaginitis (in girls), occasional appendicitis (worm in the appendix).

Toxic Shock Syndrome

– An acute, life-threatening illness caused by bacterial exotoxins. Staphylococcal TSS: associated with tampon use, surgical wounds, and nasal packing; presents with fever, hypotension, diffuse rash, and multi-organ involvement. Streptococcal TSS: associated with invasive GAS infections. Diagnosis: CDC case definition. Treatment: source control, IV fluids, vasopressors, and antibiotics (clindamycin + a beta-lactam for Strep; vancomycin for MRSA). IVIG may be beneficial.

Toxoid vaccines

– Vaccines that protect against harmful bacterial toxins. These vaccines contain toxins that have been detoxified and rendered inactive.

Toxoplasmosis

– Infection caused by the intracellular protozoan Toxoplasma gondii. Definitive host: cats (shed oocysts in feces). Humans are intermediate hosts, infected via ingestion of undercooked meat (containing tissue cysts), cat feces, or transplacentally. Immunocompetent: usually asymptomatic or mild flu-like illness with lymphadenopathy. Immunocompromised (HIV, transplant): toxoplasma encephalitis (brain abscesses, headache, confusion, seizures, focal neurology), myocarditis, pneumonitis. Congenital toxoplasmosis: infection in pregnancy can cause chorioretinitis, intracranial calcifications, hydrocephalus, and developmental delay. Diagnosis: serology (IgM, IgG, PCR), imaging (CT/MRI). Treatment: pyrimethamine + sulfadiazine + folinic acid.

Trachoma

– A chronic keratoconjunctivitis caused by Chlamydia trachomatis (serovars A-C), the leading infectious cause of blindness worldwide. Repeated infection causes conjunctival scarring, trichiasis (inturned eyelashes), corneal abrasion, opacification, and blindness. Transmission: direct contact, flies, fomites. Active trachoma: follicular conjunctivitis, papillary hypertrophy. Cicatricial trachoma: scarring, entropion, trichiasis. Diagnosis: clinical WHO grading system. Treatment: SAFE strategy (Surgery for trichiasis, Antibiotics: azithromycin mass drug administration, Facial cleanliness, Environmental improvement). Global elimination program is ongoing.

Travel Medicine

– Pre-travel consultation: vaccines (routine, hepatitis A/B, typhoid, yellow fever, meningococcal, rabies pre-exposure, Japanese encephalitis, cholera), malaria prophylaxis (atovaquone-proguanil daily, doxycycline daily, mefloquine weekly, tafenoquine.

Trichinosis

– A parasitic infection caused by Trichinella spiralis, acquired by eating undercooked pork or wild game (bear, boar) containing encysted larvae. Larvae are released in the intestine (enteric phase: diarrhea, abdominal pain), then migrate to skeletal muscle (parenteral phase: myositis, periorbital edema, fever, eosinophilia). Classic presentation: periorbital edema + myalgia + eosinophilia after eating undercooked meat. Diagnosis: serology (ELISA, Western blot), muscle biopsy. Treatment: albendazole or mebendazole (enteric phase); corticosteroids for severe myositis.

Trichomonas

– A protozoan parasite, Trichomonas vaginalis, causing trichomoniasis—the most common non-viral sexually transmitted infection worldwide (156 million cases annually). T. vaginalis is a flagellated, motile protozoan that infects the urogenital tract. Transmission: sexual intercourse. Incubation: 4–28 days. Clinical: women—vaginal discharge (frothy, yellow-green, malodorous), vulvovaginal itching, dysuria, dyspareunia, punctate cervical haemorrhages ('strawberry cervix' on colposcopy). Men—often asymptomatic (70%), may cause urethritis (thin, white discharge), balanitis, prostatitis. Complications: preterm birth, low birth weight, and increased HIV acquisition/transmission. Diagnosis:

wet mount microscopy (motile trichomonads, sensitivity 60–70%), culture, nucleic acid amplification test (NAAT—gold standard, sensitivity >95%). Treatment: metronidazole 2 g PO single dose (or tinidazole). Treat sexual partners. Metronidazole resistance occurs in 5–10% (treat with higher dose or tinidazole). Avoid alcohol during and for 24 hours after treatment (disulfiram-like reaction).

Trichomoniasis

– A sexually transmitted infection caused by the protozoan Trichomonas vaginalis. Often asymptomatic in men; women present with vaginal discharge (frothy, yellow-green, malodorous), itching, dysuria, and inflammation. Associated with preterm birth and increased HIV acquisition risk. Diagnosis: wet mount microscopy, culture, NAAT. Treatment: metronidazole or tinidazole (single dose) for patient and partner(s).

Trypanosomiasis

– Two forms: African trypanosomiasis (sleeping sickness, transmitted by tsetse fly, caused by Trypanosoma brucei gambiense – chronic West African form, or T. b. rhodesiense – acute East African form) and American trypanosomiasis (Chagas disease, transmitted by triatomine bugs, caused by Trypanosoma cruzi). Clinical features: African – chancre at bite site, fever, lymphadenopathy (Winterbottom sign), headache, followed by meningoencephalitis (sleep disturbance, neurological deterioration, coma). Chagas disease: acute phase (fever, Romana sign – unilateral periorbital edema), chronic phase (cardiomyopathy, megaoesophagus, megacolon decades later). Diagnosis: microscopy of blood/CSF, serology, PCR. Treatment: African – pentamidine/suramin (early), melarsoprol/eflornithine (late). Chagas – benznidazole, nifurtimox.

Tuberculosis

– TB: Mycobacterium tuberculosis, airborne transmission. Primary TB: Ghon focus + hilar lymphadenopathy (Ghon complex). Reactivation TB: apical/posterior upper lobe cavitary lesions. Clinical: cough >2 weeks, fever, night sweats, weight loss, hemoptysis. Diagnosis: AFB smear (3 sputums, sensitivity 50-60.

Tularemia

– A zoonotic bacterial infection caused by Francisella tularensis, transmitted through tick bites, handling infected animals (rabbits, rodents), contaminated water, or aerosolization. Ulceroglandular (most common): ulcer at inoculation site + tender lymphadenopathy. Other forms: glandular (lymphadenopathy only), oculoglandular (conjunctivitis + preauricular adenopathy), oropharyngeal (pharyngitis), typhoidal (fever without localizing signs), pneumonic (cough, chest pain). Diagnosis: serology, PCR, culture (biohazard). Treatment: streptomycin, gentamicin, doxycycline, or ciprofloxacin.

Typhus

– A group of acute febrile illnesses caused by Rickettsia species, transmitted by arthropod vectors. Three main forms: (1) Epidemic (louse-borne) typhus—Rickettsia prowazekii, transmitted by the human body louse (Pediculus humanus corporis). Associated with overcrowding, poverty, cold climate, and refugees ('war typhus'). Incubation 10–14 days. Clinical: sudden high fever, severe headache, myalgia, prostration, a macular rash starting in the axillae and spreading to the trunk (spares face, palms, soles), and meningismus. Mortality 10–60% without treatment (highest in older, malnourished patients). Brill–Zinsser disease: reactivation of latent R. prowazekii years after primary infection, milder. (2) Endemic (murine/flea-borne) typhus—Rickettsia typhi, transmitted by rat fleas (Xenopsylla cheopis). Milder than epidemic typhus; rash is maculopapular, centripetal. (3) Scrub typhus (tsutsugamushi disease)—Orientia tsutsugamushi (formerly Rickettsia tsutsugamushi), transmitted by chigger mites (Leptotrombidium). Endemic in the Asia-Pacific region ('tsutsugamushi triangle'). An eschar (tache noire) at the bite site is characteristic. Diagnosis: serology (IFA—gold standard; Weil–Felix test is obsolete), PCR on blood or eschar biopsy. Treatment: doxycycline 100 mg BD for 7–14 days (first-line). Azithromycin and chloramphenicol are alternatives (pregnancy, doxycycline allergy). Prevention: louse control (epidemic typhus—delousing with insecticides, improved hygiene), flea control (endemic typhus—rodent control), mite avoidance (scrub typhus—insect repellent DEET, protective clothing). No vaccine is currently available for typhus.

Undulant Fever

– See Brucellosis.

Urinary Tract Infection

– Infection of the urinary tract, classified as lower (cystitis, urethritis) or upper (pyelonephritis). Most common pathogen: Escherichia coli (80%). Cystitis: dysuria, urinary frequency, urgency, suprapubic pain. Pyelonephritis: fever, flank pain, nausea/vomiting, costovertebral angle tenderness. Complicated UTI: structural/functional abnormality, catheter, pregnancy, male sex, immunocompromised. Diagnosis: urinalysis with nitrites and leukocyte esterase, urine culture. Treatment: TMP-SMX, nitrofurantoin, fosfomycin for uncomplicated; fluoroquinolones or cephalosporins for complicated/pyelonephritis.

Urinary Tract Infections

– UTI: significant bacteriuria (>105 CFU/mL) with symptoms. Uncomplicated cystitis: women, dysuria, frequency, urgency, suprapubic pain. Nitrite/leukocyte esterase dipstick. Pathogens: E. coli (75-90.

Urinary Tract Infections (UTIs)

– Urinary tract infections are bacterial infections affecting any part of the urinary system, including the urethra, bladder, ureters, and kidneys. They are much more common in women due to the shorter urethra, and typical symptoms include urinary urgency, frequency, dysuria, and suprapubic pain, while upper tract infections may cause flank pain, fever, and systemic illness. Diagnosis is confirmed by urinalysis and culture, and treatment involves antibiotic therapy tailored to the causative organism, with a emphasis on adequate hydration and preventive measures.

UTI

– See Urinary Tract Infection.

Variola

– See Smallpox.

Venereal Diseases

– See Sexually Transmitted Infections (STIs).

Viral Infections

– Illnesses caused by viruses, which are intracellular pathogens that hijack host cellular machinery to replicate. They range from self-limited conditions such as the common cold and gastroenteritis to serious diseases including influenza, hepatitis, HIV, and COVID-19. Treatment is primarily supportive for most viral infections, though antiviral medications are available for specific viruses, and vaccination remains the most effective strategy for preventing many viral diseases.

Weil's Disease

– See Leptospirosis.

West Nile Virus

– A mosquito-borne flavivirus, endemic in the US, Africa, Europe, and Asia. Most infections (80%) are asymptomatic. West Nile fever (20%): fever, headache, myalgia, arthralgia, rash. Neuroinvasive disease (<1%): meningitis, encephalitis, acute flaccid paralysis (anterior horn cell involvement). Risk factors for severe disease: age >50, immunosuppression, diabetes, hypertension. Diagnosis: serum/CSF WNV IgM, RT-PCR. No specific antiviral; supportive care. Prevention: mosquito avoidance and vector control.

Whipworm (Trichuriasis)

– Intestinal infection caused by the soil-transmitted helminth Trichuris trichiura (whipworm), endemic in tropical and subtropical regions with poor sanitation. Humans acquire infection by ingesting embryonated eggs from contaminated soil, food, or water. Larvae hatch in the small intestine and migrate to the large intestine (caecum and colon), where adult worms (3-5 cm) embed their anterior ends into the mucosa. Light infections are often asymptomatic. Heavy infections cause abdominal pain, diarrhea (sometimes bloody), tenesmus, rectal prolapse (especially in children with massive worm burden), anemia (iron-deficiency from chronic blood loss), and growth retardation. Whipworm potentiates the severity of other infections (malaria, bacterial dysentery). Diagnosis: microscopic identification of characteristic bipolar-plugged eggs in stool (Kato-Katz technique, sedimentation). Treatment: albendazole (400 mg daily for 3 days) or mebendazole (100 mg twice daily for 3 days); ivermectin is less effective alone but synergistic in combination. Mass drug administration (MDA) with albendazole targets whipworm in deworming programmes. Prevention: improved sanitation, hand washing, and avoiding soil-contaminated food.

Yaws (Frambesia)

– A chronic, non-venereal treponematosis caused by Treponema pallidum subspecies pertenue, closely related to the syphilis spirochete (T. pallidum pallidum). Yaws is endemic in warm, humid tropical regions of Africa, Southeast Asia, and the Pacific Islands, affecting primarily children aged 2-15 years living in rural, resource-limited areas. Transmission: direct skin-to-skin contact with an infectious lesion (abrasion, cut, or insect bite), not sexually transmitted and not vertically transmitted. Yaws is classified into stages based on clinical progression: primary yaws (mother yaw): a single, painless, papillomatous papule at the inoculation site (typically on the lower extremities, buttocks, or face) that enlarges into a raspberry-like lesion (frambesia, from French framboise = raspberry) with a characteristic yellow crust; heals spontaneously over 3-6 months, leaving a hypopigmented scar. Secondary yaws (disseminated): occurs weeks to months after the primary lesion, consisting of multiple, painless, papillomatous or ulcerated lesions (daughter yaws) on the trunk, extremities, perioral and perianal areas; hyperkeratosis of the palms and soles (crab yaws or dry yaws — painful, fissured hyperkeratotic plaques causing a crab-like gait), periostitis, and lymphadenopathy. Late/latent yaws: occurs in 10% of untreated individuals, after 5-10 years of latency, causing destructive gummatous lesions of the skin and bones (gondou/bossed nose — hypertrophic periostitis of the maxilla), palatal and nasopharyngeal destruction (gangosa/rhinopharyngitis mutilans — destructive ulceration of the nose, palate, and pharynx), and juxta-articular nodules. Diagnosis: darkfield microscopy of serous exudate from early lesions shows motile spirochetes; serology (VDRL, RPR, TPHA, FTA-ABS — identical to syphilis, cannot distinguish between T. pallidum subspecies). Treatment: single-dose intramuscular benzathine penicillin G (1.2 million units for adults, 0.6 million units for children <10 years) is highly effective (>95% cure rate). Alternative: oral azithromycin (single 30 mg/kg dose, maximum 2 g) — azithromycin-based mass drug administration (MDA) is the cornerstone of the WHO yaws eradication campaign. Prevention: MDA, improved hygiene, and contact tracing.

Yellow Fever

– A mosquito-borne viral hemorrhagic fever caused by yellow fever virus (Flavivirus), endemic in tropical Africa and South America. Incubation 3-6 days. Presents with fever, myalgia, headache, and vomiting. A proportion of patients progress to toxic phase with jaundice, hemorrhagic fever, and multi-organ failure (hepatorenal). Case fatality rate 20-60% in severe cases. Diagnosis: RT-PCR, serology. No specific antiviral treatment; supportive care. Prevention: live-attenuated yellow fever vaccine (highly effective, required for travel to endemic areas).

Yersinia Enterocolitica

– A gram-negative coccobacillus of the family Enterobacteriaceae, a common cause of enterocolitis and mesenteric lymphadenitis, especially in children and young adults in temperate climates. Transmission: ingestion of contaminated food (undercooked pork — the most common source, unpasteurised milk, contaminated water), direct contact with infected animals (pigs are the primary reservoir), and person-to-person spread (rare). Pathogenicity: Y. enterocolitica carries a 70-kb virulence plasmid (pYV) encoding a type III secretion system (T3SS, injectisome) that injects Yersinia outer proteins (Yops) into host cells, disrupting actin cytoskeleton, inhibiting phagocytosis, and triggering apoptosis of macrophages. Clinical: acute enterocolitis (watery or bloody diarrhea, fever, abdominal pain, vomiting, most common in young children, self-limited over 5-14 days), pseudoappendicitis syndrome (mesenteric lymphadenitis and terminal ileitis mimicking acute appendicitis — fever, right lower quadrant pain, leukocytosis, common in older children and adolescents), and post-infectious complications (reactive arthritis, erythema nodosum, and uveitis, associated with HLA-B27). Diagnosis: stool culture (requires selective media — cefsulodin-irgasan-novobiocin/CIN agar), serology (antibodies to Yersinia antigens). Treatment: self-limited in immunocompetent patients; supportive care (rehydration) is the mainstay. Antibiotics are indicated for severe, prolonged, or invasive infections and immunocompromised patients: doxycycline + aminoglycoside, TMP-SMX, fluoroquinolones (ciprofloxacin), or third-generation cephalosporins (ceftriaxone). Y. enterocolitica produces beta-lactamase and is resistant to ampicillin, amoxicillin, and first-generation cephalosporins.

Zoonoses

– Infectious diseases that are naturally transmitted between vertebrate animals and humans. Zoonotic infections account for approximately 60% of all human infectious diseases and 75% of emerging infectious diseases (including HIV, SARS, MERS, Ebola, Nipah, and COVID-19). Classification by transmission route: direct zoonoses (transmission by direct contact — rabies from animal bites, anthrax from handling infected animal products), vector-borne zoonoses (transmitted by arthropod vectors — Lyme disease/Borrelia from ticks, West Nile fever from mosquitoes, plague/Yersinia pestis from fleas), food-borne zoonoses (ingestion of contaminated animal products — salmonellosis from eggs/poultry, campylobacteriosis from undercooked poultry, brucellosis from unpasteurised dairy, toxoplasmosis from undercooked meat), and water-borne zoonoses (leptospirosis from water contaminated with rodent urine). Reservoir hosts: mammals (rodents — hantavirus, plague, leptospirosis; bats — rabies, Nipah, Hendra, Ebola, Marburg, SARS, MERS; primates — HIV, yellow fever, monkeypox; pigs — influenza, Nipah, Japanese encephalitis), birds (avian influenza, West Nile virus, psittacosis), and arthropods (ticks, mosquitoes, fleas). One Health approach: integrates human health, animal health, and environmental health for surveillance, prevention, and control of zoonotic infections. Prevention: animal vaccination (rabies in dogs, brucellosis in cattle), vector control (mosquito nets, insect repellents, insecticide spraying, environmental management), food safety (pasteurisation, thorough cooking), personal protective equipment (for veterinarians, slaughterhouse workers, laboratory personnel), and surveillance of both animal and human populations for early detection of emerging zoonoses., endemic in tropical and subtropical regions with poor sanitation. Humans acquire infection by ingesting embryonated eggs from contaminated soil, food, or water. Larvae hatch in the small intestine and migrate to the large intestine (caecum and colon), where adult worms (3-5 cm) embed their anterior ends into the mucosa. Light infections are often asymptomatic. Heavy infections cause abdominal pain, diarrhea (sometimes bloody), tenesmus, rectal prolapse (especially in children with massive worm burden), anemia (iron-deficiency from chronic blood loss), and growth retardation. Whipworm potentiates the severity of other infections (malaria, bacterial dysentery). Diagnosis: microscopic identification of characteristic bipolar-plugged eggs in stool (Kato-Katz technique, sedimentation). Treatment: albendazole (400 mg daily for 3 days) or mebendazole (100 mg twice daily for 3 days); ivermectin is less effective alone but synergistic in combination. Mass drug administration (MDA) with albendazole targets whipworm in deworming programmes. Prevention: improved sanitation, hand washing, and avoiding soil-contaminated food.

Chapter 15

Medical Implants

Artificial Urinary Sphincter (AUS)

– A fully implantable device for treating severe stress urinary incontinence, most commonly in men after radical prostatectomy. Components: an inflatable periurethral cuff, a pressure-regulating balloon (placed in the space of Retzius), and a control pump (placed in the scrotum/labium). The patient activates the pump to deflate the cuff, allowing voiding; the cuff automatically refills over 3-5 minutes. The AMS 800 (Boston Scientific) is the most widely used device. Complications: erosion (cuff into urethra, 5-10% at 10 years), infection, mechanical failure, and persistent incontinence. Revision rates: 25-40% at 10 years.

Artificial Ventricle (Total Artificial Heart)

– A pulsatile, pneumatically or electrically driven biventricular replacement that removes the native ventricles. The SynCardia temporary Total Artificial Heart (TAH-t) is FDA-approved as a bridge to cardiac transplantation. The Carmat TAH is a bioprosthetic option available in Europe. Indications: severe biventricular failure, massive myocardial infarction, and end-stage heart disease when a donor heart is unavailable. Implantation involves full median sternotomy, cardiopulmonary bypass, and excision of the native ventricles. Pneumatic drivers (Big Blue, Companion 2) power the system externally. Complications: stroke, bleeding, infection (mediastinitis, driveline site), hemolysis, and device malfunction. Longest survival: >4 years as bridge to transplant.

BAC (Bone Anchored Cochlear) – See Bone-Anchored Hearing Aid (BAHA)

– An auditory implant that uses a titanium screw implanted in the temporal bone behind the ear with a sound processor attached. The implant bypasses the outer and middle ear by conducting sound through bone directly to the cochlea. Widely known as BAHA (Bone-Anchored Hearing Aid). The modern term is BCI (Bone Conduction Implant).

BAHA (Bone-Anchored Hearing Aid)

– A surgically implantable bone conduction hearing system. Components: a titanium abutment (fixture) placed in the mastoid bone, and an external sound processor that clips onto the abutment. The fixture osseointegrates with bone (takes 3-12 weeks). Indications: conductive hearing loss (chronic otitis media, aural atresia, otosclerosis), mixed hearing loss, and single-sided deafness (SSD). Contraindications: poor bone quality, uncontrolled diabetes, prior radiation to the temporal bone, and age <5 years. The Cochlear Baha system and Oticon Medical Ponto are the major brands. Complications: skin infection (peri-abutment dermatitis, most common), fixture failure (loss of osseointegration), and soft tissue overgrowth. Implant survival: >95% at 5 years.

Biodegradable Stent (Bioresorbable Stent)

– A coronary stent made of a bioabsorbable material (poly-L-lactic acid, magnesium alloy) that gradually dissolves over 2-3 years, theoretically restoring vascular vasomotion and avoiding the long-term risks of permanent metallic stents (e.g., late stent thrombosis, neoatherosclerosis). The Absorb BVS (Abbott) was the first FDA-approved bioresorbable stent but was withdrawn due to higher target lesion failure and scaffold thrombosis rates compared to drug-eluting stents. Second-generation scaffolds (e.g., Magmaris magnesium stent, Fantom sirolimus-eluting scaffold) aim to improve outcomes. Complications: scaffold thrombosis (higher than DES, especially in small vessels), malapposition, and fracture. Current use: selective, in clinical trials or specific anatomical settings.

Biologic Heart Valve (Bioprosthetic Valve)

– A prosthetic heart valve made from animal tissue (bovine pericardium or porcine aortic valve) mounted on a stent. Types: stented (most common: Carpentier-Edwards Perimount, Medtronic Hancock II), stentless (e.g., Medtronic Freestyle, for aortic root replacement), and sutureless/rapid-deployment (e.g., Edwards Intuity, for minimally invasive surgery). Tissue is treated with glutaraldehyde to reduce immunogenicity and often with anticalcification treatments (ThermaFix, Linx AC). Advantage: no need for lifelong anticoagulation (unlike mechanical valves). Disadvantage: structural valve degeneration (SVD) over 10-20 years, especially in younger patients and in the mitral position. Durable about 15-20 years in patients >65, and 8-12 years in patients <60. Transcatheter valve-in-valve (ViV) can treat SVD without redo surgery. Porcine valves (vs. bovine pericardial) have slightly higher reoperation rates.

Bone Graft (Implantable)

– Natural or synthetic bone material implanted to promote bone healing, fill defects, or achieve spinal fusion. Types: autograft (from the patient — iliac crest, fibula, rib, distal radius — the gold standard for osteogenesis, osteoconduction, and osteoinduction), allograft (cadaveric — cortical strut, cancellous chips, demineralized bone matrix DBM), synthetic (calcium phosphate hydroxyapatite, tricalcium phosphate, bioactive glass, calcium sulfate), and growth factor-enhanced (BMP-2 on an absorbable collagen sponge, e.g., Infuse Bone Graft). Formulations: putty, paste, granules, sheets, and structural blocks. Autograft limitations: donor site morbidity (pain, infection, hematoma, fracture) and limited quantity. Allograft risks: infection (very low with modern processing), and fracture. DBM: osteoinductive but inconsistent quality. BMP-2: potent but associated with ectopic bone formation, radiculitis, and cancer risk.

Bone Plate and Screws–(Implantable Fracture Fixation)

– Metal plates and screws used to stabilize bone fractures or osteotomies. Materials: stainless steel (strong, cheap, corrodes), titanium (biocompatible, MRI-safe, elastic modulus closer to bone), and bioabsorbable (PLLA, PGA — for children, no removal needed). Locking vs. non-locking screws: locking screws thread into the plate, creating a fixed-angle construct (like an external fixator internally), ideal for osteoporotic bone. Compression plating: dynamic compression plate (DCP) and limited-contact DCP (LC-DCP) for anatomic reduction and axial compression. Neutralization plate: protects lag screws from rotational forces. Buttress plate: supports a metaphyseal fragment (e.g., tibial plateau). Bridge plate: spans a comminuted fracture zone without direct reduction. Anatomical plates: pre-contoured for specific bones (distal radius, proximal humerus, acetabulum, calcaneus). The AO/ASIF principles guide fixation: anatomic reduction (articular), stable fixation, preservation of blood supply, and early mobilization. Complications: nonunion/malunion, infection (hardware removal if chronic), screw breakage, plate loosening, stress shielding (bone resorption under rigid plates), and refracture after removal.

Breast Implant (Prosthesis)

– A silicone- or saline-filled device placed in the subglandular or submuscular plane for breast augmentation (cosmetic) or reconstruction (post-mastectomy). Silicone gel implants: cohesive gel ("gummy bear") that retains shape even if the shell ruptures. Saline implants: filled intraoperatively, less natural feel. Surface types: smooth (less friction, higher capsular contracture) and textured (lower capsular contracture, but associated with BIA-ALCL — breast implant-associated anaplastic large cell lymphoma). Shape: round (most common) and anatomical (teardrop, requires textured surface to prevent rotation). FDA-approved manufacturers: Allergan (Natrelle), Mentor (MemoryGel, MemoryShape), Sientra, and Motiva (ergonomix). Implant placement: subglandular (on top of pectoralis, more animation deformity), subpectoral (dual plane, partial under muscle), and prepectoral (on top of muscle, for reconstruction). Complications: capsular contracture (most common, Baker grade I-IV), rupture (silent in silicone gel, MRI recommended at 5-6 years post-op then q2-3 years), implant malposition, rippling, infection, hematoma, and BIA-ALCL (textured implants, 1:3,000 to 1:30,000). Implant life: 10-20 years (not lifetime devices).

Cochlear Implant

– An electronic medical device that bypasses damaged hair cells in the cochlea and directly stimulates the auditory nerve. Components: internal (receiver-stimulator placed in a bony well behind the ear, electrode array inserted into the scala tympani of the cochlea) and external (microphone, speech processor worn behind the ear or body-worn, and transmitter coil held in place by a magnet). Indications: severe-to-profound bilateral sensorineural hearing loss with insufficient benefit from hearing aids (aided word recognition <50-60% in the best-aided condition). Adult candidacy: post-lingual onset. Pediatric candidacy: FDA-approved down to 9 months of age; earlier implantation improves language outcomes. Major brands: Cochlear (Nucleus series), Advanced Bionics (HiRes Ultra, Naída CI), and MED-EL (SYNCHRONY, RONDO). Speech processors replaceable/upgradable externally. Surgery: mastoidectomy, facial recess approach, cochleostomy or round window insertion, NRT/telemetry to confirm placement. Complications: facial nerve injury (rare, <1%), meningitis (pneumococcal vaccination required pre-op), flap infection, electrode migration, device failure (1-3%), and vestibular symptoms (vertigo). Outcomes: most achieve open-set speech understanding; factors are age at implantation, duration of deafness, and adherence to aural rehabilitation.

Continuous Glucose Monitor (CGM Implant)

– A subcutaneous sensor that measures interstitial glucose levels every 5-15 minutes. The Dexcom G7 and G6 (FDA-approved for non-adjunctive use, i.e., can dose insulin without fingersticks), Abbott FreeStyle Libre (flash glucose monitoring, reader on demand with 8-hour continuous data), and Medtronic Guardian (integrated with insulin pumps). Inserted subcutaneously in the abdomen (or arm for Libre) and worn for 7-14 days. Fully implantable CGM (Eversense, Senseonics) with a subcutaneous fluorescence-based sensor lasting 90-180 days; requires surgical insertion in an office procedure. MARD (mean absolute relative difference): <10%. Alarms: hypo/hyperglycemia, rate of change, and predictive alerts. Benefits: reduced HbA1c, less hypoglycemia, improved time-in-range (TIR, 70-180 mg/dL), and integration with automated insulin delivery (AID) systems. Limit: lag behind blood glucose (5-10 minutes interstitial delay).

Coronary Stent

– A metallic mesh tube deployed in a coronary artery to treat atherosclerotic narrowing. Bare metal stent (BMS): first generation, high restenosis rate (20-40%) due to neointimal hyperplasia. Drug-eluting stent (DES): releases an antiproliferative drug (sirolimus, paclitaxel, everolimus, zotarolimus) from a polymer coating to inhibit restenosis. Second-generation DES (everolimus: Xience, Promus; zotarolimus: Resolute): thinner struts, biocompatible polymers, 5-10% restenosis, and lower late stent thrombosis (LST) than first-generation (Cypher, Taxus). Third-generation: bioabsorbable polymer (Orsiro, Synergy) or polymer-free (BioFreedom) to reduce inflammation. Ultra-thin strut DES (<65 µm): improved deliverability. Cobalt-chromium and platinum-chromium alloys provide radial strength with thinner struts. Current standard: DES is used in >95% of PCI. Dual antiplatelet therapy (DAPT: aspirin + P2Y12 inhibitor) is required after deployment to prevent in-stent thrombosis, typically for 3-6 months (new-generation DES) to 12 months (high-risk ACS), then aspirin lifelong. In-stent restenosis (ISR): treated with drug-coated balloon (SeQuent Please, Lutonix) or another DES. Very late stent thrombosis (>1 year): rare with modern DES (0.2-0.5%/year).

Deep Brain Stimulator (DBS)

– A neurostimulator that delivers electrical pulses to targeted deep brain structures via implanted electrodes. Components: intracranial leads (quadripolar electrodes) placed stereotactically, extension cables tunneled subcutaneously, and an implantable pulse generator (IPG, "battery") placed in the infraclavicular or abdominal subcutaneous pocket. Targets: subthalamic nucleus (STN — for Parkinson's disease, most common), globus pallidus internus

(GPi — for Parkinson's and dystonia), ventral intermediate nucleus (VIM — for essential tremor), and anterior nucleus of the thalamus (ANT — for epilepsy). The Medtronic Percept PC (with sensing capability, BrainSense), Abbott Infinity (directional leads), and Boston Scientific Vercise (multiple independent current control) are current IPG platforms. DBS is MRI-conditional (specific conditions). Stimulation parameters: amplitude (voltage), pulse width (60-90 µs), frequency (130-180 Hz for Parkinson's), and active electrode configuration. Indications: Parkinson's disease (motor fluctuations, dyskinesias, tremor refractory to medication), essential tremor, dystonia (primary generalized), obsessive-compulsive disorder (OCD, under HDE), and epilepsy. Surgery: frame-based (CRW, Leksell) or frameless (NextFrame, StealthStation) stereotaxy, microelectrode recording (MER) for physiological mapping, intraoperative test stimulation for symptom improvement and side effect assessment. Complications: hemorrhage (1-2%, intraparenchymal or along the lead tract), infection (2-5%, often IPG pocket), lead fracture/migration, hardware erosion, stimulation-induced side effects (paresthesias, dysarthria, mood changes), and IPG battery depletion (replaced every 2-5 years, rechargeable IPGs are emerging).

Dental Implant (Endosseous)

– A titanium or titanium-alloy screw-shaped post that osseointegrates with the alveolar bone to support a crown, bridge, or denture. Materials: commercially pure titanium (Grade 4, rough surface for osseointegration) and zirconia (esthetic, metal-free, but less evidence for posterior teeth). Two-stage: implant placed, submerged under gingiva for 3-6 months of osseointegration, then uncovered and loaded. Single-stage: transgingival healing abutment placed at the time of implant insertion, avoiding a second surgery. Immediate loading: crown placed within 48 hours (for anterior teeth, ideal bone quality). Delayed loading: standard after 3-6 months. The Branemark protocol (1965) established the principles: atraumatic extraction, sterile surgical technique, and uninterrupted healing. Brands: Straumann (SLA, Roxolid), Nobel Biocare (Replace, Brånemark), Zimmer (Tapered Screw-Vent), Dentsply (Astra Tech, OsseoSpeed), and BioHorizons. Surface modifications: SLA (sandblasted + acid-etched), TiUnite (oxidized), and SLActive (hydrophilic, faster osseointegration). Contraindications: uncontrolled diabetes, bisphosphonate therapy (MRONJ risk), heavy smoking, and history of head/neck radiation. Complications: peri-implantitis (inflammation and bone loss, 20-30% of patients), implant failure (early: failed osseointegration, 1-5%; late: due to overloading or infection), nerve injury (inferior alveolar nerve, mental nerve), sinus perforation (maxillary sinus), and esthetic problems (gum recession visible metal margin). 10-year survival: >95% in well-selected patients.

Glaucoma Drainage Device (Tube Shunt)

– An implant for reducing intraocular pressure (IOP) in glaucoma by draining aqueous humor from the anterior chamber to an equatorial bleb. Components: a silicone tube (placed in the anterior chamber through a scleral track) connected to an end plate (positioned under the rectus muscles on the scleral surface). The Ahmed valve (New World Medical) has a flow restrictor (venturi-based or silicone membrane) that limits hypotony immediately post-op. The Baerveldt implant (Johnson & Johnson) has no valve; flow is restricted by a ligation suture or dissolvable Vicryl tie for a 4-6 week delay. The Molteno implant (single- or double-plate) is the original. Indications: neovascular glaucoma, uveitic glaucoma, post-penetrating keratoplasty glaucoma, prior failed trabeculectomy, congenital glaucoma, and high-risk glaucoma. Contraindications: corneal endothelial disease (tube can cause corneal decompensation) and active infection. Surgery: superotemporal quadrant, peritomy, quadrant dissection, tube trimmed bevel-up, scleral patch graft (donor sclera, pericardium, or cornea) over the tube entry site, and conjunctival closure. Complications: hypotony (especially Baerveldt without ligation), tube erosion (conjunctival breakdown), corneal edema, tube obstruction (iris, fibrin, vitreous), choroidal effusion, endophthalmitis, and diplopia from the plate interfering with extraocular muscles. Tube shunt survival: >80% at 2 years, decreasing to about 50% at 5 years.

Implantable Cardioverter-

Defibrillator (ICD)

– An electronic device that monitors heart rhythm and delivers therapy (antitachycardia pacing ATP, cardioversion, or defibrillation) for ventricular arrhythmias. Components: pulse generator (placed in the left infraclavicular subcutaneous pocket) and one or more leads (transvenous: single-coil or dual-coil, placed in the RV apex or septum; or entirely subcutaneous S-ICD). Indications (primary prevention): ischemic cardiomyopathy (EF $\leq$35% >40 days post-MI), non-ischemic cardiomyopathy (EF $\leq$35%, on GDMT for $\geq$3 months), and certain channelopathies (long QT, Brugada, HCM, ARVC). Secondary prevention: survivors of SCD due to VF/VT. Transvenous ICD: pectoral implant, 1-3 leads (RA, RV, LV for CRT-D). Subcutaneous ICD (S-ICD, Boston Scientific Emblem): no intravascular leads, left parasternal electrode and left lateral pocket, for patients without need for pacing/bradycardia pacing. Leadless ICD: entirely within the RV (currently only a pacemaker is available commercially leadless). Shock: up to 41 J delivered biphasic waveform. Defibrillation threshold (DFT): <10 J difference from maximum output is acceptable. Programming: optimizing VT detection zones (monitor, ATP, shock), S-ICD with SMART Pass algorithm reduces inappropriate shocks. Complications: infection (pocket, endocarditis, highest with system revision), lead failure (fracture, insulation break), inappropriate shocks (SVT, AF, lead noise, T-wave oversensing), pneumothorax/hemothorax (transvenous), and device erosion. Device generators last 5-10 years (transvenous), longer with S-ICD (no pacing drain). Remote monitoring is standard (CareLink, Merlin, Latitude) and reduces clinic visits and improves survival.

Implantable Loop Recorder (ILR)

– A subcutaneous cardiac monitoring device that continuously records a single-lead ECG for up to 3 years. The size of a thumb drive, inserted in the left parasternal area (4th intercostal space) in a 1-cm pocket under local anesthesia. The Medtronic LINQ II (micra-like, injectable), Abbott Confirm Rx (Bluetooth-enabled, smartphone-based transmission), and Biotronik BioMonitor 3 are current models. Autotriggered recordings: asystole (>3 sec), bradycardia, tachycardia, and AF detection (P-wave analysis, AF burden%). Patient-activated recordings: for symptoms (palpitations, syncope, presyncope, chest pain). Indications: syncope of unclear etiology (after negative workup — most common indication), cryptogenic stroke (to detect occult AF, 20-30% detection rate), and AF management (AF burden monitoring after ablation or cardioversion). Ill at TIA/CVA workup: monitor 3 years, detects AF in up to 20-30% of patients. Battery life: 2-3 years. The device is explained at the end of monitoring or battery depletion. Complications: infection (1-3%), pocket hematoma, device migration, and discomfort.

Insulin Pump (CSII – Continuous Subcutaneous Insulin Infusion)

– A battery-powered device that delivers rapid-acting insulin (analog) subcutaneously through a cannula, providing a continuous basal rate and patient-activated boluses. Types: tethered pump (Medtronic MiniMed 780G, Tandem t:slim X2 — with touchscreen, tubed, Control-IQ/Advanced Hybrid Closed Loop), patch pump (Omnipod DASH/Omnipod 5 — tubeless, fully disposable, worn for 3 days, controlled via a PDM or smartphone), and the Ypsomed mylife YpsoPump (tubed, color touchscreen). Hybrid closed-loop (HCL, automated insulin delivery AID): integrates CGM data to automatically adjust basal insulin; the Medtronic 780G (SmartGuard, targets BG 100-120 mg/dL) and Tandem Control-IQ (target 112.5-160 mg/dL) are FDA-approved. The Omnipod 5 closed-loop is the first tubeless HCL. Indications: type 1 diabetes (all patients considered), poorly controlled type 2 diabetes, frequent/severe hypoglycemia, dawn phenomenon, and gastroparesis. Cannula: inserted subcutaneously (abdomen, arm, hip, thigh) with an automatic inserter, changed every 2-3 days. Insulin: only rapid-acting (Humalog, Novolog, Fiasp, Lyumjev). Reservoirs: 180-300 units. Contraindications: unwillingness/unable to manage the system, untreated eating disorder, and severe visual impairment without caregiver support. Complications: infusion set failure (kinked cannula, dislodgment, occlusion, bleeding) leading to DKA within 4-6 hours (no long-acting insulin on board), lipohypertrophy, infection at the insertion site, and pump malfunction.

Intraocular Lens (IOL)

– An artificial lens implanted in the eye after cataract extraction (phacoemulsification). One-piece acrylic (monofocal, aspheric, with blue-blocking chromophore) is the standard of care placed in the capsular bag. Monofocal IOL: single focal point (set for distance), requires reading glasses. Multifocal IOL: diffractive rings on the surface divide light into multiple focal points (distance, near, intermediate), reduces spectacle dependence (e.g., Alcon Restor, Johnson & Johnson Tecnis Symfony). Depth-of-focus (EDOF) IOL: extends the focal range with less dysphotopsia than multifocal (e.g., Tecnis Symfony, Alcon Vivity). Toric IOL: corrects pre-existing corneal astigmatism ($\geq$0.75 D); requires careful alignment (axis marking). Accommodating IOL: changes shape with ciliary muscle contraction (e.g., Crystalens, no longer commonly used). Light-adjustable lens (LAL, RxSight): can be adjusted postoperatively with UV light to fine-tune refractive outcome. Blue-blocking IOLs: filter UV and blue light (theoretical protection against macular degeneration, though controversial). Biometry: optical (IOLMaster, Lenstar) measures axial length, K-readings, and ACD; formulas (SRK/T, Holladay 2, Barrett Universal II, Hill-RBF) calculate IOL power. Target refraction: 0.0 D (plano) for distance, -0.25 D to -0.50 D for most patients. Contraindications: zonular weakness (pseudoexfoliation, trauma), capsular tear, and uveitis (active). Complications: posterior capsule opacification (PCO, 20-40% at 5 years, treated with YAG capsulotomy), IOL dislocation/decentration, cystoid macular edema (Irvine-Gass syndrome), dysphotopsia (positive: arc-like light streaks; negative: temporal shadow), and refractive error. Phakic IOL (ICL, implantable collamer lens): placed in the sulcus (EVO ICL by STAAR Surgical) in front of the natural lens for high myopia correction, without removing the natural lens.

Intrathecal Pump (Baclofen Pump)

– An implantable pump that delivers baclofen directly into the intrathecal space for the treatment of severe spasticity. The Medtronic SynchroMed II is the only FDA-approved implantable pump for intrathecal baclofen (ITB). A catheter is tunneled subcutaneously from the pump (placed in the lower abdominal wall) to the lumbar spine and threaded into the intrathecal space (typically at L2-L3). The pump reservoir (20 mL or 40 mL) is refilled percutaneously every 1-3 months through the central fill port. Indications: severe spasticity refractory to oral therapy (max dose of oral baclofen: 80 mg/day) due to multiple sclerosis, spinal cord injury (most common indication), cerebral palsy, traumatic brain injury, and stroke. Efficacy: reduces spasticity (Ashworth score decrease of 1-2 points) and improves function/pain. Screening test: 50-100 µg intrathecal bolus to assess response before implantation. Baclofen concentration: 500-2000 µg/mL. Flow rate: adjustable (continuous infusion, simple or complex dosing). Contraindications: active infection, small body habitus, and coagulopathy. Complications: pump pocket infection (most common, 5-10%), catheter kink/occlusion/dislodgement, pump malfunction, baclofen overdose (respiratory depression, hypotonia, sedation — treat with physostigmine), baclofen withdrawal (severe rebound spasticity, fever, confusion, rhabdomyolysis — treat with oral baclofen/benzodiazepines), and CSF leak. Catheter-related failures account for 50% of ITB system revisions.

Intrauterine Device (IUD)

– A small T-shaped device inserted into the uterine cavity for long-acting reversible contraception (LARC). Hormonal IUD (levonorgestrel-releasing): Mirena (52 mg, approved 8 years), Kyleena (19.5 mg, 5 years), Skyla (13.5 mg, 3 years). Copper IUD (non-hormonal): Paragard (T380A, approved 10-12 years, contains copper wire on arms and stem). Mechanism: hormonal IUD thickens cervical mucus, suppresses endometrial proliferation, and may inhibit ovulation; copper IUD creates a sterile inflammatory reaction toxic to sperm and ova. Insertion: after sterile speculum exam, uterine sound, the IUD is placed via inserter tube; strings trimmed to 2-3 cm from the external cervical os. Best time: during menses (cervix slightly open, pregnancy excluded). Pain management: NSAIDs, para/paracervical block (for nulliparous, highly anxious). Efficacy: >99% (typical use failure rate <1%), same as tubal ligation and vasectomy. Contraindications: pregnancy, current PID, purulent cervicitis, Wilson disease (copper IUD), uterine anomaly, and unexplained vaginal bleeding. Complications: perforation (1:1000, usually at insertion), expulsion (2-10%, most in the first year), cramping, irregular bleeding (heavy menses with copper, spotting with hormonal), string problems (lost or retracted), and PID (primarily in the first 3 weeks post-insertion; higher risk with STI exposure). Return to fertility: immediate upon removal. Non-contraceptive benefits: reduces heavy menstrual bleeding (hormonal), endometrial cancer protection (hormonal), and emergency contraception (copper, up to 5 days after unprotected sex).

Mastopexy (Breast Lift Implant)

– A surgical procedure to lift and reshape sagging breasts (ptosis). Often combined with breast augmentation (implants) or reduction. Indications: involutional ptosis after pregnancy/breastfeeding, weight loss, aging. The NAC (nipple-areolar complex) is repositioned superiorly. The augmentation-mastopexy is a complex operation with a higher revision rate than augmentation alone.

Mechanical Heart Valve

– A prosthetic heart valve made of pyrolytic carbon (strong, thromboresistant, durable). The St. Jude Medical (now Abbott) bileaflet valve is the most common: two semicircular hinged leaflets that open centrally, creating three flow orifices with minimal obstruction. The Medtronic-Hall tilting disc valve (a single tilting disc suspended by a central strut) is less common. The Omnicarbon valve is another single-tilting disc. The Starr-Edwards ball-in-cage valve (caged-ball) is historical (no longer implanted but sometimes still seen in elderly patients). Advantage: excellent durability (>30 years). Disadvantage: lifelong therapeutic anticoagulation with warfarin (INR 2.0-3.0 for aortic, 2.5-3.5 for mitral) due to thrombogenicity. Risk of thromboembolism (1-3%/year with adequate anticoagulation) and bleeding (1-3%/year). Indications: age <50-60 (especially in the mitral position), kidney disease (bioprosthetic valves degenerate faster), and patient preference (willing to take warfarin). Sound: a distinct clicking/ticking sound audible to the patient and sometimes to others.

Mid-Urethral Sling (MUS)

– A polypropylene mesh tape placed under the mid-urethra to treat stress urinary incontinence (SUI) by providing dynamic urethral support during increases in intra-abdominal pressure. Retropubic (TVT, tension-free vaginal tape): a monofilament polypropylene mesh tape inserted via a vaginal incision and passed through the retropubic space with trocars exiting the suprapubic skin (risk of bladder perforation, bowel injury). Transobturator (TOT, TVT-O): the tape is passed through the obturator foramen exiting the groin crease (less bladder injury, more groin pain). Single-incision mini-sling (SIMS, e.g., Altis, MiniArc): shorter, the tape is anchored directly to the obturator internus fascia through a single vaginal incision (less invasive, but higher failure rates for severe SUI). Indications: urodynamic stress incontinence (USI, SUI confirmed on urodynamic studies) with hypermobility of the urethra. Contraindications: pregnancy (future childbearing), active UTI, untreated UTI, and desire to void. Complications: mesh erosion into the vagina (3-5%), voiding dysfunction (retention, de novo urgency), groin/thigh pain (TOT), bladder perforation (TVT, 2-5%), mesh infection, and de novo urgency urinary incontinence. Success rate: 80-90% subjective cure at 5 years. The FDA 2011 transvaginal mesh safety communication specifically warned against mesh for POP repair; slings remain the standard of care for SUI.

Pacemaker (Cardiac Implantable Electronic Device CIED)

– An implantable device that delivers electrical pulses to the heart to maintain adequate heart rate. Single-chamber: VVI (RV lead, for rate support in AF). Dual-chamber: DDD (RA + RV leads, for AV block and sinus node dysfunction). Biventricular (CRT-P): RV + LV (coronary sinus lead) for cardiac resynchronization in heart failure. Indications: symptomatic sinus node dysfunction (sick sinus syndrome, HR <40), third-degree (complete) AV block, Mobitz II second-degree AV block, bifascicular block with intermittent AV block, and chronotropic incompetence. Pacing modes (NBG code): first letter = chamber paced (A, V, D), second = chamber sensed, third = response (I, T, D), fourth = rate modulation (R). Rate-responsive (DDDR): uses an accelerometer or minute ventilation sensor to increase rate during exercise. MRI-conditional (most modern): labeled MR Conditional, safe under specified conditions. Endocardial leads: passive fixation (tines) or active fixation (screw-in helix), placed via subclavian or axillary venous access. Left subpectoral pocket is standard. Leadless pacemaker (Micra AV, Medtronic; Aveir, Abbott): all-in-one device implanted in the RV via a femoral approach (no pocket, no lead). Dual-chamber leadless pacing is emerging (Aveir DR). Generator replacement: every 6-12 years (CRT-P shorter). Remote monitoring: standard (MyCareLink, Merlin, Latitude) reduces clinic visits. Complications: infection (pocket, endocarditis, 1-2% for new implant, 5-7% for system revision), lead failure (fracture, insulation break, recall-prone), pneumothorax (subclavian puncture), pocket hematoma, and twiddler syndrome (patient rotates generator, dislodging leads).

Patent Ductus Arteriosus (PDA) Occlusion Device

– An implant for transcatheter closure of a patent ductus arteriosus, a congenital heart defect where the fetal ductus arteriosus fails to close after birth. The Amplatzer Duct Occluder II (Abbott) and the Cook detachable coil are the most common. The device is delivered through a 4-6 Fr sheath via the femoral vein or artery. The Amplatzer device is a self-expanding nitinol mesh with a retention disc on the aortic side and a plug on the pulmonary side. Indications: hemodynamically significant PDA (left-to-right shunt, enlarged left heart, continuous murmur). Closure is typically performed >6 months of age (weight >6 kg). Absolute contraindication: pulmonary vascular disease (Eisenmenger syndrome with right-to-left shunt, PDA-dependent pulmonary/systemic circulation). Complications: device embolization (most common), residual shunt, hemolysis, and obstruction of the LPA or descending aorta.

Penile Prosthesis (Implant)

– A surgically implanted device for treating erectile dysfunction (ED) refractory to oral therapy (PDE5 inhibitors) and vacuum/ICI therapies. Malleable (semi-rigid) rods: AMS 600M/650M (Boston Scientific), Coloplast Genesis. These are always semirigid and can be bent downward for urination/intercourse; simpler and cheaper, but less natural. Inflatable prostheses (two-piece: AMS Ambicor; three-piece: AMS 700 (Boston Scientific) with LGX (length-expanding) or CX (controlled expansion) cylinders, Coloplast Titan with One-Touch Release): cylinders placed in the corpora cavernosa, a scrotal pump (in the scrotum), and a fluid reservoir (retropubic space, collapsible for Coloplast; pre-connected for AMS 700). Three-piece is the standard for the best results (natural erection and flaccidity). Indications: organic ED (diabetes, post-radical prostatectomy, Peyronie disease, severe vascular disease) refractory to medical therapy. Contraindications: active infection, untreated UTI, and severe penile fibrosis (scarring). Surgery: penoscrotal (most common) or infrapubic approach. Cylinder size is measured intraoperatively (dilatation with Hegar dilators, Furlow inserter). Complication rates: infection (1-3% in naive patients; higher in revision/revision, 5-15%), mechanical failure (5-10% at 5 years, higher for 3-piece), cylinder erosion, pump migration, and auto-inflation (Correction: the Coloplast Titan has a lock-out valve to prevent auto-inflation). Satisfaction rates: >90% (patient and partner). Antibiotic impregnation: InhibiZone (rifampin/minocycline) on AMS devices reduces infection.

Port-a-Cath (Implantable Venous Access Port)

– A fully implanted central venous access device consisting of a silicone catheter (inserted into the superior vena cava via the internal jugular or subclavian vein) connected to a titanium or plastic reservoir (port) placed in a subcutaneous pocket on the chest wall (typically 2-3 cm below the clavicle, over the pectoralis major fascia). Access is through the skin via a non-coring Huber needle (right-angle or straight). The power-injectable port (e.g., Bard PowerPort, Cook Vital-Port) can handle CT contrast injection at up to 5 mL/sec. Indications: chemotherapy (most common), long-term IV antibiotics (osteomyelitis, endocarditis), TPN, frequent blood transfusions, and frequent blood draws (rare). Double-lumen ports are available for incompatible infusions. Placement: interventional radiology or surgeon, under ultrasound and fluoroscopic guidance. Tip position: cavoatrial junction. Flush: heparinized saline (100 U/mL, 5 mL) monthly when not in use, or saline-only ports (e.g., BD Q-Syte, neutral displacement). Complication rate: 5-15%.

Immediate: pneumothorax, hemothorax, arterial puncture, air embolism. Short-term: pin-hole infection, catheter malposition. Long-term: catheter-related bloodstream infection (CRBSI, most common, 1-3 per 1000 catheter-days, treat with antibiotics + port removal if tunnel infection), catheter occlusion (thrombotic: alteplase), fibrin sheath formation (retraction of blood, resistant aspiration but easy flushing), venous thrombosis (upper extremity DVT, catheter-related thrombosis), and port erosion/exposure. Port longevity: indefinite (until no longer needed or complication occurs).

Responsive Neurostimulation (RNS System)

– An implantable closed-loop neurostimulation system for treating drug-resistant focal epilepsy. The NeuroPace RNS System is the only FDA-approved device. Components: a cranially implanted neurostimulator (placed in a skull defect, similar to a cranial bone flap), one or two depth or subdural strip electrodes placed at the seizure focus(es), and a wireless telemetry wand for programming and data download. The device continuously records electrocorticography (ECoG) data and delivers stimulation (brief electrical pulses) when it detects a pattern that matches a pre-programmed seizure-onset pattern (detection and stimulation cycles). It is a responsive, adaptive, closed-loop system (not continuous like DBS). Indications: adults with drug-resistant focal epilepsy (with 1-2 seizure foci) who are not candidates for resective surgery. Contraindications: multifocal epilepsy with >2 foci, generalized epilepsy, and MRI incompatibility (RNS is MRI-conditional). Efficacy: 50-70% median seizure reduction at 2 years, continues to improve over time (possibly due to neuromodulation effects). Advantages over VNS and DBS: delivers therapy only when needed, provides long-term ECoG monitoring for seizure tracking, and is adjustable without surgery. Complications: infection (5-10%, often requires device removal), implant site pain, hemorrhage, and lead fracture. Device battery: about 4-6 years.

Shunt (Ventriculoperitoneal VP Shunt)

– A device that diverts cerebrospinal fluid (CSF) from the cerebral ventricles to the peritoneal cavity for the treatment of hydrocephalus. Components: ventricular catheter (inserted through a frontal or occipital burr hole into the lateral ventricle, typically the right frontal horn), a valve (programmable: Codman Hakim, Medtronic Strata, Sophysa Polaris; or fixed-pressure: low/medium/high), and a peritoneal catheter (tunneled subcutaneously from the head to the upper abdomen, inserted into the peritoneal cavity). Programmable valves: adjusted externally with a magnetic tool (Codman Hakim: 30-200 mm H_2O; Medtronic Strata: 0.5-2.5; Aesculap Miethke proGAV 2.0: gravitational valve that changes with body position). Indications: congenital hydrocephalus, normal pressure hydrocephalus (NPH), post-hemorrhagic hydrocephalus (IVH, SAH), post-meningitic hydrocephalus, and obstructive hydrocephalus from tumors. Contraindications: active peritonitis, infected CSF (VP shunt), and presence of a ventriculostomy (EVD) that is infected. Complications: shunt obstruction (most common, 30-40% at 2 years, most frequent in the proximal catheter), infection (5-10%, most within 3 months, typically Staphylococcus epidermidis), overdrainage (subdural hematoma, slit ventricle syndrome, orthostatic headaches), underdrainage (persistent hydrocephalus), and shunt migration (peritoneal catheter can erode into the bowel, bladder, or through the skin). Overdrainage in NPH may lead to subdural hygromas/hematomas from negative pressure. Antibiotic-impregnated catheters (Bactiseal, Codman) reduce infection (OR 0.5-0.6).

Spinal Cord Stimulator (SCS)

– An implantable neuromodulation device for treating chronic neuropathic pain. Components: epidural leads (percutaneous cylindrical leads or surgical paddle leads placed in the dorsal epidural space) and an implantable pulse generator (IPG, battery) placed in the lower back/buttock/subcostal pocket. Trial: temporary external system for 3-7 days to evaluate efficacy before permanent implantation (>50% pain reduction needed for proceeding). Indications: failed back surgery syndrome (FBSS, most common), complex regional pain syndrome (CRPS, type I and II), diabetic peripheral neuropathy, and refractory angina. Traditional SCS: paresthesia-based (tonic, 40-60 Hz) covers the painful area with a tingling sensation (paresthesia overlap). Newer waveforms: high-frequency (10 kHz, HF10 therapy) — paresthesia-free, superior for back pain; burst stimulation (40 Hz burst with 5 spikes) — targets affective component of pain; and closed-loop SCS (Abbott Proclaim, closed-loop feedback from evoked compound action potentials ECAPs) — maintains consistent activation and improves outcomes. Brands: Boston Scientific (Spectra WaveWriter, paresthesia-based + burst + combination), Abbott (Proclaim DRG, for SCS and dorsal root ganglion stimulation), Nevro (Senza HFX, high-frequency 10 kHz), and Medtronic (Intellis with differential target multiplexed programming). DRG stimulation (Abbott Axium): lead placed at the dorsal root ganglion in the epidural space at the L1-S2 levels for focal neuropathic pain (e.g., post-herniorrhaphy pain, CRPS of a limb) with better paresthesia coverage and positional stability. Complications: lead migration (most common, 5-15%), infection (2-5%, often IPG pocket), IPG site pain, lead fracture, dural puncture, and CSF leak.

Subdermal Contraceptive Implant (Nexplanon)

– A single-rod, non-biodegradable, etonogestrel-releasing implant (68 mg, released over 3 years) placed subdermally on the inner upper arm. The applicator is pre-loaded (loaded disposable inserter, changed from Implanon). Mechanism: inhibits ovulation (by suppressing LH surge), thickens cervical mucus, and suppresses endometrial proliferation. Insertion: after local anesthesia—the device is inserted subdermally at the medial aspect of the non-dominant arm, 8-10 cm above the medial epicondyle, in the sulcus between the biceps and triceps. The new Nexplanon uses a single-use, preloaded, auto-disable applicator (prevents reinsertion). The device is radiopaque (barium sulfate added). Efficacy: >99% (Pearl Index 0.05). Duration: 3 years (FDA-approved), but evidence suggests it is effective up to 5 years. Advantages: long-acting reversible contraception (LARC), 100% reversible (returns to fertility immediately after removal), excellent for adolescents and women who cannot use estrogen. Disadvantages: irregular bleeding/spotting (most common side effect, 50% after 1 year), amenorrhea (20%), weight gain, breast tenderness, and acne. Contraindications: active VTE, liver disease, breast cancer, and unexplained abnormal vaginal bleeding.

Transcatheter Aortic Valve Replacement (TAVR Valve)

– A bioprosthetic valve crimped onto a catheter and delivered percutaneously to replace a stenotic aortic valve without open heart surgery. Two main platforms: balloon-expandable (Edwards SAPIEN 3 Ultra, SAPIEN 3 — bovine pericardial leaflets, cobalt-chromium stent frame) and self-expanding (Medtronic Evolut PRO+, PRO — porcine pericardial leaflets, nitinol frame with external pericardial wrap for sealing). Access: transfemoral (most common), transapical (left lateral thoracotomy, direct LV apex), transaortic (direct ascending aorta), transaxillary/subclavian, and transcarotid. Indications: symptomatic severe aortic stenosis (reduced survival if untreated) — currently approved for all risk categories (extreme, high, intermediate, and low risk). The PARTNER trials (Edwards) and CoreValve US trials (Medtronic) validated TAVR across risk strata. Pre-procedural CT: annulus sizing (perimeter, area), coronary height (risk of coronary occlusion), and iliofemoral access assessment. Annulus measurement is crucial: oversizing by 5-15% prevents paravalvular leak; severe oversizing risks annular rupture. Valve-in-valve: TAVR within a failed surgical bioprosthesis (viable TAVR) is safe and effective. Complications: paravalvular leak (PVL, 5-10% moderate or greater for new-generation valves, less with improved sealing skirts), permanent pacemaker requirement (PPM, 15-30% for self-expanding — due to conduction disturbances LBBB/AV block, 5-10% for balloon-expandable), stroke (2-5%, lowered with embolic protection devices like Sentinel CPS), vascular access complications (bleeding, pseudoaneurysm, dissection, 5-10%), coronary obstruction (rare but fatal if undetected), and annular rupture (rare).

Transjugular Intrahepatic Portosystemic Shunt (TIPS)–See TIPS in Surgery.

– See also Intrathecal Pump for CSF shunting.

Tympanostomy Tube (Grommet)

– A small ventilation tube inserted through the tympanic membrane (eardrum) to equalize pressure and drain middle ear fluid. Materials: silicone, fluoroplastic, and titanium; with or without antibiotic coating. Types: short-term (Paparella type 1, Armstrong, Shepard, Donaldson — extrude spontaneously in 6-18 months) and long-term (T-tube, Goode T-tube — stay in 2-4+ years, may require surgical removal). Indications: persistent otitis media with effusion (OME, >3 months in children with hearing loss >20 dB), recurrent acute otitis media (AOM, >3 episodes/6 months or >4/12 months), chronic eustachian tube dysfunction, and adult barotrauma (flyers, divers). Contraindications: active otorrhea, cholesteatoma, and prior history of TM perforation. Procedure: myringotomy (anterior-inferior quadrant, avoiding the chorda tympani nerve) with aspiration of effusion, then tube insertion. Follow-up: keep water out of ears (earplugs for swimming are controversial; generally recommended to avoid deep water diving but surface swimming is acceptable). Tubes extrude spontaneously as the TM regenerates. Complications: tube otorrhea (most common, 10-30%, treat with topical antibiotics), obstruction (crust, dried effusion), early extrusion, persistent TM perforation after tube extrusion (1-5%, higher with T-tubes), tympanosclerosis (TM scarring, 20-30%), and cholesteatoma.

Vagus Nerve Stimulator (VNS)

– An implantable neurostimulator that delivers intermittent electrical pulses to the left vagus nerve for the treatment of drug-resistant epilepsy and depression. The LivaNova VNS Therapy System (Demipulse, AspireHC) is the primary device. A spiral electrode is wrapped around the left vagus nerve in the carotid sheath (left is chosen to minimize cardiac effects, as the right vagus heavily innervates the SA node). The pulse generator is placed in a left infraclavicular subcutaneous pocket. Stimulation parameters: output current (0.25-3.0 mA), frequency (20-30 Hz), pulse width (250-500 µs), on-time (30 sec), and off-time (5 minutes). Indications: drug-resistant partial-onset epilepsy (≥12 years, FDA-approved 1997) and treatment-resistant depression (FDA-approved 2005). It is not for patients with prior left cervical vagotomy. Efficacy: 30-50% seizure reduction at 3 months, improving to 50-60% at 2 years. Magnets: patients/caregivers can swipe a magnet over the generator to trigger on-demand stimulation to abort seizures. Vagal nerve stimulation for depression: NIMH STAR*D post-hoc analysis showing benefit, but insurance approval varies. Complications: voice alteration/hoarseness (most common, 30-60%, due to stimulation of the laryngeal nerve, resolves when stimulator off), cough, dysphagia, paresthesias, infection, and device malfunction. It is MRI conditional. Battery life: 4-10 years.

Vascular Graft (Synthetic)

– A prosthetic conduit used to replace or bypass diseased arteries or veins. Materials: Dacron (polyethylene terephthalate PET, woven or knitted, used for large-diameter high-flow vessels: aorta, iliac) and ePTFE (expanded polytetrafluoroethylene, Gore-Tex, used for medium-diameter vessels: femoral, popliteal, AV grafts for dialysis). Heparin-bonded grafts (e.g., Gore Propaten, GORE-TEX with heparin bioactive surface) reduce early thrombosis. Linings: tissue-engineered and autologous endothelial cell-seeded grafts (experimental). Bifurcated grafts: for aortoiliac occlusion or AAA repair (body + left/right limbs). Ringed grafts: external ePTFE rings prevent kinking at the knee (fem-pop bypass). Biological grafts: autologous (saphenous vein — best patency for infrainguinal bypass, 60-80% at 5 years ; internal mammary artery for CABG), allograft (cadaveric vein), and xenograft (bovine carotid artery, bovine mesenteric vein). Indications: peripheral arterial disease (aortobifemoral, femoropopliteal, femorotibial bypass), aortic aneurysm repair (open AAA repair or hybrid repair), and hemodialysis access (AV graft when native AV fistula is not feasible, e.g., basilic vein transposition, forearm loop). Complications: infection (most dreaded, especially if graft is in the groin, treatment includes excision and extra-anatomic bypass), thrombosis (pseudointimal hyperplasia, especially at low-flow anastomoses), anastomotic pseudoaneurysm, and graft-enteric fistula (aortoduodenal fistula after AAA repair). Patency: aortobifemoral 80-90% at 5 years; femoropopliteal 50-70%.

Vena Cava Filter (IVC Filter)

– An endovascular device placed in the inferior vena cava to prevent pulmonary embolism (PE) in patients with venous thromboembolism (VTE) who cannot tolerate anticoagulation. The filter is a metal cage (cone-shaped, with struts and hooks) that traps emboli while allowing blood flow. The most commonly used is the Greenfield filter (stainless steel or titanium, 4-6 struts). Retrievable filters: Bard Denali (conical, retrievable, nitinol), Cook Celect (conical, retrievable, with timers), Cook Gunther Tulip (conical, retrievable), and Cordis OptEase (double-basket, retrievable). Permanent filters: Greenfield (stainless steel) and TrapEase. Indications: contraindication to anticoagulation (active bleeding, recent surgery, hemorrhagic stroke), failure of anticoagulation (recurrent PE despite therapeutic anticoagulation), and prophylactic in high-risk trauma (severe pelvic/long bone fractures, TBI). Contraindications: complete IVC thrombosis (no room to deploy), severe IVC stenosis, and uncorrectable coagulopathy. The ideal position is infrarenal IVC (below the renal veins, above the IVC bifurcation at L2-L3). The filter is placed under fluoroscopic guidance via the right femoral vein (most common) or right internal jugular vein. Complications: IVC penetration (struts through the IVC wall, estimated 15-40% in retrievable filters, often asymptomatic), filter migration (to the heart, pulmonary artery — can be fatal), IVC thrombosis (5-15%, causing phlegmasia, edema, DVT), retrieval failure (filter tilting, endothelialization, strut penetration, significant clot in filter, time since implant >1 year reduces likelihood of successful retrieval — 60-90% retrieval success in experienced centers). DVT at the insertion site (5-10%). The FDA in 2010 and 2014 recommended that retrievable filters be removed as soon as PE protection is no longer needed (ideally within 29-54 days). Long-term complications: filter fracture, systemic embolization, and chronic IVC occlusion.

Ventricular Assist Device (VAD/LVAD)

– A mechanical pump that supports the failing left ventricle. The HeartMate 3 (Abbott) is the current state-of-the-art LVAD: a fully magnetically levitated centrifugal pump with artificial pulse (designed to reduce pump thrombosis and shear stress). The HeartWare HVAD (Medtronic, previously HeartWare — withdrawn from market due to mortality/morbidity concern but still used in some patients) is a smaller centrifugal pump with a hybrid suspension. Impella (Abiomed): a microaxial percutaneous VAD placed via the femoral artery (Impella CP, Impella 5.5) for short-term support (up to 30 days for Impella 5.5). The TandemHeart (LivaNova) is a percutaneous left atrial-to-femoral artery bypass system. Indications: bridge to transplant (BTT, keep patient alive until a donor heart becomes available), bridge to candidacy (BTC, improve end-organ function to become transplant-eligible), destination therapy (DT, permanent support for patients ineligible for heart transplant), and bridge to recovery (BTR, support until the native heart recovers, e.g., myocarditis, post-cardiotomy). Surgery: median sternotomy or thoracotomy, inflow cannula in the LV apex, outflow graft to the ascending aorta (LVAD). A driveline exits the abdominal wall and connects to an external controller and batteries. Complications: pump thrombosis (reduced with HeartMate 3 full mag-lev design), stroke (ischemic and hemorrhagic, 10-20% at 1-2 years), driveline infection (15-30% at 1 year, 30-50% at 2 years), GI bleeding (30-50% from acquired von Willebrand syndrome type 2A, especially with axial flow HM II), right heart failure (20-30%), aortic insufficiency (from continuous flow, 20-30% at 2-3 years), and hemolysis. Life: HM3 is approved for up to 2 years as DT but is known to run much longer.

Chapter 16

Medical Instruments

Anoscope

– A short, rigid, hollow tube used to examine the anal canal and distal rectum. Sizes: adult (7-9 cm length, 1.5-2 cm diameter). Illuminated via integrated light source or headlamp. Inserted with obturator after lubrication and gentle dilation. Indications: hemorrhoids (internal/external), anal fissures, fistula-in-ano, anorectal abscess, and rectal foreign body. Patient positioned in left lateral decubitus (Sims) or prone jackknife position.

Arthroscope

– A fiberoptic endoscope used to visualize the interior of a joint through small incisions (portals). Diameter: 2.7-4.0 mm with 30° or 70° lens angle. Introduced through a trocar/cannula system. Continuous saline irrigation distends the joint and clears debris. Common joints: knee (most common), shoulder, hip, wrist, ankle, and elbow. Diagnostic arthroscopy has been largely replaced by MRI; therapeutic arthroscopy is used for meniscectomy, ACL reconstruction, rotator cuff repair, and labral debridement.

Aspiration Needle

– A thin, hollow needle used to aspirate fluid or biopsy tissue. Types: fine needle aspiration (FNA, 22-25 gauge, for cytology), core needle (14-18 gauge, for histology), and aspiration for drainage (larger bore, for cysts, abscesses, and effusions). Ultrasound or CT guidance improves accuracy. Indications: thyroid nodules, breast masses, lymphadenopathy, liver lesions, pancreatic cysts, and pleural/peritoneal effusions. Complications: bleeding, infection, and pneumothorax (for chest procedures).

Auroscope (Otoscope)

– A handheld instrument with a light source and magnifying lens used to examine the external auditory canal and tympanic membrane. Speculum sizes: 2-5 mm (pediatric to adult). Features: pneumatic attachment for insufflation (assesses TM mobility, middle ear effusion), and cerumen removal loop. Normal TM: translucent, pearly gray, with visible landmarks (umbo, manubrium of malleus, light reflex). Abnormal: retraction, bulging, perforation, effusion (air-fluid level), cholesteatoma, and tympanosclerosis.

Bandage Scissors

– Scissors designed for cutting bandages and dressings without injuring the skin. Blades: angled with a blunt probe tip on one blade that slides between the bandage and the patient. Types: Lister bandage scissors (most common, angled), trauma shears (heavy-duty, serrated blade for cutting through clothing, belts, and boots). Disposable or reusable (stainless steel). Essential in emergency departments and surgical wards. Not intended for suture removal or tissue dissection.

BiliLight (Phototherapy Unit)

– A device emitting blue light in the 460-490 nm wavelength range used to treat neonatal hyperbilirubinemia. Mechanism: converts unconjugated bilirubin to water-soluble isomers (photobilirubin) that can be excreted without conjugation. Types: conventional (overhead lamps), fiberoptic blanket (BiliBed, wrapped around infant), and LED phototherapy. Intensity: measured in microW/cm^2/nm (target >30 for intensive therapy). Used with protective eye shields. Complications: loose stools, dehydration, and hyperthermia.

Bilirubinometer

– A handheld device that measures transcutaneous bilirubin (TcB) in neonates without blood sampling. Uses spectrophotometry to measure yellow color of subcutaneous tissue. Placed on the infant's forehead or sternum. Correlates well with serum bilirubin levels. Used as a screening tool; elevated values are confirmed with serum bilirubin measurement. Reduces the need for painful heel-stick blood draws. Affected by skin pigmentation and phototherapy exposure.

Biopsy Punch

– A cylindrical blade used to obtain a full-thickness skin sample for histopathology. Sizes: 2-8 mm diameter (4 mm most common for shave biopsy of inflammatory lesions, 6 mm for neoplasms). Technique: local anesthesia, rotate punch through epidermis and dermis to subcutaneous fat, lift specimen with forceps, cut base with scissors. Closure depends on size: small (2-3 mm) heal by secondary intention; larger require suture (4-0 or 5-0 nylon). Indications: skin cancer diagnosis, inflammatory dermatoses, and nail bed biopsy.

Bougie

– A flexible or rigid instrument used to dilate strictures in the esophagus, urethra, or other hollow organs. Esophageal bougienage: Maloney (mercury-filled, flexible), Savary-Gilliard (wire-guided, polyvinyl, tapered tip), and balloon dilators. Urethral bougie: sounds for stricture dilation. Sizing: French (Fr) scale (circumference in mm = 3x diameter in mm). Indications: esophageal stricture (peptic, radiation-induced, caustic, eosinophilic esophagitis), urethral stricture.

BP Cuff (Sphygmomanometer)

– A device for non-invasive blood pressure measurement. Components: inflatable cuff, bulb/valve, and manometer (mercury, aneroid, or digital). Cuff size: bladder width 40% of arm circumference, length 80% (pediatric, adult, large adult, thigh). Technique: cuff at heart level, inflate 30 mmHg above palpable brachial pulse, deflate at 2-3 mmHg/sec. Systolic: first Korotkoff sound. Diastolic: disappearance (fifth phase). Automated oscillometric cuffs (Dinamap) are standard in hospitals.

Bronchoscope

– A flexible or rigid endoscope for examining the tracheobronchial tree. Flexible bronchoscope: fiberoptic or video, diameter 3.5-6.0 mm, working channel 2.0-3.2 mm for instruments (brush, biopsy forceps, needle). Allows BAL, transbronchial biopsy, endobronchial biopsy, and EBUS-TBNA. Rigid bronchoscope: metal tube, used for massive hemoptysis, large foreign body removal, and stent placement. Indications: lung mass/cancer diagnosis, persistent cough, hemoptysis, interstitial lung disease, and foreign body.

Cannula

– A flexible tube inserted into a vessel, duct, or body cavity for fluid delivery or drainage. IV cannula: sizes 14-26 gauge (larger gauge = smaller diameter), color-coded: gray (16G), green (18G), pink (20G), blue (22G), yellow (24G). Nasal cannula: delivers 1-6 L/min O2 (24-44% FiO2). Surgical cannula: used for cardiopulmonary bypass, liposuction, and arthroscopy. Complications: phlebitis, infection, infiltration, and air embolism.

Cardiac Monitor (Telemetry)

– Continuous real-time ECG monitoring for arrhythmia detection. Standard 3-lead or 5-lead configuration. Displays heart rate, rhythm, and ST segments. Alarms for bradycardia, tachycardia, asystole, and ventricular arrhythmias. Central monitoring station enables surveillance of multiple patients. Indications: acute coronary syndrome, post-MI, arrhythmia evaluation, post-cardiac surgery, and critically ill patients. Cannot replace 12-lead ECG for diagnosis; limited lead views.

Cardiopulmonary Bypass Machine

– A heart-lung machine used during open-heart surgery to temporarily replace cardiac and pulmonary function. Components: venous reservoir (receives deoxygenated blood), oxygenator (gas exchange), heat exchanger (temperature control), arterial pump (roller or centrifugal, delivers oxygenated blood to aorta), and cardioplegia delivery system (arrests heart with potassium-rich solution). Circuit primes 1.5-2 L. Cannulae placed in right atrium (venous) and ascending aorta (arterial). Complications: systemic inflammatory response, coagulopathy, hemolysis, and air embolism.

Catheter

– A thin, flexible tube for drainage, injection, or access. Urinary catheters: Foley (balloon retention, indwelling), straight (intermittent catheterization), and three-way (continuous bladder irrigation for hematuria). Sizes: 8-30 Fr (pediatric to adult male). Central venous catheters: internal jugular, subclavian, or femoral; used for CVP monitoring, vasopressors, TPN, and hemodialysis. PICC lines: peripherally inserted, tip in superior vena cava. Swan-Ganz catheter: pulmonary artery pressure monitoring. Complications: infection, thrombosis, and vessel injury.

Cautery (Electrocautery)

– A device that uses electrical current to cut tissue and coagulate blood vessels. Monopolar: current travels from active electrode (pen, spatula, needle) through the patient to a grounding pad (dispersive electrode). Bipolar: current flows between two tips of the forceps (safer near delicate structures, less tissue damage). Modes: cut (continuous, higher voltage for vaporizing tissue), coag (pulsed, higher voltage for hemostasis), and blend. Uses: surgical dissection, hemostasis, and lesion ablation. Smoke plume contains toxic compounds.

Chest Tube (Thoracostomy Tube)

– A flexible plastic tube inserted into the pleural space to drain air, fluid, or blood. Sizes: 8-14 Fr (pneumothorax), 16-24 Fr (effusion, empyema), 28-40 Fr (hemothorax, postoperative). Insertion: triangle of safety (4th-5th intercostal space, midaxillary line), tunnel over the rib, directed apically (air) or basally (fluid). Connected to closed drainage system (water seal chamber, suction, collection chamber). Pleur-evac or Atrium systems. Indications: pneumothorax, hemothorax, empyema, pleural effusion, chylothorax, and postoperative thoracic surgery.

Colonoscope

– A flexible endoscope for examination of the entire large intestine (cecum to rectum). Length: 160-180 cm. Diameter: 12-13 mm. Features: tip controls (up/down, left/right angulation), working channel (3.2-4.2 mm for biopsy forceps, snare, polypectomy tools), water jet, and light source. Video colonoscope: CCD chip at the tip for high-resolution imaging. High-definition (HD) and narrow-band imaging (NBI) enhance mucosal detail. Used for CRC screening, polyp detection/removal, and diagnosis of colitis.

Colposcope

– A binocular microscope with a light source used to magnify the cervix, vagina, and vulva. Magnification: 5x-40x (most common 10-15x). Green filter enhances vascular patterns. Used after application of 5% acetic acid (aceto-white lesions indicate dysplasia) and Lugol iodine (Schiller test). Indications: abnormal Pap smear, abnormal vaginal bleeding, and cervical dysplasia surveillance. Allows targeted cervical biopsy. Digital colposcopy improves documentation and telemedicine consultation.

Cryotherapy Device

– A device that uses extreme cold to destroy abnormal tissue. Agents: liquid nitrogen (-196°C, spray or cotton-tip applicator), nitrous oxide (-89°C, cryoprobe via Joule-Thomson effect), and argon gas (-150°C, used in cryoablation for renal/liver tumors). Mechanism: rapid freeze-slow thaw cycles cause intracellular ice crystal formation, cell lysis, and vascular thrombosis. Used for: skin lesions (warts, actinic keratosis, seborrheic keratosis, basal cell carcinoma), cervical dysplasia (cryocautery), and prostate cancer ablation.

Cystoscope

– An endoscope for examination of the urethra and bladder. Flexible cystoscope: 15-17 Fr, office-based with local anesthesia. Rigid cystoscope: 19-26 Fr, requires anesthesia, has larger working channels for instruments (biopsy forceps, stone baskets, resectoscope). Lens angles: 0°, 30°, 70°. Irrigation: saline (not water, which could cause hemolysis). Indications: hematuria evaluation, recurrent UTI, bladder tumor surveillance, ureteral stent placement/removal, and diagnosis of interstitial cystitis.

Defibrillator

– A device that delivers a controlled electrical shock to the heart to terminate ventricular fibrillation or pulseless ventricular tachycardia. External: biphasic waveform (150-200 J), self-adhesive pads placed anterolateral or anteroposterior. Automated external defibrillator (AED): analyzes rhythm automatically, voice prompts for CPR and shock delivery. Implantable cardioverter-defibrillator (ICD): transvenous or subcutaneous leads, detects VT/VF and delivers shock, anti-tachycardia pacing (ATP), or pacing. Indications: sudden cardiac arrest, ischemic and non-ischemic cardiomyopathy with low EF.

Dermatoscope (Dermoscope)

– A handheld magnifying instrument with a light source and polarized/non-polarized lenses for skin surface examination. Magnification: 10x (standard) to 100x. Uses: evaluation of pigmented skin lesions (melanoma vs. benign nevi vs. seborrheic keratosis), nailfold capillaroscopy (connective tissue disease), and assessment of hair and scalp disorders (trichoscopy). Enables visualization of structures below the stratum corneum (pigment network, dots, globules, streaks, regression structures, and blue-white veil). Uses the 2-step dermoscopy algorithm (melanocytic vs. non-melanocytic, benign vs. malignant). Improves melanoma diagnostic accuracy by 25-50% compared to naked eye.

Dialysis Machine (Hemodialysis)

– A device that filters blood when the kidneys fail. Components: blood pump (roller pump, 200-500 mL/min), dialyzer (hollow fiber artificial kidney with semipermeable membrane), dialysate delivery system (mixes concentrate with water at 37°C, rate 500-800 mL/min), and safety monitors (air detector, pressure monitors, blood leak detector). Water treatment: reverse osmosis removes contaminants. Vascular access: AV fistula (gold standard), AV graft, or tunneled dialysis catheter. Sessions: 3-4 hours, 3 times weekly. Removes urea, creatinine, potassium, phosphate, and excess fluid.

Doppler Probe

– A handheld device using the Doppler effect to detect blood flow. Continuous wave Doppler: emits and receives ultrasound continuously, detects velocity but not depth. Used for: ankle-brachial index, fetal heart rate detection (10 MHz), and peripheral vascular assessment. Pulsed wave Doppler: measures velocity at a specific depth. Spectral Doppler: waveform analysis. Color Doppler: color-encodes direction and velocity on ultrasound images. Indications: DVT, PAD, carotid stenosis, and fetal heart rate.

Duodenoscope

– A specialized side-viewing endoscope used for ERCP. Length: 124 cm, diameter: 12 mm. Side-viewing optics provide en face view of the ampulla of Vater. Features: elevator (a hinged bridge that deflects cannulas, guidewires, and stents), working channel (4.2 mm for 10 Fr stents). Indications: common bile duct stone extraction, biliary stent placement for malignant obstruction, sphincterotomy, and pancreatic duct drainage. High-level disinfection critical due to complex design and elevator mechanism (risk of CRE transmission).

ECG Machine (Electrocardiograph)

– A device that records the heart's electrical activity. Standard 12-lead: 10 electrodes (4 limb: RA, LA, RL, LL; 6 precordial: V1-V6). Filters: AC filter (60 Hz), muscle tremor, and baseline drift. Paper speed: 25 mm/sec (standard), 50 mm/sec (for tachycardia). Calibration: 10 mm/mV. Acquisition: simultaneous vs. sequential lead recording. Digital ECG machines provide computerized interpretation (Marquette, Glasgow, or Cabrera algorithms) with storage and networking. Telemetry and portable devices (iPhone ECG, Holter) extend ambulatory monitoring.

Esophageal Stethoscope

– A flexible tube inserted into the esophagus with a microphone at the tip to monitor breath sounds and heart sounds during general anesthesia. Placed at 28-32 cm from the incisors. Provides continuous monitoring of ventilation (air entry, bronchospasm, breath holding) and cardiac activity (muffled heart sounds suggest hypovolemia). Largely replaced by capnography and pulse oximetry in modern anesthesia.

Eye Chart (Snellen Chart)

– A standardized chart for measuring visual acuity. Snellen: rows of progressively smaller letters (20/200 at top, 20/20 at bottom). Patient stands 20 feet (6 meters) from the chart. Visual acuity expressed as a fraction: 20/20 (normal), 20/40 (can read at 20 feet what normal sees at 40 feet). Legal blindness: 20/200 or worse in the better eye with correction. Tumbling E chart: for non-readers and children. Rosenbaum pocket chart: near vision at 14 inches. Pinhole occluder: distinguishes refractive error from organic disease.

Eye Speculum

– An instrument to hold the eyelids open during eye examination or surgery. Types: wire speculum (Lancaster, most common, self-retaining), solid blade speculum (for fornix exposure), and pediatric speculum. Used in: cataract surgery, corneal transplantation, vitrectomy, and laser procedures. Also used in cranial surgery for corneal protection. Must be placed carefully under the eyelids to avoid corneal abrasion or globe perforation.

Fetal Doppler

– A handheld ultrasound device (2-5 MHz) used to detect and auscultate fetal heart tones. Uses continuous wave Doppler with a transducer placed on the maternal abdomen with ultrasound gel. FHR detectable as early as 10-12 weeks (typically 12+). Normal: 110-160 bpm. Used in prenatal visits, labor triage, and home monitoring. Replaces fetoscope (Pinard horn). Cannot replace electronic fetal monitoring for high-risk pregnancies.

Forceps (Surgical)

– Instruments for grasping, holding, or extracting tissue. Thumb forceps: pickups with spring action (1x2 teeth: tissue forceps; Adson: fine teeth for skin; DeBakey: atraumatic cardiovascular). Dressing forceps: non-toothed, for gauze and wound packing. Ring forceps: Kelly, Mosquito, Crile, Rochester-Pean (hemostatic clamps with ratchet lock). Hemostatic: curved or straight, sizes from delicate (fine ophthalmic) to large (obstetric). Used for clamping bleeding vessels, holding sutures, and blunt dissection.

Glucose Meter (Glucometer)

– A handheld device for measuring capillary blood glucose from a fingerstick. Uses glucose oxidase, glucose dehydrogenase, or hexokinase reaction on a test strip. Blood volume: 0.3-1.0 microliter. Results in 5-15 seconds. Normal fasting: 70-100 mg/dL. Diabetes management: pre-meal 80-130, postprandial <180. Continuous glucose monitor (CGM): subcutaneous sensor measures interstitial glucose every 5 minutes (Dexcom, Medtronic, Abbott Libre). Trend arrows predict direction of change. Alarms for hypo/hyperglycemia. Essential for intensive insulin therapy.

Handpiece (Dental)

– A powered instrument used in dentistry for cutting, grinding, and polishing teeth. High-speed handpiece: air-driven turbine, up to 400,000 rpm, used for cutting enamel and dentin (caries removal, crown preparation). Uses copious water spray to prevent heat generation. Low-speed handpiece: electric motor, <40,000 rpm, with contra-angle and straight attachments, for caries excavation, polishing, and rotary endodontics. Surgical handpiece: for implant placement, bone cutting. Maintenance: sterilization, lubrication, and regular inspection.

Headlamp (Surgical Headlight)

– A wearable light source worn on the surgeon's head to illuminate the operative field. Components: LED or xenon light source, fiberoptic cable connecting to a headband-mounted lens, or battery-powered LED on the headband. Coaxial: light beam aligned with the surgeon's line of sight. Battery packs provide 2-8 hours of operation. Indications: ENT surgery (tonsillectomy, mastoidectomy, sinus surgery), neurosurgery, urology, and any deep cavity surgery where overhead lights are inadequate.

Heart-Lung Machine

– See Cardiopulmonary Bypass Machine.

Heat Lamp

– An infrared lamp used to provide localized heat therapy to body surfaces. Mechanism: infrared radiation penetrates to warm subcutaneous tissues. Indications: muscle relaxation and pain relief (strains, sprains, fibromyalgia), warming neonates (radiant warmer), and promoting wound healing (increase local blood flow). Contraindications: acute inflammation, bleeding, malignancy, and insensate skin (burn risk).

Infusion Pump

– An electronic device that delivers fluids, medications, or nutrients intravenously at controlled rates. Types: volumetric (large volume, mL/hr, for maintenance fluids, TPN), syringe pump (small volume, mL/min, for critical medications like vasopressors, insulin, heparin), and patient-controlled analgesia (PCA: patient self-administers preset bolus doses with lockout interval). Smart pumps: drug library with dose error reduction software (alerts for limits). Programming: rate, volume to be infused, dose (mcg/kg/min). Complications: infiltration, extravasation, infection, and air embolism.

Insulin Pen

– A reusable or disposable device for subcutaneous insulin injection. Contains a prefilled insulin cartridge (3 mL, 300 units) or may be fully disposable. Dial selects dose (1-60 units in 1-unit or 0.5-unit increments). Needle: 4-8 mm, 30-34 gauge (fine and short for comfort). Types: rapid-acting (NovoLog, Humalog, Apidra), short-acting (Novolin R), long-acting (Lantus, Levemir, Toujeo, Tresiba). Ease of use improves adherence compared to vial and syringe. Air shot before each injection to eliminate air bubbles.

Joint Aspiration Needle

– A sterile needle used for arthrocentesis (joint aspiration). Sizes: 18-22 gauge for larger joints (knee, shoulder), 22-25 gauge for smaller joints (wrist, ankle, elbow). Knee: medial or lateral approach with the joint slightly flexed. Synovial fluid analysis: cell count/differential, Gram stain, culture, and crystal analysis (urate vs. CPPD). Contraindications: overlying cellulitis (risk of septic arthritis from introduction), severe coagulopathy. Therapeutic drainage for symptomatic relief.

Laparoscope

– A rigid endoscope for minimally invasive (keyhole) abdominal surgery. Diameter: 5-10 mm, lens angle: 0° or 30°. High-definition video camera attached to the eyepiece. Fiberoptic cable delivers light from a xenon source. CO_2 insufflation (pressure 12-15 mmHg) creates working space (pneumoperitoneum). Access: Veress needle or Hasson open technique. Indications: cholecystectomy, appendectomy, hernia repair, colectomy, sleeve gastrectomy, diagnostic laparoscopy (abdominal pain, cancer staging). Advantages: reduced pain, shorter stay, faster recovery.

Laryngoscope

– An instrument used to visualize the larynx and vocal cords for endotracheal intubation. Direct laryngoscopy: Macintosh blade (curved, tip placed in vallecula, lifts epiglottis indirectly) or Miller blade (straight, lifts epiglottis directly). Video laryngoscopy: GlideScope, C-MAC, McGrath — camera on the blade shows glottic view on a screen, improving first-pass success. Indications: general anesthesia, emergency airway management, and difficult airway. Grades: Cormack-Lehane I (full view) to IV (no view).

Lithotripter

– A device that uses shock waves to fragment urinary tract stones. Extracorporeal shock wave lithotripsy (ESWL): focused, high-energy acoustic waves generated by an electrohydraulic, electromagnetic, or piezoelectric source, transmitted through water into the body. Indications: renal calculi <2 cm (upper and middle calyces) and proximal ureteral stones. Contraindications: pregnancy, coagulopathy, distal obstruction, and renal artery aneurysm. Intracorporeal: laser lithotripsy (holmium:YAG, thulium fiber), pneumatic (Swiss Lithoclast), and ultrasonic lithotripsy, used through a ureteroscope or nephroscope.

Loupe (Surgical Loupe)

– Binocular magnifying glasses worn by surgeons, dentists, and ENT specialists for enhanced visualization. Magnification: 2.5x-8x (most common 2.5x-3.5x for general surgery, 4x-8x for microsurgery). Types: through-the-lens (TTL, custom, most comfortable), flip-up (adjustable, modular). Features: Galilean vs. prismatic optics (prismatic provides wider field at higher magnification). Working distance: 340-500 mm. Used in: microvascular anastomosis, nerve repair, dental endodontics, and fine plastic surgery.

Mechanical Ventilator

– A machine that supports or replaces spontaneous breathing. Modes: volume control (set tidal volume, e.g., 6-8 mL/kg IBW), pressure control (set inspiratory pressure), pressure support (spontaneous breaths supported), SIMV (synchronized intermittent mandatory ventilation), and CPAP (continuous positive airway pressure). Settings: FiO2 (21-100%), PEEP (5-20 cmH2O), respiratory rate, I:E ratio (1:2 to 1:1). High-frequency oscillatory ventilation (HFOV) and airway pressure release ventilation (APRV) for refractory hypoxemia. Complications: VILI (ventilator-induced lung injury), VAP (ventilator-associated pneumonia), and hemodynamic compromise.

Microscope (Operating Microscope)

– A binocular microscope providing magnification and illumination for microsurgery. Magnification: 5x-40x adjustable, with oculars 12.5x or 10x and objective lens 200-400 mm. Coaxial illumination: light path aligned with the surgeon's line of sight. Foot pedal controls: zoom, focus, and positioning (motorized stand). Used in: ophthalmic surgery (cataract, vitrectomy, corneal transplant), neurosurgery (tumor resection, aneurysm clipping, spinal surgery), ENT (stapedectomy, cochlear implant), microvascular surgery, and dental endodontics.

Nasogastric Tube

– A flexible tube inserted through the nose into the stomach. Sizes: 8-18 Fr. Types: Salem sump (double lumen: larger for drainage, smaller for air vent), Levin (single lumen), and feeding tubes (Dobhoff: small bore, weighted tip, for enteral nutrition).

Indications: gastric decompression (ileus, obstruction), lavage (GI bleeding, overdose), enteral feeding (when oral intake is inadequate), and diagnostic (aspiration for analysis). Confirmation: X-ray (most reliable), pH testing (gastric pH <5), and auscultation (less reliable). Complications: sinusitis, trauma, aspiration, and tube malposition (lung placement).

Needle Holder

– A surgical instrument used to hold a suturing needle while passing it through tissue. Jaws: smooth or diamond-dusted (for microsurgery), cross-hatched or tungsten carbide inserts for grip. Lock: ratchet mechanism. Sizes: Mayo-Hegar (general, heavy), Crile-Wood (medium), Webster (fine, for ophthalmic/vascular), Castroviejo (spring-loaded, for microsurgery). Needle handling: hold at the junction of middle and distal third of the needle, perpendicular to the holder. The type of needle holder matches the needle size and tissue type.

Opthalmoscope (Direct)

– A handheld instrument for examining the fundus (retina, optic disc, blood vessels). Lens wheel: +20 to -20 diopters (focus). Light source: halogen or LED, with filters (red-free for nerve fiber layer, cobalt blue for fluorescein). Technique: right-right eye-right hand, left-left-left. Normal: optic disc (pink, sharp margins, cup-to-disc ratio <0.5), retinal vessels (A:V ratio 2:3), macula (avascular, 2 disc diameters temporal to disc). Findings: papilledema, optic atrophy, diabetic retinopathy, hypertensive retinopathy, and glaucoma.

Oxygen Mask

– A device for delivering supplemental oxygen. Simple mask: 5-10 L/min, FiO2 35-55%. Non-rebreather: mask + reservoir bag + one-way valves; 10-15 L/min, FiO2 60-90%. Partial rebreather: similar without expiratory valve. Venturi mask: delivers precise FiO2 (24%, 28%, 31%, 35%, 40%) via color-coded entrainment ports; used for COPD patients requiring controlled oxygen (avoid CO2 retention). Nasal cannula: 1-6 L/min, FiO2 24-44%. High-flow nasal cannula: heated, humidified, up to 60 L/min, FiO2 up to 100%, used in respiratory failure.

Pacemaker

– An implantable device that delivers electrical impulses to the heart to maintain adequate heart rate. Types: single chamber (ventricular VVI or atrial AAI), dual chamber (DDD, most physiologic), biventricular (CRT: cardiac resynchronization therapy for heart failure with wide QRS), and leadless (Micra, directly in RV). Indications: sick sinus syndrome, AV block (third-degree, symptomatic second-degree), and heart failure with reduced EF and LBBB. Pacing modes: NBG code (e.g., DDD = dual chamber paced/sensed, dual response). Complications: infection, lead dislodgement, and generator pocket hematoma.

Peak Flow Meter

– A handheld device used to measure peak expiratory flow rate (PEFR), the maximum speed of expiration after maximal inspiration. The patient takes a deep breath and blows as hard and fast as possible into the device. PEFR is measured in litres per minute. Used in asthma management: monitoring (diurnal variation, response to treatment), diagnosis (reversibility testing—improvement >20% after bronchodilator), and identification of exacerbations (PEFR <60% of personal best indicates severe exacerbation). Predicted PEFR values are based on age, sex, and height (normal healthy adult: 400–700 L/min). A zone system (green/yellow/red) is used for action plans: green (≥80% of personal best—well controlled), yellow (50–79%—caution,), red (<50%—medical emergency). Limitations: effort-dependent, poor technique reduces reliability. Electronic peak flow meters are available with digital recording and Bluetooth connectivity.

Pulse Oximeter

– A non-invasive device measuring SpO2 (peripheral oxygen saturation). Principle: spectrophotometry at 660 nm (red, deoxygenated hemoglobin) and 940 nm (infrared, oxygenated hemoglobin). Probe: finger, toe, earlobe, or forehead (for low perfusion). Normals: SpO2 95-100%. Desaturation: <90% (hypoxemia, PaO2 60 mmHg). Limitations: motion artifact, poor perfusion (cold, shock), dyshemoglobinemias (COHb: falsely high, metHb: tends toward 85%), dark nail polish, and ambient light interference. Essential for oxygen therapy titration and anesthesia monitoring.

Reflex

– An involuntary, stereotyped, rapid response to a specific stimulus, mediated by a reflex arc without conscious thought. In clinical medicine, reflexes are assessed as part of the neurological examination to localise lesions within the central and peripheral nervous system. Deep tendon reflexes (muscle stretch reflexes): elicited by tapping a tendon with a reflex hammer, activating muscle spindles, and causing a monosynaptic spinal reflex. Tested: biceps (C5–C6), brachioradialis (C5–C6), triceps (C7–C8), patellar (L3–L4), and ankle (S1). Graded 0–4+: 0 (absent), 1+ (diminished), 2+ (normal), 3+ (brisk, may be normal), 4+ (hyperactive, with clonus). Hyperreflexia (with clonus) suggests upper motor neuron lesion (stroke, spinal cord injury, multiple sclerosis). Hyporeflexia/areflexia suggests lower motor neuron lesion (peripheral neuropathy, radiculopathy, motor neuron disease). Superficial reflexes: abdominal (T8–T12), cremasteric (L1–L2), plantar (Babinski sign—L5–S1, normal is flexor; extensor/Babinski sign is abnormal after infancy, indicates upper motor neuron lesion). Primitive reflexes (frontal release signs): grasp, palmomental, snout, glabellar—emerge with frontal lobe pathology. Hoffman sign (finger flexor reflex) and Tromner sign: hyperactive deep tendon reflexes suggesting upper motor neuron lesion.

Reflex Hammer

– A tool for testing deep tendon reflexes by striking a tendon. Types: Queen Square (most common, triangular rubber head), Troemner (circular head, weighted), Babinski (whip-like for plantar response). Scoring: 0 (absent), 1+ (hypoactive), 2+ (normal), 3+ (hyperactive), 4+ (clonus). Reflexes tested: biceps (C5-6), brachioradialis (C6), triceps (C7), patellar (L3-4), Achilles (S1). Hyperreflexia suggests upper motor neuron lesion; hyporeflexia suggests lower motor neuron lesion, neuropathy, or neuromuscular junction disorder.

Retractor

– A surgical instrument used to hold back edges of tissue or organs to expose the operative field. Handheld: Army-Navy (small, versatile), Richardson (large abdominal), Deaver (deep abdominal), Senn (double-ended: rake and blade), and Roux (breast/thyroid). Self-retaining: Balfour (abdominal, blades retract laterally), Bookwalter (ring system with multiple retractor blades, for complex abdominal surgery), Gelpi (sharp points, for spinal or deep small incisions), Weitlaner (toothed, for superficial wounds), and O'Sullivan-O'Connor (obstetric, for cesarean section).

Rhinoscope

– An instrument for examining the nasal cavity. Anterior rhinoscopy: standard nasal speculum with a headlight, examines the anterior nasal cavity (inferior turbinate, septum, vestibule). Nasal endoscopy (rigid rhinoscope): 0°, 30°, or 45° Hopkins rod telescopes, 2.7-4.0 mm diameter. Provides detailed examination of the middle meatus, ostia of the sinuses, sphenoethmoidal recess, and posterior choana. Indications: sinusitis, nasal obstruction, epistaxis, polyps, and nasal masses. Used for office-based diagnosis and endoscopic sinus surgery.

Robot (Surgical Robot)

– A robotic system for minimally invasive surgery (da Vinci Surgical System). Components: surgeon console (3D high-definition view, wrist controls, foot pedals), patient-side cart (4 robotic arms with articulated EndoWrist instruments), and vision cart (dual 3D camera system). Advantages: 7 degrees of freedom, tremor filtration, motion scaling, and 3D magnified visualization. Uses: prostatectomy, hysterectomy, colectomy, cholecystectomy, thoracic surgery, and transoral robotic surgery (TORS). Cost and steep learning curve are limitations.

Scalpel

– A small, sharp blade used for surgical incisions. Handles: 3 (small, for 10, 11, 12, 15 blades), 4 (large, for 20, 21, 22, 23, 24 blades), 7 (long, delicate, for 15 blade). Blade shapes: 10 (curved, general incisions), 11 (pointed, stab incisions), 12 (curved, tonsil), 15 (small, precise, ophthalmic, plastic, hand), 20 (larger curved, abdominal). Disposable scalpel: single-piece plastic handle with pre-attached blade. Safety handles: retractable blade for sharps prevention.

Sclerometer (Tonometer)

– An instrument for measuring intraocular pressure. Goldmann applanation tonometer: gold standard, mounted on a slit lamp, flattens a 3.06 mm diameter area of cornea, pressure reading in mmHg. Indentation tonometry (Schiotz): less accurate, portable. Non-contact (air puff): convenient for screening, less accurate. Normal IOP: 10-21 mmHg. Elevated: glaucoma (primary open angle, angle closure). Used for glaucoma screening, monitoring, and management.

Slit Lamp (Biomicroscope)

– A binocular microscope with an adjustable light source (slit beam) for examining the anterior and posterior segments of the eye. Magnification: 6x-40x. Slit beam width and height adjustable. Filters: cobalt blue (fluorescein examination), red-free. Uses: cornea (abrasion, ulcer, keratitis, dystrophy), anterior chamber (cells, flare, depth), iris (synechiae, neovascularization), lens (cataract), vitreous (hemorrhage, cells), and retina (limited view without fundus lens). Combined with tonometer and gonioscope lens.

Speculum (Vaginal)

– An instrument for opening the vaginal walls to examine the cervix and upper vagina. Types: Graves (most common, two blades, screw or ratchet adjustment, sizes: neonatal, pediatric, small, medium, large), Pederson (narrower blades, for nulliparous or narrow introitus), and Cusco (bivalve, self-retaining, used outpatient). Inserted at 45-degree angle, rotated to transverse, opened by turning the screw. Used for Pap smear collection, cervical culture, IUD insertion, and colposcopy. Lubricate sparingly (affects Pap cytology).

Sphygmomanometer

– See BP Cuff.

SpO2 Probe

– See Pulse Oximeter.

Stapler (Surgical Stapler)

– A device for placing rows of staples to close tissue, divide organs, or reconnect segments. Types: linear cutter (GIA: gastrointestinal anastomosis, fires two triple-staggered rows and cuts between), linear stapler (TA: thoracic-abdominal, fires two rows, no knife), circular stapler (EEA: end-to-end anastomosis, for bowel and esophageal surgery), and skin stapler (disposable, rapid skin closure). Staples: titanium or stainless steel. Uses: lung resection, colectomy, gastrectomy, bowel anastomosis, and wound closure. Bioabsorbable and laparoscopic staplers available.

Steam Sterilizer (Autoclave)

– A device for sterilizing medical instruments and supplies with saturated steam under pressure. Conditions: 121°C at 15 psi for 30 minutes (gravity displacement) or 134°C at 30 psi for 3-15 minutes (pre-vacuum). Kills all microorganisms including spores. Indicators: chemical (autoclave tape, Bowie-Dick test) and biological (Geobacillus stearothermophilus spore vials). Flashing: short-cycle sterilization of unwrapped instruments for immediate use. Not for heat-sensitive items (use ethylene oxide or hydrogen peroxide plasma).

Stethoscope

– An acoustic device for auscultation of heart, lung, and bowel sounds. Components: chestpiece (bell for low-frequency sounds — heart murmurs; diaphragm for high-frequency sounds — breath sounds, normal heart sounds), tubing (Y-shaped, single- or dual-lumen), and earpieces (angled to match ear canal direction). Electronic stethoscope: amplifies sound (up to 40x), filters ambient noise, and can record/transmit sounds for telemedicine. Types: Littmann (Cardiology, Classic), Welch Allyn, and pediatric. Essential for physical diagnosis.

Suction Device

– A device that creates negative pressure to remove fluids, secretions, or debris from a surgical field or airway. Components: vacuum source (wall suction or portable pump), collection canister, tubing, and suction tips. Suction tips: Yankauer (tonsil tip, rigid, large-bore, for oral and surgical suction), Poole (abdominal, multi-perforated to prevent tissue grasping), Frazier (fine, flexible, for ENT/neurosurgery), and catheter suction (flexible, for endotracheal suction). Wall suction: 80-120 mmHg (low), 120-200 (high).

Surgical Drill

– A powered instrument for cutting bone and drilling holes for screws. Powered: pneumatic (nitrogen-driven, high torque, used in orthopedic and neurosurgery) or electric (battery-powered, variable speed). Attachments: burr (round, acorn, for burr holes and shaping), reamer (for acetabular preparation in hip arthroplasty), and wire driver (for K-wire placement). Used in: craniotomy (burr holes), orthopedic surgery (joint replacement, fracture fixation, spinal fusion), and ENT (mastoidectomy, cochlear implant). Sterilization between uses.

Suture Material

– Thread used to approximate tissues. Absorbable: surgical gut (10-14 days), polyglactin 910 (Vicryl, 60-90 days), poliglecaprone (Monocryl, 90-120 days), polydioxanone (PDS, 180 days). Non-absorbable: nylon (Ethilon, permanent, monofilament), polypropylene (Prolene, permanent, monofilament), silk (braided, handles well, permanent but can degrade), polyester (Ethibond, braided, permanent). Sizes: 1-0 (large, orthopedic) to 12-0 (microvascular). Needle types: cutting (skin), reverse cutting, taper (soft tissue), blunt (liver). Coated vs. uncoated.

Syringe

– A device for injecting or withdrawing fluids. Parts: barrel (graduated), plunger (rubber-tipped), and tip (Luer-Lok for secure connection, Luer-Slip for friction fit). Sizes: 1 mL (tuberculin, insulin), 3 mL (most common), 5 mL, 10 mL, 20 mL, 50-60 mL (irrigation, tube feeding). Insulin syringe: 0.3-1.0 mL, 29-31 gauge needle, marked in units (U-100). Tuberculin syringe: 1 mL, for PPD, allergy testing. Safety syringes: retractable needle or shield to prevent needlestick injury. Needles: gauge (larger number = smaller diameter), length (3/8 to 3 inches).

Thermometer

– An instrument for measuring body temperature. Types: (1) Mercury-in-glass thermometer—traditional, accurate (0.1°C precision), but mercury is toxic; phased out in many countries (banned in EU since 2009). (2) Digital electronic thermometer—uses a thermistor or thermocouple, fast (10–60 seconds), accurate, disposable probe covers. (3) Infrared tympanic thermometer—measures infrared heat from the tympanic membrane, fast (1–2 seconds), approximates core temperature, widely used in emergency departments and outpatient settings. (4) Temporal artery thermometer—infrared scan across the forehead, non-invasive, fast. (5) Liquid crystal forehead strip—least accurate, for screening only. (6) Indwelling temperature probes—esophageal, nasopharyngeal, bladder, rectal, pulmonary artery catheter (gold standard for core temperature monitoring in ICU/theatre). Normal temperature varies by site: rectal (36.6–38.0°C), oral (36.0–37.4°C), axillary (35.5–37.0°C), tympanic (35.8–38.0°C). Factors: diurnal variation, exercise, eating, age, and menstrual cycle. Clinically significant: fever (pyrexia) >38°C, hyperpyrexia >41.5°C, hypothermia <35°C.

Thoracentesis Kit

– A sterile tray containing instruments for pleural fluid aspiration. Components: 18-22 gauge thoracentesis needle/catheter, three-way stopcock, 50-60 mL syringe, extension tubing, specimen collection tubes (culture, chemistry, cytology, cell count), and sterile drape. Ultrasound guidance is standard of care (reduces pneumothorax). Insertion: at the superior aspect of the rib (to avoid intercostal neurovascular bundle), posteriorly (for effusion) or anteriorly (for pneumothorax). Maximum safe aspiration: 1-1.5 L (risk of re-expansion pulmonary edema).

Tongue Depressor

– A thin, flat wooden or plastic blade used to depress the tongue for examination of the oropharynx and oral cavity. Adult size: 15 cm x 2 cm. Indications: oropharyngeal examination for tonsillitis, pharyngitis, and oral ulcers. Also used as an emergency airway tool (simple jaw thrust in CPR) and for mixing dental materials. Disposable (single-use). Can be broken to create a sharp edge for emergency cricothyroidotomy.

Tourniquet

– A device that constricts blood flow to a limb. Types: surgical tourniquet (pneumatic cuff, controlled pressure, used in orthopedic surgery to create a bloodless field: upper arm 250 mmHg, thigh 300 mmHg, max time 1-2 hours), emergency tourniquet (CAT, SOFT-T: windlass mechanism for battlefield hemorrhage control), and venous tourniquet (elastic band for venipuncture). Complications of prolonged use: nerve injury, muscle ischemia, rhabdomyolysis, and compartment syndrome.

Tracheostomy Tube

– A curved tube inserted into the trachea through a surgical tracheostomy. Parts: outer cannula (with flange and ties), inner cannula (removable for cleaning), and obturator (for insertion). Cuffed tube: balloon inflated for mechanical ventilation and aspiration prevention. Uncuffed: for patients with spontaneous breathing and low aspiration risk. Fenestrated tube: allows speech when capped. Sizes: 4-10 (inner diameter in mm). Shiley (most common), Portex, and Bivona brands. Indications: prolonged ventilation, upper airway obstruction, and severe secretions management.

Tuning Fork

– A metal two-pronged instrument that vibrates at a specific frequency when struck, used for hearing and vibration sensation testing. Frequencies: 128 Hz and 256 Hz (vibration sensation testing), 256 Hz and 512 Hz (hearing tests). Rinne test: tuning fork on mastoid (bone conduction) then next to ear (air conduction); normal: air > bone (Rinne positive). Conductive loss: bone > air (Rinne negative). Weber test: tuning fork on vertex; lateralizes to affected ear in conductive loss, to unaffected ear in sensorineural loss.

Ultrasound Machine

– A device using high-frequency sound waves (2-18 MHz) to create real-time images of internal structures. Components: transducer (piezoelectric crystals generate and receive sound waves), processing unit (converts echo signals to images), and display. Modes: B-mode (brightness, 2D grayscale), M-mode (motion, for cardiac function), Doppler (color, spectral, power for blood flow), and 3D/4D. Transducers: curvilinear (abdominal, obstetric, 2-5 MHz), linear (vascular, breast, MSK, 7-15 MHz), phased array (cardiac, 2-4 MHz), and endocavitary (TVUS, TRUS). Indications: abdominal, obstetric, cardiac, vascular, MSK, and emergency (FAST, POCUS).

Urinary Catheter

– See Catheter.

Ventilator

– See Mechanical Ventilator.

Wheelchair

– A chair with wheels for mobility-impaired patients. Types: manual (self-propelled or attendant-propelled: standard, transport, lightweight, sports), power/electric (joystick or sip-and-puff control, for patients unable to self-propel). Components: frame (folding vs. rigid), seat and back cushion (pressure redistribution), armrests (full, half, or desk-length), footrests (swing-away, elevating leg rest), and wheels (24-inch rear with push rims, 8-inch front casters). Measurements: seat width (16-20 inches), seat depth, back height. Used for: spine cord injury, amputation, stroke, multiple sclerosis, and severe deconditioning.

Chapter 17

Medical Tests

6-Minute Walk Test

– A simple, inexpensive test measuring the distance a patient walks in 6 minutes on a flat, hard surface. Assesses functional exercise capacity and submaximal cardiopulmonary performance. Normal: 400-700 m (age, sex, height-dependent). Uses: evaluating response to therapy in pulmonary hypertension (PAH), COPD (pulmonary rehab), heart failure (cardiac rehab), and interstitial lung disease; assessing disability; determining prognosis (desaturation during test is a poor prognostic sign). Per patient on their own pace; encouragement standardized (Borg scale for dyspnea and leg fatigue).

Abdominal Aortic Aneurysm Screening

– Ultrasound screening for AAA recommended for men aged 65-75 with smoking history (USPSTF). A one-time abdominal ultrasound detects infrarenal aortic diameter ≥3 cm. Surveillance: 3.0-3.9 cm q3 years, 4.0-4.4 cm q12 months, 4.5-5.4 cm q6 months. Surgical referral at ≥5.5 cm or rapid enlargement (>1 cm/yr). Screening reduces AAA-related mortality by 50%.

Abdominal X-Ray

– Plain radiograph of the abdomen (KUB: kidneys, ureters, bladder). Indications: suspected bowel obstruction (air-fluid levels, dilated loops), perforation (free air under diaphragm), constipation, foreign body, and renal calculi (90% of stones are radiopaque). Sensitivity low for most intra-abdominal pathology. Uses: limited to specific clinical scenarios; CT is preferred for most acute abdominal conditions.

ABO Blood Typing

– Determination of a patient's ABO and Rh blood group prior to transfusion or organ transplantation. ABO system: A (A antigen, anti-B antibody), B (B antigen, anti-A antibody), AB (both antigens, no antibodies), O (no antigens, both antibodies). Rh: D antigen positive (85%) or negative (15%). Universal donor: O negative. Universal recipient: AB positive. Crossmatching confirms compatibility between donor and recipient.

Acid-Fast Bacillus (AFB) Smear

– Microscopic examination of sputum for acid-fast bacilli (Mycobacterium tuberculosis), using Ziehl-Neelsen or auramine-rhodamine stain. Positive smear indicates active pulmonary TB and correlates with infectiousness. Sensitivity: 50-80% in culture-positive TB. Three specimens collected on consecutive mornings. Grading: negative, 1+ (1-9 per 100 fields), 2+ (1-9 per 10 fields), 3+ (1-9 per field), 4+ (>9 per field). Culture and NAAT are more sensitive.

AFP (Alpha-Fetoprotein)

– A tumor marker and prenatal screening analyte. Tumor marker: hepatocellular carcinoma (elevated in 60-80%, >400 ng/mL highly specific), yolk sac tumors (germ cell tumors). Non-specific: cirrhosis, hepatitis, pregnancy. Prenatal: elevated in neural tube defects (maternal serum AFP). Screening for HCC: AFP + ultrasound q6 months in high-risk patients (cirrhosis, chronic HBV/HCV). AFP-L3 (isoform) improves specificity.

Allergy Testing

– Identification of IgE-mediated allergic sensitization. Skin prick test (SPT): allergen extract applied to skin via lancet, read after 15 min (wheal >3mm = positive). Intradermal test: more sensitive, used when SPT negative but suspicion remains. Specific IgE (formerly RAST): blood test measuring allergen-specific IgE levels; unaffected by antihistamines. Total IgE: elevated in atopic disease and parasitic infection. Indications: allergic rhinitis, asthma, food allergy, drug allergy, and anaphylaxis evaluation.

Alvarado Score

– Clinical scoring system for acute appendicitis. Components (MANTRELS): Migration of pain (1), Anorexia (1), Nausea/vomiting (1), Tenderness in RLQ (2), Rebound pain (1), Elevated temperature (1), Leukocytosis (2), Left shift (1). Scores: 1-4 (low probability, observe), 5-6 (moderate probability, CT or ultrasound), 7-10 (high probability, surgical consult). MASS (Modified Alvarado) removes left shift and uses 0-9.

Ammonia

– Measurement of serum ammonia (NH3). Normal: 15-45 mcg/dL (9-33 micromol/L). Elevated in hepatic encephalopathy (monitoring and diagnosis), urea cycle disorders (very high, >200 in neonates), Reye syndrome, and valproate therapy. Timing: must be collected in pre-chilled tube on ice and processed within 20 minutes (falsely elevated by delayed processing, smoking, exercise, and hemolysis). Used in initial evaluation of altered mental status with liver disease and suspected metabolic disorders.

Amniocentesis

– Sampling of amniotic fluid for prenatal diagnosis. Typically performed at 15-20 weeks gestation. Indications: advanced maternal age (≥35), abnormal quad screen, family history of genetic disorders, and fetal lung maturity assessment. Analyzed for karyotype, FISH (aneuploidy of 13, 18, 21, X, Y), microarray, and AFP (neural tube defects). Procedure: ultrasound-guided needle aspiration of 15-20 mL of fluid. Risks: miscarriage (0.1-0.3%), infection, and fetal injury.

Amylase and Lipase

– Pancreatic enzymes, elevated in acute pancreatitis. Lipase is more specific (pancreas-only) and remains elevated longer than amylase. Normal: amylase 30-110 U/L, lipase 10-140 U/L. Diagnosis: elevation >3x upper limit of normal, with compatible clinical presentation (epigastric pain, nausea/vomiting). Amylase also elevated in salivary gland disease, intestinal obstruction, perforated ulcer, and ectopic pregnancy. Lipase is the preferred single test for pancreatitis.

Analysis of Urine

– See Urinalysis.

Angiography

– Invasive imaging of blood vessels by injecting contrast material under fluoroscopy. Coronary angiography (cardiac catheterization): assesses coronary artery stenosis, guides PCI. Cerebral angiography: evaluates aneurysms, AVMs, and vasculitis. Peripheral angiography: evaluates PAD and guides intervention. Requires arterial access (femoral, radial). Complications: bleeding, pseudoaneurysm, contrast nephropathy, atheroembolism, and dissection. CTA and MRA have largely replaced diagnostic angiography.

Angioplasty

– Percutaneous coronary intervention (PCI) to open stenotic coronary arteries. Balloon angioplasty: inflation of balloon at the lesion to compress plaque. Bare metal stent (BMS): metal scaffold, restenosis 20-30%. Drug-eluting stent (DES): elutes antiproliferative (sirolimus, paclitaxel, everolimus), restenosis 5-10%. Dual antiplatelet therapy (DAPT: aspirin + P2Y12 inhibitor) required after stent placement. Indications: STEMI, NSTEMI, unstable angina, and stable angina refractory to medical therapy.

Ankle-Brachial Index

– Ratio of systolic BP at the ankle (dorsalis pedis or posterior tibial artery) to the brachial artery, measured with Doppler. Normal: 1.0-1.4. PAD: ABI ≤0.90 (mild 0.7-0.9, moderate 0.4-0.7, severe <0.4). Non-compressible vessels (ABI >1.4, shows calcified vessels in diabetes). Uses: screening for PAD, wound healing assessment, and cardiovascular risk stratification. Toe-brachial index (TBI) if ABI unreliable.

Antibiogram

– A periodic summary of antimicrobial susceptibility patterns of bacterial isolates from a hospital or laboratory. Guides empiric antibiotic therapy by showing local resistance patterns (e.g., % of E. coli susceptible to TMP-SMX). Updates annually. Clinicians use antibiograms to select appropriate empiric antibiotics while awaiting culture results. Cumulative antibiograms report susceptibilities for the most common pathogens.

Anti-CCP Antibody

– Anti-cyclic citrullinated peptide antibody, a highly specific serologic marker for rheumatoid arthritis (specificity >95%, sensitivity 70%). Positive anti-CCP is associated with more aggressive erosive RA.

Appears early in disease and may precede clinical symptoms. Used in the ACR/EULAR classification criteria with RF and acute phase reactants. Testing recommended in any patient with unexplained inflammatory polyarthritis.

Anti-dsDNA Antibody

– Anti-double-stranded DNA antibody, highly specific for systemic lupus erythematosus (specificity 95%, sensitivity 60-70%). Elevated titers correlate with active disease, particularly lupus nephritis and systemic vasculitis. Serial monitoring used to assess disease activity and treatment response. In the ACR classification criteria for SLE. Other autoantibodies in SLE: anti-Smith (specific but low sensitivity), anti-Ro/SSA, anti-La/SSB, anti-RNP.

Anti-Mullerian Hormone

– AMH is a glycoprotein produced by granulosa cells of ovarian follicles, reflecting the ovarian reserve (quantity of remaining oocytes). Uses: assessment of ovarian reserve in infertility evaluation, predicting response to ovarian stimulation in IVF, diagnosis of PCOS (elevated), and monitoring for granulosa cell tumors. AMH declines with age and is undetectable after menopause. Unlike FSH, AMH is cycle-independent and reliable any day of the menstrual cycle.

Antinuclear Antibody (ANA)

– A screening test for autoimmune connective tissue diseases. Positive in SLE (sensitivity 95%), Sjogren syndrome, systemic sclerosis, mixed connective tissue disease, and many other autoimmune conditions. Titer: 1:40 (low positive, common in healthy), 1:160 (clinically significant), 1:320 (strongly positive). Pattern: homogeneous (SLE, drug-induced), speckled (SLE, Sjogren, MCTD), nucleolar (scleroderma), centromere (limited scleroderma). Positive ANA requires further testing for specific autoantibodies.

Arterial Blood Gas (ABG)

– Direct measurement of arterial blood for pH, PaCO2, PaO2, HCO3, base excess, and oxygen saturation. Indications: respiratory failure assessment, acid-base disorder evaluation, ventilator management, and critical care. Normal values: pH 7.35-7.45, PaCO2 35-45 mmHg, PaO2 80-100 mmHg, HCO3 22-26 mEq/L, SaO2 95-100%. Sample: radial artery (Allen test for collateral flow). Complications: hematoma, arterial spasm, thrombosis, and infection.

Arthrocentesis

– Aspiration of joint fluid for diagnostic or therapeutic purposes. Indications: septic arthritis (urgent, crystals), inflammatory arthritis (gout, pseudogout), hemorrhagic effusion (trauma, coagulopathy), and therapeutic drainage for symptomatic relief. Synovial fluid analysis: cell count/differential, Gram stain/culture, crystal analysis (polarized light microscopy: monosodium urate = needle-shaped, negatively birefringent; CPPD = rhomboid, weakly positively birefringent), and glucose/protein.

Audiometry

– Objective measurement of hearing thresholds across frequencies. Pure tone audiometry: air conduction (headphones) and bone conduction (mastoid oscillator) thresholds at 250-8000 Hz. Conductive hearing loss: abnormal air, normal bone (air-bone gap). Sensorineural: abnormal both. Speech audiometry: speech reception threshold (SRT) and word recognition score (WRS). Tympanometry: middle ear function (type A normal, B effusion, C negative pressure). Used for hearing loss diagnosis and hearing aid candidacy.

Barium

– A metallic element (atomic number 56, symbol Ba) whose sulfate salt (barium sulfate) is used as a radiocontrast agent due to its high atomic number, which effectively absorbs X-rays. Barium sulfate is insoluble and passes through the gastrointestinal tract without absorption, allowing radiographic visualization of the esophagus (barium swallow), stomach and duodenum (upper GI series), small bowel (barium follow-through), and colon (barium enema). Adverse effects include constipation and, rarely, barium impaction. Barium should not be used if perforation is suspected.

Barium Enema

– Fluoroscopic imaging of the large intestine after retrograde filling with barium contrast. Double-contrast (barium + air) provides better mucosal detail. Indications: colorectal cancer screening (historical, now CT colonography preferred), evaluation of colonic obstruction, diverticulosis, and postsurgical anatomy. Findings: filling defects (polyp, cancer), strictures, diverticula, and fistulas. Contraindications: suspected perforation (use water-soluble contrast). Largely replaced by colonoscopy.

Barium Swallow

– Fluoroscopic examination of the esophagus (pharynx to gastroesophageal junction) with barium contrast. Modified barium swallow: speech therapist evaluates oral and pharyngeal phases of swallowing. Indications: dysphagia, esophageal dysmotility (achalasia, diffuse esophageal spasm), hiatal hernia, GERD, stricture, and esophageal cancer. Findings: aspiration, cricopharyngeal bar, webs, rings (Schatzki), strictures, and diverticula (Zenker, epiphrenic).

Basic Metabolic Panel

– A panel of blood tests measuring serum glucose, sodium, potassium, chloride, bicarbonate, BUN, and creatinine. Also includes calcium in some institutions. Uses: assess hydration status, renal function, acid-base balance, and glucose control. Abnormalities suggest electrolyte disturbances, renal impairment, diabetes, or acid-base disorders. BMP is used for monitoring hospitalized patients and annual health screening.

BCG Vaccine

– Bacille Calmette-Guerin, a live attenuated vaccine derived from Mycobacterium bovis, used for TB prevention. Efficacy: 60-80% against disseminated TB and TB meningitis in children; variable efficacy against pulmonary TB. Indications: part of routine childhood immunization in high-TB-burden countries. Not routinely used in low-risk countries (interferes with TST/IGRA screening). Also used as intravesical immunotherapy for non-muscle invasive bladder cancer.

Biophysical Profile (BPP)

– Antepartum fetal assessment combining ultrasound and NST. Components (2 points each): NST reactivity (reactive), fetal breathing movements ($\geq$1 episode of 30 seconds in 30 minutes), fetal movement ($\geq$3 discrete movements in 30 minutes), fetal tone ($\geq$1 episode of extension/flexion), and amniotic fluid volume (AFI $\geq$5 cm). Score: 8-10 (normal), 6 (equivocal, deliver if term, repeat if preterm), 4 (suspect asphyxia, deliver), 0-2 (probable asphyxia, deliver). Modified BPP: NST + AFI.

Biopsy

– Sampling of tissue for histopathologic examination. Types: needle biopsy (core needle: breast, liver, kidney, prostate; fine needle: thyroid, lymph node), excisional biopsy (entire lesion removal), incisional biopsy (partial sampling), endoscopic biopsy (GI, bronchoscopic), punch biopsy (skin), shave biopsy (skin), and stereotactic biopsy (breast guided by mammography). Essential for distinguishing benign from malignant, staging, and identifying inflammatory or infectious pathology.

Blood Alcohol Level

– Measurement of ethanol concentration in blood. Legal limit for driving: 0.08 g/dL (80 mg/dL) in most US states. Impaired: 0.04-0.08 g/dL. Severe intoxication: 0.15-0.30 g/dL. Potentially fatal: >0.30 g/dL. With chronic alcoholism: marked tolerance (may appear functional at 0.30+). Elimination rate: 15-25 mg/dL/hour. Uses: legal (DUI assessment), medical (alcohol intoxication, withdrawal management), and screening. Serum: whole blood ratio approximately 1.18:1.

Blood Culture

– Inoculation of blood into culture media to detect bacteremia or fungemia. Two sets (two bottles each: aerobic + anaerobic) from separate venipuncture sites, 20-30 mL total. Indications: fever of unknown origin, suspected sepsis, infective endocarditis, and immunocompromised with fever. Positive cultures with identification and susceptibility (MIC) guide targeted antibiotic therapy. Negative cultures do not rule out infection (especially with prior antibiotics, fastidious organisms, or intracellular pathogens).

Blood Gas (Venous)

– Sampling of venous blood (typically from a peripheral vein or central line) for pH, PvCO2, HCO3, and base excess. Less painful than ABG, avoids arterial puncture risks. Limitations: no PaO2 or SpO2 data; peripheral venous gases in shock may not reflect arterial pH. VBG pH is typically 0.03-0.05 lower than ABG; PvCO2 is 3-8 mmHg higher. Used for acid-base assessment and CO2 monitoring when oxygenation is not the primary concern.

Bone Marrow Aspiration and Biopsy

– Sampling of bone marrow, typically from the posterior iliac crest. Aspiration: liquid marrow for cytology (differential, morphology), flow cytometry, cytogenetics, and MRD. Biopsy: core of bone for cellularity, architecture, fibrosis, and infiltrative disease (lymphoma, metastatic cancer, granulomas). Indications: unexplained cytopenias, leukemia, lymphoma, multiple myeloma, myelodysplasia, myelofibrosis, and staging of certain malignancies. Contraindication: severe coagulopathy.

Bone Scan (Nuclear Scintigraphy)

– Nuclear medicine imaging using Tc-99m MDP (methylene diphosphonate) that accumulates in areas of increased bone turnover. Three phases: perfusion (immediate), blood pool (5 min), and delayed (2-3 hours). Indications: occult fracture, metastatic disease (breast, prostate, lung), osteomyelitis, complex regional pain syndrome (CRPS), avascular necrosis, and Paget disease. Findings: increased uptake in osteoblastic activity. Limitations: poor specificity (cannot distinguish tumor from infection or fracture). Used with SPECT for better localization.

Brain Natriuretic Peptide (BNP)

– A cardiac neurohormone released from the ventricles in response to volume overload and wall stress. BNP <100 pg/mL makes heart failure unlikely; >400 pg/mL suggests heart failure. NT-proBNP: <125 pg/mL (<75 years) or <450 pg/mL (>75 years) excluding HF. Also prognostic: rising levels indicate worsening HF and higher mortality. Elevated in HF (systolic and diastolic), pulmonary embolism, CKD, and sepsis. Used for diagnosis, severity assessment, and monitoring therapy.

Breast MRI

– Magnetic resonance imaging of the breast with IV gadolinium contrast. Indications: screening high-risk women (BRCA mutation, lifetime risk >20%, prior chest radiation), evaluating extent of disease after breast cancer diagnosis, assessing response to neoadjuvant chemotherapy, implant evaluation, and problem-solving for equivocal mammogram/ultrasound. High sensitivity (>90%) but lower specificity, leading to false positives. BI-RADS lexicon: kinetics (wash-in, washout) and morphology.

Breast Ultrasound

– Ultrasound imaging of the breast. Indications: evaluating palpable masses (distinguishing solid vs. cystic), characterizing mammographic abnormalities, guiding biopsy (core needle, fine needle, or vacuum-assisted), evaluating implant integrity, and imaging young/dense breast tissue. BI-RADS US classification: simple cyst (benign), complicated cyst, complex cystic and solid, and solid mass (assessed by shape, margins, orientation, echogenicity, and posterior features). Used with mammography for comprehensive breast evaluation.

Bronchoscopy

– Endoscopic examination of the tracheobronchial tree. Flexible bronchoscopy: most common, for diagnosis (lung mass, persistent cough, hemoptysis, suspected infection, ILD) and therapy (foreign body removal, bronchial washing, biopsy, BAL). Rigid bronchoscopy: for massive hemoptysis, large foreign bodies, and stent placement. Samples: BAL (bronchoalveolar lavage: cell count, culture, cytology), transbronchial biopsy (lung parenchyma), endobronchial biopsy (visible lesions), and EBUS-TBNA (mediastinal lymph node sampling).

CA-125

– Cancer antigen 125, a tumor marker most commonly used for ovarian cancer. Elevated in 80% of epithelial ovarian cancers (less in mucinous). Non-specific: elevated in endometriosis, PID, pregnancy, fibroids, pancreatitis, cirrhosis, and other malignancies (pancreatic, breast, lung, endometrial). Normal: <35 U/mL. Uses: monitoring treatment response in known ovarian cancer, detecting recurrence, and differential diagnosis of pelvic mass (with Risk of Malignancy Index). Not recommended for screening (low specificity, high false positives).

CA 19-9

– A tumor marker for pancreatic cancer. Normal: <37 U/mL. Sensitivity: 70-80% for pancreatic cancer (lower for early-stage). Specificity: limited (elevated in cholangitis, pancreatitis, cholecystitis, biliary obstruction, cirrhosis, and other GI malignancies). Uses: monitoring treatment response in known pancreatic cancer, detecting recurrence, and prognosis. Not recommended for screening (low sensitivity/specificity). Non-secretor phenotype (Lewis a-b-, 5-10% of population) cannot produce CA 19-9, limiting its use.

Capsule Endoscopy

– A wireless, ingestible camera capsule (11 x 26 mm) that takes 50,000+ images as it traverses the GI tract. Used to visualize the small intestine (beyond reach of standard endoscopy). Indications: obscure GI bleeding, Crohn disease (small bowel involvement), suspected small bowel tumors, and surveillance in familial polyposis. Images transmitted to a data recorder worn on a belt, downloaded for review (3-4 hour study). Limitations: cannot biopsy or treat, capsule retention (1-2% risk, higher in Crohn strictures), and poor visualization of the stomach and colon. PillCam (Given Imaging) and EndoCapsule (Olympus) are common systems.

Cardiac Catheterization

– Invasive procedure to evaluate cardiac anatomy, hemodynamics, and coronary arteries. Right heart catheterization: Swan-Ganz catheter measures RA, RV, PA pressures, PCWP, cardiac output (thermodilution), and SVR/PVR. Left heart catheterization: coronary angiography and ventriculography. Indications: CAD evaluation (stable angina with high-risk features, positive stress test, acute coronary syndromes), valvular disease (assess severity), cardiomyopathy, and pre-transplant evaluation. Complications: hematoma, bleeding, CVA, MI, arrhythmia.

Cardiac Stress Test

– Evaluation of cardiac function and ischemia under increased cardiac workload. Exercise stress test (treadmill, Bruce protocol): ECG monitoring for ST-segment changes, symptoms, BP response, and arrhythmias. Sensitivity 65-75% for CAD. Pharmacologic stress test: dobutamine (inotrope/chronotrope), regadenoson/adenosine/dipyridamole (vasodilator) — used for patients unable to exercise. Imaging: stress echo (wall motion), nuclear perfusion (SPECT, PET), and stress MRI.

Cardiopulmonary Exercise Testing (CPET)

– Comprehensive exercise test with breath-by-breath analysis of respiratory gases (VO2, VCO2, VE). Indications: unexplained dyspnea, pre-operative risk assessment (lung resection, cardiac surgery), evaluation of exercise tolerance in heart failure (peak VO2 predicts prognosis and guides transplant listing), and differentiating cardiac vs. pulmonary limitation. Key parameters: peak VO2 (normal >80% predicted), anaerobic threshold (AT), VEVCO2 slope (<30 normal, >34 poor prognosis in HF), and oxygen pulse (VO2/HR). Maximum exercise test with 12-lead ECG and BP monitoring.

Cardioversion (Synchronized)

– Delivery of a low-energy electrical shock synchronized to the QRS complex to terminate arrhythmias. Synchronization: shock delivered on the R wave (not T wave, which induces VF). Energy: atrial fibrillation (100-200 J biphasic), atrial flutter (50-100 J), SVT (50-100 J), stable VT (100 J). Success: 70-90% for AF, >95% for atrial flutter. Pre-treatment: anticoagulation for >3 weeks or TEE-guided to exclude LA thrombus (for AF >48 hours). Complications: thromboembolism, skin burns, arrhythmias, and sedation risks.

Caries Risk Assessment

– Systematic evaluation of an individual's likelihood of developing dental caries. Low risk: no active caries, good oral hygiene, regular fluoride exposure. Moderate risk: one or more active lesions in past 3 years, fair oral hygiene. High risk: multiple active lesions, poor oral hygiene, frequent sugar intake, low fluoride, xerostomia, smoking, or special healthcare needs. Based on clinical exam (caries, restorations), radiographs (incipient lesions), plaque assessment, and dietary/medical history. Guides recall interval (3-12 months) and preventive interventions.

Cariogram

– A computer-based program that illustrates the interaction of caries-related factors and provides an overall caries risk assessment. Factors: diet (content, frequency), bacteria (mutans streptococci, lactobacilli), fluoride exposure, saliva (flow rate, buffering capacity), plaque amount, and clinical/social history. Output: pie chart showing the relative weight of different etiological factors and the probability of avoiding new caries. Used for patient education and preventive planning.

Carotid Duplex Ultrasound

– Ultrasound evaluation of the carotid arteries for atherosclerotic plaque and stenosis. Measures peak systolic velocity (PSV) and end-diastolic velocity (EDV) in the internal carotid artery (ICA). Criteria: normal (<50% stenosis: PSV <125 cm/s), 50-69% (PSV 125-230, EDV 40-100), 70-99% (PSV >230, EDV >100), and occlusion (no detectable flow). Indications: TIA/stroke evaluation, asymptomatic bruit, pre-CABG screening, and surveillance of known stenosis. Plaque morphology: echolucent (high risk) vs. calcified.

Carpal Tunnel Syndrome Tests

– Provocative tests for diagnosing carpal tunnel syndrome. Tinel sign: percuss over median nerve at the carpal tunnel → tingling in median distribution. Phalen sign: hold wrists in full flexion for 60 seconds → reproduces symptoms. Durkan test (carpal compression): direct compression over carpal tunnel for 30 seconds → paresthesias. Thenar atrophy suggests advanced disease. Confirmatory: nerve conduction studies show prolonged distal latencies and decreased conduction velocity across the carpal tunnel.

Catecholamines (Urine/Plasma)

– Measurement of catecholamines (epinephrine, norepinephrine, dopamine) and their metabolites (metanephrines, VMA) for diagnosis of pheochromocytoma and paraganglioma. Plasma free metanephrines: sensitivity 96-100%. 24-hour urinary fractionated metanephrines: sensitivity 90-95%. Elevated levels require tumor localization (CT/MRI, MIBG scan). False positives: stress, exercise, smoking, tricyclic antidepressants, MAOIs, and levodopa.

CD4 Count

– Absolute count of CD4+ T-helper lymphocytes, essential for monitoring HIV disease progression and treatment response. Normal: 500-1500 cells/uL. HIV staging: >500 (asymptomatic, start ART regardless), 200-500 (consider prophylaxis for some OIs), <200 (AIDS-defining: start PCP prophylaxis, TB screening, high risk of OIs). <100: risk of CMV, MAC, toxoplasmosis, cryptococcosis. Used with HIV viral load (HIV RNA) to guide ART initiation and monitor response (goal: undetectable viral load, CD4 recovery).

CEA (Carcinoembryonic Antigen)

– A tumor marker primarily for colorectal cancer. Normal: <3 ng/mL (non-smokers), <5 ng/mL (smokers). Uses: monitoring treatment response in metastatic colorectal cancer, surveillance for recurrence after curative resection (q3-6 months for 2 years, then q6-12 months for 5 years), and prognosis (preoperative level). Non-specific elevations: smoking, pancreatitis, IBD, cirrhosis, hepatitis, and other malignancies (lung, gastric, pancreatic, breast). Not recommended for screening.

Central Line Placement

– Insertion of a catheter into a central vein (internal jugular, subclavian, or femoral), with the tip in the superior (or inferior) vena cava. Ultrasound guidance is standard. Indications: administration of vasopressors, TPN, chemotherapy, long-term antibiotics, central venous pressure monitoring, hemodialysis, and plasmapheresis. Complications: arterial puncture, pneumothorax (subclavian > IJ), hemothorax, air embolism, catheter-related bloodstream infection, thrombosis, and malposition. Confirm position with chest X-ray (tip at SVC-RA junction, 2-3 cm above RA). Suture and sterile dressing.

Cerebrospinal Fluid Analysis

– Laboratory examination of CSF obtained via lumbar puncture. Indications: meningitis, encephalitis, subarachnoid hemorrhage, demyelinating disease (MS: oligoclonal bands, elevated IgG index), and carcinomatous meningitis. Components: opening pressure (normal 10-20 cmH2O), appearance (clear vs. cloudy vs. bloody), cell count/differential (normal <5 WBC, all monocytes; neutrophils in bacterial), glucose (CSF:serum ratio >0.6), protein (normal 15-45 mg/dL), Gram stain, culture, and PCR.

Cervical Screening (Pap Smear)

– Screening for cervical cancer by collecting cells from the transformation zone of the cervix. Liquid-based cytology (ThinPrep) or conventional Pap smear. HPV co-testing (high-risk HPV 16, 18, others) improves sensitivity. Screening guidelines: start at age 21; ages 21-29: Pap q3 years; ages 30-65: Pap + HPV q5 years or Pap q3 years; over 65: discontinue if adequate prior negative screening. Results: ASCUS, LSIL, HSIL, AGC, and carcinoma. Management per ASCCP risk-based guidelines.

Chest Tube Thoracostomy

– Insertion of a tube into the pleural space to drain air, fluid, or blood. Indications: pneumothorax (large, tension, or on mechanical ventilation), hemothorax, pleural effusion (empyema, malignant, transudative), and postoperative (cardiac, thoracic surgery). Tube sizes: small (8-14F for air), medium (16-24F for effusion), large (28-40F for hemothorax/empyema). Placement: triangle of safety (4th-5th intercostal space, midaxillary line). Complications: bleeding, infection, lung laceration, and re-expansion pulmonary edema.

Chorionic Villus Sampling

– Prenatal diagnostic test at 10-13 weeks gestation, earlier than amniocentesis. Transabdominal or transcervical biopsy of chorionic villi (placental tissue). Indications: advanced maternal age, abnormal first-trimester screen, carrier screening, or family history of genetic disorders. Analyzed for karyotype, FISH, and microarray. Advantages: earlier diagnosis than amniocentesis (allows earlier termination if desired). Risks: miscarriage (0.5-1%, slightly higher than amniocentesis) and limb defects if performed <10 weeks. Does not detect neural tube defects (requires AFP at 16-18 weeks).

Coagulation Profile

– Panel of tests assessing the coagulation cascade. PT (prothrombin time, 11-14 sec): measures extrinsic and common pathways; elevated in vitamin K deficiency, warfarin, liver disease, DIC. INR standardizes PT for warfarin monitoring. PTT (partial thromboplastin time, 25-35 sec): measures intrinsic and common pathways; elevated in hemophilia A/B, heparin, lupus anticoagulant, factor deficiencies. Fibrinogen (200-400 mg/dL): acute phase reactant; low in DIC, liver disease. Mixing studies: corrects in factor deficiency, does not correct in inhibitor.

Colonoscopy

– Endoscopic examination of the entire colon from cecum to rectum. Indications: colorectal cancer screening (starting age 45 for average risk), surveillance (polyps: q3-5y; cancer: q1y; IBD: q1-2y for dysplasia), diagnostic (hematochezia, chronic diarrhea, iron deficiency anemia, change in bowel habits), and therapeutic (polypectomy, hemostasis, stricture dilation, stent placement). Requires full bowel preparation (polyethylene glycol, split-dose). Complications: perforation (0.03-0.3%), bleeding (0.1-1%), and sedation-related.

Colorectal Cancer Screening

– See Colonoscopy, FIT, FIT-DNA, CT Colonography.

Colorectal Cancer Screening Tests

– Multiple screening modalities. Stool-based: fecal immunochemical test (FIT, annual), high-sensitivity guaiac FOBT (annual), and multitarget stool DNA (FIT-DNA/Cologuard, q3y). Direct visualization: colonoscopy (q10y preferred), CT colonography (q5y), flexible sigmoidoscopy (q5-10y), and capsule colonoscopy (q5y). Guidelines: average risk start at age 45; stop at age 75 if adequate prior screening and negative; age 76-85: individual decision based on life expectancy and comorbidities. FIT is the most cost-effective option.

Colposcopy

– Magnified visualization of the cervix, vagina, and vulva after application of acetic acid. Indication: abnormal Pap smear (ASCUS with high-risk HPV, LSIL, HSIL, AGC). Acetic acid causes acetowhite lesions in dysplastic epithelium. Lugol iodine (Schiller test): normal glycogenated epithelium stains brown; abnormal epithelium does not. Targeted biopsy (punch biopsies and endocervical curettage) of abnormal areas to confirm CIN (cervical intraepithelial neoplasia). Grading: CIN 1 (mild), CIN 2 (moderate), CIN 3 (severe/carcinoma in situ).

Complete Blood Count (CBC)

– A panel of blood tests evaluating cellular components: WBC (leukocytes: normal 4.5-11.0 K/uL), RBC (erythrocytes: male 4.5-5.9, female 4.1-5.1 M/uL), hemoglobin (male 13.5-17.5, female 12.0-16.0 g/dL), hematocrit (male 41-53%, female 36-46%), platelets (150-450 K/uL), RBC indices (MCV, MCH, MCHC), and differential (neutrophils, lymphocytes, monocytes, eosinophils, basophils). Uses: anemia, infection, bleeding, inflammation, and hematologic malignancy screening.

Comprehensive Metabolic Panel

– Extended version of BMP including albumin, total protein, alkaline phosphatase, ALT, AST, and bilirubin (total and direct). Evaluates liver function, kidney function, electrolytes, glucose, calcium, and protein status. Uses: preoperative assessment, medication monitoring, and liver/kidney disease evaluation. Flags acute and chronic conditions: transaminitis (hepatitis, NAFLD), elevated ALP (obstruction, bone disease), low albumin (cirrhosis, malnutrition).

Coronary Calcium Score

– Non-contrast CT measurement of coronary artery calcium (Agatston score). Score 0: no plaque, very low event risk (near 0% over 5 years). 1-100: mild plaque, moderate risk. 101-400: moderate plaque, increased risk. >400: extensive plaque, high risk. Used in ASCVD risk assessment for intermediate-risk patients when risk-based decision is uncertain. Independent predictor of cardiovascular events beyond traditional risk factors. Zero calcium score is an excellent negative predictor.

COVID-19 Test (SARS-CoV-2)

– Detection of SARS-CoV-2. NAAT (RT-PCR): gold standard, nasopharyngeal/nasal swab, highly sensitive/specific, results in 1-48 hours. Rapid antigen test: lateral flow, results in 15 minutes, sensitive in high viral load (early symptomatic), less sensitive in asymptomatic. Serology: anti-S and anti-N antibodies, indicates prior infection or vaccination, not used for acute diagnosis. Used for diagnosis, infection control, and public health surveillance. Antiviral therapy (nirmatrelvir/ritonavir, remdesivir) for high-risk patients.

C-Reactive Protein (CRP)

– An acute-phase reactant produced by the liver in response to inflammation (IL-6). CRP rises within 4-6 hours of inflammatory stimulus and peaks at 36-50 hours. Uses: monitoring inflammatory disease activity (RA, vasculitis), detecting infection postoperatively, and cardiovascular risk stratification (hs-CRP: high-sensitivity). CRP >10 mg/L suggests significant inflammation or infection. More sensitive than ESR for acute changes.

Creatine Kinase (CK)

– An enzyme found in muscle, brain, and heart. CK-MB: cardiac isoenzyme (used historically for MI diagnosis, largely replaced by troponin). CK-MM: skeletal muscle (elevated in rhabdomyolysis, trauma, statins, myopathy). Total CK: elevated in muscle injury, myocardial infarction, seizures, hypothyroidism, and statin myopathy. Strikingly elevated in rhabdomyolysis (>5000-100,000 U/L). Used to monitor rhabdomyolysis (risk of AKI) and inflammatory myopathies.

Cryoglobulins

– Immunoglobulins that precipitate at cold temperatures (<37°C) and redissolve on warming. Types: Type I (monoclonal IgM, in MGUS, Waldenstrom, myeloma), Type II (mixed: monoclonal + polyclonal, in HCV infection), Type III (mixed polyclonal, in autoimmune disease). Presents with palpable purpura, arthralgias, neuropathy, and glomerulonephritis (membranoproliferative). Diagnosis: serum collected and transported at 37°C, cold-precipitated, measured as cryocrit. Treatment: address underlying cause (HCV: DAAs), immunosuppression.

CT Angiography (CTA)

– CT imaging with IV contrast timed to visualize arteries. Coronary CTA: assesses coronary artery stenosis and plaque, calcium scoring (Agatston). CTA chest: pulmonary embolism (PE protocol), aortic dissection. CTA head/neck: stroke evaluation, aneurysm, dissection, and vasculitis. CTA abdomen: mesenteric ischemia, renal artery stenosis, and AAA. 64-slice or better scanners required. Advantages: non-invasive, three-dimensional reconstruction, and rapid acquisition. Disadvantage: radiation and contrast exposure.

CT Colonography (Virtual Colonoscopy)

– CT imaging of the colon after bowel preparation and air insufflation. Indications: colorectal cancer screening (alternative to colonoscopy), incomplete colonoscopy due to obstructing mass, and patients unable to undergo colonoscopy. Sensitivity >90% for polyps >10mm. Requires same bowel prep as colonoscopy. Limitations: cannot biopsy or remove polyps; positive findings require optical colonoscopy. No sedation required.

CT Scan (Computed Tomography)

– Cross-sectional imaging using X-rays and computer reconstruction. Non-contrast: evaluate hemorrhage, stones, and bone. IV contrast (iodinated): evaluate inflammation, infection, tumor, and vascular structures. Oral contrast: opacify bowel. Window settings: bone (fractures, calcifications), soft tissue (organs, masses), lung (pulmonary parenchyma), and brain (gray-white differentiation). Radiation dose varies (2-8 mSv for head, 5-15 mSv for body). Hounsfield units quantify tissue density.

Culture and Sensitivity

– Microbiological culture of a clinical specimen (blood, urine, sputum, wound swab, CSF) to identify pathogens and determine antimicrobial susceptibility. Culture media: blood agar (most bacteria), MacConkey (Gram-negative), chocolate agar (fastidious: N. gonorrhoeae, H. influenzae), and selective media. Automated systems (VITEK, MALDI-TOF) speed identification. MIC (minimum inhibitory concentration) reported as susceptible, intermediate, or resistant per CLSI breakpoints. Guides targeted antimicrobial therapy.

Cystoscopy

– Endoscopic examination of the urethra, bladder, and ureteral orifices. Flexible cystoscopy: office-based, local anesthesia. Rigid cystoscopy: OR, allows larger instruments for interventions. Indications: hematuria (gross or microscopic), recurrent UTIs, bladder tumor surveillance, lower urinary tract symptoms (obstruction), ureteral stent placement/removal, and foreign body removal. Findings: tumors, stones, strictures, trabeculation, diverticula, and inflammation. Complications: infection, hematuria, and perforation.

Cytogenetics (Karyotype)

– Visualization of chromosomes under light microscopy from dividing cells (bone marrow, blood, amniotic fluid, tissue). Staining: G-banding (Giemsa, light and dark bands). Resolution: 400-550 bands per haploid genome. Detects: aneuploidy (trisomy 21, 18, 13, Turner, Klinefelter), structural abnormalities (translocations, deletions, inversions, duplications), and marker chromosomes. Essential in: prenatal diagnosis, hematologic malignancy (t(9;22) BCR-ABL in CML, t(15;17) PML-RARA in APL, t(8;14) MYC in Burkitt), and constitutional genetic disorders.

D-Dimer

– A fibrin degradation product, elevated in thrombosis (PE, DVT, DIC). High sensitivity but low specificity. D-dimer <500 ng/mL (ELISA) has high negative predictive value for VTE; used to rule out PE/DVT in patients with low or moderate pretest probability (Wells criteria). Age-adjusted D-dimer (>50 years: age x 0.1 mg/L) improves specificity in older patients. Also elevated in infection, inflammation, surgery, trauma, pregnancy, and malignancy.

Deep Vein Thrombosis Ultrasound

– Venous duplex ultrasound for DVT diagnosis. Complete compression ultrasound (CUS): venous segments are compressed; non-compressible = DVT (96% sensitivity for proximal DVT). Whole-leg ultrasound (proximal + distal): for symptomatic patients. Findings: intraluminal thrombus, absent flow, lack of compressibility, and abnormal Doppler signals. Distal DVT (calf) is less clinically significant; management varies.

Dental Radiographs

– Imaging for dental diagnosis. Intraoral: periapical (tooth root and surrounding bone), bitewing (interproximal caries detection, crestal bone level), and occlusal (larger area, localization). Panoramic (extraoral): wide view of jaws, teeth, sinuses, and TMJ; used for impacted teeth, fractures, pathology (cysts, tumors), and implant planning. CBCT (cone-beam CT): 3D imaging for implant planning and complex surgical cases. Frequency based on caries risk.

Direct Coombs Test (DAT)

– Direct antiglobulin test: detects antibodies or complement (C3) bound to the surface of RBCs in vivo. Positive in: autoimmune hemolytic anemia (warm: IgG, cold: C3), drug-induced hemolytic anemia, hemolytic transfusion reaction, and hemolytic disease of the newborn (HDN). Grading: 1+ to 4+. Used with indirect Coombs test (antibody screen for transfusion).

Doppler Ultrasound

– Ultrasound technique using the Doppler effect to evaluate blood flow. Color Doppler: direction and velocity of flow (red = toward transducer, blue = away). Spectral Doppler: waveform analysis (arterial: pulsatile; venous: continuous). Pulsed-wave: measures flow at a specific depth. Continuous-wave: measures high velocities (valvular stenosis). Uses: assessment of DVT, carotid stenosis, PAD, renal artery stenosis, valvular heart disease, and pregnancy (fetal umbilical artery, middle cerebral artery).

DXA Scan (Dual-Energy X-ray Absorptiometry)

– The gold standard for measuring bone mineral density (BMD) and diagnosing osteoporosis. Measures at the lumbar spine, total hip, and femoral neck. T-score: normal (≥-1.0), osteopenia (-1.0 to -2.5), osteoporosis (≤-2.5). Z-score (compared to age-matched) used in premenopausal women and men <50. Indications: women ≥65, men ≥70, younger adults with risk factors (prior fragility fracture, steroid use, hypogonadism, malabsorption). FRAX calculates 10-year fracture risk. Used for monitoring treatment response (repeat q2y).

Echocardiography

– Ultrasound imaging of the heart. Transthoracic (TTE): standard, assesses LV/RV function, wall motion, valves (stenosis, regurgitation), pericardium (effusion), and chamber sizes. Transesophageal (TEE): higher resolution of valves (mitral, aortic), left atrial appendage, and for endocarditis evaluation. Stress echo: images before and after stress (exercise or dobutamine) for inducible wall motion abnormalities (ischemia). Doppler: measures blood flow velocities across valves (pressure gradients) and estimates pulmonary artery pressure.

Electrocardiogram (ECG / EKG)

– A recording of the heart's electrical activity from 12 leads (I, II, III, aVR, aVL, aVF, V1-V6). P wave: atrial depolarization. PR interval (120-200 ms): conduction through AV node. QRS (80-110 ms): ventricular depolarization. ST segment: early repolarization. T wave: ventricular repolarization. QT interval: corrected for heart rate (QTc, normal <450 ms in men, <460 ms in women). Uses: chest pain (ischemia/infarction), arrhythmia diagnosis, syncope, palpitations, and pre-surgical evaluation.

Electroencephalogram (EEG)

– Recording of the brain's electrical activity from scalp electrodes. Background rhythms: alpha (8-13 Hz, relaxed, eyes closed, occipital), beta (13-30 Hz, alert/active), theta (4-8 Hz, drowsy/light sleep, pathological in awake adults), delta (<4 Hz, deep sleep, pathological in awake). Uses: seizure diagnosis and classification, epilepsy monitoring, altered mental status, brain death determination, and sleep disorders. Activation: hyperventilation, photic stimulation, and sleep deprivation.

Electromyography (EMG)

– Recording of electrical activity from muscles at rest and during contraction. Needle EMG: insertion activity, spontaneous activity (fibrillations, fasciculations), motor unit action potentials, and recruitment pattern. Indications: differentiate myopathic from neuropathic disorders, identify nerve root or peripheral nerve pathology (radiculopathy, neuropathy), and evaluate neuromuscular junction disorders. Abnormal spontaneous activity (fibrillation potentials, positive sharp waves): denervation.

Electrophysiology Study (EPS)

– Invasive cardiac catheterization with multielectrode catheters placed in the heart to map the cardiac conduction system. Measures: sinus node function (SNRT, SACT), AV node function (AH interval, HV interval), and inducibility of arrhythmias (programmed electrical stimulation). Indications: evaluation of syncope, sustained palpitations, wide-complex tachycardia, supraventricular tachycardia (AVNRT, AVRT, atrial flutter), ventricular tachycardia, and risk stratification in Brugada syndrome or hypertrophic cardiomyopathy. Often followed by catheter ablation.

Endoscopic Retrograde Cholangiopancreatography

– Endoscopic cannulation of the ampulla of Vater for retrograde injection of contrast into the biliary and pancreatic ducts. Combined diagnostic and therapeutic: ERCP with sphincterotomy, stone extraction, stent placement, and brush cytology. Indications: choledocholithiasis (common bile duct stones), malignant biliary obstruction (palliative stenting), biliary stricture, and sphincter of Oddi dysfunction. Risks: pancreatitis (5-10%), cholangitis, perforation, and bleeding post-sphincterotomy.

Endoscopic Ultrasound (EUS)

– Combined endoscopy and ultrasound for high-resolution imaging of the GI wall and adjacent structures. Linear echoendoscope: allows FNA of mediastinal, pancreatic, and peripancreatic lesions. Radial echoendoscope: 360-degree view for staging GI cancers (TNM staging). Indications: pancreatic cancer staging, mediastinal lymph node sampling (EBUS-TBNA or EUS-FNA), subepithelial GI lesions, choledocholithiasis, and celiac plexus neurolysis for pain. Complications: perforation (0.03%), pancreatitis (from pancreatic FNA), and infection.

Erythrocyte Sedimentation Rate (ESR)

– A non-specific inflammatory marker measuring the rate at which erythrocytes settle in 1 hour. Normals: males age/2, females (age+10)/2. Markedly elevated >100 mm/hr: suggests infection (endocarditis), autoimmune (giant cell arteritis, polymyalgia rheumatica, SLE, RA), malignancy (multiple myeloma), and chronic kidney disease. Faster in anemia (rouleaux formation). Slower in polycythemia and spherocytosis. Uses: monitoring disease activity in inflammatory conditions. Less sensitive than CRP for acute changes.

Esophageal Manometry

– Measurement of esophageal pressures at rest and during swallows. Components: upper esophageal sphincter (UES), esophageal body (amplitude, peristalsis), and lower esophageal sphincter (LES, resting pressure and relaxation). High-resolution manometry (HRM) with Chicago Classification: diagnoses achalasia (incomplete LES relaxation, absent peristalsis), esophagogastric junction outflow obstruction, distal esophageal spasm, hypercontractile (jackhammer) esophagus, and ineffective esophageal motility. Indications: dysphagia, noncardiac chest pain, and pre-operative evaluation for anti-reflux surgery.

Esophageal pH Monitoring

– 24-hour ambulatory monitoring of esophageal acid exposure. Catheter-based (transnasal pH probe placed 5 cm above LES) or wireless (Bravo capsule endoscopically attached to esophageal mucosa). Measures: % time pH <4 (total, upright, supine), DeMeester score (>14.7 abnormal), and symptom-reflux correlation. Indications: refractory GERD, atypical GERD symptoms (chest pain, cough, asthma), pre-operative evaluation for anti-reflux surgery, and monitoring of therapy. Off acid suppression for diagnostic testing.

Esophagogastroduodenoscopy (EGD)

– Upper endoscopy: flexible endoscopic examination of the esophagus, stomach, and duodenum. Indications: dysphagia, GERD (surveillance of Barrett esophagus), dyspepsia (alarm symptoms: age >60, weight loss, anemia, dysphagia, bleeding), upper GI bleeding (variceal and non-variceal: ulcers, Mallory-Weiss tears), and surveillance for polyps (FAP, gastric intestinal metaplasia). Therapeutic: hemostasis (cautery, clips, sclerotherapy), polypectomy, dilation (strictures), and PEG tube placement.

Evoked Potentials (VEP, BAER, SSEP)

– Electrophysiologic tests measuring the CNS electrical response to specific sensory stimuli. Visual evoked potentials (VEP): checkboard pattern reversal, measures P100 latency; delayed in optic neuritis/MS. Brainstem auditory evoked responses (BAER): click stimuli, waves I-V; delayed in acoustic neuroma, MS, brainstem stroke. Somatosensory evoked potentials (SSEP): electrical stimulation of peripheral nerves (median, tibial); used intraoperatively for spinal cord monitoring (scoliosis surgery) and prognostic in spinal cord injury and hypoxic-ischemic encephalopathy.

Exercise Stress Test

– Treadmill exercise with ECG monitoring using the Bruce protocol (3-minute stages increasing speed and grade). Indications: diagnosis of CAD (chest pain evaluation), functional capacity assessment, and prognosis after MI. Contraindications: acute MI (within 2 days), unstable angina, decompensated heart failure, severe aortic stenosis, and uncontrolled arrhythmias. Positive test: ≥1 mm horizontal or downsloping ST depression at 80 ms after the J point. Duke Treadmill Score (DTS) provides prognostic information (duration, ST deviation, angina).

Extraocular Muscle Function Testing

– Assessment of the six extraocular muscles innervated by CN III (medial rectus: adduction; superior rectus: elevation; inferior rectus: depression; inferior oblique: extorsion), CN IV (superior oblique: intorsion and depression), and CN VI (lateral rectus: abduction). Tested by asking the patient to follow a target in the six cardinal directions of gaze. Abnormalities: strabismus (misalignment), diplopia (double vision), and palsy. Cover-uncover test and alternate cover test identify phoria vs. tropia.

Fecal Calprotectin

– A protein released by neutrophils in the stool, a marker of intestinal inflammation. Normal: <50 mcg/g. Elevated in IBD (>150-200 mcg/g for active Crohn, ulcerative colitis). Distinguishes IBD from IBS (IBS: normal). Monitoring: correlates with endoscopic disease activity; reduction indicates treatment response. Also elevated in: NSAID enteropathy, infectious colitis, and colorectal cancer. False negatives: skip lesions in Crohn. Used to avoid unnecessary colonoscopies and guide treatment escalation.

Fecal Immunochemical Test (FIT)

– A stool-based screening test for colorectal cancer that detects human hemoglobin in a single stool sample using antibodies. Annual screening. Sensitivity: 74-80% for cancer, 25-30% for advanced adenomas.

Specificity: 94-96%. No dietary restrictions required. Positive test requires follow-up colonoscopy. More specific than guaiac FOBT. Cost-effective and non-invasive. Compared to colonoscopy: lower sensitivity but more accessible and less invasive.

FeNO (Fractional Exhaled Nitric Oxide)

– A non-invasive biomarker of type 2 airway inflammation (eosinophilic). Normal: <25 ppb (adults), <35 ppb (children). Elevated (>50 ppb): eosinophilic asthma, good response to inhaled corticosteroids. Low (<25 ppb): non-eosinophilic asthma, neutrophilic inflammation. Used to: diagnose asthma (when spirometry is equivocal), predict steroid responsiveness, monitor adherence to ICS, and guide treatment (adjust ICS dose to normalize FeNO). Affected by smoking (falsely low), atopy, and certain foods.

Ferritin

– Intracellular iron storage protein reflecting total body iron stores. Low ferritin: iron deficiency anemia (<30 ng/mL, absolute; <100 can be consistent with iron deficiency in inflammation). High ferritin: iron overload (hemochromatosis, repeated transfusions), inflammation (acute phase reactant), liver disease, malignancy, and hemophagocytic lymphohistiocytosis (HLH). Used with serum iron, TIBC, and transferrin saturation for complete iron studies. Transferrin saturation >45% suggests iron overload.

Fetal Ultrasound

– Ultrasound imaging during pregnancy. First trimester (6-12 weeks): confirm pregnancy, gestational age (crown-rump length), number of fetuses, viability (heart activity), nuchal translucency (screening for aneuploidy at 11-13 weeks). Second trimester (18-22 weeks): anatomy survey (fetal head, chest, heart, abdomen, spine, extremities, placenta, amniotic fluid), gender determination. Third trimester: growth assessment (biometric parameters: BPD, HC, AC, FL), fetal weight estimation, placental location, amniotic fluid index, and biophysical profile. Doppler: umbilical artery, middle cerebral artery.

FISH (Fluorescence In Situ Hybridization)

– A molecular cytogenetic technique using fluorescent DNA probes that hybridize to specific chromosomal regions. Faster than karyotype (24-48 hours, vs. 7-14 days). Indications: rapid prenatal aneuploidy testing (trisomy 13, 18, 21, X, Y), detection of microdeletion syndromes (22q11.2 DiGeorge, TP53 in CLL), gene rearrangements in cancer (BCR-ABL, PML-RARA, MLL rearrangements, MYC, IGH), and HER2 amplification in breast cancer. Can be performed on interphase nuclei (does not require dividing cells). Higher resolution than karyotype.

Flexible Sigmoidoscopy

– Endoscopic examination of the rectum and sigmoid colon (up to 60 cm). Indications: colorectal cancer screening (q5-10 years, combined with FIT), evaluation of rectal bleeding, chronic diarrhea, and surveillance of known distal colonic disease (ulcerative colitis). Advantages over colonoscopy: no sedation required in most cases, less bowel prep (enema only), lower cost and risk. Limitations: cannot visualize the proximal colon; positive findings require full colonoscopy. Largely replaced by colonoscopy for screening in the US.

Flow Cytometry

– Technique that measures physical and chemical characteristics of cells or particles in a fluid stream, using laser-based detection. Commonly used in hematology: immunophenotyping of leukemia and lymphoma (CD markers: CD3, CD19, CD20, CD34, CD33, CD13), detection of minimal residual disease, and evaluation of immunodeficiency disorders. Analysis of DNA content (ploidy) in neoplasms. Also used in stem cell enumeration (CD34+) for transplant.

Fluoroscopy

– Real-time X-ray imaging used to visualize dynamic processes. Common applications: upper GI series (barium swallow, barium meal), lower GI series (barium enema), voiding cystourethrogram (VCUG for VUR), hysterosalpingography (tubal patency), angiography, and interventional procedures (ERCP, PICC line placement, joint injection, pacemaker implantation, and catheterization). Utilizes continuous or pulsed X-ray beam. Radiation exposure is a concern for both patients and operators.

Genetic Testing (Sequencing)

– Analysis of DNA to identify pathogenic variants associated with genetic disorders. Sanger sequencing: gold standard for single-gene testing (CFTR for cystic fibrosis, BRCA1/2 for hereditary breast/ovarian cancer). Next-generation sequencing (NGS): massively parallel sequencing enabling panel testing (multiple genes), exome sequencing (all exons, 1-2% of genome), and genome sequencing (whole genome). Detects SNPs, indels, and copy number variants. Indications: confirm a suspected genetic diagnosis, carrier screening, prenatal diagnosis, pharmacogenomics, and cancer genomics. Variant interpretation: pathogenic, likely pathogenic, VUS (variant of uncertain significance), likely benign, benign.

Glucose Tolerance Test

– Testing for diabetes mellitus and gestational diabetes. 75g OGTT (oral glucose tolerance test): measure plasma glucose at 0, 60, and 120 minutes. Diagnostic: fasting $\geq$126 mg/dL, 2-hour $\geq$200 mg/dL. Gestational diabetes screening: 50g glucose challenge test (GCT) at 24-28 weeks; if >140 mg/dL at 1 hour, proceed to 100g 3-hour OGTT. Carpenter-Coustan criteria: $\geq$2 values elevated (fasting >95, 1h >180, 2h >155, 3h >140). Glucose tolerance categories: normal, impaired fasting glucose (IFG 100-125), impaired glucose tolerance (IGT 140-199), and diabetes.

Gram Stain

– A differential staining technique that classifies bacteria as Gram-positive (purple, thick peptidoglycan) or Gram-negative (pink/red, thin peptidoglycan + outer membrane). Also reports morphology (cocci, rods, spirals) and arrangement (clusters, chains, pairs). Used to guide empiric antibiotic therapy while awaiting culture. Quality: epithelial cells suggest contamination; polymorphonuclear leukocytes suggest true infection. Foundational in microbiology.

Haptoglobin

– An alpha-2 globulin that binds free hemoglobin, preventing oxidative damage and renal excretion. Low/undetectable: acute hemolysis (intravascular and extravascular), liver disease (decreased production), and hereditary deficiency. Elevated: acute phase reaction (inflammation, infection, malignancy). Used with LDH and bilirubin to confirm hemolytic anemia. Normal: 30-200 mg/dL.

hCG (Human Chorionic Gonadotropin)

– A glycoprotein hormone produced by the placenta, detected in urine and serum for pregnancy diagnosis. Qualitative: positive $\geq$25 mIU/mL (urine). Quantitative: serum hCG levels — <5 mIU/mL (negative), 5-25 (indeterminate), >25 (positive). In early pregnancy: hCG doubles every 48-72 hours until 8-10 weeks. Abnormal levels: ectopic pregnancy (slow rise), miscarriage (declining), or molar pregnancy (very high). Also a tumor marker for gestational trophoblastic disease and germ cell tumors.

Hemoglobin A1c

– Glycated hemoglobin reflecting average blood glucose over the preceding 2-3 months. Normal: <5.7%. Prediabetes: 5.7-6.4%. Diabetes: $\geq$6.5% (diagnostic, confirmed with repeat). Targets: generally <7% for most adults with diabetes (individualized: <6.5% for some, <8% for frail/elderly). Used for diabetes diagnosis and monitoring glycemic control. Factors affecting accuracy: hemoglobinopathies, anemia, renal failure, pregnancy, and recent transfusions.

Hemoglobin Electrophoresis

– Separation of hemoglobin variants by electrophoresis or HPLC. Normal adult: HbA (95-98%), HbA2 (2-3%), HbF (<1%). Sickle cell disease (HbSS): HbS only, no HbA. Sickle cell trait (HbAS): HbA > HbS. Thalassemia: elevated HbA2 or HbF. Beta thalassemia trait: HbA2 3.5-7%. HbF elevated in beta thalassemia, sickle cell disease, HPFH, and juvenile myelomonocytic leukemia. Used for diagnosis of hemoglobinopathies.

Hepatitis Serology

– Panel of serologic markers for hepatitis A, B, C, and E. HAV: anti-HAV IgM (acute), anti-HAV IgG (past infection/vaccination). HBV: HBsAg (active infection), anti-HBs (immunity/vaccination), anti-HBc IgM (acute), anti-HBc IgG (past), HBeAg (high replication), anti-HBe (low replication). HBV DNA: quantifies viral load. HCV: anti-HCV (screening, past or present), HCV RNA (active infection, viral load). HEV: anti-HEV IgM/IgG. Used for diagnosis, determination of phase/chronicity, and treatment monitoring.

HIV Test

– Screening and confirmatory testing for HIV. Fourth-generation HIV antigen/antibody combo test (p24 antigen + HIV-1/2 antibodies): window period 2-4 weeks. Positive screening requires confirmatory HIV-1/2 antibody differentiation assay. Rapid point-of-care tests (fingerstick, oral swab): results in 20 minutes. Nucleic acid amplification test (HIV RNA): detects virus as early as 10 days post-infection. Recommended: annual screening for sexually active individuals, all pregnant women, and high-risk populations.

Holter Monitor

– Continuous ECG recording for 24-48 hours to detect arrhythmias, quantify burden (atrial fibrillation, PVCs, PACs), and correlate symptoms with rhythm. Patient diary logs symptoms. Indications: palpitations, syncope, presyncope, and assessment of arrhythmia control. Reveals paroxysmal arrhythmias not captured on standard 12-lead ECG. Extended monitoring (event recorder, loop recorder, mobile cardiac telemetry) for up to 30 days for infrequent symptoms. Implantable loop recorder (Reveal) for annual monitoring.

Hydrogen Breath Test

– Non-invasive test for carbohydrate malabsorption (lactose, fructose) and small intestinal bacterial overgrowth (SIBO). Patient ingests a substrate (lactose, glucose, or lactulose). Breath hydrogen is measured at baseline and q15-30 minutes for 2-3 hours. Rise in H2 >20 ppm from baseline: positive (bacterial fermentation). Lactose intolerance: elevated H2 after lactose challenge, with symptoms. SIBO: early peak (within 60-90 minutes) after glucose or lactulose. False positives: recent antibiotics, colonic fermentation (rapid transit). Antibiotics, colonoscopy prep, and dietary restrictions required before testing.

Hysterosalpingography (HSG)

– Fluoroscopic imaging after injection of iodinated contrast into the uterine cavity through a cervical cannula. Used for infertility evaluation: assesses tubal patency (normal: contrast spills freely into peritoneal cavity), uterine cavity shape (fibroids, polyps, synechiae, Mullerian anomalies), and proximal/distal tubal occlusion. Performed in the follicular phase (days 7-10). Contraindications: pregnancy, active PID, and contrast allergy. Complications: pain, vasovagal reaction, and infection.

IGRA (Interferon-Gamma Release Assay)

– Blood test for Mycobacterium tuberculosis infection (latent or active). T-SPOT.TB (ELISPOT) or QuantiFERON-TB Gold Plus (ELISA). Measures IFN-gamma released by T cells after exposure to MTB-specific antigens (ESAT-6, CFP-10, TB7.7). Advantages over PPD: no cross-reactivity with BCG or most nontuberculous mycobacteria, single patient visit, no booster phenomenon. Preferred for BCG-vaccinated individuals and serial testing. Cannot distinguish latent from active TB.

Immunohistochemistry

– Technique using antibodies to detect specific antigens in tissue sections. Uses: tumor classification and subtyping (Carcinoma: CK7, CK20; Lymphoma: CD markers; Melanoma: S100, HMB45, MelanA; Sarcoma: vimentin, desmin, CD34), determination of tumor origin (CK7+/CK20-: lung, breast; CK7-/CK20+: colorectal), assessment of prognostic markers (ER/PR, HER2 in breast cancer), and identification of infectious agents. Essential in modern pathology for diagnosis and treatment planning.

Influenza Test

– Detection of influenza A and B viruses. Rapid antigen test: nasal/throat swab, results in 15 minutes, specificity >95%, sensitivity 50-70% (high false negative rate). RT-PCR: gold standard, highly sensitive and specific, results in 1-4 hours, distinguishes influenza A subtypes (H1N1, H3N2). Used during influenza season for patients with ILI (fever + cough/sore throat). Antiviral therapy (oseltamivir, baloxavir) is most effective when started within 48 hours.

Intravenous Pyelogram (IVP)

– Radiographic imaging of the urinary tract after IV contrast administration. Used to evaluate the kidneys, ureters, and bladder for obstruction, stones, tumors, and anatomic abnormalities. Sequential films show nephrogram (renal parenchyma), pyelogram (collecting system), and ureterogram (ureters). Largely replaced by CT urography and MR urography, which provide better anatomic detail and sensitivity.

IV Fluids

– Intravenous fluid therapy for volume resuscitation and maintenance. Crystalloids: normal saline (0.9% NaCl, increases chloride, risk of hyperchloremic metabolic acidosis), lactated Ringer (balanced, contains K+, Ca2+, lactate buffer, safer in large volumes), Plasma-Lyte (balanced, closest to plasma). Colloids: albumin (5%, 25%), hydroxyethyl starch (HES, nephrotoxic, no longer recommended). Resuscitation: 30 mL/kg bolus for sepsis, hemorrhage. Maintenance: 1.5-2.5 L/day with D5 1/2NS + 20 mEq KCl.

Lactate

– A product of anaerobic metabolism, elevated in tissue hypoperfusion and hypoxia. Normal: <2.0 mmol/L. Lactate >2.0: hyperlactatemia. Lactate >4.0: severe sepsis/septic shock (associated with increased mortality). Types: type A (tissue hypoxia: shock, sepsis, hypoperfusion, seizures, exercise) and type B (no hypoxia: liver disease, thiamine deficiency, metformin, mitochondrial dysfunction, malignancy). Used for sepsis screening (qSOFA), monitoring resuscitation response, and prognostication in critical illness.

Lactic Acid

– See Lactate.

Lactose Tolerance Test

– See Hydrogen Breath Test.

Lipid Panel

– Measurement of total cholesterol, LDL-C, HDL-C, and triglycerides. Fasting (9-12 hours) or non-fasting acceptable for screening. Desirable levels: total cholesterol <200 mg/dL, LDL <100 mg/dL (<70 for very high risk), HDL >40 (men) or >50 (women) mg/dL, triglycerides <150 mg/dL. Used for ASCVD risk assessment (Pooled Cohort Equations), guiding statin therapy, and monitoring treatment. Non-HDL cholesterol (total minus HDL) provides additional risk information.

Liver Function Tests

– A panel of blood tests that assess the health and function of the liver, used to screen for, diagnose, and monitor liver disease and to monitor response to treatment with hepatotoxic drugs. The typical LFT panel includes: (1) Alanine aminotransferase (ALT): a cytosolic enzyme found predominantly in the liver, released into the bloodstream when hepatocytes are damaged. Elevated in: viral hepatitis (all types), alcoholic hepatitis, non-alcoholic fatty liver disease (NAFLD), drug-induced liver injury (DILI), and ischemic hepatitis. ALT is more specific for liver injury than AST (aspartate aminotransferase). AST is present in the liver, heart, muscle, kidney, and brain. The AST/ALT ratio helps distinguish etiology: alcoholic hepatitis (AST:ALT >1.5, usually 2:1), viral hepatitis (AST:ALT <1), and cirrhosis (AST:ALT >1). (2) Alkaline phosphatase (ALP): an enzyme found in bile ducts (canalicular membrane), bone (osteoblasts), placenta, and intestine. Elevated in: cholestatic liver disease (obstructive jaundice, primary biliary cholangitis, primary sclerosing cholangitis), bone disease (Paget, metastasis, fracture, healing), infiltrative disease (sarcoidosis, amyloidosis), and third trimester of pregnancy. Gamma-glutamyl transferase (GGT) is often measured with ALP to distinguish hepatic (GGT elevated) from bone (GGT normal) sources. GGT is induced by alcohol and medications (phenytoin, carbamazepine). (3) Bilirubin: total, conjugated (direct) and unconjugated (indirect). See Jaundice. (4) Albumin: synthesised exclusively by the liver, half-life 20 days. Decreased in chronic liver disease (cirrhosis) indicating impaired synthetic function, nephrotic syndrome, and malnutrition. (5) Prothrombin time (PT) and international normalised ratio (INR): vitamin K-dependent clotting factors (II, VII, IX, X) are synthesised in the liver. PT/INR is prolonged in severe liver disease (hepatic synthetic dysfunction) and vitamin K deficiency. PT/INR is the most sensitive acute measure of liver function in acute liver failure. (6) Total protein: decreased in chronic liver disease, increased in monoclonal gammopathies and chronic inflammation.

Liver Function Tests (LFTs)

– A panel of blood tests assessing liver health: ALT (alanine aminotransferase, primarily hepatocyte injury), AST (aspartate aminotransferase, also released by muscle/heart), ALP (alkaline phosphatase, biliary obstruction, bile duct injury), GGT (gamma-glutamyl transferase, confirms hepatobiliary source of elevated ALP), total and direct bilirubin (jaundice, conjugation/excretion), albumin (synthetic function), and PT/INR (clotting factor synthesis). Patterns: hepatocellular (ALT/AST high), cholestatic (ALP/GGT/bilirubin high), and mixed.

Lumbar Puncture

– See Spinal Tap.

Lumbar Puncture (Spinal Tap)

– Insertion of a needle into the subarachnoid space at L3-L4 or L4-L5 to obtain CSF. Indications: suspected meningitis/encephalitis, subarachnoid hemorrhage, demyelinating disease, and intrathecal therapy. Contraindications: increased intracranial pressure (papilledema, mass lesion on CT), coagulopathy, and infection at the puncture site. Opening pressure: normal 10-20 cm H2O. Complications: post-dural puncture headache (volume depletion, caffeine, blood patch), infection, and bleeding.

Mammography

– X-ray imaging of the breast for cancer screening and diagnosis. Screening mammography: annual for women ≥40 (USPSTF: biennial 50-74). Views: CC (craniocaudal) and MLO (mediolateral oblique). Digital mammography with CAD (computer-aided detection) improves sensitivity. Tomosynthesis (3D): reduces tissue overlap, improves cancer detection and recall rates. BI-RADS classification: 0 (incomplete), 1 (negative), 2 (benign), 3 (probably benign, short interval), 4 (suspicious, biopsy), 5 (highly suggestive of malignancy), 6 (known biopsy-proven malignancy).

Methacholine Challenge Test

– A bronchoprovocation test for diagnosing asthma when spirometry is normal. Inhaled methacholine (a cholinergic agonist) induces bronchoconstriction. Spirometry is repeated after each escalating dose. Positive: ≥20% fall in FEV1 (PC20) at ≤16 mg/mL. Sensitivity >90% for asthma. Contraindications: FEV1 <50% predicted, recent MI, uncontrolled hypertension, and pregnancy. Bronchodilator reversibility and asthma treatment affect results.

Monospot (EBV Heterophile Antibody)

– Rapid test for infectious mononucleosis caused by Epstein-Barr virus. Detects heterophile antibodies (IgM) that agglutinate horse or sheep red blood cells. Sensitivity: 70-95% in symptomatic patients. False negative in early disease (<1 week) and children <5. False positive in autoimmune disease, HIV, lymphoma, and other viral infections. Positive: confirms EBV infection. Negative with high suspicion: order EBV-specific serology (VCA IgM, VCA IgG, EBNA IgG).

MRCP (Magnetic Resonance Cholangiopancreatography)

– Non-invasive MRI of the biliary and pancreatic ducts using heavily T2-weighted sequences that highlight static fluid. Used for: choledocholithiasis (CBD stones), biliary strictures, primary sclerosing cholangitis, pancreatic duct anatomy, and congenital biliary anomalies. Advantages over ERCP: non-invasive, no contrast, no radiation, no pancreatitis risk. Limitations: lower spatial resolution than ERCP, cannot treat (stone removal, stenting). Sensitivity for CBD stones: >90%.

MRI (Magnetic Resonance Imaging)

– Imaging using strong magnetic fields and radiofrequency pulses to generate detailed cross-sectional images. No ionizing radiation. Superior soft-tissue contrast compared to CT. Sequences: T1-weighted (anatomy, fat bright), T2-weighted (pathology, water bright), FLAIR (suppresses CSF, for brain lesions), DWI (acute stroke, abscess, cellular tumors), and STIR (suppresses fat). Gadolinium contrast: highlights inflammation, infection, and tumors. Contraindications: ferromagnetic implants (some pacemakers, aneurysm clips, metal fragments).

MRV (Magnetic Resonance Venography)

– MRI technique to visualize veins, using time-of-flight (TOF) or contrast-enhanced sequences. Indications: cerebral venous sinus thrombosis, dural sinus stenosis, abdominal/pelvic venous thrombosis, portal vein thrombosis, and renal vein thrombosis. Differentiates acute vs. chronic thrombus. Non-invasive alternative to conventional venography.

MUGA Scan (Multigated Acquisition Scan)

– Radionuclide ventriculography using Tc-99m-labeled RBCs to assess left ventricular function. ECG-gated acquisition synchronized to the QRS complex. Measures: LVEF (most accurate non-invasive method), wall motion, and ventricular volumes. Gold standard for LVEF assessment prior to cardiotoxic chemotherapy (doxorubicin, trastuzumab). Normal LVEF >55%. Decline >10% to <50% indicates cardiotoxicity. Also used for pre-operative risk assessment and heart failure evaluation.

Multiplex PCR (BioFire, FilmAr-

ray)

– A rapid molecular diagnostic system using nested multiplex PCR to detect multiple pathogens from a single sample. Panels: respiratory (adenovirus, coronavirus, hMPV, influenza, parainfluenza, RSV, rhinovirus, Bordetella, Chlamydia, Mycoplasma), GI (Campylobacter, Salmonella, Shigella, E. coli, norovirus, rotavirus, etc.), blood culture ID, and meningitis/encephalitis. Results in 45-60 minutes. Sensitivity 90-100%. Guides early targeted therapy and reduces unnecessary antibiotics.

Nerve Conduction Studies

– Measurement of the speed and amplitude of electrical impulses along peripheral nerves. Sensory NCS: stimulate at one point, record distally. Motor NCS: stimulate at two or more points along the nerve, record from muscle. Parameters: conduction velocity (m/s), amplitude (microvolts/millivolts), and distal latency. Abnormalities: axonal loss (low amplitude, normal velocity), demyelination (slow velocity, prolonged latency, conduction block). Used with EMG to diagnose neuropathy, radiculopathy, and myopathy.

Newborn Hearing Screen

– Universal screening for congenital hearing loss, performed before hospital discharge. Two-step protocol: otoacoustic emissions (OAE: measures cochlear outer hair cell response to click stimulus) and automated auditory brainstem response (AABR: measures neural response from the cochlea to the brainstem). Pass/refer. Refer: requires outpatient diagnostic audiology (full ABR) by 3 months. Early intervention by 6 months improves language outcomes. Risk factors for late-onset/progressive HL: NICU stay >5 days, congenital CMV, family history, and certain syndromes.

Newborn Screening

– A public health program testing newborns for treatable genetic, metabolic, and endocrine disorders 24-48 hours after birth. Blood spot (heel stick) on filter paper (Guthrie card). Conditions vary by state (Recommended Uniform Screening Panel: 35 core conditions). Common: phenylketonuria (PKU), congenital hypothyroidism, cystic fibrosis, sickle cell disease, medium-chain acyl-CoA dehydrogenase (MCAD) deficiency, galactosemia, biotinidase deficiency, congenital adrenal hyperplasia, and hearing loss (by otoacoustic emissions). Positive screens require confirmatory testing and prompt specialist referral. Early treatment prevents morbidity and mortality.

Non-Stress Test (NST)

– Antepartum fetal monitoring assessing fetal heart rate reactivity. Reactive NST: ≥2 accelerations of 15 bpm for 15 seconds (≥32 weeks) or 10 bpm for 10 seconds (<32 weeks) within 20 minutes. Non-reactive: insufficient accelerations after 40 minutes (may indicate fetal sleep, acidosis, or CNS depression). False positive rate high. Used for fetal well-being assessment in high-risk pregnancies (preeclampsia, IUGR, diabetes, postterm, decreased fetal movement). Often combined with AFI (biophysical profile).

Pancreatic Elastase

– Measurement of fecal elastase-1 (FE-1) as a non-invasive test for pancreatic exocrine insufficiency. Values: >200 mcg/g (normal), 100-200 (mild-moderate insufficiency), <100 mcg/g (severe insufficiency). Unaffected by pancreatic enzyme replacement therapy (does not cross-react with porcine enzymes). More sensitive and stable than fecal fat quantification. False positives: watery stool (dilution effect). Used to diagnose chronic pancreatitis, cystic fibrosis, and pancreatic insufficiency.

Paracentesis

– Aspiration of fluid from the peritoneal cavity (ascites). Indications: diagnostic (evaluate new ascites: SAAG, cell count, culture, cytology, albumin, total protein; rule out SBP: PMN >250) and therapeutic (relief of tense ascites, 4-6 L). Site: left lower quadrant (avascular zone), lateral to rectus sheath, ultrasound-guided. Contraindications: overlying cellulitis, severe coagulopathy, and pregnancy. Serum-ascites albumin gradient (SAAG): >1.1 g/dL (portal hypertension: cirrhosis, heart failure), <1.1 g/dL (peritoneal disease: malignancy, TB, pancreatitis, nephrotic). Large-volume paracentesis requires albumin infusion (6-8 g/L removed) to prevent post-paracentesis circulatory dysfunction.

PEG Tube (Percutaneous Endoscopic Gastrostomy)

– Endoscopic placement of a feeding tube through the abdominal wall directly into the stomach. Indications: long-term enteral feeding (>4 weeks) for dysphagia (stroke, ALS, dementia, head/neck cancer) and malnutrition. Contraindications: gastric outlet obstruction, severe ascites, coagulopathy, gastric cancer, and inability to transilluminate the abdominal wall. Procedure: EGD with transillumination, needle puncture under direct endoscopic visualization, wire passage, and tube placement (Pull or Push technique). Replacements: 8-12 months later (balloon or button type). Complications: aspiration, infection (peristomal cellulitis), bleeding, perforation, and tube dislodgement. Buried bumper syndrome: overgrowth of gastric mucosa over the internal bumper.

Pericardiocentesis

– Aspiration of pericardial fluid using a needle inserted through the subxiphoid or apical approach, under echocardiographic or fluoroscopic guidance. Indications: cardiac tamponade (therapeutic emergency), diagnostic evaluation of pericardial effusion (malignancy, infection, autoimmune). Complications: myocardial puncture/laceration, coronary artery injury, pneumothorax, arrhythmia, and infection. A small catheter can be left in place for continuous drainage. Fluid analysis: cell count, cytology, culture, TB PCR/ culture, and rheumatologic studies. High-risk procedure; should be performed with surgical backup.

Periodontal Probing

– Measurement of the depth of the gingival sulcus/pocket using a periodontal probe with millimeter markings. Normal: 1-3 mm. Periodontal pockets: $\geq$4 mm (indicates attachment loss and periodontitis). Full-mouth probing at 6 sites per tooth: mesio-buccal, mid-buccal, disto-buccal, mesio-lingual, mid-lingual, disto-lingual. Bleeding on probing (BOP): sign of active inflammation. Clinical attachment level (CAL): distance from CEJ to base of pocket. Used for periodontal disease diagnosis, staging (Stage I-IV), and monitoring treatment response.

Peripheral Blood Smear

– Microscopic examination of a stained blood smear (Wright-Giemsa stain). Evaluates: RBC morphology (size: macrocytic, microcytic; shape: spherocytes, sickle cells, schistocytes, target cells, elliptocytes; color: hypochromic, polychromasia; inclusions: basophilic stippling, Howell-Jolly bodies, Pappenheimer bodies), WBC morphology (blast cells, band forms, toxic granulation, Doehle bodies, smudge cells, Auer rods), and platelet morphology (size, granules). Performed when automated CBC shows abnormalities.

PET Scan (Positron Emission Tomography)

– Nuclear medicine imaging using radioactive tracers (most commonly FDG, a glucose analog) to detect metabolically active tissues. Indications: cancer staging and restaging (lung, lymphoma, melanoma, breast, colon), evaluation of solitary pulmonary nodule, myocardial viability (FDG uptake in hibernating myocardium), and neurologic (Alzheimer disease: hypometabolism; epilepsy: seizure focus). Standardized uptake value (SUV) quantifies tracer activity. Combined PET/CT provides anatomic localization. False positives in infection/inflammation.

Polysomnography (Sleep Study)

– Overnight monitoring for sleep-disordered breathing. Parameters: EEG (sleep staging: N1, N2, N3, REM), EOG (eye movements), EMG (chin and leg movements), airflow (nasal pressure, thermistor), respiratory effort (chest/abdomen belts), snore microphone, body position, oximetry (SpO2), and ECG. Calculates AHI (apnea-hypopnea index): normal <5, mild 5-15, moderate 15-30, severe >30. Used to diagnose obstructive sleep apnea, central sleep apnea, periodic limb movement disorder, and narcolepsy (MSLT following PSG).

Pregnancy Test

– See hCG (Human Chorionic Gonadotropin).

Preoperative Evaluation

– Medical assessment before elective surgery to optimize outcomes and minimize perioperative risk. Components: history and physical (comorbidities: CAD, CHF, COPD, diabetes, CKD), medications (anticoagulants, antiplatelets, antihyperglycemics), functional status (METs), and cardiac risk assessment (RCRI: Revised Cardiac Risk Index, NSQIP MICA risk calculator). Testing per RCRI and age/symptoms: ECG (age >65, cardiac risk factors, known CAD), CBC, BMP, coagulation studies, type and screen (for planned transfusions), CXR (age >70, cardiac/pulmonary disease), and PFTs (lung resection, COPD exacerbation). Nothing routine; testing guided by specific risk factors and surgical severity.

Procalcitonin

– A biomarker for bacterial infection and sepsis, produced by thyroid C-cells and peripheral tissues in response to bacterial toxins. Levels: <0.1 ng/mL (normal), 0.1-0.25 (unlikely bacterial), 0.25-0.5 (possible), >0.5 (probable bacterial infection), >2.0 (severe sepsis/septic shock). More specific for bacterial infection than CRP. Used to guide antibiotic initiation and discontinuation (procalcitonin algorithms reduce antibiotic duration). Elevated in bacterial pneumonia, meningitis, UTI. Not elevated in viral infections or non-infectious inflammation.

PSA (Prostate-Specific Antigen)

– A serine protease produced by the prostate, used for prostate cancer screening. Normal: <4.0 ng/mL, but age-specific ranges exist. PSA >10 ng/mL: high probability of cancer. PSA 4-10: gray zone; percent free PSA (<25% suggests cancer) and PSA density help. Screening controversy: reduces prostate cancer mortality (1.3 per 1000) but leads to overdiagnosis/overtreatment. Shared decision-making for men 55-69. Annual screening recommended for high-risk (African American, family history).

Pulmonary Function Tests

– Spirometry: FVC (forced vital capacity), FEV1 (forced expiratory volume in 1 second), FEV1/FVC ratio. Obstructive pattern: FEV1/FVC <0.7 (asthma, COPD). Restrictive pattern: FVC <80% with normal or increased FEV1/FVC (pulmonary fibrosis, chest wall disorders). Lung volumes: TLC (total lung capacity), RV (residual volume). Diffusion capacity (DLCO): measures gas transfer across the alveolar-capillary membrane. Indications: dyspnea evaluation, COPD staging (GOLD), preoperative assessment, and monitoring lung disease progression.

Quad Screen (Maternal Serum Screening)

– Second-trimester prenatal screening (15-22 weeks) for fetal aneuploidy and neural tube defects. Measures 4 markers: AFP (alpha-fetoprotein: elevated in NTD, low in Down syndrome), hCG (elevated in Down syndrome), uE3 (unconjugated estriol: low in Down syndrome), and inhibin A (elevated in Down syndrome). Detection rates: trisomy 21 (80%), trisomy 18 (60-70%), NTD (80%). Replaced in many settings by cell-free DNA (cfDNA, NIPT) with higher detection rates.

Rapid Strep Test

– Immunoassay for Group A Streptococcus (Streptococcus pyogenes) from throat swab. Results: 5-10 minutes. Specificity: >95%. Sensitivity: 80-90% compared to culture. Positive: treat. Negative: confirm with throat culture (especially in children). Used for diagnosis of GAS pharyngitis to guide antibiotic therapy (prevents unnecessary antibiotics for viral pharyngitis). First-line treatment for confirmed GAS: penicillin or amoxicillin.

Reticulocyte Count

– Percentage of young RBCs in the peripheral blood, reflecting bone marrow erythroid activity. Normal: 0.5-2.5% (absolute 25-100 K/uL). Reticulocyte production index (RPI) corrects for anemia severity: RPI = reticulocyte % x (Hct/45) / maturation correction factor. High RPI (>2-3): bone marrow responding appropriately (hemolysis, acute blood loss). Low RPI (<2): bone marrow not responding (iron deficiency, aplastic anemia, marrow infiltration). Used to classify anemia as hyperproliferative vs. hypoproliferative.

Rheumatoid Factor (RF)

– An autoantibody directed against the Fc portion of IgG, positive in 70-80% of rheumatoid arthritis patients. Also positive in other autoimmune diseases (SLE, Sjogren, mixed cryoglobulinemia), chronic infections (TB, endocarditis, hepatitis C), and 5% of healthy elderly. Low sensitivity (may be negative in seronegative RA) and moderate specificity. Used in ACR/EULAR classification criteria for RA with anti-CCP. Serial monitoring is not useful. Higher titers associated with more severe, erosive disease.

RPR/VDRL (Syphilis Serology)

– Non-treponemal screening tests for syphilis. Non-treponemal: RPR (rapid plasma reagin) or VDRL (venereal disease research laboratory). Positive in primary syphilis (1-4 weeks after chancre), secondary (100% positive), and early latent. Quantitative: titer correlates with disease activity and treatment response (fourfold decline indicates successful treatment). False positives: pregnancy, SLE, TB, IV drug use, hepatitis, HIV. Positive screening requires confirmatory treponemal test (FTA-ABS, TP-PA, EIA). Titer monitoring after treatment.

Serum Protein Electrophoresis

(SPEP)

– Separation of serum proteins by electrophoresis into 5 fractions: albumin, alpha-1, alpha-2, beta, and gamma. Monoclonal gammopathy: M-spike in gamma region (multiple myeloma, MGUS, Waldenstrom macroglobulinemia). Immunofixation: identifies heavy chain (IgG, IgA, IgM, IgD, IgE) and light chain (kappa, lambda) isotype. Serum free light chain assay: kappa/lambda ratio (abnormal in light chain myeloma, AL amyloidosis). Used to screen for, diagnose, and monitor plasma cell dyscrasias.

SPECT Scan (Single Photon Emission CT)

– Nuclear medicine imaging using gamma-emitting radioisotopes (Tc-99m, Tl-201, I-123) with a rotating gamma camera to create 3D images. Indications: myocardial perfusion imaging (MPI with sestamibi, dobutamine or regadenosone stress), bone scan (metastases, stress fracture, osteomyelitis), brain perfusion (dementia, seizure focus), and VQ scan (Tc-99m MAA for perfusion, Xe-133 or Tc-DTPA for ventilation). Combined with CT (SPECT/CT) improves anatomic localization.

Sweat Chloride Test

– The gold standard diagnostic test for cystic fibrosis. Measures chloride concentration in sweat after pilocarpine iontophoresis. Normal: <30 mmol/L. Intermediate: 30-59. Positive: ≥ 60 mmol/L. False positives: malnutrition, adrenal insufficiency, hypothyroidism, and ectodermal dysplasia. False negatives: rare CFTR mutations with normal sweat chloride. Requires adequate sweat collection (>75 mg). Part of CF diagnostic algorithm with clinical features and CFTR genetic testing.

Thyroid Function Tests

– Measurement of TSH (most sensitive screening test), free T4, and free T3. TSH high, T4 low $\rightarrow$ primary hypothyroidism. TSH low, T4 high $\rightarrow$ primary hyperthyroidism. TSH low, T4 normal $\rightarrow$ subclinical hyperthyroidism or T3-toxicosis (measure T3). TSH high, T4 normal $\rightarrow$ subclinical hypothyroidism. TSH normal, T4 low $\rightarrow$ central hypothyroidism (pituitary/hypothalamic). Anti-TPO antibodies: positive in Hashimoto thyroiditis. TSH receptor antibodies (TRAb): positive in Graves disease.

Tilt Table Test

– Evaluation of unexplained syncope by reproducing orthostatic stress. Patient is strapped to a motorized table tilted from supine to 60-80 degrees for 20-45 minutes. Positive: reproduction of symptoms with hypotension and/or bradycardia (vasovagal syncope, delayed orthostatic hypotension). Also assesses for postural orthostatic tachycardia syndrome (POTS: HR increase >30 bpm without hypotension). Isoproterenol or nitroglycerin provocation increases sensitivity. Used when initial evaluation for syncope is inconclusive.

TORCH Panel

– Serologic screening for congenital infections: Toxoplasma, Rubella, CMV, Herpes simplex (and others: syphilis, parvovirus B19, VZV, Zika). IgM: indicates recent/active infection (maternal or fetal). IgG: past infection or passive immunity. Paired maternal and neonatal testing. Indications: pregnancy with febrile illness or rash, fetal abnormalities on ultrasound (hydrops, microcephaly, calcifications), and neonatal evaluation for congenital infection. Management depends on the specific pathogen identified.

Troponin

– Cardiac-specific troponin I and T, the preferred biomarkers for myocardial injury (MI). Highly sensitive and specific. Rise 2-4 hours after symptom onset, peak at 12-24 hours, remain elevated for 7-10 days. High-sensitivity troponin (hs-cTn) allows earlier detection (1-2 hours). Fourth Universal Definition: detection of a rise/fall of troponin with at least one value above the 99th percentile upper reference limit, plus clinical evidence of myocardial ischemia. Used for diagnosis of NSTEMI, risk stratification in ACS, and prognosis in many cardiac conditions.

Tuberculin Skin Test (PPD)

– Intradermal injection of 0.1 mL of purified protein derivative (PPD) on the forearm (Mantoux method). Read at 48-72 hours measuring induration (not erythema). Positive thresholds: ≥ 5 mm (HIV, close contacts, transplant, immunosuppressed), ≥ 10 mm (recent immigrants, IV drug users, healthcare workers, children <4, high-risk settings), ≥ 15 mm (no risk factors). Does not differentiate latent from active TB. False positives: BCG vaccination, nontuberculous mycobacteria. Positive test requires chest

X-ray and symptom evaluation.

Ultrasonography

– See Ultrasound (Ultrasonography).

Ultrasound

– Imaging using high-frequency sound waves (2-18 MHz). No ionizing radiation. Right upper quadrant (RUQ): gallbladder (gallstones, cholecystitis), liver (steatosis, cysts), kidneys, and pancreas. Pelvic: uterus, ovaries, adnexa. Renal: hydronephrosis, stones, cysts, tumors. Scrotal: testicular torsion, epididymitis, varicocele, hydrocele. Thyroid: nodules (TI-RADS classification), goiter. Soft tissue: abscess, cellulitis, hematoma, lipoma. Vascular: DVT, carotid stenosis, AAA. Point-of-care (POCUS): FAST, cardiac, lung.

Umbilical Cord Blood Gas

– Analysis of arterial and venous cord blood collected immediately after delivery. Umbilical artery (UA): reflects fetal (baby's) acid-base status. Normal UA pH >7.20, base excess < -10. Umbilical vein (UV): reflects maternal/placental status. Indications: suspected fetal distress, low Apgar scores, and medicolegal documentation of birth asphyxia. Fetal acidosis: pH <7.0 (severe) or pH 7.0-7.19 (moderate). Mixed acidosis (respiratory + metabolic component) indicates prolonged hypoxic event.

Upper Endoscopy

– See Esophagogastroduodenoscopy (EGD).

Upper GI Series

– Fluoroscopic examination of the esophagus, stomach, and duodenum after oral barium contrast. Single-contrast: looks for gross abnormalities (obstruction, masses). Double-contrast (barium + gas): better mucosal detail for ulcers, erosions, and early malignancies. Indications: dysphagia, dyspepsia, nausea/vomiting, weight loss, and GI bleeding. Can assess for hiatal hernia, ulcers, gastric cancer, and outlet obstruction. Largely replaced by EGD.

Urinalysis

– Analysis of urine. Dipstick: specific gravity, pH, leukocyte esterase (WBCs), nitrite (bacteria), protein, glucose, ketones, bilirubin, urobilinogen, blood, and hemoglobin. Microscopy: RBCs, WBCs, casts (hyaline, granular, RBC, WBC), crystals (calcium oxalate, uric acid, triple phosphate), bacteria, yeast, squamous cells, and epithelial cells. Indications: UTI, hematuria, renal disease (proteinuria, casts), and screening. Clean-catch midstream specimen for culture reduces contamination.

Urine Drug Screen

– Immunoassay-based screening for common drugs of abuse in urine. Detects: amphetamines, benzodiazepines, cocaine (benzoylecgonine), opiates (morphine, codeine, heroin), oxycodone, methadone, fentanyl, barbiturates, PCP, THC (cannabinoids), and alcohol. Pros: rapid (minutes at POC), inexpensive. Cons: false positives (poppy seeds → opiates, certain antihistamines → PCP, NSAIDs → barbiturates), false negatives (synthetic cannabinoids/spice, synthetic cathinones/bath salts, LSD). Positive screen requires confirmatory GC-MS or LC-MS/MS.

Urine hCG (Pregnancy Test)

– Rapid point-of-care test detecting human chorionic gonadotropin in urine. Sensitivity: detects as low as 25 mIU/mL. Positive as early as 12-14 days after conception (before missed period). Results in 3-5 minutes. False negative: dilute urine, very early pregnancy, ectopic pregnancy. False positive: recent pregnancy, hCG-producing tumors (GTN, germ cell), or rarely due to cross-reacting substances. Most sensitive and specific when performed on first morning urine specimen.

Venesection (Phlebotomy)

– The controlled removal of blood from a vein for therapeutic or diagnostic purposes. Therapeutic phlebotomy: treatment for haemochromatosis (removal of excess iron by reducing red cell mass and stimulating erythropoiesis, which mobilises iron stores), polycythaemia vera, and porphyria cutanea tarda; typical removal of 450-500 mL per session, frequency determined by haematocrit or ferritin levels. Diagnostic phlebotomy: blood collection (venepuncture) for laboratory testing, using evacuated tube system (Vacutainer) with appropriate anticoagulant or additive tubes (EDTA for hematology, citrate for coagulation studies, SST/gel for chemistry, fluoride oxalate for glucose). Site: median cubital vein (antecubital fossa) most common. Complications: haematoma, nerve injury (rare), infection, vasovagal syncope, and excessive bleeding in coagulopathic patients.

Venipuncture

– Puncture of a vein to obtain blood specimens or administer IV therapy. Common sites: median cubital vein (most common), cephalic vein, basilic vein, and dorsal hand veins. Steps: tourniquet application (30-60 seconds), vein palpation, skin antisepsis (alcohol, chlorhexidine), needle insertion at 15-30 degree angle, blood collection into appropriate tubes (order of draw: culture first, then serum, then citrate, then heparin, then EDTA, then tubes with additives), and proper needle disposal. Complications: hematoma, hemolysis, nerve injury, infection, and needlestick injury.

Viral Load (HIV RNA)

– Quantitative measurement of HIV-1 RNA in plasma by RT-PCR (Roche COBAS, Abbott RealTime). Normal: undetectable (<20-40 copies/mL). Treatment goal: undetectable viral load (U=U: Undetectable = Untransmittable). Initiation: start ART regardless of viral load or CD4. Monitoring: at baseline, 4-6 weeks after ART start, then q3-6 months. Virologic failure: persistently detectable or rebound >200 copies/mL. Resistance testing before ART switch.

Vitamin B12

– Cobalamin, essential for DNA synthesis, red blood cell production, and neurologic function. Normal: 200-900 pg/mL. Deficiency: <200 pg/mL (but borderline 200-300 may still be symptomatic). Causes: pernicious anemia (autoimmune, most common), gastrectomy, Crohn disease, ileal resection, strict vegan diet, proton pump inhibitors, and metformin. Symptoms: megaloblastic anemia, neuropathy (paresthesias, ataxia), cognitive decline, and glossitis. Methylmalonic acid and homocysteine are more sensitive markers for B12 deficiency.

Vitamin D

– 25-hydroxyvitamin D, the storage form of vitamin D. Normal: >30 ng/mL (75 nmol/L). Insufficiency: 20-30 ng/mL. Deficiency: <20 ng/mL (<50 nmol/L). Extremely deficient: <10 ng/mL (high risk of osteomalacia/rickets). Sources: sun exposure (80%), diet, and supplements. Deficiency causes secondary hyperparathyroidism, increased bone turnover, osteoporosis, and increased fall/fracture risk. Associated with autoimmune disease, cardiovascular disease, and cancer (causal role unproven). Supplementation: 800-2000 IU/day for deficiency.

Voiding Cystourethrogram (VCUG)

– Fluoroscopic imaging of the bladder and urethra during voiding, after filling the bladder with contrast via a catheter. Indications: evaluation of vesicoureteral reflux (VUR) in children with recurrent UTIs, posterior urethral valves in males, and post-surgical assessment of the bladder/urethra. Grading of VUR: I (ureter only), II (ureter and pelvis, no dilation), III (mild dilation), IV (moderate dilation, tortuous ureter), V (severe dilation, tortuous ureter, intrarenal reflux). Also evaluates bladder capacity, wall thickness, and post-void residual.

X-Ray (Plain Radiography)

– The most common form of medical imaging, using X-rays to create a 2D projection image. Uses: chest (pneumonia, pneumothorax, effusion, cardiomegaly, mass), abdomen (obstruction, perforation, stones), musculoskeletal (fractures, dislocation, arthritis, osteomyelitis), and spine (fracture, alignment, degeneration). Views: PA (posteroanterior) and lateral chest, AP and lateral for spine, AP and frog-leg for hips. Density: air (black), fat (dark gray), soft tissue/water (gray), calcium/bone (white), metal (bright white).

X-Rays (Röntgen Rays)

– See X-Ray (Plain Radiography).

Chapter 18

Microbiology

Adaptive immune evasion

– Mechanisms by which pathogens evade the adaptive immune system. Strategies include antigenic variation, which changes surface proteins to avoid antibody recognition. Some viruses downregulate MHC molecules to avoid T cell recognition. Others establish latent infections to avoid immune detection. Understanding adaptive immune evasion is essential for developing new vaccines and immunotherapies. Pathogens that evade adaptive immunity can cause chronic infections.

Adenovirus

– A non-enveloped DNA virus that causes respiratory infections, conjunctivitis, gastroenteritis, and cystitis. Adenoviruses are highly stable in the environment and can cause outbreaks in closed populations. The virus has over 50 serotypes with different tissue tropisms. Diagnosis is made by PCR, antigen detection, or viral culture. There is no specific antiviral treatment, and management is supportive. Severe disseminated infection can occur in immunocompromised patients. Adenoviruses are used as vectors in gene therapy and vaccine development due to their ability to efficiently deliver DNA to target cells.

Airborne precautions

– Infection control measures used for pathogens transmitted through airborne droplet nuclei. Precautions include negative pressure rooms, N95 respirators, and restricted access. Airborne precautions are used for tuberculosis, measles, varicella, and disseminated herpes zoster. The duration of precautions depends on the pathogen and clinical situation. Understanding airborne precautions is essential for preventing transmission of airborne infections in healthcare settings.

Antibiotic resistance genes

– Genes that encode mechanisms conferring resistance to antibiotics. These genes can be located on chromosomes, plasmids, or transposons. Major resistance genes include mecA (methicillin resistance in S. aureus), vanA/vanB (vancomycin resistance in Enterococcus), and blaKPC (carbapenem resistance). Resistance genes can be spread between bacteria through horizontal gene transfer. The acquisition of resistance genes is a major public health concern. Understanding resistance genes is essential for developing effective treatment strategies and infection control policies, and for guiding antimicrobial stewardship programmes.

Antibodies

– Immunoglobulin proteins produced by B cells that bind to specific antigens. There are five classes: IgG (most abundant), IgM (first produced), IgA (mucosal immunity), IgE (allergic responses), and IgD. Antibodies neutralise pathogens, opsonise for phagocytosis, activate complement, and mediate antibody-dependent cellular cytotoxicity. Monoclonal antibodies are engineered antibodies used as therapeutics. Antibody testing is used to diagnose infections and assess immune status. Understanding antibody structure and function is fundamental to immunology, vaccine development, and the design of monoclonal antibody therapies.

Antifungal drug classes

– Major classes of antifungal agents include polyenes (amphotericin B), azoles (fluconazole, voriconazole), echinocandins (caspofungin, micafungin), and allylamines (terbinafine). Polyenes bind to ergosterol in fungal cell membranes. Azoles inhibit ergosterol synthesis. Echinocandins inhibit cell wall synthesis. Allylamines inhibit ergosterol synthesis at a different step. The choice of antifungal depends on the fungus, site of infection, and patient factors. Understanding antifungal drug classes is essential for selecting appropriate therapy, managing resistance, and optimising treatment outcomes in patients with invasive fungal infections.

Antifungal prophylaxis

– The use of antifungal agents to prevent invasive fungal infections in high-risk patients. Common indications include prolonged neutropenia, hematopoietic stem cell transplantation, and advanced HIV infection. Fluconazole, posaconazole, and micafungin are used for prophylaxis. Understanding antifungal prophylaxis is essential for preventing invasive fungal infections in immunocompromised patients. The decision to use prophylaxis should be based on individual risk assessment.

Antifungal resistance

– The ability of fungi to withstand the effects of antifungal agents. Resistance can be intrinsic or acquired. Azole resistance occurs through efflux pumps and target mutations. Echinocandin resistance occurs through FKS gene mutations. Amphotericin B resistance is rare. Candida auris is a multidrug-resistant species causing healthcare-associated outbreaks. Aspergillus fumigatus azole resistance is emerging, partly due to environmental azole use. Understanding antifungal resistance is essential for guiding treatment decisions, implementing antifungal stewardship, and developing new antifungal agents.

Antigenic variation

– The process by which pathogens change their surface proteins to evade immune recognition. Antigenic drift: gradual changes in surface proteins. Antigenic shift: major changes through reassortment of genome segments. Influenza virus demonstrates both mechanisms. Trypanosomes and Plasmodium also use antigenic variation. Understanding antigenic variation is essential for developing effective vaccines and predicting epidemic patterns.

Antimicrobial adverse effects

– Side effects associated with antimicrobial agents. Beta-lactams can cause allergic reactions. Fluoroquinolones can cause tendon rupture and QT prolongation. Aminoglycosides can cause nephrotoxicity and ototoxicity. Azole antifungals can cause hepatotoxicity. Understanding adverse effects is essential for monitoring patients and preventing harm. The benefits of antimicrobial therapy must be weighed against the risks.

Antimicrobial dosing optimization

– Strategies to ensure antimicrobial agents achieve therapeutic concentrations at the site of infection. Optimization includes appropriate drug selection, dosing interval, route of administration, and duration of therapy. Extended or continuous infusion of beta-lactams may improve outcomes. Weight-based dosing is important for many agents. Understanding dosing optimization is essential for maximizing efficacy and minimizing toxicity.

Antimicrobial drug interactions

– Interactions between antimicrobial agents and other medications. Fluoroquinolones chelate divalent cations. Macrolides inhibit CYP3A4. Rifampin is a potent enzyme inducer. Azole antifungals inhibit CYP3A4. Understanding drug interactions is essential for safe prescribing and preventing adverse effects. Careful review of concomitant medications is essential when prescribing antimicrobials.

Antimicrobial pharmacodynamics

– The relationship between antimicrobial drug concentrations and their effect on microorganisms. Time-dependent antibiotics (beta-lactams) are most effective when concentrations remain above the MIC. Concentration-dependent antibiotics (aminoglycosides) are most effective with high peak concentrations. Understanding pharmacodynamics is essential for optimizing dosing regimens and improving clinical outcomes. Pharmacodynamic principles guide antimicrobial dosing strategies.

Antimicrobial pharmacokinetics

– The absorption, distribution, metabolism, and elimination of antimicrobial agents. Pharmacokinetics varies by drug class, patient factors, and disease state. Renal impairment affects elimination of many antibiotics. Hepatic impairment affects metabolism of many antifungals. Understanding pharmacokinetics is essential for dosing optimization in special populations including critically ill patients and those with organ dysfunction.

Antimicrobial resistance surveillance

– The systematic monitoring of antimicrobial resistance patterns in clinical isolates. Surveillance programmes track resistance rates over time and guide empiric therapy. National and international networks share resistance data. Understanding resistance surveillance is essential for effective antimicrobial stewardship and infection control. Surveillance data inform treatment guidelines and public health interventions.

Antimicrobial susceptibility testing

– Laboratory methods used to determine the susceptibility of microorganisms to antimicrobial agents. Methods include disk diffusion, broth microdilution, and automated systems. MIC (minimum inhibitory concentration) is the gold standard measure. Results guide antimicrobial therapy and are interpreted using standardised breakpoints. Resistance testing is essential for managing infections caused by multidrug-resistant organisms. Understanding susceptibility testing is essential for appropriate antimicrobial prescribing and for monitoring resistance trends at local, national, and global levels.

Antiretroviral therapy

– Treatment of HIV infection with combinations of antiretroviral agents. ART targets multiple steps in the viral life cycle including entry, reverse transcription, integration, and maturation. Combination ART suppresses viral replication and restores immune function. ART is lifelong therapy. Understanding ART is essential for managing HIV infection and preventing transmission. ART has transformed HIV from a fatal disease to a chronic manageable condition.

Antiseptics

– Chemical agents applied to living tissue to reduce microbial load. Antiseptics are used for hand hygiene, skin preparation before procedures, and wound care. Common antiseptics include alcohols, chlorhexidine, povidone-iodine, and hydrogen peroxide. Antiseptics have lower efficacy than disinfectants due to toxicity concerns. The choice of antiseptic depends on the application and microbial spectrum. Hand hygiene with alcohol-based rubs is the most effective way to prevent healthcare-associated infections and is the cornerstone of infection control programmes worldwide.

Antiviral drug classes

– Major classes of antiviral agents include nucleoside analogues (acyclovir, tenofovir), protease inhibitors (ritonavir, atazanavir), reverse transcriptase inhibitors (efavirenz, sofosbuvir), and neuraminidase inhibitors (oseltamivir). Each class targets a specific step in the viral replication cycle. Understanding antiviral drug classes is essential for selecting appropriate therapy and managing drug resistance. Combination therapy is often used to prevent resistance.

Bacterial adherence

– The process by which bacteria attach to host cell surfaces, a critical first step in colonisation and infection. Adherence is mediated by adhesins, which are surface structures that bind to specific receptors on host cells. Adhesins include pili, fimbriae, outer membrane proteins, and capsules. The specificity of adhesin-receptor interactions determines the tissue tropism of bacterial pathogens. For example, uropathogenic E. coli express adhesins that bind to uroepithelial cells. Anti-adhesion strategies, including competitive inhibitors of adhesin-receptor binding, are being investigated as novel therapeutic approaches to prevent bacterial colonisation.

Bacterial biofilms

– Structured communities of bacteria attached to surfaces and embedded in a self-produced matrix of extracellular polymeric substances. Biofilms are a major cause of chronic infections, particularly on medical devices including catheters, prosthetic joints, and heart valves. Bacteria in biofilms are more resistant to antibiotics (up to 1000-fold) and host immune responses. The biofilm matrix consists of polysaccharides, proteins, and extracellular DNA. Quorum sensing regulates biofilm formation and dispersal, making it a potential target for anti-biofilm therapies.

Bacterial capsule

– A polysaccharide layer surrounding the bacterial cell wall that protects against phagocytosis and complement-mediated killing. Capsules are major virulence factors for many pathogens including Streptococcus pneumoniae, Haemophilus influenzae, and Klebsiella pneumoniae. The capsule inhibits opsonisation and prevents engulfment by immune cells. Anti-capsular antibodies are protective and form the basis of conjugate vaccines. Capsule production can be demonstrated by the quellung reaction, which shows capsular swelling when specific antibodies bind to the capsule.

Bacterial cell wall

– The rigid outer layer of bacteria that provides structural integrity and determines cell shape. The cell wall is composed primarily of peptidoglycan, a polymer of sugars cross-linked by short peptides. Gram-positive bacteria have a thick peptidoglycan layer (20-80 nm) while gram-negative bacteria have a thin layer (5-10 nm) surrounded by an outer membrane. The cell wall is essential for bacterial survival and is the target of many antibiotics including beta-lactams and vancomycin. The unique structure of the bacterial cell wall, particularly peptidoglycan synthesis enzymes, provides an excellent target for antimicrobial drugs that have minimal toxicity to human cells.

Bacterial classification

– The organisation of bacteria into taxonomic groups based on genetic, phenotypic, and evolutionary relationships. Bacteria are classified using a hierarchical system: domain, phylum, class, order, family, genus, and species. Classification methods include phenotypic characterisation (morphology, staining, biochemical tests) and genotypic methods (16S rRNA sequencing, DNA-DNA hybridisation). The 16S rRNA gene is the primary marker for bacterial phylogenetics. Modern classification emphasises molecular phylogenetic approaches, particularly whole-genome sequencing, which provides more accurate taxonomic resolution than traditional methods.

Bacterial conjugation

– The transfer of genetic material between bacteria through direct cell-to-cell contact via a pilus. Conjugation is mediated by conjugative plasmids that encode the genes for transfer. The donor cell (F+) forms a conjugative pilus that attaches to the recipient cell (F-). A single strand of plasmid DNA is transferred through the pilus, and both cells synthesise the complementary strand. Conjugation is the most efficient mechanism of horizontal gene transfer and can spread antibiotic resistance genes across bacterial species and genera, contributing to the rapid emergence of multidrug-resistant pathogens.

Bacterial dormancy

– A state of reduced metabolic activity that allows bacteria to survive unfavourable conditions. Dormant forms include spores, persisters, and viable but non-culturable cells. Spores are highly resistant structures formed by Bacillus and Clostridium. Persisters are phenotypic variants that are tolerant to antibiotics. Understanding dormancy is essential for treating chronic infections and developing new antimicrobial strategies.

Bacterial endospores

– Highly resistant dormant structures formed by certain gram-positive bacteria including Bacillus and Clostridium species. Endospores form during nutrient deprivation and can survive for decades. They are resistant to heat, radiation, chemicals, and desiccation. Sporulation involves asymmetric cell division, engulfment of the forespore, cortex formation, and coat assembly. Germination occurs when conditions improve, with the spore returning to vegetative growth. Endospore formation is a significant challenge for infection control because spores resist standard disinfection methods and can survive for extended periods in the healthcare environment.

Bacterial flagella

– Thread-like appendages composed of flagellin protein that provide motility to bacteria. Flagella are composed of three parts: the filament, hook, and basal body. The basal body is embedded in the cell wall and functions as a rotary motor. Bacteria are classified by flagellar arrangement: monotrichous (single flagellum), lophotrichous (tuft at one pole), amphitrichous (flagella at both poles), and peritrichous (flagella distributed over entire surface). Flagella are important for chemotaxis, allowing bacteria to move towards nutrients and away from harmful substances, and they also contribute to pathogenesis by enabling bacteria to reach host tissues.

Bacterial genetics

– The study of genes and genetic processes in bacteria. Bacteria have a single circular chromosome and may contain plasmids, which are extrachromosomal DNA elements. Bacterial genes are organized into operons, which are clusters of genes regulated as a unit. Genetic variation occurs through mutation, transformation, transduction, and conjugation. Horizontal gene transfer allows rapid spread of antibiotic resistance and virulence factors. The bacterial genome is compact with little non-coding DNA, and its analysis provides insights into metabolic capabilities, virulence potential, and evolutionary relationships between strains.

Bacterial growth phases

– The stages of bacterial growth in culture. The lag phase is when bacteria adapt to their environment. The log (exponential) phase is when bacteria multiply at their maximum rate. The stationary phase is when nutrient depletion and waste accumulation limit growth. The death phase is when bacteria die. Understanding growth phases is essential for antimicrobial susceptibility testing and understanding pathogenesis. Many antibiotics are most effective during the log phase.

Bacterial metabolism

– The chemical processes that occur within bacteria to maintain life. Bacteria have diverse metabolic capabilities including aerobic and anaerobic respiration, fermentation, and photosynthesis. Metabolic pathways are targets for antimicrobial drugs. Understanding bacterial metabolism is essential for developing new antibiotics and understanding pathogenesis. Metabolic flexibility allows bacteria to survive in diverse environments.

Bacterial morphology

– The shape and arrangement of bacteria as observed under microscopy. Basic shapes include cocci (spheres), bacilli (rods), and spirilla (spirals). Cocci can be arranged in pairs (diplococci), chains (streptococci), clusters (staphylococci), or tetrads. Bacilli can be single, paired, or arranged in chains. Spiral bacteria include vibrios (curved), spirilla (rigid spirals), and spirochetes (flexible spirals). Morphology is a fundamental characteristic for bacterial identification. The size of bacteria typically ranges from 0.5 to 5 micrometres, making them significantly smaller than eukaryotic cells.

Bacterial pathogenesis

– The mechanisms by which bacteria cause disease. Pathogenesis involves adhesion, invasion, toxin production, and immune evasion. Adhesins mediate attachment to host cells. Invasins allow penetration of tissues. Toxins damage host cells. Immune evasion mechanisms include capsules, biofilms, and intracellular survival. Understanding pathogenesis is essential for developing new antimicrobial therapies and vaccines. The study of pathogenesis has identified virulence factors that are targets for drug development, including anti-virulence therapies that disarm pathogens without killing them, potentially reducing selective pressure for resistance.

Bacterial pili

– Hair-like appendages on the bacterial surface composed of pilin protein. Pili are shorter and thicker than flagella. Type I pili mediate adherence to host cells and are important virulence factors. Type IV pili are involved in twitching motility and DNA uptake during transformation. F-pili (sex pili) facilitate conjugation between bacteria. Pili are antigenic (K antigens) and can stimulate immune responses. They are too thin to be visible with light microscopy. Pili are essential for colonisation and biofilm formation, making them important targets for vaccine development and anti-adhesion therapies.

Bacterial toxins

– Substances produced by bacteria that cause damage to host cells. There are two major categories: exotoxins and endotoxins. Exotoxins are proteins secreted by bacteria that have specific toxic effects. Endotoxins are lipopolysaccharide components of gram-negative bacterial outer membranes released during cell lysis. Exotoxins are classified into types based on mechanism of action: A-B toxins, membrane-disrupting toxins, and superantigens. Endotoxin causes fever, hypotension, and disseminated intravascular coagulation through activation of the innate immune system, and is a key mediator of septic shock in gram-negative bacterial infections.

Bacterial transduction

– The transfer of bacterial DNA from one bacterium to another through bacteriophages (bacterial viruses). There are two types: generalized transduction, which occurs during the lytic cycle when random bacterial DNA is packaged into phage particles, and specialized transduction, which occurs during the lysogenic cycle when specific genes adjacent to the prophage are transferred. Transduction is an important mechanism of horizontal gene transfer and can spread antibiotic resistance genes and virulence factors between bacterial strains, contributing to the evolution of pathogenic bacteria.

Bacterial transformation

– The uptake of free DNA from the environment by competent bacteria. Transformation occurs naturally in some bacterial species including Streptococcus pneumoniae, Haemophilus influenzae, and

Neisseria. Competence is the ability to take up DNA and is regulated by specific genes. The DNA can be incorporated into the bacterial chromosome through homologous recombination. Transformation is an important mechanism of horizontal gene transfer and can spread antibiotic resistance genes. Artificial transformation is widely used in molecular biology to introduce engineered plasmids into bacteria for research and industrial applications.

Bacterial virulence factors

– Molecules produced by bacteria that contribute to their ability to cause disease. Virulence factors include adhesins, invasins, toxins, capsules, and enzymes. Adhesins mediate attachment to host cells. Toxins damage host cells directly. Capsules protect against phagocytosis. Enzymes degrade host tissues. Understanding virulence factors is essential for developing new antimicrobial therapies and vaccines. Virulence factors are potential targets for drug and vaccine development.

Bactericide

– An agent that kills bacteria rather than merely inhibiting their growth (bacteriostatic). Bactericidal antibiotics include beta-lactams (penicillins, cephalosporins, carbapenems), aminoglycosides, fluoroquinolones, metronidazole, and vancomycin. The distinction is concentration-dependent and organism-specific; some drugs are bactericidal for certain organisms and bacteriostatic for others. Bactericidal activity is preferred for endocarditis, meningitis, and neutropenic fever.

Bacteriology

– The branch of microbiology concerned with the study of bacteria, including their morphology, genetics, physiology, ecology, and clinical significance. Encompasses bacterial taxonomy, identification methods (Gram stain, culture, biochemical tests, MALDI-TOF, 16S rRNA sequencing), pathogenesis mechanisms, antibiotic susceptibility testing, and the role of bacteria in health (microbiome) and disease. Foundational to clinical infectious disease practice and infection control.

Bacteriophages

– Viruses that specifically infect and replicate within bacteria, also known as phages. Each phage type typically infects a narrow range of bacterial strains. Phages attach to bacterial surface receptors and inject their genome, hijacking bacterial machinery to produce progeny phages that lyse the host cell. Phage therapy, using lytic phages to treat multidrug-resistant bacterial infections, is a re-emerging antimicrobial strategy. Phages are also essential tools in molecular biology (lambda, M13, T4).

Biofilm-associated infections

– Infections caused by bacteria growing in biofilms on medical devices or tissues. Biofilms are structured communities of bacteria embedded in extracellular polymeric substances. Bacteria in biofilms are more resistant to antibiotics and host immune responses. Common biofilm-associated infections include catheter-associated urinary tract infections, ventilator-associated pneumonia, and prosthetic joint infections. Prevention through anti-biofilm strategies and device management is essential. Understanding biofilm-associated infections is essential for developing effective prevention and treatment strategies, including antimicrobial lock therapy and biofilm-disrupting agents.

C. difficile

– Clostridioides difficile, a gram-positive, spore-forming bacterium that causes antibiotic-associated diarrhea and colitis. C. difficile produces toxins A and B that cause colonic inflammation. Risk factors include antibiotic exposure, proton pump inhibitor use, and advanced age. Treatment includes vancomycin or fidaxomicin. Recurrence occurs in approximately 25% of cases. fecal microbiota transplantation is highly effective for multiply recurrent infections. Prevention through antibiotic stewardship, environmental cleaning with sporicidal agents, and isolation of affected patients is essential to control C. difficile transmission in healthcare settings.

Chikungunya virus

– A positive-sense RNA virus belonging to the Togaviridae family that causes fever and severe joint pain. The virus is transmitted by Aedes mosquitoes. Symptoms include fever, rash, and polyarthralgia that can be debilitating. Most patients recover fully, but some develop chronic joint pain lasting months to years. Diagnosis is made by PCR or serology. Treatment is supportive with pain management. There is no specific antiviral treatment. Prevention through mosquito control is important. The virus continues to cause outbreaks in tropical and subtropical regions where vector control is inadequate, and travellers returning from endemic areas may import the disease.

Clostridioides difficile pathogenesis

– The mechanisms by which C. difficile causes disease. The organism produces toxins A and B that damage the colonic mucosa. Antibiotic disruption of normal gut flora allows C. difficile to colonize and produce toxins. The organism forms spores that are resistant to disinfectants and can persist in the environment. Understanding C. difficile pathogenesis is essential for developing new treatments and prevention strategies. The organism is a major cause of healthcare-associated diarrhea.

Clostridium botulinum

– A gram-positive, spore-forming, anaerobic rod that produces botulinum toxin, the most potent toxin known. The toxin causes flaccid paralysis by blocking acetylcholine release at neuromuscular junctions. Infant botulism occurs from ingestion of spores in honey or soil. Wound botulism occurs from injection of contaminated heroin. Food-borne botulism occurs from ingestion of preformed toxin in improperly preserved food. Diagnosis is clinical with toxin detection. Treatment includes botulinum antitoxin to neutralise circulating toxin and intensive supportive care, including mechanical ventilation if respiratory muscle paralysis occurs.

Clostridium perfringens

– A gram-positive, spore-forming, anaerobic rod that causes gas gangrene (clostridial myonecrosis) and food poisoning. The organism produces alpha-toxin, which causes tissue necrosis. Gas gangrene presents with crepitus, tissue necrosis, and systemic toxicity. Food poisoning presents with acute onset of diarrhea and abdominal cramps. Diagnosis involves culture and toxin detection. Treatment includes surgical debridement, hyperbaric oxygen, and penicillin. The organism is part of the normal gastrointestinal flora in some individuals, but when introduced into deep tissues through trauma or surgery, it can cause rapidly progressive necrotising infection.

Clostridium tetani

– A gram-positive, spore-forming, anaerobic rod that causes tetanus. The organism produces tetanospasmin, a neurotoxin that causes spastic paralysis. Spores are found in soil and enter through wound contamination. The toxin blocks inhibitory neurotransmitters causing uncontrolled muscle contractions. Symptoms include trismus (lockjaw), opisthotonus, and respiratory failure. Diagnosis is clinical. Treatment includes metronidazole, tetanus immune globulin, and supportive care. Prevention through vaccination with tetanus toxoid and appropriate wound management is highly effective.

Contact precautions

– Infection control measures used for patients colonised or infected with multidrug-resistant organisms. Precautions include gown and glove use, patient placement in single rooms, and dedicated equipment. Contact precautions are used for MRSA, VRE, CRE, and other resistant organisms. Hand hygiene remains essential in addition to contact precautions. The duration of precautions depends on the organism and clinical situation. Understanding contact precautions is essential for preventing transmission of multidrug-resistant organisms in healthcare settings and reducing the burden of healthcare-associated infections.

Coxiella burnetii

– An obligate intracellular bacterium that causes Q fever, a zoonotic infection transmitted from animals to humans. The organism is transmitted through inhalation of contaminated aerosols or ingestion of contaminated dairy products. Acute Q fever presents with fever, headache, and pneumonia. Chronic Q fever can cause endocarditis. Diagnosis involves serology and PCR. Treatment includes doxycycline for acute disease and doxycycline with hydroxychloroquine for chronic endocarditis. Prevention through pasteurisation of dairy products, protective measures for abattoir workers and veterinarians, and awareness of the disease in endemic regions is essential for reducing transmission.

CRE

– Carbapenem-resistant Enterobacteriaceae, resistant to carbapenem antibiotics through production of carbapenemases including KPC, NDM, and OXA-48-like enzymes. CRE are a major public health threat due to limited treatment options. Infections are associated with high mortality rates. Treatment options include ceftazidime-avibactam, meropenem-vaborbactam, and aminoglycosides. Infection prevention and antimicrobial stewardship are essential to control spread. Active surveillance may be necessary in high-risk settings such as intensive care units to detect colonised patients and implement timely infection control measures.

Cytomegalovirus

– A herpesvirus that causes significant disease in immunocompromised patients including transplant recipients and HIV-infected individuals. CMV infects a wide range of cell types and establishes latent infections in myeloid progenitor cells. Primary infection is usually asymptomatic in immunocompetent individuals. Reactivation can cause retinitis, colitis, oesophagitis, and pneumonitis in immunocompromised patients. Congenital CMV infection is the most common viral cause of birth defects. Diagnosis involves PCR, antigen detection, viral culture, and serology, with treatment including ganciclovir, valganciclovir, or foscarnet for active disease in immunocompromised patients.

Dengue virus

– A positive-sense RNA virus belonging to the Flaviviridae family that causes dengue fever. The virus has four serotypes, and infection with one serotype provides lifelong immunity to that serotype but only temporary cross-protection against others. Secondary infection with a different serotype can cause severe dengue hemorrhagic fever or dengue shock syndrome due to antibody-dependent enhancement. The virus is transmitted by Aedes mosquitoes. Diagnosis is made by PCR or antigen detection. Treatment is supportive with careful fluid management and avoidance of non-steroidal anti-inflammatory drugs due to the risk of bleeding complications.

Diagnostic microbiology

– The laboratory identification of microorganisms causing disease. Methods include microscopy, culture, biochemical testing, serology, and molecular methods. Rapid diagnostic tests provide same-day results. MALDI-TOF mass spectrometry revolutionized bacterial identification. PCR and nucleic acid amplification tests are essential for diagnosing infections caused by fastidious organisms. Understanding diagnostic microbiology is essential for appropriate antimicrobial therapy.

Disinfection

– The elimination of most pathogenic microorganisms on inanimate surfaces. Disinfection does not eliminate all forms of microbial life, particularly spores. High-level disinfection eliminates all microorganisms except spores. Intermediate-level disinfection eliminates mycobacteria, most viruses, and bacteria. Low-level disinfection eliminates most bacteria and some viruses. Chemical disinfectants include alcohols, bleach, quaternary ammonium compounds, and phenolics. The level of disinfection required depends on the intended use of the medical device, with critical items requiring sterilisation, semi-critical items requiring high-level disinfection, and non-critical items requiring intermediate or low-level disinfection.

Droplet precautions

– Infection control measures used for pathogens transmitted through respiratory droplets. Precautions include surgical masks, patient placement in single rooms, and spatial separation. Droplet precautions are used for influenza, pertussis, and meningococcal infections. The distance for spatial separation is typically 3-6 feet. Understanding droplet precautions is essential for preventing transmission of respiratory infections in healthcare settings.

Ebola virus

– A filovirus that causes Ebola virus disease, a severe hemorrhagic fever. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage. Diagnosis is made by PCR and antigen detection. Treatment is supportive with rehydration and management of complications. The rVSV-ZEBOV vaccine is available. Outbreaks have occurred primarily in Africa with significant public health impact. Prevention through infection control measures, including isolation of affected patients, use of personal protective equipment, and safe burial practices, is critical to containing Ebola outbreaks.

Echinococcus granulosus

– A tapeworm that causes hydatid disease (cystic echinococcosis). The adult worm infects dogs, and humans are accidental intermediate hosts. Infection occurs through ingestion of eggs from dog feces. Larval cysts form in the liver and lungs, causing space-occupying lesions. Diagnosis involves imaging showing characteristic cysts and serology. Treatment includes surgical removal or percutaneous aspiration with albendazole coverage. Puncture of cysts can cause anaphylaxis. Prevention through regular deworming of dogs and proper disposal of dog feces is essential in endemic areas to break the transmission cycle.

Ehrlichia and Anaplasma

– Obligate intracellular bacteria transmitted by ticks that cause ehrlichiosis and anaplasmosis. Ehrlichia chaffeensis causes human monocytic ehrlichiosis. Anaplasma phagocytophilum causes human granulocytic anaplasmosis. The organisms infect white blood cells (Ehrlichia) or granulocytes (Anaplasma). Symptoms include fever, headache, and myalgia. Diagnosis involves PCR and serology. Treatment includes doxycycline. Prevention through tick avoidance is essential. These infections can be severe in immunocompromised patients and may require prolonged antibiotic therapy to prevent relapse.

Endemic fungi

– Fungi that are found in specific geographic regions and cause systemic infections in immunocompetent individuals. Endemic fungi include Histoplasma capsulatum (Ohio and Mississippi River valleys), Blastomyces dermatitidis (Great Lakes region), Coccidioides immitis (southwestern United States), and Paracoccidioides brasiliensis (Central and South America). These fungi are dimorphic, existing as moulds in the environment and yeasts in tissue. Infections are acquired through inhalation of spores. Diagnosis involves culture, histopathology, and antigen detection in tissue specimens, with treatment including azole antifungals or amphotericin B depending on the species and severity.

Endotoxins

– Lipopolysaccharide components of gram-negative bacterial outer membranes released during cell lysis or bacterial growth. Endotoxin is composed of lipid A (the toxic component), core oligosaccharide, and O-antigen. It activates macrophages through toll-like receptor 4, leading to cytokine release. Clinical effects include fever, hypotension, and disseminated intravascular coagulation. Endotoxin is heat-stable and difficult to remove. All gram-negative bacteria contain endotoxin, but potency varies between species, with Neisseria meningitidis and Escherichia coli producing particularly potent endotoxin.

Entamoeba histolytica

– A protozoan parasite that causes amoebiasis, a gastrointestinal and extra-intestinal infection transmitted via the fecal-oral route. The organism exists in two forms: the infectious cyst (transmitted in contaminated food or water) and the invasive trophozoite. Trophozoites colonise the large intestine, where they can adhere to colonic epithelium via a galactose-binding lectin and cause tissue invasion, often producing characteristic flask-shaped ulcers. In most cases, infection is asymptomatic; however, invasive disease can cause amoebic dysentery (bloody diarrhea with mucus, abdominal pain, and tenesmus). Extraintestinal spread most commonly results in amoebic liver abscess. Diagnosis involves microscopy demonstrating trophozoites with ingested red blood cells, antigen detection, PCR, and serology for extra-intestinal disease. Treatment includes tinidazole or metronidazole followed by a luminal agent (paromomycin or diloxanide furoate) to eradicate cyst carriage.

Epstein-Barr virus

– A herpesvirus that causes infectious mononucleosis and is associated with several malignancies including Burkitt lymphoma, nasopharyngeal carcinoma, and Hodgkin lymphoma. EBV infects B cells and establishes latent infections. The virus encodes several oncoproteins including LMP1 and EBNA2. Primary infection is often asymptomatic in childhood but can cause mononucleosis in adolescents and adults. Diagnosis is confirmed by heterophile antibody test or EBV-specific serology. There is no specific antiviral treatment for EBV, and management of infectious mononucleosis is supportive, with corticosteroids reserved for airway compromise or severe complications.

ESBL

– Extended-spectrum beta-lactamase, enzymes that hydrolyse penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae, particularly Klebsiella pneumoniae and E. coli. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems. Infections are associated with increased morbidity and mortality. Treatment typically requires carbapenems or other agents active against ESBL-producing organisms. Stewardship strategies are essential to limit the spread of ESBL-producing organisms and preserve the efficacy of remaining treatment options.

Escherichia coli

– A gram-negative rod that is part of the normal intestinal flora but can cause various infections. Uropathogenic E. coli causes urinary tract infections. Enterotoxigenic E. coli causes traveller's diarrhea. Enterohemorrhagic E. coli (O157:H7) causes bloody diarrhea and hemolytic uraemic syndrome. ESBL-producing E. coli is an increasing cause of healthcare-associated infections. The organism is identified by lactose fermentation and biochemical tests. Treatment depends on the infection site and susceptibility profile, with increasing resistance to commonly used antibiotics necessitating tailored therapy based on local resistance patterns.

Exotoxins

– Proteins secreted by bacteria that have specific toxic effects on host cells. Exotoxins are generally more potent than endotoxins and have specific mechanisms of action. A-B toxins consist of an active (A) component and a binding (B) component. Examples include diphtheria toxin, cholera toxin, and botulinum toxin. Membrane-disrupting toxins include haemolysins and leukocidins that damage cell membranes. Superantigens including toxic shock syndrome toxin-1 cause non-specific T cell activation. Exotoxins are a major focus of vaccine development, as demonstrated by toxoid vaccines for diphtheria and tetanus, and they are also used in biomedical research for their specific cellular targets.

Extracellular pathogens

– Microorganisms that survive and replicate outside host cells. Extracellular bacteria include Streptococcus, Staphylococcus, and Escherichia coli. These pathogens are typically killed by antibodies and complement. They produce toxins and enzymes to damage host tissues. Understanding extracellular pathogenesis is essential for developing vaccines that generate protective antibodies. Extracellular pathogens are typically susceptible to antibiotics that can reach high concentrations in extracellular fluids and are effectively cleared by antibody-mediated immune mechanisms.

Filariasis

– A group of nematode infections transmitted by mosquitoes and other vectors. The major species causing lymphatic filariasis are Wuchereria bancrofti, Brugia malayi, and Brugia timori. Adult worms reside in lymphatic vessels causing lymphoedema and elephantiasis. Diagnosis involves microscopy of blood collected at night when microfilariae are circulating. Treatment includes diethylcarbamazine (DEC). Prevention through mosquito control is essential. Onchocerciasis (river blindness) is caused by Onchocerca volvulus, transmitted by black flies, and is treated with ivermectin.

Filoviruses

– A family of negative-sense RNA viruses that includes Ebola and Marburg viruses. These viruses cause severe hemorrhagic fever with high mortality rates. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage. Diagnosis is made by PCR and antigen detection. Treatment is supportive with rehydration and management of complications. The rVSV-ZEBOV vaccine is available for Ebola virus. Prevention through infection control measures, including isolation and barrier nursing, is essential for managing filovirus outbreaks in healthcare settings.

Francisella tularensis

– A gram-negative coccobacillus that causes tularemia, a zoonotic infection transmitted by ticks, deer flies, or direct contact with infected animals. The organism can be acquired through the skin, respiratory tract, or gastrointestinal tract. Ulceroglandular tularemia presents with skin ulcer and regional lymphadenopathy. Pneumonic tularemia can be severe. Diagnosis involves serology and culture (requires special media). Treatment includes streptomycin or gentamicin. Tularemia is a rare infection that is considered a potential bioterrorism agent due to its low infectious dose and ease of aerosolisation.

Fungal pathogenesis

– The mechanisms by which fungi cause disease. Fungal pathogenesis involves adhesion, invasion, and evasion of host defences. Adhesins mediate attachment to host cells. Invasive hyphae penetrate tissues. Capsules and melanin protect against immune responses. Biofilm formation provides protection against antifungal agents. Immunocompromised individuals are at increased risk for invasive fungal infections. Understanding fungal pathogenesis is essential for developing new antifungal therapies and vaccines, particularly given the increasing population of immunocompromised patients at risk for invasive fungal disease.

Germ

– A colloquial term for a microorganism that causes disease.

Giardia lamblia

– A protozoan parasite that causes giardiasis, a diarrhoeal illness transmitted through the fecal-oral route. The organism has a trophozoite and cyst stage. Infection occurs through ingestion of cysts in contaminated water. Symptoms include foul-smelling diarrhea, abdominal cramps, and bloating. Diagnosis involves stool microscopy showing trophozoites or cysts, and antigen detection. Treatment includes metronidazole or tinidazole. Prevention through water treatment and sanitation is essential. Giardia lamblia is a common cause of waterborne outbreaks and is highly prevalent in areas with poor sanitation and inadequate water treatment.

Gram stain

– A differential staining technique that classifies bacteria into two major groups based on cell wall structure. The procedure involves crystal violet staining, iodine mordant, alcohol decolorisation, and safranin counterstain. Gram-positive bacteria retain the crystal violet-iodine complex due to their thick peptidoglycan layer and appear purple. Gram-negative bacteria lose the crystal violet and take up the counterstain, appearing pink or red. The Gram stain is the most important initial step in bacterial identification and provides critical information for guiding empiric antimicrobial therapy while cultures are pending.

Haemophilus influenzae

– A gram-negative coccobacillus that causes meningitis, epiglottitis, pneumonia, and other infections. The encapsulated type b (Hib) strain was the most virulent, causing invasive disease in children. The Hib conjugate vaccine has dramatically reduced the incidence of invasive Hib disease. Non-typeable strains cause otitis media, sinusitis, and exacerbations of chronic bronchitis. The organism requires X factor (haemin) and V factor (NAD) for growth. Diagnosis involves culture on chocolate agar. Treatment includes broad-spectrum antibiotics such as third-generation cephalosporins, and prevention of invasive Hib disease has been achieved through routine childhood vaccination.

Hand hygiene

– The most effective measure for preventing healthcare-associated infections. Hand hygiene includes handwashing with soap and water and hand rubbing with alcohol-based rubs. Alcohol-based rubs are preferred for routine hand hygiene as they are faster and more effective than soap and water. Soap and water should be used when hands are visibly soiled or when caring for patients with C. difficile. The WHO has established five moments for hand hygiene in healthcare settings. Hand hygiene compliance is monitored in healthcare settings as a key quality indicator, and improvement strategies include education, feedback, and the availability of alcohol-based hand rub at the point of care.

Helminths

– Parasitic worms that cause disease in humans. The major groups are nematodes (roundworms), cestodes (tapeworms), and trematodes (flukes). Helminths have complex life cycles often involving intermediate hosts. Infection occurs through ingestion of eggs or larvae, penetration of skin, or insect vectors. Diagnosis involves stool microscopy for eggs and larvae, serology, and imaging. Treatment includes anthelmintic agents such as albendazole, mebendazole, and praziquantel. Helminth infections are a major cause of morbidity in tropical and subtropical regions, contributing to malnutrition, anemia, and impaired cognitive development in children.

Herpes simplex virus

– A double-stranded DNA virus belonging to the Herpesviridae family that causes oral and genital herpes. HSV has two serotypes: HSV-1, which typically causes oral herpes, and HSV-2, which typically causes genital herpes. The virus establishes latent infections in sensory ganglia and can reactivate periodically. Primary infection is often asymptomatic. Reactivation causes recurrent outbreaks of vesicular lesions. Antiviral treatment with acyclovir, valacyclovir, or famciclovir can reduce the frequency and severity of recurrent episodes and is recommended for patients with frequent or severe outbreaks.

Hookworms

– Nematodes including Ancylostoma duodenale and Necator americanus that cause hookworm infection. Infection occurs through skin penetration by filariform larvae, typically when walking barefoot on contaminated soil. Adult worms attach to the intestinal mucosa and feed on blood, causing iron deficiency anemia. Diagnosis involves stool microscopy for eggs. Treatment includes albendazole or mebendazole. Prevention through sanitation, wearing shoes, and avoiding soil contact is essential. Hookworm infection remains a significant public health problem in resource-limited settings, particularly among children and agricultural workers in tropical regions.

Horizontal gene transfer

– The transfer of genetic material between bacteria through mechanisms other than parent-to-offspring transmission. The three mechanisms are transformation (uptake of free DNA), transduction (transfer via bacteriophages), and conjugation (direct cell-to-cell transfer). Horizontal gene transfer allows rapid spread of antibiotic resistance genes and virulence factors. Understanding this process is essential for tracking the spread of resistance and developing strategies to combat it.

Hospital-acquired infections

– Infections that develop during or after hospitalisation. Major types include central line-associated bloodstream infections, catheter-associated urinary tract infections, ventilator-associated pneumonia, and surgical site infections. Risk factors include invasive devices, prolonged hospitalisation, and antimicrobial use. Prevention through infection control measures including hand hygiene, device bundles, and antimicrobial stewardship is essential. Hospital-acquired infections are associated with increased morbidity, mortality, length of stay, and healthcare costs, and represent a major patient safety concern.

Human immunodeficiency virus

– A retrovirus belonging to the Lentivirus genus that causes acquired immunodeficiency syndrome (AIDS). HIV has two species: HIV-1 and HIV-2. The virus targets CD4+ T cells, macrophages, and dendritic cells. HIV has a complex genome encoding structural proteins, enzymes (reverse transcriptase, integrase, protease), and regulatory proteins. The virus integrates into the host genome as a provirus and can establish latent reservoirs. Without treatment, HIV infection leads to progressive CD4+ T cell depletion, increasing susceptibility to opportunistic infections and malignancies, and ultimately resulting in AIDS and death.

Human microbiome

– The collection of microorganisms that live on and in the human body. The microbiome includes bacteria, viruses, fungi, and archaea. The gut microbiome is the most studied and plays a role in digestion, immune function, and disease. Dysbiosis (alteration of the microbiome) is associated with inflammatory bowel disease, obesity, and infections. Understanding the microbiome is essential for developing new therapeutic approaches including fecal microbiota transplantation and probiotics.

Immunodeficiency and infections

– Immunocompromised individuals are at increased risk for opportunistic infections. Primary immunodeficiencies are genetic disorders affecting immune function. Secondary immunodeficiencies include HIV/AIDS, organ transplantation, chemotherapy, and corticosteroid use. Common opportunistic infections include Pneumocystis jirovecii, CMV, Aspergillus, and Cryptococcus. Prevention through antimicrobial prophylaxis and vaccination is essential. Early diagnosis and treatment of opportunistic infections is critical for improving outcomes in immunocompromised patients, and prophylactic antimicrobial therapy should be tailored to the specific immune defect.

Influenza virus

– A segmented, negative-sense RNA virus belonging to the Orthomyxoviridae family. Influenza causes seasonal epidemics and occasional pandemics of respiratory illness. The virus has two surface glycoproteins: hemagglutinin (HA), which mediates attachment, and neuraminidase (NA), which facilitates release. Influenza A infects humans, birds, and other animals, while influenza B infects only humans. Antigenic drift (point mutations in HA and NA) causes seasonal epidemics, while antigenic shift (reassortment of genome segments) can generate novel pandemic strains to which the population has little pre-existing immunity.

Innate immune evasion

– Mechanisms by which pathogens evade the innate immune system. Strategies include capsules that prevent phagocytosis, biofilms that protect against immune cells, and intracellular survival within macrophages. Some bacteria produce enzymes that degrade complement components. Others modify their surface molecules to avoid recognition. Understanding immune evasion is essential for developing new therapeutic approaches and vaccines. Pathogens that evade innate immunity can establish infections before adaptive immune responses are mounted, making early recognition and rapid diagnosis critical for effective treatment.

Integrons

– Genetic elements that capture and express gene cassettes through site-specific recombination. Integrons contain an integrase gene and a recombination site for gene cassette insertion. They are important vehicles for the spread of antibiotic resistance genes, particularly in gram-negative bacteria. Class 1 integrons are the most clinically significant and are associated with multidrug resistance. Gene cassettes can contain multiple resistance genes, allowing accumulation of resistance. Integrons play a key role in the dissemination of antibiotic resistance in clinical settings, particularly among gram-negative pathogens where they contribute to the emergence of multidrug-resistant strains.

Intracellular pathogens

– Microorganisms that survive and replicate within host cells. Intracellular bacteria include Mycobacterium, Listeria, and Chlamydia. Viruses are obligate intracellular pathogens. Intracellular survival protects pathogens from antibiotics and immune responses. Strategies for intracellular survival include avoiding lysosomal fusion and surviving within phagosomes. Understanding intracellular pathogenesis is essential for developing new therapeutic approaches. Intracellular pathogens require treatment with antimicrobial agents that can penetrate host cell membranes, such as macrolides, tetracyclines, and fluoroquinolones, and often require prolonged therapy to achieve eradication.

Legionella pneumophila

– A gram-negative rod that causes Legionnaires' disease, a severe pneumonia, and Pontiac fever, a milder illness. The organism is found in aquatic environments and is transmitted through inhalation of contaminated water droplets. Risk factors include age, smoking, and immunosuppression. Diagnosis involves urine antigen testing and culture on BCYE agar. Treatment includes fluoroquinolones or azithromycin. Prevention through water system management is essential. Legionella can colonise building water systems, including cooling towers, hot water tanks, and decorative fountains, and outbreaks require identification and decontamination of the environmental source.

Leishmania donovani

– A protozoan parasite that causes visceral leishmaniasis (kala-azar), transmitted by the bite of infected female phlebotomine sandflies. The organism exists as flagellated promastigotes in the sandfly gut and as non-flagellated amastigotes within human macrophages. After inoculation, promastigotes are phagocytosed by macrophages and transform into amastigotes, which multiply intracellularly and disseminate throughout the reticuloendothelial system. Visceral leishmaniasis presents with fever, hepatosplenomegaly, pancytopenia, weight loss, and hypergammaglobulinaemia; it is fatal if untreated. Cutaneous leishmaniasis is caused by other Leishmania species (L. major, L. tropica, L. mexicana), presenting as skin ulcers. Diagnosis involves microscopy of tissue smears (amastigotes in macrophages), culture, and PCR. Treatment includes liposomal amphotericin B and miltefosine.

Listeria monocytogenes

– A gram-positive, facultatively anaerobic rod that causes listeriosis, a foodborne infection. The organism is transmitted through contaminated food, particularly deli meats, soft cheeses, and ready-to-eat foods. It can grow at refrigeration temperatures. Listeriosis is most severe in pregnant women, neonates, and immunocompromised patients. The organism crosses the intestinal barrier and can cross the placenta. Diagnosis involves culture and molecular methods. Treatment includes ampicillin or trimethoprim-sulfamethoxazole, and prompt therapy is critical given the high mortality rate of invasive listeriosis, particularly in neonates and immunocompromised patients.

Measles virus

– A negative-sense RNA virus that causes measles, a highly contagious respiratory illness. The virus is transmitted by respiratory droplets and is one of the most infectious pathogens known. Symptoms include fever, cough, coryza, conjunctivitis, and rash. Complications include pneumonia, encephalitis, and subacute sclerosing panencephalitis (SSPE). Diagnosis involves serology and PCR. Treatment is supportive. Prevention through vaccination with MMR vaccine is highly effective. Measles elimination remains a global health goal, but outbreaks continue to occur in communities with suboptimal vaccination coverage, highlighting the importance of maintaining high immunisation rates.

Mobile genetic elements

– Genetic elements that can move within and between genomes. These include plasmids, transposons, and integrons. Mobile genetic elements can carry antibiotic resistance genes and virulence factors. They facilitate horizontal gene transfer and contribute to the rapid evolution of bacteria. Understanding mobile genetic elements is essential for tracking the spread of resistance and developing new antimicrobial strategies.

MRSA

– Methicillin-resistant Staphylococcus aureus, resistant to beta-lactam antibiotics through acquisition of the mecA gene encoding PBP2a. MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, and linezolid. MRSA infections are associated with increased morbidity and mortality. Infection prevention includes hand hygiene, contact precautions, and decolonization protocols. The prevalence of MRSA varies by healthcare setting and geographic region, with some areas reporting endemic levels requiring ongoing surveillance and control efforts.

MRSA decolonization

– Strategies to eliminate MRSA carriage from the nares and skin of colonized individuals. Decolonization reduces the risk of infection and transmission. The most common regimen is intranasal mupirocin for 5 days. Chlorhexidine bathing is used for body decolonization. Decolonization is recommended for preoperative patients, those with recurrent infections, and during outbreaks. Eradication is not always successful, and recolonization can occur. Understanding decolonisation strategies is essential for reducing MRSA infection rates in high-risk populations and for preventing surgical site infections in colonised patients.

Mucormycosis

– A life-threatening fungal infection caused by Mucorales order fungi including Rhizopus, Mucor, and Rhizomucor species. The infection primarily affects immunocompromised patients, particularly those with uncontrolled diabetes mellitus and hematologic malignancies. The fungi invade blood vessels causing thrombosis and tissue infarction. Rhino-orbital-cerebral mucormycosis is the most common form. Diagnosis involves culture and microscopy showing broad, ribbon-like, pauciseptate hyphae. Treatment involves surgical debridement of necrotic tissue and antifungal therapy with amphotericin B or posaconazole, with early intervention critical for survival given the high mortality rate.

Mumps virus

– A paramyxovirus that causes mumps, a contagious respiratory illness. The virus is transmitted by respiratory droplets. Symptoms include fever, headache, and parotitis. Complications include orchitis, oophoritis, meningitis, and deafness. Diagnosis involves serology and PCR. Treatment is supportive. Prevention through vaccination with MMR vaccine is highly effective. Mumps outbreaks can occur in vaccinated populations, particularly in close-contact settings.

Mycobacterium avium complex

– A group of slowly growing mycobacteria that cause pulmonary disease and disseminated infection in immunocompromised patients. MAC is the most common nontuberculous mycobacterium causing pulmonary disease. Treatment for pulmonary MAC requires a macrolide (azithromycin or clarithromycin) combined with ethambutol and a rifamycin. Disseminated MAC in HIV patients requires macrolide therapy with antiretroviral therapy. The organism is ubiquitous in the environment. Diagnosis requires isolation of MAC from respiratory or tissue specimens, with molecular methods providing faster identification and susceptibility testing guiding treatment selection.

Mycobacterium leprae

– An acid-fast bacillus that causes leprosy (Hansen's disease), a chronic infection affecting the skin and peripheral nerves. The organism is transmitted through respiratory droplets, though the exact mechanism is not fully understood. The incubation period is long (2-5 years). Tuberculoid leprosy presents with few, well-defined skin lesions. Lepromatous leprosy presents with widespread skin involvement and nerve damage. Diagnosis involves skin biopsy showing acid-fast bacilli. Treatment includes a multi-drug regimen of dapsone, rifampicin, and clofazimine, with therapy duration ranging from 6 months for paucibacillary disease to 12 months or longer for multibacillary disease.

Mycobacterium tuberculosis

– An acid-fast bacillus that causes tuberculosis, a chronic respiratory infection. The organism is transmitted by respiratory droplets and primarily affects the lungs but can disseminate to other organs. Latent TB infection is asymptomatic and non-transmissible. Active TB presents with cough, haemoptysis, fever, and weight loss. Diagnosis involves sputum smear for acid-fast bacilli, culture, and nucleic acid amplification testing. Treatment requires multi-drug therapy for 6-9 months. Isoniazid, rifampicin, pyrazinamide, and ethambutol are used in the intensive phase, followed by isoniazid and rifampicin in the continuation phase, though the emergence of multidrug-resistant TB requires alternative regimens.

Mycology

– The branch of microbiology that studies fungi, including yeasts, moulds, and mushrooms. Medical mycology focuses on fungal pathogens that cause disease in humans. Fungal infections (mycoses) are classified as superficial, cutaneous, subcutaneous, or systemic. Immunocompromised individuals are at increased risk for invasive fungal infections. Common fungal pathogens include Candida, Aspergillus, Cryptococcus, and endemic fungi. Diagnosis involves culture, microscopy, antigen detection, and molecular methods, with rapid molecular tests improving time to diagnosis and facilitating early targeted therapy.

Neisseria gonorrhoeae

– A gram-negative diplococcus that causes gonorrhea, a sexually transmitted infection. The organism has developed resistance to many antibiotics including fluoroquinolones and macrolides. Current treatment recommendations include dual therapy with ceftriaxone and azithromycin, though monotherapy with ceftriaxone is now preferred in some guidelines. The organism colonises mucous membranes of the genitourinary tract, pharynx, and rectum. Complications include pelvic inflammatory disease, infertility, ectopic pregnancy, and disseminated infection, and antimicrobial resistance continues to threaten treatment efficacy.

Neisseria meningitidis

– A gram-negative diplococcus that causes meningitis and meningococcemia. The organism has a polysaccharide capsule that is the target of conjugate vaccines. Serogroups A, B, C, W, X, and Y cause most disease. Meningitis presents with fever, headache, neck stiffness, and altered mental status. Meningococcemia presents with petechial rash and septic shock. Diagnosis involves blood culture and PCR. Treatment includes ceftriaxone or penicillin. Close contacts require prophylaxis with rifampicin, ciprofloxacin, or ceftriaxone to prevent secondary cases following exposure to invasive meningococcal disease.

Norovirus

– A single-stranded RNA virus that is the leading cause of gastroenteritis outbreaks worldwide. The virus is highly contagious and can cause large outbreaks in closed settings including hospitals, cruise ships, and nursing homes. Norovirus causes acute onset of vomiting and diarrhea. Diagnosis is made by PCR or antigen detection in stool. Treatment is supportive with rehydration. The virus is highly stable in the environment and resistant to many disinfectants. There is no vaccine or specific antiviral treatment for norovirus, and management focuses on preventing dehydration through oral or intravenous rehydration.

Onchocerca volvulus

– A filarial nematode that causes onchocerciasis (river blindness), transmitted by repeated bites of infected blackflies (Simulium species) breeding near fast-flowing rivers. Adult worms reside in subcutaneous nodules (onchocercomata), where they can survive for years, producing microfilariae that migrate through the skin and eyes. The host inflammatory response to dying microfilariae causes the pathological manifestations: intense pruritus, dermatitis, depigmentation (leopard skin), lichenification, and, most importantly, progressive ocular damage leading to blindness (keratitis, iridocyclitis, chorioretinitis, optic atrophy). Diagnosis involves skin snip biopsy (microfilariae emerge when skin is placed in saline), slit-lamp examination of the cornea, and PCR. Treatment with ivermectin (annual or semi-annual mass drug administration) kills microfilariae but not adult worms. No safe, effective macrofilaricide currently exists.

Parasitology

– The branch of microbiology that studies parasites, including protozoa, helminths, and ectoparasites. Parasitic infections are a major global health problem, particularly in tropical and subtropical regions. Protozoan parasites include Plasmodium (malaria), Trypanosoma (sleeping sickness, Chagas disease), and Leishmania (leishmaniasis). Helminths include roundworms, tapeworms, and flukes. Diagnosis involves microscopy, serology, and molecular methods. Treatment depends on the parasite and includes antiparasitic agents such as metronidazole, praziquantel, and ivermectin, with prevention focusing on vector control, sanitation, and health education.

Plasmids

– Small, circular, extrachromosomal DNA elements that replicate independently of the bacterial chromosome. Plasmids carry non-essential genes that provide selective advantages including antibiotic resistance, virulence factors, and metabolic capabilities. R-plasmids carry antibiotic resistance genes and can spread resistance between bacteria. F-plasmids carry genes for conjugation. Plasmids are transferred between bacteria through conjugation, transformation, and transduction. They are important vehicles for the dissemination of antibiotic resistance in bacterial populations and are key targets for molecular surveillance of resistance mechanisms.

Poliovirus

– An enterovirus that causes poliomyelitis, a paralytic illness. The virus is transmitted by the fecal-oral route. Most infections are asymptomatic, but a small percentage cause acute flaccid paralysis. The virus attacks motor neurons in the spinal cord. Diagnosis involves stool culture and PCR. There is no specific treatment. Prevention through vaccination with inactivated (Salk) or oral (Sabin) polio vaccine is highly effective. Polio has been eliminated from most countries through vaccination efforts, but transmission continues in Afghanistan and Pakistan, and importation into polio-free areas remains a risk.

Polymicrobial infections

– Infections caused by multiple microbial species. Polymicrobial infections are common in wound infections, intra-abdominal infections, and respiratory infections. The interactions between organisms can be synergistic or antagonistic. Understanding polymicrobial infections is essential for selecting appropriate antimicrobial therapy. Culture and identification of all organisms in a polymicrobial infection can be challenging.

Post-exposure prophylaxis

– The administration of antimicrobial agents after exposure to a pathogen to prevent infection. PEP is used for HIV, hepatitis B, hepatitis C, rabies, and other infections. The timing of PEP is critical, with early administration being most effective. Understanding PEP is essential for managing occupational exposures and outbreaks. PEP guidelines vary by pathogen and exposure type.

Rabies

– A fatal viral encephalitis caused by rabies virus, transmitted through the bite of infected animals. After inoculation, the virus travels to the CNS via peripheral nerves. Symptoms include hydrophobia, aerophobia, and progressive neurological deterioration. Diagnosis is made by direct fluorescent antibody testing of brain tissue. Post-exposure prophylaxis with rabies vaccine and immunoglobulin is highly effective if administered promptly. There is no specific treatment once symptoms develop. Prevention through pre-exposure vaccination of at-risk individuals and animal control programmes, including vaccination of domestic dogs, is essential for reducing the global burden of rabies.

Respiratory viruses

– Viruses that cause respiratory tract infections. Major respiratory viruses include influenza, respiratory syncytial virus (RSV), parainfluenza virus, adenovirus, and human metapneumovirus. SARS-CoV-2 causes COVID-19. Symptoms range from mild upper respiratory illness to severe pneumonia. Diagnosis involves PCR, rapid antigen testing, and viral culture. Treatment is supportive with antiviral agents available for influenza and SARS-CoV-2. Prevention through vaccination and hygiene is essential. Respiratory viruses are responsible for significant morbidity and mortality worldwide, particularly in young children, older adults, and immunocompromised individuals.

Retrovirus

– An RNA virus that replicates by reverse transcribing its RNA genome into DNA, which is then integrated into the host cell genome. Family Retroviridae. Key enzyme: reverse transcriptase (RNA-dependent DNA polymerase) converts viral RNA into double-stranded DNA; integrase inserts the viral DNA (provirus) into the host chromosome. The provirus is replicated with the host cell genome and transcribed into new viral RNA and proteins. Genera: Lentivirus (HIV-1, HIV-2, SIV), Deltaretrovirus (HTLV-1, HTLV-2), Gammaretrovirus (murine leukemia virus), and Spumavirus (foamy viruses). Clinical significance: HIV causes AIDS (depletion of CD4+ T cells, opportunistic infections, malignancies). HTLV-1 causes adult T-cell leukemia/lymphoma and tropical spastic paraparesis (HAM/TSP). Antiretroviral therapy targets reverse transcriptase (NRTIs, NNRTIs), protease, integrase (integrase strand transfer inhibitors, INSTIs), and entry/fusion. Retroviruses are also used as gene therapy vectors (lentiviral vectors) due to their ability to stably integrate into the host genome.

Rhinovirus

– A genus of non-enveloped, single-stranded positive-sense RNA viruses in the Picornaviridae family, the most common cause of the common cold. Rhinoviruses have >160 identified serotypes, each with distinct surface proteins (viral capsid proteins VP1–VP4) that determine antigenicity and host immune response. Transmission: respiratory droplets and direct contact (contaminated hands, fomites). Incubation: 2–3 days. Clinical: nasal congestion, rhinorrhoea, sore throat, cough, sneezing, headache, and mild malaise—symptoms peak at 2–3 days and resolve in 7–10 days. Rhinoviruses replicate optimally at 33–35°C (cooler temperature of the nasal passages) and bind to ICAM-1 (intercellular adhesion molecule 1—major receptor for most serotypes) or LDL receptor (minor group). Immunity is serotype-specific and short-lived, explaining recurrent infections throughout life. Complications: acute sinusitis, acute otitis media (especially in children), exacerbation of asthma and COPD, and pneumonia (in immunocompromised patients). Diagnosis is clinical; PCR or viral culture is rarely needed. Treatment: supportive (rest, hydration, decongestants, NSAIDs, antihistamines). No specific antiviral is approved; pleconaril (viral capsid binder) is investigational. Prevention: hand hygiene, avoiding close contact with infected individuals.

Rickettsia

– A genus of obligate intracellular Gram-negative coccobacillary bacteria transmitted by arthropod vectors (ticks, fleas, lice, mites). Rickettsia are the causative agents of the spotted fever group (SFG) and typhus group (TG) rickettsioses. SFG: Rickettsia rickettsii (Rocky Mountain spotted fever—most severe, mortality 20–30% untreated), R. conorii (Mediterranean spotted fever, Boutonneuse fever), R. akari (rickettsialpox). TG: R. prowazekii (epidemic typhus—louse-borne, associated with war and poverty, mortality 10–60% untreated), R. typhi (murine/endemic typhus—flea-borne, milder). Pathophysiology: Rickettsia invade vascular endothelial cells, causing systemic vasculitis, increased vascular permeability, and microthrombi. Clinical: fever, severe headache, myalgia, and a characteristic rash (maculopapular, petechial, or eschar at the bite site—tache noire in SFG; centripetal spread in Rocky Mountain spotted fever; macular rash on the trunk spreading centrifugally in typhus). Diagnosis: serology (IFA—gold standard, Weil–Felix test—obsolete), PCR on skin biopsy or blood, immunohistochemistry of skin biopsy. Treatment: doxycycline 100 mg BD for 7–14 days (first-line for all rickettsioses, including children—short course does not cause tooth staining). Chloramphenicol is an alternative (pregnancy, doxycycline allergy). Prevention: tick avoidance, protective clothing, insect repellent (DEET), and prompt tick removal. No vaccine is available for most rickettsioses.

Rotavirus

– A double-stranded RNA virus that is the leading cause of severe gastroenteritis in children worldwide. The virus has a segmented genome and infects enterocytes of the small intestine. Rotavirus causes profuse watery diarrhea, vomiting, and dehydration.

Diagnosis is made by antigen detection in stool. Treatment is supportive with oral rehydration. Vaccination is available and recommended for infants. The virus is highly stable in the environment and transmitted through the fecal-oral route, with vaccination being the most effective preventive measure against severe disease.

Rubella virus

– A togavirus that causes rubella (German measles), a mild febrile illness with rash. The virus is transmitted by respiratory droplets. Rubella is most dangerous during pregnancy, causing congenital rubella syndrome (CRS) with deafness, cataracts, and cardiac defects. Diagnosis involves serology. Treatment is supportive. Prevention through vaccination with MMR vaccine is highly effective. Rubella has been eliminated from the Americas through vaccination. Congenital rubella syndrome is a devastating but preventable consequence of maternal rubella infection, and maintaining high vaccination coverage is essential for elimination.

Salmonella

– See Salmonella in Infectious Diseases.

Sarcoptes scabiei

– A mite that causes scabies, a contagious skin infestation transmitted by prolonged skin-to-skin contact. The female mite burrows into the stratum corneum, laying eggs along the burrow, which appears as a greyish, serpiginous line. The life cycle from egg to adult takes approximately 10-14 days. The host immune response to the mites, their eggs, and their feces (scybala) causes intense pruritus, particularly at night. The typical distribution includes interdigital finger webs, wrists, elbows, axillae, waist, and genitalia, sparing the face in adults. Crusted (Norwegian) scabies is a hyperinfestation form occurring in immunocompromised patients, characterized by thick crusted plaques containing thousands of mites. Diagnosis is clinical, confirmed by microscopy of skin scrapings from burrows (identification of mites, eggs, or scybala). Treatment involves permethrin 5% cream or oral ivermectin.

SARS-CoV-2

– A coronavirus belonging to the Coronaviridae family that causes COVID-19. SARS-CoV-2 has a positive-sense single-stranded RNA genome and a spike glycoprotein that mediates attachment to ACE2 receptors. The virus replicates in the cytoplasm using its own RNA-dependent RNA polymerase. Transmission occurs through respiratory droplets and aerosols. The disease ranges from mild respiratory symptoms to severe pneumonia and multi-organ failure. Vaccines targeting the spike protein have been developed and deployed globally, with mRNA, viral vector, and protein subunit platforms demonstrating high efficacy against severe disease. The rapid emergence of variants of concern necessitates ongoing vaccine updates and booster strategies.

Staphylococcus aureus

– A gram-positive, catalase-positive, coagulase-positive coccus that causes a wide range of infections including skin infections, pneumonia, endocarditis, and osteomyelitis. S. aureus produces numerous virulence factors including toxins, enzymes, and surface proteins. MRSA strains carry the mecA gene encoding PBP2a. The organism is part of the normal nasal flora. Diagnosis involves culture and coagulase testing. Treatment depends on susceptibility and includes beta-lactams for methicillin-susceptible strains and vancomycin, daptomycin, or linezolid for MRSA, with the choice guided by local epidemiology and patient factors.

Streptococcus agalactiae

– A gram-positive, beta-hemolytic streptococcus (Group B Streptococcus) that is a leading cause of neonatal sepsis, meningitis, and pneumonia. The organism colonises the gastrointestinal and genital tracts. Transmission occurs during birth. Risk factors include prematurity, prolonged rupture of membranes, and maternal colonisation. Screening and intrapartum antibiotic prophylaxis have reduced the incidence of early-onset GBS disease. The organism is identified by Lancefield Group B antigen and hippurate hydrolysis testing, and intrapartum antibiotic prophylaxis has been highly effective in preventing vertical transmission during labour.

Streptococcus pyogenes

– A gram-positive, beta-hemolytic streptococcus (Group A Streptococcus) that causes pharyngitis, skin infections, and post-infectious complications including rheumatic fever and glomerulonephritis. The organism produces streptolysins, erythrogenic toxin, and streptokinase. It is identified by bacitracin sensitivity and Lancefield Group A antigen. Pharyngitis presents with fever, sore throat, and exudates. Skin infections include impetigo and cellulitis. Severe invasive infections include necrotising fasciitis and streptococcal toxic shock syndrome, which require prompt recognition, aggressive surgical debridement, and appropriate antimicrobial therapy.

Toxoplasma gondii

– An obligate intracellular protozoan parasite that causes toxoplasmosis. The definitive host is the cat (family Felidae), which sheds environmentally resistant oocysts in feces. Humans are intermediate hosts, acquiring infection through ingestion of undercooked meat containing tissue cysts, ingestion of oocysts from contaminated soil or cat litter, or transplacentally from mother to fetus. In immunocompetent individuals, infection is usually asymptomatic or causes a mild flu-like illness with lymphadenopathy. Immunocompromised patients (HIV/AIDS, transplant recipients) are at risk of severe disease, most commonly toxoplasma encephalitis presenting with brain abscesses. Congenital toxoplasmosis can cause chorioretinitis, intracranial calcifications, hydrocephalus, and developmental delay. Diagnosis relies on serology (IgM, IgG) and PCR. Treatment involves pyrimethamine with sulfadiazine and folinic acid.

Transposons

– Mobile genetic elements that can move between different locations within the bacterial genome or between the chromosome and plasmids. Transposons carry genes including antibiotic resistance genes and can facilitate the spread of resistance. They are sometimes called "jumping genes." Transposons insert into DNA through a cut-and-paste or copy-and-paste mechanism. They can inactivate genes by inserting into coding regions. Composite transposons consist of insertion sequences flanking antibiotic resistance genes and facilitate their movement between replicons, contributing to the assembly of multidrug-resistant plasmids.

Travel medicine

– The branch of medicine dealing with health problems associated with international travel. Pre-travel consultation includes vaccinations, malaria prophylaxis, and travel advice. Common travel-related illnesses include traveller's diarrhea, malaria, and hepatitis A. High-risk destinations require specific vaccinations including yellow fever, typhoid, and Japanese encephalitis. Post-travel evaluation should consider tropical infections in returning travellers. Understanding travel medicine is essential for providing appropriate pre-travel advice, recognising imported diseases, and preventing the international spread of infectious diseases.

Treponema pallidum

– A spirochete that causes syphilis, a sexually transmitted infection. The organism is transmitted through sexual contact or transplacentally (congenital syphilis). Primary syphilis presents with a painless genital ulcer (chancre). Secondary syphilis presents with rash, mucous patches, and condylomata lata. Tertiary syphilis can involve the cardiovascular system, CNS, and skin. Diagnosis involves serology (RPR, FTA-ABS) and darkfield microscopy. Treatment includes penicillin G. Congenital syphilis remains a preventable cause of stillbirth and neonatal morbidity, and routine antenatal screening and treatment of maternal infection are essential for prevention.

Trypanosoma brucei

– A protozoan parasite that causes African trypanosomiasis (sleeping sickness), transmitted by the bite of the tsetse fly (Glossina species). Two subspecies cause human disease: T. b. gambiense (West African form, chronic, accounting for over 95% of reported cases) and T. b. rhodesiense (East African form, acute). The parasite is inoculated during a tsetse fly blood meal, multiplies in the blood and lymphatics (haemolymphatic stage), and eventually crosses the blood-brain barrier to invade the central nervous system (meningoencephalitic stage). The haemolymphatic stage presents with intermittent fever, headache, lymphadenopathy (classic Winterbottom sign – enlarged posterior cervical nodes), and chancre at the bite site. The meningoencephalitic stage causes progressive neurological deterioration, daytime somnolence with nighttime insomnia, ataxia, and coma. Diagnosis involves microscopy of blood and CSF, serology, and PCR. Treatment depends on stage: pentamidine or suramin for early disease; eflornithine or melarsoprol for late-stage CNS involvement.

Varicella-zoster virus

– A herpesvirus that causes chickenpox (varicella) and shingles (herpes zoster). VZV establishes latent infections in dorsal root ganglia and can reactivate decades later. Chickenpox is highly contagious and typically occurs in childhood. Reactivation as shingles causes painful dermatomal vesicular rash. Complications include postherpetic neuralgia and disseminated infection in immunocompromised patients. Antiviral treatment with acyclovir or valacyclovir is effective. Vaccination is available for varicella (chickenpox) and herpes zoster (shingles), with live attenuated vaccines recommended for immunocompetent individuals and recombinant zoster vaccine recommended for adults aged 50 years and older.

Vector-borne diseases

– Diseases transmitted by arthropod vectors including mosquitoes, ticks, fleas, and sandflies. Major vector-borne diseases include malaria, dengue, Zika, chikungunya, Lyme disease, and leishmaniasis. Prevention through vector control, personal protection, and chemoprophylaxis is essential. Climate change is affecting the distribution of vectors and vector-borne diseases. Understanding vector biology is essential for developing control strategies. Vector-borne diseases are a major global health problem, accounting for more than 17% of all infectious diseases and causing over 700,000 deaths annually worldwide, with the burden falling disproportionately on tropical and subtropical regions.

Vibrio cholerae

– A gram-negative, curved rod that causes cholera, a severe diarrhoeal illness. The organism produces cholera toxin, which causes profuse watery diarrhea (rice-water stools). Transmission occurs through contaminated water and food. The infectious dose varies depending on stomach acidity. Diagnosis involves stool culture on selective media (TCBS agar). Treatment is aggressive fluid replacement with oral rehydration solution or intravenous fluids. Antibiotics (doxycycline, azithromycin) reduce the duration and volume of diarrhea and should be given in moderate to severe cases alongside aggressive fluid resuscitation.

Viral assembly

– The process of packaging newly synthesised viral genomes and proteins into complete virions. Assembly typically occurs in the nucleus or cytoplasm, depending on the virus type. The process involves self-assembly of capsid proteins around the viral genome. Enveloped viruses acquire their envelope from host cell membranes during assembly or release. The efficiency of assembly varies by virus type and can be affected by cellular factors. Assembly is a potential target for antiviral drugs. Understanding viral assembly mechanisms provides opportunities for developing antiviral drugs that disrupt capsid formation or packaging of the viral genome.

Viral attachment

– The initial step in viral infection where viral surface proteins bind to specific receptors on host cells. Attachment determines viral tropism and host range. The viral attachment protein is called the ligand, and the host cell receptor is typically a protein or glycan. Examples include HIV gp120 binding to CD4 receptors, influenza hemagglutinin binding to sialic acid, and SARS-CoV-2 spike protein binding to ACE2 receptors. Attachment can be blocked by neutralising antibodies or receptor antagonists, making viral attachment a key target for vaccines and antiviral therapies.

Viral culture

– The growth of viruses in cell culture for identification and characterisation. Viral culture remains the gold standard for many viral infections. The technique involves inoculating cell lines with clinical specimens and observing for cytopathic effects. Shell vial culture uses centrifugation to enhance viral adsorption and provides faster results. Viral culture is more sensitive than antigen detection but less sensitive than PCR. Culture is essential for antiviral susceptibility testing. Understanding viral culture techniques remains important for virology laboratories, particularly for antiviral susceptibility testing and the isolation of novel viruses.

Viral cytopathic effects

– Observable changes in host cells caused by viral infection. CPE include cell rounding, detachment, syncytia formation (fusion of infected cells), inclusion bodies, and cell lysis. CPE are used in viral diagnosis and can be observed by light microscopy. Some viruses cause rapid cell lysis while others cause chronic infection without obvious CPE. The mechanism of CPE varies by virus type and can involve viral protein expression, host cell shutoff, or immune-mediated damage. CPE are important for viral diagnosis and quantification, and the pattern of CPE can provide clues to the identity of the infecting virus.

Viral entry

– The process by which viruses cross the host cell membrane to deliver their genome into the cytoplasm. Entry occurs through membrane fusion (enveloped viruses) or endocytosis (enveloped and non-enveloped viruses). For membrane fusion, viral fusion proteins mediate merger of the viral envelope with the host cell membrane. For endocytosis, the virus is internalised in vesicles and then escapes into the cytoplasm. The mechanism of entry varies by virus type. Entry can be blocked by fusion inhibitors such as enfuvirtide for HIV, making viral entry an important target for antiviral drug development.

Viral gastroenteritis

– Gastrointestinal inflammation caused by viral infections. The major causes include rotavirus, norovirus, adenovirus, and astrovirus. Symptoms include watery diarrhea, vomiting, and abdominal cramps. Diagnosis is made by antigen detection or PCR in stool. Treatment is supportive with rehydration. Prevention through vaccination (rotavirus) and hygiene is essential. Viral gastroenteritis is the most common cause of diarrhoeal illness worldwide. Most cases are self-limited but can cause severe dehydration in young children, older adults, and immunocompromised patients, requiring prompt oral or intravenous rehydration therapy.

Viral hepatitis

– Inflammation of the liver caused by viral infections. The major hepatitis viruses include A, B, C, D, and E. Hepatitis A and E are transmitted through the fecal-oral route and cause acute, self-limited infections. Hepatitis B and C are transmitted through blood and sexual contact and can cause chronic infections leading to cirrhosis and hepatocellular carcinoma. Hepatitis D requires hepatitis B for replication. Diagnosis involves serology and viral load testing. Prevention through vaccination is available for hepatitis A and B, and antiviral therapy can effectively control hepatitis B and C, though access to treatment remains limited in many resource-constrained settings.

Viral immune evasion

– Mechanisms by which viruses evade host immune responses. Strategies include antigenic variation, inhibition of interferon signalling, downregulation of MHC molecules, and encoding viral homologues of cytokines. Understanding viral immune evasion is essential for developing effective vaccines and antiviral therapies. Viruses that evade immune responses can establish persistent infections.

Viral latency

– A state in which viral genomes persist in host cells without producing infectious particles. Latent viruses can reactivate under certain conditions including immunosuppression. Herpesviruses establish latent infections in specific cell types (HSV in neurons, EBV in B cells, CMV in myeloid cells). HIV integrates into the host genome as a provirus and can establish latent reservoirs. Latency is a major barrier to viral eradication. Reactivation can cause disease years after initial infection. Understanding viral latency is critical for developing curative therapies for chronic viral infections, as eradication of latent reservoirs remains one of the greatest challenges in virology.

Viral latency mechanisms

– The strategies by which viruses establish and maintain latent infections. Herpesviruses maintain their genomes as episomes in the nucleus. HIV integrates into the host genome as a provirus. Latent viruses express minimal proteins to avoid immune detection. Reactivation can be triggered by immunosuppression, stress, or other stimuli. Understanding latency mechanisms is essential for developing strategies to cure chronic viral infections. Reactivation can cause disease years after initial infection, and novel therapeutic approaches including latency-reversing agents are being investigated to eliminate persistent viral reservoirs.

Viral pathogenesis

– The mechanisms by which viruses cause disease. Pathogenesis involves attachment, entry, replication, and spread within the host. Viral damage can be direct (cytopathic effects) or indirect (immune-mediated). Direct effects include cell death, altered cell function, and cell transformation. Indirect effects include inflammation, autoimmunity, and cytokine storm. The incubation period, viral load, and host immune response determine disease severity. Some viruses cause acute infections (influenza, norovirus) that are cleared by the immune system, while others establish persistent or latent infections (HIV, HBV, herpesviruses) that require long-term management.

Viral release

– The process by which newly assembled virions exit the host cell to infect other cells. Release occurs through budding (enveloped viruses) or cell lysis (non-enveloped viruses). Budding allows the virus to acquire an envelope from host cell membranes without killing the cell. Cell lysis releases large numbers of virions but destroys the host cell. Neuraminidase inhibitors block release of influenza viruses. Understanding release mechanisms is important for developing antiviral drugs that target the final stages of the viral life cycle, such as neuraminidase inhibitors for influenza and HIV protease inhibitors.

Viral replication

– The process of synthesising new viral genomes and proteins within host cells. The mechanism varies by virus type. DNA viruses typically replicate in the nucleus using host cell DNA polymerases (except poxviruses). RNA viruses typically replicate in the cytoplasm using viral RNA-dependent RNA polymerase. Retroviruses use reverse transcriptase to convert RNA to DNA, which integrates into the host genome. The replication strategy determines susceptibility to antiviral drugs. Viral replication requires exploitation of host cell machinery, and the specific replication strategy of each virus determines its susceptibility to antiviral agents and its capacity for genetic variation.

Viral replication cycle

– The series of steps by which viruses reproduce within host cells. The cycle includes attachment to host cell receptors, entry through membrane fusion or endocytosis, uncoating of the viral genome, replication of viral nucleic acids, transcription and translation of viral proteins, assembly of new virions, and release by budding or cell lysis. The replication cycle varies by virus type and can take hours to days. Some viruses integrate into the host genome (proviruses) and can establish latent infections that persist for the lifetime of the host, requiring lifelong suppressive therapy.

Viral structure

– The basic components of viral particles (virions). Viruses consist of a nucleic acid genome (DNA or RNA) surrounded by a protein coat called a capsid. Some viruses have an outer lipid envelope derived from host cell membranes. The capsid is composed of structural units called capsomeres. Viral genomes can be single-stranded or double-stranded, linear or circular. Enveloped viruses include influenza, HIV, and hepatitis B. Non-enveloped viruses include adenoviruses, noroviruses, and papillomaviruses, and are generally more resistant to environmental inactivation and disinfectants than enveloped viruses.

Viral uncoating

– The process by which the viral capsid is disassembled to release the viral genome into the host cell cytoplasm. Uncoating occurs after entry and before replication. The mechanism varies by virus type and can involve pH-dependent conformational changes, proteolytic cleavage, or disassembly of capsid proteins. For some viruses, uncoating is triggered by the acidic environment of endosomes. For others, uncoating occurs at the cell surface or in the cytoplasm. Uncoating is essential for viral genome release and is a potential target for antiviral drugs that prevent capsid disassembly.

VRE

– Vancomycin-resistant Enterococcus, resistant to vancomycin through acquisition of vanA or vanB gene clusters. VRE is a significant cause of healthcare-associated infections, particularly in immunocompromised patients. Treatment options include linezolid, daptomycin, and quinupristin-dalfopristin (for E. faecium only). VRE infections are associated with increased mortality and healthcare costs. Infection prevention includes hand hygiene, contact precautions, and environmental cleaning.

Yeast

– Unicellular fungi that reproduce by budding (blastoconidia formation) or fission. The clinically most important yeasts are Candida species (Candida albicans, C. glabrata, C. parapsilosis, C. tropicalis, C. krusei, C. auris) and Cryptococcus species (C. neoformans, C. gattii). Candida albicans is a commensal of the human gastrointestinal tract, oral cavity, and vagina, but causes opportunistic infections in immunocompromised patients: oral thrush (white, curd-like plaques on the oral mucosa that scrape off, leaving an erythematous base), vulvovaginal candidiasis (vaginal itching, burning, thick white discharge), esophageal candidiasis (dysphagia, odynophagia, seen in HIV/AIDS with CD4 <100), cutaneous candidiasis (intertrigo in skin folds), and invasive candidiasis/candidaemia (fever, sepsis in ICU patients, neutropenic patients, post-surgery, associated with central venous catheters, treated with echinocandins — caspofungin, micafungin, anidulafungin — or fluconazole for susceptible strains). Cryptococcus neoformans is an encapsulated yeast that causes cryptococcal meningitis in immunocompromised patients (HIV/AIDS with CD4 <100, organ transplant recipients). Saccharomyces cerevisiae (baker's/brewer's yeast) is rarely pathogenic but can cause fungemia in immunocompromised patients. Malassezia species are lipophilic yeasts associated with seborrhoeic dermatitis, pityriasis versicolor, and folliculitis. Diagnosis: KOH preparation (wet mount of clinical specimen shows budding yeasts with or without pseudohyphae), culture (Saboraud dextrose agar, CHROMagar Candida for species identification), and biochemical tests (germ tube test for C. albicans, API 20C, MALDI-TOF). Treatment: topical (clotrimazole, miconazole, nystatin) and systemic (fluconazole, voriconazole, echinocandins, amphotericin B) antifungals, depending on species, site of infection, and susceptibility.

Yersinia pestis

– A gram-negative rod that causes plague, a severe and potentially fatal infection. The organism is transmitted by flea bites or direct contact with infected animals. Bubonic plague presents with painful lymphadenopathy (buboes). Pneumonic plague involves the lungs and is transmitted by respiratory droplets. Septicemic plague causes widespread dissemination. Diagnosis involves culture and PCR. Treatment includes streptomycin, gentamicin, or doxycycline. Plague is a rare but serious infection, with natural reservoirs in rodent populations in various parts of the world, and it remains a public health concern due to its epidemic potential and use as a bioterrorism agent.

Zika virus

– A positive-sense RNA virus belonging to the Flaviviridae family that causes mild febrile illness. Zika virus is transmitted by Aedes mosquitoes and can also be transmitted sexually. Infection during pregnancy can cause congenital Zika syndrome including microcephaly. Most infections are asymptomatic or mild. Diagnosis is made by PCR or serology. There is no specific treatment or vaccine. Prevention through mosquito control and avoiding travel to endemic areas during pregnancy is important. The virus caused a major epidemic in the Americas in 2015-2016, and surveillance continues to monitor its spread and the risk of congenital infections.

Zoonotic infections

– Infections that are naturally transmitted between animals and humans. Major zoonotic infections include rabies, brucellosis, leptospirosis, and psittacosis. The animal reservoir can be domestic or wild animals. Transmission occurs through direct contact, ingestion, or vectors. Prevention through animal vaccination, food safety, and avoiding animal contact is essential. Zoonotic infections are a significant public health problem worldwide. Understanding zoonotic transmission is essential for preventing emerging infectious diseases, as the majority of newly identified pathogens originate from animal reservoirs.

Chapter 19

Nephrology

24-Hour Urine Collection

– Complete urine collection for quantitative measurement of protein, creatinine, calcium, oxalate, uric acid, and electrolytes. Essential for nephrolithiasis evaluation and proteinuria quantification. Inconvenient but provides comprehensive metabolic information guiding preventive therapy.

Acute Interstitial Nephritis

– Inflammatory infiltrate in the renal interstitium and tubules most commonly caused by drug reactions including NSAIDs, antibiotics particularly penicillins and cephalosporins, proton pump inhibitors, and diuretics. Also caused by infections and autoimmune diseases. Presents with AKI, low-grade fever, rash, and eosinophilia though the classic triad is present in less than one third of cases. Urinalysis shows white blood cell casts, sterile pyuria, and eosinophiluria. Diagnosis requires renal biopsy showing interstitial inflammation with eosinophils. Management includes discontinuation of the offending drug and supportive care; corticosteroids may accelerate recovery in severe cases.

Acute Kidney Injury

– Rapid decline in renal function over hours to days. KDIGO criteria define AKI as creatinine increase by 0.3 mg/dL within 48 hours, 1.5 times baseline within 7 days, or urine output less than 0.5 mL/kg/hr for 6 hours. Classified into stages 1 through 3. Causes include prerenal, intrinsic renal, and postrenal etiologies. Early recognition and treatment of reversible causes is essential to prevent permanent kidney damage.

Acute Tubular Necrosis

– Most common cause of intrinsic AKI resulting from ischemic or nephrotoxic injury to tubular epithelial cells. Ischemic ATN develops from prolonged prerenal states. Nephrotoxic ATN results from direct injury by aminoglycosides, amphotericin B, cisplatin, contrast, and myoglobin. Histology shows tubular cell flattening, brush border loss, and intratubular casts. Oliguric phase may be followed by polyuric phase during tubular regeneration. Recovery takes one to three weeks but may be prolonged in severe cases with comorbidities and advanced age.

Alport Syndrome

– Hereditary nephritis caused by mutations in type four collagen genes leading to progressive sensorineural hearing loss, ocular anomalies, and renal failure. X-linked form from COL4A5 mutations is the most common. Autosomal recessive and dominant forms exist. Kidney biopsy shows characteristic thinning and lamellation of the glomerular basement membrane on electron microscopy. Hearing loss is bilateral sensorineural and progressive. Ocular findings include anterior lenticonus and retinal flecks. There is no specific cure; management includes ACE inhibitors or angiotensin receptor blockers to reduce proteinuria and slow progression to end-stage renal disease. Hearing aids for hearing loss and genetic counseling are important.

ANCA-Associated Vasculitis

– Small vessel vasculitis with antineutrophil cytoplasmic antibodies causing pauci-immune crescentic GN. GPA features PR3-ANCA with granulomatous inflammation. MPA features MPO-ANCA without granulomas. EGPA features MPO-ANCA with asthma and eosinophilia. Treatment with corticosteroids and rituximab or cyclophosphamide for induction followed by rituximab maintenance.

Anti-GBM Disease

– Autoimmune disease with antibodies against alpha-3 chain of type IV collagen in glomerular and alveolar basement membranes. Presents with crescentic GN and pulmonary hemorrhage (Goodpasture syndrome). Linear IgG deposits on immunofluorescence. Treatment requires emergent plasmapheresis, cyclophosphamide, and corticosteroids. Dramatically improved outcomes compared to near-uniform fatality before these treatments.

Apparent Mineralocorticoid Excess

– Rare autosomal recessive disorder caused by deficiency of 11-beta-hydroxysteroid dehydrogenase type 2 normally converting cortisol to cortisone in the collecting duct. Cortisol activates mineralocorticoid receptors causing severe hypertension, hypokalemia, metabolic alkalosis, and low renin with low aldosterone. Presents in childhood with growth retardation. Exogenous liquorice and carbenoxolone inhibit the same enzyme causing an acquired form. Treatment is spironolactone and dexamethasone to suppress ACTH and reduce cortisol production.

ARBs

– Block angiotensin II AT1 receptors providing renoprotection similar to ACE inhibitors. Alternative for ACE inhibitor cough. Do not combine ACE inhibitor and ARB due to increased adverse effects without benefit. Similar renoprotective effects to ACE inhibitors.

Arteriovenous Fistula

– Surgical anastomosis between artery and vein for hemodialysis. Preferred access due to lower infection and thrombosis rates. Radiocephalic at wrist or brachiocephalic at elbow. Takes 6 to 8 weeks to mature. Failure to mature occurs in 30%.

Arteriovenous Graft

– Synthetic conduit connecting artery to vein for dialysis. Used when veins inadequate for fistula. Thrombosis and infection more common than AVF. Can be used earlier than fistula. Declotting with thrombectomy or angioplasty.

Atheroembolic Renal Disease

– Cholesterol crystal embolization to kidneys after vascular procedures. AKI with livedo reticularis and peripheral eosinophilia. Renal biopsy shows cholesterol clefts. Poor prognosis with progressive renal failure.

Autosomal Dominant PKD

– Most common hereditary kidney disease affecting 1 in 400 to 1000. PKD1 chromosome 16 more severe with earlier ESRD than PKD2 chromosome 4. Progressive cyst growth causes hypertension, pain, hematuria, infections, and eventually ESRD. Screen family members with ultrasound. Tolvaptan may slow cyst growth. Transplant may require native nephrectomy if very large.

Autosomal Recessive PKD

– Rare childhood disorder from PKHD1 mutation. Bilateral enlarged kidneys with fusiform dilation of collecting ducts. Associated with congenital hepatic fibrosis. Severe forms cause pulmonary hypoplasia in utero. Variable severity from perinatal lethal to presentation in childhood or adolescence. Management is supportive with treatment of complications.

Bartter Syndrome

– Inherited autosomal recessive disorder of the thick ascending limb of the loop of Henle causing salt wasting, hypokalemic metabolic alkalosis, and hypercalciuria. Results from mutations in genes encoding the NKCC2 transporter or its regulatory subunits. Presents in childhood with failure to thrive, polyuria, and polydipsia. Distinguished from Gitelman syndrome by normal or elevated urinary calcium. Complications include nephrocalcinosis and chronic kidney disease. Treatment includes indomethacin to reduce prostaglandin-mediated salt wasting, and potassium and magnesium supplementation as needed.

Benign Prostatic Hyperplasia (BPH)

– Benign prostatic hyperplasia is a noncancerous enlargement of the prostate gland that commonly affects aging men, compressing the urethra and obstructing urine flow. Symptoms include urinary frequency, urgency, hesitancy, weak stream, and nocturia. Management ranges from watchful waiting and alpha-blockers or 5-alpha-reductase inhibitors to minimally invasive procedures and surgery.

Berger Disease

– Historical name for IgA nephropathy named after Jean Berger who first characterized mesangial IgA deposition in 1968. Characterized by recurrent gross hematuria triggered by mucosal infections with intervals of microscopic hematuria and proteinuria. Generally benign course in most patients but subset progresses to CKD. Complement activation through alternative and lectin pathways plays important pathogenic role leading to investigation of complement-targeted therapies.

BK Virus Nephropathy

– Polyomavirus infection in transplant recipients. Viral cytopathic changes on biopsy. Reduce immunosuppression as primary treatment. Can lead to graft loss. Monitor viral loads in blood for early detection.

Bladder Health and Disorders

– Bladder health involves the proper storage and voluntary release of urine through coordinated function of the detrusor muscle and sphincters. Common disorders include urinary tract infections, overactive bladder, stress incontinence, and interstitial cystitis. Maintaining bladder health includes adequate hydration, pelvic floor exercises, and prompt treatment of infections.

Bladder Problems

– a range of conditions affecting urine storage and voiding, including incontinence, overactive bladder, urinary retention, and recurrent infections. Causes may include neurological damage, pelvic floor weakness, prostate enlargement, or bladder outlet obstruction. Evaluation includes urinalysis, urodynamics, and imaging; treatment varies from behavioral modifications and medications to surgery.

Bowman Capsule

– Double-walled epithelial cup surrounding the glomerular capillaries forming the beginning of the nephron. The visceral layer is composed of podocytes whose foot processes interdigitate to form filtration slits. The parietal layer is composed of simple squamous epithelium. Between the layers is Bowman space which collects the glomerular ultrafiltrate. The urinary pole of Bowman capsule is continuous with the proximal tubule. Injury to the podocyte layer results in proteinuria, while damage to the epithelial cells of the parietal layer can contribute to crescent formation in rapidly progressive glomerulonephritis.

Calciphylaxis

– Calcific uremic arteriolopathy causing painful skin necrosis in CKD. Risk factors include warfarin use, elevated calcium-phosphorus product, and obesity. High mortality rate. Treat with sodium thiosulfate, wound care, and discontinuation of warfarin. Avoid calcium-based binders.

Calcitriol and Analogues

– Active vitamin D forms suppressing PTH. Calcitriol, paricalcitol, doxercalciferol. Monitor calcium and phosphorus to avoid hypercalcemia and hyperphosphatemia. Part of CKD-MBD management.

Calcium Stones

– Most common kidney stone type composed of calcium oxalate or calcium phosphate. Risk factors include hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Treat with thiazides for hypercalciuria, dietary modification, and hydration. Potassium citrate for hypocitraturia. Avoid excessive dietary calcium restriction as it increases oxalate absorption.

Central Venous Catheter

– Temporary or tunneled catheter for dialysis. Internal jugular or femoral placement. Tunneled preferred for long-term with lower infection risk. Last resort access with highest complication rates.

Cerebral Salt Wasting

– Renal sodium wasting causing hyponatremia and volume depletion occurring after intracranial surgery or in setting of intracranial disease. Distinguished from SIADH by the presence of volume depletion with negative sodium balance and compensatory elevation of renin and aldosterone. Results from release of natriuretic peptides in response to intracranial disease. Treatment is saline replacement and fludrocortisone to promote sodium retention. Distinguishing from SIADH is critical because fluid restriction is harmful in cerebral salt wasting and may worsen hypovolaemia.

Chronic Allograft Nephropathy

– Progressive transplant kidney function decline over months to years. Interstitial fibrosis and tubular atrophy on histology. Can be immunologic or non-immunologic. Optimize immunosuppression and manage risk factors including hypertension and recurrent disease.

Chronic Kidney Disease

– Progressive loss of kidney function over months to years, defined by GFR <60 mL/min/1.73 m^2 or markers of kidney damage for ≥ 3 months. Stages 1-5 based on GFR. Leading causes: diabetes mellitus (diabetic nephropathy) and hypertension. Complications: anemia, mineral bone disorder, metabolic acidosis, hyperkalemia, cardiovascular disease, and uremia. Management: blood pressure control (ACEi/ARB), glycemic control, dietary protein restriction, phosphate binders, vitamin D analogs, erythropoiesis-stimulating agents, and preparation for renal replacement therapy (dialysis or transplant).

Chronic Kidney Disease (CKD)

– Chronic kidney disease is the progressive loss of kidney function over months to years, typically from diabetes, hypertension, or glomerular diseases. It is staged by estimated glomerular filtration rate and albuminuria, with end-stage renal disease requiring dialysis or transplantation. Early detection and management of underlying causes, blood pressure control, and dietary modifications slow progression.

CKD

– Abbreviation for Chronic Kidney Disease: a progressive loss of kidney function over months to years, defined by a glomerular filtration rate (GFR) less than 60 mL/min/1.73m^2 or markers of kidney damage (albuminuria, haematuria, structural abnormalities) persisting for at least 3 months. CKD is staged by GFR (G1-G5) and albuminuria category (A1-A3). Major causes include diabetes mellitus and hypertension. Complications: anemia, metabolic acidosis, hyperphosphataemia, hypocalcaemia, secondary hyperparathyroidism, hypertension, and cardiovascular disease. Management: blood pressure control (ACEi/ARB), glycaemic control, dietary protein restriction, phosphate binders, erythropoiesis-stimulating agents, and vitamin D analogues. Progression leads to end-stage renal disease requiring renal replacement therapy (dialysis or transplantation).

CKD-Mineral Bone Disorder

– Systemic disorder of mineral and bone metabolism in CKD. Includes calcium, phosphorus, PTH, and vitamin D abnormalities with vascular calcification and bone disease. Renal osteodystrophy encompasses osteitis fibrosa, osteomalacia, adynamic bone disease, and mixed uremic osteodystrophy. Management includes phosphate binders, active vitamin D analogues, and calcimimetics.

Collecting Duct

– Final segment of the nephron where fine-tuning of urine concentration and composition occurs. Principal cells respond to antidiuretic hormone by inserting aquaporin-2 water channels into the luminal membrane, enabling water reabsorption and urine concentration. Aldosterone acts on principal cells to promote sodium reabsorption and potassium secretion through ENaC channels. Intercalated cells regulate acid-base balance through hydrogen ion secretion and bicarbonate reabsorption. The collecting duct is the site of action of aldosterone, vasopressin, and loop diuretics, and is critical in the regulation of blood pressure, electrolyte balance, and acid-base homeostasis.

Contrast-Induced Nephropathy

– AKI following iodinated contrast. Risk factors include CKD, diabetes, dehydration. Prevent with IV saline hydration. Monitor creatinine 48 to 72 hours post-contrast. Hold metformin in high-risk patients.

Cystine Stones

– Rare stones from inherited cystinuria with defective cystine transport. Hexagonal crystals on urine microscopy. Difficult to dissolve. Treat with aggressive hydration, urine alkalinization to pH greater than 7.5, and thiol agents including tiopronin and D-penicillamine that form soluble complexes with cystine.

Diabetic Nephropathy

– Kidney disease caused by diabetes mellitus, the leading cause of ESRD worldwide. Pathophysiology: hyperglycemia-induced glomerular hyperfiltration, basement membrane thickening, mesangial expansion, and glomerulosclerosis. Presents with progressive albuminuria (microalbuminuria → macroalbuminuria) and declining GFR. Often accompanied by diabetic retinopathy. Screening: annual urine albumin-to-creatinine ratio and serum creatinine in all diabetic patients. Treatment: strict glycemic control, ACE inhibitors or ARBs (renoprotective independent of blood pressure), blood pressure control, and SGLT2 inhibitors for additional renoprotection.

Dialysis Disequilibrium Syndrome

– Neurological complications during hemodialysis in severely uremic patients. Headache, nausea, confusion, seizures, and coma. Results from rapid solute removal causing cerebral edema. Prevent with gradual dialysis initiation and shorter initial sessions.

Distal Tubule

– Segment of the nephron extending from the thick ascending limb of the loop of Henle to the collecting duct. The early distal tubule reabsorbs sodium and chloride through the NCC cotransporter, which is the target of thiazide diuretics. The late distal tubule and connecting tubule contain principal cells that respond to aldosterone for sodium reabsorption and potassium secretion, and intercalated cells that regulate acid-base balance. The macula densa cells at the junction with the thick ascending limb, where it forms the juxtaglomerular apparatus that regulates renin release, glomerular filtration rate, and renal blood flow through tubuloglomerular feedback.

DMSA Scan

– Nuclear medicine scan evaluating renal cortical function and scarring. Gold standard for detecting pyelonephritis and renal scars in children. Shows photopenic areas corresponding to cortical damage. Important for long-term follow-up of childhood UTI.

Dysuria

– Painful or difficult urination, typically described as burning or stinging during micturition. Most commonly caused by lower urinary tract infection (cystitis, urethritis) in women, and by urethritis or prostatitis in men. Other causes: urinary tract stones, interstitial cystitis, sexually transmitted infections (Chlamydia trachomatis, Neisseria gonorrhoeae, Trichomonas vaginalis, herpes simplex), chemical irritants (soaps, spermicides, catheterisation), and urologic malignancy. Diagnosis: urinalysis, urine culture, and STI testing if indicated. In uncomplicated UTI with typical symptoms, empiric antibiotics (nitrofurantoin, TMP-SMX) are appropriate. Persistent or atypical cases require further evaluation (cystoscopy, imaging).

Emphysematous Pyelonephritis

– Severe necrotizing infection of renal parenchyma with gas formation. Almost exclusively in diabetic patients. Presents with high fever, flank pain, and sepsis. CT shows gas within renal parenchyma. A urological emergency requiring aggressive antibiotics and usually nephrectomy. Mortality is high even with treatment.

End-Stage Renal Disease

– The final stage of chronic kidney disease (CKD stage 5), defined by GFR <15 mL/min/1.73 m^2 or the need for renal replacement therapy. Uremia develops with accumulation of toxins causing nausea, anorexia, pruritus, pericarditis, encephalopathy, and bleeding diathesis. Life-sustaining treatments: hemodialysis, peritoneal dialysis, or kidney transplantation. Management also includes anemia correction, phosphate binding, vitamin D analogs, and blood pressure control.

Epididymitis

– Inflammation of the epididymis, a common cause of acute scrotal pain in adults, typically resulting from infection. In sexually active men under 35 years, the most common pathogens are sexually transmitted: Chlamydia trachomatis (most common) and Neisseria gonorrhoeae, with epididymitis often accompanied by urethritis. In men over 35 years and in children, the causative organisms are enteric gram-negative bacilli (Escherichia coli, Pseudomonas, Klebsiella) from urinary tract infection, prostatitis, or following urological instrumentation. Presents with gradual onset of scrotal pain and swelling, often with fever and dysuria. Diagnosis is clinical, supported by urine analysis and NAAT testing for sexually transmitted infections. Treatment includes appropriate antibiotics targeting the causative organism, scrotal elevation, and NSAIDs for symptom control.

Erythropoiesis-Stimulating Agents

– Epoetin alfa and darbepoetin alfa stimulate red blood cell production in CKD anemia. Target hemoglobin 10 to 11.5 g/dL. Avoid exceeding 13 g/dL due to increased cardiovascular risk. Ensure adequate iron stores before initiating therapy.

ESKD

– End-stage kidney disease defined as GFR less than 15 or need for dialysis. Medicare eligibility at GFR less than 15 or on dialysis. Palliative care option for those not wanting renal replacement therapy. Comprehensive advance care planning essential.

Fanconi Syndrome

– Generalized proximal tubular dysfunction with impaired reabsorption of glucose, amino acids, phosphate, uric acid, and bicarbonate. Causes glycosuria, aminoaciduria, phosphaturia, and metabolic acidosis. Associated with proximal RTA. Causes include myeloma, heavy metal toxicity, Wilson disease, and genetic disorders. Treatment addresses underlying cause and replaces depleted substances.

Fibromuscular Dysplasia

– Non-atherosclerotic arterial disease affecting renal arteries. Causes renovascular hypertension in young women. String-of-beads on angiography from medial fibroplasia. Treated with angioplasty alone which is highly effective. Renal function normalizes after successful procedure.

Focal Segmental Glomerulosclerosis

– Pattern of glomerular injury with sclerosis in some focal glomeruli and segments. Can be primary idiopathic or secondary to HIV, obesity, heroin use, or adaptive responses to nephron loss. Histology shows segmental sclerosis with hyalinosis. Less responsive to steroids than minimal change with approximately 50% achieving remission. Secondary FSGS treated by addressing underlying cause. Progressive disease leads to ESRD. Recurrence after transplantation occurs in approximately 30%.

Gitelman Syndrome

– Inherited autosomal recessive disorder of the distal convoluted tubule caused by mutations in the NCC sodium-chloride cotransporter gene. Milder than Bartter syndrome typically presenting in adolescence or adulthood. Characterized by hypokalemic metabolic alkalosis, hypomagnesemia, and hypocalciuria distinguishing it from Bartter syndrome. Patients may have muscle cramps, fatigue, and chondrocalcinosis from hypomagnesemia. Treatment includes magnesium and potassium supplementation and NSAIDs or eplerenone for symptom management.

Glomerular Filtration Rate

– Volume of plasma filtered by the glomeruli per unit time normally ninety to one hundred twenty milliliters per minute per one point seven three square meters body surface area. Estimated by the CKD-EPI equation using serum creatinine, age, sex, and race. Key determinants include renal plasma flow, hydrostatic pressure across the glomerular capillary, and the ultrafiltration coefficient reflecting glomerular permeability and surface area. GFR is the single best indicator of kidney function and decreases progressively with age and nephron loss. The slope of eGFR decline over time is a key prognostic indicator in chronic kidney disease progression.

Glomerulonephritis

– Inflammation of the glomeruli, classified by clinical presentation (nephritic vs. nephrotic syndrome) and histopathology (e.g., IgA nephropathy, post-streptococcal, membranoproliferative, crescentic). Presents with hematuria, proteinuria, hypertension, and reduced GFR. Diagnosis: urinalysis, complement levels, autoantibodies (ANA, ANCA, anti-GBM), and renal biopsy. Treatment depends on etiology: immunosuppression for immune-mediated forms, blood pressure control, and supportive care.

Glomerulus

– Network of capillaries enclosed within Bowman's capsule in the renal corpuscle. Blood is filtered through the glomerular filtration barrier consisting of fenestrated endothelium, glomerular basement membrane, and podocytes with their interdigitating foot processes. The filtration barrier is selectively permeable based on size and charge, normally preventing passage of large proteins and blood cells while allowing free filtration of water, electrolytes, and small solutes. Injury to any component of the filtration barrier results in proteinuria, haematuria, and progressive loss of renal function.

Goodpasture Syndrome

– Anti-GBM disease affecting both kidneys and lungs. Pulmonary hemorrhage with hemoptysis and crescentic GN. Linear IgG deposits on renal biopsy. Emergent plasmapheresis, cyclophosphamide, and steroids. Early treatment essential to prevent irreversible renal and pulmonary damage.

Granular Casts

– Degenerated cellular debris within tubular lumen appearing as coarse or fine-grained structures. Coarse casts represent more acute process, fine casts indicate chronic degenerative process. Common in acute tubular necrosis reflecting tubular epithelial cell damage. Also seen in chronic kidney disease, nephrotic syndrome, and hypertension. Nonspecific finding of tubular injury.

Hemodialysis

– Extracorporeal blood purification using semipermeable membrane. Three sessions per week for 3 to 4 hours. Requires vascular access with AV fistula preferred. Adequacy assessed by Kt/V greater than 1.4. Complications include hypotension, access infection, and dialyzer reactions.

Hyaline Casts

– Transparent colorless cylindrical structures composed primarily of Tamm-Horsfall protein secreted by the thick ascending limb. Most common type of urine cast. Normal in concentrated urine and after exercise. Increased in dehydration, prerenal states, and volume depletion. Generally benign reflecting increased Tamm-Horsfall concentration or mild tubular dysfunction.

Hydronephrosis

– Dilation of renal collecting system proximal to obstruction. Causes include stones, strictures, malignancy, and BPH. Graded on ultrasound from mild to severe. Prolonged obstruction leads to cortical thinning and renal failure. Bilateral more clinically significant. Urgent decompression required for obstructive uropathy with infection.

Hyperkalemia

– Elevated serum potassium >5.0 mEq/L, a potentially life-threatening electrolyte abnormality. Causes: renal failure (most common), medications (ACE inhibitors, ARBs, potassium-sparing diuretics, NSAIDs), adrenal insufficiency, metabolic acidosis (shift), massive tissue breakdown (rhabdomyolysis, tumor lysis), and excessive intake. ECG changes: peaked T waves, widened QRS, loss of P wave, sine wave pattern leading to VFib/asystole. Management: stabilize myocardium (IV calcium gluconate or chloride), shift K+ intracellularly (insulin + glucose, albuterol, sodium bicarbonate if acidotic), and remove K+ (loop diuretics, patiromer, hemodialysis for severe or refractory cases).

Hypertensive Nephropathy

– Kidney damage caused by chronic hypertension, leading to nephrosclerosis (arteriolar hyalinosis and glomerulosclerosis). Presents with slowly progressive proteinuria (usually subnephrotic), hypertension, and gradual decline in kidney function. More common in African American patients. Diagnosis of exclusion. Prevention and treatment: strict blood pressure control, ACE inhibitors/ARBs are renoprotective.

Hypertensive Nephrosclerosis

– Kidney damage from chronic hypertension. Benign form with hyaline arteriolosclerosis. Malignant form with hyperplastic arteriolosclerosis and onion-skinning. Second leading cause of ESRD after diabetes. Tight BP control with ACE inhibitors or ARBs essential to slow progression.

Hypokalemia

– Serum potassium less than 3.5 mEq/L. Causes include gastrointestinal losses from vomiting and diarrhea, renal losses from diuretics, RTA, and hyperaldosteronism, intracellular shift from alkalosis and insulin, and inadequate intake. Clinical manifestations include muscle weakness, cramps, cardiac arrhythmias with characteristic ECG changes including U waves, flattened T waves, and ST depression, and rhabdomyolysis in severe cases. Treatment depends on severity and symptoms ranging from oral potassium supplementation to IV potassium replacement at rates not exceeding ten milliequivalents per hour through a peripheral line or twenty milliequivalents per hour through central access, with continuous ECG monitoring during IV replacement.

IgA Nephropathy

– The most common primary glomerulonephritis worldwide, characterized by mesangial IgA deposition. Presents with hematuria (macroscopic with mucosal infections, especially upper respiratory), proteinuria, and hypertension. Most common in young adults, with male predominance. Diagnosis: renal biopsy showing mesangial IgA deposits with mesangial proliferation on light microscopy and electron-dense deposits. Prognosis: 30-40% progress to ESRD over 20 years. Risk factors: hypertension, proteinuria >1g/day, impaired GFR, and biopsy findings of fibrosis. Treatment: BP control with ACEi/ARB; fish oil; corticosteroids for progressive disease; immunosuppression reserved for crescentic IgA.

Immunosuppression in Transplant

– Medications to prevent rejection. Induction with thymoglobulin or basiliximab. Maintenance with tacrolimus, mycophenolate, and prednisone. Monitor drug levels and renal function. Calcineurin inhibitor nephrotoxicity is major concern requiring dose optimization.

Intravenous pyelogram

– Also known as IVP or excretory urography, a radiographic imaging study of the urinary tract following intravenous administration of iodinated contrast material. The contrast is filtered by the glomeruli and excreted through the collecting system, allowing sequential imaging of the renal parenchyma, calyces, renal pelvis, ureters, and bladder. Indications include evaluation of hematuria, suspected ureteral obstruction (stones, strictures, tumors), congenital anomalies of the urinary tract, and pre-operative assessment of ureteral anatomy. CT urography has largely replaced IVP in modern practice due to superior sensitivity for small masses and stones without the need for compression.

Intrinsic Renal AKI

– AKI caused by damage to renal parenchyma. Major categories include acute tubular necrosis from ischemic or nephrotoxic injury, acute interstitial nephritis from drug reactions, glomerulonephritis, and vascular causes including thrombotic microangiopathy. FENa typically greater than 2% indicating tubular dysfunction. Urine sediment findings help differentiate specific causes: muddy brown granular casts in ATN, WBC casts in AIN, RBC casts in glomerulonephritis.

Juxtaglomerular Apparatus

– Specialized structure located at the vascular pole of the glomerulus where the distal tubule comes into contact with the afferent arteriole. Composed of macula densa cells in the tubular wall, juxtaglomerular cells in the afferent arteriolar wall that produce renin, and extraglomerular mesangial cells. Senses sodium and chloride delivery through the macula densa and renal perfusion pressure through the juxtaglomerular cells. Regulates renin secretion initiating the renin-angiotensin-aldosterone system and thereby blood pressure, fluid volume, and electrolyte homeostasis.

Kidney and Urinary Health

– Kidney and urinary health encompasses the function of the kidneys, ureters, bladder, and urethra in filtering waste, regulating fluid/electrolyte balance, and eliminating urine. Chronic kidney disease (CKD)—defined by reduced glomerular filtration rate (GFR $<$60 mL/min/1.73 m^2) or kidney damage for $>$3 months—affects approximately 15% of adults and is most commonly caused by diabetes and hypertension. Urinary tract infections (UTIs) are bacterial infections (usually *Escherichia coli*) of the bladder (cystitis) or kidney (pyelonephritis) presenting with dysuria, frequency, urgency, and flank pain. Diagnostic evaluation includes serum creatinine/eGFR, urinalysis, urine culture, renal ultrasound, and urodynamic studies. Management emphasizes blood pressure control, glucose management, dietary sodium/protein restriction, avoidance of nephrotoxins, and pharmacotherapy (ACE inhibitors, SGLT2 inhibitors for CKD; antibiotics for UTIs).

Kidney stones

– Also known as nephrolithiasis or renal calculi, crystalline mineral deposits that form within the renal pelvis or collecting system, affecting approximately 10 percent of the population over a lifetime, with a male-to-female ratio of 2:1 and peak incidence in the 4th-6th decades. Stone types and causes: calcium oxalate stones (most common, 60-70 percent, associated with hypercalciuria, hyperoxaluria, hyperuricosuria, hypocitraturia, and low urine volume), calcium phosphate stones (10-15 percent, associated with renal tubular acidosis, hyperparathyroidism, and urinary tract infections with urease-producing organisms), struvite stones (15-20 percent, associated with urease-producing bacteria such as Proteus mirabilis, forming staghorn calculi), uric acid stones (5-10 percent, radiolucent, forming in acidic urine pH less than 5.5, associated with gout, high purine intake, and chronic diarrhea), and cystine stones (1-2 percent, from cystinuria, hexagonal crystals).

Kidney Transplantation

– Best treatment for ESRD with improved survival over dialysis. Living donor superior to deceased donor. Lifelong immunosuppression required. Rejection types include hyperacute, acute, and chronic. Complications include rejection, infection, vascular complications, and recurrent disease.

Kimmelstiel-Wilson Nodules

– Characteristic round eosinophilic nodular mesangial expansions in diabetic nephropathy. Represent advanced mesangial matrix accumulation. Found in peripheral mesangium associated with diffuse mesangial expansion, GBM thickening, and arteriolar hyalinosis. Indicate advanced diabetic nephropathy correlating with heavy proteinuria and declining GFR.

Kt/V

– Dialysis adequacy measure representing the fractional clearance of urea. K is dialyzer clearance, t is treatment time, V is volume of distribution of urea. Target Kt/V greater than 1.4 for thrice-weekly hemodialysis. Urea reduction ratio greater than 70% is alternative adequacy measure.

Liddle Syndrome

– Autosomal dominant condition caused by gain-of-function mutations in the ENaC sodium channel subunits leading to constitutive sodium reabsorption and potassium secretion in the collecting duct. Presents with severe hypertension, hypokalemia, and metabolic alkalosis occurring in childhood or young adulthood. Aldosterone levels are suppressed reflecting volume expansion. Unlike primary hyperaldosteronism, the hypertension is resistant to spironolactone because the ENaC mutation is constitutively active. Diagnosis is by low aldosterone and low renin levels, which distinguishes it from primary hyperaldosteronism. Treatment is with amiloride or triamterene, which directly block the ENaC channel.

Lithotripsy

– A medical procedure that uses shock waves to fragment kidney stones (renal calculi) into smaller pieces that can pass through the urinary tract. See Shock Wave Lithotripsy.

Loop of Henle

– U-shaped tubular segment consisting of a thin descending limb, thin ascending limb, and thick ascending limb. The descending limb is permeable to water but not solutes, allowing water reabsorption into the hypertonic medullary interstitium. The thick ascending limb actively transports sodium, potassium, and chloride through the NKCC2 transporter, which is the target of loop diuretics, and is impermeable to water creating the dilute tubular fluid essential for the countercurrent multiplication system that generates the medullary osmotic gradient essential for urine concentration.

MAG3 Renal Scan

– Nuclear medicine scan evaluating renal function and obstruction. Differential function between kidneys. Non-obstructive dilation does not show delayed excretion. Useful for assessing split renal function and planning surgical interventions.

Malignant Hypertension

– Severely elevated BP greater than 180/120 with end-organ damage. Renal findings include hyperplastic arteriolosclerosis with onion-skinning and fibrinoid necrosis. Treat with IV antihypertensives. Poor prognosis if untreated. May cause AKI, encephalopathy, and retinopathy.

May-Thurner Syndrome

– Compression of left common iliac vein by right common iliac artery causing left lower extremity DVT. Stenting is mainstay of treatment.

Medullary Sponge Kidney

– Congenital malformation characterized by cystic dilatation of the collecting ducts in the renal papillae predisposing to nephrolithiasis and nephrocalcinosis. Presents with recurrent kidney stones, hematuria, and urinary tract infections. Diagnosis is by CT urogram showing dilated collecting ducts in the renal papillae with or without nephrocalcinosis. Usually bilateral. Treatment is preventive with increased fluid intake and thiazide diuretics for calcium stone prevention. Most patients have no reduction in life expectancy.

Membranoproliferative Glomerulonephritis

– Pattern of glomerular injury with mesangial proliferation and capillary wall thickening creating tram-track appearance on silver stain. Hepatitis C is common cause of immune complex MPGN. C3 glomerulopathy represents complement-mediated disease. Associated with monoclonal gammopathies. Treatment addresses underlying cause with antiviral therapy for hepatitis C and immunosuppression for immune-mediated forms.

Membranous Nephropathy

– A common cause of nephrotic syndrome in adults, characterized by immune complex deposition in the subepithelial space of the glomerular basement membrane. Primary (idiopathic, 75%): associated with anti-PLA2R antibodies (diagnostic). Secondary: infections (hepatitis B, syphilis), autoimmune (SLE), drugs (NSAIDs, penicillamine), and malignancy (lung, colon). Presents with nephrotic-range proteinuria, edema, and normal renal function initially. Diagnosis: renal biopsy with granular IgG staining along capillary loops; anti-PLA2R antibody testing. Treatment: aggressive BP control, RAAS blockade; immunosuppression (cyclophosphamide + steroids, rituximab, or calcineurin inhibitors) for high-risk progressive disease.

Microalbuminuria

– Excretion of 30 to 300 mg of albumin daily undetectable by standard dipstick. Earliest marker of diabetic nephropathy and cardiovascular risk predictor. Detected by urine albumin-to-creatinine ratio on spot specimen. Normal less than 30 mg/g creatinine. Signals early glomerular damage in diabetes warranting RAAS blockade initiation. Screen all diabetics beginning 5 years after type 1 or at diagnosis of type 2 diabetes.

Mineralocorticoid Receptor Antagonists

– Spironolactone and eplerenone block aldosterone receptors. Antifibrotic effects in CKD. Finerenone is non-steroidal MRA with cardiovascular and renal benefits in diabetic CKD. Monitor for hyperkalemia especially when combined with RAAS inhibitors.

Minimal Change Disease

– Most common cause of nephrotic syndrome in children accounting for approximately 80%. Glomeruli appear normal on light microscopy but podocyte foot processes are diffusely effaced on electron microscopy. Circulating factors from dysregulated T cells may increase glomerular permeability. Presents with sudden onset nephrotic syndrome. Responds excellently to corticosteroids with complete remission in over 90% of children. Relapse is common but long-term prognosis is excellent.

Myeloma Cast Nephropathy

– Light chain cast formation in tubules causing AKI in multiple myeloma. Friable waxy casts with giant cells. Bence Jones proteinuria. Treat myeloma and consider plasmapheresis to reduce light chain burden.

Nephritic Syndrome

– Characterized by hematuria with dysmorphic RBCs and RBC casts, oliguria, hypertension, and mild proteinuria less than 3.5 g/day. Results from glomerular inflammation increasing permeability to cells and reducing GFR. Hypertension from sodium and water retention. Caused by post-streptococcal GN, IgA nephropathy, lupus nephritis, and anti-GBM disease. Management includes blood pressure control, diuretics, and immunosuppression when indicated.

Nephrolithiasis

– Stone formation in the urinary tract. Calcium oxalate most common at 80%, followed by struvite, uric acid, and cystine. Risk factors include dehydration, hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Evaluation includes 24-hour urine analysis. Treatment includes hydration, dietary modification, and targeted medical therapy based on stone composition.

Nephron

– Functional unit of the kidney comprising the glomerulus, proximal tubule, loop of Henle, distal tubule, and collecting duct. Each kidney contains approximately one million nephrons responsible for filtering blood, reabsorbing solutes and water, and excreting waste to form urine. Nephron loss is irreversible and leads to compensatory hypertrophy of remaining nephrons, which may eventually contribute to progressive kidney disease through glomerular hyperfiltration and hypertension. The juxtamedullary nephrons with long loops of Henle are essential for producing concentrated urine.

Nephrotic Range Proteinuria

– Protein excretion greater than 3.5 g/day or urine protein-to-creatinine ratio greater than 3.5 g/g. Reflects severe glomerular damage with significant protein loss. Degree correlates with severity of injury and risk of complications including thromboembolism, infection, and progression to ESRD. Causes include minimal change disease, FSGS, membranous nephropathy, diabetic nephropathy, and amyloidosis.

Nephrotic Syndrome

– Clinical constellation of heavy proteinuria greater than 3.5 g/day, hypoalbuminemia less than 3.0 g/dL, generalized edema, hyperlipidemia, and lipiduria. Results from increased glomerular permeability to albumin. Reduced oncotic pressure causes fluid extravasation. Hepatic lipoprotein synthesis increases causing hyperlipidemia. Loss of antithrombin III increases thromboembolic risk. Treatment addresses underlying cause with diuretics for edema and RAAS blockade to reduce proteinuria.

Nutcracker Syndrome

– Compression of left renal vein between aorta and SMA causing hematuria and flank pain. Managed conservatively. Stenting or surgery for severe cases.

Oliguria

– Reduced urine output, defined as <400 mL/day or <0.5 mL/kg/hour in adults. Oliguria is an important sign of acute kidney injury (AKI) and prerenal azotaemia. Causes: prerenal (hypovolaemia—hemorrhage, dehydration, sepsis, heart failure, cirrhosis, nephrotic syndrome), renal (acute tubular necrosis—ischemic or nephrotoxic, acute interstitial nephritis, glomerulonephritis, hemolytic uraemic syndrome), and postrenal (obstruction—benign prostatic hyperplasia, stones, tumors, retroperitoneal fibrosis). Assessment: urine output monitoring (hourly in ICU), urinalysis, urine sodium and osmolality, fractional excretion of sodium (FENa <1% suggests prerenal, >2% suggests ATN), serum creatinine and BUN, renal ultrasound (rule out obstruction). Management: treat underlying cause; fluid challenge for prerenal causes, diuretics once volume replete, renal replacement therapy for refractory oliguric AKI with hyperkalaemia, acidosis, or uraemia. Anuria (<100 mL/day) suggests complete obstruction, bilateral renal cortical necrosis, or rapidly progressive glomerulonephritis.

Osmotic Demyelination Syndrome

– Demyelinating process affecting the pons and extrapontine regions caused by overly rapid correction of chronic hyponatremia. The brain adapts to chronic hyponatremia by extruding osmoles and when sodium is corrected too quickly water shifts out of brain cells causing osmotic injury to oligodendrocytes. Classically presents with quadriparesis, dysarthria, dysphagia, and locked-in syndrome. Risk increases when sodium correction exceeds ten milliequivalents per liter in twenty-four hours. Prevention is the most important strategy, with a target rate of correction not exceeding eight milliequivalents per litre per day in high-risk patients.

Parathyroidectomy

– Surgical removal of parathyroid glands for refractory secondary hyperparathyroidism unresponsive to medical therapy. Indicated when calcium-phosphorus product persistently elevated despite maximal medical management, symptomatic bone disease, or calciphylaxis. Subtotal or total parathyroidectomy with autotransplantation.

Percutaneous Nephrolithotomy

– Minimally invasive procedure for removal of large renal stones through a nephrostomy tract created in the flank. A tract is created from the skin to the collecting system under fluoroscopic guidance and dilated to allow passage of a nephroscope. Stones are fragmented and removed using ultrasonic lithotripsy, pneumatic lithotripsy, or laser lithotripsy. Preferred for stones greater than two centimeters or staghorn calculi. Success rates exceed ninety percent. Complications include bleeding requiring transfusion, pneumothorax, and infection, which are less common with experienced operators.

Peritoneal Dialysis

– Dialysis using peritoneum as semipermeable membrane. CAPD with four manual exchanges daily or APD using cycler overnight. Advantages include home-based therapy and hemodynamic stability. Complications include peritonitis and catheter infection. Tenckhoff catheter is standard access.

Peritonitis in PD

– Most common complication of peritoneal dialysis. Cloudy effluent, abdominal pain, and fever. Diagnosis by effluent cell count with neutrophils greater than 50/mm3. Treat with intraperitoneal antibiotics. Recurrent peritonitis may require catheter removal.

Phosphate Binders

– Calcium acetate, sevelamer, lanthanum. Bind dietary phosphate in GI tract. Target normal serum phosphorus in CKD-MBD. Avoid calcium-based binders in patients with vascular calcification.

Polyarteritis Nodosa

– Systemic necrotizing vasculitis affecting medium-sized arteries without glomerular involvement or ANCA positivity. Causes aneurysms, stenosis, and occlusion leading to organ ischemia. Renal involvement causes renovascular hypertension. Other manifestations include mononeuritis multiplex and mesenteric ischemia. Angiography shows microaneurysms. Treatment with corticosteroids and cyclophosphamide.

Polycystic Kidney Disease

– Inherited disorder with progressive renal cyst formation. ADPKD most common from PKD1 or PKD2 mutations. Bilateral enlarged kidneys with countless cysts. PKD1 more severe with earlier ESRD. Extrarenal manifestations include hepatic cysts and berry aneurysms. Screen family members. ARPKD is rare childhood disorder from PKHD1 mutation with congenital hepatic fibrosis.

Polycystic Liver Disease

– Dominant polycystic liver disease is an autosomal dominant condition characterized by multiple hepatic cysts that may cause massive hepatomegaly, abdominal pain, and compressive symptoms. Usually associated with autosomal dominant polycystic kidney disease. Liver function is preserved even with extensive cyst involvement. Diagnosis is by imaging showing multiple hepatic cysts. Treatment of symptoms includes cyst fenestration or liver transplantation for severe cases. Somatostatin analogues may reduce cyst volume and symptoms in selected patients.

Polyuria

– Excessive urine output, defined as >3 L/day in adults. Causes are divided into water diuresis and solute diuresis. Water diuresis: central diabetes insipidus (ADH deficiency), nephrogenic diabetes insipidus (ADH resistance from lithium, hypercalcemia, hypokalemia, genetic causes), and primary polydipsia (psychogenic or dipsogenic). Solute diuresis: hyperglycemia (diabetes mellitus, the most common cause), excess salt intake (IV saline, tube feeding), urea diuresis (high-protein feeding, post-obstruction), and mannitol. Evaluation: 24-hour urine volume, serum and urine osmolality, and water deprivation test. Treatment: address underlying cause; desmopressin (DDAVP) for central DI, thiazides for nephrogenic DI.

Postrenal AKI

– AKI caused by obstruction of urinary outflow at any level from renal tubules to urethra. Causes include BPH, nephrolithiasis, malignancy, retroperitoneal fibrosis, and neurogenic bladder. Bilateral ureteral obstruction or unilateral in a solitary kidney is required for significant impairment. Early diagnosis with renal ultrasound showing hydronephrosis is essential. Prompt decompression with stenting or nephrostomy can restore renal function if initiated before permanent damage.

Post-Streptococcal Glomerulonephritis

– Immune-mediated glomerulonephritis following infection with nephritogenic strains of group A streptococcus typically 1 to 3 weeks after pharyngitis or 3 to 6 weeks after skin infection. Subepithelial humps on electron microscopy representing immune complex deposits. Presents with nephritic syndrome including hematuria, edema, hypertension, and oliguria. Low complement levels (C3) that normalize within 6 to 8 weeks. Supportive management with most patients recovering completely.

Potassium Binders

– Sodium zirconium cyclosilicate and patiromer. Exchange potassium in GI tract for sodium or calcium. Treat chronic hyperkalemia allowing continued RAAS inhibitor use in CKD.

Prerenal Azotemia

– Most common cause of AKI resulting from decreased effective circulating volume. Causes include dehydration, hemorrhage, heart failure, sepsis, and hepatorenal syndrome. Tubular function is intact with urine sodium less than 20 mEq/L and FENa less than 1%. Urine osmolality greater than 500 reflects avid water reabsorption. BUN-to-creatinine ratio typically greater than 20:1. Usually rapidly reversible with restoration of adequate renal perfusion.

Proteinuria

– Excretion of more than 150 mg of protein in urine daily. Causes include glomerular disease, tubulointerstitial disease, overflow proteinuria from multiple myeloma, and functional proteinuria from fever or exercise. Quantification by 24-hour urine or spot protein-to-creatinine ratio. Significant proteinuria warrants evaluation to identify underlying cause and assess for nephrotic syndrome.

Proximal Tubule

– Segment of the nephron extending from Bowman capsule to the loop of Henle with a convoluted proximal straight tubule and convoluted proximal tubule. Responsible for reabsorbing approximately sixty-five percent of filtered sodium, water, bicarbonate, glucose, amino acids, and phosphate through an extensive brush border. Also secretes organic anions, drugs, and toxins. The high metabolic activity of proximal tubular cells makes them particularly vulnerable to ischemic and toxic injury, which is why proximal tubular injury predominates in ischemic acute tubular necrosis. The proximal tubule is also the primary site for renal drug elimination and plays a key role in drug-induced nephrotoxicity.

Pruritus in CKD

– Severe itching in dialysis patients from uremic toxin accumulation, elevated calcium-phosphorus product, and secondary hyperparathyroidism. Difficult to treat. Include phosphate binders, dialysis optimization, gabapentin, and UV phototherapy.

Rapidly Progressive Glomerulonephritis

– Rapid renal function loss over days to weeks with crescent formation on biopsy. Type I anti-GBM with linear deposits. Type II immune complex with granular deposits. Type III pauci-immune with ANCA vasculitis. Requires aggressive immunosuppression. Without treatment progresses rapidly to ESRD.

Rasburicase

– Recombinant urate oxidase enzyme that converts uric acid to allantoin which is more soluble and easily excreted. Highly effective for prevention and treatment of hyperuricemia in tumor lysis syndrome. More effective than allopurinol for established hyperuricemia. Contraindicated in G6PD deficiency due to risk of hemolytic anemia from oxidative stress. Reduces need for dialysis in tumor lysis syndrome.

Renal Anemia

– Normocytic normochromic anemia in CKD from decreased erythropoietin production. Hemoglobin less than 10 g/dL usually requires treatment with ESAs and iron supplementation. Target hemoglobin 10 to 11.5 g/dL. HIF-PHI prolyl hydroxylase inhibitors are newer oral agents for CKD anemia.

Renal Artery Aneurysm

– Dilatation of renal artery, most common visceral artery aneurysm. Usually asymptomatic. Risk of rupture if greater than 2 cm or symptomatic. Treat with surgical or endovascular repair. Higher rupture risk in pregnant women.

Renal Artery Embolism

– Acute occlusion of renal artery from cardiac emboli typically from atrial fibrillation or mural thrombus. Presents with acute flank pain and hematuria. CT angiography shows filling defect. Treat with anticoagulation and possibly thrombolysis or embolectomy if caught early.

Renal Artery Stenosis

– Narrowing of renal artery causing renovascular hypertension. Atherosclerotic most common in elderly at ostium. Fibromuscular dysplasia in young women causing mid-distal stenosis with string-of-beads appearance. Flash pulmonary edema and declining function on ACE inhibitors indicate intervention. Treatment with angioplasty and stenting for atherosclerotic disease.

Renal Biopsy

– Percutaneous needle biopsy for histological diagnosis. Indicated for unexplained AKI, nephrotic syndrome, hematuria with renal involvement, and transplant dysfunction. Ultrasound-guided for accuracy. Complications include bleeding, arteriovenous fistula, and rarely loss of kidney.

Renal Calculi (Kidney Stones)

– See Kidney Stones.

Renal Failure

– Decline in kidney function leading to accumulation of waste products (azotemia) and electrolyte abnormalities. Acute kidney injury (AKI): rapid decline over hours to days, potentially reversible. Chronic kidney disease (CKD): progressive decline over months to years. Uremic symptoms: nausea, fatigue, pruritus, confusion, and pericarditis. Management: treat underlying cause, dialysis for severe acute or chronic renal failure, and kidney transplant evaluation.

Renal Leukemia Infiltration

– Kidney involvement by chronic lymphocytic leukemia or other hematological malignancies causing bilateral renomegaly and renal dysfunction. Leukemic cells infiltrate the interstitium and peritubular capillaries. May cause AKI from compression of tubules and vessels or chronic kidney disease. Diagnosis is by renal biopsy showing leukemic infiltrates or non-invasive imaging showing renomegaly. Treatment is directed at the underlying hematological malignancy with chemotherapy.

Renal Osteodystrophy

– Bone disease in CKD including osteitis fibrosa, osteomalacia, adynamic bone disease, and osteoporosis. Results from calcium-phosphate imbalance, secondary hyperparathyroidism, and vitamin D deficiency. Manage with phosphate binders, vitamin D analogues, and parathyroidectomy for refractory hyperparathyroidism.

Renal Tubular Acidosis

– Group of disorders with normal anion gap metabolic acidosis from impaired renal acid excretion or bicarbonate reabsorption. Type 1 distal with impaired H+ secretion and urine pH greater than 5.5. Type 2 proximal with impaired bicarbonate reabsorption. Type 4 hyperkalemic from aldosterone deficiency or resistance. Each type has distinct pathophysiology and treatment approach.

Renal Ultrasound

– Noninvasive imaging evaluating kidney size, structure, and hydronephrosis. First-line imaging for CKD evaluation. Shows cortical thickness, echogenicity, and cysts. Resistive index estimates vascular resistance. Essential for evaluating AKI to exclude obstruction.

Rhabdomyolysis and AKI

– Muscle breakdown releases myoglobin causing tubular obstruction and AKI. Elevated CK and dark urine. Aggressive IV hydration is key prevention and treatment. Alkalinize urine to increase myoglobin solubility.

Secondary Hyperparathyroidism

– Compensatory PTH increase in CKD from decreased phosphate excretion, low calcitriol, and hypocalcemia. Leads to renal osteodystrophy and vascular calcification. Managed with dietary phosphorus restriction, phosphate binders, active vitamin D, and calcimimetics. Refractory cases may require parathyroidectomy.

Shock Wave Lithotripsy

– Non-invasive procedure using extracorporeal shock waves to fragment renal and ureteral stones. The patient is positioned over a water cushion coupling the shock wave generator to the body. Shock waves are focused on the stone under fluoroscopic or ultrasound guidance. Most effective for stones less than two centimeters in the renal pelvis or upper ureter. Success rates decrease with larger stones and lower pole location. Complications include renal hematoma, stone street from multiple fragments, residual stones, and need for multiple sessions, though the non-invasive nature makes it the preferred first-line treatment for suitable stones.

SIADH

– Syndrome of Inappropriate Antidiuretic Hormone secretion characterized by hyponatremia (dilutional) with inappropriately concentrated urine (>100 mOsm/kg) and elevated urine sodium (>40 mEq/L). Euvolemic hyponatremia. Causes: CNS disorders (meningitis, stroke, tumors), pulmonary disorders (pneumonia, TB, small cell lung cancer), medications (SSRIs, carbamazepine, vincristine), and post-operative state. Diagnosis: euvolemia, hyponatremia, low BUN/uric acid, normal TSH/cortisol. Treatment: fluid restriction (first-line), demeclocycline, vaptans (tolvaptan), and IV hypertonic saline for severe symptomatic hyponatremia with risk of osmotic demyelination.

Sodium Bicarbonate Therapy

– Alkalinizing agent for metabolic acidosis in CKD. Target serum bicarbonate greater than 22 mEq/L. Slows CKD progression and improves muscle wasting. Dose titrated to serum levels.

Specific Gravity

– Ratio of urine density to water density reflecting solute concentration. Normal 1.005 to 1.030. Less precise than osmolality but faster to measure. Elevated in dehydration, glycosuria, proteinuria, and contrast. Decreased in diabetes insipidus and fluid overload. Affected by molecular weight of dissolved substances and influenced by glucose and protein in urine.

Staghorn Calculus

– A large renal stone that occupies the renal pelvis and extends into at least two calyces, creating a cast of the collecting system. Composition: struvite (magnesium ammonium phosphate), associated with urease-producing bacteria (Proteus mirabilis, Klebsiella, Pseudomonas) causing urinary tract infection. More common in women and patients with recurrent UTIs, neurogenic bladder, or urinary diversion. Presents with recurrent UTIs, flank pain, hematuria, and can lead to renal failure if untreated. Diagnosis: CT demonstrates branching stone filling the collecting system. Treatment: PCNL (percutaneous nephrolithotomy) is the gold standard; complete stone clearance essential to prevent recurrence; antibiotics for concurrent infection.

Struvite Stones

– Infection stones composed of magnesium ammonium phosphate. Form in alkaline urine with urease-producing organisms including Proteus and Klebsiella. Staghorn calculi fill renal pelvis almost exclusively from Proteus. Treatment requires complete stone removal with antibiotics and urinary acidification. High recurrence if infection persists.

Thrombotic Microangiopathy

– Endothelial damage with platelet-rich microthrombi causing MAHA with schistocytes, thrombocytopenia, and organ ischemia. TTP from ADAMTS13 deficiency treated with plasmapheresis. HUS from Shiga toxin managed supportively. Atypical HUS from complement dysregulation responds to eculizumab.

Tram-Track Appearance

– Histological finding on silver stain of MPGN representing GBM duplication. Results from mesangial cell interposition into capillary wall with new basement membrane formation. Creates double-contour pattern on Jones methenamine silver stain. Indicates chronic glomerular injury with capillary wall remodeling seen in immune complex-mediated and complement-mediated glomerulonephritis.

Uraemia

– The clinical syndrome resulting from accumulation of urea and other nitrogenous waste products in the blood in advanced chronic kidney disease (CKD stage 4-5) and acute kidney injury. Uraemia is characterized by nausea, vomiting, anorexia, fatigue, pruritus, insomnia, hiccups, muscle cramps, neuropathy (restless legs, peripheral neuropathy), pericarditis, platelet dysfunction (prolonged bleeding time, easy bruising), metabolic acidosis, hyperkalemia, hyperphosphatemia, and encephalopathy (lethargy, confusion, asterixis, coma). The uraemic syndrome results from accumulation of many organic waste compounds (middle molecules such as beta-2-microglobulin, advanced glycation endproducts, p-cresol). Dialysis reverses most symptoms. Uraemic pericarditis is an indication for urgent dialysis. Uraemic fetor is the characteristic ammonia-like odour of the breath.

Uremic Encephalopathy

– Neurological syndrome in severe renal failure. Confusion, asterixis, seizures, coma. Results from uremic toxin accumulation. Diagnosis is clinical in context of advanced renal failure. Emergent dialysis indicated with rapid neurological improvement typically.

Uremic Frost

– Urea crystallization on skin in severe untreated uremia. Rare finding indicating extremely advanced uremia. White crystalline deposits on skin. Indicates immediate need for dialysis. Extremely rare in developed countries due to early detection.

Uremic Pericarditis

– Pericardial inflammation in advanced uremia. Pleuritic chest pain, friction rub, diffuse ST elevation. Indicates need for emergent dialysis. Does not respond to anti-inflammatory therapy. Complications include tamponade. Emergent hemodialysis produces resolution within 2 to 3 weeks.

Uremic Platelet Dysfunction

– Impaired platelet adhesion and aggregation in uremia causing bleeding tendency. Easy bruising, epistaxis, GI bleeding. Improves with dialysis and desmopressin. Cryoprecipitate provides temporary benefit before procedures.

Ureteral Obstruction

– Blockage of urine flow causing proximal dilation and increased pressure. Acute causes renal colic with flank pain. Chronic leads to hydronephrosis and CKD. Emergent decompression with stent or nephrostomy for infected obstructed kidney. Diagnosis by CT or ultrasound showing hydronephrosis.

Ureteroscopy

– Endoscopic procedure using a flexible or rigid ureteroscope passed through the urethra and bladder into the ureter and renal pelvis. Stones are visualized directly and fragmented with holmium laser lithotripsy. Dust and fragments are extracted with baskets or left to pass spontaneously. Effective for ureteral stones at all locations and renal stones less than two centimeters. Advantages include direct visualization and ability to treat stones in one setting. Complications include ureteral injury, perforation, stricture formation, and infection. Ureteroscopy has largely replaced shock wave lithotripsy for distal and mid-ureteral stones and is increasingly used for renal stones given its high single-session stone-free rate.

Urethritis

– Inflammation of the urethra, most commonly caused by sexually transmitted infections (Chlamydia trachomatis, Neisseria gonorrhoeae) or noninfectious causes. Presents with dysuria, urethral discharge, and meatal itching. Diagnosis: urine nucleic acid amplification testing (NAAT). Treatment: empiric antibiotics covering gonorrhea and chlamydia; partner treatment essential. Complications: epididymitis, prostatitis, pelvic inflammatory disease.

Uric Acid Nephropathy

– Acute kidney injury from uric acid crystal precipitation in tubules. Occurs in tumor lysis syndrome and severe hyperuricemia. Prevent with hydration and rasburicase. Alkalinization of urine may be harmful as uric acid precipitates at alkaline pH unlike other stone types.

Uric Acid Stones

– 5 to 10% of kidney stones. Form in acidic urine with pH less than 5.5 and hyperuricosuria. Risk factors include gout, high purine diet, and chronic diarrhea. Radiolucent but visible on CT. Treat with urine alkalinization with potassium citrate and allopurinol. May dissolve with alkalinization which is unique among stone types.

Urinary Bladder

– See Bladder Health and Disorders / Bladder Problems.

Urinary Incontinence

– Involuntary loss of urine, classified as stress (leakage with cough, sneeze, exertion), urge (overactive bladder with sudden strong urge and leakage), mixed, overflow, and functional. Common causes: pelvic floor weakness, prostate enlargement, neurologic disorders, and medications. Diagnosis: history, voiding diary, post-void residual, urodynamics. Treatment: pelvic floor exercises, anticholinergics, beta-3 agonists, and surgery for stress incontinence.

Urinary Retention

– Inability to completely empty the bladder, which can be acute (sudden, painful, inability to void) or chronic (gradual, often painless, with overflow incontinence). Causes: benign prostatic hyperplasia, urethral stricture, neurogenic bladder, medications (anticholinergics), and pelvic organ prolapse. Diagnosis: post-void residual >100 mL, ultrasound, cystoscopy. Acute treatment: urethral catheterization. Chronic: alpha-blockers, 5-alpha-reductase inhibitors, and surgery.

Urine Osmolality

– Concentration of urine relative to plasma normally 50 to 1200 mOsm/kg. High greater than 500 in prerenal states reflecting intact concentrating ability. Low less than 350 indicates impaired concentrating ability in diabetes insipidus, chronic kidney disease, and excessive fluid intake. More precise than specific gravity. In prerenal azotemia, intact ADH action produces concentrated urine while damaged tubules cannot concentrate effectively.

Urine Sediment

– Microscopic examination of centrifuged urine providing diagnostic information about renal pathology. Normal findings include few RBCs, WBCs, and hyaline casts. Abnormal findings include RBC casts indicating glomerulonephritis, WBC casts indicating pyelonephritis or interstitial nephritis, granular casts in ATN, waxy casts in advanced CKD, and oval fat bodies in nephrotic syndrome. Essential component of AKI and proteinuria evaluation.

Urine Sodium

– Concentration of sodium in urine measured in mEq/L. Low less than 20 in prerenal states reflecting avid tubular reabsorption. High greater than 40 in intrinsic renal disease, diuretic use, and salt-wasting nephropathies. Should be interpreted in context of volume status, medications, and concurrent therapies. Key component in differentiating prerenal from intrinsic causes of AKI.

Urolithiasis

– Stone formation in the urinary tract (kidneys, ureters, bladder, urethra). Types: calcium oxalate (most common, 60-70%), calcium phosphate (15-20%), struvite (infection-related, 10-15%), uric acid (5-10%), and cystine (1-2%). Risk factors: dehydration, hypercalciuria, hyperoxaluria, hyperuricosuria, hypocitraturia, and urinary tract infections. Ureteral stones present with colicky flank pain radiating to groin, hematuria, nausea/vomiting. Diagnosis: non-contrast CT (gold standard), ultrasound. Treatment: medical expulsive therapy (tamsulosin) for distal <10mm stones; ESWL, ureteroscopy with laser lithotripsy, or PCNL for larger/complex stones.

Vesicoureteral Reflux

– Retrograde flow of urine from the bladder into the ureters and potentially the kidneys, most commonly due to incompetent ureterovesical valve. Predisposes to pyelonephritis and renal scarring. More common in children, presenting with recurrent febrile UTIs. Diagnosis: voiding cystourethrogram (VCUG). Grading I-V. Management: observation with prophylactic antibiotics for low-grade, ureteral reimplantation or endoscopic injection for high-grade.

Vitamin D Deficiency

– Serum 25-hydroxyvitamin D less than 20 ng/mL. Causes osteomalacia, secondary hyperparathyroidism, and increased fracture risk. Supplement with cholecalciferol and monitor levels. Important in CKD management as part of mineral bone disorder treatment.

Waxy Casts

– Broad refractile cylindrical structures with smooth appearance indicating severe chronic kidney disease. Form from degeneration of granular casts in dilated tubules. Broad diameter reflects formation in dilated collecting ducts. Relatively specific for advanced CKD with significant tubular atrophy. Along with broad casts represent end-stage tubular dysfunction.

Wilms Tumor

– Most common renal malignancy in children. Large abdominal mass. Associated with WAGR, Beckwith-Wiedemann, and hemihypertrophy. Treat with surgery and chemotherapy. Excellent prognosis exceeding 90%.

Xanthogranulomatous Pyelonephritis

– Chronic destructive renal infection with replacement of parenchyma by lipid-laden macrophages. Associated with obstructing stones and Proteus infection. Enlarged kidney with scattered low-attenuation areas on CT resembling a bear paw. Treated with nephrectomy as the kidney is nonfunctional and harbors chronic infection.

Chapter 20

Neurology

Acoustic Neuroma

– A benign, slow-growing schwannoma arising from the Schwann cells of the vestibular portion of the vestibulocochlear nerve (CN VIII), most commonly at the junction of the internal auditory canal and the cerebellopontine angle. Also known as vestibular schwannoma. Accounts for approximately 8 percent of intracranial tumors. Most are unilateral and sporadic; bilateral acoustic neuromas are pathognomonic of neurofibromatosis type 2 (NF2), an autosomal dominant condition caused by mutations in the NF2 gene on chromosome 22q12.2. Clinical presentation includes ipsilateral sensorineural hearing loss (typically high-frequency), tinnitus, vertigo or imbalance, and facial numbness or weakness from compression of the trigeminal or facial nerve. Diagnosis is confirmed by MRI with gadolinium, demonstrating an enhancing mass at the cerebellopontine angle with extension into the internal auditory canal. Treatment options include observation with serial imaging for small or slow-growing tumors, stereotactic radiosurgery (Gamma Knife), or microsurgical resection via a translabyrinthine, retrosigmoid, or middle cranial fossa approach.

Agenesis of the Corpus Callosum

– Complete or partial absence of the corpus callosum, the major commissural fiber bundle connecting the cerebral hemispheres. May be isolated or associated with other CNS anomalies (Dandy-Walker, Chiari, heterotopias) and genetic syndromes (Aicardi, Andermann, acrocallosal syndromes). Presents with a variable phenotype ranging from asymptomatic to developmental delay, intellectual disability, seizures, and impaired interhemispheric transfer. Diagnosis by prenatal or postnatal MRI (coronal and sagittal views show absent cingulate gyrus, colpocephaly, and radial orientation of medial sulci). No specific treatment; management addresses associated conditions.

Age-Related Hearing Loss

– Progressive, bilateral, symmetrical sensorineural hearing loss associated with aging, also known as presbycusis, affecting approximately one-third of adults between 65 and 74 years and nearly half of those over 75. The pathogenesis involves cumulative damage to cochlear structures including degeneration of outer hair cells (beginning at the basal turn affecting high-frequency sound transduction), loss of spiral ganglion neurons, stria vascularis atrophy (reducing endocochlear potential), and basilar membrane stiffness. Diagnosis is clinical, based on history of progressive bilateral high-frequency hearing loss with preserved speech discrimination in early stages. Audiometry demonstrates bilateral sensorineural hearing loss, typically with a downward-sloping configuration affecting higher frequencies more than lower frequencies. Management includes hearing aids as the mainstay of treatment, assistive listening devices, cochlear implantation for severe-to-profound loss, and avoidance of ototoxic medications.

Agoraphobia

– Anxiety disorder characterized by fear and avoidance of situations where escape might be difficult or help unavailable during a panic-like episode. Common feared situations include public transportation, open spaces, enclosed spaces, standing in line, being in crowds, and being outside the home alone. Often develops after panic attacks and becomes self-reinforcing through avoidance behavior. Severe cases result in housebound patients unable to leave home. Diagnosis: DSM-5 criteria. Treatment: cognitive behavioural therapy (CBT) as first-line psychological intervention, SSRIs (particularly sertraline, paroxetine, escitalopram) as first-line pharmacotherapy, and combination therapy for moderate-to-severe cases. Prognosis is variable, but with appropriate treatment most patients achieve significant improvement.

Allodynia

– A form of neuropathic pain where a normally non-painful stimulus evokes pain. It is a key feature of central and peripheral sensitization and can be categorized into tactile allodynia (pain from light touch or pressure), mechanical allodynia (pain from movement or stroking of the skin), and thermal allodynia (pain from normally non-painful temperature changes). Allodynia is commonly seen in postherpetic neuralgia, where the skin remains exquisitely sensitive to clothing or light touch. Diagnosis is clinical, supported by quantitative sensory testing and response to treatments. Management includes addressing the underlying cause, with pharmacological options including gabapentinoids (gabapentin, pregabalin), tricyclic antidepressants (amitriptyline, nortriptyline), serotonin-noradrenaline reuptake inhibitors (duloxetine, venlafaxine), lidocaine patches, capsaicin cream, and opioid analgesics as a last resort.

Alzheimer Disease

– The most common neurodegenerative cause of dementia, accounting for approximately sixty to seventy percent of all dementia cases. It is characterized by progressive cognitive decline, memory loss, and functional impairment, with pathologic hallmarks of extracellular amyloid-beta (A-beta) plaques and intracellular neurofibrillary tangles composed of hyperphosphorylated tau protein. The pathophysiology begins decades before clinical symptoms, with accumulation of amyloid-beta (A-beta) plaques, followed by tau hyperphosphorylation, neurofibrillary tangle formation, synaptic dysfunction, neuronal loss, and brain atrophy, particularly affecting the hippocampus, entorhinal cortex, and temporoparietal association areas. Clinical presentation includes progressive impairment of short-term memory as the earliest and most prominent feature, followed by deficits in language (anomia, fluent aphasia), visuospatial function (topographical disorientation), executive function (poor judgement, impaired planning), and apraxia. Diagnosis is based on clinical criteria, cognitive testing (MMSE, MoCA), and supportive biomarkers including CSF analysis (reduced A-beta-42, elevated total tau and phospho-tau) and amyloid PET imaging. Management includes acetylcholinesterase inhibitors (donepezil, rivastigmine, galantamine) for mild-to-moderate disease and the NMDA receptor antagonist memantine for moderate-to-severe disease, along with behavioural interventions and caregiver support.

Amyotrophic Lateral Sclerosis

– Progressive neurodegenerative disorder affecting both upper and lower motor neurons, causing weakness, atrophy, fasciculations, and eventually respiratory failure. Most cases are sporadic (90 percent); 10 percent are familial. Average age of onset 55-60 years. Presents with asymmetric limb weakness (limb-onset) or bulbar symptoms (bulbar-onset: dysarthria, dysphagia). Upper motor neuron signs: spasticity, hyperreflexia, extensor plantar responses. Lower motor neuron signs: weakness, atrophy, fasciculations (visible muscle twitching), and cramps. Diagnosis is clinical (El Escorial criteria) with supportive electrodiagnostic studies (EMG showing active denervation and chronic reinnervation in multiple regions), neuroimaging to exclude mimics, and genetic testing for familial cases. Riluzole, a glutamate antagonist, prolongs survival by approximately 2-3 months by reducing excitotoxicity. Edaravone, a free radical scavenger, may slow functional decline in some patients. Multidisciplinary care including respiratory support (non-invasive ventilation for respiratory insufficiency), nutritional support (percutaneous endoscopic gastrostomy for dysphagia), speech therapy, and palliative care is essential for optimising quality of life. Median survival is 3-5 years from symptom onset, with respiratory failure as the most common cause of death.

Anencephaly

– A fatal neural tube defect where the cerebral hemispheres and cranial vault fail to develop due to failed closure of the rostral neuropore during the fourth week of gestation. The brainstem and spinal cord are present, and reflex movements may occur, but there is no cognitive function. Associated with polyhydramnios (due to impaired swallowing) and elevated maternal serum alpha-fetoprotein. Diagnosis by prenatal ultrasound. Most cases are stillborn or die within hours to days after birth. Prevention with folic acid supplementation.

Anxiety Disorders

– Group of psychiatric disorders characterized by excessive fear, worry, and anxiety that cause significant distress and impairment. Types include generalized anxiety disorder (persistent, excessive worry about multiple topics), panic disorder (recurrent unexpected panic attacks with palpitations, sweating, trembling), social anxiety disorder (fear of social situations), specific phobias (irrational fear of specific objects/situations), and agoraphobia (fear of situations where escape may be difficult or embarrassing. Treatment includes CBT as first-line psychological treatment and SSRIs (sertraline, escitalopram, paroxetine) as first-line pharmacotherapy. Prognosis is generally favourable with treatment, though relapse rates are significant.

Aphasia

– Impairment of language comprehension or production resulting from brain damage, most commonly from stroke affecting the left hemisphere (dominant) language areas. Types: Broca (expressive) aphasia from left inferior frontal gyrus damage—non-fluent speech, intact comprehension, frustration with communication; Wernicke (receptive) aphasia from left superior temporal gyrus damage—fluent but nonsensical speech, impaired comprehension, poor repetition; global aphasia from large left hemisphere stroke involving both anterior and posterior circulation territories —non-fluent speech, severely impaired comprehension, and poor repetition. Additional types include conduction aphasia (arcuate fasciculus lesion—fluent speech with impaired repetition but relatively intact comprehension), transcortical motor aphasia (disconnection of supplementary motor area—non-fluent speech with intact repetition), transcortical sensory aphasia (temporoparietal lesion—fluent speech with impaired comprehension but intact repetition), and anomic aphasia (angular gyrus lesion—word-finding difficulty with fluent speech). Diagnosis is through formal language assessment using the Boston Diagnostic Aphasia Examination or the Western Aphasia Battery. Treatment involves speech and language therapy, with compensatory strategies and communication aids for severe cases. Prognosis depends on the size and location of the lesion; Broca aphasia generally has a better recovery than Wernicke or global aphasia.

Apraxia

– Inability to perform learned, skilled motor movements despite intact motor function, sensation, coordination, and comprehension. Types: ideomotor apraxia (inability to perform movements on command, e.g., waving goodbye or using a hammer, due to parietal lobe lesions, typically left hemisphere), ideational apraxia (inability to sequence multi-step tasks, e.g., making a cup of tea, due to frontal or temporal lobe dysfunction), limb-kinetic apraxia (loss of fine motor dexterity, e.g., difficulty with finger tapping, due to premotor cortex lesions), buccofacial/oral apraxia (difficulty performing facial movements on command, e.g., blowing out a candle), and gait apraxia (difficulty initiating walking, common in normal pressure hydrocephalus). Apraxia is most commonly caused by stroke affecting the dominant (left) hemisphere, but can also result from Alzheimer disease, corticobasal degeneration, brain tumors, and traumatic brain injury. Diagnosis is clinical; there is no specific pharmacological treatment. Management: occupational therapy, task-specific training, and compensatory strategies.

Ataxia

– Lack of voluntary coordination of muscle movements, resulting from dysfunction of the cerebellum, posterior columns, or peripheral nerves. Types: cerebellar (gait and limb ataxia, dysarthria, nystagmus), sensory (impaired proprioception causing sensory ataxia, positive Romberg sign), and vestibular (vertigo, nystagmus, nausea). Cerebellar causes include stroke, tumors, alcohol, medications (phenytoin), degenerative diseases (spinocerebellar ataxias), and multiple sclerosis. Sensory causes include peripheral neuropathy, tabes dorsalis (neurosyphilis), and vitamin B12 deficiency causing posterior column dysfunction. Diagnosis is clinical, supported by neuroimaging (MRI brain for cerebellar causes, MRI spine for spinal causes), nerve conduction studies, and blood tests. Treatment is directed at the underlying cause, with symptomatic management including physical therapy, occupational therapy, and adaptive devices.

Autism Spectrum Disorder

– Neurodevelopmental disorder characterized by persistent deficits in social communication and social interaction across multiple contexts, restricted and repetitive patterns of behavior, interests, or activities. Symptoms present in early developmental period and cause clinically significant impairment. Severity levels: Level 1 (requiring support), Level 2 (requiring substantial support), Level 3 (requiring very substantial support). Associated features include intellectual disability (30-40 percent of cases), seizures, sleep disturbances, and gastrointestinal symptoms. Diagnosis is clinical using DSM-5 criteria, supported by multidisciplinary assessment including developmental history, cognitive testing, and speech and language evaluation. Management involves early intensive behavioural intervention (applied behaviour analysis), speech and language therapy, occupational therapy, social skills training, pharmacological treatment of associated conditions (SSRIs for anxiety, stimulants for inattention, antipsychotics for irritability and aggression), and educational support. Prognosis is highly variable, with early intervention being the most important predictor of functional outcome.

Babinski Sign

– The Babinski sign is a pathological reflex elicited by stroking the lateral aspect of the sole of the foot from the heel to the metatarsal heads with a blunt object. A normal response (flexor plantar response) is plantar flexion (curling downward) of all toes. An abnormal Babinski sign (extensor plantar response) is dorsiflexion (upgoing) of the great toe with fanning (abduction) of the other toes, indicating an upper motor neuron lesion affecting the corticospinal tract. The Babinski sign is a sign of corticospinal tract dysfunction rather than a specific disease. In infants, an extensor plantar response (upgoing toes) is normal until approximately 12-24 months of age due to incomplete myelination of the corticospinal tract, after which a flexor plantar response becomes established. A persistent Babinski sign beyond this age indicates upper motor neuron pathology and can be caused by stroke, multiple sclerosis, spinal cord injury, traumatic brain injury, cerebral palsy, amyotrophic lateral sclerosis, and other neurodegenerative conditions affecting the corticospinal tract.

Balance

– The ability to maintain the body's center of mass over its base of support, relying on input from the vestibular system, vision, and proprioception. It is essential for walking, standing, and performing daily activities without falling. Age-related decline, medications, and neurological disorders can impair balance, increasing fall risk.

Balance Disorders and Vertigo

– Balance disorders encompass conditions that cause dizziness, unsteadiness, or a sensation of motion, often arising from dysfunction in the inner ear, brain, or sensory pathways. Vertigo specifically describes the false sensation that the environment is spinning, commonly due to benign paroxysmal positional vertigo, vestibular neuritis, or Ménière's disease. Diagnosis involves history, physical exam, and vestibular testing; treatment includes repositioning maneuvers, medications, and vestibular rehabilitation.

Becker Muscular Dystrophy

– An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene, similar to Duchenne muscular dystrophy (DMD), but with in-frame mutations that allow production of some functional dystrophin protein, albeit reduced in quantity or abnormal in size. The result is a milder clinical phenotype compared to DMD. The incidence is approximately one in 18,000 to 30,000 male births. Clinical features include progressive proximal muscle weakness, typically beginning in the hip girdle and thighs (difficulty climbing stairs, rising from chairs, and walking), calf hypertrophy (pseudohypertrophy due to fatty infiltration, which is characteristic), waddling gait, Gowers manoeuvre (using hands to push off the floor and climb up the legs to stand), and reduced or absent deep tendon reflexes. Progression is slower than DMD, with loss of ambulation typically occurring in the fourth decade or later. Cardiac involvement (dilated cardiomyopathy) is common and is the leading cause of death, requiring regular cardiac surveillance. Respiratory function declines with disease progression. Serum creatine kinase levels are elevated (typically 5-10 times normal). Diagnosis is confirmed by genetic testing showing in-frame dystrophin gene mutations, with muscle biopsy reserved for cases where genetic testing is inconclusive. Management involves multidisciplinary care including physiotherapy to maintain flexibility, corticosteroids to slow progression (although less evidence than in DMD), cardiac monitoring with ACE inhibitors or beta-blockers, respiratory support, and orthopaedic management for contractures and scoliosis.

Bell Palsy

– Acute, idiopathic, unilateral facial nerve (CN VII) paralysis of lower motor neuron type, with an incidence of approximately 20-30 per 100,000 person-years. The cause is thought to be herpes simplex virus type 1 reactivation in the geniculate ganglion, leading to inflammation, demyelination, and edema of the facial nerve within the narrow confines of the facial canal (fallopian canal) of the temporal bone. Onset is acute and progressive over hours to days, with maximal weakness reached within 48 hours. Clinical features include acute-onset unilateral facial weakness that involves both the upper and lower face (forehead, eye closure, and mouth), inability to close the eye (lagophthalmos), which can lead to corneal exposure and ulceration, hyperacusis (due to stapedius muscle paralysis), loss of taste in the anterior two-thirds of the tongue (due to chorda tympani nerve involvement), and decreased tearing and salivation. Pain behind the ear is common and may precede the weakness. Diagnosis is clinical, requiring exclusion of other causes of facial paralysis such as stroke (upper motor neuron pattern sparing the forehead), Ramsay Hunt syndrome (herpes zoster oticus), Lyme disease, Guillain-Barre syndrome, otitis media, and parotid tumors. Treatment includes oral corticosteroids (prednisolone 60 mg daily for 7 days with rapid taper) started within 72 hours of symptom onset to improve outcomes, and eye protection with artificial tears, lubricating ointment, and taping the eye closed at night to prevent corneal damage. Antiviral agents (acyclovir or valacyclovir) are added if Ramsay Hunt syndrome is suspected. Prognosis is excellent, with approximately 70 percent of patients achieving near-complete recovery within 3-6 months, 15 percent developing some residual weakness, and 15 percent having moderate-to-severe permanent weakness.

Bell's Palsy

– See Bell Palsy.

Brain Abscess

– A focal, purulent infection within the brain parenchyma, most commonly occurring in the frontal or temporal lobes. Sources of infection include direct contiguous spread (otitis media and mastoiditis causing temporal lobe abscess; sinusitis causing frontal lobe abscess; dental infection), hematogenous dissemination from a distant source (bacterial endocarditis, lung abscess, congenital heart disease with right-to-left shunt), penetrating head trauma, or post-neurosurgical infection. Common causative organisms include Streptococcus milleri group (most common), anaerobic bacteria (Bacteroides, Fusobacterium), Staphylococcus aureus, and aerobic Gram-negative bacilli. Clinical presentation includes headache, fever, focal neurological deficits (hemiparesis, aphasia, visual field defects), seizures, and signs of increased intracranial pressure (nausea, vomiting, decreased level of consciousness). Diagnosis is by contrast-enhanced CT or MRI brain showing a ring-enhancing lesion with surrounding edema and central necrosis. Management includes empiric broad-spectrum antibiotics (third-generation cephalosporin plus metronidazole covering aerobic and anaerobic organisms), stereotactic aspiration or surgical excision for definitive diagnosis and source control, corticosteroids for vasogenic edema, and anticonvulsants for seizure prophylaxis. Mortality is approximately 10-20 percent, with prompt treatment improving outcomes.

Brain Health

– Brain health encompasses the ability to perform cognitive functions such as memory, learning, decision-making, and emotional regulation across the lifespan. Key factors include cardiovascular health, nutrition, sleep, social engagement, cognitive stimulation, and management of chronic conditions like hypertension and diabetes. Protecting brain health reduces the risk of dementia, stroke, and other neurological disorders.

Brain Injuries

– Brain injuries range from mild concussions to severe traumatic brain injury caused by falls, motor vehicle accidents, sports impacts, or penetrating trauma. Symptoms depend on injury location and severity, including headache, confusion, memory loss, seizures, and altered consciousness. Recovery varies widely and may involve rest, rehabilitation, cognitive therapy, and long-term supportive care.

Brain Metastases

– The most common intracranial neoplasms, occurring in approximately ten to forty percent of patients with systemic cancer, and are more common than primary brain tumors. They result from hematogenous spread of tumor cells to the brain, with the most common primary sources being lung cancer (the most common source overall, particularly non-small cell lung cancer and small cell lung cancer), breast cancer, melanoma (highest propensity for brain metastases), renal cell carcinoma, and colorectal cancer. Brain metastases typically occur at the gray-white matter junction, where the vascular supply changes from larger to smaller calibre vessels, and are most frequently located in the cerebral hemispheres (80 percent), cerebellum (15 percent), and brainstem (5 percent). Clinical presentation depends on the location and number of lesions but commonly includes headache, focal neurological deficits, seizures, cognitive impairment, and signs of increased intracranial pressure. Diagnosis is by contrast-enhanced MRI, which is more sensitive than CT for detecting small lesions. Management includes corticosteroids (dexamethasone) for vasogenic cerebral edema, whole-brain radiotherapy for multiple metastases, stereotactic radiosurgery for oligometastatic disease (typically 1-3 lesions), and surgical resection for single accessible lesions in patients with good performance status and controlled systemic disease.

Brain Tumor

– An abnormal growth of cells within the brain or its surrounding structures, classified as primary (originating from brain tissue or its coverings) or secondary/metastatic (spread from a cancer elsewhere in the body, which are more common, accounting for approximately 50 percent of all brain tumors). Primary brain tumors are classified by cell type and grade: meningiomas (most common primary brain tumor in adults, arising from the meninges, typically benign WHO grade 1), gliomas (arising from glial cells—including astrocytomas, oligodendrogliomas, glioblastomas (the most common and most malignant primary brain tumor, WHO grade 4), ependymomas, and mixed gliomas. Clinical presentation depends on the location, size, and rate of growth of the tumor but commonly includes progressive headache (often worse in the morning and worsened by Valsalva manoeuvre), seizures (focal or generalised), focal neurological deficits (hemiparesis, aphasia, visual field defects, sensory loss), cognitive and personality changes, and signs of increased intracranial pressure. Diagnosis is by neuroimaging: MRI with gadolinium is the imaging modality of choice, providing superior soft tissue resolution and demonstrating contrast enhancement, edema, mass effect, and relationship to adjacent structures. Definitive diagnosis requires histopathological examination of tissue obtained by stereotactic biopsy or surgical resection. Management is multidisciplinary and depends on tumor type, grade, and location, and includes surgical resection (gross total resection when safely achievable), radiotherapy, and chemotherapy (temozolomide for glioblastoma). Prognosis varies widely by tumor type and grade: benign meningiomas have excellent prognosis with surgical resection, while glioblastoma multiforme carries a poor prognosis with median survival of approximately 15 months despite aggressive multimodality treatment.

Brain Ventricles

– The four interconnected fluid-filled cavities within the brain that produce and contain cerebrospinal fluid (CSF). The lateral ventricles (first and second, paired, C-shaped, within each cerebral hemisphere, each with anterior/frontal horn, body, atrium, posterior/occipital horn, and temporal/inferior horn) communicate with the third ventricle via the interventricular foramina (foramina of Monro). The third ventricle (midline, within the diencephalon, between the thalami) communicates with the fourth ventricle via the cerebral aqueduct (aqueduct of Sylvius, within the midbrain). The fourth ventricle (between the brainstem and cerebellum) is continuous with the central canal of the spinal cord and drains CSF into the subarachnoid space via the midline foramen of Magendie and paired lateral foramina of Luschka. Hydrocephalus: accumulation of CSF causing ventricular enlargement, classified as communicating (impaired CSF absorption) or non-communicating/obstructive (blockage within the ventricular system). CSF is produced by the choroid plexus (modified ependymal cells) within the ventricles at a rate of 500 mL/day., which encodes a copper-transporting ATPase in hepatocytes. The defective ATP7B protein impairs incorporation of copper into apoceruloplasmin and biliary excretion of copper, leading to progressive copper accumulation in the liver, brain, cornea, and other organs. Clinical manifestations typically present between ages five and thirty-five and include hepatic disease (ranging from asymptomatic elevation of liver enzymes to acute hepatitis, cirrhosis, and fulminant hepatic failure), neurological disease (dysarthria, ataxia, dystonia, parkinsonism, tremor—particularly a wing-beating tremor of the arms, chorea, and cognitive impairment with executive dysfunction), psychiatric symptoms (depression, personality changes, psychosis), and Kayser-Fleischer rings (golden-brown discolouration at the limbus of the cornea due to copper deposition in Descemet membrane, visible on slit-lamp examination and pathognomonic for Wilson disease when present). Diagnosis is based on a combination of clinical features, Kayser-Fleischer rings on slit-lamp examination, low serum caeruloplasmin (less than 0.2 g/L), elevated 24-hour urinary copper excretion (greater than 100 micrograms), elevated hepatic copper content on liver biopsy, and genetic testing for ATP7B mutations. Serum free copper (non-caeruloplasmin-bound copper) is elevated. Treatment involves lifelong copper chelation therapy with D-penicillamine (first-line, acts by chelating copper and promoting urinary excretion) or trientine, zinc acetate (which blocks intestinal absorption of copper and induces hepatic metallothionein, used as maintenance therapy or first-line for presymptomatic patients), and dietary restriction of copper-rich foods (shellfish, nuts, chocolate, mushrooms, organ meats). Liver transplantation is curative for severe hepatic failure or refractory neurological disease. Prognosis is excellent with early diagnosis and lifelong treatment; without treatment, Wilson disease is progressive and fatal. Patients who stop chelation therapy are at risk of irreversible neurological deterioration and hepatic decompensation.

Brown Sequard Syndrome

– Brown-Sequard syndrome (hemisection of the spinal cord) results from a lesion affecting one half of the spinal cord, producing a characteristic pattern of ipsilateral and contralateral neurological deficits. The ipsilateral side below the lesion demonstrates upper motor neuron paralysis (corticospinal tract), loss of vibration and proprioception (dorsal column), and loss of tactile sense. The contralateral side below the lesion demonstrates loss of pain and temperature sensation (spinothalamic tract). The syndrome is most commonly caused by penetrating trauma (knife or gunshot wounds), but may also result from spinal cord tumors, haematomas, disc herniation, demyelinating disease (multiple sclerosis), or infarction of the spinal cord (anterior spinal artery syndrome affecting the spinothalamic tract on the contralateral side). Diagnosis is clinical, confirmed by MRI of the spinal cord showing the lateralised lesion. Treatment is directed at the underlying cause and includes high-dose corticosteroids for inflammatory or demyelinating causes, surgical decompression for compressive lesions, and supportive care including pain management and rehabilitation. Prognosis depends on the etiology and severity of the cord injury, with some recovery possible over months.

Carpal Tunnel Syndrome

– The most common entrapment neuropathy, caused by compression of the median nerve as it passes through the carpal tunnel, a narrow fibro-osseous canal bounded by the carpal bones and the flexor retinaculum (transverse carpal ligament) at the wrist. The median nerve provides sensory innervation to the palmar aspect of the thumb, index finger, middle finger, and lateral half of the ring finger, and motor innervation to the thenar muscles (abductor pollicis brevis, opponens pollicis, and flexor pollicis brevis). The tunnel also contains nine flexor tendons (flexor digitorum profundus, flexor digitorum superficialis, and flexor pollicis longus) surrounded by synovial sheaths. Compression of the median nerve within the carpal tunnel results from any condition that reduces the volume of the tunnel or increases the volume of its contents: repetitive strain, obesity, pregnancy, diabetes, rheumatoid arthritis, hypothyroidism, acromegaly, amyloidosis, and wrist fractures. Clinical presentation includes numbness, tingling, and paraesthesiae in the median nerve distribution, typically occurring at night or with repetitive hand use. Patients may report waking with numb hands and shaking them for relief (flick sign). Motor weakness and thenar atrophy occur in advanced cases. Diagnosis is clinical, confirmed by nerve conduction studies showing prolonged distal motor and sensory latencies across the carpal tunnel. Treatment includes conservative measures (wrist splinting in neutral position at night, activity modification, NSAIDs, corticosteroid injection into the carpal tunnel) and surgical release (carpal tunnel release) for refractory or severe cases.

Cauda Equina Syndrome

– A neurosurgical emergency caused by compression of the cauda equina nerve roots below the level of the conus medullaris (typically at L1-L2 and below), resulting in bilateral lower extremity weakness, sensory loss, and bladder and bowel dysfunction. The most common cause is large lumbar disc herniation, but other causes include spinal tumors, epidural abscess, epidural hematoma, spinal stenosis, and trauma. Clinical features include severe low back pain, bilateral lower extremity weakness, bilateral sciatica, saddle anesthesia (loss of sensation in the perineal and perianal regions), loss of anal sphincter tone, urinary retention (painless, with overflow incontinence), and fecal incontinence. Urinary retention is an early and critical sign that is often missed; post-void residual volume greater than 100-200 mL is suggestive. Diagnosis is clinical, confirmed by urgent MRI of the lumbar spine showing compression of the cauda equina nerve roots. Management is a surgical emergency requiring urgent decompressive laminectomy and discectomy, ideally within 24-48 hours of symptom onset to maximise the chance of neurological recovery. Prognosis is better if surgery is performed early, particularly for urinary function, but long-term deficits may persist even with timely intervention.

Central Nervous System (CNS)

– The part of the nervous system comprising the brain and spinal cord, enclosed within the skull and vertebral column. The brain includes the cerebrum (consciousness, voluntary movement, sensation, language, cognition), cerebellum (coordination, balance, motor learning), and brainstem (vital autonomic centers, cranial nerve nuclei, ascendingdescending pathways). The spinal cord extends from the foramen magnum to L1–L2, transmitting motor and sensory information between brain and periphery via ascending (spinothalamic, dorsal column) and descending (corticospinal) tracts, and housing reflex arcs. The CNS is protected by the skull and vertebral column, meninges (dura mater, arachnoid mater, pia mater), and cerebrospinal fluid. The blood-brain barrier regulates molecular exchange between blood and neural tissue.

Cerebellar Ataxia

– A neurological sign characterized by impaired coordination of voluntary movements, resulting from dysfunction of the cerebellum or its connections. The cerebellum coordinates the timing, force, and accuracy of movements, and cerebellar dysfunction produces a characteristic constellation of findings. Gait ataxia presents with a wide-based, unsteady, staggering gait with difficulty performing tandem (heel-to-toe) walking. Limb ataxia includes dysmetria (overshooting or undershooting targets), dysdiadokokinesia (impaired rapid alternating movements), intention tremor (tremor that worsens as the target is approached), and rebound phenomenon. Speech is affected with scanning dysarthria (slow, irregular, slurred speech with variable volume and tempo). Ocular findings include nystagmus and impaired smooth pursuit. Cerebellar ataxia can be caused by stroke (ischemic or hemorrhagic), tumors, alcohol toxicity and nutritional deficiencies (thiamine), medications (phenytoin, carbamazepine, lithium, benzodiazepines), autoimmune conditions (anti-GAD, anti-glutamate receptor antibodies), paraneoplastic syndromes, neurodegenerative diseases (spinocerebellar ataxias, multiple system atrophy), multiple sclerosis, infectious and post-infectious causes, and hypothyroidism. Diagnosis is clinical with MRI brain to identify structural, inflammatory, or degenerative causes. Management focuses on treating the underlying cause and providing symptomatic relief with physical and occupational therapy, and assistive devices for gait instability.

Cerebral Aneurysm

– An abnormal focal dilation of a cerebral artery, most commonly at bifurcations of the Circle of Willis. Saccular (berry) aneurysms account for 90% of intracranial aneurysms. Risk factors: smoking, hypertension, female sex, polycystic kidney disease, Ehlers-Danlos, family history. Most are asymptomatic (incidental finding). Rupture causes subarachnoid hemorrhage: thunderclap headache, nuchal rigidity, photophobia, and altered mental status. Diagnosis: CTA or MRA for detection; DSA is gold standard. Treatment: microsurgical clipping or endovascular coiling for ruptured and high-risk unruptured aneurysms.

Cerebral Cortex

– The outermost layer of the cerebrum, composed of gray matter (neuronal cell bodies) folded into gyri (ridges) and sulci (grooves), greatly increasing surface area. Divided into four lobes: frontal (motor cortex, prefrontal cortex–executive function, personality, Broca area–speech production), parietal (somatosensory cortex, spatial orientation, integration of sensory modalities), temporal (auditory cortex, Wernicke area–language comprehension, memory formation via hippocampus), and occipital (primary visual cortex, visual processing). The cortex is organized into six horizontal layers (molecular, external granular, external pyramidal, internal granular, internal pyramidal, multiform) with distinct cell types and connectivity. Cortical functions are topographically organized (homunculus in motor and sensory cortices). Damage causes focal deficits corresponding to the affected region.

Cerebrospinal Fluid (CSF)

– A clear, colorless fluid that fills the ventricles of the brain, the central canal of the spinal cord, and the subarachnoid space around the brain and spinal cord. Produced primarily by the choroid plexus of the lateral, third, and fourth ventricles (500 mLday, total volume 150 mL in adults). Functions: mechanical cushioning of the brain (buoyancy reduces effective brain weight by 97%), removal of metabolic waste, transport of hormones and nutrients, and maintenance of stable intracranial pressure. Normal composition: protein 15–45 mgdL, glucose 50–80 mgdL (two-thirds of serum glucose), WBC 0–5mcL (mononuclear). CSF analysis by lumbar puncture is essential for diagnosing meningitis, subarachnoid hemorrhage, multiple sclerosis, and other neurologic conditions.

Cerebrovascular Accident (Stroke)

– Stroke is a neurological deficit caused by focal cerebral ischemia (ischemic stroke, 87%) or hemorrhage (hemorrhagic stroke, 13%). Ischemic stroke results from thrombotic or embolic occlusion of a cerebral artery. Risk factors: hypertension, atrial fibrillation, diabetes, smoking. Management: acute thrombolysis (tPA within 4.5 hours), mechanical thrombectomy for large vessel occlusion, secondary prevention with antiplatelets and risk factor control.

Cervical Disc Herniation

– A common cause of neck pain and radiculopathy, occurring when the nucleus pulposus of an intervertebral disc in the cervical spine herniates through the annulus fibrosus, compressing adjacent nerve roots or the spinal cord. The most commonly affected levels are C5-C6 and C6-C7, accounting for approximately ninety-five percent of cervical disc herniations. Cervical radiculopathy presents with dermatomal pain radiating from the neck into the shoulder, arm, or hand, along with sensory changes (numbness, tingling) and motor weakness in the corresponding myotome. C5-C6 disc herniation compresses the C6 nerve root, causing pain radiating to the lateral arm, thumb, and index finger, weakness of biceps, brachioradialis, and wrist extensors, and decreased biceps reflex. C6-C7 disc herniation compresses the C7 nerve root, causing pain radiating to the middle finger, weakness of triceps and finger extensors, and decreased triceps reflex. Diagnosis is by MRI of the cervical spine, which demonstrates the herniated disc and its relationship to the nerve root or spinal cord. Electromyography and nerve conduction studies may confirm the radiculopathy and exclude other causes. Management includes conservative treatment (analgesics, NSAIDs, muscle relaxants, physical therapy, cervical immobilisation) for most cases, with surgical intervention (anterior cervical discectomy and fusion, cervical disc replacement) indicated for progressive motor deficit, severe refractory pain, or spinal cord compression with myelopathy.

Cervical Radiculopathy

– A condition caused by compression, irritation, or inflammation of a spinal nerve root in the cervical spine, resulting in pain, numbness, tingling, and weakness in the distribution of the affected nerve root. The most commonly affected nerve roots are C6 and C7, accounting for approximately seventy to eighty percent of cervical radiculopathies. C5 radiculopathy causes shoulder pain, weakness of deltoid and biceps, and decreased biceps reflex. C6 radiculopathy causes pain radiating from the neck into the lateral arm, thumb, and index finger, weakness of biceps and brachioradialis, and decreased biceps reflex. C7 radiculopathy causes pain radiating into the middle finger, weakness of triceps and wrist and finger extensors, and decreased triceps reflex. C8 radiculopathy causes pain radiating into the medial arm, ring and little fingers, weakness of hand intrinsics, and decreased finger jerk. Diagnosis is by MRI of the cervical spine, with EMG/NCS confirming the level of nerve root involvement and excluding peripheral nerve entrapment. Management is initially conservative with analgesics, NSAIDs, physical therapy, and cervical epidural steroid injections. Surgical options include anterior cervical discectomy and fusion (ACDF) or cervical disc replacement for persistent or progressive symptoms despite conservative therapy, or for cases with myelopathy.

Chiari Malformation

– Chiari malformations are structural abnormalities of the posterior fossa and cerebellum characterized by downward displacement of the cerebellar tonsils through the foramen magnum. Type I is the most common variant, defined as cerebellar tonsillar ectopia of five millimeters or more below the foramen magnum. It may be asymptomatic or present with occipital headache (worsened by Valsalva maneuver, coughing, or straining), neck pain, lower cranial nerve palsies, cerebellar ataxia, myelopathy, and syringomyelia (a fluid-filled cavity within the spinal cord). Type II (Arnold-Chiari malformation) is a more severe variant associated with myelomeningocoele and hydrocephalus. Diagnosis is made by MRI of the brain and cervical spine in the sagittal plane, which demonstrates tonsillar herniation. Surgical decompression via suboccipital craniectomy and cervical laminectomy is indicated for symptomatic patients with brainstem compression, syringomyelia, or progressive neurological deficits.

Chronic Fatigue Syndrome

– A complex, debilitating disorder characterized by profound fatigue lasting >6 months that is not relieved by rest and is worsened by physical or mental exertion (post-exertional malaise). Associated symptoms: unrefreshing sleep, cognitive impairment (brain fog), orthostatic intolerance, joint pain, headaches, and sore throat. Also known as myalgic encephalomyelitis (ME/CFS). Diagnosis: clinical (CDC or Canadian Consensus criteria); exclusion of other causes. No FDA-approved treatment; management focuses on symptom relief, graded activity pacing, cognitive behavioral therapy for coping, and treating comorbid conditions (sleep disorders, orthostatic intolerance).

Chronic Inflammatory Demyelinating Polyneuropathy

– An acquired, immune-mediated peripheral neuropathy with a relapsing-remitting or progressive course. Presents with symmetric proximal and distal weakness, sensory loss, areflexia, and elevated CSF protein. Distinguishing from Guillain-Barre: progressive course >8 weeks (vs. rapid onset in GBS). Diagnosis: EMG/NCS shows demyelinating features (conduction block, temporal dispersion). Treatment: corticosteroids, IVIG, or plasmapheresis. Immunosuppressants (azathioprine, mycophenolate, rituximab) for refractory cases.

CIDP

– Chronic inflammatory demyelinating polyneuropathy (CIDP) is a chronic, immune-mediated, peripheral neuropathy characterized by progressive or relapsing symmetrical weakness and sensory loss in all four limbs, with areflexia or hyporeflexia, lasting for more than eight weeks. It is the chronic counterpart of GBS and is considered to be caused by an antibody-mediated attack on peripheral nerve myelin. The pathophysiology involves both humoral and cellular immune mechanisms targeting myelin and Schwann cells. Clinical presentation includes symmetrical proximal and distal weakness in all four limbs, with early loss of deep tendon reflexes, sensory loss (impaired vibration and proprioception), and symptoms developing progressively over weeks to months. Diagnosis is made using the European Federation of Neurological Societies (EFNS) criteria, supported by nerve conduction studies showing demyelinating features (prolonged distal latencies, reduced conduction velocities, conduction block, temporal dispersion, prolonged F-wave latencies), CSF analysis showing elevated protein with normal cell count (cytoalbuminological dissociation), and MRI of the brachial or lumbosacral plexus showing nerve root thickening and gadolinium enhancement. Management includes corticosteroids (prednisolone), intravenous immunoglobulin (IVIG), and plasma exchange as first-line immunomodulatory therapies. Immunosuppressive agents (azathioprine, mycophenolate mofetil, cyclophosphamide, rituximab) are used for refractory or relapsing cases. Prognosis is variable; most patients require long-term immunomodulatory therapy to maintain function and prevent disability progression.

Cluster Headache

– A primary headache disorder characterized by severe, unilateral, strictly orbital or periorbital pain attacks lasting fifteen to one hundred eighty minutes, occurring with one of three autonomic symptoms on the same side as the pain: conjunctival injection, lacrimation, nasal congestion or rhinorrhea, eyelid edema, forehead or facial sweating, miosis, or ptosis. Attacks occur in clusters or cycles, typically one to eight attacks per day during a cluster period lasting weeks to months followed by remission periods lasting months to years. Cluster headache is the most severe primary headache disorder, often described as the most painful condition known to medicine, and is nicknamed suicide headache due to its intensity. The headache is accompanied by restlessness and agitation—patients typically pace the floor, rock back and forth, or bang their head during attacks, in contrast to migraine patients who prefer to lie still in a dark room. Diagnosis is clinical using ICHD-3 criteria. Acute treatment includes high-flow oxygen (100% at 12-15 L/min via non-rebreather mask) and subcutaneous sumatriptan or intranasal zolmitriptan, which cause vasoconstriction of dilated orbital vessels. Preventive treatment includes verapamil (first-line), lithium, topiramate, galcanezumab (a CGRP monoclonal antibody specifically approved for cluster headache), and greater occipital nerve blockade. Transitional therapy with oral or intravenous corticosteroids is used at the onset of a cluster period to break the cycle while long-term preventatives take effect.

Cognitive Fitness

– The active maintenance of mental sharpness and brain health through stimulating activities, physical exercise, social engagement, and proper nutrition. Regular challenges such as learning new skills, puzzles, reading, and social interaction support neuroplasticity and cognitive reserve. Maintaining cognitive fitness throughout life may delay age-related cognitive decline and reduce dementia risk.

Cognitive Health and Memory

– Cognitive health encompasses the ability to think clearly, learn, remember, and make decisions, with memory being a core component that involves encoding, storing, and retrieving information. Age-related changes in processing speed and recall are normal, but persistent decline may signal mild cognitive impairment or dementia. Protective factors include cardiovascular health, mental stimulation, physical activity, and social engagement.

Coma

– A state of profound unconsciousness in which the patient cannot be aroused and shows no awareness of self or environment, with eyes closed and no sleep-wake cycles. Results from widespread bilateral cerebral hemisphere dysfunction or brainstem reticular formation damage. Causes: structural (traumatic brain injury, stroke, intracranial hemorrhage, brain tumor), metabolic (hypoglycemia, hepatic encephalopathy, uremia, hyperhyponatremia, hypothyroidism), toxic (drug overdose–opioids, benzodiazepines, alcohol, barbiturates), infectious (meningitis, encephalitis), and hypoxic-ischemic (cardiac arrest). Assessment: Glasgow Coma Scale (eye, motor, verbal responses; range 3–15). Management: stabilize ABCs, treat reversible causes (naloxone, flumazenil, glucose, thiamine), neuroimaging, EEG to rule out nonconvulsive status epilepticus. Prognosis depends on cause, duration, and neurologic examination findings.

Communicating Hydrocephalus

– A type of hydrocephalus in which there is free flow of cerebrospinal fluid (CSF) between the ventricles, but impaired absorption of CSF at the arachnoid granulations or increased CSF production. The ventricles are uniformly dilated, including the fourth ventricle. The most common cause is impaired CSF absorption due to prior subarachnoid hemorrhage, meningitis, or intracranial surgery, which causes scarring and fibrosis of the arachnoid granulations. Other causes include idiopathic intracranial hypertension, cerebral venous sinus thrombosis, and overproduction of CSF by a choroid plexus papilloma.

Clinical presentation includes headache, nausea and vomiting, gait disturbance, cognitive slowing, and papilloedema from increased intracranial pressure. In contrast to obstructive (non-communicating) hydrocephalus, the ventricles are uniformly dilated and there is no obstruction to CSF flow within the ventricular system. Diagnosis is by neuroimaging (CT or MRI) showing ventricular dilation with no obstructive lesion, supported by CSF drainage or infusion studies. Management includes CSF diversion procedures such as ventriculoperitoneal (VP) shunting or endoscopic third ventriculostomy (ETV) in selected cases where an obstructive component is present. The underlying cause must also be addressed (e.g., treatment of meningitis, management of subarachnoid hemorrhage complications).

Complex Regional Pain Syndrome

– A chronic pain condition characterized by regional pain and sensory changes disproportionate to the inciting event. Type I (reflex sympathetic dystrophy): no identifiable nerve injury. Type II (causalgia): follows a major nerve injury. Presents with burning pain, allodynia, hyperalgesia, edema, vasomotor changes (temperature and color changes), sudomotor changes (sweating), and trophic changes (skin, hair, nail changes). Diagnosis: Budapest clinical criteria. Treatment: early physical therapy and occupational therapy; NSAIDs, gabapentinoids, tricyclic antidepressants, bisphosphonates; sympathetic nerve blocks for severe pain.

Concussion

– A form of mild traumatic brain injury caused by a biomechanical force (direct blow to the head, face, neck, or elsewhere on the body with impulsive force transmitted to the head), resulting in a transient disturbance of brain function without structural damage visible on conventional neuroimaging. Concussion is a functional rather than structural injury, involving neurometabolic dysfunction: neuronal membrane disruption, ionic shifts (potassium efflux, calcium influx), and increased glucose metabolism (initially hyperglycolysis followed by metabolic depression), lactate accumulation, and impaired mitochondrial function. The result is a state of metabolic vulnerability in which the brain is less able to tolerate a second impact, which is the basis for the principle of avoiding repeat head injury during the recovery period. Clinical features include headache, dizziness, nausea, balance problems, blurred vision, sensitivity to light and noise, fatigue, sleep disturbance, difficulty concentrating, memory impairment, and emotional lability (irritability, anxiety, depression). Loss of consciousness occurs in fewer than 10 percent of concussions; post-traumatic amnesia is common. Diagnosis is clinical, based on history of head trauma and symptom assessment using standardised tools such as the Sport Concussion Assessment Tool (SCAT-5) or the Standardised Assessment of Concussion (SAC). Neuroimaging (CT or MRI) is normal. Management includes physical and cognitive rest during the acute phase (24-48 hours), followed by gradual stepwise return to activity guided by symptom tolerance. Most patients recover fully within 7-14 days in adults and within 2-4 weeks in children and adolescents. Persistent symptoms beyond 4 weeks are termed post-concussion syndrome and require multidisciplinary management.

Conus Medullaris Syndrome

– A clinical condition resulting from damage to the conus medullaris, the tapered, terminal end of the spinal cord located at approximately the L1-L2 vertebral level. The conus medullaris contains the sacral spinal cord segments (S2-S5) and the lowest lumbar segments, which innervate the bladder, bowel, sexual function, and the lower extremities. Clinical features include acute or subacute bilateral lower extremity weakness (usually symmetric), saddle anesthesia (loss of sensation in the perineal, perianal, and inner thigh regions), and early bladder and bowel dysfunction (urinary retention, overflow incontinence, and fecal incontinence). The presence of a bulbocavernosus reflex and anal wink reflex helps distinguish conus medullaris syndrome from cauda equina syndrome, and the early onset of bladder and bowel dysfunction (often before significant lower extremity weakness) is a distinguishing feature. Causes include compression from tumors (metastases, ependymomas), trauma, lumbar disc herniation, spinal cord infarction from anterior spinal artery occlusion, and inflammatory conditions such as transverse myelitis. Diagnosis is by urgent MRI of the thoracolumbar spine. Management requires prompt treatment of the underlying cause, including surgical decompression for compressive lesions. Prognosis is often poor for neurological recovery, particularly for bladder and bowel function.

Corticobasal Degeneration

– A rare, progressive neurodegenerative disorder characterized by asymmetric cortical and basal ganglia dysfunction. Presents with asymmetric parkinsonism (rigidity, bradykinesia), apraxia (loss of learned motor skills), alien limb phenomenon (involuntary, purposeless movements), cortical sensory loss, myoclonus, and dystonia. Later: cognitive decline, behavioral changes, and aphasia. Diagnosis: clinical; MRI may show asymmetric frontoparietal atrophy. No disease-modifying treatment; symptomatic therapy with levodopa (poor response), botulinum toxin for dystonia, and supportive care.

Creutzfeldt Jakob Disease

– Creutzfeldt-Jakob disease (CJD) is a rare, fatal, transmissible spongiform encephalopathy caused by the accumulation of misfolded prion protein (PrPSc) in the brain, leading to rapid, progressive dementia and myoclonus. Sporadic CJD is the most common form, accounting for approximately eighty-five percent of cases, with an incidence of approximately one to two per million per year, typically affecting patients in their sixties. The clinical presentation includes rapidly progressive cognitive decline, myoclonus (brief, involuntary muscle jerks that are pathognomonic when elicited by startle), ataxia, visual disturbances (cortical blindness, visual hallucinations), and pyramidal and extrapyramidal signs. In the late stages, patients become akinetic and mute. The classic EEG finding is generalised periodic sharp-wave complexes (PSWCs) at 1-2 Hz, which are present in approximately two-thirds of cases. MRI brain may show hyperintensity in the basal ganglia (particularly the caudate and putamen) and cortical ribboning on diffusion-weighted imaging. CSF analysis may detect 14-3-3 protein and elevated tau protein, and real-time quaking-induced conversion (RT-QuIC) testing of CSF has high sensitivity and specificity for detecting PrPSc. The disease is uniformly fatal, usually within 6-12 months of symptom onset. Variant CJD (vCJD), linked to bovine spongiform encephalopathy, affects younger patients and has a longer disease course. Management is supportive, with no effective disease-modifying treatment available.

CVA

– See Cerebrovascular Accident (Stroke).

Dandy-Walker Malformation

– A congenital hindbrain malformation characterized by hypoplasia or aplasia of the cerebellar vermis, cystic dilation of the fourth ventricle, and posterior fossa enlargement with upward displacement of the tentorium, transverse sinuses, and torcular herophili. Presents with hydrocephalus, developmental delay, ataxia, and cranial nerve palsies. Associated with agenesis of the corpus callosum and other CNS anomalies. Diagnosis by prenatal or postnatal MRI. Treatment: VP shunt or endoscopic third ventriculostomy for hydrocephalus. Prognosis depends on associated anomalies.

Dementia

– A clinical syndrome characterized by progressive cognitive decline sufficient to interfere with independent function, affecting memory, language, executive function, visuospatial skills, and personality. The most common cause is Alzheimer disease (60-80 percent of cases), followed by vascular dementia (10-20 percent), Lewy body dementia (5-10 percent), and frontotemporal dementia (5-10 percent). Diagnosis requires comprehensive assessment: history from the patient and a reliable informant, cognitive testing (MMSE, MoCA, detailed neuropsychological assessment), laboratory studies to rule out reversible causes (B12, thyroid function, syphilis serology, HIV), and neuroimaging (CT or MRI) to evaluate for structural causes such as brain tumors, subdural haematoma, and white matter disease. Management is directed at the underlying cause when possible and symptomatic treatment for progressive dementias including cognitive enhancement, behavioural management, and caregiver support. Prognosis depends on the specific etiology; most progressive dementias are incurable and lead to functional decline and eventual death.

Diabetic Peripheral Neuropathy

– The most common complication of diabetes mellitus, affecting approximately fifty percent of patients with long-standing disease. It is a length-dependent, sensorimotor polyneuropathy that affects the distal lower extremities first, in a glove-and-stocking distribution, and progresses proximally over time. The pathophysiology involves multiple mechanisms including metabolic injury from hyperglycemia (polyol pathway, advanced glycation end products, oxidative stress, mitochondrial dysfunction, activation of the poly(ADP-ribose) polymerase pathway, and impaired axonal transport), microvascular disease (endoneurial capillary damage causing endoneurial hypoxia), and inflammatory mechanisms. Clinical presentation begins with numbness, tingling, and paraesthesiae in the toes and feet, progressing proximally over years. Pain is a common and disabling feature, often described as burning, shooting, or stabbing. Motor weakness (foot drop, intrinsic hand muscle weakness) develops in advanced stages. Physical examination reveals reduced vibration sense (the earliest finding), impaired pinprick and temperature sensation in a stocking distribution, loss of ankle reflexes, and eventually proprioceptive loss causing sensory ataxia. Autonomic involvement can cause orthostatic hypotension, gastroparesis, erectile dysfunction, and abnormal sweating. Diagnosis is clinical, supported by nerve conduction studies showing a length-dependent axonal sensorimotor polyneuropathy. Management involves optimising glycaemic control (the only intervention that slows progression), foot care to prevent ulceration, and symptomatic treatment of neuropathic pain with gabapentinoids, tricyclic antidepressants, or SNRIs (duloxetine).

Duchenne Muscular Dystrophy

– An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene (DMD gene) on chromosome Xp21.2, resulting in absent or severely reduced dystrophin protein, which is essential for maintaining the structural integrity of the sarcolemma during muscle contraction. Most mutations are frameshift mutations (deletions, duplications, or point mutations) that introduce premature stop codons, leading to truncated, non-functional dystrophin. The incidence is approximately one in 3,500 male births, making it the most common muscular dystrophy in children. Onset of symptoms typically occurs before age five, with delayed motor milestones (late walking), difficulty running and climbing stairs, toe walking, and a waddling gait. Gowers manoeuvre (using the hands to push against the thighs to stand from a sitting position) is a classic finding. Calf pseudohypertrophy (enlargement due to fatty infiltration) is characteristic. Progression is rapid: most boys lose ambulation by age 12, develop respiratory insufficiency requiring non-invasive ventilation in adolescence, and die from respiratory or cardiac failure in their twenties or thirties. Cardiac involvement (dilated cardiomyopathy) is universal and requires regular surveillance with echocardiography and ECG. Diagnosis is confirmed by genetic testing showing out-of-frame dystrophin gene deletions, duplications, or point mutations, or by muscle biopsy showing absent dystrophin staining. Management includes corticosteroids (prednisolone or deflazacort) to prolong ambulation and reduce the need for scoliosis surgery, angiotensin-converting enzyme inhibitors for cardiomyopathy management, respiratory support, physiotherapy, and orthopaedic management for contractures and scoliosis. Gene therapy (delandistrogene moxeparvovec) has recently been approved for ambulatory patients with specific DMD mutations, and exon skipping agents (eteplirsen, golodirsen, viltolarsen) are available for specific mutation subtypes.

DWI FLAIR Mismatch

– Diffusion-weighted imaging (DWI) and fluid-attenuated inversion recovery (FLAIR) MRI sequences are used together to estimate the age of an ischemic stroke lesion and guide treatment decisions. DWI demonstrates restricted diffusion in acute ischemic tissue within minutes of stroke onset, appearing as a hyperintense (bright) lesion with corresponding hypointensity on the apparent diffusion coefficient (ADC) map. FLAIR demonstrates increased signal from vasogenic edema, which takes approximately four to six hours to become visible on FLAIR after stroke onset. A DWI-positive, FLAIR-negative lesion, indicating that the stroke is within the window where the patient may benefit from thrombolysis even if the onset time is unknown (e.g., wake-up stroke). If both DWI and FLAIR are positive, the lesion is estimated to be greater than approximately 4.5 hours old and thrombolysis is relatively contraindicated. This mismatch paradigm has been validated in the WAKE-UP trial, which demonstrated that patients with DWI-FLAIR mismatch who received alteplase had better functional outcomes at 90 days compared to placebo, without increased mortality.

Dysarthria

– A motor speech disorder resulting from neurological impairment of the muscular components of speech production, including the lips, tongue, palate, vocal cords, and respiratory muscles. It is caused by lesions affecting the central or peripheral nervous system at various levels: upper motor neuron (stroke, traumatic brain injury, multiple sclerosis), lower motor neuron (bulbar palsy, myasthenia gravis, Guillain-Barre syndrome), extrapyramidal (Parkinson disease, Huntington disease), cerebellar (ataxic dysarthria from cerebellar lesions), and neuromuscular (myasthenia gravis, muscular dystrophies, motor neuron disease). The type of dysarthria can help localise the neurological lesion and guide diagnosis. Flaccid dysarthria (lower motor neuron) presents with hypernasal, breathy, imprecise speech with audible inspiration. Spastic dysarthria (upper motor neuron) presents with strained, strangled, slow speech with reduced range of movement. Ataxic dysarthria (cerebellar) presents with scanning, irregular, and explosive speech. Hypokinetic dysarthria (extrapyramidal, e.g., Parkinson disease) presents with quiet, rapid, monotonous, mumbling speech with festination. Hyperkinetic dysarthria (e.g., Huntington disease) presents with variable speech rate, involuntary vocalisations, and variable volume. Diagnosis is by speech-language pathology assessment and identification of the underlying neurological cause. Management includes speech and language therapy, treatment of the underlying neurological condition, and augmentative and alternative communication devices for severe cases.

Dysesthesia

– An unpleasant, abnormal sensation produced by a normally non-painful stimulus, or an unpleasant, exaggerated response to a normal stimulus. It is distinguished from paresthesia (which is simply an abnormal sensation that may or may not be unpleasant) by its consistently unpleasant quality. Dysesthesia is a feature of neuropathic pain and is caused by dysfunction of the somatosensory nervous system, including peripheral nerve damage, spinal cord lesions, or central brain lesions. Peripheral causes include compression neuropathies, radiculopathies, and polyneuropathies. The distinction between dysesthesia and other sensory symptoms is important for diagnosis: dysesthesia is consistently unpleasant (a key distinguishing feature from paresthesia), allodynia is pain from a normally non-painful stimulus, hyperalgesia is an exaggerated response to a normally painful stimulus, and hyperpathia is a delayed, explosive, and unpleasant response to a stimulus, often with radiation beyond the stimulus site. Diagnosis is clinical, with neurological examination and nerve conduction studies to identify the underlying somatosensory pathway lesion. Management includes treating the underlying cause and symptomatic treatment of the dysesthetic pain with gabapentinoids, tricyclic antidepressants, or SNRIs.

Dysphasia

– A language disorder caused by damage to the language-dominant hemisphere of the brain (typically left), involving difficulty with expression, comprehension, reading, or writing. Dysphasia (often used interchangeably with aphasia): partial loss of language function. Types: expressive dysphasia (Broca area–non-fluent, effortful speech with preserved comprehension, agrammatism), receptive dysphasia (Wernicke area–fluent but meaningless speech, poor comprehension, word substitutions (paraphasias)), global dysphasia (both hemispheres extensively affected, all language modalities impaired), conduction dysphasia (arcuate fasciculus lesion–good comprehension and fluency but poor repetition), and anomic dysphasia (word-finding difficulty with otherwise fluent speech). Most common cause: ischemic stroke. Management: speech and language therapy.

Dystonia

– A movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive, movements and postures. It can be classified by the body region affected (focal, segmental, multifocal, or generalized), by age of onset (childhood, adolescent, or adult), and by etiology (primary/genetic or secondary/symptomatic). Focal dystonia includes cervical dystonia (torticollis, the most common adult-onset focal dystonia), blepharospasm (involuntary closure of the eyelids (blepharospasm), oromandibular dystonia (jaw opening, closing, or deviation), laryngeal dystonia (spasmodic dysphonia), and task-specific dystonias such as writer's cramp and musician's dystonia. Secondary dystonia can be caused by drugs (tardive dystonia from antipsychotics), stroke, traumatic brain injury, cerebral palsy, Wilson disease, and neurodegenerative disorders. Diagnosis is clinical, supported by genetic testing for primary dystonia (DYT genes including TOR1A, THAP1, GCH1, GNAL) and neuroimaging to exclude structural causes. Management includes botulinum toxin injections for focal dystonia (the treatment of choice), oral medications (anticholinergics, benzodiazepines, baclofen, tetrabenazine), and deep brain stimulation of the globus pallidus internus for severe, medically refractory generalised or segmental dystonia.

Echolalia

– The involuntary, automatic repetition of words or phrases spoken by another person, a speech phenomenon observed in various neurologic and psychiatric conditions. Immediate echolalia involves repeating sounds immediately after hearing them; delayed echolalia involves repeating phrases heard hours, days, or weeks earlier. Most commonly associated with autism spectrum disorder (where it may serve communicative or self-regulatory functions), Tourette syndrome, and transcortical sensory aphasia. Also occurs in Alzheimer disease, frontotemporal dementia, catatonia, and schizophrenia. Echolalia can be distinguished from palilalia (repetition of one's own words) and coprolalia (involuntary utterance of obscenities).

Electroencephalography

– A non-invasive neurophysiological test that records the electrical activity of the brain using electrodes placed on the scalp according to the international 10-20 system. The EEG measures the summated postsynaptic potentials of cortical pyramidal neurons and displays them as waveforms characterized by frequency (delta 0.5-4 Hz, theta 4-8 Hz, alpha 8-13 Hz, beta 13-30 Hz, gamma greater than 30 Hz), amplitude, and morphology. The normal waking EEG shows a posterior dominant alpha rhythm (8-13 Hz) that attenuates with eye opening. EEG is essential for the diagnosis of epilepsy, classification of seizure types and syndromes, and monitoring of status epilepticus. Abnormal findings include epileptiform discharges (spikes, sharp waves, spike-wave complexes) indicating an epileptic focus or generalised epilepsy. EEG is also used in the evaluation of encephalopathy (generalised slowing), brain death (electrocerebral silence), and sleep disorders. Prolonged video-EEG monitoring is used for presurgical evaluation of drug-resistant epilepsy, capturing ictal and interictal EEG patterns to localise the seizure onset zone. Standard EEG records from 21 scalp electrodes at the standard 10-20 positions, with additional electrodes (T1/T2, zygomatic, sphenoidal) used for better localisation of temporal lobe foci. Limitations include low spatial resolution, sensitivity to movement artifact, and inability to detect deep or mesial structures.

Encephalitis

– Inflammation of the brain parenchyma, most commonly caused by viral infections (herpes simplex virus, arboviruses, enteroviruses). Presents with fever, headache, altered mental status, seizures, and focal neurological deficits. CSF shows lymphocytic pleocytosis. Diagnosis: CSF PCR, MRI, EEG. Treatment: acyclovir for HSV, supportive care. Complications: permanent neurological deficits, cognitive impairment.

Encephalocele

– A neural tube defect where brain tissue and meninges herniate through a skull defect, most commonly in the occipital region. The herniated sac contains cerebrospinal fluid, meninges, and often brain tissue. Associated with other CNS anomalies including hydrocephalus and Chiari malformation. Presents as a visible, pulsatile midline mass. Treatment: surgical repair with reduction of herniated contents and dural closure. Prognosis depends on the amount of herniated brain tissue.

Encephalopathy

– Any diffuse disease of the brain that alters its structure or function, presenting with a global change in mental status (confusion, lethargy, personality change, coma) rather than focal neurologic deficits. Causes are numerous and often reversible: metabolic (hepatic encephalopathy–ammonia accumulation, uremic encephalopathy–uremic toxins, hypoglycemia, hyper/hyponatremia, hypercapnia, Wernicke encephalopathy–thiamine deficiency), toxic (drug overdose, alcohol intoxication/withdrawal, heavy metals), infectious (sepsis-associated encephalopathy, meningitis, encephalitis), hypoxic-ischemic (cardiac arrest), hypertensive (posterior reversible encephalopathy syndrome–PRES), and neurodegenerative (chronic traumatic encephalopathy, prion disease). Diagnosis: clinical assessment, metabolic panel, toxicology screen, neuroimaging (CT/MRI), EEG. Treatment: address underlying cause.

Epidural Hematoma

– Accumulation of blood between the skull and the dura mater, typically from arterial bleeding (middle meningeal artery) following temporal bone fracture. Presents with loss of consciousness, then a lucid interval (hours), followed by rapid neurological deterioration due to mass effect. Head CT: biconvex (lentiform), hyperdense collection that does not cross suture lines. Midline shift is common. Treatment: emergency burr hole or craniotomy for evacuation within 2-4 hours. Delay increases mortality. Outcome is excellent with prompt surgical intervention.

Epilepsy

– A neurological disorder characterized by recurrent, unprovoked seizures due to abnormal electrical activity in the brain. Classification includes focal, generalized, and unknown onset seizures. Etiology: genetic, structural, metabolic, infectious, immune. Diagnosis: clinical history, EEG, neuroimaging. Treatment: antiseizure medications, epilepsy surgery, vagus nerve stimulation, ketogenic diet for refractory cases.

Epilepsy Surgery

– Indicated for patients with drug-resistant epilepsy who have a concordant, resectable seizure focus identified on comprehensive presurgical evaluation. Failure of two or more appropriately chosen and adequately dosed antiepileptic drugs to achieve sustained seizure freedom. The presurgical evaluation includes scalp EEG with video monitoring, high-resolution MRI, PET scan (typically showing hypometabolism in the seizure focus), SPECT scan (ictal SPECT showing hyperperfusion and interictal SPECT showing hypoperfusion in the epileptic focus), neuropsychological assessment, and functional MRI for language and motor mapping. Invasive EEG monitoring using subdural grid electrodes or stereoelectroencephalography (SEEG) may be required when non-invasive data are discordant or the seizure focus is in or near eloquent cortex. The type of surgery depends on the location and nature of the epileptic focus: temporal lobectomy (the most common and most successful procedure, with 60-80 percent seizure freedom rates), extratemporal resection, hemispherotomy or hemispherectomy (for large hemispheric pathologies such as Rasmussen encephalitis or Sturge-Weber syndrome), corpus callosotomy (for drop attacks in generalised epilepsy), and vagus nerve stimulation (for drug-resistant epilepsy not amenable to resective surgery). Responsive neurostimulation (RNS) and deep brain stimulation of the anterior nucleus of the thalamus are newer neuromodulation options.

Essential Tremor

– The most common movement disorder, characterized by a bilateral, symmetrical, postural and kinetic tremor predominantly affecting the upper extremities, with a frequency of four to twelve hertz. It typically begins insidiously in the second or third decade of life and slowly progresses over years. The tremor is most apparent when the arms are held in a sustained posture (such as arms outstretched) and during intentional movement (such as reaching for an object), and it usually improves temporarily with alcohol consumption (a characteristic feature). The tremor can also affect the head (titubation), voice (vocal tremor, producing a quavering quality), legs, and trunk. Advanced cases may cause significant disability in writing, drinking from a cup, eating, and fine motor tasks. Essential tremor is often misdiagnosed as Parkinson disease, but key distinguishing features include the bilateral symmetrical presentation, predominant postural and kinetic tremor (rather than the resting tremor of Parkinson disease), lack of other parkinsonian signs such as bradykinesia and rigidity, and positive family history (autosomal dominant inheritance in up to 50 percent of cases, with incomplete penetrance). The pathophysiology is not fully understood but likely involves abnormal oscillatory activity in the cerebello-thalamo-cortical circuit, with the cerebellum playing a central role. Diagnosis is clinical, based on the Movement Disorder Society consensus criteria. Treatment includes beta-blockers (propranolol is first-line) and primidone as first-line medications, with topiramate, gabapentin, and benzodiazepines as second-line options. For medically refractory cases, botulinum toxin injections can reduce tremor amplitude, and deep brain stimulation of the ventral intermediate nucleus of the thalamus is highly effective. Focused ultrasound thalamotomy is a newer, incisionless ablative option.

Facial paralysis

– Loss of voluntary motor function of the facial muscles, most commonly from peripheral facial nerve (CN VII) dysfunction. The most frequent cause is Bell palsy (idiopathic peripheral facial paralysis), accounting for 60-70 percent of cases, with an incidence of 20-30 per 100,000 annually. Bell palsy presents as acute onset (over hours to 2-3 days) of unilateral lower motor neuron facial weakness involving both the upper and lower face (forehead, eye closure, mouth), often preceded by pain behind the ear (mastoid pain). Other causes of facial paralysis include Ramsay Hunt syndrome (herpes zoster oticus presenting with vesicles in the external auditory canal and severe pain), Lyme disease (erythema migrans and systemic symptoms), Guillain-Barre syndrome (bilateral facial weakness), otitis media, parotid tumors (slowly progressive), and stroke (upper motor neuron pattern sparing the forehead). Diagnosis involves careful history and physical examination to localise the lesion and identify the cause, with additional testing (MRI, nerve conduction studies, serology) as indicated. Management is directed at the underlying cause: Bell palsy is treated with corticosteroids, Ramsay Hunt syndrome with antivirals plus corticosteroids, Lyme disease with antibiotics, and tumors with surgical resection. Eye protection is critical in all cases of incomplete eye closure to prevent corneal abrasion and ulceration. Prognosis depends on the cause: Bell palsy has 70 percent complete recovery, while facial nerve tumors and trauma have variable outcomes.

Friedreich Ataxia

– The most common inherited ataxia, an autosomal recessive neurodegenerative disorder caused by a GAA trinucleotide repeat expansion in intron one of the frataxin gene (FXN) on chromosome 9q21.1. Normal individuals have five to thirty-three GAA repeats, while patients with Friedreich ataxia have typically sixty-six to one thousand seven hundred repeats on both alleles. The expanded repeat causes transcriptional silencing of the frataxin gene, leading to reduced frataxin protein, a mitochondrial protein involved in iron-sulphur cluster biogenesis and iron homeostasis. Reduced frataxin leads to mitochondrial iron accumulation, oxidative stress, and impaired mitochondrial function, particularly in neurons and cardiac myocytes. Clinical onset is typically between ages 10 and 15, with progressive gait and limb ataxia, dysarthria, loss of deep tendon reflexes (areflexia), loss of vibration and proprioception, extensor plantar responses (upper motor neuron signs due to corticospinal tract involvement), and scoliosis. Non-neurological features include hypertrophic cardiomyopathy (the most common cause of death), diabetes mellitus, and pes cavus. Diagnosis is clinical, supported by nerve conduction studies showing absent sensory nerve action potentials with preserved motor conduction velocities, and confirmed by genetic testing demonstrating biallelic GAA repeat expansions in the FXN gene. Management is supportive and multidisciplinary, including physical and occupational therapy, speech therapy, management of cardiomyopathy, treatment of diabetes, and orthopaedic management of scoliosis. The antioxidant idebenone has not shown consistent benefit in clinical trials. Prognosis is variable, with most patients becoming wheelchair-bound within 10-15 years of onset; mean survival is approximately 40-50 years, with cardiac failure as the leading cause of death.

Frontotemporal Dementia

– A group of neurodegenerative disorders characterized by progressive degeneration of the frontal and temporal lobes, presenting with personality changes, behavioral abnormalities, language impairment, and cognitive decline. FTD is the most common cause of dementia in patients under sixty years of age and accounts for approximately five to ten percent of all dementia cases. The three main clinical variants are behavioral variant FTD (bvFTD), which presents with progressive personality changes such as apathy, disinhibition, loss of empathy, perseverative behaviours, hyperorality, and dietary changes (bvFTD), semantic variant primary progressive aphasia (svPPA) with loss of word meaning and object knowledge, and non-fluent/agrammatic variant PPA (nfvPPA) with effortful, agrammatic speech. The pathophysiology involves accumulation of abnormal proteins: about 50 percent of FTD cases have tau inclusions (tauopathy), 40 percent have TDP-43 inclusions (TDP-43 proteinopathy), and 10 percent have FUS inclusions. Approximately 30-50 percent of FTD cases have a family history, with autosomal dominant mutations in the MAPT (tau), GRN (progranulin), and C9orf72 genes being the most common. C9orf72 hexanucleotide repeat expansions are the most common genetic cause of both FTD and ALS, and these conditions may co-occur. Diagnosis is by clinical criteria (Rascovsky criteria for bvFTD, Gorno-Tempini criteria for PPA variants), neuropsychological testing, and MRI brain showing focal atrophy of the frontal and/or temporal lobes. There is no disease-modifying treatment; management is symptomatic with SSRIs for behavioural symptoms, antipsychotics for agitation, and speech therapy for language deficits. Prognosis is poor, with median survival of 6-10 years from diagnosis.

GABA (Gamma-Aminobutyric Acid)

– The primary inhibitory neurotransmitter in the mammalian central nervous system, synthesised from glutamate via glutamic acid decarboxylase (GAD) with pyridoxal phosphate (vitamin B6) as cofactor. GABA acts on two classes of receptors: GABA-A (ionotropic, chloride ion channel, target of benzodiazepines, barbiturates, and ethanol) and GABA-B (metabotropic, G-protein coupled, modulates calcium and potassium channels). Functions: reduces neuronal excitability, regulates muscle tone, influences mood and anxiety. Implicated in: anxiety disorders (reduced GABAergic tone), epilepsy (GABA deficiency may contribute to seizures), Huntington disease (reduced GABA levels in basal ganglia), and spasticity (target of baclofen, a GABA-B agonist).

Gait

– The pattern or style of walking. Gait is a complex motor function involving coordination of the central and peripheral nervous systems, muscles, and joints. Gait assessment evaluates stance, stride length, cadence, arm swing, truncal stability, and turning. Abnormal gait patterns include: hemiplegic gait (circumduction, stroke), Parkinsonian gait (festinating, shuffling, reduced arm swing), ataxic gait (wide-based, unsteady, cerebellar or sensory; Romberg test positive in sensory ataxia), steppage gait (foot drop, e.g., common peroneal nerve palsy or Charcot-Marie-Tooth disease), waddling gait (hip girdle weakness, myopathy), antalgic gait (pain avoidance, shortened stance phase on affected side), and Trendelenburg gait (pelvic drop on the swing side, gluteus medius weakness).

Gait Assessment

– A crucial component of the neurological examination that provides significant diagnostic information about the underlying neurological condition. Normal gait requires intact motor, sensory, cerebellar, and vestibular systems, as well as intact cognition. Several characteristic gait patterns are associated with specific neurological disorders. The hemiplegic gait is seen in stroke and other upper motor neuron lesions, with the affected leg stiffly circumducted and the arm flexed and adducted at the side. The parkinsonian gait (festinating gait) is characterized by small, shuffling steps with reduced arm swing, flexed posture, and difficulty initiating or stopping movement, sometimes with freezing of gait. The sensory ataxic gait is broad-based with high-stepping (foot slap) and a positive Romberg sign, caused by impaired proprioception from peripheral neuropathy, tabes dorsalis, or posterior column disease. The cerebellar ataxic gait is broad-based, staggering, and lurching, with difficulty performing tandem gait, and is caused by cerebellar vermis pathology. The myopathic gait (waddling gait) features a broad-based stance, exaggerated lateral trunk sway, and Trendelenburg lurch due to weak hip girdle muscles, seen in muscular dystrophies. The apraxic gait (magnetic gait) is characterized by difficulty lifting the feet from the floor, as if glued to the ground, with initiation difficulty, seen in normal pressure hydrocephalus and frontal lobe lesions. Diagnosis of the gait pattern and the underlying neurological cause directs further investigation and management.

Gargalesthesia

– The sensation of tickling. A complex sensory experience mediated by light tactile stimulation of sensitive skin areas (axillae, soles, flanks, ribcage). Involves both spinal reflexes (goosebumps, withdrawal) and emotional processing in the anterior cingulate cortex and somatosensory cortex. Ticklishness requires an intact somatosensory system and a social/emotional context; self-tickling does not produce the sensation due to cerebellar cancellation of self-generated movement. Loss of gargalesthesia may occur in peripheral neuropathy or spinal cord lesions.

Generalized Tonic-Clonic Seizure

– A generalized tonic-clonic seizure (formerly grand mal seizure) is a seizure involving bilateral tonic stiffening followed by rhythmic clonic jerking of the extremities, with loss of consciousness. The tonic phase consists of sustained muscle contraction causing initial extension or flexion of the limbs, opisthotonus, and jaw clenching, lasting ten to twenty seconds. The clonic phase follows with rhythmic jerking of the limbs at a frequency of approximately two to four hertz, gradually slowing over time. The postictal phase follows with confusion, drowsiness, headache, and muscle soreness, lasting minutes to hours. During the generalised seizure, there is loss of consciousness, autonomic changes (tachycardia, hypertension, diaphoresis, pupillary dilation), and sometimes tongue biting, urinary incontinence, and cyanosis during the tonic phase. Generalised tonic-clonic seizures can be primary generalised (arising from both hemispheres simultaneously, often with a genetic basis) or secondary generalised (focal seizure spreading to become bilateral). Diagnosis is clinical, supported by EEG which may show generalised spike-wave discharges interictally. First-line treatment includes sodium valproate (for primary generalised) or levetiracetam, lamotrigine, and topiramate. Status epilepticus is a medical emergency defined as a generalised tonic-clonic seizure lasting longer than 5 minutes or recurrent seizures without return to baseline, requiring urgent management with benzodiazepines, followed by second-line anticonvulsants and anesthetic agents.

Grand Mal

– A type of generalised seizure characterized by bilateral synchronous epileptic activity involving the entire brain. Formerly called grand mal, now classified as bilateral tonic-clonic seizure according to the ILAE 2017 classification. Phases: (1) tonic phase (10-30 s): sudden loss of consciousness, sustained muscle contraction causing arching of the back (opisthotonos), extension of limbs, respiratory arrest with cyanosis; (2) clonic phase (30-60 s): rhythmic alternating contraction and relaxation of all muscle groups, initially rapid then slowing; (3) postictal phase (variable): confusion, headache, drowsiness, muscle soreness, sometimes Todd paralysis (transient focal neurological deficit). Complications: tongue biting, urinary incontinence, aspiration, injury from falls. Status epilepticus: continuous seizure activity >5 minutes or >2 seizures without full recovery. Emergency management: airway, benzodiazepines (lorazepam IV or midazolam IM), then levetiracetam/phenytoin/valproate if needed.

Guillain-Barre Syndrome

– An acute, immune-mediated polyneuropathy typically presenting with ascending paralysis following an antecedent infection (Campylobacter jejuni, CMV, EBV, Zika). Variants: acute inflammatory demyelinating polyneuropathy (AIDP, most common in North America), acute motor axonal neuropathy (AMAN, more common in Asia). Presents with symmetric ascending weakness, areflexia, and paresthesias. Respiratory failure occurs in 20-30%. Diagnosis: CSF shows albuminocytologic dissociation (elevated protein, normal cell count); EMG/NCS. Treatment: IVIG or plasmapheresis (equally effective). Prognosis: most recover, but may have residual weakness.

Gyrus

– A ridge or convolution on the surface of the cerebral cortex, separated by sulci (grooves). The folded structure triples the surface area of the brain (approximately 2500 cm^2) while fitting within the cranial vault. Major gyri: precentral gyrus (primary motor cortex, Brodmann area 4), postcentral gyrus (primary somatosensory cortex, areas 3,1,2), superior temporal gyrus (auditory cortex, area 41/42, includes Heschl gyrus), middle and inferior frontal gyri (prefrontal cortex, Broca area in dominant hemisphere), fusiform gyrus (face and object recognition), angular gyrus (reading and arithmetic), supramarginal gyrus (sensory integration), cingulate gyrus (limbic system, pain and emotion), hippocampal gyrus (parahippocampal, memory encoding). Gyral abnormalities may be seen in developmental disorders (polymicrogyria, pachygyria) and neurodegenerative diseases (gyral atrophy in Alzheimer disease).

Headache

– One of the most common neurologic symptoms, classified as primary (migraine, tension-type, cluster, and trigeminal autonomic cephalalgias) or secondary (attributed to an underlying structural, infectious, vascular, or metabolic cause). Tension-type headache is the most prevalent primary headache disorder, presenting as bilateral, pressing or tightening quality, mild to moderate intensity, lasting 30 minutes to 7 days, without nausea or photophobia/phonophobia. Migraine is the second most common primary headache disorder after tension-type headache. Secondary headaches have numerous causes, with red flags (SNOOP4: systemic symptoms, neurological signs, onset sudden, older age at onset, pattern change, prior headache history, precipitating factors, pregnancy, and postural aggravation) prompting investigation. Diagnosis is by careful history and neurological examination, with neuroimaging (CT or MRI) and lumbar puncture as indicated to exclude secondary causes. Management depends on the specific headache type: acute treatment includes simple analgesics, triptans, and antiemetics; preventive treatment includes beta-blockers, tricyclic antidepressants, antiepileptics, CGRP monoclonal antibodies, and nerve blocks.

Headache and Migraine

– Headaches are pain in the head or upper neck region, classified as primary (tension-type, migraine, cluster) or secondary to underlying conditions. Migraines are a neurological disorder characterized by recurrent episodes of moderate to severe throbbing pain, often accompanied by nausea, photophobia, and aura. Management includes acute treatments such as triptans and NSAIDs, plus preventive therapies including medications and lifestyle modifications.

Hemifacial Spasm

– Involuntary, unilateral, intermittent twitching of the facial muscles innervated by the facial nerve (CN VII). Typically begins around the eye (orbicularis oculi) and spreads to the lower face. Most commonly caused by vascular compression of the facial nerve at the root exit zone (anterior inferior cerebellar artery or posterior inferior cerebellar artery). Other causes: facial nerve injury, tumors, MS. Diagnosis: clinical; MRI/MRA to rule out structural cause. Treatment: botulinum toxin injections (first-line), microvascular decompression for refractory cases.

Hemiplegia

– Paralysis of one side of the body (the face, arm, and leg on the same side), caused by damage to the corticospinal tract (pyramidal tract) at any level from the cerebral cortex to the spinal cord. The most common cause of acute hemiplegia in adults is ischemic stroke affecting the contralateral cerebral hemisphere, particularly lesions of the internal capsule or motor cortex. Other causes include hemorrhagic stroke, brain tumors, traumatic brain injury, cerebral palsy (in children), and brain tumors. The clinical presentation includes contralateral facial weakness (lower face only, with forehead sparing due to bilateral cortical innervation of the upper face), contralateral arm and leg weakness (typically more severe in the arm and face than the leg in cortical strokes, or equal involvement in internal capsule or brainstem lesions), and associated neurological deficits depending on the location of the lesion (aphasia, neglect, visual field defects, sensory loss, dysarthria). The weakness is initially flaccid in the acute phase, with spasticity and hyperreflexia developing over days to weeks. Diagnosis is by neuroimaging (CT for acute hemorrhage, MRI for ischemic stroke) to identify the cause. Management is directed at the underlying cause: acute stroke management, rehabilitation, and prevention of secondary complications including contractures, pressure ulcers, DVT, and shoulder pain. Functional recovery depends on the extent of damage and the patient's age and pre-morbid status.

Hemorrhagic Stroke

– A neurological deficit caused by rupture of a cerebral blood vessel, accounting for 13% of all strokes but 40% of stroke mortality. Intracerebral hemorrhage: hypertensive (basal ganglia, thalamus, pons, cerebellum), amyloid angiopathy (lobar, elderly), vascular malformations, anticoagulation. Subarachnoid hemorrhage: aneurysm rupture. Presents with sudden severe headache, neurological deficits, decreased consciousness, and hypertension. Diagnosis: non-contrast CT is first-line. Management: ABCs, reversal of anticoagulation (vitamin K, FFP, PCC, idarucizumab for dabigatran, andexanet alfa for factor Xa inhibitors), blood pressure control (SBP <140-160), neurosurgical evacuation for cerebellar >3cm or lobar >30cc, external ventricular drain for hydrocephalus.

Hereditary Spastic Paraplegia

– A group of genetic disorders characterized by progressive spasticity and weakness of the lower extremities, resulting from degeneration of the corticospinal tracts. Over seventy different genetic loci have been identified (SPG1 through SPG70-plus). The most common form is SPG4, caused by mutations in the spastin gene (ATL4), accounting for approximately forty percent of autosomal dominant HSP. HSP is classified as uncomplicated (pure) when spastic paraplegia is the only finding (most cases), and complicated when accompanied by additional neurological features such as ataxia, intellectual disability, neuropathy, epilepsy, or extrapyramidal signs. Onset is typically insidious in childhood or early adulthood, with slow progression over decades. Clinical features include progressive spasticity and weakness in the legs, causing a stiff-legged, scissoring gait, hyperreflexia, extensor plantar responses, and urinary urgency. Sensation, cognition, and cranial nerves are generally spared in pure HSP. Diagnosis is clinical with family history, supported by neuroimaging (MRI spine and brain to exclude structural causes such as compressive myelopathy or multiple sclerosis), neurophysiology to detect subclinical neuropathy or denervation, and genetic testing using multi-gene panels or whole-exome sequencing for the numerous causative genes. Management is symptomatic and includes physiotherapy and stretching to maintain flexibility, treatment of spasticity with oral baclofen, tizanidine, and dantrolene, botulinum toxin injections for focal spasticity, and assistive devices for gait. Disease-modifying therapies are not yet available. Prognosis is variable but progression is generally slow, and most patients remain ambulatory into later life.

Herpes Simplex Encephalitis

– The most common cause of sporadic fatal encephalitis, caused by herpes simplex virus type 1 (HSV-1) in over ninety percent of cases. The virus typically infects the temporal lobes, orbitofrontal cortex, and insular cortex, causing hemorrhagic necrosis. Clinical presentation includes fever, headache, altered mental status, behavioral changes, seizures, focal neurological deficits (aphasia, hemiparesis), and personality changes. The temporal lobe predilection explains the common presenting features of personality changes, aphasia, and temporal lobe seizures. MRI brain is the imaging modality of choice, showing characteristic T2 hyperintensity and swelling of the temporal lobes (especially the medial temporal lobe including the hippocampus and amygdala), insular cortex, and orbitofrontal region, sometimes with areas of hemorrhage. The diagnosis is confirmed by detection of HSV-1 DNA in CSF by polymerase chain reaction (PCR), which has high sensitivity and specificity. CSF analysis typically shows a lymphocytic pleocytosis, elevated protein, and normal or slightly low glucose. EEG may show periodic lateralising epileptiform discharges (PLEDs) originating from the temporal region. Treatment with high-dose intravenous acyclovir (10 mg/kg every 8 hours for 14-21 days) should be started immediately when the diagnosis is suspected, even before confirmation by PCR, as early treatment significantly reduces mortality and improves neurological outcomes. Mortality is approximately 10-20 percent with treatment, and survivors often have long-term neurological deficits including memory impairment, personality changes, and epilepsy.

Holoprosencephaly

– A structural brain malformation resulting from incomplete or failed division of the forebrain (prosencephalon) into the cerebral hemispheres during the fifth week of gestation. Classified by severity: alobar (single ventricle, fused thalami, no interhemispheric fissure, most severe), semilobar (partial division), and lobar (nearly complete division, mildest). Presents with developmental delay, seizures, spasticity, and hypothalamic/pituitary dysfunction. Craniofacial anomalies correlate with severity: cyclopia, ethmocephaly, cebocephaly, or cleft lip/palate. Causes include trisomy 13 and mutations in SHH, ZIC2, SIX3, or TGIF genes. Treatment is supportive.

Horner Syndrome

– A clinical triad caused by disruption of the sympathetic pathway to the eye and face, resulting in ipsilateral miosis (constricted pupil), ptosis (drooping of the upper eyelid due to paralysis of the superior tarsal muscle), and anhidrosis (loss of sweating on the affected side of the face). The sympathetic pathway to the eye is a three-neuron chain: the first-order neuron descends from the hypothalamus through the brainstem to the ciliospinal center of Budge at C8-T2; the second-order neuron exits the spinal cord at C8-T2, ascends in the sympathetic chain, and synapses in the superior cervical ganglion; the third-order neuron travels along the internal carotid artery to innervate the pupillary dilator muscle, Muller muscle of the upper eyelid, and sweat glands of the face. Causes of Horner syndrome are classified by the site of the lesion along this three-neuron chain. Central (first-order) causes include brainstem stroke (Wallenberg syndrome), multiple sclerosis, syringobulbia, and spinal cord lesions above T1. Preganglionic (second-order) causes include Pancoast tumor (apical lung cancer), cervical rib, thoracic aortic aneurysm, thyroid carcinoma, neck trauma or surgery, and mediastinal masses. Postganglionic (third-order) causes include carotid artery dissection (a critical cause to identify), carotid aneurysm or thrombosis, cavernous sinus lesions, cluster headache, and otitis media. Diagnosis involves pharmacological testing with topical apraclonidine or cocaine drops, which confirm Horner syndrome, and topical hydroxyamphetamine to differentiate preganglionic from postganglionic lesions. Urgent imaging of the brain, neck, and chest (CT, MRI, CTA, or MRA) is required to identify the underlying cause, with carotid dissection being a medical emergency requiring antithrombotic therapy.

Huntington Disease

– An autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion in the huntingtin (HTT) gene on chromosome 4p16.3. The normal number of CAG repeats is ten to thirty-five, while forty or more repeats are fully penetrant, causing the disease. There is an inverse correlation between the number of CAG repeats and the age of onset: larger repeat expansions cause earlier onset. The disease is characterized by a classic triad of motor, cognitive, and psychiatric abnormalities. The motor features include chorea (rapid, involuntary, dance-like movements) as the most characteristic feature, which initially may be subtle—mild fidgetiness, clumsiness, or involuntary facial movements—and progresses to involve the trunk and limbs, affecting gait, speech, and swallowing. Dystonia and parkinsonism can also occur, particularly in juvenile-onset cases. Cognitive decline begins with executive dysfunction (reduced flexibility, poor planning, and impulsivity), progressing to global dementia. Psychiatric manifestations are common and include depression (the most common), apathy, irritability, obsessive-compulsive symptoms, and psychosis. Diagnosis is clinical, confirmed by genetic testing demonstrating 40 or more CAG repeats in the HTT gene. Presymptomatic testing is available for individuals at risk, requiring careful genetic counseling. MRI brain shows atrophy of the caudate nucleus and putamen (striatal atrophy) with corresponding enlargement of the frontal horns of the lateral ventricles. There is no disease-modifying treatment; management is symptomatic with antipsychotics (particularly tetrabenazine and deutetrabenazine for chorea), antidepressants, and anxiolytics. Mean survival is 15-20 years from symptom onset.

Huntington's Disease (Chorea)

– An autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion (>36 repeats, normally <26) in the HTT gene on chromosome 4 (4p16.3), which encodes the huntingtin protein. The expanded repeat produces an elongated polyglutamine tract causing toxic protein aggregation and neuronal death, especially in the caudate nucleus, putamen, and cerebral cortex. Anticipation: repeat length tends to increase in successive generations, especially with paternal transmission (daughters of affected fathers are more likely to develop earlier onset). Prevalence: 5-10 per 100,000. Age of onset: typically 35-50 years; juvenile onset (<20 years, large repeats >60 CAG) presents with rigidity and seizures. Clinical triad: (1) Movement disorder: chorea (irregular, rapid, jerky, purposeless movements of face, trunk, limbs; initially subtle may be masked by voluntary movement), dystonia, parkinsonism (rigidity, bradykinesia, especially in later stages), impaired voluntary movement (incoordination, dysarthria, dysphagia). (2) Cognitive decline: executive dysfunction (impaired planning, organising, multitasking, cognitive flexibility), memory deficits, psychomotor slowing, progressing to subcortical dementia. (3) Psychiatric symptoms: depression (most common, 30-50%, may precede motor onset), irritability, apathy, anxiety, obsessive-compulsive symptoms, psychosis (hallucinations, delusions, 5-15%). Diagnosis: clinical (positive family history in 90%, diagnostic criteria: motor symptoms in at-risk individual), genetic testing (CAG repeat length, confirmatory). Presymptomatic testing offered with genetic counseling. Imaging: MRI/CT shows caudate atrophy (bicaudate ratio increased) and ex vacuo ventricular enlargement. Management: no disease-modifying treatment. Chorea: tetrabenazine (VMAT2 inhibitor, depletes dopamine), deutetrabenazine, valbenazine; antipsychotics (olanzapine, risperidone) for chorea and psychiatric symptoms. Depression: SSRIs. Multidisciplinary care: physiotherapy, speech therapy, occupational therapy, nutritional support (high-calorie diet). Disease progression: inexorable, median survival 15-20 years from onset, death typically from pneumonia or complications of immobility. Research: antisense oligonucleotides (ASO) targeting HTT mRNA are in clinical trials.

Hydranencephaly

– A rare congenital condition where the cerebral hemispheres are nearly completely absent, replaced by thin-walled, fluid-filled sacs, while the brainstem, basal ganglia, and cerebellum are relatively preserved. Caused by bilateral internal carotid artery occlusion in utero (vascular insult, infection, or twin-twin transfusion). Presents with normal head size at birth, rapid head growth (mimicking hydrocephalus), preserved brainstem reflexes (sucking, swallowing, eye movements), but no cognitive development, visual tracking, or voluntary movement. Diagnosis by transillumination (glows brightly) and MRI/CT showing absent cerebral tissue with preserved posterior fossa structures. Differentiated from severe hydrocephalus (which has a thin cortical rim). No treatment; supportive care. Most die in infancy or childhood.

Hydrocephalus

– Abnormal accumulation of cerebrospinal fluid (CSF) within the ventricles of the brain, leading to increased intracranial pressure. Communicating hydrocephalus: impaired CSF absorption (subarachnoid hemorrhage, meningitis, leptomeningeal carcinomatosis). Non-communicating (obstructive) hydrocephalus: blockage within the ventricular system (aqueductal stenosis, mass lesions, colloid cyst). Normal pressure hydrocephalus (NPH): classic triad of gait apraxia, urinary incontinence, and cognitive decline (dementia); enlarged ventricles on imaging with normal opening pressure. In infants: rapidly enlarging head circumference, bulging fontanelles, sunsetting eyes. Treatment: ventriculoperitoneal shunt (mainstay), endoscopic third ventriculostomy for selected obstructive cases.

Insomnia

– A sleep disorder characterized by difficulty falling asleep, staying asleep, or waking too early despite adequate opportunity for sleep, with resulting daytime impairment. It may be acute or chronic and is often triggered by stress, anxiety, depression, or poor sleep habits. Treatment includes cognitive behavioral therapy for insomnia (CBT-I) as first-line therapy, with medications considered for short-term use.

Internuclear Ophthalmoplegia

– A conjugate horizontal gaze disorder caused by a lesion of the medial longitudinal fasciculus (MLF), a heavily myelinated fiber tract that connects the abducens nucleus (CN VI) on one side to the oculomotor nucleus (CN III) on the contralateral side, coordinating horizontal gaze. In a left INO (the most common type), a lesion of the left MLF impairs adduction of the left eye (the eye cannot move past the midline) and causes abducting nystagmus of the right eye (the abducting eye develops jerky nystagmus as it tries to overcome the impaired adduction signal). Convergence is preserved because the medial rectus subnucleus in CN III is also innervated via the vergence pathway, which bypasses the MLF. INO is most commonly bilateral in multiple sclerosis and unilateral in ischemic stroke affecting the brainstem (particularly the pons). The lesion is in the ipsilateral MLF relative to the eye that fails to adduct. Diagnosis is clinical by examination of horizontal gaze in both directions, with the characteristic pattern of impaired adduction on the ipsilateral side and abducting nystagmus on the contralateral side. MRI brain demonstrates the MLF lesion and any associated demyelinating or ischemic pathology. Treatment is directed at the underlying cause, with symptomatic management of oscillopsia from the nystagmus (gabapentin, memantine, clonazepam).

Intracerebral Hemorrhage

– Bleeding directly into the brain parenchyma, accounting for approximately ten to fifteen percent of all strokes and carrying a high mortality rate of approximately forty percent at thirty days. The most common cause is chronic hypertension, which causes lipohyalinosis and Charcot-Buchard microaneurysm formation in the penetrating arterioles of the basal ganglia (most commonly the putamen), thalamus, pons, and cerebellum. Other causes include cerebral amyloid angiopathy (common cause of lobar ICH in elderly patients), vascular malformations (arteriovenous malformations, cavernous malformations), intracranial aneurysms, coagulopathy (warfarin, DOACs, thrombocytopenia), brain tumors (tumoural hemorrhage), vasculitis, cerebral venous sinus thrombosis with venous infarction, and drug abuse (cocaine, amphetamines). Clinical presentation includes sudden onset of severe headache, vomiting, focal neurological deficit (hemiparesis, aphasia, sensory loss, visual field defects), altered level of consciousness, and seizures. The deficit progresses over minutes to hours as the haematoma expands. Diagnosis is by non-contrast CT head, which shows a hyperdense (white) lesion within the brain parenchyma. MRI is more sensitive for detecting the underlying cause (amyloid angiopathy, cavernous malformations, tumors). CT angiography may detect a spot sign indicating active extravasation and risk of haematoma expansion. Management includes blood pressure control (systolic <140 mmHg to reduce haematoma expansion), reversal of anticoagulation (vitamin K, prothrombin complex concentrate, protamine), intracranial pressure monitoring and management, and surgical haematoma evacuation for cerebellar hemorrhage greater than 3 cm, lobar hemorrhage with significant mass effect, or superficial hemorrhage in a deteriorating patient. The STICH trial showed no overall benefit for early surgery in spontaneous supratentorial ICH.

Intracranial Hemorrhage

– Bleeding within the cranial cavity. Epidural hematoma: arterial bleeding (middle meningeal artery), typically from temporal bone fracture; lucid interval then rapid deterioration. Subdural hematoma: venous bleeding from bridging veins; acute (<72h), subacute, or chronic (elderly, alcoholics). Subarachnoid hemorrhage: arterial bleeding (aneurysm rupture 85%); thunderclap headache, nuchal rigidity. Intracerebral hemorrhage: hypertensive (basal ganglia, thalamus, pons, cerebellum), amyloid angiopathy, vascular malformations. Intraventricular hemorrhage: extension from ICH or primary. Diagnosis: non-contrast CT is first-line. Treatment: ABCs, reversal of anticoagulation, blood pressure control (SBP <140 for ICH), surgical evacuation (cerebellar >3cm, lobar >30cc), external ventricular drain for hydrocephalus, aneurysm clipping/coiling for SAH.

Ischemic Stroke

– A neurological deficit caused by focal cerebral ischemia due to thrombotic or embolic occlusion of a cerebral artery (87% of all strokes). Thrombotic: in situ occlusion of atherosclerotic vessel. Embolic: embolus from cardiac source (atrial fibrillation, valvular disease, patent foramen ovale) or large artery (carotid/vertebral). Lacunar: small vessel occlusion due to lipohyalinosis. Presents with sudden focal deficits (hemiparesis, aphasia, neglect, ataxia) corresponding to vascular territory. Diagnosis: non-contrast CT (rule out hemorrhage), CT perfusion, MRI DWI (gold standard). Acute management: IV tPA within 4.5 hours, mechanical thrombectomy for large vessel occlusion within 24 hours. Secondary prevention: antiplatelets, statins, blood pressure control, anticoagulation for cardioembolic.

Jacksonian Epilepsy (Focal Seizures)

– A type of focal (partial) seizure originating in the primary motor cortex, characterized by clonic (rhythmic jerking) or tonic (sustained contraction) movements that spread sequentially to adjacent body parts along the motor homunculus (Jacksonian march). Named after John Hughlings Jackson (1835-1911), who first described the phenomenon. The march typically begins in the hand or face (largest cortical representation) and spreads proximally up the arm, then to the leg on the same side. Can also present as sensory seizures (somatosensory cortex, paraesthesiae) or versive seizures (frontal eye field). Consciousness is usually preserved during the seizure (focal aware seizure, ILAE 2017 classification). May evolve into a bilateral tonic-clonic seizure (secondary generalisation). Aetiologies: any focal brain lesion–cortical dysplasia, tumor (gliomas, meningiomas), stroke, arteriovenous malformation, traumatic brain injury, infection (abscess, encephalitis), and mesial temporal sclerosis. Diagnosis: EEG (may show focal spikes or sharp waves over the contralateral central region), MRI brain with epilepsy protocol. Treatment: antiseizure medications (carbamazepine, lamotrigine, levetiracetam, oxcarbazepine). Surgical resection of the epileptogenic focus may be curative in 60-80% of drug-resistant cases.

Karnofsky Performance Status

– The Karnofsky Performance Status (KPS) scale is a widely used clinical tool for assessing the functional status of patients, particularly those with cancer, and quantifies the patient's ability to perform activities of daily living and tolerate treatments. The scale ranges from zero (dead) to one hundred (normal, no complaints, no evidence of disease) in increments of ten. KPS scores of eighty to one hundred indicate that the patient is able to carry on normal activity and work, with no special care needed; KPS 70 indicates that the patient is able to care for self but cannot carry on normal activity or do active work; KPS 50 indicates that the patient requires considerable assistance and frequent medical care; and KPS 20 indicates that the patient is very sick, requiring active supportive treatment. The scale was developed by Karnofsky and Burchenal in 1949 and is used in oncology to determine prognosis, guide treatment intensity, and assess eligibility for clinical trials. The KPS has been validated as an independent predictor of survival in many cancers. A score of 70 or higher is typically required for aggressive treatments such as combination chemotherapy or major surgery, while lower scores indicate the need for palliative and supportive care. The scale has limitations, including inter-rater variability and a lack of sensitivity to small changes in function.

Lewy Body Dementia

– The second most common neurodegenerative cause of dementia after Alzheimer disease and is characterized by the presence of alpha-synuclein-containing Lewy bodies in cortical and subcortical neurons. LBD encompasses two clinical entities: dementia with Lewy bodies (DLB), where dementia occurs within one year of the onset of parkinsonism, and Parkinson disease dementia (PDD), where dementia occurs after at least one year of established Parkinson disease. The core clinical features include dementia (typically with prominent fluctuations in cognition and alertness, well-formed recurrent visual hallucinations, and parkinsonism), REM sleep behaviour disorder (acting out dreams, often a prodromal feature), and autonomic dysfunction. Other supportive features include severe neuroleptic sensitivity (a critical clinical point: antipsychotics can precipitate severe rigidity and autonomic instability in LBD), repeated falls, syncope, and depression. The diagnostic criteria include the presence of two or more core features for probable DLB and one core feature for possible DLB. Diagnosis is clinical, supported by dopamine transporter (DaT) SPECT imaging showing reduced presynaptic dopamine uptake in the basal ganglia (the most useful diagnostic test), and polysomnography may confirm REM sleep behaviour disorder. Management includes cholinesterase inhibitors (rivastigmine, donepezil) for cognition and visual hallucinations—these are more effective than in Alzheimer disease—and careful use of atypical antipsychotics (quetiapine, clozapine) for severe hallucinations or agitation, with avoidance of typical antipsychotics due to neuroleptic sensitivity. Levodopa may be used for parkinsonism but has limited efficacy. Prognosis is poor, with progressive decline over several years, parkinsonism worsening, and eventual death from complications of immobility and dysphagia.

Limbic System

– A set of interconnected brain structures located in the medial temporal lobe and around the corpus callosum, involved in emotion, memory, motivation, behaviour, and autonomic regulation. The term was coined by Paul Broca (1878, le grand lobe limbique) and expanded by Papez (1937, Papez circuit) and MacLean (1952, 'visceral brain'). Key structures: (1) Hippocampus: memory formation (spatial, episodic), encoding and consolidation from short-term to long-term memory. (2) Amygdala: emotional processing (fear, anger, anxiety, pleasure), emotional memory, social behaviour, and threat detection. (3) Cingulate gyrus: emotional processing, conflict monitoring, pain processing, decision-making. (4) Hypothalamus: autonomic (sympathetic and parasympathetic), neuroendocrine (CRH, TRH, GHRH, GnRH, ADH), hunger/satiety, thirst, thermoregulation, circadian rhythms. (5) Fornix: major output tract from hippocampus to mammillary bodies. (6) Mammillary bodies: memory (reciprocal connections with hippocampus via fornix, and with thalamus). (7) Anterior thalamic nucleus: part of the Papez circuit, spatial memory. (8) Septal nuclei: reward, pleasure, reinforcement. (9) Parahippocampal gyrus: input to hippocampus. (10) Olfactory tubercle: smell. Limbic system dysfunction causes: memory disorders (Alzheimer disease–hippocampal atrophy, Korsakoff syndrome–mammillary body and thalamic damage), emotional dysregulation (anxiety, depression, PTSD, phobias), seizure disorders (temporal lobe epilepsy), schizophrenia (disrupted limbic connectivity), and autonomic dysfunction.

Lissencephaly

– A severe brain malformation characterized by a smooth cerebral surface due to absent (agyria) or decreased (pachygyria) sulcation and gyration, caused by impaired neuronal migration during 12-16 weeks of gestation. Presents with severe developmental delay, intractable epilepsy (infantile spasms, Lennox-Gastaut), spasticity, and feeding difficulties. Facial features may include bitemporal hollowing, prominent forehead, and micrognathia. Diagnosis by MRI showing a smooth brain surface, thick cortex, and shallow or absent Sylvian fissures. Genetic causes include LIS1 (Miller-Dieker syndrome, 17p13.3 deletion) and DCX (X-linked) mutations. Prognosis is poor; most children have severe disabilities and shortened lifespan.

Locked-in Syndrome

– A neurological condition in which the patient is fully conscious and aware but cannot move or speak due to complete paralysis of all voluntary muscles except for vertical eye movements and blinking (total locked-in syndrome: loss of all eye movements). The patient is 'locked in' a body that cannot respond but has intact cognition, sensation, and consciousness. Most common cause: bilateral ventral pontine stroke (basilar artery occlusion/ thrombosis, infarction of the ventral pons, damaging the corticospinal tracts and corticobulbar tracts, sparing the reticular activating system (consciousness) and the midbrain tectum (vertical eye movements). Other causes: central pontine myelinolysis (rapid correction of hyponatraemia), pontine hemorrhage, brainstem tumor, encephalitis (enterovirus 71), trauma (basilar skull fracture), amyotrophic lateral sclerosis (advanced), and severe Guillain-Barré syndrome. Diagnosis: clinical (consciousness confirmed by eye movement communication). MRI brain shows ventral pontine lesion. EEG is normal (distinguishes from vegetative state/minimally conscious state). Communication: often established through eye movements (blinking code, eye-tracker technology, EEG-based brain-computer interfaces). Prognosis: poor for functional recovery; mortality 60-90% in the first year. Some patients survive many years with intensive support and communication devices. The condition is distinguished from: coma (no eye opening, no awareness), vegetative state (wakefulness without awareness, no purposeful responses), and minimally conscious state (inconsistent but reproducible evidence of awareness).

Locomotor Ataxia (Tabes Dorsalis)

– A late manifestation of untreated tertiary syphilis (neurosyphilis) caused by infection with Treponema pallidum, involving progressive degeneration of the dorsal columns (fasciculus gracilis and cuneatus) and dorsal roots of the spinal cord. The demyelination and neuronal loss affect the sensory pathways mediating proprioception (joint position sense) and vibration sense. Clinical features: sensory ataxia (wide-based, stamping gait, worse with eyes closed, positive Romberg sign), paraesthesiae (lightning pains–sudden, severe, stabbing pain in the legs, back, trunk), loss of deep tendon reflexes (especially knee and ankle jerks, areflexia due to dorsal root involvement), impaired proprioception and vibration sense, Charcot joints (neuropathic arthropathy, painless joint destruction, most commonly knee, ankle), Argyll Robertson pupils (bilateral small irregular pupils that accommodate but do not react to light, light-near dissociation), and optic atrophy. Onset is typically 15-30 years after primary infection. Diagnosis: clinical, serum treponemal tests

(TPPA, FTA-ABS) positive, CSF analysis (lymphocytic pleocytosis, elevated protein, positive CSF VDRL), and spinal cord MRI (dorsal column T2 hyperintensity and atrophy). Treatment: aqueous crystalline penicillin G 18-24 million units IV daily for 10-14 days. Prognosis: some improvement in symptoms, but neurological deficits are often permanent.

Lumbar Disc Herniation

– The most common cause of lumbar radiculopathy (sciatica), occurring when the nucleus pulposus of a lumbar intervertebral disc herniates through the annulus fibrosus, compressing adjacent nerve roots. The most commonly affected level is L4-L5 and L5-S1, accounting for approximately ninety-five percent of lumbar disc herniations. L4-L5 disc herniation typically compresses the L5 nerve root, causing pain radiating from the buttock down the lateral leg to the dorsum of the foot and great toe, weakness of ankle dorsiflexors and evertors (foot drop), and no reflex changes. L5-S1 disc herniation typically compresses the S1 nerve root, causing pain radiating from the buttock down the posterior leg to the heel and lateral foot, weakness of plantar flexion (gastrocnemius, soleus) and ankle eversion, and decreased ankle jerk reflex. Cauda equina syndrome is a surgical emergency requiring urgent MRI and neurosurgical referral. The majority of lumbar disc herniations resolve with conservative management including NSAIDs, activity modification, and physiotherapy. Indications for surgical intervention include cauda equina syndrome, progressive neurological deficit, and persistent radicular pain refractory to 6-12 weeks of conservative management. Surgical options include microdiscectomy, endoscopic discectomy, and minimally invasive tubular discectomy.

Lumbar puncture

– Also known as spinal tap, a procedure to obtain cerebrospinal fluid from the subarachnoid space in the lumbar spine, typically at the L3-L4 or L4-L5 interspace (below the termination of the spinal cord at L1-L2 in adults). Indications include diagnosis of meningitis and encephalitis (CSF Gram stain, culture, cell count, glucose, protein, PCR for viruses and bacteria), subarachnoid hemorrhage (xanthochromia from bilirubin breakdown products, RBC count not clearing from tube 1 to tube 4), neuroinflammatory and demyelinating diseases (multiple sclerosis, Guillain-Barre syndrome, chronic inflammatory demyelinating polyneuropathy), and neoplastic conditions (leptomeningeal carcinomatosis, lymphoma). The procedure involves inserting a hollow needle (typically 22G or 20G) into the subarachnoid space at the L3-L4 or L4-L5 interspace, after sterile preparation and local anesthesia. The patient is positioned in the lateral decubitus (fetal) position or sitting and leaning forward to widen the interlaminar space. Opening pressure is measured using a manometer before CSF is collected, with normal opening pressure ranging from 10-20 cm H2O. CSF is collected in sequential tubes for cell count and differential, glucose and protein, Gram stain and culture, and additional studies (PCR, cytology, oligoclonal bands, flow cytometry) as indicated. Contraindications include increased intracranial pressure with mass effect or intracranial mass (risk of brain herniation—the most feared complication), coagulopathy or thrombocytopenia, infection at the puncture site, and spinal cord compression. Complications include post-dural puncture headache (the most common complication, caused by CSF leak through the dural puncture, managed with bed rest, caffeine, hydration, and epidural blood patch for severe cases), bleeding (spinal epidural haematoma, rare but serious), infection (meningitis, epidural abscess), and radicular pain from nerve root irritation.

Lumbar Radiculopathy

– A condition caused by compression, irritation, or inflammation of a spinal nerve root in the lumbar spine, resulting in pain radiating into the lower extremity along the distribution of the affected nerve root (sciatica). The most commonly affected nerve roots are L4, L5, and S1. L4 radiculopathy causes pain radiating into the medial leg and medial foot, weakness of knee extension (quadriceps), and decreased knee reflex. L5 radiculopathy causes pain radiating into the lateral leg and dorsum of the foot, weakness of ankle dorsiflexion (foot drop) and great toe extension (extensor hallucis longus), with no reflex changes. S1 radiculopathy causes pain radiating into the posterior leg and lateral foot, weakness of plantar flexion (gastrocnemius, soleus), and decreased ankle jerk reflex. Diagnosis is clinical, confirmed by MRI of the lumbar spine demonstrating nerve root compression from disc herniation, spinal stenosis, or foraminal narrowing. EMG/NCS may confirm the affected nerve root and exclude peripheral neuropathy or plexopathy. Management is initially conservative with NSAIDs, activity modification, and physiotherapy. Epidural corticosteroid injections provide temporary relief. Surgical intervention (microdiscectomy or decompressive laminectomy) is indicated for cauda equina syndrome, progressive motor deficit, or persistent radicular pain despite 6-12 weeks of conservative management.

Memory

– The cognitive process of encoding, storing, and retrieving information, mediated by distributed neural networks involving the hippocampus, medial temporal lobe, prefrontal cortex, and limbic system. Types: sensory memory (brief, millisecond–second), short-term/working memory (seconds–minutes, limited capacity 7 items), and long-term memory (subdivided into explicit/episodic and semantic and implicit/procedural). Memory impairment (amnesia) can be acute (transient global amnesia, concussion, seizure, medication side effects) or progressive (Alzheimer disease, vascular dementia, frontotemporal dementia, Lewy body dementia). Evaluation includes mental status examination (MMSE, MoCA), collateral history, laboratory workup (B12, TSH, syphilis, HIV), and brain imaging (MRI for atrophy pattern, vascular disease). Management: treat reversible causes, cognitive stimulation, safety planning, and pharmacotherapy (cholinesterase inhibitors for Alzheimer disease; memantine for moderate–severe stages).

Meningioma

– The most common primary intracranial tumor, accounting for approximately thirty-seven percent of all primary brain tumors, and arises from arachnoid cap cells of the meninges. Most meningiomas are benign (WHO grade I), slow-growing, and well-circumscribed, with a whorled histopathological pattern and psammoma bodies. Atypical (grade II) and anaplastic (grade III) meningiomas are more aggressive with higher recurrence rates. Meningiomas are more common in women, with a female-to-male ratio of approximately 2-3:1, and their incidence increases with age. Most meningiomas are asymptomatic and discovered incidentally on neuroimaging. Symptomatic meningiomas present with symptoms related to mass effect on adjacent structures, including headache, seizures (focal or generalised), and focal neurological deficits. The most common locations are parasagittal/falcine region, cerebral convexities, sphenoid wing, and olfactory groove. Sphenoid wing meningiomas may present with proptosis and visual disturbances. Diagnosis is by MRI with gadolinium, which shows a dural-based, extra-axial mass with homogeneous contrast enhancement and a dural tail sign. CT may show hyperostosis of the adjacent bone. Asymptomatic or small meningiomas with no mass effect are managed with observation and serial imaging. Symptomatic, growing, or large tumors are treated with surgical resection (Simpson grade I-V, with grade I meaning complete removal including dural attachment). Adjuvant radiotherapy or stereotactic radiosurgery is used for residual, recurrent, atypical, or malignant meningiomas. WHO grade I meningiomas have excellent prognosis, with 5-year recurrence rates of 5-10 percent after complete resection; grade II (atypical) have 30-40 percent recurrence; and grade III (anaplastic/malignant) have 50-80 percent recurrence.

Meningitis

– Inflammation of the meninges most commonly caused by bacterial infection (Neisseria meningitidis, Streptococcus pneumoniae, Haemophilus influenzae, Group B Streptococcus). Presents with fever, nuchal rigidity, photophobia, and altered mental status. Diagnosis: lumbar puncture with CSF analysis. Bacterial meningitis is a medical emergency requiring prompt empiric antibiotics and dexamethasone. Vaccination prevents several causes.

Migraine

– A primary headache disorder characterized by recurrent episodes of moderate to severe, typically unilateral, pulsating headache lasting 4-72 hours, associated with nausea, vomiting, photophobia, and phonophobia. Migraine is the second most disabling condition worldwide (second only to low back pain in years lived with disability), affecting 15-18 percent of women and 6-8 percent of men globally. The pathophysiology involves a complex neurovascular cascade: cortical spreading depression (a wave of neuronal depolarisation followed by prolonged suppression of cortical electrical activity that propagates across the cortex at 2-6 mm per minute), which is thought to be the pathophysiological substrate of the migraine aura, followed by activation of the trigeminovascular system (trigeminal nerve endings surrounding intracranial blood vessels release vasoactive neuropeptides such as CGRP, substance P, and neurokinin A), causing neurogenic inflammation and vasodilation of intracranial and extracranial blood vessels, which activates nociceptive pathways to the thalamus and cortex. Triggers include stress (the most common trigger), hormonal fluctuations (menstrual migraine), lack of sleep or excessive sleep, skipped meals, dehydration, alcohol (especially red wine), caffeine withdrawal, bright lights, loud noises, strong odours, and certain foods. Migraine is classified as episodic (fewer than 15 headache days per month) or chronic (15 or more headache days per month for more than 3 months, with at least 8 days meeting migraine criteria). Chronic migraine often evolves from episodic migraine and is associated with overuse of acute medications. Management of acute attacks includes simple analgesics (ibuprofen, paracetamol, aspirin), triptans (sumatriptan, rizatriptan, eletriptan—serotonin 5-HT1B/1D receptor agonists that inhibit CGRP release and cause vasoconstriction), antiemetics (metoclopramide, domperidone), and gepants (ubrogepant, rimegepant—CGRP receptor antagonists). Preventive therapy is indicated for patients with frequent or disabling attacks and includes beta-blockers (propranolol, metoprolol), tricyclic antidepressants (amitriptyline), antiepileptics (topiramate, sodium valproate), CGRP monoclonal antibodies (erenumab, galcanezumab, fremanezumab, eptinezumab), and botulinum toxin type A (OnabotulinumtoxinA) for chronic migraine.

Migraine With Aura

– A primary headache disorder characterized by recurrent attacks of fully reversible visual, sensory, or speech symptoms that develop gradually over at least five minutes and last no more than sixty minutes, followed by headache with the characteristics of migraine. Visual aura is the most common type, affecting approximately ninety percent of patients with migraine with aura, and typically consists of a fortification spectrum (scintillating scotoma) that expands over the visual field over 20-60 minutes, leaving a scotoma (blind spot) in its wake. Sensory aura is the second most common type, presenting with unilateral tingling and numbness that spreads slowly from the hand up the arm to the face, typically lasting 5-30 minutes. Speech and language aura is less common and presents with dysphasic speech. The migraine aura is thought to be caused by cortical spreading depression (CSD)—a wave of neuronal and glial depolarisation followed by prolonged suppression of neural activity that propagates across the cerebral cortex at a rate of 2-6 mm per minute. The visual aura localises to the occipital cortex, sensory aura to the somatosensory cortex, and speech aura to the language areas. If the aura occurs without headache (migraine aura without headache, formerly called acephalgic migraine), other causes such as transient ischemic attack and seizure must be excluded, particularly in older patients. Triptans and NSAIDs are effective for acute treatment, and they are safe to take during the aura phase. Preventive treatment is indicated for frequent or disabling attacks (typically more than 4 per month) and includes the same options as for migraine without aura. The risk of ischemic stroke is slightly elevated in migraine with aura, particularly in women who smoke and use oral contraceptives, and this should be considered in management.

Migraine Without Aura

– A primary headache disorder characterized by recurrent attacks of moderate to severe, pulsating, unilateral headache lasting four to seventy-two hours, associated with nausea, vomiting, photophobia, and phonophobia. The International Classification of Headache Disorders (ICHD-3) diagnostic criteria require at least five attacks fulfilling the following: headache lasting four to seventy-two hours, with at least two of the following four characteristics: unilateral location, pulsating quality, moderate or severe pain intensity, and aggravation by or causing avoidance of routine physical activity, and during the headache attack at least one of nausea and/or vomiting or photophobia and phonophobia. The attack is not better accounted for by another diagnosis. Migraine without aura is the most common subtype of migraine, accounting for approximately

70-80 percent of all migraine attacks. The headache phase is thought to be mediated by activation of the trigeminovascular system, with release of vasoactive neuropeptides such as CGRP causing neurogenic inflammation of the meningeal blood vessels. The pathophysiology of migraine without aura is less well understood than migraine with aura, as cortical spreading depression is not thought to occur. Management of acute attacks includes stratified care based on headache severity: mild attacks are treated with simple analgesics (ibuprofen, paracetamol, aspirin, diclofenac) plus an antiemetic; moderate-to-severe attacks are treated with triptans (sumatriptan, rizatriptan, eletriptan) or gepants (rimegepant, ubrogepant). Triptans are most effective when taken early during the headache phase before central sensitisation develops. Preventive therapy is indicated for patients with frequent or disabling attacks and includes beta-blockers (propranolol, metoprolol), tricyclic antidepressants (amitriptyline), antiepileptics (topiramate, sodium valproate), and CGRP monoclonal antibodies.

Mild Cognitive Impairment

– A transitional cognitive state between normal aging and dementia, characterized by subjective cognitive complaints confirmed by objective cognitive impairment that is not severe enough to significantly interfere with daily activities. MCI is classified as amnestic MCI (aMCI), where memory is the primary domain affected, and non-amnestic MCI (naMCI), where other cognitive domains (attention, executive function, visuospatial ability, language) are primarily affected. The amnestic subtype is the most common and is considered a prodromal stage of Alzheimer disease in most cases. The prevalence of MCI in adults over 60 is approximately 15-20 percent, and the annual conversion rate from MCI to dementia is approximately 10-15 percent, although some patients remain stable or revert to normal cognition. Diagnosis requires objective cognitive testing confirming impairment one to two standard deviations below age- and education-matched norms, with intact functional abilities. Biomarkers such as CSF A-beta and tau levels, amyloid PET imaging, and FDG-PET can help determine the likelihood of progression to Alzheimer disease. Management includes addressing modifiable vascular risk factors (hypertension, diabetes, smoking), cognitive stimulation, physical exercise, and monitoring for cognitive decline. There is no pharmacological treatment approved for MCI; acetylcholinesterase inhibitors are not recommended as they do not prevent progression to dementia. Prognosis is variable, with some patients remaining stable indefinitely while others progress to dementia within a few years.

Multiple Sclerosis

– A chronic autoimmune demyelinating disease of the central nervous system, most commonly presenting in young adults. Characterized by relapsing-remitting episodes of neurological dysfunction (optic neuritis, transverse myelitis, brainstem syndromes). Diagnosis: McDonald criteria with MRI dissemination in space and time. Disease-modifying therapies reduce relapse rate and disability progression.

Multiple System Atrophy

– A progressive, fatal neurodegenerative disorder characterized by varying combinations of autonomic failure, cerebellar ataxia, parkinsonism, and pyramidal signs. It is classified into two subtypes: MSA-P (Parkinsonian variant, previously striatonigral degeneration), characterized by prominent parkinsonism with poor levodopa response, and MSA-C (cerebellar variant, previously sporadic olivopontocerebellar atrophy), characterized by prominent cerebellar ataxia. Pathologically, MSA is characterized by the accumulation of alpha-synuclein in oligodendrocytes, forming glial cytoplasmic inclusions (GCIs), which is the pathological hallmark and distinguishes MSA from other synucleinopathies such as Parkinson disease and Lewy body dementia where alpha-synuclein accumulates in neurons. Onset is typically in the 50s or 60s, with rapidly progressive course leading to severe disability within 5-10 years. Clinical features of autonomic failure (present in both subtypes) include orthostatic hypotension (symptomatic drop in blood pressure on standing, often the earliest symptom), urinary incontinence, erectile dysfunction, and impaired sweating. Other features include stridor (nocturnal and diurnal, indicating vocal cord abductor paralysis, a potentially life-threatening complication requiring CPAP or tracheostomy), REM sleep behaviour disorder, and respiratory dysfunction. The parkinsonism in MSA-P is typically symmetrical, with poor or absent response to levodopa, early postural instability, and craniocervical dystonia. The cerebellar features in MSA-C include gait and limb ataxia, dysarthria, and oculomotor abnormalities. Diagnosis is clinical (second consensus criteria for MSA) with supportive features including MRI findings: putaminal atrophy, putaminal rim sign (hyperintense rim on T2-weighted MRI in MSA-P), hot cross bun sign (cruciform hyperintensity in the pons on T2-weighted MRI in MSA-C), and middle cerebellar peduncle atrophy. The putaminal rim sign on susceptibility-weighted imaging is characteristic. There is no disease-modifying treatment; management is symptomatic and supportive, including treatment of orthostatic hypotension (fludrocortisone, midodrine, droxidopa), physical therapy, speech therapy, and in advanced cases, CPAP for stridor and feeding tube placement for dysphagia. Prognosis is poor, with median survival of 6-10 years from diagnosis.

Myasthenia Gravis

– An autoimmune disorder caused by antibodies against the nicotinic acetylcholine receptor (AChR, 85%) or MuSK at the neuromuscular junction, leading to fluctuating muscle weakness that worsens with use (fatigability). Presents with ptosis, diplopia, dysphagia, dysarthria, and proximal limb weakness. Ocular MG: limited to eye muscles. Generalized MG: bulbar and limb involvement. Myasthenic crisis: respiratory failure requiring ventilation. Diagnosis: AChR antibodies, repetitive nerve stimulation (decrement), single-fiber EMG, edrophonium test. Treatment: acetylcholinesterase inhibitors (pyridostigmine), corticosteroids, immunosuppressants (azathioprine, mycophenolate), thymectomy for thymoma, IVIG/plasmapheresis for crisis.

Myelopathy

– A general term for any neurological deficit resulting from spinal cord pathology, encompassing a wide range of causes and clinical presentations. The clinical features depend on the level and completeness of the spinal cord lesion and may include motor weakness (upper motor neuron pattern below the lesion, with spasticity, hyperreflexia, and extensor plantar responses), sensory loss (with a sensory level corresponding to the lesion), bladder and bowel dysfunction, and gait disturbance, spastic gait, sensory ataxia, or a combination. The sensory level is a key finding in myelopathy—the dermatomal level at which sensation changes from normal to abnormal below the level of the lesion. This helps localise the spinal cord lesion. Diagnosis is by MRI of the spinal cord, which demonstrates the underlying pathology (compression, inflammation, infarction, tumor, or demyelination). Management depends on the cause and may include surgical decompression for extramedullary compression (disc, tumor, abscess, haematoma), corticosteroids for inflammatory or demyelinating myelopathy, immunosuppression for autoimmune causes, and rehabilitation for functional recovery. Prognosis depends on the etiology, severity of the initial deficit, and timeliness of treatment.

Myotonic Dystrophy

– The most common adult-onset muscular dystrophy, an autosomal dominant multisystem disorder caused by CTG trinucleotide repeat expansions in the DMPK gene (type 1, DM1) or CCTG tetranucleotide repeat expansions in the CNBP gene (type 2, DM2) on chromosome 19q13.3. DM1 is more common and more severe than DM2. The pathophysiology involves a toxic gain-of-function mechanism, where the expanded repeat RNA forms hairpin structures that sequester RNA-binding proteins (particularly muscleblind-like proteins MBNL1 and MBNL2), leading to dysregulation of alternative splicing of many downstream genes (spliceopathy). The expanded repeats can also be transcribed into toxic proteins through repeat-associated non-ATG (RAN) translation. Clinical features of DM1 include myotonia (delayed relaxation after muscle contraction, demonstrable by grip myotonia or percussion myotonia of the thenar eminence and tongue), progressive distal weakness (worse than proximal), facial weakness with a characteristic hatchet face and ptosis, frontal balding, cataracts (posterior subcapsular iridescent cataracts), cardiac conduction abnormalities (first-degree heart block, bundle branch block, complete heart block—a leading cause of death), insulin resistance, and excessive daytime somnolence. DM2 (proximal myotonic myopathy) is milder, with predominantly proximal weakness, less severe myotonia, fewer cardiac complications, and later onset. Myotonic dystrophy exhibits anticipation (earlier and more severe disease in successive generations) due to expansion of the CTG repeat during transmission, particularly through maternal inheritance. Diagnosis is clinical, confirmed by genetic testing showing expanded CTG repeats in DMPK (DM1) or CCTG repeats in CNBP (DM2). There is no disease-modifying treatment. Management is symptomatic and includes mexiletine for myotonia, cardiac monitoring with pacemaker insertion for conduction abnormalities, treatment of cataracts and endocrine dysfunction, and respiratory support for sleep-disordered breathing. Avoid depolarising muscle relaxants (suxamethonium) due to risk of malignant hyperthermia. Prognosis in DM1 is variable; patients with severe congenital DM1 have poor outcomes, while milder adult-onset forms allow near-normal lifespan with appropriate surveillance.

Neglect Syndrome

– Hemispatial neglect (unilateral neglect, hemineglect) is a neurological condition characterized by failure to attend to, respond to, or orient toward stimuli on the side of space contralateral to the brain lesion, in the absence of primary sensory or motor deficits sufficient to explain the neglect. It is most commonly caused by damage to the right hemisphere, particularly the right parietal lobe (especially the inferior parietal lobule and temporoparietal junction), right thalamus, or right basal ganglia. The patient fails to respond to stimuli on the contralateral (left) side, which may include visual, auditory, tactile, or even olfactory stimuli. Patients may fail to eat food on the left side of their plate, bump into objects on the left, shave only the right side of their face, and may even deny ownership of their left limbs (somatoparaphrenia). Neglect is more severe and persistent for left-sided stimuli after right hemisphere damage due to the right hemisphere's dominant role in spatial attention (both hemispheres attend to the right, but only the right attends to the left). The contralesional side is the left side. Diagnosis includes bedside tests such as line bisection (patient marks the middle of a horizontal line, bisecting it further to the right of center), cancellation tasks (patient fails to cancel targets on the left side of the page), and drawing tasks (patient omits left-sided details when drawing a clock or a daisy). The Behavioural Inattention Test (BIT) provides a standardised assessment. Management includes visual scanning training, limb activation therapy, prism adaptation, and dopamine agonists. The condition has a significant negative impact on functional recovery and rehabilitation outcomes, with less severe forms resolving over weeks to months.

Neurocysticercosis

– The most common parasitic infection of the central nervous system worldwide and is caused by the larval stage (cysticercus) of the pork tapeworm Taenia solium. It is endemic in Latin America, Asia, and Africa, and is an important cause of epilepsy in endemic regions. The larvae are ingested through fecal-oral transmission (ingestion of food or water contaminated with tapeworm eggs from an infected individual), and the oncospheres cross the intestinal wall, disseminate hematogenously to the brain, where they develop into cysticerci within the brain parenchyma, subarachnoid space, ventricles, or spinal cord. Clinical presentation depends on the location and number of cysts and the host immune response. Parenchymal neurocysticercosis commonly presents with seizures (the most common manifestation in endemic regions, accounting for up to 30 percent of epilepsy in endemic areas), headache, and focal neurological deficits. Ventricular cysts can cause obstructive hydrocephalus, and subarachnoid cysts can cause arachnoiditis and communicating hydrocephalus. Diagnosis is by neuroimaging: CT shows calcified dead cysts (small hyperdense nodules) and viable cysts (hypodense lesions with a scolex visible as a hyperdense dot within the cyst). MRI is more sensitive, showing the cyst fluid intensity similar to CSF and the scolex as a mural nodule on T1-weighted imaging. Serological testing (enzyme-linked immunoelectrotransfer blot) in serum and CSF supports the diagnosis. Management includes antiparasitic drugs (albendazole, praziquantel) for active parenchymal cysts, corticosteroids to control inflammation and cerebral edema during antiparasitic treatment, anticonvulsants for seizure control, and surgical management of hydrocephalus (CSF shunting) and intraventricular or spinal cysts. Prognosis is generally good for parenchymal disease with treatment; ventricular and subarachnoid disease have worse outcomes.

Prevention focuses on improved hygiene, sanitation, and meat inspection.

Neuromyelitis Optica

– Neuromyelitis optica spectrum disorder (NMOSD), formerly known as Devic disease, is an autoimmune, inflammatory demyelinating disorder of the central nervous system that preferentially targets the optic nerves and spinal cord. It is caused by pathogenic aquaporin-4 (AQP4) immunoglobulin G autoantibodies in approximately seventy-five to eighty percent of cases, which bind to AQP4 water channels on astrocyte foot processes, activating complement-mediated astrocyte damage and secondary demyelination. Clinical presentation of NMOSD includes optic neuritis (acute or subacute vision loss, often bilateral or sequential, and more severe than in multiple sclerosis, with poor recovery), longitudinally extensive transverse myelitis (LETM, defined as spinal cord lesion extending three or more vertebral segments on MRI, causing paraplegia, sensory loss, and bladder dysfunction), area postrema syndrome (intractable nausea, vomiting, and hiccoughs due to involvement of the medullary area postrema, which is a highly characteristic presentation), and acute brainstem syndrome. The distinction from multiple sclerosis is critical because treatments differ: MS disease-modifying therapies (interferon-beta, fingolimod, natalizumab) are ineffective or may exacerbate NMOSD. Diagnosis is based on clinical criteria (IPND 2015 criteria) requiring at least one core clinical characteristic plus positive AQP4-IgG serology. In seronegative cases, myelin oligodendrocyte glycoprotein (MOG) antibody-associated disease should be considered. MRI findings include longitudinally extensive optic nerve involvement, LETM with central cord involvement, and periaqueductal and area postrema brain lesions. Acute treatment includes high-dose intravenous corticosteroids as first-line, with plasma exchange for steroid-resistant cases. Long-term immunosuppression to prevent relapses includes mycophenolate mofetil, azathioprine, rituximab, eculizumab (a complement C5 inhibitor), inebilizumab (anti-CD19 monoclonal antibody), and satralizumab (anti-IL-6 receptor monoclonal antibody). Prognosis is relapse-dependent, with most patients accumulating disability from recurrent attacks. Attacks are typically more severe than in MS, but progression is attack-related rather than gradual.

Neuropathic Pain

– Chronic pain caused by damage or disease affecting the somatosensory nervous system, including the peripheral nerves, dorsal root ganglia, spinal cord, or brain. It is characteristically described as burning, shooting, electric shock-like, tingling, or stabbing, and may be accompanied by allodynia (pain from a normally non-painful stimulus) and hyperalgesia (exaggerated pain from a normally painful stimulus). Common causes include diabetic peripheral neuropathy, postherpetic neuralgia (the most common cause), painful diabetic peripheral neuropathy, trigeminal neuralgia, spinal cord injury pain, post-stroke central pain, chemotherapy-induced peripheral neuropathy, and HIV-associated neuropathy. Diagnosis is clinical, based on the characteristic pain descriptors and neurological examination findings. Screening tools such as the Douleur Neuropathique 4 (DN4) questionnaire and the Leeds Assessment of Neuropathic Symptoms and Signs (LANSS) scale help differentiate neuropathic from nociceptive pain. Nerve conduction studies and skin biopsy (quantifying intraepidermal nerve fibre density) may confirm small fibre neuropathy. Management follows a stepwise approach: first-line treatments include gabapentinoids (gabapentin, pregabalin), tricyclic antidepressants (amitriptyline, nortriptyline), and serotonin-noradrenaline reuptake inhibitors (duloxetine, venlafaxine). Topical treatments include lidocaine 5 percent patches and capsaicin 8 percent patches for localised neuropathic pain. Second-line options include tramadol and other opioid analgesics, which should be used cautiously due to the risk of dependence. Non-pharmacological treatments include transcutaneous electrical nerve stimulation (TENS), cognitive behavioural therapy, and spinal cord stimulation for refractory cases. Management of the underlying condition (e.g., glycaemic control in diabetes, antiviral therapy for herpes zoster) is also important.

Neuropathy (Peripheral Neuropathy)

– Damage to peripheral nerves causing sensory, motor, or autonomic dysfunction. Common causes: diabetes mellitus, alcohol use disorder, vitamin

B12 deficiency, chemotherapy, autoimmune diseases, and hereditary conditions (Charcot-Marie-Tooth disease). Presents with numbness, tingling, burning pain, weakness, and gait instability. Treatment addresses the underlying cause, with symptomatic management including gabapentin, duloxetine, and tricyclic antidepressants.

NIH Stroke Scale

– The National Institutes of Health Stroke Scale (NIHSS) is a standardized, validated clinical assessment tool used to quantify the severity of acute ischemic stroke. It consists of fifteen items assessing level of consciousness, gaze, visual fields, facial palsy, motor strength in the arms and legs, limb ataxia, sensory function, language, dysarthria, and extinction/inattention. Each item is scored with a specific number of points, and the total score ranges from zero (no stroke symptoms) to forty-two (most severe stroke). Each item is assessed by a certified examiner, and the scale can be completed in approximately 5-8 minutes. The NIHSS is used to assess baseline stroke severity, guide treatment decisions (e.g., thrombolysis is generally recommended for NIHSS 4-25, though patients with severe strokes may still benefit), monitor neurological changes during the acute phase, and predict outcomes. Higher scores predict poorer outcomes at 90 days. The NIHSS has well-established reliability and validity but has limitations, including poor sensitivity for posterior circulation stroke and right hemisphere stroke, and language barriers for aphasic patients. It is also less accurate for very mild (NIHSS 0-3) or very severe strokes.

Non communicating Hydrocephalus

– Non-communicating (obstructive) hydrocephalus is a type of hydrocephalus in which there is obstruction to the flow of cerebrospinal fluid (CSF) within the ventricular system, preventing CSF from reaching the arachnoid granulations for absorption. The obstruction can occur at any level of the ventricular system, causing dilation of the ventricles proximal to the obstruction. Common causes include tumors (particularly those obstructing the fourth ventricle or the aqueduct of Sylvius, such as posterior fossa tumors such as medulloblastoma, ependymoma, and brainstem glioma), aqueductal stenosis (congenital or acquired), intraventricular hemorrhage (particularly in premature infants causing post-hemorrhagic hydrocephalus), and intraventricular cysts or masses. Clinical presentation depends on the severity and rate of progression of the obstruction. Acute obstruction presents with severe headache, nausea and vomiting, papilloedema, and rapidly decreasing level of consciousness (life-threatening). Chronic obstruction presents with headache, developmental delay (in children), macrocephaly in infants, gait disturbance, cognitive decline, and the sunsetting eye sign (upward gaze palsy due to pressure on the tectal plate from a dilated third ventricle). Diagnosis is by neuroimaging (CT or MRI) showing dilation of the ventricles proximal to the obstruction (lateral and third ventricles above the aqueduct, or fourth ventricle below the aqueduct if the obstruction is at the foramen of Luschka and Magendie), with normal or small ventricles distal to the obstruction. Aqueductal flow studies on MRI can confirm obstruction. Management includes urgent CSF diversion via endoscopic third ventriculostomy (ETV) for aqueductal stenosis and selected obstructive causes, or ventriculoperitoneal (VP) shunt placement, with removal of the obstructing lesion if possible (e.g., tumor resection).

Normal Pressure Hydrocephalus

– A potentially reversible cause of dementia characterized by the classic triad of gait disturbance, urinary incontinence, and cognitive impairment, with enlarged cerebral ventricles on imaging despite normal cerebrospinal fluid pressure on lumbar puncture. NPH is classified as idiopathic (primary) or secondary to conditions such as subarachnoid hemorrhage, meningitis, or traumatic brain injury. The pathophysiology is thought to involve impaired CSF absorption at the arachnoid granulations due to reduced CSF conductance, leading to increased ventricular pressure that is not reflected by a single normal lumbar puncture opening pressure but may be detected by prolonged intracranial pressure monitoring. The classic clinical triad (Wet, Wacky, Wobbly) includes gait apraxia (the earliest and most prominent feature—difficulty initiating walking with a magnetic, shuffling gait described as feet feeling stuck to the floor), cognitive impairment (subcortical dementia with psychomotor slowing, inattention, and executive dysfunction rather than the prominent memory loss of Alzheimer disease), and urinary incontinence (urinary urgency and frequency progressing to complete incontinence, typically occurring last in the triad). Diagnosis is clinical with supportive imaging: MRI brain shows enlarged ventricles out of proportion to cortical atrophy (ventriculomegaly with Evan index greater than 0.3), periventricular white matter changes, and tight high-convexity sulci (disproportionately enlarged subarachnoid space hydrocephalus). Additional diagnostic tests include a positive response to CSF drainage via lumbar puncture (Tap test) or extended lumbar CSF drainage (ELD), where improvement in gait is the most predictive finding. CSF infusion studies may show reduced CSF conductance. Management is CSF diversion via a ventriculoperitoneal (VP) shunt. Response rates are variable, with gait improvement occurring in 70-80 percent of appropriately selected patients, cognitive improvement in 50-70 percent, and urinary improvement in 50-60 percent. The best outcomes are seen in patients with dominant gait disturbance, known etiology (secondary NPH), and a positive response to CSF drainage.

Occipital Neuralgia

– A paroxysmal, severe, stabbing or electric shock-like pain in the distribution of the greater, lesser, or third occipital nerves (posterior scalp, from the nuchal line to the vertex). The pain is unilateral or bilateral, often described as 'ice pick' or 'stabbing', with tenderness over the occipital nerve and paraesthesias in the affected dermatome. Attacks last seconds to minutes but may be followed by a dull ache. Causes: entrapment or irritation of the occipital nerves (trauma, whiplash, cervical spine disease—C1–C3 facet arthropathy, tumors, vascular compression). Differential diagnosis: cervicogenic headache, migraine (especially occipital migraine), giant cell arteritis (must be excluded—check ESR, CRP, temporal artery biopsy), and tension-type headache. Diagnosis is clinical; occipital nerve block (injection of lidocaine and corticosteroid at the midpoint between the occipital protuberance and mastoid) is both diagnostic and therapeutic. Treatment: nerve blocks, NSAIDs, muscle relaxants, neuropathic pain agents (gabapentin, amitriptyline), and in refractory cases, pulsed radiofrequency ablation or occipital nerve stimulation.

Oligoclonal Bands

– Immunoglobulin G (IgG) proteins produced by a limited number of B-cell clones in the cerebrospinal fluid (CSF), detected by isoelectric focusing of CSF and serum samples. Oligoclonal bands represent intrathecal IgG synthesis and are present in approximately ninety to ninety-five percent of patients with multiple sclerosis, making them one of the most sensitive laboratory markers for the disease. The detection of OCBs in CSF but not in serum (type 2 pattern) indicates intrathecal IgG synthesis and supports the diagnosis of multiple sclerosis (MS). The presence of identical OCBs in both CSF and serum (type 3 pattern) indicates systemic IgG production with passive transfer into the CSF, which is less specific for MS. OCBs are detected in approximately 90-95 percent of MS patients, making them the most sensitive laboratory marker for the disease. They are also present in other inflammatory and infectious CNS conditions such as neuromyelitis optica, neurosarcoidosis, neurosyphilis, and neuroborreliosis, but with lower frequency. The absence of OCBs in a patient with typical MS clinical and MRI findings is associated with a milder disease course. OCB typing by isoelectric focusing of paired CSF and serum samples is considered the gold standard method.

Pachymeningitis

– Inflammation of the dura mater (pachymeninx), the outermost layer of the meninges. Pachymeningitis is less common than leptomeningitis (inflammation of pia and arachnoid mater). Causes: infection (tuberculosis—most common in developing countries, syphilis, fungi—Cryptococcus, bacteria—direct spread from sinusitis/otitis/osteomyelitis), autoimmune (IgG4-related disease—hypertrophic pachymeningitis, granulomatosis with polyangiitis, sarcoidosis, rheumatoid arthritis), and neoplastic (meningeal carcinomatosis, lymphoma). Presentation: chronic headache (most common), cranial nerve palsies (II, III, VI, VII, VIII), radicular pain, and myelopathy (spinal pachymeningitis). Hypertrophic pachymeningitis: focal or diffuse dural thickening on MRI with contrast enhancement (pachymeningeal enhancement). CSF: lymphocytic pleocytosis, elevated protein. Diagnosis: MRI brain with contrast, CSF analysis, and biopsy if autoimmune or neoplastic cause suspected. Treatment: corticosteroids for autoimmune causes, antitubercular therapy for TB, antibiotics for bacterial infection. Prognosis depends on the underlying etiology.

Paranoia

– A thought process characterized by excessive or irrational suspiciousness and distrust of others, often manifesting as delusions of persecution, reference, or grandeur. Paranoia occurs across a spectrum: paranoid personality disorder (pervasive, lifelong pattern of distrust without the severity of delusions), delusional disorder (persecutory type—non-bizarre delusions in the absence of other psychotic symptoms), paranoid schizophrenia (prominent persecutory delusions with hallucinations and disorganised thinking), and secondary paranoia (due to dementia, substance use—cocaine, amphetamines, alcohol withdrawal, traumatic brain injury, or brain tumors). Paranoia may also occur in mood disorders (psychotic depression, bipolar mania). Treatment: antipsychotic medication, cognitive behavioural therapy, and treatment of underlying cause.

Paraplegia

– Paralysis of the lower extremities, with or without loss of bladder, bowel, and sexual function, caused by damage to the spinal cord. The most common causes include spinal cord injury (traumatic, from motor vehicle accidents, falls, and sports injuries), spinal cord compression (from tumors, epidural abscess, or disc herniation), spinal cord infarction (anterior spinal artery occlusion), transverse myelitis, and demyelinating diseases such as multiple sclerosis. The level of the spinal cord lesion determines the amount of residual function, with lower thoracic lesions preserving arm and hand function but causing paraplegia, and higher lesions causing more extensive paralysis. Complete spinal cord injury results in complete loss of motor and sensory function below the level of the lesion (ASIA A), while incomplete injury preserves some function below the level (ASIA B-D), with central cord syndrome, anterior cord syndrome, Brown-Sequard syndrome, and conus medullaris syndrome being recognised incomplete spinal cord injury syndromes. Management includes acute stabilisation and surgical decompression for traumatic causes, treatment of the underlying cause (radiotherapy for tumors, antibiotics for infection, corticosteroids for inflammation), and lifelong rehabilitation to maximise functional independence. Prognosis depends on the level and completeness of injury, with the best recovery in incomplete injuries.

Paresthesia

– An abnormal sensation in the skin, described as tingling, prickling, numbness, burning, or a pins-and-needles sensation, caused by dysfunction of the sensory nerves. It can be caused by any condition that affects the peripheral nerves, nerve roots, or central sensory pathways. Common causes include peripheral neuropathy (diabetes, alcohol, vitamin deficiencies), nerve compression (carpal tunnel syndrome, ulnar neuropathy, radiculopathy), multiple sclerosis, spinal cord lesions, stroke, transient ischemic attack, migraine with aura (sensory aura), anxiety and hyperventilation, and certain medications. Paresthesia is a symptom rather than a diagnosis and may be transient (e.g., pressure on a nerve causing the limb to fall asleep) or chronic (e.g., diabetic peripheral neuropathy). The distribution of the paresthesia is important for localising the lesion: peripheral nerve distribution suggests a mononeuropathy, stocking-and-glove distribution suggests a length-dependent polyneuropathy, dermatomal distribution suggests a radiculopathy, a hemi-body distribution suggests a central (brain) lesion, and bilateral leg involvement suggests a spinal cord lesion. Diagnosis involves identifying the underlying cause through history, neurological examination, electrodiagnostic studies, and appropriate laboratory tests. Management is directed at the underlying cause, with symptomatic treatment of neuropathic pain if the paresthesia is painful.

Parinaud Syndrome

– A clinical condition caused by lesions of the dorsal midbrain, particularly the pretectal area and superior colliculi, resulting in a characteristic triad of vertical gaze palsy (particularly impaired upward gaze), convergence-retraction nystagmus (involuntary retraction of the eyeballs into the orbits with attempted upward gaze), and light-near dissociation (pupils that do not react to light but do react to accommodation). Other features include Collier sign (eyelid retraction, also known as the lid-lag sign or setting-sun sign), convergence-retraction nystagmus, and papilloedema. Parinaud syndrome is most commonly caused by compression of the dorsal midbrain by a pineal gland tumor (pinealoma, pineal germinoma, pineocytoma, pineoblastoma), which accounts for the classic presentation in young males. Other causes include midbrain infarction, hemorrhage, multiple sclerosis, arteriovenous malformations, neurocysticercosis, and hydrocephalus (which can cause dilation of the third ventricle with upward pressure on the tectum). Diagnosis is by MRI brain with particular attention to the pineal region and dorsal midbrain, including CSF cytology and tumor markers (alpha-fetoprotein, beta-human chorionic gonadotropin) for suspected germ cell tumors. Management is directed at the underlying cause: surgical resection of pineal tumors, radiotherapy or chemotherapy for germinomas, ventriculoperitoneal shunting for hydrocephalus, and treatment of vascular or demyelinating causes. Prognosis depends on the underlying etiology and response to treatment.

Parkinson Disease

– A progressive neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra pars compacta of the midbrain, leading to reduced dopamine levels in the striatum and the cardinal motor features of bradykinesia, resting tremor, rigidity, and postural instability. The cardinal motor features include bradykinesia (slowness of movement, the hallmark feature, with progressive reduction in amplitude and speed of repetitive movements), resting tremor (a pill-rolling tremor of the hand at 4-6 Hz that is present at rest and diminishes with voluntary movement, typically asymmetrical at onset), rigidity (lead-pipe or cogwheel resistance to passive movement), and postural instability (impaired balance and tendency to fall, typically occurring later in the disease course). Non-motor features are increasingly recognised and include hyposmia (reduced sense of smell, often a prodromal feature that may precede motor symptoms by years), REM sleep behaviour disorder (acting out dreams), constipation, depression (the most common neuropsychiatric symptom), anxiety, apathy, cognitive impairment (executive dysfunction and bradyphrenia, progressing to dementia in up to 80 percent of patients with long-standing disease), autonomic dysfunction (orthostatic hypotension, urinary frequency, erectile dysfunction), and pain. Diagnosis is clinical using the UK Parkinson Disease Society Brain Bank criteria, which require bradykinesia plus at least one of rigidity, resting tremor, or postural instability. Dopamine transporter (DaT) SPECT scanning can confirm presynaptic dopaminergic denervation when the diagnosis is uncertain. The gold standard treatment is levodopa combined with a peripheral decarboxylase inhibitor (carbidopa or benserazide), which is the most effective symptomatic therapy, but long-term use is complicated by motor fluctuations (wearing-off phenomena, on-off fluctuations) and dyskinesias (involuntary choreiform movements). Other medications include dopamine agonists (pramipexole, ropinirole, rotigotine), monoamine oxidase B inhibitors (selegiline, rasagiline), catechol-O-methyltransferase inhibitors (entacapone, opicapone), anticholinergics (for tremor in younger patients), and amantadine (for dyskinesias). Deep brain stimulation of the subthalamic nucleus or globus pallidus internus is effective for advanced disease with motor fluctuations and dyskinesias. Multidisciplinary management including physiotherapy, occupational therapy, speech therapy, and specialist nursing is essential.

Parkinsonism

– A clinical syndrome characterized by bradykinesia, rigidity, tremor (resting, pill-rolling), and postural instability. Idiopathic Parkinson disease: most common cause, responds well to levodopa. Atypical parkinsonian disorders: multiple system atrophy (Parkinsonism plus cerebellar/autonomic), progressive supranuclear palsy (vertical gaze palsy, early falls), corticobasal degeneration (asymmetric apraxia, alien limb), dementia with Lewy bodies (Parkinsonism plus early dementia). Drug-induced parkinsonism: antipsychotics, resolves with drug withdrawal. Vascular parkinsonism: lower body predominance, poor levodopa response.

Petit Mal

– See Absence Seizure.

Pituitary Adenoma

– Pituitary adenomas are benign neoplasms arising from the adenohypophysis (anterior pituitary gland), accounting for approximately ten to fifteen percent of all intracranial neoplasms. They are classified by size as microadenomas (less than 10 millimeters) or macroadenomas (10 millimeters or greater), and by functional status as functioning (hormone-secreting) or non-functioning. Functioning adenomas include prolactinomas (most common functioning adenoma, causing galactorrhea, amenorrhea, and infertility. Non-functioning adenomas present with mass effect: visual field deficits (bitemporal hemianopia from compression of the optic chiasm), headache, hypopituitarism (from compression of the normal pituitary gland), and cranial nerve palsies from cavernous sinus invasion. Prolactinomas are treated medically with dopamine agonists (cabergoline is first-line, bromocriptine is second-line), which are highly effective in reducing prolactin levels and tumor size. Surgery (trans-sphenoidal adenomectomy) is indicated for non-functioning adenomas causing mass effect, functioning adenomas not responsive to medical therapy (except prolactinomas), and apoplexy (hemorrhagic infarction of a pituitary adenoma causing acute headache, visual loss, ophthalmoplegia, and altered consciousness—a pituitary emergency requiring urgent corticosteroid replacement and surgical decompression). Radiotherapy or radiosurgery (Gamma Knife) is used for residual, recurrent, or aggressive adenomas. Endocrine assessment and replacement of deficient hormones are essential before and after treatment. Prognosis is excellent for microadenomas, and macroadenomas have good outcomes with surgical resection, though long-term endocrine follow-up is required.

Polymicrogyria

– A brain malformation characterized by an excessive number of abnormally small, fused gyri, resulting in an irregular, bumpy cortical surface with shallow sulci. Caused by abnormal late neuronal migration or post-migrational cortical organization. May be focal or diffuse, unilateral or bilateral, and symmetric or asymmetric. Presents with variable severity depending on extent: seizures (often focal epilepsy), developmental delay, motor deficits, and language impairment. Bilateral perisylvian polymicrogyria presents with pseudobulbar palsy, dysarthria, and cognitive impairment. Diagnosis by MRI (thin-section T1-weighted and T2-weighted imaging). No specific treatment; management is symptomatic.

Primary Lateral Sclerosis

– A rare neurodegenerative disorder confined to the upper motor neurons (corticospinal tract) of the motor cortex and brainstem. Presents with slowly progressive spasticity, hyperreflexia, Babinski sign, and pseudobulbar affect (emotional lability). Unlike ALS, there is no significant lower motor neuron involvement (no muscle atrophy or fasciculations) and survival is longer (10-15 years vs. 2-5 years for ALS). Diagnosis: clinical, EMG/NCS to exclude ALS mimics. No disease-modifying therapy; symptomatic management with baclofen, tizanidine, and speech therapy.

Progressive Supranuclear Palsy

– A neurodegenerative tauopathy characterized by the accumulation of tau protein in the basal ganglia, brainstem, and cerebral cortex, belonging to the family of Parkinson plus syndromes. The cardinal clinical features include progressive vertical supranuclear gaze palsy (particularly impairment of voluntary downward gaze, which is the hallmark feature), early postural instability with recurrent unexplained falls (typically backwards), axial rigidity greater than in the limbs (axial rigidity), bradykinesia, and early cognitive decline (including frontal lobe dysfunction with apathy, disinhibition, and executive impairment, which helps distinguish PSP from Parkinson disease). Other common features include dysphagia, dysarthria (hypokinetic or spastic), pseudobulbar affect (emotional lability with pathological laughing or crying), and eyelid opening apraxia (difficulty opening the eyes voluntarily). The classic PSP phenotype is Richardson syndrome (RS), but other variants include PSP-parkinsonism (PSP-P, resembling Parkinson disease with moderate levodopa response), PSP-progressive gait freezing (PSP-PGF), and PSP-corticobasal syndrome (PSP-CBS). The pathophysiology involves abnormal aggregation of the microtubule-associated protein tau, predominantly the 4-repeat (4R) tau isoform, in neurons and glia, forming globose neurofibrillary tangles and tufted astrocytes. The H1 haplotype of the MAPT gene is the major genetic risk factor. Diagnosis is clinical (MDS-PSP criteria) with supportive MRI findings including the hummingbird sign (midbrain atrophy on sagittal T1-weighted MRI showing a flat or concave superior profile of the midbrain resembling a hummingbird) and the Mickey Mouse sign (preserved pons with atrophied midbrain on axial MRI, giving the appearance of Mickey Mouse ears). Midbrain-to-pons area ratio of less than 0.2 is highly suggestive. There is no disease-modifying treatment; management is symptomatic with levodopa for parkinsonism (limited response, usually modest and transient), botulinum toxin for blepharospasm and eyelid apraxia, speech therapy, swallowing assessment, and fall prevention strategies. Prognosis is poor, with median survival of 6-9 years from symptom onset, with falls, dysphagia, and aspiration pneumonia as common causes of death.

Quadriplegia

– Paralysis of all four extremities, with or without loss of trunk, bladder, bowel, and sexual function, caused by damage to the cervical spinal cord. the most common causes include traumatic spinal cord injury (motor vehicle accidents, diving accidents, falls), cervical disc herniation, cervical spinal stenosis with myelopathy, spinal cord tumors, demyelinating diseases, and inflammatory myelopathies. the level of the cervical lesion determines the extent of functional impairment. c1-c4 lesions cause quadriplegia with respiratory compromise (diaphragmatic paralysis if c3-c5 is affected, requiring mechanical ventilation), and patients may need a phrenic nerve pacemaker. c5 lesions preserve shoulder abduction and elbow flexion. c6 lesions allow wrist extension. c7 lesions allow elbow extension and finger extension. c8-t1 lesions allow hand intrinsic function. incomplete spinal cord injury syndromes include central cord syndrome (most common incomplete syndrome, typically from hyperextension injury in cervical spondylosis, affecting the arms more than the legs due to the somatotopic organisation of the corticospinal tract), anterior cord syndrome (loss of motor function and pain/temperature sensation with preserved proprioception due to anterior spinal artery infarction), and brown-sequard syndrome (hemisection with ipsilateral motor loss and contralateral pain/temperature loss). management includes acute stabilisation, surgical decompression for compressive causes, rehabilitation, and lifelong multidisciplinary care to prevent complications such as pressure ulcers, urinary tract infections, autonomic dysreflexia, spasticity, contractures, and respiratory infections. prognosis depends on the level and completeness of injury, with complete cervical injuries having the poorest prognosis for functional recovery.

Restless Legs Syndrome (RLS) and Periodic Limb Movement Disorder

– Restless legs syndrome is a neurological disorder characterized by an uncontrollable urge to move the legs, often accompanied by uncomfortable sensations such as tingling, crawling, or aching, typically worsening during periods of rest or at night. Periodic limb movement disorder involves repetitive, involuntary jerking or twitching of the legs during sleep, which can fragment sleep and cause daytime fatigue. Both conditions are related and often occur together; treatment may include lifestyle modifications, iron supplementation if deficient, and medications such as dopamine agonists or gabapentin.

Rippling Muscle Disease

– A rare, benign myopathy characterized by mechanically-induced muscle contractions that spread across the muscle as a visible wave (rippling), typically triggered by stretch, percussion, or movement. Rippling muscle disease is caused by mutations in the CAV3 gene (encoding caveolin-3), inherited as an autosomal dominant trait with variable expressivity. Caveolin-3 is a structural protein in the sarcolemma membrane of skeletal muscle, involved in membrane stabilisation and signalling. Onset can occur at any age from childhood to adulthood. Clinical features: muscle rippling (percussion-induced or stretch-induced visible wave of muscle contraction, lasting 1–5 seconds), muscle mounding (percussion-induced localised muscle swelling, lasting up to 30 seconds), myoedema (localised dimpling on percussion), muscle stiffness/cramps after exercise, and mild proximal weakness in some patients. Serum creatine kinase is elevated (2–10 times normal). EMG shows no myotonic discharges (distinguishing from myotonic dystrophy and channelopathies). Muscle biopsy: normal or mild nonspecific changes; caveolin-3 immunostaining shows reduced sarcolemmal caveolin-3. Treatment: symptomatic—avoidance of triggering stimuli, mexiletine or carbamazepine may reduce muscle stiffness. Prognosis: benign, non-progressive, normal life expectancy. Consider genetic counseling.

Romberg Test

– The Romberg test is a clinical neurological test used to assess the integrity of the dorsal column-medial lemniscus pathway (proprioception) and the cerebellum in maintaining balance. The test is performed by asking the patient to stand with feet together and eyes open, and then to close the eyes. A positive Romberg test is indicated by loss of balance (increased sway or falling) when the eyes are closed, with the patient being stable with eyes open. A positive result indicates impaired proprioception (dorsal column dysfunction) causing sensory ataxia, with the patient able to maintain balance when visual input compensates for the loss of proprioception. A negative Romberg test (patient remains stable with eyes closed) suggests a cerebellar cause of ataxia rather than a sensory cause. The test should be performed with the patient standing on a firm, level surface with adequate lighting and within arm's reach of a wall or examiner to prevent falls. The patient should be allowed to stand with eyes open initially, and if stable, then asked to close the eyes. The test is considered positive if the patient sways or falls within approximately 20-30 seconds of eye closure. A positive Romberg test indicates dysfunction of the dorsal columns of the spinal cord or peripheral nerves that carry proprioceptive information. Causes include tabes dorsalis (neurosyphilis), vitamin B12 deficiency (subacute combined degeneration of the cord), diabetic peripheral neuropathy, Friedreich ataxia, and other causes of sensory neuropathy.

Schizencephaly

– A rare brain malformation characterized by a full-thickness, gray matter-lined cleft extending from the pial surface to the lateral ventricle. Caused by early focal neuronal migration abnormalities, often associated with prenatal vascular disruption or COL4A1 mutations. Closed-lip (Type I): the cleft walls are apposed. Open-lip (Type II): there is a CSF-filled space between the cleft walls. May be unilateral or bilateral. Presents with seizures, motor deficits (hemiparesis, spasticity), developmental delay, and hydrocephalus. Diagnosis by MRI. Treatment includes seizure management, physical/occupational therapy, and hydrocephalus management if needed.

Sciatica

– Pain radiating along the distribution of the sciatic nerve (L4-S3), from the lower back through the buttock and down the posterior thigh to the foot. Most commonly caused by lumbar disc herniation compressing the nerve root (L4-L5 or L5-S1). Other causes: spinal stenosis, spondylolisthesis, piriformis syndrome, and epidural abscess. Presents with sharp, burning pain, paresthesias, and sometimes motor weakness (foot drop). Positive straight leg raise test. Diagnosis: clinical; MRI if red flags (cauda equina, progressive weakness, trauma, infection). Management: conservative (NSAIDs, physical therapy, activity modification); epidural steroid injections; surgical discectomy for refractory or severe cases.

Seizure

– A transient occurrence of signs and/or symptoms due to abnormal, excessive, or synchronous neuronal activity in the brain. Provoked seizures have an identifiable acute cause (fever, electrolyte disturbance, drug toxicity, withdrawal). Unprovoked seizures occur without an acute trigger. Classification: focal onset, generalized onset, unknown onset. Evaluation: EEG, neuroimaging, laboratory studies. Management depends on etiology and recurrence risk.

Sleep

– A naturally recurring physiological state characterized by altered consciousness, reduced sensory responsiveness, and distinct brain wave patterns that is essential for physical restoration, memory consolidation, and metabolic regulation. It is divided into non-rapid eye movement sleep, which includes light and deep restorative stages, and rapid eye movement sleep, which is associated with dreaming and emotional processing. Chronic sleep deprivation is linked to increased risks of obesity, cardiovascular disease, cognitive impairment, and weakened immune function, underscoring the critical role of adequate sleep in overall health.

Sleep Studies (Polysomnography)

– Sleep tests, most commonly polysomnography, are diagnostic studies used to evaluate sleep architecture and identify sleep disorders. Polysomnography records brain waves, eye movements, heart rate, breathing patterns, blood oxygen levels, and limb movements throughout the night in a controlled laboratory setting. Home sleep apnea tests are simpler devices that primarily measure breathing and oxygen saturation to screen for obstructive sleep apnea, while multiple sleep latency tests assess daytime sleepiness by measuring how quickly a person falls asleep in a quiet environment.

Spasm

– A sudden, involuntary, sustained or repetitive contraction of a muscle or group of muscles. Types: (1) Skeletal muscle spasm—muscle cramps (common, benign—nocturnal leg cramps, exercise-associated), muscle strain, tetany (hypocalcaemia—perioral paraesthesias, carpopedal spasm, Trousseau sign, Chvostek sign), tetanus (generalised rigidity, risus sardonicus, opisthotonos), dystonia (sustained muscle contractions causing twisting/repetitive movements—cervical dystonia, blepharospasm, writer's cramp), and myoclonus (brief, shock-like jerks). (2) Smooth muscle spasm—bronchospasm (asthma, COPD exacerbation), esophageal spasm (chest pain, dysphagia), ureteric colic (kidney stone—severe flank pain radiating to groin), and biliary colic (gallstone obstruction of cystic duct). Treatment: directed at the underlying cause; muscle relaxants (baclofen, tizanidine, benzodiazepines) for skeletal muscle spasm; antispasmodics (hyoscine butylbromide/Buscopan) for smooth muscle spasm; calcium channel blockers for esophageal spasm; analgesics and NSAIDs for ureteric colic.

Spina Bifida Meningocele

– A form of spina bifida where the meninges (but not neural tissue) herniate through a vertebral defect, forming a CSF-filled sac. The spinal cord remains in its normal position and is not involved. Presents as a visible, fluid-filled, translucent sac on the back, typically without neurologic deficits. Surgical repair involves reduction and closure. Prognosis is excellent with normal neurologic function expected.

Spina Bifida Myelomeningocele

– The most severe form of spina bifida, where both the meninges and neural tissue (spinal cord and nerve roots) herniate through a vertebral defect. Presents as a visible, flat or sac-like lesion on the back with neurologic deficits below the level of the lesion (paralysis, sensory loss, bowel/bladder dysfunction). Associated with hydrocephalus (in 80-90% of cases) and Arnold-Chiari malformation type II. Diagnosis by prenatal ultrasound and elevated maternal serum alpha-fetoprotein. Treatment: surgical closure within 24-48 hours of birth, ventriculoperitoneal shunt for hydrocephalus, and lifelong multidisciplinary care.

Spinal Shock

– A state of transient physiological areflexia and flaccid paralysis occurring immediately after acute spinal cord injury, below the level of the lesion. It is characterized by loss of all somatic and autonomic nervous system function, including motor paralysis, sensory loss, areflexia, hypotension (from loss of sympathetic tone), bradycardia (from unopposed vagal tone), and loss of bladder and bowel function. The spinal shock phase typically begins immediately after injury and lasts days to several weeks, with gradual recovery of reflexes and autonomic function as spinal cord edema resolves and neuronal function returns. The duration of spinal shock is longer in more severe injuries. The development of hyperreflexia and spasticity marks the transition from spinal shock to the chronic phase of spinal cord injury. Management during the spinal shock phase includes aggressive hemodynamic support (maintaining mean arterial pressure >85 mmHg with IV fluids and vasopressors to prevent secondary ischemic injury to the spinal cord), immobilisation of the spine, and early surgical decompression for correctable compressive causes. The return of the bulbocavernosus reflex is the earliest sign of the end of spinal shock and is used as a clinical milestone.

Spinocerebellar Ataxias

– A group of autosomal dominant neurodegenerative disorders characterized by progressive cerebellar ataxia, often with additional neurological features. Over forty different SCA types have been identified, with SCA1, SCA2, SCA3 (Machado-Joseph disease, the most common worldwide), SCA6, and SCA7 being the most prevalent. Most SCAs are caused by CAG trinucleotide repeat expansions encoding polyglutamine (polyQ) tracts in the affected proteins, similar to Huntington disease, with anticipation (earlier onset in successive generations) observed in some SCA types due to repeat instability, especially with paternal transmission. Clinical features include progressive gait and limb ataxia, dysarthria, and eye movement abnormalities (nystagmus, saccadic intrusions). Additional features vary by SCA type: SCA1 includes pyramidal signs, peripheral neuropathy, and dysphagia; SCA2 includes slow saccades, peripheral neuropathy, and cognitive decline; SCA3 (Machado-Joseph disease) includes dystonia, parkinsonism, and spasticity in addition to ataxia; SCA6 is a pure cerebellar ataxia with later onset; and SCA7 includes retinal degeneration causing progressive vision loss. Diagnosis is based on clinical presentation, family history (autosomal dominant inheritance), and genetic testing identifying the specific CAG repeat expansion in the relevant gene. MRI brain shows progressive cerebellar atrophy, particularly of the vermis and cerebellar hemispheres, with brainstem atrophy in some types. There is no disease-modifying treatment; management is symptomatic and supportive, including physical and occupational therapy, speech therapy, and treatment of specific complications. Prognosis is variable but most SCAs lead to severe disability and reduced lifespan over 10-30 years.

Stroke

– A medical emergency caused by the interruption of blood flow to the brain, resulting in neuronal death and neurological deficits. Ischemic stroke, accounting for approximately 87 percent of cases, occurs when a thrombus or embolus blocks a cerebral artery, while hemorrhagic stroke results from rupture of a blood vessel causing bleeding into or around the brain. Rapid recognition using the FAST acronym and immediate medical intervention with thrombolytic therapy or endovascular thrombectomy can significantly reduce disability and mortality, followed by rehabilitation and secondary prevention with antiplatelet agents, anticoagulation, and risk factor modification.

Stupor

– A state of reduced consciousness in which the patient is unresponsive to most environmental stimuli but can be temporarily aroused by vigorous, repeated stimulation (painful stimuli, loud shouting). Stupor is distinguished from coma (no response to any stimuli, no arousal), obtundation (reduced alertness but responds to verbal stimuli), and delirium (impaired attention and cognition with fluctuating consciousness). Causes: structural brain lesions (brainstem stroke—basilar artery occlusion, thalamic stroke, intracerebral hemorrhage, subdural haematoma, brain tumor, cerebral abscess), metabolic (hypoglycaemia, hyperosmolar hyperglycaemic state, hepatic encephalopathy, uraemia, hyponatraemia, hypercalcaemia, hypothyroidism, Wernicke encephalopathy), drug intoxication (opioids, benzodiazepines, barbiturates, alcohol, antipsychotics, antidepressants, carbon monoxide poisoning), CNS infection (meningitis, encephalitis, cerebral malaria), psychiatric (catatonia, dissociative stupor—rare). Assessment: ABCs, Glasgow Coma Scale (GCS <8 indicates severe brain injury, consider airway protection), pupillary response, eye movements (doll's eye, oculocephalic reflex, cold caloric testing), motor responses, and deep tendon reflexes. Emergent CT head, blood glucose, electrolytes, toxicology screen, ABG, and lumbar puncture if indicated. Management: stabilise airway, breathing, circulation; administer naloxone if opioid overdose suspected, thiamine (before glucose) for possible Wernicke, and 50% dextrose for hypoglycaemia. Treat underlying cause.

Subarachnoid Hemorrhage

– Bleeding into the subarachnoid space, the area between the arachnoid membrane and the pia mater surrounding the brain, most commonly caused by rupture of a cerebral (berry) aneurysm. Approximately eighty-five percent of non-traumatic SAH cases are caused by aneurysm rupture, with the remaining cases due to perimesencephalic (benign) SAH, arteriovenous malformations, mycotic aneurysms, and other vascular pathologies. The classic presentation is thunderclap headache—the sudden onset of the worst headache of the patient's life, reaching maximal intensity within seconds to minutes. Associated symptoms include nausea and vomiting, photophobia, nuchal rigidity (meningismus from blood in the subarachnoid space), loss of consciousness, seizures, and focal neurological deficits. The Ottawa SAH Rule and the classic presentation guide the need for urgent investigation. Diagnosis begins with a non-contrast CT head, which has 100 percent sensitivity for SAH within the first 6 hours and over 93 percent sensitivity within the first 24 hours; sensitivity declines to approximately 50 percent after one week. If CT is normal but clinical suspicion is high, a lumbar puncture is performed to detect CSF xanthochromia (yellow discolouration from bilirubin breakdown products), which is the gold standard for confirming SAH. CT angiography (CTA) or digital subtraction angiography (DSA) is used to identify the causative aneurysm. Management includes initial stabilisation in a neurosurgical critical care unit, securing the aneurysm via endovascular coiling or surgical clipping (to prevent rebleeding), management of delayed cerebral ischemia (vasospasm, which peaks at days 7-10 post-SAH, treated with nimodipine and hypertensive therapy), and management of complications including hydrocephalus (requiring external ventricular drain or VP shunt), rebleeding, and electrolyte disturbances. Prognosis is guarded, with mortality of approximately 50 percent and significant disability among survivors.

Subdural Hematoma

– A collection of blood between the dura mater and the arachnoid mater, caused by tearing of bridging veins that traverse the subdural space. It is classified as acute (symptoms within hours to three days), subacute (symptoms within three days to three weeks), or chronic (symptoms after three weeks), depending on the rate of bleeding and time from injury. Acute subdural hematoma is most commonly caused by severe head trauma, particularly in the elderly and alcoholics, and carries a high mortality rate (30-50 percent). Subacute and chronic subdural haematomas are more common in the elderly, alcoholics, epileptics, and patients on anticoagulation, where even minor or unrecognised head trauma can cause bleeding. Chronic subdural haematoma features a liquefied collection surrounded by an inflammatory neomembrane, which can rebleed and progressively enlarge, causing subacute or slowly progressive neurological deterioration. Clinical presentation varies: acute SDH presents with decreased consciousness, focal neurological deficit (hemiparesis), and ipsilateral pupillary dilation (from uncal herniation). Chronic SDH presents with headache, cognitive decline, gait disturbance, and personality changes, mimicking dementia or normal pressure hydrocephalus. Diagnosis is by CT head without contrast: acute SDH appears as a hyperdense, crescent-shaped collection along the convexity of the brain that does not cross the midline but can cross suture lines. Chronic SDH appears as a hypodense crescentic collection. Management: large acute SDH with mass effect requires emergency surgical evacuation via craniotomy or burr hole drainage. Chronic SDH is managed with burr hole drainage under local anesthesia or conservative observation for small asymptomatic collections. Anticoagulation reversal is critical in all cases. Prognosis: acute SDH has 30-50 percent mortality, while chronic SDH has good prognosis with treatment, though recurrence after burr hole drainage is approximately 10-20 percent.

Syringomyelia

– A condition characterized by the formation of a fluid-filled cyst (syrinx) within the spinal cord, which can expand and elongate over time, compressing and destroying the surrounding spinal cord tissue. The most common cause is Chiari malformation type I (see separate entry), where cerebellar tonsillar ectopia obstructs CSF flow at the foramen magnum, leading to syrinx formation. Other causes include spinal cord trauma, spinal cord tumors, arachnoiditis, and spinal dysraphism. The syrinx typically expands within the central canal of the spinal cord (dilating it) in a segmental pattern, most commonly in the cervical region, and can extend rostrally into the brainstem (syringobulbia) or caudally down the entire cord. Clinical presentation depends on the location and extent of the syrinx. Typical symptoms include a capelike distribution of dissociated sensory loss (loss of pain and temperature sensation with preserved touch and vibration sensation) over the shoulders and upper back (due to crossing spinothalamic fibers being affected first in the central cord), segmental weakness and atrophy in the arms and hands (from damage to anterior horn cells), and in advanced cases, spastic paraparesis, bladder dysfunction, and Horner syndrome (if the syrinx expands laterally). Pain is common, often described as a deep, aching, or burning sensation in the neck, shoulders, and arms. Diagnosis is by MRI of the spinal cord, which demonstrates a CSF-filled cavity within the spinal cord and any associated Chiari malformation or other pathology. Management is directed at the underlying cause: for Chiari-related syringomyelia, posterior fossa decompression (suboccipital craniectomy and cervical laminectomy with duraplasty) is the treatment of choice. For post-traumatic syringomyelia, the treatment is syrinx shunting or untethering of the spinal cord. Patients with stable, asymptomatic syringomyelia can be managed conservatively with serial MRI monitoring. The natural history is variable: some syringes remain stable while others progress insidiously over years, leading to increasing neurological deficit.

Syrinx

– A syrinx is a fluid-filled cavity within the spinal cord, typically containing cerebrospinal fluid or a similar fluid, that develops as a result of abnormal CSF flow dynamics within the spinal cord. Syringomyelia: a syrinx in the central spinal cord. Hydromyelia: dilation of the central canal. In practice, the terms are often used interchangeably. The most common cause of syrinx formation is Chiari malformation type I, where cerebellar tonsillar ectopia obstructs CSF flow at the foramen magnum, causing abnormal CSF pressure dynamics within the central canal of the spinal cord, leading to cyst formation. Other causes include spinal cord trauma, spinal cord tumors (especially intramedullary tumors such as ependymomas and haemangioblastomas, which can cause syrinx formation through altered CSF flow dynamics), and arachnoiditis (inflammation of the arachnoid mater due to infection, hemorrhage, or previous surgery). The syrinx may be asymptomatic if small, or may cause progressive myelopathy depending on its size and location. Symptoms include central cord syndrome with dissociated sensory loss (loss of pain and temperature with preserved touch and vibration in a cape-like distribution over the shoulders and upper back), segmental weakness and atrophy (from anterior horn cell damage), spastic paraparesis (from corticospinal tract involvement), and neuropathic pain. Diagnosis is by MRI of the spinal cord, which shows the CSF-filled cavity. Management depends on the cause and symptoms: asymptomatic syringes may be observed with serial imaging, while symptomatic syringes may require surgical treatment including posterior fossa decompression for Chiari malformations, cyst drainage or shunting, or tumor resection. Prognosis is variable; timely treatment of the underlying cause may stabilise or improve symptoms, but neurological deficits may be irreversible if the syrinx has caused significant cord damage.

Temporal Lobe

– One of the four major lobes of the cerebral hemisphere, located beneath the lateral fissure (Sylvian fissure) on the lateral surface of the brain, extending medially to the parahippocampal gyrus. The temporal lobe is involved in auditory processing, language comprehension, memory, and emotion. Key structures: primary auditory cortex (Heschl gyrus, Brodmann area 41/42—superior temporal gyrus), Wernicke area (posterior superior temporal gyrus—language comprehension, dominant hemisphere; lesions cause receptive aphasia/Wernicke aphasia), medial temporal lobe structures (hippocampus—memory formation, consolidation, spatial navigation; amygdala—emotion processing, fear, aggression; parahippocampal gyrus—memory and visuospatial processing). Functions: auditory perception, phonemic and semantic processing (language), declarative memory (episodic and semantic—hippocampus-dependent), facial recognition (fusiform gyrus—inferior temporal), and emotional regulation (amygdala). Pathology: temporal lobe epilepsy (most common focal epilepsy), temporal lobe tumors (glioblastoma, meningioma), herpes simplex encephalitis (hemorrhagic necrosis of the medial temporal lobes), dementia (Alzheimer disease—hippocampal atrophy is an early feature), and traumatic brain injury (temporal lobe contusions are common).

Temporal Lobe Epilepsy

– The most common form of focal (partial) epilepsy, arising from the medial temporal lobe structures (hippocampus, amygdala, parahippocampal gyrus). Cause: hippocampal sclerosis (most common—neuronal loss and gliosis in the CA1 and CA3 regions of the hippocampus, associated with febrile seizures in childhood), tumors (low-grade gliomas, DNET), cortical dysplasia, vascular malformations, infections (herpes simplex encephalitis, neurocysticercosis), and trauma. Seizures: focal impaired awareness seizures (complex partial seizures)—characterized by an aura (epigastric rising sensation, déjà vu, jamais vu, olfactory/gustatory hallucinations, fear), followed by behavioural arrest, automatisms (lip smacking, chewing, fumbling), and postictal confusion/amnesia. Duration: 30 seconds to 2 minutes. Seizures may secondarily generalise to bilateral tonic-clonic seizures. Interictal EEG: anterior temporal sharp waves or spikes. MRI: high-resolution coronal T2/FLAIR images show hippocampal atrophy and increased T2 signal (hippocampal sclerosis). Treatment: antiseizure medications (first-line: lamotrigine, levetiracetam, carbamazepine, oxcarbazepine). Drug-resistant epilepsy (30%): anterior temporal lobectomy or selective amygdalohippocampectomy—seizure-free outcome in 60–80%. Vagus nerve stimulation and responsive neurostimulation are alternatives.

Tethered Cord Syndrome

– A congenital or acquired condition where the spinal cord is abnormally attached (tethered) to surrounding tissues, restricting its normal upward movement within the spinal canal. Caused by a thickened filum terminale, lipomeningomyelocele, diastematomyelia, or previous myelomeningocele repair. Presents with back pain, leg weakness, sensory loss, gait abnormalities, urinary incontinence, and foot deformities. Symptoms may worsen with growth, flexion, or trauma. Diagnosis by MRI showing a low-lying conus medullaris (below L2-L3). Treatment: surgical detethering to prevent progressive neurologic deterioration.

Tetraplegia (Quadriplegia)

– Paralysis of all four limbs and the trunk resulting from injury to the cervical spinal cord (C1–C8/T1). Causes: trauma (motor vehicle accidents—most common, falls, sports injuries, violence), ischemia (spinal cord infarction), hemorrhage, infection (epidural abscess, transverse myelitis), demyelination (multiple sclerosis), and tumors. The level of injury determines functional impairment: C1–C3 (requires mechanical ventilation, minimal movement), C4 (phrenic nerve intact—diaphragmatic breathing, but requires 24-hour care, can use chin- or mouth-controlled wheelchair), C5 (deltoid and biceps—can feed self with adaptive equipment, assist with transfers), C6 (wrist extension—tenodesis grasp, can self-propel wheelchair with adaptations), C7–C8 (triceps, finger flexors/extensors—independent transfers, self-catheterisation). The American Spinal Injury Association (ASIA) Impairment Scale grades completeness (A = complete, E = normal). Complications: respiratory compromise, autonomic dysreflexia (hypertensive crisis from noxious stimuli below the level of injury), neurogenic shock, pressure ulcers, spasticity, neuropathic pain, deep vein thrombosis, osteoporosis, and psychological adjustment. Management: acute—spinal immobilisation, high-dose methylprednisolone (controversial), surgical decompression/stabilisation. Chronic—multidisciplinary rehabilitation (physiotherapy, occupational therapy, speech therapy, psychological support), management of spasticity (baclofen, botulinum toxin), bowel and bladder programme, and assistive technology.

Thrombolysis for Stroke

– Thrombolysis with intravenous alteplase (tPA) is the primary acute treatment for ischemic stroke when administered within the time window of symptom onset. Alteplase is a recombinant tissue plasminogen activator that converts plasminogen to plasmin, which degrades fibrin in the thrombus, restoring blood flow to the ischemic brain. The standard dose is 0.9 milligrams per kilogram (maximum 90 milligrams), with ten percent given as an intravenous bolus over one minute and the remainder infused over one hour. The therapeutic window is 0-4.5 hours from symptom onset for intravenous thrombolysis, with stricter criteria applied after 3 hours (age over 80, NIHSS >25, prior stroke plus diabetes, and anticoagulant use becoming relative contraindications). Treatment within 3 hours has the best benefit-risk profile (number needed to treat of 8 for improved outcome). Exclusion criteria include intracranial hemorrhage, recent major surgery, gastrointestinal or urinary tract bleeding, recent head trauma, uncontrolled hypertension (systolic >185 mmHg, diastolic >110 mmHg), current anticoagulation with INR >1.7, thrombocytopenia (platelets <100,000), blood glucose <2.8 or >22.2 mmol/L, and clinical presentation suggestive of subarachnoid hemorrhage even if CT is normal. The risk of symptomatic intracranial hemorrhage is approximately 6-7 percent within the first 36 hours, which limits the net benefit in patients treated later in the window. Endovascular thrombectomy with stent retrievers is indicated for large vessel occlusion (internal carotid artery, M1 segment of middle cerebral artery, basilar artery) within 6 hours of onset, and may be extended to 24 hours in selected patients based on perfusion imaging (DAWN and DEFUSE-3 trial criteria).

Tic Douloureux

– See Trigeminal Neuralgia.

Transient Ischemic Attack

– A transient ischemic attack (TIA) is a temporary episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia without acute infarction. Historically defined by an arbitrary time threshold of twenty-four hours, the modern tissue-based definition recognizes TIA as a brief episode of neurological symptoms caused by focal brain ischemia that resolves without evidence of acute infarction on neuroimaging (typically diffusion-weighted MRI). Most TIAs resolve within one hour, with the majority resolving within 10-60 minutes. The clinical presentation depends on the affected vascular territory and can include transient hemiparesis, hemisensory loss, aphasia, visual field defects, diplopia, vertigo, dysarthria, ataxia, and amaurosis fugax (transient monocular vision loss, the classic retinal TIA, which is a harbinger of ipsilateral carotid artery stenosis). The ABCD2 score is used to stratify the short-term risk of stroke after TIA: Age over 60 (1 point), Blood pressure >140/90 (1 point), Clinical features (unilateral weakness 2 points, speech impairment without weakness 1 point), Duration (10-59 minutes 1 point, 60+ minutes 2 points), and Diabetes (1 point). Patients with ABCD2 scores of 4 or higher or with cerebral ischemia on MRI (DWI-positive) are at higher risk and typically require urgent evaluation. The urgency of evaluation and treatment is critical because the risk of stroke is highest in the first 48 hours after a TIA, with approximately 10 percent of patients having a stroke within 90 days. Emergency management includes urgent neuroimaging (MRI brain with DWI, MRA of the head and neck), ECG and cardiac monitoring to detect atrial fibrillation, carotid Doppler ultrasound for significant carotid stenosis, and laboratory investigations for vascular risk factors. Secondary prevention includes antiplatelet therapy (aspirin plus clopidogrel for 21 days in high-risk patients, followed by single antiplatelet therapy), statin therapy (high-intensity), blood pressure control, smoking cessation, dietary modification, and carotid endarterectomy or stenting for significant ipsilateral carotid artery stenosis (70-99 percent). Anticoagulation is indicated for TIA associated with atrial fibrillation.

Transverse Myelitis

– An inflammatory disorder of the spinal cord causing acute or subacute motor, sensory, and autonomic dysfunction. It results from immune-mediated inflammation of the spinal cord parenchyma and can occur as a isolated condition or as part of a systemic autoimmune disease or demyelinating disorder such as multiple sclerosis or neuromyelitis optica. Clinical features include bilateral lower extremity weakness, a sensory level (a dermatome below which sensation is impaired), urinary retention (more common than incontinence in the acute phase, progressing to bladder and bowel incontinence), and a sensory level (a transverse band-like boundary below which there is loss of pain, temperature, and light touch sensation). The sensory level is typically in the mid-thoracic region, as the thoracic spinal cord is the longest and most vulnerable segment. Fever or a recent history of infection or vaccination may precede the onset. If the inflammation occurs as part of multiple sclerosis, it is typically shorter (less than one vertebral segment) and more asymmetrical, whereas NMOSD-related transverse myelitis is typically longitudinally extensive (three or more vertebral segments). Diagnosis is by MRI of the spinal cord, which shows a central T2-hyperintense intramedullary lesion with gadolinium enhancement in the acute phase. CSF analysis typically shows a lymphocytic pleocytosis (elevated white cell count) and elevated protein, with positive oligoclonal bands if associated with multiple sclerosis. Treatment of acute attacks includes high-dose intravenous corticosteroids (methylprednisolone 1 gram daily for 5 days), with plasma exchange for steroid-resistant cases. Long-term treatment depends on the underlying cause: immunosuppression for NMOSD, disease-modifying therapy for MS, and observation for monophasic idiopathic cases. Prognosis is variable: approximately one-third of patients recover fully, one-third have moderate residual disability, and one-third have severe disability.

Trigeminal Neuralgia

– A chronic pain condition characterized by recurrent, sudden, severe, electric shock-like pain in the distribution of the trigeminal nerve (typically V2/V3). Often triggered by light touch, chewing, or brushing teeth. Most cases are caused by vascular compression of the trigeminal nerve root. First-line treatment: carbamazepine or oxcarbazepine. Surgical options: microvascular decompression, gamma knife radiosurgery, percutaneous rhizotomy.

Tuberous Sclerosis Complex

– An autosomal dominant genetic disorder caused by mutations in TSC1 (hamartin) or TSC2 (tuberin) genes, leading to mTOR pathway activation. Features: cortical tubers, subependymal nodules, cardiac rhabdomyomas, angiomyolipomas, facial angiofibromas, and seizures (often infantile spasms). Intellectual disability and autism spectrum disorder are common. Treatment: mTOR inhibitors (everolimus) for angiomyolipomas and subependymal giant cell astrocytomas.

Vascular Dementia

– The second most common cause of dementia after Alzheimer disease and results from cerebrovascular disease causing sufficient brain injury to impair cognitive function. It is caused by large vessel cerebrovascular disease (multi-infarct dementia from multiple cortical and subcortical infarcts), small vessel cerebrovascular disease (subcortical ischemic vascular dementia from lacunar infarcts and white matter changes), strategic single infarct (infarction in a cognitively critical area such as the dominant thalamus, angular gyrus, or medial temporal lobe), or global hypoperfusion injury from cardiac arrest or hypotension causing watershed infarcts between vascular territories. Clinical presentation of vascular dementia is characterized by a stepwise deterioration and a patchy pattern of cognitive deficits, in contrast to the gradual progressive decline of Alzheimer disease. The cognitive profile includes executive dysfunction (impaired planning, organisation, and abstraction) more than memory loss, slowness of processing (bradyphrenia), and subcortical features (apathy, emotional lability, depression). Focal neurological signs (hemiparesis, sensory loss, visual field deficits, pseudobulbar palsy, gait disturbance) are common. The Hachinski ischemic Score helps differentiate vascular dementia from Alzheimer disease. Diagnosis is by clinical assessment, cognitive testing, and neuroimaging (CT or MRI) showing multiple infarcts, extensive white matter disease (leukoaraiosis), or a strategically located infarct. Management includes control of vascular risk factors (hypertension, diabetes, hyperlipidaemia, smoking cessation, antiplatelet therapy), and symptomatic treatment of cognitive and behavioural symptoms. Acetylcholinesterase inhibitors have limited benefit in vascular dementia, unlike their use in Alzheimer disease. Prognosis is variable and depends on the progression of underlying cerebrovascular disease.

Vertebrobasilar Insufficiency

– A condition of reduced blood flow through the vertebral arteries or basilar artery to the posterior circulation of the brain, including the brainstem, cerebellum, occipital lobes, and thalami. It is most commonly caused by atherosclerosis of the vertebral arteries or basilar artery, with risk factors including hypertension, diabetes, smoking, hyperlipidemia, and advanced age. The posterior circulation supplies vital structures including the brainstem (containing cranial nerve nuclei and vital autonomic centers), the cerebellum (controlling coordination and balance), the occipital lobes (vision), and the thalami (sensory and motor relay). Clinical presentation of VBI includes dizziness and vertigo (the most common symptom), diplopia, dysarthria, dysphagia, ataxia, bilateral weakness or sensory loss, drop attacks (sudden loss of postural tone without loss of consciousness), visual disturbances (blurred vision, visual field defects, cortical blindness), and tinnitus. The diagnosis of VBI is distinguished from benign paroxysmal positional vertigo (which has positional triggers), Meniere disease (which has episodic vertigo with hearing loss and tinnitus), and migraine-associated vertigo. Diagnosis is by clinical assessment of posterior circulation symptoms and signs, with imaging including MRA or CTA of the vertebral and basilar arteries to assess for stenosis, occlusion, or dissection. Ultrasound of the vertebral arteries (extracranial segment) and transcranial Doppler of the basilar artery may demonstrate flow reduction. Management includes aggressive risk factor modification, antiplatelet therapy (aspirin or clopidogrel), and revascularisation (vertebral artery angioplasty and stenting, or vertebral artery endarterectomy) for selected patients with symptomatic severe stenosis refractory to medical therapy. Anticoagulation is indicated for embolism from the heart or proximal vertebral artery. Prognosis is poorer than for anterior circulation disease, with a higher risk of disabling or fatal stroke if large vessel disease is present.

Vestibular Neuritis

– Acute vestibular dysfunction due to inflammation of the vestibular nerve, typically viral in origin. Presents with acute onset of vertigo, nausea, vomiting, and nystagmus lasting days. Hearing is normal (unlike labyrinthitis). Diagnosis: clinical, HINTS exam (Head Impulse, Nystagmus, Test of Skew) to differentiate from central causes. Treatment: corticosteroids if early, vestibular suppressants acutely, and vestibular rehabilitation.

Vestibular Schwannoma

– Vestibular schwannoma, also known as acoustic neuroma, is a benign, slow-growing tumor arising from the Schwann cells of the vestibular portion of the eighth cranial nerve (vestibulocochlear nerve), typically from the superior vestibular nerve. It is the most common tumor of the cerebellopontine angle, accounting for approximately eighty percent of tumors in this location. The most common presenting symptom is unilateral sensorineural hearing loss, which develops gradually over months to years and is often accompanied by tinnitus, vertigo or imbalance, and a sensation of fullness in the ear. Larger tumors compress the trigeminal nerve (causing facial numbness and paraesthesiae), the facial nerve (causing facial weakness or spasm), and the brainstem and cerebellum (causing ataxia, long tract signs, and hydrocephalus from compression of the fourth ventricle). The diagnosis is confirmed by MRI of the internal auditory meatus and brain with gadolinium showing an enhancing mass in the cerebellopontine angle that extends into and expands the internal auditory canal. The differential diagnosis includes meningioma, facial nerve schwannoma, epidermoid cyst, and trigeminal schwannoma. Audiometry shows unilateral sensorineural hearing loss with poor speech discrimination. Auditory brainstem response testing shows delayed latencies. Management options include observation with serial imaging for small tumors in elderly or medically unfit patients, fractionated stereotactic radiosurgery (Gamma Knife or linear accelerator) for small to medium tumors (less than 3 cm), and microsurgical resection via a translabyrinthine, retrosigmoid, or middle cranial fossa approach for large tumors causing significant mass effect or brainstem compression. The goal of radiosurgery is tumor control (prevention of growth) rather than complete elimination. Hearing preservation is possible in selected cases with small tumors and good pre-treatment hearing, but the likelihood of hearing preservation decreases with tumor size.

Wilson Disease

– An autosomal recessive inherited disorder of copper metabolism caused by mutations in the ATP7B gene on chromosome 13, which encodes a copper-transporting ATPase in hepatocytes. The defective ATP7B protein impairs incorporation of copper into apoceruloplasmin and biliary excretion of copper, leading to progressive copper accumulation in the liver, brain, cornea, and other organs. Clinical manifestations typically present between ages five and thirty-five and include hepatic disease (ranging from asymptomatic elevation of liver enzymes to acute hepatitis, cirrhosis, and fulminant hepatic failure), neurological disease (dysarthria, ataxia, dystonia, parkinsonism, tremor—particularly a wing-beating tremor of the arms, chorea, and cognitive impairment with executive dysfunction), psychiatric symptoms (depression, personality changes, psychosis), and Kayser-Fleischer rings (golden-brown discolouration at the limbus of the cornea due to copper deposition in Descemet membrane, visible on slit-lamp examination and pathognomonic for Wilson disease when present). Diagnosis is based on a combination of clinical features, Kayser-Fleischer rings on slit-lamp examination, low serum caeruloplasmin (less than 0.2 g/L), elevated 24-hour urinary copper excretion (greater than 100 micrograms), elevated hepatic copper content on liver biopsy, and genetic testing for ATP7B mutations. Serum free copper (non-caeruloplasmin-bound copper) is elevated. Treatment involves lifelong copper chelation therapy with D-penicillamine (first-line, acts by chelating copper and promoting urinary excretion) or trientine, zinc acetate (which blocks intestinal absorption of copper and induces hepatic metallothionein, used as maintenance therapy or first-line for presymptomatic patients), and dietary restriction of copper-rich foods (shellfish, nuts, chocolate, mushrooms, organ meats). Liver transplantation is curative for severe hepatic failure or refractory neurological disease. Prognosis is excellent with early diagnosis and lifelong treatment; without treatment, Wilson disease is progressive and fatal. Patients who stop chelation therapy are at risk of irreversible neurological deterioration and hepatic decompensation.

Chapter 21

Obstetrics and Gynecology

Abdominal Cerclage

– A cerclage placed abdominally (laparoscopically or open) at the cervicouterine junction, typically for patients with failed transvaginal cerclage, extremely short or absent cervix, or congenital cervical anomaly. Left in place permanently; future deliveries by cesarean section. Robotic-assisted laparoscopic approach is becoming more common (reduced morbidity compared to open laparotomy). Indications: prior failed McDonald/Shirodkar cerclage, deeply notched or absent cervix, and cervical agenesis. Pregnancy loss rate: reduced from 80-90% to 10-15%.

Abdominal Circumference (AC)

– The circumference of the fetal abdomen measured at the level of the stomach, umbilical vein (portal sinus), and spine (ribs not visible). The best single parameter for estimating fetal weight. AC is also the most sensitive measurement for detecting IUGR (the liver is the first organ affected by placental insufficiency). AC growth: 1-1.5 cm per week. A declining AC percentile (or growth arrest) strongly suggests placental insufficiency.

Abdominal Pregnancy

– A very rare ectopic pregnancy (1%) implanted in the peritoneal cavity outside the fallopian tubes, ovaries, and uterus. Risk factors: prior ectopic, tubal damage, ART. Diagnosis: ultrasound (empty uterus, gestational sac outside the uterus, fetal parts adjacent to maternal organs). High maternal mortality (hemorrhage from placental separation). Treatment: surgical removal. The placenta may be left in situ (risks of infection, hemorrhage) if adherence to vital organs prevents safe removal.

Abnormal Uterine Bleeding

– Uterine bleeding that is abnormal in frequency, regularity, duration, or volume, occurring outside of normal menstruation or in women who are not pregnant. The PALM-COEIN classification system categorizes causes into structural (PALM: polyps, adenomyosis, leiomyomas, malignancy/hyperplasia) and non-structural (COEIN: coagulopathy, ovulatory dysfunction, endometrial, iatrogenic, not otherwise classified). Ovulatory dysfunction is the most common cause in reproductive-age women, resulting from anovulation due to polycystic ovary syndrome, thyroid disorders, hyperprolactinemia, hypothalamic dysfunction, or perimenopause. Structural causes include endometrial polyps (benign overgrowths of endometrial glands and stroma), uterine leiomyomas (fibroids, benign smooth muscle tumors), adenomyosis (endometrial tissue within the myometrium), and endometrial hyperplasia or carcinoma (particularly in postmenopausal women).

Abortion

– The termination of a pregnancy by medical or surgical means before the fetus reaches viability (typically before 24 weeks gestation in the UK). Medical abortion uses mifepristone (antiprogestogen) followed by misoprostol (prostaglandin analog) to induce uterine contractions and expel pregnancy tissue. Surgical abortion involves vacuum aspiration (suction evacuation) or dilatation and evacuation (D&E). In the UK, abortion is regulated under the Abortion Act 1967, requiring two doctors to certify that the grounds for abortion are met. Complications are rare: infection, hemorrhage, uterine perforation, and incomplete evacuation. Psychological outcomes are generally positive with appropriate support.

Abruptio Placentae

– Premature separation of the placenta from the uterine wall before delivery, occurring in approximately 0.5-1% of pregnancies. Classified as revealed (vaginal bleeding present, blood tracks downward) or concealed (bleeding remains behind the placenta, no visible bleeding, higher risk of coagulopathy). Risk factors include hypertension, preeclampsia, maternal smoking, cocaine use, abdominal trauma, thrombophilia, prior abruption, and advanced maternal age. Clinical features include vaginal bleeding (may be absent in concealed abruption), abdominal pain, uterine tenderness and hypertonicity (board-like abdomen), and fetal distress or demise. Complications include maternal hemorrhage, disseminated intravascular coagulation (DIC) from release of thromboplastin into the maternal circulation, fetal hypoxia, preterm delivery, and perinatal death. Diagnosis is primarily clinical; ultrasound has limited sensitivity for detecting retroplacental clot. Management depends on gestational age and severity: immediate delivery (usually by caesarean) is indicated for maternal or fetal compromise irrespective of gestational age, with aggressive blood product replacement and coagulopathy management.

Active Management of Third Stage

– A bundle of interventions to reduce the risk of postpartum hemorrhage: (1) administration of oxytocin (10 IU IM/IV) immediately after delivery of the baby, (2) controlled cord traction (Brandt-Andrews maneuver) to deliver the placenta, and (3) uterine massage after placental delivery. Reduces the incidence of PPH by 50-60%, reduces mean blood loss, and shortens the third stage. The WHO recommends active management for all vaginal deliveries. Misoprostol (600-800 mcg sublingual/rectal) is an alternative in settings where oxytocin is unavailable.

Active Phase of Labor

– The part of the first stage of labor from 6 cm to full dilatation (10 cm). Characterized by more rapid cervical change (minimum: 1 cm/hour in nulliparas, 1.2 cm/hour in multiparas). The cervix dilates more rapidly as labor progresses. Arrest of active phase: no cervical change for ≥4 hours with adequate contractions (≥200 Montevideo units for 2 hours) or ≥6 hours with inadequate contractions. Management: oxytocin augmentation for inadequate contractions, cesarean delivery for failed progress despite adequate contractions.

Acute Fatty Liver of Pregnancy (AFLP)

– A rare (1:7000-1:15,000) but life-threatening pregnancy complication characterized by microvesicular fatty infiltration of the liver leading to hepatic failure. Occurs in the third trimester (most commonly 35-37 weeks). Risk factors: primiparity, multiple gestation, male fetus, and disorders of fatty acid oxidation (LCHAD deficiency in the fetus, transmitted to the mother). Presents: nausea, vomiting, abdominal pain, jaundice, and encephalopathy. Lab findings: markedly elevated LFTs (though often mild relative to disease severity), elevated bilirubin, hypoglycemia, elevated ammonia, leukocytosis, DIC (prolonged PT/aPTT, low fibrinogen), and renal impairment. The Swansea criteria are used for diagnosis: ≥6 features in the setting of no other cause. Treatment: immediate delivery (regardless of gestational age), aggressive supportive care (glucose, coagulation factors, dialysis, and ICU), and liver transplantation if not improving within days. Maternal mortality: 10-20%. Fetal mortality: 10-20%.

Adenomyosis

– A condition in which endometrial glands and stroma are present within the myometrium (uterine muscle). Most common in multiparous women in their 40s and 50s. Presents with heavy menstrual bleeding (menorrhagia), severe dysmenorrhea, chronic pelvic pain, and uterine enlargement. Diagnosis: transvaginal ultrasound shows diffuse or focal myometrial thickening with heterogeneous echogenicity; MRI is more specific (junctional zone thickening >12 mm). Treatment: NSAIDs, hormonal contraceptives, levonorgestrel IUD, GnRH agonists, endometrial ablation, and hysterectomy for definitive management.

Adhesions

– Bands of fibrous scar tissue that form between organs or tissues, most commonly following abdominal or pelvic surgery. In gynecology, adhesions can develop after cesarean section, myomectomy, ovarian cystectomy, or treatment of endometriosis. Adhesions can cause pelvic pain, bowel obstruction, and female infertility by distorting tubo-ovarian anatomy and impairing oocyte pickup. Diagnosis is made at laparoscopy (the gold standard). Prevention: gentle tissue handling, meticulous hemostasis, minimizing operative time, and use of adhesion barriers (e.g., Interceed, Seprafilm). Treatment: laparoscopic adhesiolysis for symptomatic adhesions, though adhesions can reform.

Advanced Maternal Age

– Maternal age ≥35 years at the time of delivery. Associated with increased risks: aneuploidy (trisomy 21: 1:350 at age 35, 1:100 at age 40, 1:25 at age 45), miscarriage, gestational diabetes, preeclampsia, preterm birth, IUGR, placenta previa, and stillbirth. Management: offers of genetic counseling and prenatal diagnosis (NIPT, CVS, amniocentesis). Older mothers also have increased medical comorbidities that require management.

Amenorrhea

– Absence of menstruation. Primary amenorrhea: no menarche by age 15 with normal secondary sexual development, or by age 13 without. Secondary amenorrhea: absence of menses for ≥3 cycles or ≥6 months in women with prior menstruation. Causes: pregnancy (most common), hypothalamic suppression (stress, exercise, weight loss), PCOS, hyperprolactinemia, premature ovarian insufficiency, thyroid disorders, outflow obstruction. Evaluation: pregnancy test, FSH, LH, prolactin, TSH, and imaging.

Amniocentesis

– An invasive prenatal diagnostic procedure involving the aspiration of amniotic fluid from the uterine cavity under ultrasound guidance, typically performed between 15 and 20 weeks of gestation for diagnostic indications and after 32 weeks for fetal lung maturity assessment. A thin needle is inserted through the maternal abdomen and uterus into the amniotic sac, avoiding the fetus and placenta, to collect 20-30 milliliters of fluid. The procedure is indicated for advanced maternal age (greater than 35 years), abnormal maternal serum screening, abnormal ultrasound findings (structural anomalies, nuchal translucency greater than 3 mm), previous child with chromosomal abnormality or neural tube defect, and fetal lung maturity testing. Fetal cells in the amniotic fluid are cultured for chromosomal analysis (karyotype or chromosomal microarray) to detect aneuploidies (trisomy 21, 18, 13, sex chromosome abnormalities) and structural chromosomal rearrangements. Amniotic fluid alpha-fetoprotein (AFP) levels are measured to screen for neural tube defects (elevated in open spina bifida and anencephaly).

Amnion

– The innermost fetal membrane, lining the amniotic cavity and containing the amniotic fluid. A thin, avascular membrane composed of a single layer of cuboidal epithelium overlying a basement membrane and connective tissue. The amnion produces and resorbs amniotic fluid. It fuses with the chorion by 12-14 weeks. Amniotic membranes rupture spontaneously during labor (spontaneous ROM) or are artificially ruptured (AROM). Amniotic bands are rare complications of amnion rupture.

Amniotic Fluid Embolism

– A rare, catastrophic obstetric emergency where amniotic fluid enters the maternal circulation, causing acute cardiovascular collapse. Characterized by sudden respiratory distress, hypotension, coagulopathy, and seizure. Mortality rate 20-60% among survivors; 50% of affected mothers die. Treatment is supportive: cardiopulmonary resuscitation, mechanical ventilation, and management of disseminated intravascular coagulation. High association with mortality and severe maternal morbidity.

Amniotic Fluid Index

– A semi-quantitative assessment of amniotic fluid volume using ultrasound. The uterus divided into four quadrants, and the single deepest vertical pocket measured in each quadrant, then summed. Normal range is 5-24 cm at 16-40 weeks. Less than 5 cm indicates oligohydramnios; greater than 24 cm indicates polyhydramnios. Used in conjunction with fetal growth assessment and Doppler studies to evaluate fetal well-being.

Anatomy Survey (Level II Ultrasound)

– A comprehensive fetal ultrasound performed at 18-22 weeks to evaluate fetal anatomy and screen for structural anomalies. Standard views: fetal head (BPD, HC, lateral ventricles, choroid plexus, cerebellum, cisterna magna), face (lips, orbits, nasal bone), chest (heart rate, rhythm, four-chamber view, outflow tracts, lungs), abdomen (stomach, AC, kidneys, bladder, bowel, anterior abdominal wall), spine (longitudinal and transverse), extremities (femur, humerus, hands, feet), and placenta (location, grade, cord insertion) and amniotic fluid

(AFI or single deepest pocket). Sensitivity for major anomalies: 60-80% depending on BMI, gestational age, and operator experience.

Antenatal

– The period before birth, also called prenatal. The medical supervision and screening provided to pregnant women to monitor the health of the mother and fetus, detect complications early, and provide health education. Antenatal appointments typically follow a schedule: booking appointment (8-12 weeks), anomaly scan (18-21 weeks), and regular follow-ups. Key components: blood pressure monitoring, urine testing, fundal height measurement, fetal heart rate auscultation, and screening for gestational diabetes, pre-eclampsia, and fetal anomalies.

Antepartum Fetal Surveillance

– The monitoring of fetal well-being in the third trimester for high-risk pregnancies. Methods: fetal kick counts (daily maternal counting of fetal movements), non-stress test (NST), biophysical profile (BPP, AFI, fetal tone, movement, breathing), and Doppler (umbilical artery, MCA). Indications: postterm pregnancy, IUGR, preeclampsia, diabetes (pregestational), decreased fetal movement, maternal medical conditions (SLE, antiphospholipid syndrome), and prior stillbirth. Timing: weekly or twice-weekly starting at 32-34 weeks (earlier for severe conditions).

Antihypertensives in Pregnancy

– Medications used to treat hypertension in pregnancy. First-line agents: labetalol (combined alpha/beta blocker, 100-1200 mg BID-TID PO, or IV push for acute hypertension), nifedipine (calcium channel blocker, immediate-release 10-20 mg PO q30min for urgent, or XL 30-60 mg daily for maintenance), and methyldopa (central alpha agonist, 250-1000 mg BID, historically safest, first-line for chronic hypertension). Second-line: hydralazine (direct vasodilator, 5-10 mg IV q20min for acute hypertension). Contraindicated: ACE inhibitors and ARBs (fetal renal agenesis, oligohydramnios, fetal death), atenolol (IUGR), and nitroprusside (cyanide toxicity).

APGAR Score

– A scoring system developed by Virginia Apgar in 1952 to assess the newborn's immediate condition at 1 and 5 minutes after birth. Five components (each scored 0, 1, or 2): Appearance (skin color), Pulse (heart rate), Grimace (reflex irritability), Activity (muscle tone), and Respiration (breathing effort). Score 7-10: normal; 4-6: moderately depressed (may require stimulation or oxygen); 0-3: severely depressed (requires full resuscitation). The 5-minute score is more predictive of neonatal outcome. A low APGAR score does not equate to birth asphyxia.

Artificial Rupture of Membranes (AROM)

– Also called amniotomy. The intentional rupture of the amniotic sac with a small hook (Amnihook) during a cervical exam. Indications: induction or augmentation of labor, placement of internal monitors (fetal scalp electrode, intrauterine pressure catheter), and assessment of amniotic fluid (meconium staining). Risks: cord prolapse (especially with high presenting part), chorioamnionitis (with prolonged ROM), fetal injury, umbilical cord compression, and variable decelerations. AROM alone may augment labor by releasing prostaglandins.

Assisted Breech Delivery

– Vaginal breech delivery with operator assistance after spontaneous delivery to the umbilicus. Maneuvers: delivery of the legs (pinch them at the knees and deliver one at a time), delivery of the arms (Løvset maneuver: rotate the body to deliver the posterior arm, then the anterior arm), and delivery of the aftercoming head (Mauriceau-Smellie-Veit maneuver: the operator's hand supports the head flexed while traction is applied to the shoulders; or Piper forceps). The head must be delivered flexed to enter the pelvis with its smallest diameter. Episiotomy is generally performed.

Asymptomatic Bacteriuria in Pregnancy

– The presence of >100,000 CFU/mL of bacteria on a clean-catch urine culture without urinary symptoms. Occurs in 2-10% of pregnancies. Risk factors: prior UTI, diabetes, sickle cell trait, and multiparity. If untreated, 20-40% develop pyelonephritis. Screening: urine culture at the first prenatal visit. Treatment: nitrofurantoin (100 mg BID x 5-7 days, avoid near term), cephalexin (500 mg QID x 5-7 days), or amoxicillin (500 mg TID x 5-7 days). Recurrence is common; monthly surveillance cultures and suppression may be needed.

Augmentation of Labor

– The stimulation of spontaneous contractions that are judged to be inadequate to achieve progressive cervical dilatation. Most commonly uses oxytocin (Pitocin) infusion. Also includes artificial rupture of membranes (AROM, amniotomy). Indications: arrest of active phase (no cervical change for ≥4 hours with adequate MVUs), or prolonged latent phase. Pitocin is started at 1-2 mU/min and titrated every 15-30 minutes (up to 40 mU/min) until achieving adequate uterine activity (200-250 Montevideo units). Hyperstimulation (>5 contractions in 10 minutes or contractions lasting >2 minutes) is managed by reducing or stopping oxytocin.

Betamethasone

– A corticosteroid administered to pregnant women at risk of preterm delivery to accelerate fetal lung maturation. Indications: all pregnancies between 24-36 weeks 6 days at risk for preterm delivery within 7 days. Also for late preterm (34-36 weeks) if not previously given and delivery anticipated in ≤7 days. Dose: two doses of 12 mg IM, 24 hours apart. Mechanism: stimulates surfactant production in type II pneumocytes, induces structural maturation of the lungs (thinner alveolar walls), and reduces risk of RDS (by 40-50%), IVH, and NEC. Optimal benefit: 24 hours to 7 days after the first dose. Can be repeated (rescue course) if delivery is still imminent >14 days after the initial course.

Bile Acids

– Substances produced in the liver that aid digestion of fats and absorption of fat-soluble vitamins. In pregnancy, elevated bile acids (≥10 μmol/L) indicate intrahepatic cholestasis of pregnancy (obstetric cholestasis), a liver disorder characterized by pruritus (itching, typically palms and soles) and elevated serum bile acids. ICP is associated with increased risks of preterm birth, meconium-stained amniotic fluid, and stillbirth. Treatment: ursodeoxycholic acid reduces pruritus and improves bile acid levels. Delivery is typically planned at 37-39 weeks to reduce stillbirth risk.

Biophysical Profile

– A non-stress test combined with ultrasound assessment of five fetal parameters, each scored 0 or 2 for a total of 0-10. Parameters include non-stress test (fetal heart rate reactivity), fetal breathing movements, fetal tone, fetal movement, and amniotic fluid volume. Score of 8-10 is normal; 6 is equivocal; 4 or less is abnormal and may warrant delivery. Used to assess fetal well-being in high-risk pregnancies.

Biparietal Diameter (BPD)

– The transverse diameter of the fetal skull measured at the level of the thalami and cavum septum pellucidum. Used for dating (14-26 weeks, ±7-10 days) and assessing growth. Normal growth: 2-4 mm per week. BPD is less reliable after 26 weeks and in abnormal fetal head shapes (dolichocephaly, brachycephaly). The cephalic index (BPD/OFD ratio, normal 0.74-0.84) helps identify head shape abnormalities.

Bishop Score

– A pre-induction cervical scoring system that predicts the success of labor induction. Five components: cervical dilatation (0-3), effacement (0-3), station (-3 to +2, scored 0-3), consistency (firm, medium, soft; scored 0-2), and position (posterior, mid, anterior; scored 0-2). Total score: 0-13. Favorable (≥6-8): high likelihood of successful induction. Unfavorable (≤5): cervical ripening needed (prostaglandins or mechanical dilation with a Foley balloon). The Bishop score is used to choose the induction method and predict the risk of failed induction.

Bladder Exstrophy

– A rare congenital abdominal wall defect where the bladder is exposed on the outside of the body due to failure of the lower abdominal wall and bladder to close during embryonic development. Part of the exstrophy-epispadias complex. Associated with pubic bone diastasis, external genital anomalies, and potential bladder dysfunction. Treatment involves staged surgical repair beginning in the neonatal period, followed by long-term urologic and orthopedic follow-up.

B-Lynch Suture

– A surgical technique for controlling postpartum hemorrhage due to uterine atony unresponsive to medical management. A compressive suture placed around the uterus to mechanically compress the uterine walls and reduce blood flow. The suture creates a brace-like effect. Can be combined with other compression sutures (Cho, Hayman). Often successful in avoiding hysterectomy. Requires laparotomy and is performed under general anesthesia.

Braxton Hicks Contractions

– Irregular, painless uterine contractions occurring throughout pregnancy, becoming more frequent in the third trimester. Usually last 30-60 seconds and are felt as a tightening of the abdomen. Distinguished from true labour contractions by their irregularity, lack of cervical change, and resolution with rest or hydration. Named after John Braxton Hicks, the English physician who first described them in 1872. They do not cause progressive cervical dilatation or effacement. Often confused with prodromal labour or preterm labour, especially in primiparous women. Increased frequency at the end of the day or with physical activity. No treatment is necessary; reassurance and hydration are sufficient.

BRCA

– BReast CAncer gene: BRCA1 and BRCA2 are tumor suppressor genes involved in DNA double-strand break repair via homologous recombination. Mutations in these genes confer a significantly increased lifetime risk of breast cancer (up to 70-80% for BRCA1, 40-70% for BRCA2) and ovarian cancer (up to 40-50% for BRCA1, 10-20% for BRCA2). In obstetrics and gynecology, BRCA testing is offered to women with a strong family history of breast or ovarian cancer. Management options for mutation carriers: enhanced breast surveillance (MRI and mammography), risk-reducing salpingo-oophorectomy (RRSO, typically after childbearing is complete, around age 35-40 for BRCA1, 40-45 for BRCA2), and risk-reducing mastectomy.

Breast Abscess

– A localized collection of pus within the breast tissue, most commonly a complication of lactational mastitis. Presents as a fluctuant, painful mass with overlying erythema and systemic symptoms. Diagnosis by ultrasound. Treatment requires incision and drainage plus antibiotics. Non-lactational breast abscesses may be associated with smoking or duct ectasia and may require biopsy to rule out underlying malignancy.

Breast Engorgement

– Overfilling of the breasts with milk, causing bilateral pain, firmness, warmth, and nipple flattening. Occurs 3-5 days postpartum. Pathophysiology: vascular congestion and edema before milk secretion is established. Management: frequent feeding or pumping, cold compresses between feedings, warm compresses before feeding, and NSAIDs for pain relief. Differentiate from mastitis (usually unilateral, with fever and erythema). Engorgement is a common cause of early breastfeeding cessation if not managed properly.

Breast Health

– Breast health involves routine self-awareness, clinical examinations, and mammographic screening to detect abnormalities such as lumps, pain, or skin changes that may indicate benign conditions or breast cancer. Risk factors include genetics, age, hormone exposure, and lifestyle factors. Early detection through regular screening significantly improves treatment outcomes and survival.

Breech Delivery

– Vaginal delivery of a breech presentation. Types: assisted breech (the baby delivers spontaneously to the umbilicus, then the operator assists delivery of the shoulders and head) and total breech extraction (the operator delivers the entire baby by grasping the feet and pulling). Total breech extraction is reserved for the second twin or extreme prematurity. Prerequisites for planned breech delivery: frank or complete breech, estimated fetal weight 2000-3800 g, adequate pelvis, head flexed, and no fetal anomalies. The Term Breech Trial (2000) showed increased perinatal mortality with planned vaginal breech delivery, leading to a sharp decline in vaginal breech births.

Breech Presentation

– A fetal presentation where the buttocks or feet are positioned to deliver first. Types include frank breech (hips flexed, knees extended, most common), complete breech (hips and knees flexed), and footling breech (one or both feet presenting). Associated with increased risk of cord prolapse, head entrapment, and birth trauma. External cephalic version attempted at 36-38 weeks. Cesarean delivery recommended for persistent breech at term.

Brow Presentation

– A fetal presentation where the fetal head is in a military position between full flexion (vertex) and hyperextension (face). The presenting diameter is the largest (occipitofrontal, 13.5 cm). Incidence: 1:1000-1:2000. Vaginal delivery is impossible with a persistent brow in a term fetus unless it converts to vertex or face. Diagnosis: palpation of the anterior fontanelle and orbital ridges side by side. Management: conversion to vertex by applying fundal pressure and flexing the head or conversion to face by encouraging hyperextension. If conversion fails, cesarean delivery.

CA125

– Cancer Antigen 125, a protein tumor marker used primarily in the evaluation of ovarian cancer. CA125 is elevated in approximately 80% of epithelial ovarian cancers but can also be elevated in benign conditions (endometriosis, pelvic inflammatory disease, uterine fibroids, pregnancy, menstruation) and other cancers (pancreatic, breast, lung). In premenopausal women, the positive predictive value of CA125 for ovarian cancer is low due to frequent false positives. The Risk of Malignancy Index combines CA125, ultrasound findings, and menopausal status to stratify ovarian cancer risk. CA125 is also used to monitor response to ovarian cancer treatment and detect recurrence.

Caesarean Birth

– An operative procedure in which a baby is delivered through an incision in the abdominal wall and uterus. Indications: fetal distress, failure to progress in labor, breech presentation, placenta praevia, multiple pregnancy, previous cesarean, and maternal request. Types: planned (elective) cesarean, typically performed at 39 weeks; emergency cesarean, performed when urgent delivery is needed for maternal or fetal compromise. Procedure: Joel-Cohen or Pfannenstiel skin incision, lower uterine segment transverse incision (isthmic), delivery of the baby, delivery of the placenta, and closure in layers. Regional anesthesia (spinal or epidural) is preferred. Recovery: 2-4 day hospital stay, 6-week activity restriction.

Cardiotocography (CTG)

– A continuous electronic recording of the fetal heart rate and uterine contractions used to assess fetal well-being during pregnancy and labor. The CTG machine uses two transducers placed on the maternal abdomen: an ultrasound transducer records the fetal heart rate, and a tocodynamometer records uterine contractions. CTG interpretation uses the four features of the fetal heart rate: baseline rate (normal 110-160 bpm), baseline variability (normal 5-25 bpm), accelerations (increase of ≥15 bpm for ≥15 seconds, reassuring), and decelerations (early, late, or variable). CTG is categorized as normal (reassuring), suspicious (one non-reassuring feature), or pathological (two or more non-reassuring or abnormal features). Fetal blood sampling may be used to confirm acidosis in suspicious/pathological CTG.

Cardiovascular Changes in Pregnancy

– Plasma volume increases 40-50% (total volume 5 L), red cell mass increases 20-30% (physiologic anemia of pregnancy due to dilution). Cardiac output increases 30-50% by 32 weeks (stroke volume increases initially, then heart rate). Systemic vascular resistance decreases (progesterone-mediated vasodilation) causing a slight decrease in blood pressure despite increased cardiac output. Supine hypotensive syndrome: the gravid uterus compresses the IVC, reducing venous return and cardiac output. Auscultation: S3 gallop, systolic flow murmur. ECG: left axis deviation.

Category I Fetal Heart Tracing

– A normal fetal heart rate tracing by NICHD 2008 criteria. All three characteristics present: baseline 110-160 bpm, moderate variability (6-25 bpm), and no late or variable decelerations. May have accelerations. Category I tracings are associated with normal fetal acid-base status. Routine management: continue labor management; no intervention needed. Predictive of normal fetal oxygenation.

Category II Fetal Heart Tracing

– An indeterminate fetal heart rate tracing that falls between Category I and III. Includes a wide range of patterns: baseline tachycardia or bradycardia (100-110 bpm), minimal or absent variability without decelerations, absent variability with recurrent decelerations, marked variability, variable decelerations with slow return to baseline or with other concerning features, and prolonged decelerations. Category II tracings are the most common in labor (70-80%). Management: intrauterine resuscitation (position change, maternal oxygen, IV fluids, reduce or stop oxytocin), further evaluation (fetal scalp stimulation, scalp pH), and continued close observation. Progression to Category III warrants intervention.

Category III Fetal Heart Tracing

– An abnormal fetal heart rate tracing indicating potential fetal hypoxia or acidosis. Two specific patterns: (1) absent baseline variability with recurrent late decelerations, (2) absent baseline variability with recurrent variable decelerations, and (3) sinusoidal pattern. Category III tracings require prompt evaluation and intervention: intrauterine resuscitation measures, and if the pattern does not resolve, immediate delivery (operative vaginal or cesarean) to prevent fetal acidemia and neurologic injury.

Cell-Free Fetal DNA (cfDNA) Screening

– Also known as noninvasive prenatal testing (NIPT). Analyzes cell-free fetal DNA (cffDNA) circulating in maternal blood, derived from placental trophoblast apoptosis. Performed from 10 weeks onward. Highly sensitive and specific for trisomy 21 (>99%, FPR <0.1%), trisomy 18 (>97%), and trisomy 13 (>90%). Also screens for sex chromosome aneuploidies and microdeletions (22q11.2, 1p36, Prader-Willi/Angelman). False positives can occur from placental mosaicism, vanishing twin, or maternal chromosomal abnormalities. NIPT is a screening test, not diagnostic; positive results require confirmatory diagnostic testing (CVS or amniocentesis).

Cephalhaematoma

– A collection of blood between the periosteum and the skull bone, caused by rupture of blood vessels during childbirth, most commonly associated with ventouse (vacuum) delivery. Unlike caput succedaneum, a cephalhaematoma does not cross suture lines and is limited by the periosteal attachments. It presents as a firm, unilateral swelling on the newborn's scalp, appearing within hours to days after birth. It resolves spontaneously over weeks to months as the blood is reabsorbed. Complications: jaundice (from breakdown of red blood cells), anemia, and rarely infection or calcification. No treatment is required unless complications develop.

Cerclage

– A surgical placement of a suture around the cervix to prevent preterm delivery in patients with cervical insufficiency. Types include transvaginal (McDonald or Shirodkar) and transabdominal. Usually placed at 12-14 weeks gestation and removed at 36-37 weeks. Indicated for history of painless second trimester loss, cervical dilatation on exam, or short cervix on ultrasound. Emergency cerclage performed when dilatation occurs in the second trimester.

Cerebroplacental Ratio (CPR)

– The ratio of the middle cerebral artery (MCA) pulsatility index to the umbilical artery (UA) pulsatility index. CPR <5th percentile (or <1.0-1.1) suggests fetal brain-sparing effect (vasodilation of cerebral vessels in response to hypoxemia) due to placental insufficiency. A low CPR in IUGR fetuses predicts adverse perinatal outcomes (stillbirth, acidosis, NICU admission) better than UA Doppler alone. CPR may also be abnormal in small-for-gestational-age fetuses before UA Doppler becomes abnormal.

Cervical Cerclage

– Surgical placement of a non-absorbable suture around the cervix to provide mechanical support and prevent preterm delivery in patients with cervical insufficiency. Most commonly performed using the McDonald technique (purse-string suture at the cervicovaginal junction without bladder dissection) or the Shirodkar technique (with bladder and rectal dissection to place the suture at the internal os). Elective cerclage is placed at 12-14 weeks gestation, with removal at 36-37 weeks or at the onset of labour. Emergency or rescue cerclage may be performed in the second trimester if cervical dilation is detected before 24 weeks. Contraindications include chorioamnionitis, active vaginal bleeding, preterm labour, preterm prelabour rupture of membranes, and fetal demise. Risks include membrane rupture, chorioamnionitis, cervical trauma, and preterm labour.

Cervical Dilatation

– Progressive opening of the cervical os during labor, measured in centimeters from 0 (closed) to 10 cm (fully dilated). The cervix must also efface (thin out) before significant dilatation occurs. Cervical status assessed by digital vaginal examination during labor.

Cervical Dysplasia

– Abnormal growth of cervical epithelial cells, representing a premalignant condition that may progress to cervical cancer if untreated. Also known as cervical intraepithelial neoplasia (CIN). CIN is caused by persistent infection with high-risk human papillomavirus (HPV) types, most commonly HPV 16 and 18. Grading: CIN 1 (mild dysplasia, involves the lower one-third of the epithelium, high rate of spontaneous regression), CIN 2 (moderate dysplasia, lower two-thirds), and CIN 3 (severe dysplasia/carcinoma in situ, full-thickness involvement). Most CIN 1 lesions regress spontaneously; CIN 2 and 3 require treatment. Diagnosis: abnormal Pap smear followed by colposcopy with directed biopsy and endocervical curettage. Treatment: CIN 1 may be observed; CIN 2/3 is treated with excisional procedures (loop electrosurgical excision procedure, cold knife conization) or ablative therapy (cryotherapy, laser ablation) depending on lesion size, location, and patient factors. Follow-up is essential as recurrence can occur. Prevention: HPV vaccination.

Cervical Effacement

– Thinning and shortening of the cervix during labor, measured as a percentage from 0% (thick, normal length) to 100% (paper thin). Effacement precedes and accompanies cervical dilatation. Nulliparous women often efface before dilating; multiparous women may efface and dilate simultaneously. Effacement occurs as the internal os and external os are pulled upward into the lower uterine segment. Assessed by digital vaginal examination. In nulliparous women, effacement typically occurs before significant dilatation; in multiparous women, effacement and dilatation often occur concurrently. Contributes to the Bishop score for predicting labor induction success.

Cervical Exam in Labor

– Serial digital vaginal examinations to assess cervical dilatation (cm), effacement (%), station (relationship to ischial spines, -5 to +5), consistency (firm, medium, soft), and position (posterior, mid, anterior). Performed every 2-4 hours in active labor. In the second stage, assessing fetal head position and asynclitism. Risks: infection with frequent exams (especially with ruptured membranes). Current guidelines: minimize exams after ROM, limit to q4h in active labor unless clinically indicated.

Cervical Incompetence

– The inability of the cervix to maintain pregnancy in the second trimester due to painless cervical dilatation. Results in recurrent second trimester pregnancy loss or preterm delivery. Risk factors include prior cervical trauma, cone biopsy, and congenital anomalies. Management includes cerclage (McDonald or Shirodkar) typically placed at 12-14 weeks and removed at 36-37 weeks. Cervical length screening by transvaginal ultrasound in high-risk patients.

Cervical Length Screening

– Transvaginal ultrasound measurement of the cervix between 18-24 weeks to assess risk of spontaneous preterm birth. Normal cervix measures greater than 25 mm. Short cervix (less than 25 mm) increases preterm delivery risk. Cervical funneling and dilatation are additional concerning findings. Interventions for short cervix include progesterone supplementation, cerclage, and cervical pessary. Most beneficial in asymptomatic women with prior preterm birth.

Cervical Mucus

– Secretions from the cervical glands that change throughout the menstrual cycle under hormonal influence. Estrogen produces thin, clear, stretchy mucus favorable for sperm transport around ovulation (spinnbarkeit). Progesterone produces thick, sticky, opaque mucus that forms a plug in the cervical canal during pregnancy. The mucus plug (operculum) protects against ascending infection throughout gestation.

Cervical Pregnancy

– A rare ectopic pregnancy (0.1-1%) implanted in the cervical canal. Presents with painless vaginal bleeding in the first trimester. Diagnosis: ultrasound (hourglass-shaped uterus, gestational sac in the cervix with absent sliding sign, empty uterine cavity). High risk of severe hemorrhage. Treatment: methotrexate (systemic or local injection), uterine artery embolization, dilation and curettage with Foley balloon tamponade, or hysterectomy for uncontrolled bleeding.

Cervical Ripening Agents

– Medications or mechanical devices used to soften and dilate the cervix before induction of labor when the Bishop score is unfavorable (≤5). Pharmacologic: prostaglandin E1 (misoprostol/Cytotec, 25 mcg PV q4h), prostaglandin E2 (dinoprostone/Cervidil vaginal insert 10 mg over 12 hours, or Prepidil gel 0.5 mg intracervical q6h). Mechanical:

transcervical Foley catheter (16-30 Fr, 30-60 mL balloon), osmotic dilators (laminaria, Dilapan-S). Membrane stripping (sweeping) is a low-risk office procedure. Combination of pharmacologic and mechanical methods is common. Contraindication: prior classical cesarean (risk of uterine rupture with prostaglandins). Misoprostol has the highest risk of tachysystole.

Cervicitis

– Inflammation of the uterine cervix, most commonly caused by sexually transmitted infections (Chlamydia trachomatis and Neisseria gonorrhoeae) but also by other organisms (Mycoplasma genitalium, Trichomonas vaginalis, herpes simplex virus), or non-infectious causes (chemical irritation, foreign body, allergy). Presents with mucopurulent vaginal discharge, cervical friability (bleeding on touch), intermenstrual or postcoital bleeding, and dyspareunia. Many cases are asymptomatic. Diagnosis: microscopic examination of cervical secretions showing increased polymorphonuclear leukocytes, nucleic acid amplification tests (NAATs) for Chlamydia and Neisseria gonorrhoeae, and cervical cultures. Treatment includes appropriate antibiotics (azithromycin or doxycycline for Chlamydia, ceftriaxone for gonorrhea). Sexual partners require treatment and screening. Untreated cervicitis can ascend to cause pelvic inflammatory disease, leading to tubal factor infertility, ectopic pregnancy, and chronic pelvic pain. Barrier contraception reduces transmission risk.

Cervix

– The lower, narrow part of the uterus that opens into the vagina. The cervix is approximately 3-4 cm long in the non-pregnant state and consists of two parts: the ectocervix (visible portion protruding into the vagina, covered by stratified squamous epithelium) and the endocervical canal (lined by columnar epithelium). The transformation zone (squamocolumnar junction) is where cervical cell changes and cancer typically arise. Functions: produces cervical mucus that changes in consistency during the menstrual cycle to facilitate or impede sperm transport; dilates during labor to allow passage of the baby; and provides mechanical support to maintain pregnancy.

Cesarean Delivery

– Surgical delivery of the fetus through abdominal and uterine incisions. Performed when vaginal delivery is contraindicated or fails. Indications include malpresentation, placenta previa, previous cesarean, fetal distress, and failed induction. Most common uterine incision is low transverse (classical for very preterm or transverse lie). Regional anesthesia preferred. Complications include infection, hemorrhage, thromboembolism, and injury to adjacent organs.

Chignon

– A temporary swelling (edema) on the baby's scalp caused by the vacuum cap used during ventouse (vacuum) assisted delivery. The chignon forms as the vacuum creates negative pressure pulling the scalp tissue into the cup. It settles within 24-48 hours after birth without any treatment. It should be distinguished from cephalhaematoma (subperiosteal bleed) and caput succedaneum (generalized scalp edema). No specific management is needed; parents should be reassured.

Childbirth

– See Parturition (Childbirth).

Chorioamnionitis

– Infection of the chorion and amnion, typically due to ascending bacterial infection from the genital tract. Risk factors include prolonged rupture of membranes, multiple vaginal exams, and preterm labor. Presents with maternal fever, uterine tenderness, and fetal tachycardia. Diagnosis is clinical; amniocentesis for confirmation. Treatment includes intravenous antibiotics (ampicillin and gentamicin) and delivery. Neonatal sepsis is a major concern.

Chorion

– The outermost fetal membrane, derived from the trophoblast. The chorion frondosum (villous chorion) develops into the fetal portion of the placenta. The chorion laeve (smooth chorion) is the non-villous portion. Chorionic villi contain fetal blood vessels and are bathed in maternal blood. Chorionic villus sampling (CVS) biopsies these villi for prenatal genetic diagnosis. The chorion is separated from the amnion by the extraembryonic coelom until the amnion fuses with it at 12-14 weeks.

Chorionic Villus Sampling (CVS)

– A first-trimester invasive prenatal diagnostic procedure (10-13 weeks) that samples chorionic villi for genetic analysis. Transcervical (catheter through the cervix into the placenta under ultrasound guidance) or transabdominal (needle through the abdominal wall). Advantages over amniocentesis: earlier results (24-48 hours for rapid FISH, 7-14 days for karyotype). Indications: advanced maternal age, abnormal NIPT, prior aneuploidy, family history of genetic disorders, and abnormal ultrasound findings. Risks: miscarriage (0.2-0.5% above background), and limb reduction defects (rare, associated with CVS before 10 weeks). Confined placental mosaicism (CPM, 1-2%): karyotype differs between placenta and fetus, requiring amniocentesis for confirmation.

Choroid Plexus Cyst (CPC)

– A sonolucent fluid-filled cyst within the choroid plexus of the lateral cerebral ventricle, seen in 1-3% of second-trimester fetuses. In isolation, it is a normal variant with no clinical significance. CPC is associated with trisomy 18 (30-50% of trisomy 18 fetuses have CPC, but most fetuses with CPC are normal). If isolated, the risk of trisomy 18 is very low, particularly if NIPT is normal. No further workup is needed for isolated CPC after 22 weeks gestation.

Chromosomal Abnormality

– Any deviation from the normal number (46 chromosomes, 23 pairs) or structure of chromosomes. Numerical abnormalities: aneuploidy (trisomy 21/Down syndrome, trisomy 18/Edwards syndrome, trisomy 13/Patau syndrome, monosomy X/Turner syndrome), polyploidy (triploidy, tetraploidy). Structural abnormalities: deletions (Cri-du-chat, 22q11.2 deletion), duplications, translocations (Robertsonian, reciprocal), inversions, and ring chromosomes. Screening: combined test (nuchal translucency, hCG, PAPP-A), NIPT (cell-free fetal DNA), quadruple test. Diagnosis: chorionic villus sampling (11-14 weeks), amniocentesis (15+ weeks).

Chromosomes

– The thread-like structures within the cell nucleus that carry genetic information in the form of DNA. Humans have 46 chromosomes arranged in 23 pairs: 22 pairs of autosomes and 1 pair of sex chromosomes (XX in females, XY in males). Each parent contributes one chromosome to each pair. Chromosomes are visible under a light microscope during cell division and are arranged in a standardized format called a karyotype (ordered by size, centromere position, and banding pattern). Chromosomal abnormalities can involve number (aneuploidy) or structure (deletions, duplications, translocations, inversions).

Chronic Hypertension in Pregnancy

– Hypertension (BP ≥140/90 mmHg) that predates pregnancy or is diagnosed before 20 weeks gestation. Affects 1-5% of pregnancies. Increased risk of superimposed preeclampsia (20-50%), IUGR, preterm birth, and placental abruption. Management: continue or adjust antihypertensives (avoid ACE inhibitors and ARBs during pregnancy; labetalol, nifedipine, and methyldopa are safe). Low-dose aspirin (81 mg daily) from 12-36 weeks reduces the risk of preeclampsia. Delivery timing depends on blood pressure control and fetal status.

Clavicle Fracture (Intentional)

– Intentional fracture of the fetal clavicle as a last resort to reduce the bisacromial diameter in shoulder dystocia. The operator applies upward pressure on the midportion of the anterior fetal clavicle with a finger, fracturing it. The clavicle heals without long-term sequelae. Usually unnecessary because the posterior arm delivery and Gaskin maneuver are more effective and less traumatic. Accidental clavicle fracture occurs in 0.5-2% of normal deliveries.

Clear Margins

– A histological finding indicating that no abnormal cells (e.g., cervical intraepithelial neoplasia or cancer cells) are present at the edge of a tissue specimen removed during a procedure such as LLETZ, cone biopsy, or surgical excision. Clear margins suggest that the entire abnormal area has been removed, reducing the risk of persistent or recurrent disease. Positive margins (abnormal cells at the edge) indicate incomplete excision and require closer follow-up or repeat treatment. Clear margins are the goal of excisional treatment for cervical cell changes and other gynecologic conditions.

Clitoris

– A small, sensitive erectile organ located at the anterior junction of the labia minora, beneath the clitoral hood (prepuce). The clitoris is the primary source of female sexual pleasure and is homologous to the male penis. It consists of the external glans (visible, about the size of a pea), the body (corpus), and the internal crura (two elongated erectile structures that extend along the pubic rami). The clitoris contains approximately 8,000 nerve endings and becomes engorged with blood during sexual arousal. The only known function of the clitoris is sexual pleasure.

Colostrum

– The thick, yellowish fluid produced by the mammary glands in late pregnancy and the first few days postpartum. Rich in immunoglobulins (especially IgA), proteins, leukocytes, and growth factors.

Combined Spinal-Epidural (CSE) Analgesia

– A neuraxial analgesia technique for labor that combines a spinal injection (short-acting opioid or opioid + low-dose local anesthetic) with an epidural catheter. Advantages: rapid onset of pain relief (3-5 minutes from the spinal component), minimal motor block, and the ability to dose through the epidural catheter for prolonged labor. The spinal block provides immediate relief for women in active labor; the epidural provides sustained relief. The CSE may be associated with more maternal pruritus (from intrathecal opioids) and a higher rate of fetal bradycardia than epidural alone.

Compound Presentation

– A fetal presentation where an extremity (hand, arm, foot) presents alongside the vertex or breech. Most common: hand alongside the head. Incidence: 1:300-1:1000. Fetal prognosis is generally good; the prolapsed extremity often retracts as labor progresses. Management: expectant management in most cases. The prolapsed hand may be gently pushed back after descent. The cord should be assessed for prolapse. Cesarean if labor fails to progress or the extremity causes obstruction.

Conception (Fertilization)

– The fusion of a sperm and an ovum in the ampulla of the fallopian tube, forming a zygote. The sperm penetrates the corona radiata and zona pellucida (acrosome reaction). After fertilization, the zygote undergoes cleavage (mitotic division) as it travels toward the uterus. Implantation occurs approximately 6-10 days after ovulation. Conception timing is critical for dating pregnancy and determining the window of fertility.

Conjoined Twins

– Monozygotic, monochorionic, monoamniotic twins who are physically connected at birth, resulting from incomplete fission of the fertilized ovum after day 13 of fertilization. Incidence: 1 in 50,000-100,000 births. Classification by the site of union: thoracopagus (chest, most common, 40%, often with shared heart — poor prognosis), omphalopagus (abdomen, 33%), pygopagus (sacrum/buttocks, 19%), ischiopagus (pelvis, 6%), craniopagus (head, 2%), and parapagus (lateral fusion). Determination of shared vital organs (especially the heart) is critical for prognosis and surgical planning. Diagnosis by prenatal ultrasound and MRI. Delivery is typically by planned cesarean at 34-36 weeks. Surgical separation after birth is complex and requires a multidisciplinary team. Prognosis depends on the degree of organ sharing.

Continuous Electronic Fetal Monitoring (EFM)

– The continuous recording of fetal heart rate (FHR) and uterine contractions using external transducers (Doppler ultrasound and tocodynamometer) or internal devices (fetal scalp electrode and intrauterine pressure catheter). Used intrapartum to detect fetal hypoxia. The NICHD 2008 classification categorizes tracings as Category I (normal), II (indeterminate), or III (abnormal). EFM is associated with an increased rate of cesarean and operative vaginal delivery without clear reduction in cerebral palsy (false positives are common). Recommended for high-risk labor and women who choose augmentation. Intermittent auscultation is appropriate for low-risk labor.

Contraception and Birth Control

– Contraception encompasses methods used to prevent pregnancy, including hormonal options like oral contraceptives, patches, rings, injections, and implants, as well as non-hormonal methods like copper IUDs, barrier methods, diaphragms, and permanent sterilization. Emergency contraception is available for use shortly after unprotected intercourse. Method selection considers effectiveness, side effects, reversibility, and individual health factors.

Contraction Stress Test

– An antepartum fetal surveillance technique assessing fetal heart rate response to induced uterine contractions. Oxytocin or nipple stimulation produces contractions. A negative test shows no late decelerations. A positive CST demonstrates late decelerations with more than 50% of contractions. Equivocal results show intermittent late decelerations. Contraindications include preterm labour, placenta previa, previous classical caesarean, multiple gestation, and preterm prelabour rupture of membranes. The CST has largely been replaced by the biophysical profile and modified BPP as first-line antepartum surveillance due to its higher false-positive rate, greater time requirement, and need for intravenous access. However, it remains useful when BPP is equivocal or when fetal status is uncertain.

Contraction Stress Test (CST)

– An antepartum test that evaluates the fetal heart rate response to uterine contractions. Contractions are induced with oxytocin or nipple stimulation (until 3 contractions in 10 minutes). Negative (normal): no late or significant variable decelerations. Positive (abnormal): late decelerations with $\geq$50% of contractions. Equivocal: intermittent late decelerations. The CST is less commonly used due to the higher success and lower risk of the BPP. Indications: when BPP is equivocal or maternal perception of decreased movement.

Controlled Cord Traction (CCT)

– A technique for active management of third-stage labor: gentle, steady traction on the umbilical cord with one hand while the other hand applies suprapubic counter-pressure (to prevent uterine inversion). Performed after signs of placental separation. The Brandt-Andrews maneuver and the Finocchietti technique are CCT methods. CCT reduces the duration of the third stage and the risk of postpartum hemorrhage. Contraindication: signs of uterine inversion.

Cord Presentation

– The presence of the umbilical cord between the presenting fetal part and the cervix, with intact membranes. Risk factors: malpresentation, unstable lie, low-lying placenta, and long cord. If the membranes rupture, the cord may prolapse (umbilical cord prolapse). Diagnosis: digital exam (palpation of a pulsating cord through intact membranes) or ultrasound. Management: admit to the hospital, strict bed rest, avoid vaginal exams, and cesarean delivery before membrane rupture.

COVID-19 in Pregnancy

– Pregnant women with COVID-19 are at increased risk of severe illness (ICU admission, mechanical ventilation, death) compared to non-pregnant women. Risks to the fetus: preterm birth, cesarean delivery, and possibly stillbirth. Vertical transmission is rare but possible. Management: vaccination is recommended (no increased risk of miscarriage or adverse outcomes). Treatment: remdesivir, dexamethasone (for oxygen requirement), and anticoagulation as indicated. Pregnant women should be offered vaccination at any stage of pregnancy.

Crown-Rump Length (CRL)

– The measurement from the top of the fetal head (crown) to the bottom of the buttocks (rump), the most accurate method for establishing gestational age in the first trimester. Measured at 7-13 weeks. Accuracy: $\pm$3-5 days. The EDD is set using CRL if the difference from LMP dating exceeds 7 days. CRL growth: approximately 1 mm per day. CRL <7 mm without cardiac activity: uncertain viability, repeat in 1 week. Absent cardiac activity at CRL $\geq$7 mm is diagnostic of pregnancy failure.

Decidua

– The transformed endometrium of pregnancy, shed after delivery. Three regions: decidua basalis (under the placenta, site of trophoblast invasion), decidua capsularis (over the gestational sac, covering the embryo), and decidua parietalis (the remainder of the uterine lining). The decidua provides nutrition to the early embryo and produces hormones (prolactin, relaxin). Decidual cells and natural killer cells play key roles in immune tolerance of the fetus.

Delayed Cord Clamping

– The practice of delaying umbilical cord clamping for $\geq$30-60 seconds after delivery, allowing placental transfusion of blood to the newborn. Benefits: increases neonatal blood volume by 30%, improves iron stores (ferritin, hemoglobin) for the first 6 months, reduces anemia, and improves transitional circulation. For preterm infants (<34 weeks), delayed clamping for 30-60 seconds reduces intraventricular hemorrhage, necrotizing enterocolitis, and need for blood transfusion. Contraindications: need for immediate resuscitation, cord avulsion, placental abruption with hemorrhage, and maternal instability.

Delivery of Posterior Arm

– A shoulder dystocia maneuver: the operator's hand enters the posterior vagina along the fetal chest, finds the fetal hand, grasps the forearm, sweeps it across the chest and face, and delivers the posterior arm and hand over the perineum. This reduces the bisacromial diameter by 20% and is often highly effective. Flexing the elbow helps retrieve the hand. Risk: humerus fracture (treat with immobilization; heals well). Often the definitive maneuver after McRoberts and rotational maneuvers fail.

Dexamethasone

– An alternative corticosteroid for fetal lung maturation, similar to betamethasone. Dose: four doses of 6 mg IM, 12 hours apart. While betamethasone is the preferred agent in the US (more evidence for RDS reduction), dexamethasone is used in some settings and is effective. Both cross the placenta and induce fetal lung surfactant production. Dexamethasone has slightly more evidence for long-term neurodevelopmental outcomes in certain studies. WHO recommends either agent.

Diathermy

– The use of high-frequency electrical current to generate heat for surgical purposes, including cutting tissue, coagulating blood vessels to stop bleeding (hemostasis), and destroying abnormal tissue (electrocautery, electrocoagulation). In gynecologic surgery, diathermy is used in laparoscopic procedures (ovarian drilling for PCOS, endometriosis ablation, myomectomy), hysteroscopic surgery (endometrial ablation, polypectomy, fibroid resection), and colposcopy (treatment of cervical intraepithelial neoplasia). Monopolar diathermy: current flows from electrode through patient to grounding pad. Bipolar diathermy: current flows between two forceps tips, safer for use near delicate structures.

Dichorionic Twins

– Twins each having their own chorion and amnion, occurring in approximately 30% of monozygotic twins and all dizygotic twins. Each twin has its own placenta, though placentas may be adjacent. Complications include growth restriction, preterm delivery, and preeclampsia, but not twin-to-twin transfusion syndrome. Require regular ultrasound monitoring for growth discordance.

Dinoprostone (Cervidil, Prepidil)

– A synthetic prostaglandin E2 (PGE2) used for cervical ripening and induction. Forms: Cervidil (10 mg vaginal insert, slow-release over 12 hours, removed if tachysystole occurs), Prepidil (0.5 mg intracervical gel, may be repeated q6h), and Prostin E2 (vaginal suppository). Indications: unfavorable cervix (Bishop ≤ 5). Side effects: tachysystole (5-10%), fever, nausea, vomiting, and diarrhea. Contraindications: prior uterine surgery (including classical cesarean), active vaginal bleeding, and fetal compromise. Remove promptly if tachysystole or nonreassuring FHR occurs.

Donor Insemination

– A fertility treatment in which sperm from a donor (anonymous or known) is placed into a woman's vagina, cervix, or uterus to achieve pregnancy. Indications: male factor infertility (azoospermia, severe oligospermia), genetic disorders in the male partner, single women, and same-sex female couples. In the UK, donor insemination is regulated by the Human fertilization and Embryology Authority, requiring donor screening for infectious diseases, genetic disorders, and medical conditions. Recipients may require ovulation induction or intrauterine insemination (IUI) to optimize timing.

Doula Support

– A trained non-medical birth companion who provides continuous emotional, physical, and informational support to the laboring mother. Services: massage, positioning guidance, breathing techniques, advocacy, and reassurance. Doula support has Level 1 evidence for improved outcomes: reduced duration of labor (by 40 minutes), reduced need for cesarean (by 25%), reduced epidural requests, reduced Pitocin use, and higher satisfaction. Doulas do not perform clinical tasks (vaginal exams, FHR monitoring). They complement but do not replace midwives or physicians.

Dysmenorrhea

– Painful menstruation with cramping pelvic pain. Primary dysmenorrhea: no underlying pathology, associated with prostaglandin release, starts within

3 years of menarche. Secondary dysmenorrhea: due to underlying pathology (endometriosis, fibroids, adenomyosis, PID). Treatment: NSAIDs (ibuprofen, naproxen), hormonal contraceptives (pills, IUD, implant), and treatment of underlying cause. Severe cases may require GnRH agonists or surgical intervention.

Dysmenorrhoea

– Painful menstruation, classified as primary (no identifiable pelvic pathology) or secondary (underlying structural or inflammatory condition). Primary dysmenorrhoea is caused by prostaglandin-induced uterine contractions and occurs in the first 2-3 years after menarche, typically starting 6-12 hours before menstruation and lasting 1-3 days. Secondary dysmenorrhoea is associated with endometriosis, adenomyosis, fibroids, or pelvic inflammatory disease. Treatment: NSAIDs (inhibit prostaglandin synthesis), combined oral contraceptives (reduce menstrual volume and prostaglandin production), Mirena IUS, and treatment of underlying cause.

Dyspareunia

– Painful sexual intercourse.

Early Decelerations

– A fetal heart rate pattern with a gradual decrease (onset to nadir >30 seconds) that mirrors the contraction and returns to baseline as the contraction ends. Caused by fetal head compression during contractions (vagal response). A benign, physiologic finding associated with normal fetal acid-base status. No intervention required. Early decelerations are characteristic of active labor and descent of the fetal head.

Echogenic Bowel

– Increased echogenicity of the fetal bowel relative to adjacent bone (iliac wing). Identified in 0.1-0.5% of second-trimester ultrasounds. Associations: aneuploidy (trisomy 21, 18, 13), cystic fibrosis (echogenic bowel may be the first sign of meconium ileus), congenital infection (CMV, toxoplasmosis, parvovirus B19), IUGR, and swallowing of intra-amniotic blood. Workup: NIPT, cystic fibrosis carrier screening, infection serologies (TORCH), and serial growth scans.

Echogenic Intracardiac Focus (EIF)

– A small bright spot (calcification) in the fetal heart, typically in the left ventricle (papillary muscle). Seen in 3-5% of normal pregnancies. In isolation, it is a normal variant with minimal aneuploidy risk. However, EIF is more common in trisomy 21 fetuses (10-30%), particularly when combined with other soft markers. If isolated and NIPT/low-risk serum screen, no further action is needed.

Eclampsia

– The occurrence of generalized tonic-clonic seizures in a patient with preeclampsia, representing a medical emergency. Complications include aspiration, placental abruption, disseminated intravascular coagulation, and maternal/fetal death. Prevention and treatment involves intravenous magnesium sulfate (4-6 g IV bolus followed by 1-2 g/hour maintenance), which reduces the risk of seizures by 50%. Management includes airway protection, seizure control (magnesium sulfate first-line; benzodiazepines if magnesium fails), blood pressure control with antihypertensives (labetalol or hydralazine for BP $\geq$160/110 mmHg), and delivery once maternal stability is achieved. Eclampsia can occur antepartum (50%), intrapartum (25%), or postpartum (25%, most within 48 hours). Recurrence risk in future pregnancies is approximately 2%. Maternal mortality is approximately 2-5% in high-resource settings.

Ectopic Pregnancy

– Implantation of a fertilized ovum outside the uterine cavity, most commonly in the fallopian tube (95%). Risk factors: prior ectopic, PID, tubal surgery, IUD, smoking, and assisted reproductive technology. Presents with lower abdominal pain, amenorrhea, and vaginal bleeding. Rupture causes hemodynamic instability. Diagnosis: transvaginal ultrasound (no intrauterine pregnancy with hCG >discriminatory zone), serial hCG monitoring. Treatment: methotrexate for early, unruptured ectopic; salpingectomy or salpingostomy for surgical management.

Electrocoagulation

– A surgical technique using high-frequency electrical current to destroy tissue or stop bleeding by coagulating proteins, also known as diathermy or cauterization. In gynecology, electrocoagulation is used in laparoscopic treatment of endometriosis (ablation of implants), polycystic ovary syndrome (ovarian drilling), and hysteroscopic endometrial ablation. It may be monopolar or bipolar. Complications are rare but include thermal injury to adjacent organs (bowel, ureter, bladder) and infection.

Embryo

– The developing human organism from fertilization (conception) through the eighth week of gestation (10th week by last menstrual period). After fertilization, the zygote undergoes cleavage (cell division) as it travels through the fallopian tube, reaching the morula stage (16 cells) and then the blastocyst stage (5–6 days), which implants in the endometrium. During the embryonic period, all major organ systems develop (organogenesis): gastrulation forms the three germ layers (ectoderm, mesoderm, endoderm), neurulation forms the neural tube, and the heart begins beating at 5–6 weeks. The embryonic period is the most vulnerable to teratogens (medications, infections, radiation, alcohol), which can cause major structural malformations. After 8 weeks, the developing human is called a fetus.

Endometrial Cancer

– The most common gynecologic malignancy in developed countries, arising from the lining of the uterine cavity, typically in postmenopausal women. Type I (endometrioid) carcinomas account for 80 percent of cases and are estrogen-driven, associated with endometrial hyperplasia, obesity, diabetes, polycystic ovary syndrome, unopposed estrogen therapy, and tamoxifen use; they are generally low-grade with a favorable prognosis. Type II (non-endometrioid) carcinomas (serous, clear cell, carcinosarcoma) are not estrogen-driven, arise in atrophic endometrium, and are aggressive with a poor prognosis. The lifetime risk is approximately 3 percent in the general population. Clinical presentation: postmenopausal bleeding is the cardinal symptom (occurring in 90 percent of cases), along with abnormal uterine bleeding in premenopausal women, watery or bloody vaginal discharge, and pelvic pain in advanced disease. Diagnosis is by endometrial biopsy or dilation and curettage, with transvaginal ultrasound showing endometrial thickness greater than 4-5 mm in postmenopausal women requiring further evaluation. Histologic grading (FIGO grade 1-3 based on glandular architecture), subtype determination, and molecular classification (POLE ultramutated, microsatellite instability-high, copy number low, copy number high) guide prognosis and treatment. Staging is surgical according to the FIGO system, based on depth of myometrial invasion, cervical stromal involvement, adnexal spread, lymph node metastasis, and peritoneal cytology. Treatment: total hysterectomy with bilateral salpingo-oophorectomy is the mainstay; lymph node assessment is performed for high-risk histologies. Adjuvant radiotherapy and/or chemotherapy are indicated for high-stage or high-grade disease. Prognosis is generally favourable due to early presentation with postmenopausal bleeding; five-year survival exceeds 95% for stage I disease.

Endometriosis

– A chronic, estrogen-dependent gynecologic disorder characterized by the presence of endometrial-like tissue (glands and stroma) outside the uterine cavity, most commonly implanted on the pelvic peritoneum, ovaries, uterosacral ligaments, and rectovaginal septum. The ectopic endometrial tissue responds to hormonal stimulation, proliferating and shedding cyclically, causing local inflammation, adhesion formation, and fibrosis. The prevalence is 6-10 percent of women of reproductive age, rising to 35-50 percent in women with infertility and chronic pelvic pain. The exact pathogenesis remains unclear, but leading theories include retrograde menstruation (Sampson theory, with menstrual debris transported through the fallopian tubes into the peritoneal cavity), coelomic metaplasia (transformation of peritoneal mesothelium into endometrial tissue), lymphatic or hematogenous dissemination, and immune dysregulation (impaired clearance of endometrial cells from the peritoneal cavity). Genetic factors contribute, with a 6-9 fold increased risk in first-degree relatives. Clinical presentation includes chronic pelvic pain (often cyclic but may become continuous), severe dysmenorrhea, dyspareunia (painful intercourse), infertility, and dyschezia (painful defecation). Physical examination may reveal tender nodularity in the posterior vaginal fornix, fixed retroverted uterus, and adnexal masses (endometriomas). Diagnosis is definitively made by direct visualization at laparoscopy, with histological confirmation of endometrial glands and stroma in biopsy specimens. Treatment includes analgesics (NSAIDs), hormonal therapy (combined oral contraceptives, progestins, GnRH agonists, levonorgestrel IUS), and surgical excision or ablation of implants.

Fertility management ranges from expectant management to assisted reproductive technologies.

Endometritis

– Infection of the endometrium, the most common cause of puerperal fever after cesarean delivery. Risk factors include prolonged labor, multiple vaginal examinations, premature rupture of membranes, and cesarean section. Presents with fever, uterine tenderness, foul-smelling lochia, and leukocytosis. Treatment with broad-spectrum IV antibiotics (ampicillin and gentamicin plus clindamycin) for polymicrobial infection including anaerobes.

Endometrium

– The lining of the uterus.

Engagement

– The descent of the fetal presenting part into the pelvis, typically occurring 2-3 weeks before labor in nulliparas or at the onset of labor in multiparas.

Epidural Analgesia for Labor

– A neuraxial technique for labor analgesia: a catheter is placed in the epidural space (L3-L4 or L2-L3 interspace) and local anesthetic (bupivacaine 0.0625-0.125%) with opioid (fentanyl 2 µg/mL) is infused continuously or in patient-controlled epidural analgesia (PCEA) boluses. Produces segmental analgesia (T10-L1 for the first stage, T10-S4 for the second stage). Benefits: excellent pain relief, mother can rest, and can be converted to surgical anesthesia for cesarean. Side effects: hypotension, motor block (reduced with low-dose solutions), fever (can increase the diagnosis of chorioamnionitis), pruritus, and urinary retention. Contraindications: maternal coagulopathy, infection at the insertion site, and untreated maternal hypovolemia.

Episiotomy

– A surgical incision of the perineum to enlarge the vaginal outlet during delivery. Two types: mediolateral (most common worldwide) and median/midline. Previously performed routinely; current evidence supports selective use. Indications include operative vaginal delivery, fetal distress requiring expedited delivery, and shoulder dystocia. Classified with perineal lacerations as first degree (vaginal mucosa only), second degree (mucosa and perineal muscles), third degree (involving the anal sphincter complex), or fourth degree (extending through the rectal mucosa). Healing occurs over 2-4 weeks. Complications include pain, infection, dyspareunia, and wound dehiscence. The mediolateral episiotomy is preferred over midline because it is associated with a lower risk of third- and fourth-degree tears involving the anal sphincter. Modern obstetrics recommends restrictive rather than routine use of episiotomy.

Episiotomy Repair

– Surgical repair of an episiotomy incision after delivery. Technique: continuous suturing of the vaginal mucosa (2-0 chromic), interrupted sutures for the perineal muscles, and subcuticular closure of the perineal skin. The rectum should be examined after repair (suture inadvertently through the rectal mucosa). Mediolateral episiotomy is repaired in the same fashion. Pain management: ice packs, topical anesthetics, sitz baths, NSAIDs. Avoid prolonged sitting and constipation.

Erythromycin Ophthalmic Prophylaxis

– Erythromycin 0.5% ophthalmic ointment administered to all newborns within 1 hour of birth to prevent ophthalmia neonatorum (neonatal conjunctivitis) from Neisseria gonorrhoeae and Chlamydia trachomatis. Instilled into both conjunctival sacs. Silver nitrate (Credé's method) was used historically but is less effective for chlamydia. The prophylaxis is required by law in most US states. Does not prevent nasopharyngeal colonization.

Estimated Due Date (EDD)

– The expected date of delivery, calculated as 40 weeks (280 days) from the first day of the last menstrual period. Confirmed or adjusted by first-trimester ultrasound crown-rump length measurement (most accurate ±5-7 days). Second-trimester ultrasound is less accurate (±10-14 days). Third-trimester is least accurate (±21-30 days). Only 5% of women deliver on their exact EDD; most deliver within 2 weeks before or after. Used to determine gestational age for prenatal screening, timing of interventions, and postterm management.

Estimated Fetal Weight (EFW)

– An ultrasound-derived estimate of fetal weight, calculated using formulas that combine BPD, HC, AC, and FL. The Hadlock formula (using HC, AC, FL) is the most commonly used. Accuracy: ±10-15%. Indications: assess fetal growth, determine size for gestational age, and plan delivery. EFW <10th percentile: SGA/IUGR. EFW >90th percentile: LGA/macrosomia. EFW >4500 g may influence delivery decisions (consider cesarean).

External Cephalic Version (ECV)

– A procedure to manually rotate a breech or transverse fetus to a cephalic (head-down) presentation from outside the abdomen. Performed after 36-37 weeks (term, with adequate amniotic fluid). Success rate: 30-80% (higher with prior vaginal delivery, transverse lie, adequate fluid, and non-anterior placenta). Contraindications: non-reassuring fetal status, oligohydramnios, uterine anomaly, multiple gestation, and contraindications to vaginal delivery. Fetal monitoring before and after. RhoGAM for Rh-negative patients. Tocolysis (terbutaline) may increase success.

Extrauterine Pregnancy

– A pregnancy in which the fertilized ovum implants outside the uterine cavity, an alternative term for ectopic pregnancy. By far the most common site is the fallopian tube (ampulla 70%, isthmus 12%, fimbria 11%, cornual/interstitial 2–4%). Other extrauterine sites: ovary, cervix, cesarean scar, abdominal cavity, and the rudimentary horn of a bicornuate uterus. Risk factors: prior ectopic pregnancy, tubal surgery, pelvic inflammatory disease, endometriosis, smoking, assisted reproductive technology, and IUD in situ. Presentation: abdominal pain, amenorrhea, vaginal bleeding–classic triad; rupture causes hemoperitoneum, hypovolemic shock. Diagnosis: quantitative beta-hCG and transvaginal ultrasound. Treatment: medical (methotrexate for unruptured, low hCG, no fetal cardiac activity) or surgical (salpingectomy or salpingostomy by laparoscopy).

Face Presentation

– A fetal presentation where the neck is hyperextended and the face is the presenting part. Incidence: 1:500-1:1000. Types: mentum anterior (chin toward pubis, delivers vaginally), mentum posterior (chin toward the sacrum, cannot deliver vaginally unless rotates to anterior). Diagnosis: ultrasound or palpation (feel nose, eyes, mouth, and chin during vaginal exam). Management: if mentum anterior, labor may proceed; if mentum posterior and persistent, cesarean. Preterm and anencephalic fetuses have a higher rate of face presentation.

Failed Induction of Labor

– The inability to achieve active labor and progressive cervical change after an adequate trial of induction. Definition: no active labor (cervical change) after ≥12-18 hours of oxytocin with ruptured membranes and adequate contractions. Management: the latent phase may be prolonged (up to 24 hours or more). If the cervix does not change after adequate contractions with a favorable Bishop score and ROM, cesarean delivery is indicated. Failed induction is more common in nulliparas and with unfavorable cervical status at the beginning.

Fecundity

– The biological ability to conceive and produce a live birth. Fecundity is influenced by age (declines significantly after 35), ovarian reserve, tubal patency, uterine factors, and male fertility (sperm count, motility, morphology). Fecundability: the probability of achieving pregnancy in a single menstrual cycle. Normal fecundability is approximately 20-25% per cycle in healthy couples. Infertility is defined as failure to conceive after 12 months of regular unprotected intercourse.

Femur Length (FL)

– The length of the fetal femur (diaphysis), measured from the greater trochanter to the distal femoral metaphysis. Used for dating (14-26 weeks) and growth assessment. Normal growth: 2 mm per week. Short femur: can be constitutional (especially with short parents), associated with aneuploidy (trisomy 21, skeletal dysplasias), or IUGR. The FL/AC ratio helps differentiate IUGR (increased ratio) from constitutional smallness.

Fertilisation

– The union of a sperm and an egg (oocyte) to form a zygote, the first cell of a new individual. Fertilization occurs in the ampulla of the fallopian tube and involves: capacitation of sperm, binding of sperm to the zona pellucida, acrosome reaction, fusion of sperm and oocyte plasma membranes, cortical reaction (prevents polyspermy), completion of the second meiotic division of the oocyte, formation of the male and female pronuclei, and syngamy (fusion of pronuclei). The resulting zygote then begins cell division (cleavage) as it travels down the fallopian tube to implant in the uterus.

Fertility

– The biological capacity to conceive and produce offspring. Fertility depends on the proper functioning of the reproductive system in both males and females, including ovulation, sperm production, and hormonal regulation. Infertility affects a significant proportion of couples and may require medical evaluation and assisted reproductive technologies such as in vitro fertilization (IVF).

Fetal Acid-Base Status

– The assessment of fetal oxygenation and acid-base balance during labor, reflecting fetal metabolic condition. Normal: umbilical artery pH 7.25-7.35, pCO2 40-50 mmHg, base excess 0 to -5. Fetal acidosis: pH <7.20 (mild), pH <7.00 (severe, associated with neonatal encephalopathy). Respiratory acidosis: elevated pCO2 from cord compression (rapidly reversible). Metabolic acidosis: decreased base excess from prolonged hypoxia (associated with neurologic injury). Fetal scalp pH: direct measurement from fetal blood; pH >7.25 = normal, <7.20 = acidosis.

Fetal Alcohol Spectrum Disorder (FASD)

– A broad spectrum of physical, behavioral, and cognitive impairments caused by prenatal alcohol exposure. The most severe form is fetal alcohol syndrome (FAS): characteristic facial features (smooth philtrum, thin vermillion border, small palpebral fissures), growth deficiency (prenatal or postnatal), and central nervous system abnormalities (microcephaly, developmental delay, intellectual disability). No amount of alcohol is considered safe in pregnancy. Diagnosis: based on confirmed maternal alcohol use, dysmorphology exam, and neurodevelopmental assessment. Prevention: screening and counseling in pregnancy.

Fetal Alcohol Syndrome

– A collection of birth defects resulting from exposure of the fetus to alcohol during pregnancy, representing the most severe form of fetal alcohol spectrum disorder (FASD). Characteristic features include distinctive facial dysmorphology (smooth philtrum, thin vermillion border, small palpebral fissures), prenatal and postnatal growth deficiency, and central nervous system abnormalities (microcephaly, developmental delay, intellectual disability, behavioral problems). No amount of alcohol is considered safe during pregnancy. Diagnosis requires confirmed prenatal alcohol exposure, characteristic facial features, growth deficits, and neurodevelopmental impairment. Management is supportive with early intervention services.

Fetal Attitude

– The relationship of fetal body parts to one another, describing the degree of flexion or extension of the fetal head and extremities. Normal attitude is full flexion (vertex presentation with chin tucked to chest). attitudes include deflexion (less flexion), military (neutral position), brow, face, or complete extension. Deflexed attitudes result in larger presenting diameters, potentially complicating vaginal delivery.

Fetal Bradycardia

– A baseline fetal heart rate <110 bpm for ≥10 minutes. Moderate (100-110 bpm): often benign, especially in post-term fetuses. Severe (<100 bpm): concerning for fetal hypoxia, acidosis, congenital heart block (SLE, anti-Ro/anti-La antibodies), cord accident, placental abruption, or maternal seizures. A prolonged deceleration lasting >2 minutes is managed as an emergency. Congenital heart block: fixed bradycardia 50-80 bpm, unrelated to contractions, diagnosed prenatally by fetal echocardiography.

Fetal Heart Rate

– Normal fetal heart rate ranges from 110-160 beats per minute. Monitored via auscultation with a fetoscope, Doppler ultrasound, or continuous electronic fetal monitoring. Baseline FHR is determined over a 10-minute window and is modulated by the autonomic nervous system. Variability (fluctuations in baseline) is classified as absent, minimal (≤5 bpm), moderate (6-25 bpm), or marked (>25 bpm) and reflects fetal acid-base status. Accelerations (transient increases of ≥15 bpm for ≥15 seconds in fetuses beyond 32 weeks) are reassuring signs of fetal well-being. Decelerations are classified as early (head compression, benign), variable (cord compression, most common), or late (uteroplacental insufficiency, concerning). Fetal tachycardia (>160 bpm) may indicate maternal fever, infection, or hypoxia; bradycardia (<110 bpm) may suggest fetal distress, congenital heart block, or maternal hypotension.

Fetal Kick Counts (Fetal Movement Counting)

– A method of antepartum fetal surveillance where the mother counts fetal movements (kicks, rolls, stretches) daily. Cardiff count (10 kicks in 2 hours): lie on the left side, record the time until 10 movements are felt. Normal: 10 movements in ≤2 hours. Decreased fetal movement is associated with increased risk of stillbirth and IUGR. A single episode of decreased fetal movement warrants NST/BPP. Maternal perception of fetal movement is generally reliable and may be the earliest sign of fetal compromise.

Fetal Lie

– The relationship of the long axis of the fetus to the long axis of the mother. Longitudinal lie (parallel, 99%) or transverse lie (perpendicular, 1%). Most common is left occiput transverse or left occiput anterior. Transverse lie at term is abnormal and requires cesarean delivery if not corrected. Lie determined by palpation and ultrasound, particularly important for planning delivery route and management of labor.

Fetal Presentation

– The relationship of the fetal presenting part to the maternal pelvis. Includes cephalic (head first, most common at 97%), breech (buttocks or feet first), shoulder/oblique, and transverse lie. Cephalic presentations include vertex, face, brow, and compound. Breech presentations include complete, frank, and footling. Presentation assessed by Leopold maneuvers and confirmed with ultrasound. External cephalic version may be attempted for breech at 36-38 weeks.

Fetal Scalp Electrode (FSE)

– A spiral wire electrode placed on the fetal scalp through the cervix to directly record the fetal heart rate. Provides a more accurate FHR signal than external Doppler, particularly in obese patients or when external monitoring is inadequate. Indications: poor-quality external tracing, need for internal monitoring. Contraindications: HIV and hepatitis C (risk of vertical transmission), placenta previa, and scalp infection at the electrode site. The electrode is placed after membrane rupture, with cervical dilatation ≥2 cm.

Fetal Scalp Stimulation

– A test of fetal well-being during labor by digitally rubbing or pinching the fetal scalp during a vaginal exam. A reactive acceleration (increase of ≥15 bpm for ≥15 seconds) in the FHR in response to stimulation is a reassuring sign of normal fetal acid-base status (negative predictive value 95% for fetal acidosis). If no acceleration, fetal scalp pH may be indicated. The test is noninvasive and can be performed repeatedly.

Fetal Tachycardia

– A baseline fetal heart rate >160 bpm for ≥10 minutes. Mild (161-180 bpm) or severe (>180 bpm). Causes: maternal fever (most common, often from chorioamnionitis), medications (terbutaline, atropine, scopolamine), maternal hyperthyroidism, fetal infection (myocarditis), fetal hypoxia (compensatory response), and fetal tachyarrhythmia (SVT, atrial flutter, 200-300 bpm). Management: identify and treat the cause (maternal cooling, antibiotics for chorioamnionitis). Persistent tachycardia with minimal variability is concerning.

Fetal Viability

– The gestational age at which a fetus has the potential to survive outside the uterus with medical support, generally considered to be 23-24 weeks of gestation. Survival rates increase significantly with each additional week: approximately 30-50% at 23 weeks, 50-70% at 24 weeks, and over 90% by 27-28 weeks. Factors influencing viability include birth weight, sex, multiple gestation, access to neonatal intensive care, and presence of congenital anomalies. The limit of viability is defined by each country's legal and ethical framework. In the UK, the upper limit for termination of pregnancy is 24 weeks for most grounds. Prognosis for intact survival without severe neurodevelopmental impairment remains guarded at the earliest gestational ages, with approximately 40% of survivors at 23 weeks having moderate to severe disability.

Fetomaternal Hemorrhage

– Transplacental passage of fetal red blood cells into the maternal circulation. Occurs in small amounts during normal pregnancy but can be massive during trauma, placental abruption, or cesarean delivery.

Fetus

– An unborn offspring from the 9th week of gestation (after the embryonic period) until birth. The fetal period is characterized by rapid growth, maturation of organ systems, and increasing complexity of function. Key fetal milestones: 12 weeks — sex distinguishable, kidneys begin producing urine; 20 weeks — fetal movements felt by mother (quickening); 24 weeks — viability (potential to survive outside the womb with intensive neonatal support); 28-32 weeks — lung surfactant production increases; 37-42 weeks — term.

Fibrocystic Breast Changes

– A benign condition characterized by lumpy, painful breast tissue, most common in women aged 20-50, driven by hormonal fluctuations during the menstrual cycle (estrogen and progesterone). Microscopic features include stromal fibrosis, cyst formation (macrocysts and microcysts), and adenosis (increased glandular tissue). Palpable breast lumps may be tender and fluctuate in size with the menstrual cycle. Diagnosis relies on clinical breast exam, mammography (showing dense breast tissue and cysts), and ultrasound (anechoic simple cysts), with biopsy (core needle or fine-needle aspiration) reserved for suspicious or complex cysts or solid areas to exclude malignancy. Fibrocystic changes do not increase breast cancer risk except when associated with proliferative changes with atypia (columnar cell change with atypia or atypical ductal hyperplasia), which warrants increased surveillance. Management includes supportive bras, NSAIDs, oral contraceptives (suppressing hormonal fluctuations), and avoiding dietary methylxanthines (caffeine) if symptomatic, though evidence for dietary modification is weak.

Fibroids, Cysts, and Polyps

– Benign growths that can develop in the reproductive organs. Fibroids are smooth muscle tumors of the uterus; cysts are fluid-filled sacs that commonly form on the ovaries; and polyps are tissue overgrowths in the endometrium or cervix. Although typically noncancerous, they can cause symptoms such as pain, abnormal bleeding, and fertility issues, often requiring monitoring or surgical removal.

Fimbriae

– The finger-like projections at the distal end of each fallopian tube that sweep over the ovary during ovulation to capture the released egg and guide it into the fallopian tube. The fimbriae are lined with ciliated epithelium that creates a current to draw the egg into the tube. At the center of the fimbriae is the abdominal ostium, the opening into the fallopian tube. Damage to the fimbriae from infection (pelvic inflammatory disease, chlamydia), endometriosis, or surgery can impair oocyte pickup and cause infertility.

First Stage of Labor

– From the onset of regular contractions to full cervical dilatation. Two phases: latent (0-6 cm, slow progress, 6-12 hours in nulliparas) and active (6-10 cm, faster progress, 1-2 cm/hour in nulliparas). Active phase begins at 6 cm (the Friedman curve was revised: the old 4 cm threshold was updated to 6 cm by ACOG/NICHD in 2014). Arrest of first stage: no cervical change for ≥4 hours with adequate contractions or ≥6 hours with inadequate contractions.

First Trimester

– The period from conception to 12 weeks gestational age. Characterized by rapid embryonic organogenesis, rising hCG levels, and establishment of the placenta. Common symptoms include nausea, vomiting, fatigue, and breast tenderness. Dating ultrasound performed to confirm gestational age. Risk of miscarriage highest during this trimester. Prenatal vitamins with folic acid recommended.

Folic Acid Supplementation

– Daily folic acid (400-800 mcg) taken before conception and during the first trimester to prevent neural tube defects (NTDs: spina bifida, anencephaly, encephalocele). Folic acid is a B vitamin (folate) essential for DNA synthesis and cell division. Higher doses (4-5 mg daily) are recommended for women with a prior NTD-affected pregnancy, diabetes, obesity, or those taking antiepileptic medications (valproate, carbamazepine). NTDs occur at 21-28 days post-conception, often before pregnancy is recognized, making preconceptional supplementation critical.

Forceps-Assisted Delivery

– Operative vaginal delivery using forceps (two curved blades applied to the fetal head). Types: outlet (forceps applied at the introitus, fetal scalp visible), low (station ≥+2, not at outlet), midpelvic (station ≥+2 but head above ischial spines), and rotational (forceps used to rotate from OP to OA). Most common types: Simpson (for molded heads), Elliot (for unmolded heads), and Piper (for aftercoming head in breech delivery). Prerequisites: full dilatation, ROM, engaged head, known position, adequate pelvis, and consent. Neonatal risks: facial nerve palsy, brachial plexus injury, bruising, and cephalohematoma. Maternal risks: third/fourth-degree lacerations (higher than vacuum).

Fourth-Degree Tear

– The most severe type of perineal tear during childbirth, extending through the anal sphincter (both internal and external) and into the rectal mucosa. Fourth-degree tears occur in approximately 0.5-1% of vaginal deliveries and are associated with forceps delivery, shoulder dystocia, large baby, and episiotomy. Repair requires expert surgical reconstruction in the operating room: layered closure of rectal mucosa, internal anal sphincter, external anal sphincter, and perineal muscles. Complications: anal incontinence (flatus and fecal), rectovaginal fistula, and dyspareunia.

Friedman Curve

– The classic labor curve described by Emanuel Friedman in the 1950s, plotting cervical dilatation against time. Defined the latent and active phases, with active phase beginning at 4 cm. The Friedman curve was the standard for labor management for decades. Modern revisions (Consortium on Safe Labor, Zhang 2010) showed that labor progresses more slowly than Friedman described, particularly in nulliparas and with epidural anesthesia. The active phase now starts at 6 cm. The new paradigm recommends more patience before diagnosing arrest of labor.

Fundal Height

– The distance from the symphysis pubis to the top of the uterus measured in centimeters. Used to estimate gestational age and fetal growth. Generally correlates with weeks of gestation between 20-36 weeks (e.g., 24 cm at 24 weeks). Discrepancies may suggest fetal growth restriction, macrosomia, polyhydramnios, or multiple gestation. Measured with patient supine using a tape measure.

Galactocele

– A milk-filled retention cyst of the breast, occurring during lactation or after weaning. Results from obstruction of a milk duct. Presents as a painless, fluctuant breast mass. Diagnosis by ultrasound showing a well-defined, hypoechoic or anechoic mass. Treatment includes aspiration. May become infected, requiring antibiotics. Recurrence is uncommon after complete drainage.

Galactogogue

– A substance that promotes breast milk production.

Gaskin Maneuver (All-Fours Position)

– A maneuver for shoulder dystocia resolution: the mother is rolled onto her hands and knees (all-fours position). Gravity and the change in pelvic dimensions (increased anteroposterior diameter of the pelvic outlet) may dislodge the impacted shoulder. The posterior arm may also be delivered more easily. Success rate: 25-50% after failed McRoberts. If the posterior shoulder delivers, the anterior shoulder usually follows.

Genetic Sonogram

– A targeted ultrasound performed at 18-22 weeks that combines standard anatomic survey with specific soft markers for aneuploidy. Findings assessed: nuchal fold thickened ($\geq$6 mm), echogenic intracardiac focus (EIF), echogenic bowel, renal pyelectasis ($\geq$4 mm anterior-posterior), short femur/humerus, choroid plexus cyst, and sandal gap. The presence of multiple soft markers increases the risk of aneuploidy. The genetic sonogram is used for risk adjustment when combined with maternal age and serum screening (likelihood ratio for each marker). Markers are nonspecific and many are normal variants in euploid fetuses.

Genitals

– The external and internal reproductive organs. Female genitals: external (vulva — mons pubis, labia majora, labia minora, clitoris, vaginal opening, Bartholin glands) and internal (vagina, cervix, uterus, fallopian tubes, ovaries). Male genitals: external (penis, scrotum) and internal (testes, epididymis, vas deferens, seminal vesicles, prostate, bulbourethral glands). The genitals develop from common embryological structures (genital tubercle, urogenital sinus, Wolffian and Müllerian ducts) under the influence of sex-determining genes and hormones.

Gestation

– The period of development of a fetus from fertilization to birth. Human gestation averages 280 days (40 weeks) from the first day of the last menstrual period (LMP) or 266 days (38 weeks) from conception. Trimesters: first (weeks 1-12: organogenesis, highest vulnerability to teratogens), second (weeks 13-27: fetal growth and maturation, quickening at 18-22 weeks), third (weeks 28-40: rapid weight gain, lung maturation, acquisition of immune competence). Gestational age is assessed by: LMP, early ultrasound (crown-rump length at 11-13 weeks most accurate), symphysis-fundal height measurement after 20 weeks, and clinical assessment. Gestational age categories: preterm (<37 weeks), term (37-41 weeks), post-term (>41 weeks).

Gestational Age

– The age of pregnancy measured in weeks from the first day of the last menstrual period (LMP). Also called menstrual age. Typically 2 weeks longer than fertilization age. Calculated using LMP or firsttrimester ultrasound. Used to estimate due date (40 weeks from LMP). Preterm is less than 37 weeks; post-term is beyond 42 weeks. Key metric for fetal development milestones and management decisions.

Gestational Diabetes

– Glucose intolerance first diagnosed during pregnancy, affecting 6-9% of pregnancies. Risk factors include obesity, advanced maternal age, family history, and prior gestational diabetes. Complications include macrosomia, birth trauma, neonatal hypoglycemia, and preeclampsia. Screening by 50g glucose challenge test at 24-28 weeks, with confirmatory 100g oral glucose tolerance test. Management includes dietary modification, glucose monitoring, and insulin therapy when diet fails.

Gestational Hypertension

– New-onset hypertension (BP ≥140/90 mmHg) after 20 weeks of gestation in a woman with previously normal blood pressure, without proteinuria or other features of preeclampsia. Affects 6-8% of pregnancies. Management: monitoring blood pressure, screening for preeclampsia progression (proteinuria, labs, symptoms). Delivery at term (37-40 weeks) is typical. Some patients progress to preeclampsia. Treatment with antihypertensives (labetalol, nifedipine, methyldopa) is recommended for BP ≥160/110 mmHg.

Gestational Sac

– The first sonographic sign of an intrauterine pregnancy, visible on transvaginal ultrasound at approximately 4.5-5 weeks gestational age. A fluid-filled structure within the decidua, surrounded by the chorionic ring. Mean sac diameter (MSD) correlates with gestational age. The yolk sac is visible within the gestational sac at 5-6 weeks, and the embryo with cardiac activity at 6-7 weeks. Abnormal findings: empty gestational sac without yolk sac (pseudogestational sac in ectopic, anembryonic pregnancy).

Gestational Trophoblastic Disease

– A spectrum of conditions arising from abnormal trophoblast proliferation, including complete and partial hydatidiform moles and gestational choriocarcinoma. Complete mole has 46,XX karyotype (all paternal origin); partial mole is triploid (69 chromosomes). Presents with elevated hCG, uterus large for dates, and vaginal bleeding. Diagnosis by ultrasound (snowstorm pattern) and histology. Managed by uterine evacuation and hCG monitoring.

Gonadotrophins

– Hormones that stimulate the gonads (ovaries and testes). In women, the pituitary gonadotrophins FSH (follicle-stimulating hormone) and LH (luteinizing hormone) regulate the menstrual cycle: FSH stimulates follicular growth and estrogen production; LH triggers ovulation and supports the corpus luteum. In men, FSH supports spermatogenesis and LH stimulates testosterone production. GnRH (gonadotropin-releasing hormone) from the hypothalamus controls pituitary gonadotrophin secretion. Exogenous gonadotrophins are used in assisted reproduction for controlled ovarian hyperstimulation (e.g., recombinant FSH, hMG, hCG).

Group B Streptococcus

– A gram-positive bacterium (Streptococcus agalactiae) colonizing the genitourinary and gastrointestinal tracts. Asymptomatic vaginal colonization occurs in 20-30% of pregnant women. Screening is performed at 36-37 weeks gestation via rectovaginal swab culture. Intrapartum antibiotic prophylaxis (IV penicillin G or ampicillin) is indicated for women with positive GBS screening, prior infant with GBS disease, or GBS bacteriuria during the current pregnancy. Penicillin-allergic women receive cefazolin (if low risk) or clindamycin/vancomycin (if high risk). Prophylaxis reduces early-onset neonatal GBS disease (sepsis, pneumonia, meningitis) by

80%. Early-onset GBS disease occurs in the first 7 days of life and has a mortality rate of 5-10% in term infants.

Growth Scan

– An ultrasound performed in the third trimester (usually 28-32 weeks and/or 36-38 weeks) to assess fetal growth. Measurements: BPD, HC, AC, FL, and estimated fetal weight (EFW). Growth percentiles plotted on a growth curve. Abnormal: <10th percentile (SGA/IUGR) or >90th percentile (LGA/macrosomia. Amniotic fluid volume and Doppler studies (umbilical artery, MCA) often included. Indications: suspected growth abnormality, multiple gestation, maternal diabetes, hypertension, prior IUGR, and postterm surveillance.

Gynaecologist

– A medical doctor who specializes in the health of the female reproductive system (vagina, cervix, uterus, fallopian tubes, ovaries) and breasts. Gynecologists diagnose and treat conditions including menstrual disorders, pelvic pain, endometriosis, fibroids, ovarian cysts, pelvic organ prolapse, incontinence, infections, and gynecologic cancers. They perform procedures including colposcopy, hysteroscopy, laparoscopy, and surgery. Many gynecologists also practice obstetrics (obstetrician-gynecologists, OB/GYNs). Subspecialties: gynecologic oncology, reproductive endocrinology and infertility, urogynecology, and pediatric/adolescent gynecology.

gynecology

– The medical specialty concerned with the health of the female reproductive system (uterus, ovaries, fallopian tubes, cervix, vagina, and vulva).

Haemolytic Disease of the Newborn

– See blood-related entries in hematology.

Hand Prolapse

– Prolapse of a fetal hand or arm alongside or below the presenting part. May occur spontaneously or after manual rotation. The prolapsed arm may be mistaken for a cord or foot. Management: if the hand is above the presenting part (compound presentation), it may be gently pushed back during contractions. If it presents beside the head after full dilatation, spontaneous delivery may occur, but rotation may be impeded. If vaginal delivery is obstructed, cesarean is performed.

Head Circumference (HC)

– The circumference of the fetal skull, measured at the same level as BPD (thalami, cavum septum pellucidum). Used for dating (second trimester) and for assessing head growth. HC is more reliable than BPD for dating when head shape is abnormal. Microcephaly: HC <3rd percentile (workup for infection, aneuploidy, genetic syndromes). Macrocephaly: HC >97th percentile.

HELLP Syndrome

– A severe variant of preeclampsia characterized by Hemolysis, Elevated Liver enzymes, and Low Platelets. Occurs in 10-20% of patients with severe preeclampsia. Diagnostic criteria include haemolysis (LDH >600 U/L, schistocytes on peripheral smear), elevated liver enzymes (AST ≥70 U/L), and low platelets (<100,000/µL). HELLP can develop before or after delivery and may present without classic preeclampsia features (15-20% of cases). Symptoms include right upper quadrant or epigastric pain, nausea, vomiting, headache, and malaise. Complications include DIC, placental abruption, hepatic rupture or infarction, acute kidney injury, pulmonary edema, and maternal death. Management: delivery is indicated regardless of gestational age. Corticosteroids (dexamethasone) may improve platelet count but are not recommended as routine treatment. Plasma exchange may be considered for persistent thrombocytopenia postpartum.

HELPERR Mnemonic

– A mnemonic for the sequential management of shoulder dystocia: H — Help (call for additional personnel, anesthesia, and pediatrics). E — Evaluate for episiotomy (generous episiotomy to provide more room). L — Legs (McRoberts maneuver: hyperflex maternal legs onto the abdomen). P — Suprapubic pressure (downward pressure on the maternal suprapubic area to dislodge the anterior shoulder). E — Enter internal rotation (Rubin II or Woods screw maneuver: rotate the fetal shoulders 180 degrees). R — Remove the posterior arm (sweep the posterior arm across the chest and deliver it). R — Roll the patient (Gaskin maneuver: onto all fours). If all maneuvers fail: Zavanelli maneuver (replace the fetal head and proceed with emergency cesarean).

Hematologic Changes in Pregnancy

– Plasma volume increases 40-50% while red cell mass increases 20-30%, resulting in physiologic anemia (Hb 10-11 g/dL). Leukocytosis (WBC up to 15,000-16,000/µL, predominantly neutrophils) is normal. The coagulation cascade shifts to a prothrombotic state: increased fibrinogen (factor I, 300-600 mg/dL), factors VII, VIII, IX, X, XII, and von Willebrand factor. Decreased protein S activity (free S decreases). The overall effect is an increased risk of venous thromboembolism (5-10 fold) and the utility of DVT prophylaxis in certain populations.

hemorrhage

– Excessive or severe bleeding. In obstetrics, hemorrhage is classified by timing: antepartum hemorrhage (bleeding after 24 weeks of pregnancy before birth, causes include placenta praevia, placental abruption, vasa praevia) and postpartum hemorrhage (bleeding >500 mL after vaginal delivery or >1000 mL after cesarean, causes include uterine atony, retained placenta, genital tract trauma, coagulopathy). Postpartum hemorrhage is a leading cause of maternal mortality worldwide. Management: uterine massage, oxytocin, ergometrine, carboprost, tranexamic acid, intrauterine balloon tamponade, uterine compression sutures (B-Lynch), and emergency hysterectomy.

Hepatitis B Vaccine (Birth Dose)

– The first dose of the hepatitis B vaccine is recommended within 24 hours of birth for all medically stable infants. Administered IM in the vastus lateralis. Prevents perinatal transmission of HBV. For infants born to HBsAg-positive mothers: HBIG (0.5 mL IM) is given within 12 hours of birth along with the vaccine. The three-dose series (0, 1-2, and 6-18 months) provides >95% protection. Universal vaccination has dramatically reduced childhood HBV infection.

Herpetic Whitlow

– A painful viral infection of the finger caused by the herpes simplex virus (typically HSV-1), characterized by swelling, redness, and painful vesicular lesions on the finger. In obstetrics, healthcare workers (midwives, obstetricians) are at risk when attending deliveries of women with genital herpes, as the virus can enter through breaks in the skin of the gloved hand. Treatment: oral antiviral medication (acyclovir, valacyclovir). Prevention: wearing double gloves during vaginal examinations and deliveries of women with active genital herpes lesions.

Heterotopic Pregnancy

– Simultaneous intrauterine and ectopic pregnancy. Rare in spontaneous conception (1:30,000) but more common with ART (1:100-1:500). Risk factors: ovulation induction, IVF with multiple embryo transfer. The presence of an intrauterine pregnancy does not exclude an ectopic pregnancy. Diagnosis: ultrasound showing intrauterine and extrauterine gestational sacs. Treatment: surgical removal of the ectopic pregnancy (salpingectomy) with preservation of the intrauterine pregnancy (which can proceed normally).

High-Dependency Unit

– A ward or area within a hospital that provides a level of care between a general ward and an intensive care unit, for patients who require intensive observation or treatment but not mechanical ventilation. In obstetrics, women may be admitted to HDU for: severe pre-eclampsia, postpartum hemorrhage, sepsis, or medical complications in pregnancy. HDU provides continuous monitoring (pulse oximetry, ECG, invasive blood pressure), one-to-one or one-to-two nursing, and the ability to administer vasoactive drugs or blood products.

HRT

– Hormone Replacement Therapy, the use of estrogen (with or without progesterone) to treat menopausal symptoms caused by declining ovarian hormone production. HRT is the most effective treatment for vasomotor symptoms (hot flushes, night sweats), prevents bone loss (osteoporosis), and improves urogenital atrophy. Routes: oral tablets, transdermal patches/gels, vaginal preparations. Estrogen alone is used in women after hysterectomy; combined estrogen-progestogen is used in women with an intact uterus to prevent endometrial hyperplasia/cancer. The timing hypothesis: HRT initiated within 10 years of menopause has cardiovascular benefits; late initiation increases cardiovascular and thromboembolic risks.

Human Chorionic Gonadotropin

– A glycoprotein hormone produced by trophoblast cells after implantation. Detectable in serum 8-11 days post-conception and in urine 12-14 days. Levels double approximately every 48-72 hours in early pregnancy, peaking at 8-11 weeks then plateauing. Used in pregnancy testing, ectopic pregnancy diagnosis (abnormal rise), and molar pregnancy monitoring. Beta-hCG subunit measured quantitatively for clinical decisions.

Hydatidiform Mole

– A type of gestational trophoblastic disease characterized by abnormal proliferation of trophoblastic tissue. Complete mole: 46,XX (all paternal), no fetal tissue, diffuse vesicular swelling; risk of malignant transformation (15-20%). Partial mole: triploid (69,XXX), some fetal tissue, focal swelling; lower risk. Presents with vaginal bleeding in first trimester, excessive uterine size, theca lutein cysts, hyperemesis, and very high hCG. Diagnosis: ultrasound (snowstorm appearance, no fetal parts), markedly elevated hCG. Treatment: suction curettage with careful hCG monitoring until undetectable. Chemotherapy (methotrexate, actinomycin D) for persistent or malignant disease.

Hydrocele

– An accumulation of fluid within the tunica vaginalis surrounding the testicle, causing scrotal swelling.

Hydrops Fetalis

– Abnormal accumulation of fluid in two or more fetal compartments, including pleural effusion, pericardial effusion, ascites, and subcutaneous edema. Causes include Rh isoimmunization, chromosomal abnormalities, cardiac defects, infections (parvovirus B19), and twin-to-twin transfusion syndrome. Often associated with severe fetal anemia or cardiac failure. Mortality rate high; management depends on etiology. May require intrauterine transfusion.

Hymen

– A thin fold of mucous membrane that partially covers the external opening of the vagina. The hymen is a remnant of fetal development, with no known physiological function in postnatal life. Shape and configuration vary widely: annular (most common), crescentic, cribriform (multiple small openings), septate, redundant, imperforate (complete closure, requiring surgical incision). The hymen typically has a central or crescentic opening of variable size that allows passage of menstrual blood. Rupture (tearing) of the hymen was traditionally associated with first vaginal intercourse (defloration), but the hymen can be stretched or torn by: tampon use, vigorous exercise, horseback/bicycle riding, gynaecological examination, or trauma. Its appearance does not reliably indicate virginity or sexual activity. Clinical significance: (1) Imperforate hymen: presents in adolescents with primary amenorrhoea and cyclic pelvic pain due to haematocolpos (accumulated menstrual blood in the vagina), treated by hymenotomy (cruciate incision). (2) Hymenal tags or polyps. (3) Hymenal tears in sexual abuse cases (forensic examination). (4) Hymenal stenosis or scarring. The concept of the hymen as a marker of virginity is a cultural and social construct, not a reliable anatomical indicator.

Hyperemesis Gravidarum

– Severe, persistent nausea and vomiting during pregnancy resulting in weight loss greater than 5%, dehydration, electrolyte imbalances, and ketosis. Occurs in 0.3-2% of pregnancies, typically presenting in the first trimester. Differentiated from normal morning sickness by severity and metabolic derangement. Management includes IV fluids, antiemetics (ondansetron, promethazine), thiamine supplementation to prevent Wernicke encephalopathy, and dietary modification. May require hospitalisation for IV hydration and electrolyte correction. Risk factors include female fetus, multiple gestation, molar pregnancy, and history of hyperemesis in a prior pregnancy. Complications include Wernicke encephalopathy (from thiamine deficiency), hypokalaemia, hyponatraemia, and vitamin deficiencies. Prognosis is generally good with symptoms improving by 16-20 weeks, though some women experience symptoms throughout pregnancy.

Hyperprolactinaemia

– Elevated levels of the hormone prolactin in the blood, above the normal range (typically >500 mIU/L in women). Causes: physiological (pregnancy, breastfeeding, stress, exercise), pharmacological (antipsychotics, metoclopramide, SSRIs, verapamil), and pathological (prolactinoma — pituitary adenoma, hypothyroidism, polycystic ovary syndrome, chronic renal failure). In women, hyperprolactinaemia causes galactorrhoea (milk production), oligomenorrhea or amenorrhea, anovulation, and infertility. Diagnosis: serum prolactin measurement (fasting, morning, avoid venipuncture stress), thyroid function tests, and pituitary MRI for macroprolactinoma. Treatment: dopamine agonists (cabergoline first-line due to better tolerability, bromocriptine second-line).

Hysterectomy

– The surgical removal of the uterus, the second most common major surgical procedure in women after cesarean section in the United States. The types of hysterectomy are classified as total (complete removal of the uterus and cervix, the most common), subtotal or supracervical (removal of the uterine body, leaving the cervix in place, preserving pelvic support and potentially reducing operative complications but requiring continued cervical cancer screening), radical (removal of the uterus, cervix, upper vagina, parametrial tissue, and pelvic lymph nodes, reserved for cervical and uterine malignancies), and total hysterectomy with bilateral salpingooophorectomy (removal of the uterus, cervix, fallopian tubes, and ovaries, inducing surgical menopause).

Hysterectomy and Surgical Menopause

– Hysterectomy is the surgical removal of the uterus, and when the ovaries are also removed (oophorectomy), it induces surgical menopause regardless of age. Surgical menopause causes an abrupt drop in estrogen levels, leading to more severe menopausal symptoms than natural menopause, including hot flashes, bone loss, and increased cardiovascular risk. Hormone replacement therapy is often recommended for younger women to mitigate these effects.

Hysterosalpingo-Contrast-Sonography

– An ultrasound-based technique for evaluating the uterine cavity and fallopian tube patency. Sterile saline or contrast medium is injected through the cervix while transvaginal ultrasound is performed. Advantages over HSG: no ionizing radiation, can be performed in clinic, assesses the uterine cavity (polyps, fibroids, adhesions) and tubal patency simultaneously. Disadvantages: less detailed tubal assessment than X-ray HSG, operator-dependent. Used in infertility workup to screen for tubal occlusion before proceeding to IVF.

Hysterosalpingogram (HSG)

– An X-ray procedure used to evaluate the uterine cavity and fallopian tube patency. A radiopaque contrast medium is injected through the cervix, and serial X-ray images are taken as the contrast fills the uterine cavity and spills from the fallopian tubes. Indications: infertility workup (tubal factor), recurrent miscarriage (uterine cavity assessment), and evaluation after tubal sterilization or tubal surgery. Findings: normal (smooth uterine cavity, free spill from both tubes), tubal occlusion (proximal or distal), hydrosalpinx (dilated, blocked tube), uterine filling defects (polyps, fibroids, adhesions, septate uterus).

Implantation

– The embedding of the blastocyst into the endometrium, occurring approximately 6-10 days after ovulation. The blastocyst hatches from the zona pellucida, and the trophoblast invades the decidua. Implantation triggers hCG production, which maintains the corpus luteum. Failure of implantation is a common cause of early pregnancy loss. Ectopic implantation occurs outside the uterus, most commonly in the fallopian tube.

Induction of Labor

– The artificial initiation of uterine contractions before the onset of spontaneous labor. Indications: postterm pregnancy (>41-42 weeks), hypertensive disorders (preeclampsia, gestational hypertension), maternal medical conditions (diabetes, renal disease), fetal compromise (IUGR, oligohydramnios), PROM, chorioamnionitis, and elective reasons (shared decision-making after 39 weeks). Methods: cervical ripening agents (prostaglandins, mechanical dilation with Foley balloon), followed by oxytocin if needed. Contraindications: placenta previa, vasa previa, transverse lie, prior classical cesarean, active genital herpes, and absolute cephalopelvic disproportion.

Infant Sleep Position

– The supine sleep position (back to sleep) is recommended by the AAP to reduce the risk of sudden infant death syndrome (SIDS). The Back to Sleep campaign (1994) reduced SIDS rates by >50%. Always place the baby on the back for sleep (naps and nighttime). Use a firm sleep surface with no soft bedding, pillows, bumpers, or stuffed animals. Room-sharing (same room, separate bed) is recommended for the first 6-12 months. Prone and side positions increase SIDS risk. Tummy time when awake promotes motor development and prevents positional plagiocephaly.

Infertility

– Inability to conceive after 12 months of regular unprotected intercourse (6 months if woman >35). Affects 15% of couples. Female causes: ovulation disorders (PCOS, hypothalamic amenorrhea), tubal factor (blockage, PID, endometriosis), uterine (fibroids, polyps, adhesions), and diminished ovarian reserve. Male causes: low sperm count/motility, varicocele, and hormonal dysfunction. Evaluation: ovarian reserve testing, semen analysis, hysterosalpingogram, and hormonal profiling. Treatment: ovulation induction, IUI, IVF, and donor gametes/embryos.

Influenza Vaccination in Pregnancy

– Annual inactivated influenza vaccine recommended for all pregnant women. Pregnancy increases the risk of severe influenza (pneumonia, ICU admission, death) due to changes in immune function and cardiopulmonary physiology. Maternal vaccination also protects the newborn during the first 6 months of life (transplacental antibody transfer) and reduces the risk of influenza-like illness in the infant by 50-70%. The vaccine is safe at any trimester.

Intensive Care Unit

– A specialist hospital unit providing continuous monitoring and life support for critically ill patients. In obstetrics, women may require ICU admission for: severe pre-eclampsia or eclampsia, massive postpartum hemorrhage, amniotic fluid embolism, sepsis, cardiac disease in pregnancy, or complications of anesthesia. ICU provides mechanical ventilation, continuous hemodynamic monitoring (invasive arterial pressure, central venous pressure), renal replacement therapy, and vasoactive drug support. Multidisciplinary care involves intensivists, obstetricians, and midwives.

Intermittent Auscultation (IA)

– The periodic auscultation of the fetal heart rate with a fetoscope or Doppler ultrasound during labor, without continuous electronic monitoring. Protocol: auscultate after a contraction for 60 seconds, during the active phase every 15-30 minutes in the first stage and every 5-15 minutes in the second stage. Used for low-risk pregnancies (no meconium, normal labor progress, no maternal risk factors). Advantages: less restricted mobility, lower intervention rate. Disadvantages: requires one-to-one nursing, may miss subtle changes. ACOG supports IA for low-risk patients.

Internal Podalic Version

– A maneuver where the operator inserts a hand into the uterus, grasps one or both fetal feet, and converts a transverse lie or cephalic presentation to a breech, then delivers by total breech extraction. Indication: delivery of the second twin (transverse lie) or severe fetal distress in the second stage. Rare in modern obstetrics. Contraindications: contracted pelvis, uterine scar, ruptured uterus, and fetal distress from other causes.

Interstitial (Cornual) Pregnancy

– A rare ectopic pregnancy (2-4%) implanted in the intramural portion of the fallopian tube within the uterine wall. Higher morbidity than tubal ectopic because the myometrial layer expands, allowing later rupture (12-16 weeks) with massive hemorrhage. Risk factors: prior ipsilateral salpingectomy, IVF. Diagnosis: ultrasound (eccentric gestational sac surrounded by <5 mm myometrium), interstitial line sign. Treatment: cornual resection (wedge resection of the uterine horn) or hysterectomy for rupture. Methotrexate may be considered for early diagnosis with low hCG.

Intimate Partner Violence (IPV) in Pregnancy

– Physical, sexual, or psychological abuse by a current or former partner during pregnancy. Prevalence: 3-9% of pregnancies. Risk factors: unplanned pregnancy, late prenatal care, substance use, and young age. IPV increases the risk of miscarriage, preterm birth, low birth weight, placental abruption, and maternal death. Screening: universal screening with validated tools (HITS, AAS, PRISMA). All positive screens require safety assessment, referral to domestic violence resources (hotline, shelter), and documentation. Mandatory reporting varies by state.

Intrahepatic Cholestasis of Pregnancy (ICP)

– A pregnancy-specific liver disorder characterized by pruritus (palms and soles, often nocturnal, with no rash) and elevated serum bile acids (>10-40 µmol/L). Incidence: 0.5-2% of pregnancies. Risk factors: prior ICP, multiple gestation, hepatitis C, and high estrogen states. Onset: third trimester (after 28 weeks), resolves within 48 hours of delivery. Diagnosis: elevated total bile acids (fasting), elevated liver enzymes (AST, ALT, GGT). Fetal risks: preterm birth (spontaneous and iatrogenic), meconium-stained amniotic fluid, and stillbirth (with bile acids >40-100 µmol/L). Management: ursodeoxycholic acid (UDCA, 10-15 mg/kg/day) for pruritus, vitamin K supplementation (coagulopathy risk from fat malabsorption), and planned delivery at 36-39 weeks depending on bile acid level. Fetal surveillance advised.

Intrapartum

– The period during labor and childbirth, from the onset of regular uterine contractions leading to cervical dilation until delivery of the baby and placenta. Intrapartum care includes: monitoring maternal vital signs and fetal wellbeing (CTG), pain relief (epidural, gas and air, opioids), management of labor progress (partogram), assisted delivery if needed (forceps, ventouse), and active management of the third stage (oxytocin for placental delivery). Intrapartum complications include fetal distress, failure to progress, shoulder dystocia, and postpartum hemorrhage.

Intrauterine Fetal Demise (IUFD) / Stillbirth

– Fetal death before complete expulsion or extraction from the mother, at ≥20 weeks gestation. Affects 1:160 pregnancies in the US (25,000 annually). Causes: placental abruption, umbilical cord accidents, infection, genetic abnormalities, maternal medical conditions (diabetes, hypertension, SLE), and unexplained (>25%). Evaluation: fetal autopsy, placental pathology, karyotype/microarray, and maternal testing (Kleihauer-Betke, lupus anticoagulant, anticardiolipin, thrombophilia, HbA1c, infection screen, thyroid). Management: induction of labor or D&E. Grief support and counseling for subsequent pregnancy.

Intrauterine Pressure Catheter (IUPC)

– A fluid-filled catheter inserted into the uterine cavity alongside the fetus to measure the intensity and frequency of contractions in mm Hg (Montevideo units, MVU). Indications: inadequate labor progress despite oxytocin, need to assess contraction adequacy, determining if tachysystole is present, and when external tocodynamometer cannot provide adequate data. MVU: sum of peak contraction pressures (minus resting tone) over 10 minutes. Adequate labor: ≥200-250 MVU. Placed through the cervix after ROM, typically during active labor.

Intrauterine Transfusion

– Ultrasonically guided intravascular or intraperitoneal transfusion of fetal red blood cells to treat severe fetal anemia, most commonly from Rh isoimmunization. Performed through the umbilical vein using a 22-25 gauge needle. Packed red blood cells, CMV-negative and crossmatched, are transfused slowly. Risks include bradycardia, infection, preterm labor, and fetal death. May need repeated transfusions until delivery.

In Vitro Fertilisation (IVF)

– An assisted reproductive technology in which eggs are retrieved from the ovaries, fertilized with sperm in a laboratory, and resulting embryos are transferred to the uterus. Stages: ovarian stimulation (FSH injections for 10-14 days), oocyte retrieval (transvaginal ultrasound-guided aspiration under sedation), insemination (conventional or ICSI), embryo culture (3-6 days), and embryo transfer (1-2 embryos placed in the uterine cavity, typically at blastocyst stage). Success rates: approximately 30-40% live birth per cycle for women under 35, declining with age.

Involution

– The process by which the uterus returns to its prepregnancy size after delivery. The uterus weighs approximately 1000 grams immediately postpartum and returns to approximately 60 grams by 6 weeks. Fundal height decreases approximately 1 cm per day.

Kangaroo Care (Skin-to-Skin Contact)

– Holding a newborn against the mother's bare chest, typically for at least 60 minutes, promoting physiologic stability, thermoregulation, bonding, and breastfeeding initiation. Especially beneficial for preterm and low-birth-weight infants: reduces mortality (by 40%), hypothermia, and hospital-acquired infections. Improves maternal confidence and reduces postpartum depression. Now standard practice for all newborns, including term infants in the immediate postpartum period.

Karyotyping

– A laboratory technique that produces an image (karyotype) of an individual's complete set of chromosomes, arranged in pairs by size, centromere position, and banding pattern. Karyotyping is used to detect chromosomal abnormalities: aneuploidy (trisomy 21, 18, 13), sex chromosome abnormalities (Turner, Klinefelter), structural abnormalities (deletions, duplications, translocations), and marker chromosomes. In obstetrics, karyotyping is performed on samples obtained by chorionic villus sampling or amniocentesis, or on products of conception after miscarriage.

Kegel

– An exercise that helps prevent and treat incontinence by strengthening pelvic floor muscles.

Ketones

– Acidic byproducts produced when the body breaks down fat for energy instead of glucose (ketogenesis). Ketones can be detected in blood and urine. In pregnancy, ketones may be elevated due to: hyperemesis gravidarum (prolonged vomiting and starvation), gestational diabetes or poorly controlled pre-existing diabetes (diabetic ketoacidosis), and prolonged labor with inadequate caloric intake. Testing for ketones (urine dipstick or blood ketone meter) is part of the assessment of women with vomiting, diabetes, or reduced oral intake in pregnancy. Significant ketonuria may indicate dehydration or metabolic decompensation requiring IV fluids and glucose.

Labour

– The process of childbirth, consisting of regular uterine contractions that cause progressive cervical effacement and dilation, leading to delivery of the baby and placenta. Three stages: First stage — from onset of labor to full cervical dilation (10 cm), divided into latent (slow dilation to 4-6 cm) and active (rapid dilation from 4-6 cm to 10 cm) phases. Second stage — from full dilation to delivery of the baby, with passive descent and active pushing. Third stage — delivery of the placenta and membranes, active management (oxytocin + controlled cord traction) recommended. The partogram records labor progress and alerts to abnormalities.

Lactation

– The physiological process of milk production and secretion from the mammary glands, occurring after childbirth (puerperium). See Lactation (Breastfeeding).

Lactation (Breastfeeding)

– The production of milk by the mammary glands stimulated by the hormone prolactin. Colostrum is produced for the first 2-5 days (rich in IgA, proteins, and growth factors). Mature milk appears at day 5-7. The milk ejection reflex (let-down) is mediated by oxytocin. Benefits to the baby: optimal nutrition, passive immunity (IgA, lactoferrin, lysozyme), reduced risk of infections (otitis, gastroenteritis, pneumonia), SIDS, and long-term protection against obesity and diabetes. Benefits to mother: reduced risk of breast and ovarian cancer, postpartum weight loss, and uterine involution from oxytocin release.

Lanugo

– Fine, soft hair that grows all over the body of a fetus and is typically shed before birth.

Laparoscopy

– A minimally invasive surgical technique using a laparoscope (a thin, lighted telescope) inserted through a small incision (typically at the umbilicus) to visualize the abdominal and pelvic organs. Carbon dioxide gas is insufflated to create a working space. Additional small incisions (ports) allow insertion of instruments. In gynecology, laparoscopy is used for diagnosis and treatment of: endometriosis, ovarian cysts, ectopic pregnancy, pelvic adhesions, and tubal infertility. Also used for sterilization (tubal occlusion) and hysterectomy.

Laparotomy

– A surgical incision into the abdominal cavity, typically a midline vertical or transverse (Pfannenstiel) incision, providing wide access to the abdominal and pelvic organs. In obstetrics and gynecology, laparotomy is performed for: emergency cesarean delivery, ectopic pregnancy (ruptured), ovarian tumor (suspected malignant), pelvic abscess, hysterectomy for large fibroids or cancer, and repair of uterine rupture. Laparotomy carries greater morbidity than laparoscopy: longer recovery, more pain, higher infection risk, and increased adhesion formation.

Last Menstrual Period (LMP)

– The first day of the last normal menstrual period, used to estimate gestational age and the due date. Requires regular menstrual cycles and reliable recall. Limitations: irregular cycles, recent oral contraceptive use, early pregnancy bleeding mistaken for menses, and poor recall. Discriminatory zone: if ultrasound dating differs from LMP dating by more than 7 days in the first trimester, 14 days in the second trimester, or 21 days in the third trimester, ultrasound dating is preferred.

Late Decelerations

– A fetal heart rate pattern with a gradual decrease (onset to nadir >30 seconds) that starts at or after the peak of the contraction and returns to baseline after the contraction ends. Associated with uteroplacental insufficiency (reduced oxygen transfer across the placenta). Recurrent late decelerations (>50% of contractions) with moderate variability require intrauterine resuscitation. Late decelerations with absent variability are Category III and indicate potential fetal acidosis. Management: maternal oxygen, position change (left lateral), IV fluids, reduce oxytocin.

Latent Phase of Labor

– The early part of the first stage of labor, from the onset of regular contractions to 6 cm cervical dilatation. Characterized by slow, progressive cervical change (usually <1 cm/hour). Duration is highly variable: up to 20 hours in nulliparas and 14 hours in multiparas. Active-phase labor arrest cannot be diagnosed during the latent phase. Prolonged latent phase (>20 hours nullipara, >14 hours multipara) is managed with rest, hydration, and observation or oxytocin augmentation.

Late Preterm (34-36 weeks)

– Preterm births between 34 weeks 0 days and 36 weeks 6 days, accounting for 70% of all preterm births. Late preterm infants are at increased risk compared to term infants: respiratory distress (TTN, RDS), temperature instability, hypoglycemia, jaundice, feeding difficulties, and hospital readmission. The risk of adverse outcomes is inversely related to gestational age. Antenatal corticosteroids (betamethasone) are recommended for all preterm births <37 weeks within 7 days of expected delivery.

Laxatives

– Medications used to treat constipation by stimulating bowel movements or softening stool. In obstetrics, constipation is common in pregnancy (progesterone slows bowel motility, iron supplements, and later mechanical compression) and postpartum (perineal pain, fear of damaging sutures, opioid pain relief). Types: bulk-forming (psyllium, Fybogel) — increase stool water content; osmotic (lactulose, polyethylene glycol/macrogol) — draw water into the bowel; stimulant (senna, bisacodyl) — increase peristalsis; and stool softeners (docusate). Safe in pregnancy and breastfeeding.

Leiomyoma (Fibroid)

– A benign smooth muscle tumor of the uterus, arising from the myometrium. See Fibroids (Uterine Leiomyomas).

Leopold Maneuvers

– A systematic abdominal palpation technique used to determine fetal position and presentation. Four maneuvers: (1) fundal grip identifies fetal lie (head vs breech), (2) umbilical grip locates the fetal back, (3) Pawlik grip identifies the presenting part and its position, (4) pelvic grip confirms presentation and attitude. Performed after 36 weeks with an empty bladder. Helps determine need for ultrasound confirmation.

Levator Ani

– The primary pelvic floor muscle complex supporting the pelvic organs, forming a hammock from the pubic symphysis to the coccyx. Composed of the pubococcygeus, puborectalis, and iliococcygeus muscles. Innervated by the pudendal nerve (S2-S4).

Libido

– Sexual desire or drive, influenced by biological, psychological, and social factors. In women, libido is affected by hormonal changes (menstrual cycle, pregnancy, menopause), medical conditions (depression, thyroid disorders, PCOS), medications (SSRIs, hormonal contraceptives), and life circumstances (stress, relationship issues, fatigue). Low libido (hypoactive sexual desire disorder) is the most common female sexual complaint. Management: address underlying causes, relationship counseling, psychosexual therapy, and consider testosterone therapy in selected cases (postmenopausal women).

Linea Nigra

– A dark brown or black vertical line appearing on the abdomen during pregnancy, extending from the pubic symphysis to the umbilicus or beyond. Caused by hormonal stimulation of melanocytes, resulting in hyperpigmentation. More prominent in darker-skinned women. Usually appears in the second trimester and fades weeks to months after delivery. One of many normal dermatologic changes of pregnancy along with melasma and striae gravidarum.

Listeriosis in Pregnancy

– Infection with Listeria monocytogenes, a foodborne Gram-positive rod. Pregnant women are 10-20 times more susceptible than the general population. Ingestion of contaminated food (deli meats, soft cheeses, unpasteurized dairy, raw sprouts). Maternal symptoms: fever, myalgias, GI symptoms (nausea, diarrhea). Fetal/neonatal infection: chorioamnionitis, preterm labor, miscarriage, stillbirth, and neonatal sepsis/granulomatosis infantiseptica. Diagnosis: blood culture, placental culture. Treatment: IV ampicillin (high dose) + gentamicin. Prevention: avoid high-risk foods during pregnancy.

LLETZ

– Large Loop Excision of the Transformation Zone, also called LEEP (Loop Electrosurgical Excision Procedure). A common outpatient procedure used to diagnose and treat cervical intraepithelial neoplasia (CIN). A thin wire loop carrying an electrical current is used to remove the transformation zone of the cervix (where cell changes typically occur). The procedure takes approximately 5-10 minutes under local anesthesia. The removed tissue is sent for histology. Complications: bleeding, infection, and cervical stenosis. LLETZ does not significantly affect fertility but may slightly increase the risk of preterm birth in future pregnancies.

Local anesthetic

– A medication that causes reversible loss of sensation in a specific area of the body without affecting consciousness. Local anesthetics work by blocking voltage-gated sodium channels in nerve cell membranes, preventing the propagation of action potentials and transmission of pain signals. Common agents: lidocaine (1-2%, most widely used), bupivacaine (longer-acting), and ropivacaine. Onset: 2-5 minutes (infiltration). Duration: 30 minutes to several hours depending on agent, dose, and use of vasoconstrictor (epinephrine prolongs action and reduces bleeding). In obstetrics: used for perineal infiltration before episiotomy repair, pudendal block, and as part of epidural/spinal anesthesia.

Lochia

– Vaginal discharge following delivery that undergoes characteristic changes over the puerperium. Lochia rubra (days 1-4): red, bloody discharge containing blood, decidual tissue, and fetal membranes. Lochia serosa (days 5-10): pinkish-brown discharge with serous fluid, leukocytes, and organisms. Lochia alba (days 11-42): yellowish-white discharge containing mucus, leukocytes, and bacteria. The sequence reflects progressive uterine involution and healing of the placental site. Foul-smelling lochia suggests endometritis or retained products of conception. Prolonged or heavy lochia warrants evaluation for retained tissue or subinvolution. Normal total volume is approximately 200-500 mL over the first 2 weeks.

Macrosomia

– Birth weight exceeding 4000 grams (8 pounds 13 ounces) regardless of gestational age. Risk factors include maternal diabetes, obesity, post-term pregnancy, and prior macrosomia. Complications include shoulder dystocia, brachial plexus injury, clavicular fracture, and neonatal hypoglycemia. Cesarean delivery may be considered for estimated fetal weight over 4500 grams in diabetic patients or over 5000 grams in non-diabetic patients.

Magnesium Sulfate for Neuroprotection

– Intravenous magnesium sulfate administered to women in imminent preterm labor (<32 weeks) to reduce the risk of cerebral palsy and major motor dysfunction in the child. Dose: 4-6 g IV bolus, followed by 1-2 g/hour maintenance for up to 24 hours or until delivery. Mechanism: thought to stabilize cerebral blood flow, prevent excitotoxic injury, and reduce inflammation. The BEAM trial showed a 30-45% reduction in the risk of moderate-to-severe cerebral palsy. Side effects: flushing, warmth, nausea, dizziness, and respiratory depression (monitor deep tendon reflexes and respiratory rate). Contraindication: myasthenia gravis.

Male Infertility

– Male infertility—the inability to conceive after 12 months of unprotected intercourse—contributes to approximately 50% of couple infertility cases. Causes include varicocele (most common correctable cause, 40%), obstructive azoospermia (congenital bilateral absence of the vas deferens from CFTR mutations, vasectomy, infection), hormonal disorders (hypogonadotropic hypogonadism, hyperprolactinemia), spermatogenic failure (Y-chromosome microdeletions, Klinefelter syndrome, chemotherapy, cryptorchidism), and idiopathic oligoasthenoteratozoospermia. Evaluation: semen analysis per WHO criteria (volume ≥1.4 mL, concentration ≥16 million/mL, motility ≥30%, morphology ≥4%), hormonal profile (FSH, LH, testosterone), genetic testing (karyotype, Y-deletion, CFTR), and scrotal ultrasound. Treatment: varicocelectomy, hormonal therapy (hCG, clomiphene), surgical sperm extraction (TESE, micro-TESE) for IVF/ICSI, and lifestyle optimization (weight loss, smoking cessation, reducing scrotal heat).

Malpresentation

– Any fetal presentation other than vertex (cephalic) at term. Includes breech (buttocks or feet first), face, brow, and shoulder presentations. Breech presentation occurs in 3-4% of term pregnancies and increases the risk of cord prolapse and difficult delivery. External cephalic version may be attempted at 36-38 weeks. Persistent breech presentation typically managed with cesarean delivery.

Manual Removal of Placenta

– The manual extraction of a retained placenta from the uterine cavity. Indication: retained placenta beyond 30-60 minutes. Technique: under adequate anesthesia (regional or general), one hand inserts into the uterine cavity with fingers together (like a knife hand), and the placenta is gently separated from the uterine wall using a sweeping motion. The other hand is placed on the abdomen to stabilize the uterus. After removal, the uterus is explored to ensure complete evacuation. Oxytocin is given. Complications: hemorrhage, infection, perforation, and Asherman syndrome.

Mastitis

– Inflammation of the breast tissue, most commonly occurring during lactation (lactational mastitis, affecting 5-20 percent of breastfeeding women), typically within the first 6-12 weeks postpartum. Lactational mastitis results from milk stasis combined with bacterial infection, most frequently Staphylococcus aureus (including MRSA) entering through cracked or fissured nipples. Clinical presentation includes localized breast pain, erythema, warmth, swelling, induration, and systemic symptoms of fever, chills, and malaise. Diagnosis is primarily clinical. Treatment includes milk removal through continued breastfeeding or pumping (emptying the affected breast is essential), supportive measures (cold compresses between feedings, warm compresses before feedings, NSAIDs for pain), and antibiotics if symptoms are severe or persist beyond 12-24 hours of conservative management (dicloxacillin or cephalexin for 10-14 days; clindamycin for penicillin-allergic or MRSA-suspected cases). The mother should continue breastfeeding or pumping to maintain milk flow, as abrupt cessation worsens the condition and increases the risk of breast abscess. Preventive measures include proper latch technique, frequent feeding, and hand hygiene. Recurrent mastitis may require investigation for underlying ductal abnormalities or milk supply issues.

Maternal Physiologic Changes in Pregnancy

– Extensive adaptive changes across all organ systems to support the growing fetus. Key changes: plasma volume increases 40-50% (hemodilution), cardiac output increases 30-50%, heart rate increases 10-15 bpm, systemic vascular resistance decreases, tidal volume and minute ventilation increase (chronic respiratory alkalosis), renal blood flow increases 50% (increased GFR, decreased creatinine and BUN), and insulin resistance increases (mediated by human placental lactogen, cortisol, and growth hormone). These changes impact medication dosing, interpretation of lab values, and response to illness.

Maternal Pushing (Second Stage)

– Maternal expulsive efforts during the second stage of labor to facilitate fetal descent. Directed (closed-glottis, Valsalva) pushing: sustained breath-hold for 10 seconds with three pushes per contraction. Spontaneous (open-glottis) pushing: shorter pushes with the mother pushing when she feels an urge. Evidence: spontaneous pushing results in less perineal trauma, less fatigue, and better fetal acid-base status. Directed pushing may slightly shorten the second stage but increases perineal injury and is associated with lower umbilical cord pH.

MCA Peak Systolic Velocity (PSV)

– Doppler measurement of the peak systolic velocity in the fetal middle cerebral artery. An indicator of fetal anemia: elevated MCA PSV (>1.5 MoM) correlates with moderate-to-severe anemia. The most important application is surveillance of Rh-sensitized pregnancies (detect anemia before hydrops develops). Also used to assess fetal anemia due to parvovirus B19 infection or fetomaternal hemorrhage. MCA PSV is also used as part of the cerebroplacental ratio (CPR = MCA PI / umbilical artery PI) to assess brain sparing in IUGR.

McDonald Cerclage

– A transvaginal cervical cerclage using a purse-string suture (Mersilene tape or Prolene) placed at the cervicovaginal junction, without bladder dissection. The most common cerclage technique. Indications: cervical insufficiency (history-indicated or ultrasound-indicated short cervix <25 mm before 24 weeks). Placed at 12-14 weeks. The suture is removed at 36-37 weeks or at the onset of labor. Suture around the cervix, not through it. No flap of the bladder or rectum is created. Success rate: 80-90% for term delivery in selected patients.

McRoberts Maneuver

– A positioning maneuver for managing shoulder dystocia where the mother's legs are hyperflexed onto her abdomen, with thighs against the abdomen. This flattens the sacrum, increases the anteroposterior diameter of the pelvis, and rotates the symphysis pubis superiorly. Often combined with suprapubic pressure. First-line maneuver for shoulder dystocia resolution, with a reported success rate of approximately 40%.

Melasma

– A common hyperpigmentation disorder characterized by brown or gray-brown patches on the face, particularly the cheeks, forehead, nose, and upper lip. Also called chloasma or the mask of pregnancy.

Menopause

– The point marking the end of menstruation, officially designated as one year after a woman's final period.

Menorrhagia

– Excessive or prolonged menstrual bleeding (>80 mL blood loss or >7 days duration). Causes: uterine fibroids, adenomyosis, endometrial polyps, coagulation disorders, thyroid dysfunction, and anovulation. Presents with heavy bleeding, flooding, passage of clots, and iron deficiency anemia. Diagnosis: CBC, ferritin, TSH, transvaginal ultrasound, saline infusion sonography, and endometrial biopsy if indicated. Treatment: NSAIDs, tranexamic acid, hormonal contraceptives, levonorgestrel IUD, endometrial ablation, and hysterectomy in severe cases.

Mesh

– A synthetic or biological implant used to provide structural support in pelvic organ prolapse and stress urinary incontinence surgery. Types: polypropylene (most common, mid-urethral slings for SUI, vaginal mesh for prolapse), biological (porcine, bovine, or cadaveric dermis). Controversy: transvaginal mesh for prolapse has been associated with complications including mesh erosion (exposure into vagina), chronic pain, dyspareunia, infection, and organ perforation. The NHS stopped routine use of transvaginal mesh for prolapse in 2018. Mid-urethral slings for SUI (retropubic and transobturator tapes) remain in use with careful patient selection and counseling.

Metrorrhagia

– Uterine bleeding at irregular intervals, particularly between expected menstrual periods. Causes: anovulation, endometrial polyps, submucosal fibroids, endometrial hyperplasia/cancer, cervical pathology, and breakthrough bleeding with hormonal contraceptives. Diagnosis: transvaginal ultrasound, endometrial biopsy, and hysteroscopy. Treatment depends on the underlying cause and patient age.

Midtrimester

– The middle third of pregnancy, from 14 to 28 weeks gestation. The midtrimester is when many women feel their best: nausea and fatigue of the first trimester resolve, and the physical discomforts of the third trimester have not yet developed. Key events: fetal movements felt (quickening, around 18-20 weeks), anomaly scan (18-21 weeks), maternal serum screening, and the fetus grows rapidly. By 24 weeks, the fetus reaches viability. Midtrimester complications include miscarriage (late), cervical insufficiency, and preterm labor.

Miscarriage

– Also known as spontaneous abortion, the spontaneous loss of a pregnancy before 20 weeks of gestation, occurring in 15-20 percent of clinically recognized pregnancies. The majority (80 percent) occur in the first trimester, with risk decreasing significantly after visualization of fetal cardiac activity. Chromosomal abnormalities account for 50-60 percent of first-trimester miscarriages, most commonly autosomal trisomies (trisomy 16, trisomy 22, trisomy 21, trisomy 18, trisomy 13), monosomy X, and triploidy. Risk increases with advancing maternal age: 10 percent at age 20, 15-20 percent at age 35, 25-40 percent at age 40, and greater than 50 percent at age 45. Other causes include uterine anatomic abnormalities (septate uterus, fibroids, adhesions), maternal disorders (diabetes, thyroid disease, antiphospholipid syndrome, thrombophilias), infections (Listeria, Toxoplasma, rubella, parvovirus B19), cervical insufficiency, and environmental factors (smoking, alcohol, high caffeine). Diagnosis is classified as threatened (bleeding, closed cervix, viable pregnancy), inevitable (open cervix with bleeding), incomplete (partial tissue passage), complete (empty uterine cavity), or missed (fetal demise without expulsion). Management is expectant, medical (misoprostol), or surgical (suction evacuation) depending on clinical presentation and patient preference.

Misoprostol (Cytotec)

– A synthetic prostaglandin E1 analog used off-label for cervical ripening and labor induction. Dose: 25-50 mcg vaginally or 25-100 mcg orally q4-6h. Advantages: inexpensive, stable at room temperature, and multiple routes of administration. Risks: uterine tachysystole (with and without FHR changes), the highest rate among induction agents. Contraindicated in patients with prior cesarean or uterine scar (increased risk of uterine rupture). The oral route has less tachysystole than vaginal. The buccal/sublingual route is also used.

Modified Biophysical Profile (Modified BPP)

– A simplified antepartum test combining the nonstress test (NST) with the amniotic fluid index (AFI). The NST assesses acute fetal status (heart rate reactivity over 20-40 minutes) and the AFI assesses chronic uteroplacental function (oligohydramnios suggests chronic hypoxia or placental insufficiency). Scoring: NST (reactive: 2, nonreactive: 0), AFI (>5 cm: 2, ≤ 5 cm: 0). The total modified BPP score is 4 (normal), 2 (equivocal, repeat in 24 hours), or 0 (abnormal, consider delivery). Equivalent to the full BPP for outcome prediction in most settings.

Molar Pregnancy

– See Hydatidiform Mole.

Montevideo Units (MVU)

– A quantitative measure of uterine contraction intensity, calculated by summing the peak contraction pressures (in mm Hg) minus the resting tone for each contraction in a 10-minute window. Adequate labor: ≥ 200 MVU (spontaneous) or ≥ 250 MVU (augmented). Measured with an intrauterine pressure catheter (IUPC). Low MVU may prompt oxytocin augmentation. The concept helps distinguish true arrest of labor from inadequate contractions.

Montgomery Tubercles

– Small, raised sebaceous glands visible on the areola of the breast during pregnancy and lactation. They appear as small bumps surrounding the nipple and secrete an oily substance that lubricates the nipple and areola, protecting against infection. Their prominence increases during pregnancy under hormonal influence. Often mistaken for pathological lesions but are normal anatomic structures. First described by William Fetherston Montgomery in 1837.

Naegele Rule

– A formula for estimating the due date (estimated date of delivery, EDD): add 7 days to the first day of the last menstrual period (LMP), then add 9 months (or subtract 3 months). Example: LMP January 1 yields EDD October 8. Assumes a 28-day cycle with ovulation on day 14. The rule is most accurate with regular cycles and confirmed LMP. First-trimester ultrasound dating is more accurate and preferred when LMP is uncertain or cycles are irregular.

Neonatal Resuscitation

– The systematic approach to stabilizing a newborn after delivery, guided by the Neonatal Resuscitation Program (NRP). Steps: (1) initial steps (warm, dry, stimulate, position, clear airway if needed), (2) assess breathing, heart rate, and color, (3) positive pressure ventilation (PPV) with a bag-mask if gasping or HR <100 bpm, (4) chest compressions (3:1 ratio with ventilations) if HR <60 bpm despite PPV, (5) endotracheal intubation if needed, (6) epinephrine (IV or ET) if HR remains <60 bpm, and (7) volume expansion if suspected hypovolemia. The transition from fetal to neonatal circulation requires successful lung aeration and closure of fetal shunts.

Newborn Hearing Screen

– Universal screening for hearing loss in newborns before hospital discharge. Two methods: otoacoustic emissions (OAE, measures cochlear outer hair cell function) and automated auditory brainstem response (AABR, measures the auditory nerve pathway). Both are noninvasive and take 5-10 minutes. Infants who fail the screen are referred for diagnostic audiology before 3 months of age. Early detection and intervention (before 6 months) significantly improve language outcomes.

Newborn Metabolic Screen

– A dried blood spot test (heel stick) collected 24-48 hours after birth to screen for congenital disorders before symptoms develop. The panel varies by country/state but typically includes: phenylketonuria (PKU), congenital hypothyroidism, cystic fibrosis, sickle cell disease, medium-chain acyl-CoA dehydrogenase deficiency (MCADD), maple syrup urine disease (MSUD), galactosemia, severe combined immunodeficiency (SCID), and many others (50-80 conditions in some states). Positive results require confirmatory testing.

Nitrous Oxide for Labor

– A 50:50 mixture of nitrous oxide and oxygen (Entonox) inhaled by the mother during labor for analgesia. Self-administered via a mask or mouthpiece; the woman controls her own dosing, taking a breath 30 seconds before a contraction. Advantages: noninvasive, rapid onset and offset, no effect on the baby at low concentrations, and the mother can ambulate. Disadvantages: incomplete pain relief (does not eliminate pain, only modulates it), can cause nausea, dizziness, and drowsiness. Contraindications: pneumothorax, bowel obstruction, and vitamin B12 deficiency. Widely used in the UK and Canada; increasingly available in US hospitals.

Noninvasive Prenatal Testing (NIPT)

– See Cell-Free Fetal DNA (cfDNA) Screening.

Non-Stress Test

– A fetal heart rate monitoring technique assessing fetal heart rate reactivity over 20-40 minutes. Normal (reactive) tracing shows at least 2 fetal heart rate accelerations of 15 beats per minute above baseline lasting at least 15 seconds within a 20-minute period. Non-reactive tracings may indicate fetal hypoxia or sleep cycle. Used for antepartum surveillance in high-risk pregnancies, including post-dates pregnancy and maternal conditions affecting fetal well-being.

Nuchal Translucency (NT) Screening

– An ultrasound measurement of the fluid-filled space at the posterior fetal nuchal region, performed at 11-14 weeks (crown-rump length 45-84 mm). Increased NT (>95th percentile or >3.0-3.5 mm) is associated with trisomies 21, 18, 13, Turner syndrome, and cardiac defects. Combined with maternal serum biomarkers (PAPP-A, free β-hCG) in the first-trimester combined screen, it detects 85-90% of Down syndrome. NT measurement requires certified training and quality assurance (Fetal Medicine Foundation).

Nutrition in Pregnancy

– A balanced diet providing adequate calories, protein, vitamins, and minerals to support maternal and fetal health. Additional caloric needs: 300 kcal/day in the second and third trimesters. Key nutrients: folic acid (400 mcg/day), iron (27 mg/day), calcium (1000 mg/day), vitamin D (600 IU/day), iodine (220 mcg/day), DHA (200-300 mg/day). Foods to avoid: raw/undercooked meat and fish (listeriosis, toxoplasmosis), high-mercury fish (shark, swordfish, king mackerel), unpasteurized dairy, and excess caffeine (<200 mg/day).

Obstetric Anal Sphincter Injury (OASIS)

– Third- and fourth-degree perineal lacerations involving the anal sphincter complex. Incidence: 0.5-5% of vaginal deliveries. Risk factors: midline episiotomy (increases risk 2-3 fold), operative vaginal delivery, primiparity, macrosomia, and occiput posterior position. Repair: immediate repair in the delivery room under adequate anesthesia (regional or general). End-to-end or overlap repair of the EAS (3-0 PDS), repair of the IAS if involved. Post-operative: broad-spectrum antibiotics, stool softeners (lactulose, docusate), and avoid constipation. Follow-up: pelvic floor physiotherapy, ultrasound, or MRI for persistent symptoms.

Obstetrician

– A medical doctor who specializes in the care of women during pregnancy, labor, and the postpartum period (obstetrics). Obstetricians manage normal and high-risk pregnancies, perform cesarean deliveries and assisted vaginal deliveries, and treat pregnancy complications (pre-eclampsia, gestational diabetes, hemorrhage, preterm labor). In the UK, obstetrics and gynecology are combined as a single specialty (O&G). Training: 2 years foundation training + 7 years specialty training to become a consultant obstetrician and gynecologist.

Obstetrics

– The branch of medicine and surgery concerned with the care of women during pregnancy, childbirth, and the postpartum period. Obstetricians manage normal and high-risk pregnancies, perform antenatal screening and diagnosis, conduct deliveries (spontaneous vaginal, assisted vaginal—forceps, ventouse, and caesarean section), and manage obstetric emergencies (pre-eclampsia/eclampsia, hemorrhage, sepsis, shoulder dystocia, cord prolapse, uterine rupture). Antenatal care in the UK follows a schedule of 10 appointments for nulliparous women and 7 for multiparous, including ultrasound dating (10–13 weeks), anomaly scan (18–21 weeks), screening for aneuploidy, gestational diabetes, and infections. Maternal medicine deals with pre-existing medical conditions in pregnancy (diabetes, hypertension, cardiac disease, epilepsy). The specialty has close links with midwifery (midwife-led care for low-risk pregnancies), neonatology, and fetal medicine.

Occiput Posterior (OP) Position

– The fetal head presenting with the occiput oriented toward the maternal sacrum (back-to-back position). Occurs in 15-20% of labors, about half rotate spontaneously to OA by delivery. Persistent OP is associated with prolonged labor (especially the second stage), arrest of descent, severe back pain (back labor), and higher rate of operative vaginal delivery and cesarean. Management: maternal positioning (hands and knees, pelvic rocking), manual rotation (hands or forceps), and vacuum or forceps delivery if needed.

Occiput Transverse (OT) Position

– The fetal head presenting with the sagittal suture oriented transversely. Usually the normal descent position; most rotate to OA with contractions. When persistent, it can cause protracted descent and prolonged second stage. Management: maternal position changes, and if persistent, manual rotation to OA or direct OP extraction with forceps/vacuum. Deep transverse arrest: OT at the outlet, more common with an android pelvis and requires rotation before delivery.

Oestrogen

– The primary female sex hormone, produced mainly by the ovaries (granulosa cells of developing follicles) and in smaller amounts by adipose tissue, the adrenal glands, and the placenta during pregnancy. The three major forms: estradiol (E2, most potent, predominant in reproductive-age women), estrone (E1, predominant after menopause), and estriol (E3, produced by the placenta). Functions: development of secondary sexual characteristics, regulation of the menstrual cycle, endometrial proliferation, breast development, bone density maintenance, and cardiovascular health. Used in HRT, combined oral contraceptives, and treatment of anovulation.

Oligohydramnios

– Insufficient amniotic fluid volume, defined as amniotic fluid index less than 5 cm or single deepest pocket less than 2 cm. Causes include fetal renal anomalies, post-term pregnancy, premature rupture of membranes, maternal dehydration, and placental insufficiency. Complications include umbilical cord compression, fetal growth restriction, and pulmonary hypoplasia. Management includes amniotic fluid infusion, maternal hydration, and delivery when indicated.

One-Hour Glucose Challenge Test (GCT)

– A screening test for gestational diabetes mellitus (GDM) performed at 24-28 weeks. The patient drinks 50 g of glucose (regardless of fasting state), and plasma glucose is measured at 1 hour. Threshold: ≥130-140 mg/dL (7.2-7.8 mmol/L) is considered positive, warranting diagnostic 3-hour oral glucose tolerance test (OGTT). Sensitivity: 80-90% depending on the threshold used. A negative test (glucose <130 mg/dL) excludes GDM. Earlier testing (first trimester) is performed in high-risk women (obesity, prior GDM, family history of diabetes).

Oocyte

– The female gamete (egg cell) produced in the ovary through oogenesis. Oocytes develop from primordial germ cells, which undergo mitosis to form oogonia, then enter meiosis I before birth and arrest in prophase I until puberty. Each menstrual cycle, a cohort of follicles is recruited; one dominant follicle matures and releases its oocyte at ovulation (meiosis I completed, arrested in metaphase II). The oocyte is arrested in metaphase II until fertilization, which triggers completion of meiosis II. The oocyte is the largest cell in the human body (100–120 μm diameter) and contains maternal mRNA, proteins, organelles, and nutrients (yolk) necessary for early embryonic development until the embryonic genome activates (8-cell stage). Oocyte quality declines with maternal age (increased aneuploidy risk, reduced mitochondrial function). Oocyte cryopreservation (egg freezing) and in vitro fertilization (IVF) are key assisted reproductive technologies.

Oophorectomy

– Surgical removal of one (unilateral) or both (bilateral) ovaries. Indications: ovarian masses (benign or malignant), ovarian torsion, endometriosis, pelvic inflammatory disease (abscess), risk-reducing surgery (BRCA1/BRCA2 mutation carriers—risk-reducing salpingo-oophorectomy reduces ovarian cancer risk by 80–96% and breast cancer risk by 50%), and as part of treatment for hormone-sensitive cancers (breast cancer, endometrial cancer). Bilateral oophorectomy causes surgical menopause (immediate loss of oestrogen, progesterone, and testosterone production), with symptoms (hot flushes, vaginal atrophy, mood changes, decreased libido) and long-term health effects (increased cardiovascular disease, osteoporosis, cognitive decline). Hormone replacement therapy mitigates these risks in women aged <50 without contraindications. Oophorectomy is often performed with salpingectomy (salpingo-oophorectomy); bilateral salpingo-oophorectomy plus hysterectomy is common for benign indications and ovarian cancer staging.

Operative Vaginal Delivery

– Vaginal delivery assisted by vacuum or forceps. Indications: prolonged second stage, suspected fetal compromise, need to shorten the second stage for maternal benefit (exhaustion, cardiac disease, hypertensive crisis). Prerequisites: full dilatation, ruptured membranes, engaged head (station $\geq$+2), known fetal position, adequate pelvis, and consent. Vacuum: soft or rigid cup applied to the fetal occiput. Forceps: two blades applied to the fetal head (outlet, low, or mid-pelvic). Success rate: 80-95%. Complications: maternal (perineal lacerations, anal sphincter injury, vaginal lacerations, hematoma) and neonatal (cephalohematoma, subgaleal hemorrhage with vacuum, facial nerve palsy with forceps).

Oral Contraceptives

– Hormonal medications taken orally to prevent pregnancy. Combined oral contraceptive pills (COCP) contain oestrogen (ethinylestradiol 20–35 μg) and a progestogen (levonorgestrel, norethisterone, drospirenone, desogestrel). Mechanisms: inhibit ovulation (suppression of LH surge), thicken cervical mucus (impedes sperm penetration), and alter endometrial receptivity. Perfect-use failure rate: 0.3% per year; typical-use failure rate: 7–9%. Benefits: cycle regulation, reduced dysmenorrhoea, decreased ovarian/endometrial cancer risk (protective effect persists >30 years after stopping), reduced acne, and improved bone density. Contraindications (WHO categories 3–4): age $\geq$35 and smoking $\geq$15 cigarettes/day, uncontrolled hypertension, migraine with aura, history of VTE, breast cancer, liver disease. Cardiovascular risks: increased VTE risk (3–9 per 10,000 woman-years vs 1–5 in non-users). Progestogen-only pill (POP, 'mini-pill') contains desogestrel 75 μg or norethisterone 350 μg; strict timing required (within 3-hour window); safe for use during breastfeeding and in women with contraindications to oestrogen.

Ovarian Cyst

– A fluid-filled sac within or on the ovary, most commonly functional (follicular or corpus luteum cysts) related to the menstrual cycle. Usually asymptomatic; can present with pelvic pain, pressure, and bloating. Complications: rupture, hemorrhage, torsion (sudden severe pain with nausea/vomiting). Diagnosis: transvaginal ultrasound (assess size, morphology, septations, solid components, and vascularity). Simple cysts <5 cm in reproductive-age women are usually benign. Treatment: observation for simple cysts; surgical removal for complex/malignant features or persistent large cysts.

Ovarian Pregnancy

– A rare ectopic pregnancy (0.5-3%) implanted on the ovary. Diagnosis: Spiegelberg criteria (ipsilateral fallopian tube intact, gestational sac on the ovary, connected to the ovary by the uterovarian ligament, ovarian tissue in the sac wall). Treatment: ovarian wedge resection or oophorectomy.

Ovariectomy

– Surgical removal of one or both ovaries.

Ovulation

– The release of a mature egg from the ovary, at which time it is available to be fertilized by sperm.

Ovum

– The female reproductive cell (egg) after ovulation and before fertilization. The ovum is a large, nonmotile cell (100 μm diameter) surrounded by the zona pellucida (glycoprotein layer that mediates sperm binding and prevents polyspermy after fertilization) and the corona radiata (follicular cells). The ovum is arrested in metaphase II of meiosis at ovulation. Upon sperm penetration, the ovum completes meiosis II (producing a mature oocyte and a second polar body), and the male and female pronuclei fuse (syngamy) to form a diploid zygote. The ovum is viable for approximately 12–24 hours after ovulation; fertilization must occur within this window. Ovum donation is used in assisted reproduction for women with poor oocyte quality, premature ovarian insufficiency, or genetic disorders.

Oxytocics

– A class of medications that stimulate uterine contractions, used for induction and augmentation of labor and prevention/treatment of postpartum hemorrhage. Oxytocin (Syntocinon) — synthetic analog of endogenous oxytocin, administered IV infusion for labor induction or IM/IV for PPH prevention. Ergometrine — causes sustained tetanic uterine contraction, used IM for PPH, contraindicated in hypertension and preeclampsia. Carboprost (Hemabate) — prostaglandin F2-alpha analog, IM for refractory PPH. Misoprostol (Cytotec) — PGE1 analog, oral/sublingual/rectal for PPH prevention and treatment in resource-limited settings.

Oxytocin Infusion

– Synthetic oxytocin (Pitocin) administered intravenously for induction or augmentation of labor. Standard dosing: start at 1-2 mU/min, increase by 1-2 mU/min every 15-30 minutes to achieve 200-250 Montevideo units (MVU, measured by IUPC). Maximum dose: 40 mU/min. After achieving adequate labor, the dose may be maintained or decreased. Complications: uterine tachysystole (>5 contractions/10 min, leading to fetal hypoxia), hyperstimulation (tachysystole with nonreassuring FHR), water intoxication (antidiuretic effect at high doses), and uterine rupture (rare, increased risk with prior cesarean). Stop or reduce oxytocin if hyperstimulation occurs.

Partograph (Partogram)

– A graphic record of cervical dilatation, fetal descent, and uterine contractions over time, used to monitor labor progress and identify labor abnormalities. The WHO partograph has an alert line (1 cm/hour dilatation) and an action line (4 hours after the alert line, indicating need for intervention). Components: cervical dilatation (plotted against time), fetal descent (station), contractions (frequency and duration), FHR, and maternal vital signs. The partograph reduces prolonged labor, augmentation rates, and intrapartum stillbirths in low-resource settings. Modern use: most centers now use the Zhang labor curve (active phase at 6 cm).

Parturition (Childbirth)

– The process of giving birth, encompassing three stages: (1) First stage: from the onset of regular uterine contractions (cervical effacement and dilation) to full dilation (10 cm). Latent phase—slow dilation (0–4 cm) over hours, variable duration. Active phase—more rapid dilation (4–10 cm), typically 1–1.5 cm/hour in nulliparous women. (2) Second stage: from full dilation to delivery of the baby. Passive phase—descent of the presenting part without active pushing. Active phase—maternal pushing with contractions. Duration: up to 2 hours in nulliparous (3 hours with epidural), 1 hour in multiparous (2 hours with epidural). (3) Third stage: delivery of the placenta and membranes. Active management (oxytocin, controlled cord traction) reduces the risk of postpartum hemorrhage. Signs of labour: regular painful contractions, cervical dilation and effacement, show (blood-stained mucus), and spontaneous rupture of membranes (SROM). Induction of labour (IOL) is indicated for post-term pregnancy, pre-eclampsia, diabetes, intrauterine growth restriction, and premature rupture of membranes. Methods: prostaglandins (dinoprostone, misoprostol), balloon catheter, amniotomy, and oxytocin infusion. Assisted vaginal delivery: forceps or ventouse (vacuum) for prolonged second stage, fetal distress, or maternal exhaustion. Caesarean section: for failed IOL, fetal distress, malpresentation, placenta praevia, and previous caesarean section (VBAC may be attempted).

Pelvic Floor Health and Disorders

– The pelvic floor is a musculofascial sling (levator ani, coccygeus, and surrounding fascia) spanning the pelvic outlet, supporting the pelvic organs (bladder, uterus, rectum), maintaining continence, and facilitating urination, defecation, and sexual function. Pelvic floor disorders include pelvic organ prolapse (cystocele, rectocele, uterine prolapse, vaginal vault prolapse—affecting up to 50% of parous women), stress urinary incontinence (leakage with cough, sneeze, exercise—treated with pelvic floor muscle training/Kegel exercises, pessaries, or midurethral sling), overactive bladder (urgency, frequency, urge incontinence—treated with behavioral modification, anticholinergics, beta-3 agonists), and pelvic pain syndromes (levator ani spasm, pudendal neuralgia). Risk factors: pregnancy/childbirth (vaginal delivery, especially with forceps or prolonged second stage), aging, obesity, chronic constipation, and heavy lifting. Diagnosis: Pelvic Organ Prolapse Quantification (POP-Q) system, urodynamic studies, and MRI defecography for complex prolapse. Treatment progresses from conservative (pelvic floor physical therapy, biofeedback, vaginal pessaries, lifestyle modification) to surgical (native tissue repair, sacrocolpopexy, colpocleisis for frail patients, sling procedures for SUI).

Pelvic Inflammatory Disease

– Infection and inflammation of the upper female genital tract (uterus, fallopian tubes, ovaries, and peritoneum), most commonly caused by sexually transmitted infections (Chlamydia trachomatis, Neisseria gonorrhoeae). Presents with lower abdominal pain, cervical motion tenderness, adnexal tenderness, and fever. Complications: tubo-ovarian abscess, infertility, ectopic pregnancy, and chronic pelvic pain. Diagnosis: clinical criteria (CDC minimal criteria: cervical motion, uterine, or adnexal tenderness). Treatment: empiric broad-spectrum antibiotics (ceftriaxone + doxycycline +/- metronidazole).

Pelvic Inlet

– The superior opening of the true pelvis, bounded by the sacral promontory posteriorly, the arcuate lines of the ilia laterally, and the symphysis pubis anteriorly. Its anteroposterior diameter (true conjugate) normally measures at least 11 cm. The transverse diameter is typically 13 cm. Contraction of the pelvic inlet (cephalopelvic disproportion) may prevent engagement of the fetal head, necessitating cesarean delivery.

Pelvic Outlet

– The inferior opening of the true pelvis, bounded by the pubic symphysis, ischial tuberosities, sacrotuberous ligaments, and coccyx. Normal anteroposterior diameter is 11-13 cm (from the lower border of the pubic symphysis to the tip of the coccyx), and the transverse diameter (between the ischial tuberosities) is approximately 10-11 cm. A narrow pelvic outlet can cause shoulder dystocia or obstructed labour. Assessment of the pelvic outlet is part of clinical pelvimetry examination during prenatal care and is relevant when planning vaginal delivery, particularly in cases of suspected cephalopelvic disproportion.

Pelvic Pain

– Pain localized to the lower abdomen, pelvis, or perineum, lasting <3 months (acute) or >3–6 months (chronic), and can arise from gynecologic, urologic, gastrointestinal, musculoskeletal, or psychologic causes. Acute causes: ectopic pregnancy (life-threatening emergency), pelvic inflammatory disease (PID—sexually transmitted, polymicrobial; treated with antibiotics), ovarian torsion (sudden severe pain, adnexal mass; surgical emergency), acute cystitis, appendicitis, and diverticulitis. Chronic pelvic pain (CPP) affects 15% of women, with common etiologies: endometriosis (endometrial glands outside uterus, cyclic pain, dyspareunia, dysmenorrhea; laparoscopy for diagnosis; medical or surgical management), interstitial cystitis/bladder pain syndrome (bladder pressure/pain with filling, urgency, frequency), irritable bowel syndrome, pelvic floor myofascial pain, vulvodynia, and adenomyosis. Evaluation: bimanual exam, pregnancy test, imaging (transvaginal ultrasound, MRI, CT), laparoscopy for suspected endometriosis, and multidisciplinary assessment. Multimodal treatment: NSAIDs, hormones (OCPs, GnRH agonists for endometriosis), physical therapy, trigger point injections, CBT, and nerve blocks.

Percutaneous Umbilical Blood Sampling (PUBS)

– Also called cordocentesis. An invasive procedure for fetal blood sampling (cardocentesis) from the umbilical vein under ultrasound guidance, typically performed after 18-20 weeks. Indications: fetal anemia (Rh isoimmunization, parvovirus B19, hemoglobin Bart), fetal platelet disorders (immune thrombocytopenia), rapid karyotyping (late gestation), and fetal infection assessment (toxoplasmosis, CMV, parvovirus). Risks: fetal loss (1-2%), bleeding from the puncture site, bradycardia, and cord hematoma.

Perimenopause

– The transition time in a woman's life that begins when ovaries produce less estrogen and menstruation becomes less frequent, and ends when the ovaries no longer produce eggs and menstruation stops.

Perineal Laceration

– Tearing of the perineum during vaginal delivery. First degree: skin only (perineal skin, fourchette). Second degree: perineal muscles (bulbocavernosus, transverse perineal) without anal sphincter involvement. Third degree: anal sphincter injury—3a (<50% of EAS), 3b (>50% of EAS), 3c (both EAS and IAS). Fourth degree: extends through the anal sphincter into the rectal mucosa. Risk factors: primiparity, operative vaginal delivery, midline episiotomy, large baby, and precipitous delivery. Repair: absorbable sutures, layers corresponding to the injury.

Perineal Protection

– Techniques used during the second stage to minimize perineal trauma (lacerations, episiotomy). Manual protection: the attendant's hand applies gentle pressure to the fetal head to control the speed of delivery (not too fast), supporting the perineum and flexing the fetal head. Warm compresses applied to the perineum during crowning may reduce third- and fourth-degree lacerations. Maternal pushing efforts should be gentle during crowning to allow the perineum to stretch gradually. Perineal massage in the second stage may reduce trauma.

Perineal Tear

– Laceration of the perineum during vaginal delivery, classified by degree: first degree (skin only), second degree (perineal muscles), third degree (sphincter complex involvement), and fourth degree (rectal mucosa). Most deliveries involve first or second degree tears. Third and fourth degree tears risk fecal incontinence. Risk factors include nulliparity, operative delivery, and midline episiotomy. Repair under local anesthesia with absorbable suture.

Perineum

– The diamond-shaped area between the pubic symphysis, coccyx, and ischial tuberosities, containing the urogenital and anal triangles. The urogenital triangle contains the external genitalia and urethral opening; the anal triangle contains the anus and coccygeus muscles. During delivery, the perineum stretches to allow passage of the fetus. Perineal tears classified as first through fourth degree.

Peripartum Cardiomyopathy

– See Postpartum Cardiomyopathy.

Pessaries

– Devices inserted into the vagina to support pelvic organs in cases of pelvic organ prolapse (cystocele, rectocele, uterine prolapse). Types: ring pessary (most common, with or without supportive membrane), shelf pessary (for severe prolapse), cube pessary (suction-based), and Gellhorn pessary (for advanced prolapse). Pessaries are fitted by a trained clinician and can be self-managed for removal, cleaning, and reinsertion. Indications: women who decline or are unsuitable for surgery, women awaiting surgery, or those who wish to conceive (pessary avoids surgical risks).

Pessary

– A removable device placed in the vagina to provide structural support for pelvic organ prolapse (uterine prolapse, cystocele, rectocele) or stress urinary incontinence. Pessaries are available in various shapes and sizes: ring pessary (most common—used for mild to moderate prolapse), Gellhorn pessary (for severe prolapse—has a knob that sits above the levator plate), cube pessary (suction-cup shaped, for advanced prolapse), and incontinence pessary (with a knob that supports the urethrovesical junction). Pessaries are fitted by a trained clinician; the correct size prevents discomfort or expulsion. Patients are taught to remove, clean, and reinsert the pessary (if they can) or return every 3–6 months for cleaning and inspection. Complications: vaginal discharge, ulceration, bleeding, and rarely fistula formation. Oestrogen cream may reduce vaginal erosion in postmenopausal women.

Physiologic Jaundice of Newborn

– A common, benign condition characterized by unconjugated hyperbilirubinemia appearing after 24 hours of age (peak at 3-5 days) in healthy term newborns. Caused by increased bilirubin production (high RBC turnover, short RBC lifespan) and immature hepatic conjugation (low UDP-glucuronyl transferase activity). Total bilirubin typically <12-15 mg/dL. Resolves spontaneously within 1-2 weeks. Management: monitor bilirubin, assess risk factors, and treat with phototherapy if bilirubin exceeds thresholds on the Bhutani nomogram.

Placental Abruption

– Premature separation of the placenta from the uterine wall before delivery, occurring in 0.5-1% of pregnancies. Risk factors: hypertension, preeclampsia, smoking, cocaine use, trauma, prior abruption, and thrombophilia. Presents with vaginal bleeding (concealed or revealed), abdominal pain, uterine tenderness, and uterine hypertonicity (board-like abdomen). Fetal distress may be present. Complications: maternal hemorrhage, DIC (due to release of thromboplastin), fetal hypoxia, and fetal death. Diagnosis: clinical; ultrasound (retroplacental clot, sensitivity limited). Management: immediate delivery regardless of gestational age if maternal or fetal compromise. Severe abruption: emergency cesarean, blood transfusion, and coagulopathy management.

Placental Accreta Spectrum

– Abnormal placental attachment to the myometrium, ranging from accreta (adherent to decidua basalis) to increta (invading myometrium) to percreta (invading through serosa). Major risk factor is prior caesarean section with anterior placenta previa. Diagnosis by prenatal ultrasound or MRI, with findings including loss of the clear zone between the placenta and myometrium, abnormal placental lacunae, turbulent flow on Doppler, and increased vascularity. Management requires a multidisciplinary team approach with planned caesarean hysterectomy, as attempted manual removal of the placenta leads to massive hemorrhage. Conservative management (leaving the placenta in situ) may be considered in carefully selected cases but carries risks of infection, delayed hemorrhage, and sepsis. Maternal mortality is approximately 5-10%, primarily from hemorrhage. The rising incidence parallels the increasing caesarean delivery rate.

Placental Separation Signs

– Clinical signs indicating the placenta has separated from the uterine wall and is ready for delivery. Classic signs: lengthening of the umbilical cord (cord lengthens as the placenta descends), a sudden gush of blood, and uterine elevation and firming (the uterus rises and becomes globular as the placenta moves into the lower segment). The mother may feel an urge to push. Manual exam may show the placenta in the vagina. The Brandt-Andrews maneuver (controlled cord traction) is performed after signs of separation are observed.

Placenta Praevia

– Placental implantation over or near the internal cervical os, affecting approximately 0.5% of pregnancies at term. Risk factors: previous caesarean section (risk increases with number of CS), placenta praevia in a prior pregnancy, multiple gestation, advanced maternal age (>35), smoking, cocaine use, and IVF. Classification per ultrasound: complete (placenta covers the os completely), partial (covers the os partially when dilated), marginal (placental edge reaches but does not cover the os), and low-lying (placental edge within 20 mm of the os). Presentation: painless, bright red vaginal bleeding in the second half of pregnancy (typically after 28 weeks—sentinel bleed). Diagnosis is by transvaginal ultrasound (accurate, safe—no risk of provoking bleeding). Management: (1) Antepartum: hospitalisation for significant bleeding, corticosteroids (betamethasone 12 mg IM $\times 2$, 24 hours apart) for fetal lung maturity if preterm, anti-D immunoglobulin for Rh-negative women, avoid vaginal examination until placenta praevia is excluded. (2) Delivery: elective caesarean section at 36–37 weeks (increased bleeding risk), or earlier if bleeding is heavy. If the placenta is anterior and there is a prior CS scar, assess for placenta accreta spectrum (MRI or Doppler ultrasound). Complications: massive obstetric hemorrhage (blood transfusion often required), placenta accreta/increta/percreta, preterm delivery, and disseminated intravascular coagulation.

Placenta Previa

– A condition where the placenta implants in the lower uterine segment, partially or completely covering the internal cervical os. Classified as complete (total), partial, marginal, or low-lying. Major risk factor is prior cesarean section. Presents with painless, bright red vaginal bleeding in the third trimester. Diagnosis by transvaginal ultrasound. Management includes pelvic rest, corticosteroids for lung maturity, and cesarean delivery for complete previa.

Polyhydramnios

– Excess amniotic fluid volume, defined as amniotic fluid index greater than 24 cm or single deepest pocket greater than 8 cm. Occurs in 1-2% of pregnancies. Causes include gestational diabetes, fetal anomalies (GI atresia, neural tube defects), chromosomal abnormalities, and idiopathic. Complications include preterm labor, preterm premature rupture of membranes, cord prolapse, and malpresentation. Mild cases managed expectantly; severe cases may require amnioreduction.

Postmenopausal osteoporosis

– Bone loss caused by lower estrogen levels associated with menopause. Sometimes called type I osteoporosis.

Postmenopause

– The period in a woman's life lasting from the end of perimenopause until the end of life.

Postpartum Blues

– A transient, self-limited condition affecting 50-80% of women, characterized by mood swings, tearfulness, irritability, and anxiety beginning 2-4 days after delivery and resolving within 2 weeks. No psychomotor disturbance or suicidal ideation. Treatment: reassurance, support, and rest. No pharmacotherapy needed. Differentiated from PPD by timing, duration, and lack of functional impairment. Women with postpartum blues are at increased risk of developing PPD and should be followed closely.

Postpartum Cardiomyopathy (Peripartum Cardiomyopathy)

– An idiopathic dilated cardiomyopathy presenting with heart failure in the last month of pregnancy or within 5 months of delivery. Incidence: 1:1000-1:4000 live births. Risk factors: advanced maternal age, African descent, multiple gestation, hypertension, and preeclampsia. Presents: dyspnea, orthopnea, edema, fatigue, and chest pain. Diagnosis: echocardiography (LVEF <45%, LV dilatation). Management: standard heart failure therapy (beta-blockers, diuretics, ACE inhibitors/ARBs after delivery). Bromocriptine may improve outcomes (prolactin inhibition). Recovery: 50-60% recover LVEF within 6 months. Prognosis: worse in those with LVEF <30% at diagnosis.

Postpartum Depression (PPD)

– A major depressive episode occurring within 4 weeks of delivery, affecting 10-15% of postpartum women. Symptoms persist >2 weeks and include depressed mood, anhedonia, sleep disturbance (insomnia despite exhaustion), appetite changes, fatigue, impaired concentration, guilt, and thoughts of harming self or the baby. Onset: most common within the first 3 months. Risk factors: prior depression, PPD in a previous pregnancy, stressful life events, lack of social support, and infant difficulties. Screening: Edinburgh Postnatal Depression Scale (EPDS, cutoff ≥10). Treatment: cognitive behavioral therapy (CBT), SSRIs (sertraline, citalopram, fluoxetine — compatible with breastfeeding), and social support.

Postpartum Hemorrhage

– Blood loss >500 mL after vaginal delivery or >1000 mL after cesarean section, the leading cause of maternal mortality worldwide. Causes: uterine atony (most common, 80%), retained placenta, genital tract trauma (lacerations, episiotomy), and coagulopathy. Presents with heavy vaginal bleeding, hypotension, tachycardia, and pallor. Management: uterine massage, bimanual compression, uterotonic medications (oxytocin, misoprostol, methylergonovine, carboprost), balloon tamponade, uterine artery embolization, and surgical interventions (B-lynch suture, hysterectomy) for refractory cases.

Postpartum Hemorrhage (Secondary)

– Excessive bleeding occurring between 24 hours and 12 weeks postpartum. Causes: retained products of conception (most common), endometritis, subinvolution of the placental site, and coagulopathy. Presents with heavy or prolonged bleeding, cramping, and passing clots. Diagnosis: ultrasound (retained tissue, thickened endometrium). Management: uterine evacuation (D&C), antibiotics, uterotonics, and rarely, uterine artery embolization. Retained products of conception can be missed if the ultrasound is performed too early (endometrial stripe not reliable before 2 weeks).

Postpartum Infection

– Infections occurring after delivery, a leading cause of maternal morbidity. Types: endometritis (most common, especially after cesarean), wound infection (cesarean incision, episiotomy, perineal laceration), urinary tract infection, mastitis, septic pelvic thrombophlebitis, and pneumonia. Risk factors: prolonged ROM, multiple vaginal exams, cesarean section, chorioamnionitis, and retained products of conception. Presentation: fever (≥38°C after the first 24 hours postpartum), uterine tenderness, foul lochia, and wound erythema/drainage. Treatment: broad-spectrum antibiotics (gentamicin + clindamycin for endometritis, cephalexin for wound infection).

Postpartum Psychosis

– A rare but severe psychiatric emergency (1-2 per 1000 deliveries) characterized by the acute onset of delusions, hallucinations, disorganized behavior, confusion, and mood swings (mania or depression) within 1-4 weeks postpartum. Risk factors: bipolar disorder (50% risk of recurrence in the postpartum period), prior postpartum psychosis, and a family history of bipolar disorder. High risk of infanticide and suicide requiring immediate psychiatric hospitalization. Treatment: mood stabilizers (lithium), antipsychotics (olanzapine, haloperidol), and ECT for severe cases. Breastfeeding considerations: lithium is relatively contraindicated; antipsychotics are preferred.

Postpartum Thyroiditis

– Autoimmune thyroid inflammation presenting within the first year postpartum (typically 2-6 months). Incidence: 5-10% of women, especially those with positive TPO antibodies (50-70% risk). Triphasic pattern: thyrotoxic phase (2-4 months, due to destructive release of stored thyroid hormone), followed by hypothyroid phase (4-8 months), followed by return to euthyroid (80%). The thyrotoxic phase may be mistaken for postpartum depression (fatigue, palpitations, heat intolerance). Treatment: beta-blockers for thyrotoxic symptoms, levothyroxine for hypothyroidism. Up to 30% develop permanent hypothyroidism. Recurrence in future pregnancies is common (70%).

Postterm Pregnancy

– Pregnancy that extends beyond 42 weeks (294 days) of gestation, occurring in 5-10% of pregnancies. Risk factors: prior postterm pregnancy, inaccurate dating, male fetus, and obesity. Complications: increased perinatal mortality (2-3 fold), meconium aspiration syndrome, macrosomia with shoulder dystocia, oligohydramnios (leading to cord compression), and placental insufficiency. Management: accurate dating (first-trimester ultrasound), antepartum surveillance (BPP, NST, AFI) starting at 41 weeks. Induction of labor is recommended by 41-42 weeks to reduce perinatal complications.

Preconception Counseling

– Counseling provided to women of reproductive age before pregnancy to optimize outcomes. Components: folic acid supplementation (400 mcg, 4-5 mg for high risk), confirmation of immunity (rubella, varicella), management of chronic conditions (diabetes HbA1c <6.5%, hypertension, epilepsy, thyroid disease), medication review (stop teratogens: isotretinoin, valproate, ACE inhibitors), vaccinations (MMR, varicella, hepatitis B, Tdap, influenza), healthy weight, smoking/alcohol cessation, genetic carrier screening (cystic fibrosis, spinal muscular atrophy, hemoglobinopathies per ethnicity), and reproductive life planning.

Preeclampsia

– A pregnancy-specific hypertensive disorder defined as new-onset hypertension (blood pressure 140/90 mmHg or higher) after 20 weeks gestation with proteinuria or other end-organ damage. Risk factors include nulliparity, advanced maternal age, obesity, chronic hypertension, diabetes, multiple gestation, preeclampsia in a prior pregnancy, and thrombophilia. Pathophysiology involves abnormal placentation, endothelial dysfunction, and systemic inflammatory response. Diagnosis requires BP $\geq$140/90 on two occasions at least 4 hours apart with proteinuria $\geq$300 mg/24 hours or protein/creatinine ratio $\geq$0.3. In the absence of proteinuria, preeclampsia is diagnosed if new-onset hypertension is accompanied by thrombocytopenia, renal insufficiency, impaired liver function, pulmonary edema, or cerebral/visual symptoms. Management includes antihypertensives for BP $\geq$160/110 mmHg (labetalol, nifedipine, hydralazine), magnesium sulfate for seizure prophylaxis, and delivery for gestational age $\geq$37 weeks or for severe features. Low-dose aspirin (81 mg daily) from 12-36 weeks reduces the risk in high-risk women.

Preeclampsia with Severe Features

– Preeclampsia (BP $\geq$140/90 + proteinuria/end-organ dysfunction) with one or more severe criteria: BP $\geq$160/110 mmHg, thrombocytopenia (<100,000/µL), impaired liver function (AST/ALT >2x normal, epigastric/RUQ pain), progressive renal insufficiency (creatinine >1.1 mg/dL), pulmonary edema, new-onset cerebral or visual symptoms (headache, scotomata, blurred vision). Management: antihypertensives for BP $\geq$160/110 (labetalol, hydralazine, nifedipine), magnesium sulfate for seizure prophylaxis, and delivery (regardless of gestational age if maternal/fetal instability, otherwise at 34-37 weeks with corticosteroids).

Pregnancy

– The physiological state in which a developing embryo or fetus grows within the female uterus, typically lasting approximately 40 weeks from the last menstrual period to childbirth. It involves complex hormonal and anatomical changes that support fetal development, including increased blood volume, altered metabolism, and growth of the placenta. Regular prenatal care is essential for monitoring maternal health and fetal growth, as well as identifying potential complications such as gestational diabetes, preeclampsia, or preterm labor.

Pregnancy-Induced Hypertension

– Hypertension developing after 20 weeks gestation without other features of preeclampsia. Blood pressure 140/90 mmHg or higher on two readings at least 4 hours apart. Affects 6-8% of pregnancies. Managed with antihypertensive therapy (labetalol, nifedipine, methyldopa) when blood pressure exceeds 160/110 mmHg. Close monitoring for progression to preeclampsia. May resolve postpartum or persist as chronic hypertension.

Premenstrual Syndrome

– A recurrent pattern of physical, emotional, and behavioral symptoms during the luteal phase of the menstrual cycle that resolve with menses. Affects up to 50% of reproductive-age women. Symptoms: irritability, mood swings, depression, anxiety, bloating, breast tenderness, fatigue, and appetite changes. Premenstrual dysphoric disorder (PMDD) is a severe form with marked mood symptoms and functional impairment. Treatment: lifestyle modification, SSRIs (continuous or luteal-phase), oral contraceptives, spironolactone, and CBT.

Prenatal Care

– Regular medical care during pregnancy to monitor maternal and fetal health. Includes history and physical exam, weight and blood pressure monitoring, urine testing, blood work, fundal height measurements, fetal heart rate auscultation, and ultrasound. Schedules visits monthly in first trimester, biweekly in second, and weekly in third trimester. Screens for gestational diabetes, preeclampsia, anemia, infections, and fetal anomalies.

Prenatal Vitamins

– Multivitamin supplements formulated for pregnancy, containing folic acid (400-800 mcg), iron (27-60 mg), calcium (200-300 mg), vitamin D (400-600 IU), DHA (200 mg), and iodine (150-220 mcg). Started preconceptionally or at the first prenatal visit. Evidence supports NTD reduction from folic acid. Iron prevents maternal anemia. Additional screening detects deficiencies (vitamin D, B12, iron). Gummy vitamins may lack iron and calcium.

Preterm Birth

– Delivery before 37 weeks gestation, occurring in 10% of pregnancies worldwide. Subtypes: spontaneous preterm labor (50%), preterm PROM (30%), and medically indicated preterm birth (20%, for maternal or fetal indications). Risk factors: prior preterm birth (strongest), short cervix, multiple gestation, smoking, infection, periodontal disease, and socioeconomic factors. Complications: respiratory distress syndrome (RDS), intraventricular hemorrhage (IVH), necrotizing enterocolitis (NEC), sepsis, and neurodevelopmental impairment. Prevention: progesterone supplementation, cervical cerclage, and screening for short cervix.

Preterm Labor

– Regular uterine contractions causing cervical change before 37 weeks gestation. Risk factors include prior preterm delivery, multiple gestation, infection, and cervical insufficiency. Diagnosis requires regular contractions with documented cervical change. Tocolytic therapy may delay delivery to allow corticosteroid administration for fetal lung maturity. Magnesium sulfate for neuroprotection if delivery anticipated before 32 weeks. Antenatal corticosteroids reduce respiratory distress syndrome.

Preterm Premature Rupture of Membranes

– Rupture of the amniotic membranes before the onset of labor at <37 weeks gestation. Causes: infection (subclinical chorioamnionitis), cervical insufficiency, and prior PPROM. Presents with sudden gush or leakage of fluid per vagina. Diagnosis: sterile speculum exam (pooling of amniotic fluid), ferning pattern on microscopy, and nitrazine test (turns blue, pH >6.5). Management: depends on gestational age. <24 weeks: counseling for expectant management or induction. 24-34 weeks: expectant management with corticosteroids (betamethasone for fetal lung maturity), antibiotics (ampicillin + erythromycin to prolong latency and reduce neonatal complications), and monitoring for chorioamnionitis. At term: induction of labor.

Progesterone Supplementation

– Administration of progesterone to prevent preterm birth in high-risk women. Natural progesterone (17-alpha hydroxyprogesterone caproate) shown to reduce preterm birth rate in women with prior spontaneous preterm delivery. Also beneficial for short cervix detected on screening. Typically started in the second trimester. Route options include weekly intramuscular injections or vaginal suppositories.

Progestogen

– A synthetic form of progesterone, the hormone that maintains the uterine lining in the secretory phase of the menstrual cycle and supports early pregnancy. Progestogens are used in: combined oral contraceptives (with estrogen), progestogen-only contraceptives (mini-pill, Depo-Provera injection, implant, Mirena IUS), HRT (with estrogen to protect the endometrium), treatment of menstrual disorders (heavy bleeding, endometriosis), and luteal phase support in IVF. Examples: norethisterone, levonorgestrel, medroxyprogesterone acetate, and micronized progesterone.

Prolapse of the Uterus (Uterine Prolapse)

– Descent of the uterus into the vaginal canal due to weakening of pelvic floor supports, most commonly the uterosacral and cardinal ligaments. Graded by the POP-Q system (Pelvic Organ Prolapse Quantification): Stage 0 (no prolapse), Stage I (descent >1 cm above hymen), Stage II (descent to within 1 cm of hymen), Stage III (descent >1 cm below hymen), Stage IV (complete eversion of vagina). Risk factors: vaginal childbirth (especially multiple, prolonged, or traumatic deliveries), aging, menopause (oestrogen deficiency), obesity, chronic cough, constipation, and heavy lifting. Symptoms: vaginal bulge or pressure, sensation of something falling out, urinary incontinence or retention, incomplete bladder emptying, and difficulty with defecation. Diagnosis: clinical, confirmed on pelvic examination with Valsalva manoeuvre. Treatment: observation for asymptomatic mild prolapse; pelvic floor muscle training (Kegel exercises); vaginal pessary (ring or shelf pessary for symptomatic relief); surgery (vaginal hysterectomy with uterosacral ligament suspension, sacrospinous fixation, or sacrocolpopexy for apical prolapse). Concomitant anti-incontinence procedures may be performed.

Prolonged Deceleration

– A fetal heart rate deceleration lasting ≥2 minutes but <10 minutes. If the deceleration persists ≥10 minutes, it is a baseline change. Causes: maternal hypotension (epidural-induced aortocaval compression), uterine hyperstimulation, umbilical cord prolapse, placental abruption, and maternal seizure. Management: immediate intrauterine resuscitation (position change, maternal oxygen, IV fluids, stop oxytocin, treat hypotension). Vaginal exam to exclude cord prolapse. If not resolved in minutes, emergency delivery (cesarean or operative vaginal).

Prolonged Second Stage

– Second stage of labor exceeding the time limits:

nulliparas — 3 hours with regional anesthesia (2 hours without); multiparas — 2 hours with regional (1 hour without). Management: reassess position, contractions, and fetal status. For inadequate contractions: oxytocin augmentation. For malposition (OP, OT): manual rotation. For inadequate maternal effort: coaching or rest. If vaginal delivery is not imminent after a reasonable trial, operative vaginal delivery or cesarean is indicated. Prolonged second stage increases maternal (hemorrhage, chorioamnionitis, lacerations) and neonatal (acidosis, low APGAR, NICU admission) risks.

Prostaglandins

– Lipid compounds involved in labor initiation and cervical ripening. Endogenous prostaglandins (PGE2, PGF2alpha) stimulate uterine contractions and promote cervical softening. Synthetic analogs used clinically: misoprostol (PGE1) and dinoprostone (PGE2 gel or insert) for cervical ripening and labor induction. Also used for postpartum hemorrhage management. Contraindicated in patients with previous cesarean section due to uterine rupture risk.

Pudendal Block

– A local anesthetic block of the pudendal nerve (S2-S4) used for perineal anesthesia during the second stage of labor, outlet forceps delivery, or perineal repair. Technique: transvaginal approach — a needle guide is placed on the ischial spine, and the needle is advanced through the sacrospinous ligament. 1% lidocaine (10 mL on each side) is injected. Provides perineal and lower vaginal analgesia. Does not provide uterine pain relief. Rarely used today because epidural analgesia provides superior pain relief. Complications: intravascular injection (seizures, cardiac toxicity), ischiorectal hematoma, and sciatic nerve block.

Puerperal Fever

– Fever higher than 38 degrees Celsius occurring during the first 10 days postpartum, not including the first 24 hours. Most common cause is endometritis, particularly after cesarean delivery. Other causes include urinary tract infection, wound infection, mastitis, and thrombophlebitis. Evaluation includes blood cultures, urinalysis, wound assessment, and pelvic exam. Treatment depends on suspected source; broad-spectrum antibiotics for endometritis.

Puerperium

– The period following delivery during which the reproductive organs return to their prepregnancy state, approximately 6 weeks in duration. Characterized by uterine involution, cervical closure, lochial discharge changes, and hormonal shifts. Physical recovery includes perineal healing, breast engorgement, and metabolic readjustment. Emotional changes common; screening for postpartum depression recommended. Return of ovulation variable, particularly with breastfeeding.

Quickening

– The first perception of fetal movement by the mother, typically felt between 16-20 weeks in multiparous women and 18-22 weeks in nulliparous women. Described initially as a subtle fluttering sensation that becomes stronger and more regular as pregnancy progresses. Factors affecting the timing include maternal body habitus, placental location (anterior placenta may delay perception), and gestational age. Quickening was historically used to estimate gestational age before ultrasound became available. The relationship between maternal perception and fetal activity patterns informs kick-count protocols for antepartum surveillance. Absence of perceived movement by 24 weeks warrants ultrasound evaluation for fetal wellbeing.

Rectocele

– A weakening of the vaginal wall that allows the rectum to bulge into the vagina.

Rectum

– The final straight section of the large intestine, approximately 12-15 cm long, extending from the sigmoid colon to the anal canal. The rectum stores feces before defecation. In obstetrics, the rectum is relevant for: perineal tears (third- and fourth-degree involve the anal sphincter and rectal mucosa), rectocele (prolapse of the rectum into the posterior vaginal wall), and rectal examination (assessing descent of the fetal head or performing an enema).

Renal Changes in Pregnancy

– Renal blood flow increases 50-80% by the second trimester, and GFR increases 50% (creatinine decreases to 0.4-0.6 mg/dL, BUN to 8-10 mg/dL). Glycosuria and aminoaciduria are normal due to decreased tubular reabsorption. The ureters dilate (hydronephrosis of pregnancy, more prominent on the right) due to progesterone and mechanical compression by the gravid uterus. Proteinuria up to 300 mg/day is normal. The kidney increases in size (1-1.5 cm). Serum uric acid decreases due to increased clearance.

Reproductive Organs

– The organs involved in reproduction. Female: ovaries (produce eggs and hormones), fallopian tubes (transport eggs, site of fertilization), uterus (nurtures the developing fetus), cervix (lower part of uterus, dilates during labor), vagina (passage for sperm and baby), and vulva (external genitalia). Male: testes (produce sperm and testosterone), epididymis (sperm maturation), vas deferens (sperm transport), seminal vesicles and prostate (produce seminal fluid), and penis (delivers sperm). Reproductive years: the period from menarche to menopause during which a woman is capable of conceiving.

Reproductive Years

– The period of a woman's life during which she is capable of conceiving, typically from menarche (first menstrual period, average age 12-13) to menopause (final menstrual period, average age 51). During the reproductive years, women experience regular menstrual cycles, ovulation, and fertility that peaks in the mid-20s and declines after age 35. The most fertile window is approximately 5-6 days before ovulation. Contraception may be used to prevent pregnancy during the reproductive years.

Respiratory Changes in Pregnancy

– Functional residual capacity (FRC) decreases by 20% due to elevated diaphragm from the gravid uterus. Tidal volume increases 30-40%, minute ventilation increases 50% (due to progesterone-mediated increase in respiratory drive). Result: chronic respiratory alkalosis (PaCO2 28-32 mmHg) with compensatory renal bicarbonate excretion (serum HCO3 18-22 mEq/L). PO2 is slightly increased (100-110 mmHg). Oxygen consumption increases 20%. The decreased FRC means pregnant women desaturate faster during apnea (important for airway management).

Retained Placenta

– Failure of the placenta to deliver spontaneously within 30 minutes after delivery of the baby. Incidence: 1-3% of deliveries. Types: placenta adherens (failure of separation due to inadequate myometrial contractions) and trapped placenta (separated but trapped behind a closed cervix). Risk factors: prior retained placenta, uterine anomaly, placenta previa, and preterm delivery. Management: manual removal of the placenta under anesthesia, oxytocin, and gentle cord traction. Retained placenta increta/percreta may require hysterectomy. Complications: hemorrhage, infection, and Asherman syndrome.

Retained Products of Conception (RPOC)

– Fragments of placental or fetal tissue retained in the uterus after delivery, miscarriage, or abortion. Causes: incomplete placental separation, succenturiate lobe, and placenta accreta. Presents: persistent or heavy bleeding, pain, and fever in the postpartum period. Diagnosis: ultrasound (echogenic mass in the uterine cavity, Doppler shows vascularity). Management: dilation and curettage (D&C) or hysteroscopic resection. Antibiotics if febrile. RPOC is a common cause of secondary postpartum hemorrhage.

Rhesus (Rh) Factor

– See Rhesus (Rh) Factor in Hematology.

Rh Isoimmunization

– Maternal alloantibody production against fetal Rh D-positive red blood cells in an Rh D-negative mother, occurring when fetal RBCs enter the maternal circulation (fetomaternal hemorrhage). Sensitisation typically occurs at delivery of an Rh-positive infant, but may also follow miscarriage, ectopic pregnancy, amniocentesis, chorionic villus sampling, abdominal trauma, or antepartum hemorrhage. The maternal immune system produces anti-D antibodies (IgG) that cross the placenta and cause haemolysis of fetal RBCs, leading to fetal anemia, hydrops fetalis, and potentially fetal death in subsequent pregnancies. Prevention with RhoGAM (anti-D immune globulin) at 28 weeks gestation and within 72 hours of potential sensitising events has reduced the incidence from 10-15% to less than 0.1%. Affected pregnancies are managed with serial middle cerebral artery Doppler monitoring and intrauterine transfusions if fetal anemia develops.

RhoGAM

– Rh immune globulin containing anti-D antibodies, used to prevent Rh isoimmunization in Rh-negative mothers. Administered intramuscularly at 28 weeks gestation and within 72 hours after delivery of an Rh-positive infant. Also given after amniocentesis, chorionic villus sampling, miscarriage, ectopic pregnancy, or abdominal trauma. Works by coating and clearing fetal Rh-positive red blood cells before the maternal immune system can respond.

Rubin Maneuver

– An internal rotational maneuver for shoulder dystocia: the operator inserts a hand and applies pressure to the posterior aspect of the anterior fetal shoulder, rotating it toward the fetal chest (adducting the shoulders). This reduces the bisacromial diameter and disimpacts the anterior shoulder from the pubic symphysis. Often performed in sequence: Rubin I (external suprapubic pressure), Rubin II (internal pressure on the posterior aspect of the anterior shoulder). Performed after McRoberts and suprapubic pressure.

Salpingectomy

– Surgical removal of a fallopian tube, the definitive treatment for tubal ectopic pregnancy and for hydrosalpinx. Laparoscopic approach is standard (three-port technique). Indications: ruptured ectopic, hemodynamic instability, failed medical management, contralateral tube is healthy, and patient desires sterilization. Salpingectomy preserves ovarian function if the blood supply (ovarian artery anastomosis) is preserved. Methotrexate is an alternative for early, unruptured ectopic pregnancy.

Salpingitis

– Inflammation of the fallopian tubes, most commonly caused by ascending infection from the lower genital tract. Salpingitis is a component of pelvic inflammatory disease and is often bilateral. etiology: sexually transmitted infections—Chlamydia trachomatis (most common), Neisseria gonorrhoeae, and bacterial vaginosis-associated organisms (Gardnerella vaginalis, anaerobes—Bacteroides, Peptostreptococcus, Atopobium vaginae). Clinical: bilateral lower abdominal pain, abnormal vaginal/cervical discharge, fever, dysmenorrhoea, deep dyspareunia, intermenstrual bleeding, and cervical motion tenderness (chandelier sign) on bimanual examination. Diagnosis: clinical (CDC diagnostic criteria—minimum: cervical motion, uterine, or adnexal tenderness + one of: cervical mucopurulent discharge, increased WBC on saline microscopy of vaginal fluid, positive chlamydia/gonorrhea NAAT). Ultrasound may show thickened, fluid-filled tubes (tubo-ovarian complex or abscess). Complications: tubo-ovarian abscess, Fitz–Hugh–Curtis syndrome (perihepatitis—right upper quadrant pain), infertility (tubal factor—risk increases with number of episodes: 10% after 1 episode, 25% after 2, 50% after 3), ectopic pregnancy (6–10-fold increased risk), and chronic pelvic pain. Treatment: antibiotics (regimen: ceftriaxone 500 mg IM single dose + doxycycline 100 mg PO BD for 14 days + metronidazole 500 mg PO BD for 14 days). Hospitalise if severe disease, pregnancy, tubo-ovarian abscess, or failure of oral therapy. Sexual partner(s) must be treated. Prevention: barrier contraception (condoms), chlamydia/gonorrhea screening in sexually active women <25.

Salpingostomy (Salpingotomy)

– A fertility-preserving surgical procedure for tubal ectopic pregnancy: a linear incision on the antimesenteric border of the fallopian tube to remove the ectopic gestation, leaving the tube in situ. Indications: unruptured ectopic, desire for future fertility, contralateral tubal damage or absence. Risk: persistent trophoblast (up to 15%, requires hCG monitoring and possibly methotrexate). The tube may heal with scarring, reducing patency (50-60% patency rate). Fertility rates after salpingostomy are similar to salpingectomy if the contralateral tube is normal.

Second Stage of Labor

– From full cervical dilatation (10 cm) to delivery of the baby. The mother feels the urge to push as the presenting part descends. Pushing: closed-glottis (Valsalva) or open-glottis. Duration: up to 3 hours in nulliparas with regional anesthesia (2 hours without), and 2 hours in multiparas (1 hour without). Prolonged second stage: longer than these limits. Management: continued maternal effort, operative vaginal delivery (vacuum or forceps), or cesarean delivery if vaginal delivery is not imminent.

Second Trimester

– The period from 13 to 27 weeks gestational age. Fetal growth accelerates, mother begins to feel quickening (fetal movement) between 16-20 weeks.

Anatomy ultrasound at 18-22 weeks screens for structural anomalies. Maternal serum screening offered. Nausea typically resolves. Uterus becomes palpable above the symphysis pubis. Amniocentesis may be performed between 15-20 weeks. Risk of miscarriage decreases substantially after the first trimester. Cervical length screening by transvaginal ultrasound is performed in high-risk women between 18-24 weeks to assess preterm birth risk.

Semen

– The fluid ejaculated from the male reproductive tract, containing sperm (produced in the testes) suspended in seminal plasma (produced by the seminal vesicles, prostate, and bulbourethral glands). Semen provides nutrients (fructose, amino acids), protective agents (zinc, antioxidants), and a medium for sperm transport. Normal semen parameters (WHO 2021): volume $\geq$1.4 mL, sperm concentration $\geq$16 million/mL, total sperm count $\geq$39 million, progressive motility $\geq$30%, normal morphology $\geq$4%.

Sheehan Syndrome (Postpartum Pituitary Necrosis)

– Ischemic necrosis of the anterior pituitary gland due to severe postpartum hemorrhage (PPH) and hypotension. The pituitary enlarges during pregnancy and is vulnerable to ischemia from hypotensive episodes. Clinical features: failure of lactation (first sign, most common), amenorrhea, loss of axillary and pubic hair, breast atrophy, fatigue, hypoglycemia, and hypotension. Diagnosis: low serum cortisol, low TSH/T4, low LH/FSH, low prolactin, and MRI (empty sella). Treatment: lifelong hormone replacement (glucocorticoids first, thyroid hormone, estrogen, and growth hormone as needed).

Shirodkar Cerclage

– A transvaginal cervical cerclage technique that involves dissection of the bladder and rectum to place the suture (Mersilene tape) as high as possible at the level of the internal os. Technically more difficult than McDonald cerclage with higher risk of hemorrhage and bladder injury. Indications: cervical insufficiency when the cervix is too short for McDonald placement, or failed McDonald cerclage. The suture is buried below the vaginal mucosa. Removed at 36-37 weeks.

Shoulder Dystocia

– An obstetric emergency where the fetal head delivers but the anterior shoulder is impacted behind the maternal symphysis pubis. Defined as head-to-body delivery time exceeding 60 seconds or need for ancillary maneuvers. Risk factors include macrosomia, diabetes, prior shoulder dystocia, and operative vaginal delivery. Management follows HELPERR mnemonic: call for help, episiotomy, legs (McRoberts), suprapubic pressure, internal rotation (Woods screw), delivery of posterior arm, and clavicle fracture.

Sinusoidal Fetal Heart Rate Pattern

– A rare, ominous FHR pattern characterized by a smooth, sine-wave-like undulation with a fixed frequency (2-5 cycles/minute) and absent variability. Strongly associated with severe fetal anemia (Rh isoimmunization, fetomaternal hemorrhage, parvovirus B19), fetal hypoxia, and fetal acidosis. Differential: pseudo-sinusoidal pattern (from maternal Demerol administration, usually resolves). A true sinusoidal pattern is a Category III tracing requiring immediate evaluation, intrauterine resuscitation, and urgent delivery.

Soft Markers for Aneuploidy

– Sonographic findings in the second trimester that are common in normal fetuses but have a statistically higher prevalence in aneuploid fetuses (especially trisomy 21). Common soft markers: echogenic intracardiac focus (EIF), echogenic bowel, renal pyelectasis (4-7 mm), short femur/humerus, nuchal fold $\geq$6 mm, choroid plexus cyst, and sandal gap. Each marker has a likelihood ratio (LR) for trisomy 21 (1.5-5). In isolation and with low-risk serum screening, no further workup is needed. Multiple markers increase risk and warrant genetic counseling.

Sonographer

– A healthcare professional trained to perform ultrasound scans (sonography). In obstetrics, sonographers perform: dating scans (first trimester), nuchal translucency measurement (11-14 weeks), anomaly scans (18-21 weeks), growth scans (serial assessment of fetal size and amniotic fluid), and Doppler studies (umbilical artery, middle cerebral artery). Sonographers also perform gynecological scans (pelvic ultrasound for ovarian cysts, fibroids, endometriosis). They work closely with obstetricians and radiologists to interpret findings.

Special Care Baby Unit

– A ward within a hospital that provides specialist care for newborn babies who need additional support but not intensive care. SCBU provides: monitoring of premature babies (usually born at 32-36 weeks), phototherapy for jaundice, treatment of hypoglycemia, intravenous fluids or nasogastric tube feeding, and treatment of mild respiratory distress. SCBU differs from a Neonatal Intensive Care Unit (NICU), which provides full intensive care including mechanical ventilation for the sickest or most premature infants.

Sperm

– The male gamete (reproductive cell), produced in the seminiferous tubules of the testis through spermatogenesis. Sperm consists of: head (contains the haploid nucleus and acrosome containing enzymes for egg penetration), midpiece (contains mitochondria for energy), and tail (flagellum for motility). Approximately 1,500 sperm are produced per second. Sperm travel from the testis through the epididymis (maturation), vas deferens, seminal vesicles, and urethra during ejaculation.

Spinal Anesthesia for Cesarean Delivery

– A neuraxial anesthetic technique used for elective and many emergency cesarean deliveries. A small dose of local anesthetic (hyperbaric bupivacaine 12-15 mg) with opioid (fentanyl 15-25 mcg, morphine 0.15-0.25 mg) is injected into the subarachnoid space. Produces rapid surgical anesthesia (T4 sensory level for cesarean). Advantages: minimal fetal drug exposure (compared to general anesthesia), the mother is awake for delivery, and excellent postoperative analgesia with intrathecal morphine. Disadvantages: maternal hypotension (from sympathectomy — prevent with left uterine displacement, IV fluids, vasopressors), pruritus, nausea, and PDPH (post-dural puncture headache, 1-2% with pencil-point needles).

Sporadic

– Occurring in isolated, random, or scattered instances, without a predictable pattern or clear cause. In medicine, sporadic cases of a disease occur in individuals with no family history of the condition (as opposed to familial or hereditary cases). Sporadic cancers arise from acquired (somatic) mutations rather than inherited germline mutations. Most cases of ovarian, breast, and endometrial cancers are sporadic. Sporadic miscarriage is usually due to random chromosomal abnormalities in the embryo.

Stages of Labor

– Four stages describing the progression from uterine contractions to delivery of the placenta. First stage: onset of regular contractions to full cervical dilatation (10 cm). Second stage: full dilatation to delivery of the baby. Third stage: delivery of the baby to delivery of the placenta. Fourth stage: the first 1-2 hours after placental delivery (immediate postpartum, highest risk of hemorrhage). Normal labor requires coordinated uterine contractions, progressive cervical change, and descent of the fetal presenting part.

Station

– The relationship of the fetal presenting part to the level of the ischial spines, measured in centimeters.

Sterilisation

– A permanent method of contraception. Female sterilization (tubal occlusion): the fallopian tubes are blocked or cut to prevent the egg from reaching the uterus, performed by laparoscopy (clip or diathermy) or at the time of cesarean delivery. Male sterilization (vasectomy): the vas deferens is cut or sealed to prevent sperm from entering the ejaculate. Both are highly effective (>99%) and intended to be permanent, though reversal is sometimes possible through microsurgery.

Striae Gravidarum

– Stretch marks developing on the abdomen, breasts, thighs, and buttocks during pregnancy due to mechanical stretching of the dermis combined with hormonal effects on collagen. Initially appear as red or purple striae (striae rubrae) and eventually fade to silver-white (striae albae). Occur in 50-90% of pregnancies. Prevention with oils and creams has limited evidence. Do not affect fetal development.

Stripping of Membranes (Membrane Sweep)

– A procedure performed during a cervical exam where the examiner inserts a finger through the internal os and sweeps the membranes circumferentially, separating them from the lower uterine segment. Releases endogenous prostaglandins, promoting cervical ripening and potentially initiating labor. Performed at 38-41 weeks to reduce the need for formal induction. Discomfort: moderate cramping and spotting. Contraindications: low-lying placenta, placenta previa, and active bleeding. Studies show reduced postterm pregnancy and induction rates with serial sweeping. Nipple stimulation may also augment endogenous oxytocin.

Substance Use in Pregnancy

– Use of alcohol, tobacco, marijuana, opioids, cocaine, and other drugs during pregnancy. Tobacco: increased risk of IUGR, preterm birth, placental abruption, SIDS, and low birth weight. Opioids: neonatal abstinence syndrome (NAS, withdrawal in 60-80% of exposed newborns), treated with morphine or methadone. Alcohol: FASD. Marijuana: limited evidence but possible impact on fetal neurodevelopment. Cocaine: abruption, IUGR, preterm birth. Screening: urine drug screen with patient consent (universal screening recommended, but risk of legal/social consequences).

Succenturial Lobe

– An accessory lobe of the placenta, connected to the main placenta by blood vessels that traverse the membranes. If undetected after delivery, the succenturial lobe may be retained in the uterus, causing postpartum hemorrhage or infection. Diagnosis: careful inspection of the placenta and membranes after delivery. A succenturial lobe can also be associated with vasa praevia (fetal vessels running across the internal cervical os), which can cause life-threatening fetal hemorrhage when membranes rupture.

Superimposed Preeclampsia

– Preeclampsia developing in a woman with preexisting chronic hypertension. Diagnosis: sudden worsening of hypertension, new-onset proteinuria or a sudden increase in proteinuria, or new-onset thrombocytopenia, elevated liver enzymes, or symptoms (headache, visual changes). Occurs in 20-50% of women with chronic hypertension. Higher risk of adverse maternal and fetal outcomes compared to preeclampsia alone. Requires close monitoring and early delivery if severe features develop.

Supine Hypotensive Syndrome

– Compression of the inferior vena cava by the gravid uterus when the pregnant woman lies flat on her back after 20 weeks gestation. Results in decreased venous return, reduced cardiac output, and maternal hypotension, dizziness, pallor, and syncope. Uteroplacental perfusion may be compromised, potentially causing fetal bradycardia and distress. The syndrome typically occurs after 20 weeks gestation when the uterus is large enough to compress the great vessels. Prevention involves avoiding the supine position and maintaining left lateral tilt during labour, surgery, and transport. Treatment includes repositioning to the left lateral decubitus position and IV fluid bolus. The syndrome is especially relevant during regional anesthesia (spinal or epidural) where loss of sympathetic tone compounds the hemodynamic effect.

Sutures

– Surgical threads or stitches used to close wounds or surgical incisions. In obstetrics, sutures are used for: episiotomy repair, perineal tear repair (1st-4th degree), cesarean section closure (uterine, fascial, and skin layers), and cervical cerclage. Absorbable sutures (Vicryl, Monocryl, chromic catgut) dissolve over time and do not require removal. Nonabsorbable sutures (nylon, Prolene) require removal. Continuous suturing is faster; interrupted sutures may provide better wound edge apposition.

Tachysystole (Uterine Hyperstimulation)

– More than 5 contractions in 10 minutes averaged over 30 minutes, or contractions lasting ≥ 2 minutes. Tachysystole with fetal heart rate changes (late decelerations, prolonged deceleration, loss of variability) is more concerning. Causes: oxytocin (most common), prostaglandins (misoprostol > dinoprostone), and cocaine. Management: stop oxytocin (if running), administer a tocolytic (terbutaline 0.25 mg SQ for persistent tachysystole with nonreassuring FHR), maternal position change, IV fluids, and oxygen. Uterine hypertonus: increased resting tone between contractions.

Tdap Vaccination in Pregnancy

– Administration of tetanus, diphtheria, and acellular pertussis (Tdap) vaccine during each pregnancy, optimally at 27-36 weeks. The goal: boost maternal antibodies to protect the newborn from pertussis (whooping cough) before the infant can be vaccinated at 2 months. Transplacental transfer of anti-pertussis antibodies provides passive immunity to the infant. Tdap is safe in pregnancy. The cocooning strategy (vaccinating family members) is no longer recommended as a primary strategy; maternal vaccination is more effective.

Teratoma

– A type of germ cell tumor composed of tissues derived from two or three germ layers (ectoderm, endoderm, mesoderm), reflecting pluripotent differentiation. Teratomas are classified as mature (benign—well-differentiated tissues, most common in children and young adults) or immature (malignant—primitive/embryonal tissues, higher risk of recurrence and metastasis). Sites: ovaries (most common—dermoid cyst/mature cystic teratoma, usually benign), testes (prepubertal boys—benign; postpubertal men—may contain malignant germ cell elements), sacrococcygeal region (most common congenital tumor in neonates, 1 in 35,000 births), mediastinum (anterior mediastinal germ cell tumor), and intracranial (pineal region, suprasellar). Ovarian dermoid cyst: contains hair, teeth, sebum, bone, and thyroid tissue, often discovered incidentally on ultrasound. Treatment: surgical resection (ovarian cystectomy or oophorectomy for ovarian teratoma). Mature teratoma has an excellent prognosis; immature teratoma requires staging, chemotherapy (BEP: bleomycin, etoposide, cisplatin), and surveillance. Sacrococcygeal teratoma in neonates requires early resection (risk of malignant transformation and bleeding).

Third Stage of Labor

– From delivery of the baby to delivery of the placenta. The placenta separates from the uterine wall and is expelled spontaneously or with gentle cord traction. Normal duration: 5-30 minutes. Active management of the third stage: administration of oxytocin immediately after delivery of the baby, controlled cord traction, and uterine massage. This reduces the risk of postpartum hemorrhage by 60%. Retained placenta: failure to deliver the placenta within 30 minutes; management includes manual removal or curettage.

Third Trimester

– The period from 28 weeks to delivery. Rapid fetal weight gain occurs, lungs mature with surfactant production. Fetal movements become regular and countable. Braxton Hicks contractions begin. Group B Streptococcus screening at 36 weeks. Fetal positioning assessed for delivery. Viability generally considered at 23-24 weeks. Cesarean section rate highest in this trimester.

Thrush

– A common vaginal infection caused by the yeast Candida albicans (also called candidiasis, vaginal thrush, or candida vulvovaginitis). Symptoms: itching, soreness, white discharge (like cottage cheese), dyspareunia, and external dysuria. Risk factors: pregnancy, antibiotic use, diabetes, immunosuppression, and high estrogen states. Diagnosis: clinical and confirmed by vaginal swab (microscopy and culture showing budding yeast and pseudohyphae). Treatment: topical clotrimazole or oral fluconazole (single dose). Recurrent thrush (>4 episodes/year) may require maintenance therapy.

Tocolysis

– The use of medications to suppress uterine contractions and delay preterm labor, providing time for corticosteroid administration (fetal lung maturity) and in utero transfer to a unit with appropriate neonatal facilities. First-line: nifedipine (calcium channel blocker) or atosiban (oxytocin receptor antagonist). Second-line: indomethacin (NSAID, short-term). Beta-agonists (terbutaline, ritodrine) are no longer first-line due to maternal side effects. Maintenance tocolysis beyond 48 hours is not recommended.

Tocolytic Therapy

– Medications used to inhibit uterine contractions temporarily to prolong pregnancy, allowing time for corticosteroid administration and transport to a tertiary care center. Indications: preterm labor with intact membranes, 24-34 weeks. First-line: nifedipine (calcium channel blocker, 20-30 mg PO q3-4h) or indomethacin (NSAID, 50-100 mg PR/PO then 25-50 mg PO q6h, limited to 48 hours due to oligohydramnios and premature ductus arteriosus closure). Second-line: terbutaline (beta-agonist, SQ, not for prolonged use). Atosiban (oxytocin receptor antagonist) is used in Europe. Tocolysis should not delay delivery in the setting of chorioamnionitis, severe preeclampsia, or fetal compromise.

Total Breech Extraction

– A maneuver where the operator grasps the fetal feet and delivers the entire baby in breech presentation. The most common indication is delivery of the second twin (malpresentation). Contraindicated for term singleton breech due to high risk of head entrapment, nuchal arms, and birth trauma. Technique: identify the feet, grasp them at the ankles, and apply gentle traction. Deliver the legs, trunk, shoulders, and head sequentially. Anesthesia: adequate relaxation (regional or general) is needed.

Toxemia

– An older term for pre-eclampsia (also called toxemia of pregnancy), reflecting the now-outdated theory that the condition was caused by toxins in the blood. Pre-eclampsia is a multisystem disorder of pregnancy characterized by hypertension and proteinuria developing after 20 weeks gestation, affecting 2-8% of pregnancies. Features: elevated blood pressure (≥140/90 mmHg), proteinuria (≥300 mg/24h), and potentially headache, visual disturbances, epigastric pain, and seizures (eclampsia). The term toxemia has been largely replaced by pre-eclampsia in modern practice.

Transabdominal Scan

– An ultrasound scan performed with the transducer placed on the maternal abdomen. A full bladder improves visualization of pelvic structures. Used in early pregnancy to confirm viability and gestational age, and throughout pregnancy to assess fetal growth, anatomy, amniotic fluid volume, and placental location. Transabdominal scans provide a wider field of view than transvaginal scans but have lower resolution for early pregnancy (before 10-12 weeks) and detailed assessment.

Transcervical Foley Catheter for Cervical Ripening

– A mechanical method for cervical ripening: a Foley catheter (16-30 Fr) is inserted through the cervix, the balloon is inflated with 30-60 mL of saline, and traction is applied (or the catheter is taped to the thigh with gentle tension). The balloon applies direct pressure to the internal os, stimulating endogenous prostaglandin release. Advantages: no increased risk of uterine tachysystole (unlike pharmacologic agents), safe in patients with prior cesarean, and minimal systemic side effects. Often combined with oxytocin after the balloon falls out. Typically <12 hours from insertion to expulsion.

Transverse Lie

– A fetal lie where the long axis of the fetus is perpendicular to the long axis of the mother. The presenting part is a shoulder or arm. Occurs in approximately 1% of term pregnancies. Risk of cord prolapse and obstructed labor. Cesarean delivery required as vaginal delivery is not possible. Version may be attempted to convert to longitudinal lie but often recurs. More common in multiparous women with lax abdominal walls.

TRAP Sequence

– Twin Reversed Arterial Perfusion sequence, a complication of monochorionic twin gestations where one twin (the pump twin) perfuses the other (the acardiac twin) via arterioarterial anastomoses. The acardiac twin receives retrograde, deoxygenated blood through its umbilical artery, resulting in severe anomalies (acardia, acephaly, rudimentary limbs) and no functional heart. The pump twin is structurally normal but at risk for high-output cardiac failure, polyhydramnios, and hydrops from the hemodynamic burden of perfusing the acardiac twin. Diagnosis by ultrasound showing a structurally abnormal twin with absent cardiac activity lacking the visible heartbeat. Doppler demonstrates reverse umbilical artery flow to the acardiac twin. Risk to the pump twin: mortality up to 50% without intervention. Treatment: fetoscopic laser coagulation or radiofrequency ablation of the acardiac twin's umbilical cord to stop perfusion. Outcomes for the pump twin are excellent with early intervention.

Trial of Labor After Cesarean (TOLAC)

– The attempt at vaginal birth in a woman with a prior cesarean, which may result in VBAC or a failed TOLAC (requiring repeat cesarean). Candidates: one prior low-transverse cesarean, clinically adequate pelvis, no contraindications. Counseling: success rate 60-80%, uterine rupture risk 0.5-0.9%, perinatal mortality from rupture <0.1%. Induction of labor increases rupture risk (especially prostaglandins, avoid misoprostol). Oxytocin may be used with caution. Patients must deliver at hospitals capable of emergency cesarean within 30 minutes.

Tubal Pregnancy

– An ectopic pregnancy implanted in the fallopian tube, accounting for >95% of ectopic pregnancies.

Most common site: ampulla (70%). Risk factors: prior ectopic, PID (chlamydia), tubal surgery, smoking, IUD, and ART. Presents with lower abdominal pain, amenorrhea, and vaginal bleeding. Diagnosis: transvaginal ultrasound (no intrauterine pregnancy with hCG above discriminatory zone, typically 1500-3000 IU/L), adnexal mass, or tubal ring. Unruptured: treated with methotrexate (single-dose regimen: 50 mg/m² IM) or salpingectomy/salpingostomy. Ruptured: surgical emergency with hemorrhagic shock requiring salpingectomy.

Twin-to-Twin Transfusion Syndrome

– See complete definition in the Multiple Gestation section above.

Twin-Twin Transfusion Syndrome

– A complication of monochorionic diamniotic twin pregnancies where placental vascular anastomoses (arteriovenous connections) cause unbalanced blood flow from one twin (the donor) to the other (the recipient). The donor twin becomes hypovolemic, anemic, and growth-restricted with oligohydramnios ("stuck twin"). The recipient twin becomes hypervolemic, polycythemic, and develops polyhydramnios, cardiac dysfunction, and hydrops. Quintero staging (I-V): I — oligohydramnios/polyhydramnios sequence without Doppler abnormalities; II — bladder not visible in donor; III — abnormal Doppler studies (absent/reversed end-diastolic flow in donor or reverse flow in ductus venosus in recipient); IV — hydrops in recipient; V — intrauterine fetal demise of one or both twins. Diagnosis by ultrasound in monochorionic twins. Treatment: fetoscopic laser photocoagulation of the placental anastomoses (first-line for Quintero II-IV). Severe TTTS without treatment has a mortality rate exceeding 80%.

Umbilical Artery Doppler

– Ultrasound assessment of blood flow in the umbilical arteries to evaluate placental resistance. Normal flow shows a pulsatility index decreasing throughout pregnancy with forward diastolic flow. Absent or reversed end-diastolic flow indicates severe placental dysfunction and is associated with increased perinatal morbidity and mortality. Used in management of intrauterine growth restriction to guide timing of delivery.

Umbilical Cord

– The connection between the fetus and the placenta, containing two umbilical arteries (carrying deoxygenated blood from the fetus to the placenta) and one umbilical vein (carrying oxygenated blood from the placenta to the fetus). The vessels are embedded in Wharton jelly, a mucopolysaccharide matrix that prevents compression. Normal length: 55-60 cm. The cord inserts centrally into the placenta in most cases. Abnormalities: velamentous insertion (vessels insert into the membranes), vasa previa (vessels cross the internal os), true knots, nuchal cord (around the fetal neck), and single umbilical artery (two-vessel cord).

Umbilical Cord Prolapse

– Descent of the umbilical cord through the cervix alongside or past the presenting fetal part, occurring when the membranes rupture. Risk factors: malpresentation (transverse lie, breech), preterm/low birth weight, polyhydramnios, multiparity, and artificial ROM with high presenting part. Presents as cord palpable on vaginal exam or visible at the introitus after ROM, often with sudden fetal bradycardia. Management: immediate cesarean delivery. While preparing for delivery, relieve cord compression by elevating the presenting part (knee-chest position, Trendelenburg, or manual elevation). Keep the cord inside the vagina to prevent vasospasm. Tocolysis may be used.

Upright Labor Positions

– Maternal positions during labor that use gravity to promote fetal descent: walking, standing, squatting, kneeling, and sitting upright on a birth stool. Benefits of upright vs. recumbent positions: shorter first stage (1 hour), less epidural use, fewer operative deliveries, less perineal trauma, and less severe pain. The supine position is discouraged because it decreases uterine blood flow and cardiac output. Upright positions increase pelvic diameters and contraction efficiency. Water immersion (hydrotherapy) in the first stage is also beneficial.

Urethracele

– A herniation or prolapse of the urethra into the anterior vaginal wall, often occurring together with a cystocele (bladder prolapse). Urethracele is caused by weakening of the pubocervical fascia, often from childbirth, aging, or increased intra-abdominal pressure. Symptoms: stress urinary incontinence (leakage with cough, sneeze, exercise), urinary urgency, and a vaginal bulge. Diagnosis: clinical (pelvic examination). Treatment: pelvic floor physiotherapy, pessary, or surgical repair (anterior colporrhaphy).

Urinary Tract Infection in Pregnancy

– Asymptomatic bacteriuria (2-10% of pregnancies) and symptomatic UTI (cystitis, pyelonephritis). Pregnancy increases the risk of pyelonephritis due to hormonally mediated ureteral dilatation and stasis. Asymptomatic bacteriuria: screening and treatment (nitrofurantoin, cephalexin, amoxicillin) reduce the risk of pyelonephritis and preterm birth. Cystitis: dysuria, frequency, urgency. Pyelonephritis: fever, flank pain, nausea, higher risk of preterm labor/preterm delivery. Treatment: IV antibiotics (ceftriaxone, gentamicin, ampicillin+gentamicin) for pyelonephritis.

Urodynamics

– A set of tests that assess the function of the lower urinary tract (bladder and urethra). Urodynamic studies include: uroflowmetry (flow rate), filling cystometry (bladder pressure during filling, assesses detrusor overactivity and compliance), pressure-flow study (voiding phase, assesses obstruction), and leak point pressure (stress incontinence). Indicated when surgery is planned for stress incontinence, when the diagnosis is unclear, or when mixed incontinence is suspected. Urodynamics helps distinguish genuine stress incontinence from detrusor overactivity.

Uterine Atony

– Failure of the uterus to contract adequately after delivery, resulting in postpartum hemorrhage. The most common cause of PPH. Risk factors include prolonged labor, multiple gestation, macrosomia, oxytocin augmentation, and uterine fibroids. Diagnosis is clinical: boggy, soft uterus with excessive bleeding. Management: uterine massage, oxytocin, methylergonovine, carboprost, misoprostol, balloon tamponade, and surgical interventions (B-Lynch suture, uterine artery ligation) for refractory cases.

Uterine Balloon Tamponade

– A minimally invasive technique for managing postpartum hemorrhage due to uterine atony when uterotonics fail. A balloon (Bakri Postpartum Balloon, 500-1000 mL capacity) is inserted transvaginally into the uterine cavity and inflated with saline until bleeding stops. The balloon provides mechanical compression against the uterine wall. Success rate: 70-90%. The balloon is left in place for 12-24 hours, then gradually deflated. If bleeding continues despite balloon tamponade, surgical intervention (uterine artery embolization, B-Lynch suture, hysterectomy) is indicated. Uterine rupture must be excluded before balloon placement.

Uterine Fibroid (Leiomyoma)

– Benign smooth muscle tumors of the uterus, very common (70-80% of women by age 50). Submucosal: most symptomatic, cause heavy bleeding. Intramural: most common. Subserosal: cause pressure symptoms. Symptoms: menorrhagia, pelvic pain/pressure, urinary frequency, and reproductive dysfunction (infertility, miscarriage). Diagnosis: ultrasound, MRI. Treatment: symptomatic management (NSAIDs, tranexamic acid), hormonal contraceptives, levonorgestrel IUD, GnRH agonists, myomectomy, uterine artery embolization, and hysterectomy.

Uterine Fibroids

– See Uterine Fibroid (Leiomyoma).

Uterine Inversion

– A rare but life-threatening obstetric emergency where the uterus turns inside out, protruding through the cervix or vaginal introitus. Occurs in 1:2000-1:20,000 deliveries. Causes: fundal pressure, excessive cord traction (especially with fundal placenta), uterine atony, and fundal fibroids. Presents: sudden massive hemorrhage, shock out of proportion to blood loss, and a mass at the introitus or vagina. Management: Johnson maneuver (push the fundus back through the cervix with an open palm), tocolytics (terbutaline, magnesium) to relax the uterus, and if manual replacement fails, a surgical approach (Huntington or Hautlain procedure). IV fluids, blood products, and uterotonics after replacement.

Uterine Rupture

– A full-thickness tear through the uterine wall, potentially life-threatening to both mother and fetus. Risk factors include prior caesarean section (especially classical incision), oxytocin augmentation, and operative vaginal delivery. Presents with sudden abdominal pain, loss of fetal heart rate, and maternal hemorrhage. Requires emergent caesarean delivery and surgical repair. Risk is approximately 0.5% in women with a prior low-transverse caesarean attempting trial of labour after caesarean (TOLAC). The risk is higher with classical or T-shaped uterine incisions (5-10%), multiple prior caesareans, short interpregnancy interval, labour induction with prostaglandins (especially misoprostol), and traumatic delivery. Complete rupture involves full-thickness separation through serosa; incomplete rupture (dehiscence) involves separation of the myometrium with intact serosa. Management: emergency caesarean delivery and surgical repair or hysterectomy. Maternal mortality is less than 1%, but perinatal mortality ranges from 5-20%.

Vacuum-Assisted Delivery

– Operative vaginal delivery using a vacuum extractor applied to the fetal scalp. Types: soft cup (Kiwi OmniCup, Mityvac, for outlet and low deliveries) and rigid cup (metal Kiwi, for posterior positions and midpelvic deliveries). Steps: confirm position, cup placed over the flexion point (3 cm anterior to the posterior fontanelle), vacuum is created to 600-650 mm Hg, traction is applied during contractions with maternal pushing. Descent should be progressive; if no descent after three pulls, abandon. Cup pops-off: reposition after inspection. Contraindications: face presentation, <34 weeks (fragile skull), unengaged head. Neonatal risks: cephalohematoma, subgaleal hemorrhage (life-threatening, monitor), retinal hemorrhage.

Vaginal Birth After Cesarean (VBAC)

– Vaginal delivery in a woman who has had a prior cesarean delivery. Success rate: 60-80% in appropriately selected candidates. Contraindications: prior classical or T-incision, prior uterine rupture, vertical uterine incision, more than 2 prior cesareans, and other contraindications to vaginal delivery. Factors favoring success: prior vaginal delivery (especially VBAC), spontaneous labor, non-recurrent indication for prior cesarean. Risk: uterine rupture (0.5-0.9%, higher with labor induction). Resources: continuous EFM, in-hospital capability for emergency cesarean.

Vaginal Delivery (Spontaneous)

– The spontaneous passage of the fetus through the birth canal without the use of operative instruments (vacuum or forceps). Typically occurs in the second stage of labor with maternal pushing. Includes crowning of the fetal head, delivery of the head (with restitution and external rotation), delivery of the anterior shoulder, delivery of the posterior shoulder, and delivery of the body. The umbilical cord is clamped and cut. The baby is placed on the mother's abdomen or chest for skin-to-skin contact. APGAR scores are assigned at 1 and 5 minutes.

Vaginal Dryness and Atrophy (Genitourinary Syndrome of Menopause)

– Vaginal dryness and atrophy, collectively termed genitourinary syndrome of menopause, result from declining estrogen levels that cause thinning, drying, and loss of elasticity of the vaginal tissues. Symptoms include vaginal dryness, irritation, burning, dyspareunia, and urinary symptoms such as frequency and recurrent infections. Treatment options include over-the-counter vaginal lubricants and moisturizers, prescription topical estrogen therapy in various formulations, and nonhormonal options such as ospemifene or vaginal laser therapy.

Vaginismus

– Involuntary spasms of the vaginal muscles.

Vaginitis

– Inflammation of the vaginal mucosa, one of the most common gynecologic complaints. Types: bacterial vaginosis (BV: thin, fishy-odor discharge, clue cells, pH >4.5), vulvovaginal candidiasis (thick, white, cottage-cheese discharge, pruritus, pH normal), and trichomoniasis (frothy, green-yellow discharge, strawberry cervix, pH >4.5, motile trichomonads). Atrophic vaginitis: postmenopausal, due to estrogen deficiency, thin vaginal mucosa, and dyspareunia. Diagnosis: vaginal pH, wet mount microscopy, KOH preparation, and NAAT for trichomonas. Treatment: metronidazole or clindamycin for BV; topical azoles or oral fluconazole for candidiasis; metronidazole or tinidazole for trichomoniasis; topical estrogen for atrophic vaginitis.

Variable Decelerations

– A common fetal heart rate pattern characterized by an abrupt decrease in FHR (onset to nadir <30 seconds) with a variable shape, typically caused by umbilical cord compression. Variable decelerations are classified as mild (<30 sec duration, >80 bpm), moderate (30-60 sec, 70-80 bpm), or severe (>60 sec, <70 bpm). Recurrent variable decelerations (>50% of contractions) with moderate variability are reassuring. Management for repetitive/severe variable decels: maternal position change, amnioinfusion, and reduction of oxytocin.

VEAL CHOP Mnemonic

– A memory aid for interpreting fetal heart rate patterns. VEAL: Variable decelerations → Cord compression, Early decelerations → Head compression, Accelerations → OK (fetal well-being), Late decelerations → Placental insufficiency. CHOP: Cord compression → Change position, Head compression → (no intervention needed), OK → (reassuring), Placental insufficiency → Oxygen. A quick reference for nurses and residents to correlate FHR patterns with their causes and guide initial interventions.

Venous Thromboembolism in Pregnancy

– Pregnancy increases the risk of VTE (DVT and PE) by 5-10 fold due to a hypercoagulable state (increased fibrinogen, factors VII-XII, vWF, decreased protein S, and venous stasis from the gravid uterus). Risk factors: prior VTE, thrombophilia (factor V Leiden, prothrombin G20210A, antithrombin deficiency, protein C/S deficiency), obesity, age >35, and cesarean section. DVT: most commonly left leg (by 3:1), proximal (iliofemoral rather than calf). Diagnosis: compression ultrasonography (DVT), CT pulmonary angiography (PE). Treatment: LMWH (enoxaparin 1 mg/kg BID for full-dose, 40 mg daily for prophylaxis), anticoagulate for entire pregnancy and at least 6 weeks postpartum.

Vitamin K Prophylaxis

– Intramuscular administration of vitamin K (1 mg, 0.5 mg for preterm) to all newborns at birth to prevent vitamin K deficiency bleeding (VKDB, formerly hemorrhagic disease of the newborn). Newborns have low stores of vitamin K (limited placental transfer, sterile gut, low in breast milk). VKDB presents with intracranial, GI, or umbilical bleeding. Oral vitamin K is less effective, especially for late VKDB (2-12 weeks). The IM route is superior and standard in most countries.

Vulvar vestibulitis

– Pain and inflammation of the vulvar vestibule.

Vulvitis

– Inflammation of the vulva (external female genitalia), often presenting with pruritus, burning, irritation, erythema, edema, and discharge. Vulvitis is rarely isolated and is most commonly part of vulvovaginitis (inflammation of both vulva and vagina). Causes: infections (Candida albicans — most common, Trichomonas vaginalis, bacterial vaginosis, herpes simplex, human papillomavirus), irritants (soaps, detergents, douches, perfumed hygiene products, sanitary pads, synthetic underwear, chlorinated water), allergens (latex condoms, spermicides, topical medications), dermatologic conditions (lichen sclerosus, lichen planus, psoriasis, atopic dermatitis, contact dermatitis), and systemic conditions (diabetes mellitus, immunosuppression, oestrogen deficiency in menopause causing atrophic vulvovaginitis). Diagnosis: history (symptom onset, triggers, sexual history), examination (vulvar erythema, fissures, discharge, atrophy, excoriations, lichenification), swabs for culture, pH testing, and KOH preparation. Management: treat underlying cause (antifungals for Candida, antibiotics for bacterial infection, topical corticosteroids for inflammatory dermatoses, oestrogen cream for atrophic vulvitis). General measures: avoidance of irritants, sitz baths, loose cotton underwear, and good vulvar hygiene.

Vulvodynia

– Pain in the vulva that may or may not be brought on by touch or pressure.

Water Birth (Underwater Delivery)

– Delivery of a baby in a tub of warm water. Proponents claim: reduced pain, fewer perineal lacerations, and a gentler transition for the baby. Water birth is used in many birth centers and some hospitals. Criteria: low-risk pregnancy, term, clear fluid, and no signs of fetal distress. Risks: maternal or neonatal infection, newborn aspiration (the baby does not breathe until exposed to air), umbilical cord avulsion, difficulty assessing blood loss, and water embolism. The AAP and ACOG state that water birth should be offered only in the context of clinical trials.

Weight Gain in Pregnancy

– Recommended gestational weight gain according to the Institute of Medicine (IOM) guidelines, based on prepregnancy BMI. Underweight (BMI <18.5): 28-40 lbs (12.5-18 kg). Normal weight (BMI 18.5-24.9): 25-35 lbs (11.5-16 kg). Overweight (BMI 25-29.9): 15-25 lbs (7-11.5 kg). Obese (BMI $\geq$30): 11-20 lbs (5-9 kg). Twin gestations: higher weight gain recommended. Inadequate weight gain is associated with IUGR and preterm birth; excessive gain is associated with macrosomia, gestational diabetes, cesarean delivery, and postpartum weight retention.

Wharton Jelly

– A gelatinous substance (mucopolysaccharide matrix) surrounding the umbilical cord vessels, composed primarily of hyaluronic acid and chondroitin sulfate. It protects the umbilical vessels from compression and kinking during fetal movement and uterine contractions. Its elastic properties allow the cord to maintain blood flow despite external pressure. Depletion of Wharton jelly is associated with oligohydramnios and cord accidents.

Whites (Leucorrhoea)

– A non-bloody, physiological or pathological vaginal discharge. Physiological leucorrhoea: normal, clear to whitish, odourless discharge that varies with the menstrual cycle (increased at ovulation, premenstrually, during pregnancy, sexual arousal, and with oral contraceptive use). It consists of desquamated epithelial cells, cervical mucus (from endocervical glands), vaginal flora (lactobacilli producing lactic acid, pH 3.8-4.5), and transudate. Pathological leucorrhoea: abnormal discharge in color, odour, or consistency. Causes: vulvovaginal candidiasis (white, thick, curd-like discharge, pruritus, erythema, pH normal 4-4.5, KOH-positive for pseudohyphae), bacterial vaginosis (thin, gray-white, fishy odour, elevated pH >4.5, clue cells on microscopy, positive whiff test with KOH), Trichomonas vaginalis (yellow-green, frothy, malodorous, pruritus, dysuria, strawberry cervix, pH >5, motile trichomonads on wet mount), Chlamydia trachomatis (mucopurulent cervicitis, yellow discharge, cervical friability), Neisseria gonorrhoeae (purulent discharge, PID risk), and atrophic vaginitis (thin, watery, sometimes blood-tinged, pH >5, in menopause). Diagnosis: history, speculum examination, pH testing, wet mount microscopy, KOH preparation, NAAT for Chlamydia/gonorrhea/Trichomonas. Management: treat underlying infection (antifungals for Candida, metronidazole for BV and Trichomonas, antibiotics for STIs). Recurrent or persistent leucorrhoea requires evaluation for cervical pathology and retained foreign body (tampon).

Womb

– The common term for the uterus, the muscular organ in the female pelvis where a fetus develops during pregnancy. The womb is pear-shaped, approximately 7.5 cm long, 5 cm wide, and 2.5 cm thick in the non-pregnant state. It consists of three layers: perimetrium (outer serosa), myometrium (thick muscular layer, responsible for uterine contractions during labor), and endometrium (inner lining, shed during menstruation and the site of implantation). The womb expands dramatically during pregnancy, increasing in capacity from 10 mL to 5 L.

Woods Corkscrew Maneuver

– An internal rotation maneuver for shoulder dystocia where the operator inserts a hand and rotates the fetal shoulders 180 degrees by applying pressure to the posterior aspect of the anterior shoulder. This disimpacts the shoulder from behind the symphysis pubis. Can be repeated if the shoulder re-impacts in the new position. Used when McRoberts maneuver and suprapubic pressure fail to resolve the dystocia.

Woods Screw Maneuver

– See Woods Corkscrew Maneuver in the main section above.

Yolk Sac

– The first extraembryonic structure visible on ultrasound at 5-6 weeks gestation. It provides early nutrition to the embryo before the placental circulation is established, produces blood cells until the fetal liver takes over, and forms the primitive gut. A normal yolk sac is 3-6 mm in diameter. Abnormal size (large >6 mm or small <2 mm) or shape is associated with early pregnancy failure. The yolk sac degenerates by the end of the first trimester.

Zavanelli Maneuver

– The final, rarely performed maneuver for an irreducible shoulder dystocia: the fetal head is replaced into the vagina (cephalic replacement) and the patient undergoes emergency cesarean delivery. Technique: the head is flexed and pushed back into the vagina, then into the birth canal under tocolysis (terbutaline, magnesium, or general anesthesia for uterine relaxation). High rates of fetal morbidity and mortality. The maneuver is used only when all other maneuvers have failed.

Zygote

– The single diploid cell formed by the fusion of male and female gametes (spermatozoon and secondary oocyte) at fertilization. The zygote contains 46 chromosomes (23 from each parent) and represents the beginning of a new, genetically unique individual. After fertilization, the zygote undergoes a series of mitotic cell divisions (cleavage) as it travels through the fallopian tube toward the uterine cavity. The first cleavage produces two blastomeres, then four, eight, and sixteen cells (morula), which then develops into a blastocyst (with inner cell mass → embryo and outer trophoblast → placenta) and implants in the endometrium approximately 5-6 days after fertilization. The zygote stage lasts approximately 24-30 hours, after which it becomes a 2-cell embryo. Abnormal zygote formation can result from polyspermy (fertilization by multiple sperm, resulting in triploidy), parthenogenesis (activation of the oocyte without sperm, not viable in humans), or chromosomal nondisjunction (leading to aneuploidy). In assisted reproductive technology (IVF/ICSI), the zygote is assessed for pronuclear morphology (2 pronuclei = normal fertilization) before embryo culture and transfer. Physiological leucorrhoea: normal, clear to whitish, odourless discharge that varies with the menstrual cycle (increased at ovulation, premenstrually, during pregnancy, sexual arousal, and with oral contraceptive use). It consists of desquamated epithelial cells, cervical mucus (from endocervical glands), vaginal flora (lactobacilli producing lactic acid, pH 3.8-4.5), and transudate. Pathological leucorrhoea: abnormal discharge in color, odour, or consistency. Causes: vulvovaginal candidiasis (white, thick, curd-like discharge, pruritus, erythema, pH normal 4-4.5, KOH-positive for pseudohyphae), bacterial vaginosis (thin, gray-white, fishy odour, elevated pH >4.5, clue cells on microscopy, positive whiff test with KOH), Trichomonas vaginalis (yellow-green, frothy, malodorous, pruritus, dysuria, strawberry cervix, pH >5, motile trichomonads on wet mount), Chlamydia trachomatis (mucopurulent cervicitis, yellow discharge, cervical friability), Neisseria gonorrhoeae (purulent discharge, PID risk), and atrophic vaginitis (thin, watery, sometimes blood-tinged, pH >5, in menopause). Diagnosis: history, speculum examination, pH testing, wet mount microscopy, KOH preparation, NAAT for Chlamydia/gonorrhea/Trichomonas. Management: treat underlying infection (antifungals for Candida, metronidazole for BV and Trichomonas, antibiotics for STIs). Recurrent or persistent leucorrhoea requires evaluation for cervical pathology and retained foreign body (tampon).

Chapter 22

Oncology

Acute Lymphoblastic Leukemia

– A clonal malignancy of lymphoid progenitor cells, most common in children (peak age 2-5 years) but also occurring in adults with a worse prognosis with increasing age. B-cell ALL (85 percent of childhood, 75 percent of adult ALL) is classified by immunophenotype: early pre-B (CD19+, CD10-, TdT+), common ALL (CD19+, CD10+, TdT+), pre-B (CD19+, cytoplasmic mu heavy chain+), and mature B-cell (Burkitt-type, CD19+, surface immunoglobulin+, CD10+). T-cell ALL (15 percent of cases) is associated with a worse prognosis and presents more commonly in adolescents and young adults, often with a mediastinal mass.

Acute Myeloid Leukemia

– A heterogeneous clonal disorder of hematopoietic stem cells characterized by the accumulation of immature myeloid blasts in the bone marrow and peripheral blood. Diagnostic criteria require 20 percent or more marrow or blood blasts (or blasts with AML-defining genetic abnormalities regardless of blast count). The WHO 2022 classification integrates morphologic (FAB subtypes M0-M7), immunophenotypic (flow cytometry for myeloid markers: CD13, CD33, CD34, CD117, MPO), and cytogenetic/molecular abnormalities (NPM1, CEBPA, FLT3-ITD, IDH1/2, RUNX1, TP53) for risk stratification and targeted therapy selection.

Adjuvant Therapy

– Treatment given after definitive surgery or radiation to eliminate micrometastatic disease and reduce the risk of recurrence. Chemotherapy: breast (anthracycline + taxane, dose-dense AC $\rightarrow$ T), colon (FOLFOX for stage III and high-risk stage II), lung (cisplatin-based for stage II-IIIA NSCLC), pancreas (FOLFIRINOX or gemcitabine + capecitabine), ovarian (carboplatin + paclitaxel). Radiation: post-mastectomy chest wall RT (for T3-4 or node-positive breast cancer), post-operative RT for head and neck cancer with high-risk features. Hormonal therapy: tamoxifen or aromatase inhibitor for ER+ breast cancer for 5-10 years. Targeted therapy: trastuzumab + pertuzumab for HER2+ breast cancer for 1 year. Duration: based on the natural history of the disease and drug tolerance. Timing: typically starts within 4-8 weeks of surgery.

Anthracyclines

– A class of potent chemotherapy drugs derived from the bacterium Streptomyces, used to treat a wide range of cancers including breast cancer (doxorubicin, epirubicin), leukemias (daunorubicin, idarubicin), lymphomas (doxorubicin), sarcomas (doxorubicin), and many others. Mechanism: intercalation into DNA (disrupts replication and transcription), inhibition of topoisomerase II (prevents DNA repair), and generation of free radicals (oxidative damage). A major dose-limiting side effect is dose-dependent cardiotoxicity: acute (arrhythmias, pericarditis) and chronic (dilated cardiomyopathy, heart failure). Cumulative lifetime dose limits: doxorubicin 450-550 mg/m^2, epirubicin 900 mg/m^2. Cardioprotection: dexrazoxane (iron chelator, reduces free radical formation). Other toxicities: myelosuppression, mucositis, alopecia, nausea/vomiting, and extravasation (tissue necrosis). Extravasation from IV administration is a medical emergency requiring immediate cold packs, dexrazoxane, and surgical consultation.

Bladder Cancer

– The most common malignancy of the urinary tract, ranking as the ninth most common cancer worldwide, with a male-to-female ratio of 4:1 and peak incidence in the seventh decade of life. The most significant risk factor is cigarette smoking (accounting for 50-65 percent of cases in men and 20-30 percent in women, with a 2-4 fold increased risk). Occupational exposures to aromatic amines (aniline dyes, in rubber, leather, and textile industries) are also important. The overwhelming majority (90-95 percent) are urothelial carcinoma (transitional cell carcinoma), with squamous cell carcinoma (3-5 percent, more common in regions with schistosomiasis) and adenocarcinoma (1-2 percent) comprising the remainder.

Bone Marrow Transplantation

– Hematopoietic stem cell transplantation (HSCT) replaces diseased or ablated bone marrow with healthy donor or autologous stem cells, used to treat hematologic malignancies, bone marrow failure syndromes, hemoglobinopathies, and immune deficiencies. Autologous HSCT uses the patient's own stem cells collected after mobilization (G-CSF with or without plerixafor), providing hematopoietic rescue after high-dose chemotherapy without graft-versus-host disease risk; indications include relapsed lymphoma and multiple myeloma.

Bone Tumor

– A neoplasm arising from bone tissue or its supporting structures, classified as primary (originating from bone, cartilage, or other connective tissue cells) or secondary/metastatic (spread from a distant primary cancer, which is the most common type). Primary bone tumors are rare, accounting for less than 1 percent of all cancers, while metastatic bone disease is common, particularly from primary cancers of the breast, prostate, lung, kidney, and thyroid. Primary bone tumors are classified by the type of matrix produced (osteoid, cartilage, or collagen) and the degree of cellular atypia and malignancy.

Brachytherapy

– A form of radiation therapy where a radioactive source is placed inside or immediately adjacent to the tumor. Low-dose-rate (LDR): seeds (I-125, Pd-103, Cs-131) implanted permanently (prostate cancer, delivers radiation over months). High-dose-rate (HDR): a single source (Ir-192) is moved through catheters into the tumor for brief periods (minutes), then removed (for cervical, endometrial, breast, and prostate cancer). Pulsed-dose-rate (PDR): a source delivers radiation in pulses, mimicking LDR radiobiology with HDR equipment. Applications: cervical cancer (tandem and ovoid, intracavitary; interstitial needles for advanced), prostate cancer (LDR permanent seeds monotherapy for low-risk; HDR as boost for intermediate/high-risk), endometrial cancer (vaginal cylinder HDR), breast cancer (accelerated partial breast irradiation with balloon or multicatheter), and head and neck (interstitial HDR). Advantages: high radiation dose to the tumor with rapid fall-off, sparing surrounding normal tissues. Disadvantages: invasive placement under anesthesia, radiation exposure to staff (mitigated by afterloading techniques), and limited to accessible tumor sites.

Breast Cancer

– The most common cancer in women worldwide, arising from breast tissue (ductal or lobular epithelium). Risk factors: female sex, age, BRCA1/BRCA2 mutations, family history, early menarche, late menopause, nulliparity, and hormone therapy. Presents with a breast mass, skin changes (peau d'orange), nipple retraction, or axillary lymphadenopathy. Diagnosis: mammography, ultrasound, biopsy with hormone receptor and HER2 testing. Treatment: surgery, radiation, chemotherapy, endocrine therapy, and targeted therapy (trastuzumab for HER2+).

Cancer

– A group of diseases characterized by uncontrolled cell growth and spread to nearby tissues or distant organs through the bloodstream or lymphatic system. It results from genetic mutations that disrupt normal cell cycle regulation, often influenced by environmental exposures, lifestyle factors, and inherited predisposition. Treatment modalities include surgery, radiation, chemotherapy, immunotherapy, targeted therapy, and hormone therapy.

Cancer Genetics

– The study of inherited and acquired genetic alterations that contribute to cancer development, progression, and treatment response. Hereditary cancer syndromes: hereditary breast and ovarian cancer (BRCA1, BRCA2), Lynch syndrome (MLH1, MSH2, MSH6, PMS2, EPCAM — colorectal, endometrial, ovarian, gastric), familial adenomatous polyposis (APC), Li-Fraumeni syndrome

(TP53), MEN1/MEN2 (RET, MEN1), von Hippel-Lindau (VHL), neurofibromatosis (NF1, NF2), and retinoblastoma (RB1). Indications for genetic testing: early-onset cancer (<50), multiple primaries, bilateral disease, multiple family members with the same or related cancer, and specific tumor types (triple-negative breast, serous ovarian, medullary thyroid, male breast cancer). Testing: germline (blood or saliva — inherited mutations) and somatic (tumor tissue — acquired mutations for targeted therapy). Multigene panel testing (NGS) is now standard. Management of mutation carriers: enhanced screening, risk-reducing surgery (prophylactic bilateral mastectomy for BRCA, colectomy for FAP), chemoprevention (tamoxifen for BRCA), and targeted therapy (PARP inhibitors for BRCA-mutated cancers, PD-1 inhibitors for MSI-high).

Cancer Pain

– Cancer pain can arise from the tumor itself compressing nerves or organs, from metastases, or as a side effect of treatments like surgery, chemotherapy, or radiation. It may be acute or chronic and is managed with a multimodal approach including analgesics, nerve blocks, palliative radiation, and integrative therapies. Effective pain control is a core component of cancer care and improves quality of life.

Cancer Prevention

– Cancer prevention involves strategies to reduce the risk of developing cancer, including lifestyle modifications such as avoiding tobacco, limiting alcohol, maintaining a healthy weight, exercising, and protecting against carcinogens like UV radiation. Vaccination against hepatitis B and HPV prevents infection-related cancers, and regular screenings catch precancerous changes early. Genetic testing and prophylactic surgery are options for high-risk individuals.

Cancer Screening

– Systematic testing of asymptomatic individuals to detect cancer at an early, curable stage. Principles of screening (WHO Wilson-Jungner): the disease should be an important health problem, an acceptable treatment should be available, facilities for diagnosis and treatment should be available, a recognizable latent or early symptomatic stage should exist, a suitable screening test should exist (safe, acceptable, high sensitivity/specificity, low cost), and the natural history should be understood. Recommended screenings: breast (mammography q2 years ages 50-74, ages 40-49 shared decision), cervical (Pap and HPV co-testing q5 years ages 25-65, or Pap alone q3 years), colorectal (colonoscopy q10 years, FIT q1 year, CT colonography q5 years, starting age 45), lung (low-dose CT annually ages 50-80 with 20 pack-year smoking history), and prostate (PSA, shared decision for ages 55-69). Overdiagnosis: detection of cancers that would never have caused symptoms (a risk of screening). False positives: require additional testing (imaging, biopsy) with associated anxiety and complications.

Cancer Staging

– The classification of cancer extent using the TNM system (Tumor, Node, Metastasis) from the AJCC (American Joint Committee on Cancer). T: size and local invasion of the primary tumor (Tx-T4). N: regional lymph node involvement (N0-N3). M: distant metastasis (M0-M1). Stage grouping: Stage 0 (carcinoma in situ), Stage I (localized, smaller tumors), Stage II-III (locally advanced, nodal involvement), Stage IV (metastatic). Clinical staging (cTNM): based on physical exam, imaging, and biopsy. Pathological staging (pTNM): based on surgical pathology, more precise. Other staging systems: FIGO (gynecologic cancers), BCLC (HCC), Ann Arbor (lymphoma), Dukes (colorectal cancer, historical), Breslow thickness and Clark level (melanoma), Gleason score (prostate cancer). Staging determines prognosis, guides treatment selection (surgery, radiation, systemic therapy), and standardizes clinical trial eligibility. Stage migration: improved imaging shifts patients to higher stages (Will Rogers phenomenon).

Cancer Survivorship

– Cancer survivorship encompasses the physical, psychosocial, and economic effects of cancer diagnosis and treatment from the point of diagnosis through the balance of life. Over 18 million cancer survivors live in the United States, with prevalence increasing due to improved treatments and earlier detection. Survivorship care addresses treatment-related late effects and long-term complications: cardiovascular disease (anthracyclines causing cardiomyopathy, radiation causing coronary artery disease), second malignancies, endocrine dysfunction (hypothyroidism, gonadal failure, osteoporosis), cognitive impairment (chemo brain), chronic fatigue, peripheral neuropathy, and psychosocial issues (depression, anxiety, fear of recurrence, financial toxicity).

Cancer Treatment

– Cancer treatment encompasses surgery, radiation therapy, systemic therapies (chemotherapy, targeted therapy, immunotherapy, hormone therapy), and emerging modalities like CAR-T cell therapy. The choice depends on cancer type, stage, biomarkers, and patient health, with many patients receiving multimodal regimens. Advances in precision medicine tailor treatments to the genetic profile of individual tumors.

Cervical Cancer

– The fourth most common cancer in women worldwide, with approximately 600,000 new cases annually, caused by persistent infection with high-risk human papillomavirus (HPV) types, most commonly HPV 16 and 18 (accounting for approximately 70 percent of all cervical cancers). Squamous cell carcinoma accounts for approximately 80 percent of cervical cancers, followed by adenocarcinoma (15-20 percent) and adenosquamous carcinoma. Risk factors include early sexual debut, multiple sexual partners, smoking, immunosuppression (HIV), long-term oral contraceptive use, and multiparity.

Chemoradiation (Chemoradiotherapy)

– Concurrent or sequential administration of chemotherapy and radiation therapy for synergistic anti-tumor effect. Chemotherapy acts as a radiosensitizer (increases radiation damage to DNA, inhibits repair, synchronizes cells in radiosensitive phases of the cell cycle). Indications: locally advanced cervical cancer (cisplatin + RT), head and neck squamous cell carcinoma (cisplatin + RT — definitive or adjuvant), anal cancer (5-FU + mitomycin + RT — definitive), NSCLC stage III (platinum doublet + RT — definitive), esophageal cancer (carboplatin + paclitaxel + RT — definitive; or FLOT + RT — perioperative), glioblastoma (temozolomide + RT — Stupp protocol), and rectal cancer (5-FU/capecitabine + RT — neoadjuvant). Sequential: chemotherapy first, then radiation. Concurrent: both given simultaneously (more effective, more toxic). Toxicity: enhanced acute effects (mucositis, esophagitis, enteritis, dermatitis, myelosuppression), less effect on late toxicity if doses are appropriate.

Chemotherapy

– The use of cytotoxic drugs to kill rapidly dividing cancer cells. Classes: alkylating agents (cyclophosphamide, cisplatin, oxaliplatin — crosslink DNA), antimetabolites (5-FU, capecitabine, gemcitabine, methotrexate — interfere with DNA/RNA synthesis), antitumor antibiotics (doxorubicin, daunorubicin, bleomycin — intercalate DNA, generate free radicals), plant alkaloids (paclitaxel, docetaxel — microtubule stabilizers; vincristine, vinblastine — microtubule inhibitors), topoisomerase inhibitors (irinotecan, etoposide, topotecan), and miscellaneous (L-asparaginase, procarbazine). Administered IV (push, infusion), oral, intrathecal, intraperitoneal, or intra-arterial. Dosing: body surface area (mg/m^2), dose-limiting toxicity varies by drug (myelosuppression, cardiotoxicity, neurotoxicity, nephrotoxicity). Combination regimens are standard (e.g., CHOP, R-CHOP, FOLFOX, ABVD). Response assessment: RECIST 1.1 criteria (complete response, partial response, stable disease, progressive disease). Side effects: myelosuppression (neutropenia, thrombocytopenia, anemia), alopecia, nausea/vomiting, mucositis, and organ-specific toxicity.

Chemotherapy Extravasation

– Accidental leakage of chemotherapy drugs from the vein into the surrounding tissues, a potential medical emergency. Classification: vesicants (cause blisters, tissue necrosis — doxorubicin, daunorubicin, epirubicin, vincristine, vinblastine, paclitaxel, docetaxel, dactinomycin, mitomycin C, mechlorethamine), irritants (cause inflammation/pain — carboplatin, cisplatin, etoposide, topotecan, ifosfamide), and non-vesicants (benign — bleomycin, cyclophosphamide, 5-FU, methotrexate). Signs: immediate pain, burning, erythema, swelling at the site; later blistering, ulceration, necrosis. Management: STOP infusion immediately, leave the IV needle in place (aspirate residual drug), apply cold (anthracyclines, paclitaxel) or warm compresses (vinca alkaloids, etoposide). Specific antidotes: dexrazoxane (for anthracycline extravasation, IV over days 1-3), hyaluronidase (for vinca alkaloids and taxanes, subcutaneous injection), DMSO topical (for anthracyclines), and sodium thiosulfate (for mechlorethamine and cisplatin). Surgical debridement may be required for severe necrosis.

Chemotherapy-Induced Nausea and Vomiting (CINV)

– A common and feared side effect of chemotherapy, classified by timing: acute (<24 hours after chemo), delayed (>24 hours), anticipatory (conditioned response before treatment), and breakthrough (occurs despite prophylaxis). High emetic risk (>90%): cisplatin, cyclophosphamide >1500 mg/m^2, dacarbazine, carmustine. Moderate (30-90%): carboplatin, oxaliplatin, doxorubicin, irinotecan. Low (10-30%): paclitaxel, docetaxel, 5-FU, gemcitabine. Minimal (<10%): vincristine, bleomycin, bevacizumab. Antiemetic agents: NK1 receptor antagonists (aprepitant, fosaprepitant, netupitant), 5-HT3 receptor antagonists (ondansetron, granisetron, palonosetron), corticosteroids (dexamethasone), and dopamine antagonists (metoclopramide, prochlorperazine). Guidelines (ASCO, NCCN): high risk — NK1 RA + 5-HT3 RA + dexamethasone (day 1), then dexamethasone ± NK1 RA (days 2-4). Moderate risk — 5-HT3 RA + dexamethasone. Anticipatory: behavioral therapy and optimal acute/delayed control. Breakthrough: add a different class antiemetic.

Chronic Lymphocytic Leukemia

– The most common leukemia in Western countries, characterized by the accumulation of mature but immunologically incompetent monoclonal B lymphocytes in the blood, bone marrow, lymph nodes, and spleen. Diagnosis requires a peripheral blood monoclonal B-lymphocyte count greater than or equal to 5000 per microliter with a characteristic immunophenotype (CD5+, CD19+, CD23+, CD200+, weak surface immunoglobulin, weak CD20). CLL has a highly variable clinical course, with some patients remaining asymptomatic for years while others progress rapidly.

Chronic Myeloid Leukemia

– A myeloproliferative neoplasm characterized by the Philadelphia chromosome [t(9;22)(q34;q11.2)], which creates the BCR-ABL1 fusion gene encoding a constitutively active tyrosine kinase. CML has three phases: chronic phase (most common at diagnosis, 85-90 percent, characterized by low blast count less than 10 percent, thrombocytosis, and basophilia), accelerated phase (increasing blasts 10-19 percent, basophilia greater than 20 percent, or additional cytogenetic abnormalities), and blast phase (more than 20 percent blasts, resembling acute leukemia).

Colon Cancer / Colorectal Cancer

– Malignancy of the colon or rectum, typically arising from adenomatous polyps. Third most common cancer worldwide. Risk factors: age >50, family history, IBD, Lynch syndrome, smoking, obesity, and red meat consumption. Presents with change in bowel habits, hematochezia, abdominal pain, weight loss, and iron deficiency anemia. Screening: colonoscopy (starting at age 45), FIT, CT colonography. Treatment: surgical resection, chemotherapy (FOLFOX, FOLFIRI), radiation for rectal cancer, and targeted therapy.

Colorectal Cancer

– The third most common cancer worldwide, arising from abnormal growths called polyps in the colon or rectum that become malignant over time. Risk factors include age, family history, inflammatory bowel disease, diet high in red and processed meats, obesity, and smoking. Regular screening through colonoscopy can detect and remove precancerous polyps, dramatically reducing incidence and mortality.

Disseminated Intravascular Coagulation

– A pathological state of simultaneous systemic activation of coagulation and fibrinolysis, leading to widespread microvascular thrombosis that causes organ ischemia and dysfunction, paradoxically accompanied by consumptive coagulopathy and increased bleeding risk. DIC is triggered by conditions that expose tissue factor to the circulation: sepsis (most common trigger, especially gram-negative and fungal), trauma (tissue factor release), malignancy (especially acute promyelocytic leukemia and pancreatic cancer), and obstetric complications (placental abruption, amniotic fluid embolism, pre-eclampsia).

Electrochemotherapy

– A local cancer treatment that combines chemotherapy (bleomycin or cisplatin) with electroporation — brief, high-voltage electrical pulses applied to the tumor that create transient pores in cell membranes, dramatically increasing the intracellular concentration of chemotherapeutic drugs. Indications: cutaneous and subcutaneous metastases (melanoma, breast cancer, head and neck cancer), basal cell carcinoma, squamous cell carcinoma, and Kaposi sarcoma. The drug is injected IV or intratumorally, then electrical pulses are delivered through needle electrodes placed around the tumor under ultrasound guidance. Response rates: 70-85% for cutaneous lesions. Advantages: minimally invasive, treats irregularly shaped tumors, spares surrounding normal tissue. Limitations: requires specialized equipment, general anesthesia for larger treatments, limited to accessible lesions.

Esophageal Cancer

– Malignancy of the esophagus with two main histologic types: squamous cell carcinoma (upper/mid esophagus, associated with smoking and alcohol) and adenocarcinoma (lower esophagus, associated with GERD and Barrett esophagus). Presents with progressive dysphagia, weight loss, odynophagia, and regurgitation. Diagnosis: upper endoscopy with biopsy, EUS for staging. Treatment depends on stage: endoscopic resection for early, neoadjuvant chemoradiation followed by esophagectomy for locally advanced, and palliative chemo/palliation for metastatic.

Febrile Neutropenia

– An oncologic emergency defined as a single oral temperature ≥38.3°C (101°F) or ≥38.0°C (100.4°F) for ≥1 hour, with an absolute neutrophil count (ANC) <500/µL or expected to fall <500/µL within 48 hours. Most common dose-limiting toxicity of chemotherapy. Highest risk: after cytotoxic chemotherapy (especially anthracyclines, platinum, taxanes), typically 7-14 days after treatment. Common pathogens: gram-positive (coagulase-negative staphylococci, viridans group streptococci, Staph aureus) and gram-negative (E. coli, Klebsiella, Pseudomonas aeruginosa). MASCC risk index: low risk (≥21, safe for outpatient oral antibiotics), high risk (<21, inpatient IV antibiotics). Empiric antibiotics: anti-pseudomonal beta-lactam (cefepime, meropenem, piperacillin-tazobactam) monotherapy. Add vancomycin for suspected line infection, skin/soft tissue infection, or MRSA colonization. Duration: treat until ANC >500/µL and afebrile for 48 hours. G-CSF (filgrastim, pegfilgrastim) for primary prophylaxis (risk >20%) or secondary after febrile neutropenia episode.

Gastric Cancer

– Malignancy of the stomach, with intestinal (gland-forming, more common in high-risk regions) and diffuse (limits plastica, associated with CDH1 mutations) types. Risk factors: H. pylori infection, smoking, diet (pickled/smoked foods), and family history. Presents with dyspepsia, early satiety, weight loss, and epigastric pain. Often diagnosed at advanced stage. Diagnosis: upper endoscopy with biopsy, EUS, and CT for staging. Treatment: surgical resection, perioperative chemotherapy, and trastuzumab for HER2+ metastatic disease.

Glioma

– A tumor arising from glial cells (the supportive cells of the central nervous system). Gliomas account for approximately 25% of primary brain tumors and 80% of malignant brain tumors. WHO classification (2021, integrates histology and molecular markers): (1) Adult-type diffuse gliomas: astrocytoma, IDH-mutant (CNS WHO grade 2-4); oligodendroglioma, IDH-mutant and 1p/19q-codeleted (grade 2-3); glioblastoma, IDH-wildtype (grade 4). (2) Paediatric-type diffuse low-grade and high-grade gliomas. (3) Circumscribed astrocytic gliomas (pilocytic astrocytoma, grade 1, most common childhood glioma). Clinical features: headaches worse in morning, seizures, focal neurological deficits (weakness, sensory loss, speech difficulty, visual changes), cognitive changes, personality changes, signs of raised ICP (nausea, papilloedema). Diagnosis: MRI brain with contrast (T1 post-gadolinium, T2/FLAIR, diffusion, perfusion, spectroscopy). Treatment: maximal safe surgical resection, followed by radiotherapy and chemotherapy (temozolomide for glioblastoma–Stupp protocol, 60 Gy + 6 weeks temozolomide, then 6 cycles maintenance). Median survival: glioblastoma 12-15 months; anaplastic astrocytoma (grade 3) 2-5 years; low-grade glioma (grade 2) 5-15 years depending on IDH and 1p/19q status.

Gynecologic Cancers

– Cancers that originate in the female reproductive organs, including ovarian, cervical, uterine (endometrial), vulvar, and vaginal cancers. Risk factors vary by type but may include HPV infection, genetic mutations, hormonal factors, and age. Screening, such as Pap smears and HPV testing, is essential for early detection, and treatment often involves surgery, chemotherapy, and radiation.

Hemophilia

– Hemophilia A (factor VIII deficiency, X-linked, 1 in 5000 males) and hemophilia B (factor IX deficiency, X-linked, 1 in 25,000 males) are inherited bleeding disorders caused by deficiency of intrinsic pathway coagulation factors, resulting in impaired thrombin generation and delayed clot formation. Severity correlates with factor activity level: severe (less than 1 percent, spontaneous hemarthrosis and soft tissue bleeding), moderate (1-5 percent, bleeding with minor trauma), and mild (5-40 percent, bleeding after surgery or significant injury).

Hepatocellular Carcinoma

– The most common primary liver malignancy, arising predominantly in the setting of chronic liver disease and cirrhosis. Risk factors include hepatitis B virus (HBV, the leading cause worldwide, especially in Asia and sub-Saharan Africa), hepatitis C virus (HCV, most common in developed countries), alcohol-related liver disease, non-alcoholic steatohepatitis (NASH), and hereditary hemochromatosis. Diagnosis in cirrhotic patients can be made non-invasively on imaging (CT or MRI) showing arterial hyperenhancement with venous washout.

Hodgkin Lymphoma

– A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells. Classical HL (cHL) accounts for 95 percent of cases and is subdivided into nodular sclerosis (most common, mediastinal predominance, young adults), mixed cellularity, lymphocyte-rich, and lymphocyte-depleted subtypes. Nodular lymphocyte-predominant HL (NLPHL, 5 percent) features lymphocyte-predominant cells (popcorn cells) with a nodular growth pattern and a more indolent clinical course.

Hormonal Therapy (Endocrine Therapy)

– Cancer treatments that block hormone signaling to inhibit growth of hormone-sensitive tumors. Breast cancer: selective estrogen receptor modulators (tamoxifen, raloxifene — block ER in breast, agonize in bone/uterus), aromatase inhibitors (anastrozole, letrozole, exemestane — block estrogen synthesis in postmenopausal women), selective estrogen receptor degraders (fulvestrant — downregulates ER), and ovarian suppression (GnRH agonists: leuprolide, goserelin — for premenopausal women). Prostate cancer: androgen deprivation therapy (GnRH agonists: leuprolide, goserelin — chemical castration; GnRH antagonist: degarelix), antiandrogens (bicalutamide, flutamide, nilutamide), and next-generation androgen receptor inhibitors (enzalutamide, apalutamide, darolutamide). Androgen synthesis inhibitor: abiraterone (inhibits CYP17, blocks adrenal and intratumoral androgen synthesis). Duration: adjuvant tamoxifen for 5-10 years, ADT for months to years depending on risk. Side effects: tamoxifen (hot flashes, VTE, endometrial cancer risk), aromatase inhibitors (arthralgias, osteoporosis, fractures), ADT (hot flashes, fatigue, loss of libido, erectile dysfunction, osteoporosis, metabolic syndrome, cardiovascular risk).

Immunotherapy Adverse Events

– Immune-related adverse events (irAEs) are inflammatory complications arising from checkpoint inhibitor-mediated immune activation against normal tissues, occurring in 60-80 percent of patients on combination therapy and 20-40 percent on monotherapy. The pathophysiology involves loss of immune tolerance to self-antigens due to blockade of inhibitory checkpoints that normally maintain peripheral tolerance. irAEs can affect virtually any organ system and occur at any time during treatment (median onset of 2-16 weeks after treatment initiation, depending on the agent and organ system involved.

Immunotherapy (Cancer)

– Cancer treatments that harness the immune system to recognize and destroy tumor cells. Checkpoint inhibitors (most impactful): anti-PD-1 (pembrolizumab, nivolumab, cemiplimab), anti-PD-L1 (atezolizumab, durvalumab, avelumab), anti-CTLA-4 (ipilimumab). Indications: melanoma, NSCLC, RCC, bladder, head and neck, Hodgkin lymphoma, MSI-high colorectal, and many others. Combination: PD-1 + CTLA-4 (nivolumab + ipilimumab) in melanoma, RCC, mesothelioma. Chimeric antigen receptor T-cell therapy (CAR-T): autologous T cells engineered to express a synthetic receptor targeting a tumor antigen (CD19 for ALL, DLBCL; BCMA for multiple myeloma). Approved products: tisagenlecleucel (Kymriah), axicabtagene ciloleucel (Yescarta), brexucabtagene autoleucel (Tecartus), lisocabtagene maraleucel (Breyanzi), idecabtagene vicleucel (Abecma). Cytokine therapy: IL-2 (high-dose for metastatic RCC, melanoma) and interferon-alpha (melanoma, RCC, CML). Oncolytic virus therapy: talimogene laherparepvec (T-VEC, modified herpes simplex virus injected into melanoma lesions). Bispecific T-cell engagers (BiTEs): blinatumomab (CD3-CD19 for ALL). Immune-related adverse events (irAEs): colitis, pneumonitis, hepatitis, dermatitis, endocrinopathies (thyroiditis, hypophysitis), and myocarditis (rare but fatal). Response patterns: pseudoprogression (transient enlargement before regression), durable responses, and hyperprogression (uncommon).

Kidney Cancer (Renal Cell Carcinoma)

– The most common type of kidney cancer, arising from renal tubular epithelium. Subtypes: clear cell (75%), papillary (15%), and chromophobe (5%). Risk factors: smoking, obesity, hypertension, and genetic syndromes (VHL, Birt-Hogg-Dube, hereditary papillary RCC). Classic triad: hematuria, flank pain, and palpable mass (now less common with incidental detection on imaging). Diagnosis: CT or MRI with contrast. Treatment: partial or radical nephrectomy; targeted therapy and immunotherapy for advanced disease.

Laryngeal Cancer

– Malignancy of the larynx, most commonly squamous cell carcinoma. Risk factors: smoking, alcohol, HPV infection. Presents with hoarseness (glottic tumors), dysphagia, stridor, and referred ear pain. Diagnosis: laryngoscopy with biopsy, CT/MRI for staging. Treatment: early stage (radiation or transoral laser surgery), advanced stage (total laryngectomy with chemoradiation, voice rehabilitation). Prognosis depends on stage and location.

Liver Cancer

– Malignant tumors originating in the liver (primary liver cancer). The most common type is hepatocellular carcinoma (HCC), accounting for 85-90% of primary liver cancers. See Hepatocellular Carcinoma.

Liver Cancer (Hepatocellular Carcinoma)

– The most common primary liver malignancy, typically arising in the setting of cirrhosis (hepatitis B/C, alcohol, NAFLD/NASH). Presents with right upper quadrant pain, hepatomegaly, ascites, and elevated alpha-fetoprotein (AFP). Diagnosis: multiphasic CT or MRI shows arterial hyperenhancement with washout. Treatment: surgical resection, liver transplantation, ablation (RFA, MWA), TACE, and systemic therapy (atezolizumab-bevacizumab, sorafenib, lenvatinib).

Living with Cancer

– Living with cancer encompasses the continuum from diagnosis through active treatment to survivorship or advanced illness, addressing physical, emotional, social, and practical challenges. Active treatment involves surgery, chemotherapy, radiation, immunotherapy, targeted therapy, or combinations thereof, tailored to cancer type, stage, and biomarkers. Common treatment side effects include fatigue, pain, nausea, myelosuppression, neuropathy, and immunotoxicity requiring proactive supportive care (antiemetics, growth factors, pain management). Survivorship care focuses on monitoring for recurrence, managing late effects (cardiotoxicity, secondary malignancies, cognitive impairment), and promoting health behaviors. For advanced/metastatic disease, palliative care integrated early improves quality of life, symptom control, and survival; goals-of-care discussions and advance care planning are essential.

Lung Cancer

– The leading cause of cancer death worldwide. Non-small cell lung cancer (NSCLC, 85%) includes adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. Small cell lung cancer (SCLC, 15%) is highly aggressive. Risk factor: tobacco smoking (85% of cases). Presents with cough, hemoptysis, dyspnea, chest pain, and weight loss. Diagnosis: CT chest, biopsy (bronchoscopy, needle aspiration), molecular testing (EGFR, ALK, ROS1, PD-L1). Treatment: surgical resection, chemotherapy, radiation, targeted therapy (osimertinib, alectinib), and immunotherapy (pembrolizumab, nivolumab).

Myelodysplastic Syndromes

– Clonal hematopoietic stem cell disorders characterized by ineffective hematopoiesis, peripheral blood cytopenias, dysplastic morphology, and risk of transformation to acute myeloid leukemia. The WHO classification categorizes MDS by percentage of blasts (MDS with single lineage dysplasia, MDS with multilineage dysplasia, MDS with ring sideroblasts, MDS with excess blasts-1 [5-9 percent], MDS with excess blasts-2 [10-19 percent], and MDS/AML [20-29 percent]) and cytogenetic abnormalities (deleted 5q, -7, deleted 7q, +8, deleted 20q, complex karyotype) for risk stratification using the Revised International Prognostic Scoring System (IPSS-R).

Neoadjuvant Therapy

– Treatment given before definitive surgery to downstage tumors, convert unresectable to resectable, and assess response. Chemotherapy: breast (doxorubicin + cyclophosphamide → paclitaxel, or TCHP for HER2+), esophageal/gastric (FLOT), rectal (5-FU/capecitabine + RT), bladder (MVAC or gemcitabine + cisplatin), and pancreatic (FOLFIRINOX or gemcitabine + nab-paclitaxel for borderline resectable). Hormonal therapy: aromatase inhibitor for 3-6 months in postmenopausal ER+ breast cancer. Targeted/immunotherapy: checkpoint inhibitors in melanoma, NSCLC. Benefits: allows in vivo assessment of response (pathologic complete response pCR strongly predicts survival), facilitates breast-conserving surgery, treats micrometastases early. Disadvantages: initial treatment delay if the diagnosis is uncertain, treatment-related toxicity before surgery.

Neuroblastoma

– The most common extracranial solid tumor in children, arising from neural crest cells of the sympathetic nervous system. Median age at diagnosis: 17 months. Most common site: adrenal medulla (40%). Presents with abdominal mass, bone pain (from metastases), proptosis (periorbital metastases), and Horner syndrome. Favorable prognosis: age <18 months, localized disease, MYCN amplification absent. Unfavorable: MYCN amplification, older age, metastatic disease. Urine catecholamines (VMA/HVA) elevated in 90%. Treatment: risk-stratified: surgery alone for low-risk, chemotherapy + surgery ± radiation for intermediate-risk, intensive chemotherapy + stem cell transplant + immunotherapy + isotretinoin for high-risk.

Non-Hodgkin Lymphoma

– A diverse group of lymphoproliferative malignancies originating from B cells (85%), T cells, or NK cells. Common subtypes: diffuse large B-cell lymphoma (DLBCL), follicular lymphoma, marginal zone lymphoma, mantle cell lymphoma, and peripheral T-cell lymphoma. Presents with painless lymphadenopathy, fever, night sweats, and weight loss (B symptoms). Diagnosis: lymph node biopsy with flow cytometry, cytogenetics, and FISH. Treatment: chemoimmunotherapy (R-CHOP for DLBCL), targeted therapy, radiation, and stem cell transplantation for relapsed/refractory disease.

Operable

– A term describing a tumor or condition that is amenable to surgical removal or treatment, typically implying that the disease is localised (no distant metastases) and that the patient's general health and fitness (performance status, cardiopulmonary reserve, comorbidities) are adequate to tolerate the planned surgical procedure and anesthesia. In oncology, 'operable' vs 'inoperable' is a critical distinction: operable cancers are treated with curative-intent resection, whereas inoperable cancers (locally advanced or metastatic) are managed with non-surgical modalities (chemotherapy, radiotherapy, immunotherapy, palliative care). Resectability criteria vary by tumor type—e.g., pancreatic cancer is considered borderline resectable if there is abutment of the superior mesenteric artery or celiac axis; liver metastases are resectable if adequate functional liver remnant remains after resection. Assessment includes staging investigations (CT, PET-CT, MRI), multidisciplinary team discussion, and preoperative risk assessment (cardiopulmonary exercise testing, ASA grade).

Oral Cancer

– Malignancy of the oral cavity, most commonly squamous cell carcinoma. Risk factors: tobacco (smoking, chewing), alcohol, betel nut, and HPV infection. Presents with non-healing ulcer, erythroplakia, leukoplakia, or mass in the mouth. Diagnosis: biopsy. Treatment: surgical resection with or without radiation and chemotherapy. Early detection significantly improves prognosis. Prevention: tobacco cessation, alcohol moderation, and HPV vaccination.

Other Cancers

– malignancies beyond the most common types (breast, lung, prostate, colorectal), including pancreatic (high lethality, 5-year survival 12%), liver/hepatocellular (rising incidence, linked to HBV/HCV, MASLD, alcohol), ovarian ("silent killer," often diagnosed at late stage), gastric, esophageal, bladder, kidney (renal cell carcinoma), thyroid (generally favorable prognosis), head and neck (squamous cell, linked to HPV and tobacco/alcohol), brain (glioblastoma, meningioma), sarcoma (bone and soft tissue), lymphoma (Hodgkin and non-Hodgkin), leukemia, multiple myeloma, and melanoma. Each has distinct biology, risk factors, screening modalities (where applicable), and treatment paradigms. General principles: staging (TNM classification) dictates treatment—surgery, radiation, systemic therapy (chemotherapy, molecular targeted therapy, immunotherapy, hormonal therapy)—and prognosis. Emerging approaches: tumor molecular profiling (next-generation sequencing, liquid biopsy), cancer immunotherapy (immune checkpoint inhibitors, CAR-T cells), and personalized medicine based on biomarkers.

Ovarian Cancer

– Epithelial ovarian cancer is the leading cause of death from gynecologic malignancies, with 70-80 percent of patients diagnosed at advanced stage due to the absence of reliable screening tests. High-grade serous carcinoma (the most common subtype, 70 percent) originates from the fallopian tube epithelium (serous tubal intraepithelial carcinoma, STIC, is the precursor lesion), not the ovary as previously thought. BRCA1/2 germline mutations are found in 15-20 percent of ovarian cancers, conferring sensitivity to platinum-based chemotherapy and PARP inhibitors.

Paget Disease of the Nipple

– A rare presentation of breast cancer involving the nipple–areola complex, accounting for 1–4% of all breast cancers. Paget disease of the nipple is characterized by infiltration of the nipple epidermis by malignant glandular cells (Paget cells—large, pale, vacuolated cells with prominent nucleoli, positive for CK7, HER2/neu, and GATA3). Clinical presentation: an erythematous, eczematous, scaling, or crusted lesion of the nipple and areola, often associated with itching, burning, or tenderness. The lesion may be mistaken for eczema or dermatitis; failure to resolve with topical corticosteroids should prompt biopsy. Underlying breast cancer is present in >90% of cases: either an in situ component (ductal carcinoma in situ—DCIS) or invasive ductal carcinoma, which may be palpable or detected on imaging. Diagnosis: full-thickness skin punch biopsy of the nipple–areola lesion, mammogram and ultrasound of the ipsilateral breast, and MRI if indicated. Treatment: total mastectomy (with or without sentinel lymph node biopsy) is standard; breast-conserving surgery (central lumpectomy with nipple–areola complex excision and whole-breast radiotherapy) is an option for localised disease without palpable mass. Prognosis depends on the stage of the underlying breast cancer; Paget disease without associated invasive cancer has an excellent prognosis.

Palliative Care in Oncology

– An interdisciplinary approach to improving quality of life for patients with serious illness, focused on symptom management, communication about goals of care, and caregiver support. Not limited to end-of-life; early concurrent palliative care (alongside active treatment) improves symptom burden, quality of life, and survival in metastatic NSCLC (Temel 2010). Symptom management: pain (WHO analgesic ladder, opioids, neuropathic agents, palliative radiation), nausea, dyspnea, fatigue, cachexia, depression, and anxiety. Advance care planning: goals of care discussions, code status (DNR/DNI, DNR-CC), living will, and healthcare proxy. Hospice: palliative care for patients with a life expectancy of ≤6 months; focuses on comfort rather than disease-modifying therapy. Referral to palliative care: recommended for all patients with advanced cancer, regardless of prognosis.

Pancreatic Ductal Adenocarcinoma

– One of the most lethal solid organ malignancies, with 5-year overall survival rates of approximately 12 percent overall and less than 3 percent for metastatic disease, due to late diagnosis and chemoresistance. Risk factors include smoking (2-3 fold increased risk), obesity, chronic pancreatitis, hereditary pancreatitis, family history, BRCA1/2 and PALB2 germline mutations, Lynch syndrome, and Peutz-Jeghers syndrome. The majority (85 percent) are located in the pancreatic head, causing early obstructive jaundice, while body and tail tumors present at a more advanced stage.

Paraneoplastic Syndromes

– Systemic manifestations of cancer that occur distant from the primary tumor or metastases, mediated by hormones, cytokines, antibodies, or immune cross-reactivity between tumor and normal tissue antigens. Endocrine paraneoplastic syndromes include ectopic ACTH production (small cell lung cancer causing Cushing syndrome with hypokalemia, metabolic alkalosis), syndrome of inappropriate antidiuretic hormone secretion (SIADH, small cell lung cancer causing hyponatraemia), hypercalcaemia (squamous cell lung cancer producing PTHrP), and hypoglycaemia (insulinoma or IGF-2-producing sarcomas).

Paroxysmal Nocturnal Hemoglobinuria

– An acquired clonal hematopoietic stem cell disorder caused by somatic mutations in the PIGA gene (phosphatidylinositol glycan class A), which encodes an enzyme essential for the biosynthesis of glycosylphosphatidylinositol (GPI) anchors that attach complement regulatory proteins (CD55 and CD59) to the cell surface. Loss of CD55 (accelerator of C3 convertase decay) and CD59 (inhibitor of membrane attack complex formation) renders red blood cells, white blood cells, and platelets susceptible to complement-mediated destruction, resulting in intravascular haemolysis, thrombosis, and bone marrow failure.

Pediatric Solid Tumors

– Pediatric solid tumors differ from adult cancers in histology, biology, treatment response, and long-term survivorship considerations. Wilms tumor (nephroblastoma) is the most common renal malignancy in children (peak age 2-4 years), presenting as a large, smooth, flank mass with staging by National Wilms Tumor Study (NWTS) criteria; treatment includes nephrectomy and chemotherapy (vincristine, dactinomycin, doxorubicin for unfavorable histology). Neuroblastoma arises from neural crest cells (most commonly adrenal), presenting with an abdominal mass, bone pain, proptosis, Horner syndrome, and elevated urinary catecholamines.

Photodynamic Therapy (PDT)

– A treatment that combines a photosensitizing agent with light to produce reactive oxygen species that destroy cancer cells. Mechanism: the photosensitizer (porfimer sodium Photofrin, aminolevulinic acid ALA) is administered IV or topically and accumulates preferentially in tumor tissue. A laser or LED light of a specific wavelength activates the photosensitizer, generating singlet oxygen that causes direct tumor cell kill, vascular damage, and immune activation. Indications: actinic keratosis and superficial BCC (topical ALA-PDT), Barrett esophagus with high-grade dysplasia (porfimer sodium — Photofrin PDT), early-stage endobronchial NSCLC, and malignant pleural mesothelioma (intraoperative PDT). Side effects: photosensitivity (skin and eye sensitivity for 4-6 weeks), local pain, edema, and scarring. Limited depth of penetration ($<$1 cm for most photosensitizers) restricts PDT to superficial tumors.

Precision Oncology (Molecular Oncology)

– The tailoring of cancer treatment based on the unique molecular, genomic, and immunologic characteristics of each patient's tumor. Approaches: next-generation sequencing (NGS) of tumor tissue or liquid biopsy (ctDNA from blood) to identify actionable mutations. Biomarker-driven therapy: EGFR mutations (osimertinib in NSCLC), ALK fusions (alectinib), BRAF V600E (vemurafenib, dabrafenib + trametinib in melanoma), HER2 amplification (trastuzumab, T-DXd in breast, gastric, lung), MSI-high/dMMR (pembrolizumab in any solid tumor — tissue-agnostic approval), NTRK fusions (larotrectinib, entrectinib — tissue-agnostic), and BRCA mutations (olaparib, talazoparib). Tumor mutational burden (TMB): high TMB ($\geq$10 mut/Mb) predicts immunotherapy response. Homologous recombination deficiency (HRD): predicts PARP inhibitor and platinum sensitivity. Minimal residual disease (MRD): ctDNA detection after curative-intent treatment predicts recurrence. Challenges: tumor heterogeneity, acquired resistance, access to testing, and cost.

Prostate Cancer

– The most common cancer in men, typically adenocarcinoma arising from the peripheral zone of the prostate. Risk factors: age, family history, African American race, and BRCA2 mutations. Often asymptomatic early; advanced disease causes urinary symptoms, hematuria, and bone pain (from metastases). Screening: PSA blood test and digital rectal exam. Diagnosis: transrectal ultrasound-guided biopsy with Gleason scoring. Treatment: active surveillance for low-risk, prostatectomy, radiation, androgen deprivation therapy, and novel hormonal agents for advanced disease.

Radiation Sickness

– A constellation of symptoms resulting from exposure to high doses of ionizing radiation, typically >1 gray (Gy) whole-body or significant partial-body exposure. Acute radiation syndrome (ARS) follows a predictable dose-dependent course: prodromal phase (within minutes to hours—nausea, vomiting, diarrhea, anorexia, fatigue), latent phase (hours to days—apparent recovery), manifest illness phase (days to weeks—depending on the dose and organ system affected), and recovery or death. Three main ARS subsyndromes: (1) hematopoietic (2–6 Gy)—pancytopenia from bone marrow destruction, leading to infection, bleeding, and anemia; mortality without supportive care is high. (2) Gastrointestinal (6–10 Gy)—destruction of intestinal crypt cells, leading to severe diarrhea, dehydration, electrolyte imbalance, and sepsis from gut barrier breakdown. (3) Neurovascular (»10 Gy)—rapid onset of confusion, ataxia, seizures, coma, and cardiovascular collapse; universally fatal within hours to days. Chronic radiation effects: carcinogenesis (leukemia, solid tumors), cataracts, fibrosis, and hypothyroidism. Cutaneous radiation syndrome: erythema, desquamation, ulceration, necrosis. Management of ARS: remove from source, decontaminate, supportive care (transfusions, growth factors—G-CSF, GM-CSF, antibiotics, antiemetics), and potentially bone marrow transplant for severe hematopoietic syndrome. Potassium iodide for radioactive iodine exposure. Radiation sickness is rare and most commonly associated with nuclear accidents (Chernobyl, Fukushima), radiation therapy accidents, and potential nuclear events.

Radiation Therapy (Radiotherapy)

– The use of ionizing radiation to kill cancer cells by inducing DNA damage (double-strand breaks). External beam radiation therapy (EBRT): linear accelerator delivers photons (X-rays) or electrons from outside the body. Techniques: 3D conformal (3D-CRT), intensity-modulated (IMRT), volumetric modulated arc (VMAT), stereotactic radiosurgery (SRS, for brain lesions — Gamma Knife, CyberKnife), and stereotactic body RT (SBRT, for lung, liver, spine, pancreas — high-dose, 1-5 fractions). Brachytherapy: radioactive source placed inside or near the tumor (prostate I-125 seeds, cervical cancer tandem and ovoid, breast accelerated partial breast irradiation). Proton therapy: protons deposit most energy at a specific depth (Bragg peak), sparing deeper tissues; used for pediatric, skull base, ocular, and prostate cancers. Fractionation: total dose divided into daily fractions (1.8-2.0 Gy) to allow normal tissue repair. Standard total doses: 45-50.4 Gy for microscopic disease, 60-70 Gy for gross disease. SBRT: 24-60 Gy in 1-5 fractions. Side effects: acute (skin erythema, mucositis, diarrhea, fatigue — during treatment) and late (fibrosis, xerostomia, secondary malignancy, cardiac toxicity, pulmonary fibrosis — months to years after). Radioprotectors (amifostine) and radiosensitizers (cisplatin, 5-FU) modulate radiation effects.

Radioembolization (Y-90 Selective Internal Radiation Therapy SIRT)

– A liver-directed therapy where yttrium-90 (Y-90) microspheres (glass: TheraSphere; resin: SIR-Spheres) are infused into the hepatic artery and lodge in tumor microvasculature, delivering high-dose beta radiation (30-60 Gy average, up to 200 Gy to tumor) while sparing normal liver tissue. Indications: HCC (BCLC B, C with PVT), colorectal liver metastases (chemoresistant), and other liver-dominant tumors. Pre-treatment: mesenteric angiography with Tc-99m MAA scan to calculate lung shunt fraction (<20%) and identify extrahepatic flow. Lung shunt >20%: contraindicated (risk of radiation pneumonitis). Advantages over TACE: less post-embolization syndrome, applicable with portal vein thrombosis. Complications: radiation-induced liver disease (RILD, rare with proper dosimetry), radiation pneumonitis, gastric/duodenal ulceration (from non-target embolization), and biliary toxicity.

Radioisotope

– A radioactive isotope of an element that emits radiation (alpha particles, beta particles, gamma rays) as it decays to a stable form. Radioisotopes are used extensively in medicine for diagnostic imaging, therapeutic procedures, and research. Diagnostic radioisotopes: technetium-99m (most common, used in SPECT imaging—bone scans, myocardial perfusion, renal scans, lung ventilation/perfusion scans), iodine-123 (thyroid imaging), thallium-201 (myocardial perfusion), gallium-67 (infection/inflammation imaging), indium-111 (white blood cell scanning), and fluorine-18 (FDG-PET—most common PET tracer, glucose analogue for oncology, neurology, cardiology). Therapeutic radioisotopes: iodine-131 (hyperthyroidism, thyroid cancer), yttrium-90 (selective internal radiation therapy—SIRT for liver tumors, radioembolisation), lutetium-177 (PRRT for neuroendocrine tumors), radium-223 (bone metastases in castration-resistant prostate cancer), strontium-89 and samarium-153 (palliative therapy for painful bone metastases). Production: nuclear reactors (neutron activation) or cyclotrons (proton bombardment). Half-life: time for 50% of atoms to decay—ranges from minutes (technetium-99m: 6 hours) to years (cobalt-60: 5.3 years). Safety: radioisotopes require careful handling, shielding, and disposal; ALARA principle (as low as reasonably achievable).

Retinoblastoma

– The most common intraocular malignancy in children, arising from retinal precursor cells. Presents with leukocoria (white pupillary reflex, most common), strabismus, and vision loss. Bilateral in 40% (associated with germline RB1 mutation). Unilateral: median age 24 months; bilateral: 12 months. Diagnosis: fundoscopic examination under anesthesia, ultrasound, MRI. Treatment: enucleation for large unilateral tumors; chemoreduction (vincristine, carboplatin, etoposide) plus focal therapy (cryotherapy, laser photocoagulation) for bilateral and globe-sparing approaches. External beam radiation reserved due to second malignancy risk.

Skin Cancer

– The most common cancer overall, including basal cell carcinoma (BCC, most common), squamous cell carcinoma (SCC), and melanoma (most dangerous). BCC: pearly papule with telangiectasias, rarely metastasizes. SCC: scaly, hyperkeratotic lesion, can metastasize. Risk factors: UV radiation exposure, fair skin, immunosuppression, and genetic disorders. Prevention: sun protection, avoiding tanning beds. Treatment: surgical excision, Mohs surgery for cosmetically sensitive areas, and topical therapies for superficial lesions.

Stem Cell Transplantation (Hematopoietic)

– See Bone Marrow Transplantation.

Targeted Therapy

– Drugs that interfere with specific molecules required for tumor growth and progression, as opposed to non-specific cytotoxic chemotherapy. Tyrosine kinase inhibitors (TKIs): imatinib (BCR-ABL in CML, KIT in GIST), erlotinib and osimertinib (EGFR in NSCLC), crizotinib and alectinib (ALK in NSCLC), sorafenib and lenvatinib (VEGFR in HCC, RCC), sunitinib (VEGFR, PDGFR in RCC, GIST), and ibrutinib (BTK in CLL, MCL). Monoclonal antibodies: trastuzumab (HER2 in breast/gastric cancer), cetuximab (EGFR in colorectal/head and neck), bevacizumab (VEGF in colorectal, NSCLC, RCC — antiangiogenic), and rituximab (CD20 in B-cell lymphomas). Small molecule inhibitors: vemurafenib and dabrafenib (BRAF V600E in melanoma), palbociclib (CDK4/6 in breast cancer), enzalutamide (androgen receptor in prostate cancer), and olaparib (PARP inhibitor in BRCA-mutated breast, ovarian, pancreatic, prostate cancer). Mechanism: TKIs bind the ATP-binding pocket of the kinase domain, inhibiting phosphorylation and downstream signaling. Resistance mechanisms: secondary mutations (T790M for first-generation EGFR inhibitors, T315I for imatinib), bypass pathway activation, and drug efflux. Side effects: fatigue, diarrhea, rash (EGFR inhibitors), hand-foot syndrome (sorafenib, sunitinib), hypertension (VEGF inhibitors), and pneumonitis (mTOR inhibitors, EGFR inhibitors).

Teratoma

– See Teratoma in Obstetrics and Gynecology.

Testicular Cancer

– The most common solid tumor in young men (age

15-35), with an excellent prognosis with cure rates exceeding 95 percent even for advanced disease. Risk factors include cryptorchidism, history of contralateral testicular cancer, and germ cell neoplasia in situ (GCNIS, formerly intratubular germ cell neoplasia). Pathology classification includes seminoma (50 percent, excellent prognosis, sensitive to radiation and chemotherapy) and non-seminomatous germ cell tumors (NSGCT, 50 percent including embryonal carcinoma, yolk sac tumor, choriocarcinoma, and teratoma).

Thrombotic Thrombocytopenic Purpura

– A life-threatening thrombotic microangiopathy caused by severe deficiency of ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin type 1 motif member 13), a von Willebrand factor-cleaving protease. ADAMTS13 deficiency (activity less than 10 percent) leads to accumulation of ultra-large von Willebrand factor multimers that cause spontaneous platelet adhesion and aggregation in the microcirculation, forming platelet-rich thrombi that consume platelets and cause microangiopathic hemolytic anemia with characteristic schistocytes on peripheral blood smear.

Thyroid Cancer

– Malignancy of the thyroid gland, with subtypes: papillary (80%, best prognosis), follicular, medullary (associated with MEN2), and anaplastic (most aggressive). Usually presents as a thyroid nodule. Diagnosis: thyroid ultrasound and fine-needle aspiration biopsy (Bethesda classification). Treatment: thyroidectomy, radioactive iodine ablation for differentiated types, and TSH suppression with levothyroxine. Prognosis is excellent for papillary and follicular, poor for anaplastic.

Transarterial Chemoembolization (TACE)

– A liver-directed therapy for hepatocellular carcinoma (HCC) that delivers chemotherapy directly to the tumor through its hepatic artery blood supply while blocking blood flow. Procedure: a catheter is advanced through the femoral artery into the hepatic artery, then selectively into tumor-feeding vessels. Chemotherapy (doxorubicin, cisplatin, or mitomycin C, often emulsified in Lipiodol) is infused, followed by embolic agents (gelatin sponge particles, microspheres, or drug-eluting beads DEB-TACE). Indications: BCLC stage B (intermediate) HCC, also used as bridging therapy to transplant, downstaging, or palliation. Contraindications: decompensated cirrhosis (Child-Pugh C), main portal vein thrombosis, extensive extrahepatic disease, and poor performance status. Response: assessed by mRECIST (modified RECIST) — viability based on arterial enhancement. Side effects: post-embolization syndrome (fever, pain, nausea — 80%, self-limited), liver abscess, biliary injury, and tumor lysis. Repeat: every 6-12 weeks as needed.

Tumor

– An abnormal mass of tissue resulting from uncontrolled, excessive cell proliferation. tumors are classified as benign or malignant. Benign tumors: well-circumscribed, slow-growing, non-invasive, do not metastasise, composed of well-differentiated cells resembling the tissue of origin. Often cured by surgical excision. Examples: lipoma (adipose tissue), leiomyoma/fibroid (smooth muscle—uterus), adenoma (glandular epithelium—colon, pituitary, thyroid), papilloma (epithelial—skin, breast ducts), meningioma (meninges), fibroadenoma (breast), osteochondroma (bone). Malignant tumors (cancer): invasive, rapid growth, poorly differentiated (anaplasia), can metastasise via lymphatics or blood vessels, can recur after treatment. Named by tissue of origin: carcinoma (epithelial—most common, 80%: adenocarcinoma, squamous cell carcinoma), sarcoma (mesenchymal—bone, muscle, fat, connective tissue), leukemia (hematopoietic—bone marrow), lymphoma (lymphoid tissue—nodes, spleen, mucosa-associated lymphoid tissue), melanoma (melanocytes), and germ cell tumors (testis, ovary). Grading: histological degree of differentiation (well, moderate, poorly, undifferentiated). Staging: TNM system (tumor size/extent, nodal involvement, metastases). Cachexia: weight loss, muscle wasting, anorexia in advanced malignancy. Paraneoplastic syndromes: remote effects of cancer not due to direct invasion (e.g., hypercalcaemia in squamous cell lung cancer, SIADH in small cell lung cancer, myasthenia gravis in thymoma).

Tumor Ablation

– Minimally invasive techniques that destroy tumors using extreme temperatures or chemicals. Radiofrequency ablation (RFA): an electrode delivers alternating current, generating heat (>60°C) that causes coagulative necrosis. Microwave ablation (MWA): electromagnetic waves produce rapid heating with larger and more uniform ablation zones; advantages over RFA: faster, less heat-sink effect, no grounding pads needed. Cryoablation: argon gas-based freezing (to -40°C) with freeze-thaw cycles, creating an ice ball that destroys cells by membrane rupture and ischemia. Ethanol ablation: 95% ethanol injection causes cellular dehydration and protein denaturation. Irreversible electroporation (IRE): non-thermal, high-voltage electrical pulses create nanopores in cell membranes, inducing apoptosis; preserves extracellular matrix (safe near bile ducts, ureters). Indications: HCC (RFA/MWA for early-stage, BCLC 0-A), colorectal liver metastases (small, <3 cm), RCC (cryoablation for T1a tumors, <4 cm), small lung tumors, bone metastases (cryoablation for pain palliation). Guidance: ultrasound, CT, or MRI. Efficacy: local control 80-95% for small tumors (<3 cm). Advantages: less invasive than surgery, shorter recovery, preserves parenchyma. Limitations: size dependent (<3 cm ideal), proximity to vessels (heat-sink effect), proximity to critical structures.

Tumor Lysis Syndrome (TLS)

– An oncologic emergency caused by the rapid release of intracellular contents from lysing cancer cells, occurring most commonly after initiation of chemotherapy for highly proliferative tumors (ALL, Burkitt lymphoma, AML, DLBCL). Laboratory TLS: ≥2 of hyperuricemia, hyperkalemia, hyperphosphatemia, and hypocalcemia (corrected calcium, from calcium-phosphate precipitation). Clinical TLS: lab TLS + acute kidney injury, cardiac arrhythmia (hyperkalemia), seizure, or death. Risk stratification: high-risk (ALL with WBC >100K, Burkitt lymphoma), intermediate (DLBCL, AML with WBC >25K), low-risk (CLL, solid tumors). Prevention: aggressive IV hydration, rashburicase (recombinant urate oxidase, rapidly lowers uric acid) or allopurinol (prevents new uric acid formation — not for established TLS). Management: cardiac monitoring, treat hyperkalemia (calcium gluconate, insulin + glucose, albuterol, kayexalate, dialysis), correct hypocalcemia only if symptomatic, and hemodialysis for refractory hyperkalemia/azotemia. Cairo-Bishop grading: Grade 0 (no TLS), Grade I (lab TLS), Grade II (lab + mild Cr elevation), Grade III (lab + Cr >3x ULN, arrhythmia, seizure), Grade IV (life-threatening), Grade V (death).

Tumor Markers

– Substances produced by cancer cells or the body in response to cancer, measured in blood, urine, or tissue to aid diagnosis, prognosis, monitoring treatment response, and detecting recurrence. Common tumor markers: AFP (HCC, nonseminomatous germ cell tumors), CA-125 (ovarian cancer), CA 19-9 (pancreatic cancer), CA 15-3 and CA 27.29 (breast cancer), CEA (colorectal, pancreatic, lung, gastric cancer), hCG (choriocarcinoma, germ cell tumors), PSA (prostate cancer), thyroglobulin (thyroid cancer — post-thyroidectomy), LDH (germ cell tumors, lymphoma — prognostic), beta-2 microglobulin (CLL, multiple myeloma), and calcitonin (medullary thyroid carcinoma). Limitations: not diagnostic alone (false positives from benign conditions), not sufficiently sensitive for screening (except PSA), and many cancers lack specific markers. Role: monitoring response to treatment (declining levels indicate response, rising levels suggest recurrence) and surveillance after definitive therapy.

Uterine Cancer (Endometrial Cancer)

– The most common gynecologic cancer in developed countries, arising from the endometrium. Risk factors: unopposed estrogen exposure, obesity, nulliparity, late menopause, tamoxifen use, and Lynch syndrome. Presents with postmenopausal bleeding. Diagnosis: endometrial biopsy. Treatment: total abdominal hysterectomy with bilateral salpingo-oophorectomy; adjuvant radiation and/or chemotherapy for high-risk features (high grade, deep myometrial invasion, advanced stage).

Vitamin B12 Deficiency

– Vitamin B12 (cobalamin) deficiency causes megaloblastic anemia and neurologic dysfunction through impaired DNA synthesis and abnormal fatty acid metabolism in myelin. Dietary sources include animal products (meat, fish, dairy), with daily requirement of 2-3 mcg. Absorption requires adequate gastric acid and pepsin to release B12 from food proteins, binding to intrinsic factor (IF) produced by gastric parietal cells, and absorption in the terminal ileum via the cubilin-amnionless receptor complex in the terminal ileum. Causes of deficiency include pernicious anemia (autoimmune destruction of parietal cells), gastric surgery, ileal resection or disease (Crohn disease), pancreatic insufficiency, dietary deficiency (vegans), and malabsorption syndromes.

Chapter 23

Ophthalmology

Aflibercept

– A recombinant fusion protein consisting of the extracellular domains of human vascular endothelial growth factor receptors 1 and 2 fused to the Fc portion of human immunoglobulin G1, functioning as a soluble decoy receptor that binds to vascular endothelial growth factor A, vascular endothelial growth factor B, and placental growth factor with high affinity, thereby inhibiting angiogenesis and reducing vascular permeability. The drug is administered by intravitreal injection everyevery 4 weeks initially, with extension of intervals up to 12 or 16 weeks depending on treatment response and disease activity as assessed by optical coherence tomography and visual acuity. Aflibercept has a higher binding affinity for VEGF-A than ranibizumab or bevacizumab and a longer intravitreal half-life of 7 to 8 days, allowing a more durable treatment effect. The VIEW 1 and VIEW 2 clinical trials demonstrated that aflibercept 2 mg every 8 weeks after 3 monthly loading doses was non-inferior to monthly ranibizumab for wet AMD, establishing it as a standard of care for neovascular AMD, DME, and macular edema secondary to retinal vein occlusion.

Age-Related Macular Degeneration

– The leading cause of irreversible central vision loss in individuals over 50 in developed countries, affecting approximately 200 million people worldwide with prevalence increasing with age (10 percent over age 65, 30 percent over age 75). Pathogenesis involves progressive degeneration of the macula with accumulation of drusen (extracellular deposits of lipids, proteins, and complement factors between the retinal pigment epithelium and Bruch membrane) anand loss of retinal pigment epithelium cells, leading to accumulation of lipofuscin and reactive oxygen species that damage photoreceptors and RPE. AMD is classified into early (medium drusen greater than 63 micrometres without pigmentary changes), intermediate (large drusen greater than 125 micrometres with pigmentary abnormalities), and advanced AMD, which is subdivided into dry AMD (geographic atrophy, accounting for 85 to 90 percent of advanced AMD cases) and wet AMD (neovascular or exudative AMD, accounting for 10 to 15 percent of advanced AMD but responsible for 80 to 90 percent of severe vision loss, characterized by the growth of abnormal new blood vessels from the choroid through Bruch membrane into the sub-RPE or subretinal space, which leak fluid, lipid, and blood). Risk factors include increasing age, smoking (the strongest modifiable risk factor, increasing risk 2- to 4-fold), family history, Caucasian race, and genetic susceptibility variants in complement factor H (CFH), ARMS2, and HTRA1 genes.

Allergic Conjunctivitis

– An inflammatory condition of the conjunctiva mediated by IgE-dependent mast cell degranulation in response to environmental allergens, characterized by bilateral ocular itching, tearing, conjunctival injection, and eyelid edema. The condition is classified into two main forms: seasonal allergic conjunctivitis, which occurs during specific pollen seasons and presents with acute symptoms of itching, tearing, and conjunctival injection; and perennial allergic conjunctivitconjunctivitis, which may be less symptomatic but persistent throughout the year, with indoor allergens such as dust mites, mould, and animal dander as common triggers. The ocular surface reaction is characterized by vasodilation of conjunctival blood vessels (causing injection), increased vascular permeability (leading to chemosis), and stimulation of the trigeminal nerve (causing intense itching). Diagnosis is primarily clinical, and treatment consists of topical antihistamine-mast cell stabiliser combination drops (olopatadine, ketotifen, azelastine, epinastine, bepotastine), systemic antihistamines for associated allergic rhinitis, artificial tears, and cold compresses.

Alpha-Agonists

– Alpha-adrenergic agonists are a class of topical medications used in the treatment of glaucoma that lower intraocular pressure through a dual mechanism: reduction of aqueous humor production by the ciliary body through alpha-2 adrenergic receptor stimulation and enhancement of uveoscleral outflow through remodeling of the extracellular matrix. Available agents include brimonidine, apraclonidine, and dipivefrin, with brimonidine being the most commonly used due to its superior tolerability profilprofil. Brimonidine is the most widely prescribed topical alpha-2 agonist for chronic glaucoma therapy, reducing intraocular pressure by 20 to 30 percent. Brimonidine is often used as adjunctive therapy in patients inadequately controlled on prostaglandin analogs and beta-blockers. Common side effects include ocular allergy and conjunctival folliculosis (allergic conjunctivitis develops in up to 10 to 15 percent of patients within 9 to 12 months of initiation), dry mouth, ocular burning and stinging, blurred vision, and fatigue. Apraclonidine is primarily used for acute short-term control of intraocular pressure spikes, such as before, during, and after laser procedures.

Amblyopia

– Amblyopia, commonly known as lazy eye, is a neurodevelopmental disorder of visual acuity that results from abnormal visual experience during the critical period of visual development (birth to approximately eight years of age), leading to reduced best-corrected visual acuity that cannot be attributed to structural ocular pathology. The condition is classified based on the underlying cause: strabismic amblyopia from ocular misalignment, anisometropic amblyopia from unequal refractive errors betwebetween the two eyes, and deprivation amblyopia from obstruction of the visual axis by cataract, ptosis, or corneal opacity. The pathophysiology involves abnormal development of the visual cortex, particularly the lateral geniculate nucleus and primary visual cortex (V1), where the input from the amblyopic eye is suppressed by the cortical dominance of the normal eye. The critical period of susceptibility to amblyopia spans from birth to approximately 8 years of age. The diagnosis is made by detecting a 2-line or greater difference in best-corrected visual acuity between the two eyes. Management consists of correction of any underlying refractive error (the first and most important step), occlusion therapy (patching of the non-amblyopic eye for 2 to 6 hours per day), and pharmacological penalisation (atropine 1 percent drops in the non-amblyopic eye once daily as an alternative to patching).

Anterior Chamber Angle

– The anterior chamber angle is the anatomical region where the cornea and iris meet, and it contains the trabecular meshwork, the primary drainage structure for aqueous humor from the eye. The angle is bounded anteriorly by Schwalbe line (the termination of Descemet membrane), posteriorly by the anterior surface of the iris root, and laterally by the scleral spur and ciliary body band. The trabecular meshwork consists of three regions: the uveal meshwork adjacent to the iris, the corneoscleral memeshwork, which is the outermost layer and the primary site of aqueous humor outflow resistance in primary open-angle glaucoma, and the juxtacanalicular meshwork adjacent to Schlemm canal. The trabecular meshwork is drained by Schlemm canal, a circular, lymphatic-like channel lined by endothelial cells, and from there aqueous humor drains through 25 to 30 collector channels into the episcleral and conjunctival venous system. The anterior chamber angle is evaluated clinically using gonioscopy, which allows direct visualisation of the angle structures and determination of whether the angle is open, narrow, or closed.

Anterior Uveitis

– Anterior uveitis, also known as iritis or iridocyclitis when the ciliary body is involved, is inflammation of the anterior segment of the eye involving the iris and/or ciliary body, and it is the most common form of uveitis. Patients typically present with acute onset of unilateral eye pain, photophobia, blurred vision, and conjunctival injection, and slit-lamp examination reveals cells and flare in the anterior chamber, which represent inflammatory cells and leaked protein from inflamed blood vblood vessels. The etiology of anterior uveitis can be classified as idiopathic (the most common, accounting for 50 to 60 percent of cases), HLA-B27-associated (20 to 30 percent, including patients with ankylosing spondylitis, reactive arthritis, psoriatic arthritis, and inflammatory bowel disease), sarcoidosis (5 to 10 percent, typically with a chronic, bilateral, granulomatous course), and herpes simplex or herpes zoster virus (5 to 10 percent). Treatment consists of topical corticosteroids (prednisolone acetate 1 percent or difluprednate 0.05 percent, with hourly dosing during the acute phase, tapering slowly over weeks to months), cycloplegic agents (homatropine 2 or 5 percent or cyclopentolate 1 percent, to dilate the pupil, relieve ciliary spasm pain, and reduce the risk of posterior synechiae), and systemic therapy for refractory or severe cases.

Aqueous Humor

– A clear, watery fluid that fills the anterior and posterior chambers of the eye, maintaining intraocular pressure and providing nutrients and oxygen to the avascular cornea and lens while removing metabolic waste products. The fluid is produced by the nonpigmented epithelium of the ciliary processes at a rate of approximately 2.5 microliters per minute, through a combination of active secretion (involving carbonic anhydrase and sodium-potassium ATPase), ultrafiltration, and difand diffusion of fluid and electrolytes across the ciliary epithelium, and it drains from the eye through the trabecular meshwork and Schlemm canal into the venous system (conventional outflow pathway, responsible for 80 to 90 percent of outflow resistance and the site of pathology in primary open-angle glaucoma) and through the uveoscleral pathway (unconventional outflow, responsible for 10 to 20 percent of outflow, which is enhanced by prostaglandin analogs).

Astigmatism

– A refractive error caused by an irregular curvature of the cornea or lens, resulting in light rays being focused at multiple points rather than a single point on the retina, leading to blurred or distorted vision at all distances. The most common form is regular astigmatism, in which the cornea has two principal meridians with different curvatures at right angles to each other, creating a shape similar to a football rather than a basketball. The steepest meridian is called the agaagain or the flattest meridian, and light rays are refracted unequally in the two principal meridians. The diagnosis is made by refraction, keratometry, corneal topography, and autorefraction. Correction is achieved with toric spectacle or contact lenses (which have different powers in the two principal meridians), or refractive surgery including LASIK, PRK, and SMILE, which can reshape the cornea to reduce or eliminate the irregular curvature.

Bacterial Conjunctivitis

– An infection of the conjunctival mucous membrane caused by various bacterial pathogens, presenting with a purulent or mucopurulent discharge that causes the eyelids to become matted together, particularly upon waking, conjunctival injection, and a gritty or foreign body sensation. Acute bacterial conjunctivitis is most commonly caused by Staphylococcus aureus, Streptococcus pneumoniae, and Haemophilus influenzae, and it is usually self-limited but may be treated with with topical antibiotics, most commonly a fluoroquinolone (ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, or gatifloxacin) or polymyxin B-trimethoprim combination, instilled 4 to 6 times daily for 5 to 7 days. Gonorrhoeal conjunctivitis, caused by Neisseria gonorrhoeae, is a sight-threatening emergency presenting with a profuse, purulent discharge, marked chemosis, eyelid edema, and preauricular lymphadenopathy, requiring systemic antibiotic treatment with intramuscular or intravenous ceftriaxone. Chlamydial conjunctivitis, caused by Chlamydia trachomatis serotypes D to K, presents with a chronic, mucopurulent discharge, conjunctival follicles, and keratitis, and requires systemic therapy with azithromycin or doxycycline.

Beta-Blockers

– Beta-adrenergic blocking agents, commonly known as beta-blockers, are topical medications used in the treatment of glaucoma that lower intraocular pressure by reducing the production of aqueous humor by the ciliary body, primarily through blockade of beta-2 adrenergic receptors. Available agents include timolol, levobunolol, metipranolol, and carteolol, typically administered once or twice daily, and they reduce intraocular pressure by approximately twenty to twenty-five percent. Beta-blockers acontraindicated in patients with asthma (due to the risk of bronchospasm), chronic obstructive pulmonary disease, heart block, bradycardia, and congestive heart failure, and they should be used with caution in patients with diabetes mellitus and depression. The alpha-2 agonist brimonidine and the carbonic anhydrase inhibitors dorzolamide and brinzolamide are often used as second-line agents or as adjunctive therapy.

Bevacizumab

– A full-length recombinant humanized monoclonal antibody that binds to all isoforms of vascular endothelial growth factor A, originally developed and approved for the treatment of various cancers including colorectal cancer, non-small cell lung cancer, and glioblastoma, but widely used off-label in ophthalmology for the treatment of wet age-related macular degeneration, diabetic macular edema, and retinal vein occlusion through intravitreal injection. The CATT trial demonstrated thdemonstrated that bevacizumab is non-inferior to ranibizumab (the first FDA-approved anti-VEGF agent) for the treatment of wet age-related macular degeneration at one year, leading to widespread off-label use due to its significantly lower cost (approximately 1/40th the cost of ranibizumab per dose). The drug is administered by intravitreal injection at a typical dose of 1.25 mg in 0.05 mL, originally every 4 weeks. The risk of serious ocular adverse events, including endophthalmitis (0.05 to 0.1 percent per injection), retinal detachment, and traumatic cataract, is low but requires sterile technique and careful patient monitoring.

Blepharitis

– Chronic inflammation of the eyelid margins, one of the most common ophthalmic conditions, classified into anterior blepharitis (affecting the eyelash follicle and base) and posterior blepharitis (affecting the meibomian glands). Anterior blepharitis is further subdivided into staphylococcal blepharitis (caused by Staphylococcus aureus colonization, presenting with collarettes and crusting at the base of lashes) and seborrheic blepharitis (associated with seborrheic dermatitis, presenting with grgreasy scales and crusts along the eyelid margin with associated seborrheic dermatitis of the scalp, eyebrows, and nasolabial folds. The pathophysiology involves bacterial colonisation (particularly Staphylococcus epidermidis and Staphylococcus aureus, which produce exotoxins and enzymes that destabilise the tear film), meibomian gland dysfunction (with altered meibum composition, thickened and turbid meibum that obstructs the gland orifices, leading to reduced tear film lipid layer, increased tear evaporation, and tear hyperosmolarity), and ocular surface inflammation. Diagnosis is clinical, based on slit lamp examination of the lid margins, and treatment includes eyelid hygiene (warm compresses for 5 to 10 minutes twice daily followed by lid margin massage and scrubbing), topical antibiotics (bacitracin or erythromycin ointment applied to the lid margins at bedtime, or azithromycin 1 percent solution twice daily), and artificial tears for dry eye symptoms.

Bulbar Conjunctiva

– The bulbar conjunctiva is the portion of the conjunctival membrane that covers the anterior surface of the sclera, extending from the limbus (the junction of the cornea and sclera) to the fornices where it reflects onto the inner surface of the eyelids. It is a thin, transparent, loosely attached membrane through which the underlying white sclera is visible, and it contains blood vessels, lymphatics, and sensory nerve fibers. The bulbar conjunctiva is normally smooth and moist, and its vessels cvessels can become dilated and tortuous in conditions such as conjunctivitis and uveitis, providing important diagnostic information on slit-lamp examination. The bulbar conjunctiva is freely movable over the underlying sclera, to which it is only loosely attached by a thin layer of connective tissue, and this mobility allows for the formation of conjunctival folds or chemosis (edema of the conjunctiva) in inflammatory and allergic conditions. The blood supply to the bulbar conjunctiva arises from two main sources: the anterior ciliary arteries, which are branches off the muscular arteries supplying the extraocular muscles, and the posterior conjunctival arteries from the palpebral arcades.

Carbonic Anhydrase Inhibitors

– A class of medications used in the treatment of glaucoma that lower intraocular pressure by reducing the production of aqueous humor through inhibition of the enzyme carbonic anhydrase, which catalyzes the conversion of carbon dioxide and water to bicarbonate and hydrogen ions, a process essential for the active secretion of bicarbonate into the aqueous humor by the ciliary epithelium. Available topical agents include dorzolamide and brinzolamide, administered tthree times daily. Systemic carbonic anhydrase inhibitors, including acetazolamide and methazolamide, are reserved for acute pressure elevation in angle-closure glaucoma, uveitic glaucoma, and as a temporising measure before surgery, due to the systemic side effects of paresthesias, metallic taste, fatigue, depression, weight loss, and the risk of nephrolithiasis with prolonged use.

Cataract

– A cataract is an opacity or clouding of the crystalline lens that impairs the transmission of light to the retina, resulting in blurred vision, glare, decreased contrast sensitivity, and progressive vision loss, and it is the leading cause of preventable blindness worldwide. The most common form is age-related cataract, which is classified by the location of the opacity within the lens: nuclear sclerosis, characterized by yellowing and hardening of the lens nucleus; cortical cataract, charactericharacterised by radial, spoke-like opacities that extend from the periphery toward the center of the lens; and posterior subcapsular cataract, characterized by granular, plaque-like opacities located at the posterior pole of the lens, just anterior to the posterior capsule, which are particularly visually disabling due to their location in the visual axis and are strongly associated with corticosteroid use, diabetes, trauma, and radiation exposure.

Cataracts

– A clouding of the eye's natural lens that develops gradually with aging, causing blurred vision, glare sensitivity, faded colors, and difficulty seeing at night. They are the leading cause of vision loss worldwide but are treatable with surgical removal of the cloudy lens and replacement with an artificial intraocular lens. Surgery is safe and highly effective, restoring vision in the vast majority of cases.

Chalazion

– A chronic, non-infectious granulomatous inflammation of a meibomian gland caused by duct obstruction. Presents as a firm, painless, well-circumscribed nodule in the eyelid, usually away from the lid margin. Different from a stye (which is acute and tender). Most resolve spontaneously over weeks to months. Treatment: warm compresses, lid massage. For persistent chalazia: intralesional corticosteroid injection or incision and curettage (I&C).

Choroidal Neovascularization

– The pathological growth of new, fragile blood vessels from the choroid through Bruch membrane into the subretinal or sub-retinal pigment epithelium space, and it is the hallmark of wet age-related macular degeneration and the primary cause of severe central vision loss in this condition. The neovascular vessels are stimulated by vascular endothelial growth factor, which is produced in response to hypoxia, inflammation, and oxidative stress in the aging retina, andand they are fragile and prone to leakage and bleeding, leading to sudden central vision loss from hemorrhage as well as gradual vision loss from exudation and fibrosis. The diagnosis of choroidal neovascularization is confirmed by fluorescein angiography (which shows the classic lacy, petaloid hyperfluorescence in the subretinal space with early leakage and late staining in the classic type) and optical coherence tomography (which shows subretinal or sub-RPE fluid, intraretinal cystic spaces, and a hyperreflective lesion above or below the RPE).

Color Blindness

– Inability or decreased ability to perceive color differences under normal lighting conditions. Most commonly X-linked recessive, affecting 8% of males and 0.5% of females. Types: deuteranopia (green deficiency, most common), protanopia (red deficiency), tritanopia (blue-yellow deficiency, rare), and achromatopsia (complete color blindness, very rare). Most congenital color blindness is non-progressive and does not affect visual acuity. Diagnosis: Ishihara color plates. No treatment available; patients adapt with compensatory strategies.

Color Vision Deficiency

– Color vision deficiency, commonly known as color blindness, is a condition in which the ability to perceive and discriminate certain colors is reduced or absent, most commonly affecting the perception of red and green hues, and it is caused by abnormalities in the cone photoreceptors responsible for color vision. The most common form is red-green color deficiency, which is an X-linked recessive condition affecting approximately eight percent of males and 0.5 percent of females, caused by mutatiomutations in the opsin genes (OPN1LW and OPN1MW on the X chromosome encoding the long-wavelength and middle-wavelength cone photopigments, respectively, which cause the common red-green color deficiencies through deletions, fusions, or point mutations). Blue-yellow color deficiency (tritanopia) is much rarer and is caused by mutations in the OPN1SW gene on chromosome 7 encoding the short-wavelength cone photopigment, and it is inherited as an autosomal dominant condition. Complete color blindness (achromatopsia) is a rare autosomal recessive condition characterized by the absence of all three types of cone function, resulting in severe photophobia, nystagmus, and poor visual acuity.

Congenital Cataract

– Opacification of the crystalline lens present at birth, a leading cause of treatable childhood blindness worldwide. Causes: genetic (autosomal dominant most common, mutations in crystallin genes or PAX6), infectious (congenital rubella syndrome, toxoplasmosis, CMV, syphilis, HSV), metabolic (galactosemia, hypoglycemia, Wilson disease), and syndromic (Down syndrome, Lowe syndrome, Marfan syndrome). Bilateral cataracts are more commonly genetic or metabolic; unilateral cataracts suggest structural eye anomalies or asymmetric intrauterine infection. Presents with leukocoria (white pupillary reflex), nystagmus, strabismus, or poor visual tracking. Screening: red reflex examination (absent or abnormal). Diagnosis: slit-lamp examination after dilation. Treatment: early surgical extraction (within 4-6 weeks for dense bilateral cataracts to prevent deprivation amblyopia), primary or secondary intraocular lens implantation (depending on age), and immediate postoperative aphakic correction with contact lenses or glasses. Lifelong monitoring for glaucoma, strabismus, and amblyopia.

Congenital Glaucoma

– A rare form of glaucoma (1 in 10,000-20,000 births) caused by developmental anomalies of the trabecular meshwork and anterior chamber angle, impairing aqueous humor outflow. Most cases are sporadic; familial forms are autosomal recessive (CYP1B1 gene mutations). Presents in the first year of life with the classic triad: epiphora (excessive tearing, often the earliest sign), photophobia, and blepharospasm. Corneal edema causes corneal enlargement (buphthalmos, or ox eye, from elevated IOP stretching the immature collagen), Haab striae (linear breaks in Descemet membrane), and corneal clouding. Increased axial length leads to progressive myopia. Diagnosis: examination under anesthesia — elevated intraocular pressure, increased corneal diameter (>12 mm by 1 year), and optic nerve cupping (reversible in early stages). Treatment: surgical angle surgery (goniotomy or trabeculotomy) is first-line (success rate 80-90% if performed early). Trabeculectomy or drainage implants for failed angle surgery. Lifelong IOP monitoring is essential.

Conjunctiva

– The conjunctiva is a thin, transparent, mucous membrane that lines the inner surface of the eyelids (palpebral conjunctiva) and covers the anterior surface of the sclera (bulbar conjunctiva), forming a continuous sheet that protects the eye and facilitates smooth eyelid movement. The palpebral conjunctiva is further divided into the tarsal conjunctiva, which is firmly adherent to the tarsal plates of the eyelids, and the forniceal conjunctiva, which forms loose folds at the junction between the tarsal and forniceal conjunctiva. The conjunctiva is composed of a non-keratinising stratified columnar epithelium with goblet cells that secrete mucin, a critical component of the tear film that stabilises the tear film by lowering its surface tension, and an underlying substantia propria, a layer of loose connective tissue containing blood vessels, lymphatics, nerves, and immune cells. The conjunctiva is a highly vascularised, immunologically active tissue that protects the eye from pathogens and environmental insults.

Conjunctivitis

– Inflammation of the conjunctiva, the thin, transparent mucous membrane lining the inner surface of the eyelids and the anterior surface of the sclera. Conjunctivitis is the most common cause of red eye and is classified by etiology. Infectious conjunctivitis: viral (most common overall, caused by adenovirus serotypes 3, 4, 7, 8, 19, 37, presenting with watery discharge, preauricular lymphadenopathy, and highly contagious, often involving both eyes with one eye affected first), bacterial (purulenpurulent discharge, often unilateral, treated with topical antibiotics), and chlamydial (inclusion conjunctivitis, with a chronic course, mucopurulent discharge, and follicles, requiring systemic antibiotics). Infectious conjunctivitis is distinguished from non-infectious causes: allergic conjunctivitis (intense itching, bilateral, seasonal or perennial, with diffuse conjunctival injection and chemosis, treated with antihistamine-mast cell stabiliser drops) and chemical or irritant conjunctivitis (acute exposure to chemicals or irritants, with immediate onset of redness, tearing, and pain, treated with copious irrigation and lubrication).

Corneal Cross-Linking

– Corneal collagen cross-linking is a minimally invasive therapeutic procedure that strengthens the corneal stroma by creating additional chemical bonds between collagen fibrils, thereby stiffening the cornea and halting or slowing the progression of corneal ectatic disorders, particularly keratoconus and post-refractive surgery ectasia. The procedure involves debriding the central corneal epithelium, applying riboflavin (vitamin B2) drops to the cornea for approximately thirty minutes to saturatesaturate the corneal stroma with riboflavin, and then exposing the cornea to ultraviolet A light (365 nm, 3 mW/cm2) for 30 minutes, which causes the riboflavin to undergo a photochemical reaction that generates reactive oxygen species that induce new covalent cross-links between collagen molecules, increasing the biomechanical rigidity of the cornea by 3 to 4 times. The epithelium-on (epi-on) and epithelium-off (epi-off) techniques exist, with epi-off being the traditional Dresden protocol and still considered the gold standard.

Corneal Ulcer

– A corneal ulcer is a defect in the corneal epithelium with associated stromal inflammation and necrosis, typically caused by infectious or non-infectious mechanisms, and it is a sight-threatening condition that requires prompt diagnosis and treatment. Bacterial corneal ulcers, the most common infectious type, are associated with contact lens wear, ocular surface disease, and corneal trauma, and common causative organisms include Pseudomonas aeruginosa, Staphylococcus aureus, and Streptococcus pn-Streptococcus pneumoniae. Fungal corneal ulcers, more common in tropical regions and associated with agricultural trauma, contact lens wear, and topical corticosteroid use, are caused by filamentous fungi (Fusarium, Aspergillus) and yeasts (Candida), and they present with a dry, elevated infiltrate with a feathery border and satellite lesions. Viral corneal ulcers, most commonly caused by herpes simplex virus and varicella-zoster virus, present with a characteristic dendritic epithelial defect on fluorescein staining, and they are treated with topical or oral antivirals (acyclovir, ganciclovir, valacyclovir) and cycloplegia.

Cortical Cataract

– A type of age-related lens opacity that begins in the peripheral lens cortex as wedge-shaped or spoke-like opacities that extend centripetally toward the visual axis, caused by disruption of the normally orderly arrangement of lens fibers and the accumulation of fluid between them. The opacities appear as white, wedge-shaped streaks in the lens cortex on slit-lamp examination, with the narrow end of the wedge pointing toward the lens center, and they may progress centrally ttoward the visual axis, eventually involving the central visual axis and causing significant visual impairment. The cortical spokes may form a wedge-shaped opacity with a clear center (cuneiform cataract) or may extend centrally to create a radial, star-shaped pattern. Patients with cortical cataract often complain of glare and difficulty with night vision.

Cup-to-Disc Ratio

– The cup-to-disc ratio is a clinical measurement used to assess the severity of glaucomatous optic nerve damage, comparing the diameter of the optic cup (the central depression within the optic nerve head where neural tissue is absent) to the diameter of the optic disc (the entire optic nerve head). A normal cup-to-disc ratio ranges from 0.1 to 0.3, while progressive enlargement of the cup, known as cupping, indicates loss of retinal ganglion cell axons and is a hallmark of glaucomatous optic neuneuropathy. The cup-to-disc ratio is assessed during fundoscopic examination of the optic nerve head, and its measurement is essential for the diagnosis and monitoring of glaucoma, with a focal or diffuse enlargement of the cup (increased cup-to-disc ratio), asymmetry of cup-to-disc ratio between the two eyes (a difference of greater than 0.2 is suspicious for glaucoma), and notching or thinning of the neuroretinal rim being characteristic of glaucomatous optic nerve damage.

Dacryoadenitis

– Inflammation of the lacrimal gland, presenting as tender swelling in the superolateral aspect of the upper eyelid with characteristic S-shaped ptosis. It can be acute or chronic in presentation. Acute dacryoadenitis is most commonly caused by viral infection (Epstein-Barr virus, mumps, adenovirus, herpes zoster, enterovirus), with bacterial causes (Staphylococcus aureus, Streptococcus, Haemophilus influenzae, gonococcus) less common but more severe. Chronic dacryoadenitis is associated with autoautoimmune conditions including sarcoidosis, rheumatoid arthritis, Sjogren syndrome, granulomatosis with polyangiitis, systemic lupus erythematosus, IgG4-related disease, and thyroid eye disease. Diagnosis is clinical, with slit-lamp examination showing an enlarged, erythematous lacrimal gland. Imaging with CT or MRI may be needed to confirm the diagnosis and rule out orbital mass lesions. Treatment of acute viral dacryoadenitis is supportive, while bacterial dacryoadenitis requires systemic antibiotics. Chronic dacryoadenitis requires treatment of the underlying systemic condition.

Dacryocystitis

– Inflammation of the lacrimal sac, most commonly caused by obstruction of the nasolacrimal duct leading to stagnation of tears and secondary bacterial infection (Staphylococcus aureus, Streptococcus pneumoniae, Haemophilus influenzae). Presents with pain, swelling, erythema, and tenderness over the medial canthus, with epiphora (excessive tearing) and purulent discharge expressed from the punctum. Acute dacryocystitis requires systemic antibiotics and warm compresses; chronic cases may require dacryocystorhinostomy (DCR) to bypass the obstruction.

Dendritic Ulcer

– A characteristic branching (dendritic) epithelial ulcer of the cornea caused by herpes simplex virus (HSV) infection. It is the most common infectious cause of corneal blindness in developed countries. Presents with unilateral red eye, photophobia, tearing, foreign body sensation, and decreased vision. Slit-lamp examination with fluorescein staining shows a branching linear ulcer with terminal bulbs (dendrite). Treatment: topical antivirals (trifluridine drops, ganciclovir gel) or oral antivirals (acyclovir, valacyclovir). Topical corticosteroids are contraindicated in active epithelial disease.

Detached Retina

– Separation of the neurosensory retina from the underlying retinal pigment epithelium (RPE), a medical emergency requiring prompt surgical repair to prevent permanent vision loss. Types: rhegmatogenous (most common–a tear or hole in the retina allows vitreous fluid to accumulate in the subretinal space, associated with posterior vitreous detachment, high myopia, trauma, and aphakia), tractional (vitreoretinal adhesions from proliferative diabetic retinopathy, retinopathy of prematurity, or penetrating trauma pull the retina away), and exudative (fluid accumulation beneath the retina without a tear, from inflammation, tumors, or choroidal disorders). Presents with sudden onset of flashes (photopsia), floaters, and a curtain-like or shadowy visual field defect. Treatment: pneumatic retinopexy, scleral buckle, or pars plana vitrectomy with gas or silicone oil tamponade.

Diabetic Macular Edema

– The accumulation of fluid within the macula due to breakdown of the blood-retinal barrier in diabetic retinopathy, resulting in thickening of the macula and central vision loss, and it can occur at any stage of diabetic retinopathy, from mild nonproliferative to proliferative disease. The pathogenesis involves hyperglycemia-induced damage to retinal vascular endothelial cells and pericytes, leading to increased vascular permeability, leakage of plasma components into tinto the retinal tissue, and thickening of the macula. The severity of DME is graded on clinical examination based on the distance of the retinal thickening from the fovea. The diagnosis of DME is confirmed and quantified by optical coherence tomography. The treatment of center-involving DME is primarily with intravitreal anti-VEGF agents (ranibizumab, aflibercept, bevacizumab, and faricimab), which have revolutionised the management of DME.

Diplopia (Double Vision)

– The perception of two images of a single object, classified by onset, duration, and whether it resolves with covering one eye. Monocular diplopia (persists with one eye covered): caused by ocular surface abnormalities (dry eye, irregular astigmatism, cataract, corneal scar, lens dislocation)–usually benign. Binocular diplopia (resolves with either eye covered): results from misalignment of the visual axes due to cranial nerve palsy (CN III, IV, VI), neuromuscular junction disease (myasthenia gravis), restrictive orbital disease (thyroid eye disease), or strabismus. Evaluation: cover test in nine gaze positions, forced duction testing, and neuroimaging if nerve palsy suspected.

Drusen

– Yellowish-white deposits of extracellular material that accumulate beneath the retinal pigment epithelium between the retinal pigment epithelium basement membrane and Bruch membrane, and they are the hallmark clinical finding in dry age-related macular degeneration. Drusen are composed of lipids, complement components, inflammatory mediators, and other proteins, and they represent localized abnormalities of the retinal pigment epithelium basement membrane. Small hard drusen (less thanthan 63 micrometres in diameter) are common in the aging retina and are not associated with an increased risk of progression to advanced AMD, while medium (63 to 125 micrometres) and large (greater than 125 micrometres) drusen, particularly when associated with pigmentary abnormalities, are the hallmark of early and intermediate AMD and are associated with a significantly increased risk of progression to advanced AMD.

Dry Eye Disease

– Dry eye disease, also known as keratoconjunctivitis sicca, is a multifactorial disorder of the ocular surface characterized by loss of homeostasis of the tear film, resulting in symptoms of ocular discomfort, visual disturbance, and tear film instability, with potential damage to the ocular surface. The disease is classified into two major subtypes: aqueous-deficient dry eye, caused by decreased lacrimal gland production of the aqueous layer of the tear film, as seen in Sjogren syndrome and aginaging in non-Sjogren dry eye), and evaporative dry eye, caused by meibomian gland dysfunction (the most common cause of evaporative dry eye, accounting for 80 percent of dry eye cases in some series), where a reduction in the quantity or quality of the lipid layer of the tear film leads to increased tear evaporation and tear hyperosmolarity. The diagnosis of dry eye disease is made by a combination of patient symptom assessment using validated questionnaires and objective clinical tests including the Schirmer test, tear film breakup time, corneal and conjunctival staining, and evaluation of the meibomian glands.

Dry Eye Syndrome

– A multifactorial disease of the ocular surface characterized by tear film instability, hyperosmolarity, and inflammation. Aqueous-deficient dry eye: Sjogren syndrome (autoimmune), non-Sjogren (lacrimal gland dysfunction, age). Evaporative dry eye: meibomian gland dysfunction (most common), incomplete blink, environmental factors. Symptoms: dryness, grittiness, burning, photophobia, blurred vision that improves with blinking. Diagnosis: Schirmer test, tear breakup time, corneal fluorescein staining, meibomian gland evaluation. Treatment: artificial tears (preservative-free), warm compresses, lid hygiene, topical cyclosporine, punctal plugs, and omega-3 supplements.

Endophthalmitis

– A severe, sight-threatening intraocular inflammation involving the vitreous cavity, aqueous humor, and adjacent structures, representing an ophthalmologic emergency. It is classified as exogenous (most common, caused by direct inoculation from intraocular surgery particularly cataract extraction, intravitreal injections, penetrating trauma, or corneal infection) or endogenous (metastatic seeding from a distant infection source, more common in immunocompromised patients, endocarditis, intravenousintravenous drug use, indwelling catheters, and septicemia, with the most common causative organisms being Streptococcus species, Staphylococcus aureus, and gram-negative bacilli). The condition presents with rapid onset of severe, progressive eye pain, decreased visual acuity, conjunctival injection, chemosis, corneal edema, hypopyon (the hallmark of endophthalmitis, present in 80 to 90 percent of cases), and vitreous inflammation. The Endophthalmitis Vitrectomy Study established the management protocol: immediate intravitreal injection of antibiotics (vancomycin 1 mg/0.1 mL for gram-positive coverage and ceftazidime 2.25 mg/0.1 mL for gram-negative coverage), with vitreous tap for culture, and vitrectomy for patients with light perception vision only.

Entropion

– An eyelid malposition in which the eyelid margin rolls inward, causing the eyelashes and skin to rub against the ocular surface, resulting in corneal irritation, conjunctival injection, tearing, photophobia, and a foreign body sensation. The condition most commonly affects the lower eyelid and is classified into four types. Involutional entropion (most common, age-related, due to horizontal laxity from canthal tendon weakening, dehiscence or attenuation of the lower eyelid retractors, and overrioverriding of the preseptal orbicularis oculi muscle, allowing the lower eyelid to roll inward), cicatricial entropion (caused by scarring and contraction of the posterior lamella, most commonly following chemical burns, Stevens-Johnson syndrome, ocular cicatricial pemphigoid, or chronic trachoma), congenital entropion (rare, due to hypertrophic marginal bundle of the orbicularis oculi muscle), and spastic entropion (temporary, due to sustained orbicularis oculi contraction). The diagnosis is clinical, based on slit-lamp examination.

Epiphora

– Excessive tearing caused by impaired tear drainage or reflex hypersecretion. Causes: nasolacrimal duct obstruction (most common in infants and elderly), punctal stenosis, eyelid malposition (ectropion, entropion), and reflex tearing from dry eye or ocular surface irritation. Congenital NLDO: presents with tearing and mucoid discharge in infants; most resolve spontaneously by 12 months. Diagnosis: dye disappearance test, lacrimal irrigation, dacryocystography. Treatment: lacrimal massage for infants; probing and irrigation; dacryocystorhinostomy (DCR) for adults with structural obstruction.

Episcleritis

– A benign, self-limited inflammatory condition of the episclera, the thin layer of connective tissue between the sclera and conjunctiva. It is most common in young to middle-aged women and is typically idiopathic, though it is associated with systemic inflammatory conditions in approximately 30 percent of cases (rheumatoid arthritis, systemic lupus erythematosus, Sjogren syndrome, granulomatosis with polyangiitis, inflammatory bowel disease, psoriatic arthritis, and ankylosing spondylitis). The pThe patient presents with acute or subacute onset of unilateral or bilateral eye redness, a dull ache or irritation, and sectoral injection of the episcleral vessels (which appear as bright red, radially oriented, tortuous engorged vessels that blanch with topical phenylephrine 2.5 percent, distinguishing episcleritis from scleritis). The diagnosis is clinical. The condition is self-limited and resolves spontaneously within 2 to 3 weeks in most cases.

Esotropia

– A type of strabismus in which one eye deviates inward (toward the nose). Congenital/infantile esotropia: large-angle, constant, onset before 6 months; requires early surgical correction. Accommodative esotropia: associated with hyperopia, onset age 2-3 years; treatment with corrective glasses improves alignment. Intermittent esotropia can be controlled with fusion. Complications: amblyopia (lazy eye) and loss of binocular vision. Diagnosis: cover-uncover test, alternate cover test, cycloplegic refraction. Management: glasses, patching, prism therapy, and strabismus surgery.

Exophthalmos

– Abnormal protrusion of the eyeball (globe) from the orbit, most commonly caused by thyroid eye disease (Graves ophthalmopathy–an autoimmune condition associated with hyperthyroidism). The proptosis results from inflammation, edema, and fibrosis of the extraocular muscles and proliferation of orbital fat within the fixed bony orbit. Bilateral exophthalmos is classic for Graves disease. Unilateral exophthalmos suggests orbital tumor (hemangioma, meningioma, optic nerve glioma, lymphoma), orbital pseudotumor, cellulitis, or carotid-cavernous fistula. Measurement: Hertel exophthalmometry (normal <20 mm, or asymmetry >2 mm). Symptoms: proptosis, lid retraction, lid lag (lagophthalmos), exposure keratopathy, diplopia, and optic nerve compression. Treatment: glucocorticoids, orbital decompression surgery, and management of the underlying thyroid disease.

Exotropia

– A type of strabismus in which one eye deviates outward (toward the temple). Intermittent exotropia: most common type, often noticed when tired or daydreaming, can be controlled with fusion. Constant exotropia: requires treatment to prevent amblyopia. Congenital exotropia: rare, requires early evaluation. Diagnosis: cover test, angle measurement. Treatment: part-time patching, orthoptic exercises (convergence exercises), prism glasses, and surgical correction for persistent deviations.

Eye Disorders

– See Vision.

Eye Health

– The maintenance of optimal vision and prevention of ocular disease through regular eye examinations, protective measures, and timely treatment. Common conditions affecting eye health include refractive errors, cataracts, glaucoma, diabetic retinopathy, and age-related macular degeneration. Protective strategies include wearing UV-blocking sunglasses, managing systemic health conditions, and avoiding eye strain.

Fluorescein Angiography

– An ophthalmological diagnostic procedure that involves intravenous injection of sodium fluorescein dye and sequential photography of the retinal vasculature using a specialized fundus camera fitted with excitation and barrier filters, providing detailed information about retinal blood flow, vascular permeability, and retinal pigment epithelium integrity. The dye is excited by blue light at approximately 490 nanometers and emits yellow-green fluorescence at approximatelapproximately 520 to 530 nanometres. The procedure involves a series of timed images captured in rapid succession following the injection: the early arterial phase (10 to 15 seconds), the arteriovenous phase (15 to 20 seconds), the venous phase (20 to 30 seconds), and the late phase (5 to 10 minutes). The normal fluorescein angiogram shows the retinal vessels fluoresce brightly against the dark background of the choroidal fluorescence.

Fornix

– The conjunctival fornix is the redundant, dome-shaped fold of conjunctiva where the palpebral conjunctiva transitions to the bulbar conjunctiva, forming a deep recess superiorly (superior fornix), inferiorly (inferior fornix), and medially and laterally (medial and lateral fornices). The fornices are important anatomical landmarks for contact lens fitting, conjunctival sampling for diagnosis of chlamydial and viral infections, and as reservoirs for topical medications applied to the eye. The supThe superior fornix is the deepest, extending approximately 8 to 10 mm above the superior corneal limbus, while the inferior fornix is approximately 6 to 8 mm deep. The fornices are also the site of lymphatic drainage of the conjunctiva, which drains to the preauricular and submandibular lymph nodes, and they contain accessory lacrimal glands (the glands of Krause and Wolfring) that contribute to the basal secretion of the aqueous component of the tear film.

Fovea Centralis

– The fovea centralis is the small, central depression within the macula of the retina that represents the area of highest visual acuity, containing the highest concentration of cone photoreceptors and no rod photoreceptors. The unique anatomy of the fovea, where the inner retinal layers are displaced laterally in a process called foveal pit formation, allows light to reach the photoreceptors with minimal scattering, resulting in the sharpest possible image formation. The foveola, the central 0.3535 mm in diameter, contains only cone photoreceptors (no rod photoreceptors), with the highest concentration of cones in the entire retina, providing the highest possible spatial resolution for central, sharp vision (visual acuity of 20/20 or better). The central 0.5 mm of the fovea, the foveola, is avascular and has no inner retinal layers, ensuring that light reaches the cone outer segments with minimal scattering.

Fundus Photography

– The documentation of the appearance of the retina, optic nerve, and retinal vasculature using a specialized camera that illuminates and photographs the posterior segment of the eye, providing a permanent, objective record of retinal findings that can be used for comparison over time, patient education, and telemedicine consultations. Standard fundus photography captures images of the optic disc, macula, and major retinal vessels, and widefield fundus photography can image uup to 200 degrees of the retina in a single image. Fundus photography can be performed with or without pupillary dilation. The images are taken in color, red-free (using green light to enhance the visibility of retinal vessels and haemorrhages), and autofluorescence (using blue light to excite lipofuscin in the RPE).

Geographic Atrophy

– The advanced form of dry age-related macular degeneration, characterized by well-demarcated, progressively enlarging areas of complete loss of the retinal pigment epithelium, choriocapillaris, and overlying photoreceptors in the macula, resulting in irreversible central vision loss. The atrophic areas appear as well-defined, depigmented regions on fundus examination, with visible underlying choroidal vessels, and they typically begin in the parafoveal region and expand centcentrally over time, leading to a progressive, bilateral, irreversible, and painless loss of central vision, with the preservation of peripheral vision until late in the disease process. The diagnosis of geographic atrophy is based on fundoscopic examination (showing well-demarcated, depigmented areas with visible choroidal vessels), fundus autofluorescence (showing areas of absent autofluorescence), and optical coherence tomography (showing the loss of the RPE and outer retinal layers).

Glaucoma

– A group of progressive optic neuropathies characterized by the degeneration of retinal ganglion cells and their axons, resulting in characteristic optic nerve head changes and visual field loss, with elevated intraocular pressure being the most important modifiable risk factor. Primary open-angle glaucoma, the most common form, involves gradual obstruction of the trabecular meshwork with chronically elevated intraocular pressure, progressive cupping of the optic nerve head, and characharacteristic visual field defects (paracentral scotomas, arcuate scotomas, and nasal steps in the early stages, progressing to central vision loss and tunnel vision in advanced disease). Other forms of glaucoma include normal-tension glaucoma (glaucomatous optic neuropathy and visual field loss with intraocular pressure consistently within the normal range), primary angle-closure glaucoma (caused by pupillary block, leading to an acute, painful, red eye with fixed, mid-dilated pupil, corneal edema, and elevated IOP), and secondary glaucomas resulting from other ocular or systemic disease.

Glea

– A thick, tenacious, mucopurulent discharge from the eye. An older term for the viscous crusting and discharge seen in purulent conjunctivitis, especially gonococcal ophthalmia neonatorum. The term gummous gleet referred to a urethral discharge in gonorrhea. Modern usage: the medical term is purulent discharge or mucopurulent discharge.

Goldmann Tonometry

– Goldmann applanation tonometry is the gold standard method for measuring intraocular pressure, the most important modifiable risk factor for glaucoma, and it involves applanating (flattening) a fixed area of the cornea using a biprism attached to a slit lamp, measuring the force required to achieve this applanation. The principle is based on the Imbert-Fick law, which states that the pressure inside a sphere is equal to the force needed to flatten a given area of its surface, and the Goldmann tothe Goldmann tonometer is calibrated to measure the intraocular pressure based on the force required to applanate a 3.06 mm diameter area of the central cornea. The Goldmann tonometer is mounted on a slit lamp, topical anesthesia is applied to the cornea, and fluorescein dye is added to the tear film to stain the tear meniscus.

Gonioscopy

– An ophthalmological examination technique that allows direct visualization of the anterior chamber angle, the anatomical region where the cornea and iris meet and where the trabecular meshwork is located, and it is essential for distinguishing between open-angle and angle-closure forms of glaucoma. The procedure uses a specialized goniolens placed on the surface of the eye that overcomes total internal reflection at the corneal-air interface, allowing light to enter and illuminate the angle structures for gonioscopic evaluation. Gonioscopy is performed using either a direct goniolens (such as the Koeppe lens, providing a direct view of the angle) or an indirect goniolens (such as the Goldmann three-mirror lens or the Zeiss four-mirror lens, providing an indirect, mirrored view of the angle using mirrors at various angles).

Hordeolum (Stye)

– See Stye.

Hyperopia

– Hyperopia, commonly known as farsightedness, is a refractive error in which parallel light rays from distant objects are focused posterior to the retina rather than directly on it, resulting in blurred near vision that may be accompanied by blurred distance vision in moderate to high degrees. The condition is caused by an insufficient axial length of the globe relative to the refractive power of the cornea and lens, or by a flat corneal curvature, and it is corrected with convex (plus-powered) cconvex (plus-powered) converging lenses (spectacles or contact lenses) that move the focal point of light from behind the retina back onto the retina. Hyperopia is quantified by the refractive error in dioptres, with the degree classified as low (less than +2.00 D), moderate (+2.00 D to +5.00 D), and high (greater than +5.00 D).

Hypertensive Retinopathy

– Retinal changes caused by chronic or acute hypertension. Grading (Keith-Wagener-Barker): Grade I: generalized arteriolar narrowing; Grade II: arteriovenous nicking; Grade III: cotton-wool spots, flame hemorrhages, hard exudates; Grade IV: papilledema (malignant hypertension). Severe hypertension can cause retinal artery occlusion, vitreous hemorrhage, and vision loss. Management: blood pressure control; ophthalmology referral for advanced grades.

Hyphema

– Bleeding into the anterior chamber of the eye, between the cornea and iris, typically resulting from blunt ocular trauma that tears the blood vessels of the iris or ciliary body. Less common causes include spontaneous hyphema (from iris neovascularization in diabetes, retinal vein occlusion, or ocular ischemic syndrome), intraocular surgery (post-cataract or glaucoma surgery), and coagulopathy (anticoagulant or antiplatelet therapy, hemophilia, von Willebrand disease, thrombocytopenia). The blooThe blood in the anterior chamber layers dependently due to gravity, forming a visible level (the hyphema) that is graded based on the height of the blood level relative to the total anterior chamber height. The immediate management includes strict bed rest with the head elevated at 30 to 45 degrees, shielding the eye with a protective patch, and cycloplegia with atropine 1 percent or homatropine 5 percent.

Intermediate Uveitis

– Inflammation primarily affecting the vitreous body, peripheral retina, and ciliary body, with the classic presentation being pars planitis, which is characterized by the presence of snowbanking (a white, inflammatory exudate on the inferior pars plana) and snowballs (vitreous inflammatory aggregates). Patients typically present with floaters and blurred vision, often with minimal pain or redness, and slit-lamp examination reveals vitreous cells and haze, with a quiet antequiet anterior chamber. The main differential diagnosis includes multiple sclerosis, sarcoidosis, tuberculosis, Lyme disease, and cat-scratch disease, as well as the masquerade syndrome (vitreoretinal lymphoma). The diagnosis is made by slit-lamp examination showing vitreous cells and haze, with OCT used to detect cystoid macular edema, the most common cause of vision loss in intermediate uveitis.

Intraocular Lens

– An intraocular lens is an artificial, biocompatible, transparent lens implanted during cataract surgery to replace the natural crystalline lens that has been removed, restoring the eye's refractive power and enabling clear vision. Intraocular lenses are typically made of polymethylmethacrylate, silicone, or acrylic materials, and they are available in various designs including monofocal (correcting vision at a single distance), multifocal (providing vision at multiple distances), and toric (corrcorrecting astigmatism. The power of the IOL is calculated using biometric measurements of the eye, including axial length, corneal curvature, and anterior chamber depth, with formulas such as the SRK/T, Holladay, and Haigis formulas. The most common IOL implantation technique is in-the-bag placement within the intact capsular bag.

Intravitreal Injection

– An ophthalmological procedure in which medication is injected directly into the vitreous cavity of the eye through the pars plana, a safe entry zone located approximately 3.5 to 4 millimeters posterior to the limbus, to deliver therapeutic agents directly to the retina and vitreous for the treatment of various retinal diseases. The most commonly injected medications are anti-vascular endothelial growth factor agents for wet age-related macular degeneration, diabetic macmacular edema secondary to retinal vein occlusion. The procedure is performed in the clinic under sterile conditions using a topical anesthetic, and the eye is prepared with 5 percent povidone-iodine. The injection is performed through the pars plana, located 3.5 to 4 mm posterior to the limbus, using a 30-gauge or 31-gauge needle.

Iridotomy

– Laser peripheral iridotomy is a minimally invasive laser procedure used to treat or prevent angle-closure glaucoma by creating a small full-thickness hole in the peripheral iris, allowing aqueous humor to flow directly from the posterior chamber to the anterior chamber, bypassing the pupil and equalizing the pressure between the two chambers. The procedure is indicated for acute angle-closure glaucoma, where it is performed as an emergency to relieve pupillary block and lower intraocular pressuras an emergency to relieve pupillary block and lower intraocular pressure within minutes, and for eyes with occludable angles at high risk of angle closure. The iridotomy is created at the 10 or 2 o'clock position to avoid the visual axis and is typically 100 to 200 micrometres in diameter.

Iritis

– Inflammation of the iris, the most common form of anterior uveitis. Presents with acute onset eye pain, photophobia, blurred vision, and redness (ciliary flush). Signs: cells and flare in the anterior chamber, keratic precipitates (KP), synechiae (adhesions between iris and lens). Causes: idiopathic, trauma, HLA-B27-associated (ankylosing spondylitis, reactive arthritis, inflammatory bowel disease), sarcoidosis, herpes simplex/zoster. Diagnosis: slit lamp examination. Treatment: topical corticosteroids (prednisolone acetate) and cycloplegics (homatropine, atropine) to reduce inflammation and prevent synechiae.

Juvenile angiofibroma

– A rare, benign but locally aggressive vascular tumor of the nasopharynx, accounting for 0.05-0.5 percent of all head and neck tumors, occurring almost exclusively in adolescent males (mean age 14 years, 80-90 percent male predominance). The tumor originates from the posterolateral wall of the nasal cavity at the sphenopalatine foramen and is composed of vascular channels (irregular, thin-walled vessels lacking smooth muscle and elastic fibers) embedded in a fibrous stroma. Although histologicallhistologically benign, the tumor is locally aggressive, eroding bone and extending into the paranasal sinuses, orbit, and cranial cavity. Clinical presentation is with progressive, recurrent, typically unilateral epistaxis (the most common presenting symptom), followed by nasal obstruction, facial swelling, proptosis, and anosmia. Diagnosis is by nasal endoscopy, CT, and angiography.

Keratitis

– Inflammation of the cornea. Infectious keratitis: bacterial (contact lens wear, trauma: Pseudomonas, Staph aureus), viral (HSV, VZV, adenovirus), fungal (Candida, Fusarium, Aspergillus), and Acanthamoeba (contact lens-related). Presents with eye pain, photophobia, tearing, conjunctival injection, and corneal opacity/ulcer. Diagnosis: slit lamp exam with fluorescein staining reveals corneal epithelial defect. Treatment: topical antibiotics (fluoroquinolones, fortified aminoglycosides) for bacterial; antivirals (acyclovir, gancyclovir) for herpetic; antifungals for fungal. Corneal perforation requires urgent ophthalmology referral and possible penetrating keratoplasty.

Keratoconus

– A progressive, non-inflammatory ectatic corneal disorder characterized by bilateral (90 percent of cases) thinning and cone-like protrusion of the cornea, leading to irregular astigmatism, myopia, and decreased visual acuity. The condition typically presents in adolescence or early adulthood (peak incidence at age 15-25) and progresses until the 4th-5th decade of life, when it usually stabilizes. The prevalence is approximately 1 in 2000 in the general population. The pathophysiology involves weweakening of the corneal stroma due to reduced structural integrity of the collagen framework, with a genetic predisposition (association with Down syndrome, Ehlers-Danlos syndrome, Marfan syndrome, and mutations in the VSX1 and LOX genes). The diagnosis is made by slit-lamp examination (showing corneal thinning, Vogt striae, Fleischer ring, and apical scarring), corneal topography, and pachymetry.

Lacrimal Apparatus

– The anatomical structures responsible for tear production, distribution, and drainage. Components: (1) Lacrimal gland—situated in the superolateral orbit (frontal bone lacrimal fossa); produces the aqueous layer of the tear film. Divided into orbital and palpebral parts; innervated by parasympathetic fibers from CN VII (via pterygopalatine ganglion). Accessory lacrimal glands (Krause, Wolfring) contribute to basal tear secretion. (2) Tear film—three layers: outer lipid layer (meibomian glands—reduces evaporation), middle aqueous layer (lacrimal gland—nutrients, antimicrobials—lysozyme, IgA), inner mucin layer (goblet cells—spreads tears evenly over the cornea). (3) Lacrimal drainage: tears flow from the lateral eye across the cornea to the puncta (upper and lower, medial eyelid margins), through the canaliculi, lacrimal sac, and nasolacrimal duct, draining into the inferior meatus of the nose. Blinking spreads tears and pumps them through the drainage system. Disorders: dry eye disease/keratoconjunctivitis sicca (aqueous deficiency—Sjögren syndrome, or evaporative—meibomian gland dysfunction, MGD), epiphora (excessive tearing—blocked nasolacrimal duct, punctal stenosis, ectropion, reflex tearing from dry eye), dacryocystitis (infection of the lacrimal sac—pain, swelling at the medial canthus, purulent discharge, treated with warm compresses, massage, antibiotics, and dacryocystorhinostomy).

Laser Trabeculoplasty

– A minimally invasive laser procedure used in the treatment of open-angle glaucoma that lowers intraocular pressure by improving the outflow of aqueous humor through the trabecular meshwork. Selective laser trabeculoplasty, the current standard technique, uses a frequency-doubled Q-switched Nd:YAG laser to deliver non-thermal laser burns to the pigmented trabecular meshwork cells, stimulating biological changes that enhance aqueous humor outflow through the conventional tthe conventional outflow pathway, enhancing the outflow of aqueous humor through the trabecular meshwork without damaging adjacent tissues. SLT is most effective in primary open-angle glaucoma, pseudoexfoliation glaucoma, and pigmentary glaucoma, with minimal post-operative pain and inflammation.

LASIK

– LASIK, or laser-assisted in situ keratomileusis, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser, after creating a thin corneal flap with either a microkeratome (a mechanical blade) or a femtosecond laser. The procedure begins with the creation of a hinged corneal flap approximately 100 to 160 micrometers thick, which is folded back to expose the underlying corneal stroma, and the excimer laser then ablateablates the stromal tissue according to a pre-programmed treatment profile that reshapes the corneal curvature to correct the refractive error, with sub-micron precision in the removal of stromal tissue, with the depth and pattern determined by the refractive error, the optical zone, and the transition zone.

LASIK (Laser-Assisted In Situ Keratomileusis)

– A refractive surgical procedure that reshapes the cornea using an excimer laser to correct myopia, hyperopia, and astigmatism, thereby reducing dependence on glasses or contact lenses. A corneal flap is created with a femtosecond laser or microkeratome, the underlying stroma is ablated per preoperative topography, and the flap is repositioned. Ideal candidates are adults aged >21 with stable refraction for ≥1 year, adequate corneal thickness, and no active ocular disease (keratoconus, severe dry eye, autoimmune disorders). Most patients achieve 20/20 or better vision; temporary side effects include dry eye, glare, halos, and fluctuating vision. Serious complications (flap striae, infection, ectasia, epithelial ingrowth) are rare with appropriate screening and technique.

Leber Congenital Amaurosis

– The most severe inherited retinal dystrophy, presenting at birth or within the first few months of life with profound visual impairment, nystagmus, and pupillary light-near dissociation. Caused by mutations in over 20 genes (most commonly GUCY2D, CEP290, CRB1, RPE65, and AIPL1), encoding proteins critical for phototransduction, retinal development, or ciliary function. Inheritance is autosomal recessive. Presents with poor visual behavior (inability to fix and follow), roving nystagmus, oculodigital sign (eye poking, rubbing), and photophobia or night blindness. Fundus examination may appear normal initially, but electroretinography (ERG) is the diagnostic gold standard — shows markedly reduced or absent rod and cone responses. Treatment: gene therapy (voretigene neparvovec, Luxturna) for RPE65-associated LCA — the first FDA-approved gene therapy for a retinal disease. Low-vision rehabilitation, orientation and mobility training, and genetic counseling. Most patients have severe visual impairment from birth.

Macula

– The macula is a specialized region of the central retina approximately five and a half millimeters in diameter that is responsible for high-acuity central vision, color perception, and fine detail discrimination, and it contains the highest density of cone photoreceptors in the eye. The central-most region of the macula is the fovea centralis, a depression approximately one and a half millimeters in diameter where the inner retinal layers are displaced laterally, allowing light to strike the phophotoreceptors directly, maximising the efficiency of photon capture. The fovea is surrounded by the parafovea and perifovea, where the retinal layers transition to the normal architecture of the peripheral retina.

Macular degeneration

– Also known as age-related macular degeneration (AMD), a progressive degenerative disease of the central retina (macula) and the leading cause of irreversible vision loss in individuals over age 50 in developed countries, affecting approximately 10 percent of those over 65 and 30 percent over 80. The pathophysiology involves oxidative stress, chronic inflammation, and complement system dysregulation at the interface between the retinal pigment epithelium (RPE) and Bruch membrane, resulting in accaccumulation of drusen, lipofuscin, and inflammatory debris that trigger a chronic, complement-mediated immune response and progressive RPE and photoreceptor atrophy. The dry form accounts for 85-90 percent of AMD, with no effective treatment to reverse or slow the atrophy. The wet form accounts for 10-15 percent but causes the majority of severe vision loss, treated with anti-VEGF intravitreal injections.

Myopia

– Myopia, commonly known as nearsightedness, is a refractive error in which parallel light rays from distant objects are focused anterior to the retina rather than directly on it, resulting in blurred distance vision while near vision remains relatively clear. The condition is caused by an excessive axial length of the globe relative to the refractive power of the cornea and lens, or by excessive corneal curvature, and it is corrected with concave (minus-powered) diverging lenses that move the focfocal point of light from in front of the retina back onto the retina, enabling clear distance vision. The degree of myopia is quantified in dioptres (D), with low (less than -3.00 D), moderate (-3.00 D to -6.00 D), and high (greater than -6.00 D) categories, with high myopia being associated with increased risk of retinal detachment, choroidal neovascularization, macular atrophy, glaucoma, and cataract.

Night Blindness

– Impaired vision in dim light (nyctalopia). Causes: vitamin A deficiency (most common cause worldwide, leading to retinol deficiency and rod dysfunction), retinitis pigmentosa (progressive rod-cone degeneration, genetic), congenital stationary night blindness, advanced glaucoma, and cataract. Vitamin A deficiency: also causes xerophthalmia (dry cornea) and Bitot spots (foamy conjunctival deposits); treat with vitamin A supplementation. Retinitis pigmentosa: no cure; vitamin A, lutein supplementation, and low-vision aids.

Non-Contact Tonometry

– Non-contact tonometry, commonly known as the air puff test, is a method of measuring intraocular pressure that uses a calibrated air pulse to applanate the cornea, with the force of the air pulse determined by a sensor that detects the moment of applanation, providing a rapid, non-invasive measurement without requiring direct corneal contact. The technique eliminates the need for topical anesthesia and reduces the risk of cross-infection between patients, making it particularly useful for screenscreening in high-volume settings and for patients unable to tolerate contact tonometry. The device measures the force of the air puff at the moment of corneal applanation, but it is generally less accurate than Goldmann applanation tonometry, especially in patients with irregular corneas.

Non-Proliferative Diabetic Retinopathy

– The early stage of diabetic retinopathy, characterized by structural abnormalities of the retinal vasculature resulting from chronic hyperglycemia-induced damage to the endothelial cells and pericytes of retinal capillaries, without neovascularization. The clinical features include microaneurysms, which are the earliest clinically visible sign and appear as small, red dots on fundus examination; dot and blot hemorrhages, representing deeper retinal hemorhemorrhages, representing deeper retinal hemorrhages; hard exudates, which are lipid deposits in the outer plexiform layer; cotton-wool spots, representing nerve fibre layer infarcts; and retinal vascular changes, including venous beading, arteriolar narrowing, and intraretinal microvascular abnormalities.

Nuclear Sclerosis

– The most common type of age-related cataract, characterized by progressive yellowing and hardening of the lens nucleus due to the accumulation and aggregation of insoluble crystallin proteins, resulting in a gradual decrease in vision that is often noticed as increased difficulty with distance vision and a phenomenon called second-sight vision, where temporarily improved near vision occurs as the lens becomes more myopic from its increased refractive index. The lens nucleus becomes progressively harder and more yellow, and the resulting increase in the refractive index of the nucleus causes the lens power to increase, leading to a shift toward myopia. The diagnosis is made by slit-lamp examination after dilation, showing the yellowing and increased density of the lens nucleus, graded from 1+ to 4+.

Nystagmus

– A rhythmic, involuntary oscillation of the eyes that can be horizontal, vertical, rotational, or mixed in pattern, and it is classified based on the underlying cause as infantile (developing in the first six months of life), acquired (developing later in life), or periodic alternating. Infantile nystagmus is the most common form, often associated with albinism, congenital cataracts, or other conditions that impair visual development, and it typically has a null point (a position of the specific direction of gaze, or a head turn or tilt that reduces the intensity of the nystagmus). In acquired nystagmus, the onset is typically subacute and oscillopsia is prominent, and neurological evaluation with imaging is essential to rule out an underlying structural or metabolic cause, including vestibular disorders, brainstem or cerebellar lesions, and drug-induced causes.

Opacity

– Lack of transparency of a normally transparent ocular structure (cornea, lens, vitreous, or aqueous humour). Corneal opacity results from scarring (trauma, infection—herpes simplex keratitis, trachoma), dystrophies (Fuchs endothelial dystrophy, granular/lattice/macular dystrophies), or edema. Lenticular opacity is cataract—age-related, congenital, traumatic, or metabolic (diabetes). Vitreous opacity arises from hemorrhage (diabetic retinopathy, retinal tear), inflammation (vitritis, posterior uveitis), or asteroid hyalosis (calcium-lipid deposits in the vitreous, benign). Assessment: slit-lamp examination, visual acuity, and optical coherence tomography. Opacity causes decreased visual acuity, glare, and contrast sensitivity loss. Management depends on cause: corneal transplant (penetrating keratoplasty, DSEK) for irreversible corneal opacity; cataract surgery (phacoemulsification with intraocular lens implantation) for visually significant cataract; vitrectomy for non-clearing vitreous hemorrhage.

Optical Coherence Tomography

– A non-invasive imaging technique that uses low-coherence interferometry to produce high-resolution, cross-sectional images of the retina and other ocular structures, providing quantitative measurements of retinal layer thickness and detecting structural abnormalities with micrometer-level resolution. In ophthalmology, OCT is the gold standard for diagnosing and monitoring macular diseases including age-related macular degeneration, diabetic macular edema, and macumacular hole, epiretinal membrane, and central serous chorioretinopathy, and for measuring retinal nerve fibre layer thickness in glaucoma. The principle of OCT is analogous to ultrasound imaging but using light rather than sound, achieving an axial resolution of 2 to 7 micrometres.

Optic Atrophy

– End-stage degeneration of the optic nerve fibers resulting from any optic neuropathy. Causes: glaucoma, ischemic optic neuropathy, multiple sclerosis, compressive lesions (pituitary tumor, aneurysm), toxins (methanol, ethambutol), and hereditary (Leber hereditary optic neuropathy). Fundoscopic findings: pale optic disc with sharp margins (primary atrophy) or blurred margins (secondary atrophy). Presents with visual acuity loss, color vision deficits, and visual field defects. Irreversible; treatment addresses underlying cause.

Optic Neuritis

– An inflammatory demyelinating condition of the optic nerve that typically presents with acute or subacute unilateral vision loss, pain with eye movement, and a relative afferent pupillary defect, and it is most commonly associated with multiple sclerosis, occurring in approximately fifty percent of patients at some point during their disease course. The pathophysiology involves autoimmune-mediated demyelination of the optic nerve, with T lymphocyte infiltration and antibody-medantibody-mediated demyelination. The afferent visual dysfunction is characterized by reduced best-corrected visual acuity, decreased contrast sensitivity, dyschromatopsia (impaired color vision, particularly red color desaturation), and a relative afferent pupillary defect. A central or paracentral scotoma is found on formal visual field testing.

Optometry

– A healthcare profession concerned with examining the eyes and visual system for defects and abnormalities, prescribing corrective lenses (spectacles, contact lenses), diagnosing eye disease, and providing pre- and post-operative care for surgical eye treatments. Optometrists (in the UK) perform sight tests (refraction, visual field assessment, tonometry, retinal examination), detect ocular disease (glaucoma, diabetic retinopathy, macular degeneration, cataracts), and refer to ophthalmologists when medical or surgical intervention is required. They also manage low vision, binocular vision disorders (strabismus, amblyopia), and provide advice on visual ergonomics. In the UK, optometrists train for 4 years (BSc Optometry) followed by a pre-registration year and professional qualifying examinations.

Orbital Cellulitis

– A serious infection of the orbital tissues posterior to the orbital septum, potentially sight-threatening and life-threatening due to the risk of optic nerve compression, cavernous sinus thrombosis, and intracranial extension. The condition is classified by the Chandler system into five stages: preseptal cellulitis (infection confined to the eyelid and tissues anterior to the orbital septum), orbital cellulitis (infection of the orbital fat and muscles without abscess formaformation), and postseptal cellulitis (the true orbital cellulitis, involving the orbital fat, muscles, optic nerve, and neurovascular structures). Preseptal cellulitis presents with eyelid erythema, edema, warmth, and tenderness, with no proptosis, no limitation of extraocular movements, no relative afferent pupillary defect, and no visual loss, and it is managed with oral antibiotics.

Palpebral Conjunctiva

– The palpebral conjunctiva is the portion of the conjunctival membrane that lines the inner surface of the upper and lower eyelids, and it is subdivided into the tarsal conjunctiva, which is firmly adherent to the tarsal plates and appears smooth and pink, and the forniceal conjunctiva, which forms loose, redundant folds at the superior and inferior fornices where the palpebral conjunctiva reflects to become the bulbar conjunctiva. The palpebral conjunctiva contains meibomian glands (sebaceous glmeibomian glands that produce the lipid layer of the tear film. The tarsal conjunctiva is normally pink and smooth, and abnormalities such as follicles (characteristically seen in viral and chlamydial conjunctivitis), papillae (characteristically seen in allergic conjunctivitis), and membranes provide important diagnostic clues.

Panretinal Photocoagulation

– A laser treatment for proliferative diabetic retinopathy that involves applying laser burns to the peripheral retina (typically 1,200 to 2,000 burns over three to four sessions) to ablate ischemic retinal tissue, reducing the production of vascular endothelial growth factor and other angiogenic factors that drive pathological neovascularization. The laser burns destroy the oxygen-consuming photoreceptor cells in the peripheral retina, improving oxygenation of the rremaining retina, reducing the VEGF drive for neovascularization, thereby causing regression of existing new vessels and preventing the growth of new abnormal vessels. The specific indications for PRP are high-risk proliferative diabetic retinopathy.

Panuveitis

– Inflammation involving all segments of the uveal tract, including the anterior chamber, vitreous cavity, and choroid, resulting in a severe, potentially sight-threatening condition that requires aggressive systemic immunosuppressive therapy. The clinical presentation combines features of anterior, intermediate, and posterior uveitis, with patients experiencing significant eye pain, photophobia, blurred vision, and decreased vision, and examination revealing cells and flare in the aanterior chamber, vitreous haze and cells, and active choroiditis or chorioretinitis, often with exudative retinal detachment, cystoid macular edema, and papillitis. The etiology includes infectious causes (toxoplasmosis, tuberculosis, syphilis, cytomegalovirus), non-infectious causes (sarcoidosis, Behcet disease, Vogt-Koyanagi-Harada syndrome, birdshot chorioretinopathy), and masquerade syndrome (intraocular lymphoma).

Papilledema

– Swelling of the optic disc caused by elevated intracranial pressure, and it is a critical clinical sign that requires urgent investigation to identify the underlying cause, which may be life-threatening. The pathophysiology involves transmission of elevated cerebrospinal fluid pressure through the optic nerve sheath to the optic nerve head, causing axoplasmic stasis, disc edema, and potentially progressive optic nerve damage. Clinical findings include disc blurring, elevation, venvenous congestion, retinal venous pulsation loss, and peripapillary haemorrhages. The urgency mandates immediate neuroimaging (CT head without contrast) to rule out a space-occupying lesion or hydrocephalus, followed by lumbar puncture with opening pressure measurement if CT is negative.

Phacoemulsification

– The most commonly performed modern cataract surgery technique, involving the use of an ultrasonic handpiece with a titanium tip that vibrates at high frequency to fragment and emulsify the opaque lens nucleus while simultaneously irrigating and aspirating the lens material through a coaxial or bimanual system, leaving the posterior lens capsule intact for implantation of an artificial intraocular lens. The procedure is performed through a small self-sealing incision, typictypically 2.2 to 3.2 mm in width, through the clear cornea near the limbus. Phacoemulsification has a low rate of wound leak, post-operative inflammation, and induced astigmatism, with rapid visual recovery and a return to full activity within days to weeks.

Posterior Subcapsular Cataract

– A type of lens opacity located at the posterior aspect of the lens, just anterior to the posterior lens capsule, in the visual axis, and it is particularly visually disabling because of its central location. The opacity appears as a granular, plaque-like whitish area on the posterior lens surface on slit-lamp examination, and it may have a flower-petal or vacuolar pattern. This form of cataract is more common in younger patients and is strongly associated with ccorticosteroid use (both topical, inhaled, and systemic, which is the strongest modifiable risk factor, with risk increasing with dose and duration of use), diabetes mellitus, trauma, intraocular inflammation, and radiation exposure. The treatment is surgical removal with phacoemulsification and IOL implantation.

Posterior Uveitis

– Inflammation primarily affecting the choroid (choroiditis), retina (retinitis), or both (chorioretinitis), and it can be caused by infectious agents including Toxoplasma gondii, cytomegalovirus, herpes simplex virus, tuberculosis, and syphilis, as well as non-infectious causes including sarcoidosis, Behcet disease, and Vogt-Koyanagi-Harada syndrome. The clinical presentation depends on the underlying cause and may include floaters, blurred vision, scotomas, and decreased visdecreased vision with a relative afferent pupillary defect in severe or unilateral cases. The diagnosis is based on fundoscopic examination showing active choroidal or retinal lesions, with ancillary testing to characterise the disease and laboratory evaluation to identify the underlying cause.

Presbyopia

– The age-related, progressive loss of the accommodative ability of the crystalline lens, resulting in difficulty focusing on near objects, typically becoming symptomatic around age forty to forty-five and progressing until approximately age sixty-five. The condition is caused by age-related changes in the lens, including increased rigidity of the lens capsule, decreased elasticity of the lens substance, and continued growth of the lens throughout life, which collectively reduce the the amplitude of accommodation from approximately 15 dioptres in childhood to less than 1 dioptre by age 65. The treatment is with convex (plus-powered) converging lenses for near vision, either as reading glasses, bifocal glasses, progressive addition lenses, or contact lenses (bifocal, multifocal, or monovision designs).

Preseptal Cellulitis

– Preseptal cellulitis, also known as periorbital cellulitis, is an infection of the eyelid and periorbital tissues anterior to the orbital septum, without involvement of the orbital contents, and it is the most common orbital infection, particularly in children. The condition typically results from contiguous spread from adjacent skin infections, insect bites, or trauma, and common causative organisms include Staphylococcus aureus and Streptococcus pyogenes in community-acquired cases, and HaemopHaemophilus influenzae in children and Staphylococcus aureus and Streptococcus pyogenes in adults. Preseptal cellulitis presents with eyelid edema, erythema, warmth, and tenderness, with no proptosis, no limitation of extraocular movements, no relative afferent pupillary defect, and no visual loss. Management is with oral antibiotics covering the common pathogens.

PRK

– Photorefractive keratectomy, commonly known as PRK, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser after mechanical removal of the corneal epithelium, without creating a corneal flap. The procedure begins with the removal of the corneal epithelium using an alcohol solution, mechanical debridement, or the excimer laser itself, and the excimer laser then ablates the exposed stromal surface according to a coa customized laser treatment profile, removing the Bowman layer and anterior stroma without creating a flap. After ablation, a bandage contact lens is placed to promote epithelial healing. The epithelial defect typically heals over 3 to 5 days, and visual recovery is more gradual than LASIK.

Proliferative Diabetic Retinopathy

– The advanced stage of diabetic retinopathy, characterized by the growth of new, fragile blood vessels (neovascularization) on the retinal surface, optic disc, or iris in response to severe retinal ischemia and the release of vascular endothelial growth factor. These new vessels are abnormal in structure, lacking tight junctions and pericyte support, making them prone to hemorrhage into the vitreous cavity, which can cause sudden, painless, severe vision lossloss due to vitreous hemorrhage. The neovascular vessels also grow on the iris (rubeosis iridis) and in the anterior chamber angle, leading to neovascular glaucoma, a severe, refractory form of glaucoma requiring prompt panretinal photocoagulation to induce regression of the new vessels.

Prostaglandin Analogs

– A class of topical medications used in the treatment of open-angle glaucoma and ocular hypertension that lower intraocular pressure by increasing the uveoscleral outflow of aqueous humor, reducing intraocular pressure by approximately twenty-five to thirty-five percent. The mechanism involves binding to prostaglandin F2-alpha receptors in the ciliary muscle and ciliary body, which activates matrix metalloproteinases that remodel the extracellular matrix of the uveosclerthe uveoscleral pathway through enhanced matrix metalloproteinase activity, reducing the outflow resistance. Available agents include latanoprost (the most commonly prescribed), travoprost, tafluprost, and bimatoprost, and they are generally safe and well tolerated, with the most common side effect being conjunctival hyperaemia.

Ptosis

– The drooping of the upper eyelid, which can be congenital or acquired, and can range from mild drooping that is primarily a cosmetic concern to severe drooping that obstructs the visual axis and causes amblyopia in children or visual impairment in adults. Congenital ptosis is most commonly caused by developmental abnormalities of the levator palpebrae superioris muscle, including myogenic dysgenesis, and is typically present at birth with poor levator function and an absent lid crease.congenital ptosis is most commonly caused by developmental abnormalities of the levator palpebrae superioris muscle, including myogenic dysgenesis, and is typically present at birth with poor levator function and an absent lid crease. Acquired ptosis is further classified by etiology into aponeurotic (most common in older adults, caused by dehiscence of the levator aponeurosis), myogenic, neurogenic, and mechanical ptosis.

Pupil

– The central aperture of the iris that regulates the amount of light entering the eye. Pupil size is controlled by two smooth muscles: the sphincter pupillae (circular fibers, innervated by parasympathetic fibers from CN III—oculomotor nerve, via the ciliary ganglion, muscarinic M_3 receptors) causes constriction (miosis); the dilator pupillae (radial fibers, innervated by sympathetic fibers from the superior cervical ganglion, α_1 receptors) causes dilation (mydriasis). Normal pupil diameter: 2–4 mm in bright light, 4–8 mm in dim light. Pupillary light reflex: light in one eye causes direct (same eye) and consensual (opposite eye) constriction (afferent: CN II—optic nerve; efferent: CN III—oculomotor nerve). Near reflex: convergence, accommodation, and pupillary constriction. Abnormal pupils: anisocoria (unequal size—physiological in 20%, Horner syndrome, CN III palsy, Adie tonic pupil), afferent pupillary defect (relative afferent pupillary defect—Marcus Gunn pupil, in optic nerve disease—optic neuritis, ischemic optic neuropathy, compression), and Argyll Robertson pupils (bilateral small irregular pupils that accommodate but do not react to light—classically in neurosyphilis).

Ranibizumab

– A recombinant humanized monoclonal antibody Fab fragment that binds to and neutralizes all isoforms of vascular endothelial growth factor A, inhibiting the growth of pathological choroidal neovascular blood vessels and reducing vascular permeability in wet age-related macular degeneration. The drug is administered by intravitreal injection, typically monthly initially with the possibility of extending intervals based on treatment response, and it was the first anti-vascular endothendothelial growth factor agent approved for wet AMD, receiving FDA approval in 2006 based on the MARINA and ANCHOR clinical trials, which demonstrated that monthly intravitreal injections led to a mean improvement in visual acuity of 7 to 10 letters at 12 months.

Refraction

– The bending of light rays as they pass through the eye's optical system (cornea, aqueous humour, crystalline lens, vitreous humour) to focus a sharp image on the retina. The refractive power of the eye is measured in dioptres (D). Emmetropia: normal refractive state—parallel light rays focus exactly on the retina without accommodation. Ametropia (refractive error): (1) Myopia (short-sightedness)—light focuses in front of the retina, corrected by concave (negative) lenses; axial myopia (elongated eyeball) is most common. (2) Hypermetropia (long-sightedness)—light focuses behind the retina, corrected by convex (positive) lenses; young hypermetropes may compensate by accommodation. (3) Astigmatism—unequal curvature of the cornea or lens causes light to focus on multiple points, corrected by cylindrical lenses. (4) Presbyopia—age-related loss of accommodation (lens hardening), typically becomes symptomatic after age 40, corrected by reading glasses or bifocals. Refraction is measured objectively (retinoscopy, autorefractor) and subjectively (the patient's response to lens choices during a sight test). Refractive surgery (LASIK, PRK, SMILE) reshapes the cornea to correct refractive errors.

Refractive Errors and Astigmatism

– Refractive errors are vision disorders caused by the eye's inability to properly focus light onto the retina, resulting in blurred vision. Common types include myopia (nearsightedness), hyperopia (farsightedness), presbyopia (age-related loss of near vision), and astigmatism, which occurs when the cornea or lens has an irregular curvature that causes distorted vision at all distances. These conditions are typically diagnosed through a comprehensive eye examination and are correctable with eyeglasses, contact lenses, or refractive surgery such as LASIK.

Retinal Detachment

– A separation of the neurosensory retina from the underlying retinal pigment epithelium, creating a potential space that fills with fluid and resulting in progressive vision loss in the affected area, and it is classified into three main types based on the underlying mechanism. Rhegmatogenous retinal detachment, the most common type, is caused by a full-thickness break or tear in the retina that allows liquefied vitreous to enter the subretinal space, separating the retina ffrom the underlying RPE. Tractional retinal detachment is caused by the contraction of fibrovascular membranes on the retinal surface, most commonly in proliferative diabetic retinopathy. Exudative retinal detachment occurs when subretinal fluid accumulates due to leakage from blood vessels, without a retinal break, most commonly associated with inflammatory conditions.

Retinal Layers

– The retina consists of ten distinct layers that process visual information through a complex neural circuit. From innermost to outermost, these layers include the internal limiting membrane (the boundary between retina and vitreous), nerve fiber layer (containing retinal ganglion cell axons), ganglion cell layer (containing retinal ganglion cell bodies), inner plexiform layer (synapses between bipolar and ganglion cells), inner nuclear layer (cell bodies of bipolar, horizontal, amacrine, and MllMuller cells), outer plexiform layer (synapses between photoreceptors and bipolar cells), outer nuclear layer (cell bodies of rod and cone photoreceptors), external limiting membrane, and the photoreceptor layer. The retinal pigment epithelium (RPE) is a single layer of pigmented cells beneath the photoreceptors that is essential for photoreceptor health and function.

Retinitis Pigmentosa

– A group of inherited retinal dystrophies characterized by progressive degeneration of rod and cone photoreceptors, leading to visual field loss and eventual blindness. Prevalence: 1 in 4,000. Genetics: autosomal dominant, autosomal recessive, X-linked, and mitochondrial inheritance; >60 genes implicated (RHO, RPGR, RP2, USH2A, PDE6B, ABCA4). Pathophysiology: rod photoreceptors degenerate first, followed by cones. Characteristic fundus findings: 'bone-spicule' pigmentary deposits in the midperipheral retina (perivascular), waxy pallor of the optic disc, and attenuated retinal arterioles. Clinical: night blindness (nyctalopia—earliest symptom, often in childhood), progressive constriction of the visual field (tubular vision—preserves central vision until late stages), and eventually loss of central vision (legal blindness by age 40–60). Associated systemic conditions: Usher syndrome (RP plus congenital sensorineural hearing loss), Bardet–Biedl syndrome (RP, obesity, polydactyly, intellectual disability, renal abnormalities). Diagnosis: electroretinography (ERG—reduced or extinguished rod and cone responses), visual field testing (Goldmann perimetry—constricted fields), optical coherence tomography (OCT—thinning of outer retinal layers), and genetic testing. Treatment: no cure. Management: low vision aids, vitamin A palmitate (15,000 IU/day—may slow progression in some forms, controversial), and avoidance of bright light. Gene therapy: voretigene neparvovec (Luxturna) approved for RPE65-related RP (restores visual function). Retinal prostheses (Argus II—epiretinal implant) for end-stage disease. Research: stem cell therapy, optogenetics. Prognosis: progressive, most patients are legally blind by age 60.

Rhegmatogenous Retinal Detachment

– The most common type of retinal detachment, caused by a full-thickness break or tear in the retina that allows liquefied vitreous humor to enter the subretinal space, separating the neurosensory retina from the underlying retinal pigment epithelium. The condition typically occurs in association with posterior vitreous detachment, where vitreoretinal traction at sites of abnormal adhesion causes retinal tears, most commonly at the equator of the eye. Patients typically present with sudden onset of unilateral, painless, progressive visual field loss, often described as a curtain or shade over their vision, with or without preceding photopsias and floaters. On examination, there is a relative afferent pupillary defect, decreased visual acuity if the macula is detached, and the neurosensory retina is seen to be elevated and billowing on fundoscopy.

Slit Lamp Biomicroscopy

– A fundamental ophthalmological examination technique that uses a high-intensity, adjustable slit beam of light combined with a binocular microscope to provide a magnified, stereoscopic view of the anterior segment and, with the use of auxiliary lenses, the posterior segment of the eye. The slit beam can be adjusted in width, height, and angle, allowing the examiner to create an optical section through the transparent ocular media and visualize individual corneal layersand structures in fine detail, including the corneal epithelium, Bowman layer, stroma, Descemet membrane, and endothelium, as well as allowing the detection of cells and flare in the anterior chamber, cataract in the lens, and vitreous cells. The slit lamp is also used for Goldmann applanation tonometry and for performing diagnostic and therapeutic procedures.

SMILE

– Small incision lenticule extraction, commonly known as SMILE, is a minimally invasive refractive surgical procedure that corrects myopia and myopic astigmatism by creating and removing a thin, disc-shaped piece of corneal stroma called a lenticule through a small incision, without creating a corneal flap. The procedure uses a femtosecond laser to create two intrastromal cuts: the posterior surface of the lenticule and the anterior cap, and a small peripheral incision approximately 2 to 4 millimemillimetres in length creating a side-entry channel for extraction of the lenticule. SMILE offers advantages over LASIK, including the absence of a corneal flap, preservation of corneal biomechanical stability, and a lower risk of dry eye syndrome.

Squint

– See Strabismus.

Strabismus

– A misalignment of the visual axes of the two eyes, where one eye is deviated inward (esotropia), outward (exotropia), upward (hypertropia), or downward (hypotropia) relative to the other, disrupting binocular vision and potentially leading to amblyopia in children. The condition can be constant or intermittent, unilateral or alternating, and it may be comitant (the angle of deviation is the same in all directions of gaze) or incomitant (the angle varies depending on the direction odirection of gaze, indicating a paretic or restrictive cause). The diagnosis is made by the cover-uncover test and the alternate cover test, which detect manifest or latent strabismus and measure the angle of deviation in prism dioptres. Additional tests include measurement of visual acuity, evaluation of ductions and versions, and the Hess chart.

Stye

– A common infection of the eyelid. External hordeolum: acute bacterial infection of the Zeis or Moll glands at the eyelash follicle, typically caused by Staphylococcus aureus. Presents as a tender, red, well-localized swelling at the lid margin. Internal hordeolum: infection of the meibomian gland, larger and more painful than external. Treatment: warm compresses 3-4 times daily, lid hygiene. Incision and drainage for persistent cases. Avoid squeezing to prevent spread of infection.

Trabeculectomy

– A conventional glaucoma filtration surgery that creates a new drainage pathway for aqueous humor by removing a small piece of the trabecular meshwork and creating a partial-thickness scleral flap under which aqueous humor filters from the anterior chamber into the subconjunctival space, forming a filtering bleb. The procedure is indicated for glaucoma that is uncontrolled with maximal medical and laser therapy, and it achieves intraocular pressure reduction to target levels in target levels in up to 80 to 90 percent of patients at 2 years. The procedure involves creating a conjunctival flap, a partial-thickness scleral flap, excising a trabecular and deep scleral block, and creating a peripheral iridectomy. The scleral flap is then sutured to control the rate of aqueous filtration.

Uvea

– The middle vascular coat of the eye, consisting of three parts from anterior to posterior: the iris (the coloured part of the eye, controlling pupil size), the ciliary body (producing aqueous humour and controlling lens shape via the zonular fibers), and the choroid (the vascular layer supplying the outer retina, between the sclera and the retinal pigment epithelium). The uvea is the site of inflammation in uveitis. Uveal melanoma is the most common primary intraocular malignancy in adults, arising from melanocytes in the choroid, ciliary body, or iris. Benign uveal tumors include naevi.

Uveitis

– An inflammatory condition of the uveal tract, which includes the iris, ciliary body, and choroid, and it is classified anatomically based on the location of the inflammation: anterior uveitis (iritis and iridocyclitis), intermediate uveitis (pars planitis and periphlebitis), posterior uveitis (choroiditis and chorioretinitis), and panuveitis (involvement of all three segments). Anterior uveitis is the most common form, presenting with eye pain, photophobia, blurred vision, and rednessredness with ciliary flush. The diagnosis is confirmed by slit-lamp examination revealing cells and flare in the anterior chamber, keratic precipitates, and synechiae. Treatment depends on location and severity, with topical corticosteroids for anterior uveitis and systemic immunosuppression for posterior and panuveitis.

Viral Conjunctivitis

– The most common form of infectious conjunctivitis, caused predominantly by adenoviruses (serotypes 3, 4, 7, 8, 19, and 37), and it is highly contagious, spreading through direct contact with infected ocular secretions or contaminated surfaces. The condition typically begins unilaterally and spreads to the contralateral eye within one to two days, presenting with watery discharge, foreign body sensation, tearing, and conjunctival injection, often accompanied by preauriculapreauricular lymphadenopathy in 30 to 50 percent of cases. The diagnosis is clinical, and treatment is largely supportive (cool compresses, lubricating drops, and careful hand hygiene to prevent spread), as the disease is self-limited and resolves within 2 to 3 weeks.

Vision

– The sensory ability to perceive light and interpret visual information processed by the eyes and brain, enabling recognition of shapes, colors, depth, and motion. Normal vision depends on the proper function of the cornea, lens, retina, and optic nerve, as well as intact neural pathways to the visual cortex. Age-related changes such as presbyopia and cataracts commonly affect vision, while conditions like glaucoma, macular degeneration, and diabetic retinopathy can cause permanent vision loss if not detected and treated early.

Vision and Hearing

– The two primary sensory modalities through which humans perceive and interact with their environment. Age-related declines in both senses are common, with presbyopia and cataracts affecting vision and presbycusis affecting hearing, often beginning as early as the fourth decade of life. Regular sensory screening is important for early detection of impairments, and corrective measures including eyeglasses, contact lenses, hearing aids, and cochlear implants can substantially improve quality of life and reduce the risk of falls, social isolation, and cognitive decline.

Vision Loss

– A decrease in the ability to see that cannot be fully corrected with eyeglasses, contact lenses, or medical treatment. It ranges from partial impairment to complete blindness and can result from conditions such as macular degeneration, glaucoma, diabetic retinopathy, cataracts, and traumatic injury. Low vision rehabilitation services, assistive technologies including magnifiers and screen readers, and environmental modifications help individuals with vision loss maintain independence and quality of life.

Visual Field Defects

– Areas of reduced or absent vision within the visual field that correspond to damage at specific points along the visual pathway, and their pattern provides crucial localizing information for diagnosis. A scotoma is an area of diminished or absent vision surrounded by normal vision, with the most common types being central scotomas (from macular disease or optic neuropathy) and paracentral scotomas (from glaucoma or retinal disease). Homonymous hemianopia is loss of the sthe same side of the visual field in both eyes, which localizes to the optic tract or optic radiation contralateral to the side of the field loss, most commonly caused by stroke. Bitemporal hemianopia is loss of the temporal half of the visual field in both eyes, which localizes to the optic chiasm, most commonly caused by a pituitary adenoma.

Visual Field Perimetry

– A systematic method of measuring the sensitivity of the retina across the entire visual field using standardized light stimuli, providing a quantitative map of visual function that can detect, characterize, and monitor visual field loss caused by diseases of the retina, optic nerve, and visual pathways. Automated static perimetry, most commonly performed using the Humphrey Field Analyzer, presents points of light at varying intensities throughout the visual field and rereports the threshold sensitivity at each tested point. The Humphrey 24-2 and 30-2 protocols are the most commonly used. The results are displayed as a gray-scale map, a total deviation probability map, and a pattern deviation probability map.

Vitreous Body

– See Vitreous Humor.

Vitreous Humor

– The vitreous humor, or vitreous body, is a transparent, gel-like substance that fills the space between the lens and the retina, occupying approximately eighty percent of the volume of the eye, and it consists primarily of water (approximately ninety-nine percent) with a network of collagen fibrils and hyaluronic acid that give it its gel-like consistency.

The vitreous is attached to the retina at several sites, including the vitreous base anteriorly (the strongest attachment), the optic disc mamargin (the ring of attachment around the optic disc). The vitreous is subject to age-related changes, with liquefaction (synchysis senilis) and posterior vitreous detachment, which can be asymptomatic or cause flashes and floaters.

Xerophthalmia

– A medical condition of progressive drying of the conjunctiva and cornea caused by vitamin A deficiency, one of the leading causes of preventable blindness worldwide, particularly in developing countries where vitamin A deficiency is endemic. Xerophthalmia encompasses a spectrum of ocular manifestations: night blindness (nyctalopia, earliest symptom, due to impaired rhodopsin regeneration in rod photoreceptors), conjunctival xerosis (dry, lustreless, thickened conjunctiva with loss of transparency and wrinkling on movement), Bitot spots (superficial, foamy, white, cheesy patches on the bulbar conjunctiva, typically temporal, composed of desquamated epithelial cells and keratin debris, pathognomonic but not always present), corneal xerosis (dry, hazy, dull cornea with loss of corneal lustre), corneal ulceration and keratomalacia (liquefactive necrosis of the cornea, the most severe form, leading to corneal perforation, prolapse of intraocular contents, and permanent blindness). Diagnosis: clinical (WHO classification: XN = night blindness, X1A = conjunctival xerosis, X1B = Bitot spots, X2 = corneal xerosis, X3A = corneal ulceration <1/3 corneal surface, X3B = keratomalacia), serum retinol <0.35 micromol/L. Treatment: high-dose vitamin A supplementation (WHO regimen: 200,000 IU vitamin A orally immediately, repeated the next day and at least 2 weeks later for severe xerophthalmia; children under 12 months: 50,000 IU; 12-36 months: 100,000 IU). Prevention: vitamin A supplementation (in measles-endemic areas, postpartum women), dietary diversification, and food fortification.

YAG Capsulotomy

– YAG laser posterior capsulotomy is a minimally invasive laser procedure used to treat posterior capsule opacification, the most common late complication of cataract surgery, in which the posterior lens capsule becomes opacified by proliferation of residual lens epithelial cells, causing decreased vision. The procedure uses a neodymium:yttrium-aluminum-garnet laser to create a central opening in the opacified posterior capsule, allowing light to pass unobstructed to the retina and restoring clear vision, often within 24 to 48 hours. The procedure is performed on an outpatient basis at the slit lamp, with the pupil dilated and topical anesthetic applied. The laser creates an opening in the posterior capsule typically measuring 3 to 5 mm in diameter, with a total of 20 to 50 laser applications of 1 to 5 mJ each.

Chapter 24

Orthopedics

Achilles Tendinopathy

– A spectrum of conditions affecting the Achilles tendon, including tendinitis (acute inflammation) and tendinosis (chronic degeneration), most commonly occurring in the midsubstance 2-6 cm above the insertion on the calcaneus (the watershed area with reduced blood supply). Risk factors include overuse (running, jumping sports), sudden increase in activity, tight calf muscles, poor biomechanics, fluoroquinolone antibiotics, corticosteroid use, and advanced age. Pathology: degenerative changes in tthe tendon, including collagen disorganisation, mucoid degeneration, neovascularization, and tenocyte apoptosis (tendinosis), with little to no inflammation. The condition presents as gradual onset of Achilles tendon pain and stiffness with activity, and the tendon may be diffusely or focally thickened.

Achilles Tendonitis

– Inflammation and degeneration of the Achilles tendon, most common in runners and middle-aged athletes. Insertional: at the calcaneal insertion, associated with bone spurs and Haglund deformity. Non-insertional (mid-portion): 2-6 cm above insertion, due to repetitive microtrauma and tendinosis. Presents with posterior heel pain and morning stiffness, improving with activity and worsening with activity (later stages). Diagnosis: clinical; ultrasound/MRI for confirmation. Treatment: eccentric calf stretching (most effective), activity modification, heel lifts, NSAIDs, and shockwave therapy. Refractory cases: PRP, tenotomy, or surgical debridement.

Achilles Tendon Rupture

– Complete tear of the Achilles tendon, most common in middle-aged men during recreational sports (weekend warrior). Mechanism: sudden push-off with knee extended. Presents with hearing/feeling a pop, acute posterior ankle pain, inability to stand on tiptoes, and palpable gap in the tendon. Positive Thompson test (no plantarflexion with calf squeeze). Diagnosis: clinical; MRI/ultrasound for confirmation. Treatment: non-operative (functional bracing in equinus with early weight-bearing, complication: rerupture 5-10%) vs. operative repair (open vs. percutaneous, complication: wound healing, infection 2-5%). Both have similar long-term outcomes.

ACL Injuries

– Anterior cruciate ligament (ACL) tears: non-contact (pivot, deceleration) > contact. Clinical: Lachman (most sensitive), pivot shift (most specific), anterior drawer. MRI confirms (bone bruise lateral femoral condyle + lateral tibial plateau, ACL discontinuity). Treatment: nonoperative (rehab, bracing) for sedentary, low-demand; ACL reconstruction (autograft: BTB - gold standard, hamstring, quadriceps tendon; allograft: higher failure in young active) for active, instability. Timing: acute (<3 wacute (<3 weeks): nonoperative vs operative (controversial, no difference in outcomes); subacute to chronic: reconstruction. Post-op: brace locked 0-90, NWB 6 weeks, WBAT 6-12 weeks, return to sports 6-9 months.

ACL Reconstruction

– Surgical reconstruction of the anterior cruciate ligament (ACL) following rupture, one of the most common knee ligament injuries. The ACL is a primary restraint to anterior tibial translation and rotatory instability, and its rupture frequently occurs during non-contact pivoting sports activities (skiing, soccer, basketball). Risk factors include female sex (2-8 times higher incidence than males, attributed to hormonal, anatomic, and neuromuscular factors), family history, participation in highhigh-risk sports with cutting, pivoting, and jumping. The surgical technique involves harvesting an autograft (bone-patellar tendon-bone or hamstring tendon graft are the most common) or using an allograft for multi-ligament knee injuries.

Acromioclavicular Joint Injuries

– AC joint injuries: Rockwood classification: I (sprain AC ligaments), II (AC ligaments torn, CC intact), III (AC+CC torn, 25-100% displacement), IV (posterior displacement into trapezius), V (superior displacement >100%), VI (inferior displacement subcoracoid/subacromial). Treatment: I/II: nonoperative (sling, PT); III: controversial (nonoperative for most, surgical for high-demand overhead athletes, laborers); IV-VI: surgical (anatomic CC reconstruction: allograft/autograft, synthetic tape, sutusutures, cerclage wires). Complications: AC joint arthritis, distal clavicle osteolysis, coracoclavicular ossification, fixation failure, and infection.

Adhesive Capsulitis

– A condition of progressive shoulder stiffness and pain caused by inflammation and fibrosis of the glenohumeral joint capsule. Common name: frozen shoulder. Stages: freezing (painful, 2-9 months), frozen (stiff, 4-12 months), thawing (gradual improvement, 5-24 months). Most common ages 40-60. Risk factors: diabetes (strongest), thyroid disease, female sex, and prior shoulder trauma/surgery. Bilateral in up to 40% of diabetics. Diagnosis: clinical (restricted active and passive ROM in all planes, especially external rotation). Treatment: physical therapy, NSAIDs, intra-articular corticosteroid injection, hydrodilatation (capsular distension), and arthroscopic capsular release for refractory cases.

Adult Spinal Deformity

– Adult spinal deformity (ASD): scoliosis (coronal Cobb >10 deg), kyphosis (sagittal), sagittal imbalance. SRS-Schwab classification: PI-LL mismatch, SVA (sagittal vertical axis), PT (pelvic tilt). Clinical: back pain, radiculopathy, neurogenic claudication, disability (ODI, SRS-22). Imaging: full-length standing AP/lateral, bending, CT for bone quality. Treatment: nonoperative (PT, core, epidural injections); operative: decompression + fusion (PSF, ALIF, TLIF, XLIF, OLIF, anterior column realignmanterior column realignment, three-column osteotomy, PSO). Complications: pseudarthrosis, implant failure, proximal junctional kyphosis (PJK), infection, neurologic deficit. Outcomes: reliable pain relief, functional improvement, but high complication rates (30-40 percent).

Amelia

– A rare congenital anomaly characterized by complete absence of one or more limbs. Unilateral upper limb amelia is the most common form. Causes include vascular disruption, amnion rupture sequence, and teratogens. May be isolated or syndromic (Roberts syndrome, ADAM sequence, amniotic band syndrome). Diagnosis by prenatal ultrasound. Management: prosthetic fitting as early as 6 months of age to promote functional adaptation and normal development. Multidisciplinary care includes physical and occupational therapy. Children with amelia can achieve high levels of independence with adaptive technology.

Ankle Arthroplasty

– Total ankle arthroplasty (TAA): indications: end-stage ankle OA (post-traumatic, primary, inflammatory), preserved hindfoot alignment (<10-15 deg varus/valgus), no severe osteoporosis, no active infection. Contraindications: severe deformity, Charcot, prior infection, young high-demand. Implants: 3-component (mobile bearing: STAR, Salto, Hintegra) vs 2-component (fixed bearing: Infinity, Vantage). Approaches: anterior (standard), anterolateral. Alignment: neutral mechanical axis. Complications: Complications: infection, wound healing problems, deep vein thrombosis, aseptic loosening, impingement, gutter pain, subsidence, and polyethylene wear.

Ankle Fractures

– Ankle fractures: Weber classification (fibular fracture relative to syndesmosis): A (below syndesmosis, stable), B (at syndesmosis, variable stability), C (above syndesmosis, unstable). Lauge-Hansen: supination-external rotation (SER, most common), supination-adduction, pronation-external rotation, pronation-abduction. Treatment: stable (Weber A, SER stage I/II): closed reduction, short leg cast/boot, NWB 6 weeks; unstable (Weber B/C, SER III/IV): ORIF (lateral plate for fibula, medial screws/temedial screws/tension band for medial malleolus), syndesmotic screw fixation (if unstable, 3.5-4.5 mm quadricortical screw 2-3 cm above tibiotalar joint, weightbearing at 8 weeks with screw removal at 3-4 months; alternatively, suture button fixation). Post-op: splint, NWB 6-8 weeks, then progressive weightbearing. Complications: wound breakdown, infection, malunion (most common), nonunion, post-traumatic osteoarthritis (most common late complication).

Ankle Sprain

– An injury to the ligaments surrounding the ankle joint, most commonly involving the lateral ligament complex (anterior talofibular ligament [ATFL, most commonly injured], calcaneofibular ligament [CFL], and posterior talofibular ligament [PTFL]) from an inversion mechanism. Medial (deltoid) ligament sprains occur less commonly from eversion injuries and syndesmotic (high ankle) sprains from external rotation. Classification: Grade I (mild, microscopic ligament fibers, minimal sweswelling and ecchymosis with a stable ankle joint), Grade II (moderate, partial tearing of ligament fibers, with moderate swelling, ecchymosis, and joint laxity with a firm end point), and Grade III (severe, complete ligament rupture, with profound swelling, ecchymosis, and joint instability with no firm end point).

Ankylosing Spondylitis

– A chronic, progressive inflammatory disease primarily affecting the axial skeleton (sacroiliac joints and spine), characterized by inflammation at entheses (where tendons, ligaments, and joint capsules insert into bone), leading to new bone formation, syndesmophyte development, and eventual vertebral fusion (bamboo spine). It is the prototype of the spondyloarthritis family of diseases and is strongly associated with HLA-B27 (present in 80-90 percent of patients, though HLA-B27 is present in 6-86-8 percent of the general population, with the risk of developing AS in HLA-B27-positive individuals being 1 to 2 percent per year. The disease typically begins in early adulthood with insidious onset of low back pain and stiffness that is worse in the morning and with rest, relieved by activity.

Anterior Cruciate Ligament Injury

– Tear of the ACL, a common sports injury (noncontact deceleration, pivoting, or hyperextension). Presents with a pop, immediate swelling (hemarthrosis), and giving way of the knee. Positive Lachman test (most sensitive), anterior drawer test, and pivot shift test. Associated injuries: MCL (O'Donoghue triad: ACL, MCL, medial meniscus), meniscal tears. Diagnosis: MRI. Treatment: functional rehab for low-demand patients; ACL reconstruction (hamstring autograft, patellar tendon autograft, quadriceps tendon, or allograft) for high-demand athletes or those with persistent instability.

Arthrogryposis

– A group of congenital disorders characterized by multiple joint contractures present at birth, caused by decreased fetal movement (fetal akinesia) during development. Etiology includes neuropathic (most common, spinal anterior horn cell degeneration), myopathic (congenital muscular dystrophies), and mechanical (oligohydramnios, uterine abnormalities). Amyoplasia is the most common type (1 in 3,000 births): symmetric limb involvement, no internal organ involvement, and normal cognition. Physical exam shows fixed joint contractures (clubfeet, flexed wrists, extended knees), smooth, shiny skin over affected joints, and decreased muscle bulk. Joints are typically rigid with no skin creases. Treatment: aggressive physical therapy, serial casting, and orthopedic surgery (tendon releases, osteotomies) to improve function. Most children achieve ambulation with assistive devices. Intelligence is typically normal.

Articular Cartilage Injuries

– Chondral/osteochondral lesions: Outerbridge grading (I: softening, II: fibrillation <50%, III: fibrillation >50%, IV: subchondral bone exposed). OCD (osteochondritis dissecans): subchondral bone necrosis, fragment separation (knee: lateral femoral condyle, elbow: capitellum, ankle: talar dome). Treatment: microfracture (marrow stimulation, fibrocartilage, <2-3cm2), OATS (osteochondral autograft transfer, mosaicplasty), allograft transplantation (large defects), MACI (matrix-induced autologous chchondrocyte implantation). Treatment: repair if possible (young, active, large fragment) vs excision (small, loose).

Backache

– Pain or discomfort located anywhere in the posterior trunk from neck to buttocks, a symptom rather than a disease. Causes include musculoskeletal strain (most common), disc herniation, spinal stenosis, facet arthropathy, sacroiliac joint dysfunction, and referred pain from renal, pancreatic, or vascular sources. Acute backache often resolves with conservative care; chronic backache benefits from exercise, physical therapy, and multidisciplinary pain management.

Back Pain

– One of the most common medical complaints, affecting up to 80 percent of the population at some point in life, and the leading cause of disability worldwide. Back pain is classified by duration: acute (less than 4 weeks), subacute (4-12 weeks), and chronic (greater than 12 weeks). It is further divided into mechanical (non-specific) back pain (90 percent of cases, caused by muscle strain, ligament sprain, degenerative disc disease, facet joint arthropathy), radicular pain (from nerve root comprecompression (from herniated disc, spinal stenosis, or foraminal stenosis), and referred pain (from conditions such as pancreatitis, renal colic, aortic aneurysm). The diagnosis is clinical, based on history and physical examination. Imaging is indicated when red flags are present.

Baker Cyst

– A fluid-filled cyst in the popliteal fossa, resulting from herniation of the synovial membrane through the posterior knee joint capsule or from a communication between the knee joint and the gastrocnemius-semimembranosus bursa. A Baker cyst is not a true cyst but rather an effusion of the gastrocnemius-semimembranosus bursa that communicates with the knee joint, containing synovial fluid. It occurs in association with knee joint effusion from any underlying cause, most commonly osteoarthritis, rrheumatoid arthritis, or trauma, and it presents as a palpable, firm, mildly tender mass in the popliteal fossa that is more prominent with knee extension (Foucher sign). Diagnosis is confirmed by ultrasound or MRI.

Bone Metastases

– Bone metastases: most common from breast, prostate, lung, kidney, thyroid. Solitary vs multiple. Clinical: pain (worse at night), pathologic fracture, hypercalcemia, spinal cord compression. Mirels score for impending pathologic fracture: site (upper limb 1, lower limb 2, peritrochanteric 3), pain (mild 1, moderate 2, functional 3), lesion (blastic 1, mixed 2, lytic 3), size (<1/3 1, 1/3-2/3 2, >2/3 3). Score >=9: prophylactic fixation. Spinal metastases: Bilsky epidural spinal cord compression compression (Bilsky grade 0-3). Grade 1: external beam radiotherapy + bisphosphonates/denosumab; grade 2/3: surgical decompression plus stabilisation followed by radiotherapy. Pathologic fracture: surgical fixation or arthroplasty.

Bones and Joints

– Bones and joints form the skeletal system, providing structural support, protection of organs, and enabling movement through articulations between bones. Joints are classified as fibrous, cartilaginous, or synovial, with synovial joints offering the greatest range of motion. Common conditions include osteoarthritis, rheumatoid arthritis, fractures, and osteoporosis.

Brachial Plexus Injuries

– Brachial plexus injuries: traction (motorcycle accidents, sports), compression (crutches, thoracic outlet). Classification: supraclavicular (roots/trunks), infraclavicular (divisions/cords). Narakas classification: I (proximal C5-C6, Erb's palsy: waiter's tip), II (C5-C7), III (C5-T1), IV (complete C5-T1), V (complete + root avulsions). Root avulsion: pseudomeningocele on MRI myelography, Horner syndrome (T1 avulsion), SER test (sensory evoked potentials). Treatment: early (3-6 months): neurolysneurolysis vs reconstruction (sural nerve graft, nerve transfer). Late (>6 months): tendon transfer, free muscle transfer, arthrodesis. Prognosis: better for lower trunk/infraclavicular, worse for upper trunk/supraclavicular.

Bunion

– A bony prominence (exostosis) at the medial aspect of the first metatarsal head, causing lateral deviation of the great toe (hallux valgus). The deformity is characterized by a structural abnormality at the first metatarsophalangeal joint, with the first metatarsal deviating medially and the proximal phalanx deviating laterally. The bunion itself is the medial eminence of the first metatarsal head, which becomes prominent and painful due to chronic pressure from footwear. Risk factors include nanarrow-toed shoes, high heels, female sex (10:1), family history, and inflammatory arthritis. Diagnosis is made on physical examination with weight-bearing radiographs confirming the hallux valgus angle and intermetatarsal angle.

Bursitis

– Inflammation of a bursa, a fluid-filled sac that reduces friction between tendons, muscles, bones, and skin. Bursae are lined by synovial cells and normally contain a thin layer of lubricating synovial fluid. Bursitis most commonly affects the subacromial (shoulder), olecranon (elbow), trochanteric (hip), prepatellar (knee), infrapatellar, and retrocalcaneal (ankle) bursae. Causes include repetitive microtrauma or overuse (the most common mechanism), acute trauma, infection (septic bursitis, moscommonly Staphylococcus aureus (80 to 90 percent of cases), which may complicate superficial skin infections, rheumatoid arthritis, and sporting injuries, with the olecranon and prepatellar bursae being the most common sites.

Carpal Tunnel Syndrome

– Carpal tunnel syndrome: median nerve compression at wrist (carpal tunnel: transverse carpal ligament roof, carpal bones floor). Clinical: nocturnal paresthesias (thumb, index, middle, radial ring), thenar wasting late, Tinel wrist, Phalen (60 sec wrist flexion), Durkan (carpal compression). NCS/EMG: prolonged DML, DSL, CMAP amplitude. Treatment: mild/mod: night splinting (neutral), corticosteroid injection (diagnostic/therapeutic), PT; severe/failed nonop: carpal tunnel release (open, endoscopicendoscopic). Prognosis: excellent, recovery 90 percent plus.

Cervical Spine Trauma

– Cervical spine trauma: Subaxial (C3-C7) injury classification (SLIC - Subaxial Cervical Spine Injury Classification): morphology (1-4), discoligamentous complex (0-2), neurologic (0-3). Total <4 nonop, >4 op, =4 surgeon discretion. Upper cervical: occiput-C1 (occipital condyle fracture, occipitocervical dissociation), C1 (Jefferson burst fracture: lateral mass displacement >6.9mm on CT), C2 (odontoid fracture: Type I tip, Type II base - most common, nonunion risk >5mm displacement/age >70/smokismoking/diabetes); Type III (odontoid body/synchrondrosis, less than 3 percent, surgical). Treatment: nonoperative (collar, Halo vest) for stable; C1 lateral mass screws (Harms), C1-2 transarticular screws (Magerl), occipitocervical fusion for unstable.

Chondrosarcoma

– The second most common primary malignant bone tumor (after osteosarcoma) and the most common primary malignant bone tumor in adults (peak age 40-70 years). It arises from cartilage-forming cells and is classified by grade (Grade 1: low-grade/conventional; Grade 2: intermediate; Grade 3: high-grade) and by location (central/intramedullary, surface/peripheral, or secondary). Conventional chondrosarcoma typically arises in the proximal femur, pelvis, and proximal humerus. Secondarary chondrosarcoma may arise from a pre-existing osteochondroma (in multiple hereditary exostosis when the cartilage cap thickness exceeds 1.5 to 2 cm, the risk of malignant transformation is approximately 2 to 5 percent) or from Paget disease of bone. The clinical presentation is with localised pain, a palpable mass, and pathological fracture in approximately 10 to 15 percent of cases.

Clavicle Fractures

– Clavicle fractures: 80% midshaft (Group I), 15% lateral (Group II: Neer I nondisplaced, II displaced with CC ligament disruption, III intra-articular), 5% medial (Group III). Midshaft: nondisplaced/shortening <2cm: sling, figure-8; displaced/comminuted/shortening >2cm/skin tenting: ORIF (superior plate, anteroinferior plate). Lateral: Neer I: nonoperative; Neer II: ORIF (hook plate, anatomic plate, CC screw + plate); Neur III: ORIF. Medial: rare, check for mediastinal injury. Complications: nonunonunion (5-15 percent), malunion, hardware irritation, shoulder stiffness.

Clubfoot (Talipes Equinovarus)

– A congenital foot deformity with an incidence of approximately 1 in 1,000 live births, characterized by four components: equinus (ankle plantarflexion), varus (hindfoot inversion), adductus (forefoot adduction), and cavus (high arch). Clubfoot is classified as idiopathic (most common, no identifiable cause), postural (positional, passively correctable), or syndromic (associated with neuromuscular conditions like myelomeningocele, arthrogryposis, or chromosomal abnormalities). The male-to-female male-to-female ratio is 2:1. The Ponseti method is the gold standard for treatment, involving serial casting, Achilles tenotomy, and bracing.

Compartment Syndrome

– A surgical emergency caused by increased pressure within a closed osteofascial compartment, leading to tissue ischemia, necrosis, and permanent functional loss. The lower leg (anterior, lateral, deep posterior, and superficial posterior compartments) and forearm (flexor and extensor compartments) are the most commonly affected sites. Causes include tibial fractures (most common cause of lower leg compartment syndrome), forearm fractures (Colles, Smith, radial neck), crushcrush injuries, vascular injuries, burns, limb compression, and reperfusion injury after revascularisation. The earliest and most important symptom is severe, unremitting pain out of proportion to the injury, worse with passive stretching. Diagnosis is clinical, with compartment pressure measurement confirming the diagnosis.

Congenital Scoliosis

– A lateral curvature of the spine caused by congenital vertebral anomalies that arise during embryogenesis (failure of formation or segmentation). Failure of formation: hemivertebra (wedge-shaped vertebra causing progressive curvature). Failure of segmentation: unilateral unsegmented bar (block of bone on one side). Mixed: combination of hemivertebra and unsegmented bar (most progressive). Presents with a progressive spinal curvature noted in infancy or early childhood. Associated with intraspinal anomalies (tethered cord, syringomyelia, diastematomyelia in 20-30%), renal anomalies (10-20%), and cardiac defects (10-15%). Diagnosis: spine X-ray and MRI (to evaluate intraspinal anatomy) and renal ultrasound. Treatment: observation for mild curves; bracing is generally ineffective for congenital scoliosis. Surgical correction: in situ fusion, hemivertebra excision, or growth-friendly techniques (growing rods, VEPTR) for young children with progressive curves.

Cubital Tunnel Syndrome

– Cubital tunnel syndrome: ulnar nerve compression at elbow (cubital tunnel: Osborne's ligament, FCU aponeurosis, medial epicondyle, olecranon). Clinical: paresthesias ulnar 1.5 digits, nocturnal, Tinel at elbow, Froment sign (IP flexion of thumb with key pinch), Wartenberg sign (abduction of small finger), thenar/hypothenar wasting late. McGowan classification: I (intermittent), II (constant sensory, no motor), III (motor weakness). NCS/EMG: slowing across elbow. Treatment: I/II: nonoperative (ninight orthosis (elbow splint at 45-60 degrees of flexion), activity modification, ergonomics; Grade III or failed nonop: cubital tunnel release (in situ decompression, medial epicondylectomy, or transposition).

Degenerative Spine Disease

– Lumbar spinal stenosis: central (neurogenic claudication: bilateral leg pain with walking, relieved by sitting/flexion), lateral recess, foraminal. Imaging: MRI (central canal AP diameter <10mm, cross-sectional area <70mm2). Treatment: nonoperative (PT, epidural steroid, prostaglandin); operative: laminectomy (open, minimally invasive, tubular), decompression + fusion if instability (spondylolisthesis, scoliosis). Herniated nucleus pulposus: posterolateral (radiculopathy), central (cauda equina:cauda equina syndrome: surgical emergency; treated with urgent decompression.

De Quervain Tenosynovitis

– De Quervain tenosynovitis: first dorsal compartment (APL, EPB) stenosis. Clinical: radial wrist pain, Finkelstein test (thumb in palm, ulnar deviation), swelling over radial styloid. Treatment: splint (thumb spica), NSAIDs, corticosteroid injection (80-90% cure); surgical: first dorsal compartment release (identify radial sensory nerve branches, watch for subcompartments).

Developmental Dysplasia of the Hip

– A spectrum of hip abnormalities in infants and children, ranging from mild acetabular dysplasia to complete hip dislocation. The femoral head is partially or completely displaced from the acetabulum due to capsular laxity and acetabular underdevelopment. Risk factors: female sex (4:1), breech presentation, family history, first-born, and oligohydramnios. Screening: clinical examination at every well-child visit (Ortolani test for reduction, Barlow test for dislocation, Galeazzi sign, asymmetric thigh/gluteal folds). Imaging: hip ultrasound for infants <4-6 months (dynamic and static views); X-ray for older infants and children (Hilgenreiner line, Perkin line, acetabular index). Treatment: Pavlik harness for infants <6 months (success rate 85-95%); closed reduction and spica casting for harness failure or older infants; open reduction and pelvic/femoral osteotomy for children >18 months or failed closed reduction. Complications: avascular necrosis of the femoral head, redislocation, and early osteoarthritis.

Diskectomy

– The surgical removal of herniated intervertebral disk material that compresses nerve roots or the spinal cord, causing radiculopathy or myelopathy. Indicated in severe disk herniation causing unremitting pain, progressive motor weakness, or cauda equina syndrome.

Dislocation

– Complete displacement of the articular surfaces of a joint, resulting in loss of normal bony alignment and often associated with ligamentous, capsular, and neurovascular injury. The most commonly dislocated joints are the glenohumeral shoulder joint (accounting for 50 percent of all dislocations, anterior in 95 percent of cases), followed by the interphalangeal joints of the fingers, patellofemoral joint, elbow, and hip. The mechanism typically involves traumatic force applied to the joint beyonbeyond the normal range of motion, usually with significant force, causing capsular and ligamentous tearing, and often associated fractures, nerve injuries (axillary nerve in anterior shoulder dislocation, radial nerve in posterior shoulder dislocation), vascular injuries, and soft tissue injuries. Treatment involves urgent reduction followed by immobilization and rehabilitation.

Distal Radius Fractures

– The most common upper extremity fracture. Classification: AO/OTA: A (extra-articular), B (partial articular), C (complete articular); Frykman. Colles fracture: dorsal angulation/displacement (fall on outstretched hand). Smith fracture: volar angulation (fall on flexed wrist). Barton fracture: intra-articular fracture-dislocation (dorsal or volar). Chauffeur fracture: radial styloid. Treatment: nondisplaced/extra-articular: closed reduction + cast/splint; displaced intdisplaced intra-articular: ORIF (volar locking plate, dorsal plate, external fixation, fragment-specific fixation). Post-op: splint, early motion (for ORIF). Complications: malunion, nonunion (rare), carpal tunnel syndrome, post-traumatic arthritis, EPL rupture (3-5 percent), and complex regional pain syndrome (CRPS type I).

Dupuytren contracture

– A benign, progressive fibroproliferative disorder of the palmar fascia causing nodules, cords, and contractures of the fingers, most commonly affecting the ring and small fingers. The condition is caused by proliferation of myofibroblasts and deposition of type III collagen in the palmar aponeurosis, leading to formation of pathologic cords that contract and flex the metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints. Prevalence is highest in individuals of Northern European desNorthern European descent (up to 30 percent in some Scandinavian populations), with men affected 3 to 4 times more than women. The condition is associated with diabetes mellitus, epilepsy, alcoholism, and manual labour. The diagnosis is clinical.

Ectrodactyly

– A congenital limb malformation characterized by a deep median cleft of the hand or foot with missing central digits, resulting in a claw-like or lobster-claw appearance. Also known as split hand/foot malformation (SHFM). Most cases are autosomal dominant with incomplete penetrance, linked to mutations in TP63, DLX5, or WNT10B genes. May be isolated or syndromic (EEC syndrome — ectrodactyly, ectodermal dysplasia, cleft lip/palate; Split hand/foot malformation with long bone deficiency). Physical exam shows variable absence of the central rays (second, third, and/or fourth digits), ranging from a mild cleft to complete absence of central structures. Treatment: surgical reconstruction to improve pinch and grip function; prosthetics for severe cases. Genetic counseling is recommended.

Ewing Sarcoma

– A highly malignant small round blue cell tumor of bone and soft tissue, representing the second most common primary bone malignancy in children and adolescents (peak age 10-20 years), with a strong predilection for males and Caucasian populations. It arises from the EWSR1-FLI1 fusion gene (t(11;22)(q24;q12)) present in approximately 85 percent of cases, with other EWSR1 translocations (EWSR1-ERG, EWSR1-WT1) in the remainder. Unlike osteosarcoma, Ewing sarcoma most commonly involinvolves the diaphysis of long bones (femur, tibia, humerus) and the pelvis, with a predilection for the femur (25 percent), pelvis (20 percent), and tibia (15 percent). The typical presentation is painful swelling, often with fever, warmth, and erythema. Diagnosis is by biopsy and molecular genetics showing the EWSR1-FLI1 fusion transcript.

Femoral Shaft Fractures

– Femoral shaft fractures: high energy (MVC, fall from height) or low energy (elderly, pathologic). Winquist-Hansen classification of comminution: I (none), II (butterfly <50% width), III (butterfly >50%), IV (segmental). Treatment: antegrade intramedullary nailing (gold standard, reamed, statically locked), retrograde nailing (pregnant, bilateral, ipsilateral tibia fracture, morbid obesity), plate fixation (distal metaphyseal, nonunion). Early fixation (<24h) reduces ARDS, MODS, mortality (EPIC).(EPIC, early versus delayed stabilisation of the femur in multiply injured patients, with damage control orthopedics involving initial temporary external fixation followed by delayed definitive nailing at 48 to 72 hours, being the current standard of care for the polytrauma patient).

Fracture

– A break in the continuity of bone, classified by location, pattern (transverse, oblique, spiral, comminuted), displacement, and whether the fracture communicates with the external environment (open vs. closed). Fracture healing occurs through inflammatory, reparative, and remodeling phases. Management: reduction, immobilization (cast, splint, or surgical fixation with plates/screws/intramedullary nails), and rehabilitation. Open fractures require urgent debridement and antibiotics. Complications: nonunion, malunion, infection, compartment syndrome, and avascular necrosis.

Ganglion Cysts

– Ganglion cysts: mucin-filled cysts from joint capsule/tendon sheath. Dorsal wrist (60-70%, scapholunate interval), volar wrist (radial artery), flexor tendon sheath (volar retinaculum, "seed ganglion"), DIP (mucous cyst, Heberden node association). Clinical: fluctuant mass, transilluminates, may cause nerve compression (volar: median/ulnar). Treatment: observation (spontaneous resolution 30-50%), aspiration (high recurrence), surgical excision (capsule stalk removal, recurrence 5-10%). Mucous cyMucous cyst (ganglion at DIP joint): associated with Heberden node, risk of nail deformity. Treatment: decompression, excision, pinning of DIP joint.

Golfer Elbow

– Medial epicondylitis, a tendinopathy of the common flexor/pronator tendon at the medial epicondyle. Less common than tennis elbow. Caused by repetitive wrist flexion and forearm pronation. Presents with medial elbow pain worsened by wrist flexion and gripping. Tenderness over the medial epicondyle. Positive Golfer elbow test. Risk of ulnar nerve compression. Diagnosis: clinical. Treatment: rest, ice, activity modification, NSAIDs, physical therapy (eccentric exercises), and corticosteroid injection (avoid ulnar nerve). Refractory cases: surgical release.

Hallux Valgus

– Lateral deviation of the great toe (hallux) with medial prominence of the first metatarsal head (bunion). More common in women, associated with narrow-toed shoes, high heels, and genetic predisposition. Presents with pain over the medial eminence, difficulty with shoe wear, and progressive deformity. Second toe can become hammer toe from crowding. Diagnosis: weight-bearing foot X-ray (hallux valgus angle >15°, intermetatarsal angle >9°). Treatment: wide-toed shoes, orthotics, toe spacers, and NSAIDs. Surgery (osteotomy, arthrodesis, or arthroplasty) for severe deformity or failed conservative treatment.

Hallux Valgus (Bunion)

– A progressive deformity of the first metatarsophalangeal (MTP) joint, characterized by lateral deviation of the great toe (hallux valgus) and medial deviation of the first metatarsal head (metatarsus primus varus), resulting in a bony prominence on the medial side of the forefoot (the bunion). Epidemiology: 3 times more common in women, prevalence increases with age (up to 35% in adults >65). Risk factors: narrow-toed shoes with high heels (most significant modifiable risk factor), family history, heredity, pes planus (flatfoot), hypermobile first ray, ligamentous laxity, and inflammatory arthritis (rheumatoid arthritis, gout). Pathophysiology: the proximal phalanx rotates and drifts laterally, the first metatarsal head drifts medially, the sesamoids sublux laterally, and the MTP joint capsule stretches medially. Symptoms: pain over the medial eminence (bursitis), difficulty fitting shoes, cosmetic concern, and progression of deformity. May be associated with hammer toe of the second toe and transfer metatarsalgia. Treatment: (1) Conservative: footwear modification (wide-toed shoes, soft uppers, bunion pads/orthoses/splints; NSAIDs for acute pain). (2) Surgical: over 150 described techniques - distal chevron osteotomy (mild-moderate), Scarf osteotomy (moderate-severe), Lapidus arthrodesis (severe with first TMT instability), Akin osteotomy (proximal phalanx). Post-op: 6 weeks non-weight-bearing in a postoperative shoe or boot, gradual return to normal footwear.

Hammer Toe

– A deformity of the second, third, or fourth toe characterized by flexion at the PIP joint and extension at the MTP and DIP joints. Caused by muscle imbalance (intrinsic foot muscle weakness) and narrow-toed shoes. Presents with pain over the dorsal PIP joint (corn/callus) and the tip of the toe. Flexible hammer toe: correctable passively. Rigid: fixed deformity. Diagnosis: clinical. Treatment: conservative: wide toe box shoes, toe crests/pads, and stretching exercises. Surgical: flexor tenotomy (flexible) or PIP arthroplasty/arthrodesis (rigid).

Hamstring

– One of three muscles forming the posterior compartment of the thigh: semitendinosus, semimembranosus, and biceps femoris (long and short heads). Common origin: ischial tuberosity (except short head of biceps femoris which originates from the linea aspera of the femur). Insertion: semitendinosus and semimembranosus to the medial tibial condyle (pes anserinus area); biceps femoris to the fibular head. Actions: hip extension, knee flexion, and (with knee flexed) tibial rotation (medial by semitendinosus/semimembranosus, lateral by biceps femoris). Innervation: tibial branch of sciatic nerve (L5-S2) for all except short head of biceps femoris (common peroneal nerve, L5-S2). Hamstring strain is the most common muscle injury in athletes (especially sprinters, football, rugby), typically involving the proximal musculotendinous junction of biceps femoris. Grades: grade 1 (mild, microscopic tears), grade 2 (partial tear), grade 3 (complete rupture). Risk factors: eccentric overload, fatigue, poor flexibility, previous hamstring injury, muscle imbalance (quadriceps dominance). Management: RICE (rest, ice, compression, elevation) in acute phase, followed by progressive rehabilitation (eccentric strengthening, Nordic hamstring curl for prevention, core stability, sport-specific training). Return to sport when full range of motion, strength >90% of unaffected side, and sport-specific movements are pain-free.

Herniated Disc

– Protrusion of the nucleus pulposus through a tear in the annulus fibrosus of the intervertebral disc, causing compression or inflammation of adjacent nerve roots. Most common at L4-L5 and L5-S1 (lumbar) and C5-C6, C6-C7 (cervical). Lumbar: sciatica (radiating leg pain), paresthesias, sensory loss, and motor weakness (foot drop). Cervical: neck pain radiating to arm with sensory/motor deficits. Cauda equina syndrome: surgical emergency with saddle anesthesia, bowel/bladder dysfunction, and bilateral leg weakness. Diagnosis: MRI. Treatment: conservative first (NSAIDs, PT, activity modification, epidural steroid injections). Surgical: microdiscectomy for refractory pain, progressive weakness, or cauda equina syndrome.

Herniated disk

– Also known as intervertebral disc herniation, displacement of nucleus pulposus material through a tear in the annulus fibrosus, most commonly occurring in the lumbar spine (L4-L5 and L5-S1 account for 95 percent of lumbar herniations) and cervical spine (C5-C6 and C6-C7). The pathophysiology involves age-related degenerative changes in the intervertebral disc: desiccation of the nucleus pulposus, fissuring of the annulus fibrosus, and eventual extrusion of nuclear material into the spinal canal or neural foramen, where the herniated disc material can compress the spinal cord (central herniation) or the nerve roots (posterolateral herniation, the most common), causing the characteristic radicular pain, sensory and motor deficits, and reflex changes in a dermatomal and myotomal distribution.

Hip Disorders

– The hip joints are ball-and-socket joints connecting the femur to the pelvis, providing stability and a wide range of motion for walking, running, and weight-bearing activities. Common hip conditions include osteoarthritis, fractures, bursitis, and labral tears. Management involves physical therapy, pain management, and joint replacement surgery for advanced degenerative disease.

Hip Fractures

– Common in elderly (osteoporosis, falls), classified by location: femoral neck (intracapsular), intertrochanteric (extracapsular), subtrochanteric. Garden classification for femoral neck: I (incomplete/impacted), II (complete nondisplaced), III (complete partially displaced), IV (complete displaced). Treatment: Garden I/II: cannulated screws; Garden III/IV in elderly: hemiarthroplasty (bipolar preferred) or total hip arthroplasty (if active, pre-existing arthritis); young patienpatients (<65, good bone): ORIF (cannulated screws or sliding hip screw, blade plate, or cephalomedullary nail depending on fracture pattern). Subtrochanteric fractures: high nonunion rate, IM nail (long, statically locked, reamed) versus plate fixation.

Hip Replacement

– A surgical procedure in which the diseased hip joint is replaced with a prosthetic implant (arthroplasty). Total hip replacement (THR) replaces both the femoral head and the acetabulum. Indications: osteoarthritis (most common, 60%), rheumatoid arthritis, avascular necrosis (AVN) of the femoral head, femoral neck fracture (especially displaced in elderly), hip dysplasia, ankylosing spondylitis, Paget disease, and failed previous hip surgery. Components: acetabular component (metal shell with polyethylene or ceramic liner), femoral component (metal stem with modular head–cobalt-chrome or ceramic). Fixation: cemented (using polymethylmethacrylate bone cement, preferred in elderly with osteoporotic bone) or cementless (press-fit, with porous coating for bone ingrowth, preferred in younger patients with good bone quality). Surgical approaches: posterior (most common, risk of dislocation 2-5% in first 3 months, repairs capsule/external rotators), lateral (Hardinge, gluteus medius split, lower dislocation risk), anterolateral (Watson-Jones), and direct anterior (muscle-sparing, faster recovery, less postoperative pain). Outcomes: excellent pain relief and functional improvement in 90-95% at 10 years, 80-85% implant survival at 20 years. Complications: dislocation (2-5%, highest in first 3 months), periprosthetic fracture, infection (1-2%, devastating, requires two-stage revision: removal of implant + antibiotic cement spacer + 6 weeks IV antibiotics + reimplantation), aseptic loosening (most common cause of late failure, due to wear debris-induced osteolysis, polyethylene wear particles stimulate macrophage activation and RANKL-mediated osteoclastic bone resorption), leg length discrepancy (subjective or objective), DVT/PE (routine thromboprophylaxis), nerve injury (sciatic, femoral).

Joint Diseases

– A broad category of pathological conditions affecting the structure and function of joints. Classification by etiology: (1) Degenerative: osteoarthritis (most common, non-inflammatory, affects loadbearing joints–knee, hip, spine). (2) Inflammatory: rheumatoid arthritis (autoimmune, symmetric small-joint synovitis), psoriatic arthritis, ankylosing spondylitis, reactive arthritis, gout (crystal arthropathy, uric acid, podagra), pseudogout (CPPD). (3) Infectious: septic arthritis (bacterial, urgent, joint destruction within 24 hours), Lyme arthritis, tuberculosis arthritis, viral arthritis (parvovirus, rubella, hepatitis B/C). (4) Metabolic: gout, pseudogout, haemochromatosis (chondrocalcinosis, calcium pyrophosphate), ochronosis (alkaptonuria, homogentisic acid accumulation, pigmented cartilage, spondylosis). (5) Trauma: intra-articular fracture, dislocation, ligament tear, meniscal tear, haemarthrosis. (6) Neoplastic: synovial sarcoma, pigmented villonodular synovitis (benign, recurrent haemarthrosis). (7) Childhood: juvenile idiopathic arthritis (JIA). Diagnosis: history (mechanical vs inflammatory pain, stiffness, swelling), examination (warmth, effusion, crepitus, range of motion, instability), imaging (X-ray, ultrasound, MRI), synovial fluid analysis (cell count, crystals, Gram stain, culture). Diagnosis: medical history, physical examination, imaging (X-ray, ultrasound, MRI), and synovial fluid analysis. Treatment directed at the underlying cause.

Joint Pain

– Discomfort arising from articular structures including cartilage, synovium, ligaments, and periarticular soft tissues due to inflammation, degeneration, injury, or infection. Common causes: osteoarthritis (degenerative, most prevalent), rheumatoid arthritis (autoimmune inflammatory), gout (crystal arthropathy), septic arthritis (bacterial infection requiring urgent intervention), and trauma. Clinical features include pain worsened by movement, stiffness, swelling, warmth, and decreased range of motion. Diagnosis requires history, physical exam, imaging (X-ray, MRI), and laboratory studies (inflammatory markers, autoantibodies, uric acid, synovial fluid analysis). Treatment targets the underlying cause: NSAIDs or corticosteroids for inflammation, disease-modifying antirheumatic drugs (DMARDs) for inflammatory arthritis, analgesics, physical therapy, and joint replacement for advanced disease.

Joint Replacement (Arthroplasty)

– A surgical procedure in which a damaged or arthritic joint surface is removed and replaced with a prosthetic implant, most commonly performed for the hip (total hip arthroplasty) and knee (total knee arthroplasty). Indications include end-stage osteoarthritis, rheumatoid arthritis, avascular necrosis, and severe joint trauma causing debilitating pain and functional limitation refractory to conservative therapy. The procedure involves resurfacing the articular ends of bone with metal, polyethylene, or ceramic components fixated with cement or press-fit technique. Postoperative management includes early mobilization, physical therapy, thromboprophylaxis, and activity restrictions to prevent dislocation. Complications: infection, loosening, fracture, nerve injury, venous thromboembolism, and prosthetic wear requiring revision.

Joints

– The sites where two or more bones meet, providing structural support and enabling movement. Classification by structure and function: (1) Fibrous joints (synarthroses, immovable): sutures of the skull, gomphoses (tooth in socket), syndesmoses (tibia-fibula). (2) Cartilaginous joints (amphiarthroses, slightly movable): synchondroses (costochondral, growth plates), symphyses (pubic symphysis, intervertebral discs). (3) Synovial joints (diarthroses, freely movable): characterized by a joint cavity lined by synovial membrane containing synovial fluid, articular cartilage (hyaline), and a fibrous joint capsule. Further classified by shape: hinge (elbow, knee), ball-and-socket (hip, shoulder), pivot (atlantoaxial, proximal radioulnar), condylar (temporomandibular, knee–also hinge), saddle (first carpometacarpal–thumb), gliding (intercarpal, intertarsal), and ellipsoid (wrist). Synovial fluid: plasma dialysate with hyaluronic acid (lubrication, shock absorption, nutrition for avascular articular cartilage). Joint health is maintained by low-impact exercise, weight-bearing activity, and avoidance of excessive repetitive loading. Joint pathology: osteoarthritis, rheumatoid arthritis, gout, septic arthritis, trauma (dislocation, fracture), and haemarthrosis.

Kienbock Disease

– Kienbock disease (lunatomalacia): AVN of lunate. Lichtman classification: I (normal XR, MRI edema), II (lunate sclerosis), IIIA (lunate collapse, no scaphoid rotation), IIIB (lunate collapse, fixed scaphoid rotation, DISI), IV (perilunate arthritis). Etiology: negative ulnar variance, trauma, vascular variation. Treatment: I/II: radial shortening (negative ulnar variance), capitate shortening, vascularized bone graft (4,5-extensor compartments artery); IIIA: radial shortening, capitolunate fusiocapitolunate fusion, capitate-lunate-triquetrum fusion, lunate core decompression; IIIB-IV: salvage (proximal row carpectomy, scaphoidectomy plus 4-corner fusion, total wrist arthrodesis).

Klippel-Feil Syndrome

– A congenital condition characterized by fusion of two or more cervical vertebrae, resulting from failure of normal segmentation of the cervical somites during weeks 3-8 of gestation. Classic triad: short neck, low posterior hairline, and limited cervical range of motion (present in <50% of cases). Patients are at risk for atlantoaxial instability, cervical stenosis, and the CRUS sequence (Cervical fusion, Renal anomalies, Uterine anomalies, and Sensorineural hearing loss). Associated anomalies: Sprengel deformity (30%), scoliosis (60%), renal agenesis/ectopia, congenital heart disease (VSD, ASD), hearing loss, and synkinesia (mirror movements). Diagnosis: cervical spine X-ray or CT shows fused vertebral bodies and posterior elements; MRI for spinal cord and brainstem assessment; renal ultrasound and audiometry for associated anomalies. Treatment: activity restriction to prevent cervical injury; surgical stabilization for instability or neurologic compromise.

Kneecap

– See Patella (Kneecap).

Knee Disorders

– The knee is a complex hinge-type synovial joint connecting the femur, tibia, and patella, stabilized by ligaments (ACL, PCL, MCL, LCL), menisci (medial and lateral), and surrounding muscles/tendons. Common knee disorders include osteoarthritis (degenerative cartilage loss most prevalent in older adults), anterior cruciate ligament (ACL) tears (sports-related pivot injury), meniscal tears (twisting injury with mechanical locking or clicking), patellofemoral pain syndrome (anterior knee pain in active young adults), and Baker's cyst (popliteal swelling from fluid extrusion). Diagnosis relies on history, physical examination (Lachman test, McMurray test, joint line tenderness), and imaging (weight-bearing X-rays, MRI for soft-tissue pathology). Treatment ranges from conservative (activity modification, physical therapy, bracing, NSAIDs) to surgical (arthroscopic meniscectomy/repair, ligament reconstruction, partial or total knee arthroplasty).

Kyphosis

– An excessive anterior-posterior curvature of the thoracic spine, measured by the Cobb angle on lateral spine x-ray (normal thoracic kyphosis is 20-40 degrees). Types: postural kyphosis (most common, flexible, correctable with hyperextension, in adolescents, benign, not progressive, no structural vertebral abnormalities), Scheuermann kyphosis (rigid, structural kyphosis in adolescents, caused by anterior wedging of greater than 5 degrees in three or more consecutive thoracic vertebrae, with endplendplate irregularities, Schmorl nodes, and decreased disc height), and congenital kyphosis (rare, due to failure of formation or segmentation of vertebral bodies). The diagnosis is made by standing lateral x-ray with the Cobb angle measured from the superior endplate of one vertebra to the inferior endplate of the lowest involved vertebra.

Lateral Collateral Ligament Injury

– Tear of the LCL, less common than MCL injury but more significant. Mechanism: varus stress to the knee (direct blow to medial knee). Presents with lateral knee pain, swelling, and varus laxity. Part of the posterolateral corner (PLC) — when injured, causes significant rotational instability. Grade III LCL injury is often associated with PCL injury and peroneal nerve injury. Diagnosis: varus stress test at 0° and 30°; MRI. Treatment: Grade I-II: bracing. Grade III or multiligament: surgical repair or reconstruction.

Lordosis

– The normal inward curvature of the cervical and lumbar spine. Excessive lumbar lordosis (hyperlordosis): increased pelvic tilt, commonly due to hip flexor tightness, weak abdominals, or pregnancy. Presents with low back pain and forward protrusion of the abdomen and buttocks. Causes: postural, congenital (achondroplasia), neuromuscular (cerebral palsy), and compensatory (kyphosis). Diagnosis: clinical, standing lateral X-ray. Treatment: postural exercises, core strengthening, hamstring/hip flexor stretching. Surgery rarely needed.

Lumbago

– A non-specific term for low back pain, typically referring to acute or chronic pain in the lumbar region (between the lower ribs and the gluteal folds). Lumbago is a symptom, not a diagnosis. Causes: mechanical low back pain (most common, 90%, includes lumbar strain/sprain, discogenic pain, facet joint pain, sacroiliac joint dysfunction), disc herniation with radiculopathy (sciatica, 5%), spinal stenosis, spondylolisthesis, vertebral compression fracture (osteoporotic or traumatic), infection (discitis, osteomyelitis, epidural abscess), malignancy (bone metastases, multiple myeloma, primary bone tumor), inflammatory (ankylosing spondylitis, psoriatic arthritis), and visceral referred pain (pancreatitis, nephrolithiasis, AAA, endometriosis). Red flags (cauda equina syndrome, infection, malignancy, fracture): saddle anesthesia, bowel/bladder incontinence, bilateral sciatica, progressive neurological deficit, fever, night pain, weight loss, age >50, history of cancer, IV drug use, immunosuppression. Yellow flags (psychosocial risk factors for chronicity): fear-avoidance beliefs, catastrophising, depression, job dissatisfaction, pending litigation. Most acute lumbago resolves within 4-6 weeks. Management: conservative–analgesics (paracetamol, NSAIDs–ibuprofen, naproxen, diclofenac), muscle relaxants (short-term, diazepam, cyclobenzaprine), heat/ice, activity modification (avoid bed rest, stay active), physiotherapy (core stability, McKenzie exercises, manual therapy). For radicular pain: epidural corticosteroid injections. Surgery: discectomy for refractory sciatica, decompression for spinal stenosis, fusion for instability.

Medial Collateral Ligament Injury

– Tear of the MCL, the most common knee ligament injury. Mechanism: valgus stress to the knee (direct blow to lateral knee). Presents with medial knee pain, swelling, and valgus laxity. Grading: I (pain, no laxity), II (partial tear, some laxity), III (complete tear, significant laxity). Diagnosis: valgus stress test at 0° and 30° of flexion; MRI. Treatment: almost always conservative — functional bracing with early motion. Grade III injuries with associated ACL/PCL/PLC injuries may require surgical repair.

Meniscal Injuries

– Meniscal tears: traumatic (vertical longitudinal, bucket handle, radial, horizontal cleavage, flap) vs degenerative (complex, horizontal). Clinical: joint line tenderness, McMurray, Apley, Thessaly test. MRI: sensitivity 90%+. Treatment: repairable (vertical longitudinal, peripheral 100 3mm, red-red/red-white zone, young): inside-out, all-inside, outside-in; irreparable (degenerative, white-white, complex): partial meniscectomy (preserve rim). Root tears: radial tears at posterior horn attachmenattachment site (medial meniscus). Treatment: repair (transtibial pull-out, suture anchor) vs meniscectomy.

Meniscal Tear

– Tear of the C-shaped fibrocartilage (medial or lateral meniscus) that provides shock absorption and stability in the knee. Medial meniscus tears more common (attached to capsule). Mechanism: twisting injury with weight-bearing. Presents with joint line tenderness, pain with deep squatting, catching, locking, and giving way. Positive McMurray test. Diagnosis: MRI. Treatment: conservative for small, peripheral (red zone) tears; arthroscopic partial meniscectomy for large, complex tears that cannot be repaired; meniscal repair for peripheral tears in young patients.

Muscle Pain (Myalgia)

– Myalgia—muscle pain—is a common symptom arising from skeletal muscle injury, inflammation, overuse, infection, or systemic disease. Acute myalgia is often due to trauma (muscle strain/tear, contusion), eccentric exercise (delayed-onset muscle soreness peaking 24–72 hours post-exertion), or viral infections (influenza, COVID-19, enterovirus). Chronic diffuse myalgia characterizes fibromyalgia (widespread pain with tender points, fatigue, sleep disturbance, cognitive fog—affecting 2–4% of population, 80% female) and polymyalgia rheumatica (morning stiffness and aching in shoulder/pelvic girdles in adults >50, ESR/CRP markedly elevated). Diagnosis: CK, ESR, CRP, TSH, vitamin D, autoimmune serologies, EMG, and muscle biopsy when indicated. Treatment: symptomatic (acetaminophen, NSAIDs, heat/ice, massage), treatment of underlying cause (antibiotics for infectious myositis, corticosteroids for polymyalgia rheumatica/idiopathic inflammatory myopathies, exercise and tricyclic antidepressants/gabapentinoids for fibromyalgia).

Nonunion and Malunion

– Nonunion is failure of a fractured bone to heal after an adequate period (generally 6-9 months for long bones) without clinical or radiographic evidence of progression toward union for 3 months. Classification: hypertrophic (viable ends, inadequate stability), atrophic (devitalized ends, poor biology), oligotrophic (intermediate). Risk factors: open fractures, severe comminution, inadequate fixation, infection, smoking, NSAIDs, steroids, diabetes, vascular disease, radiation. Diagnosis: persistepersistent pain, implant failure. Treatment: hypertrophic (increase stability with compression plating, exchange nailing, dynamization); atrophic (improve biology with bone graft, vascularized bone graft).

Open Fractures

– Fractures with communication between the fracture site and the external environment, classified by Gustilo-Anderson: Grade I (<1 cm wound, clean), Grade II (1-10 cm wound, moderate soft tissue injury), Grade IIIA (>10 cm wound, adequate soft tissue coverage), Grade IIIB (>10 cm wound, extensive soft tissue loss requiring flap coverage), Grade IIIC (any open fracture with arterial injury requiring repair). Management: ATLS protocol, antibiotics within 1 hour (Grade I: cefazolincefazolin 1-2g IV q8h; Grade II/III: cefazolin plus gentamicin; Grade III: penicillin for farm injuries, tetanus prophylaxis, thorough irrigation and debridement within 6 hours, fracture stabilisation (external fixation for severe contamination), wound closure by 7 days.

Orthopedic Biologics

– Bone graft substitutes: autograft (iliac crest gold standard, osteogenic/osteoinductive/osteoconductive, donor site morbidity), allograft (osteoconductive, some osteoinductive if DBM, disease transmission risk), DBM (demineralized bone matrix, osteoinductive), synthetic (calcium phosphate, calcium sulfate, bioactive glass - osteoconductive), BMPs (rhBMP-2, rhBMP-7, potent osteoinductive, swelling/ectopic bone cost), stem cells (BMAC, MSCs, adipose-derived), PRP (platelet-rich plasma, growth factgrowth factors, cytokines). Complications: infection, graft resorption, nonunion, heterotopic ossification. Applications: nonunion, delayed union, spinal fusion, osteochondral defects.

Orthopedic Infections

– Septic arthritis: see Infectious Disease. Osteomyelitis: see Infectious Disease. Prosthetic joint infection: see Periprosthetic Joint Infection. Diabetic foot infection: Wagner classification (0-5), University of Texas (grade + ischemia + infection). Treatment: offloading (total contact cast), debridement, vascular assessment (ABI, TBI, TcPO2), revascularization (endovascular, bypass), antibiotics (mild: oral; moderate/severe: IV), amputation (toe, ray, transmetatarsal, BKA, AKA). Charcot neuroaosteoarthropathy (Charcot foot): midfoot collapse (rocker bottom), neuropathic, treated with Charcot reconstruction (superconstruct fixation, intramedullary beam, exostectomy, TCC, and AFO).

orthopedics

– The branch of surgery concerned with the musculoskeletal system, including bones, joints, ligaments, tendons, muscles, and nerves. Orthopaedic surgeons diagnose and treat conditions such as fractures and dislocations, arthritis (osteoarthritis, inflammatory arthritis), spinal disorders (disc herniation, spinal stenosis, scoliosis), sports injuries (ACL tear, rotator cuff tear, meniscal injury), paediatric orthopaedic conditions (developmental dysplasia of the hip, clubfoot, Perthes disease), bone tumors, and infections (osteomyelitis, septic arthritis). Treatment modalities: conservative (cast/splint, physiotherapy, medications), minimally invasive (arthroscopy, percutaneous fixation) and open surgery (joint replacement—hip, knee, shoulder; spinal fusion; osteotomy; limb reconstruction). Subspecialties include trauma, arthroplasty, hand surgery, foot and ankle, spine, paediatric orthopedics, sports medicine, and orthopaedic oncology.

Orthotics

– The field of healthcare concerned with the design, manufacture, and application of orthoses—external devices that support, align, prevent deformity, or improve function of the musculoskeletal system. Orthoses include ankle-foot orthoses (AFO—for foot drop, stroke, cerebral palsy), knee braces (ACL/PCL injury, patellofemoral syndrome), spinal braces (lumbar support, scoliosis—Boston brace, Milwaukee brace), wrist splints (carpal tunnel syndrome, rheumatoid arthritis), shoe inserts/orthotics (plantar fasciitis, flat feet), and fracture braces. Orthotists assess patients, prescribe appliances, and fit custom-made orthoses. Orthotics differs from prosthetics (artificial limbs). Evidence supports orthotic use in conditions such as diabetic foot ulcer prevention, post-stroke gait rehabilitation, and scoliosis bracing (reduces curve progression in adolescents).

Osgood-Schlatter Disease

– A traction apophysitis of the tibial tuberosity at the insertion of the patellar tendon, caused by repetitive microtrauma during periods of rapid growth. It is a common cause of anterior knee pain in adolescents aged 10–15 years, more common in boys (ratio 3:1) and in active/sporty children (running, jumping sports—basketball, football, volleyball). Presentation: gradual onset of unilateral (30% bilateral) anterior knee pain just below the kneecap, worsened by activity (running, jumping, kneeling, climbing stairs), improved by rest. Physical exam: tenderness and swelling over the tibial tuberosity, pain on resisted knee extension or squatting. X-ray may show irregularity or fragmentation of the tibial tuberosity apophysis, but diagnosis is clinical. Management: activity modification (reduce high-impact sports, not complete rest), ice after activity, NSAIDs, quadriceps and hamstring stretching/strengthening, patellar tendon strap, and reassurance (self-limiting condition that resolves with skeletal maturity—apophyseal closure at age 14–18). Surgery (ossicle excision, tibial tuberosity reduction) is rarely needed for persistent symptoms.

Osteochondritis Dissecans

– A condition characterized by separation of a fragment of articular cartilage and subchondral bone from the joint surface, most commonly in the knee (lateral aspect of the medial femoral condyle). Most common in adolescents and young adults. Etiology: repetitive microtrauma, ischemia, or genetic predisposition. Presents with vague joint pain, swelling, catching, locking, and giving way. Diagnosis: X-ray (lucent crater with loose fragment), MRI (fragment stability and viability). Treatment: stable lesions in skeletally immature: activity modification, immobilization, or drilling. Unstable or loose bodies: arthroscopic fixation or fragment excision.

Osteochondroma

– The most common benign bone tumor, accounting for approximately 20-35 percent of all benign bone tumors. It is characterized by a bony projection with a cartilage cap arising from the surface of a bone, continuous with the underlying cortex and medullary cavity. Solitary osteochondromas typically present in the second decade of life (ages 10-20 years) as a painless, firm, immobile mass near the knee (distal femur or proximal tibia, most common), proximal humerus, oror the proximal tibia or humerus. The characteristic radiographic appearance is a pedunculated or sessile bony excrescence arising from the metaphysis, pointing away from the adjacent joint, with a cartilage cap continuous with the host bone. Malignant transformation to secondary chondrosarcoma occurs in approximately 1 to 2 percent of solitary osteochondromas but in 2 to 5 percent of patients with multiple hereditary exostosis.

Osteoid Osteoma

– A benign bone-forming tumor that accounts for approximately 10-12 percent of all benign bone tumors, predominantly affecting adolescents and young adults (ages 5-25 years, male-to-female ratio 2:1). It is characterized by a small (< 1.5-2 cm) radiolucent nidus surrounded by reactive sclerotic bone. The nidus contains osteoblasts, osteoid, and woven bone with prominent vascular channels. Osteoid osteomas produce high levels of prostaglandin E2, which causes the characteristic scharacteristic severe, localised, deep, boring pain that is worse at night and dramatically relieved by NSAIDs, which inhibit COX-2 and reduce the production of prostaglandin E2, the key mediator of pain in this tumor.

Osteopenia

– Bone mineral density (BMD) that is lower than normal but not low enough to be classified as osteoporosis. T-score between -1.0 and -2.5 on DXA scan. A risk factor for osteoporosis and fractures. Management: lifestyle modifications (weight-bearing exercise, adequate calcium [1200 mg/day] and vitamin D [800-1000 IU/day]), smoking cessation, fall prevention, and reassess BMD every 1-2 years. Pharmacotherapy considered if FRAX score indicates high fracture risk.

Osteopetrosis

– A rare inherited bone disease characterized by defective osteoclast function, leading to impaired bone resorption and abnormally dense, sclerotic bones. The bone is structurally abnormal—brittle despite increased density—and prone to fracture. Two main forms: (1) Autosomal recessive ('malignant'): presents in infancy with failure to thrive, anemia, thrombocytopenia (bone marrow failure from marrow space obliteration), hepatosplenomegaly (extramedullary haematopoiesis), cranial nerve palsies (optic atrophy, deafness, facial palsy due to nerve compression), and recurrent infections. Life expectancy is poor without treatment. (2) Autosomal dominant ('benign'): milder, presents in adolescence or adulthood with fractures, osteomyelitis of the mandible, and sometimes asymptomatic (incidental finding on X-ray). Diagnosis: X-ray shows generalised osteosclerosis ('sandwich vertebrae', 'bone-within-bone' appearance, Erlenmeyer flask deformity of femurs). Treatment: hematopoietic stem cell transplantation (HSCT) for the malignant form (restores osteoclast function, improves survival), supportive care for fractures and anemia, and interferon gamma-1b. Gene therapy is experimental.

Osteoporosis

– A systemic skeletal disease of low bone mass and microarchitectural deterioration of bone tissue, leading to increased bone fragility and fracture risk. It is most common in postmenopausal women (due to estrogen deficiency accelerating bone resorption) and older adults; peak bone mass is achieved by age 30 and declines with age. Fragility fractures—vertebral compression fractures (often asymptomatic or causing height loss, kyphosis), hip fracture (associated with 20–30% 1-year mortality), and distal radius (Colles') fracture—are hallmark clinical events. Screening: DXA (dual-energy X-ray absorptiometry) at hip and lumbar spine; T-score

≤ -2.5 defines osteoporosis, -1.0 to -2.5 defines osteopenia. FRAX tool estimates 10-year fracture risk to guide treatment decisions. Pharmacotherapy: first-line bisphosphonates (alendronate, risedronate, zoledronic acid), anabolic agents (teriparatide, romosozumab) for high-risk or severe disease, denosumab (RANKL inhibitor), and raloxifene (SERM). Nonpharmacologic: adequate calcium (1000–1200 mg/day from diet or supplements), vitamin D (800–1000 IU/day), weight-bearing and resistance exercise, fall prevention, and smoking cessation.

Osteosarcoma

– The most common primary malignant bone tumor in children and adolescents (excluding leukemia), with a peak incidence during the adolescent growth spurt (ages 10-20 years), and a second smaller peak in older adults (usually secondary to Paget disease or prior radiation). It most commonly arises in the metaphysis of long bones, particularly the distal femur (most common, 30-40 percent), proximal tibia, and proximal humerus. Histologically, it is characterized by malignant osteoid pproduction (malignant osteoid) by malignant tumor cells. Clinical presentation is with localised pain (the most common symptom, worse at night), swelling, a palpable mass, and pathological fracture in 10 to 15 percent of cases.

Osteotomy

– A surgical procedure involving cutting and repositioning of bone to correct deformity, realign joint surfaces, alter weight-bearing forces, or change limb length. Indications: (1) Lower limb—high tibial osteotomy (HTO) for medial compartment knee osteoarthritis in young, active patients (realigns the mechanical axis, delaying arthroplasty), distal femoral osteotomy for lateral compartment or valgus deformity, and femoral/tibial osteotomy for congenital deformities (blount disease, leg length discrepancy). (2) Hip—pelvic osteotomy (periacetabular osteotomy—Ganz procedure) for developmental dysplasia of the hip in young adults, and femoral varus/valgus osteotomy for Perthes disease or hip dysplasia. (3) Spine—spinal osteotomy (Smith-Petersen, pedicle subtraction osteotomy, vertebral column resection) for sagittal imbalance (kyphosis, flatback syndrome). (4) Foot—calcaneal osteotomy for hindfoot deformity, first metatarsal osteotomy for hallux valgus. Osteotomy is also used in orthognathic surgery (jaw osteotomy for malocclusion). Fixation: plates, screws, external fixators. Recovery involves non-weight-bearing or protected weight-bearing for 6–12 weeks post-operatively.

Paget Disease of Bone

– A chronic bone disorder characterized by excessive, disorganized bone remodeling with increased osteoclastic bone resorption followed by compensatory osteoblastic bone formation, producing structurally abnormal bone that is enlarged, weak, and prone to fracture. Often asymptomatic; can present with bone pain, deformity (bowing of tibia, skull enlargement), fractures, and arthritis. Diagnosis: elevated serum alkaline phosphatase with normal calcium, phosphate, and PTH; X-ray findings (lytic lesions, bone enlargement). Treatment: bisphosphonates (alendronate, risedronate, zoledronic acid).

Paget's Disease

– See Paget Disease of Bone.

Patella (Kneecap)

– The largest sesamoid bone in the human body, embedded within the quadriceps tendon anterior to the knee joint. The patella increases the lever arm of the quadriceps muscle (improving mechanical advantage for knee extension by 30%), protects the anterior knee joint, and distributes compressive forces across the femoral trochlea. Articular surface: the posterior patella has a V-shaped ridge that articulates with the femoral trochlear groove (medial and lateral facets). Patellofemoral joint: transmits up to 3–5 times body weight during stair climbing and 7–8 times during squatting. Patellar dislocation: typically lateral (vastus medialis obliquus disruption, trochlear dysplasia, patella alta, increased Q-angle), reduced by extending the knee. Patellar fracture: caused by direct trauma or forceful quadriceps contraction (transverse fracture most common); non-displaced: cast immobilisation; displaced/disrupted extensor mechanism: tension band wiring. Patellofemoral pain syndrome (anterior knee pain): common in young active individuals and runners, treated with quadriceps strengthening, activity modification, and patellar taping.

Patellofemoral Instability

– Patellar dislocation: lateral dislocation (95%), often reduces spontaneously. Risk factors: trochlear dysplasia (Dejour classification), patella alta (Caton-Deschamps >1.2), increased TT-TG distance (>20mm), MPFL insufficiency, femoral anteversion, genu valgum. First-time: nonoperative (brace, PT, MPFL healing) if no loose body; recurrent: MPFL reconstruction (gold standard), tibial tubercle osteotomy (Fulkerson anteromedialization) for patella alta/TT-TG >20mm, trochleoplasty (deepening) for sesevere trochlear dysplasia (Dejour type B/D).

Patellofemoral Pain Syndrome

– Anterior knee pain related to the patellofemoral joint, common in adolescents and young adults (especially runners). Etiology: multifactorial (malalignment, quadriceps imbalance, overuse, tight lateral structures, weak VMO, and patellar instability). Presents with retropatellar or peripatellar pain with activities (squatting, running, stair climbing, prolonged sitting with knees bent — theater sign). Diagnosis: clinical; X-ray and MRI to rule out other pathology. Treatment: quadriceps and VMO strengthening, IT band and hamstring stretching, patellar taping/bracing, and activity modification. Most improve with non-operative management.

PCL Injuries

– Posterior cruciate ligament (PCL) tears: dashboard injury (posterior tibial force), hyperflexion. Clinical: posterior drawer (most sensitive), quadriceps active test, posterior sag sign. Grade I (1-5mm), II (6-10mm), III (>10mm). Isolated PCL: nonoperative (brace, rehab) for Grade I/II; surgical for Grade III or combined. PCL reconstruction: transtibial or tibial inlay (open posterior approach), single or double bundle. Combined with PLC/ACL/MCL for multi-ligament.

Peripheral Nerve Injuries

– Peripheral nerve injury classification (Sunderland): 1st degree (neuropraxia, conduction block, full recovery), 2nd degree (axonotmesis, Wallerian degeneration, endoneurium intact, recovery), 3rd degree (endoneurium disrupted, perineurium intact, partial recovery), 4th degree (perineurium disrupted, neuroma-in-continuity, no recovery without surgery), 5th degree (neurotmesis, complete transection). Management: acute sharp transection: primary epineurial repair (9-0/10-0 nylon); blunt/crush: waitwait 3-6 weeks, then repair; chronic (>6-12 months): nerve transfer, tendon transfer, free functional muscle transfer. Prognosis: better for distal, sharp, clean injuries; worse for proximal, blunt, contaminated.

Periprosthetic Fractures

– Periprosthetic fractures: Vancouver classification (hip): A (trochanteric), B (stem level: B1 stable, B2 loose, B3 loose + bone loss), C (distal to stem). Treatment: A: ORIF if displaced; B1: ORIF (cable plate); B2: revision stem (long, cementless, distal fixation); B3: revision + allograft/augment. UCS classification (knee): femoral, tibial, patellar. Treatment: ORIF if stable implant; revision if loose.

Periprosthetic Joint Infection

– Periprosthetic joint infection (PJI): MSIS criteria (major: sinus tract, two positive cultures; minor: elevated serum CRP/ESR, synovial WBC/PMN, positive histology, single positive culture). Acute postop (<4 weeks) or acute hematogenous: debridement, antibiotics, implant retention (DAIR) with polyethylene exchange. Chronic: one-stage (debridement, exchange, antibiotics) or two-stage (resection, spacer, antibiotics 6 weeks, reimplantation) exchange. Antibiotics: organism-specific, biofilm-active biofilm-active (rifampin for staph, ciprofloxacin for gram-negative). Prevention: preoperative antibiotics, skin preparation, laminar flow, antibiotic cement, optimisation of medical comorbidities. Outcomes: 80-90 percent success for two-stage exchange, 70-80 percent for one-stage, 60-80 percent for DAIR.

Phocomelia

– A severe congenital limb deficiency where the proximal portion of a limb (humerus/femur, radius/tibia) is absent or severely hypoplastic, and the hand or foot attaches directly to the trunk. Most famously associated with thalidomide exposure during the first trimester of pregnancy (days 20-36 after conception). Can also occur in Roberts syndrome, Holt-Oram syndrome, and TAR syndrome. Complete phocomelia: hand/foot directly attached to the trunk. Proximal phocomelia: absent humerus/femur with forearm/leg present. Distal phocomelia: absent radius/ulna with humerus present. Treatment: prosthetics and adaptive devices to optimize function; reconstructive surgery for selected cases. Genetic counseling and prenatal diagnosis are important.

Plantar Fasciitis

– The most common cause of heel pain, caused by degeneration and microtears of the plantar fascia at its calcaneal origin. Presents with sharp, stabbing heel pain with the first steps in the morning (post-static dyskinesia), improving after a few minutes but worsening with prolonged standing or activity. Risk factors: tight Achilles/gastrocnemius, obesity, prolonged standing, and high-impact activities. Diagnosis: clinical. Treatment: calf and plantar fascia stretching (first-line), orthotics (arch support, heel cup), night splints, NSAIDs, and corticosteroid injection. Refractory: shockwave therapy, PRP, or fasciotomy.

Polydactyly

– A congenital anomaly characterized by supernumerary digits on the hands or feet. Postaxial polydactyly (ulnar/fibular side, most common): more frequent in African American and Native American populations, often isolated and autosomal dominant. Preaxial polydactyly (radial/tibial side): more common in Asian populations, may be associated with syndromes (Holt-Oram, Fanconi anemia, VACTERL, TAR syndrome). Central polydactyly: rare, often syndromic. Classification: Type A (well-formed, articulating digit) vs. Type B (rudimentary, pedunculated skin tag). Physical exam and X-ray to assess for bony connection, joint articulation, and functional impact. Treatment: surgical excision for aesthetic and functional concerns; Type B can be ligated in the nursery. Preaxial polydactyly may require more complex reconstruction.

Posterior Cruciate Ligament Injury

– Tear of the PCL, less common than ACL injury. Mechanism: dashboard injury (tibia striking dashboard), hyperflexion, or fall on a flexed knee with foot plantarflexed. Presents with posterior knee pain, mild swelling, and posterior sag of the tibia. Positive posterior drawer test (most sensitive), posterior sag sign (Godfrey test). Often associated with PLC injury (posterolateral corner: popliteus, popliteofibular ligament, LCL). Diagnosis: MRI. Treatment: Grade I-II: conservative with quadriceps strengthening. Grade III or combined injuries: PCL reconstruction.

Proximal Humerus Fractures

– Proximal humerus fractures: Neer classification (4 parts: articular surface, greater tuberosity, lesser tuberosity, shaft; displaced >1 cm or angulated >45 degrees = part). 1-part (nondisplaced): sling, early motion. 2-part: surgical neck (ORIF plate, IM nail) or greater tuberosity (if displaced >5mm or rotator cuff involvement). 3-part: ORIF (locking plate) or hemiarthroplasty (elderly, osteoporotic). 4-part: high risk of AVN; hemiarthroplasty or reverse total shoulder arthroplasty (RSA, preferpreferred, particularly for elderly patients with 4-part fractures). Post-op: sling, early passive ROM, active assist at 6 weeks, strengthening at 12 weeks. Complications: AVN (up to 30 percent for 4-part), malunion, nonunion, stiffness, hardware failure, infection.

Radial Club Hand

– A congenital longitudinal deficiency of the radial ray, ranging from thumb hypoplasia to complete absence of the radius. The hand is radially deviated due to the absent radial support, with the carpus displaced proximally and radially. Associated with Holt-Oram syndrome (TBX5 mutation, with VSD or ASD), TAR syndrome (thrombocytopenia-absent radius, bilateral radial aplasia with thumbs present — key distinguishing feature), Fanconi anemia (progressive pancytopenia, radial anomalies), and VACTERL association. Physical exam shows radial deviation of the hand, short forearm, thumb anomalies (absent, hypoplastic, or triphalangeal), and limited elbow extension. X-ray confirms radius deficiency and associated skeletal anomalies. Treatment: centralization or radialization of the wrist (surgical repositioning over the ulna), thumb reconstruction (pollicization), and splinting. Genetic testing for Fanconi anemia is essential due to associated bone marrow failure and malignancy risk.

Rickets

– A childhood disorder of bone mineralization due to vitamin D deficiency, calcium deficiency, or phosphate deficiency (see also endocrine chapter). X-ray findings: widened, irregular metaphyseal plates (cupping, fraying, splaying) at sites of rapid bone growth (wrists, knees). Clinical features: bone pain, deformities (bowing of legs, frontal bossing, rachitic rosary, Harrison groove), growth retardation, and motor delay. Treatment: high-dose vitamin D (600,000 IU total over several months), calcium supplementation, and orthopedic bracing/surgery for severe deformities.

Rotator Cuff Tears

– Rotator cuff tears: partial (articular-sided PASTA, bursal-sided, intrasubstance) vs full-thickness. Ellman classification for PASTA: Grade I (<3mm), II (3-6mm), III (>6mm). Cofield sizing: small (<1cm), medium (1-3cm), large (3-5cm), massive (>5cm). Clinical: Neer/Hawkins impingement, drop arm, lag signs (external rotation lag, internal rotation lag), weakness. MRI: tear size, retraction, fatty infiltration (Goutallier classification: 0-4), muscle atrophy. Treatment: partial tears <50%: nonopernonoperative (PT, corticosteroid injection, NSAIDs); full-thickness: repair (arthroscopic primary repair: single-row vs double-row, suture bridge, transosseous equivalent). Massive tears: nonop vs repair vs tendon transfer (latissimus dorsi for posterosuperior, pectoralis major for subscapularis) vs RTSA (reverse total shoulder arthroplasty, for massive retracted tears with rotator cuff arthropathy).

Sacroiliac Joint

– See Sacroiliac Joint in Anatomy.

Scaphoid Fractures

– Scaphoid fractures: most common carpal fracture (young males, FOOSH). Classification: Herbert: A (incomplete), B (complete: B1 distal waist, B2 proximal waist, B3 proximal pole, B4 distal tubercle, B5 transverse). Vascularity: retrograde from distal radius -> proximal pole at risk for AVN. Clinical: snuffbox tenderness, scaphoid compression test, pain with axial thumb loading. Imaging: PA/lateral/oblique/scaphoid views; MRI/CT if occult (high suspicion, negative XR). Treatment: distal pole/waistdistal pole/waist fractures: closed reduction and cast immobilization (below elbow thumb spica or scaphoid cast for 6-12 weeks). Proximal pole fractures: ORIF (headless compression screw) due to high risk of nonunion and AVN. Nonunion: bone graft with or without fixation.

Scoliosis

– A lateral spinal curvature >10 degrees on standing radiograph (Cobb angle). Idiopathic scoliosis (most common, 80%): adolescent onset (≥10 years), more common in girls. Congenital: due to vertebral anomalies. Neuromuscular: associated with cerebral palsy, muscular dystrophy. Degenerative: from asymmetric disc/ facet degeneration in older adults. Screening: Adam forward bend test (rib hump). Management: observation for mild curves (<25°), bracing for moderate (25-45°) in skeletally immature, surgical correction for severe (>45-50°) or progressive curves.

Shoulder Instability

– Anterior instability (95%): Bankart lesion (anterior-inferior labral tear), Hill-Sachs lesion (posterolateral humeral head impression fracture), ALPSA (anterior labroligamentous periosteal sleeve avulsion), HAGL (humeral avulsion of glenohumeral ligament), GLAD (glenoid labral articular disruption). Recurrence: <20 years 80-90%, >40 years <20%. Treatment: first-time: immobilization (controversial, may not reduce recurrence), early PT; recurrent: arthroscopic Bankart repair (suture anchors), rempremplissage (Hill-Sachs remplissage), coracoid transfer (Latarjet, Bristow) for glenoid bone loss >20-25 percent, bone grafting (distal tibia allograft, iliac crest autograft).

SLAP Tears

– Superior labrum anterior-posterior (SLAP) tears: Snyder classification: I (fraying), II (detachment of biceps anchor, most common), III (bucket-handle tear of labrum, stable biceps), IV (bucket-handle extending into biceps). Clinical: O'Brien active compression test, Speed test, Yergason test, crank test. MRI arthrogram: sensitivity >90%. Treatment: Type I: debridement; Type II: repair (young, overhead athletes), biceps tenodesis (older, >35-40, non-overhead); Type III: debridement; Type IV: debdebridement with biceps tenodesis or tenotomy; Type IV: repair plus biceps tenodesis/tenotomy.

Soft Tissue Tumors

– Lipoma: most common benign soft tissue tumor, subcutaneous, encapsulated. Liposarcoma: most common malignant STS in adults, deep, subtypes (well-differentiated/dedifferentiated, myxoid/round cell, pleomorphic). Synovial sarcoma: t(X;18), SS18-SSX, young adults, deep extremity, "triple sign" on MRI. Rhabdomyosarcoma: most common pediatric STS, embryonal (favorable), alveolar (PAX3/7-FOXO1, unfavorable). Malignant peripheral nerve sheath tumor (MPNST): NF1 association, painful, rapid growth. DesmoDesmoid tumor: locally aggressive, not metastatic, watch and wait, surgery (high recurrence), cryotherapy, NSAIDs, tyrosine kinase inhibitor. Myxoma: intramuscular, benign. Nodular fasciitis: reactive, self-limited. Elastofibroma dorsi: subscapular, benign. Imaging: MRI with contrast. Biopsy: core needle (preferred), incisional, excisional. Staging: AJCC, Enneking classification.

Spinal Cord Injury

– Spinal cord injury: ASIA impairment scale (A-E). Complete (ASIA A) vs incomplete. Syndromes: central cord (cervical hyperextension, upper > lower extremity weakness, burning hands), Brown-Sequard (hemisection: ipsilateral motor/proprioception loss, contralateral pain/temp loss), anterior cord (motor + pain/temp loss, proprioception spared), posterior cord (proprioception loss), conus medullaris (L1-L2, saddle anesthesia, bowel/bladder, mixed UMN/LMN), cauda equina (LMN, radicular pain, asymmetriasymmetric neurologic involvement). Treatment: high-dose methylprednisolone (controversial), blood pressure augmentation (MAP >85 mmHg for 7 days), surgical decompression and stabilisation. Timing: early (<24h) for cervical injury with incomplete deficit. Rehabilitation, bowel/bladder management, and psychosocial support.

Spinal Stenosis

– Narrowing of the spinal canal causing compression of the spinal cord or nerve roots. Lumbar stenosis (most common): neurogenic claudication (leg pain/weakness with walking, relieved by sitting/leaning forward), often bilateral. Cervical stenosis: myelopathy (hyperreflexia, Babinski, Hoffman, gait disturbance, hand clumsiness, bowel/bladder dysfunction). Risk factors: aging, degenerative disc disease, facet hypertrophy, ligamentum flavum hypertrophy, and congenital stenosis. Diagnosis: MRI. Treatment: physical therapy, NSAIDs, epidural steroid injections, and decompressive laminectomy with or without fusion for refractory or severe cases.

Spine Disorders

– The spine, or vertebral column, is a complex bony and soft-tissue structure that provides axial support, protects the spinal cord, and enables flexible movement of the trunk. It consists of 33 vertebrae divided into cervical, thoracic, lumbar, sacral, and coccygeal regions, separated by intervertebral discs that act as shock absorbers. Common spinal conditions include degenerative disc disease, herniated discs, spinal stenosis, scoliosis, and vertebral fractures, which can cause pain, neurological symptoms, and impaired mobility; management ranges from physical therapy and medications to surgical intervention.

Spondylolisthesis

– Forward slippage of one vertebral body relative to the vertebra below it. Isthmic (Type II, most common in children/adolescents): defect in pars interarticularis (spondylolysis), often L5-S1. Degenerative (Type III, most common in adults): facet joint and disc degeneration, often L4-L5. Presents with low back pain, radicular leg pain, and hamstring tightness. High-grade slips: may cause postural deformity and neurologic deficits. Diagnosis: standing lateral X-ray (Myerding grading: I-V based on percentage slip), MRI. Treatment: conservative (PT, bracing, activity modification) for low-grade. Surgical: reduction and fusion for high-grade, progressive slip, or persistent pain.

Sprain

– An injury to a ligament (connective tissue connecting bone to bone) caused by excessive stretching or tearing. Grading: Grade I (mild stretching, microscopic tears), Grade II (partial tear with laxity), Grade III (complete tear with instability). Common sites: ankle (lateral ligaments most common), knee (ACL, MCL), wrist, and thumb (gamekeeper's thumb). Presents with pain, swelling, bruising, and joint instability (Grades II-III). Diagnosis: clinical, stress testing, MRI for confirmation. Treatment: RICE (rest, ice, compression, elevation), NSAIDs, immobilization, physical therapy. Grade III injuries may require surgical reconstruction.

Sprengel Deformity

– A congenital elevation of the scapula due to failure of the scapula to descend from its embryonic cervical position to its normal thoracic position during weeks 9-12 of gestation. The scapula is higher, smaller, and rotated inferiorly compared to the normal side. Range of motion is limited, especially in shoulder abduction. Often associated with Klippel-Feil syndrome, scoliosis, and cervical ribs. An omovertebral bone (fibrous, cartilaginous, or bony connection between the scapula and cervical spine) is present in

30-50% of cases. Physical exam shows an elevated scapula with a webbed-appearing neck on the affected side. X-ray confirms the elevated scapula and identifies associated cervical anomalies. Treatment: observation for mild cases; surgical correction (Woodward or Green procedure) for moderate to severe cases with functional or cosmetic concerns, ideally performed at 3-8 years of age.

Strain

– An injury to a muscle or tendon (connective tissue connecting muscle to bone) caused by excessive stretching, contraction, or overload. Grading: Grade I (mild stretching with few muscle fibers torn), Grade II (moderate partial tear), Grade III (complete muscle/tendon rupture). Common sites: hamstring, quadriceps, calf (gastrocnemius), lower back, and rotator cuff. Presents with acute pain, swelling, muscle spasm, and loss of function. Diagnosis: clinical, ultrasound or MRI for Grade II-III. Treatment: RICE (rest, ice, compression, elevation), NSAIDs, gradual return to activity with physical therapy. Grade III ruptures (e.g., Achilles tendon) may require surgical repair.

Syndactyly

– A congenital anomaly where two or more digits are fused together by skin (simple syndactyly) or by both skin and bone (complex syndactyly). Most common congenital hand anomaly, often bilateral. May be isolated (autosomal dominant, incomplete penetrance) or syndromic (Poland, Apert, Crouzon, Holt-Oram syndromes). The third web space is most commonly affected. Physical exam assesses the extent of fusion, nail involvement, and functional limitation. X-ray evaluates the degree of bony fusion and associated anomalies. Treatment: surgical separation (syndactyly release) performed at 6-18 months to prevent angular growth deformity. Z-plasties or skin grafts are used to reconstruct the web space. Separation of multiple digits requires staging to avoid vascular compromise.

Tendinitis

– Inflammation of a tendon, typically at its insertion (enthesis) or at the musculotendinous junction. Tendinitis is often caused by repetitive microtrauma/overuse but the term is increasingly replaced by 'tendinopathy' as histopathology shows tendinosis (collagen degeneration, neovascularisation, mucoid change) rather than acute inflammation. Common sites: rotator cuff (supraspinatus—shoulder impingement, subacromial pain), lateral epicondylitis (tennis elbow—extensor carpi radialis brevis origin), medial epicondylitis (golfer's elbow—common flexor origin), patellar tendinitis (jumper's knee—inferior pole of patella), Achilles tendinitis (insertional or mid-portion), de Quervain tenosynovitis (first dorsal compartment of the wrist—abductor pollicis longus, extensor pollicis brevis), and bicipital tendinitis (long head of biceps). Diagnosis: clinical (tenderness over the affected tendon, pain on resisted contraction, and passive stretch), ultrasound or MRI to confirm. Treatment: relative rest, activity modification, ice, NSAIDs, physiotherapy (eccentric exercises—proven effective for Achilles and patellar tendinopathy), corticosteroid injection (short-term relief, but may increase rupture risk—avoid for Achilles and patellar tendons), platelet-rich plasma (PRP) injection (evidence mixed), extracorporeal shock wave therapy, and surgery (tendon debridement, repair) for refractory cases.

Tennis Elbow

– Lateral epicondylitis, a tendinopathy of the common extensor tendon at the lateral epicondyle (extensor carpi radialis brevis). Caused by repetitive wrist extension and overuse, not limited to tennis. Presents with lateral elbow pain worsened by wrist extension and grip. Tenderness over the lateral epicondyle. Positive Cozen test and Mill test. Diagnosis: clinical; MRI or ultrasound to confirm. Treatment: rest, ice, activity modification, NSAIDs, physical therapy (eccentric exercises), and corticosteroid injection. Refractory cases: PRP injection, extracorporeal shock wave therapy, or surgical release.

Thoracolumbar Spine Trauma

– Thoracolumbar trauma: T10-L2 (thoracolumbar junction), L3-L5. AO classification: A (compression), B (distraction), C (translation/rotation). TLICS (Thoracolumbar Injury Classification and Severity): morphology (1-3), PLC (0-2), neurologic (0-3). <4 nonop, >4 op. Neurologic: complete (ASIA A) vs incomplete. Treatment: nonop (TLSO brace, early mobilization); op: posterior instrumentation (pedicle screws 1 above/1 below), decompression (laminectomy if retropulsed fragment, transpedicular or costotrcostotransversectomy for anterolateral compression). Post-op: TLSO brace, early mobilisation. Complications: infection, hardware failure, pseudarthrosis, proximal junctional kyphosis.

Thumb CMC Arthritis

– Thumb carpometacarpal (CMC) arthritis: Eaton-Littler classification: I (normal), II (slight narrowing, osteophytes <2mm), III (narrowing, osteophytes >2mm, subluxation), IV (pantrapezial arthritis). Clinical: base of thumb pain, pinch weakness, thenar wasting, squaring (subluxation). Treatment: I/II: splint, NSAIDs, corticosteroid injection, PT; III/IV: surgical: trapeziectomy (LRTI - ligament reconstruction tendon interposition with FCR/APL, suture button suspension, tightrope), total joint arttotal joint arthroplasty with hemi or total (trapezium-sparing, resurfacing).

Tibial Shaft Fractures

– Tibial shaft fractures: most common long bone fracture. AO/OTA: 42-A (simple), 42-B (wedge), 42-C (complex). Soft tissue critical: Gustilo for open. Treatment: IM nailing (gold standard, reamed, anteromedial or medial parapatellar entry, suprapatellar), plating (medial, anterolateral - for proximal/distal metaphyseal), external fixation (severe open, damage control). Compartment syndrome risk high (4 compartments). Complications: malunion (valgus/procurvatum), nonunion (exchange nail, plate + bobone grafting, exchange nailing, plate + bone graft).

Torticollis (Wryneck)

– A condition in which the head is tilted to one side (ear toward the shoulder) and rotated to the opposite side (chin toward opposite shoulder), due to contraction or shortening of the sternocleidomastoid muscle. Congenital muscular torticollis (most common): presents in infancy (2–3 weeks of age), caused by fibrosis of the sternocleidomastoid (possibly from intrauterine positioning, birth trauma, or compartment syndrome). A palpable neck mass (pseudotumour of infancy) may be present. Diagnosis: clinical, ultrasound to exclude other causes. Management: passive stretching exercises (95% resolve with physiotherapy by 12 months). If persistent after 12–18 months: surgical release of the sternocleidomastoid. Acquired torticollis: causes—infection (retropharyngeal abscess, cervical lymphadenitis—Grisel syndrome, tonsillitis, mastoiditis—inflammatory torticollis), trauma (atlantoaxial rotatory subluxation/fixation—C1–C2 rotary fixation—most common in children after minor trauma or upper respiratory infection, presents with Cock Robin position), neurological (cervical dystonia/spasmodic torticollis—idiopathic, adult onset, treated with botulinum toxin injections), ocular (superior oblique palsy—head tilt to compensate for diplopia), and drug-induced (dystonic reaction to antipsychotics—acute, responsive to anticholinergics). Red flags: fever, meningism, neurological deficit, trauma, pain, or fixed deformity.

Total Hip Arthroplasty

– THA indications: end-stage OA, inflammatory arthritis, AVN, fracture, dysplasia. Approaches: posterior (most common, dislocation risk), direct anterior (DAA, muscle-sparing, fluoroscopy), anterolateral (Watson-Jones), lateral (Hardinge). Bearings: ceramic-on-poly (standard), ceramic-on-ceramic (young, squeaking risk), metal-on-poly (historical). Fixation: cemented (elderly, osteoporotic), cementless (porous coated, hydroxyapatite, young), hybrid (cemented stem + cementless cup). Complications: ddislocation (posterior approach 1-3 percent, DAA <1 percent), infection (1-2 percent), periprosthetic fracture, aseptic loosening, leg length discrepancy, nerve injury, venous thromboembolism.

Total Knee Arthroplasty

– TKA indications: end-stage OA, inflammatory arthritis, post-traumatic. Components: femoral (cobalt-chrome), tibial (titanium tray + polyethylene insert), patellar (polyethylene, resurfacing vs not). Alignment: mechanical (neutral), kinematic (patient-specific), restricted kinematic. Ligament balancing: measured resection, gap balancing. Patellar resurfacing: selective vs routine. Complications: stiffness (manipulation under anesthesia, arthroscopic lysis), instability (collateral, PCL), patellofpatellofemoral complications, aseptic loosening, periprosthetic fracture, infection (1-2 percent), venous thromboembolism, arthrofibrosis, and extensor mechanism disruption.

Trigger Finger

– Trigger finger (stenosing tenosynovitis): A1 pulley thickening, flexor tendon nodule -> catching/locking. Clinical: triggering, palpable nodule at distal palmar crease, morning stiffness. Grades: I (pain only), II (catching, active extension), III (catching, passive extension), IV (locked flexed). Treatment: corticosteroid injection (60-70% cure, less in diabetics); percutaneous release (needle); open A1 pulley release (gold standard for refractory/recurrent). Thumb: MP flexion contracture risk.

Whiplash Injury

– An acceleration-deceleration injury to the cervical spine, most commonly resulting from rear-end motor vehicle collisions. Mechanism: sudden hyperextension followed by hyperflexion of the neck causes stretching or tearing of cervical muscles, ligaments, facet joint capsules, and annulus fibrosus of intervertebral discs. The S-shaped curve of the cervical spine during whiplash (initial straightening of the cervical lordosis followed by flexion at the lower segments and extension at the upper segments) produces shear stress on the facet joints. Clinical presentation: neck pain and stiffness (onset within 24 hours, may be delayed up to 72 hours), headache (typically occipital or suboccipital, cervicogenic origin), interscapular pain, upper limb pain and paraesthesiae (referred or radicular, from nerve root stretching), dizziness, fatigue, and temporomandibular joint pain. Quebec Task Force Classification: Grade 0 (no neck complaint, no physical sign), Grade I (neck pain, stiffness, tenderness only, no neurological signs), Grade II (neck complaint AND musculoskeletal signs — decreased range of motion, point tenderness), Grade III (neck complaint AND neurological signs — decreased or absent deep tendon reflexes, weakness, sensory deficits), Grade IV (neck complaint AND fracture or dislocation on imaging). Chronic whiplash syndrome: persistent pain and disability beyond 6 months, associated with high initial pain intensity, female sex, older age, and psychological factors (catastrophizing, fear avoidance). Diagnosis: clinical; imaging (cervical spine X-ray, CT, MRI) indicated if red flags present (fracture suspicion, neurological deficit, age >65, high-energy mechanism). Management: active treatment (early mobilisation, neck exercises, NSAIDs, muscle relaxants) is superior to immobilisation (soft collar). Cervical collars should be discontinued within 72 hours. Physiotherapy, manual therapy, and acupuncture for persistent pain. Prognosis: 50% recover within 3 months, 75% by 6 months. Chronic pain develops in 25%.

Wrist Drop

– See Radial Nerve Palsy (Wrist Drop).

Chapter 25

Pathology

Abscess

– An abscess is a localized collection of pus within a tissue or organ, resulting from pyogenic bacterial infection that has been walled off by the immune system. The pus consists of dead neutrophils, liquefied necrotic tissue, bacteria, and cellular debris. Abscesses form when the body's immune system attempts to contain an infection, creating a cavity lined by granulation tissue and surrounded by a fibrous capsule. The most common causative organism is Staphylococcus aureus for skin and soft tissue abscesses, and Escherichia coli for intra-abdominal abscesses. Treatment requires surgical drainage and appropriate antibiotic therapy.

Acute Inflammation

– A rapid, predominantly non-specific immune response to tissue injury or infection that typically lasts hours to days and involves vascular and cellular reactions. The cardinal signs are redness, heat, swelling, pain, and loss of function, resulting from increased blood flow, vascular permeability, and cellular infiltration. The process begins with vasodilation mediated by histamine, prostaglandins, and nitric oxide, followed by increased vascular permeability allowing plasma proteins and leukocytes to exit the circulation. Neutrophils are the first leukocytes recruited, followed by monocytes that transform into tissue macrophages, working together to eliminate the inciting stimulus and initiate tissue repair.

Adrenal Cortical Carcinoma

– A rare, aggressive malignancy arising from the adrenal cortex, with an incidence of approximately one to two per million population per year. The tumor may be functional, producing excessive hormones, or non-functional. Functional tumors may produce cortisol causing Cushing syndrome, aldosterone causing Conn syndrome, or androgens causing virilization in women or feminization in men. Non-functional tumors present with abdominal pain, early satiety, or are discovered incidentally on abdominal imaging. Complete surgical resection is the treatment of choice, but prognosis is poor, with five-year survival rates under 40 percent due to advanced stage at presentation and high recurrence rates.

Adult Still's Disease

– Adult Still's disease, also known as systemic-onset juvenile idiopathic arthritis when occurring in children, is a systemic inflammatory disorder characterized by high-spiking daily fevers, evanescent salmon-colored rash, arthritis, and multiorgan involvement. The pathogenesis involves excessive activation of the innate immune system with overproduction of IL-1, IL-6, IL-18, and TNF. The characteristic fever pattern is quotidian, meaning it occurs once daily typically in the evening, with rapid return to baseline between spikes. The rash is typically salmon-coloured, transient, and associated with fever spikes. Arthritis may be severe and destructive. Laboratory findings include markedly elevated ferritin levels, leukocytosis, and elevated inflammatory markers. Treatment involves NSAIDs, corticosteroids, and biologic agents targeting IL-1 or IL-6.

Air Embolism

– Air embolism occurs when air enters the venous or arterial circulation, potentially causing cardiovascular collapse and death. Venous air embolism is more common, typically occurring during neurosurgery in the sitting position, central venous catheterization or removal, chest trauma, or underwater diving. Approximately 100 milliliters of air introduced rapidly into the venous circulation can be fatal, as air bubbles coalesce in the right ventricle and create an air lock that obstructs pulmonary outflow, causing right heart strain and cardiovascular collapse. Treatment includes immediate placement in the left lateral decubitus position, administration of 100 percent oxygen, and aspiration of air from the right ventricle via a central venous catheter.

Alzheimer's Disease

– A progressive neurodegenerative disorder and the most common cause of dementia, accounting for 60 to 80 percent of cases. It is characterized pathologically by accumulation of extracellular amyloid-beta plaques and intracellular tau-containing neurofibrillary tangles in the cerebral cortex and hippocampus, along with progressive neuronal loss and cerebral atrophy. The disease typically begins with insidious memory impairment, particularly of recent events, and progresses over years with declining cognitive function, behavioural changes, and loss of independence. APOE epsilon-4 is the strongest genetic risk factor for late-onset Alzheimer disease. Definitive diagnosis requires pathological examination, though clinical criteria and biomarkers have high diagnostic accuracy.

Amyloidosis

– A group of disorders characterized by the extracellular deposition of misfolded proteins that polymerize into insoluble fibrils with a distinctive beta-pleated sheet configuration. These fibrils bind Congo red dye and show characteristic apple-green birefringence under polarized light microscopy. AA amyloidosis results from chronic inflammatory conditions with serum amyloid A protein depositing predominantly in the kidneys, spleen, and liver. AL amyloidosis involves immunoglobulin light chain deposition from plasma cell dyscrasias, affecting the heart, kidneys, and peripheral nerves. Diagnosis requires tissue biopsy with Congo red staining showing apple-green birefringence under polarised light. Treatment targets the underlying plasma cell disorder.

Anaphylaxis

– A severe, potentially life-threatening systemic allergic reaction that occurs rapidly after exposure to an allergen, typically within minutes. The mechanism involves massive systemic mast cell and basophil degranulation, releasing histamine, leukotrienes, prostaglandins, and other mediators that cause widespread vasodilation, increased vascular permeability, bronchoconstriction, and cardiac dysfunction. Clinical features include urticaria and angioedema, wheezing and stridor from laryngeal edema, hypotension, tachycardia, and potential cardiovascular collapse. First-line treatment is intramuscular epinephrine, followed by antihistamines, corticosteroids, and intravenous fluids. Prompt recognition and treatment are essential to prevent fatal outcomes.

Anaplasia

– A lack of differentiation, describing cells that have reverted to a more primitive, embryonic phenotype with loss of the specialized structural and functional characteristics of their tissue of origin. Anaplastic cells exhibit marked pleomorphism with variation in cell size and shape, hyperchromatic nuclei with coarse irregular chromatin, prominent and irregular nucleoli, and a high nuclear-to-cytoplasmic ratio exceeding 80 percent. Mitotic figures are abundant, frequently atypical, including tripolar and quadripolar forms. Anaplasia is a hallmark of malignancy and correlates with aggressive clinical behaviour, high histological grade, and poor prognosis. The most anaplastic tumors are classified as poorly differentiated or undifferentiated (grade IV).

Angiogenesis

– The formation of new blood vessels from preexisting vasculature, a process essential for wound healing, embryonic development, and pathological conditions including tumor growth. It is regulated by the balance between pro-angiogenic factors, particularly vascular endothelial growth factor, and anti-angiogenic factors like thrombospondin and endostatin. During wound healing, hypoxia in the injured tissue stimulates VEGF production through HIF-1 alpha, which activates endothelial cell proliferation, migration, and tube formation. The new vessels are immature, leaky, and lack pericyte coverage, contributing to the abnormal tumor microenvironment. Anti-angiogenic therapies targeting VEGF have been developed as anti-cancer treatments, though resistance is common.

APC Tumor Suppressor

– The adenomatous polyposis coli protein is a large multidomain tumor suppressor that functions as a critical negative regulator of the canonical Wnt signaling pathway. In the absence of Wnt ligands, APC forms a destruction complex with axin, glycogen synthase kinase-3 beta, and casein kinase 1, which phosphorylates beta-catenin, targeting it for ubiquitination and proteasomal degradation. When APC is inactivated by mutation, the destruction complex cannot degrade beta-catenin, which accumulates in the nucleus and activates transcription of pro-proliferative genes including MYC and cyclin D1. Germline mutations in APC cause familial adenomatous polyposis, with hundreds of colorectal adenomas and near-100 percent risk of colorectal cancer by age 40.

Apoptosis

– A tightly regulated form of programmed cell death that serves as a critical homeostatic mechanism in multicellular organisms, removing damaged or unnecessary cells without inciting inflammation. The intrinsic pathway is initiated by intracellular stressors such as DNA damage, oxidative stress, or growth factor withdrawal, leading to mitochondrial outer membrane permeabilization through BAX and BAK pore formation. This releases cytochrome c into the cytoplasm, which binds APAF-1 to form the apoptosome, which activates caspase-9, initiating a caspase cascade leading to DNA fragmentation and cell dismantling. The extrinsic pathway is triggered by death receptor ligation. Both pathways converge on effector caspases-3 and -7, which execute the final phases of apoptosis.

Apoptosis Extrinsic Pathway

– The extrinsic pathway of apoptosis is triggered by extracellular ligands binding to death receptors belonging to the tumor necrosis factor receptor superfamily, including Fas (CD95), TNFR1, and TRAIL receptors DR4 and DR5. Upon ligand binding, these receptors trimerize and recruit intracellular adaptor proteins including FADD and TRADD through their death domains, forming the death-inducing signaling complex. FADD then recruits procaspase-8 through death effector domain interactions, leading to autocatalytic activation of caspase-8, which then cleaves and activates effector caspases-3 and -7. The extrinsic pathway can also engage the intrinsic pathway through BID cleavage, amplifying the apoptotic signal. This pathway is critical for immune surveillance and lymphocyte homeostasis.

Apoptosis Intrinsic Pathway

– The intrinsic, or mitochondrial, pathway of apoptosis is initiated by intracellular stress signals including DNA damage, growth factor withdrawal, endoplasmic reticulum stress, and hypoxia. These signals activate BH3-only proteins such as BIM, BAD, BID, and PUMA, which neutralize anti-apoptotic BCL-2 family members or directly activate the pro-apoptotic effectors BAX and BAK. Activated BAX and BAK oligomerize and form pores in the outer mitochondrial membrane, a process called mitochondrial outer membrane permeabilisation, releasing cytochrome c and other pro-apoptotic factors. Cytochrome c binds APAF-1 to form the apoptosome, activating caspase-9. This pathway is tightly regulated by the BCL-2 family and is disrupted in many cancers, contributing to chemotherapy resistance.

Atherosclerotic Plaque

– An atherosclerotic plaque is a lesion within the arterial intima composed of a necrotic lipid core covered by a fibrous cap, representing the pathological substrate of atherosclerotic cardiovascular disease. The necrotic core contains cholesterol crystals, calcium deposits, cellular debris, and foam cells derived from lipid-laden macrophages. The fibrous cap consists of smooth muscle cells, collagen, elastin, and proteoglycans that separate the thrombogenic core from the flowing blood. The plaque may be stable, with a thick fibrous cap and small lipid core, or vulnerable, with a thin cap, large lipid core, and dense inflammation. Rupture of a vulnerable plaque exposes thrombogenic core material, triggering acute thrombosis responsible for myocardial infarction and stroke.

Autoimmune Disease

– Autoimmune diseases result from loss of immunological self-tolerance, leading to immune-mediated attack on the body's own tissues. The mechanisms include molecular mimicry, where microbial antigens share epitopes with self-antigens triggering cross-reactive immune responses; bystander activation, where tissue damage releases sequestered self-antigens; epitope spreading, where immune responses extend to additional self-epitopes; and defective regulatory T cell function. Systemic lupus erythematosus, rheumatoid arthritis, type 1 diabetes, multiple sclerosis, and inflammatory bowel disease are common examples. Treatment typically involves immunosuppressive agents including corticosteroids, disease-modifying anti-rheumatic drugs, and biologic therapies targeting specific immune pathways.

Bacterial Endocarditis Vegetations

– Vegetations in bacterial endocarditis are complex masses composed of bacteria, fibrin, and platelets that form on heart valve surfaces or endocardium. The pathogenesis begins with bacteremia seeding a site of endothelial damage or a previously abnormal valve surface. The damaged endothelium exposes subendothelial collagen and von Willebrand factor, promoting platelet adhesion and fibrin deposition, creating a non-bacterial thrombotic endocarditis nidus. Bacteria, particularly Staphylococcus aureus, Streptococcus viridans, and Enterococcus species, adhere to the thrombotic nidus and proliferate, embedded in layers of fibrin and platelets. The vegetations can embolise to distant organs and destroy valve tissue, leading to acute or subacute endocarditis with fever, heart murmur, and peripheral stigmata.

Behcet Disease

– A chronic, relapsing systemic vasculitis characterized by recurrent oral ulcers, genital ulcers, and ocular inflammation, with a distribution along the ancient Silk Road from the Mediterranean to East Asia. The pathogenesis involves inappropriate neutrophil and T-cell mediated vascular inflammation, with genetic susceptibility associated with HLA-B51. The hallmark oral aphthous ulcers are painful, recurrent, and heal without scarring. Genital ulcers typically affect the scrotum and vulva, healing with scarring. Ocular involvement manifests as anterior or posterior uveitis and retinal vasculitis, potentially causing blindness. Neurological and large-vessel involvement carries the highest morbidity. Diagnosis is clinical. Treatment includes corticosteroids, immunosuppressants, and biologic agents.

Benign Neoplasm

– Benign neoplasms are proliferations of cells that remain confined to their site of origin, grow by expansion rather than invasion, and do not metastasize to distant sites. They typically grow slowly, pushing aside adjacent normal tissue and often becoming enclosed by a fibrous pseudocapsule formed from compressed connective tissue. Histologically, benign neoplasms show well-differentiated cells with regular nuclear morphology, low mitotic activity, no atypical mitotic figures, and growth pattern that preserves normal tissue architecture. They are usually encapsulated or well-circumscribed. Clinical effects result from local compression, hormone production, or obstruction of adjacent structures. Complete surgical excision is generally curative with an excellent prognosis.

Bilirubin Metabolism

– Bilirubin is the breakdown product of heme from senescent red blood cells, metabolized primarily by the liver through a series of enzymatic steps. Heme is converted to biliverdin by heme oxygenase, then to unconjugated bilirubin by biliverdin reductase. Unconjugated bilirubin is lipophilic and transported bound to albumin to the liver. In hepatocytes, unconjugated bilirubin is conjugated with glucuronic acid by UDP-glucuronosyltransferase, forming water-soluble conjugated bilirubin that is excreted into bile. In the intestine, conjugated bilirubin is deconjugated by bacterial enzymes and reduced to urobilinogen. Most urobilinogen is excreted in feces, but a portion undergoes enterohepatic circulation and is excreted in urine. Elevation of unconjugated or conjugated bilirubin causes clinical jaundice.

Biopsy

– A medical procedure involving the removal of a small sample of tissue from the body for microscopic examination to establish a diagnosis, most commonly to differentiate benign from malignant pathology and to characterize disease processes. Biopsies are classified by technique: excisional biopsy (the entire lesion is removed, providing the most complete specimen), incisional biopsy (a portion of a larger lesion is sampled, often used for large tumors), needle biopsy (core needle using a hollow needle to obtain a core of tissue for histological analysis), fine needle aspiration (extracting cells for cytological examination), and endoscopic biopsy (during colonoscopy or bronchoscopy). Proper tissue handling, fixation, and orientation are essential for accurate pathological diagnosis.

Carcinoid Tumor

– Carcinoid tumors are well-differentiated neuroendocrine neoplasms that arise from enterochromaffin cells throughout the gastrointestinal tract and bronchial system. The most common primary sites are the appendix, small intestine, and rectum. The tumors produce biogenic amines and peptides, but clinically significant hormone secretion occurs mainly with hepatic metastases, which allow vasoactive substances to bypass hepatic first-pass metabolism. Carcinoid syndrome occurs in approximately 10 percent of patients, presenting with flushing, diarrhea, wheezing, and right-sided valvular heart disease from serotonin and other vasoactive substances. Diagnosis involves urinary 5-HIAA measurement. Surgical resection is curative for localised disease; somatostatin analogues control symptoms in metastatic disease.

Carcinoma

– A malignant neoplasm arising from epithelial cells, representing the most common class of cancer. Carcinomas originate from any epithelial tissue and are classified based on cell type, architectural pattern, and anatomical location. Squamous cell carcinoma arises from stratified squamous epithelium and commonly affects the skin, lung, esophagus, cervix, and head and neck regions. Adenocarcinoma develops from glandular epithelium and commonly affects the colon, breast, prostate, stomach, and lung. Subtypes include small cell carcinoma, large cell carcinoma, transitional cell carcinoma, and basal cell carcinoma. Carcinomas are graded by differentiation and staged by tumor size, lymph node involvement, and metastasis. Prognosis varies widely by type, stage, and grade.

Cardiomyopathy

– Diseases of the myocardium that result in cardiac dysfunction, excluding coronary artery disease, valvular disease, and congenital heart disease as primary causes. The three main types are dilated, hypertrophic, and restrictive cardiomyopathy. Dilated cardiomyopathy is characterized by ventricular dilation and impaired systolic function with reduced ejection fraction. It may be idiopathic, familial with mutations in genes encoding cytoskeletal and sarcomeric proteins, or secondary to causes such as alcoholic cardiomyopathy, chemotherapy-induced cardiotoxicity, or viral myocarditis. Hypertrophic cardiomyopathy is characterized by left ventricular hypertrophy without dilation, typically due to sarcomeric protein mutations. Restrictive cardiomyopathy involves impaired diastolic filling with preserved systolic function.

Caseous Necrosis

– A distinctive form of cell death that characteristically occurs in granulomatous infections, most notably tuberculosis. The term derives from the cheese-like, friable, whitish-yellow appearance of the necrotic tissue on gross examination. Histologically, caseous necrosis appears as amorphous, granular, eosinophilic debris without recognizable cellular outlines or tissue architecture, distinct from the structural preservation seen in coagulative necrosis. The process results from the combination of macrophage activation, T-cell mediated immunity, and bacterial products creating a hypoxic microenvironment. The necrotic material contains dead immune cells and bacterial debris. It may undergo liquefaction and cavitation, facilitating spread of infection to other sites.

Cellular Adaptation

– Cellular adaptations are reversible changes in cell morphology and function that occur in response to altered physiological demands or mild pathological conditions. Hypertrophy involves an increase in cell size through increased protein synthesis and organelle production, resulting in increased organ size, as seen in cardiac hypertrophy from chronic hypertension or skeletal muscle growth with resistance exercise. Hyperplasia involves an increase in cell number through mitotic division, as seen in the breast and endometrium in response to hormonal stimulation. Atrophy involves decreased cell size and organ mass due to reduced functional demand, inadequate blood supply, or ageing. Metaplasia is a reversible change where one differentiated cell type is replaced by another more resilient to environmental stress.

Cerebral Edema

– The accumulation of excess fluid within the brain parenchyma, leading to increased intracranial pressure and potentially fatal brain herniation. There are three main types with different pathophysiological mechanisms. Vasogenic edema results from disruption of the blood-brain barrier, allowing protein-rich fluid to enter the extracellular space; it is seen in brain tumors, abscesses, hypertensive encephalopathy, and posterior reversible encephalopathy syndrome. Cytotoxic edema results from energy failure and impaired ion pumps, causing intracellular swelling; it is seen in cerebral ischemia, hypoxia, and metabolic disorders. Interstitial edema occurs in obstructive hydrocephalus from transependymal CSF flow. Management focuses on reducing intracranial pressure through osmotic therapy, corticosteroids, or surgical decompression.

Cerebral Infarction

– Cerebral infarction, or ischemic stroke, results from occlusion of a cerebral artery causing irreversible neuronal damage. The most common cause is thromboembolism from atherosclerotic carotid or intracranial arteries, or cardioembolism from atrial fibrillation or cardiac thrombus. The brain is particularly vulnerable to ischemia due to its high metabolic rate and limited energy reserves, with neurons dying within minutes of complete ischemia. The ischemic core is surrounded by a penumbra of potentially salvageable tissue where blood flow is reduced but above the threshold for cell death. Clinical presentation depends on the affected vascular territory. Treatment includes thrombolysis within the therapeutic window, endovascular thrombectomy, and secondary prevention with antiplatelet agents and risk factor modification.

Cholangiocarcinoma

– A malignant tumor arising from the biliary epithelium, classified based on anatomical location as intrahepatic, perihilar, or distal extrahepatic. Risk factors include primary sclerosing cholangitis, particularly in association with inflammatory bowel disease, parasitic liver fluke infections, hepatolithiasis, and congenital biliary anomalies including choledochal cysts. Intrahepatic cholangiocarcinoma presents as a liver mass with abdominal pain and weight loss. Perihilar (Klatskin) tumors are the most common type and present with obstructive jaundice. Distal extrahepatic tumors have the best resectability rate. Overall prognosis is poor, with five-year survival under 20 percent. Surgical resection with negative margins offers the only chance of cure; liver transplantation may be considered in selected cases.

Cholecystitis

– Inflammation of the gallbladder, most commonly caused by obstruction of the cystic duct by a gallstone, which occurs in approximately 95 percent of cases. The obstruction leads to bile stasis, increased intraluminal pressure, and ischemia of the gallbladder wall, which may progress to bacterial infection. Acute cholecystitis presents with right upper quadrant pain, fever, nausea, and vomiting, with physical findings including Murphy's sign, arrest of inspiration during palpation of the right upper quadrant. Ultrasound findings include gallbladder wall thickening, distension, pericholecystic fluid, and a positive sonographic Murphy sign. Treatment involves intravenous antibiotics and cholecystectomy, typically performed laparoscopically either early or delayed depending on clinical severity.

Chronic Granulomatous Disease

– A primary immunodeficiency disorder caused by defects in the NADPH oxidase enzyme complex of phagocytes, resulting in inability to generate the respiratory burst needed for intracellular killing of microorganisms. The disease is most commonly X-linked, caused by mutations in the CYBB gene encoding gp91phox, or autosomal recessive, involving mutations in other components of the NADPH oxidase complex. Without the respiratory burst, phagocytes can ingest but cannot effectively kill catalase-positive organisms, particularly Staphylococcus aureus, Aspergillus species, and gram-negative enteric bacteria. Patients present with recurrent infections and granuloma formation in the lungs, skin, and lymph nodes. Management includes prophylactic antibiotics, antifungals, and interferon-gamma. hematopoietic stem cell transplantation may be curative.

Chronic Inflammation

– A prolonged inflammatory response characterized by simultaneous tissue destruction and attempts at repair, lasting weeks to years. Unlike acute inflammation, it is predominantly mononuclear, involving macrophages, lymphocytes, and plasma cells rather than neutrophils. The process is driven by persistent infections such as tuberculosis, autoimmune reactions as in rheumatoid arthritis, prolonged exposure to toxic agents including silica, or non-degradable foreign bodies. Chronic inflammation is characterized by mononuclear cell infiltration, tissue destruction, and fibrosis. Granuloma formation represents a specialized pattern seen in tuberculosis, sarcoidosis, and foreign body reactions. Chronic inflammation underlies many common diseases including rheumatoid arthritis and atherosclerosis.

Cirrhosis

– The end stage of chronic liver disease, characterized by diffuse fibrosis with distortion of the hepatic architecture and formation of regenerative nodules. The most common causes include chronic hepatitis C infection, alcoholic liver disease, non-alcoholic steatohepatitis, chronic hepatitis B infection, and autoimmune hepatitis. The pathogenesis involves repeated cycles of hepatocyte injury, inflammation, and repair, with activation of hepatic stellate cells that produce excessive extracellular matrix. The resulting fibrosis disrupts normal blood flow and hepatocyte function, causing portal hypertension and hepatic synthetic dysfunction. Clinical features include jaundice, ascites, variceal bleeding, hepatic encephalopathy, and coagulopathy. Cirrhosis is irreversible and increases the risk of hepatocellular carcinoma.

Coagulation Cascade

– The coagulation cascade is a series of enzymatic reactions leading to fibrin clot formation, involving serine protease activation cascades. The intrinsic pathway is initiated by contact activation when factor XII binds exposed collagen, sequentially activating factors XI, IX, and the cofactor VIII. The extrinsic pathway is initiated when tissue factor released from damaged cells binds factor VII, forming the TF-VIIa complex that activates factor X. Both pathways converge at factor X activation, forming the prothrombinase complex that converts prothrombin to thrombin. Thrombin cleaves fibrinogen to fibrin, which polymerises to form a stable clot. The cascade is regulated by natural anticoagulants including antithrombin III, protein C, and protein S. Laboratory assessment uses prothrombin time and activated partial thromboplastin time.

Coagulative Necrosis

– The most common form of necrosis, typically resulting from ischemic injury or infarction in most solid organs except the brain. The fundamental process involves denaturation of both structural proteins and enzymatic proteins, which prevents the immediate liquefaction of the dead tissue. The affected tissue maintains its basic structural outline for several days, a phenomenon known as structural preservation that allows histological identification of the original organ architecture. Over time, the necrotic tissue is removed by phagocytes and replaced by granulation tissue, which matures into a fibrous scar. Coagulative necrosis is the typical pattern in myocardial infarction, renal infarction, and splenic infarction, where the organ outline is preserved histologically for days to weeks.

Colorectal Adenocarcinoma

– The third most common cancer worldwide and the second leading cause of cancer death, arising from the colonic epithelium through the adenoma-carcinoma sequence over 10 to 15 years. The pathogenesis involves sequential accumulation of genetic mutations beginning with APC loss, followed by KRAS activation, TP53 loss, and other alterations. Approximately 15 percent of cases are associated with hereditary syndromes including Lynch syndrome with mismatch repair gene mutations, and familial adenomatous polyposis with APC mutation. Sporadic cases account for the majority. Screening via colonoscopy with polypectomy reduces both incidence and mortality. Treatment includes surgical resection, chemotherapy, and targeted therapies including anti-EGFR and anti-VEGF agents based on tumor molecular profile.

Cyanosis

– A bluish discoloration of the skin and mucous membranes resulting from excessive amounts of deoxygenated hemoglobin in the capillary blood. It becomes clinically apparent when the concentration of deoxygenated hemoglobin exceeds 5 grams per deciliter, which corresponds to an oxygen saturation of approximately 85 percent. Central cyanosis affects the entire body and results from arterial hypoxemia, as seen in congenital heart disease with right-to-left shunting, severe pulmonary disease, and severe pneumonia. Peripheral cyanosis affects the extremities and results from reduced blood flow with increased oxygen extraction, as seen in cold exposure, shock, and peripheral vascular disease. Differential cyanosis, affecting only the lower body, can indicate specific congenital heart defects such as patent ductus arteriosus with Eisenmenger syndrome.

Cyst

– An abnormal closed cavity or sac lined by epithelium, containing liquid, semisolid, or gaseous material, and distinguished from an abscess (which contains pus) and a pseudocyst (which lacks an epithelial lining). Cysts occur in virtually every organ and tissue and are classified by pathogenesis: retention cysts (from obstruction of a duct, e.g., sebaceous cyst, ovarian follicular cyst, breast cyst), developmental cysts (arising from embryonic remnants, e.g., branchial cleft cyst, thyroglossal duct cyst, bronchogenic cyst), parasitic cysts (e.g., hydatid cyst from Echinococcus), and neoplastic cysts (e.g., cystic teratoma). Diagnosis is by imaging. Treatment depends on the type, size, location, and symptoms, ranging from observation to complete surgical excision.

Dysplasia

– Disordered cellular growth and maturation characterized by structural abnormalities within a tissue, representing a pre-neoplastic condition that may progress to invasive carcinoma. At the cellular level, dysplasia shows nuclear enlargement, hyperchromasia with clumped chromatin, irregular nuclear contours, increased nuclear-to-cytoplasmic ratio, and prominent nucleoli. Architecturally, there is loss of normal tissue organization, with cells failing to mature properly as they migrate from the basal layer to the surface. Dysplasia is graded as mild, moderate, or severe based on the extent of cytologic and architectural abnormalities. Severe dysplasia may progress to carcinoma in situ, where abnormal cells involve the full epithelial thickness but have not breached the basement membrane.

Dystrophic Calcification

– The deposition of calcium phosphate salts in damaged, necrotic, or degenerating tissues despite normal serum calcium levels. It occurs in areas of tissue injury, inflammation, or chronic degeneration where the local tissue environment favors calcium precipitation. The mechanism involves release of calcium-binding phospholipids and matrix vesicles from damaged cell membranes, which serve as nucleation sites for hydroxyapatite crystal formation. Common sites include atherosclerotic plaques, heart valves in degenerative aortic stenosis, and areas of previous trauma or infection. Dystrophic calcification can impair organ function, for example by restricting valve leaflet motion. It is distinct from metastatic calcification, which occurs in normal tissues in the setting of hypercalcaemia.

Ecchymosis

– An ecchymosis is a flat, purple or blue discoloration of the skin larger than 10 millimeters caused by extravasation of blood from damaged blood vessels into the subcutaneous tissue. Unlike petechiae, which result from platelet disorders, ecchymoses may result from coagulopathies, trauma, or vascular fragility. They are commonly seen in the elderly due to skin thinning and increased vascular fragility, as well as in patients taking anticoagulant or antiplatelet medications. The color of an ecchymosis evolves through characteristic changes over days to weeks: initially red-purple from deoxygenated hemoglobin, then blue-black, green from biliverdin, and finally yellow-brown from haemosiderin before complete resolution. The color progression helps estimate the age of the bruise.

Edema

– The accumulation of excess fluid in the interstitial tissue spaces, resulting from disturbances in Starling forces, increased vascular permeability, or impaired lymphatic drainage. Transudates form when hydrostatic pressure increases or oncotic pressure decreases, producing fluid with low protein content and few inflammatory cells. Common causes of transudative edema include congestive heart failure, liver cirrhosis with hypoalbuminemia, and nephrotic syndrome. Exudates form when vascular permeability increases, producing fluid with high protein content and inflammatory cells, as seen in inflammation, infection, and malignancy. Lymphoedema results from impaired lymphatic drainage, commonly after lymphatic obstruction from surgery, radiation, or filariasis. edema impairs tissue perfusion, delays wound healing, and increases infection risk.

Embolism

– An embolism is the occlusion of a blood vessel by a detached mass that has traveled through the bloodstream from its site of origin. The most common form is thromboembolism, where a fragment of a thrombus dislodges and travels to a distant site through the venous or arterial circulation. Other types of emboli include fat emboli from bone fractures, amniotic fluid emboli during pregnancy, air emboli from trauma or surgery, tumor emboli from malignancy, foreign body emboli from injected materials, and cholesterol emboli from atherosclerotic plaque rupture. Clinical consequences depend on the size, composition, and location of the embolus. Pulmonary embolism is the most common life-threatening form. Treatment includes anticoagulation, thrombolysis, or embolectomy in selected cases.

Exudate

– An exudate is an inflammatory fluid with high protein content and abundant cells, resulting from increased vascular permeability during inflammation. Exudates are typically caused by bacterial infections, autoimmune diseases, viral infections, or malignancy. The fluid is often cloudy or turbid due to the presence of inflammatory cells, particularly neutrophils in bacterial infections. Exudates have high specific gravity above 1.020 and contain more than 300 cells per microliter. They are rich in fibrin and inflammatory mediators and are distinguished from transudates by Light's criteria: pleural fluid protein-to-serum protein ratio greater than 0.5, pleural fluid LDH-to-serum LDH ratio greater than 0.6, or pleural fluid LDH greater than two-thirds the upper limit of normal for serum.

Fat Embolism

– Fat embolism occurs when fat globules enter the bloodstream, typically following long bone fractures, orthopedic surgery, or severe trauma involving adipose tissue. The fat emboli travel through the venous circulation to the lungs, where they cause mechanical obstruction of pulmonary capillaries and chemical inflammation as lipases break down neutral fat into toxic free fatty acids. The pulmonary manifestations range from mild hypoxia to acute respiratory distress syndrome with severe hypoxemia and acute respiratory distress syndrome. Systemic embolisation causes petechial rash, neurological symptoms including confusion and focal deficits, and retinal changes. Diagnosis is clinical, supported by imaging. Treatment is supportive with oxygenation, ventilation, and hemodynamic support.

Fat Necrosis

– A specialized form of necrosis that occurs in adipose tissue, most commonly following acute pancreatitis when activated pancreatic lipases escape into the surrounding tissues and digest triglycerides in mesenteric or subcutaneous fat cells. The released fatty acids combine with calcium ions to form chalky-white deposits known as soap-stone lesions or saponification, which are visible on gross examination as firm, white, opaque areas. Histologically, fat necrosis shows shadowy outlines of necrotic adipocytes with basophilic calcium deposits. Over time, the necrotic fat is replaced by fibrosis and chronic inflammation. In the breast, fat necrosis can mimic carcinoma clinically and radiographically, often requiring biopsy for definitive diagnosis.

Fever

– An elevation of body temperature above the normal daily variation, typically defined as a core temperature exceeding 38 degrees Celsius or 100.4 degrees Fahrenheit. It results from resetting of the hypothalamic thermoregulatory set point upward, mediated by pyrogenic substances. Exogenous pyrogens include microbial products such as lipopolysaccharide, which stimulate immune cells to produce endogenous pyrogens. The major endogenous pyrogens are IL-1, IL-6, TNF, and interferon, which act on the hypothalamus to increase the thermoregulatory set point via prostaglandin E2 synthesis. Fever enhances immune function by increasing neutrophil and T-cell activity and inhibiting pathogen growth. Antipyretic treatment is reserved for patients with significant discomfort or underlying cardiovascular or neurological disease.

Fibrinoid Necrosis

– A specialized pattern of vascular necrosis characterized by the deposition of immune complexes, fibrin, and plasma proteins within the walls of blood vessels, creating an amorphous, bright pink, smudgy eosinophilic appearance on hematoxylin and eosin staining. This pattern is characteristic of immune-mediated vascular injury, occurring in several important clinical conditions. In malignant hypertension, severely elevated blood pressure above 200/120 millimeters of mercury causes direct arterial wall damage, with plasma proteins leaking into the damaged vessel wall. In polyarteritis nodosa, immune complex deposition in small and medium arteries triggers fibrinoid necrosis. In systemic lupus erythematosus, similar immune complex-mediated vascular damage occurs. Affected vessels may thrombose, leading to tissue ischemia or rupture.

Fibrinous Pericarditis

– Inflammation of the pericardium characterized by deposition of fibrin on the pericardial surfaces, giving them a rough, shaggy appearance described as bread-and-butter pericarditis. It is the most common form of acute pericarditis, typically occurring after myocardial infarction, uremia, systemic lupus erythematosus, radiation therapy, or post-cardiac surgery syndrome. The fibrinous exudate consists of fibrin strands deposited on the visceral and parietal pericardial surfaces. The fibrin may organize into fibrous adhesions if the underlying process persists, potentially obliterating the pericardial space and causing constrictive pericarditis. Clinical presentation includes sharp retrosternal chest pain that improves with sitting forward, pericardial friction rub on auscultation, and diffuse ST-segment elevation on ECG.

Fibrosis

– The excessive deposition of extracellular matrix components, primarily collagen types I and III, in response to chronic tissue injury. It represents a pathological wound healing response that leads to organ scarring and dysfunction. The process is driven by activated fibroblasts and myofibroblasts, which are stimulated by growth factors including TGF-beta, PDGF, and CTGF produced by inflammatory cells and injured epithelium. Fibrosis can occur in any organ, but is particularly problematic in the liver (cirrhosis), lungs (pulmonary fibrosis), heart (myocardial fibrosis), and kidneys (renal fibrosis). Unlike normal wound healing, fibrosis involves excessive extracellular matrix that distorts tissue architecture and impairs organ function. Fibrosis is often progressive and currently lacks effective treatments beyond addressing the underlying cause.

Fistula

– A fistula is an abnormal epithelial-lined connection between two body surfaces or organs that normally do not communicate. Fistulas may be congenital, such as tracheoesophageal fistula, or acquired, resulting from inflammation, infection, surgery, trauma, or malignancy. Common acquired types include enterocutaneous fistulas between the intestine and skin, seen after surgery or in Crohn's disease; perianal fistulas following anorectal abscesses; and arteriovenous fistulas between arteries and veins from traumatic injury or congenital malformations. Fistulas may cause continuous leakage of contents, infection, electrolyte disturbances, and malnutrition. Diagnosis involves imaging including fistulography, CT, or MRI. Treatment depends on the site and cause, ranging from medical therapy to surgical closure.

Free Radical Injury

– Free radicals are highly reactive molecules with unpaired electrons that can damage cellular components through oxidative stress. Reactive oxygen species include the superoxide anion, hydrogen peroxide, and the hydroxyl radical generated through the Fenton reaction. Sources include normal mitochondrial oxidative metabolism, the NADPH oxidase respiratory burst in neutrophils, cytochrome P450 reactions, and external sources including ionizing radiation and ultraviolet light. Free radical damage to lipids initiates lipid peroxidation of polyunsaturated fatty acids in cell membranes, disrupting membrane integrity and function. Protein damage causes enzyme inactivation and receptor dysfunction. DNA damage leads to strand breaks and base modifications. Antioxidant defences include superoxide dismutase, catalase, glutathione peroxidase, and vitamins C and E.

Gangrene

– Death of tissue due to loss of blood supply, typically affecting the extremities and bowel, and represents a form of coagulative necrosis. Dry gangrene occurs in tissues with gradual arterial occlusion, where the dead tissue becomes dry, shrunken, and dark brown or black. It is commonly seen in the toes and feet of patients with diabetes mellitus and peripheral arterial disease. The black color results from hemoglobin degradation and iron sulfide deposition in the necrotic tissue. A clear line of demarcation typically separates viable from non-viable tissue. Wet gangrene results from bacterial superinfection of ischemic tissue, producing a swollen, foul-smelling, liquefied mass with systemic toxicity and a less distinct line of demarcation. Gas gangrene is a specific form caused by Clostridium species, particularly

C. perfringens, producing gas within tissues and representing a surgical emergency requiring urgent debridement and antibiotics.

Gas Gangrene

– A life-threatening bacterial infection characterized by the presence of gas in tissues (subcutaneous emphysema) and rapid myonecrosis. Caused by Clostridium perfringens (type A, 80% of cases), C. novyi, C. septicum, and C. histolyticum–Gram-positive, spore-forming, obligate anaerobic bacilli that produce exotoxins (alpha-toxin: phospholipase C, destroys cell membranes; theta-toxin: cytolysin; collagenase, hyaluronidase, DNase). Predisposing factors: contaminated wounds (soil, foreign bodies), trauma (crush injuries, deep lacerations), devascularised tissue, immunosuppression, and abdominal surgery. Incubation period: 6 hours to 6 days. Clinical features: sudden onset of severe pain at wound site, discoloured (bronze or purplish) skin, bullae filled with serosanguinous fluid, crepitus (gas in tissue, palpable), foul-smelling discharge, tachycardia (out of proportion to fever), hypotension, renal failure, hemolytic anemia, sepsis, and rapidly progressive necrosis. Diagnosis: clinical (crepitus, gas on X-ray/CT, Gram stain of exudate showing Gram-positive rods with absent white cells), culture, and PCR. Treatment: emergent surgical debridement (fasciotomy and excision of all necrotic tissue, often requiring amputation), high-dose intravenous penicillin G plus clindamycin (metronidazole as alternative), hyperbaric oxygen therapy (adjunctive, suppresses toxin production by raising tissue oxygen tension). Mortality: 20-30% overall, >50% with trunk involvement or delay in treatment.

Glomerulonephritis

– An inflammatory disease of the glomeruli, typically immune-mediated. Classification by clinical presentation: nephritic syndrome (acute onset: haematuria, red cell casts, hypertension, oliguria, edema) vs nephrotic syndrome (insidious: heavy proteinuria >3.5 g/day, hypoalbuminaemia, hyperlipidaemia, generalised edema). Classification by histology (light microscopy, immunofluorescence, electron microscopy): (1) IgA nephropathy (Berger disease): most common worldwide, mesangial IgA deposits, post-infectious triggers, variable progression. (2) Post-streptococcal GN: immune complex deposition after group A streptococcal infection, subepithelial hump-like deposits. (3) Membranous GN: subepithelial IgG deposits, PLA2R antibodies, 80% cause of nephrotic syndrome in adults. (4) Minimal change disease: podocyte foot process effacement on EM, most common nephrotic syndrome in children. (5) Focal segmental glomerulosclerosis (FSGS): scarring of some glomeruli, progressive to ESRD. (6) Membranoproliferative GN (MPGN): mesangial proliferation and basement membrane thickening. (7) Rapidly progressive GN (RPGN/crescentic GN): crescents in Bowman's space, medical emergency. (8) Lupus nephritis: immune complex deposits in SLE, WHO/ISN-RPS classification (I-VI).

Granuloma

– A granuloma is an organized collection of macrophages, often modified into epithelioid cells, typically surrounded by a cuff of lymphocytes and occasionally by fibroblasts and multinucleated giant cells. Granulomas represent a distinctive pattern of chronic inflammation that forms in response to persistent stimuli that resist elimination by ordinary phagocytosis and acute inflammation. The two main types are caseating granulomas, which have a central area of necrosis resembling cheese on gross examination (tuberculoma), and non-caseating granulomas, which lack central necrosis and are characteristic of sarcoidosis, Crohn's disease, and foreign body reactions. Granuloma formation requires T-cell mediated immunity, with interferon-gamma playing a central role in macrophage activation and epithelioid transformation.

Granulomatous Inflammation

– A distinctive pattern of chronic inflammation characterized by organized collections of activated macrophages that have undergone morphological transformation into epithelioid cells, often surrounded by a rim of lymphocytes and occasionally fibroblasts. Epithelioid cells have abundant eosinophilic cytoplasm and cell borders that blend together, resembling epithelial cells. Multinucleated giant cells, formed by fusion of macrophages under the influence of cytokines including interferon-gamma, form around persistent stimuli such as Mycobacterium tuberculosis, fungi, foreign bodies, and in sarcoidosis. The granuloma walls off the inciting agent and contains the infection, but the inflammatory response also causes tissue destruction and fibrosis in surrounding tissues.

Hemorrhage

– Extravasation of blood from the cardiovascular system (heart, arteries, veins, capillaries). Classification by severity: class 1 (<15% blood loss, compensated, minimal symptoms), class 2 (15-30%, mild tachycardia, decreased pulse pressure), class 3 (30-40%, significant tachycardia, hypotension, pallor, oliguria, altered mental status–decompensated), class 4 (>40%, severe hypotension, anuria, loss of consciousness–immediately life-threatening). Classification by location: external (visible from wound, or body orifice–epistaxis, haemoptysis, hematemesis, melaena, haematuria, vaginal) vs internal (concealed, into body cavities–intracranial, intrathoracic/ haemothorax, intra-abdominal/ haemoperitoneum, retroperitoneal, haemarthrosis, muscle haematoma). Causes: trauma (most common cause of preventable death from exsanguination), ruptured aneurysm (AAA, cerebral berry aneurysm, splenic artery), gastrointestinal bleeding (peptic ulcer, varices, diverticulosis, angiodysplasia, cancer), obstetric hemorrhage (postpartum hemorrhage–atomic uterus, placental abruption, placenta praevia), surgery, anticoagulation, coagulopathy (hemophilia, DIC, vitamin K deficiency). Management: ABC (airway, breathing, circulation), direct pressure for external bleeding, IV access (two large-bore cannulae), fluid resuscitation (crystalloids, blood products–massive transfusion protocol: 1:1:1 ratio of PRBCs:FFP:platelets), pressure control (tourniquet for life-threatening extremity hemorrhage, pelvic binder for pelvic fracture), reversal of anticoagulation (vitamin K, fresh frozen plasma, PCC–prothrombin complex concentrate, protamine for heparin, tranexamic acid), damage control surgery (abbreviated laparotomy, packing to control hemorrhage, angioembolisation, endovascular stenting).

Hemosiderin

– A golden-brown, granular, iron-containing pigment derived from the degradation of hemoglobin and stored within macrophages as an insoluble complex. It forms when red blood cells are phagocytosed and degraded, typically in conditions of hemorrhage, hemolysis, or excessive red blood cell destruction. The iron from hemoglobin is initially stored as soluble ferritin, but under conditions of iron overload, ferritin aggregates to form hemosiderin, which is more insoluble and visible on routine haematoxylin and eosin sections. The Prussian blue stain specifically identifies iron in haemosiderin deposits. Common sites include the lungs in chronic left heart failure (heart failure cells), the spleen and liver in chronic hemolytic anemia, and at sites of old hemorrhage.

Huntington's Disease

– An autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion in the huntingtin gene on chromosome 4p16.3. The disease is characterized by the triad of progressive chorea, cognitive decline, and psychiatric manifestations. The pathogenesis involves accumulation of mutant huntingtin protein with expanded polyglutamine tract in neurons, particularly in the caudate nucleus and putamen, leading to transcriptional dysregulation, mitochondrial dysfunction, excitotoxicity, and oxidative stress. The disease typically manifests in mid-adulthood with progressive chorea, cognitive decline, and psychiatric symptoms. The number of CAG repeats correlates inversely with age of onset. There is no cure; treatment is supportive and symptom-focused, with tetrabenazine for chorea.

Hyperaldosteronism

– Hyperaldosteronism, also known as Conn syndrome when caused by an adrenal adenoma, is characterized by excessive aldosterone production causing hypertension, hypokalemia, and metabolic alkalosis. Primary hyperaldosteronism results from autonomous aldosterone production by the adrenal gland, most commonly from a unilateral aldosterone-producing adenoma or bilateral adrenal hyperplasia. The excessive aldosterone acts on the distal nephron, promoting sodium reabsorption and potassium and hydrogen ion secretion, leading to hypokalaemia and metabolic alkalosis. Hypertension results from sodium and water retention. Diagnosis involves measuring the plasma aldosterone-to-renin ratio, followed by confirmatory suppression testing. Treatment includes surgical resection for adenomas and mineralocorticoid receptor antagonists for bilateral hyperplasia.

Hypersensitivity Type I

– Type I hypersensitivity, also known as immediate hypersensitivity or allergic reactions, is mediated by IgE antibodies and mast cells, producing symptoms within minutes of allergen re-exposure. During sensitization, initial allergen exposure activates Th2 cells that secrete IL-4 and IL-13, promoting B cell class switching to IgE production. IgE antibodies bind with high affinity to Fc epsilon RI receptors on mast cells and basophils, priming these cells for re-exposure. Upon subsequent allergen exposure, IgE cross-linking on mast cells triggers degranulation, releasing histamine, tryptase, leukotrienes, and prostaglandins that cause vasodilation, bronchoconstriction, and increased vascular permeability. Clinical manifestations range from mild allergic rhinitis and urticaria to life-threatening anaphylaxis. Treatment includes antihistamines, corticosteroids, and epinephrine.

Hypersensitivity Type II

– Type II hypersensitivity, also known as cytotoxic hypersensitivity, is mediated by IgG or IgM antibodies directed against antigens on cell surfaces or the extracellular matrix, causing tissue damage through complement activation, antibody-dependent cellular cytotoxicity, and phagocytosis. When antibodies bind to cell-surface antigens, the classical complement pathway is activated, generating C3b for opsonization and the membrane attack complex for direct cell lysis. Antibody-coated cells are also killed by NK cells through antibody-dependent cellular cytotoxicity. Clinical examples include autoimmune hemolytic anemia, Goodpasture syndrome, myasthenia gravis, and immune thrombocytopenia. In drug-induced hemolytic anemia, drugs such as penicillin act as haptens, binding to red cell surfaces and triggering antibody-mediated destruction.

Hypersensitivity Type III

– Type III hypersensitivity is caused by deposition of antigen-antibody immune complexes in tissues and blood vessel walls, activating complement and recruiting neutrophils that release destructive enzymes and reactive oxygen species. This reaction occurs when there is a slight excess of antigen relative to antibody, forming small, soluble immune complexes that are not efficiently cleared by the reticuloendothelial system. The complexes deposit in vessel walls, particularly in the kidneys, joints, and skin, causing vasculitis, arthritis, and glomerulonephritis. Clinical examples include systemic lupus erythematosus (DNA-anti-DNA complexes), post-streptococcal glomerulonephritis, serum sickness, and the Arthus reaction. Chronic immune complex deposition leads to persistent inflammation and progressive tissue damage.

Hypersensitivity Type IV

– Type IV hypersensitivity, also known as delayed-type hypersensitivity, is mediated by T lymphocytes rather than antibodies and takes 48 to 72 hours to develop after antigen exposure. The reaction involves sensitized CD4 Th1 cells that, upon re-exposure to the antigen presented by dendritic cells or macrophages, release pro-inflammatory cytokines including interferon-gamma and TNF, which activate macrophages and cause tissue damage. Contact dermatitis is the most common clinical example, occurring after contact with poison ivy, nickel, and other sensitising chemicals. The tuberculin reaction (PPD test) is another classic example. Type IV hypersensitivity is also involved in allograft rejection and many autoimmune diseases. It is characterized by a delayed onset of 24 to 72 hours.

Hypovolemic Shock

– A life-threatening condition resulting from inadequate circulating blood volume, leading to decreased venous return, reduced cardiac output, and tissue hypoperfusion. The most common causes are hemorrhage from trauma, gastrointestinal bleeding, or ruptured aortic aneurysm, and severe dehydration from vomiting, diarrhea, burns, or third-space fluid sequestration. The pathophysiology involves activation of the sympathetic nervous system, causing tachycardia, peripheral vasoconstriction, and increased heart rate to maintain blood pressure and vital organ perfusion. As shock progresses, compensatory mechanisms fail, leading to hypotension, oliguria, altered mental status, and metabolic acidosis. Treatment involves rapid volume resuscitation with crystalloids and blood products, control of bleeding, and treatment of the underlying cause.

Increased Vascular Permeability

– A fundamental feature of acute inflammation that allows plasma proteins, fluid, and leukocytes to exit the vascular space and enter the extravascular tissue.

The most common mechanism involves endothelial cell contraction caused by histamine, leukotrienes C4 and D4, and complement components C3a and C5a, creating intercellular gaps between endothelial cells that permit passage of protein-rich fluid. This response occurs predominantly in postcapillary venules and is immediate but short-lived, lasting 15 to 30 minutes. Other mechanisms include direct endothelial damage (burns, radiation, toxins) causing immediate and prolonged leakage, leukocyte-mediated damage via toxic oxygen species, and increased transcytosis through vesicular transport.

Inflammatory Myopathies

– A group of autoimmune disorders characterized by chronic muscle inflammation and progressive proximal muscle weakness. The three main types are dermatomyositis, polymyositis, and inclusion body myositis. Dermatomyositis is characterized by perifascicular atrophy of muscle fibers, perivascular inflammation, and characteristic skin findings including heliotrope rash on the eyelids, Gottron papules over the knuckles, and mechanic hands. It is associated with complement-mediated microangiopathy with membrane attack complex deposition in endomysial capillaries. Polymyositis shows endomysial inflammation with CD8 T cells invading non-necrotic muscle fibers. Inclusion body myositis, the most common type in patients over 50, features rimmed vacuoles and protein aggregates. Diagnosis involves EMG, muscle biopsy, and elevated creatine kinase.

Interstitial Nephritis

– Inflammation of the renal interstitium and tubules, which can be acute or chronic. Acute interstitial nephritis is most commonly caused by medications, particularly NSAIDs, antibiotics including penicillins and cephalosporins, and proton pump inhibitors. The pathogenesis involves immune-mediated hypersensitivity reaction to the offending drug, with T-cell and eosinophil infiltration of the renal interstitium. Clinical features include acute kidney injury occurring days to weeks after starting the offending medication, with fever, rash, eosinophilia, and sterile pyuria. Renal biopsy shows interstitial inflammation with eosinophils. Treatment involves discontinuation of the causative drug and corticosteroids in severe cases. Most patients recover renal function fully.

Ischemia-Reperfusion Injury

– The paradoxical tissue damage that occurs when blood flow is restored to previously ischemic tissue, a phenomenon with major clinical significance in myocardial infarction treatment, stroke management, organ transplantation, and shock resuscitation. During ischemia, ATP depletion causes failure of the sodium-potassium pump leading to cellular swelling, calcium accumulation, activation of phospholipases and proteases, and anaerobic metabolism with lactic acid accumulation. Upon reperfusion, the sudden reintroduction of oxygen triggers a burst of reactive oxygen species from mitochondria and neutrophils, causing lipid peroxidation, protein damage, and DNA injury. Inflammatory mediators are released, recruiting more neutrophils and amplifying the damage. Therapeutic strategies focus on minimising this injury through controlled reperfusion.

Lipofuscin

– A yellow-brown granular pigment composed of oxidized lipids and cross-linked proteins that accumulates progressively within cells with age, particularly in post-mitotic cells such as neurons, cardiac myocytes, and hepatocytes. It represents the residues of lysosomal digestion of autophagocytosed organelles and damaged cellular components that cannot be further degraded or exocytosed. Lipofuscin accumulation is a hallmark of cellular aging and is found in increased amounts in tissues with high oxidative stress. It is often called the 'wear and tear' pigment of ageing. Lipofuscin granules are autofluorescent and stain with periodic acid-Schiff and Sudan black. Its accumulation does not directly harm cells but serves as a marker of cumulative oxidative damage and cellular senescence.

Liquefactive Necrosis

– Characterized by the complete enzymatic digestion of dead cells, resulting in a liquid, viscous mass that eventually liquefies and forms a cystic cavity. This form of necrosis occurs most commonly in focal bacterial or fungal infections, where the inflammatory response releases abundant hydrolytic enzymes from neutrophils and macrophages that digest both dead and viable tissue. In the central nervous system, liquefactive necrosis follows ischemic stroke because the brain has abundant hydrolytic enzymes and limited connective tissue, so necrotic tissue liquefies rapidly, forming a cystic cavity. In bacterial infections, accumulated neutrophils release proteolytic enzymes that digest dead tissue, producing pus. The resulting cavity may persist as a cyst or be replaced by fibrous tissue during healing.

Liver Failure

– A life-threatening condition resulting from severe impairment of hepatic synthetic, metabolic, and detoxification functions. Acute liver failure occurs in previously healthy individuals, most commonly from acetaminophen toxicity accounting for approximately 50 percent of cases in the United States, viral hepatitis, drug-induced liver injury, or autoimmune hepatitis. The pathogenesis involves massive hepatocyte necrosis leading to loss of synthetic function with coagulopathy, impaired bilirubin clearance with jaundice, and hepatic encephalopathy from failure to detoxify ammonia and other neurotoxins. Chronic liver failure develops more slowly on a background of cirrhosis, with portal hypertension and synthetic dysfunction. Liver transplantation is the definitive treatment for end-stage liver disease.

Lupus Erythematosus

– Systemic lupus erythematosus is a chronic autoimmune disease characterized by production of autoantibodies against nuclear antigens, leading to immune complex-mediated inflammation and tissue damage in multiple organ systems. The disease predominantly affects women of childbearing age, with a striking female predominance of 9 to 1. The pathogenesis involves loss of self-tolerance to nuclear antigens, with formation of immune complexes that deposit in tissues and activate complement, causing inflammation in the kidneys, skin, joints, and serosal surfaces. The malar butterfly rash is characteristic. Key laboratory findings include positive ANA, anti-dsDNA, and anti-Smith antibodies. Disease flares and remissions are typical. Treatment includes NSAIDs, corticosteroids, hydroxychloroquine, and immunosuppressive agents for severe organ involvement.

Malignant Neoplasm

– Malignant neoplasms, commonly called cancers, are uncontrolled proliferations of cells that invade surrounding tissues and have the capacity to metastasize to distant sites through lymphatic, hematogenous, or transcoelomic routes. They exhibit infiltrative growth, penetrating basement membranes and spreading along tissue planes, nerve sheaths, and vascular structures. Malignant cells show varying degrees of anaplasia, increased and often atypical mitotic activity, and loss of normal tissue architecture capable of invasion and metastasis. Malignant tumors grow rapidly, have poorly defined margins, and can recur after resection. They cause morbidity through local tissue destruction, obstruction, hemorrhage, and systemic effects including cachexia, paraneoplastic syndromes, and metastatic spread.

Mediators of Inflammation

– Inflammatory mediators are soluble molecules that initiate, amplify, or regulate the inflammatory response. They include lipid mediators derived from arachidonic acid such as prostaglandins and leukotrienes, vasoactive amines like histamine and serotonin, cytokines including TNF, IL-1, and IL-6, chemokines that direct leukocyte migration, complement components, and kinins like bradykinin. These mediators are produced by activated macrophages, mast cells, endothelial cells, and platelets. Many mediators have multiple overlapping functions and are tightly regulated to prevent excessive tissue damage. Understanding their roles has enabled targeted therapies including NSAIDs (inhibiting prostaglandins), leukotriene inhibitors, and biologic agents blocking TNF-alpha, IL-1, and IL-6 in chronic inflammatory diseases such as rheumatoid arthritis and inflammatory bowel disease.

Metaplasia

– A reversible adaptive change in which one differentiated adult cell type is replaced by another, representing the body's attempt to better withstand a chronic stress or irritation. The process involves reprogramming of tissue stem cells to differentiate along a different lineage that is more resilient to the new environment. The most common example is squamous metaplasia in the respiratory tract of chronic smokers, where the normal ciliated pseudostratified columnar epithelium is replaced by non-ciliated stratified squamous epithelium that is more resistant to smoke damage. Another key example is Barrett's esophagus, where the normal squamous epithelium of the lower esophagus is replaced by intestinal-type columnar epithelium due to chronic gastro-esophageal reflux, increasing the risk of esophageal adenocarcinoma.

Metastatic Calcification

– The deposition of calcium salts in otherwise normal tissues due to systemic hypercalcemia, where elevated serum calcium levels exceed the tissue ability to maintain calcium homeostasis. Causes include hyperparathyroidism with excess PTH causing bone resorption, vitamin D intoxication increasing intestinal calcium absorption, milk-alkali syndrome, bone destruction from malignancy or Paget's disease, sarcoidosis through extrarenal production of 1,25-dihydroxyvitamin D, and renal failure with phosphate retention. Metastatic calcification primarily affects tissues that excrete acid or generate alkaline environments, including the gastric mucosa, kidneys, lungs, and blood vessels. The deposits are intracellular and extracellular and may impair organ function over time.

Mural Thrombus

– A mural thrombus is a blood clot that forms on the wall of a heart chamber or large blood vessel, typically in areas of damaged endothelium, stasis, or turbulent flow. In the heart, mural thrombi commonly form in the left ventricle following myocardial infarction, where the akinetic or dyskinetic segment promotes stasis of blood, or in the left atrium in patients with atrial fibrillation or mitral stenosis where blood stasis favors clot formation. The thrombus is attached to the wall by a broad base. In large vessels, mural thrombi form over atherosclerotic plaques or aneurysms. They can embolise to distant sites, causing infarction. Treatment includes anticoagulation to prevent propagation and embolisation. Over time, mural thrombi may organize via fibroblast ingrowth and become incorporated into the vessel wall or recanalise.

Myocardial Infarction

– The irreversible necrosis of myocardial tissue resulting from prolonged ischemia, most commonly due to atherosclerotic plaque rupture with superimposed thrombosis in a coronary artery. The process begins with platelet adhesion to the disrupted plaque, followed by activation of the coagulation cascade and formation of an occlusive thrombus. The resulting complete cessation of blood flow causes coagulative necrosis of the supplied myocardium within 20 to 40 minutes. The infarct evolves through stages: coagulation necrosis in the first 24 hours, neutrophil infiltration at days 1 to 3, granulation tissue formation at days 7 to 14, and eventual fibrous scar formation over 4 to 6 weeks. Diagnosis relies on ECG changes, elevated cardiac troponin, and imaging. Treatment includes reperfusion, antiplatelet therapy, and risk factor modification.

Necroptosis

– A regulated, programmed form of necrosis that shares features of both classical necrosis and apoptosis, serving as a backup cell death mechanism when apoptosis is inhibited. It is particularly important during viral infections, as many viruses encode proteins that block caspase-mediated apoptosis to ensure their own replication. The process is mediated by receptor-interacting protein kinase 1 and 3, known as RIPK1 and RIPK3, which form a complex called the necrosome. When TNF binds its receptor, RIPK1 and RIPK3 are recruited and activated. RIPK3 phosphorylates MLKL, which oligomerises and inserts into the plasma membrane, forming pores that cause cell swelling and membrane rupture, releasing damage-associated molecular patterns that trigger inflammation. Necroptosis is implicated in ischemia-reperfusion injury and neurodegenerative diseases.

Necrosis

– A pathological, unregulated form of cell death resulting from severe, irreversible injury that invariably provokes an intense inflammatory response. Unlike apoptosis, necrosis occurs when cells are exposed to extreme stresses such as profound hypoxia, toxins, trauma, burns, or extreme temperatures that overwhelm cellular repair mechanisms. The hallmark features include cellular swelling due to failure of the sodium-potassium pump, membrane disruption, organelle breakdown, and leakage of cytoplasmic contents into the extracellular space, triggering an inflammatory response. The morphologic patterns include coagulative (most common, in solid organs), liquefactive (brain and abscesses), caseous (tuberculosis), fat (adipose tissue), and fibrinoid (blood vessels).

Neurodegeneration

– The progressive loss of structure and function of neurons, culminating in their death, and represents the underlying mechanism of many devastating neurological disorders. It encompasses a heterogeneous group of diseases including Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis, and frontotemporal dementia. The mechanisms of neuronal death are multifactorial and include protein aggregation and misfolding, mitochondrial dysfunction, and oxidative stress. Common themes across neurodegenerative diseases include aberrant protein aggregation, mitochondrial impairment, excitotoxicity, neuroinflammation, and defective axonal transport. Despite intensive research, most neurodegenerative diseases remain incurable, with current treatments focused on symptom management.

Neurotransmitter

– Neurotransmitters are endogenous chemical messengers that transmit signals across synapses from one neuron to another or to target cells such as muscle or gland cells. They are synthesized in the presynaptic neuron from precursor molecules through specific enzymatic pathways, stored in synaptic vesicles, and released upon arrival of an action potential and calcium influx. Major neurotransmitters include acetylcholine synthesized from choline and acetyl-CoA, dopamine synthesized from tyrosine through tyrosine hydroxylase, and serotonin from tryptophan through tryptophan hydroxylase. After release, neurotransmitters bind to specific receptors on the postsynaptic membrane, either ionotropic receptors that directly open ion channels or metabotropic G-protein-coupled receptors. Signalling is terminated by reuptake, enzymatic degradation, or diffusion.

Neutrophil Recruitment

– A multistep process essential for acute inflammation and host defense, involving sequential adhesive interactions between neutrophils and endothelial cells. The process begins with the rolling phase, where neutrophils loosely attach to endothelial cells through selectin-mediated interactions. Activated endothelial cells express P-selectin stored in Weibel-Palade bodies and newly synthesized E-selectin, which bind to sialyl-Lewis X carbohydrates on neutrophils. The next step is firm adhesion, mediated by integrins including LFA-1 and Mac-1 on neutrophils binding to ICAM-1 on activated endothelial cells. Finally, neutrophils transmigrate through the endothelium via the paracellular route between endothelial cells (diapedesis) and migrate along chemotactic gradients to reach the site of injury.

Obstructive Shock

– A form of shock caused by physical obstruction of blood flow outside the heart, reducing cardiac output despite normal cardiac function. Common causes include cardiac tamponade, where fluid accumulation in the pericardial space compresses the heart and prevents adequate diastolic filling; tension pneumothorax, where air in the pleural space causes mediastinal shift and compression of the great veins reducing venous return; and massive pulmonary embolism, where thrombus in the pulmonary arteries obstructing right ventricular outflow. In all cases, cardiac output falls and compensatory vasoconstriction may be insufficient. Treatment involves relieving the obstruction: pericardiocentesis for tamponade, needle decompression for tension pneumothorax, and thrombolysis or embolectomy for massive pulmonary embolism.

Obstructive Uropathy

– Urinary tract obstruction at any level from the renal pelvis to the urethra, leading to impaired urine flow and potential renal damage. Causes include nephrolithiasis, benign prostatic hyperplasia, ureteral strictures, tumors of the bladder, prostate, or cervix, and external compression from lymphadenopathy or retroperitoneal fibrosis. The pathogenesis involves increased intratubular pressure proximal to the obstruction, leading to decreased glomerular filtration, tubular dysfunction, and ultimately irreversible renal parenchymal damage (obstructive nephropathy). Acute complete obstruction causes severe flank pain, anuria, and rapidly progressive renal failure. Chronic partial obstruction leads to hydronephrosis and gradual loss of renal function. Treatment involves relieving the obstruction.

Ochronosis

– A blue-black or brownish pigmentation of connective tissues (cartilage, skin, sclerae, tendons) caused by accumulation of homogentisic acid and its oxidation products. Ochronosis is the characteristic feature of alkaptonuria (also spelled alcaptonuria), a rare autosomal recessive disorder of tyrosine metabolism due to deficiency of homogentisate 1,2-dioxygenase (HGD gene). Homogentisic acid accumulates, polymerises, and deposits in collagen-rich tissues. Clinical: urine turns dark on standing (alkalinisation or oxidation—classic sign), ochronotic pigmentation (ear cartilage—blue-black, sclerae—triangular brown patches, skin over nose and knuckles, costochondral junctions), and ochronotic arthropathy (progressive osteoarthritis, especially of the spine and large joints—lumbar kyphosis, disc degeneration with calcification on X-ray, joint space narrowing). Other complications: aortic valve stenosis, kidney stones, and prostate stones. Onset of arthropathy in the 4th–5th decade. Treatment: nitisinone (inhibits 4-hydroxyphenylpyruvate dioxygenase upstream, reduces homogentisic acid production—licensed for alkaptonuria), low-protein diet, vitamin C (may delay polymerisation), and joint replacement for advanced arthropathy.

Oncogenes

– Mutated, amplified, or overexpressed versions of normal cellular genes called proto-oncogenes that encode proteins involved in cell growth, proliferation, and survival. Proto-oncogenes encode growth factors, growth factor receptors, signal transduction molecules, and transcription factors that normally regulate controlled cell growth. When activated by mutation, gene amplification, chromosomal translocation, or viral insertion, they become constitutively active, driving uncontrolled cell proliferation and survival. Common oncogenes include RAS (mutated in many cancers), MYC (transcription factor), HER2/ERBB2 (amplified in breast cancer), and BCR-ABL (translocation causing chronic myeloid leukemia). Targeted therapies against specific oncogenic drivers, such as tyrosine kinase inhibitors, have revolutionised cancer treatment.

Oncosis

– A form of cell death characterized by cellular swelling, derived from the Greek word onkos meaning swelling, representing a pre-lethal state that precedes necrosis. It is typically caused by ischemic injury leading to ATP depletion, which impairs the sodium-potassium pump and causes osmotic water influx. Unlike apoptosis, oncosis involves progressive cellular and organellar swelling, increased membrane permeability, cellular fragmentation, and eventual membrane rupture. Histologically, oncotic cells appear swollen with pale, vacuolated cytoplasm and clumped chromatin. The term is often used interchangeably with necrosis in the context of ischemic injury. Oncosis is distinct from apoptosis, which involves cell shrinkage rather than swelling, and is considered an unregulated, energy-independent form of cell death.

p53 Tumor Suppressor

– The p53 protein, encoded by the TP53 gene on chromosome 17p, is a transcription factor and the most commonly mutated gene in human cancer, with mutations found in over 50 percent of all malignancies. Known as the guardian of the genome, p53 is a short-lived protein normally kept at low levels by MDM2-mediated ubiquitination and proteasomal degradation. In response to DNA damage, oncogene activation, hypoxia, or nutrient deprivation, p53 is stabilized through phosphorylation and acetylation that increase its half-life and transcriptional activity. Activated p53 transactivates target genes involved in cell cycle arrest (p21), DNA repair (GADD45), apoptosis (BAX, PUMA), and senescence. Mutant p53 not only loses tumor suppressor function but often gains oncogenic properties. Germline TP53 mutations cause Li-Fraumeni syndrome.

Pancreatic Adenocarcinoma

– A highly lethal malignancy arising from the pancreatic ductal epithelium, with a five-year survival rate of less than 10 percent, making it one of the deadliest cancers. The most common risk factors include smoking, obesity, chronic pancreatitis, diabetes mellitus, and hereditary syndromes including BRCA2 mutations and Lynch syndrome. The pathogenesis involves sequential genetic mutations including KRAS activation, TP53 loss, SMAD4 loss, and CDKN2A inactivation. The majority of tumors arise in the head of the pancreas, presenting with painless obstructive jaundice. CA 19-9 is a useful serum tumor marker, though not specific for screening. Surgical resection (pancreaticoduodenectomy, Whipple procedure) offers the only chance of cure, but less than 20 percent of patients are resectable at presentation.

Pancreatic Pseudocyst

– A pancreatic pseudocyst is a localized collection of pancreatic fluid surrounded by a wall of granulation tissue without an epithelial lining, typically occurring as a complication of acute or chronic pancreatitis. Unlike true cysts, pseudocysts lack an epithelial lining, which is why they are called pseudocysts. The pathogenesis involves disruption of the pancreatic duct with leakage of pancreatic enzymes, causing enzymatic digestion of surrounding tissues and formation of a fluid collection that may reach large sizes. Pseudocysts present with epigastric pain, fullness, and tenderness. Complications include infection, rupture, and hemorrhage. Diagnosis is confirmed by CT or MRI showing a thick-walled fluid collection. Treatment options include observation, endoscopic drainage, or surgical cystogastrostomy.

Parkinson's Disease

– A progressive neurodegenerative disorder caused by selective loss of dopaminergic neurons in the substantia nigra pars compacta, leading to dopamine depletion in the striatum. The pathological hallmark is the presence of Lewy bodies, intracellular inclusions composed of aggregated alpha-synuclein protein, within surviving neurons. The disease typically manifests after age 60 with a male predominance. Clinical features include the cardinal motor symptoms of resting tremor, typically a pill-rolling tremor, bradykinesia, rigidity, and postural instability, as well as non-motor symptoms including depression, dementia, and autonomic dysfunction. Treatment is based on dopamine replacement with levodopa-carbidopa, dopamine agonists, and deep brain stimulation for advanced disease with motor fluctuations.

Pathologic Hypertrophy

– The enlargement of an organ resulting from an increase in the size of its constituent cells without an increase in cell number, typically occurring as a response to chronically increased workload or hormonal stimulation. Unlike physiologic hypertrophy, pathologic hypertrophy occurs under abnormal conditions and may eventually decompensate. The most important example is cardiac hypertrophy in response to chronic pressure overload from systemic hypertension or aortic stenosis. Initially compensatory and beneficial, sustained pathologic hypertrophy leads to maladaptive changes including myocardial fibrosis, diastolic dysfunction, and eventual systolic failure. The transition from compensated hypertrophy to heart failure involves altered calcium handling, energy metabolism, and apoptosis of cardiomyocytes.

Petechiae

– Small, pinpoint, non-blanching red or purple spots on the skin or mucous membranes caused by extravasation of red blood cells from capillaries into the surrounding tissue. They are typically less than 2 millimeters in diameter and result from decreased platelet count, platelet dysfunction, or increased vascular fragility. The most common cause is thrombocytopenia, where platelet counts fall below 20,000 to 30,000 per microliter, impeding primary hemostasis. Other causes include vascular fragility from vasculitis, connective tissue disorders, or corticosteroid use. Petechiae are non-blanching and extremely small. Their distribution provides diagnostic clues: dependent areas suggest thrombocytopenia, while distribution above the nipple line suggests superior vena cava syndrome. Diagnosis involves evaluating platelet count and coagulation studies.

Platelet Plug Formation

– The initial hemostatic response to vascular injury, involving platelet adhesion, activation, and aggregation. When the endothelium is disrupted, subendothelial von Willebrand factor and collagen are exposed to flowing blood. Platelet glycoprotein Ib-IX-V complex binds to vWF under high shear stress, tethering and decelerating platelets on the damaged surface. Collagen binding to glycoprotein VI and integrin alpha-2-beta-1 triggers intracellular signaling cascades through activation of phospholipase C and increased cytosolic calcium, leading to conformational change of integrin alpha-IIb-beta-3 which binds fibrinogen, cross-linking platelets into an aggregated plug. The plug is stabilised by the coagulation cascade generating a fibrin mesh. This primary hemostatic response is critical for stopping bleeding but can contribute to pathological thrombosis.

Post-Mortem

– An examination of a body after death to determine the cause of death, identify disease processes, and evaluate the effects of medical or surgical treatment. Also called autopsy, necropsy. Types: hospital (clinical) post-mortem — performed with consent to investigate the cause of death, assess treatment, or advance medical knowledge; medico-legal (coroner's/forensic) post-mortem — ordered by a coroner or legal authority when the death is sudden, unexpected, suspicious, or unnatural (required by law; consent not needed). Procedure: external examination, internal examination (thoracic, abdominal, and cranial cavities), histopathology (tissue sampling), toxicology (blood, urine, vitreous humor), microbiology, and other specialized tests. The report includes: cause of death (immediate, underlying, and contributing factors), and relevant pathological findings. Post-mortem examination is essential for quality assurance in medicine (verification of clinical diagnosis, identification of missed diagnoses, and audit of surgical and medical care). In obstetrics, perinatal post-mortem is crucial for understanding stillbirth, congenital anomalies, and maternal death.

Pressure Atrophy

– The gradual wasting of tissue resulting from sustained external compression that impairs blood supply and causes cellular shrinkage and eventual death. The mechanism involves compression of blood vessels, which reduces perfusion and causes ischemia of the dependent tissue. Prolonged ischemia leads to cellular atrophy with reduced cell size and organ function, and if compression continues, progresses to necrosis. Clinical examples include decubitus ulcers that develop over bony prominences in bedridden patients, and atrophy of normal tissue adjacent to an expanding tumor. Affected cells show reduced size, decreased protein synthesis, and increased autophagic vacuoles. Unlike necrosis, pressure atrophy is a reversible adaptive change if the compressive force is removed before irreversible cell damage occurs.

Prion Disease

– Prion diseases, also known as transmissible spongiform encephalopathies, are a group of invariably fatal neurodegenerative disorders caused by misfolding of the normal cellular prion protein PrPC into the abnormal scrapie form PrPSc. The misfolded protein acts as an infectious agent by inducing normal PrPC to adopt the pathological conformation, propagating the misfolding process through an autocatalytic mechanism. PrPSc accumulates in the brain, causing neuronal death and characteristic spongiform change in the brain. The most common human prion disease is sporadic Creutzfeldt-Jakob disease, presenting with rapidly progressive dementia, myoclonus, and ataxia. Variant CJD is linked to bovine spongiform encephalopathy. Diagnosis is based on clinical features, MRI, EEG, and detection of 14-3-3 protein in CSF. All prion diseases are uniformly fatal.

Prostate Adenocarcinoma

– The most common non-cutaneous malignancy in men and the second leading cause of cancer death in men after lung cancer. The disease typically arises in the peripheral zone of the prostate and is hormone-dependent, requiring androgens for growth. Risk factors include age, family history, and race, with African American men having higher incidence and mortality. Most tumors are slow-growing and may never cause symptoms, while a subset are aggressive and metastasize. The Gleason grading system, based on histological glandular architecture, is the most important prognostic factor, with scores ranging from 6 to 10. PSA screening has improved early detection but remains controversial due to overdiagnosis of indolent disease. Treatment options include active surveillance, radical prostatectomy, radiation therapy, and androgen deprivation therapy.

Purpura

– Purple discoloration of the skin or mucous membranes caused by extravasation of red blood cells from blood vessels into the surrounding tissue. Purpura can be classified by size: petechiae are less than 2 millimeters, purpura are 2 to 10 millimeters, and ecchymoses are greater than 10 millimeters. Causes include thrombocytopenia from decreased production or increased destruction, platelet dysfunction from medications or uremia, vascular fragility from aging or vitamin C deficiency (scurvy), and Henoch-Schonlein purpura with IgA immune complex deposition. Palpable purpura suggests underlying vasculitis. Diagnosis involves complete blood count, coagulation studies, and skin biopsy if vasculitis is suspected. Treatment targets the underlying cause; corticosteroids are used in vasculitic causes.

Pyelonephritis

– An infection of the renal parenchyma and pelvicalyceal system, typically resulting from ascending urinary tract infection. Acute pyelonephritis most commonly affects women and is caused by gram-negative enteric organisms, particularly Escherichia coli, which account for approximately 80 percent of cases. The pathogenesis involves bacterial ascent from the bladder to the kidneys, facilitated by vesicoureteral reflux, urinary obstruction, and bacterial virulence factors including adhesins and toxins. Acute pyelonephritis presents with fever, chills, flank pain, nausea, and vomiting, with pyuria and bacteriuria on urinalysis. Treatment involves antibiotics guided by culture sensitivity. Complications include renal abscess, emphysematous pyelonephritis, sepsis, and chronic pyelonephritis leading to renal scarring and hypertension.

Red Infarction

– Red infarction, also known as hemorrhagic infarction, occurs in tissues with dual blood supply, loose tissue architecture, or venous obstruction, allowing blood to accumulate in the necrotic area and produce a dark red appearance. The hemorrhagic appearance results from extravasation of red blood cells from congested vessels into the necrotic tissue. Red infarcts characteristically occur in the lung, where the dual pulmonary arterial and bronchial arterial circulation provides collateral flow, allowing blood to seep into the necrotic area from adjacent patent vessels. In the liver and small intestine, red infarction can also occur due to dual blood supply. The affected tissue is dark red, wedge-shaped, and poorly demarcated. Organisation and scar formation follow, with functional impact depending on the size and location of the infarct.

Renal Infarction

– Renal infarction results from occlusion of the renal artery or its branches, causing coagulative necrosis of the renal parenchyma. The kidney has end-arterial supply with limited collateral circulation, making it susceptible to ischemic injury even from small vessel occlusion. Common causes include thromboembolism from the heart, aortic dissection extending into the renal arteries, atherosclerotic disease, and fibromuscular dysplasia. Clinical features include sudden onset severe flank pain, haematuria, nausea, and vomiting. Hypertension may develop due to renin release from the ischemic kidney. Diagnosis is confirmed by CT angiography or renal scintigraphy. Treatment depends on the cause and may include anticoagulation, revascularisation, or conservative management with analgesics and blood pressure control.

Sarcoma

– A malignant neoplasm arising from mesenchymal cells, including bone, cartilage, fat, muscle, blood vessels, and fibrous tissue. Sarcomas are rare, accounting for approximately one percent of all adult cancers but are more common in children and adolescents, where they represent approximately 15 percent of pediatric malignancies. They are classified based on the tissue of origin: osteosarcoma from bone, chondrosarcoma from cartilage, liposarcoma from adipose tissue, leiomyosarcoma from smooth muscle, and rhabdomyosarcoma from skeletal muscle. Histological grading and staging are critical for prognosis. Wide surgical resection with negative margins is the primary treatment. Sarcomas are relatively resistant to chemotherapy, though specific subtypes show variable sensitivity to radiation and cytotoxic agents.

Septic Embolism

– Septic embolism occurs when infected material, typically thrombi containing bacteria, embolizes through the bloodstream to distant sites, causing metastatic infection. The most common source is infective endocarditis, where infected vegetations on heart valves break off and lodge in peripheral vessels. Other sources include infected central venous catheters, septic pelvic thrombophlebitis, and infected arterial thrombi. Septic emboli typically lodge in the lungs, causing multiple bilateral wedge-shaped pulmonary infarcts with cavitation, presenting with chest pain, haemoptysis, and fever. Systemic embolisation can cause abscesses in the brain, spleen, kidneys, and skin. Diagnosis involves blood cultures and imaging. Treatment requires prolonged antibiotics and often surgical intervention for source control.

Septic Shock

– A form of distributive shock caused by overwhelming infection and the host dysregulated immune response, leading to systemic vasodilation, increased vascular permeability, myocardial depression, and multiorgan dysfunction. It is the most common cause of death in intensive care units worldwide. The pathogenesis involves recognition of microbial products including lipopolysaccharide and peptidoglycan by pattern recognition receptors on immune cells, triggering massive release of pro-inflammatory cytokines including TNF-alpha, IL-1, and IL-6, causing widespread vasodilation, increased capillary permeability, and myocardial depression. This results in refractory hypotension, tissue hypoperfusion, and multiorgan failure. Management includes early broad-spectrum antibiotics, fluid resuscitation, vasopressors, and source control of infection.

Shock

– A state of generalized tissue hypoperfusion resulting from inadequate cardiac output or effective circulating blood volume, leading to cellular hypoxia, metabolic acidosis, and potentially irreversible organ damage if not promptly treated. The four main categories are hypovolemic, cardiogenic, distributive, and obstructive shock, each with distinct pathophysiological mechanisms but sharing the common endpoint of inadequate oxygen delivery to meet metabolic demands. Compensatory mechanisms include activation of the sympathetic nervous system, the renin-angiotensin-aldosterone axis, and release of vasopressin, all attempting to maintain perfusion to vital organs. However, prolonged shock leads to irreversible cellular damage, multiorgan failure, and death. Early recognition and treatment targeted at the specific type of shock are critical to survival.

Splenic Infarction

– Splenic infarction occurs when the splenic artery or its branches are occluded, typically by emboli from the heart. The spleen has end-arterial supply with limited collateral circulation, making it vulnerable to infarction even with small vessel occlusion. The most common cause is cardioembolism, particularly in patients with atrial fibrillation, infective endocarditis, or left ventricular thrombus. Other causes include hematologic disorders like sickle cell disease and myeloproliferative disorders such as polycythaemia vera. Splenic infarction presents with left upper quadrant pain, often pleuritic, sometimes radiating to the left shoulder. Fever and splenomegaly may be present. Most cases are managed conservatively with analgesics and treatment of the underlying cause, though complications include abscess formation and splenic rupture.

Starling Forces

– Starling forces describe the balance of hydrostatic and oncotic pressures that govern fluid movement across capillary walls, determining the distribution of fluid between the intravascular and extravascular compartments. The Starling equation states that net filtration equals the filtration coefficient times the difference between capillary and interstitial hydrostatic pressures minus the difference between capillary and interstitial oncotic pressures. Capillary hydrostatic pressure, generated by arterial blood pressure, pushes fluid out of the capillary at the arteriolar end. Plasma oncotic pressure, generated primarily by albumin, pulls fluid back into the capillary at the venular end. The net result is that approximately 90 percent of filtered fluid is reabsorbed; the remainder drains via the lymphatic system as lymph.

Storage Diseases

– A group of inherited metabolic disorders characterized by accumulation of substrates within cells due to specific enzyme deficiencies. Lysosomal storage diseases result from defects in lysosomal hydrolytic enzymes, leading to accumulation of undigested macromolecules within lysosomes. Gaucher disease results from glucocerebrosidase deficiency, causing glucocerebroside accumulation in macrophages that become enlarged Gaucher cells with characteristic wrinkled tissue paper cytoplasm. Niemann-Pick disease results from sphingomyelinase deficiency causing sphingomyelin accumulation in macrophages. Tay-Sachs disease results from hexosaminidase A deficiency causing GM2 ganglioside accumulation in neurons, leading to neurodegeneration. Mucopolysaccharidoses result from deficiencies of enzymes degrading glycosaminoglycans.

Synaptic Transmission

– The fundamental process by which neurons communicate with each other through specialized junctions called synapses. At chemical synapses, an action potential arriving at the presynaptic terminal opens voltage-gated calcium channels, allowing calcium influx that triggers fusion of synaptic vesicles with the presynaptic membrane through SNARE protein complexes. Neurotransmitters are released into the synaptic cleft by exocytosis and bind to specific receptors on the postsynaptic membrane, generating either excitatory postsynaptic potentials (EPSPs) or inhibitory postsynaptic potentials (IPSPs). Signal termination occurs via neurotransmitter reuptake by specific transporters, enzymatic degradation in the synaptic cleft, or diffusion away from the synapse. This process forms the basis of all neural communication and information processing.

Systemic Sclerosis

– Systemic sclerosis, also known as scleroderma, is a chronic autoimmune disease characterized by progressive fibrosis of the skin and internal organs, vasculopathy, and immune dysfunction. The pathogenesis involves an aberrant wound healing response with excessive collagen production by activated fibroblasts, triggered by endothelial injury and immune activation. The disease exists in two main forms: limited cutaneous systemic sclerosis, formerly called CREST syndrome, with skin involvement distal to the elbows and knees, and diffuse cutaneous systemic sclerosis with widespread skin thickening and early internal organ involvement including interstitial lung disease, pulmonary hypertension, and renal crisis. Anti-centromere antibodies are associated with limited disease; anti-Scl-70 (anti-topoisomerase I) is associated with diffuse disease.

Takayasu Arteritis

– A chronic granulomatous vasculitis primarily affecting the aorta and its major branches, most commonly diagnosed in women under 50 years of age, particularly in Asian populations. The disease involves transmural inflammation of the aortic wall and its branches, leading to intimal fibrosis, luminal narrowing, and sometimes aneurysmal dilation. The pathogenesis involves inappropriate immune-mediated vascular inflammation with CD4 T cell and macrophage infiltration. Clinical features include absent or diminished pulses, blood pressure discrepancies between arms, vascular bruits, claudication, and syncope. Diagnosis is based on imaging showing vascular wall thickening, stenosis, or aneurysm. Treatment includes high-dose corticosteroids and other immunosuppressive agents to control inflammation and prevent disease progression.

Temporal Arteritis

– A vasculitic disorder affecting medium and large arteries, particularly the temporal artery, predominantly occurring in adults over 50 years of age. The disease is characterized by granulomatous inflammation of the arterial wall with fragmentation of the internal elastic lamina, intimal proliferation, and luminal narrowing that can lead to ischemia of affected tissues. The most feared complication is anterior ischemic optic neuropathy, which can cause sudden, irreversible vision loss. Other symptoms include headache, jaw claudication, scalp tenderness, and constitutional symptoms such as fever and weight loss. ESR and CRP are typically markedly elevated. Temporal artery biopsy shows granulomatous inflammation with giant cells and elastic lamina fragmentation. Prompt high-dose corticosteroid therapy protects against vision loss.

Teratoma

– See Teratoma in Obstetrics and Gynecology.

Thrombosis

– The formation of a blood clot within the cardiovascular system during life, potentially obstructing blood flow and causing tissue ischemia or infarction. Arterial thrombi typically form on ruptured atherosclerotic plaques, are composed primarily of platelets and fibrin, and appear pale and grossly attached to the vessel wall. Venous thrombi form in areas of stasis, are rich in fibrin and trapped red blood cells, and appear red with a characteristic Lines of Zahn layering pattern. Venous thrombi commonly cause deep vein thrombosis with leg swelling and pain, and may embolise to the lungs causing pulmonary embolism. Arterial thrombosis leads to myocardial infarction, stroke, or peripheral arterial occlusion. Risk factors are described by Virchow's triad. Treatment involves anticoagulants, thrombolytics, or mechanical thrombectomy.

Thrombus Organization

– The process by which a thrombus is gradually replaced by granulation tissue and eventually fibrous scar tissue. The process begins when inflammatory cells, particularly macrophages, infiltrate the thrombus and begin digesting the fibrin and dead cells through fibrinolysis. Fibroblasts then migrate into the thrombus and begin producing collagen and other extracellular matrix components, gradually replacing the original clot. New blood vessels grow into the thrombus through the process of angiogenesis, a phenomenon called recanalisation. The organized thrombus becomes a fibrous scar that may contract and narrow the vessel lumen or, in the case of recanalisation, restore partial blood flow. This process is part of the natural healing response to thrombosis.

Thyroid Follicular Carcinoma

– approximately 10 to 15 percent of thyroid cancers and is the second most common differentiated thyroid malignancy. Unlike papillary carcinoma, follicular carcinoma typically spreads hematogenously to distant sites including bone, lung, and liver, rather than to regional lymph nodes. The diagnosis requires demonstration of capsular invasion or vascular invasion on histological examination, distinguishing it from follicular adenoma which lacks these features. Prognosis is generally good, with ten-year survival exceeding 85 percent, though worse in the presence of distant metastasis. Treatment includes total thyroidectomy followed by radioactive iodine ablation and TSH suppression therapy.

Thyroid Papillary Carcinoma

– The most common type of thyroid cancer, accounting for approximately 80 percent of all thyroid malignancies, and has an excellent prognosis with over 95 percent 10-year survival. The tumor arises from follicular thyroid cells and is characterized by papillary architectural growth pattern and characteristic nuclear features including nuclear grooves, intranuclear inclusions, and orphan eye nuclei. Risk factors include childhood radiation exposure, particularly from therapeutic radiation to the head and neck in childhood. The tumor has a propensity for lymphatic spread, with cervical lymph node metastases present in up to 50 percent of cases at diagnosis. However, the prognosis remains excellent. Treatment involves thyroidectomy with or without radioactive iodine ablation and TSH suppression.

TNF Alpha

– Tumor necrosis factor alpha is a potent proinflammatory cytokine produced primarily by activated macrophages, as well as by T cells, NK cells, mast cells, and endothelial cells. TNF-alpha exists as a 26-kilodalton transmembrane protein that is cleaved by the metalloprotease TACE to release a soluble 17-kilodalton active form. It acts through two receptors: TNFR1, which contains a death domain and can activate both NF-kappaB survival signaling and caspase-8-mediated apoptosis, and TNFR2, which lacks a death domain and preferentially activates NF-kappaB, promoting cell survival and inflammation. TNF-alpha is a key therapeutic target in chronic inflammatory diseases including rheumatoid arthritis, psoriasis, and inflammatory bowel disease, where monoclonal antibodies (infliximab, adalimumab) and fusion proteins (etanercept) are used.

Transudate

– A transudate is a type of edema fluid with low protein content and few cells, resulting from imbalances in hydrostatic and oncotic pressures across capillary walls rather than from inflammation or increased vascular permeability. Transudates are typically caused by conditions that increase capillary hydrostatic pressure, such as congestive heart failure causing pulmonary edema, or decrease plasma oncotic pressure, such as hypoalbuminemia from liver cirrhosis or nephrotic syndrome. The fluid is clear, straw-coloured, and has low cellularity. Laboratory criteria for transudate include pleural fluid protein-to-serum protein ratio less than 0.5, pleural fluid LDH-to-serum LDH ratio less than 0.6, and pleural fluid LDH less than two-thirds the upper limit of normal for serum.

Tumor Differentiation

– The degree to which neoplastic cells morphologically and functionally resemble their normal counterparts of origin. Well-differentiated tumors retain many structural features of the original tissue, including organized architectural patterns, specialized cellular structures, and sometimes functional capabilities such as hormone production or mucin secretion. Poorly differentiated or anaplastic tumors show minimal resemblance to the normal tissue and often require immunohistochemistry and molecular profiling for accurate classification. tumor grade, a measure of differentiation, is a powerful independent prognostic factor in most cancers. Well-differentiated tumors generally have a better prognosis than poorly differentiated ones, though notable exceptions exist.

Tumor Embolism

– Tumor embolism occurs when malignant cells, tumor fragments, or tumor thrombi enter the venous circulation and travel to distant sites. The pulmonary vasculature is the most common site of embolization, as venous blood from most organs drains to the lungs through the pulmonary arteries. Tumor emboli can also reach the arterial circulation through pulmonary capillary shunts or a patent foramen ovale, causing systemic embolization to the brain, kidneys, or other organs. The clinical presentation ranges from asymptomatic pulmonary emboli to acute respiratory failure and cor pulmonale. tumor embolism is an under-recognised complication of malignancy and may mimic thromboembolic disease. Diagnosis requires a high index of suspicion and histopathological confirmation.

Tumor Suppressor Genes

– Tumor suppressor genes encode proteins that normally inhibit cell proliferation, promote DNA repair, or induce programmed cell death in response to cellular damage. They function as brakes on the cell cycle, and loss of their function removes critical checkpoints that normally prevent uncontrolled growth. According to Knudson's two-hit hypothesis, both alleles must be inactivated for loss of function, which can occur through mutation, deletion, or epigenetic silencing. Hereditary cancer syndromes, such as Li-Fraumeni syndrome (TP53), familial adenomatous polyposis (APC), and hereditary breast and ovarian cancer (BRCA1, BRCA2), involve germline mutations of tumor suppressor genes. Epigenetic silencing through promoter hypermethylation is another common mechanism of inactivation in sporadic cancers.

Urothelial Carcinoma

– Urothelial carcinoma, previously called transitional cell carcinoma, is the most common malignancy of the urinary tract, arising from the urothelium that lines the renal pelvis, ureters, bladder, and proximal urethra. The most important risk factor is cigarette smoking, which accounts for approximately 50 percent of cases, with other risk factors including occupational exposure to aromatic amines, cyclophosphamide therapy, and chronic schistosomiasis. The pathogenesis involves field cancerization, where the entire urothelium is exposed to carcinogens, leading to multifocal and recurrent tumors. Gross haematuria is the most common presenting symptom. Diagnosis is by cystoscopy and biopsy. Treatment depends on stage: transurethral resection and intravesical therapy for non-muscle-invasive disease, and radical cystectomy for muscle-invasive disease.

Valvular Heart Disease

– Valvular heart disease involves dysfunction of one or more heart valves, leading to stenosis, regurgitation, or a combination of both. The most common causes are degenerative calcification of aortic and mitral valves, rheumatic heart disease, infective endocarditis, and ischemic papillary muscle dysfunction. Aortic stenosis results from calcification of a congenitally bicuspid or tricuspid aortic valve, causing left ventricular outflow obstruction with pressure overload. Aortic regurgitation results from incomplete valve closure, causing volume overload and left ventricular dilation. Mitral stenosis is most commonly rheumatic in origin, causing left atrial enlargement and pulmonary hypertension. Treatment includes medical management and valve replacement or repair for severe symptomatic disease.

Vasodilation

– The widening of blood vessels during inflammation, resulting in increased blood flow to the affected area. This process is mediated primarily by histamine released from mast cell granules acting on H1 receptors, nitric oxide produced by endothelial nitric oxide synthase, and prostaglandins particularly PGE2. The increased blood flow is responsible for the redness and heat characteristic of acute inflammation. Vasodilation occurs within seconds to minutes of tissue injury and represents the earliest vascular response in the inflammatory cascade. The increased blood flow brings oxygen, nutrients, and immune cells to the affected tissue and dilutes toxins. Vasodilation is accompanied by increased vascular permeability, allowing exudation of protein-rich fluid and formation of tissue edema, contributing to the cardinal signs of inflammation.

Vegetation

– A vegetation is a mass composed of platelets, fibrin, and microorganisms that forms on a heart valve or endocardial surface in the setting of infective endocarditis. The vegetation typically attaches to the valve leaflet on the low-pressure side and may progressively destroy the underlying valve tissue, causing valvular regurgitation. The most common causative organisms include Staphylococcus aureus in acute endocarditis, which can affect normal valves, and viridans streptococci in subacute endocarditis, which typically affects previously damaged valves. Diagnosis relies on blood cultures and echocardiography, with transoesophageal echocardiography being more sensitive. Treatment requires prolonged intravenous antibiotics and sometimes surgical valve debridement or replacement.

Virchow's Triad

– Virchow's triad describes three broad categories of factors that contribute to thrombosis: endothelial injury or dysfunction, stasis or turbulent blood flow, and hypercoagulability. Endothelial injury exposes subendothelial collagen and tissue factor, initiating platelet adhesion and the coagulation cascade. Causes include atherosclerosis, hypertension, vasculitis, and direct trauma. Stasis or turbulent flow disrupts laminar blood flow, bringing platelets into contact with the endothelium, preventing dilution of activated clotting factors and causing endothelial hypoxia. Hypercoagulability encompasses inherited conditions such as factor V Leiden, prothrombin gene mutation, and deficiencies of protein C, protein S, or antithrombin III, as well as acquired states including pregnancy, malignancy, and oral contraceptive use.

Wegener's Granulomatosis

– Wegener's granulomatosis, now called granulomatosis with polyangiitis, is a systemic vasculitis characterized by necrotizing granulomatous inflammation of the upper and lower respiratory tracts and pauci-immune necrotizing glomerulonephritis. It is associated with anti-neutrophil cytoplasmic antibodies, specifically c-ANCA targeting proteinase 3. The disease affects small to medium-sized vessels and can involve virtually any organ system. Upper respiratory tract involvement causes chronic sinusitis, nasal mucosal ulceration, and saddle nose deformity. Pulmonary involvement causes cough, haemoptysis, and cavitating lung nodules. Renal involvement, if untreated, rapidly progresses to end-stage renal disease. Treatment includes high-dose corticosteroids with cyclophosphamide or rituximab, which has dramatically improved outcomes.

White Infarction

– White infarction, also known as anemic infarction, occurs in solid organs with end-arterial circulation where collateral blood supply is limited, resulting in pale, well-demarcated areas of coagulative necrosis. The affected tissue appears pale because compression of the capillary bed by interstitial edema limits secondary hemorrhage into the necrotic area. White infarcts characteristically occur in the heart, kidney, spleen, and other solid organs with single-source arterial supply. The mechanism involves sudden arterial occlusion producing a wedge-shaped area of coagulative necrosis, with the apex pointing toward the occlusion site. The infarct appears pale because the necrotic tissue loses its blood content and surrounding vessels constrict. Organisation and fibrous scar formation follow over several weeks.

Xeroderma Pigmentosum

– A rare autosomal recessive disorder of nucleotide excision repair (NER) characterized by extreme sun sensitivity and a 1000-fold increased risk of skin cancers (basal cell carcinoma, squamous cell carcinoma, and melanoma, often in childhood). Mutations occur in one of eight genes (XPA-XPG and XP-variant/polymerase eta). Defective NER leads to accumulation of UV-induced pyrimidine dimers and other photoproducts, causing mutations in tumor suppressor genes (p53, PTCH). Clinical features: severe sunburn with minimal sun exposure, poikiloderma (hyperpigmentation, hypopigmentation, telangiectasias, atrophy), xerosis (dry skin), actinic keratoses, and multiple skin cancers. Ocular involvement (photophobia, conjunctivitis, keratitis, eyelid skin cancers) and neurological involvement (progressive neurodegeneration in XP-A, -B, -D, -G: microcephaly, spasticity, chorea, sensorineural hearing loss, peripheral neuropathy, cognitive impairment). Diagnosis: clinical criteria with confirmatory DNA repair assay (unscheduled DNA synthesis/UDS) showing reduced NER capacity or gene sequencing. Management: rigorous sun protection (sunscreen SPF 50+, UV-protective clothing, hats, sunglasses), regular dermatological surveillance (every 3-6 months) with early treatment of skin cancers (cryotherapy, photodynamic therapy, surgical excision, topical 5-fluorouracil, imiquimod). Oral retinoids (acitretin) reduce skin cancer incidence by 50-70% in XP patients. Prognosis: reduced life expectancy primarily from metastatic skin cancer or progressive neurodegeneration. Neurological complications are the major determinant of poor prognosis.

Chapter 26

Pediatrics

Achondroplasia

– The most common form of short-limbed dwarfism, caused by an autosomal dominant FGFR3 gene mutation. Features: rhizomelic limb shortening, frontal bossing, midface hypoplasia, and trident hand configuration. Intelligence normal. Complications include spinal stenosis, obstructive sleep apnea, and recurrent otitis media. Management: growth hormone therapy has limited benefit; limb-lengthening surgery in selected cases. Genetic counseling essential.

Acne

– A chronic inflammatory disorder of the pilosebaceous unit, extremely common during adolescence. Pathogenesis involves sebum production, follicular hyperkeratinization, Cutibacterium acnes colonization, and inflammation. Grading: comedonal, papulopustular, nodulocystic. Treatment: topical retinoids, benzoyl peroxide, and antibiotics for mild to moderate; isotretinoin for severe or refractory disease. Early treatment prevents scarring.

Acute Otitis Media

– Infection of the middle ear space with acute onset of symptoms and signs of middle ear inflammation. Diagnosis requires middle ear effusion plus acute symptoms (ear pain, fever, irritability) or signs (bulging tympanic membrane, erythema). Most common pathogen: Streptococcus pneumoniae. Treatment: amoxicillin for 10 days; amoxicillin-clavulanate for treatment failure or concurrent sinusitis.

Alagille Syndrome

– An autosomal dominant disorder caused by mutations in JAG1 (95%) or NOTCH2 (<5%), affecting the Notch signaling pathway. Features: chronic cholestasis from paucity of intrahepatic bile ducts (90-100%), cardiac anomalies (peripheral pulmonary stenosis most common, also tetralogy of Fallot), butterfly vertebrae (50%), posterior embryotoxon (a ring in the anterior chamber of the eye, 90%), and characteristic facies (broad forehead, deep-set eyes, pointed chin, 90-95%). Presents in infancy with prolonged jaundice, pruritus, and failure to thrive. Other findings: renal anomalies (30-50%), hypercholesterolemia, and coagulopathy from fat malabsorption. Diagnosis: clinical criteria (Alagille score) and JAG1/NOTCH2 genetic testing. Management: ursodeoxycholic acid for cholestasis, fat-soluble vitamin supplementation, nutritional support, cardiac repair as needed, and liver transplantation for end-stage liver disease (15-20%).

Angelman Syndrome

– A genetic disorder caused by loss of function of the UBE3A gene on chromosome 15 (maternal deletion or paternal uniparental disomy). Features: severe intellectual disability, absent speech, seizures, ataxia, and frequent laughter. Characteristic happy demeanor with hand-flapping movements. Management: seizure control, physical therapy, and communication support.

Annular Pancreas

– A congenital anomaly where a ring of pancreatic tissue completely or partially encircles the second portion of the duodenum, causing duodenal obstruction. It results from failure of the ventral pancreatic bud to completely rotate around the duodenum during embryogenesis. Associated with Down syndrome, duodenal atresia (coexisting in 50%), and other GI anomalies. Presents in the newborn period with bilious or non-bilious vomiting (depending on the position relative to the ampulla) and polyhydramnios (prenatal). In adults, may present with recurrent pancreatitis or peptic ulcer disease. Diagnosis: upper GI series shows a narrowed, eccentric duodenum with obstructive findings; CT or MRCP demonstrates the pancreatic tissue ring encasing the duodenum. Treatment: surgical bypass (duodeno-duodenostomy or duodeno-jejunostomy) without resection of the annular tissue (risk of pancreatitis and fistula). Prognosis: excellent after surgical repair.

Anorectal Malformation

– A spectrum of congenital anomalies involving the anus and rectum, classified by the location of the rectal pouch relative to the puborectalis sling: Low (passes through the levator muscle, good prognosis) — perineal fistula, vestibular fistula, or covered anus. High (ends above the levator muscle, poor prognosis, 60%) — rectourethral fistula (boys), rectovaginal fistula (girls), or rectobladder neck fistula. Cloaca (girls): a single common channel for the urethra, vagina, and rectum. Presents with absence of a normal anal opening on perineal examination and failure to pass meconium within 24 hours. Associated anomalies (VACTERL workup): vertebral, cardiac (VSD), tracheoesophageal, renal (VUR, renal agenesis), and limb. Diagnosis: invertogram (Wangensteen-Rice X-ray) and cross-table lateral X-ray to determine the level. MRI for complex cases. Treatment: low lesions — primary perineal anoplasty. High lesions — colostomy in the newborn period, followed by posterior sagittal anorectoplasty (PSARP, deVries-Peña procedure) at 3-6 months. Prognosis: low lesions — excellent bowel control. High lesions — variable fecal continence, requiring bowel management programs (enemas, laxatives, antegrade continence enema).

Anorexia Nervosa

– An eating disorder characterized by restriction of energy intake, intense fear of weight gain, and distorted body image. Highest mortality of any psychiatric disorder. Features: emaciation, lanugo, bradycardia, hypotension, amenorrhea, and electrolyte abnormalities. Diagnosis by DSM-5 criteria. Treatment: nutritional rehabilitation, cognitive behavioral therapy, and family-based therapy. Medical stabilization when severely malnourished.

Apert Syndrome

– An autosomal dominant craniosynostosis syndrome caused by FGFR2 mutations (most commonly Ser252Trp or Pro253Arg). Features: craniosynostosis (bilateral coronal, resulting in turribrachycephaly — tower-shaped skull), midface hypoplasia, proptosis, hypertelorism, and syndactyly of the hands and feet (mitten or spade-like hands, complete fusion of digits). Cleft palate occurs in 30%. May have intellectual disability (variable). Other associations: cervical spine fusion, congenital heart disease, and hydrocephalus. Diagnosis: clinical and genetic testing. Management: staged surgical reconstruction — cranial vault remodeling (3-6 months), midface advancement (Le Fort III, 6-12 years), and syndactyly release (6-18 months). Lifelong neurosurgical and dental monitoring.

Apgar Score

– Rapid clinical assessment of newborn well-being performed at 1 and 5 minutes after birth. Five criteria scored 0-2 each: Appearance (skin color), Pulse (heart rate), Grimace (reflex irritability), Activity (muscle tone), and Respiration. Total score 0-10. Score 7-10 normal; 4-6 moderately depressed; 0-3 severely depressed. Determines need for immediate resuscitation. May extend to 10 and 20 minutes in severely depressed infants.

Apnea of Prematurity

– Cessation of breathing for more than 20 seconds or shorter episodes with bradycardia or desaturation in infants less than 34-37 weeks gestation. Three types: central (no respiratory effort), obstructive (airway collapse), and mixed. Treatment: caffeine citrate (first-line), continuous positive airway pressure, and methylxanthine therapy. Monitor with cardiorespiratory monitors. Most resolve by 34-44 weeks corrected gestational age.

Arnold-Chiari Malformation

– Herniation of cerebellar tonsils through the foramen magnum into the upper cervical spinal canal. Type II (myelomeningocele-associated) is the most common form in pediatrics. Associated with hydrocephalus, syringomyelia, and spinal cord dysfunction. Symptoms include headache, neck pain, and lower cranial nerve palsies. Treatment: posterior fossa decompression with duraplasty. Often coexists with myelomeningocele requiring concurrent repair.

Babinski Reflex

– An extensor plantar response present from birth to approximately 2 years. When the lateral sole is stroked, the great toe dorsiflexes and other toes fan outward. Disappears as the corticospinal tract myelinates. Persistence beyond age 2 may indicate neurologic abnormality. Pathologic in older children and adults, suggesting upper motor neuron lesion. One of the deep tendon reflexes assessed in infants.

Bacterial Meningitis

– Infection of the meninges by bacteria, a medical emergency. In neonates: Group B Streptococcus, E. coli, Listeria. In older children: Neisseria meningitidis, Streptococcus pneumoniae. Presents with fever, headache, neck stiffness, altered consciousness, and Kernig/Brudzinski signs. Diagnosis by lumbar puncture. Treatment: immediate empiric antibiotics (ceftriaxone and vancomycin) without delaying for imaging.

Ballard Score

– Clinical assessment of gestational age in newborns using 6 physical and 6 neuromuscular maturity criteria. Physical parameters: skin texture, ear form and recoil, eye appearance, breast tissue, and genitalia. Neuromuscular: posture, arm and leg recoil, popliteal angle, scarf sign, and heel-to-ear. Each scored 0-5; total converted to gestational age in weeks. Most accurate near term; used when prenatal dating is uncertain.

Beckwith-Wiedemann Syndrome

– An overgrowth disorder associated with 11p15.5 imprinting defects. Features: macrosomia, macroglossia, omphalocele, visceromegaly, and neonatal hypoglycemia. Increased risk of embryonal tumors (Wilms tumor, hepatoblastoma). Management: glucose monitoring, tumor surveillance with abdominal ultrasound and AFP levels, and surgical correction of structural anomalies.

Binge Eating Disorder

– Recurrent episodes of eating large quantities of food with a sense of lack of control, without regular compensatory behavior. Associated with obesity, depression, and anxiety. Diagnosis by DSM-5 criteria. Treatment: cognitive behavioral therapy, interpersonal therapy, and SSRIs. Weight management with dietary counseling and exercise.

Biotinidase Deficiency

– An autosomal recessive inborn error of biotin recycling caused by BTD gene mutations, resulting in inability to cleave biotin from biocytin (proteolytic degradation product of biotin-dependent carboxylases). This leads to multiple carboxylase deficiency affecting four biotin-dependent enzymes: pyruvate carboxylase, propionyl-CoA carboxylase, 3-methylcrotonyl-CoA carboxylase, and acetyl-CoA carboxylase. Presents between 1 week and 10 years of age (mean 3 months). Features: neurological — seizures (myoclonic, infantile spasms), hypotonia, ataxia, hearing loss, and optic atrophy; cutaneous — periorificial dermatitis, alopecia, and candidal infections; metabolic — metabolic acidosis, ketolactic acidosis, and organic aciduria (3-hydroxyisovalerate, 3-methylcrotonylglycine, lactate, propionate). Diagnosis: biotinidase enzyme assay in serum and BTD genetic testing. Included in newborn screening in many countries. Treatment: oral biotin (5-20 mg/day) — a dramatically effective and inexpensive therapy. Starting biotin promptly reverses most symptoms except sensorineural hearing loss and optic atrophy, which may be irreversible if delayed.

Bochdalek Hernia

– A congenital diaphragmatic hernia (CDH) occurring through the posterolateral foramen of Bochdalek, resulting from failure of the pleuroperitoneal membrane to close by the 8th week of gestation. The left side is affected in 85% of cases (the right pleuroperitoneal canal closes earlier). Abdominal organs (stomach, spleen, small intestine, colon, and sometimes the left lobe of the liver) herniate into the chest, causing pulmonary hypoplasia and pulmonary hypertension. Presents at birth with severe respiratory distress, cyanosis, scaphoid abdomen, and decreased breath sounds on the affected side with bowel sounds heard in the chest. Diagnosis: prenatal ultrasound (polyhydramnios, mediastinal shift, herniated organs), postnatal chest X-ray shows abdominal organs in the thorax. Treatment: endotracheal intubation (avoid bag-mask ventilation — distends the stomach), nasogastric decompression, and surgical repair via an abdominal approach once pulmonary hypertension is managed. Delayed surgical repair after physiologic stabilization improves outcomes. High-frequency ventilation, inhaled nitric oxide, and ECMO for severe pulmonary hypertension. Prognosis depends on the degree of pulmonary hypoplasia; LHR (lung-to-head ratio) is the best prenatal predictor.

Brachial Plexus Injury

– Damage to the brachial plexus nerves during birth, most commonly from shoulder dystocia or excessive traction. Erb's palsy (C5-C6) presents with waiter's tip position (arm adducted, elbow extended, forearm pronated). Klumpke's palsy (C8-T1) affects the hand and may cause Horner syndrome. Most cases resolve spontaneously. Physical therapy initiated early; surgical exploration considered if no recovery by 3-6 months.

Bradycardia of Prematurity

– Heart rate less than 100 beats per minute in a premature infant, often associated with apnea and oxygen desaturation. May be caused by vagal stimulation, reflux, or immaturity of the cardiac conduction system. Treatment: gentle stimulation, caffeine, and addressing underlying causes. Usually self-limited and resolves as the infant matures. Monitoring with continuous cardiorespiratory monitoring.

Breast Engorgement

– Swelling and tenderness of neonatal breasts occurring in both male and female infants at 2-5 days of age, caused by maternal estrogen withdrawal. May produce milky secretions (witch's milk). A normal, self-limited condition. No treatment required; typically resolves within 2-4 weeks. Infection is rare. Historically, traditional practices of pressing the breasts are harmful and should be avoided.

Breath-Holding Spells

– A benign paroxysmal phenomenon in young children (6 months to 6 years), provoked by frustration, pain, or fear. Cyanotic type (more common): child cries, forcefully exhales, becomes cyanotic, and may lose consciousness briefly. Pallid type: triggered by sudden pain/fright, child becomes pale and limp, may have brief seizure-like activity due to reflex anoxic seizure. Pathophysiology: autonomic dysregulation, vagal overactivity. Diagnosis: clinical; EEG normal (distinguishing from epilepsy). Treatment: parental reassurance, management of triggers, iron supplementation if anemic. Spells resolve spontaneously by age 6. No increased risk of epilepsy or other neurological problems.

Bronchiolitis

– Lower respiratory tract infection causing inflammation and obstruction of small airways. Most common cause is respiratory syncytial virus. Affects infants under 12 months. Presents with wheezing, tachypnea, nasal flaring, and retractions. Diagnosis is clinical. Treatment: supportive care with nasal suctioning, supplemental oxygen, and respiratory support. Severe cases may require CPAP or mechanical ventilation. No effective antiviral therapy.

Bronchopulmonary Dysplasia

– A chronic lung disease of premature infants, resulting from lung injury from mechanical ventilation, oxygen toxicity, and inflammation. Most commonly occurs in infants born at less than 28 weeks gestation with birth weight less than 1500 grams. Pathophysiology involves arrested alveolar and vascular development with simplified alveolar structures, decreased alveolar number, and pulmonary vascular remodeling. The National Institutes of Health consensus definition classifies severity at 36 weeks postmenstrual age or at discharge: mild (no supplemental oxygen), moderate (oxygen less than 30%), or severe (oxygen at least 30% or positive pressure ventilation). Pathogenesis is multifactorial: ventilator-induced lung injury, oxygen toxicity, infection, patent ductus arteriosus, and fluid overload. Management: lung-protective ventilation strategies, judicious oxygen therapy targeting saturations 90-95%, fluid restriction, diuretics, caffeine, and systemic corticosteroids for severe cases not weaning from ventilation. Nutritional support with high-calorie formulas to promote lung growth. Long-term complications include pulmonary hypertension, reactive airway disease, and neurodevelopmental impairment. Most infants improve with lung growth, but pulmonary function abnormalities may persist into adulthood.

Bulimia Nervosa

– An eating disorder characterized by recurrent episodes of binge eating followed by compensatory behaviors (purging, laxatives, excessive exercise). Self-esteem overly influenced by body shape and weight. Complications: electrolyte abnormalities (hypokalemia, metabolic alkalosis), dental erosion, esophageal tears, and parotid gland enlargement. Treatment: cognitive behavioral therapy and SSRIs (fluoxetine).

Caput Succedaneum

– Edema of the scalp presenting as a soft, boggy swelling that crosses suture lines. Caused by pressure of the vaginal walls on the fetal head during labor. Resolves within hours to days. Differentiated from cephalohematoma (subperiosteal, does not cross suture lines) and subgaleal hemorrhage (life-threatening, crosses suture lines). No treatment required; spontaneous resolution.

Cerebral Palsy

– A group of permanent disorders of movement and posture development causing activity limitation, attributed to non-progressive disturbances occurring in the developing fetal or infant brain. It is the most common motor disability in childhood, affecting approximately 1 in 500 live births. Classification by motor type: spastic (80 percent, most common, includes hemiplegia, diplegia, and quadriplegia), dyskinetic (athetoid, choreoathetoid, dystonic), ataxic, hypotonic, and mixed. Spastic cerebral palsy is subclassified by distribution: hemiplegia (unilateral arm and leg), diplegia (lower limbs more affected, common in premature infants with periventricular leukomalacia), and quadriplegia (all four limbs, often with intellectual disability and seizures). Causes: prematurity, intrauterine infection, perinatal hypoxia-ischemia, intracranial hemorrhage, and genetic abnormalities. Diagnosis is clinical, based on delayed motor milestones, abnormal muscle tone, and persistent primitive reflexes. Management is multidisciplinary: physical therapy, occupational therapy, speech therapy, orthotics, spasticity management (botulinum toxin, oral baclofen, intrathecal baclofen pump), orthopaedic surgery for contractures, and management of associated conditions including epilepsy, feeding difficulties, and learning disabilities. Prognosis depends on severity and associated impairments; most children survive to adulthood with appropriate support.

CHARGE Syndrome

– A genetic disorder caused by CHD7 gene mutations (autosomal dominant, most cases de novo) affecting multiple organ systems. The mnemonic CHARGE describes: Coloboma (coloboma of the eye, 80-90%), Heart defects (tetralogy of Fallot, VSD, ASD, PDA, 75-85%), Atresia choanae (choanal atresia/stenosis, 50-60%), Retarded growth and development (growth hormone deficiency, intellectual disability), Genital hypoplasia (cryptorchidism, micropenis, 50-60% in males), and Ear anomalies (characteristic cup-shaped ears with prominent antihelix, hearing loss, 90-95%). Presents with life-threatening respiratory distress at birth from choanal atresia. Other common features: facial palsy, cleft lip/palate, and feeding difficulties. Diagnosis: clinical criteria (Verloes or Blake) and CHD7 genetic testing. Management: multidisciplinary — airway management, cardiac repair, hearing aids/cochlear implants, growth hormone therapy, and developmental support.

Chickenpox

– Primary infection with Varicella-Zoster Virus, characterized by pruritic, vesicular rash in different stages (macules, papules, vesicles, crusts). Highly contagious via respiratory droplets and contact with lesions. Incubation 10-21 days. Complications include secondary bacterial infection, pneumonia, encephalitis, and Reye syndrome. Prevention by varicella vaccine. Treatment: supportive care; acyclovir for severe cases.

Child Abuse

– Physical, sexual, or emotional harm or neglect of a child by a caregiver. Recognized by inconsistent histories, injury patterns inconsistent with mechanism, and behavioral indicators. Types: physical abuse, sexual abuse, neglect, and emotional abuse. Mandatory reporting requirements for healthcare providers. Evaluation includes full skeletal survey for suspected physical abuse, laboratory tests, and social work involvement.

Child Neglect

– The failure of a caregiver to provide adequate food, shelter, clothing, supervision, education, or medical care. The most common form of child maltreatment. Types: physical neglect, educational neglect, emotional neglect, and medical neglect. May result in failure to thrive, developmental delay, and chronic illness. Identified through failure to attend appointments, untreated medical conditions, and inadequate living conditions.

Chlamydia

– The most common reportable bacterial STI, caused by Chlamydia trachomatis. Often asymptomatic in women, causing cervicitis, urethritis, and pelvic inflammatory disease. Complications include tubal factor infertility and ectopic pregnancy. Diagnosis by nucleic acid amplification test. Treatment: azithromycin or doxycycline. Partner treatment essential to prevent reinfection. Screening recommended for all sexually active women under 25.

Choanal Atresia

– A congenital obstruction of the posterior nasal passage (choana) due to failure of the bucconasal membrane to rupture during embryogenesis. Bilateral choanal atresia presents as a respiratory emergency in the newborn, who is an obligate nasal breather — cyclic cyanosis that improves with crying. Unilateral atresia presents later with persistent unilateral nasal discharge and obstruction. 70% are bony, 30% are membranous. Associated with CHARGE syndrome (coloboma, heart defects, choanal atresia, retardation, genital hypoplasia, ear anomalies) and other anomalies. Diagnosis: inability to pass a nasogastric tube through the nose, confirmed by CT. Treatment: immediate airway management (oral airway, McGovern nipple), followed by surgical repair via transnasal endoscopic choanoplasty with stenting. Unilateral cases are repaired electively.

Clavicle Fracture

– The most common birth-related fracture, occurring in approximately 1-2% of vaginal deliveries. Risk factors include shoulder dystocia, macrosomia, and breech presentation. Presents with asymmetric arm movements, palpable crepitus, and reluctance to move the affected arm. Diagnosis by X-ray. Treatment: arm immobilization against the body (pinned diaper or arm sling). Heals rapidly in newborns with excellent prognosis.

Cloacal Exstrophy

– The most severe congenital anomaly of the ventral abdominal wall, also called the OEIS complex (Omphalocele, Exstrophy of the bladder, Imperforate anus, Spinal defects). Incidence 1 in 200,000-400,000 births. Features: an exstrophic central bowel segment (cecum, between two hemi-bladders), omphalocele, imperforate anus, and spinal anomalies (myelomeningocele, tethered cord). Girls have a duplicated vagina and uterus; boys have an epispadiac, short phallus. There is near-universal association with Chiari malformation and hydrocephalus. Diagnosis: prenatal ultrasound shows a large midline abdominal wall defect with cystic structures (hemi-bladders) and spinal abnormalities. Treatment: staged surgical reconstruction — immediate bowel coverage (silo), separation of bowel from bladder, closure of the abdominal wall; later — bladder closure, genital reconstruction, and spinal cord repair. Prognosis: survival >90% with modern care. Bowel and bladder function outcomes are guarded; most require intermittent catheterization and bowel management.

Congenital CMV

– Cytomegalovirus infection acquired in utero, the most common congenital viral infection. Maternal primary infection poses highest risk of fetal transmission and disease. Can cause microcephaly, sensorineural hearing loss, chorioretinitis, hepatosplenomegaly, and petechiae. Diagnosis by CMV PCR in urine or saliva. Treatment: ganciclovir or valganciclovir for symptomatic congenital CMV. Long-term sequelae include hearing loss and developmental delay.

Congenital Diaphragmatic Hernia

– A defect in the diaphragm allowing abdominal organs to herniate into the thoracic cavity, causing pulmonary hypoplasia and respiratory distress at birth. Most commonly a posterolateral (Bochdalek) defect on the left side. Presents with cyanosis, scaphoid abdomen, and bowel sounds in the chest. Treatment: intubation, nasogastric decompression, stabilization, and surgical repair once pulmonary hypertension is controlled.

Congenital Heart Disease

– Structural heart defects present at birth, the most common congenital anomaly (1% of live births). Common defects: ventricular septal defect (VSD, 30%), atrial septal defect (ASD), patent ductus arteriosus (PDA), tetralogy of Fallot (TOF), transposition of great arteries (TGA), coarctation of aorta, and hypoplastic left heart syndrome. Presentation depends on shunt direction and severity: cyanotic (TOF, TGA, tricuspid atresia) vs. acyanotic (VSD, ASD, PDA). Cyanotic: blue spells, clubbing, polycythemia. Acyanotic: heart murmur, heart failure, failure to thrive. Detected by prenatal ultrasound, newborn pulse oximetry screening, or clinical signs. Diagnosis: echocardiography. Treatment: surgical repair or palliation, catheter-based interventions for selected defects.

Congenital Heart Murmur

– A heart murmur detected in the newborn period, present in approximately 50-60% of healthy newborns. Most are innocent (functional) murmurs that disappear by 6 months of age. Pathologic murmurs may indicate congenital heart defects. Differentiation based on murmur characteristics, timing, and associated findings. Echocardiography recommended for pathologic murmurs. Innocent murmurs require no treatment.

Congenital Pneumonia

– Lung infection acquired in utero, often from ascending chorioamnionitis. Presents with respiratory distress, fever or hypothermia, and sepsis. Common organisms include Group B Streptococcus, E. coli, and Ureaplasma. Diagnosis by chest X-ray showing infiltrates and blood culture. Treatment: broad-spectrum IV antibiotics (ampicillin and gentamicin) plus respiratory support.

Congenital Rubella Syndrome

– Fetal infection with rubella virus, particularly dangerous in the first trimester. Classic triad: congenital heart disease (patent ductus arteriosus, pulmonary stenosis), cataracts, and sensorineural hearing loss. Also causes hepatosplenomegaly, thrombocytopenia, and blueberry muffin rash. Prevention by maternal vaccination before pregnancy. No specific antiviral treatment; supportive care.

Congenital Syphilis

– Treponema pallidum infection acquired transplacentally. Can cause stillbirth, prematurity, hepatosplenomegaly, rash, snuffles, osteochondritis, and Hutchinson teeth. Diagnosed by darkfield microscopy, VDRL/RPR in neonatal serum, or PCR. Treatment: IV penicillin G for 10 days. Prevention through maternal screening and treatment during pregnancy.

Congenital Zika Syndrome

– Fetal brain infection with Zika virus, causing microcephaly, brain abnormalities, eye defects, hearing loss, and limb contractures. Transmission occurs via mosquito bite or sexual contact. Diagnosis by Zika virus PCR in maternal serum or amniotic fluid. No specific treatment. Prevention by avoiding endemic areas during pregnancy. Surveillance and developmental monitoring recommended for exposed infants.

Contraception for Adolescents

– Methods to prevent unintended pregnancy. Options: combined oral contraceptives, progestin-only pills, depot medroxyprogesterone acetate, contraceptive implant (Nexplanon), intrauterine devices (IUDs), condoms, and abstinence. Long-acting reversible contraceptives (LARC) such as IUDs and implants have highest efficacy and are first-line recommendations. Dual protection against STIs requires barrier methods.

Cornelia de Lange Syndrome

– A genetic disorder caused by mutations in genes of the cohesin complex (NIPBL 60%, SMC1A, SMC3, RAD21, HDAC8). Features: characteristic facies — synophrys (confluent eyebrows), long eyelashes, low-set ears, depressed nasal bridge with anteverted nares, thin lips, and micrognathia; prenatal and postnatal growth restriction; hirsutism; microcephaly; limb defects (oligodactyly, ectrodactyly, phocomelia, 30%); and congenital heart disease (VSD, ASD). Intellectual disability ranges from moderate to severe. Gastroesophageal reflux and feeding difficulties are near-universal. Diagnosis: clinical and genetic testing. Management: growth and feeding support (gastrostomy tube often required), antireflux therapy, developmental therapies, and surveillance for hearing loss, vision problems, and scoliosis.

Craniosynostosis

– Premature fusion of one or more cranial sutures, restricting skull growth perpendicular to the affected suture. Presents with abnormal head shape (scaphocephaly from sagittal fusion, plagiocephaly from coronal fusion). May increase intracranial pressure. Diagnosis by physical examination and CT with 3D reconstruction. Treatment: surgical cranioplasty to release fused suture and allow normal brain growth, typically between 3-12 months of age.

Cri-du-Chat Syndrome

– A genetic disorder caused by a deletion of the short arm of chromosome 5 (5p-), with the size of the deletion correlating with severity. Incidence 1 in 15,000-50,000. Features: a characteristic high-pitched, cat-like cry in infancy (due to laryngeal hypoplasia), microcephaly, round face, hypertelorism, epicanthal folds, micrognathia, and hypotonia. Intellectual disability ranges from moderate to severe. Other findings: congenital heart disease (VSD, PDA), scoliosis, and hearing loss. Diagnosis: karyotype, FISH, or chromosomal microarray showing 5p deletion. Management: early intervention programs, speech therapy, physical therapy, and treatment of associated anomalies. Patients have a characteristic friendly, sociable personality. Life expectancy is near normal for mild cases.

Croup

– Acute laryngotracheobronchitis, most commonly caused by parainfluenza virus. Affects children 6 months to 3 years. Presents with barking cough, stridor, hoarseness, and inspiratory distress, worse at night. Diagnosis is clinical; X-ray shows subglottic narrowing (steeple sign). Treatment: single dose of oral dexamethasone and nebulized epinephrine for moderate to severe cases. Self-limited; resolves in 3-7 days.

Crouzon Syndrome

– An autosomal dominant craniosynostosis syndrome caused by mutations in the FGFR2 or FGFR3 genes, leading to premature fusion of multiple cranial sutures. Features: brachycephaly (short, broad head from bilateral coronal synostosis), proptosis (shallow orbits), hypertelorism, midface hypoplasia, beaked nose, and Class III malocclusion. Unlike Apert syndrome, there are no hand or foot anomalies. Intelligence is typically normal. Increased intracranial pressure can occur from progressive synostosis. Diagnosis: clinical and genetic testing; CT for craniosynostosis confirmation. Management: multidisciplinary — cranial vault remodeling surgery for craniosynostosis (3-6 months), midface advancement (Le Fort III osteotomy, 6-12 years), and orthognathic surgery for malocclusion. Lifelong monitoring for elevated ICP, sleep apnea, and vision changes.

Developmental Milestones

– Skills and abilities that children typically acquire at predictable ages. Examples: social smile at 2 months, rolling over at 4-6 months, sitting independently at 6-8 months, walking at 12 months, and speaking two-word phrases at 2 years. Delays in reaching milestones may indicate developmental delay, autism, or neurologic impairment. Screening at each well-child visit using age-appropriate tools.

DiGeorge Syndrome

– A 22q11.2 microdeletion syndrome affecting the third and fourth pharyngeal pouches. Features: cardiac defects (tetralogy of Fallot, truncus arteriosus), hypocalcemia (parathyroid hypoplasia), immune deficiency (thymic hypoplasia), and characteristic facial features. Diagnosed by FISH testing or chromosomal microarray. Management: cardiac repair, calcium supplementation, immune monitoring, and speech therapy.

Down Syndrome

– Trisomy 21, the most common chromosomal abnormality in live births, occurring in approximately 1 in 700 births. Risk increases with maternal age. Features: characteristic facies (epicanthal folds, flat nasal bridge, protruding tongue), hypotonia, intellectual disability, congenital heart defects (AV canal), and increased risk of leukemia and early-onset Alzheimer's. Diagnosis by karyotype; prenatal screening with nuchal translucency and maternal serum markers.

DTaP Vaccine

– Diphtheria, Tetanus, and acellular Pertussis vaccine given at 2, 4, 6, and 15-18 months, and 4-6 years. Protects against three serious bacterial diseases. Diphtheria causes respiratory and cardiac complications. Tetanus causes lockjaw and muscle rigidity. Pertussis causes whooping cough, most severe in young infants. Acellular formulation reduces side effects compared to whole-cell pertussis vaccine.

Ductal-Dependent Lesions

– Congenital heart defects that require the ductus arteriosus to remain patent for adequate systemic or pulmonary blood flow. Includes pulmonary atresia, critical aortic stenosis, coarctation, hypoplastic left heart syndrome, and transposition of the great arteries. Present with cyanosis or shock when the ductus closes (typically within 24-48 hours). Treatment: prostaglandin E1 to maintain ductal patency until surgical intervention.

Duodenal Atresia

– A congenital obstruction of the duodenum caused by failure of recanalization during the 8th-10th week of gestation. Associated with Down syndrome (30%), polyhydramnios (due to impaired fetal swallowing of amniotic fluid), and other anomalies (VACTERL, congenital heart disease, annular pancreas, malrotation). Presents within hours of birth with bilious vomiting (90%, if distal to the ampulla of Vater), epigastric distension, and failure to pass meconium. The classic X-ray finding is the double bubble sign (air in the stomach and proximal duodenum, with no distal gas). Diagnosis: prenatal ultrasound (polyhydramnios, dilated stomach and duodenum), postnatal abdominal X-ray, and upper GI series to confirm and exclude malrotation. Treatment: nasogastric decompression, IV fluids, and surgical duodeno-duodenostomy (diamond-shaped anastomosis, Kimura technique). Prognosis: excellent with surgical repair (>95% survival), but depends on associated anomalies.

Dyslexia

– A specific learning disorder with neurobiological origin, characterized by difficulties with accurate and fluent word recognition, spelling, and decoding ability, despite adequate intelligence, motivation, and educational opportunity. Affects approximately 5–10% of the population. The underlying cause involves impaired phonological processing (difficulty recognizing and manipulating the sounds of spoken language) and deficits in rapid automatized naming. Functional neuroimaging shows reduced left temporoparietal and occipitotemporal activation during reading. Diagnosis: comprehensive educational assessment evaluating reading accuracy, fluency, comprehension, and phonological processing. Management: structured, explicit, multisensory phonics-based reading instruction (Orton-Gillingham, Wilson Reading System, Barton), assistive technology, and educational accommodations.

Edwards Syndrome

– Trisomy 18, the second most common autosomal trisomy in live births (1 in 5,000). Caused by complete trisomy 18 (94%), mosaicism, or partial trisomy (translocation). Features: severe intrauterine growth restriction, clenched fists with overlapping fingers (index over middle, fifth over fourth), rocker-bottom feet, low-set malformed ears, micrognathia, prominent occiput, and congenital heart defects (VSD, PDA, polyvalvular disease, 90-95%). Other findings: horseshoe kidney, esophageal atresia, omphalocele, and neural tube defects. Diagnosis: prenatal screening (low PAPP-A, low hCG, high inhibin A), NIPT, and karyotype/CMA. Management: comfort care is the standard (median survival 5-15 days; 5-10% survive to 1 year). Surgical intervention for life-threatening anomalies is controversial. Survivors have severe intellectual disability.

Ehlers-Danlos Syndrome

– A group of hereditary connective tissue disorders caused by defects in collagen synthesis or processing. The most common types: Classic (COL5A1, COL5A2) — skin hyperextensibility, atrophic scarring, and joint hypermobility; Hypermobile (hEDS, unknown genetic cause, most common) — generalized joint hypermobility, chronic pain, and easy bruising; Vascular (COL3A1, most dangerous) — thin translucent skin, easy bruising, arterial rupture (carotid, aortic, visceral), uterine rupture, and sigmoid colon perforation; Kyphoscoliotic (PLOD1) — congenital hypotonia, scoliosis, and joint laxity; and Arthrochalasia (COL1A1, COL1A2) — severe joint hypermobility and congenital hip dislocation. Diagnosis: clinical (Beighton score for hypermobility), collagen typing on skin biopsy (vascular type), and genetic testing. Management: activity modification and joint protection, vascular surveillance for vascular EDS (carotid duplex, aortic imaging, avoidance of invasive procedures), high-dose vitamin C, and genetic counseling. No cure; management is symptomatic and preventive.

Emergency Contraception

– Postcoital contraception to prevent pregnancy after unprotected intercourse or contraceptive failure. Levonorgestrel (Plan B): effective within 72 hours, available over the counter. Ulipristal acetate (Ella): effective within 120 hours, prescription only. Copper IUD: most effective emergency contraception, placed within 5 days. Not abortifacients; prevent or delay ovulation.

Enuresis (Bedwetting)

– Involuntary urination during sleep in a child aged 5 years or older, affecting approximately 5–10% of 7-year-olds. Primary enuresis: the child has never achieved sustained nighttime dryness. Secondary enuresis: wetting recurs after at least 6 months of dryness, often triggered by psychosocial stress, urinary tract infection, or diabetes. Pathophysiology involves a mismatch between nocturnal bladder capacity, nighttime urine production (lack of circadian ADH rhythm), and inability to arouse from sleep in response to bladder fullness. Evaluation: history (wetting pattern, fluid intake, constipation), urinalysis (glucose, infection). Management: behavioral (timed voiding, reward systems, fluid restriction before bed), enuresis alarm (first-line, 70% success), and pharmacotherapy (desmopressin–synthetic ADH–reduces nighttime urine production).

Epiglottitis

– Inflammation of the epiglottis, a life-threatening emergency. Most commonly caused by Haemophilus influenzae type b (now rare due to vaccination). Presents with high fever, drooling, dysphagia, tripod position, and rapid progression to airway obstruction. Do not examine the throat. Treatment: secure airway in operating room, IV antibiotics (ceftriaxone). Requires immediate airway management; mortality high if untreated.

Epispadias

– A congenital anomaly where the urethral meatus opens on the dorsal (top) surface of the penis in boys or in a split, open position on the clitoral hood in girls. Caused by abnormal development of the genital tubercle and cloacal membrane. In boys, the opening may be at the glans (glandular, mildest), penile shaft, or penopubic (most severe, associated with bladder exstrophy). In girls, the urethra is short with a bifid clitoris and separated labia. Epispadias presents as part of the exstrophy-epispadias complex spectrum. The penis is short, wide, and upwardly curved (dorsal chordee). Diagnosis: clinical. Treatment: surgical repair (Mitchell or Cantwell-Ransley technique) between 6-12 months — urethral reconstruction, penile straightening, and urethroplasty. Prognosis: urinary continence is achieved in 70-80% with repaired epispadias.

Erb's Palsy

– Upper brachial plexus injury (C5-C6) causing weakness or paralysis of the shoulder and arm muscles. Presents with the characteristic waiter's tip position: arm adducted, elbow extended, forearm pronated, wrist flexed. Most commonly caused by shoulder dystocia during delivery. Approximately 80-90% recover spontaneously. Management includes physical therapy to prevent contractures; surgical intervention if no improvement by 3-6 months.

Esophageal Atresia

– A congenital anomaly where the esophagus ends in a blind pouch, often with a tracheoesophageal fistula connecting the distal esophagus to the trachea. Presents with drooling, inability to pass a nasogastric tube, and choking with feedings. The 3 Es association: Esophageal atresia, Esophageal anomalies, and Extremity anomalies. Treatment: surgical ligation of fistula and esophageal anastomosis.

Exchange Transfusion

– Emergent procedure to rapidly lower severe hyperbilirubinemia by removing small volumes of the infant's blood and replacing with compatible donor blood. Used when phototherapy fails or bilirubin approaches toxic levels (greater than 25 mg/dL in term infants or above exchange threshold on Bhutani curve). Risks include infection, electrolyte imbalances, and air embolism. Double-volume exchange (160-200 mL/kg) replaces approximately 85% of circulating red blood cells.

Failure to Thrive

– Inadequate growth in children, defined as weight below the 5th percentile or crossing two major percentile lines on growth charts. Causes include caloric insufficiency (neglect, poor feeding), organic disease (congenital heart disease, renal disease), and psychosocial factors. Evaluation includes comprehensive history, physical examination, laboratory studies, and calorie counts. Treatment addresses underlying cause and nutritional rehabilitation.

Fanconi Anemia

– An autosomal recessive (or X-linked) bone marrow failure syndrome caused by mutations in at least 22 FANC genes involved in DNA repair (the FA/BRCA pathway). Features: progressive pancytopenia (onset typically at 5-10 years), radial ray anomalies (thumb hypoplasia or aplasia, radial club hand), microcephaly, short stature, café-au-lait spots, and renal anomalies. Increased risk of myelodysplastic syndrome, acute myeloid leukemia, and solid tumors (head and neck squamous cell carcinoma, gynecological cancer). Diagnosis: chromosomal breakage test (diepoxybutane or mitomycin C — increased chromosome breaks) and FANC gene sequencing. Management: androgens (oxymetholone) for cytopenias, hematopoietic stem cell transplant for bone marrow failure (with reduced-intensity conditioning), and lifelong cancer surveillance.

Febrile Seizures

– Seizures occurring in children 6 months to 5 years triggered by fever (temperature greater than 38 degrees Celsius), without intracranial infection or defined cause. Occurs in 2-5% of children. Simple:

generalized, single seizure lasting less than 15 minutes, no recurrence within 24 hours. Complex: focal, prolonged (greater than 15 minutes), or recurrent within 24 hours. Diagnosis of exclusion after ruling out meningitis and intracranial infection. In simple febrile seizures, there is no increased risk of epilepsy. Management: parental education on seizure first aid, antipyretics for comfort (do not prevent recurrence), and reassurance. No routine antiepileptic medication is indicated. Lumbar puncture should be considered when meningeal signs are present or in infants under 12 months with incomplete vaccination.

Fetal Distress

– Abnormal fetal status during labor, indicated by abnormal fetal heart rate patterns (late or prolonged decelerations, minimal variability, absent accelerations). May indicate fetal hypoxia, cord compression, or placental insufficiency. Management includes intrauterine resuscitation: change maternal position, stop oxytocin, give IV fluids and oxygen, and amnioinfusion. If persistent, emergent delivery by cesarean or operative vaginal delivery.

Fifth Disease

– Erythema infectiosum caused by Parvovirus B19. Characterized by bright red cheeks (slapped cheek appearance) followed by a lacy, reticulated rash on the trunk and extremities. Mild febrile illness in children. Concern in pregnancy (hydrops fetalis risk) and immunocompromised patients (pure red cell aplasia). Diagnosis by serology. Self-limited; no specific treatment.

Fontanelle

– Soft membranous spaces between skull bones that allow brain growth and skull molding during delivery. Anterior fontanelle diamond-shaped, closes by 12-18 months. Posterior fontanelle triangular, closes by 2-3 months. Bulging fontanelle suggests increased intracranial pressure; sunken fontanelle indicates dehydration. Palpation is part of the routine infant physical examination.

Formula Feeding

– Alternative to breastfeeding using commercially prepared infant formula. Types include cow's milk-based, soy-based, hydrolyzed protein, and amino acid-based formulas. Iron-fortified formulas standard to prevent iron deficiency anemia. Preparation requires clean water and proper sterilization of bottles. Feeding on demand approximately every 3-4 hours. Provides adequate nutrition when breastfeeding is not possible or desired.

Fragile X Syndrome

– X-linked dominant disorder caused by CGG trinucleotide repeat expansion in the FMR1 gene. Most common inherited cause of intellectual disability. Features: long face, prominent ears, macroorchidism in post-pubertal males, and behavioral problems including autism spectrum features. Diagnosis by DNA testing for CGG repeat expansion. No cure; behavioral and educational interventions.

Functional Asplenia

– Loss of splenic function due to sickling and infarction in sickle cell disease, typically by age 5 years. Increases risk of overwhelming bacterial infection, particularly from encapsulated organisms (Streptococcus pneumoniae, Haemophilus influenzae). Management: prophylactic penicillin, pneumococcal vaccines, and parental education about fever as a medical emergency.

Galactocele

– A milk-filled cyst in the breast, typically occurring in lactating women. Caused by obstruction of a milk duct. Presents as a palpable, tense, fluctuant mass that may be tender. Diagnosis is clinical and by ultrasound (simple cyst, may have fat-fluid level). Milky fluid may be aspirated. Differentiated from breast abscess (fever, erythema, tenderness, purulent fluid). Management: conservative (continued breastfeeding, warm compresses), aspiration if symptomatic or for diagnosis. Rarely, galactocele can occur in the breast bud of a neonate (maternal hormone stimulation).

Galant Reflex

– A primitive reflex present from birth to 4 months, elicited by stroking the skin along the side of the infant's back. The infant curves toward the stimulated side. Assessed by running a finger from the lateral back toward the hip while the infant is in ventral suspension. Absence may indicate neurologic abnormality. One of the trunk reflexes evaluated during newborn neurologic examination.

Gastroschisis

– A full-thickness abdominal wall defect, usually to the right of the umbilicus, through which bowel herniates without a covering membrane. Not associated with chromosomal abnormalities. Presents at birth with exposed bowel loops that may be edematous, thickened, and discolored from amniotic fluid exposure. Management: immediate coverage with saline-moistened dressings, nasogastric decompression, and surgical repair (primary closure or staged reduction).

German Measles (Rubella)

– See Rubella.

Glandular Fever (Infectious Mononucleosis)

– See Mononucleosis.

Glycogen Storage Disease

– A group of inherited metabolic disorders caused by defects in glycogen synthesis, breakdown, or regulation, resulting in abnormal glycogen accumulation in tissues. Major types: Type I (von Gierke, G6PC deficiency) — fasting hypoglycemia, lactic acidosis, hepatomegaly, hyperuricemia, and hypertriglyceridemia. Type II (Pompe disease, GAA deficiency) — infantile form: hypertrophic cardiomyopathy, hypotonia, and death by 1 year; late-onset: proximal myopathy and respiratory weakness. Type III (Cori disease, AGL deficiency) — hypoglycemia, hepatomegaly, and myopathy. Type IV (Andersen disease, GBE1 deficiency) — liver failure, cirrhosis, and cardiomyopathy. Type V (McArdle disease, PYGM deficiency) — exercise intolerance, myoglobinuria, and second-wind phenomenon. Type VI (Hers disease, PYGL deficiency) — mild fasting hypoglycemia and hepatomegaly. Type VII (Tarui disease, PFKM deficiency) — exercise intolerance similar to McArdle with hemolytic anemia. Diagnosis depends on the type: enzyme assay in liver or muscle biopsy, genetic testing, and newborn screening for Pompe. Treatment varies by type: Type I — continuous glucose therapy (cornstarch), allopurinol, and lipid-lowering agents. Type II (Pompe) — enzyme replacement therapy (alglucosidase alfa). Types III, VI — high-protein diet. Types V, VII — exercise modification and creatine supplementation.

Goldenhar Syndrome

– A variant of hemifacial microsomia characterized by the triad: epibulbar dermoids (benign conjunctival tumors), preauricular skin tags or sinuses, and vertebral anomalies (hemivertebrae, fused vertebrae, scoliosis). Also involves facial asymmetry, microtia, conductive hearing loss, and macrostomia. Other associations: congenital heart disease (VSD, tetralogy of Fallot), renal anomalies, and cleft lip/palate. Most cases are sporadic. Diagnosis: clinical and imaging (CT for skeletal anomalies, echocardiogram, renal ultrasound). Management: multidisciplinary — ophthalmology (epibulbar dermoid excision), audiology (hearing aids), plastic surgery (auricular reconstruction, mandibular distraction), and orthopedic monitoring for scoliosis.

Gonorrhea

– A bacterial STI caused by Neisseria gonorrhoeae, the second most common reportable infection. Can cause cervicitis, urethritis, pharyngeal infection, and disseminated gonococcal infection. Diagnosis by nucleic acid amplification test. Treatment: dual therapy with ceftriaxone and azithromycin (or doxycycline for uncomplicated cases). Increasing antibiotic resistance is a major public health concern.

Growing Pains

– Recurrent, bilateral leg pain in children ages 3-12, typically occurring in the late afternoon or evening and resolving by morning. Pain is deep, aching in the calves, shins, and thighs (not joints). No limping, fever, swelling, or functional impairment. Etiology unclear: may relate to increased activity, musculoskeletal overuse, or decreased pain threshold. Diagnosis: clinical (diagnosis of exclusion). Red flags suggesting pathology: unilateral pain, joint involvement, morning stiffness, limping, fever, night sweats, nocturnal pain waking the child. Treatment: reassurance, massage, stretching, warm baths, and acetaminophen/ibuprofen as needed. Benign and self-limited.

Gynecomastia

– Enlargement of breast tissue in males, often bilateral, occurring physiologically in neonates, during puberty (40-60% of boys), and in aging men. Pathologic causes include Klinefelter syndrome, testicular tumors, medications (spironolactone, cannabis), and liver disease. Persistent or asymmetric gynecomastia warrants evaluation. Management: observation for physiologic cases; treatment of underlying cause; mastectomy for severe persistent cases.

Hand-Foot-Mouth Disease

– A viral illness caused by Coxsackievirus A16 or Enterovirus 71, common in children under 5. Presents with fever, malaise, and characteristic vesicular rash on hands, feet, and mouth. Oral lesions are painful ulcers on the tongue and buccal mucosa. Self-limited in 7-10 days. Complications are rare but include encephalitis, myocarditis, and nail loss (onychomadesis) weeks later.

Head Circumference

– Measured at the largest circumference above the ears and eyebrows, reflecting brain growth. Plotted on growth charts to detect microcephaly (below 3rd percentile) or macrocephaly (above 98th percentile). Rapid increase may indicate hydrocephalus or intracranial hemorrhage; failure to grow may indicate microcephaly or brain atrophy. Measured at each well-child visit until age 2.

Head lice

– Also known as pediculosis capitis, an infestation of the scalp by the obligate human ectoparasite Pediculus humanus capitis, a wingless, blood-sucking insect that feeds on human blood. It is extremely common, especially in school-aged children aged 3-11 years, with an estimated 6-12 million infestations annually in the United States. Transmission is primarily through direct head-to-head contact (the most common route), and less commonly through shared combs, brushes, hats, headphones, pillows, or other personal items. Lice do not jump or fly; they crawl. Diagnosis is confirmed by finding live lice or nits attached to hair shafts within 1 cm of the scalp using a fine-toothed detection comb. Treatment: topical pediculicides containing permethrin 1% or dimeticone; oral ivermectin for treatment-resistant cases; wet combing (bug busting) as a non-insecticidal alternative. Bedding and clothing should be washed in hot water (60 degrees Celsius) and non-washable items sealed in a plastic bag for 2 weeks. Prophylactic treatment of household contacts is not recommended. School exclusion policies vary; the American Academy of Pediatrics and the UK National Institute for Health and Care Excellence recommend children remain in school after first treatment, as no-nit policies are unnecessary and promote stigma.

Hemifacial Microsomia

– A congenital craniofacial disorder involving asymmetric hypoplasia of the structures derived from the first and second branchial arches, most commonly affecting the mandible, ear, and soft tissues on one side. Also known as craniofacial microsomia or Goldenhar syndrome when epibulbar dermoids and vertebral anomalies are present. Presents with asymmetry: mandibular hypoplasia (Pruzansky-Kaban classification I-III), microtia (small or absent pinna), preauricular tags or sinuses, conductive hearing loss, and macrostomia (enlarged oral commissure). Severity ranges from mild asymmetry to severe facial deformity. Diagnosis: clinical, CT for skeletal assessment, and audiometry. Management: reconstructive surgery (mandibular distraction, costochondral graft, or orthognathic surgery), auricular reconstruction, hearing rehabilitation, and speech therapy.

Henoch-Schonlein Purpura

– A systemic vasculitis (IgA-mediated) affecting small vessels, primarily in children. Classic tetrad: palpable purpura (lower extremities), arthritis/arthralgia, abdominal pain, and renal involvement. May follow upper respiratory infection. Diagnosis is clinical. Renal involvement monitored with urinalysis. Most cases self-limited; corticosteroids for severe abdominal or renal disease.

Hepatitis B Vaccine

– Recombinant vaccine given at birth, 1-2 months, and 6-18 months. Protects against hepatitis B virus, which can cause chronic liver disease, cirrhosis, and hepatocellular carcinoma. Also prevents perinatal transmission when given to infants of HBsAg-positive mothers within 12 hours of birth, plus hepatitis B immune globulin.

Holt-Oram Syndrome

– An autosomal dominant disorder caused by TBX5 gene mutations, characterized by congenital heart disease and upper limb malformations. Cardiac anomalies: atrial septal defect (most common), VSD, and conduction abnormalities (heart block, atrial fibrillation). Limb defects are bilateral and asymmetric: thumb anomalies (absent, hypoplastic, triphalangeal, or finger-like thumb, 100%), radial dysplasia (radial club hand), and phocomelia. Diagnosis: clinical (heart defect + preaxial radial ray anomaly) and TBX5 genetic testing. Management: cardiac repair (ASD/VSD closure, pacemaker for heart block), reconstructive hand surgery (pollicization for absent thumb, centralization for radial club hand), and genetic counseling.

Homocystinuria

– An autosomal recessive inborn error of methionine metabolism caused most commonly by deficiency of cystathionine beta-synthase (CBS), leading to accumulation of homocysteine and methionine. Features (develop in childhood if untreated): ectopia lentis (lens dislocation, typically downward, differentiating from Marfan syndrome), severe myopia, intellectual disability, thromboembolism (the major cause of morbidity/mortality, affecting arteries and veins), osteoporosis, and a Marfanoid habitus (tall stature, pectus deformities, arachnodactyly). Diagnosis: elevated plasma homocysteine and methionine, homocystinuria, and CBS gene sequencing. Newborn screening includes homocystinuria in expanded panels. Treatment: pyridoxine (vitamin B6) responsiveness testing (approximately 50% are responsive), betaine (to remethylate homocysteine to methionine), methionine-restricted diet (especially for non-responders), and anticoagulation (aspirin for thromboprophylaxis). Lifelong monitoring for thromboembolism and lens dislocation.

Horseshoe Kidney

– The most common renal fusion anomaly (1 in 400-500), where the lower poles of both kidneys are fused across the midline by an isthmus of renal parenchyma or fibrous tissue. The fused kidney is located lower in the abdomen than normal (at L3-L5, anterior to the aorta and IVC) due to the isthmus being trapped under the inferior mesenteric artery during ascent. Associated with Turner syndrome, Edwards syndrome, Patau syndrome, and other urogenital anomalies. Often asymptomatic and discovered incidentally. May present (30%) with hydronephrosis from UPJ obstruction (the ureter passes anterior to the isthmus, causing kinking), recurrent UTIs, nephrolithiasis, and increased risk of Wilms tumor. Diagnosis: ultrasound, CT, or IV pyelogram. Treatment: surgical management of complications (pyeloplasty for UPJ obstruction, and nephrolithotomy for stones). Surveillance imaging for Wilms tumor is controversial.

HPV Vaccine

– Human Papillomavirus vaccine given at 11-12 years (can start at 9), with series completion by 15 years. Two-dose series if started before 15; three-dose series if started at 15 or older. Protects against HPV types causing cervical cancer, oropharyngeal cancer, and genital warts. Gardasil 9 covers nine HPV types. Expected to prevent over 90% of HPV-associated cancers.

Human Papillomavirus

– The most common STI, with over 100 types. High-risk types (16, 18) cause cervical, oropharyngeal, anal, and genital cancers. Low-risk types (6, 11) cause genital warts. Prevention: HPV vaccine (Gardasil 9) protects against 9 HPV types. Screening: Pap smear and HPV testing for cervical cancer prevention. Most infections clear spontaneously.

Hunter Syndrome

– Mucopolysaccharidosis type II (MPS II), an X-linked recessive lysosomal storage disorder caused by iduronate-2-sulfatase deficiency (IDS gene mutations). Accumulation of dermatan sulfate and heparan sulfate (same as Hurler). Two forms: severe (neuronopathic, 2/3 of cases) — progressive intellectual disability, neurodegeneration, and death by 10-15 years; attenuated (no or minimal neurologic involvement) — survival into adulthood. Features: coarse facies, corneal clouding (less prominent than Hurler), hepatosplenomegaly, recurrent hernias, dysostosis multiplex, joint stiffness, short stature, and cardiac valvular disease. A characteristic feature is the pebbled, ivory-white skin lesions (papules) over the back and shoulders (pathognomonic). Diagnosis: urine GAG analysis, iduronate-2-sulfatase enzyme assay, and IDS gene sequencing. Treatment: enzyme replacement therapy (idursulfase, Elaprase) improves somatic features, and intrathecal idursulfase is under investigation for neurologic disease. HSCT is less effective than in MPS I.

Hurler Syndrome

– Mucopolysaccharidosis type I (MPS I), the severe form, an autosomal recessive lysosomal storage disorder caused by alpha-L-iduronidase deficiency (IDUA gene mutations). Accumulation of dermatan sulfate and heparan sulfate leads to progressive multisystem involvement. Features: coarse facies (gargoyle-like appearance), corneal clouding, hepatosplenomegaly, recurrent hernias, dysostosis multiplex (kyphoscoliosis, J-shaped sella, paddle ribs, claw-hand deformity, hip dysplasia), aortic valve disease, and severe intellectual disability. Hearing loss, obstructive airway disease, and hydrocephalus (communicating) are common. Diagnosis: urine glycosaminoglycan (GAG) analysis (elevated dermatan and heparan sulfate) and IDUA enzyme assay in leukocytes. Newborn screening is available. Treatment: hematopoietic stem cell transplant (HSCT) performed early (<2.5 years) preserves cognitive function and improves somatic features but does not fully correct skeletal or corneal disease. Enzyme replacement therapy (laronidase, iduronidase) improves somatic features but does not cross the blood-brain barrier. Prognosis: untreated, death by 6-10 years from cardiorespiratory failure.

Hydrocephalus

– Abnormal accumulation of cerebrospinal fluid within the ventricular system, causing ventricular dilation and increased intracranial pressure. In neonates, presents with rapid head circumference growth, bulging fontanelle, and irritability. Causes include neural tube defects, intraventricular hemorrhage, aqueductal stenosis, and infection. Treatment: ventriculoperitoneal shunt placement or endoscopic third ventriculostomy.

Hypertelorism

– Increased distance between the orbits (the bony eye sockets), measured as an increased interorbital distance on imaging. Can result from premature fusion of the cranial sutures (craniosynostosis syndromes), frontonasal dysplasia, facial clefting, or can be associated with genetic syndromes (Apert, Crouzon, Noonan, 22q11.2 deletion, and CHARGE). Presents with widely spaced eyes (telecanthus). Diagnosis: clinical measurement of the intercanthal distance and orbital imaging (CT or MRI) to confirm increased interorbital distance. Management: surgical correction by orbital box osteotomy or facial bipartition for severe cases; treatment of the underlying syndrome. Not typically a functional concern.

Immune Thrombocytopenic Purpura

– Autoimmune destruction of platelets by antiplatelet antibodies, causing isolated thrombocytopenia. Most common in children 2-6 years, often following viral infection. Presents with petechiae, purpura, and mucosal bleeding. Diagnosis of exclusion. Treatment: observation for mild cases; IVIG or corticosteroids for severe thrombocytopenia or active bleeding. Spontaneous resolution in 80% within 6 months.

Imperforate Anus

– A congenital anomaly where the anus is completely absent or fails to open at the perineum. Classified as low (distal to the levator ani complex) or high (proximal to the levator ani). Often associated with other anomalies including vertebral, anal, cardiac, tracheoesophageal, renal, and limb defects. Presents with absence of anal opening on physical examination. Management: colostomy for high lesions; perineal anoplasty for low lesions.

Impetigo

– A superficial bacterial skin infection, most commonly caused by Staphylococcus aureus or Group A Streptococcus. Presents with honey-colored crusted lesions (nonbullous) or fluid-filled blisters (bullous). Common in children. Diagnosis is clinical. Treatment: topical mupirocin for localized disease; oral antibiotics for extensive disease. Prevention by good hand hygiene and avoiding sharing personal items.

Infantile Colic

– A behavioral syndrome in healthy infants characterized by episodes of excessive, inconsolable crying for >3 hours per day, >3 days per week, for >3 weeks (Wessel criteria). Onset typically at 2-3 weeks, peaks at 6-8 weeks, and resolves by 3-4 months of age. Etiology poorly understood: possible GI (gas, gut microbiota, cow milk protein intolerance), neurodevelopmental, or behavioral. Diagnosis: clinical; must rule out organic causes (intussusception, GERD, infection, hair tourniquet, corneal abrasion, testicular torsion). Treatment: parental reassurance, carrying/rocking, white noise, and swaddling. Probiotics, simethicone, and hypoallergenic formula may help some infants. Prognosis: excellent, self-limited.

Influenza Vaccine

– Inactivated influenza vaccine given annually starting at 6 months of age. Two doses needed in the first season for children under 9. Live attenuated nasal spray vaccine (FluMist) available for children over 2. Updated annually to match circulating strains. Reduces influenza complications, hospitalizations, and deaths. Recommended for all children.

Intestinal Atresia

– A congenital obstruction of the small intestine (jejunum or ileum) caused by an intrauterine vascular accident (mesenteric ischemia) leading to aseptic necrosis and resorption of a segment of bowel. In contrast to duodenal atresia, it is not associated with Down syndrome. Presents in the first 24-48 hours with bilious vomiting and progressive abdominal distension. Failure to pass meconium is universal with distal atresia. Classification (Grosfeld): Type I — mucosal web with intact bowel wall; Type II — blind ends connected by a fibrous cord; Type IIIa — complete separation of blind ends with a V-shaped mesenteric defect; Type IIIb — apple-peel or Christmas-tree deformity (extensive atresia with a spiral-shaped distal bowel); Type IV — multiple atresias. Diagnosis: abdominal X-ray shows dilated proximal bowel loops (number indicates level of atresia) with no distal gas; contrast enema shows an unused microcolon. Treatment: primary anastomosis after resection of the dilated proximal segment; tapering enteroplasty may be needed for severe proximal dilation. Prognosis: excellent for Type I-IIIa; guarded for IIIb and IV.

Intraventricular Hemorrhage

– Bleeding into the germinal matrix and ventricular system, primarily in premature infants less than 32 weeks gestation. Graded I-IV by severity on cranial ultrasound. Grade I (germinal matrix only) and Grade II (intraventricular without dilation) have good prognosis. Grade III (with ventricular dilation) and Grade IV (parenchymal involvement) risk neurodevelopmental impairment. Prevention: antenatal corticosteroids, avoiding rapid fluid shifts.

Jaundice in newborns

– Also known as neonatal hyperbilirubinemia, a common condition in newborns caused by increased bilirubin production from the breakdown of fetal red blood cells (high haematocrit, shortened red cell life span of 70-90 days vs. 120 days in adults) and immature hepatic conjugation and excretion. Physiologic jaundice of the newborn appears after 24 hours of life, peaks at 3-5 days in term infants (bilirubin typically less than 12-15 mg/dL) and 5-7 days in preterm infants, and resolves spontaneously within 2 weeks (term) or 3-4 weeks (preterm) without treatment. Pathologic jaundice is characterized by onset within 24 hours of life, rapid rise of bilirubin (greater than 5 mg/dL/day), levels above the physiologic range, or jaundice persisting beyond 2 weeks. Causes include ABO or Rh incompatibility, G6PD deficiency, hereditary spherocytosis, cephalohematoma, breast milk jaundice, and biliary atresia (conjugated hyperbilirubinemia). Assessment involves total and direct bilirubin levels plotted on the Bhutani nomogram to determine intervention thresholds. Phototherapy is first-line treatment, using blue light (460-490 nm) to convert bilirubin into water-soluble isomers excreted in urine and stool. Exchange transfusion is reserved for severe hyperbilirubinemia approaching neurotoxic levels to prevent kernicterus. Adequate feeding and frequent wet nappies facilitate bilirubin elimination. All newborns should be monitored before and after discharge with appropriate follow-up.

Juvenile Idiopathic Arthritis (JIA)

– A heterogeneous group of autoimmune and inflammatory conditions causing persistent arthritis (joint swelling, pain, limited range of motion) in children aged <16 years, lasting >6 weeks. Classification: oligoarticular (most common, 50%, <5 joints, high risk of uveitis), polyarticular (RF-positive or RF-negative, 30%), systemic-onset (Still disease, high spiking fevers, salmon-pink evanescent rash, serositis, hepatosplenomegaly), enthesitis-related (enthesitis, sacroiliitis, HLA-B27 association), and psoriatic arthritis. etiology: complex genetic predisposition (HLA class II associations) plus environmental triggers. Management: NSAIDs (first-line), intra-articular corticosteroid injections, methotrexate (first-line DMARD), biologic agents (anti-TNF–etanercept, adalimumab; anti-IL-6–tocilizumab for systemic JIA; anti-IL-1–anakinra). Uveitis screening is essential (slit-lamp every 3-4 months) as most cases are asymptomatic and can cause blindness.

Kawasaki Disease

– An acute vasculitis of childhood affecting medium-sized arteries, particularly coronary arteries. Diagnosis requires fever greater than 5 days plus 4 of 5 criteria: bilateral conjunctivitis, oral mucosal changes, cervical lymphadenopathy, extremity changes, and polymorphous rash. Complications include coronary artery aneurysms. Treatment: high-dose IV immunoglobulin and aspirin. Most common acquired heart disease in children in developed countries.

Kernicterus

– Bilirubin encephalopathy resulting from severe unconjugated hyperbilirubinemia (typically greater than 25 mg/dL in term infants). Bilirubin deposits in the basal ganglia, hippocampus, and brainstem nuclei. Presents with lethargy, poor feeding, high-pitched cry, opisthotonus, seizures, and choreoathetosis. May cause permanent neurologic damage including hearing loss, cerebral palsy, and intellectual disability. Prevention by treating severe jaundice with phototherapy or exchange transfusion.

Klippel-Trenaunay Syndrome

– A congenital vascular disorder characterized by the triad: capillary malformations (port-wine stains), venous malformations (varicose veins, persistent embryonic veins), and soft tissue or bone hypertrophy of the affected limb (typically a lower extremity). Caused by a somatic activating mutation in PIK3CA (part of the PIK3CA-related overgrowth spectrum). The hypertrophy is disproportionate, involving bone (increased length and girth) and soft tissues. Present at birth; progresses with growth. Complications: chronic venous insufficiency, cellulitis, deep vein thrombosis, pulmonary embolism (from anomalous veins), and life-threatening bleeding from pelvic or abdominal venous malformations. Diagnosis: clinical and MRI/MRV of the affected limb (shows abnormal venous anatomy and hypertrophy). Treatment: compression stockings for venous insufficiency; epiphyseal stapling or surgical debulking for limb overgrowth; sclerotherapy or surgical excision for symptomatic venous malformations. Anticoagulation for thromboembolic events. Differentiated from Parkes Weber syndrome by the absence of arteriovenous fistulas.

Legg-Calve-Perthes Disease

– Idiopathic avascular necrosis of the femoral head in children, typically ages 4-8 years. More common in boys (4:1) and Caucasian children. Presents with limping (antalgic gait), hip/knee pain, and limited hip range of motion (especially abduction and internal rotation). Etiology: temporary disruption of blood supply to the femoral head. Diagnosis: hip X-ray shows fragmentation, flattening, and collapse of the femoral head (staging: Waldenstrom). MRI for early diagnosis. Prognosis: better in younger children (<6), with preservation of femoral head sphericity. Treatment: observation for mild cases; containment therapy (Petrie cast, abduction brace, or femoral/pelvic osteotomy) for moderate to severe.

Malrotation

– A congenital anomaly of intestinal rotation and fixation in which the midgut fails to complete its normal 270-degree counterclockwise rotation and fixation during embryogenesis. This leaves the midgut suspended on a narrow mesenteric pedicle (Ladd bands), creating a high risk of midgut volvulus — a surgical emergency. Presents in the first month of life (most commonly in the first week) with bilious vomiting (green, 95%), abdominal distension, and rectal bleeding or shock (indicating vascular compromise with bowel ischemia). Older children may present with intermittent, chronic abdominal pain. Diagnosis: upper GI series is the gold standard — shows the duodenojejunal junction located to the right of the spine (failure to cross the midline) and a corkscrew appearance of the duodenum and proximal jejunum. Treatment: emergent laparotomy with Ladd procedure (division of Ladd bands, reduction of volvulus, widening of the mesenteric base, and appendectomy). A midgut volvulus requires immediate detorsion; irreversible ischemia requires intestinal resection. Prognosis: excellent if treated early; extensive bowel loss from delayed volvulus leads to short bowel syndrome.

Maple Syrup Urine Disease

– An autosomal recessive inborn error of branched-chain amino acid metabolism caused by deficiency of the branched-chain alpha-ketoacid dehydrogenase complex (BCKDC) — mutations in BCKDHA, BCKDHB, DBT, or DLD. Results in accumulation of leucine, isoleucine, and valine (and their ketoacids) in the blood, giving a characteristic maple syrup (burnt sugar) odor to urine and cerumen. The classic neonatal form presents at 3-5 days of life with poor feeding, vomiting, lethargy, hypertonicity (opisthotonus), seizures, and cerebral edema progressing to coma and death if untreated. Diagnosis: plasma amino acid analysis (elevated leucine, isoleucine, valine, and alloisoleucine — pathognomonic) and BCKDHA/B gene testing. Newborn screening detects MSUD in most developed countries. Treatment: acute crises — rapid removal of branched-chain amino acids via hemodialysis or hemofiltration, IV glucose, insulin, and supplementation with isoleucine and valine to promote protein synthesis. Chronic management: strict dietary restriction of leucine, isoleucine, and valine (specialized medical formulas), thiamine supplementation (thiamine-responsive forms, 5%), and metabolic monitoring. Liver transplantation can be curative.

Marfanoid Habitus

– A body habitus characterized by tall stature, long limbs, long fingers (arachnodactyly), and joint laxity. Seen in Marfan syndrome, but also in other connective tissue disorders and homocystinuria. Differentiated from Marfan syndrome by absence of aortic root dilation and lens subluxation. Requires cardiac evaluation to rule out aortic pathology.

Measles

– A highly contagious viral illness caused by Morbillivirus, transmitted by respiratory droplets. Prodrome of fever, coryza, cough, and conjunctivitis, followed by Koplik spots (white buccal mucosal lesions) and maculopapular rash spreading cephalocaudally. Complications: pneumonia, encephalitis, and subacute sclerosing panencephalitis. Prevention by MMR vaccination. Highly contagious with R0 of 12-18.

Meconium Aspiration Syndrome

– Respiratory distress in newborns born through meconium-stained amniotic fluid, caused by aspiration of meconium into the airways. Presents with respiratory distress, barrel-shaped chest, and coarse crackles. Meconium is thick, greenish-black fetal stool passed in utero. Risk factors include post-term pregnancy and fetal distress. Management: airway suctioning, mechanical ventilation, and treatment of complications including pneumothorax.

Meconium-Stained Amniotic Fluid

– Greenish or yellowish discoloration of amniotic fluid due to fetal meconium passage in utero. Associated with fetal distress, post-term pregnancy, and fetal growth restriction. Thick meconium increases risk of meconium aspiration syndrome. Management of non-vigorous newborns: tracheal suctioning before positive pressure ventilation. Vigorous newborns: routine care with close monitoring.

Meningococcal Vaccine

– Conjugate vaccine (MenACWY) and serogroup B vaccine (MenB) given at 11-12 years with booster at 16 years. Protects against Neisseria meningitidis, causing meningitis and septicemia. MenACWY booster at 16 years covers the high-risk period for serogroup C and Y disease. MenB given based on shared clinical decision-making or during outbreaks. One of the most devastating pediatric infections.

Menkes Disease

– An X-linked recessive disorder of copper metabolism caused by ATP7A mutations, resulting in impaired intestinal copper absorption and defective copper transport across the blood-brain barrier. Copper accumulates in intestinal and renal cells but is deficient in the liver and brain. Presents at 2-3 months with progressive neurologic deterioration, seizures, and characteristic hair abnormalities — stubby, white, brittle, and twisted (pili torti, pathognomonic on microscopy). Other features: hypotonia, failure to thrive, metaphyseal spurring (bladder-like metaphyses), wormian skull bones, and arterial tortuosity (cerebral and femoral arteries, prone to thrombosis and aneurysms). Diagnosis: low serum copper and ceruloplasmin (after 6-8 weeks of age), elevated copper in cultured fibroblasts, and ATP7A sequencing. Treatment: early parenteral copper-histidine supplementation (before symptoms) can improve neurologic outcomes. Established neurologic disease is irreversible. Prognosis: severe neurodegeneration and death by 3 years without treatment.

Metaphyseal Corner Fractures

– Bucket-handle or corner fractures at the metaphysis of long bones, characteristic of child abuse. Caused by twisting or pulling forces on extremities. Most common in the knee, ankle, and elbow. Seen on skeletal survey; may not be visible initially. Highly specific for non-accidental trauma in infants. Orthopedic consultation and full skeletal survey recommended.

Micrognathia

– A condition where the mandible is abnormally small, resulting in a retruded chin and decreased size of the oral cavity. May be isolated or syndromic (Pierre Robin sequence, Treacher Collins, Nager syndrome, Stickler syndrome). Presents at birth with potential airway obstruction (glossoptosis) and feeding difficulties. The severity depends on the degree of mandibular hypoplasia and tongue position. Diagnosis: clinical examination and cephalometric analysis. Management: prone positioning or lateral positioning for mild cases (gravity helps keep the tongue forward), nasopharyngeal airway for moderate obstruction, and mandibular distraction osteogenesis for severe obstruction. Most cases show significant mandibular catch-up growth during childhood.

MMR Vaccine

– Measles, Mumps, and Rubella live attenuated vaccine given at 12-15 months and 4-6 years. Two doses provide 97% protection against measles and mumps, and 98% against rubella. Contraindicated in pregnancy and immunocompromised individuals. Post-exposure prophylaxis within 72 hours. Measles remains a global health threat; outbreaks occur in under-vaccinated communities.

Morgagni Hernia

– A congenital diaphragmatic hernia occurring through the anterior retrosternal foramen of Morgagni (space of Larrey), resulting from failure of the sternal and costal portions of the diaphragm to fuse. The right side is affected in 90% of cases (the hernia sac is present in 90%). Abdominal contents (typically omentum, transverse colon, and liver) herniate into the anterior mediastinum. Often asymptomatic and incidentally discovered on chest X-ray (seen as a right cardiophrenic angle mass). Presents later in life than Bochdalek hernias (childhood to adulthood) with recurrent chest infections, dyspnea, or abdominal pain. Associated with obesity and trauma (increases intra-abdominal pressure). Diagnosis: chest X-ray shows a well-defined opacity in the right cardiophrenic angle; CT confirms the hernia, the contents, and the defect. Treatment: surgical repair (laparoscopic or open) with reduction of hernia contents and primary closure of the defect with or without mesh. Prognosis: excellent, with low recurrence rates.

Moro Reflex

– A primitive reflex present from birth to 3-6 months, elicited by suddenly startling the infant (loud noise, head drop). The infant extends both arms outward with fingers spread, then draws arms inward as if embracing. Absence at birth suggests central nervous system injury or brachial plexus palsy. Asymmetric response may indicate clavicle fracture or Erb palsy. Used as a marker of neurologic integrity.

Multicystic Dysplastic Kidney

– A congenital renal anomaly where one or both kidneys are replaced by multiple non-communicating cysts of varying sizes, with no functioning renal parenchyma. The most common cause of an abdominal mass in a newborn (unilateral). The ureter is typically atretic. The contralateral kidney may have VUR (20%) or UPJ obstruction (10%). Bilateral MCDK is incompatible with life (Potter syndrome). Most unilateral cases are asymptomatic and discovered on prenatal ultrasound. Diagnosis: postnatal ultrasound shows multiple non-communicating cysts of varying sizes, no identifiable renal sinus, and a reniform shape may be lost. A DMSA renal scan shows no function in the affected kidney. Treatment: observation is standard — most MCDKs involute (shrink) spontaneously over years. Blood pressure monitoring and urinalysis annually. Nephrectomy is rarely indicated (hypertension, recurrent UTI, or mass effect). Contralateral kidney evaluation (VCUG) for VUR and (US) for UPJ obstruction is essential.

Mumps

– Paramyxovirus infection causing bilateral parotitis, fever, and headache. Complications include orchitis (rarely causing infertility), meningitis, pancreatitis, and sensorineural hearing loss. Prevention by MMR vaccination. Outbreaks occur in unvaccinated populations. Diagnosis by serology or RT-PCR from saliva. Supportive treatment; no specific antiviral therapy.

Myelomeningocele

– The most severe form of spina bifida, where the spinal cord and meninges herniate through a vertebral defect into a sac on the back. Presents with a visible, fluid-filled sac at birth. Associated with hydrocephalus, Arnold-Chiari malformation, and lower extremity paralysis. Management: surgical closure within 24-48 hours, CSF drainage if hydrocephalus develops, and long-term rehabilitation for mobility and bowel/bladder function.

Necrotizing Enterocolitis

– An acquired necrotic inflammatory disease of the intestine, primarily affecting premature infants. Risk factors include prematurity, enteral feeding, and bacterial colonization. Presents with abdominal distension, bloody stools, bilious vomiting, and feeding intolerance. Pneumatosis intestinalis on X-ray is pathognomonic. Management: NPO, nasogastric decompression, IV antibiotics, and supportive care. Complications include perforation, peritonitis, and stricture formation.

Neonatal Abstinence Syndrome

– A withdrawal syndrome in neonates following in utero exposure to opioids (most common) or other substances (benzodiazepines, SSRIs, alcohol). Onset within 24-72 hours of birth (varies by half-life of substance). Presents with CNS hyperirritability (high-pitched crying, tremors, irritability, hypertonia, seizures), autonomic dysfunction (sweating, fever, tachypnea), and GI symptoms (poor feeding, vomiting, diarrhea). Diagnosis: Finnegan Neonatal Abstinence Scoring System quantifies severity. Treatment: non-pharmacologic (swaddling, low stimulation, skin-to-skin contact, breastfeeding if no contraindications) and pharmacologic for moderate-severe (morphine or methadone weaning protocol). Adjunctive: clonidine, phenobarbital. Prognosis: most recover fully with appropriate management.

Neonatal Cephalohematoma

– A subperiosteal hematoma of the skull, occurring in approximately 1-2% of births, usually from forceps or vacuum delivery. Presents as a boggy, fluctuant swelling over the skull that does not cross suture lines (differentiating from caput succedaneum). Resolves spontaneously over weeks to months. Complications include hyperbilirubinemia, anemia, and calcification. No treatment required in most cases.

Neonatal Herpes Simplex

– Herpes simplex virus infection in the first 28 days of life, usually HSV-2 from maternal genital infection. Three forms: skin, eye, and mouth disease; CNS disease; and disseminated disease. Disseminated form mimics sepsis with high mortality. Diagnosis by viral PCR, culture, or Tzanck smear. Treatment: IV acyclovir for 14-21 days. Cesarean delivery recommended for active genital lesions near term.

Neonatal Hypoglycemia

– Blood glucose less than 40-45 mg/dL in term infants or less than 25 mg/dL in preterm infants. Risk factors include prematurity, small for gestational age, large for gestational age, and maternal diabetes. Presents with jitteriness, lethargy, poor feeding, and seizures. Treatment: early feeding, IV dextrose infusion, and monitoring. Persistent hypoglycemia may require glucagon or hydrocortisone.

Neonatal Hypothermia

– Core body temperature below 36.5 degrees Celsius in a newborn. Classified as mild (36-36.4), moderate (32-35.9), or severe (below 32). Occurs due to rapid heat loss through evaporation, convection, conduction, and radiation. Prevention: skin-to-skin contact, drying, warm environment, and delayed bathing. Treatment: warming methods appropriate to severity; monitor glucose and electrolytes.

Neonatal Jaundice

– Yellowing of the skin and sclera due to elevated bilirubin levels, occurring in approximately 60% of term newborns. Physiologic jaundice appears at 24-48 hours, peaks at day 3-5, and resolves by 2 weeks. Pathologic jaundice appears within 24 hours or persists beyond 2 weeks. Causes include hemolysis, infection, breast milk jaundice, and conjugation defects. Severe hyperbilirubinemia risks kernicterus.

Neonatal Jaundice (Jaundice of the Newborn)

– Yellowish discolouration of the skin and sclerae in newborn infants due to elevated unconjugated bilirubin. Occurs in 60% of term infants and 80% of preterm infants. Physiological jaundice: appears after 24 hours of life, peaks at day 3-5 in term infants (day 5-7 in preterm), resolves by 2 weeks. Caused by: increased RBC mass, shorter RBC lifespan (70-90 days), immature hepatic conjugation (decreased UGT activity), increased enterohepatic circulation. Pathological jaundice: appears within 24 hours, bilirubin rising >85 micromol/L/day, or persisting >14 days (term) or >21 days (preterm). Causes: hemolytic (ABO incompatibility, Rh incompatibility, G6PD deficiency, hereditary spherocytosis), bruising/cephalhaematoma, polycythaemia, sepsis, biliary atresia, galactosaemia, Crigler-Najjar syndrome, Gilbert syndrome. Complications: acute bilirubin encephalopathy (lethargy, poor feeding, hypotonia, high-pitched cry, opisthotonos, seizures) and kernicterus (chronic, irreversible choreoathetoid cerebral palsy, upward gaze palsy, sensorineural hearing loss, enamel dysplasia). Management: phototherapy (most effective, blue light 460-490 nm, converts bilirubin to water-soluble isomers excreted without conjugation), exchange transfusion (for severe hyperbilirubinaemia failing phototherapy), and treatment of underlying cause. NICE and AAP guidelines provide phototherapy and exchange transfusion thresholds based on gestational age, birth weight, and risk factors.

Neonatal Sepsis

– Systemic bacterial infection in the first 28 days of life, classified as early-onset (within 72 hours, usually from vertical transmission) or late-onset (after 72 hours, often nosocomial). Common organisms: Group B Streptococcus, E. coli, Listeria. Presents with non-specific symptoms: poor feeding, lethargy, temperature instability, tachycardia, and respiratory distress. Diagnosis by blood culture; treatment with empiric IV antibiotics (ampicillin and gentamicin).

Neonatologist

– A pediatrician who specializes in the medical care of newborn infants, particularly premature and critically ill neonates. Neonatologists manage conditions including respiratory distress syndrome, bronchopulmonary dysplasia, intraventricular hemorrhage, necrotizing enterocolitis, neonatal sepsis, hypoxic-ischemic encephalopathy (therapeutic hypothermia), congenital anomalies, and extreme prematurity. They work in neonatal intensive care units (NICU), lead resuscitation teams at high-risk deliveries, and coordinate care with pediatric surgeons, cardiologists, and geneticists. Training: pediatric residency + 3-year neonatal-perinatal medicine fellowship.

Neural Tube Defects

– Congenital malformations resulting from failure of the neural tube to close during the third to fourth week of gestation. Include anencephaly (incomplete cranial closure), encephalocele (herniation through skull defect), and spina bifida (incomplete spinal closure). Spina bifida ranges from occulta (mild, asymptomatic) to myelomeningocele (severe, with neural tissue protrusion). Prevention with folic acid supplementation before and during pregnancy.

Neurofibromatosis Type 2

– An autosomal dominant disorder caused by NF2 gene mutation on chromosome 22. Characterized by bilateral vestibular schwannomas (acoustic neuromas), meningiomas, and ependymomas. Features: hearing loss, tinnitus, and balance problems. Diagnosis: MRI showing bilateral vestibular schwannomas. Management: hearing preservation surgery, radiation therapy, and surveillance imaging.

Newborn Circumcision

– Surgical removal of the foreskin from the penis, typically performed within the first few days of life. Medical benefits include reduced risk of urinary tract infections, sexually transmitted infections, and penile cancer. Complications include bleeding, infection, and meatal stenosis. Techniques include Plastibell, Gomco clamp, and Mogen clamp. Performed with parental informed consent. Not universally recommended; benefits versus risks should be discussed.

Niemann-Pick Disease

– A group of autosomal recessive lysosomal storage disorders caused by sphingomyelinase deficiency (ASM, types A and B, SMPD1 mutations) or NPC1/NPC2 cholesterol transport defects (type C). Type A (infantile neuropathic): severe, presents at 3-6 months with hepatosplenomegaly, cherry-red macula (50%), developmental regression, neurodegeneration, and death by 2-3 years. Type B (chronic visceral): later onset, hepatosplenomegaly, pulmonary infiltrates (foam cells in alveoli), thrombocytopenia, and dyslipidemia — no or minimal neurologic involvement; survival into adulthood. Type C: variable onset (infantile to adult) with neurologic features: vertical supranuclear gaze palsy (pathognomonic), ataxia, dystonia, seizures, dementia, and cataplexy. Diagnosis: types A/B — acid sphingomyelinase enzyme assay in leukocytes. Type C — filipin staining for cholesterol accumulation in fibroblasts and NPC1/NPC2 sequencing. Treatment: types A/B — olipudase alfa (enzyme replacement therapy, FDA-approved for type A/B non-neurologic). Type C — miglustat (substrate reduction therapy). Supportive care for neurological symptoms.

Non-Accidental Trauma

– Injuries inflicted on a child, suspected when history and examination findings are inconsistent with the reported mechanism. Red flags: multiple fractures at different stages of healing, posterior rib fractures, metaphyseal corner fractures, and retinal hemorrhages. Requires comprehensive evaluation including skeletal survey, CT head, ophthalmologic exam, and laboratory workup. Social work and child protective services involvement mandatory.

Noonan Syndrome

– An autosomal dominant multisystem disorder caused by mutations in the RAS-MAPK pathway genes (PTPN11 50%, SOS1 10-15%, RAF1 5-10%, RIT1, KRAS). Features: characteristic facies (hypertelorism, downslanting palpebral fissures, low-set posteriorly rotated ears, webbed neck, ptosis), short stature, congenital heart disease (pulmonary stenosis 50-60%, hypertrophic cardiomyopathy 20-30%), pectus deformities, and cryptorchidism. Coagulation defects (factor XI deficiency, von Willebrand disease) are common. Intelligence is usually normal, though learning disabilities and speech delays are common. Diagnosis: clinical criteria (van der Burgt) and genetic testing. Management: growth hormone therapy for short stature, cardiology follow-up (pulmonary stenosis balloon valvuloplasty, HCM monitoring), and developmental support.

Omphalitis

– Infection of the umbilical stump in neonates, typically occurring within the first 1–2 weeks of life. Omphalitis is a potentially serious condition requiring prompt treatment due to the risk of spread to the liver (portal vein thrombosis, portal hypertension) and peritoneum via the umbilical vessels. Risk factors: home delivery with unsterile cord care, low birth weight, prolonged rupture of membranes (>18 hours), umbilical catheterisation, and immunocompromised state. Presentation: periumbilical erythema, edema, tenderness, purulent or foul-smelling discharge from the umbilicus. Signs of systemic infection (fever, lethargy, poor feeding, tachycardia, hypotension) indicate severe disease. Pathogens: Staphylococcus aureus, Group A Streptococcus, Escherichia coli, Klebsiella, and anaerobes (Bacteroides fragilis). Diagnosis is clinical; wound culture, blood culture, and ultrasound (assess for abscess, thrombophlebitis, necrotising fasciitis) in severe cases. Treatment: IV antibiotics covering Gram-positive, Gram-negative, and anaerobic organisms (flucloxacillin plus gentamicin plus metronidazole). Surgical debridement for necrotising fasciitis or abscess drainage. Complications: necrotising fasciitis (abdominal wall cellulitis with crepitus, high mortality), portal vein thrombosis, peritonitis, and neonatal sepsis. Prevention: chlorhexidine cord care (reduces omphalitis in high-mortality settings per WHO recommendation).

Omphalocele

– A congenital abdominal wall defect where abdominal organs herniate into the base of the umbilical cord, covered by a peritoneal-amniotic membrane. Associated with chromosomal abnormalities (trisomies 13, 18) and other anomalies. Differentiated from gastroschisis by membrane coverage and midline location. Small omphaloceles may be repaired primarily; large defects require staged closure with silastic silo and gradual reduction.

Osteogenesis Imperfecta

– A connective tissue disorder caused by collagen type I mutations, ranging from mild to lethal. Features: fractures with minimal trauma, blue sclerae, dental abnormalities, and hearing loss. Severe forms (type II) cause multiple fractures in utero and perinatal death. Diagnosis: clinical features, genetic testing, and bone biopsy. Management: bisphosphonates, physical therapy, and surgical correction of deformities.

Otitis Media

– Middle ear infection, the most common childhood illness after the common cold. Peak incidence between 6-15 months. Risk factors include secondhand smoke, daycare attendance, formula feeding, and anatomy. Presents with ear pain, fever, irritability, and hearing loss. Diagnosis by otoscopic exam showing erythema, bulging tympanic membrane, or effusion. Treatment: antibiotics (amoxicillin first-line) for acute otitis media.

Palmar Grasp Reflex

– A primitive reflex present from birth to 5-6 months, elicited by placing a finger in the infant's palm. The infant grips tightly, strong enough to support the infant's weight briefly. Bilateral symmetry assessed; asymmetry may indicate birth injury. Disappears as voluntary grasping develops around 3-4 months. Used as a neurologic marker in the newborn examination.

Patau Syndrome

– Trisomy 13, affecting 1 in 10,000-20,000 live births.

Features: holoprosencephaly (most characteristic, incomplete forebrain division), microcephaly, microphthalmia or anophthalmia, cleft lip and palate, polydactyly (postaxial), clenched fists with overlapping fingers, and congenital heart defects (VSD, PDA, dextrocardia, 80%). Other findings: cutis aplasia (scalp defect, particularly the occipital scalp), omphalocele, and renal anomalies (polycystic kidneys). Diagnosis: prenatal screening (abnormal first-trimester ultrasound with increased nuchal translucency), NIPT, and karyotype/CMA. Management: comfort care (median survival 7-10 days; <10% survive to 1 year). Survivors have profound intellectual disability and severe growth restriction.

Pectus Carinatum

– A congenital chest wall deformity characterized by anterior protrusion of the sternum (pigeon chest), caused by excessive growth of the costal cartilages pushing the sternum forward. Less common than pectus excavatum. Associated with congenital heart disease, Marfan syndrome, Noonan syndrome, and homocystinuria. Presents in childhood and progresses during the adolescent growth spurt. Symptoms are predominantly cosmetic, but some children experience chest pain, exercise intolerance, or psychosocial distress. Diagnosis: clinical examination and CT or X-ray to evaluate symmetry and severity. Treatment: orthotic bracing (dynamic chest compressor, effective for flexible deformities in younger children) is first-line, worn 12-23 hours daily for 6-12 months. Surgical correction (Ravitch-type sternal osteotomy with costal cartilage resection) for severe or rigid deformities or bracing failure. Outcomes are excellent with both methods.

Pectus Excavatum

– The most common congenital chest wall deformity, characterized by depression of the sternum (funnel chest), caused by excessive growth of the costal cartilages that push the sternum inward. Typically sporadic, but associated with Marfan syndrome, Ehlers-Danlos syndrome, Noonan syndrome, and Poland syndrome. Presents in early childhood and progresses during the adolescent growth spurt. Symptoms are usually cosmetic, but severe cases can cause cardiopulmonary impairment (reduced stroke volume, dyspnea on exertion, exercise intolerance). The Haller index (CT ratio of transverse chest diameter to anterior-posterior diameter, >3.25 is severe) quantifies severity. Diagnosis: clinical (the depression is obvious) and CT chest for Haller index. Treatment: observation for mild cases. Surgical correction: Nuss procedure (minimally invasive, pectus bar placed retrosternally under thoracoscopic guidance, removed after 2-4 years) for moderate to severe cases. Modified Ravitch procedure (open resection of costal cartilages and sternal osteotomy) for older patients or failed Nuss. Outcomes: excellent cosmetic and functional results.

Pediatric Diabetic Ketoacidosis

– A life-threatening complication of type 1 diabetes characterized by hyperglycemia, metabolic acidosis, and ketosis. Presents with polyuria, polydipsia, vomiting, abdominal pain, Kussmaul respirations, and fruity breath odor. Diagnosis: blood glucose greater than 250 mg/dL, pH less than 7.3, bicarbonate less than 15 mEq/L, and positive ketones. Treatment: IV fluids, insulin infusion, and electrolyte replacement with careful potassium monitoring.

Pediatric Health

– Pediatric health encompasses the physical, cognitive, emotional, and social development of children from infancy through adolescence, with emphasis on preventive care, developmental surveillance, and age-appropriate management of acute and chronic conditions. Well-child care follows a schedule: immunizations (per CDC/ACIP: DTaP, IPV, MMR, varicella, HPV, meningococcal, influenza, COVID-19), developmental screening (Ages and Stages Questionnaire, M-CHAT for autism at 18 and 24 months), growth monitoring (WHO/CDC growth charts), vision/hearing screening, and anticipatory guidance (injury prevention, nutrition, sleep, media use, substance use prevention). Common pediatric conditions: otitis media, asthma, atopic dermatitis, allergic rhinitis, ADHD (prevalence 9%), autism spectrum disorder (prevalence 2.8%), anxiety/depression, obesity (affecting 20% of US children), infectious diseases (RSV, croup, gastroenteritis), and congenital anomalies. Adolescent health addresses puberty, sexual health (STI prevention, contraception), mental health (leading cause of disability in teens), substance use, and transition to adult care. Family-centered care and trauma-informed approaches improve outcomes.

Pediatrician

– A medical doctor who specializes in the health and medical care of children from birth through adolescence (typically up to age 16-18). Pediatricians focus on: growth and development, preventive care (immunizations, developmental screening, anticipatory guidance), acute illness (infections, injuries), chronic disease management (asthma, diabetes, epilepsy, allergies), and behavioral/mental health. Subspecialties: neonatology, pediatric cardiology, pediatric oncology, pediatric gastroenterology, pediatric emergency medicine, pediatric intensive care, pediatric endocrinology, pediatric neurology, and developmental pediatrics. Training: medical school + 2 years foundation training (UK) or 3 years pediatrics residency (US) + subspecialty fellowship.

Pediatric Meningitis

– Inflammation of the meninges in children, bacterial forms being the most dangerous. Presents with fever, headache, neck stiffness, photophobia, altered mental status, and petechial rash (meningococcal). Infants may present with irritability, poor feeding, bulging fontanelle, and high-pitched cry. Diagnosis by lumbar puncture showing elevated WBC, elevated protein, low glucose, and positive Gram stain/culture. Treatment: empiric ceftriaxone and vancomycin.

Pediatrics

– The branch of medicine concerned with the health and medical care of infants, children, and adolescents from birth to age 18 (or up to age 16 in the UK). Pediatric care encompasses: well-child visits (growth monitoring, developmental screening, immunisations, anticipatory guidance), management of acute illness (infections, injuries), management of chronic conditions (asthma, diabetes, epilepsy, cystic fibrosis, congenital heart disease), and behavioural/mental health (ADHD, autism, anxiety, depression). Subspecialties: neonatology, pediatric cardiology, pediatric oncology, pediatric gastroenterology, pediatric pulmonology, pediatric neurology, pediatric nephrology, pediatric endocrinology, pediatric rheumatology, pediatric infectious disease, pediatric emergency medicine, pediatric intensive care, developmental pediatrics, and pediatric surgery. Preventive care: the childhood immunisation schedule (UK: 2, 3, 4 months; 12–13 months; 3 years 4 months; 12–14 years—boosters for tetanus, diphtheria, polio, HPV, MenACWY). The UK has a comprehensive child health surveillance programme (health visitor reviews, school entry checks).

Periventricular Leukomalacia

– Ischemic white matter injury in the premature infant brain, a major cause of cerebral palsy. Risk factors include prematurity, hypotension, and inflammation. Diagnosis by cranial ultrasound showing periventricular echodensities or cystic changes. Prevention: avoiding hypotension, maintaining normothermia, and managing infection. Long-term sequelae include spastic diplegia, cognitive impairment, and visual disturbances.

Pertussis

– A highly contagious bacterial infection caused by Bordetella pertussis, producing characteristic paroxysmal cough with whooping inspiratory sound. Three stages: catarrhal (1-2 weeks of URI symptoms), paroxysmal (weeks to months of violent coughing spells), and convalescent. Most dangerous in infants under 6 months. Treatment: macrolide antibiotics (azithromycin). Prevention: DTaP vaccination.

Pfeiffer Syndrome

– An autosomal dominant craniosynostosis syndrome caused by FGFR1 or FGFR2 mutations. Features: craniosynostosis (bilateral coronal), midface hypoplasia, proptosis, hypertelorism, broad thumbs and great toes (medial deviation), and variable syndactyly. Three clinical types: Type 1 (classic, milder, normal intelligence, good prognosis with surgery), Type 2 (cloverleaf skull, severe proptosis, elbow ankylosis, poor prognosis), and Type 3 (similar to Type 2 without cloverleaf skull, severe neurological impairment). Types 2 and 3 have significant mortality. Diagnosis: clinical and genetic testing. Management: staged cranial vault and midface surgery for Type 1; palliative management for Types 2 and 3.

Pierre Robin Sequence

– A congenital anomaly triad: micrognathia (small mandible), glossoptosis (posteriorly displaced tongue causing airway obstruction), and U-shaped cleft palate. The sequence arises from early mandibular hypoplasia, which prevents the tongue from descending, thereby impairing palatal shelf fusion. Presents at birth with respiratory distress (stridor, retractions, cyanosis, worse supine) and feeding difficulties. Diagnosis: clinical examination and polysomnography to assess obstructive sleep apnea severity. Management: prone positioning (first-line, 70% improve), nasopharyngeal airway, and surgical intervention for severe cases (tongue-lip adhesion, mandibular distraction osteogenesis, or tracheostomy). Cleft palate repair is typically performed at 9-18 months.

Pneumococcal Conjugate Vaccine

– PCV13 and PCV15 vaccines given at 2, 4, 6, and 12-15 months. Protects against Streptococcus pneumoniae serotypes causing invasive disease, pneumonia, and otitis media. PCV20 now recommended for older children and adults. Has dramatically reduced rates of invasive pneumococcal disease in children since universal vaccination.

Poland Syndrome

– A congenital anomaly characterized by unilateral absence or hypoplasia of the pectoralis major muscle (specifically the sternocostal head) and ipsilateral hand anomalies (syndactyly, most commonly; also brachydactyly, ectrodactyly, or radial club hand). The right side is affected in 60-75% of cases. Males are affected 2-3 times more frequently. Associated findings: rib defects (aplasia or hypoplasia of the second to fifth ribs), chest wall depression, breast and nipple hypoplasia or absence (in females), axillary hair loss, and vertebral anomalies. The cause is thought to be an interruption of the subclavian artery blood supply during the sixth week of gestation. Diagnosis: clinical with ultrasound or CT for chest wall evaluation. Treatment: hand surgery for syndactyly release; breast reconstruction with implants or autologous tissue for females; chest wall reconstruction for severe rib defects.

Posterior Urethral Valves

– The most common cause of lower urinary tract obstruction in male infants, caused by congenital mucosal folds (valves) in the posterior urethra (prostatic urethra) that obstruct urine outflow. Presents prenatally with oligohydramnios, bilateral hydronephrosis, and a distended, thick-walled bladder, and postnatally with poor urinary stream, urinary retention, and abdominal distension. Severe cases present with the prune belly sequence (deficient abdominal wall musculature from massive bladder distension) and Potter sequence (pulmonary hypoplasia from oligohydramnios). Diagnosis: prenatal ultrasound, postnatal voiding cystourethrogram (VCUG) shows a dilated posterior urethra with a sharp cutoff. Treatment: immediate bladder decompression (urethral catheter), endoscopic valve ablation (transurethral incision of the valves, first-line). Temporary vesicostomy for small or premature infants. Prognosis: depends on the degree of renal impairment at presentation. Even with successful valve ablation, 25-30% develop end-stage renal disease in childhood or adolescence due to preexisting renal dysplasia.

Post-Term Pregnancy

– Pregnancy continuing beyond 42 weeks gestation (294 days from LMP). Risk factors include prior post-term pregnancy, nulliparity, and genetic factors. Complications include macrosomia, oligohydramnios, meconium aspiration, placental insufficiency, and stillbirth. Management: antenatal surveillance with NST and biophysical profile starting at 41 weeks. Induction of labor recommended by 41-42 weeks.

Potter Syndrome

– A sequence of anomalies resulting from severe oligohydramnios (low amniotic fluid volume), most commonly caused by bilateral renal agenesis, but also from autosomal recessive polycystic kidney disease, posterior urethral valves, or chronic amniotic fluid leak. Features: pulmonary hypoplasia (the most critical, incompatible with life), Potter facies (low-set, flattened ears, recessed chin, prominent epicanthal folds, flattened nose), limb deformities (clubfoot, hip dislocation, flexion contractures), and amnion nodosum (nodules on the amnion). Presents at birth with severe respiratory distress, inability to oxygenate, and typical facies. Diagnosis: prenatal ultrasound shows anhydramnios/oligohydramnios and absent or abnormal kidneys. Chest X-ray shows small lung volumes. Prognosis: uniformly fatal in the newborn period due to pulmonary hypoplasia. Management: perinatal palliative care.

Prader-Willi Syndrome

– A genetic disorder caused by loss of function of genes on chromosome 15 (paternal deletion or maternal uniparental disomy). Features: neonatal hypotonia and failure to thrive, followed by hyperphagia and obesity in childhood. Intellectual disability, short stature, hypogonadism, and behavioral problems. Management: controlled diet, growth hormone therapy, and behavioral interventions.

Prune Belly Syndrome

– A congenital syndrome comprising the triad: deficient abdominal wall musculature (wrinkled, prune-like appearance), bilateral cryptorchidism (undescended testes), and urinary tract anomalies (dilated ureters, dilated and trabeculated bladder, and posterior urethral valves or urethral atresia). Almost exclusively affects boys. Other associations: pulmonary hypoplasia from oligohydramnios, orthopedic anomalies (clubfoot, hip dislocation), and cardiac anomalies. The severity ranges from the lethal form (severe oligohydramnios, pulmonary hypoplasia, and renal dysplasia) to the mild form (preserved renal function and mild abdominal wall laxity). Diagnosis: prenatal ultrasound shows a distended bladder with thickened walls and oligohydramnios. Postnatal diagnosis is clinical. Treatment: depends on severity — mild cases: observation and orchiopexy. Moderate-severe: urinary tract reconstruction (reduction cystoplasty, ureteral reimplantation, and abdominoplasty). Renal transplantation for end-stage renal disease. All patients require lifelong urologic and nephrologic surveillance.

Renal Agenesis

– Congenital absence of one (unilateral, 1 in 1,000) or both (bilateral, 1 in 4,000) kidneys. Unilateral renal agenesis: usually asymptomatic, discovered incidentally — the solitary kidney undergoes compensatory hypertrophy. The contralateral kidney may have VUR, UPJ obstruction, or ectopia. Associated with VACTERL, Mayer-Rokitansky-Kuster-Hauser syndrome (Müllerian agenesis), and other GU anomalies. Bilateral renal agenesis (Potter syndrome): incompatible with life — causes severe oligohydramnios, leading to Potter sequence: pulmonary hypoplasia (the proximal cause of death), limb deformities (clubfoot, contractures), characteristic Potter facies (low-set ears, flat nose, epicanthal folds), and amnion nodosum. Diagnosis: prenatal ultrasound (absent renal tissue, empty renal fossa, severe oligohydramnios). Bilateral renal agenesis is uniformly fatal in the newborn period from pulmonary hypoplasia. Unilateral: annual blood pressure monitoring, urinalysis, and renal function surveillance.

Respiratory Distress Syndrome

– A condition primarily affecting premature infants caused by pulmonary surfactant deficiency, leading to alveolar collapse and respiratory failure. Presents with tachypnea, grunting, nasal flaring, retractions, and cyanosis. Chest X-ray shows diffuse ground-glass opacity. Treatment: surfactant replacement therapy, mechanical ventilation, and continuous positive airway pressure. Prevention with antenatal corticosteroids when preterm delivery anticipated.

Respiratory Syncytial Virus

– A paramyxovirus and the most common cause of bronchiolitis and pneumonia in infants and young children. Seasonal outbreaks in fall, winter, and spring. Transmission by respiratory droplets and fomites. Mild in older children and adults; can be severe or fatal in infants, premature infants, and those with chronic lung or heart disease. Prevention with palivizumab (Synagis) in high-risk infants during RSV season.

Retinal Hemorrhages

– Bleeding within the retina, a hallmark finding in abusive head trauma (shaken baby syndrome). Found on dilated fundoscopic examination, often bilateral and multi-layered. Also seen in severe accidental trauma, bleeding disorders, and increased intracranial pressure. In the context of suspected abuse, strongly suggestive of non-accidental injury. Ophthalmologic consultation essential for documentation.

Retinopathy of Prematurity

– Abnormal vascularization of the retina in premature infants, a leading cause of childhood blindness. Affects infants less than 32 weeks gestation and less than 1500 grams. Stages range from mild (avascular retina) to severe (retinal detachment). Screening began at 31-33 weeks corrected gestational age or 4 weeks chronological age. Treatment: laser photocoagulation or anti-VEGF injection for type 1 ROP.

Rett Syndrome

– An X-linked dominant neurodevelopmental disorder affecting almost exclusively females, caused by mutations in the MECP2 gene. Normal development for the first 6-18 months, followed by regression with loss of acquired language and motor skills, stereotypical hand-wringing movements, deceleration of head growth (acquired microcephaly), seizures, and gait ataxia. Autonomic dysfunction causes breathing irregularities (hyperventilation, breath-holding), cold extremities, and bruxism. Diagnosis: clinical criteria (Neul 2010) and MECP2 genetic testing. Treatment: symptom management — antiepileptic drugs for seizures, physical/occupational therapy, nutritional support, and scoliosis monitoring. Trofinetide is FDA-approved for Rett syndrome in adults and children >2 years.

Rheumatic Fever

– A nonsuppurative inflammatory complication of Group A Streptococcal pharyngitis, occurring 2-4 weeks after infection. Diagnosed by Jones criteria: major manifestations (carditis, polyarthritis, chorea, erythema marginatum, subcutaneous nodules) plus evidence of prior streptococcal infection. Major criterion plus two minor criteria. Treatment: anti-inflammatory agents and antibiotics. Prevention: prompt treatment of strep throat.

Rooting Reflex

– A primitive reflex present from birth to 3-4 months, triggered by stroking the infant's cheek or corner of the mouth. The infant turns toward the stimulus and opens the mouth, searching for the nipple. Facilitates breastfeeding. Absence may indicate neurologic impairment. Disappears as voluntary feeding develops. One of several primitive reflexes assessed in the newborn period to evaluate neurologic function.

Roseola Infantum

– A viral illness caused by Human Herpesvirus 6 (HHV-6), affecting children 6 months to 2 years. Characterized by high fever (39-40.5 degrees Celsius) for 3-5 days followed by defervescence and appearance of a maculopapular rash starting on the trunk and spreading outward. Rash appears as fever resolves. Diagnosis is clinical. Self-limited; treatment is supportive.

Rotavirus Vaccine

– Oral live attenuated vaccine given at 2, 4, and 6 months (RV1) or at 2, 4, and 8 months (RV5). Protects against rotavirus gastroenteritis, a leading cause of severe diarrhea in infants. Reduced rotavirus hospitalizations by over 80% since introduction. Mild side effects include fever and diarrhea. Contraindicated in infants with severe combined immunodeficiency.

Rubella

– German measles caused by Rubivirus. Mild illness with fever, rash, and lymphadenopathy in children. Dangerous in pregnancy due to congenital rubella syndrome (cataracts, heart defects, deafness). Rash similar to measles but milder, with postauricular and suboccipital lymphadenopathy. Prevention by MMR vaccination. Eradicated in the Americas since 2015.

Rubinstein-Taybi Syndrome

– A genetic disorder caused by mutations in CREBBP (50-60%) or EP300 (3-8%), encoding histone acetyltransferases involved in transcriptional regulation. Features: characteristic facies — downslanting palpebral fissures, broad nasal bridge, beaked nose, high-arched palate, and low-set ears; broad thumbs and halluces (pathognomonic, 100%); short stature; microcephaly; and moderate to severe intellectual disability. Other findings: congenital heart disease (PDA, ASD, VSD, 30-35%), cryptorchidism, renal anomalies, and increased risk of tumors (benign: meningioma, pilomatrixoma; malignant: leukemia). Diagnosis: clinical and genetic testing. Management: multidisciplinary — cardiac evaluation, developmental therapies, physical therapy for joint laxity, and tumor surveillance. Anesthesia risk: difficult airway from micrognathia and potential malignant hyperthermia.

Scarlatina

– See Scarlet Fever.

Scarlet Fever

– A toxin-mediated illness caused by Group A Streptococcus producing erythrogenic toxin. Presents with fever, pharyngitis, and a characteristic diffuse, erythematous, sandpaper-textured rash that blanches with pressure. Circumoral pallor, Pastia lines (linear petechiae in skin folds), and strawberry tongue are classic findings. Treatment: penicillin or amoxicillin for 10 days. Prevents rheumatic fever complications.

Shaken Baby Syndrome

– A form of abusive head trauma caused by violent shaking, typically in infants under 1 year. The triad: subdural hemorrhage, retinal hemorrhages, and encephalopathy. May present with irritability, lethargy, vomiting, seizures, and apnea. CT shows subdural hematoma; MRI shows diffuse axonal injury. Ophthalmologic exam reveals retinal hemorrhages. High mortality and long-term neurodevelopmental impairment.

Sickle Cell Anemia

– The most severe form of sickle cell disease (HbSS), causing chronic hemolytic anemia and vaso-occlusive crises. Features: jaundice, splenomegaly (infancy), pain crises, acute chest syndrome, stroke, and increased infection risk from functional asplenia. Diagnosis: hemoglobin electrophoresis showing HbS. Management: penicillin prophylaxis, hydroxyurea, pneumococcal vaccines, and folic acid supplementation.

Skeletal Survey

– A series of radiographs obtained when child abuse is suspected. Includes AP and lateral views of the skull, chest (including oblique views), abdomen, spine, and bilateral upper and lower extremities. Identifies fractures of varying ages, which is highly suggestive of non-accidental trauma. Initial survey may be negative; repeat at 2 weeks may reveal healing fractures. Interpretation by pediatric radiologist recommended.

Slipped Capital Femoral Epiphysis

– Displacement of the femoral head relative to the femoral neck through the growth plate (physis) in adolescents. Most common hip disorder in adolescents (10-16 years). Risk factors: obesity, male sex, African American, and endocrine disorders (hypothyroidism, growth hormone deficiency). Presents with hip, groin, or knee pain with limping (antalgic gait). On examination: limited internal rotation and obligatory external rotation with hip flexion. Diagnosis: bilateral hip X-ray (AP and frog-leg lateral) shows femoral head displaced posteriorly and medially relative to the neck (Klein line). Treatment: surgical stabilization with in situ screw fixation of the affected and contralateral hip. Urgent: non-weight-bearing to prevent further slippage.

Smith-Lemli-Opitz Syndrome

– An autosomal recessive disorder of cholesterol metabolism caused by mutations in DHCR7, encoding 7-dehydrocholesterol reductase, the final enzyme in the cholesterol biosynthetic pathway. Features: microcephaly, characteristic facies (ptosis, anteverted nares, micrognathia, low-set ears), intellectual disability (mild to profound), syndactyly of the second and third toes (pathognomonic, 95%), postaxial polydactyly, cleft palate, hypospadias (ambiguous genitalia in males), and congenital heart disease. Diagnosis: elevated serum 7-dehydrocholesterol levels and DHCR7 genetic testing. Treatment: dietary cholesterol supplementation (improves growth and behavior), and symptomatic management of associated anomalies. Prognosis correlates with severity of malformations and cholesterol levels.

Spina Bifida Occulta

– The mildest form of spina bifida, characterized by incomplete closure of the vertebral arch without herniation of neural elements. Often asymptomatic; may be detected by skin findings (dimple, hair tuft, lipoma) over the lower spine. Incidence approximately 1-2% of the population. Usually requires no treatment unless associated with tethered cord or spinal cord symptoms.

Stepping Reflex

– A primitive reflex present from birth to 2 months, elicited by holding the infant upright with feet touching a flat surface. The infant makes stepping-like movements as if walking. Disappears by 2 months as voluntary motor control develops. Reappears around 9-10 months as voluntary walking emerges. Assessed during newborn and early infant neurologic examination.

Stickler Syndrome

– An autosomal dominant connective tissue disorder caused by mutations in collagen genes (COL2A1 most common, also COL11A1, COL11A2, COL9A1). Features: Pierre Robin sequence (micrognathia, glossoptosis, cleft palate, 30%), high myopia (present in 90% by 10 years), retinal detachment (highest risk of any pediatric syndrome), sensorineural hearing loss, and early-onset osteoarthritis with mild epiphyseal dysplasia. Pierre Robin sequence in Stickler syndrome mandates ophthalmologic screening. Diagnosis: clinical criteria (major: Pierre Robin-associated cleft palate, high myopia, retinal detachment, hearing loss, family history) and genetic testing. Management: ophthalmology — annual dilated fundus exams, retinal detachment prevention; audiology — hearing aids; orthopedics — joint protection and pain management; dental palatal repair; avoidance of contact sports.

Sturge-Weber Syndrome

– A neurocutaneous syndrome (phakomatosis) characterized by a facial capillary malformation (port-wine stain) in the V1 distribution of the trigeminal nerve (forehead and upper eyelid) and ipsilateral leptomeningeal angiomatosis (venous malformations of the pia mater). Caused by a somatic activating mutation in GNAQ. Presents in infancy with the port-wine stain, which is the visible marker. Neurologic features: seizures (75-90%, often focal or infantile spasms, typically starting in the first year), hemiparesis (contralateral to the angioma, 30-50%), developmental delay, and intellectual disability. Ocular features: glaucoma (30-70%, can be congenital or develop later, ipsilateral to the angioma), and choroidal hemangioma. Diagnosis: clinical with contrast-enhanced brain MRI (leptomeningeal enhancement, cerebral atrophy, gyriform calcification, and choroid plexus enlargement). Skull X-ray may show tram-track calcifications. Treatment: anticonvulsants for seizures; early hemispherotomy or corpus callosotomy for refractory epilepsy. Regular ophthalmologic monitoring for glaucoma (lifelong). Laser therapy for the port-wine stain. Surveillance: developmental assessments and neuroimaging.

Subgaleal Hemorrhage

– A life-threatening accumulation of blood in the subgaleal space (between the periosteum and the epicranial aponeurosis). Can contain up to 50% of the infant's blood volume. Risk factors include vacuum-assisted delivery. Presents with rapidly enlarging, boggy scalp swelling that crosses suture lines, associated with anemia and hemodynamic instability. Requires urgent evaluation, blood transfusion, and resuscitation.

Sucking Reflex

– A primitive reflex present from birth, elicited by placing an object in the infant's mouth. The infant rhythmically sucks on the object. Coordinated sucking and swallowing typically develop by 34-36 weeks gestational age. Premature infants may lack coordinated suck-swallow, requiring gavage feeding. Absence in a term infant may indicate neurologic impairment. Important for successful breastfeeding and oral feeding.

Sudden Infant Death Syndrome (SIDS)

– The sudden, unexpected death of an infant less than 1 year of age that remains unexplained after thorough investigation. Peak incidence: 2-4 months. Risk factors: prone sleeping position, soft bedding, maternal smoking, prematurity, overheating, and bed-sharing. Triple risk model: vulnerable infant (brainstem abnormalities in arousal), critical developmental period, and exogenous stressors (prone sleep). Prevention: supine sleep position (Back to Sleep campaign reduced SIDS by greater than 50%), firm sleep surface, room-sharing without bed-sharing, avoiding overheating, pacifier use, and avoiding smoke exposure.

Surfactant

– A phospholipid-protein complex produced by type II pneumocytes that reduces alveolar surface tension, preventing atelectasis. Deficiency causes respiratory distress syndrome in premature infants. Exogenous surfactant (beractant, poractant alfa) administered via endotracheal tube. Natural surfactant contains phosphatidylcholine, phosphatidylglycerol, and surfactant proteins A, B, C, and D. Production begins at approximately 24-28 weeks gestation.

Syphilis

– A bacterial STI caused by Treponema pallidum. Primary: chancre at inoculation site. Secondary: rash, mucous patches, condylomata lata. Latent: asymptomatic. Tertiary: gummas, cardiovascular, neurosyphilis. Diagnosis: non-treponemal tests (RPR/VDRL) confirmed by treponemal tests. Treatment: penicillin G benzathine IM. Congenital syphilis preventable with maternal screening and treatment.

Tanner Staging

– A scale for assessing sexual maturity during puberty. Stage 1: prepubertal. Stage 2: breast buds (girls), testicular enlargement (boys). Stage 3: breast enlargement, pubic hair. Stage 4: areolar development, more pubic hair. Stage 5: adult breasts and genitalia, adult hair pattern. Used to track pubertal progression and identify precocious or delayed puberty.

Tdap Vaccine

– Tetanus, Diphtheria, and acellular Pertussis booster given at 11-12 years and during each pregnancy (27-36 weeks gestation). Replaces one dose of Td. Important for pertussis protection, particularly for protecting newborns through maternal antibody transfer. Also given to adolescents and adults to maintain tetanus and diphtheria immunity.

Thalassemia Major

– Beta-thalassemia, an autosomal recessive hemoglobin disorder with absent or minimal beta-globin chain production. Presents in infancy with severe anemia, failure to thrive, hepatosplenomegaly, and skeletal deformities. Diagnosis by hemoglobin electrophoresis showing absent HbA and elevated HbF and HbA2. Treatment: chronic blood transfusions, iron chelation, and hematopoietic stem cell transplant.

Tonic Neck Reflex

– A primitive reflex present from birth to 6 months, also called fencer's pose. When the head is turned to one side, the arm on that side extends while the opposite arm flexes. Present in both directions. Asymmetry may indicate neurologic injury. Disappears as voluntary head control develops. One of several posture-related reflexes assessed in the newborn period.

Toxoplasmosis

– Parasitic infection with Toxoplasma gondii, transmissible transplacentally. Maternal infection from undercooked meat, cat feces, or contaminated soil. Can cause hydrocephalus, intracranial calcifications, chorioretinitis, and seizures in the fetus. Congenital disease may present at birth or manifest later in childhood. Treatment: pyrimethamine and sulfadiazine with folinic acid.

Tracheoesophageal Fistula

– An abnormal connection between the trachea and esophagus, often associated with esophageal atresia. Presents with recurrent aspiration, coughing with feedings, and cyanosis during meals. The H-type fistula may present later in infancy with chronic aspiration and failure to thrive. Diagnosis by esophagram or bronchoscopy. Treatment: surgical ligation and division of the fistula.

Transfusion of Neonate

– Administration of packed red blood cells to neonates for symptomatic anemia or hematocrit below transfusion threshold. Indications include acute blood loss, severe anemia (hematocrit less than 20%), and respiratory distress. Volume: 10-20 mL/kg over 2-4 hours. Blood products must be CMV-negative, irradiated, and crossmatched. Complications include volume overload, infection, and transfusion reactions.

Transposition of the Great Arteries

– A cyanotic congenital heart defect where the aorta arises from the right ventricle and the pulmonary artery arises from the left ventricle (ventriculoarterial discordance). The result: two parallel circulations (systemic and pulmonary) with no mixing. Survival depends on associated shunts (ASD, VSD, PDA) to allow mixing of oxygenated and deoxygenated blood. Presents in the first days of life with severe cyanosis, tachypnea, and profound hypoxemia. Classic chest X-ray: egg-on-side appearance (narrow superior mediastinum). Treatment: immediate PGE1 infusion to maintain PDA patency, balloon atrial septostomy (Rashkind procedure) to improve mixing, and ultimately arterial switch operation (Jatene procedure) within the first 2 weeks of life.

Treacher Collins Syndrome

– An autosomal dominant craniofacial disorder caused by mutations in TCOF1 (most common), POLR1C, or POLR1D genes, encoding ribosomal RNA transcription factors. Bilateral, symmetric craniofacial anomalies: zygomatic hypoplasia (malar flattening), micrognathia, downward-slanting palpebral fissures, coloboma of the lower eyelid, microtia (small, malformed pinna), and conductive hearing loss (from middle ear anomalies). Intelligence is typically normal. Diagnosis: clinical and genetic testing. Management: multidisciplinary — airway assessment (possible tracheostomy for severe micrognathia), hearing aids or bone-anchored hearing aids for hearing loss, and staged reconstructive surgery (zygomatic and mandibular distraction, eyelid reconstruction, and external ear reconstruction).

Tricuspid Atresia

– Absence of the tricuspid valve, resulting in no direct communication between the right atrium and right ventricle. Blood flow depends on an atrial septal defect and ventricular septal defect for mixing. Presents with cyanosis. Staged surgical repair: modified Blalock-Taussig shunt in infancy, followed by Fontan procedure in early childhood. Oxygen and prostaglandin E1 used for stabilization.

Truncus Arteriosus

– A congenital heart defect where a single large vessel arises from both ventricles, receiving a mixed blood supply. Associated with a large ventricular septal defect and abnormal semilunar valve. Presents with cyanosis and heart failure early in life. Requires surgical repair in the neonatal period, involving VSD closure and placement of a right ventricle-to-pulmonary artery conduit.

Tuberculin Skin Test

– Intradermal injection of purified protein derivative (PPD) to test for Mycobacterium tuberculosis infection. Read at 48-72 hours measuring induration (not erythema). Positive: greater than 5 mm in HIV or close contacts; greater than 10 mm in high-risk groups; greater than 15 mm in low-risk populations. Does not differentiate latent from active infection. False positives from BCG vaccination.

Tuberculosis in Children

– Infection with Mycobacterium tuberculosis, often from household contacts. Latent TB infection (LTBI) positive tuberculin skin test or interferon-gamma release assay with no symptoms or radiographic findings. Active TB presents with cough, fever, weight loss, and lymphadenopathy. Diagnosis by PPD, chest X-ray, and sputum culture. Treatment: isoniazid, rifampin, pyrazinamide, and ethambutol for active disease.

Tuberous Sclerosis Complex

– An autosomal dominant disorder caused by TSC1 or TSC2 gene mutations, resulting in benign tumors in multiple organs. Features: cortical tubers, subependymal giant cell astrocytomas, renal angiomyolipomas, cardiac rhabdomyomas, and hypomelanotic macules (ash-leaf spots). May present with seizures in infancy. Diagnosis: clinical criteria and genetic testing. Management: seizure control, mTOR inhibitors (everolimus), and organ surveillance.

Tympanostomy Tubes

– Small tubes inserted through the tympanic membrane to ventilate the middle ear and prevent fluid accumulation. Indicated for recurrent acute otitis media (3 episodes in 6 months or 4 in 12 months) or persistent otitis media with effusion greater than 3 months with hearing loss. Performed as outpatient procedure under general anesthesia. Most tubes extrude spontaneously within 6-18 months.

Tyrosinemia

– An autosomal recessive inborn error of tyrosine metabolism. Type I (fumarylacetoacetate hydrolase deficiency, FAH mutations) is the most severe: presents in infancy with liver failure, renal tubular dysfunction (Fanconi syndrome — hypophosphatemic rickets), neurologic crises (painful paresthesias, autonomic instability, porphyria-like symptoms from succinylacetone accumulation), and hepatocellular carcinoma risk. Type II (tyrosine aminotransferase deficiency, TAT mutations): oculocutaneous tyrosinemia with painful palmar/plantar hyperkeratosis and corneal ulcers; no liver or kidney disease. Type III (4-hydroxyphenylpyruvate dioxygenase deficiency, HPD mutations): rare, mild intellectual disability and ataxia. Newborn screening shows elevated tyrosine and succinylacetone (pathognomonic for Type I). Diagnosis: urine succinylacetone (Type I), plasma amino acids, and FAH/TAT/HPD genetic testing. Treatment: Type I — nitisinone (NTBC, 2-(2-nitro-4-trifluoromethylbenzoyl)-1,3-cyclohexanedione) blocks tyrosine catabolism upstream of FAH, preventing toxic metabolite accumulation, combined with a tyrosine- and phenylalanine-restricted diet. Liver transplantation for refractory liver disease or HCC. Type II — dietary restriction of tyrosine and phenylalanine.

Urea Cycle Disorder

– A group of inborn errors of ammonia metabolism caused by deficiency of any of the six enzymes or two transporters in the urea cycle. X-linked: ornithine transcarbamylase deficiency (OTC, most common, 50%). Autosomal recessive: N-acetylglutamate synthase (NAGS), carbamoyl phosphate synthetase I (CPS1), argininosuccinate synthetase (citrullinemia), argininosuccinate lyase (argininosuccinic aciduria), and arginase 1 (argininemia). Presents with hyperammonemic encephalopathy: poor feeding, vomiting, lethargy, tachypnea/respiratory alkalosis (from central hyperventilation), ataxia, seizures, and coma. Onset varies from neonatal (complete deficiency) to late-onset (partial deficiency). Diagnosis: plasma ammonia (elevated

>150 μmol/L in neonates is symptomatic), quantitative plasma amino acids (citrulline, arginine, glutamine, alanine), urine orotic acid, and genetic testing. Treatment: acute hyperammonemia — IV sodium phenylacetate/sodium benzoate (Ammonul), IV arginine, nitrogen-free IV fluids, and hemodialysis for severe (>500 μmol/L). Chronic management — low-protein diet, ammonia-scavenging drugs (sodium phenylbutyrate, glycerol phenylbutyrate, arginine or citrulline supplementation), and liver transplantation for severe OTC deficiency. NAGS deficiency is uniquely treatable with oral N-carbamylglutamate.

VACTERL Association

– A non-random association of congenital anomalies, diagnosed when at least three of the following are present: Vertebral anomalies (hemivertebrae, fused vertebrae, scoliosis, 70%), Anorectal malformations (imperforate anus, 80%), Cardiac defects (VSD, tetralogy of Fallot, 50-80%), Tracheo-Esophageal fistula (with or without esophageal atresia, 70%), Renal anomalies (renal agenesis, horseshoe kidney, hydronephrosis, 60%), and Limb defects (radial dysplasia, thumb anomalies, polydactyly, 50-70%). The acronym also includes non-VACTERL findings: single umbilical artery and rib anomalies. The cause is unknown; most cases are sporadic. Diagnosis: clinical with systematic evaluation of involved organ systems. Management: staged surgical repair (imperforate anus colostomy, TEF repair, cardiac surgery), and long-term follow-up for renal function, growth, and development.

Varicella Vaccine

– Live attenuated vaccine given at 12-15 months and 4-6 years. Protects against chickenpox (varicella-zoster virus). Two doses provide greater than 90% protection. Contraindicated in immunocompromised and pregnant individuals. Also reduces risk of herpes zoster later in life. Universal vaccination significantly reduced varicella incidence and complications.

Viral Meningitis

– Inflammation of the meninges caused by viruses, usually Enteroviruses. Presents similarly to bacterial meningitis but generally less severe. CSF shows lymphocytic pleocytosis, normal glucose, and mildly elevated protein. Self-limited in most cases; supportive treatment. Differentiated from bacterial meningitis by CSF analysis to prevent unnecessary prolonged antibiotic therapy.

Whooping Cough

– See Pertussis.

Williams Syndrome

– A microdeletion syndrome involving the elastin gene on chromosome 7q11.23. Features: elfin facies, supravalvular aortic stenosis, hypercalcemia, intellectual disability with strong verbal skills, and overly friendly personality. Often associated with connective tissue abnormalities. Management: cardiac monitoring, calcium management, and educational support.

Wilms' tumor (Nephroblastoma)

– The most common renal malignancy in children, accounting for 95% of pediatric renal tumors (peak incidence 2-5 years). It is an embryonal tumor arising from metanephric blastemal cells. Associated syndromes: WAGR (Wilms, Aniridia, Genitourinary anomalies, mental Retardation — WT1 deletion on 11p13), Beckwith-Wiedemann syndrome (hemihypertrophy, macroglossia, omphalocele — WT2 locus 11p15), and Denys-Drash syndrome (WT1 mutation, male pseudohermaphroditism, nephrotic syndrome). Pathology: triphasic histology (blastemal, stromal, epithelial components). Anaplastic histology (focal or diffuse) confers a worse prognosis. Presents with a large, asymptomatic abdominal mass (often noted by parent while bathing the child), abdominal pain, haematuria (microscopic or gross), and hypertension (from renin secretion by the tumor or renal artery compression). Workup: abdominal ultrasound (distinguishes solid tumor from hydronephrosis, cystic lesions, neuroblastoma) and CT abdomen (staging: tumor extent, lymph node involvement, bilateral disease, IVC thrombus). Staging: Stage I (limited to kidney, capsule intact, completely resected), Stage II (extends beyond kidney, completely resected), Stage III (gross residual disease, positive lymph nodes, tumor spillage), Stage IV (haematogenous metastases — lung, liver, bone, brain), Stage V (bilateral renal involvement). Treatment: multimodal therapy based on staging and histology. Surgery: radical nephrectomy (transperitoneal approach, remove kidney + Gerota fascia + ipsilateral ureter + lymph node sampling). Chemotherapy: regimen depends on stage and histology (favourable vs. anaplastic) — vincristine, dactinomycin, doxorubicin. Radiation: for high-stage disease (Stage III-IV, anaplastic histology). Bilateral Wilms: neoadjuvant chemotherapy followed by nephron-sparing surgery. Prognosis: overall cure rate exceeds 90% (favourable histology). Anaplastic histology and metastatic disease have lower survival rates (50-70%). Long-term follow-up for renal function, cardiac function (doxorubicin), and second malignancies.

Chapter 27

Pharmacology

5-Alpha Reductase Inhibitor

– A class of medications that inhibit the enzyme 5-alpha-reductase, which converts testosterone to dihydrotestosterone (DHT), the more potent androgen responsible for prostate growth and male pattern baldness. Two isoforms: Type I (predominant in skin and scalp) and Type II (predominant in the prostate). Finasteride inhibits primarily Type II (blocks 70% of DHT); dutasteride inhibits both Type I and Type II (blocks >90% of DHT). Indications: benign prostatic hyperplasia — reduces prostate volume, improves urinary flow, reduces risk of acute urinary retention and need for surgery (requires 6-12 months for maximal effect); male androgenetic alopecia — slows hair loss and promotes regrowth. Adverse effects: decreased libido, erectile dysfunction (2-5%), ejaculatory dysfunction, and gynecomastia. Serum PSA levels decrease by approximately 50% — adjusted PSA values are needed for prostate cancer screening.

Abacavir

– A nucleoside reverse transcriptase inhibitor (NRTI) used in combination antiretroviral therapy for HIV-1 infection. Abacavir is a guanosine analogue that inhibits viral reverse transcriptase by competing with natural deoxyguanosine triphosphate and causing chain termination. A unique and potentially fatal hypersensitivity reaction occurs in approximately 5-8% of patients, presenting with fever, rash, gastrointestinal symptoms, and respiratory distress. This reaction is strongly associated with the HLA-B*5701 allele; genetic screening for this allele is mandatory before initiating therapy, and abacavir should never be used in patients who test positive due to the high risk of a potentially fatal hypersensitivity reaction.

ACE Inhibitors

– Angiotensin-converting enzyme inhibitors are a class of medications that block the conversion of angiotensin I to angiotensin II by inhibiting the angiotensin-converting enzyme (ACE, also known as kininase II). This results in decreased aldosterone secretion, reduced sodium and water retention, decreased vasoconstriction, and reduced sympathetic nervous system activity. ACE inhibitors also inhibit the breakdown of bradykinin, a vasodilator, which contributes to their antihypertensive effect but also contributes to the characteristic adverse effect of dry cough, which occurs in approximately 10-20% of patients, and the less common but more serious angioedema.

Acetaminophen Mechanism

– Acetaminophen, also known as paracetamol, is an analgesic and antipyretic drug with weak anti-inflammatory activity, differing from NSAIDs in its mechanism and side effect profile. Its precise mechanism of action remains incompletely understood, but it is thought to inhibit a COX variant in the central nervous system, possibly COX-3, and may also activate descending serotonergic pain inhibition pathways and interact with the endocannabinoid system through its metabolite AM404. Acetaminophen inhibits prostaglandin synthesis in the central nervous system, reducing pain perception and fever, and also activates descending serotonergic pathways that modulate pain signalling at the spinal cord level.

Acetaminophen (Paracetamol)

– An analgesic and antipyretic drug that inhibits central COX enzymes and may have serotonergic and endocannabinoid mechanisms. It is first-line for mild pain and fever with fewer side effects than NSAIDs. Maximum daily dose is 4 grams (reduced to 2-3 grams in liver disease or alcohol use). Acetaminophen overdose causes hepatotoxicity due to the toxic metabolite NAPQI. Antidote is N-acetylcysteine, which replenishes glutathione stores.

Acinetobacter baumannii resistance

– Acinetobacter baumannii is a gram-negative coccobacillus that has emerged as a significant cause of healthcare-associated infections, particularly in intensive care units. The organism has intrinsic and acquired resistance mechanisms, including beta-lactamases, efflux pumps, and porin mutations. Multidrug-resistant and carbapenem-resistant strains are increasingly common. Treatment options are limited and may include polymyxins, tigecycline, and minocycline. Infection prevention strategies are critical for preventing nosocomial transmission, particularly as Acinetobacter can survive on environmental surfaces for extended periods.

Adverse Drug Reaction

– An adverse drug reaction is any noxious or unintended response to a drug that occurs at doses used for prophylaxis, diagnosis, or therapy. ADRs are classified as type A reactions, which are dose-dependent and predictable extensions of the drug's pharmacological effects, such as bleeding with warfarin or hypoglycemia with insulin; and type B reactions, which are idiosyncratic, unpredictable, and not dose-related, such as drug allergies and Stevens-Johnson syndrome. ADRs are further classified as type C, dose- and time-dependent reactions such as cumulative toxicity; type D, delayed effects such as carcinogenicity and teratogenicity; and type E, withdrawal reactions upon discontinuation.

Adverse Drug Reactions

– Any harmful or unintended response to a drug at normal doses. ADRs are classified as Type A (augmented, dose-dependent, predictable: side effects, toxicity) or Type B (bizarre, dose-independent, unpredictable: allergy, idiosyncrasy). ADRs are a major cause of morbidity and mortality, accounting for 5-7% of hospital admissions. Risk factors include polypharmacy, renal/hepatic impairment, age extremes, and genetic predisposition. Prevention involves careful prescribing and monitoring.

Agonist-Antagonist

– A drug that has mixed pharmacological activity, acting as an agonist at one receptor type and an antagonist at another, or as a partial agonist at a single receptor. Mixed opioid agonist-antagonists: pentazocine — kappa agonist and mu partial antagonist, producing analgesia with less respiratory depression and lower abuse potential than pure mu agonists; nalbuphine — kappa agonist and mu antagonist; butorphanol — kappa agonist and mu antagonist. Partial agonists: buprenorphine — partial mu agonist and kappa antagonist, used for opioid use disorder due to ceiling effect on respiratory depression and lower abuse liability. The clinical advantage of agonist-antagonists is the reduced risk of respiratory depression and dependence compared to full mu agonists.

Aminoglycoside nephrotoxicity

– Kidney damage caused by aminoglycoside antibiotics, characterized by acute tubular necrosis. Nephrotoxicity is caused by accumulation of aminoglycosides in the proximal tubular cells. The risk is increased with prolonged therapy, high doses, concurrent nephrotoxic drugs, and renal impairment. Gentamicin is considered the most nephrotoxic aminoglycoside. The nephrotoxicity is usually reversible with early detection and discontinuation. Regular monitoring of serum creatinine and drug levels is essential for early detection of nephrotoxicity, and dose adjustment based on renal function is required to minimise the risk of acute kidney injury.

Aminoglycoside ototoxicity

– Damage to the inner ear caused by aminoglycoside antibiotics, resulting in hearing loss and vestibular dysfunction. Ototoxicity is caused by accumulation of aminoglycosides in the inner ear, damaging hair cells. The risk is increased with prolonged therapy, high doses, concurrent ototoxic drugs, and renal impairment. Amikacin is considered the most ototoxic aminoglycoside. Symptoms include tinnitus, hearing loss, and vertigo. Hearing loss may be irreversible. Baseline and periodic audiometry should be performed at baseline and repeated regularly during treatment, particularly in patients receiving high cumulative doses or those with pre-existing hearing impairment.

Aminoglycosides

– A class of bactericidal antibiotics that bind irreversibly to the 30S ribosomal subunit, causing misreading of messenger RNA and inhibiting protein synthesis. They include gentamicin, tobramycin, amikacin, and streptomycin. Aminoglycosides are particularly effective against aerobic gram-negative organisms including Pseudomonas aeruginosa and have synergistic activity with beta-lactams against enterococci and staphylococci. They are poorly absorbed from the gastrointestinal tract and must be administered parenterally for systemic infections, with careful monitoring of serum concentrations to balance efficacy against nephrotoxicity and ototoxicity.

Amiodarone

– A class III antiarrhythmic agent that primarily blocks potassium channels, prolonging the action potential duration and effective refractory period in cardiac tissue. It also has properties of all four Vaughan Williams classes: class I sodium channel blockade, class II beta-adrenergic blockade, class III potassium channel blockade, and class IV calcium channel blockade. Amiodarone is the most effective antiarrhythmic drug available, used for both ventricular and supraventricular arrhythmias, including atrial fibrillation and ventricular tachycardia, but its use is limited by a distinctive and extensive side effect profile including pulmonary fibrosis, hepatotoxicity, thyroid dysfunction, corneal deposits, and blue-gray skin pigmentation.

Amphotericin B

– A polyene antifungal antibiotic that binds to ergosterol in fungal cell membranes, forming pores that cause leakage of intracellular ions and cell death. It has the broadest spectrum of antifungal activity, effective against Candida, Aspergillus, Cryptococcus, Mucorales, and endemic dimorphic fungi including Histoplasma and Blastomyces. Amphotericin B is considered the gold standard for serious invasive fungal infections, though its use has been limited by significant toxicity.The major dose-limiting adverse effects include nephrotoxicity, infusion-related reactions (fever, rigors, hypotension), and electrolyte disturbances such as hypokalaemia and hypomagnesaemia.

Amprenavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection, now largely replaced by its prodrug fosamprenavir. Amprenavir inhibits the HIV aspartyl protease, preventing viral polyprotein processing and maturation. The drug has moderate oral bioavailability and requires dosing two to three times daily. Common adverse effects include gastrointestinal disturbances such as nausea, diarrhea, and vomiting, as well as perioral paresthesias. Hypertriglyceridemia and hypercholesterolaemia are common metabolic adverse effects associated with protease inhibitor therapy.

Angiotensin Receptor Blockers

– A class of cardiovascular drugs that selectively block the angiotensin II type 1 receptor, preventing the binding of angiotensin II and its effects including vasoconstriction, aldosterone secretion, and sympathetic activation. Unlike ACE inhibitors, ARBs do not affect bradykinin metabolism, resulting in lower rates of cough and angioedema. Common ARBs include losartan, valsartan, irbesartan, and olmesartan. ARBs have similar hemodynamic effects to ACE inhibitorsbut are generally better tolerated due to a lower incidence of cough and angioedema.

Anidulafungin

– An echinocandin antifungal agent used in the treatment of invasive candidiasis and esophageal candidiasis. Anidulafungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis. A unique feature is its lack of hepatic metabolism, as it is degraded chemically rather than enzymatically, making it safe in patients with hepatic impairment. The drug is available only intravenously and has a very long half-life of approximately 24-48 hours, allowing for once-daily dosing. Common aadverse effects include headache, nausea, diarrhea, and phlebitis at the infusion site, and hepatotoxicity with elevated transaminases has been reported.

Antacids

– Medications that neutralise gastric acid, providing rapid symptomatic relief in dyspepsia, gastroesophageal reflux, and peptic ulcer disease. Commonly contain aluminium hydroxide, magnesium hydroxide, calcium carbonate, or sodium bicarbonate. Aluminium-based antacids can cause constipation; magnesium-based antacids can cause diarrhea; combination preparations balance these effects. Antacids also promote gastric mucosal defence by stimulating bicarbonate and prostaglandin secretion. Drug interactions: reduce absorption of tetracyclines, iron, digoxin, and bisphosphonates (separate dosing by 2 hours). Not first-line for chronic GERD (PPIs preferred).

Antianginal

– A class of medications used to prevent or relieve angina pectoris (chest pain from myocardial ischemia). The three main classes of antianginal drugs target different mechanisms: nitrates (nitroglycerin, isosorbide mononitrate/dinitrate) — venodilators that reduce preload and myocardial oxygen demand; beta-blockers — reduce heart rate, contractility, and oxygen demand; and calcium channel blockers (verapamil, diltiazem, nifedipine) — vasodilators that reduce afterload and coronary spasm. Ranolazine is a newer antianginal agent that inhibits the late sodium current in cardiac myocytes, reducing intracellular calcium overload and improving diastolic function without affecting heart rate or blood pressure.

Antibiotic cycling

– The scheduled rotation of antimicrobial agents within a healthcare facility to reduce selective pressure for resistance. The strategy involves periodically changing the preferred antibiotics for specific infections. For example, cycling between carbapenems and fluoroquinolones for gram-negative infections. The evidence for antibiotic cycling is mixed, with some studies showing reductions in resistance and others showing no benefit. Cycling may be combined with other stewardship strategies includincluding education, surveillance, and the development of evidence-based prescribing guidelines tailored to local resistance patterns.

Antibiotic de-escalation

– The process of narrowing antimicrobial spectrum based on microbiological results and clinical response. De-escalation aims to reduce selective pressure for resistance while maintaining therapeutic efficacy. The process involves identifying the causative organism and susceptibility pattern, then switching from broad-spectrum to narrow-spectrum agents. De-escalation should be performed as soon as possible, typically within 48-72 hours when culture results are available. This strategy reduces the rrisk of antibiotic resistance, selective pressure on the microbiome, superinfection risk, and adverse drug effects, while also lowering overall treatment costs.

Antibiotic mixing

– A stewardship strategy where different patients with the same indication receive different antibiotics concurrently, with the goal of reducing resistance emergence. For example, using different agents for ventilator-associated pneumonia in different patients within the same unit. The evidence for antibiotic mixing is limited, but theoretical models suggest it may reduce resistance emergence. This strategy may be combined with other stewardshipstrategies including prospective audit and feedback, formulary restriction, and computerised decision support to achieve optimal antimicrobial use.

Antibiotics

– Medications that kill or inhibit the growth of bacteria, used to treat bacterial infections such as pneumonia, strep throat, and urinary tract infections. They are ineffective against viral infections like colds or influenza. Overuse and misuse have driven antibiotic resistance, a growing public health crisis that makes infections harder to treat.

Anticonvulsants

– Medications used to prevent or control seizures in epilepsy and other seizure disorders. Mechanisms: enhance GABA activity (benzodiazepines, barbiturates, vigabatrin, tiagabine), block voltage-gated sodium channels (phenytoin, carbamazepine, lamotrigine), block calcium channels (gabapentin, pregabalin, ethosuximide), or inhibit glutamate release (levetiracetam). Choice depends on seizure type: sodium valproate for generalised, carbamazepine/lamotrigine for focal. Side effects: sedation, dizziness, ataxia, rash, hepatotoxicity, teratogenicity (valproate). Therapeutic drug monitoring required for phenytoin and carbamazepine.

Antidiabetic

– A class of medications used to manage diabetes mellitus by lowering blood glucose levels through various mechanisms. Major classes include: biguanides (metformin) — reduce hepatic glucose production; sulfonylureas (glipizide, glyburide) — stimulate insulin secretion; meglitinides (repaglinide) — stimulate rapid, short-acting insulin secretion; thiazolidinediones (pioglitazone) — improve insulin sensitivity; DPP-4 inhibitors (sitagliptin) — increase incretin levels; GLP-1 receptor agonists (semaglutide, liraglutide) — stimulate insulin secretion, slow gastric emptying; SGLT2 inhibitors (empagliflozin, dapagliflozin) — increase urinary glucose excretion; insulin preparations — replace endogenous insulin; and alpha-glucosidase inhibitors (acarbose) — delay carbohydrate absorption.

Antidote

– An agent that counteracts the toxic effects of a poison or drug overdose by various mechanisms: chemical neutralization (e.g., naloxone reverses opioid overdose by competitive antagonism at mu-opioid receptors), chelation (e.g., deferoxamine binds iron, EDTA binds lead, dimercaprol binds arsenic and mercury), competitive inhibition (e.g., fomepizole inhibits alcohol dehydrogenase in ethylene glycol and methanol poisoning), metabolic intervention (e.g., N-acetylcysteine replenishes glutathione in acetaminophen overdose), or receptor blockade (e.g., flumazenil reverses benzodiazepine effects at GABA-A receptors). Timely administration of the appropriate antidote is critical in poisoning management and can be life-saving.

Antiemetic

– A medication that prevents or treats nausea and vomiting. Classes target different pathways and receptors: 5-HT3 receptor antagonists (ondansetron, granisetron) — block serotonin receptors in the chemoreceptor trigger zone and GI tract, first-line for chemotherapy-induced nausea and vomiting; NK1 receptor antagonists (aprepitant, fosaprepitant) — block substance P/neurokinin-1 receptors in the vomiting center, used for highly emetogenic chemotherapy; dopamine D2 antagonists (metoclopramide, prochlorperazine, droperidol) — block dopamine receptors in the CTZ, used for migraine and postoperative nausea; antihistamines (dimenhydrinate, meclizine) — block H1 receptors, effective for motion sickness; anticholinergics (scopolamine) — block muscarinic receptors, used for motion sickness; and corticosteroids (dexamethasone) — mechanism unclear, used for chemotherapy-induced nausea and postoperative nausea.

Antiemetics

– Medications that prevent or treat nausea and vomiting. Classes and mechanisms: 5-HT_3 antagonists (ondansetron — block serotonin receptors in CTZ and gut, effective for chemotherapy and postoperative nausea), dopamine antagonists (metoclopramide, prochlorperazine — block D_2 receptors in CTZ, prokinetic effect), antihistamines (cyclizine, promethazine — block H_1 receptors, effective for motion sickness), anticholinergics (hyoscine — motion sickness), neurokinin-1 antagonists (aprepitant — chemotherapy), and cannabinoids (nabilone). The cause of vomiting dictates drug choice.

Antiepileptic Drugs

– Medications used to prevent and control seizures by modulating neuronal excitability through various mechanisms. They can be classified by their primary mechanism of action. Sodium channel blockers including phenytoin, carbamazepine, and lamotrigine stabilize the inactivated state of voltage-gated sodium channels, reducing high-frequency repetitive firing. GABA-enhancing drugs including valproic acid, benzodiazepines, and vigabatrin increase inhibitory GABAergic transmissmission; glutamate receptor antagonists such as topiramate and felbamate reduce excitatory neurotransmission; and drugs with multiple mechanisms including valproic acid and levetiracetam provide broad-spectrum seizure control through combined action on GABA, sodium channels, and calcium channels.

Antifungal prophylaxis

– The use of antifungal agents to prevent the development of invasive fungal infections in high-risk patients. Common indications include prolonged neutropenia following chemotherapy, hematopoietic stem cell transplantation, solid organ transplantation, and advanced HIV infection. Fluconazole is the most widely used agent for prophylaxis in neutropenic patients, with doses of 200-400 mg daily. It is effective against Candida species but does not cover Aspergillus. Itraconazole provides broader covcoverage, including Aspergillus, and is used for prophylaxis in high-risk patients, though tolerability is limited by gastrointestinal side effects.

Antigout

– A class of medications used to treat gout by reducing uric acid levels or controlling acute inflammation. Acute gout treatment: NSAIDs (indomethacin, naproxen) — first-line for pain and inflammation; colchicine — inhibits microtubule polymerization, reducing neutrophil chemotaxis and inflammation, most effective when started within 24 hours of attack; corticosteroids (oral, intra-articular, or IV) for refractory cases. Long-term urate-lowering therapy: xanthine oxidase inhibitors (allopurinol, febuxostat) — reduce uric acid production by inhibiting xanthine oxidase; uricosuric agents (probenecid, lesinurad) — increase renal uric acid excretion by inhibiting URAT1 transporter; and recombinant uricase (pegloticase) — enzymatically degrades uric acid, reserved for severe tophaceous gout.

Antihistamines

– Medications that block histamine receptors, primarily H1 and H2 receptors, used to treat allergic conditions and other histamine-mediated disorders. H1 antihistamines are classified as first-generation (sedating, with significant anticholinergic effects) and second-generation (non-sedating, with minimal anticholinergic activity). First-generation H1 antihistamines include diphenhydramine, chlorpheniramine, hydroxyzine, and doxylamine; they cross the blood-brain barrier, causincausing sedation, cognitive impairment, and anticholinergic side effects including dry mouth, urinary retention, and constipation; second-generation H1 antihistamines include loratadine, cetirizine, and fexofenadine, which are preferred for long-term use due to their favourable safety profile.

Antimicrobial desensitization

– A procedure that involves administering gradually increasing doses of an antimicrobial to induce tolerance in patients with allergy. Desensitization is used when the causative agent is the only effective treatment and the patient has a history of hypersensitivity. The procedure is performed under close medical supervision with monitoring for anaphylaxis. It is most commonly performed for beta-lactam antibiotics in patients with severe allergy. Desensitization involves starting with very low doses, typically micrograms, and increasing gradually over several hours until the full therapeutic dose is tolerated, after which desensitisation must be maintained by continuous drug exposure.

Antimicrobial drug interactions

– Numerous drug interactions exist between antimicrobial agents and other medications. Fluoroquinolones chelate divalent cations and can interact with antacids, dairy products, and iron supplements. Macrolides inhibit CYP3A4 and can increase levels of co-administered medications. Azole antifungals inhibit CYP3A4 and have numerous drug interactions. Rifampin is a potent inducer of CYP enzymes and can decrease levels of many medications. TMP-SMX can increase levels of warfarin and phenytoin. Underststanding these interactions is critical for safe prescribing and requires careful review of the patient's complete medication list before initiating antimicrobial therapy.

Antimicrobial hypersensitivity reactions

– Hypersensitivity reactions to antimicrobial agents range from mild rash to life-threatening anaphylaxis. Beta-lactam antibiotics are the most common cause of drug allergy. Cross-reactivity between penicillins and cephalosporins is approximately 2%. Sulfonamide antibiotics can cause severe cutaneous reactions. Fluoroquinolones can cause photosensitivity and rare anaphylaxis. Vancomycin can cause red man syndrome, which is an infusion-related reaction rather than true allergy. Proper allergy assessment is essential, including skin testing when indicated, and documentation of the reaction type and severity to guide future prescribing decisions.

Antimicrobial resistance mechanisms

– Mechanisms by which bacteria develop resistance to antimicrobial agents. Major mechanisms include enzymatic inactivation (beta-lactamases, aminoglycoside-modifying enzymes), target modification (PBP alterations, ribosomal methylation), efflux pumps, and decreased permeability (porin mutations). Resistance can be intrinsic, acquired through mutation, or transferred horizontally via plasmids, transposons, and integrons. Understanding resistance mechanisms is essential for developing new antibioticantibiotics and for guiding empirical therapy based on local resistance surveillance data.

Antimicrobial Stewardship

– A coordinated programme of interventions designed to optimize antimicrobial use, improve patient outcomes, reduce adverse effects, and minimize the development of antimicrobial resistance. Core strategies include prospective audit and feedback where infectious disease specialists review antimicrobial prescriptions and recommend changes; formulary restrictions limiting the use of certain antimicrobials to specific indications; clinical practice guidelines based on loccal epidemiology and resistance patterns, with regular review and updating of prescribing policies based on surveillance data.

Antioxidants

– Substances that prevent or slow oxidative damage to cells caused by free radicals (reactive oxygen and nitrogen species). Endogenous antioxidants: glutathione, superoxide dismutase, catalase. Dietary antioxidants: vitamin C (ascorbic acid — water-soluble, scavenges aqueous-phase radicals), vitamin E (tocopherol — lipid-soluble, protects cell membranes), beta-carotene (provitamin A), selenium (cofactor for glutathione peroxidase), and flavonoids. Despite theoretical benefits, large clinical trials have not consistently demonstrated reduced cardiovascular events or cancer with antioxidant supplementation. A balanced diet rich in fruits and vegetables provides adequate antioxidants.

Antiparkinson

– A class of medications used to manage Parkinson disease symptoms by restoring dopaminergic activity or modulating other neurotransmitter systems. Dopaminergic agents: levodopa (combined with carbidopa to prevent peripheral decarboxylation) — the most effective symptomatic treatment, converted to dopamine in the brain; dopamine agonists (pramipexole, ropinirole, rotigotine patch) — directly stimulate dopamine receptors; MAO-B inhibitors (selegiline, rasagiline) — inhibit dopamine breakdown; COMT inhibitors (entacapone, tolcapone) — inhibit peripheral levodopa metabolism, extending its half-life. Anticholinergic agents (benztropine, trihexyphenidyl) — reduce tremor by blocking muscarinic receptors. Amantadine — increases dopamine release and reduces dyskinesias.

Antipruritic

– A medication that relieves itching (pruritus). First-line for most causes: topical emollients and moisturizers — restore skin barrier function; topical corticosteroids — reduce inflammation in atopic dermatitis, contact dermatitis, and psoriasis; topical antihistamines (diphenhydramine, doxepin) — block histamine H1 receptors in the skin; topical calcineurin inhibitors (tacrolimus, pimecrolimus) — for atopic dermatitis without steroid side effects. Systemic antiprurities: oral antihistamines (cetirizine, loratadine, hydroxyzine) — especially for urticaria; antidepressants (doxepin, mirtazapine) — for chronic pruritus; opioid antagonists (naltrexone) — for cholestatic pruritus; and aprepitant — an NK1 receptor antagonist for refractory pruritus.

Antipyretics

– Medications that reduce fever by inhibiting cyclo-oxygenase (COX) enzymes, reducing hypothalamic prostaglandin E_2 synthesis and resetting the thermoregulatory set point. Common antipyretics: paracetamol (acetaminophen — preferred first-line, particularly in children and patients with contraindications to NSAIDs, but hepatotoxic in overdose), ibuprofen (NSAID, also anti-inflammatory, avoid in dehydration, renal impairment, and peptic ulcer disease), and aspirin (avoid in children due to Reye syndrome risk). Antipyretics should be used for symptomatic relief, not routinely for all fevers, as fever is a physiological defence mechanism.

Antiseptic

– An antimicrobial substance applied to living tissue (skin, mucous membranes) or wounds to reduce the risk of infection by destroying or inhibiting microorganisms. Common antiseptics include: alcohol (ethanol, isopropanol, 60-90%) — denatures proteins, effective against bacteria and enveloped viruses, used for hand hygiene and skin preparation; chlorhexidine (2-4%) — disrupts cell membranes, broad-spectrum with persistent activity, used for surgical hand scrub and preoperative skin prep; povidone-iodine (10%) — releases free iodine that oxidizes cell components, broad-spectrum including spores; hydrogen peroxide (3%) — produces free radicals, mechanical debridement through effervescence, limited antimicrobial activity; and octenidine — newer antiseptic with broad activity, common in Europe. Antiseptics are distinct from antibiotics (systemic use) and disinfectants (applied to inanimate objects).

Antispasmodic

– A medication that reduces smooth muscle spasm, primarily in the gastrointestinal tract. Anticholinergic agents: hyoscyamine, dicyclomine, atropine — block muscarinic M3 receptors on GI smooth muscle, reducing contractions and peristalsis, used for irritable bowel syndrome and functional dyspepsia. Direct smooth muscle relaxants: mebeverine, papaverine, peppermint oil — act directly on smooth muscle cells without anticholinergic effects, providing relief from GI spasms. Urinary antispasmodics: oxybutynin, tolterodine, solifenacin — block M3 receptors on detrusor muscle in the bladder wall, reducing overactive bladder symptoms.

Antithyroid

– A class of medications that reduce thyroid hormone synthesis, used to treat hyperthyroidism (Graves disease, toxic nodular goiter). Thioamide derivatives: methimazole (propylthiouracil as second-line) — inhibit thyroid peroxidase, blocking iodination of tyrosine residues and coupling of iodotyrosines, reducing production of T3 and T4. PTU also inhibits peripheral conversion of T4 to T3. Methimazole is first-line due to better tolerability and once-daily dosing. Adverse effects: agranulocytosis (0.3-0.5%, the most serious, presenting with fever and sore throat), hepatotoxicity (rare with methimazole, more common with PTU), rash, arthralgia, and cholestatic jaundice. Treatment monitoring: thyroid function tests every 4-6 weeks during dose titration. Beta-blockers (propranolol, atenolol) are used adjunctively to control adrenergic symptoms (tachycardia, tremor) until euthyroid.

Antitubercular Agents

– Antimicrobial drugs used to treat Mycobacterium tuberculosis infections. The first-line agents include isoniazid, which inhibits mycolic acid synthesis; rifampicin, which inhibits RNA polymerase; pyrazinamide, which disrupts membrane energetics; ethambutol, which inhibits arabinosyl transferase; and streptomycin, which inhibits protein synthesis. The standard regimen for drug-susceptible tuberculosis is a four-drug combination of isoniazid, rifampicin, pyrazinamide, andethambutol for two months followed by a continuation phase of isoniazid and rifampicin for four months; treatment adherence is critical and directly observed therapy is recommended to prevent treatment failure and acquired drug resistance.

Antitussive

– A medication that suppresses the cough reflex, used for dry, non-productive cough. Centrally acting antitussives: dextromethorphan — acts on the cough center in the medulla oblongata, likely through sigma-1 receptor agonism, NMDA antagonism, and serotonin reuptake inhibition; codeine — a prodrug converted to morphine, acts on mu-opioid receptors in the cough center, controlled substance due to abuse potential, not recommended in children. Peripherally acting antitussives: benzonatate — a local anesthetic that reduces cough reflex by numbing stretch receptors in the respiratory tract; inhaled sodium cromoglycate and nedocromil — stabilize mast cells and reduce cough in asthma. Demulcents (honey, glycerin, lozenges) soothe irritated pharyngeal mucosa. Expectorants (guaifenesin) are not antitussives but increase mucus production to facilitate coughing up phlegm.

Antivirals

– Drugs that inhibit the replication of viruses, reducing the severity and duration of viral infections. They are used for conditions such as influenza, herpes, HIV, hepatitis B and C, and COVID-19. Unlike antibiotics, antivirals target specific viral enzymes or life cycle stages and often work best when started early in the infection.

Anxiolytic

– A medication that reduces anxiety without producing excessive sedation. First-line for generalized anxiety disorder: SSRIs (escitalopram, paroxetine, sertraline) and SNRIs (venlafaxine, duloxetine) — increase serotonin/norepinephrine activity, require 2-4 weeks for therapeutic effect, first-line for chronic anxiety disorders. Buspirone — a 5-HT1A partial agonist, effective for GAD, no sedation, no dependence, requires 2-4 weeks for effect. Benzodiazepines (alprazolam, lorazepam, diazepam, clonazepam) — enhance GABA-A receptor activity, rapid onset (minutes to hours), effective for acute anxiety and panic attacks, limited to short-term use (2-4 weeks) due to tolerance, dependence, and withdrawal risks. Beta-blockers (propranolol) — reduce somatic symptoms of performance anxiety (tachycardia, tremor) through peripheral beta-blockade.

Aromatase Inhibitor

– A class of medications that inhibit aromatase, the enzyme that converts androgens (androstenedione, testosterone) to estrogens (estrone, estradiol) in peripheral tissues (adipose, breast, muscle, bone). Used primarily in hormone receptor-positive breast cancer in postmenopausal women. Third-generation aromatase inhibitors: anastrozole (nonsteroidal, reversible), letrozole (nonsteroidal, reversible), and exemestane (steroidal, irreversible). In postmenopausal women, AIs are first-line adjuvant therapy and first-line therapy for metastatic breast cancer (superior to tamoxifen). In premenopausal women, AIs are used in combination with ovarian suppression. Adverse effects: arthralgia and myalgia (20-50%, most common cause of discontinuation), accelerated bone loss (osteoporosis, fracture risk), hot flashes, and dyslipidemia. AIs do not cause endometrial cancer or thromboembolism (unlike tamoxifen).

Atazanavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection. Atazanavir inhibits the HIV aspartyl protease, preventing viral polyprotein processing and producing immature, non-infectious viral particles. A key advantage of atazanavir is its once-daily dosing and relatively favorable lipid profile compared to other protease inhibitors, with minimal impact on triglycerides and cholesterol levels. The most characteristic adverse effect is indirect hyperbilirubinemia dueto inhibition of UGT1A1, causing unconjugated hyperbilirubinaemia and clinical jaundice in a subset of patients, which is reversible upon discontinuation but is not associated with hepatotoxicity.

Atypical Antipsychotics

– Atypical antipsychotics, also known as second-generation antipsychotics, block both dopamine D2 receptors and serotonin 5-HT2A receptors, which reduces extrapyramidal side effects compared to typical agents and may improve negative symptoms. Common atypical antipsychotics include risperidone, olanzapine, quetiapine, aripiprazole, ziprasidone, lurasidone, and clozapine. The dual D2/5-HT2A blockade in the nigrostriatal pathway reduces EPS risk, while blockade in the mesolimbic pathway treats positpositive symptoms (delusions, hallucinations) through D2 blockade and improves negative symptoms (apathy, social withdrawal) and cognitive deficits through 5-HT2A blockade; clozapine is reserved for treatment-resistant schizophrenia due to its unique efficacy but requires monitoring for agranulocytosis.

Azithromycin

– A macrolide antibiotic with an extended half-life allowing for once-daily dosing and short treatment courses. Azithromycin inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It has activity against gram-positive organisms, atypical pathogens (Mycoplasma, Chlamydophila, Legionella), and some gram-negative organisms. The drug is indicated for respiratory tract infections, sexually transmitted infections, and skin infections. Common adverse effects include gastrointestinaestinal upset, diarrhea, and nausea, and azithromycin is associated with a small increased risk of cardiovascular death due to QT prolongation.

Azole Antifungals

– A class of antifungal agents that inhibit the enzyme lanosterol 14-alpha-demethylase, a cytochrome P450 enzyme essential for ergosterol synthesis in fungal cell membranes. The depletion of ergosterol and accumulation of toxic methylated sterols disrupt membrane integrity and function. Azoles are divided into imidazoles including ketoconazole and clotrimazole, used primarily for topical infections, and triazoles including fluconazole, itraconazole, voriconazole, and posaconaposaconazole, which provide broader antifungal coverage; posaconazole and voriconazole have activity against Aspergillus species, while fluconazole has limited activity against this genus.

Aztreonam

– A monobactam antibiotic with activity only against aerobic gram-negative bacteria. Aztreonam inhibits cell wall synthesis by binding to penicillin-binding protein 3 (PBP3). It is resistant to hydrolysis by most beta-lactamases, including extended-spectrum beta-lactamases (ESBLs). Aztreonam is indicated for serious gram-negative infections including pneumonia, urinary tract infections, and bacteremia. It is an alternative for patients with beta-lactam allergy, as there is minimal cross-reactivitywith other beta-lactam antibiotics, making it a useful alternative for patients with reported penicillin allergy.

Bacteriophage therapy

– The use of bacterial viruses (bacteriophages) to treat bacterial infections. Bacteriophages are viruses that specifically infect and lyse bacteria. They have narrow specificity for target bacteria and do not affect normal flora. Phage therapy is being studied for multidrug-resistant bacterial infections, particularly those failing conventional antibiotics. Clinical trials have shown promise for chronic pseudomonal infections and prosthetic joint infections. The main challenges include regulatoryhurdles including standardised production, stability testing, pharmacokinetic characterisation, and regulatory approval pathways that vary across jurisdictions.

Barbiturate

– A class of CNS depressants derived from barbituric acid that enhance GABA-A receptor activity by increasing the duration of chloride channel opening (as opposed to benzodiazepines, which increase the frequency). At low doses: anxiolysis and sedation. At moderate doses: hypnosis and anesthesia. At high doses: respiratory depression, coma, and death. Classification by duration of action: ultra-short (thiopental, methohexital — IV induction agents), short/intermediate (pentobarbital, secobarbital — sedation, now rarely used), and long-acting (phenobarbital — anticonvulsant). Indications now limited: phenobarbital for neonatal seizures and status epilepticus when other agents fail; thiopental for rapid sequence induction historically. Adverse effects: narrow therapeutic index, respiratory depression, tolerance, physical dependence, severe withdrawal (anxiety, tremors, seizures, delirium), and enzyme induction (CYP3A4, CYP2C9). Drug interactions: reduces efficacy of oral contraceptives, warfarin, and many other drugs. Barbiturates are Schedule II-IV controlled substances.

Bedaquiline

– A diarylquinoline antibiotic used in the treatment of multidrug-resistant tuberculosis. Bedaquiline inhibits the mycobacterial ATP synthase, disrupting energy production. It is indicated for pulmonary multidrug-resistant tuberculosis in combination with other agents. The drug is administered orally three times weekly for the first 2 weeks, then twice weekly. Common adverse effects include QT prolongation, hepatotoxicity, and arthralgia. Bedaquiline is metabolized by CYP3A4 and has significant drug interactions, particularly as a CYP3A4 substrate, and requires ECG monitoring for QT interval prolongation during therapy.

Benzodiazepines

– A class of drugs that enhance the effect of the inhibitory neurotransmitter gamma-aminobutyric acid at the GABA-A receptor, increasing the frequency of chloride channel opening and producing anxiolytic, sedative, anticonvulsant, and muscle relaxant effects. They bind to an allosteric site on the GABA-A receptor, distinct from the GABA binding site, and potentiate the effect of GABA when it binds. Common benzodiazepines include diazepam, lorazepam, alprazolam, clonazepam, midamidazolam. They differ in potency, half-life, lipid solubility, and metabolic pathways, influencing their clinical applications for anxiety, insomnia, seizure disorders, and procedural sedation.

Beta-Lactam Antibiotics

– A large class of antimicrobial agents characterized by a beta-lactam ring in their molecular structure, which is essential for their bactericidal activity. They include penicillins, cephalosporins, carbapenems, and monobactams, all of which share the mechanism of inhibiting bacterial cell wall synthesis by binding to penicillin-binding proteins and preventing cross-linking of peptidoglycan chains. Beta-lactams are bactericidal, killing bacteria by weakening the cell wwall, leading to osmotic lysis and cell death; they are the most widely used class of antibiotics globally, with spectrum determined by their affinity for specific PBPs and their ability to penetrate the outer membrane of gram-negative organisms.

Beta-lactamase inhibitor combinations

– Combinations of beta-lactam antibiotics with beta-lactamase inhibitors that extend activity against beta-lactamase producing organisms. The major combinations include amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam, cefoperazone-sulbactam, and ceftolozane-tazobactam. Clavulanate, sulbactam, and tazobactam are beta-lactamase inhibitors that bind irreversibly to beta-lactamase enzymes, preventing degradation of the partner beta-lactam. Clavulanate has the strongest beta-lactactamase inhibition; tazobactam is the most potent beta-lactamase inhibitor in clinical use, while avibactam is a newer non-beta-lactam inhibitor that effectively restores activity against class A and class C beta-lactamases.

Biguanide

– A class of oral antidiabetic drugs derived from guanidine, with metformin being the only biguanide in current clinical use (phenformin and buformin were withdrawn due to lactic acidosis risk). Metformin is the first-line pharmacotherapy for type 2 diabetes mellitus. Mechanism: activates AMP-activated protein kinase, reducing hepatic gluconeogenesis and glycogenolysis; improves peripheral insulin sensitivity (increases glucose uptake in skeletal muscle); and reduces intestinal glucose absorption. Metformin does not stimulate insulin secretion and therefore does not cause hypoglycemia as monotherapy. Effects: lowers HbA1c by 1-1.5%, weight neutral or modest weight loss, and improves lipid profile. Adverse effects: GI intolerance (diarrhea, nausea, metallic taste, 20-30%), vitamin B12 deficiency with chronic use, and lactic acidosis (rare, 3-10 per 100,000 patient-years, primarily in patients with contraindications). Contraindications: eGFR <30 mL/min, acute kidney injury, decompensated liver disease, and acute medical illness with hypoxia (sepsis, MI, shock). Extended-release formulations improve GI tolerability.

Bioavailability

– The fraction of an administered drug that reaches the systemic circulation in its unchanged active form. It is expressed as a percentage and is determined by the route of administration and the extent of first-pass metabolism. For intravenous administration, bioavailability is 100 percent by definition. For oral administration, bioavailability is typically less than 100 percent due to incomplete absorption and pre-systemic metabolism by gut wall enzymes and the liver. The fraction of an oral dose reaching systemic circulation can be increased by altering the formulation, using prodrugs, or co-administering with food or other drugs that affect absorption or first-pass metabolism.

Bioequivalence

– The property of two drug products (brand-name and generic) that have the same rate and extent of absorption when administered at the same molar dose under similar conditions, resulting in comparable bioavailability and therapeutic effect. The FDA requires generic drugs to demonstrate bioequivalence to the reference listed drug through pharmacokinetic studies showing that the 90% confidence interval of the ratio of generic-to-brand for AUC and Cmax falls within 80-125%. Bioequivalence ensures that generic drugs can be substituted for brand-name drugs without dose adjustment, providing the same clinical efficacy and safety profile at a lower cost.

Biotransformation

– The process by which drugs and other xenobiotics are chemically modified by metabolic enzymes, primarily in the liver, to facilitate elimination from the body. Biotransformation occurs in two phases. Phase I reactions (functionalization): oxidation, reduction, and hydrolysis catalyzed primarily by cytochrome P450 enzymes (CYP1A2, CYP2C9, CYP2C19, CYP2D6, CYP3A4), introducing or exposing functional groups (e.g., -OH, -NH2, -COOH). Phase II reactions (conjugation): glucuronidation (UGT), sulfation (SULT), acetylation (NAT), methylation, and conjugation with glutathione (GST), attaching endogenous hydrophilic molecules to the drug or Phase I metabolite, increasing water solubility and promoting renal or biliary excretion. Biotransformation generally produces inactive, water-soluble metabolites, but some drugs undergo bioactivation (prodrugs) or produce toxic metabolites that contribute to drug toxicity.

Blood-Brain Barrier

– The blood-brain barrier is a highly selective semipermeable membrane that separates the circulating blood from the brain extracellular fluid, consisting of endothelial cells connected by tight junctions, astrocyte foot processes, and a basement membrane. The BBB restricts the passage of most molecules from the blood into the brain, protecting the central nervous system from toxins and pathogens while maintaining a stable environment for neural function. Only small, lipophilic molecules and gasescan cross the BBB via passive diffusion or active transport; the BBB is disrupted in pathological conditions such as meningitis, ischemia, and multiple sclerosis, affecting drug distribution to the brain.

Caffeine

– A naturally occurring xanthine alkaloid and central nervous system stimulant found in coffee, tea, chocolate, and many soft drinks and energy beverages. Mechanism: adenosine receptor antagonist (primarily A1 and A2A subtypes), increasing neuronal activity and promoting catecholamine release. Effects:

increased alertness, reduced fatigue, improved cognitive performance, diuresis, and bronchodilation. Half-life: 3–7 hours in adults, prolonged in pregnancy, liver disease, and with oral contraceptives. Chronic use produces tolerance; abrupt cessation causes withdrawal syndrome (headache, fatigue, irritability, dysphoria) lasting 2–9 days. Acute toxicity (>500 mg): anxiety, insomnia, tremor, tachycardia; severe toxicity (>5 g): seizures, arrhythmias, death.

Carbamazepine

– An anticonvulsant and mood-stabilizing drug used primarily for focal seizures, generalized tonic-clonic seizures, and trigeminal neuralgia. Mechanism: use-dependent blockade of voltage-gated sodium channels, stabilizing neuronal membranes and inhibiting repetitive firing. Also used as a mood stabilizer in bipolar disorder. Notable: potent CYP3A4 inducer (reduces levels of oral contraceptives, warfarin, many other drugs). Side effects: dizziness, ataxia, diplopia, hyponatremia (SIADH-like), agranulocytosis (rare but serious), rash including Stevens-Johnson syndrome (higher risk in Asian populations with HLA-B*1502 allele). Monitoring: CBC, LFTs, and serum levels (therapeutic 4–12 mcg/mL). Dose-dependent nonlinear kinetics require slow titration.

Carbapenem-resistant organisms

– Bacteria that are resistant to carbapenem antibiotics, typically through production of carbapenemases. The most important carbapenemases include KPC (Klebsiella pneumoniae carbapenemase), NDM (New Delhi metallo-beta-lactamase), and OXA-48-like enzymes. CRO are a major public health threat due to limited treatment options. Infections are associated with high mortality rates. Treatment options include ceftazidime-avibactam, meropenem-vaborbactam, and colistin-based combinations. Infection prevention and antimicrobial stewardship are critical for controlling the spread of carbapenem-resistant organisms in healthcare settings.

Carbolic Acid (Phenol)

– An aromatic organic compound (C_6H_5OH) with antiseptic, disinfectant, and caustic properties. Historically significant as the first surgical antiseptic introduced by Joseph Lister in 1867, dramatically reducing post-operative infection and mortality. Mechanism: denatures bacterial proteins and disrupts cell membranes. Phenol is bactericidal, fungicidal, and virucidal but is toxic to human tissues at high concentrations causing chemical burns and systemic toxicity (neurologic depression, cardiovascular collapse). Modern medical uses: phenol-based chemical peels (dermatology), nerve ablation for chronic pain (neurolysis), and as a preservative in some injectable medications. Safer, less toxic antiseptics (chlorhexidine, povidone-iodine) have largely replaced it for routine disinfection.

Cardioselective Beta-Blockers

– Cardioselective beta-blockers selectively block beta-1 adrenergic receptors, which are predominant in the heart, while having minimal effects on beta-2 receptors in the bronchial smooth muscle and peripheral vasculature at therapeutic doses. Common cardioselective beta-blockers include metoprolol, atenolol, bisoprolol, and nebivolol. At low to moderate doses, these agents produce cardiac effects including decreased heart rate, reduced myocardial contractility, and decreased cardiac output withouout significant bronchospasm, making them preferred in patients with asthma or COPD, though cardioselectivity is dose-dependent and may be lost at higher doses.

Caspofungin

– An echinocandin antifungal agent used in the treatment of invasive candidiasis, esophageal candidiasis, and invasive aspergillosis (as salvage therapy). Caspofungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis. The drug is available only intravenously and has a half-life of approximately 9-11 hours, allowing for once-daily dosing. Common adverse effects include headache, nausea, and phlebitis at the infusion site. Fever and rash have been reported. Hepatotoxicity has been reported, and the drug has fewer drug interactions than azole antifungals as it is not a significant CYP inhibitor.

Ceftazidime-avibactam

– A combination of the third-generation cephalosporin ceftazidime with the non-beta-lactam beta-lactamase inhibitor avibactam. Avibactam inhibits class A, class C, and some class D beta-lactamases, including KPC carbapenemases. This combination has activity against many carbapenem-resistant Enterobacteriaceae and Pseudomonas aeruginosa. It is indicated for complicated intra-abdominal infections, complicated urinary tract infections, and hospital-acquired pneumonia. The combination is administered administered intravenously and requires dose adjustment based on renal function; the most common adverse effects include nausea, diarrhea, and headache.

Cephalosporin

– A class of beta-lactam antibiotics derived from the fungus Cephalosporium acremonium, with a mechanism identical to penicillins — inhibition of bacterial cell wall synthesis by binding to penicillin-binding proteins and inhibiting transpeptidase. Cephalosporins are classified into five generations based on their antimicrobial spectrum. First generation (cefazolin, cephalexin): good gram-positive coverage (Staph aureus, Strep pyogenes), limited gram-negative coverage. Second generation (cefuroxime, cefoxitin, cefotetan): expanded gram-negative coverage including H. influenzae and anaerobes (cefoxitin). Third generation (ceftriaxone, cefotaxime, ceftazidime): broad gram-negative coverage including Enterobacteriaceae; ceftazidime also covers Pseudomonas. Fourth generation (cefepime): broad coverage including Pseudomonas and gram-positive with stability against AmpC beta-lactamases. Fifth generation (ceftaroline): anti-MRSA activity plus standard cephalosporin coverage. Adverse effects: hypersensitivity reactions (cross-reactivity with penicillins 10% based on similar beta-lactam ring structure), GI disturbances, and bleeding (cefotetan, due to N-methylthiotetrazole side chain causing hypoprothrombinemia).

Chloramphenicol

– A broad-spectrum antibiotic with activity against gram-positive and gram-negative organisms, as well as Rickettsia. Chloramphenicol inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for serious infections when other agents are not appropriate, including bacterial meningitis and typhoid fever. Common adverse effects include bone marrow suppression, gray baby syndrome in neonates, and aplastic anemia. The drug is rarely used due to its toxicity profile. Although effective, its use is limited to life-threatening infections due to the availability of safer alternatives.

Cholinesterase Inhibitor

– A class of medications that inhibit acetylcholinesterase, the enzyme that breaks down acetylcholine in the synaptic cleft, thereby increasing cholinergic neurotransmission in the central nervous system and peripheral tissues. Centrally acting cholinesterase inhibitors: donepezil, rivastigmine, and galantamine — used for symptomatic treatment of Alzheimer disease and other dementias by enhancing central cholinergic activity, providing modest cognitive and functional benefit (typically 6-12 months of stabilization before decline resumes). Rivastigmine is also approved for Parkinson disease dementia. Peripherally acting: neostigmine, pyridostigmine — used for myasthenia gravis to increase neuromuscular junction acetylcholine levels; and physostigmine — used as an antidote for anticholinergic poisoning. Adverse effects: GI symptoms (nausea, vomiting, diarrhea, most common), bradycardia, increased gastric acid secretion, muscle cramps, and insomnia.

Ciprofloxacin

– A second-generation fluoroquinolone with excellent gram-negative coverage, including Pseudomonas aeruginosa. Ciprofloxacin inhibits bacterial DNA gyrase and topoisomerase IV. It is indicated for urinary tract infections, respiratory infections, gastrointestinal infections, and bone and joint infections. The drug is available in oral and intravenous formulations. Common adverse effects include gastrointestinal upset, tendon rupture, and QT prolongation. Ciprofloxacin should not be used in childrechildren under 18 years of age due to the risk of arthropathy, except for specific indications such as anthrax exposure and complicated urinary tract infections associated with multidrug-resistant organisms.

Clarithromycin

– A macrolide antibiotic with activity against gram-positive organisms, atypical pathogens, and Helicobacter pylori. Clarithromycin inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for respiratory infections, skin infections, and H. pylori eradication. The drug is metabolized by CYP3A4 and has numerous drug interactions. Common adverse effects include gastrointestinal upset, taste disturbance, and QT prolongation. Clarithromycin is available in oral and-intravenous formulations; it is commonly used in combination therapy for H. pylori eradication as part of a triple regimen with amoxicillin and a proton pump inhibitor.

Clearance

– The volume of plasma from which a drug is completely removed per unit time, typically expressed in milliliters per minute or liters per hour. Total clearance is the sum of all clearance mechanisms, primarily hepatic and renal clearance. Hepatic clearance depends on hepatic blood flow, enzyme activity, and protein binding, with the hepatic extraction ratio determining the contribution of blood flow versus enzyme activity to clearance. High-extraction drugs like lidocaine have clearanclearance that is highly dependent on hepatic blood flow, while low-extraction drugs like warfarin have clearance that is primarily determined by intrinsic enzyme activity and protein binding.

Clindamycin

– A lincosamide antibiotic with activity against gram-positive organisms and anaerobes. Clindamycin inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for skin infections, intra-abdominal infections, dental infections, and bone infections. The drug inhibits bacterial toxin production and is used for toxic shock syndrome and necrotizing fasciitis. Common adverse effects include diarrhea and Clostridioides difficile colitis. Clindamycin achieves excellent bbone penetration and is used extensively in dental and orthopaedic infections, including chronic osteomyelitis.

Clinical Trial Phases

– The stages of drug development: Phase I (safety, dosing, pharmacokinetics in 20-100 healthy volunteers); Phase II (efficacy and side effects in 100-300 patients with the target condition); Phase III (large-scale efficacy and safety comparison with standard treatment in 1,000-3,000 patients); Phase IV (post-marketing surveillance for long-term effects and rare adverse events). Each phase has specific endpoints and success criteria before advancing to the next phase.

Clostridioides difficile infection

– An infectious diarrhea caused by the bacterium Clostridioides difficile, typically occurring after disruption of normal gastrointestinal flora by antibiotics. C. difficile produces toxins A and B that cause colonic inflammation and diarrhea. Risk factors include antibiotic exposure, proton pump inhibitor use, and advanced age. Treatment options include vancomycin, fidaxomicin, and metronidazole. Recurrence occurs in approximately 25% of cases. Fecal microbiota transplantation is highly effectivefor recurrent C. difficile infection, with cure rates exceeding 90%.

Competitive Antagonist

– A competitive antagonist is a drug that binds to the same receptor site as an agonist and competes for binding, reducing the effect of the agonist without producing an effect of its own. The antagonism is surmountable by increasing the agonist concentration, which shifts the dose-response curve of the agonist to the right without reducing the maximal response. This means that with enough agonist, the full effect can still be achieved. The degree of rightward shift depends on the concentration anand the concentration of the antagonist; the relationship is quantitated using the Schild regression, which yields a pA2 value representing the concentration of antagonist required to produce a twofold rightward shift of the agonist dose-response curve.

Controlled Substance

– A drug or chemical whose manufacture, possession, and use are regulated by government authorities due to its potential for abuse, dependence, or harm. In the United States, the Controlled Substances Act (CSA) classifies controlled substances into five schedules (I-V) based on their accepted medical use, abuse potential, and dependence liability. Schedule I: high abuse potential, no accepted medical use (heroin, LSD, marijuana federally). Schedule II: high abuse potential, severe dependence liability, accepted medical use with restrictions (morphine, oxycodone, fentanyl, cocaine, amphetamine). Schedule III: moderate to low abuse potential, moderate dependence liability (codeine-containing products, ketamine, anabolic steroids). Schedule IV: low abuse potential, limited dependence liability (benzodiazepines, zolpidem, tramadol). Schedule V: lowest abuse potential, limited dependence liability (codeine-containing antitussives, pregabalin). Healthcare providers must register with the DEA to prescribe controlled substances and comply with state and federal regulations.

Corticosteroids Mechanism

– Corticosteroids exert their anti-inflammatory and immunosuppressive effects primarily by binding to intracellular glucocorticoid receptors, which then translocate to the nucleus and modulate gene transcription. They transactivate anti-inflammatory genes including those encoding lipocortin-1, which inhibits phospholipase A2 and reduces arachidonic acid release, and they transrepress pro-inflammatory genes by inhibiting transcription factors including NF-kappaB and AP-1. This reduces the productioproduction of prostaglandins, leukotrienes, and other inflammatory mediators, which underlies their therapeutic efficacy in inflammatory conditions, autoimmune diseases, and allergic reactions.

COX Inhibitor

– A class of drugs that inhibit cyclooxygenase enzymes (COX-1 and COX-2), which catalyze the conversion of arachidonic acid to prostaglandins and thromboxanes. COX-1 is constitutively expressed in most tissues, producing prostaglandins that protect the gastric mucosa, maintain renal blood flow, and support platelet aggregation. COX-2 is induced during inflammation, producing prostaglandins that mediate pain, fever, and inflammation. Non-selective COX inhibitors (aspirin, ibuprofen, naproxen, indomethacin, diclofenac) inhibit both isoforms, providing analgesic, anti-inflammatory, and antipyretic effects but causing GI toxicity (gastritis, ulcers, bleeding) and platelet inhibition. Selective COX-2 inhibitors (celecoxib) provide anti-inflammatory effects with reduced GI toxicity but do not inhibit platelet aggregation and are associated with increased cardiovascular risk (MI, stroke). The COX hypothesis explains the therapeutic and adverse effects of NSAIDs.

CYP2C19 Polymorphism

– CYP2C19 is a cytochrome P450 enzyme involved in the metabolism of approximately 10 percent of drugs, including the antiplatelet agent clopidogrel, proton pump inhibitors, some antidepressants, and the antiepileptic drug phenytoin. Genetic polymorphisms result in poor metabolizer phenotypes in 2 to 5 percent of Caucasians and 15 to 20 percent of Asian populations. Poor metabolizers have reduced conversion of clopidogrel, a prodrug requiring CYP2C19-mediated activation, to its active metabolite, lleading to reduced antiplatelet efficacy and increased risk of stent thrombosis in poor metabolisers; alternative antiplatelet agents such as ticagrelor or prasugrel, which are not dependent on CYP2C19 activation, are preferred in such patients.

CYP2D6 Polymorphism

– CYP2D6 is a cytochrome P450 enzyme responsible for the metabolism of approximately 25 percent of commonly prescribed drugs, including codeine, tamoxifen, many antidepressants, and antipsychotics. Genetic polymorphisms in the CYP2D6 gene result in highly variable enzyme activity among individuals, classified into four phenotype groups: poor metabolizers with two non-functional alleles and absent enzyme activity; intermediate metabolizers with reduced activity; extensive metabolizers with normal aactivity, and ultrarapid metabolisers with increased enzyme activity due to gene duplication. This phenotypic variability has significant clinical implications for drugs metabolized by CYP2D6: codeine prodrug activation to morphine is reduced in poor metabolisers and enhanced in ultrarapid metabolisers, who may experience opioid toxicity at standard doses; tamoxifen efficacy is reduced in poor metabolisers; and antidepressant dosing may require individualisation based on CYP2D6 phenotype.

Cytochrome P450

– The cytochrome P450 system is a superfamily of heme-containing monooxygenase enzymes primarily located in the endoplasmic reticulum of hepatocytes, responsible for Phase I drug metabolism through oxidation, reduction, and hydrolysis reactions. The most important CYP isoforms in drug metabolism include CYP3A4, which metabolizes approximately 50 percent of all drugs; CYP2D6, responsible for approximately 25 percent of drug metabolism; CYP2C9, CYP2C19, CYP1A2, and CYP2E1. CYP enzymes can be inducedinduced or inhibited by various drugs and xenobiotics, leading to clinically significant drug interactions; enzyme induction occurs over days to weeks through increased gene transcription, while enzyme inhibition occurs rapidly through direct competition for the active site.

Dalbavancin

– A long-acting lipoglycopeptide antibiotic with an extremely long half-life allowing for once-weekly dosing. Dalbavancin inhibits cell wall synthesis and has activity against gram-positive organisms including MRSA. It is indicated for acute bacterial skin infections. The drug is administered intravenously with a loading dose followed by a single maintenance dose. The half-life is approximately 346 hours, allowing for extended dosing intervals. Common adverse effects include nausea, headache, and infusion site reactions, though the long half-life compensates for these minor tolerability concerns.

Dapsone

– A sulfone antibiotic with both antimicrobial and anti-inflammatory properties. Mechanism: inhibits bacterial folate synthesis (competitive inhibitor of dihydropteroate synthase). Used primarily for leprosy (Hansen disease) as part of multi-drug therapy, and for dermatologic conditions including dermatitis herpetiformis, erythema elevatum diutinum, and acne vulgaris. Also used for Pneumocystis jirovecii prophylaxis in HIV (alternative to TMP-SMX). Notable side effect: dose-dependent hemolysis (especially in G6PD deficiency), methemoglobinemia, agranulocytosis, and hypersensitivity syndrome (dapsone syndrome–fever, rash, eosinophilia, hepatitis). G6PD screening required before therapy.

Daptomycin

– A cyclic lipopeptide antibiotic that disrupts bacterial cell membrane integrity. Daptomycin has rapid bactericidal activity against gram-positive organisms including MRSA, VRE, and penicillin-resistant streptococci. It is indicated for complicated skin infections, bacteremia, and right-sided endocarditis. The drug is administered intravenously once daily. Common adverse effects include myopathy and rhabdomyolysis, particularly with concurrent statin therapy. Daptomycin is inactivated by pulmonarular surfactant and should not be used for the treatment of community-acquired pneumonia.

Daptomycin resistance

– Resistance to daptomycin can develop through mutations in genes encoding cell membrane phospholipid synthesis. The mechanism involves changes in the bacterial cell membrane that reduce daptomycin binding. Resistance is more common in VRE than MRSA. Resistance can emerge during therapy, particularly in endocarditis and bacteremia. Daptomycin resistance is typically accompanied by vancomycin resistance, a phenomenon known as daptomycin-vancomycin cross-resistance. Combination therapy may be requirred in selected cases and careful monitoring of creatine kinase levels is recommended during daptomycin therapy.

Darunavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for both treatment-naive and treatment-experienced HIV-1 patients, including those with resistance to other PIs. Darunavir inhibits the HIV aspartyl protease with high potency and has a high genetic barrier to resistance, requiring multiple mutations for significant resistance to develop. The drug must be co-administered with ritonavir or cobicistat as a pharmacokinetic booster to achieve therapeutic drug levels. Common adverseadverse effects include gastrointestinal disturbances, headache, and rash; hypertriglyceridaemia and hypercholesterolaemia are less common than with other protease inhibitors.

Decongestants

– Medications that reduce nasal congestion by constricting blood vessels in the nasal mucosa, decreasing swelling and mucus production. Systemic decongestants: pseudoephedrine and phenylephrine (oral), which stimulate alpha-adrenergic receptors causing vasoconstriction. Topical decongestants: oxymetazoline, xylometazoline, naphazoline, phenylephrine (nasal sprays)–more potent and rapid-acting but risk of rhinitis medicamentosa (rebound congestion) with use beyond 3–5 days. Indications: nasal congestion from allergic rhinitis, common cold, sinusitis. Contraindications: uncontrolled hypertension, severe coronary artery disease, narrow-angle glaucoma, and MAOI use.

Delamanid

– A nitroimidazole derivative used in the treatment of multidrug-resistant tuberculosis. Delamanid inhibits mycolic acid synthesis in Mycobacterium tuberculosis, disrupting the cell wall. It is indicated for pulmonary multidrug-resistant tuberculosis in combination with other agents. The drug is administered orally twice daily. Common adverse effects include headache, nausea, and QT prolongation. Delamanid is metabolized by plasma esterases and has minimal CYP450 interactions. The drug has demonstonstrated efficacy in MDR-TB treatment cohorts, with improved outcomes compared to background regimens alone.

Delavirdine

– A non-nucleoside reverse transcriptase inhibitor (NNRTI) that was previously used in combination antiretroviral therapy for HIV-1 infection but is now rarely used due to the availability of more effective and better-tolerated agents. Delavirdine binds to the HIV reverse transcriptase enzyme at a site distinct from the active site, causing a conformational change that inhibits enzymatic activity. The drug has relatively low potency compared to newer NNRTIs and requires dosing three times daily. CCommon adverse effects include rash, elevated transaminases, and nausea, and the drug has significant CYP3A4 inhibitory activity leading to numerous drug interactions, limiting its use in contemporary practice.

Diazepam

– A benzodiazepine used for its anxiolytic, sedative, muscle relaxant, anticonvulsant, and amnestic properties. Mechanism: enhances GABA-A receptor activity by increasing the frequency of chloride channel opening. Indications: anxiety disorders, alcohol withdrawal, status epilepticus (IV, first-line for benzodiazepine-responsive seizures), preoperative sedation, muscle spasm, and procedural sedation. Pharmacokinetics: highly lipid-soluble, rapid onset, long half-life (20–50 hours), metabolized by CYP2C19 and CYP3A4 to active metabolites (nordazepam, oxazepam, temazepam). Adverse effects: sedation, ataxia, tolerance, dependence, withdrawal syndrome, and paradoxic agitation (especially in children and older adults). Flumazenil is the reversal agent.

Didanosine

– A nucleoside reverse transcriptase inhibitor (NRTI) used in the treatment of HIV-1 infection, typically as part of combination antiretroviral therapy. Didanosine is an adenosine analogue that inhibits viral reverse transcriptase, leading to chain termination of viral DNA synthesis. The drug requires acidic pH for stability and must be administered with buffer tablets or antacids to prevent gastric acid degradation. Didanosine has a bioavailability of approximately 30-40% when taken with buffer, buffer, and is dosed twice daily; common adverse effects include peripheral neuropathy, pancreatitis, and lactic acidosis, with pancreatitis being a potentially fatal complication that requires immediate discontinuation.

Digoxin

– A cardiac glycoside that inhibits the sodium-potassium ATPase pump in cardiac myocytes, causing intracellular sodium accumulation, which reduces sodium-calcium exchange through the sodium-calcium exchanger, leading to increased intracellular calcium and enhanced myocardial contractility. This positive inotropic effect increases cardiac output in heart failure. Digoxin also increases vagal tone, slowing atrioventricular nodal conduction, making it useful for rate control in atrial fibrillation and atrial flutter; digoxin has a narrow therapeutic index, requiring therapeutic drug monitoring to avoid toxicity, which manifests as nausea, vomiting, visual disturbances (xanthopsia, halos), and arrhythmias including heart block and ventricular tachycardia.

DOACs

– Direct oral anticoagulants are a class of anticoagulant drugs that directly inhibit specific coagulation factors, providing an alternative to warfarin with more predictable pharmacokinetics and no routine monitoring. The factor Xa inhibitors include rivaroxaban, apixaban, and edoxaban, which directly and reversibly bind to the active site of factor Xa. The direct thrombin inhibitor dabigatran directly binds to and inhibits thrombin. DOACs have rapid onset of action, predictable pharmacokinetics,pharmacokinetics, fixed dosing, fewer drug and food interactions than warfarin, and no requirement for routine coagulation monitoring, though they are more expensive and specific reversal agents are limited.

Domperidone

– A dopamine D2 receptor antagonist used as a prokinetic agent and antiemetic. It increases gastrointestinal motility by enhancing acetylcholine release from the myenteric plexus and facilitates gastric emptying. Unlike metoclopramide, domperidone does not readily cross the blood-brain barrier, causing fewer central nervous system side effects (extrapyramidal symptoms). Indications: gastroparesis (diabetic, postsurgical), dyspepsia, and GERD. Also used for lactation induction (off-label) by stimulating prolactin release. Concerns: QT prolongation with risk of ventricular arrhythmias (requires ECG monitoring with prolonged use). Contraindicated in patients with preexisting QT prolongation or cardiac disease.

Dopamine Agonist

– A class of medications that stimulate dopamine receptors (D1-D5), mimicking the effects of endogenous dopamine. Used primarily in Parkinson disease: pramipexole and ropinirole (non-ergot dopamine agonists) are effective for early and advanced PD, delaying levodopa initiation and reducing motor fluctuations; rotigotine is available as a transdermal patch providing continuous delivery. Apomorphine is a potent, rapid-acting injectable dopamine agonist used for acute rescue in refractory motor fluctuations and off episodes. Adverse effects: impulse control disorders (pathological gambling, hypersexuality, compulsive shopping, binge eating — unique to dopamine agonists, occurring in 15-25%), nausea, orthostatic hypotension, hallucinations, and daytime somnolence (sudden sleep attacks). Ergot-derived dopamine agonists (bromocriptine, cabergoline) are rarely used due to risk of pulmonary and retroperitoneal fibrosis.

Doravirine

– A non-nucleoside reverse transcriptase inhibitor (NNRTI) used in combination antiretroviral therapy for HIV-1 infection. Doravirine binds to the HIV reverse transcriptase enzyme and inhibits its activity through a unique mechanism that maintains activity against strains resistant to first-generation NNRTIs. The drug has a favorable tolerability profile with minimal impact on lipid levels and body weight compared to other antiretrovirals. Common adverse effects include headache, nausea, and diarrdiarrhoea, and the drug has fewer drug interactions than other NNRTIs as it is not a significant inducer or inhibitor of CYP enzymes.

Dose-Response Curve

– A graphical representation of the relationship between the dose of a drug and the magnitude of its biological effect. The x-axis represents dose (typically on a logarithmic scale), and the y-axis represents the response (as percentage of maximum effect). The curve is sigmoidal in shape, characterized by: threshold (minimum dose producing a measurable effect), slope (steepness of the curve, indicating how rapidly the effect changes with dose), maximal efficacy (the plateau, Emax), and potency (the dose producing 50% of maximal effect, ED50). The therapeutic window is the range between the ED50 for the desired effect and the TD50 (toxic dose in 50% of patients). The therapeutic index (TI = TD50 / ED50) measures relative safety. Dose-response curves are essential for determining appropriate dosing regimens and comparing drug potency and efficacy.

Dose-Response Relationship

– The dose-response relationship describes the quantitative relationship between the dose of a drug and the magnitude of the pharmacological response it produces. This relationship is typically represented as a sigmoidal curve when response is plotted against the log of the dose. The curve is characterized by several key parameters: the maximal effect or efficacy represents the maximum response achievable; the EC50 or ED50 represents the dose producing 50 percent of the maximal response and indicaindicates the potency of the drug. The slope of the curve reflects the mechanism of drug action, with steeper slopes indicating a more direct relationship between dose and response, as seen with drugs that act on a single receptor type.

Doxycycline

– A tetracycline antibiotic with broad-spectrum activity against gram-positive and gram-negative bacteria, as well as atypical organisms including Mycoplasma, Chlamydophila, and Rickettsia. Doxycycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It is indicated for respiratory tract infections, skin infections, sexually transmitted infections, and malaria prophylaxis. The drug is well absorbed orally and can be taken with food. Common adverse effects include photophotosensitivity, and esophageal ulceration; it should be taken with a full glass of water and patients should remain upright for at least 30 minutes after dosing.

DPP-4 Inhibitors

– Dipeptidyl peptidase-4 inhibitors that prevent degradation of endogenous GLP-1 and GIP, enhancing incretin effects on insulin secretion. Examples include sitagliptin, saxagliptin, linagliptin, and alogliptin. They are well-tolerated with low hypoglycemia risk and weight neutrality. They have modest glucose-lowering effects. Saxagliptin was associated with increased heart failure hospitalizations in the SAVOR-TIMI 53 trial. Linagliptin does not require dose adjustment for renal impairment.

Drug Allergy

– An adverse immune-mediated reaction to a medication, representing a hypersensitivity response that is distinct from side effects, toxicities, and idiosyncratic reactions. Drug allergies can be IgE-mediated causing immediate reactions including urticaria, angioedema, and anaphylaxis within minutes to hours; T-cell mediated causing delayed reactions including contact dermatitis, drug rash with eosinophila and systemic symptoms, and Stevens-Johnson syndrome occurring days to weeks after exposure; diagnosis involves skin testing, in vitro assays for specific IgE, and oral challenge in selected cases, with management based on avoidance of the offending drug and use of alternative agents.

Drug Antagonism

– A drug that binds a receptor without activating it, blocking the effect of agonists. Competitive antagonists (e.g., naloxone vs opioids) can be overcome by increasing agonist concentration; noncompetitive antagonists (e.g., phenoxybenzamine vs norepinephrine) cannot. Physiological antagonism occurs when two drugs produce opposing effects through different receptors (e.g., histamine vs epinephrine on bronchial smooth muscle). Chemical antagonism involves direct chemical inactivation (e.g., protamine sulfate neutralises heparin).

Drug Classification Systems

– Drugs are classified using multiple complementary systems. The Anatomical Therapeutic Chemical (ATC) Classification System, maintained by the WHO, groups drugs into five levels: anatomical main group (e.g., cardiovascular system), therapeutic subgroup (e.g., antihypertensives), pharmacological subgroup (e.g., beta-blockers), chemical subgroup (e.g., selective beta-blockers), and chemical substance (e.g., metoprolol). The Therapeutic Classification groups drugs by their clinical use (e.g., antihypertensives, antidepressants, antibiotics). The Mechanism of Action Classification groups drugs by their molecular target (e.g., ACE inhibitors, calcium channel blockers, SSRIs). The Chemical Structure Classification groups drugs by their molecular composition (e.g., benzodiazepines, statins, penicillins). The Regulatory Classification includes prescription vs. over-the-counter status and controlled substance schedules. The Combination of these systems provides a comprehensive framework for understanding drug actions, interactions, and therapeutic applications.

Drug Dependence

– A state of physiological or psychological adaptation to a drug, characterized by a withdrawal syndrome upon discontinuation or dose reduction. Physical dependence: the body has adapted to the presence of the drug, and abrupt cessation produces a predictable withdrawal syndrome specific to the drug class. Opioid withdrawal: anxiety, mydriasis, lacrimation, rhinorrhea, piloerection, yawning, diarrhea, and muscle cramps. Alcohol withdrawal: anxiety, tremor, autonomic hyperactivity, seizures, and delirium tremens. Benzodiazepine withdrawal: anxiety, insomnia, tremor, and seizures. Psychological (behavioral) dependence: a compulsive pattern of drug use, often associated with craving and difficulty controlling use. Dependence is distinct from addiction (substance use disorder), which involves loss of control over use despite negative consequences.

Drug-Drug Interactions

– When one drug alters the pharmacokinetics or pharmacodynamics of another. Pharmacokinetic interactions affect absorption, distribution, metabolism, or excretion (e.g., CYP inhibitors increasing warfarin levels). Pharmacodynamic interactions involve additive, synergistic, or antagonistic effects at the site of action (e.g., two drugs prolonging QT interval). Clinically significant interactions can be identified through databases, clinical experience, and therapeutic drug monitoring.

Drug-Induced Liver Injury

– Liver damage caused by medications, herbal products, or dietary supplements, representing the most common cause of acute liver failure in the United States. DILI can be dose-dependent and predictable, as with acetaminophen hepatotoxicity from overdose, or idiosyncratic and unpredictable, as with isoniazid, nitrofurantoin, and amoxicillin-clavulanate.

The pathogenesis involves formation of reactive metabolites that bind to cellular proteins, triggering immune-mediatedimmune-mediated hepatocyte injury. DILI ranges from asymptomatic transaminase elevation to fulminant hepatic failure and death; early recognition and discontinuation of the offending agent are critical, as no specific therapy exists for most forms of DILI.

Drug Interactions

– Drug interactions occur when one drug affects the activity of another drug when both are administered together, potentially leading to therapeutic failure or toxicity. Pharmacokinetic interactions alter the absorption, distribution, metabolism, or excretion of a drug. Absorption interactions include chelation of tetracyclines with calcium and antacids reducing absorption, and altered GI motility affecting drug absorption. Distribution interactions include displacement from protein binding sites,sites, and drug interactions affecting protein binding. Pharmacodynamic interactions, where drugs have additive, synergistic, or antagonistic effects at the same receptor or physiological system, are equally important and can lead to enhanced therapeutic effects or unexpected toxicity.

Drug Metabolism Pharmacogenomics

– Pharmacogenomics of drug metabolism examines how genetic variations affect individual responses to drugs, enabling personalized medicine approaches to prescribing. The most clinically relevant pharmacogenes include CYP2D6, affecting metabolism of approximately 25 percent of drugs including codeine, tamoxifen, and many antidepressants; CYP2C19, affecting clopidogrel activation and proton pump inhibitor metabolism; CYP2C9, affecting warfarin metabolism; TPMT, affecting thiopurine metabolism; and Dand DPYD, affecting fluorouracil metabolism and risk of severe toxicity. Pre-emptive pharmacogenomic testing is increasingly being integrated into routine clinical practice to guide drug selection and dosing, with significant benefits in terms of efficacy and safety.

Drug Resistance

– The ability of microorganisms, cancer cells, or parasites to survive and proliferate despite exposure to drugs that were previously effective. In antimicrobial resistance, mechanisms include enzymatic inactivation of the drug such as beta-lactamase hydrolyzing penicillins, target modification such as PBP2a in MRSA conferring resistance to beta-lactams, efflux pumps actively pumping drugs out of the cell, decreased permeability through porin mutations, and metabolic bypapassing of the inhibited pathway, allowing the pathogen to survive drug exposure; understanding these diverse mechanisms is essential for the rational design of new therapeutic strategies and for guiding appropriate antimicrobial stewardship interventions.

Drug Tolerance

– A decreased pharmacological response to a drug after repeated administration, requiring dose escalation to achieve the same effect. Tolerance develops through multiple mechanisms: pharmacokinetic tolerance — increased drug metabolism through enzyme induction (e.g., alcohol, barbiturates); pharmacodynamic tolerance — receptor downregulation (decreased number of receptors, e.g., opioids, beta-agonists) or receptor desensitization (uncoupling of receptors from signal transduction, e.g., beta-agonists); functional tolerance — compensatory physiological adaptations; and behavioral tolerance — learned adaptations to drug effects. Tolerance is distinct from dependence and addiction. Cross-tolerance occurs when tolerance to one drug confers tolerance to another drug in the same class (e.g., morphine and heroin).

Echinocandins

– A class of antifungal agents that inhibit beta-1,3-glucan synthase, an enzyme essential for synthesis of beta-1,3-glucan, a major component of the fungal cell wall. This unique mechanism targets a structure absent in mammalian cells, resulting in favorable toxicity profiles. The available agents include caspofungin, micafungin, and anidulafungin, all administered intravenously. Echinocandins are fungicidal against Candida species and fungistatic against Aspergillus species. TheThe three available echinocandins have similar spectra of activity, though minor differences in pharmacokinetics and dosing schedules exist, and they are generally well tolerated with few drug interactions.

Efficacy

– The maximum therapeutic effect a drug or intervention can produce under ideal conditions (e.g., in a clinical trial). Efficacy is determined by the drug's mechanism of action and the receptor-effector system; it is measured by the Emax (maximal effect) on the dose-response curve. A drug with high efficacy produces a greater maximal response than one with low efficacy, regardless of dose. Efficacy is distinct from potency (the dose required to produce a given effect). Clinically, efficacy determines whether a drug can achieve the desired therapeutic outcome; a drug with insufficient efficacy will not be effective even at high doses. Not the same as effectiveness, which reflects real-world performance.

Enfuvirtide

– A fusion inhibitor used in combination antiretroviral therapy for HIV-1 infection in treatment-experienced patients with multidrug-resistant virus. Enfuvirtide is a synthetic peptide that mimics a region of the HIV-1 gp41 transmembrane protein, preventing the conformational change required for fusion of the viral envelope with the host cell membrane. The drug must be administered twice daily by subcutaneous injection, which is the primary limitation to its use. Common adverse effects include injinjection site reactions, and the drug is associated with an increased rate of lymph node enlargement and potential immune reconstitution syndromes when used as part of combination antiretroviral therapy.

Enterobacteriaceae carbapenem resistance

– Resistance to carbapenem antibiotics in Enterobacteriaceae, primarily through production of carbapenemases. The most important carbapenemases include KPC, NDM, and OXA-48-like enzymes. Carbapenem-resistant Enterobacteriaceae (CRE) are a major public health threat due to limited treatment options. Infections are associated with high mortality rates. Treatment options include ceftazidime-avibactam, meropenem-vaborbactam, and aminoglycosides. Infection prevention and antimicrobial stewardship are eessential for controlling the spread of carbapenem-resistant Enterobacteriaceae in healthcare settings.

Enzyme Induction

– A process by which a drug or xenobiotic increases the synthesis or activity of drug-metabolizing enzymes, primarily cytochrome P450 enzymes in the liver, accelerating the metabolism of itself (autoinduction) or other drugs. Enzyme induction typically occurs over days to weeks as it requires new protein synthesis. Important enzyme inducers: rifampin (potent CYP3A4, CYP2C9, CYP2C19 inducer), carbamazepine, phenytoin, phenobarbital, St. John's wort, and chronic alcohol consumption. Clinical consequences: reduced efficacy of co-administered drugs (e.g., rifampin reduces oral contraceptive efficacy, requiring alternative contraception), increased metabolism of warfarin (requiring higher doses), and reduced cyclosporine levels in transplant patients. discontinuing an inducer can lead to increased drug levels of co-administered drugs, requiring dose reduction.

Enzyme Inhibition

– A process by which a drug inhibits the activity of drug-metabolizing enzymes, leading to increased plasma concentrations of co-administered drugs that are metabolized by the same enzyme. Enzyme inhibition occurs rapidly (hours to days) as it does not require new protein synthesis. Important enzyme inhibitors and their substrates: ketoconazole and ritonavir (potent CYP3A4 inhibitors — increase levels of statins, benzodiazepines, calcium channel blockers); cimetidine (inhibits CYP1A2, CYP2D6, CYP3A4); fluoxetine and paroxetine (CYP2D6 inhibitors — increase levels of atomoxetine, TCAs, and tamoxifen); amiodarone (CYP2C9 inhibitor — increases warfarin levels); grapefruit juice (inhibits intestinal CYP3A4 — increases levels of felodipine, simvastatin, cyclosporine). Clinical consequences: increased drug toxicity (e.g., rhabdomyolysis with simvastatin when given with potent CYP3A4 inhibitors), requiring dose reduction or alternative therapy.

Epinephrin (Adrenaline)

– Alternate spelling of epinephrine (adrenaline). See Epinephrine (Adrenaline).

Epinephrine (Adrenaline)

– A catecholamine hormone and neurotransmitter produced by the adrenal medulla and sympathetic nerve terminals, acting on alpha-1, alpha-2, beta-1, beta-2, and beta-3 adrenergic receptors. Effects: alpha-1: vasoconstriction (increases blood pressure), mydriasis; beta-1: increased heart rate and contractility; beta-2: bronchodilation, vasodilation in skeletal muscle, glycogenolysis. Clinical uses: first-line treatment for anaphylaxis (0.3–0.5 mg IM, repeat every 5–15 minutes), cardiac arrest (1 mg IV every 3–5 minutes during ACLS), severe asthma exacerbation (subcutaneous or IV), septic shock (IV infusion), and local vasoconstriction (combined with local anesthetics to prolong duration and reduce bleeding). Side effects: hypertension, tachycardia, arrhythmias, tremor, anxiety, hyperglycemia. Autoinjectors (EpiPen) are prescribed for patients with severe allergies.

Eravacycline

– A fluorocycline antibiotic with activity against gram-positive and gram-negative organisms, including multidrug-resistant strains. Eravacycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against MRSA, VRE, ESBL-producing Enterobacteriaceae, and anaerobes. The drug is indicated for complicated intra-abdominal infections. Eravacycline is administered intravenously and achieves high concentrations in abdominal tissues. Common adverse effects includincluding nausea, vomiting, and infusion site reactions; eravacycline has a favourable safety profile with minimal phototoxicity compared to other tetracyclines.

Erythromycin

– The prototype macrolide antibiotic with activity against gram-positive organisms and some gram-negative organisms. Erythromycin inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for respiratory infections, skin infections, and as a prokinetic agent for gastroparesis. Common adverse effects include gastrointestinal upset, hepatotoxicity, and QT prolongation. Erythromycin is a potent inhibitor of CYP3A4 and has numerous drug interactions. The drug is nownow rarely used as a first-line agent due to the availability of better-tolerated alternatives with broader spectra of activity.

ESBL-producing organisms

– Bacteria that produce extended-spectrum beta-lactamases (ESBLs), enzymes that hydrolyze penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae, particularly Klebsiella pneumoniae and E. coli. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems. The prevalence of ESBL-producing organisms is increasing globally. Infections are associated with increased morbidity, mortality, and healthcare costs. Treatment typically requires carbapenemcarbapenems for serious infections, and antimicrobial stewardship is essential to preserve the utility of remaining effective agents.

Essential Medicines

– Drugs that satisfy the priority healthcare needs of a population, as defined by the WHO Model List of Essential Medicines. Selection criteria include efficacy, safety, comparative cost-effectiveness, and suitability for use in resource-limited settings. The list includes approximately 400 drugs covering major disease areas: analgesics, anesthetics, antibiotics, antiepileptics, antihypertensives, antiretrovirals, antimalarials, vaccines, and others. The concept guides national formulary development, procurement, and rational prescribing worldwide.

Etravirine

– A second-generation non-nucleoside reverse transcriptase inhibitor (NNRTI) used in combination antiretroviral therapy for HIV-1 infection, particularly in treatment-experienced patients with NNRTI resistance. Etravirine binds to the HIV reverse transcriptase enzyme and inhibits its activity through a unique mechanism that maintains activity against strains resistant to first-generation NNRTIs. The drug has a high genetic barrier to resistance due to its flexible structure that allows it to adaptadapt to changes in the reverse transcriptase enzyme structure, maintaining antiviral activity against a broad range of NNRTI-resistant variants.

Expectorants

– Medications that promote the clearance of mucus from the respiratory tract by increasing sputum volume and reducing its viscosity, facilitating productive cough. The most common expectorant is guaifenesin (glyceryl guaiacolate), which increases respiratory tract fluid secretion and reduces sputum adhesiveness. Guaifenesin is used for symptom relief in acute respiratory infections, chronic bronchitis, and COPD. Evidence for efficacy is moderate. Other expectorants: potassium iodide (rarely used due to side effects), ammonium chloride, and sodium citrate. Expectorants are distinguished from mucolytics (acetylcysteine, dornase alfa–break down the chemical structure of mucus) and antitussives (dextromethorphan, codeine–suppress the cough reflex). Adequate hydration is essential for expectorant efficacy.

Fecal microbiota transplantation

– The transfer of fecal material from a healthy donor to a patient with recurrent Clostridioides difficile infection. The procedure aims to restore normal gastrointestinal flora and eliminate C. difficile colonization. FMT is highly effective, with cure rates exceeding 90% for recurrent C. difficile infection. The procedure can be administered by colonoscopy, nasojejunal tube, or oral capsules. Donor screening includes testing for bloodborne pathogens and gastrointestinal pathogens. FMT has revolurevolutionised the management of recurrent C. difficile infection and is being investigated for other conditions associated with intestinal dysbiosis, including inflammatory bowel disease and metabolic syndrome.

Fidaxomicin

– A macrolide antibiotic with a narrow spectrum of activity against Clostridioides difficile. Fidaxomicin inhibits bacterial RNA polymerase. It is indicated for Clostridioides difficile colitis. The drug is administered orally twice daily for 10 days. Common adverse effects include nausea and vomiting. Fidaxomicin has minimal systemic absorption, acting locally in the gastrointestinal tract. The drug has lower rates of recurrence compared to vancomycin for C. difficile colitis. Fidaxomicin is not not recommended for use in patients with inflammatory bowel disease due to a possible increased risk of gastrointestinal bleeding.

First-Pass Effect

– The metabolism of a drug by the liver and intestinal wall before it reaches the systemic circulation after oral administration. When a drug is absorbed from the gastrointestinal tract, it travels through the portal venous system to the liver, where it may be extensively metabolized by cytochrome P450 enzymes before reaching the general circulation. This process significantly reduces the bioavailability of many drugs. The hephepatic extraction ratio quantifies this effect: drugs with high extraction ratios have clearance that depends primarily on hepatic blood flow, while low-extraction drugs have clearance that depends on intrinsic enzyme activity and protein binding; this distinction has important implications for drug dosing in patients with liver disease and for predicting drug interactions.

Fluoroquinolones

– Broad-spectrum bactericidal antibiotics that inhibit bacterial DNA gyrase and topoisomerase IV, enzymes essential for DNA replication, transcription, and repair. They are synthetic derivatives of nalidixic acid and include ciprofloxacin, levofloxacin, moxifloxacin, and others. Fluoroquinolones have excellent bioavailability, with oral dosing achieving serum concentrations comparable to intravenous administration. They distribute widely to tissues including the lungs, prostatprostate, and bone; fluoroquinolones are generally well tolerated but carry boxed warnings for tendonitis and tendon rupture, peripheral neuropathy, CNS effects, and exacerbation of myasthenia gravis, and their use is reserved for situations where alternative treatment options are limited.

Fosamprenavir

– A prodrug of amprenavir, used in combination antiretroviral therapy for HIV-1 infection. Fosamprenavir is rapidly hydrolyzed by intestinal alkaline phosphatase to amprenavir, which then inhibits the HIV aspartyl protease. The prodrug formulation improves the pharmacokinetic profile of amprenavir, allowing for twice-daily dosing with improved bioavailability. Fosamprenavir is available as tablets and oral suspension, with the suspension containing calcium carbonate and other buffering agents. ComCommon adverse effects include gastrointestinal disturbances, headache, and rash, and the drug is generally well tolerated with favourable lipid effects compared to other PIs; fosamprenavir is available as a once-daily or twice-daily formulation.

Fosfomycin

– A phosphonic acid derivative antibiotic that inhibits cell wall synthesis. Fosfomycin has broad-spectrum activity including gram-positive and gram-negative organisms. It is indicated for uncomplicated urinary tract infections and is available as a single oral dose. The drug is also available intravenously for serious infections. Common adverse effects include diarrhea and headache. Fosfomycin achieves high concentrations in urine. The drug maintains activity against many multidrug-resistant orgaorganisms, including ESBL-producing Enterobacteriaceae and carbapenem-resistant strains, making it a valuable reserve antibiotic for complicated urinary tract infections.

Fostemsavir

– A prodrug of temsavir, an attachment inhibitor that binds to HIV-1 gp120, preventing viral attachment to the CD4 receptor on human T cells. Fostemsavir is used in combination with other antiretrovirals for the treatment of multidrug-resistant HIV-1 infection in heavily treatment-experienced patients. The drug is administered orally twice daily, which is a significant advantage over injectable agents. Common adverse effects include nausea, diarrhea, headache, and rash. Elevated transaminases have been reported; fostemsavir represents a valuable option for patients with limited therapeutic options due to extensive drug resistance.

Full Agonist

– A full agonist is a drug that binds to a receptor and produces the maximum possible biological response that the receptor can generate, achieving 100 percent of the maximal effect. Full agonists have both high affinity for the receptor and high intrinsic activity, meaning they are very effective at activating the receptor once bound. The efficacy of a full agonist is inherent to the drug-receptor interaction and cannot be exceeded by increasing the dose. Examples include morphine at mu-opioid rereceptors, with full agonists producing maximal activation, and their effect is determined by both the drug concentration at the receptor site and the receptor density in the target tissue.

Fungal resistance mechanisms

– Mechanisms by which fungi develop resistance to antifungal agents. Azole resistance occurs through upregulation of efflux pumps (CDR1, CDR2, MDR1), mutations in the target enzyme ERG11, and overexpression of ERG11. Echinocandin resistance is primarily due to mutations in the FKS1 and FKS2 genes encoding beta-(1,3)-D-glucan synthase. Amphotericin B resistance is rare but can occur through alterations in ergosterol biosynthesis. Multidrug resistance is increasingly common in Candida glabrata, withwith azole resistance being the most clinically challenging, particularly in Candida glabrata and Candida auris, and susceptibility testing is increasingly important for guiding antifungal therapy.

Gamma-Globulin

– A fraction of plasma proteins containing antibodies (immunoglobulins). Historically the term referred to a group of serum proteins with the slowest electrophoretic mobility (gamma region on serum protein electrophoresis). Currently synonymous with immunoglobulin preparations used for passive immunisation and immunotherapy. Indications: (1) Replacement therapy for primary immunodeficiency (X-linked agammaglobulinaemia, common variable immunodeficiency); (2) Immunomodulation for autoimmune and inflammatory diseases (Kawasaki disease, Guillain-Barré syndrome, chronic inflammatory demyelinating polyneuropathy, ITP, myasthenia gravis, Stevens-Johnson syndrome). Administration: intravenous (IVIG) or subcutaneous (SCIG). Side effects: headache, fever, chills, myalgia, aseptic meningitis, thrombosis, anaphylaxis (rare, in IgA deficiency). Prepared by cold ethanol fractionation (Cohn method) from pooled human plasma from thousands of donors.

Ganglionic Blocker

– A class of drugs that block nicotinic acetylcholine receptors at autonomic ganglia (both sympathetic and parasympathetic), inhibiting neurotransmission in both divisions of the autonomic nervous system. Due to their non-selective blockade of all autonomic ganglia, they produce widespread and unpredictable effects including severe orthostatic hypotension, blurred vision (cycloplegia), dry mouth, constipation, urinary retention, and impotence. Prototype: hexamethonium. Clinical use: historically used for hypertensive emergencies before the development of safer agents. Currently used rarely, primarily for inducing controlled hypotension during surgery (trimethaphan). Ganglionic blockers are of more pharmacological than clinical importance, serving as tools for understanding autonomic physiology.

Gelfoam

– An absorbable hemostatic agent made from purified porcine gelatin. Used in surgical settings to control bleeding where conventional hemostasis is difficult (liver, bone, neurosurgery). Mechanism: provides a physical matrix that promotes platelet aggregation and clot formation. Absorbed within 4-6 weeks. Prepared as a sterile sponge that can be cut to size and applied dry or saturated with thrombin. Contraindicated: infected wounds, gross contamination, intravascular use (risk of embolism). Brand examples: Gelfoam (Pfizer), Surgifoam (Ethicon).

Gentamicin

– An aminoglycoside antibiotic produced by Micromonospora purpurea. Mechanism: binds 30S ribosomal subunit, inhibits protein synthesis by interfering with initiation complex and causing mRNA misreading. Bactericidal, concentration-dependent killing with a post-antibiotic effect. Spectrum: Gram-negative bacteria (E. coli, Klebsiella, Pseudomonas, Serratia, Enterobacter, Proteus, Acinetobacter), synergistic with beta-lactams for Gram-positive infections (enterococcal endocarditis, group B streptococcus in neonates). Resistance: aminoglycoside-modifying enzymes (acetylation, adenylation, phosphorylation), efflux pumps, ribosomal methylation. Clinical uses: (1) Neonatal sepsis (often with ampicillin); (2) Gram-negative bacteraemia and sepsis; (3) Infective endocarditis (enterococcal, viridans group streptococci–synergistic combination therapy); (4) Complicated Gram-negative urinary tract infections; (5) Plague (Yersinia pestis). Administration: once-daily dosing (5-7 mg/kg IV) for most indications; multiple daily dosing for enterococcal endocarditis and neonates. Monitoring: trough levels (<1-2 mg/L to avoid toxicity), peak levels (therapeutic 6-10 mg/L). Side effects: nephrotoxicity (5-25%, proximal tubular necrosis, usually reversible), ototoxicity (cochlear and vestibular, often irreversible), neuromuscular blockade (rare).

GLP-1 Receptor Agonists

– Glucagon-like peptide-1 receptor agonists are a class of injectable antidiabetic agents that mimic the effects of the incretin hormone GLP-1, which is secreted by intestinal L-cells in response to food intake. GLP-1 stimulates glucose-dependent insulin secretion, suppresses glucagon secretion, delays gastric emptying, and promotes satiety through central nervous system effects. Available agents include exenatide, liraglutide, dulaglutide, semaglutide, and tirzepatide, which is a dual GIP/GLP-1 aagonist. GLP-1 receptor agonists improve glycaemic control with a low risk of hypoglycaemia and promote weight loss, making them particularly valuable in overweight or obese patients with type 2 diabetes; gastrointestinal side effects including nausea, vomiting, and diarrhea are common but often diminish with gradual dose titration.

Glycerol (Glycerine)

– A triol compound (CH_2OH-CHOH-CH_2OH), a colourless, odourless, viscous liquid with a sweet taste. Used in medicine and pharmaceuticals as a: humectant (retains moisture in topical preparations), laxative (glycerol suppositories for constipation), sweetener (in syrups, sugar-free alternatives), solvent for drugs, and osmotherapy agent (intravenous glycerol reduces intracranial pressure). Glycerol is also a by-product of soap manufacturing and biodiesel production (transesterification).

GnRH Agonist

– Gonadotropin-releasing hormone agonists are a class of medications that initially stimulate (flare effect, 1-3 weeks) and then suppress pituitary gonadotropin (LH and FSH) secretion through receptor downregulation, leading to profound suppression of sex steroid production (testosterone in men, estradiol in women). Used in: prostate cancer — medical castration (leuprolide, goserelin, triptorelin); breast cancer — ovarian suppression in premenopausal women; endometriosis and uterine fibroids — suppression of ovarian estrogen; central precocious puberty — halting pubertal progression; and assisted reproduction — controlled ovarian hyperstimulation protocols. The initial testosterone flare in prostate cancer can cause disease exacerbation (bone pain, spinal cord compression, urinary obstruction), mitigated by adding an antiandrogen (bicalutamide) for 2-4 weeks at initiation.

GnRH Antagonist

– A class of medications that competitively block GnRH receptors on pituitary gonadotrophs, producing immediate, reversible suppression of LH and FSH secretion without the initial flare effect seen with GnRH agonists. Used in: prostate cancer — degarelix (monthly subcutaneous injection) and relugolix (oral daily) provide rapid testosterone suppression without flare; endometriosis and uterine fibroids — elagolix (oral); assisted reproduction —

cetrorelix and ganirelix prevent premature LH surge during controlled ovarian stimulation. Advantage over GnRH agonists: no initial flare, faster onset of suppression, and reversibility. Adverse effects: hot flashes, injection site reactions (degarelix), and osteoporosis with long-term use.

Half-Life

– Elimination half-life is the time required for the plasma concentration of a drug to decrease by 50 percent after reaching equilibrium. It is calculated from the elimination rate constant using the equation t1/2 equals 0.693 divided by the elimination rate constant. Half-life depends on both the volume of distribution and the clearance of the drug, with the relationship t1/2 equals 0.693 times Vd divided by clearance. Drugs with large volumes of distribution or low clearance have long half-liveslives; understanding half-life is essential for determining appropriate dosing intervals, time to reach steady state, and duration of drug action after discontinuation.

Heparin

– A naturally occurring glycosaminoglycan anticoagulant that enhances the activity of antithrombin, accelerating the inactivation of thrombin and factor Xa. Unfractionated heparin is a mixture of molecules of varying sizes administered intravenously, with rapid onset of action and short half-life of approximately one hour. Low-molecular-weight heparins including enoxaparin, dalteparin, and tinzaparin are produced by depolymerization of unfractionated heparin, providing more predictable predictable pharmacokinetics, longer half-life allowing once- or twice-daily subcutaneous dosing, and reduced risk of heparin-induced thrombocytopenia compared to unfractionated heparin, though they cannot be fully reversed with protamine and accumulate in renal impairment.

Histamine

– A biogenic amine (imidazolylethylamine) synthesised from L-histidine by histidine decarboxylase, stored in basophils, mast cells, and platelets, and released during allergic and inflammatory responses. Histamine acts via four G-protein-coupled receptors: (1) H1 receptor: bronchoconstriction (asthma), vasodilation (hypotension, flushing), increased vascular permeability (urticaria, angioedema), pruritus, pain, mucus secretion, and CNS effects (wakefulness). H1-antihistamines (cetirizine, loratadine, fexofenadine, chlorphenamine, promethazine, diphenhydramine) are used for allergic rhinitis, urticaria, pruritus, insomnia, motion sickness. (2) H2 receptor: gastric acid secretion (parietal cells). H2-antagonists (cimetidine, ranitidine, famotidine) are used for peptic ulcer disease, GORD, Zollinger-Ellison syndrome. (3) H3 receptor: presynaptic auto- and heteroreceptor in the CNS, inhibits histamine release and synthesis; inverse agonists/antagonists (pitolisant) are used for narcolepsy (increases wakefulness). (4) H4 receptor: expressed on immune cells (eosinophils, neutrophils, mast cells, dendritic cells), involved in chemotaxis and inflammation; H4 antagonists are under investigation for pruritus and asthma. Degradation: histamine N-methyltransferase and diamine oxidase. Histamine is also involved in gastric acid secretion, neurotransmission (wakefulness, appetite, learning), and smooth muscle contraction. Histamine poisoning (scombroid poisoning): from spoiled fish (tuna, mackerel) containing high levels of histamine due to bacterial decarboxylation, causes flushing, headache, urticaria, diarrhea within minutes, treated with H1-antihistamines.

Hydrocortisone

– The pharmaceutical name for cortisol, the primary endogenous glucocorticoid hormone produced by the zona fasciculata of the adrenal cortex. Hydrocortisone has both glucocorticoid and mineralocorticoid activity (ratio of approximately 1:1 at therapeutic doses). Indications: (1) Replacement therapy: adrenal insufficiency–primary (Addison disease, autoimmune, TB, bilateral adrenalectomy) and secondary (pituitary/hypothalamic failure). Standard dosing: 15-25 mg/day in divided doses (morning 10-20 mg, afternoon 5-10 mg). Stress dosing: increase dose 2-3 fold during illness or surgery to prevent adrenal crisis. (2) Acute severe inflammatory conditions: anaphylaxis (300 mg IV, then 200 mg IV 6 hourly), severe asthma exacerbation, status epilepticus, Stevens-Johnson syndrome, severe allergic reactions. (3) Topical: eczema, dermatitis, psoriasis (various potencies, from 0.5% to 2.5% cream/ointment). (4) Intra-articular injection: inflammatory arthritis, gout. Side effects: Cushing syndrome with prolonged high-dose use (central obesity, moon face, buffalo hump, striae, osteoporosis, hyperglycaemia, hypertension, immunosuppression, myopathy, cataracts, growth suppression in children).

Hypnotic

– A medication that induces, maintains, or improves sleep, used for insomnia. Major classes: benzodiazepine receptor agonists (Z-drugs: zolpidem, zaleplon, eszopiclone) — selective for GABA-A receptors containing alpha-1 subunits, promoting sleep onset with fewer next-day effects than benzodiazepines; melatonin receptor agonists (ramelteon, prolonged-release melatonin) — activate MT1 and MT2 receptors in the suprachiasmatic nucleus, regulating the sleep-wake cycle, no abuse potential; orexin receptor antagonists (suvorexant, daridorexant) — block orexin signaling that promotes wakefulness, used for sleep maintenance insomnia; sedating antidepressants (trazodone, doxepin, mirtazapine) — used in low doses for insomnia, especially with comorbid depression; and antihistamines (diphenhydramine, doxylamine) — sedating H1 antagonists available OTC, limited efficacy with chronic use due to tolerance.

Ibalizumab

– A humanized IgG4 monoclonal antibody that binds to domain 2 of the CD4 receptor on human T cells, preventing post-attachment steps required for HIV-1 infection. Ibalizumab is used in combination with other antiretrovirals for the treatment of multidrug-resistant HIV-1 infection in heavily treatment-experienced patients. The drug is administered intravenously every 2 weeks after an initial loading dose, which is a significant advantage over enfuvirtide's twice-daily injections. Common adverse effeff, and the drug has a favourable safety profile with no significant organ toxicity, though it may cause immune-related adverse events such as rash and elevated liver enzymes.

Imipenem-cilastatin

– A combination of the carbapenem imipenem with the dehydropeptidase inhibitor cilastatin. Imipenem is the broadest-spectrum beta-lactam antibiotic, with activity against most aerobic and anaerobic bacteria, including Pseudomonas aeruginosa and Bacteroides fragilis. Cilastatin inhibits renal dehydropeptidase, preventing degradation of imipenem and allowing for higher drug levels. The combination is administered intravenously every 6 hours, with dose adjustment required for renal impairment. CommonCommon adverse effects include nausea, vomiting, diarrhea, rash, and infusion-related reactions; imipenem has the greatest potential for seizure induction among carbapenems, particularly in patients with renal impairment or CNS disorders, and dose adjustment is essential.

Indinavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection. Indinavir inhibits the HIV aspartyl protease, preventing the cleavage of viral polyproteins and resulting in the production of immature, non-infectious viral particles. The drug requires dosing every 8 hours on an empty stomach with water, and patients must maintain adequate hydration to prevent nephrolithiasis, which occurs in approximately 10-15% of patients. Nephrolithiasis presents as flank pain, hematuhematura, crystalluria, and acute kidney injury; other adverse effects include gastrointestinal intolerance, hyperbilirubinaemia, and metabolic disturbances including hypertriglyceridaemia and diabetes. Indinavir is rarely used today due to its inconvenient dosing schedule and the availability of more convenient protease inhibitors.

Inverse Agonist

– An inverse agonist is a drug that binds to a receptor and produces a response opposite to that produced by an agonist, reducing the constitutive activity of the receptor. Receptor activation in the absence of an agonist, which occurs with some receptors that have spontaneous activity. Inverse agonists reduce this baseline activity, producing a negative response rather than simply blocking agonist effects as competitive antagonists do. Examples include certain antiantipsychotics, such as pimozide and haloperidol, which show inverse agonist activity at D2 receptors and reduce the baseline activity of dopamine signalling, a mechanism that may contribute to both therapeutic efficacy and the development of extrapyramidal side effects.

Isavuconazole

– A triazole antifungal agent used in the treatment of invasive aspergillosis and mucormycosis. Isavuconazole inhibits fungal lanosterol 14-alpha demethylase, disrupting ergosterol synthesis and fungal cell membrane integrity. A key advantage is its broad-spectrum activity covering Aspergillus, Candida, Cryptococcus, Mucorales, and endemic fungi. The drug has excellent oral bioavailability of approximately 98% and is available as both intravenous and oral formulations. Unlike voriconazole, isavucoisavucoazole has minimal effect on CYP3A4 and does not prolong the QT interval to the same degree as voriconazole, giving it a more favourable drug interaction profile.

Lactulose

– A synthetic disaccharide (galactose + fructose) used as an osmotic laxative and for the treatment of hepatic encephalopathy. As a laxative: it is not absorbed in the small intestine and reaches the colon intact, where it is fermented by colonic bacteria into short-chain fatty acids and gases, creating an osmotic gradient that draws water into the bowel lumen, increasing stool volume and stimulating bowel movement. Onset: 24-48 hours. Dose: 15-30 mL (10-20 g) once or twice daily. For hepatic encephalopathy: lactulose acidifies the colonic pH, favouring the conversion of ammonia ($NH3$, absorbable) to ammonium ($NH4+$, non-absorbable), reducing systemic ammonia absorption. Dose: 30-50 mL three times daily, titrated to produce 2-3 soft stools per day. Side effects: bloating, flatulence, abdominal cramps, diarrhea (excessive dose), electrolyte disturbances (hypernatraemia) with prolonged high doses. Contraindicated in galactosaemia (contains galactose).

Lenacapavir

– A first-in-class capsid inhibitor used in combination antiretroviral therapy for HIV-1 infection. Lenacapavir targets the HIV-1 capsid protein, interfering with multiple essential capsid-dependent steps in the viral lifecycle including nuclear import, viral DNA integration, capsid assembly, and virus release. The drug is administered as a subcutaneous injection every 6 months after oral loading, making it the longest-acting antiretroviral available. This extended dosing interval dramatically impimproves adherence and quality of life for patients with multidrug-resistant HIV-1 infection, though injection site reactions, including nodules and induration, are common and lenacapavir requires careful administration technique.

Levofloxacin

– A third-generation fluoroquinolone with improved gram-positive coverage, including Streptococcus pneumoniae. Levofloxacin inhibits bacterial DNA gyrase and topoisomerase IV. It is indicated for community-acquired pneumonia, urinary tract infections, and skin infections. The drug is available in oral and intravenous formulations with excellent bioavailability. Common adverse effects include gastrointestinal upset, tendon rupture, and QT prolongation. Levofloxacin is the L-isomer of ofloxacin and and has a slightly lower risk of QT prolongation than ciprofloxacin; levofloxacin is commonly used in combination therapy for community-acquired pneumonia and in the treatment of multidrug-resistant tuberculosis.

Linezolid

– An oxazolidinone antibiotic that inhibits bacterial protein synthesis by binding to the 23S rRNA of the 50S ribosomal subunit. Linezolid has activity against gram-positive organisms including MRSA, VRE, and penicillin-resistant streptococci. It is indicated for complicated skin infections, pneumonia, and VRE infections. The drug is available in both intravenous and oral formulations with excellent bioavailability. Common adverse effects include thrombocytopenia, which occurs with prolonged use, Thus, appropriate use of linezolid requires monitoring of platelet counts at least weekly in patients receiving treatment courses exceeding two weeks, and patients should be counselled regarding the need to avoid tyramine-rich foods to prevent hypertensive crises.

Liposomal amphotericin B

– A liposomal formulation of amphotericin B used in the treatment of severe systemic fungal infections, with reduced nephrotoxicity compared to conventional amphotericin B. The liposomal formulation encapsulates amphotericin B within lipid vesicles, which alters the drug's biodistribution and reduces renal tubular damage. It is the preferred formulation for neutropenic patients with fever and suspected fungal infections. The drug has broad-spectrum activity against Candida, Aspergillus, CryptococcAdditionally, liposomal amphotericin B is generally better tolerated with fewer infusion-related reactions and lower rates of nephrotoxicity, allowing for higher doses to be administered with improved therapeutic outcomes in critically ill patients with invasive fungal infections.

Loading Dose

– A higher initial dose of a drug administered to rapidly achieve therapeutic plasma concentrations, followed by lower maintenance doses to sustain the steady state. The loading dose is calculated as: Loading dose = (Vd x desired concentration) / bioavailability. Indications: drugs with long half-lives where waiting 4-5 half-lives to reach steady state would delay clinical effect (amiodarone, half-life 40-55 days; digoxin, half-life 36-48 hours); drugs requiring immediate therapeutic effect (phenytoin for status epilepticus, antibiotics in sepsis, antiarrhythmics in unstable arrhythmias). The loading dose does not depend on clearance, only on the volume of distribution and the desired concentration. Caution is required for drugs with narrow therapeutic indices to avoid toxicity from the high initial dose.

Loop Diuretics

– A class of potent diuretics that act on the thick ascending limb of the loop of Henle by inhibiting the sodium-potassium-chloride cotransporter, blocking sodium and chloride reabsorption and producing a powerful natriuresis and diuresis. The prototypical loop diuretic is furosemide, with other agents including bumetanide and torsemide. Loop diuretics are the most potent diuretics available, producing greater sodium excretion than thiazides. They also increase calcium and magneLoop diuretics are used in the treatment of hypertension, heart failure, and oedematous states including cirrhosis and nephrotic syndrome, and their dosing must be individualised based on renal function and clinical response, with careful monitoring of electrolytes and volume status to avoid hypokalaemia, hypomagnesaemia, and volume depletion.

Lopinavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection, typically co-administered with ritonavir as a fixed-dose combination. Lopinavir inhibits the HIV aspartyl protease, preventing viral polyprotein processing and maturation. Ritonavir significantly increases lopinavir plasma concentrations through CYP3A4 inhibition, allowing for twice-daily dosing. Common adverse effects include gastrointestinal disturbances such as nausea, diarrhea, and abdominal pain, as weas well as hyperlipidaemia, insulin resistance, and lipodystrophy, which are class effects of the protease inhibitor drug class.

Macrolides

– A class of bacteriostatic antibiotics that bind to the 50S ribosomal subunit and inhibit translocation of peptidyl-tRNA during protein synthesis. The major macrolides include erythromycin, clarithromycin, and azithromycin. Erythromycin has a narrow spectrum primarily against gram-positive organisms and is limited by gastrointestinal side effects from motilin receptor agonism. Clarithromycin has improved oral bioavailability and extended spectrum including Haemophilus influenzae anand azithromycin has become a widely used agent for respiratory infections due to its extended half-life and once-daily dosing. Macrolides also have immunomodulatory and anti-inflammatory properties independent of their antimicrobial activity, which contribute to their efficacy in chronic respiratory diseases including cystic fibrosis, bronchiectasis, and diffuse panbronchiolitis.

Maintenance Dose

– The dose of a drug administered at regular intervals to maintain a desired steady-state concentration, balancing the rate of drug administration with the rate of elimination. The maintenance dose is calculated as: Maintenance dose rate = (Clearance x desired concentration) / Bioavailability. At steady state, the rate of drug administration equals the rate of elimination. Maintenance doses are determined by the drug's clearance and target therapeutic concentration. Factors affecting maintenance dose requirements: changes in clearance (renal impairment, hepatic impairment, drug interactions affecting metabolism), changes in volume of distribution (obesity, edema), and patient factors (age, weight, organ function). Maintenance dosing intervals are based on the drug's half-life, typically administered every half-life for drugs with short half-lives or at longer intervals for drugs with long half-lives.

MAOIs

– Monoamine oxidase inhibitors are a class of antidepressant drugs that inhibit the enzyme monoamine oxidase, which breaks down serotonin, norepinephrine, and dopamine in the brain. By inhibiting this enzyme, MAOIs increase the levels of these neurotransmitters in the synaptic cleft. There are two forms of MAO: MAO-A preferentially metabolizes serotonin and norepinephrine, while MAO-B preferentially metabolizes dopamine and tyramine. Non-selective MAOIs including phenelzine and tranylcypromine inhibit both MAO-A and MAO-B, while selective MAO-B inhibitors such as selegiline are used in Parkinson disease. MAOIs have significant potential for drug interactions and require dietary restriction of tyramine-containing foods due to the risk of hypertensive crisis.

Maraviroc

– A CCR5 antagonist (entry inhibitor) used in combination antiretroviral therapy for HIV-1 infection in patients with CCR5-tropic virus. Maraviroc binds to the human CCR5 coreceptor on CD4+ T cells, preventing HIV-1 gp120 binding and viral entry. A critical requirement for maraviroc use is tropism testing to confirm the presence of CCR5-tropic virus, as the drug has no activity against CXCR4-tropic or dual/mixed-tropic virus. Common adverse effects include hepatotoxicity with elevated transaminases levels, which is typically reversible upon discontinuation; postural hypotension, diarrhea, nausea, and headache are also reported.

Maximum Tolerated Dose

– The highest dose of a drug that produces acceptable, manageable toxicity without causing unacceptable adverse effects, determined through dose-escalation studies, most commonly in Phase I clinical trials. The MTD identifies the dose range for subsequent efficacy studies. In oncology, the MTD is typically defined as the dose at which no more than 33% of patients experience dose-limiting toxicity (DLT) during the first treatment cycle. The 3+3 design is the classic approach: 3 patients are treated at each dose level; if 0/3 have DLT, the next cohort is escalated; if 1/3 have DLT, 3 more patients are treated at the same level; if 2/3 have DLT, the previous dose is declared the MTD. Modern adaptive designs (e.g., Bayesian continual reassessment method) are increasingly used.

Medical Cannabis and Cannabidiol (CBD)

– Cannabis (marijuana) and cannabidiol (CBD) are cannabinoid compounds derived from *Cannabis sativa* that interact with the endocannabinoid system (CB1 and CB2 receptors) to modulate pain, appetite, mood, and inflammation. THC (delta-9-tetrahydrocannabinol) is the primary psychoactive component responsible for euphoria, cognitive impairment, and dependence risk; CBD is non-intoxicating and has demonstrated anti-inflammatory, anxiolytic, and anticonvulsant properties. FDA-approved cannabinoid preparations: dronabinol and nabilone for chemotherapy-induced nausea and HIV/AIDS anorexia, and purified CBD (Epidiolex) for Dravet and Lennox-Gastaut syndromes. Evidence supports medical cannabis for chronic neuropathic pain, multiple sclerosis spasticity, and chemotherapy-induced nausea, though data for most other conditions are limited by small trials and regulatory barriers. Risks: impaired driving, psychosis vulnerability (especially high-THC products), cannabinoid hyperemesis syndrome, pulmonary harm from smoking, and addiction (9% of users).

Medication Management

– The systematic process of optimizing pharmacotherapy to achieve therapeutic goals while minimizing adverse effects, encompassing prescribing, dispensing, administration, monitoring, and deprescribing. Key components: medication reconciliation at transitions of care (comparing patient's current regimen against new orders to prevent errors), adherence assessment (pill counts, pharmacy refill data, electronic monitoring), dose adjustment for renal/hepatic impairment, therapeutic drug monitoring (e.g., warfarin INR, vancomycin trough, lithium levels), and deprescribing (tapering or discontinuing unnecessary or harmful medications). Polypharmacy—use of $\geq$5 medications—increases risk of adverse drug events, falls, cognitive impairment, and hospitalization; an annual medication review using tools like the Beers Criteria and STOPP/START criteria is recommended for older adults. Patient education on indication, dose, timing, side effects, and drug–drug interactions is essential for safe and effective therapy.

Medications and Treatments

– all pharmacologic and therapeutic interventions used to prevent, manage, or cure disease, classified by mechanism of action, therapeutic class, and route of administration. Major drug classes: analgesics (NSAIDs, opioids, acetaminophen), cardiovascular agents (statins, ACE inhibitors, beta-blockers, diuretics, anticoagulants), antimicrobials (antibiotics, antivirals, antifungals), endocrine therapies (insulin, metformin, thyroid hormone, corticosteroids), psychotropics (SSRIs, antipsychotics, benzodiazepines, mood stabilizers), antineoplastics (chemotherapy, targeted therapy, immunotherapy), and disease-modifying biologic agents. Routes include oral, intravenous, intramuscular, subcutaneous, topical, inhaled, transdermal, intrathecal, and intra-articular. The principles of pharmacotherapy: right drug, right dose, right route, right time, right patient—guided by evidence-based guidelines, contraindications, allergies, renal/hepatic function, and cost-effectiveness.

Medication Side Effects

– Unintended harmful responses to medications occurring at standard doses, ranging from mild (nausea, rash, dry mouth) to severe (anaphylaxis, Stevens-Johnson syndrome, QT prolongation, hepatotoxicity). ADRs are classified as Type A (augmented pharmacologic effect, predictable and dose-dependent, 80% of ADRs, e.g., bleeding with warfarin, hypoglycemia with insulin) and Type B (bizarre, unpredictable, immune-mediated or idiosyncratic, e.g., penicillin anaphylaxis, statin-induced myopathy). Risk factors: polypharmacy, advanced age, renal/hepatic impairment, genetic polymorphisms (CYP450 variants, HLA-B*5701 for abacavir hypersensitivity), and drug–drug interactions (e.g., simvastatin–clarithromycin). Management: prompt recognition, discontinuation or dose reduction, switching to alternative agent, and reporting to pharmacovigilance systems (FDA MedWatch). The Naranjo algorithm and causality assessment tools aid in determining likelihood of ADR.

Meropenem-vaborbactam

– A combination of the carbapenem meropenem with the cyclic boronate beta-lactamase inhibitor vaborbactam. Vaborbactam inhibits class A and class C beta-lactamases, including KPC carbapenemases, but does not inhibit class D or metallo-beta-lactamases. This combination has activity against many carbapenem-resistant Enterobacteriaceae. It is indicated for complicated urinary tract infections and is being studied for other indications. The combination is administered intravenously every 8 hours, withwith dose adjustment required for impaired renal function; the most common adverse effects include diarrhea, headache, nausea, and infusion-site reactions.

Metformin

– A biguanide oral antidiabetic agent and first-line treatment for type 2 diabetes mellitus. Its primary mechanism involves activation of AMP-activated protein kinase, which reduces hepatic glucose production through inhibition of gluconeogenesis and glycogenolysis. Metformin also improves peripheral insulin sensitivity and reduces intestinal glucose absorption. It does not stimulate insulin secretion, which is why it does not cause hypoglycemia when used as monotherapy. Metformin produces gastrointestinal side effects including diarrhea, nausea, and abdominal discomfort, which often diminish with continued use or dose adjustment; lactic acidosis is a rare but serious adverse effect that requires immediate discontinuation; long-term use is associated with vitamin B12 deficiency due to interference with calcium-dependent absorption.

Methicillin-resistant Staphylococcus aureus

– Staphylococcus aureus resistant to beta-lactam antibiotics through acquisition of the mecA gene encoding penicillin-binding protein 2a (PBP2a). MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, linezolid, and trimethoprim-sulfamethoxazole. MRSA infections are associated with increased morbidity and mortality. Infection prevention strategies include hand hygiene, contact precautions, and decolonization protocols. MRSA colonisation and infection surveillance, decolonisation strategies using intranasal mupirocin and chlorhexidine bathing, and antimicrobial stewardship to preserve the effectiveness of remaining active agents.

Methylphenidate

– A central nervous system stimulant, a piperidine derivative that blocks the reuptake of dopamine and noradrenaline (norepinephrine) by inhibiting the dopamine transporter (DAT) and noradrenaline transporter (NAT) in the prefrontal cortex, increasing their synaptic concentrations. Methylphenidate is the first-line pharmacological treatment for attention deficit hyperactivity disorder (ADHD) in children, adolescents, and adults (NICE guidelines). It improves core ADHD symptoms: inattention, hyperactivity, and impulsivity. Two main formulations: immediate-release (IR—duration 3–4 hours, multiple daily doses) and extended-release (ER/LA/Concerta—duration 10–12 hours, once-daily dosing). Dose is titrated based on response and tolerability (typically 10–60 mg/day). Side effects: decreased appetite, insomnia, headache, abdominal pain, nausea, anxiety, irritability, and increased heart rate/blood pressure (dose-dependent). Contraindications: severe hypertension, hyperthyroidism, glaucoma, tics/Tourette syndrome, severe anxiety or agitation, and monoamine oxidase inhibitor use. Methylphenidate has abuse potential (schedule II controlled substance in the US; class B in the UK). Long-term effects: growth suppression (mild, reversible) and potential for dependence.

Metronidazole

– A nitroimidazole antibiotic and antiprotozoal agent that is selectively toxic to anaerobic organisms and certain parasites. The drug is activated by reduction of its nitro group within anaerobic cells, generating toxic free radicals that damage DNA. Metronidazole is highly effective against anaerobic bacteria including Bacteroides fragilis, Clostridium species, and Helicobacter pylori, as well as protozoa including Trichomonas vaginalis, Giardia lamblia, and Entamoeba histolyticlytica. It is the drug of choice for C. difficile infection when used orally, for anaerobic infections above the diaphragm, and for bacterial vaginosis. Common adverse effects include metallic taste, nausea, and a disulfiram-like reaction with alcohol; prolonged use may cause peripheral neuropathy.

Micafungin

– An echinocandin antifungal agent used in the treatment and prophylaxis of invasive candidiasis and esophageal candidiasis. Micafungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis and leading to osmotic instability and cell death. A key advantage is its fungicidal activity against Candida species and fungistatic activity against Aspergillus. The drug is available only intravenously, as it is not absorbed orally. Common adverse effects include headache, nausea, and elevated transaminases; micafungin is also associated with an increased risk of hepatocellular tumors in animal studies, though the clinical significance in humans is uncertain.

Minimum inhibitory concentration

– The lowest concentration of an antimicrobial agent that inhibits visible growth of a microorganism under standardized in vitro conditions. MIC is the gold standard for measuring antimicrobial susceptibility and is determined by broth microdilution or agar dilution methods. The MIC value is used to categorize isolates as susceptible, intermediate, or resistant based on established breakpoints. MIC values guide dosing decisions and can predict clinical outcomes. For some infections, including endoendocarditis and osteomyelitis, maintaining drug concentrations above the MIC for the entire dosing interval is critical for achieving bacteriological cure, and time above the MIC is the key pharmacodynamic parameter for beta-lactam antibiotics.

Minocycline

– A tetracycline antibiotic with broader spectrum than tetracycline, including activity against MRSA and Nocardia. Minocycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It is indicated for acne, respiratory infections, and meningococcal prophylaxis. The drug is well absorbed orally with excellent tissue penetration, including the central nervous system. Common adverse effects include vestibular toxicity (dizziness, vertigo), skin hyperpigmentation, and photosensensitivity; minocycline has good oral bioavailability and is available in oral and intravenous formulations; it is particularly useful for infections caused by MRSA and Nocardia, and for the treatment of acne vulgaris due to its anti-inflammatory properties.

Moxifloxacin

– A fourth-generation fluoroquinolone with broad-spectrum activity including anaerobes. Moxifloxacin inhibits bacterial DNA gyrase and topoisomerase IV. It is indicated for community-acquired pneumonia, intra-abdominal infections, and sinusitis. The drug is available in oral and intravenous formulations. Common adverse effects include gastrointestinal upset, tendon rupture, and QT prolongation. Moxifloxacin has the broadest spectrum of any fluoroquinolone, including activity against Bacteroides frfragilis, and it is used as monotherapy for community-acquired pneumonia and complicated intra-abdominal infections.

Mycobacterium avium complex

– A group of closely related mycobacterial species that cause pulmonary disease and disseminated infection in immunocompromised patients. MAC is the most common NTM causing pulmonary disease. Treatment for pulmonary MAC requires a macrolide (azithromycin or clarithromycin) combined with ethambutol and a rifamycin. Disseminated MAC in HIV patients requires macrolide therapy with antiretroviral therapy. The organism is ubiquitous in the environment. Diagnosis requires isolation of MAC from clinical MAC infection from respiratory specimens in the presence of compatible clinical and radiological findings, and treatment is prolonged, typically lasting 12-18 months following culture conversion.

Narcotic

– A term derived from the Greek narkotikos (to numb or stupor), historically referring to opium-derived drugs that induce sleep or stupor and relieve pain. In modern medical usage, the term is largely replaced by the more specific term opioid. Legally, the term narcotic in the Controlled Substances Act refers broadly to any drug, including cocaine and coca leaves, that is considered addictive and abused, though cocaine and coca are pharmacologically stimulants, not narcotics. The imprecise legal definition has led to confusion. In clinical practice, the preferred term for morphine-like analgesics is opioid, which specifies drugs acting at opioid receptors.

Narrow Therapeutic Index

– A drug characteristic where there is a small difference between the therapeutic dose and the toxic dose, requiring careful dosing and therapeutic drug monitoring to ensure efficacy while avoiding toxicity. Narrow therapeutic index drugs have a therapeutic index <2-3 (the ratio of toxic dose to effective dose). Examples: warfarin (TI 2), digoxin, aminoglycosides (gentamicin, tobramycin, amikacin), vancomycin, cyclosporine, tacrolimus, lithium, phenytoin, carbamazepine, valproic acid, theophylline, and thyroid hormones. Managing NTI drugs requires: regular monitoring of drug concentrations; individualized dosing based on pharmacokinetic parameters; awareness of drug interactions that can alter levels; and patient education about signs of toxicity.

Nelfinavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection. Nelfinavir inhibits the HIV aspartyl protease, blocking viral polyprotein processing and producing immature, non-infectious viral particles. Unlike many other protease inhibitors, nelfinavir does not require ritonavir boosting and is administered twice daily, though its efficacy may be slightly lower than boosted PI regimens. The most common adverse effect is diarrhea, occurring in approximately 20-30% of pof patients, which can be managed with antidiarrhoeal agents and typically resolves with continued therapy.

Neuraminidase Inhibitors

– A class of antiviral drugs that inhibit the neuraminidase enzyme on the surface of influenza viruses, preventing the release of newly formed viral particles from infected cells and thereby limiting viral spread. The two available agents are oseltamivir, administered orally, and zanamivir, administered by inhalation. These drugs are effective against both influenza A and influenza B viruses. Neuraminidase cleaves sialic acid residues from glycoproteins on the cell surface, allowing release of newly formed viral particles; inhibition of neuraminidase prevents this release, limiting viral spread within the respiratory tract and reducing the duration and severity of influenza symptoms when started within 48 hours of symptom onset.

Neuroleptic Malignant Syndrome

– A rare but life-threatening idiosyncratic reaction to dopamine receptor-blocking agents, typically antipsychotics, characterized by hyperthermia, severe muscle rigidity, altered mental status, and autonomic dysfunction. The pathogenesis involves sudden blockade of dopamine D2 receptors in the hypothalamus causing thermoregulatory dysfunction, in the nigrostriatal pathway causing extrapyramidal symptoms, and in the mesocortical pathway causing altered mental statstatus; treatment involves immediate discontinuation of the causative agent, supportive care, and pharmacological therapy including dantrolene (a direct-acting muscle relaxant that inhibits calcium release from the sarcoplasmic reticulum), bromocriptine (a dopamine agonist), and benzodiazepines for agitation and muscle rigidity, with careful monitoring of vital signs in an intensive care setting.

Nitrates

– Vasodilator drugs that release nitric oxide, causing relaxation of vascular smooth muscle through activation of guanylyl cyclase and increased cyclic GMP. They primarily cause venodilation at low doses, reducing preload and myocardial oxygen demand, and arterial dilation at higher doses, reducing afterload. Nitroglycerin is the prototype nitrate, available in sublingual tablets for acute angina, intravenous infusion for acute coronary syndromes, topical patches for prophylaxis, and and oral sustained-release formulations for long-term prophylaxis. Nitrate tolerance develops rapidly with continuous exposure and requires a daily nitrate-free interval of 8-12 hours to maintain therapeutic efficacy; common adverse effects include headache, flushing, and postural hypotension.

Nitrofurantoin

– A nitrofuran antibiotic used for uncomplicated urinary tract infections. Nitrofurantoin is concentrated in the urine, achieving high levels in the bladder while having minimal systemic concentrations. The drug inhibits bacterial enzymes involved in aerobic metabolism. Common adverse effects include gastrointestinal upset, pulmonary toxicity with chronic use, and peripheral neuropathy. Nitrofurantoin is not effective for systemic infections due to poor tissue penetration. The drug is available inavailable in oral and intravenous formulations; nitrofurantoin is contraindicated in patients with impaired renal function (creatinine clearance below 60 mL/min) due to inadequate urine concentrations and increased risk of toxicity.

Noncompetitive Antagonist

– A noncompetitive antagonist is a drug that reduces the effect of an agonist by a mechanism that cannot be overcome by increasing the agonist concentration. This occurs either through irreversible binding to the receptor, preventing agonist binding permanently, or through binding to an allosteric site that reduces receptor function regardless of agonist binding. The dose-response curve of the agonist is shifted to the right with a decrease in maximal response, meaning the full effect of the agonionist is reduced or eliminated, distinguishing noncompetitive antagonism from competitive antagonism; this type of antagonism is often irreversible on the timescale of clinical drug therapy and requires synthesis of new receptors for recovery of full agonist responsiveness.

Nonselective Beta-Blockers

– Nonselective beta-adrenergic blocking agents block both beta-1 and beta-2 adrenergic receptors, reducing heart rate, myocardial contractility, and cardiac output through beta-1 blockade, while also blocking beta-2 receptors in the bronchial smooth muscle and peripheral vasculature. Common nonselective beta-blockers include propranolol, nadolol, timolol, and pindolol. Propranolol is the prototype agent, with lipophilic properties allowing good CNS penetration, making it useful for performance anxxiety and essential tremor, though this property also increases the risk of CNS side effects including fatigue, depression, and sleep disturbances.

Non-tuberculous mycobacteria

– A group of mycobacterial species other than M. tuberculosis complex that cause disease in humans. NTM include rapidly growing species (M. abscessus, M. chelonae, M. fortuitum) and slowly growing species (M. avium complex, M. kansasii). NTM infections are increasingly recognized, particularly in patients with chronic lung disease and immunocompromise. Treatment requires combination therapy based on species identification and susceptibility testing. Macrolides are backbone agents for many NTM infections; treatment is prolonged and requires multiple agents, with regimens tailored to species identification and susceptibility results, and outcomes vary widely depending on the species, extent of disease, and host immune status.

NSAIDs

– Nonsteroidal anti-inflammatory drugs are a class of drugs that inhibit cyclooxygenase enzymes, COX-1 and COX-2, reducing the synthesis of prostaglandins and thromboxanes from arachidonic acid. They produce analgesic, anti-inflammatory, antipyretic, and antiplatelet effects. COX-1 is constitutively expressed and maintains gastric mucosal protection, renal blood flow, and platelet aggregation. COX-2 is inducible and upregulated during inflammation. Traditional NSAIDs including ibuprofen, naproxen,and diclofenac have analgesic, antipyretic, and anti-inflammatory effects but carry risks including gastrointestinal bleeding and ulceration, renal impairment, and cardiovascular events; the risk-benefit profile must be evaluated for each patient based on their individual risk factors.

Nucleoside Analogs

– A class of antiviral drugs that structurally resemble natural nucleosides and inhibit viral replication by interfering with DNA or RNA synthesis. They require intracellular phosphorylation by viral or cellular kinases to their active triphosphate form, which then competes with natural nucleotides for incorporation into the growing viral nucleic acid chain, causing chain termination or lethal mutagenesis. Examples include acyclovir for herpes simplex virus and varicella-zososter virus, tenofovir for HIV and hepatitis B, and ribavirin for hepatitis C, with resistance developing through mutations in the target viral enzymes that reduce drug binding affinity.

Off-Label Use

– The use of a licensed medication for an indication, dose, route, or patient population not included in its marketing authorisation (product licence). Off-label prescribing is legal, common in many areas of medicine (pediatrics—up to 50% of hospital prescriptions, oncology, psychiatry, obstetrics), and may represent best clinical practice when evidence supports the use (e.g., β-blockers in heart failure were off-label before regulatory approval). The prescriber bears responsibility for the patient's safety and should: (1) be satisfied that there is a sufficient evidence base (published studies, guidelines, expert consensus) to support the off-label use; (2) explain the off-label nature to the patient and obtain informed consent; (3) document the rationale in the medical record. Regulatory frameworks: in the UK, the MHRA allows off-label prescribing under the prescriber's professional responsibility; the FDA prohibits pharmaceutical companies from promoting off-label use but does not restrict physician prescribing. Notable examples: ketamine for treatment-resistant depression, metformin for gestational diabetes, and aspirin for primary prevention of cardiovascular disease (controversial).

Ointment

– A semi-solid, greasy topical preparation for application to the skin, mucous membranes, or eyes. Ointments are typically composed of a hydrophobic base (petrolatum, lanolin, paraffin, beeswax, vegetable oils) and may contain dissolved or suspended active pharmaceutical ingredients. Ointments have high occlusivity (form a water-repellent barrier, preventing transepidermal water loss) and are preferred for dry, scaly, or lichenified skin conditions (psoriasis, chronic eczema, xerosis). They have higher drug penetration than creams (oil-in-water emulsions) or lotions due to their lipophilic nature. Disadvantages: greasy feel, staining of clothing, and less cosmetic acceptability than creams. Eye ointments (e.g., chloramphenicol, tetracycline) provide prolonged ocular contact time and are often used at bedtime. Ointment bases can be classified as hydrocarbon (petrolatum), absorption (lanolin—can absorb water), water-removable (emulsifying ointment), and water-soluble (polyethylene glycol).

Omadacycline

– A newer aminomethylcycline antibiotic with activity against gram-positive, gram-negative, and atypical organisms. Omadacycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against MRSA, vancomycin-resistant enterococci, community-acquired pneumonia pathogens, and skin infections. The drug is available in both intravenous and oral formulations, allowing for IV-to-oral step-down therapy. Omadacycline is indicated for community-acquired bacterial pneneumonia and acute bacterial skin infections, with common adverse effects including nausea, vomiting, and infusion-site reactions; photosensitivity is less common than with earlier tetracyclines.

Opiate

– A natural alkaloid derived from the opium poppy (Papaver somniferum) or a semi-synthetic derivative that binds to opioid receptors. Natural opiates: morphine (prototypical opioid analgesic, gold standard for severe pain), codeine (prodrug metabolized to morphine by CYP2D6—poor metabolisers derive minimal analgesia), and thebaine (used in synthesis of semi-synthetic opioids—oxycodone, buprenorphine, naloxone). Semi-synthetic opiates: heroin (diamorphine—more lipid-soluble than morphine, rapid CNS penetration), oxycodone, and hydromorphone. See also Opioid for the broader class including synthetic opioids (fentanyl, methadone, tramadol). Historically, opium has been used for pain relief, diarrhea, and cough suppression for millennia. Opiates produce euphoria, miosis, respiratory depression, constipation, and physical dependence. Overuse leads to opioid use disorder (addiction).

Opioid Analgesics

– A class of drugs that bind to opioid receptors in the central and peripheral nervous systems, producing analgesia and other effects. There are three main opioid receptor types: mu receptors mediating analgesia, respiratory depression, euphoria, and physical dependence; delta receptors involved in analgesia and mood; and kappa receptors mediating analgesia and sedation. Morphine is the prototype opioid, derived from opium, with other natural opioids including codeine and semsynthetic opioids including fentanyl and methadone. Common adverse effects include respiratory depression, constipation, nausea and vomiting, sedation, tolerance, physical dependence, and the potential for addiction; careful dosing and monitoring are required to balance effective pain relief with the risk of adverse events.

Oritavancin

– A lipoglycopeptide antibiotic with a long half-life allowing for single-dose dosing. Oritavancin inhibits cell wall synthesis and disrupts cell membrane integrity. It has activity against gram-positive organisms including MRSA. The drug is indicated for acute bacterial skin infections. It is administered as a single intravenous dose, providing complete therapy in one administration. The drug achieves high concentrations in skin and soft tissues. Common adverse effects include nausea, headache, and infusion-site reactions, and oritavancin has a favourable safety profile for a single-dose regimen, though it may interfere with coagulation testing for up to 24 hours after administration.

Orphan Drug

– A pharmaceutical agent developed specifically to treat a rare medical condition (affecting fewer than 200,000 people in the United States, as defined by the Orphan Drug Act of 1983). Orphan drug designation provides incentives including seven-year market exclusivity after FDA approval, tax credits for clinical trial costs, waived FDA application fees, and research grants. Examples include lumacaftor/ivacaftor for cystic fibrosis, nusinersen for spinal muscular atrophy, and eliglustat for Gaucher disease. The Orphan Drug Act has successfully stimulated development of treatments for previously neglected rare diseases.

Orphan Drugs

– Medicinal products developed for the diagnosis, prevention, or treatment of rare diseases (defined in the EU as affecting <5 per 10,000 people; in the US as affecting <200,000 people). Orphan drug designation provides incentives for pharmaceutical companies: 7 years of market exclusivity in the US (Orphan Drug Act 1983), 10 years in the EU (2000), reduced regulatory fees, tax credits for clinical research costs (25% in the US), and protocol assistance from regulatory agencies. More than 600 orphan drugs have been approved in the US since 1983. Examples: imiglucerase and miglustat (Gaucher disease), ivacaftor (cystic fibrosis—G551D mutation), eculizumab (paroxysmal nocturnal haemoglobinuria, atypical hemolytic uraemic syndrome), nusinersen (spinal muscular atrophy), and eliglustat (Gaucher disease type 1). Challenges: high drug costs (often >$200,000 per patient per year), limited natural history data, small trial populations, and ethical concerns about equitable access. The UK Rare Diseases Framework (2021) prioritises timely diagnosis and access to treatment for rare diseases.

Over-the-Counter Drug

– A medication that can be purchased without a prescription, deemed safe and effective for self-medication when used according to label instructions. OTC drugs have established safety profiles, low toxicity, and clear dosing guidelines suitable for consumer use. Examples include acetaminophen, ibuprofen, antihistamines (loratadine, diphenhydramine), antacids, and topical antifungals. The

FDA determines OTC status based on safety, labeling adequacy, and the condition being treatable without professional supervision. Some drugs are switched from prescription to OTC status after post-marketing safety data accumulate.

Over-the-Counter (OTC) Medications

– Nonprescription drugs approved as safe and effective for self-management of common symptoms, available at pharmacies and retail stores without a prescription. Major categories: analgesics/antipyretics (acetaminophen—max 3000–4000 mg/day, hepatotoxicity risk with overdose; NSAIDs—ibuprofen, naproxen—GI bleeding and cardiovascular risk with chronic use), cough and cold preparations (decongestants—pseudoephedrine, phenylephrine; antihistamines—diphenhydramine, cetirizine; dextromethorphan for cough), gastrointestinal (PPIs/omeprazole, H2 blockers/famotidine for GERD; antacids; loperamide for diarrhea; psyllium, polyethylene glycol for constipation; bismuth subsalicylate for dyspepsia), allergy (oral/loratadine, cetirizine; intranasal corticosteroids, cromolyn), dermatologic (hydrocortisone, antifungals—clotrimazole, terbinafine), and sleep aids (diphenhydramine, doxylamine, melatonin). Risks: drug–drug interactions (NSAIDs with anticoagulants, antacids with antibiotics), unintentional overdose (especially combination products with acetaminophen), and delayed diagnosis of serious conditions. Pharmacist counseling improves safe OTC selection.

Paracetamol

– A widely used analgesic and antipyretic, also known as acetaminophen. Paracetamol is a non-opioid, non-NSAID analgesic; its mechanism is not fully understood but involves inhibition of cyclooxygenase (COX) enzymes in the central nervous system (weak peripheral COX inhibition—hence no anti-inflammatory effect) and interaction with the serotonergic and cannabinoid systems. Indications: mild to moderate pain (headache, musculoskeletal pain, dental pain, dysmenorrhoea, osteoarthritis), fever reduction. Dose: adults 500–1000 mg every 4–6 hours (max 4 g/day). Children: 10–15 mg/kg every 4–6 hours (max 60 mg/kg/day). It is the analgesic of choice in children and patients with contraindications to NSAIDs (peptic ulcer disease, renal impairment, asthma, anticoagulation). Paracetamol is safe in pregnancy and breastfeeding. Toxicity: overdose (single dose >150 mg/kg or >7.5 g in adults) causes hepatocellular necrosis due to accumulation of the toxic metabolite N-acetyl-p-benzoquinone imine (NAPQI) when hepatic glutathione stores are depleted. N-acetylcysteine (NAC) is the antidote, most effective when given within 8 hours of overdose. Paracetamol is available in combination with opioids (codeine, dihydrocodeine, tramadol) and decongestants.

Partial Agonist

– A partial agonist is a drug that binds to a receptor and produces a biological response that is less than the maximum possible response, even when all receptors are occupied. Partial agonists have affinity for the receptor but lower intrinsic activity compared to full agonists, producing a submaximal response regardless of dose. The maximal effect of a partial agonist is always less than that of a full agonist acting at the same receptor. An important property of partial agonists is that they cacan act as functional antagonists in the presence of a full agonist: by occupying receptors that would otherwise be activated by the full agonist, the partial agonist reduces the net response, which is the basis for their use in maintenance therapy for opioid dependence and smoking cessation.

Penicillin

– The prototypical beta-lactam antibiotic discovered in 1928 by Alexander Fleming, derived from the Penicillium mold. Penicillins inhibit bacterial cell wall synthesis by binding to penicillin-binding proteins (transpeptidases) and inhibiting the final transpeptidation step in peptidoglycan cross-linking, leading to cell lysis. Classes: Natural penicillins (penicillin G, penicillin V) — narrow spectrum, mostly gram-positive (Strep, oral anaerobes, Treponema pallidum). Penicillinase-resistant (methicillin, nafcillin, oxacillin) — active against penicillinase-producing Staph aureus. Aminopenicillins (ampicillin, amoxicillin) — expanded activity including gram-negative (E. coli, Proteus, H. influenzae). Extended-spectrum (piperacillin, ticarcillin) — broader gram-negative coverage including Pseudomonas. Antipseudomonal penicillins combined with beta-lactamase inhibitors: piperacillin-tazobactam (Zosyn), amoxicillin-clavulanate (Augmentin), ampicillin-sulbactam (Unasyn). Adverse effects: hypersensitivity reactions (5-10%, from mild rash to anaphylaxis 0.01%), GI disturbances (diarrhea), and neurotoxicity (seizures with high doses, especially in renal impairment). Penicillin is the most common drug allergy reported (though >90% of labeled patients tolerate penicillin).

Pharmacodynamics

– The study of the biochemical and physiological effects of drugs on the body and their mechanisms of action, describing what the drug does to the body. It encompasses drug-receptor interactions (binding affinity, intrinsic activity, efficacy), signal transduction pathways (second messengers, gene expression), dose-response relationships, and therapeutic vs. adverse effects. Key concepts: receptors — proteins or macromolecules that drugs bind to exert their effects; agonists — drugs that activate receptors to produce a response; antagonists — drugs that block receptor activation; partial agonists — drugs that produce a submaximal response even at full receptor occupancy; and allosteric modulators — drugs that bind to sites distinct from the orthosteric binding site to modulate receptor activity.

Pharmacogenomics

– The study of how genetic variations influence individual responses to drugs, including drug efficacy, toxicity, and metabolism. Genetic polymorphisms in drug-metabolizing enzymes, transporters, and receptors can lead to significant inter-individual variability in drug response. Examples: CYP2D6 poor metabolizers have reduced codeine activation to morphine, leading to ineffective analgesia; CYP2C19 poor metabolizers require reduced clopidogrel dosing; TPMT deficiency increases risk of severe myelosuppression with azathioprine; HLA-B*5701 testing is required before abacavir to prevent hypersensitivity reaction; UGT1A1*28 homozygotes have increased risk of irinotecan toxicity. Pharmacogenomic testing is increasingly integrated into clinical practice, guiding drug selection and dosing to optimize efficacy and minimize adverse effects.

Pharmacokinetics

– The study of drug movement through the body, describing what the body does to the drug, defined by four processes: Absorption — the process by which a drug enters the systemic circulation from its site of administration (oral, IV, IM, subcutaneous, topical, inhalation); Distribution — the movement of drug from the circulation to body tissues, determined by blood flow, protein binding, tissue affinity, and barriers (blood-brain barrier, placental barrier); Metabolism — the chemical modification of drugs by enzymes (primarily hepatic CYP450) through Phase I (oxidation, reduction, hydrolysis) and Phase II (conjugation) reactions; and Excretion — the elimination of drugs and their metabolites from the body, primarily through the kidneys (glomerular filtration, tubular secretion, reabsorption) and also through bile, feces, lungs, and sweat. Pharmacokinetic parameters: half-life, clearance, volume of distribution, bioavailability, and steady state concentration.

Pharmacovigilance

– The science and activities relating to the detection, assessment, understanding, and prevention of adverse drug reactions and other drug-related problems. It encompasses post-marketing surveillance of drugs after they have been approved for clinical use, as pre-marketing clinical trials may not detect rare adverse effects, long-term effects, or effects in special populations. Pharmacovigilance systems collect adverse event reports from healthcare professionals, patients, andand pharmaceutical companies, and uses data from multiple sources including clinical trials, spontaneous reporting, electronic health records, and published literature to generate safety signals that may lead to regulatory actions including label changes, warnings, and market withdrawals.

Phase II Metabolism

– Phase II metabolism, also known as conjugation reactions, involves the attachment of endogenous hydrophilic molecules to drugs or their Phase I metabolites, dramatically increasing water solubility and facilitating renal excretion. The major Phase II reactions include glucuronidation catalyzed by UDP-glucuronosyltransferases, which is the most common conjugation reaction and accounts for approximately 35 percent of Phase II metabolism; sulfation catalyzed by sulfotransferases; acetylation catalycatalysed by N-acetyltransferases; methylation; and conjugation with glutathione. Phase II metabolites are generally pharmacologically inactive and are rapidly excreted in urine or bile; the efficiency of these conjugation pathways depends on the availability of the endogenous substrate and the activity of the conjugating enzymes, which can be influenced by genetic polymorphisms, drug interactions, and disease states.

Phase I Metabolism

– Phase I metabolism involves chemical modification of drug molecules through oxidation, reduction, and hydrolysis reactions, primarily catalyzed by the cytochrome P450 enzyme system in the liver. These reactions introduce or expose functional groups such as hydroxyl, amino, sulfhydryl, and carboxyl groups, typically increasing the polarity and water solubility of the parent compound. The most common Phase I reaction is oxidation, catalyzed by CYP enzymes requiring molecular oxygen and NADPH. OxidOxidation reactions introduce hydroxyl, carboxyl, or epoxide groups, while reduction reactions add hydrogen or remove oxygen, and hydrolysis reactions cleave ester or amide bonds; the resulting metabolites may be pharmacologically inactive, less active, equally active, or occasionally more active than the parent compound, and some undergo further Phase II conjugation for elimination.

Phosphodiesterase Inhibitor

– A class of drugs that inhibit phosphodiesterase enzymes, which break down cyclic nucleotides (cAMP and cGMP), thereby increasing intracellular levels of these second messengers and potentiating their downstream effects. Nonselective PDE inhibitors: theophylline (PDE3 and PDE4 inhibition) — bronchodilator for asthma and COPD, also has anti-inflammatory effects. PDE3 inhibitors: milrinone — positive inotrope and vasodilator for acute decompensated heart failure. PDE4 inhibitors: roflumilast — used for severe COPD with chronic bronchitis and frequent exacerbations; apremilast — used for psoriasis and psoriatic arthritis. PDE5 inhibitors: sildenafil, tadalafil, vardenafil — inhibit cGMP-specific PDE5 in corpus cavernosum, used for erectile dysfunction; also treat pulmonary arterial hypertension (sildenafil, tadalafil). Each PDE isozyme has distinct tissue distribution, allowing targeted therapeutic effects.

Placebo Effect

– The measurable, observable, or felt improvement in a condition that results from the patient's belief in a treatment, despite the treatment having no specific pharmacological activity for the condition being treated. The placebo effect is mediated through psychological and neurobiological mechanisms: expectation and conditioning activate endogenous opioid and dopamine pathways, leading to measurable pain relief; the nocebo effect is the opposite phenomenon — negative expectations produce adverse effects. Placebo response rates vary by condition: 20-40% in depression, 30-50% in pain, 30-60% in irritable bowel syndrome. In clinical trials, placebos are used as controls to distinguish the specific pharmacological effect from the placebo effect, natural history, and regression to the mean.

Plazomicin

– An aminoglycoside antibiotic with activity against many multidrug-resistant gram-negative organisms including carbapenem-resistant Enterobacteriaceae. Plazomicin inhibits protein synthesis by binding to the 30S ribosomal subunit. It is indicated for complicated urinary tract infections caused by resistant organisms. The drug is administered intravenously once daily. Common adverse effects include nephrotoxicity and ototoxicity. Plazomicin is designed to resist most aminoglycoside-modifying enzymzymes, maintaining activity against many aminoglycoside-resistant gram-negative pathogens; however, nephrotoxicity and ototoxicity remain significant concerns and require close monitoring, particularly in patients receiving prolonged therapy or those with pre-existing renal impairment.

Potassium-Sparing Diuretics

– Potassium-sparing diuretics act on the distal nephron to promote sodium excretion while minimizing potassium loss. They include epithelial sodium channel blockers such as amiloride and triamterene, and mineralocorticoid receptor antagonists such as spironolactone and eplerenone. Amiloride and triamterene directly block the epithelial sodium channel in the collecting duct, reducing sodium reabsorption and potassium secretion. Spironolactone and eplerenone competitively inhibit aldosterone bindingto the mineralocorticoid receptor, blocking aldosterone-mediated sodium reabsorption and potassium excretion. Spironolactone is used as a diuretic in conditions associated with secondary hyperaldosteronism, including heart failure and hepatic cirrhosis, and as an antiandrogen in the treatment of acne, hirsutism, and female pattern hair loss. Hyperkalaemia is the most important adverse effect and requires regular monitoring of serum potassium levels, particularly in patients with renal impairment or those receiving concurrent ACE inhibitors or ARBs.

Potency vs Efficacy

– Potency is the amount of drug required to produce a given effect, measured by the EC50 (the concentration producing 50% of maximal effect). Drugs with higher potency achieve effects at lower doses (e.g., fentanyl is more potent than morphine). Potency is clinically relevant for dosing but does not indicate which drug is therapeutically superior. Efficacy is the maximum effect a drug can produce at saturating concentrations, measured by the Emax (plateau of the dose-response curve). Efficacy determines the clinical utility of a drug — a drug with high efficacy can produce an effect that a low-efficacy drug cannot. For example, morphine (high efficacy mu-opioid agonist) can relieve severe pain that codeine (low efficacy partial agonist) cannot. A drug can have high potency but low efficacy and vice versa.

Prescription Drug

– A medication that can only be dispensed with a valid prescription from a licensed healthcare provider, requiring professional oversight due to the drug's complexity, toxicity, or potential for harm. Prescription drugs undergo rigorous FDA approval through clinical trials (Phase I-IV) to establish safety and efficacy before marketing. Examples include antibiotics, antihypertensives, antidepressants, and most psychotropic medications. Prescribing requires consideration of indication, dosing, drug interactions, and monitoring parameters.

Prescription Medications

– Pharmaceutical drugs that require authorization from a licensed healthcare provider before they can be dispensed to a patient. These medications are regulated because they carry risks that necessitate professional oversight, including potential side effects, drug interactions, and the need for precise dosing. Physicians prescribe them based on a specific diagnosis, and patients must follow the prescribed regimen closely to achieve therapeutic benefits while minimizing adverse outcomes.

Procalcitonin-guided antibiotic therapy

– The use of procalcitonin levels to guide initiation and discontinuation of antibiotics. Procalcitonin is a biomarker that rises in bacterial infections and falls with effective treatment. Elevated levels support the need for antibiotic therapy, while declining levels support discontinuation. Procalcitonin-guided protocols have been shown to reduce antibiotic duration and exposure in respiratory infections and sepsis. The approach is most validated for lower respiratory tract infections and sepsis; however, false-positive results can occur in non-infectious inflammatory conditions and false-negative results in localised infections, so procalcitonin should be used as an adjunct to, rather than a replacement for, clinical judgement.

Prodrug

– A prodrug is a pharmacologically inactive compound that is converted to an active drug through metabolic processes within the body. Prodrugs are designed to overcome limitations of the parent drug including poor bioavailability, rapid metabolism, lack of tissue selectivity, or unacceptable side effects. Examples include enalapril, which is hydrolyzed by hepatic esterases to the active form enalaprilat, an ACE inhibitor that is poorly absorbed orally; clopidogrel, which requires CYP2C19-mediated mediated activation to its active metabolite; and valacyclovir, which undergoes hepatic hydrolysis to acyclovir, significantly improving oral bioavailability from 20% to 55%.

Protease Inhibitors

– A class of antiviral drugs that block viral proteases, enzymes essential for processing viral polyproteins into functional components needed for viral assembly and maturation. In HIV treatment, protease inhibitors including ritonavir, lopinavir, atazanavir, and darunavir inhibit HIV-1 protease, preventing the cleavage of Gag and Gag-Pol polyproteins into structural proteins and enzymes. This results in the production of immature, non-infectious viral particles. Protease iinhibitors are also used in the treatment of hepatitis C, where NS3/4A protease inhibitors including simeprevir, grazoprevir, and glecaprevir block viral polyprotein processing and are used in combination with other direct-acting antiviral agents to achieve high rates of sustained virological response.

Protein Binding

– The reversible association of drug molecules with plasma proteins, primarily albumin for acidic drugs and alpha-1 acid glycoprotein for basic drugs. Only unbound or free drug is pharmacologically active and available for distribution to tissues, metabolism by the liver, and excretion by the kidneys. The fraction of drug bound to protein is described by the binding ratio, which depends on the total drug concentration, total protein concentration, and the associationassociation constant; changes in protein binding can significantly affect drug disposition, particularly for highly bound drugs (greater than 90% bound), where small changes in binding can lead to large changes in free drug concentration, potentially affecting both efficacy and toxicity.

Proton Pump Inhibitor

– A class of medications that irreversibly inhibit the gastric H+/K+ ATPase (proton pump) in parietal cells, the final common pathway of gastric acid secretion, producing profound and sustained suppression of both basal and stimulated acid production. PPIs are prodrugs that require activation in the acidic environment of the parietal cell canaliculus, where they form a sulfenamide that binds covalently to cysteine residues of the proton pump. Available agents: omeprazole, lansoprazole, pantoprazole, rabeprazole, esomeprazole, and dexlansoprazole. Indications: gastroesophageal reflux disease (GERD), erosive esophagitis, peptic ulcer disease, Helicobacter pylori eradication (with antibiotics), Zollinger-Ellison syndrome, and prevention of NSAID-induced ulcers. Adverse effects: increased risk of Clostridioides difficile infection, pneumonia, hypomagnesemia, vitamin B12 deficiency, and osteoporosis-related fractures with long-term use. Rebound acid hypersecretion occurs upon discontinuation after >8 weeks of use.

Pseudomonas aeruginosa resistance

– Pseudomonas aeruginosa has intrinsic and acquired resistance mechanisms making it inherently resistant to many antibiotics. Intrinsic resistance includes low outer membrane permeability, efflux pumps, and chromosomal beta-lactamase. Acquired resistance can develop through mutations in target genes, acquisition of resistance plasmids, and overexpression of efflux pumps. Treatment typically requires antipseudomonal beta-lactams (piperacillin-tazobactam, cefepime, ceftazidime, carbapenems) or fluororoquinolones (ciprofloxacin, levofloxacin), often in combination therapy due to the high rate of resistance emergence during monotherapy; susceptibility testing is essential to guide treatment decisions.

QT Prolongation

– An electrocardiographic finding characterized by prolongation of the QT interval, representing the time from ventricular depolarization to repolarization, which predisposes to the potentially fatal arrhythmia torsades de pointes. The corrected QT interval should be less than 440 milliseconds in men and 460 milliseconds in women. Numerous medications can prolong the QT interval by blocking the cardiac potassium channel encoded by the KCNH2 gene, prolonging phase 3 of the cardiacardiac action potential, prolonging repolarisation and increasing the risk of triggered activity, torsades de pointes, and sudden cardiac death; risk factors include female sex, advanced age, electrolyte disturbances (hypokalaemia, hypomagnesaemia), bradycardia, structural heart disease, and the use of multiple QT-prolonging medications.

Quinidine

– A class Ia antiarrhythmic agent, a stereoisomer of quinine. Quinidine prolongs the action potential duration and effective refractory period by blocking sodium channels (slow phase 0 upstroke) and potassium channels (prolongs repolarisation). Indications: atrial fibrillation (conversion and maintenance of sinus rhythm), supraventricular tachycardia, and ventricular arrhythmias (now rarely used due to safer alternatives). It also has vagolytic effects (increases sinus rate and AV conduction—may paradoxically increase ventricular rate in atrial flutter). Side effects: cinchonism (tinnitus, hearing loss, headache, nausea, visual disturbance—dose-related), QT prolongation (risk of torsades de pointes—monitor ECG), diarrhea (most common, up to 30%), thrombocytopenia (immune-mediated), and hepatotoxicity. Drug interactions: increases digoxin levels (P-glycoprotein inhibition), potentiates warfarin, and interacts with CYP3A4 inhibitors/inducers. Therapeutic range: 2–6 μg/mL (free). Rarely used in contemporary practice; reserved for refractory cases or Brugada syndrome (prevents VF storms).

Quinine

– An alkaloid derived from cinchona bark, with antimalarial, antipyretic, and skeletal muscle relaxant properties. Quinine is a blood schizonticide with activity against all Plasmodium species, but resistance has emerged in P. falciparum. Current indications: (1) Treatment of severe/complicated falciparum malaria (WHO—quinine IV infusion plus doxycycline or clindamycin; artesunate IV is now preferred due to lower mortality). (2) Babesiosis (combined with clindamycin). (3) Nocturnal leg cramps (used off-label—risk of serious adverse effects outweighs benefit; only if cramps are disabling and no alternative exists). Mechanism: accumulates in the parasite's food vacuole, inhibits haem polymerase, and prevents detoxification of haem (toxic to the parasite). Side effects: cinchonism (tinnitus, dizziness, headache, nausea, visual disturbances—dose-related, reversible), QT prolongation, hypoglycaemia (stimulates insulin secretion—dangerous in severe malaria), hypersensitivity (urticaria, flushing, hemolytic anemia), blackwater fever (massive intravascular haemolysis—rare, idiosyncratic). Contraindications: G6PD deficiency (hemolytic anemia), QT prolongation, myasthenia gravis, optic neuritis. Therapeutic range: 8–15 μg/mL. Quinine is also used in tonic water (bitter flavouring, subtherapeutic concentrations).

Quinupristin-dalfopristin

– A combination streptogramin antibiotic that inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. The combination is bactericidal against most gram-positive organisms including MRSA. It is indicated for vancomycin-resistant Enterococcus faecium infections. The drug is administered intravenously every 8-12 hours. Common adverse effects include arthralgia, myalgia, and infusion-related reactions including phlebitis. The drug is a potent inhibitor of CYP3A4 and can interact with numerous medications, particularly those that inhibit CYP3A4 and prolong the QT interval.

Receptor Desensitization

– A rapid decrease in the response of a receptor to continuous or repeated agonist stimulation, occurring within minutes to hours. Desensitization mechanisms: receptor phosphorylation by G protein-coupled receptor kinases which uncouples the receptor from G proteins, leading to rapid functional desensitization; beta-arrestin binding which sterically prevents G protein coupling and targets the receptor for internalization; receptor internalization (endocytosis) into intracellular compartments, reducing cell-surface receptor number; and receptor degradation (downregulation) after prolonged stimulation, reducing total receptor protein. Desensitization contributes to drug tolerance (e.g., beta-agonist desensitization in asthma, opioid receptor desensitization in chronic pain) and can be reversed by removing the agonist, allowing receptor recycling to the cell surface.

Receptor Downregulation

– A decrease in the number or sensitivity of receptors in response to chronic receptor stimulation by agonists. When receptors are chronically activated, the cells compensate by internalizing and degrading receptors, reducing receptor density on the cell surface, and decreasing receptor responsiveness. This process contributes to drug tolerance, where increasing doses are needed to achieve the same therapeutic effect. Chronic use of beta-agonists such as albuterol for asasthma leads to reduced bronchodilator response, and tolerance to opioid analgesics develops through downregulation of mu-opioid receptors in the central nervous system.

Receptor Upregulation

– An increase in the number or sensitivity of receptors in response to chronic receptor blockade or decreased neurotransmitter activity. When receptors are chronically blocked by antagonists, the cells compensate by synthesizing more receptors to maintain homeostatic signaling, leading to increased sensitivity to agonists. This phenomenon contributes to drug tolerance, where increasing doses are needed to achieve the same effect, and to withdrawal syndromes when the blockidrug is discontinued; well-recognised examples include the upregulation of beta-adrenergic receptors following chronic beta-blocker therapy, causing rebound tachycardia and hypertension upon abrupt withdrawal, and the upregulation of histamine H2 receptors after chronic proton pump inhibitor use, leading to rebound acid hypersecretion.

Renin Inhibitor

– A class of antihypertensive drugs that directly inhibit renin, the first and rate-limiting enzyme in the renin-angiotensin-aldosterone system. Renin catalyzes the conversion of angiotensinogen to angiotensin I, which is then converted to angiotensin II by ACE. Aliskiren is the only direct renin inhibitor available. It binds to the active site of renin, preventing angiotensinogen cleavage and reducing angiotensin I, angiotensin II, and aldosterone levels. Indication: hypertension. Aliskiren is used alone or in combination with other antihypertensives. Adverse effects: diarrhea (especially at higher doses), cough, hyperkalemia, and angioedema. Contraindicated in pregnancy (fetal renin-angiotensin system toxicity). Combination with ACE inhibitors or ARBs in diabetic patients with renal impairment is contraindicated due to increased risk of hyperkalemia, hypotension, and renal impairment.

Rezafungin

– A long-acting echinocandin antifungal agent used in the treatment of invasive candidiasis and candidemia. Rezafungin inhibits beta-(1,3)-D-glucan synthase with a unique pharmacokinetic profile characterized by an extremely long half-life of approximately 130 hours, allowing for once-weekly dosing after an initial loading regimen. This extended dosing interval is a significant advantage over other echinocandins and may improve adherence. The drug has excellent stability in solution and can be stored at room temperature, maintaining potency during extended infusion periods; common adverse effects include headache, nausea, diarrhea, and hypokalaemia; hepatotoxicity has been observed in clinical trials.

Rifaximin

– A rifamycin derivative with minimal systemic absorption, acting locally in the gastrointestinal tract. Rifaximin inhibits bacterial RNA polymerase. It is indicated for traveler's diarrhea caused by noninvasive E. coli and hepatic encephalopathy. The drug is administered orally twice or three times daily. Common adverse effects include nausea and flatulence. Rifaximin maintains activity against many gram-negative and gram-positive organisms in the gut. The drug is used for hepatic encephalopathyosis at a dose of 550 mg twice daily, which reduces the risk of hospitalisation and improves cognitive function in patients with recurrent hepatic encephalopathy by reducing ammonia-producing gut bacteria.

Ritalin

– See Methylphenidate.

Ritonavir

– A protease inhibitor (PI) with potent cytochrome P450 3A4 (CYP3A4) inhibitory activity, widely used as a pharmacokinetic booster in combination antiretroviral therapy for HIV-1 infection. Ritonavir inhibits viral maturation by blocking the HIV aspartyl protease, but its clinical utility primarily lies in its ability to significantly increase plasma concentrations of co-administered protease inhibitors through CYP3A4 inhibition. This pharmacokinetic boosting allows for lower doses and reduced dosing frequency, reducing pill burden, and improving adherence. Common adverse effects when used as a booster are limited, but at full therapeutic doses, ritonavir causes significant gastrointestinal intolerance, hyperlipidaemia, and insulin resistance, and it has a high potential for drug interactions due to potent CYP3A4 inhibition.

Saquinavir

– A protease inhibitor (PI) used in combination antiretroviral therapy for HIV-1 infection. Saquinavir was the first protease inhibitor approved for clinical use and inhibits the viral aspartyl protease responsible for cleaving Gag and Gag-Pol polyproteins into functional structural proteins. By blocking this essential step in viral maturation, saquinavir produces immature, non-infectious viral particles. The drug has poor oral bioavailability due to extensive first-pass metabolism by cytochrome Pp450 CYP3A4, resulting in variable plasma concentrations. Saquinavir is now rarely used due to the availability of more effective and better-tolerated protease inhibitors with improved pharmacokinetic profiles.

Saquinavir/ritonavir

– A combination of the protease inhibitor saquinavir with ritonavir as a pharmacokinetic booster, used in antiretroviral therapy for HIV-1 infection. The ritonavir component inhibits CYP3A4, significantly increasing saquinavir plasma concentrations and allowing for reduced dosing frequency and improved bioavailability. This combination allows saquinavir to be dosed twice daily instead of three times daily, improving patient adherence. The combined regimen inhibits HIV aspartyl protease, preventingviral polyprotein processing and maturation. Common adverse effects include gastrointestinal disturbances, hyperlipidaemia, and hepatotoxicity, and the combination requires careful monitoring of liver enzymes and lipid profiles throughout therapy.

Sedative

– A medication that reduces central nervous system activity, producing calming effects, drowsiness, and sleep at higher doses. Sedatives are used for anxiety, procedural sedation, insomnia, and as adjuncts in anesthesia. Classes include: benzodiazepines (midazolam, lorazepam, diazepam) — enhance GABA-A receptor activity, used for sedation, anxiolysis, and amnesia; barbiturates (phenobarbital, thiopental) — enhance GABA-A activity with broader CNS depression, largely replaced by benzodiazepines due to narrow therapeutic index and high dependence potential; propofol — an IV anesthetic that potentiates GABA-A receptors, rapid onset and offset, used for procedural sedation and general anesthesia; dexmedetomidine — an alpha-2 agonist producing sedation with minimal respiratory depression; and chloral hydrate — older sedative/hypnotic used for short-term sedation in children.

Sedatives

– A class of drugs that produce a calming or drowsy effect by depressing the central nervous system. Sedatives reduce anxiety, promote sleep, and induce relaxation. Classes: (1) Benzodiazepines—enhance $GABA_A$ receptor activity (diazepam, lorazepam, midazolam, temazepam). Used for anxiety, insomnia, seizures, muscle relaxation, procedural sedation, and alcohol withdrawal. (2) Z-drugs—non-benzodiazepine hypnotics (zolpidem, zopiclone, zaleplon)—selective for $GABA_A$ $\alpha 1$ subunit, primarily for insomnia. (3) Barbiturates—enhance $GABA_A$ activity but with a narrower therapeutic index (phenobarbital, thiopental)—now rarely used except for refractory epilepsy and anesthetic induction. (4) Antihistamines—H1 antagonists with sedative properties (diphenhydramine, hydroxyzine, promethazine)—mild sedation. (5) Antipsychotics—low-dose quetiapine, olanzapine—used as sedatives off-label. (6) Melatonin receptor agonists (ramelteon, prolonged-release melatonin)—regulate circadian rhythm. (7) Orexin receptor antagonists (suvorexant, daridorexant)—promote sleep by blocking wake-promoting orexin signalling. (8) $\alpha 2$-adrenergic agonists (clonidine, dexmedetomidine)—sedation without respiratory depression. Adverse effects: drowsiness, daytime sedation, cognitive impairment, impaired coordination, falls (especially elderly), tolerance, dependence, and withdrawal. Respiratory depression (especially with alcohol, opioids). Long-term use of benzodiazepines is discouraged (dependence, tolerance, cognitive decline, increased dementia risk—controversial).

Selective Estrogen Receptor Modulator

– A class of medications that bind to estrogen receptors (ER-alpha and ER-beta) with tissue-specific agonist or antagonist activity, depending on the target tissue and differential co-regulator recruitment. Tamoxifen: antagonist in breast tissue (first-line adjuvant therapy for hormone receptor-positive breast cancer in premenopausal women), partial agonist in bone (preserves bone density) and endometrium (increases risk of endometrial cancer and hyperplasia), and improves lipid profile. Raloxifene: antagonist in breast and agonist in bone without endometrial stimulation (used for osteoporosis prevention and treatment in postmenopausal women, reduces breast cancer risk). Toremifene: similar to tamoxifen for metastatic breast cancer. Bazedoxifene: used for postmenopausal osteoporosis, often combined with conjugated estrogens (duavee). Adverse effects: tamoxifen — hot flashes, vaginal dryness, and thromboembolism (2-3% risk). SERMs do not cause the systemic side effects of estrogen therapy.

Serotonin Syndrome

– A potentially life-threatening condition caused by excessive serotonergic activity in the central and peripheral nervous systems, typically resulting from drug interactions or overdose involving serotonergic medications. The classic triad includes neuromuscular hyperactivity with tremor, clonus, rigidity, and hyperreflexia; autonomic dysfunction with hyperthermia, diaphoresis, tachycardia, and hypertension;

and altered mental status with agitation, confusion, and delirium. Severe cases may progress to hyperpyrexia, rhabdomyolysis, multi-organ failure, and death. Treatment involves immediate discontinuation of all serotonergic agents, supportive care with intravenous fluids and cooling, and administration of cyproheptadine, a serotonin antagonist, for moderate to severe cases.

SGLT2 Inhibitors

– Sodium-glucose co-transporter 2 inhibitors are a class of oral antidiabetic agents that block the reabsorption of glucose in the proximal convoluted tubule of the kidney, promoting glycosuria and reducing blood glucose levels. The kidney normally filters approximately 180 grams of glucose daily, with SGLT2 responsible for approximately 90 percent of glucose reabsorption. By blocking this transporter, SGLT2 inhibitors increase urinary glucose excretion by 60 to 80 grams per day. Common agents incinclude dapagliflozin, empagliflozin, and canagliflozin. In addition to glycaemic control, SGLT2 inhibitors have demonstrated significant cardiovascular and renal benefits in major outcome trials, reducing the risk of heart failure hospitalisation and progression of chronic kidney disease, making them an important treatment option in patients with type 2 diabetes and established cardiovascular or renal disease.

Skeletal Muscle Relaxant

– A medication that reduces muscle spasm, spasticity, and associated pain. Centrally acting muscle relaxants: baclofen — GABA-B agonist that reduces spinal reflex activity, first-line for spasticity in cerebral palsy, multiple sclerosis, and spinal cord injury; tizanidine — alpha-2 adrenergic agonist that inhibits spinal motor neuron activity; diazepam — enhances GABA-A activity, effective for acute muscle spasm; cyclobenzaprine — structurally related to tricyclic antidepressants, reduces brainstem descending noradrenergic activity; methocarbamol and carisoprodol — centrally acting, mechanism less defined. Peripherally acting: dantrolene — inhibits calcium release from the sarcoplasmic reticulum by blocking ryanodine receptors, used for malignant hyperthermia and chronic spasticity. Botulinum toxin — blocks acetylcholine release at the neuromuscular junction, used for focal spasticity and dystonia.

Sleep Medications and Supplements

– Pharmacological agents used to treat insomnia and other sleep disorders when behavioral interventions alone are insufficient. Prescription sleep aids include benzodiazepine receptor agonists such as zolpidem, melatonin receptor agonists, and low-dose sedating antidepressants, while common over-the-counter options include antihistamines and melatonin supplements. These agents should be used with caution due to risks of dependence, tolerance, daytime drowsiness, and potential interactions with other medications, and are typically recommended for short-term use under medical supervision.

SNRIs

– Serotonin-norepinephrine reuptake inhibitors are a class of antidepressant drugs that inhibit the reuptake of both serotonin and norepinephrine from the synaptic cleft, increasing both serotonergic and noradrenergic neurotransmission. Common SNRIs include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran. They are used for major depressive disorder, generalized anxiety disorder, panic disorder, social anxiety disorder, and chronic pain conditions. Duloxetine is also approved for diabeabetic peripheral neuropathy, fibromyalgia, and chronic musculoskeletal pain. Common adverse effects include nausea, dry mouth, insomnia, sweating, and sexual dysfunction; discontinuation syndrome with dizziness and paraesthesias may occur with abrupt cessation, requiring gradual tapering.

Somatostatin Analog

– A class of synthetic peptides that mimic the actions of native somatostatin, a hormone that inhibits the secretion of growth hormone, thyroid-stimulating hormone, insulin, glucagon, gastrin, and other GI hormones. Octreotide (short-acting, subcutaneous TID) and lanreotide (long-acting, IM every 4 weeks) are somatostatin analogs with longer half-lives (1-2 hours for octreotide vs. 2-3 minutes for native somatostatin). Indications: acromegaly — suppress GH and IGF-1 levels, reduce tumor size; neuroendocrine tumors (carcinoid syndrome, VIPoma, glucagonoma) — control hormone-related symptoms and may slow tumor growth; variceal bleeding — reduce splanchnic blood flow (octreotide infusion); and refractory diarrhea from chemotherapy or short bowel syndrome. Adverse effects: gallstones (25-50% due to inhibition of gallbladder contraction), bradycardia, hyperglycemia/hypoglycemia, and GI symptoms.

SSRIs

– Selective serotonin reuptake inhibitors are a class of antidepressant drugs that selectively inhibit the reuptake of serotonin from the synaptic cleft into the presynaptic neuron by blocking the serotonin transporter, increasing serotonergic neurotransmission. Common SSRIs include fluoxetine, sertraline, paroxetine, citalopram, escitalopram, and fluvoxamine. They are first-line agents for major depressive disorder, generalized anxiety disorder, panic disorder, obsessive-compulsive disorder, posttraumatic stress disorder, premenstrual dysphoric disorder, and bulimia nervosa. Common adverse effects include nausea, insomnia, sexual dysfunction, and weight changes; SSRIs have a favourable safety profile in overdose compared to TCAs and MAOIs, making them the preferred first-line pharmacotherapy for most depressive and anxiety disorders.

Steady State

– The condition where the rate of drug administration equals the rate of elimination, resulting in constant average plasma concentrations with repeated dosing. During regular dosing at fixed intervals, steady state is achieved after approximately 4 to 5 half-lives, regardless of the dosing frequency or dose size. At steady state, the peak and trough concentrations fluctuate within a therapeutic range, with the degree of fluctuation determined by the dosing interval relative to the the drug's half-life; the concept of steady state is fundamental to therapeutic drug monitoring and ensures that drug concentrations remain within the therapeutic window for optimal efficacy and safety.

Stenotrophomonas maltophilia resistance

– Stenotrophomonas maltophilia is a gram-negative bacterium with intrinsic resistance to many antibiotics including carbapenems. The organism produces a chromosomal beta-lactamase (L1) that confers resistance to beta-lactams. Treatment options include trimethoprim-sulfamethoxazole, fluoroquinolones, and minocycline. S. maltophilia is an emerging opportunistic pathogen, particularly in immunocompromised patients and those with cystic fibrosis. The organism forms biofilms and is resistant to many disinfectants, contributing to its persistence in healthcare environments; treatment requires combination therapy based on susceptibility testing, and antimicrobial stewardship is critical for preserving the limited treatment options available for this organism.

Stevens-Johnson Syndrome

– A severe, life-threatening mucocutaneous reaction characterized by extensive necrosis and detachment of the epidermis and mucous membranes, typically triggered by medications or infections. It is considered a medical emergency requiring burn unit or intensive care management. The pathogenesis involves a cell-mediated immune reaction with cytotoxic T lymphocytes attacking keratinocytes, causing widespread apoptosis and epidermal necrolysis. Drugs are the most common trincluding sulfonamides, anticonvulsants (carbamazepine, lamotrigine, phenytoin, phenobarbital), allopurinol, NSAIDs, and antiretrovirals (nevirapine). Management involves immediate discontinuation of the suspected causative drug, supportive care in a burn unit or intensive care setting, wound care, fluid and electrolyte management, and ophthalmological evaluation to assess ocular involvement and prevent long-term visual impairment.

Sulfonamides

– A class of bacteriostatic antibiotics that competitively inhibit dihydropteroate synthase, an enzyme essential for folic acid synthesis in bacteria. They are structural analogs of para-aminobenzoic acid and compete with PABA for the active site of the enzyme. Trimethoprim is combined with sulfamethoxazole to produce synergistic bactericidal activity by blocking sequential steps in folate synthesis. Co-trimoxazole, the combination of trimethoprim and sulfamethoxazole, is used forurinary tract infections, Pneumocystis jirovecii pneumonia, and other infections; hypersensitivity reactions to sulfonamides are common, particularly in patients with HIV infection, and range from mild rash to severe cutaneous adverse drug reactions including Stevens-Johnson syndrome.

Sulfonylureas

– Oral antidiabetic agents that stimulate insulin secretion from pancreatic beta cells by blocking ATP-sensitive potassium channels on the beta cell membrane. This causes membrane depolarization, opening of voltage-gated calcium channels, calcium influx, and exocytosis of insulin-containing granules. They are effective at lowering blood glucose but carry a risk of hypoglycemia because they stimulate insulin secretion regardless of glucose levels. First-generation sulfonylureas ininclude tolbutamide and chlorpropamide; second-generation sulfonylureas including glipizide, glyburide, and glimepiride are more potent, have a more favourable side effect profile, and are dosed once or twice daily. Sulfonylureas remain an important treatment option for type 2 diabetes, particularly in resource-limited settings where cost is a significant consideration.

Sulphonamides

– A class of antimicrobial agents containing the sulfonamide group (SO_2NH_2). Sulphonamides were the first effective systemic antibacterial agents (Prontosil, 1935—Gerhard Domagk). Mechanism: competitive inhibition of dihydropteroate synthase, blocking folate synthesis (para-aminobenzoic acid analogue)—bacteriostatic. Spectrum: broad—Gram-positive (Streptococcus, Staphylococcus), Gram-negative (E. coli, Klebsiella, Haemophilus, Nocardia), and some protozoa (Toxoplasma, Plasmodium). Clinical uses: (1) Sulfamethoxazole combined with trimethoprim (co-trimoxazole)—first-line for Pneumocystis jirovecii pneumonia (PCP), Nocardia, Toxoplasma encephalitis, and UTIs. (2) Sulfadiazine—with pyrimethamine for toxoplasmosis. (3) Sulfasalazine—used in inflammatory bowel disease and rheumatoid arthritis (5-ASA prodrug). (4) Mafenide and silver sulfadiazine—topical for burn wounds. (5) Sulfacetamide—ophthalmic drops for conjunctivitis. Adverse effects: hypersensitivity reactions (rash, Stevens–Johnson syndrome/toxic epidermal necrolysis—rare but life-threatening, fever, serum sickness), crystalluria (prevented by adequate hydration and urine alkalinisation—risk with older, less soluble sulphonamides), hemolytic anemia in G6PD deficiency, bone marrow suppression, photosensitivity, and kernicterus in neonates (displaces bilirubin from albumin—contraindicated in pregnancy, breastfeeding, and infants <2 months). Drug interactions: potentiates warfarin, methotrexate, and sulfonylureas.

Suppository

– A solid, single-dose dosage form designed to be inserted into a body orifice (rectum, vagina, or urethra) where it melts, softens, or dissolves, releasing the active pharmaceutical ingredient for local or systemic effects. Rectal suppositories (the most common): base materials (cocoa butter/theobroma oil, hard fat/Witepsol, glycero-gelatin, polyethylene glycol). Indications: local—hemorrhoids (anusol, hydrocortisone), constipation (glycerol, bisacodyl); systemic—antiemetics (prochlorperazine, promethazine), antipyretics/analgesics (paracetamol, diclofenac), bronchodilators (theophylline), and antiemetics when oral route is unavailable (nausea/vomiting, post-operative). Advantages: bypasses first-pass hepatic metabolism (some drugs achieve higher bioavailability), useful when oral route is unavailable (NPO, vomiting, unconscious patients), and avoids GI irritation. Insertion: supine or lateral position, push past the internal anal sphincter, retain for 15–30 minutes. Vaginal suppositories (pessaries): for local treatment of vaginal infections (clotrimazole, metronidazole, oestrogen) and contraception.

Tachyphylaxis

– A rapid, acute decrease in response to a drug after repeated administration over a short period, distinct from tolerance which develops more gradually. Tachyphylaxis results from rapid depletion of neurotransmitter stores, receptor desensitization, or receptor downregulation. A classic example is the rapid development of tachyphylaxis to ephedrine, where repeated doses lead to depletion of norepinephrine stores in nerve terminals, reducing the sympathomimetic effect. Nicotine alalso demonstrates tachyphylaxis, and this phenomenon contributes to the need for dose escalation during the first few days of therapy. Tachyphylaxis is distinct from tolerance, which develops more slowly, and from desensitisation, which results from reduced receptor responsiveness.

TCAs

– Tricyclic antidepressants are a class of drugs that inhibit the reuptake of serotonin and norepinephrine, similar to SNRIs, but also block histamine H1 receptors, muscarinic acetylcholine receptors, and alpha-1 adrenergic receptors, producing numerous side effects. Common TCAs include amitriptyline, nortriptyline, imipramine, and clomipramine. They are effective antidepressants but are now used less frequently for depression due to their side effect profile and toxicity in overdose. TCAs are usesed for a range of other conditions including neuropathic pain (amitriptyline, nortriptyline), migraine prophylaxis (amitriptyline), chronic pain syndromes, enuresis, and insomnia (doxepin, trimipramine). Their anticholinergic effects cause dry mouth, blurred vision, constipation, urinary retention, and cognitive impairment in elderly patients; cardiotoxicity in overdose manifests as QT prolongation, conduction abnormalities, and potentially fatal arrhythmias, limiting their use in patients with cardiac disease or at risk of suicide.

Tedizolid

– An oxazolidinone antibiotic related to linezolid with improved pharmacokinetic properties and activity. Tedizolid inhibits bacterial protein synthesis by binding to the 23S rRNA of the 50S ribosomal subunit. It has activity against gram-positive organisms including MRSA, VRE, and penicillin-resistant streptococci. Tedizolid is indicated for acute bacterial skin and skin structure infections. The drug is available in both intravenous and oral formulations with once-daily dosing. A key advantage oover linezolid is once-daily dosing and a more favourable safety profile with reduced risk of thrombocytopenia and myelosuppression, as tedizolid has less inhibitory effect on mitochondrial protein synthesis than linezolid.

Tedizolid pharmacokinetics

– Tedizolid has a longer half-life than linezolid, allowing for once-daily dosing. The drug has approximately 90% oral bioavailability. Tedizolid is metabolized to an active metabolite (tzd-M1) that contributes to efficacy. The drug achieves high concentrations in skin and soft tissues. Tedizolid does not inhibit monoamine oxidase to the same degree as linezolid, reducing drug interaction potential. The drug is excreted primarily in feces.

Teicoplanin

– A glycopeptide antibiotic related to vancomycin with a longer half-life allowing for less frequent dosing. Teicoplanin inhibits cell wall synthesis by binding to D-Ala-D-Ala precursors. It has activity against gram-positive organisms including MRSA. The drug is administered intravenously or intramuscularly, with once-daily or less frequent dosing after loading. Common adverse effects include injection site reactions, rash, and nephrotoxicity. Teicoplanin achieves higher tissue concentrations thathan vancomycin, making it useful for deep-seated infections including bone and joint infections, endocarditis, and prosthetic device infections; teicoplanin is not available in the United States but is widely used in Europe and other regions.

Telavancin

– A lipoglycopeptide antibiotic derived from vancomycin with a lipophilic side chain that enhances membrane binding. Telavancin inhibits cell wall synthesis and disrupts cell membrane integrity. It has activity against gram-positive organisms including MRSA. The drug is indicated for complicated skin infections and hospital-acquired pneumonia. It is administered intravenously once daily. Common adverse effects include QT prolongation, taste disturbance, and nephrotoxicity. Telavancin achieves concentrations in pulmonary surfactant, making it effective for hospital-acquired pneumonia, and it does not require routine therapeutic drug monitoring, though renal function should be monitored during therapy.

Teratogenicity

– The ability of a drug, chemical, or infectious agent to cause birth defects or developmental abnormalities when exposure occurs during pregnancy. The risk of teratogenic effects depends on the timing of exposure, with the embryonic period from 3 to 8 weeks after fertilization being the most critical for organogenesis. Certain drugs have well-established teratogenic effects: thalidomide causes limb defects, isotretinoin causes craniofacial, cardiac, and CNS abnormalities,and sodium valproate is associated with neural tube defects; the management of drug therapy during pregnancy requires careful risk-benefit assessment, minimising exposure during organogenesis whenever possible, and using alternative agents with established safety profiles for the treatment of chronic conditions in women of childbearing potential.

Terbinafine

– An allylamine antifungal agent that inhibits squalene epoxidase, an early enzyme in the ergosterol biosynthesis pathway, causing accumulation of squalene and depletion of ergosterol in the fungal cell membrane. The accumulation of squalene is fungicidal, while ergosterol depletion disrupts membrane function. Terbinafine has excellent activity against dermatophytes including Trichophyton, Microsporum, and Epidermophyton species, making it the treatment of choice for dermatophyte ininfections of the skin and nails. Unlike azole antifungals, terbinafine does not significantly inhibit CYP450 enzymes and has a lower potential for drug interactions; adverse effects include mild gastrointestinal disturbances, headache, and taste disturbance, with rare cases of hepatotoxicity requiring periodic liver function monitoring during prolonged therapy.

Tetracycline

– The prototype tetracycline antibiotic with broad-spectrum activity against gram-positive and gram-negative bacteria, as well as atypical organisms. Tetracycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It is indicated for respiratory infections, acne, and chlamydial infections. Common adverse effects include gastrointestinal upset, photosensitivity, and tooth discoloration in children. Tetracycline chelates divalent cations, so it should not be taken with daidairy products, antacids, and iron-containing preparations; tetracycline is now rarely used due to the availability of more effective and better-tolerated alternatives, particularly doxycycline and minocycline.

Tetracyclines

– A class of bacteriostatic antibiotics that bind reversibly to the 30S ribosomal subunit and block the attachment of aminoacyl-tRNA to the acceptor site, inhibiting protein synthesis. They include tetracycline, doxycycline, and minocycline. Tetracyclines have a broad spectrum of activity against gram-positive and gram-negative bacteria, atypical organisms including Chlamydia, Mycoplasma, and Rickettsia, and the malaria parasite. Doxycycline is the most commonly used tetracyclinefor clinical use, with excellent oral bioavailability, favourable pharmacokinetics allowing once- or twice-daily dosing, and a well-established safety profile; tetracyclines also have anti-inflammatory and immunomodulatory properties that are independent of their antimicrobial activity and contribute to their efficacy in acne vulgaris and rosacea.

Thalidomide

– A synthetic glutamic acid derivative with immunomodulatory, anti-inflammatory, and anti-angiogenic properties. Thalidomide was introduced in the 1950s as a sedative and antiemetic in pregnancy, but was withdrawn in 1961 after causing one of the worst pharmaceutical disasters: over 10,000 infants born with phocomelia (severe limb shortening or absence) and other congenital malformations (cardiac, renal, ear, eye defects) due to its potent teratogenicity. Mechanism: binds to cereblon (CRBN), a substrate receptor of the E3 ubiquitin ligase complex, leading to ubiquitination and degradation of the transcription factors Ikaros (IKZF1) and Aiolos (IKZF3), which inhibits multiple myeloma cell proliferation and enhances T-cell and NK-cell anti-tumor immunity. Anti-angiogenic: inhibits VEGF and FGF-induced angiogenesis. Current indications: (1) Multiple myeloma—first-line combination therapy (with dexamethasone, lenalidomide, bortezomib). (2) Erythema nodosum leprosum (ENL) in leprosy—FDA approved. (3) Behçet disease—refractory oral/genital ulcers. Side effects: teratogenicity (strict pregnancy prevention programme required—two forms of contraception, monthly pregnancy testing), peripheral neuropathy (dose-limiting, often irreversible, 20–50% of patients), sedation, rash, constipation, venous thromboembolism (prophylactic anticoagulation recommended). Thalidomide analogues (lenalidomide, pomalidomide) have improved potency and safety profiles.

Therapeutic Drug Monitoring

– The clinical practice of measuring specific drug concentrations in blood or plasma to individualize drug dosing and ensure therapeutic effectiveness while minimizing toxicity. TDM is particularly important for drugs with narrow therapeutic indices, where small changes in concentration can lead to therapeutic failure or toxicity. Examples include aminoglycoside antibiotics where both subtherapeutic and supratherapeutic concentrations can occur, vancomycin with target trough concentrations for vancomycin requiring trough monitoring in the range of 10-20 mcg/mL depending on the infection type and severity; other drugs requiring TDM include digoxin, lithium, theophylline, cyclosporine, tacrolimus, and certain anticonvulsants including phenytoin and valproic acid.

Therapeutic Index

– The therapeutic index is a quantitative measure of the relative safety of a drug, defined as the ratio of the dose that produces toxicity in 50 percent of the population to the dose that produces the desired therapeutic effect in 50 percent of the population. A wider therapeutic index indicates a larger margin of safety, meaning there is a greater difference between effective and toxic doses. Drugs with narrow therapeutic indices require careful dosing and monitoring because small changes in doses in drug concentration produce significant clinical consequences; drugs with narrow therapeutic indices require individualised dosing based on patient factors, therapeutic drug monitoring, and careful management of drug interactions to maintain concentrations within the therapeutic window and avoid toxicity.

Therapeutics

– The branch of medicine concerned with the treatment of disease, encompassing all interventions used to prevent, manage, or cure illness. Therapeutics includes: pharmacotherapy (drug treatment—most common), surgery (therapeutic procedures), radiotherapy (ionizing radiation for cancer), physical therapy (physiotherapy, rehabilitation), psychological therapy (CBT, psychotherapy), gene and cell therapy, immunotherapy, and complementary/alternative medicine. Principles: (1) Evidence-based medicine: treatment decisions based on the best available evidence (randomised controlled trials, systematic reviews, meta-analyses, clinical guidelines). (2) Individualised therapy: patient-specific factors (age, genetics, comorbidities, concurrent medications, preferences) guide treatment choice. (3) Risk–benefit assessment: consider efficacy, safety, tolerability, cost, and quality of life. (4) Therapeutic index: the ratio of toxic to therapeutic dose; drugs with a narrow therapeutic index (warfarin, lithium, digoxin, phenytoin) require therapeutic drug monitoring. (5) Adherence/compliance: the extent to which a patient follows prescribed treatment; poor adherence (50% of chronic disease patients) is a major cause of treatment failure. (6) Polypharmacy: use of multiple medications ($\geq$5) increases risk of adverse drug reactions, drug interactions, and non-adherence. (7) Deprescribing: systematic withdrawal of inappropriate or unnecessary medications.

Thiazide Diuretics

– Thiazide diuretics act on the distal convoluted tubule by inhibiting the sodium-chloride cotransporter, reducing sodium and chloride reabsorption and producing natriuresis and diuresis. They are less potent than loop diuretics but have a longer duration of action, typically 6 to 12 hours. Common thiazides include hydrochlorothiazide, chlorthalidone, and metolazone. Thiazides are first-line antihypertensive agents, particularly effective in elderly patients with isolated systolic hypertension andand in patients of African descent; adverse effects include hypokalaemia, hyperuricaemia, hyperglycaemia, hypercalcaemia, and metabolic alkalosis, and they should be used with caution in patients with diabetes mellitus, gout, and electrolyte imbalances.

Thiopental Sodium

– An ultra-short-acting barbiturate, used for induction of general anesthesia (IV, onset 30–60 seconds, duration 5–10 minutes) and as a third-line agent for raised intracranial pressure (cerebroprotective effect: reduces cerebral metabolism, cerebral blood flow, and intracranial pressure). Mechanism: enhances $GABA_A$ receptor activity (prolongs chloride channel opening). Side effects: dose-dependent respiratory depression, hypotension (decreased cardiac output, peripheral vasodilation), laryngospasm, and tissue necrosis/extravasation injury (highly alkaline, pH 10.5—administer through a secure IV line). Contraindications: porphyria (acute intermittent porphyria—induces haem oxygenase, precipitates attack), severe shock, status asthmaticus, and hypersensitivity to barbiturates. Thiopental was also used in lethal injection protocols (US) and for barbiturate coma (refractory intracranial hypertension). Now largely replaced by propofol (faster recovery, fewer side effects) for routine anesthetic induction. Historical: 'truth serum' myth—thiopental reduces inhibition but does not produce truthful statements reliably.

Tigecycline

– A glycylcycline antibiotic derived from minocycline with expanded spectrum of activity. Tigecycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against gram-positive organisms including MRSA and VRE, gram-negative organisms including ESBL-producing Enterobacteriaceae, and anaerobes. Tigecycline is indicated for complicated intra-abdominal infections, complicated skin infections, and community-acquired pneumonia. The drug is administered intravenously twice daily, with common adverse effects including nausea, vomiting, abdominal pain, and hepatotoxicity; tigecycline carries a FDA black box warning for increased all-cause mortality compared to comparator agents, particularly in patients treated for hospital-acquired pneumonia, and should be reserved for situations where alternative treatment options are limited or inadequate.

Tipranavir

– A non-peptidic protease inhibitor (PI) used in combination antiretroviral therapy for treatment-experienced HIV-1 patients with resistance to multiple other PIs. Tipranavir inhibits the HIV aspartyl protease with a unique binding mechanism that maintains activity against strains resistant to other PIs. The drug must be co-administered with ritonavir to achieve adequate drug levels due to extensive CYP3A4 metabolism. Tipranavir has a significant drug interaction profile and is a potent inducer ofof CYP3A4 and CYP2C9, contributing to a high potential for drug interactions and requiring careful management of concomitant medications, including dose adjustment of statins and other CYP3A4 substrates.

Tocolytics

– Medications used to suppress preterm labor (contractions before 37 weeks gestation), to delay delivery for corticosteroid administration (fetal lung maturity) and in utero transport to a tertiary care center. First-line: calcium channel blockers (nifedipine) — inhibit smooth muscle contraction by blocking voltage-gated calcium channels, oral administration, fewer side effects than other tocolytics. Beta-2 adrenergic agonists (terbutaline, ritodrine) — activate beta-2 receptors on uterine smooth muscle, causing relaxation via increased intracellular cAMP, side effects include maternal tachycardia, pulmonary edema, and hypokalemia. NSAIDs (indomethacin) — inhibit prostaglandin synthesis by COX inhibition, effective for short-term tocolysis (<48 hours), side effects include oligohydramnios and premature closure of the fetal ductus arteriosus with prolonged use. Magnesium sulfate — blocks calcium entry into smooth muscle cells, used for neuroprotection in preterm infants (<32 weeks) and seizure prophylaxis in preeclampsia, but has marginal tocolytic efficacy.

Tolbutamide

– A first-generation sulfonylurea oral hypoglycaemic agent used in the management of type 2 diabetes. Mechanism: stimulates insulin secretion from pancreatic β-cells by binding to the sulfonylurea receptor (SUR1) on ATP-sensitive potassium (K_{ATP}) channels, causing membrane depolarisation, calcium influx, and exocytosis of insulin granules. Tolbutamide has a short half-life (4–7 hours) and is metabolized by the liver to inactive metabolites (CYP2C9). It is less potent and shorter-acting than second-generation sulfonylureas (glibenclamide/glyburide, gliclazide, glipizide, glimepiride). Clinical use: now rarely used (largely replaced by newer sulfonylureas and other oral agents). Indication: adjunct to diet and exercise in type 2 diabetes when metformin is contraindicated or not tolerated. Side effects: hypoglycaemia (less frequent than with longer-acting sulfonylureas), weight gain, and hypersensitivity reactions (rash, photosensitivity). The University Group Diabetes Program (UGDP) trial (1970s) suggested an increased cardiovascular mortality with tolbutamide, though this finding has been disputed. Drug interactions: potentiates effect of alcohol (disulfiram-like reaction), warfarin, and NSAIDs; diminished effect with thiazide diuretics, corticosteroids, and rifampicin.

Tolerance

– A decreased response to a drug after repeated administration, requiring dose escalation to achieve the same effect. Unlike tachyphylaxis, tolerance develops over weeks to months of chronic use. The mechanisms include pharmacokinetic tolerance from increased drug metabolism through enzyme induction, pharmacodynamic tolerance from receptor down-regulation and desensitization, and learned or behavioral tolerance from compensatory physiological responses. Opioid tolerance is a major clinical challenge in chronic pain management, leading to the need for gradual dose escalation, careful assessment of risk versus benefit, and the use of multidisciplinary approaches including non-opioid analgesics, interventional procedures, and behavioural therapies.

TPMT Polymorphism

– Thiopurine S-methyltransferase is an enzyme that metabolizes thiopurine drugs including azathioprine, 6-mercaptopurine, and thioguanine, which are used in the treatment of leukemia, inflammatory bowel disease, and autoimmune conditions. TPMT catalyzes the S-methylation of thiopurines, diverting them away from the formation of active thioguanine nucleotides that cause cytotoxicity. Genetic polymorphisms in the TPMT gene result in variable enzyme activity, with approximately 10 percent of the popuopulation having intermediate activity and approximately 0.3% having low or absent activity; patients with intermediate or low TPMT activity are at increased risk of severe myelosuppression when treated with standard doses of thiopurines, and pre-treatment TPMT genotyping or phenotyping is recommended to guide initial dosing and minimise the risk of haematological toxicity.

Tranquillisers

– A class of drugs that have a calming or sedative effect, reducing anxiety, tension, and agitation. Historically classified as major tranquillisers (antipsychotics/neuroleptics—for psychosis) and minor tranquillisers (anxiolytics—for anxiety). The term is now less common in clinical practice. Major tranquillisers (antipsychotics): first-generation/typical (chlorpromazine, haloperidol, fluphenazine—block D_2 receptors), second-generation/atypical (risperidone, olanzapine, quetiapine, aripiprazole, clozapine—block D_2 and 5-HT_{2A} receptors). Used for schizophrenia, bipolar mania, agitation, and treatment-resistant depression. Minor tranquillisers (anxiolytics): benzodiazepines (diazepam, lorazepam, alprazolam—enhance $GABA_A$ activity), buspirone (5-HT_{1A} partial agonist—non-sedating, no dependence), and β-blockers (propranolol—for performance anxiety, somatic symptoms). Adverse effects: (1) Benzodiazepines—dependence, tolerance, withdrawal (rebound anxiety, insomnia, seizures), sedation, cognitive impairment, falls. (2) Antipsychotics—extrapyramidal side effects (dystonia, parkinsonism, akathisia, tardive dyskinesia), weight gain, metabolic syndrome, hyperprolactinaemia, QT prolongation. Non-pharmacological alternatives: cognitive behavioural therapy, relaxation techniques, and mindfulness.

Triamcinolone

– A synthetic corticosteroid with potent anti-inflammatory and immunosuppressive properties. Triamcinolone is intermediate-acting (duration 12–36 hours) and is available in multiple formulations: topical (cream, ointment, lotion—0.025–0.5% for dermatological conditions: eczema, psoriasis, contact dermatitis, lichen planus); intralesional (Kenalog—intra-articular injection for inflammatory arthritis—rheumatoid, psoriatic, gout; intralesional for hypertrophic scars, keloids, alopecia areata, cystic acne); inhaled (for asthma—delivers active drug to the lungs, but less commonly used now); and intramuscular (for systemic inflammatory conditions—not recommended due to adrenal suppression risk). Triamcinolone acetonide is the most widely used formulation. Side effects: depend on dose, route, and duration. Topical: skin atrophy, striae, telangiectasias, perioral dermatitis, and systemic absorption with prolonged use over large areas. Intralesional: post-injection flare, infection, tendon rupture (avoid direct injection into tendons), fat atrophy, and depigmentation. Systemic: adrenal suppression, osteoporosis, hyperglycaemia, hypertension, immunosuppression, weight gain, Cushing syndrome, and growth suppression in children.

Trimethoprim-sulfamethoxazole

– A fixed-dose combination of trimethoprim and sulfamethoxazole that synergistically inhibits folate synthesis. TMP-SMX is indicated for urinary tract infections, Pneumocystis jirovecii pneumonia, and community-acquired pneumonia. The combination is bactericidal against many gram-positive and gram-negative organisms. Common adverse effects include hypersensitivity reactions, bone marrow suppression, and hyperkalemia. The drug is contraindicated in patients with folate deficiency and in late pregnanancy; resistance is increasingly common among many bacterial species, particularly

E. coli, and susceptibility testing is essential before prescribing TMP-SMX for urinary tract infections.

Triptan

– A class of medications used for acute migraine treatment, acting as 5-HT1B/1D receptor agonists. 5-HT1B receptors are located on cranial blood vessels; activation causes vasoconstriction of dilated meningeal arteries. 5-HT1D receptors are located on presynaptic trigeminal nerve terminals; activation inhibits the release of vasoactive neuropeptides (calcitonin gene-related peptide, substance P), reducing neurogenic inflammation and pain transmission. Available triptans: sumatriptan (first, available as oral, subcutaneous, intranasal, and rectal), rizatriptan, zolmitriptan, naratriptan, eletriptan, almotriptan, and frovatriptan. Contraindications: ischemic heart disease, Prinzmetal angina, uncontrolled hypertension, hemiplegic migraine, and basilar migraine. Adverse effects: chest tightness/pressure (may mimic cardiac ischemia but typically benign), paresthesias, dizziness, and injection site reactions. Triptans should be taken early in the migraine attack for optimal efficacy.

Typical Antipsychotics

– Typical antipsychotics, also known as first-generation antipsychotics, primarily block dopamine D2 receptors in the mesolimbic pathway, reducing positive symptoms of schizophrenia including hallucinations, delusions, and disorganized thinking. They are classified by potency: high-potency agents including haloperidol and fluphenazine require lower doses and have more extrapyramidal side effects; low-potency agents including chlorpromazine and thioridazine require higher doses and have more anticholinergic side effects including sedation, dry mouth, constipation, and postural hypotension; all typical antipsychotics carry a risk of tardive dyskinesia, a potentially irreversible movement disorder characterized by involuntary, repetitive movements of the face, mouth, and extremities, with higher potency agents posing a greater risk.

Tyrosine Kinase Inhibitor

– A class of targeted cancer therapy drugs that inhibit tyrosine kinases, enzymes that phosphorylate tyrosine residues on proteins and activate signaling pathways involved in cell growth, proliferation, differentiation, and survival. Tyrosine kinase mutations or overexpression drive many cancers. TKIs are small molecules (oral) or monoclonal antibodies (IV). Examples: imatinib — BCR-ABL inhibitor for chronic myeloid leukemia (first TKI approved, revolutionized CML treatment); erlotinib and gefitinib — EGFR inhibitors for non-small cell lung cancer with EGFR mutations; sorafenib and sunitinib — multi-kinase inhibitors (VEGFR, PDGFR, RAF) for renal cell carcinoma and hepatocellular carcinoma; osimertinib — third-generation EGFR TKI for T790M mutant NSCLC. Adverse effects vary by target: hypertension (VEGF inhibitors), rash and diarrhea (EGFR inhibitors), fatigue, and hand-foot skin reaction.

Unguentum

– See Ointment.

Uterotonic

– A medication that stimulates uterine contraction, used for induction/augmentation of labor, prevention and treatment of postpartum hemorrhage, and medical termination of pregnancy. Oxytocin — a synthetic analog of the endogenous hormone, stimulates oxytocin receptors on uterine smooth muscle, producing rhythmic contractions, administered IV for labor induction/augmentation; also used IV or IM for active management of the third stage of labor to prevent postpartum hemorrhage. Ergot alkaloids (ergometrine, methylergonovine) — cause sustained uterine tetanic contraction by acting on alpha-adrenergic and serotonin receptors, used IM/IV for postpartum hemorrhage, contraindicated in hypertension and preeclampsia. Prostaglandins: carboprost (PGF2-alpha) — IM for refractory postpartum hemorrhage; misoprostol (PGE1 analog) — oral, sublingual, rectal, or vaginal for cervical ripening, labor induction, and postpartum hemorrhage; dinoprostone (PGE2) — vaginal insert for cervical ripening.

Valproate (Sodium Valproate)

– Valproate (valproic acid, sodium valproate, divalproex sodium) is a broad-spectrum antiepileptic drug effective against all seizure types (generalised tonic-clonic, absence, myoclonic, and partial seizures). It is also used as a mood stabiliser in bipolar disorder (particularly for acute mania and maintenance) and for prophylaxis of migraine headaches. Mechanism: multiple actions including enhancement of GABAergic transmission (by increasing GABA synthesis via glutamic acid decarboxylase activation and inhibiting GABA transaminase), blockade of voltage-gated sodium channels, and inhibition of T-type calcium channels. Pharmacokinetics: well absorbed orally, highly protein-bound (90%), extensively metabolized in the liver (glucuronidation, beta-oxidation, and CYP-mediated oxidation), half-life 9-16 hours. Adverse effects: dose-related (tremor, weight gain, alopecia, sedation, thrombocytopenia, elevated liver enzymes), idiosyncratic (hepatotoxicity — highest risk in children under 2 years on polytherapy, pancreatitis), teratogenic (neural tube defects, spina bifida — absolute contraindication in pregnancy unless no alternative, requiring high-dose folic acid supplementation), and polycystic ovary syndrome with long-term use. Monitoring: serum levels (therapeutic range 50-100 mcg/mL), LFTs, CBC, and pregnancy testing. Drug interactions: inhibits CYP2C9, displaces protein-bound drugs (phenytoin).

Vancomycin

– A glycopeptide antibiotic that inhibits cell wall synthesis by binding to D-Ala-D-Ala terminal of peptidoglycan precursors. Vancomycin is the drug of choice for MRSA infections and serious gram-positive infections. It is indicated for bacteremia, endocarditis, pneumonia, and complicated skin infections. The drug is administered intravenously every 8-12 hours, with dosing guided by therapeutic drug monitoring to achieve target AUC/MIC ratios. Common adverse effects include nephrotoxicity, ototoxiototoxicity, and red man syndrome (an infusion-related reaction caused by histamine release), and therapeutic drug monitoring of trough concentrations is recommended to optimise efficacy and minimise the risk of nephrotoxicity and ototoxicity; vancomycin dosing is based on actual body weight and requires adjustment for impaired renal function.

Vancomycin-resistant Enterococcus

– Enterococci resistant to vancomycin through acquisition of vanA or vanB gene clusters encoding altered cell wall precursors. VRE is a significant cause of healthcare-associated infections, particularly in immunocompromised patients. Treatment options include linezolid, daptomycin, and quinupristin-dalfopristin (for E. faecium only). VRE infections are associated with increased mortality and healthcare costs. Infection prevention strategies include hand hygiene, contact precautions, and environmeonmental decontamination, and antimicrobial stewardship to preserve the effectiveness of remaining active agents.

Vaughan Williams Classification

– The Vaughan Williams classification categorizes antiarrhythmic drugs into four classes based on their primary mechanism of action. Class I agents block sodium channels and slow conduction velocity, further subdivided into IA with moderate sodium channel blockade and action potential prolongation including quinidine and procainamide, IB with fast sodium channel binding and action potential shortening including lidocaine and mexiletine, and IC with slow sodium channel binding and minimal effect onon action potential duration; class II agents are beta-blockers that reduce sympathetic activity and slow AV conduction; class III agents block potassium channels, prolonging repolarisation; and class IV agents are calcium channel blockers that slow AV conduction; while clinically useful, the classification has limitations as many drugs have multiple mechanisms of action and it does not account for all clinically important antiarrhythmic drugs.

Verapamil

– A phenylalkylamine calcium channel blocker used as an antiarrhythmic (class IV) and antihypertensive agent. Verapamil blocks L-type calcium channels in cardiac myocytes and vascular smooth muscle, with greater selectivity for the heart (particularly the sinoatrial node and atrioventricular node) than dihydropyridine CCBs (nifedipine, amlodipine), making it effective for supraventricular tachycardias (AV nodal reentrant tachycardia, atrial fibrillation with rapid ventricular response), angina (stable and vasospastic/Prinzmetal), and hypertension. It causes negative chronotropy (slows SA node rate), negative dromotropy (slows AV conduction — prolonged PR interval), and negative inotropy (reduced contractility), contraindicating its use in heart failure with reduced ejection fraction, sick sinus syndrome, and second/third-degree AV block (without a pacemaker). Adverse effects: constipation (most common), hypotension, bradycardia, edema, headache, and gingival hyperplasia. Drug interactions: verapamil is a CYP3A4 substrate and inhibitor, contraindicating concomitant use with strong CYP3A4 inducers; also increases levels of digoxin, statins, and immunosuppressants.

Viagra (Sildenafil)

– Sildenafil is a phosphodiesterase type 5 (PDE5) inhibitor used primarily for erectile dysfunction (ED) and pulmonary arterial hypertension (PAH, as Revatio). Mechanism: PDE5 degrades cGMP in penile corpus cavernosum smooth muscle; sildenafil inhibits PDE5, increasing cGMP levels and enhancing nitric oxide (NO)-mediated vasodilation and smooth muscle relaxation, improving penile blood flow during sexual stimulation. For PAH: sildenafil dilates pulmonary vasculature, reducing pulmonary vascular resistance. Pharmacokinetics: oral, onset within 30-60 minutes, duration 4-6 hours, metabolized by CYP3A4. Adverse effects: headache, flushing, dyspepsia, nasal congestion, visual disturbances (blue tinged vision/cyanopsia due to PDE6 inhibition in retinal photoreceptors), and myalgia. Serious: priapism (erection >4 hours — medical emergency), hypotension (especially when combined with nitrates — absolute contraindication due to risk of severe hypotension and syncope), and non-arteritic anterior ischemic optic neuropathy (NAION, rare). Contraindications: concurrent nitrate therapy, severe hepatic impairment, severe hypotension, and recent stroke or MI.

Vibrio vulnificus infections

– Infections caused by Vibrio vulnificus, a gram-negative bacterium found in warm seawater and raw seafood. V. vulnificus can cause wound infections, septicemia, and gastroenteritis. The most severe form is necrotizing fasciitis and primary septicemia, which can be fatal. Risk factors include liver disease, immunocompromise, and iron overload states. Treatment requires doxycycline combined with a third-generation cephalosporin. Early wound care and antibiotic therapy are essential. The organism hahas a high case-fatality rate, and prevention involves avoiding consumption of raw or undercooked seafood, particularly in patients with liver disease or immunocompromising conditions.

Volume of Distribution

– A theoretical pharmacokinetic parameter that relates the total amount of drug in the body to the plasma concentration, calculated as Vd equals amount of drug in the body divided by plasma concentration. It represents the apparent volume into which a drug distributes and is not a true anatomical volume. A small volume of distribution, such as 3 to 5 liters, indicates that the drug remains primarily in the plasma and does not extensively distribute into tissues, as seen with drugs that are highly protein-bound and extensively distributed to tissues; understanding volume of distribution is critical for calculating loading doses, interpreting drug concentrations, and predicting the extent of drug distribution to the site of action.

Voriconazole

– A triazole antifungal agent used in the treatment of invasive aspergillosis, candidemia, and other serious fungal infections. Voriconazole inhibits the fungal cytochrome P450 enzyme lanosterol 14-alpha demethylase, preventing ergosterol synthesis and disrupting fungal cell membrane integrity. The drug has broad-spectrum activity against Aspergillus, Candida, Cryptococcus, and other fungi. A key advantage of voriconazole over itraconazole is its excellent oral bioavailability of approximately 96%and has a long half-life allowing for twice-daily oral dosing after an initial intravenous loading regimen; voriconazole is associated with visual disturbances (photopsia, altered color perception), hepatotoxicity, and QT prolongation, and requires therapeutic drug monitoring due to non-linear pharmacokinetics and significant inter-individual variability in plasma concentrations; drug interactions are numerous due to CYP2C19 and CYP3A4 inhibition.

Warfarin

– A vitamin K antagonist that inhibits the enzyme vitamin K epoxide reductase, preventing the recycling of vitamin K and thereby blocking the gamma-carboxylation of clotting factors II, VII, IX, and X as well as proteins C and S. This post-translational modification is essential for the calcium-binding activity of these factors. Warfarin has a slow onset of action, requiring several days to achieve therapeutic anticoagulation, as existing clotting factors must be cleared before the effeffect is fully established; warfarin has a narrow therapeutic index and requires regular INR monitoring to maintain the INR within the target range of 2-3 for most indications, with dose adjustments based on numerous factors including dietary vitamin K intake, drug interactions, and genetic polymorphisms in CYP2C9 and VKORC1 that affect warfarin metabolism and pharmacodynamics; bleeding is the most important adverse effect and requires prompt reversal with vitamin K, fresh frozen plasma, or prothrombin complex concentrate depending on the severity of bleeding and the INR.

Weight Loss Medications

– Pharmacologic agents approved for the treatment of obesity when used in conjunction with lifestyle modifications. These medications work through various mechanisms, including appetite suppression through GLP-1 receptor agonists such as semaglutide, inhibition of fat absorption with orlistat, or combination therapies targeting multiple neuropeptide pathways. They are indicated for individuals with a body mass index of 30 or higher, or 27 or higher with obesity-related comorbidities, and require ongoing medical supervision due to potential side effects and the need for monitoring.

Withdrawal Symptoms

– See Withdrawal Syndrome. used primarily for erectile dysfunction (ED) and pulmonary arterial hypertension (PAH, as Revatio). Mechanism: PDE5 degrades cGMP in penile corpus cavernosum smooth muscle; sildenafil inhibits PDE5, increasing cGMP levels and enhancing nitric oxide (NO)-mediated vasodilation and smooth muscle relaxation, improving penile blood flow during sexual stimulation. For PAH: sildenafil dilates pulmonary vasculature, reducing pulmonary vascular resistance. Pharmacokinetics: oral, onset within 30-60 minutes, duration 4-6 hours, metabolized by CYP3A4. Adverse effects: headache, flushing, dyspepsia, nasal congestion, visual disturbances (blue tinged vision/cyanopsia due to PDE6 inhibition in retinal photoreceptors), and myalgia. Serious: priapism (erection >4 hours — medical emergency), hypotension (especially when combined with nitrates — absolute contraindication due to risk of severe hypotension and syncope), and non-arteritic anterior ischemic optic neuropathy (NAION, rare). Contraindications: concurrent nitrate therapy, severe hepatic impairment, severe hypotension, and recent stroke or MI.

Withdrawal Syndrome

– A predictable set of signs and symptoms that occur when a drug is discontinued or the dose is reduced after chronic use, reflecting the physiological rebound from drug adaptation. Specific withdrawal syndromes: Opioid withdrawal (not life-threatening but intensely uncomfortable): yawning, lacrimation, rhinorrhea, piloerection (cold turkey), mydriasis, abdominal cramps, diarrhea, nausea, vomiting, muscle aches, and drug craving. Alcohol withdrawal (potentially life-threatening): tremor, anxiety, autonomic hyperactivity (tachycardia, hypertension, diaphoresis), insomnia, seizures (6-48 hours after last drink), and delirium tremens (48-96 hours later: agitation, hallucinations, autonomic instability, and altered mental status). Benzodiazepine withdrawal: anxiety, insomnia, tremor, palpitations, and seizures (may be delayed with long-acting agents). SSRI discontinuation syndrome: dizziness, paresthesias, nausea, headache, and irritability (not life-threatening). Management involves gradual tapering (detoxification), supportive care, and pharmacotherapy (benzodiazepines for alcohol withdrawal, methadone/buprenorphine for opioid withdrawal).

Xanthine Oxidase Inhibitor

– A class of medications that inhibit xanthine oxidase, the enzyme that catalyzes the final steps of purine metabolism: hypoxanthine to xanthine and xanthine to uric acid. By inhibiting this enzyme, XOIs reduce uric acid production and lower serum uric acid levels, used for management of chronic gout and tumor lysis syndrome. Allopurinol (first-line, 40+ years of clinical use): a purine analog that acts as a competitive substrate and is metabolized to oxypurinol (active metabolite with a long half-life of 18-30 hours). Dosing starts at 100 mg daily, titrated to target serum urate <6 mg/dL. Febuxostat: a non-purine selective XOI, metabolized in the liver, dose 40-80 mg daily, preferred in patients with allopurinol hypersensitivity or renal impairment. Cardiotoxicity warning: febuxostat has a higher risk of cardiovascular death compared to allopurinol in patients with preexisting CVD. Adverse effects: allopurinol hypersensitivity syndrome (DRESS, Stevens-Johnson syndrome, most severe, risk increased in HLA-B*5801 carriers, especially Han Chinese and African Americans), rash, and GI upset.

Xylometazoline

– A direct-acting sympathomimetic decongestant and vasoconstrictor, available as a topical intranasal spray or drops (0.05% for children 2-12 years, 0.1% for adults). Mechanism: xylometazoline is an imidazoline derivative that acts as a potent agonist at alpha-1 and alpha-2 adrenergic receptors on nasal blood vessels. Activation of alpha-1 receptors causes vasoconstriction of the cavernous venous sinuses (erectile tissue of the nasal turbinates), reducing nasal mucosal edema and congestion. It has a rapid onset of action (within 5-10 minutes) and a duration of 8-10 hours. Indications: symptomatic relief of nasal congestion associated with the common cold, allergic rhinitis (seasonal and perennial), and sinusitis; also used as a vasoconstrictor before nasal examination or procedures. It is used as a hemostatic agent in epistaxis (anterior nasal packing with cotton wool soaked in xylometazoline). Contraindications: narrow-angle glaucoma, transsphenoidal hypophysectomy, rhinitis sicca, and concomitant use with MAOIs (risk of hypertensive crisis). Adverse effects: local (nasal dryness, burning, stinging, sneezing, epistaxis), systemic (rare at recommended doses — hypertension, palpitations, headache, insomnia, tremor), and rhinitis medicamentosa (rebound nasal congestion from prolonged use >7-10 days due to downregulation of adrenergic receptors and secondary vasodilation). Patients should be warned to limit use to 3-5 days and not to exceed 7 days.

Yohimbine

– An indole alkaloid derived from the bark of the West African yohimbe tree (Pausinystalia yohimbe), historically used as an aphrodisiac and for erectile dysfunction. Mechanism: yohimbine is a selective alpha-2 adrenergic receptor antagonist (blocks presynaptic alpha-2 receptors, increasing norepinephrine release from sympathetic nerve terminals and increasing central sympathetic outflow and peripheral norepinephrine levels). The increased norepinephrine causes vasodilation in the corpus cavernosum (through nitric oxide-independent relaxation of trabecular smooth muscle) and increased penile blood flow, but the evidence for efficacy in erectile dysfunction is weak and its use is not recommended by most guidelines (AUA, EAU). Other claimed (unproven) uses: weight loss (increases lipolysis by blocking alpha-2 receptors on adipocytes), athletic performance enhancement, and orthostatic hypotension. Adverse effects: anxiety, agitation, hypertension, tachycardia, palpitations, headache, dizziness, nausea, insomnia, and panic attacks (contraindicated in panic disorder, PTSD, and bipolar disorder). Yohimbine is not FDA-approved for any indication and is classified as a dietary supplement in the US, which means its manufacture and quality are not regulated. The therapeutic index is narrow, and serious adverse effects (hypertensive crisis, arrhythmias, myocardial infarction) have been reported at high doses or in susceptible individuals (CYP2D6 poor metabolisers). Due to lack of proven efficacy, an unfavourable safety profile, and the availability of effective, well-studied PDE5 inhibitors (sildenafil, tadalafil), yohimbine is rarely used in modern medical practice.

Zalcitabine

– A nucleoside reverse transcriptase inhibitor (NRTI) that was previously used in combination antiretroviral therapy for HIV-1 infection but is now rarely used due to the availability of more effective and better-tolerated agents. Zalcitabine is a cytidine analogue that inhibits viral reverse transcriptase by competing with deoxycytidine triphosphate and causing DNA chain termination. The drug has a relatively short half-life of approximately 1-2 hours, requiring dosing three times daily. The primprimary adverse effect was peripheral neuropathy, which occurred in a significant proportion of patients and was the dose-limiting toxicity; pancreatitis and lactic acidosis were also reported. Zalcitabine was withdrawn from the market in 2006 due to limited clinical use and the availability of safer, more effective NRTIs.

Zidovudine (AZT)

– Zidovudine (azidothymidine, AZT, ZDV) is a nucleoside analogue reverse transcriptase inhibitor (NRTI), the first antiretroviral drug approved for the treatment of HIV infection (FDA approval 1987). Mechanism: zidovudine is a thymidine analogue that is phosphorylated intracellularly to zidovudine triphosphate by cellular kinases. Zidovudine triphosphate competes with endogenous thymidine triphosphate for incorporation into the growing viral DNA chain by HIV-1 reverse transcriptase. Because zidovudine lacks the 3'-hydroxyl group needed for further chain elongation, its incorporation results in premature termination of the viral DNA chain. Zidovudine has a higher affinity for HIV-1 RT than for human DNA polymerases (particularly DNA polymerase gamma, the mitochondrial enzyme), which accounts for its relative selectivity for viral inhibition. The historical landmark ACTG 019 trial (1989) showed that zidovudine monotherapy delayed disease progression in asymptomatic HIV-infected patients. Zidovudine was the first drug shown to reduce maternal-fetal HIV transmission (ACTG 076, 1994: zidovudine given antepartum and intrapartum to the mother and then for 6 weeks to the neonate reduced transmission by 67%). Combination therapy (HAART, highly active antiretroviral therapy) replaced monotherapy in 1996 due to rapid development of resistance. Current use: zidovudine is still used in fixed-dose combinations (Combivir = zidovudine + lamivudine) and in resource-limited settings, and for prevention of mother-to-child transmission. Resistance: mutations at codons 41, 67, 70, 210, 215, and 219 of HIV-1 RT (thymidine analogue mutations, TAMs) confer zidovudine resistance. TAMs are selected by zidovudine and stavudine and confer cross-resistance to all NRTIs. Adverse effects: bone marrow suppression (macrocytic anemia, neutropenia — the dose-limiting toxicity, more common with advanced HIV, monitored by serial CBC), myopathy (mitochondrial toxicity due to inhibition of DNA polymerase gamma, causing skeletal muscle weakness and wasting, elevated CK, myalgia, and histologic findings of ragged red fibers on muscle biopsy), lactic acidosis and hepatic steatosis (mitonuchondrial toxicity, hepatic steatosis and severe hepatomegaly with lactic acidosis, a rare but potentially fatal complication), headache, nausea, vomiting, and skin rash. Zidovudine should be avoided in patients with low baseline hemoglobin (<9 g/dL) or neutrophil count (<750 cells/microlitre)., the enzyme that catalyzes the final steps of purine metabolism: hypoxanthine to xanthine and xanthine to uric acid. By inhibiting this enzyme, XOIs reduce uric acid production and lower serum uric acid levels, used for management of chronic gout and tumor lysis syndrome. Allopurinol (first-line, 40+ years of clinical use): a purine analog that acts as a competitive substrate and is metabolized to oxypurinol (active metabolite with a long half-life of 18-30 hours). Dosing starts at 100 mg daily, titrated to target serum urate <6 mg/dL. Febuxostat: a non-purine selective XOI, metabolized in the liver, dose 40-80 mg daily, preferred in patients with allopurinol hypersensitivity or renal impairment. Cardiotoxicity warning: febuxostat has a higher risk of cardiovascular death compared to allopurinol in patients with preexisting CVD. Adverse effects: allopurinol hypersensitivity syndrome (DRESS, Stevens-Johnson syndrome, most severe, risk increased in HLA-B*5801 carriers, especially Han Chinese and African Americans), rash, and GI upset.

Chapter 28

Physiology

Active Transport

– Movement of substances across cell membranes against their concentration gradient, requiring energy input, typically from ATP hydrolysis. Primary active transport directly uses ATP: the Na+/K+ ATPase pumps 3 Na+ out and 2 K+ in per ATP, maintaining the electrochemical gradient essential for neuronal signaling, cell volume regulation, and secondary active transport. Other examples include the Ca2+ ATPase, H+/K+ ATPase (gastric acid secretion), and ABC transporters. Secondary active transport uses the Na+ gradient established by the Na+/K+ ATPase to cotransport or countertransport other solutes (e.g., SGLT1 for glucose and amino acid cotransport in the intestine, Na+/Ca2+ exchanger in cardiac muscle).

Aldosterone

– A mineralocorticoid hormone produced by the zona glomerulosa of the adrenal cortex, the final effector of the renin-angiotensin-aldosterone system (RAAS). Aldosterone acts on principal cells of the collecting duct, increasing ENaC (epithelial sodium channel) expression on the apical membrane and Na+/K+ ATPase on the basolateral membrane, promoting sodium reabsorption and potassium secretion. Water follows sodium osmotically, expanding extracellular fluid volume and increasing blood pressure. Aldosterone secretion is stimulated by angiotensin II, hyperkalaemia, and ACTH, and inhibited by ANP and hypokalaemia. Excessive aldosterone (Conn syndrome) causes hypertension with hypokalaemia, metabolic alkalosis, and suppressed renin.

Allosteric Regulation

– The regulation of enzyme activity by the binding of an effector molecule at a site other than the active site (allosteric site), causing a conformational change that alters the enzyme's activity. Allosteric enzymes typically have multiple subunits with cooperative binding kinetics (sigmoidal rather than hyperbolic velocity curves). Positive allosteric effectors (activators) shift the curve leftward (increase affinity for substrate, decrease Km), while negative allosteric effectors (inhibitors) shift the curve rightward (decrease affinity, increase Km). Classic examples include oxygen binding to hemoglobin (cooperative, with 2,3-BPG as a negative allosteric effector), aspartate transcarbamoylase (ATCase) in pyrimidine synthesis, and feedback inhibition in metabolic pathways.

Amino Acid Catabolism

– The breakdown of amino acids after protein digestion, with the amino group removed and the carbon skeleton metabolized for energy or gluconeogenesis/ketogenesis. Transamination: amino groups are transferred to alpha-ketoglutarate, forming glutamate, by aminotransferases requiring pyridoxal phosphate (PLP, vitamin B6) as a cofactor. ALT (alanine aminotransferase) and AST (aspartate aminotransferase) are clinically important: elevated levels indicate liver damage (ALT is more liver-specific) or myocardial injury (AST). Following transamination, the alpha-keto acids enter gluconeogenesis or the TCA cycle, and ammonium is converted to urea in the urea cycle.

Baroreceptor Reflex

– The rapid neural mechanism for short-term blood pressure regulation, involving baroreceptors (mechanoreceptors) in the carotid sinus (innervated by CN IX) and aortic arch (innervated by CN X). When blood pressure rises, baroreceptors increase their firing rate, signaling the cardiovascular center in the medulla. The response: increased parasympathetic (vagal) output to the heart (decreasing HR) and decreased sympathetic output (decreasing HR, contractility, and peripheral resistance), lowering blood pressure towards normal. The reflex is essential for maintaining blood pressure stability during postural changes and blood volume fluctuations.

Basal Metabolic Rate (BMR)

– The minimum energy expenditure required to maintain vital physiological functions at rest in a thermoneutral environment, measured after 12 hours of fasting and 8 hours of sleep. BMR accounts for approximately 60–75% of total daily energy expenditure in sedentary individuals. It is influenced by body composition (lean mass is the primary determinant), age (declines 1–2% per decade after age 20), sex (5–10% higher in males), thyroid status, and genetics. BMR is estimated using predictive equations (Harris-Benedict, Mifflin-St Jeor) or measured by indirect calorimetry. Clinical relevance: assessing nutritional needs, managing obesity, and diagnosing hyper- or hypometabolic states.

Bile Salt Emulsification

– Bile salts (derived from cholesterol in the liver) are amphipathic molecules with hydrophobic and hydrophilic regions that act as biological detergents, emulsifying dietary lipids into smaller droplets (micelles). This increases the surface area of lipid substrate available for pancreatic lipase action. Bile salts form mixed micelles with phospholipids, cholesterol, and fat-soluble vitamins (A, D, E, K), facilitating their transport across the unstirred water layer to the enterocyte brush border for absorption. Bile salts are reabsorbed in the terminal ileum via the apical sodium-dependent bile acid transporter (ASBT) and returned to the liver through the enterohepatic circulation, recycling 6-10 times daily.

Blood Pressure

– The force exerted by blood against arterial walls, measured in millimeters of mercury (mmHg). Systolic pressure (peak during ventricular ejection, normally approximately 120 mmHg) and diastolic pressure (minimum during ventricular relaxation, normally approximately 80 mmHg). Mean arterial pressure (MAP) is approximately diastolic + one-third pulse pressure (approximately 93 mmHg). MAP is the average pressure driving blood through the systemic circulation. Blood pressure depends on cardiac output and systemic vascular resistance (MAP = CO x SVR). It is regulated by short-term mechanisms (baroreceptor reflex, chemoreceptor reflex, and hormonal responses: ADH, adrenaline) and long-term mechanisms (renin-angiotensin-aldosterone system and renal body fluid regulation by the kidneys).

Bohr Effect

– The decrease in hemoglobin's oxygen affinity (right shift of the dissociation curve) caused by decreased pH (increased H+) and increased PCO2. In metabolically active tissues, CO2 production and lactic acid accumulation lower the local pH, causing hemoglobin to release more oxygen precisely where it is needed most. The mechanism involves protonation of hemoglobin histidine residues, which stabilize the T (tense, low-affinity) state. The Bohr effect is quantified as approximately 0.48 pH units per log10 PCO2 change and is clinically significant in conditions that alter acid-base status, enhancing oxygen delivery in exercising muscle and hypoxic tissues.

Calciferol

– See Vitamin D: Calcitriol.

Calorie

– A unit of energy measurement used in nutrition to quantify the energy content of foods and energy expenditure of the body. One kilocalorie (kcal, commonly called a Calorie with capital C) equals the energy required to raise 1 kilogram of water by 1 degree Celsius. The body derives energy from macronutrients: carbohydrates and proteins provide 4 kcal/g, fats provide 9 kcal/g, and alcohol provides 7 kcal/g. Total daily energy expenditure comprises basal metabolic rate (60–75%), thermic effect of food (10%), and physical activity (15–30%). Caloric balance determines body weight: excess calories are stored as fat; caloric deficit leads to weight loss.

Capillary

– The smallest blood vessel in the body, forming a microscopic network (capillary bed) that connects arterioles to venules. Capillary walls consist of a single layer of endothelial cells (frequently fenestrated) and a basement membrane, allowing exchange of oxygen, carbon dioxide, nutrients, wastes, and other substances between blood and interstitial fluid. Types:

continuous (most common, in muscle, skin, lung, CNS–blood-brain barrier), fenestrated (kidney, intestine, endocrine glands, with pores for rapid filtration), and sinusoidaldiscontinuous (liver, bone marrow, spleen with large gaps allowing passage of cells and large molecules). Starling forces–hydrostatic and oncotic pressures–govern fluid movement across capillary walls.

Cardiac Cycle

– The sequence of mechanical and electrical events occurring during one heartbeat, lasting approximately 0.8 seconds at 75 bpm. The cycle has two main phases: systole (ventricular contraction and ejection) and diastole (ventricular relaxation and filling). Isovolumetric contraction: all valves closed, ventricles contract, pressure rises rapidly. Ventricular ejection: aortic/pulmonary valves open, blood is ejected (stroke volume approximately 70 mL). Isovolumetric relaxation: all valves closed, ventricles relax, pressure falls rapidly below atrial pressure. Ventricular filling (diastole): AV valves open, passive filling occurs (80 percent), followed by atrial contraction (20 percent, the "atrial kick"). Heart sounds: S1 (AV valve closure), S2 (semilunar valve closure), S3 (rapid ventricular filling, normal in children, abnormal in adults indicating heart failure), S4 (atrial contraction against a stiff ventricle).

Cardiac Output

– The volume of blood pumped by each ventricle per minute, calculated as heart rate multiplied by stroke volume (CO = HR x SV). Normal resting CO is approximately 5 liters per minute (70 mL stroke volume x 70 bpm). Stroke volume depends on preload (Frank-Starling), afterload (resistance to ejection), and contractility (inotropic state). Heart rate is modulated by the autonomic nervous system: sympathetic stimulation increases HR (positive chronotropy) via norepinephrine acting on beta-1 receptors; parasympathetic (vagal) stimulation decreases HR (negative chronotropy) via acetylcholine acting on M2 receptors. Cardiac output increases during exercise up to 4-5 times resting levels through combined increases in HR and stroke volume, with trained athletes achieving higher maximal CO.

Cholecystokinin

– CCK is a peptide hormone produced by I-cells in the duodenal and jejinal mucosa in response to fatty acids (long-chain fatty acids are the most potent stimuli) and amino acids in the chyme. CCK's primary actions: stimulates gallbladder contraction (promoting bile release for fat emulsification and absorption), stimulates pancreatic enzyme secretion (acinar cells), relaxes the sphincter of Oddi, slows gastric emptying (reducing the rate of acid delivery to the duodenum), and promotes satiety (acting on the hypothalamus via CCK-B receptors). CCK is also found in the brain as a neurotransmitter. Its secretion is inhibited by trypsin and chymotrypsin (negative feedback via trypsin-sensitive CCK-releasing factor). CCK analogues are used as diagnostic agents in gallbladder imaging.

Cholesterol Synthesis

– Occurs in all nucleated cells (primarily liver and intestine) through a complex 30+ step pathway beginning with acetyl-CoA. Key steps: three acetyl-CoA molecules condense to form HMG-CoA, which is reduced to mevalonate by HMG-CoA reductase (the rate-limiting enzyme, located in the endoplasmic reticulum). Mevalonate is converted through multiple steps to squalene, then lanosterol, then cholesterol. HMG-CoA reductase is the target of statins (competitive inhibitors), the most widely prescribed cholesterol-lowering drugs. Cholesterol is the precursor for bile salts, steroid hormones (cortisol, aldosterone, sex hormones), and vitamin D, and is a key component of cell membranes where it modulates fluidity.

Competitive Inhibition

– A type of enzyme inhibition where the inhibitor (I) competes with the substrate (S) for binding to the enzyme's active site. The inhibitor structurally resembles the substrate and binds reversibly to the free enzyme (E), forming an EI complex that cannot proceed to product. The apparent Km increases (lower apparent affinity for substrate) while Vmax remains unchanged (since the inhibition can be overcome by sufficiently high substrate concentrations). On a Lineweaver-Burk plot: lines intersect on the y-axis (same 1/Vmax) but have different slopes and x-intercepts (increasing apparent Km). Competitive inhibition is reversible and clinically exploited by enzyme inhibitors such as statins (HMG-CoA reductase), ACE inhibitors, and many chemotherapeutic agents that resemble natural substrates.

Dead Space

– The volume of ventilated air that does not participate in gas exchange. Anatomic dead space: the conducting airways (nose, pharynx, trachea, bronchi, terminal bronchioles) that warm, humidify, and filter inspired air but contain no alveoli for gas exchange. Approximately 150 mL in a 70 kg adult. Physiologic dead space includes anatomic dead space plus alveolar dead space (ventilated alveoli that are not perfused, as in pulmonary embolism). Normally, physiologic dead space equals anatomic dead space (approximately 150 mL), but in pulmonary embolism, hypotension, or lung disease, alveolar dead space increases, reducing effective alveolar ventilation. The dead space fraction (VD/VT) is measured by the Bohr equation and is clinically important in assessing ventilatory efficiency.

Diffusion

– The net movement of molecules from an area of higher concentration to lower concentration due to random thermal motion (Brownian motion). Fick's first law states that the rate of diffusion is proportional to the concentration gradient, surface area, and diffusion coefficient, and inversely proportional to membrane thickness: J = -P x A x (C1-C2)/d, where J is flux, P is the permeability coefficient, A is surface area, and d is thickness. Simple diffusion occurs for small, nonpolar molecules (O_2, CO_2, N_2, steroid hormones) that can dissolve in and pass through the lipid bilayer without a transporter. Factors affecting diffusion rate include: concentration gradient, temperature, molecular size, lipid solubility, membrane surface area, and membrane thickness. In biological systems, diffusion is effective over short distances (less than 100 micrometres), which is why capillaries are narrow and alveoli are thin-walled.

DNA Double Helix

– The molecular structure of DNA described by Watson and Crick in 1953, consisting of two antiparallel polynucleotide chains wound around each other in a right-handed helix. The sugar-phosphate backbone is on the exterior; nitrogenous bases project inward. Complementary base pairing: adenine (A) pairs with thymine (T) via two hydrogen bonds; guanine (G) pairs with cytosine (C) via three hydrogen bonds. The helix has a major groove (22 Angstrom) and minor groove (12 Angstrom) where proteins interact with specific base sequences without unwinding the DNA. This structure enables information storage, replication, transcription, and repair, and is the foundation of molecular genetics.

DNA Repair Mechanisms

– Cellular processes that correct DNA damage caused by replication errors, chemicals, radiation, and reactive oxygen species. Mismatch repair (MMR): corrects base-base mismatches and insertion-deletion loops that escape proofreading during replication. MutS (MSH2/MSH6 in eukaryotes) recognizes the mismatch; MutL (MLH1/PMS2) nicks the newly synthesized strand; exonuclease removes the error; DNA polymerase and ligase fill the gap. Deficiency causes Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC), the most common cause of hereditary colorectal cancer. Base excision repair (BER): repairs small, non-helix-distorting lesions such as oxidised or alkylated bases. Nucleotide excision repair (NER): repairs bulky, helix-distorting lesions such as UV-induced pyrimidine dimers. Double-strand break repair: homologous recombination (HR, error-free, uses sister chromatid as template) and non-homologous end joining (NHEJ, error-prone, active in G1 phase).

DNA Replication

– The semi-conservative process by which DNA duplicates before cell division, producing two identical daughter molecules each containing one original and one new strand. Key steps: (1) initiation at origins of replication (oriC in bacteria, multiple origins in eukaryotes), where helicase (DnaB in bacteria, MCM complex in eukaryotes) unwinds the double helix; (2) single-strand binding proteins (SSB) stabilize unwound DNA; (3) primase (DnaG) synthesizes short RNA primers; (4) DNA polymerase III (prokaryotes) or DNA polymerase delta/epsilon (eukaryotes) extends the primer, adding nucleotides to the 3' end; (5) on the leading strand, synthesis is continuous; on the lagging strand, synthesis is discontinuous, producing Okazaki fragments; (6) RNA primers are removed by DNA polymerase I (prokaryotes) or RNase H and FEN1 (eukaryotes); (7) DNA ligase seals the nicks between fragments. Telomerase extends telomeres to prevent chromosome shortening in germ cells and stem cells.

Electrolytes

– Ions in body fluids that carry an electrical charge and are essential for normal physiologic function. Major electrolytes: sodium ($Na+$, 135–145 mEq/L, primary extracellular cation, regulates fluid balance and membrane potential), potassium ($K+$, 3.5–5.0 mEq/L, primary intracellular cation, critical for cardiac and neuromuscular function), chloride (Cl-, 98–106 mEq/L, major extracellular anion, follows sodium), bicarbonate (HCO3-, 22–28 mEq/L, buffer in acid-base balance), calcium (Ca2+, 8.5–10.5 mg/dL, bone structure, muscle contraction, coagulation), magnesium (Mg2+, 1.7–2.2 mg/dL, enzyme cofactor, neuromuscular stability), and phosphate (PO4 3-, 2.5–4.5 mg/dL, energy metabolism, bone structure). Electrolyte imbalances cause a wide range of symptoms and are routinely measured by basic metabolic panel or comprehensive metabolic panel.

Electron Transport Chain

– A series of protein complexes (I-IV) and mobile carriers embedded in the inner mitochondrial membrane that transfer electrons from NADH and FADH2 to molecular oxygen (the final electron acceptor), generating a proton gradient used for ATP synthesis. Complex I (NADH dehydrogenase): accepts electrons from NADH, pumps 4 H+ across the membrane. Complex II (succinate dehydrogenase): accepts electrons from FADH2, does not pump protons. Coenzyme Q (ubiquinone) transfers electrons from Complexes I and II to Complex III; Complex III (cytochrome bc1) transfers electrons to cytochrome c (pumps 4 H+); Complex IV (cytochrome c oxidase) transfers electrons to O2, forming H2O (pumps 2 H+). The resulting proton gradient powers ATP synthase (Complex V), which produces approximately 2.5 ATP per NADH and 1.5 ATP per FADH2. Rotenone (Complex I inhibitor), antimycin A (Complex III), and cyanide/carbon monoxide (Complex IV) are potent toxins.

Endocytosis

– Process by which cells internalize extracellular material by engulfing it with the cell membrane, forming intracellular vesicles. Phagocytosis ("cell eating") involves engulfing large particles (>0.5 micrometers) such as bacteria or debris, forming a phagosome that fuses with lysosomes for digestion. This is performed by macrophages, neutrophils, and dendritic cells as part of innate immunity. Pinocytosis ("cell drinking") non-specifically engulfs extracellular fluid and dissolved solutes in small vesicles (approximately 0.1 micrometres). Receptor-mediated endocytosis (clathrin-dependent) is highly specific: ligands bind to surface receptors, which cluster in clathrin-coated pits, forming coated vesicles that are uncoated and delivered to early endosomes. Examples include LDL uptake (deficiency causes familial hypercholesterolaemia), transferrin-bound iron uptake, and growth factor internalisation.

Exocytosis

– Process by which intracellular vesicles fuse with the plasma membrane, releasing their contents into the extracellular space and incorporating membrane proteins/lipids into the cell membrane. The SNARE complex (v-SNARE on vesicles and t-SNARE on target membrane) mediates vesicle docking and fusion. Calcium-triggered exocytosis occurs in neurons (neurotransmitter release at synapses), endocrine cells (hormone secretion), and exocrine cells (enzyme secretion). Constitutive exocytosis continuously delivers membrane components and secreted proteins (collagen, matrix proteins) in all cells. Regulated exocytosis stores secretory vesicles until a signal (typically increased cytosolic Ca2+) triggers fusion. Defects in exocytosis underlie diseases such as botulism (toxin cleaves SNARE proteins) and familial haemophagocytic lymphohistiocytosis.

Facilitated Diffusion

– Passive transport of molecules across cell membranes via specific transmembrane protein carriers or channels, moving substances down their concentration gradient without energy expenditure. Channel-mediated facilitated diffusion uses ion channels (voltage-gated, ligand-gated, or mechanically-gated) for rapid ion movement. Carrier-mediated facilitated diffusion uses specific carrier proteins that undergo conformational changes to transport molecules like glucose (via GLUT transporters) and amino acids. GLUT1 (erythrocytes, brain), GLUT2 (liver, pancreas), GLUT3 (neurons), GLUT4 (muscle, adipose, insulin-responsive). The rate of facilitated diffusion is limited by transporter number (saturable, following Michaelis-Menten kinetics) and is specific, saturable, and inhibitable.

Filtration Fraction

– The fraction of renal plasma flow that is filtered at the glomerulus, calculated as GFR divided by renal plasma flow (RPF). Normal GFR is approximately 125 mL/min and RPF is approximately 625 mL/min, yielding a filtration fraction of approximately 0.20 (20 percent). This means that approximately 20 percent of the plasma entering the glomerulus is filtered; the remaining 80 percent continues through the efferent arteriole to the peritubular capillaries. Changes in filtration fraction reflect changes in glomerular capillary hydrostatic pressure, plasma oncotic pressure, or renal blood flow. Increased filtration fraction (as in diabetic nephropathy) raises peritubular capillary oncotic pressure, enhancing reabsorption. Decreased filtration fraction suggests reduced GFR relative to RPF, as seen in prerenal azotaemia.

Frank-Starling Mechanism

– The relationship between ventricular end-diastolic volume (preload) and stroke volume: within physiological limits, increased venous return stretches the ventricular wall, increasing the force of contraction and stroke volume. This occurs because stretching the myocardium optimizes the overlap between actin and myosin filaments, increasing the number of active cross-bridges and enhancing calcium sensitivity. The mechanism allows the heart to automatically match its output to the volume of blood returning to it, balancing left and right ventricular outputs over successive beats. It is the basis for the clinical use of fluid resuscitation in heart failure and the relationship between central venous pressure and cardiac output on the Starling curve.

Galactosemia

– An inborn error of galactose metabolism, most commonly caused by deficiency of galactose-1-phosphate uridylyltransferase (GALT, autosomal recessive), which converts galactose-1-phosphate to glucose-1-phosphate. Deficiency causes accumulation of galactose-1-phosphate and galactitol (from galactose reduction by aldose reductase) in tissues. Newborn screening detects elevated galactose. Symptoms appear after milk feeding (containing lactose, which is hydrolysed to glucose and galactose): vomiting, diarrhea, failure to thrive, hypoglycaemia, hepatomegaly, jaundice, and cataracts. Treatment is a galactose-restricted diet. Long-term complications despite dietary management include intellectual disability, speech delay, and premature ovarian failure in females.

Gastrin Secretion

– Gastrin is a peptide hormone produced by G-cells in the gastric antrum and duodenum, stimulating hydrochloric acid secretion by parietal cells, pepsinogen secretion by chief cells, and growth of the gastric mucosa. Stimuli for release include: gastric distension, amino acids (especially phenylalanine and tryptophan) and peptides in the stomach, and vagal stimulation (via GRP, gastrin-releasing peptide). Gastrin acts on cholecystokinin-B (CCK-B) receptors on parietal cells and histamine-releasing enterochromaffin-like (ECL) cells (which in turn stimulate parietal cells via histamine H2 receptors). Gastrin release is inhibited by low antral pH (negative feedback via somatostatin from D-cells). Zollinger-Ellison syndrome (gastrinoma) causes severe peptic ulcer disease from unregulated gastrin secretion.

Gaucher Disease

– The most common lysosomal storage disease, caused by deficiency of glucocerebrosidase (glucosylceramidase, beta-glucocerebrosidase, GBA), an autosomal recessive disorder leading to accumulation of glucocerebroside (glucosylceramide) in macrophages of the reticuloendothelial system. These lipid-laden macrophages are called "Gaucher cells" with a characteristic "crinkled tissue paper" cytoplasmic appearance. Type 1 (non-neuronopathic, most common, approximately 90 percent of cases): hepatosplenomegaly, anemia, thrombocytopenia, bone pain crises, and Erlenmeyer flask deformity of the distal femurs. Type 2 (acute neuronopathic, infantile): severe neurodegeneration with brainstem involvement and death by age 2. Type 3 (chronic neuronopathic): later onset with myoclonus, seizures, and progressive neurological deterioration.

Gene Therapy

– The therapeutic delivery of nucleic acids into cells to treat or prevent disease by altering gene expression. Two main approaches: (1) gene replacement/supplementation (adding a functional copy of a defective gene), and (2) gene editing (correcting the defective gene in situ). Viral vectors: adeno-associated virus (AAV, non-integrating, low immunogenicity, limited packaging capacity), lentivirus (integrates into the host genome, effective for dividing and non-dividing cells), and adenovirus (episomal, highly immunogenic, large packaging capacity). Non-viral methods include lipid nanoparticles (mRNA vaccines, siRNA delivery) and electroporation. Major challenges include immunogenicity, off-target effects, delivery efficiency, and long-term expression durability.

Glomerular Filtration Rate

– The volume of plasma filtered through the glomerular capillaries per unit time, normally approximately 125 mL/min or approximately 180 L/day. GFR depends on Starling forces across the glomerular capillary wall: glomerular capillary hydrostatic pressure (approximately 55 mmHg, favoring filtration) minus Bowman's capsule hydrostatic pressure (approximately 15 mmHg) minus glomerular capillary oncotic pressure (approximately 30 mmHg, opposing filtration), yielding a net filtration pressure of approximately 10 mmHg. GFR is autoregulated between MAP of approximately 80-180 mmHg by the myogenic response (afferent arteriolar constriction in response to increased pressure) and tubuloglomerular feedback (macula densa sensing NaCl delivery). Clinical GFR measurement uses creatinine clearance or cystatin C. Reduced GFR defines acute kidney injury and chronic kidney disease staging.

Gluconeogenesis

– The synthesis of glucose from non-carbohydrate precursors (lactate, glycerol, glucogenic amino acids, and propionate) primarily in the liver (90 percent) and kidneys (10 percent). It uses most glycolytic enzymes in reverse but must bypass three irreversible glycolytic reactions using different enzymes: (1) pyruvate carboxylase converts pyruvate to oxaloacetate, then PEPCK (phosphoenolpyruvate carboxykinase) converts oxaloacetate to phosphoenolpyruvate; (2) fructose-1,6-bisphosphatase converts fructose-1,6-bisphosphate to fructose-6-phosphate; (3) glucose-6-phosphatase (present in liver and kidney but not muscle) converts glucose-6-phosphate to free glucose for release into the blood. Gluconeogenesis is energy-intensive (6 ATP equivalents per glucose) and is stimulated by glucagon, cortisol, and catecholamines, and inhibited by insulin and ethanol.

Glycogenesis

– The synthesis of glycogen from glucose, primarily in liver (for blood glucose maintenance) and skeletal muscle (for local energy use). The process involves two key enzymes: glycogen synthase (adding glucose residues from UDP-glucose to the non-reducing ends of glycogen via alpha-1,4-glycosidic bonds) and the branching enzyme (creating alpha-1,6-glycosidic bonds every 8-12 residues, creating branch points). Glycogen synthase is the rate-limiting enzyme, regulated by phosphorylation (inactivated by PKA, activated by dephosphorylation via protein phosphatase 1) and allosteric activation by glucose-6-phosphate. Insulin promotes glycogenesis by activating protein phosphatase 1 and inactivating glycogen synthase kinase 3 (GSK-3). Glucagon and adrenaline inhibit glycogenesis via cAMP-mediated phosphorylation.

Glycogenolysis

– The breakdown of glycogen to glucose, primarily in the liver (to maintain blood glucose) and muscle (for local energy use). Key enzyme: glycogen phosphorylase, which cleaves alpha-1,4-glycosidic bonds by phosphorylating glucose residues to produce glucose-1-phosphate. The debranching enzyme handles branch points (alpha-1,6 bonds). In the liver, glucose-1-phosphate is converted to glucose-6-phosphate by phosphoglucomutase, then to free glucose by glucose-6-phosphatase (released into blood). In muscle, glucose-6-phosphate enters glycolysis directly (cannot be released as free glucose). Glycogenolysis is activated by glucagon (liver), adrenaline (liver and muscle), and AMP (muscle, during exercise), and inhibited by insulin and glucose.

Glycogen Storage Diseases

– A group of inherited disorders caused by deficiencies in enzymes involved in glycogen metabolism, affecting the liver, muscle, or both. Type I (von Gierke disease): glucose-6-phosphatase deficiency, causing severe fasting hypoglycemia, hepatomegaly (massive glycogen and fat accumulation), lactic acidosis, hyperuricemia, and hyperlipidemia. Type II (Pompe disease): acid alpha-glucosidase (lysosomal acid maltase) deficiency, causing accumulation of glycogen in lysosomes, leading to cardiomegaly, hypertrophic cardiomyopathy, and muscle weakness;

infantile form is fatal without enzyme replacement therapy. Type III (Cori disease): debranching enzyme deficiency, milder hypoglycaemia and hepatomegaly. Type V (McArdle disease): muscle glycogen phosphorylase deficiency, causing exercise intolerance, myoglobinuria, and second wind phenomenon.

Glycolysis

– The metabolic pathway converting glucose to pyruvate, occurring in the cytoplasm of all cells. Ten enzymatic steps divided into two phases: the energy investment phase (steps 1-5, consuming 2 ATP) and the energy payoff phase (steps 6-10, producing 4 ATP and 2 NADH). Key regulatory enzymes: hexokinase/glucokinase (step 1, phosphorylating glucose), phosphofructokinase-1 (PFK-1, step 3, the rate-limiting step, allosterically activated by AMP, fructose-2,6-bisphosphate, and inhibited by ATP and citrate). The net yield per glucose is 2 ATP and 2 NADH (under aerobic conditions); under anaerobic conditions, NAD+ is regenerated by lactate dehydrogenase reducing pyruvate to lactate, allowing continued glycolysis but with net 2 ATP per glucose.

G-Protein Coupled Receptors

– The largest family of cell surface receptors with seven transmembrane alpha-helical domains. Upon ligand binding, the receptor undergoes a conformational change that activates an associated heterotrimeric G-protein (Galpha, Gbeta, Ggamma). The Galpha subunit exchanges GDP for GTP and dissociates from the Gbeta-gamma complex. Gs-proteins activate adenylyl cyclase (increasing cAMP), Gi-proteins inhibit adenylyl cyclase (decreasing cAMP), and Gq-proteins activate phospholipase C (producing IP3 and DAG, increasing intracellular Ca2+). The system provides signal amplification and is the target of many drugs (beta-blockers, antihistamines, opioids) and bacterial toxins (cholera toxin permanently activates Gs, pertussis toxin inactivates Gi).

Graded Potentials

– Local changes in membrane potential that vary in magnitude proportional to stimulus strength, unlike all-or-none action potentials. They include receptor potentials (in sensory receptors), postsynaptic potentials (EPSPs and IPSPs at synapses), and pacemaker potentials (in cardiac/specialized cells). Graded potentials decay with distance from their source (decremental conduction) because they spread passively through the cytoplasm. They can summate temporally (rapid successive stimuli) and spatially (simultaneous stimuli from multiple sources), integrating signals at the axon hillock. If the summed depolarisation reaches threshold (approximately -55 mV), an action potential is triggered. Graded potentials are essential for sensory transduction and synaptic communication and are the basis for the all-or-none law of neural signalling.

Haldane Effect

– The decrease in hemoglobin's CO2 carrying capacity when oxygen binds to it, facilitating CO2 release at the lungs. Deoxygenated hemoglobin (deoxyhemoglobin) is a better buffer for H+ ions and forms more carbaminohemoglobin than oxyhemoglobin. In the tissues: oxygen unloading increases hemoglobin's ability to bind CO2 (as carbaminohemoglobin) and buffer H+ (from CO2 hydration to carbonic acid), enhancing CO2 loading. In the lungs: oxygen binding displaces CO2 from hemoglobin and reduces H+ buffer capacity, promoting CO2 release into the alveoli. The Haldane effect explains that approximately 20 percent of CO2 transport is via carbaminohemoglobin. Together with the Bohr effect, it optimizes O2 delivery and CO2 removal, a concept known as the "Haldane-Bohr coupling" that is essential for efficient gas exchange.

Homeostasis

– The maintenance of a stable internal environment despite external changes, achieved through physiological regulation and feedback mechanisms. The concept was introduced by Walter Cannon (1929) building on Claude Bernard's milieu intérieur. Homeostasis involves: (1) Set point: the ideal or normal value for a physiological variable (e.g., body temperature 37 C, blood pH 7.4, blood glucose 4-6 mmol/L). (2) Sensors (receptors): detect deviations from the set point (e.g., thermoreceptors in the hypothalamus for temperature, chemoreceptors for pH and O2/CO2, baroreceptors for blood pressure). (3) Integrating center: compares input to the set point and coordinates a response (e.g., hypothalamus, brainstem, endocrine glands). (4) Effectors:

muscles, glands, or organs that produce the corrective response (e.g., sweat glands and shivering for temperature; insulin and glucagon for glucose; heart, vessels, and kidneys for BP). Feedback mechanisms: (1) Negative feedback (most common): reverses the direction of change (e.g., rising blood glucose stimulates insulin release, which lowers glucose; blood pressure increase triggers baroreceptor-mediated vasodilation and bradycardia). (2) Positive feedback: amplifies the initial change, driving a process to completion (e.g., oxytocin release during childbirth intensifies uterine contractions; sodium channel opening during action potential depolarisation; blood clotting cascade). Disorders of homeostasis: failure of temperature regulation (hyperthermia, hypothermia), pH (acidosis, alkalosis), glucose (diabetes, hypoglycaemia), blood pressure (hypertension, hypotension), and water/electrolyte balance (dehydration, hyponatraemia, hyperkalaemia).

Hypoxic Pulmonary Vasoconstriction

– The unique response of pulmonary arterioles to low alveolar oxygen tension (PO2 less than approximately 60 mmHg), causing vasoconstriction. Unlike systemic vessels that dilate in response to hypoxia, pulmonary vessels constrict, redirecting blood flow away from poorly ventilated alveoli toward better-ventilated regions, optimizing ventilation-perfusion (V/Q) matching. The mechanism involves inhibition of voltage-gated K+ channels in pulmonary artery smooth muscle cells, causing membrane depolarization, Ca2+ influx via L-type Ca2+ channels, and vasoconstriction. Chronic global hypoxia (as in COPD, high altitude) causes sustained vasoconstriction leading to pulmonary hypertension, right ventricular hypertrophy (cor pulmonale), and vascular remodelling. This response is unique to the pulmonary circulation; systemic vessels dilate in hypoxia.

JAK-STAT Pathway

– A rapid signaling pathway for cytokines and growth factors (interferons, interleukins, erythropoietin, growth hormone) that lack intrinsic enzymatic activity. Ligand binding to the cytokine receptor causes receptor dimerization and activation of associated Janus kinases (JAK1, JAK2, JAK3, TYK2) by trans-phosphorylation. Activated JAKs phosphorylate tyrosine residues on the receptor cytoplasmic domain, creating docking sites for STAT (signal transducers and activators of transcription) proteins, which are phosphorylated by JAKs, form homo- or heterodimers, and translocate to the nucleus where they bind gamma-interferon activation site (GAS) or interferon-stimulated response element (ISRE) sequences to regulate gene transcription. The pathway is negatively regulated by SOCS (suppressor of cytokine signalling) proteins, PTPs (protein tyrosine phosphatases), and PIAS (protein inhibitor of activated STAT).

Ketogenesis

– The hepatic synthesis of ketone bodies (acetoacetate, beta-hydroxybutyrate, and acetone) from acetyl-CoA during periods of prolonged fasting, starvation, uncontrolled diabetes, or very low carbohydrate diets. When oxaloacetate is depleted for gluconeogenesis, accumulated acetyl-CoA cannot enter the TCA cycle and is diverted to ketogenesis. The process occurs in the mitochondrial matrix: two acetyl-CoA molecules condense to form acetoacetyl-CoA (by thiolase), which combines with another acetyl-CoA to form HMG-CoA by HMG-CoA synthase. HMG-CoA lyase cleaves HMG-CoA to acetoacetate and acetyl-CoA. Acetoacetate is reduced to beta-hydroxybutyrate by beta-hydroxybutyrate dehydrogenase (NaDH-dependent) or spontaneously decarboxylated to acetone. Ketone bodies are important alternative fuels for the brain during fasting and starvation, providing up to 70 percent of cerebral energy needs.

Lipoprotein Metabolism

– Lipoproteins transport hydrophobic lipids (triglycerides, cholesterol, cholesterol esters, fat-soluble vitamins) in the aqueous blood. Four main classes: chylomicrons (from intestine, largest, transport dietary triglycerides; ApoB-48; acted on by lipoprotein lipase), VLDL (from liver, transport endogenous triglycerides; ApoB-100; converted to IDL), LDL (from VLDL metabolism, carry cholesterol to tissues; ApoB-100; the "bad" cholesterol; taken up by LDL receptors), and HDL (from liver/intestine, reverse cholesterol transport, ApoA-I, the "good" cholesterol, protective against atherosclerosis). Lipoprotein lipase (LPL) hydrolyses triglycerides in chylomicrons and VLDL, releasing free fatty acids for tissue uptake. HDL mediates reverse cholesterol transport, delivering excess cholesterol from peripheral cells to the liver for excretion or recycling.

Membrane Potential

– The electrical potential difference across a cell membrane, approximately -70 mV in most neurons (resting membrane potential). Established by the unequal distribution of ions (primarily K+, Na+, Cl-) across the membrane and selective permeability. The Nernst equation calculates the equilibrium potential for a single ion: E = (RT/zF) ln([ion outside]/[ion inside]). At body temperature for a monovalent cation, this simplifies to E = 61.5 log([out]/[in]) mV. The Goldman equation accounts for multiple ion permeabilities to calculate the resting membrane potential: Vm = 61.5 log((PK[K+]o + PNa[Na+]o + PCl[Cl-]i)/(PK[K+]i + PNa[Na+]i + PCl[Cl-]o)). The resting membrane is most permeable to K+ (via leak channels), so Vm is close to EK (approximately -94 mV for K+). Changes in ion permeability (e.g., Na+ channels opening during depolarisation) drive electrical signalling.

Michaelis-Menten Kinetics

– The mathematical model describing the rate of enzyme-catalyzed reactions as a function of substrate concentration. The Michaelis-Menten equation: v = (Vmax x [S]) / (Km + [S]), where v is the reaction velocity, Vmax is the maximum velocity (when all enzyme is saturated with substrate), [S] is the substrate concentration, and Km is the Michaelis constant (the substrate concentration at which v = Vmax/2). Km reflects the enzyme's affinity for its substrate: a low Km indicates high affinity (the enzyme reaches half-maximal velocity at a low substrate concentration). The turnover number (kcat = Vmax/[Et]) describes the catalytic efficiency; the ratio kcat/Km is the best measure of enzyme specificity, approaching the diffusion limit for perfect enzymes. Lineweaver-Burk (double reciprocal) plots linearise the equation: 1/v = Km/Vmax x 1/[S] + 1/Vmax.

Migrating Motor Complex

– Cyclical patterns of intense gastrointestinal motility occurring during fasting (interdigestive period), sweeping residual food, bacteria, and debris from the stomach through the small intestine to the colon. Each cycle lasts approximately 90-120 minutes and has four phases: Phase I (quiescence, approximately 45-60 minutes), Phase II (irregular contractions, approximately 30 minutes), Phase III (regular high-amplitude contractions at maximum frequency, approximately 5-15 minutes, the "housekeeper" phase that propels luminal contents distally), and Phase IV (declining activity returning to Phase I). The MMC is regulated by the migrating motor complex, motilin (released during fasting), and the enteric nervous system. Disruption of the MMC is associated with small intestinal bacterial overgrowth (SIBO).

Na+/K+ ATPase

– The sodium-potassium pump is a P-type ATPase found in the plasma membrane of all animal cells, directly consuming approximately 20-30 percent of cellular ATP. It pumps 3 Na+ ions out of the cell and 2 K+ ions in per ATP hydrolyzed, establishing steep electrochemical gradients: high Na+ outside, high K+ inside. This gradient is essential for maintaining the resting membrane potential (approximately -70 mV), driving secondary active transport (e.g., glucose and amino acid absorption in the intestine and kidney, Na+/Ca2+ exchange), and regulating cell volume. The pump is inhibited by cardiac glycosides such as ouabain and digoxin (used in heart failure to increase intracellular Ca2+ and contractility).

Nephron Function

– The nephron is the functional unit of the kidney, with approximately 1 million per kidney. Each nephron consists of a renal corpuscle (glomerulus and Bowman's capsule) and a renal tubule (proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct). The nephron performs three primary functions: filtration (at the glomerulus, producing approximately 180 L/day of ultrafiltrate), reabsorption (reclaiming approximately 99 percent of filtered water and solutes), and secretion (transporting substances from the blood into the tubular lumen for elimination). The loop of Henle establishes the medullary osmotic gradient for urine concentration; the collecting duct performs final regulation of water and electrolyte balance under the control of ADH and aldosterone.

Noncompetitive Inhibition

– A type of enzyme inhibition where the inhibitor binds at a site distinct from the active site (allosteric site), and can bind to both the free enzyme (E) and the enzyme-substrate complex (ES) with equal affinity. The inhibitor does not compete with the substrate and cannot be overcome by increasing substrate concentration. The apparent Vmax decreases (fewer functional enzyme molecules) while Km remains unchanged (substrate affinity is unaffected since the inhibitor doesn't compete for the active site). On a Lineweaver-Burk plot: lines intersect on the x-axis (same -1/Km) with different slopes and y-intercepts. Noncompetitive inhibition is less common than competitive inhibition in pharmacology but is important in metal ion toxicity and certain pesticide actions.

Osmosis

– The net movement of water molecules across a semipermeable membrane from a region of lower solute concentration (higher water concentration) to a region of higher solute concentration (lower water concentration). Driven by the osmotic pressure difference, which is proportional to the number of solute particles. Osmolarity is measured in osmoles per liter (mOsm/L); normal plasma osmolarity is approximately 285-295 mOsm/L. Tonicity describes the effect of a solution on cell volume: hypotonic solutions cause cell swelling (osmotic lysis), hypertonic solutions cause cell shrinkage (crenation). The concept is clinically critical for intravenous fluid therapy: normal saline (0.9 percent NaCl, approximately 308 mOsm/L) is isotonic and does not cause red blood cell lysis; hypotonic fluids (0.45 percent NaCl) may cause haemolysis if infused rapidly.

Oxidative Phosphorylation

– The process by which ATP is synthesized using the energy stored in the proton gradient (proton motive force) across the inner mitochondrial membrane, established by the electron transport chain. ATP synthase (F1F0-ATPase) is a molecular turbine with two components: F0 (membrane-spanning proton channel) and F1 (catalytic head projecting into the matrix). Protons flow through F0 down their electrochemical gradient, causing the gamma subunit to rotate relative to the alpha3beta3 hexamer of F1, driving conformational changes that synthesise ATP from ADP and Pi. The chemiosmotic theory (Peter Mitchell, 1978 Nobel Prize) explains how the proton motive force (proton gradient + membrane potential) couples electron transport to ATP synthesis. Uncouplers (e.g., DNP) dissipate the proton gradient, generating heat instead of ATP.

Oxygen-Hemoglobin Dissociation Curve

– The sigmoidal relationship between partial pressure of oxygen (PO2) and hemoglobin oxygen saturation (SO2). The curve reflects hemoglobin's cooperative binding: oxygen binding to one subunit increases affinity for subsequent oxygen molecules. At PO2 of 100 mmHg (alveolar), hemoglobin is approximately 98 percent saturated. At PO2 of 40 mmHg (mixed venous), saturation is approximately 75 percent. The steep portion of the curve (PO2 20-60 mmHg) allows large oxygen unloading with small PO2 decreases in tissue capillaries. The curve is shifted rightward (decreased affinity) by increased temperature, decreased pH (Bohr effect), increased PCO2, and increased 2,3-BPG, enhancing oxygen unloading. Leftward shifts (increased affinity) occur with decreased temperature, increased pH, and decreased 2,3-BPG (e.g., stored blood), reducing oxygen unloading. fetal hemoglobin (HbF) has a left-shifted curve for placental oxygen transfer.

Pentose Phosphate Pathway

– A cytoplasmic glucose oxidation pathway running parallel to glycolysis, primarily in the liver, adipose tissue, adrenal cortex, and red blood cells. It has two phases: the oxidative phase (producing NADPH and ribose-5-phosphate) and the non-oxidative phase (interconverting sugar phosphates). Key enzyme: glucose-6-phosphate dehydrogenase (G6PD), the rate-limiting enzyme, producing NADPH. NADPH is essential for: fatty acid and steroid biosynthesis, maintaining reduced glutathione (protecting red blood cells from oxidative damage and preventing haemolysis), and cytochrome P450 detoxification. The non-oxidative phase recycles ribose-5-phosphate to glycolytic intermediates. G6PD deficiency (X-linked, common in Mediterranean, African, and Asian populations) causes hemolytic anemia triggered by oxidative stress from drugs (primaguine, sulphonamides) or fava beans.

Peristalsis

– Rhythmic sequential contractions of longitudinal and circular smooth muscle that propel gastrointestinal contents in an aboral (oral to anal) direction. The peristaltic reflex involves stretch activation of the intestinal wall: proximal to the stretch, longitudinal muscle contracts and circular muscle relaxes (ascending contraction); distally, circular muscle contracts and longitudinal muscle relaxes (descending relaxation). This is mediated by the myenteric (Auerbach's) plexus: excitatory neurotransmitters (acetylcholine, substance P) cause proximal contraction, and inhibitory neurotransmitters (VIP, nitric oxide) cause distal relaxation. Peristalsis propels chyme at a rate of approximately 1 cm/min in the small intestine, 5-10 cm/min in the esophagus, and is slower in the colon. Disorders include achalasia (failure of LES relaxation), intestinal pseudo-obstruction, and Hirschsprung disease.

Phenylketonuria

– The most common inborn error of amino acid metabolism, caused by deficiency of phenylalanine hydroxylase (PAH, autosomal recessive), which converts phenylalanine to tyrosine. The deficiency leads to accumulation of phenylalanine and its transamination product phenylpyruvate in the blood and brain. Newborn screening (Guthrie test or tandem mass spectrometry) detects elevated phenylalanine levels (greater than 120 micromol/L). Untreated PKU causes severe intellectual disability, seizures, behavioural problems, eczema, microcephaly, and a musty odour. Newborn screening and early dietary restriction (low phenylalanine diet with special formula) prevents neurological damage. Maternal PKU (uncontrolled hyperphenylalaninaemia during pregnancy) causes fetal intellectual disability, microcephaly, and congenital heart defects.

Phospholipase C Pathway

– A GPCR signaling pathway activated primarily by Gq-protein coupled receptors (e.g., alpha-1 adrenergic, muscarinic M1/M3, vasopressin V1, angiotensin II AT1 receptors). When the Gq-alpha subunit is activated, it stimulates phospholipase C-beta (PLC-beta), which cleaves the membrane phospholipid phosphatidylinositol 4,5-bisphosphate (PIP2) into two second messengers: inositol 1,4,5-trisphosphate (IP3) and diacylglycerol (DAG). IP3 is a water-soluble molecule that diffuses through the cytoplasm and binds to IP3 receptors on the endoplasmic reticulum, triggering Ca2+ release into the cytoplasm. DAG remains in the membrane and activates protein kinase C (PKC), which phosphorylates downstream target proteins. The Ca2+ signal activates calmodulin-dependent kinases (CaMK), and the pathway mediates smooth muscle contraction, neurotransmitter release, hormone secretion, and cell proliferation.

Phytomenadione

– See Vitamin K: Phylloquinone.

Protein Folding

– The process by which newly synthesized polypeptide chains acquire their functional three-dimensional structures. The amino acid sequence (primary structure) determines the folding pathway. Key structural levels: primary (amino acid sequence), secondary (alpha-helices and beta-sheets formed by hydrogen bonding), tertiary (overall 3D fold of a single polypeptide), and quaternary (arrangement of multiple subunits). Chaperone proteins assist proper folding: Hsp70 (heat shock protein 70) binds exposed hydrophobic regions of nascent polypeptides; Hsp60 (chaperonin, GroEL/GroES in bacteria) provides an isolated compartment for folding. Misfolded proteins are targeted for degradation by the ubiquitin-proteasome system. Protein misfolding causes conformational diseases including Alzheimer's (amyloid-beta), Parkinson's (alpha-synuclein), prion diseases (Creutzfeldt-Jakob), and cystic fibrosis (CFTR misfolding).

Proximal Tubule Reabsorption

– The proximal convoluted tubule (PCT) reabsorbs approximately 65 percent of filtered sodium, water, glucose, amino acids, bicarbonate, and phosphate. Sodium reabsorption is the driving force for most solute and water transport: the basolateral Na+/K+ ATPase creates a low intracellular Na+ concentration, allowing Na+ entry from the tubular lumen via secondary active transporters (Na+/glucose, Na+/amino acid cotransporters on the apical membrane; Na+/H+ exchanger). Water follows sodium osmotically via aquaporins (AQP1) in the proximal tubule. Glucose and amino acids are completely reabsorbed in the early PCT via sodium-coupled transporters (SGLT2 for glucose, SGLT1 for glucose and amino acids) until their transport maximum (Tm) is exceeded. Bicarbonate is reabsorbed via H+ secretion (Na+/H+ exchanger) and carbonic anhydrase in the brush border. Phosphate reabsorption is regulated by PTH (decreases reabsorption, increases phosphate excretion).

Pulmonary Surfactant

– A phospholipid-protein mixture produced by type II alveolar pneumocytes that reduces surface tension at the air-liquid interface within alveoli. The primary component is dipalmitoylphosphatidylcholine (DPPC, lecithin), which forms a monolayer that reduces surface tension to near zero during expiration. Surfactant prevents alveolar collapse (atelectasis) at end-expiration by reducing the collapsing force, increases lung compliance, and stabilizes alveoli of different sizes (preventing small alveoli from collapsing into larger ones via the LaPlace relationship). Surfactant is produced from approximately week 24 of gestation, and its deficiency in premature infants causes respiratory distress syndrome (RDS) with atelectasis, hypoxaemia, and the need for surfactant replacement therapy and mechanical ventilation.

Pyruvate

– The three-carbon alpha-keto acid produced at the end of glycolysis from phosphoenolpyruvate by pyruvate kinase. Under aerobic conditions, pyruvate enters the mitochondrial matrix via the mitochondrial pyruvate carrier (MPC) and is converted to acetyl-CoA by the pyruvate dehydrogenase complex (PDC), an irreversible reaction that also produces CO2 and NADH. The PDC requires five cofactors derived from vitamins: thiamine pyrophosphate (B1), FAD (B2), NAD+ (B3), CoA (from pantothenic acid, B5), and lipoic acid. Under anaerobic conditions, pyruvate is reduced to lactate by lactate dehydrogenase, regenerating NAD+ for continued glycolysis. Pyruvate is the branch point between aerobic (TCA cycle) and anaerobic (lactate) metabolism, and its fate is determined by oxygen availability and tissue metabolic demands.

Receptor Tyrosine Kinases

– Single-pass transmembrane receptors with intrinsic tyrosine kinase activity in their intracellular domain. Ligand binding (e.g., insulin, EGF, PDGF, FGF) induces receptor dimerization and autophosphorylation of tyrosine residues. Phosphorylated tyrosines serve as docking sites for intracellular signaling proteins containing SH2 domains, activating downstream cascades: the Ras-MAP kinase pathway (cell proliferation and differentiation), the PI3-kinase/Akt pathway (cell survival and metabolism), and the PLC-gamma pathway (Ca2+ signalling). RTKs are frequently mutated or overexpressed in cancer (e.g., HER2 in breast cancer, EGFR in lung cancer), making them important therapeutic targets for tyrosine kinase inhibitors (imatinib, gefitinib, trastuzumab).

Renal Clearance

– The volume of plasma completely cleared of a substance by the kidneys per unit time, measured in mL/min. Clearance of any substance X is calculated as: CX = (Ux x V)/Px, where Ux is urine concentration, V is urine flow rate, and Px is plasma concentration. Inulin clearance equals GFR (approximately 125 mL/min) because inulin is freely filtered but neither reabsorbed nor secreted. Para-aminohippuric acid (PAH) clearance approximates renal plasma flow (approximately 625 mL/min) because it is both filtered and completely secreted by the proximal tubule. Creatinine clearance is the most common clinical GFR estimate; cystatin C is a more accurate alternative unaffected by muscle mass. The concept is essential for determining drug dosing in renal impairment.

Renin-Angiotensin-Aldosterone System

– The RAAS is a hormonal cascade regulating blood pressure, fluid volume, and electrolyte balance. Stimulus: decreased renal perfusion pressure (detected by baroreceptors in the afferent arteriole), decreased sodium delivery to the macula densa, or increased sympathetic activity. Renin (from juxtaglomerular cells) cleaves angiotensinogen (from the liver) to angiotensin I. Angiotensin-converting enzyme (ACE, primarily in pulmonary endothelium) converts angiotensin I to angiotensin II. Angiotensin II is a potent vasoconstrictor, stimulates aldosterone secretion (Na+ and water retention), increases ADH release, enhances thirst, increases sympathetic activity, and promotes cardiac and vascular remodelling. The RAAS is a target of ACE inhibitors, angiotensin receptor blockers (ARBs), direct renin inhibitors, and mineralocorticoid receptor antagonists for treating hypertension and heart failure.

Retinol

– See Vitamin A: Retinal.

Salivation

– The secretion of saliva by the salivary glands. Salivation is controlled by the autonomic nervous system: parasympathetic stimulation (CN VII—submandibular and sublingual glands; CN IX—parotid gland) produces profuse, watery saliva via acetylcholine acting on muscarinic M_3 receptors; sympathetic stimulation produces scanty, viscous saliva via noradrenaline acting on β-adrenergic receptors. Salivation is triggered by: sight, smell, or thought of food (cephalic phase), mechanical stimulation of the mouth (food in the oral cavity), and the presence of acid or chemical irritants. The salivary center is located in the medulla oblongata (superior and inferior salivatory nuclei). Daily volume: 1–1.5 L. Hypersalivation (sialorrhoea) can be caused by oral inflammation, teething, pregnancy, neurological disorders (Parkinson disease, motor neuron disease, cerebral palsy, rabies), gastro-esophageal reflux, and medications (clozapine, anticholinesterases). Hyposalivation (xerostomia/dry mouth): Sjögren syndrome, radiation therapy to the head and neck, medications (anticholinergics, antidepressants, antihistamines, diuretics), diabetes, and dehydration.

Saltatory Conduction

– The rapid propagation of action potentials along myelinated nerve fibers, where the action potential appears to "jump" from one Node of Ranvier to the next. Myelin (produced by Schwann cells in the PNS and oligodendrocytes in the CNS) insulates the axon, preventing ion flow across the internodal membrane. Action potentials are generated only at the nodes (approximately 1 mm apart), where voltage-gated Na+ channels are concentrated. The local current from one node depolarises the next node to threshold, causing saltatory conduction at speeds up to 50 m/s in myelinated fibers (compared to approximately 0.5 m/s in unmyelinated fibers). Saltatory conduction is energy-efficient, as Na+/K+ ATPase activity is required only at the nodes. Demyelinating diseases such as multiple sclerosis (CNS) and Guillain-Barré syndrome (PNS) slow or block conduction, causing neurological deficits.

Second Messengers

– Intracellular signaling molecules generated in response to extracellular signals (first messengers like hormones or neurotransmitters). cAMP (cyclic adenosine monophosphate): produced from ATP by adenylyl cyclase, activated by Gs-protein coupled receptors, activates protein kinase A (PKA); degraded by phosphodiesterase. IP3 (inositol 1,4,5-trisphosphate): produced from PIP2 by phospholipase C, triggers Ca2+ release from the endoplasmic reticulum. DAG (diacylglycerol): remains in the membrane along with IP3, and together they activate protein kinase C (PKC). Ca2+ is a versatile second messenger that binds calmodulin, activating CaMKII (in neurons, critical for long-term potentiation and memory formation). The second messenger concept is central to understanding hormone action, drug effects, and intracellular communication.

Second Messenger Systems

– Intracellular signaling molecules that relay extracellular signals (first messengers: hormones, neurotransmitters, growth factors) to cellular responses. Major second messengers: cAMP (produced by adenylyl cyclase from ATP, activated by Gs-GPCRs; activates PKA; degraded by phosphodiesterase; example: epinephrine binding beta-1 receptors in the heart increasing heart rate and contractility). cGMP (produced by guanylyl cyclase; activated by nitric oxide in smooth muscle causing vasodilation; also activated by natriuretic peptides ANP/BNP). IP3 (produced by phospholipase C from PIP2, releases Ca2+ from ER). DAG (produced with IP3, activates PKC). Ca2+ itself acts as a second messenger, activating calmodulin and CaMK. Second messenger systems provide signal amplification: a single activated receptor can activate many G-proteins, each activating an enzyme that produces many second messenger molecules.

Secretin

– A peptide hormone produced by S-cells in the duodenal and jejunal mucosa in response to acidic chyme (pH less than 4.5) entering the duodenum from the stomach. Secretin's primary action is to stimulate the pancreatic ductal cells to secrete a bicarbonate-rich fluid, neutralizing duodenal acid and raising the pH to approximately 7-8, which optimizes pancreatic enzyme activity. Secretin also inhibits gastric acid secretion and gastric motility (slowing gastric emptying), stimulates bile secretion and gallbladder contraction, and enhances the effects of cholecystokinin (CCK). Secretin release is inhibited by H+ buffering in the duodenum (negative feedback). Secretin was the first hormone discovered (Bayliss and Starling, 1902), establishing the concept of hormonal regulation.

Segmentation

– A pattern of intestinal motility involving localized, rhythmic contractions of circular muscle that segment the intestinal contents, mixing them with digestive secretions and increasing contact with the absorptive mucosal surface. Unlike peristalsis, segmentation does not propel contents distally. It occurs predominantly in the fed state in the small intestine, at a frequency determined by the slow wave rhythm of the intestine (approximately 12 per minute in the duodenum, 8 per minute in the ileum). The contractions are mediated by the enteric nervous system and the interstitial cells of Cajal (pacemaker cells). Segmentation is essential for efficient digestion and absorption; loss of segmentation occurs in intestinal obstruction and ileus.

Sensory Nerves

– Nerves that transmit sensory information from peripheral receptors to the central nervous system. Sensory (afferent) neurons have their cell bodies in the dorsal root ganglia (spinal nerves) or cranial nerve ganglia (cranial nerves). Sensory receptors: mechanoreceptors (touch, pressure, vibration, stretch—Pacinian corpuscles, Meissner corpuscles, Merkel discs, Ruffini endings), thermoreceptors (cold—Aδ fibers; warm—unmyelinated C fibers), nociceptors (pain—Aδ fibers: fast, sharp, localised; C fibers: slow, dull, diffuse), chemoreceptors (taste, smell, O_2, CO_2, pH), and photoreceptors (rods and cones in the retina). Sensory pathways: spinothalamic tract (pain, temperature, crude touch—crosses in spinal cord), dorsal columns/medial lemniscus (fine touch, vibration, proprioception—crosses in medulla), and trigeminothalamic tract (facial sensation). Dermatomes: areas of skin supplied by a single spinal nerve (C4—shoulder, C6—thumb, C8—little finger, T4—nipple, T10—umbilicus, L4—medial foot, S1—lateral foot). Sensory deficits help localise neurological lesions at the level of the peripheral nerve, plexus, nerve root, spinal cord, brainstem, thalamus, or cortex.

Sympathetic Nervous System

– The division of the autonomic nervous system responsible for the 'fight or flight' response, preparing the body for intense physical activity or stress. Sympathetic preganglionic neurons originate in the intermediolateral column of the spinal cord gray matter (T1–L2), making it the thoracolumbar system. Preganglionic fibers are short and synapse in the paravertebral sympathetic chain ganglia (bilaterally, 22 ganglia) or prevertebral ganglia (celiac, superior mesenteric, inferior mesenteric, aorticorenal). The neurotransmitter of preganglionic fibers is acetylcholine (nicotinic receptors); postganglionic fibers release noradrenaline (norepinephrine) at target organs (adrenergic α and β receptors), except sweat glands and some blood vessels (cholinergic). Sympathetic effects: pupil dilation (mydriasis), increased heart rate and contractility (chronotropic and inotropic—$\beta 1$), bronchodilation ($\beta 2$), vasoconstriction of skin and splanchnic vessels ($\alpha 1$), vasodilation of skeletal muscle vessels ($\beta 2$), increased sweating (cholinergic), increased renin secretion ($\beta 1$), decreased GI motility and secretion ($\alpha 2$, $\beta 2$), contraction of sphincters (urinary, anal—$\alpha 1$), ejaculation, and release of glucose (glycogenolysis—$\beta 2$, gluconeogenesis). The adrenal medulla (modified sympathetic ganglion) secretes adrenaline (epinephrine, 80%) and noradrenaline (20%) into the bloodstream. The sympathetic nervous system also controls blood pressure via the baroreceptor reflex.

Synapse

– The specialized junction between a neuron and another neuron (or effector cell—muscle, gland) where information is transmitted from one cell to another. Types: (1) Chemical synapses (most common in the nervous system)—the presynaptic terminal releases neurotransmitter molecules into the synaptic cleft (20–40 nm wide) in response to an action potential. Neurotransmitters diffuse across the cleft and bind to postsynaptic receptors (ionotropic—ligand-gated ion channels, fast response; metabotropic—G-protein coupled receptors, slow, modulatory). This generates either an excitatory postsynaptic potential (EPSP—depolarisation) or inhibitory postsynaptic potential (IPSP—hyperpolarisation). Summation of many EPSPs reaching threshold generates a new action potential. (2) Electrical synapses—gap junctions (connexin proteins) allow direct flow of ions between cells, enabling very fast, bidirectional transmission (found in smooth and cardiac muscle, some CNS regions). Key neurotransmitters: glutamate (main excitatory CNS transmitter), GABA (main inhibitory CNS transmitter), acetylcholine (neuromuscular junction, autonomic ganglia), dopamine, noradrenaline, serotonin, and neuropeptides (substance P, endorphins). Synaptic plasticity: long-term potentiation (LTP) and long-term depression (LTD)—mechanisms of learning and memory. The synapse is the site of action of many drugs (anaesthetics, antidepressants, antipsychotics, anxiolytics, muscle relaxants) and toxins (botulinum toxin—blocks ACh release, tetanus toxin—blocks inhibitory interneurons).

Taste

– The sensation produced by the interaction of chemical substances with taste receptor cells (TRCs) on the tongue, palate, pharynx, epiglottis, and upper esophagus. Taste is one of the five primary senses (gustation). Five basic taste qualities: sweet (sugars, artificial sweeteners—T1R2/T1R3 receptors), sour (acids—protons, PKD2L1 receptor), salty (sodium ions—ENaC channel), bitter (alkaloids, toxins—T2R receptor family, 25 subtypes), and umami (savoury—glutamate, T1R1/T1R3 receptor). Taste buds (5,000) are housed in papillae on the tongue: fungiform (anterior), foliate (lateral), circumvallate (posterior), and filiform (no taste buds—mechanical). Each taste bud contains 50–100 TRCs with microvilli exposed to the oral cavity via a taste pore. Taste information is transmitted via cranial nerves: facial (CN VII—chorda tympani, anterior two-thirds of tongue), glossopharyngeal (CN IX—posterior one-third), and vagus (CN X—epiglottis and pharynx) to the nucleus of the solitary tract in the medulla, then to the thalamus (VPM) and primary gustatory cortex (insula and operculum). Taste disorders: ageusia (loss of taste), hypogeusia (reduced), dysgeusia (distorted), and phantom taste perception. Causes: viral infections (COVID-19—anosmia and ageusia are hallmark symptoms), medications (ACE inhibitors—metallic taste), chemotherapy, head trauma, radiation, Bell palsy, and ageing. Taste is closely linked to smell (olfaction); much of what is perceived as 'taste' is actually retronasal olfaction.

Tay-Sachs Disease

– A lysosomal storage disease caused by deficiency of hexosaminidase A (Hex A), an autosomal recessive disorder leading to accumulation of GM2 ganglioside in neurons of the central nervous system. Hex A is a heterodimer of alpha and beta subunits; mutations in the HEXA gene (encoding the alpha subunit) cause the classic infantile form. Normal carrier screening: Hex A activity is approximately 50 percent of normal. Affected infants appear normal at birth but develop progressive neurodegeneration by 3-6 months with hypotonia, loss of motor skills, exaggerated startle reflex, cherry-red macula on fundoscopy, macrocephaly, and seizures, with death typically by age 4. Carrier frequency is approximately 1 in 30 in Ashkenazi Jewish populations, and enzyme-based carrier screening has reduced the incidence through prenatal diagnosis.

TCA Cycle

– The tricarboxylic acid (Krebs) cycle occurs in the mitochondrial matrix, completing the oxidation of acetyl-CoA (from carbohydrates, fats, and proteins) to CO2 while generating reducing equivalents (NADH, FADH2) for the electron transport chain. Eight steps: (1) citrate synthase (acetyl-CoA + oxaloacetate to citrate), (2) aconitase (citrate to isocitrate), (3) isocitrate dehydrogenase (rate-limiting, isocitrate to alpha-ketoglutarate, producing NADH and CO2), (4) alpha-ketoglutarate dehydrogenase complex (alpha-ketoglutarate to succinyl-CoA, producing NADH and CO2, requiring TPP, lipoic acid, CoA, FAD, NAD+), (5) succinyl-CoA synthetase (succinyl-CoA to succinate, producing GTP), (6) succinate dehydrogenase (succinate to fumarate, producing FADH2, also Complex II of ETC), (7) fumarase (fumarate to malate), (8) malate dehydrogenase (malate to oxaloacetate, producing NADH). The cycle produces 3 NADH, 1 FADH2, and 1 GTP per acetyl-CoA, yielding approximately 10 ATP equivalents.

Transcription

– The synthesis of RNA from a DNA template by RNA polymerase, the first step of gene expression. In eukaryotes, RNA polymerase II transcribes protein-coding genes into pre-mRNA. Initiation: RNA polymerase binds to the promoter region (containing the TATA box approximately 25-30 base pairs upstream of the transcription start site), with the help of general transcription factors (TFIIA, TFIIB, TFIID/TBP, TFIIE, TFIIF, TFIIH) forming the pre-initiation complex. TFIID contains the TATA-binding protein (TBP) that binds the TATA box. Elongation: RNA polymerase II moves along the template strand, synthesising pre-mRNA in the 5' to 3' direction, adding ribonucleotides complementary to the DNA template. Termination: polyadenylation signal (AAUAAA) triggers cleavage and polyadenylation of the 3' end, followed by RNA polymerase dissociation. Transcription is regulated by promoter elements, enhancers, silencers, and transcription factors, allowing precise spatiotemporal gene expression.

Translation

– The synthesis of proteins from mRNA templates by ribosomes, the final step of gene expression. The genetic code is read in triplets (codons), each specifying an amino acid. Initiation: the small ribosomal subunit (40S in eukaryotes) binds the mRNA at the 5' cap, scans for the start codon (AUG), and the initiator tRNA (Met-tRNAi) binds. The large subunit (60S) joins to form the complete ribosome (80S), with the start codon in the P site. Eukaryotic initiation factors (eIFs) facilitate this process: eIF4E binds the 5' cap, eIF2 binds Met-tRNAi to the small subunit, and eIF5B joins the large subunit. Elongation: aminoacyl-tRNAs enter the A site via EF-Tu (eEF1A in eukaryotes), peptide bond formation occurs in the P site, and translocation moves the ribosome three nucleotides along the mRNA via EF-G (eEF2). Termination: a stop codon (UAA, UAG, UGA) in the A site is recognised by release factors, releasing the completed polypeptide.

Tubuloglomerular Feedback

– A feedback mechanism at the juxtaglomerular apparatus (JGA) that adjusts GFR based on tubular fluid NaCl concentration. The macula densa cells (specialized tubular epithelial cells in the thick ascending limb) sense NaCl concentration via the NKCC2 cotransporter. When GFR is high, increased NaCl delivery to the macula densa triggers release of ATP and adenosine, causing afferent arteriole constriction and reducing GFR. When GFR is low, decreased NaCl delivery causes afferent arteriole dilation (via increased nitric oxide and prostaglandins), increasing GFR. This feedback loop maintains a relatively constant NaCl delivery to the distal tubule and stabilises GFR despite changes in blood pressure. The JGA also releases renin in response to low NaCl, activating the RAAS. Tubuloglomerular feedback is impaired in diabetic nephropathy and during ACE inhibitor therapy.

Uncompetitive Inhibition

– A type of enzyme inhibition where the inhibitor binds only to the enzyme-substrate (ES) complex at a site distinct from the active site, forming an inactive ESI complex. The inhibitor cannot bind to the free enzyme. Both Vmax and Km decrease: Vmax decreases because the ESI complex is non-productive (fewer active ES complexes), and Km decreases because the inhibitor effectively traps substrate in the ES complex (increasing apparent substrate affinity). On a Lineweaver-Burk plot: lines are parallel, with different slopes and both x- and y-intercepts changing (same slope, different y-intercept and x-intercept). Uncompetitive inhibition is rare in single-substrate reactions but occurs in multisubstrate reactions (e.g., inhibition of lactate dehydrogenase by pyruvate at high concentrations) and is the mechanism of some drugs such as methotrexate (inhibition of dihydrofolate reductase when bound to NADPH).

Urea Cycle

– The hepatic metabolic pathway converting toxic ammonia (NH3/NH4+) to urea for renal excretion, occurring partly in the mitochondrial matrix and partly in the cytoplasm. Five steps: (1) carbamoyl phosphate synthetase I (CPS-I, mitochondrial, rate-limiting, activated by N-acetylglutamate), combining CO2, NH3, and ATP; (2) ornithine transcarbamylase (OTC, mitochondrial), combining carbamoyl phosphate with ornithine to form citrulline; (3) argininosuccinate synthetase (cytoplasmic), combining citrulline with aspartate to form argininosuccinate; (4) argininosuccinate lyase (cytoplasmic), cleaving argininosuccinate to arginine and fumarate; (5) arginase (cytoplasmic), hydrolysing arginine to ornithine and urea. Urea is the non-toxic nitrogenous waste product excreted by the kidneys. Deficiencies cause hyperammonaemia (ammonia intoxication), with symptoms including lethargy, vomiting, cerebral edema, and coma.

Vitamin A: Retinal

– Vitamin A (retinol) is a fat-soluble vitamin essential for vision, immune function, cell differentiation, and reproduction. In the eye, retinol is oxidized to retinal (retinaldehyde), which binds to opsin in the photoreceptor outer segments forming rhodopsin (in rods) and photopsins (in cones). Light absorption by 11-cis-retinal causes isomerization to all-trans-retinal, triggering the phototransduction cascade (G-protein transducin activation, phosphodiesterase activation, cGMP hydrolysis, Na+ channel closure, and hyperpolarisation of the photoreceptor cell). Vitamin A deficiency causes night blindness (nyctalopia), xerophthalmia (dry cornea), Bitot spots, keratomalacia, and increased infection risk. Excess vitamin A (hypervitaminosis A) causes hepatotoxicity, teratogenicity, and raised intracranial pressure.

Vitamin B12: Cobalamin

– Vitamin B12 (cobalamin) is a water-soluble vitamin containing cobalt, essential for two enzymatic reactions: (1) methionine synthase (cytoplasmic, requiring methylcobalamin), converting homocysteine to methionine (linking to folate metabolism), and (2) methylmalonyl-CoA mutase (mitochondrial, requiring adenosylcobalamin), converting methylmalonyl-CoA to succinyl-CoA (in odd-chain fatty acid and branched-chain amino acid metabolism). Intrinsic factor (IF, from gastric parietal cells) is essential for B12 absorption in the terminal ileum. B12 deficiency causes megaloblastic anemia (with hypersegmented neutrophils), peripheral neuropathy, subacute combined degeneration of the spinal cord (dorsal columns and corticospinal tracts), and glossitis. Pernicious anemia is an autoimmune condition causing IF deficiency.

Vitamin B1: Thiamine

– Thiamine (thiamine pyrophosphate, TPP) is a water-soluble B vitamin essential as a cofactor for several key metabolic enzymes: pyruvate dehydrogenase (linking glycolysis to the TCA cycle), alpha-ketoglutarate dehydrogenase (TCA cycle), transketolase (pentose phosphate pathway), and branched-chain alpha-ketoacid dehydrogenase (branched-chain amino acid catabolism). Thiamine deficiency causes: beriberi (dry: peripheral neuropathy and muscle wasting; wet: high-output heart failure with edema and vasodilation), and Wernicke-Korsakoff syndrome (confusion, ataxia, ophthalmoplegia, and amnesia, commonly seen in chronic alcoholism). Wernicke encephalopathy is a medical emergency treated with intravenous thiamine before glucose administration to prevent Korsakoff psychosis.

Vitamin B3: Niacin

– Niacin (nicotinic acid and nicotinamide) is a water-soluble vitamin that serves as the precursor for the coenzymes NAD+ (nicotinamide adenine dinucleotide) and NADP+ (nicotinamide adenine dinucleotide phosphate), which are essential for over 400 enzymatic reactions including glycolysis, the TCA cycle, oxidative phosphorylation, and fatty acid synthesis. NAD+ is a key electron carrier in catabolic reactions; NADP+ is important in anabolic reactions and antioxidant defense (via glutathione). Niacin deficiency causes pellagra, characterized by the 4 Ds: dermatitis (photosensitive), diarrhea, dementia, and death. Pellagra is endemic in populations consuming maize-based diets low in tryptophan (niacin precursor). Therapeutic niacin (in high doses) lowers triglycerides and LDL while raising HDL.

Vitamin B6: Pyridoxine

– Pyridoxine (pyridoxal phosphate, PLP, the active form) is a water-soluble B vitamin essential as a cofactor for over 100 enzymatic reactions, particularly in amino acid metabolism. Key reactions include transamination (ALT, AST), decarboxylation (producing neurotransmitters: serotonin from 5-HTP, dopamine/norepinephrine from L-DOPA, GABA from glutamate, histamine from histidine), and glycogen phosphorylase (glycogenolysis). PLP is also required for heme synthesis (ALA synthase step), sphingolipid synthesis, and serine racemization. PLP is also required for amino acid transport across cell membranes. Pyridoxine deficiency is rare but causes dermatitis, glossitis, microcytic anemia, peripheral neuropathy, and seizures in infants. Isoniazid (tuberculosis treatment) can cause B6 deficiency as a side effect, requiring supplementation.

Vitamin B9: Folate

– Folate (folic acid, tetrahydrofolate/THF) is a water-soluble B vitamin essential for one-carbon transfer reactions in nucleotide synthesis (thymidylate synthesis for DNA, purine synthesis), amino acid metabolism (methionine synthesis from homocysteine), and methylation reactions. THF carries one-carbon units at various oxidation states (methyl, methylene, methenyl, formyl, formimino). Key reactions: thymidylate synthase converts dUMP to dTMP (DNA synthesis, the target of 5-fluorouracil), and methionine synthase converts homocysteine to methionine (requiring methylcobalamin/B12). Folate deficiency causes megaloblastic anemia (identical to B12 deficiency), neural tube defects in pregnancy (spina bifida, anencephaly), and elevated homocysteine (cardiovascular risk factor). Folic acid supplementation before and during early pregnancy prevents neural tube defects.

Vitamin B Complex

– The vitamin B complex comprises eight water-soluble B vitamins that function primarily as coenzymes or cofactors in cellular metabolism: Vitamin B1 (thiamine, a cofactor in carbohydrate metabolism and nerve function), Vitamin B2 (riboflavin, precursor of FAD/FMN in oxidation-reduction reactions), Vitamin B3 (niacin, precursor of NAD+/NADP+), Vitamin B5 (pantothenic acid, component of coenzyme A), Vitamin B6 (pyridoxine, cofactor in amino acid metabolism and neurotransmitter synthesis), Vitamin B7 (biotin, cofactor in carboxylation reactions), Vitamin B9 (folate/folic acid, essential for one-carbon transfer in nucleotide synthesis), and Vitamin B12 (cobalamin, cofactor for methionine synthase and methylmalonyl-CoA mutase). B vitamins are obtained from a wide variety of foods (meat, eggs, dairy, legumes, leafy greens, whole grains) and are generally not stored in large quantities, requiring regular dietary intake. Deficiencies cause specific syndromes (beriberi for B1, pellagra for B3, megaloblastic anemia for B9/B12, see individual entries).

Vitamin C (Ascorbic Acid)

– A water-soluble vitamin and a powerful antioxidant, essential for collagen synthesis (required for proline and lysine hydroxylation in collagen triple helix formation), carnitine synthesis, neurotransmitter synthesis (dopamine to norepinephrine), and iron absorption (reduces ferric to ferrous iron in the gut). Humans lack L-gulonolactone oxidase and cannot synthesise vitamin C, requiring dietary intake (citrus fruits, berries, kiwi, tomatoes, peppers, broccoli). Vitamin C deficiency causes scurvy: impaired wound healing, gingival hyperplasia and bleeding (swollen, bleeding gums), perifollicular haemorrhages, ecchymoses, coiled hairs, joint pain, fatigue, and anemia. In severe cases, scurvy causes haemarthrosis, intraosseous bleeding, and death. Vitamin C is also a cofactor for dopamine beta-hydroxylase (norepinephrine synthesis). High-dose vitamin C has been used as an unproven treatment for the common cold and has been investigated in sepsis and cancer (intravenous high-dose as adjunctive therapy, limited evidence).

Vitamin D: Calcitriol

– Vitamin D is a fat-soluble vitamin (actually a prohormone) essential for calcium and phosphorus homeostasis and bone health. Cholecalciferol (D3) is synthesized in the skin from 7-dehydrocholesterol by UVB radiation, or obtained from diet (ergocalciferol/D2 from plants, D3 from animal sources). It is hydroxylated in the liver to 25-hydroxyvitamin D (calcidiol), then in the kidney to the active form 1,25-dihydroxyvitamin D (calcitriol) by 1-alpha-hydroxylase (regulated by PTH, low calcium, and low phosphate; inhibited by FGF23 and high phosphate). Calcitriol increases intestinal calcium and phosphate absorption, promotes renal calcium reabsorption, and mobilises calcium from bone. Vitamin D deficiency causes rickets in children (soft, deformed bones) and osteomalacia in adults (bone pain, weakness, fractures).

Vitamin E: Tocopherol

– Vitamin E (alpha-tocopherol) is a fat-soluble vitamin and the body's primary lipid-soluble antioxidant, protecting polyunsaturated fatty acids in cell membranes from lipid peroxidation by free radicals (reactive oxygen species). It donates a hydrogen atom to lipid peroxyl radicals, breaking the chain reaction of lipid peroxidation. Vitamin E works synergistically with vitamin C (which regenerates reduced vitamin E at the membrane-cytosol interface) and selenium (as a cofactor for glutathione peroxidase) in protecting against oxidative damage. Vitamin E deficiency is rare but occurs in fat malabsorption syndromes (cystic fibrosis, abetalipoproteinaemia) and causes peripheral neuropathy, spinocerebellar ataxia, retinopathy, and haemolysis (due to red blood cell membrane lipid peroxidation).

Vitamin K: Phylloquinone

– Vitamin K is a fat-soluble vitamin essential for the post-translational gamma-carboxylation of glutamate residues to gamma-carboxyglutamate (Gla) in specific clotting factors (II, VII, IX, X, protein C, protein S) and anticoagulant proteins (protein C, protein S, protein Z). The gamma-carboxylation reaction, catalyzed by gamma-glutamyl carboxylase, requires reduced vitamin K (vitamin K hydroquinone) as a cofactor and is essential for the calcium-binding function of these proteins, allowing them to bind calcium and phospholipid membranes. Vitamin K deficiency causes bleeding (elevated PT/INR), most commonly in newborns (given prophylactic vitamin K at birth), patients with fat malabsorption, or those on warfarin therapy (which inhibits vitamin K epoxide reductase, preventing recycling of vitamin K). The antidote is vitamin K administration.

Zymogen (Proenzyme)

– An inactive precursor of an enzyme that requires proteolytic cleavage to become catalytically active. Zymogens are synthesised and stored in cells (typically in secretory vesicles or zymogen granules) and are activated at the appropriate site and time by specific proteases, preventing premature or inappropriate enzymatic activity that could damage the cell of origin or surrounding tissues. The conversion of a zymogen to an active enzyme involves the cleavage of one or more specific peptide bonds, which induces a conformational change that exposes the active site. Major zymogen systems in human physiology: digestive zymogens (secreted by the exocrine pancreas and gastric chief cells as inactive precursors to prevent autodigestion — trypsinogen → trypsin, chymotrypsinogen → chymotrypsin, proelastase → elastase, procarboxypeptidase → carboxypeptidase, pepsinogen → pepsin in the stomach, activated by gastric acid), clotting cascade (coagulation factors circulate as inactive zymogens — prothrombin → thrombin, fibrinogen → fibrin, factor X → Xa, etc. — activated sequentially in the coagulation cascade), fibrinolytic system (plasminogen → plasmin), complement system (complement proteins C1-C9 circulate as inactive zymogens, activated by classical, lectin, or alternative pathways), and procollagen (procollagen → collagen by procollagen peptidases). The pancreas is particularly dependent on zymogen physiology: the pancreatic acinar cells synthesise and package digestive enzymes as zymogens in zymogen granules, which are secreted into the pancreatic duct and activated in the duodenum by enteropeptidase (enterokinase). Premature activation of trypsinogen to trypsin within the pancreas causes acute pancreatitis (autodigestion of pancreatic tissue)., essential for collagen synthesis (required for proline and lysine hydroxylation in collagen triple helix formation), carnitine synthesis, neurotransmitter synthesis (dopamine to norepinephrine), and iron absorption (reduces ferric to ferrous iron in the gut). Humans lack L-gulonolactone oxidase and cannot synthesise vitamin C, requiring dietary intake (citrus fruits, berries, kiwi, tomatoes, peppers, broccoli). Vitamin C deficiency causes scurvy: impaired wound healing, gingival hyperplasia and bleeding (swollen, bleeding gums), perifollicular haemorrhages, ecchymoses, coiled hairs, joint pain, fatigue, and anemia. In severe cases, scurvy causes haemarthrosis, intraosseous bleeding, and death. Vitamin C is also a cofactor for dopamine beta-hydroxylase (norepinephrine synthesis). High-dose vitamin C has been used as an unproven treatment for the common cold and has been investigated in sepsis and cancer (intravenous high-dose as adjunctive therapy, limited evidence).

Chapter 29

Psychiatry

Acute Stress Disorder

– Characterized by the development of specific symptoms following exposure to a traumatic event. The same symptom clusters as PTSD are required: intrusion, negative mood, dissociation, avoidance, and arousal. Symptoms must last for at least three days and up to one month after the trauma. If symptoms persist beyond one month, the diagnosis changes to PTSD. Early intervention may prevent progression to PTSD.

Addiction

– A chronic, relapsing brain disorder characterized by compulsive substance use or behavior despite harmful consequences. It involves changes in brain reward, motivation, and memory circuits, leading to impaired control over the behavior. Treatment often combines medication, behavioral therapy, and social support.

Adjustment Disorder

– Emotional or behavioral symptoms in response to an identifiable stressor occurring within three months of the onset of the stressor. The symptoms are disproportionate to the severity of the stressor and cause significant distress or impairment. It resolves within six months of the termination of the stressor. Subtypes include with depressed mood, with anxiety, with mixed anxiety and depressed mood, with disturbance of conduct, with mixed disturbance of emotions, and unspecified.

Affective Flattening

– A significant reduction in the range and intensity of emotional expression. It is a negative symptom of schizophrenia characterized by reduced facial expression, monotonous voice, and decreased body language. It is distinct from blunted affect (reduced range) and constricted affect (narrowed range). It is less responsive to antipsychotics.

Agitation

– A state of excessive motor activity, emotional excitement, and restlessness. In psychiatric settings, it may be a symptom of mania, psychosis, delirium, substance intoxication, or withdrawal. Acute agitation is managed with de-escalation techniques first, then pharmacological management with benzodiazepines (lorazepam), antipsychotics (haloperidol, olanzapine IM), or combination (B52: diphenhydramine + haloperidol + lorazepam). Physical restraints are used as a last resort.

Agoraphobia

– Marked fear or anxiety about two or more of five situations: using public transportation, being in open spaces, being in enclosed places, standing in line or being in a crowd, and being outside the home alone. The individual fears or avoids these situations because escape might be difficult or help unavailable if panic-like symptoms develop. The fear is out of proportion, persists for six months, and causes significant impairment.

Alcohol Use Disorder

– A problematic pattern of alcohol use leading to significant impairment or distress. Severity is classified as mild (2–3 criteria), moderate (4–5 criteria), or severe (6 or more criteria). Criteria include drinking more or longer than intended, persistent desire to cut down, spending excessive time with alcohol, craving, failure to fulfill obligations, continued use despite social or interpersonal problems, and tolerance and withdrawal. Withdrawal can include seizures, delirium tremens, and autonomic hyperactivity. Severe alcohol withdrawal can progress to delirium tremens, a life-threatening condition with a mortality rate of 5-15 percent if untreated. Management of alcohol withdrawal involves the CIWA-Ar (Clinical Institute Withdrawal Assessment for Alcohol, revised) scale to guide benzodiazepine administration. First-line treatment is symptom-triggered benzodiazepine therapy (lorazepam, diazepam, or chlordiazepoxide). Supportive care includes intravenous fluids, thiamine supplementation to prevent Wernicke-Korsakoff syndrome (100 mg IV or IM daily), correction of electrolyte disturbances (hypokalaemia, hypomagnesaemia, hypophosphataemia), and multivitamins. Carbamazepine and gabapentin are alternative treatments for mild-to-moderate withdrawal. Severe withdrawal may require ICU admission and management of delirium tremens.

Alcohol Withdrawal

– Symptoms occurring after cessation or reduction of prolonged heavy drinking. Timeline includes: tremor and anxiety at 6–12 hours, perceptual disturbances at 12–24 hours, withdrawal seizures at 24–48 hours, and delirium tremens at 48–96 hours. Delirium tremens includes autonomic hyperactivity, tremor, hallucinations, and confusion with a mortality rate of 5–15% if untreated. CIWA-Ar scale guides management. Benzodiazepines are first-line treatment.

Alogia

– Poverty of speech or poverty of content of speech, a negative symptom of schizophrenia. It manifests as brief, empty replies to questions, reduced spontaneous speech, and lack of elaboration. It reflects dysfunction in the mesocortical dopamine pathway and is less responsive to antipsychotic medications than positive symptoms.

Anhedonia

– The inability to experience pleasure from activities that were previously enjoyable. It is a core symptom of major depressive disorder and a negative symptom of schizophrenia. It is mediated by dysfunction in the brain's reward circuitry, particularly the mesolimbic dopamine system.

Anorgasmia

– Persistent or recurrent delay in, or absence of, orgasm following a normal sexual excitement phase. Causes: antidepressants (especially SSRIs, the most common iatrogenic cause), psychosocial factors (anxiety, relationship issues, past trauma), neurological disorders (spinal cord injury, multiple sclerosis, diabetic neuropathy), hormonal abnormalities (low testosterone, hypothyroidism), and pelvic surgery. Management: for SSRI-induced anorgasmia, strategies include dose reduction, drug holiday, switching to bupropion or mirtazapine, or augmentation with buspirone, cyproheptadine, or sildenafil. Psychosexual therapy and behavioural techniques are effective for psychogenic causes. Prognosis depends on etiology; medication-induced anorgasmia is usually reversible.

Anticholinergic Burden

– The cumulative effect of taking multiple medications with anticholinergic properties. It can cause dry mouth, constipation, urinary retention, blurred vision, confusion, and delirium, especially in elderly patients. Many psychiatric medications (TCAs, first-generation antipsychotics, some antihistamines) have anticholinergic effects.

Anticonvulsant Mood Stabilizers

– Antiepileptic drugs used as mood stabilizers in bipolar disorder. Valproate (divalproex) is effective for acute mania and rapid cycling. Carbamazepine is effective for acute mania but has numerous drug interactions via CYP3A4 induction. Lamotrigine is effective for bipolar depression prevention but not acute mania. All require monitoring for side effects and drug levels (except lamotrigine, which is dose-titrated).

Antisocial Personality Disorder

– A pattern of disregard for and violation of the rights of others occurring since age 15 years. At least three of seven criteria: failure to conform to social norms, deceitfulness, impulsivity, irritability and aggressiveness, reckless disregard for safety, consistent irresponsibility, and lack of remorse. Diagnosis requires evidence of conduct disorder before age 15. Associated with psychopathy, which includes affective and interpersonal features beyond the DSM criteria.

Anxiety

– A normal emotional response to perceived danger, but anxiety disorders involve excessive, persistent worry that interferes with daily life. Symptoms can include restlessness, rapid heart rate, muscle tension, and avoidance behaviors. Common types include generalized anxiety disorder, panic disorder, social anxiety, and phobias, all of which are treatable with therapy and medication.

Anxiety Disorder

– A group of psychiatric conditions characterized by excessive fear, anxiety, and related behavioral disturbances. Includes generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, specific phobias, and agoraphobia. GAD: persistent, excessive worry about multiple domains for ≥6 months, with restlessness, fatigue, difficulty concentrating, irritability, muscle tension, and sleep disturbance. Treatment: SSRIs/SNRIs (sertraline, escitalopram, venlafaxine), buspirone, benzodiazepines for short-term use, and cognitive-behavioral therapy (CBT).

Applied Behavior Analysis

– A behavioral intervention based on learning theory principles that uses systematic reinforcement to shape desired behaviors and reduce problematic behaviors. It is the primary evidence-based intervention for autism spectrum disorder, particularly in young children. Core components include positive reinforcement, prompting, fading, chaining, and generalization. It requires intensive, structured sessions.

Attention-Deficit/Hyperactivity Disorder

– A neurodevelopmental disorder characterized by persistent patterns of inattention, hyperactivity, and impulsivity that interfere with functioning or development. Three presentations: predominantly inattentive, predominantly hyperactive-impulsive, and combined. Inattention symptoms include difficulty sustaining attention, poor organization, forgetfulness, and distractibility. Hyperactivity-impulsive symptoms include fidgeting, excessive talking, difficulty waiting turns, and interrupting. Must have symptoms present before age 12. Diagnosis is clinical, using DSM-5 criteria, supported by rating scales (Conners Rating Scales, SNAP-IV, Vanderbilt Assessment Scale), collateral history from parents and teachers, and neuropsychological testing to evaluate attention and executive function. Management includes pharmacological treatment with psychostimulants (methylphenidate, dexamphetamine, lisdexamfetamine) as first-line, non-stimulant options (atomoxetine, guanfacine, clonidine), and behavioural interventions (parent training, behavioural classroom management, organisational skills training). Prognosis with treatment is good; untreated ADHD is associated with academic underachievement, social difficulties, substance misuse, and occupational impairment in adulthood.

Attention-Deficit/Hyperactivity Disorder (ADHD)

– A neurodevelopmental condition marked by persistent inattention, hyperactivity, and impulsivity that interferes with daily functioning. In children, symptoms often include difficulty staying seated, frequent interrupting, and trouble sustaining attention. Adults may experience restlessness, poor time management, and difficulty concentrating, though hyperactivity often diminishes with age.

Atypical Features

– A specifier for depressive episodes characterized by mood reactivity, significant weight gain or increased appetite, hypersomnia, leaden paralysis, and a long-standing pattern of interpersonal rejection sensitivity. Atypical depression often presents in younger patients and may respond preferentially to MAOIs or SSRIs rather than TCAs.

Autism Spectrum Disorder

– A neurodevelopmental disorder characterized by persistent deficits in social communication and social interaction across multiple contexts, and restricted, repetitive patterns of behavior, interests, or activities. Social deficits include difficulty with social-emotional reciprocity, nonverbal communicative behaviors, and developing/maintaining relationships. Restricted behaviors include stereotyped movements, insistence on sameness, and restricted interests. Severity is specified both domains. Diagnosis is based on DSM-5 criteria. Management includes early intensive behavioural intervention (applied behaviour analysis), speech and language therapy, social skills training, occupational therapy, and pharmacological treatment of associated conditions such as SSRIs for anxiety and antipsychotics for irritability. Prognosis is variable, with early intervention improving outcomes.

Avoidant Personality Disorder

– A pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation. At least four of seven criteria: avoids occupational activities involving interpersonal contact, unwilling to get involved unless certain of being liked, shows restraint within intimate relationships due to fear of shame, preoccupied with being criticized or rejected in social situations, inhibited in new interpersonal situations due to feelings of inadequacy, views self as socially inept, and is unwilling to get involved unless certain of being liked. Diagnosis is clinical using DSM-5 criteria. Treatment includes psychodynamic therapy and CBT focusing on social skills training and graded exposure to feared social situations. Pharmacotherapy with SSRIs may help with comorbid social anxiety. Prognosis is guarded, as the disorder is typically lifelong, though symptoms may improve with treatment.

Avolition

– A severe lack of motivation to initiate and perform self-directed, purposeful activities. It is a negative symptom of schizophrenia and is seen in severe depression. It manifests as neglect of personal hygiene, inability to complete tasks, and social withdrawal. It is mediated by dysfunction in the prefrontal cortex and dopamine reward pathways.

Benzodiazepine Use Disorder

– A problematic pattern of benzodiazepine use leading to significant impairment. Benzodiazepines include alprazolam, clonazepam, diazepam, and lorazepam. Dependence can develop rapidly with regular use. Withdrawal can be life-threatening, presenting with anxiety, insomnia, tremor, seizures, and psychosis. Cross-tolerance with alcohol exists. Abrupt discontinuation should be avoided; gradual taper is essential.

Benzodiazepine Withdrawal

– Symptoms from cessation or reduction of prolonged benzodiazepine use. Early symptoms include anxiety, insomnia, tremor, and diaphoresis. Severe withdrawal can include seizures, psychosis, and delirium. Withdrawal severity depends on the half-life of the benzodiazepine, duration of use, and dose. Abrupt discontinuation after chronic use can be fatal. A slow taper, often switching to a long-acting benzodiazepine, is the standard approach.

Bipolar Disorder

– A mood disorder characterized by episodes of mania/hypomania and depression. Bipolar I: at least one manic episode (elevated mood, grandiosity, decreased need for sleep, pressured speech, increased goal-directed activity, risky behavior) lasting ≥ 1 week or requiring hospitalization. Bipolar II: hypomanic episodes and major depressive episodes without full mania. Treatment: mood stabilizers (lithium, valproate, lamotrigine), atypical antipsychotics, and psychotherapy. Antidepressants used cautiously due to risk of mood switching.

Bipolar I Disorder

– A mood disorder defined by at least one manic episode lasting at least seven days or requiring hospitalization. A manic episode is a distinct period of abnormally elevated, expansive, or irritable mood and increased goal-directed activity or energy. Associated symptoms include inflated self-esteem, decreased need for sleep, pressured speech, flight of ideas, distractibility, increased activity, and risky behavior. Major depressive episodes may occur but are not required for diagnosis.

Bipolar II Disorder

– A mood disorder characterized by at least one hypomanic episode and at least one major depressive episode, with no history of a full manic episode. The depressive episodes in bipolar II tend to be more frequent and longer-lasting than in bipolar I, leading to greater cumulative disability. Patients may be misdiagnosed with unipolar depression if the hypomanic episodes are not identified.

Body Dysmorphic Disorder

– Preoccupation with one or more perceived defects or flaws in physical appearance that are not observable or appear slight to others. The individual performs repetitive behaviors in response to appearance concerns such as mirror checking, excessive grooming, skin picking, reassurance seeking, and comparing. The preoccupation causes clinically significant distress or impairment. Insight is often poor, and patients may seek cosmetic procedures that do not resolve the concern.

Borderline Personality Disorder

– A pattern of instability in interpersonal relationships, self-image, and affects, with marked impulsivity. At least five of nine criteria: frantic efforts to avoid abandonment, unstable and intense relationships, identity disturbance, impulsivity, recurrent suicidal behavior, affective instability, chronic feelings of emptiness, inappropriate intense anger, and transient paranoid ideation or dissociation. Strongly associated with childhood trauma, especially sexual abuse. DBT is the evidence-based treatment of choice. Other treatments include mentalisation-based therapy (MBT), transference-focused psychotherapy (TFP), and general psychiatric management (GPM). Pharmacotherapy targets specific symptoms: SSRIs for mood instability, low-dose antipsychotics for cognitive-perceptual symptoms, and mood stabilisers for affective dysregulation. Prognosis is guarded; symptoms typically improve with age and treatment, but functional impairment often persists.

Brief Psychotic Disorder

– A disturbance characterized by the sudden onset of at least one psychotic symptom: delusions, hallucinations, disorganized speech, or grossly disorganized or catatonic behavior. The disturbance lasts at least one day but less than one month, after which the individual fully returns to premorbid functioning. It may occur in response to a very stressful event and is diagnosed only when the disturbance is not better explained by another psychotic disorder, mood disorder, or substance use.

Caffeine Withdrawal

– A syndrome occurring after abrupt cessation or reduction of caffeine use. Symptoms include headache, fatigue, depressed mood, difficulty concentrating, and flu-like symptoms. Onset is typically 12–24 hours after last caffeine intake, peak at 20–51 hours, and duration of 2–9 days. It is one of the most commonly experienced withdrawal syndromes.

Cannabis Use Disorder

– A problematic pattern of cannabis use leading to significant impairment. Criteria include tolerance and withdrawal (irritability, anger, decreased appetite, restlessness, anxiety, sleep disturbance). Cannabis intoxication includes euphoria, impaired judgment, and time distortion. High-potency concentrates (dabbing) increase the risk of dependence. Cannabinoid hyperemesis syndrome presents with cyclic vomiting in heavy users.

Catatonia

– A neuropsychiatric syndrome characterized by motor abnormalities including immobility, mutism, posturing, waxy flexibility, stereotypy, mannerisms, grimacing, echolalia, and echopraxia. It can occur in mood disorders, psychotic disorders, medical conditions, or as catatonia due to another medical condition. Lorazepam challenge and ECT are first-line treatments. Immobility and autonomic instability suggest malignant catatonia.

Claustrophobia

– An anxiety disorder characterized by an intense, irrational fear of enclosed or confined spaces (e.g., lifts, tunnels, MRI scanners, windowless rooms). Individuals experience marked anxiety, panic attacks, and avoidance behaviour. Prevalence: approximately 5-10% of the general population. Claustrophobia is classified as a specific phobia (situational type) in DSM-5. The fear is out of proportion to the actual danger and causes significant distress or functional impairment. Pathophysiology involves conditioned fear responses, catastrophic misinterpretation of bodily sensations, and possibly a heightened sensitivity to suffocation cues. Management: CBT with graded in vivo exposure is the most effective treatment; virtual reality exposure therapy is an emerging alternative. Pharmacotherapy (SSRIs, benzodiazepines) may be used adjunctively. Prognosis is excellent with treatment; most patients achieve significant symptom reduction.

Cognitive Behavioral Therapy

– A structured, time-limited psychotherapy that identifies and modifies maladaptive thought patterns (cognitive distortions) and behaviors. It is based on the cognitive model that thoughts, feelings, and behaviors are interconnected. CBT uses techniques including cognitive restructuring, behavioral activation, exposure, and skills training. It is the most evidence-based psychotherapy for depression, anxiety disorders, OCD, PTSD, and eating disorders.

Cognitive Behavioral Therapy for Insomnia (CBT-I)

– Cognitive behavioral therapy for insomnia is a structured, evidence-based program that targets the thoughts and behaviors perpetuating poor sleep. Components include sleep restriction, stimulus control, cognitive restructuring, and sleep hygiene education. CBT-I is recommended as first-line treatment for chronic insomnia and yields durable improvements without the risks of sleep medications.

Cognitive Distortions

– Systematic errors in thinking that perpetuate negative emotional states and maladaptive behaviors. Common distortions include all-or-nothing thinking, catastrophizing, mind reading, overgeneralization, emotional reasoning, and should statements. CBT focuses on identifying and modifying these patterns.

Comorbidity

– The co-occurrence of two or more psychiatric disorders in a single patient. Psychiatric comorbidity is the rule rather than the exception. For example, depression commonly co-occurs with anxiety disorders, substance use disorders, and personality disorders. Comorbidity complicates treatment and worsens prognosis.

Compulsions

– Repetitive behaviors or mental acts that the individual feels driven to perform in response to an obsession or according to rigid rules. They are aimed at preventing or reducing anxiety or preventing some dreaded event, but are not connected in a realistic way with what they are designed to neutralize. Common compulsions include washing, checking, counting, ordering, arranging, and mental rituals. Time-consuming compulsions (more than one hour daily) are associated with greater severity.

Conduct Disorder

– A childhood/adolescent disorder characterized by a persistent pattern of violating others' basic rights or major age-appropriate norms. Behaviors: aggression to people/animals, destruction of property, deceitfulness/theft, and serious rule violations. Diagnosis requires $\geq$3 criteria in the past 12 months with functional impairment. Risk factors: abuse, neglect, parental substance use, and genetic predisposition. Treatment: parent management training, multisystemic therapy, and CBT; limited pharmacotherapy for comorbid conditions.

Consultation-Liaison Psychiatry

– The subspecialty of psychiatry focused on the interface between psychiatry and general medical illness. Psychiatrists provide consultation on medical-surgical wards for patients with psychiatric symptoms arising from or coexisting with medical conditions. Common issues include delirium, depression in medically ill patients, factitious disorder, adjustment reactions, and capacity assessments. It requires expertise in both psychiatric and medical conditions.

Conversion Disorder

– A condition in which patients present with neurological symptoms (weakness, paralysis, non-epileptic seizures, sensory loss, gait abnormalities, dystonia, speech difficulties) that cannot be explained by organic disease. Previously known as hysteria. The symptoms are not intentionally produced (unlike malingering or factitious disorder). Associated with psychological stressors and trauma. Most common in young women. Diagnosis: positive signs on neurological exam (Hoover sign for weakness, variability in dystonia), inconsistency with known neurological disease. Treatment: psychoeducation, physical therapy, CBT, and treatment of comorbid psychiatric conditions. Prognosis: variable; acute onset has better outcomes.

Countertransference

– The therapist's emotional reactions to the patient, which may be influenced by the therapist's own unresolved conflicts or may be a response to the patient's transference. Awareness of countertransference is essential for effective therapy and can provide valuable clinical information about the patient's interpersonal style.

Cross-Tolerance

– Tolerance to one substance that extends to another substance with a similar mechanism of action. For example, tolerance to one benzodiazepine extends to all benzodiazepines, and tolerance to alcohol extends partially to benzodiazepines due to shared GABA-A receptor mechanisms. Cross-tolerance is clinically relevant when switching between substances of the same class.

Cultural Psychiatry

– The study and clinical practice of how cultural factors influence mental health, illness expression, diagnosis, and treatment. Cultural concepts of distress, idioms of distress, and cultural syndromes must be considered. The DSM-5 includes a Cultural Formulation Interview. Culture influences symptom presentation, help-seeking behavior, treatment adherence, and the therapeutic relationship.

Cyclothymic Disorder

– A chronic mood disorder characterized by numerous periods of hypomanic symptoms and periods of depressive symptoms over at least two years (one year in children and adolescents). The hypomanic and depressive symptoms do not meet criteria for a hypomanic or major depressive episode. The patient is never symptom-free for more than two months at a time. It carries a 15–50% risk of developing bipolar I or II disorder.

Defense Mechanisms

– Unconscious psychological strategies used to manage anxiety and protect the ego from distressing thoughts or feelings. Mature defenses include sublimation, humor, and altruism. Immature defenses include projection, denial, splitting, and acting out. Understanding defense mechanisms helps clinicians conceptualize personality organization.

Delayed Ejaculation

– A sexual dysfunction characterized by a persistent or recurrent delay in, infrequency of, or absence of ejaculation during partnered sexual activity, lasting ≥6 months and causing marked distress. Prevalence: 1-5% of men. Lifelong vs. acquired. Causes: psychosocial (masturbatory style discrepancy, relationship issues, performance pressure, religious upbringing), medication-induced (SSRIs, antipsychotics, alpha-blockers, thiazide diuretics), neurological (spinal cord injury, multiple sclerosis, diabetic neuropathy), surgical (radical prostatectomy, retroperitoneal lymph node dissection), and hormonal (low testosterone, hyperprolactinemia). Diagnosis: clinical interview distinguishing anejaculation from retrograde ejaculation (post-orgasmic urine analysis). Treatment: address underlying cause, switch/reduce offending medications, vibratory stimulation (for neurogenic), electroejaculation (for fertility), psychotherapy, and off-label cabergoline or oxytocin (limited evidence).

Delirium

– An acute, fluctuating disturbance of attention and awareness with an underlying medical cause. It is characterized by disorganized thinking, altered level of consciousness, and perceptual disturbances. Risk factors include advanced age, dementia, infection, medications, and metabolic derangements. Prevention through hydration, sleep promotion, and minimizing restraints is essential.

Delirium Tremens

– The most severe form of alcohol withdrawal syndrome, a medical emergency with mortality of 5–15% if untreated. Typically occurs 48–96 hours after the last drink in individuals with chronic heavy alcohol use. Characterized by: altered mental status (confusion, disorientation), profound autonomic hyperactivity (tachycardia, hypertension, hyperthermia, diaphoresis), tremor, hallucinations (visual, tactile, auditory), agitation, and seizures. Risk factors: prior DTs, concurrent infection or illness, electrolyte abnormalities. Management: ICU admission, benzodiazepines (diazepam or lorazepam, high-dose titrated to symptom control), thiamine, correction of electrolyte disturbances, and supportive care (hydration, nutrition, monitoring).

Delusional Disorder

– A disorder characterized by the presence of one or more delusions lasting one month or more in the context of relatively preserved functioning. The delusions are non-bizarre (involving real-life situations that could conceivably occur, such as being followed, infected, loved from afar, or conspired against). If hallucinations are present, they are not prominent and are related to the delusional theme. The behavior is not obviously bizarre or odd beyond the impact of the delusion.

Denial

– A defense mechanism in which a person refuses to accept an unpleasant reality. It is a common initial response to traumatic events and terminal illness diagnoses. While adaptive in the short term, chronic denial can prevent appropriate medical treatment and psychological processing.

Dependent Personality Disorder

– A pattern of submissive and clinging behavior related to an excessive need to be taken care of. At least five of eight criteria: difficulty making everyday decisions without excessive advice, needing others to assume responsibility, difficulty expressing disagreement due to fear of loss of support, difficulty initiating projects, going to excessive lengths to obtain nurturance, feeling uncomfortable or helpless when alone, urgently seeking another relationship when one ends, and unrealistically preoccupied with fears of being left to care for self. Diagnosis is clinical using DSM-5 criteria. Treatment includes psychodynamic psychotherapy and CBT focusing on building autonomy and assertiveness. Pharmacotherapy is used for comorbid depression or anxiety. Prognosis is variable, with some improvement possible with long-term therapy.

Depersonalization-Derealization Disorder

– Persistent or recurrent experiences of depersonalization, derealization, or both. Depersonalization includes experiences of unreality, detachment, or being an outside observer with respect to one's thoughts, feelings, sensations, body, or actions. Derealization includes experiences of unreality or detachment with respect to surroundings. The individual maintains intact reality testing, meaning they recognize the experiences as subjective and not objectively real.

Depression

– Major depressive disorder is a common, disabling mood disorder characterized by persistent sadness and loss of interest or pleasure in activities, affecting approximately 7 percent of adults worldwide annually. According to DSM-5 criteria, diagnosis requires five or more of the following symptoms present during the same 2-week period representing a change from previous functioning, with at least one being either depressed mood or anhedonia: depressed mood most of the day nearly every day, marked diminished interest or pleasure in all or almost all activities most of the day nearly every day, significant weight loss or gain, insomnia or hypersomnia, psychomotor agitation or retardation, fatigue or loss of energy, feelings of worthlessness or excessive or inappropriate guilt, diminished ability to think or concentrate, and recurrent thoughts of death or suicidal ideation. The episode causes clinically significant distress or impairment. Depression is the leading cause of disability worldwide and is a major risk factor for suicide. Treatment includes antidepressants (SSRIs, SNRIs, bupropion, mirtazapine as first-line), psychotherapy (CBT, interpersonal therapy, behavioural activation), and combination therapy for moderate-to-severe depression. Electroconvulsive therapy (ECT) is the most effective treatment for severe depression with psychotic features or treatment resistance. Prognosis is variable; many patients achieve remission, but recurrence rates are high, particularly without maintenance treatment. Management of suicide risk is a priority.

Dialectical Behavior Therapy

– A comprehensive cognitive-behavioral treatment originally developed for borderline personality disorder. It combines cognitive-behavioral techniques with mindfulness and acceptance strategies. Core modules include mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness. DBT is the treatment of choice for borderline personality disorder and has been adapted for substance use disorders, eating disorders, and PTSD.

Dissociation

– A disruption in the normal integration of consciousness, memory, identity, emotion, perception, behavior, and sense of self. It exists on a spectrum from normal (daydreaming) to pathological (DID, depersonalization-derealization). It is strongly associated with childhood trauma and can be triggered by stress, sleep deprivation, and certain substances.

Dissociative Amnesia

– Inability to recall important autobiographical information, usually of a traumatic or stressful nature, that is inconsistent with ordinary forgetting. The amnesia may be localized, selective, generalized, or continuous. In dissociative amnesia with dissociative fugue, there is purposeful travel or bewildered wandering associated with amnesia for identity or other important autobiographical information.

Dissociative Identity Disorder

– The presence of two or more distinct personality states or an experience of possession involving marked discontinuity in sense of self, agency, affect, behavior, memory, perception, cognition, and/or sensory-motor functioning. Identity disruption is accompanied by gaps in recall of everyday events, personal information, and/or traumatic events that are inconsistent with ordinary forgetting. The disturbance causes clinically significant distress or impairment and is not part of a broadly accepted cultural or religious practice. Caused by severe childhood trauma, particularly prolonged and severe abuse in early childhood. The presentation is often misunderstood and can be confused with malingering. Diagnosis requires careful clinical assessment using structured interviews (SCID-D). Treatment focuses on trauma-informed psychotherapy, with phases of stabilisation, trauma processing, and integration. Medications treat comorbid symptoms. Prognosis is guarded.

Dysthymia (Persistent Depressive Disorder)

– A chronic form of depression with depressed mood for most of the day, more days than not, for ≥2 years (≥1 year in children/adolescents). Symptoms: poor appetite or overeating, insomnia or hypersomnia, low energy, low self-esteem, poor concentration, and hopelessness. Does not meet full criteria for major depressive episode continuously. Double depression: MDE superimposed on dysthymia. Treatment: SSRIs/SNRIs and psychotherapy (CBT, interpersonal therapy).

Eating Disorder

– A group of conditions characterized by abnormal eating habits and distorted body image, including anorexia nervosa, bulimia nervosa, binge eating disorder, and avoidant/restrictive food intake disorder (ARFID). Anorexia: severe calorie restriction, intense fear of weight gain, and body image disturbance with dangerously low weight. Bulimia: binge eating followed by purging behaviors. Binge eating: recurrent binge eating without purging. Treatment: nutritional rehabilitation, CBT, family-based therapy for adolescents, and fluoxetine for bulimia. Medical complications require monitoring.

Electroconvulsive Therapy

– A neuromodulation treatment involving the induction of a generalized seizure under general anesthesia using electrical current applied to the scalp. It is the most effective treatment for severe depression, especially with psychotic features, catatonia, or suicidal ideation. It is also effective for severe mania and treatment-resistant schizophrenia. Cognitive side effects include transient memory impairment. Bilateral electrode placement is more effective but has more cognitive side effects than unilateral placement. ECT is highly effective, with response rates of 70-90 percent for severe depression. The mechanism of action is not fully understood but involves modulation of neurotransmitter systems, neuroplasticity, and increased hippocampal neurogenesis. Modern ECT is performed under general anesthesia with muscle relaxation, using brief electrical pulses to induce a generalised seizure lasting approximately 20-60 seconds. Contraindications include raised intracranial pressure, recent myocardial infarction, and unstable vascular aneurysms. Memory loss (both retrograde and anterograde) is the most significant side effect and is usually transient. Maintenance ECT is sometimes used to prevent relapse.

Erectile Disorder

– A sexual dysfunction characterized by persistent or recurrent difficulty achieving or maintaining an erection sufficient for sexual activity, or a marked decrease in erectile rigidity, lasting ≥6 months and causing marked distress. Prevalence: 10-20% of men under 60, rising to 40-70% over age 70. Etiology: organic (vascular disease, diabetes, hypertension, hyperlipidemia, obesity, smoking, hypogonadism, neurological conditions, postsurgical) and psychogenic (performance anxiety, depression, relationship conflict). Diagnosis: clinical history, IIEF-5 questionnaire, nocturnal penile tumescence testing, penile duplex Doppler ultrasound, and hormonal profile. First-line treatment: PDE5 inhibitors (sildenafil, tadalafil, vardenafil, avanafil). Second-line: intracavernosal injections (alprostadil, papaverine, phentolamine), vacuum erection devices, and low-intensity shockwave therapy. Third-line: penile prosthesis surgery. Treat underlying medical conditions and offer psychotherapy for psychogenic factors.

Euphoria

– An intense feeling of well-being, elation, or happiness, which can be a normal emotional response or a pathologic symptom. In medical contexts, euphoria is most commonly associated with: substance use (opioids, stimulants–cocaine, amphetamines, MDMA, cannabis, alcohol), psychiatric disorders (mania in bipolar I disorder–elevated, expansive, or irritable mood with increased energy), neurologic conditions (multiple sclerosis–lesional euphoria, frontal lobe syndromes), and endocrine disorders (Cushing syndrome, hyperthyroidism). Euphoric mood is distinguished from euthymia (normal mood) and from the emotional lability of pseudobulbar affect. Pathologic euphoria may impair judgment and lead to risky behavior.

Excoriation Disorder

– Recurrent skin picking resulting in skin lesions. The individual has made repeated attempts to decrease or stop the skin picking. The skin picking causes clinically significant distress or impairment. It is not attributable to a medical condition or substance.

Common sites include face, arms, and hands. The picking may be automatic or focused and is often preceded by tension and followed by relief.

Exhibitionistic Disorder

– A paraphilic disorder involving recurrent, intense sexual arousal from exposing one's genitals to an unsuspecting person. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration ≥6 months. Onset typically before age 18. Treatment: CBT and SSRIs.

Extrapyramidal Symptoms

– Drug-induced movement disorders caused by D2 receptor blockade in the nigrostriatal pathway. They include acute dystonia (muscle spasm, typically within hours of starting), akathisia (inner restlessness), drug-induced parkinsonism (tremor, rigidity, bradykinesia after weeks of use), and tardive dyskinesia (involuntary movements after months to years). EPS is more common with high-potency typical antipsychotics. Anticholinergic agents (benztropine, diphenhydramine) treat acute dystonia and parkinsonism, and beta-blockers (propranolol) treat akathisia. Tardive dyskinesia is managed by switching to a second-generation antipsychotic with lower D2 potency (clozapine, quetiapine) and adding VMAT2 inhibitors (valbenazine, deutetrabenazine). Prevention involves using the lowest effective dose of antipsychotic and regular monitoring using the Abnormal Involuntary Movement Scale (AIMS).

Factitious Disorder

– Presentation of physical or psychological symptoms or imposing injury or disease on another, with the deceptive intention of assuming the sick role. In factitious disorder imposed on self, the individual falsifies signs or symptoms, induces injury or disease, and presents to medical providers. In factitious disorder imposed on another, the individual falsifies symptoms in another person. The deceptive behavior occurs even in the absence of obvious external rewards.

Female Orgasmic Disorder (Anorgasmia)

– A sexual dysfunction characterized by a persistent or recurrent delay in, infrequency of, or absence of orgasm, or markedly reduced intensity of orgasmic sensation, in ≥75% of sexual encounters, lasting ≥6 months and causing marked distress. Lifelong (primary, never achieved orgasm by any means) vs. acquired (previously able) vs. situational (only with certain partners or activities). Prevalence: 10-15% of women. Causes: psychosocial (anxiety, distraction, lack of knowledge, relationship issues, negative attitudes toward sex), medication-induced (SSRIs — most common, antipsychotics), medical (spinal cord injury, multiple sclerosis, diabetes, pelvic surgery), and partner factors. Diagnosis: clinical DSM-5 criteria. Treatment: education, directed masturbation, CBT, mindfulness-based sex therapy, sensate focus, and vibratory stimulation. Switching to bupropion or adding buspirone may help with SSRI-induced anorgasmia. Filbanserin and bremelanotide are approved for HSDD not anorgasmia specifically.

Female Sexual Interest/Arousal Disorder

– A sexual dysfunction in women characterized by absent or reduced sexual interest, thoughts, or receptivity, and/or absent or reduced sexual arousal (genital or nongenital), persisting for ≥6 months and causing marked distress. Combines the former DSM-IV diagnoses of hypoactive sexual desire disorder and female sexual arousal disorder. Prevalence: 10-20% of women. Etiology: multifactorial (relationship factors, partner factors, individual vulnerability, medical/psychiatric comorbidities, medication effects). Diagnosis: clinical DSM-5 criteria. Treatment: psychotherapy (CBT, mindfulness-based sex therapy), sensate focus exercises, education, addressing relationship factors, and pharmacotherapy (flibanserin, bremelanotide, off-label testosterone therapy). Lubricants and vaginal moisturizers for arousal complaints.

Fetishistic Disorder

– A paraphilic disorder characterized by recurrent, intense sexual arousal from either the use of nonliving objects (fetish) or a highly specific focus on nongenital body parts (partialism), lasting ≥6 months and causing distress or impairment or involving acting on these urges. Common fetishes: shoes, feet, footwear, leather, lingerie, and rubber. Fetishism is common in the general population (up to 30% report some fetish interest) and typically only becomes a disorder when it causes significant distress, impairment, or involves nonconsenting individuals or causes harm.

Most common in males. Treatment: CBT, aversion therapy (historical, controversial), and SSRIs.

Forensic Psychiatry

– The interface of psychiatry and the legal system. It involves evaluations for criminal responsibility, competency to stand trial, malingering assessment, risk assessment for violence, and civil commitment. Forensic psychiatrists serve as expert witnesses and provide opinions on legal questions. Malingering is the intentional production of false or grossly exaggerated symptoms for external gain.

Frotteuristic Disorder

– A paraphilic disorder involving recurrent, intense sexual arousal from touching or rubbing against a nonconsenting person, lasting ≥6 months and causing distress or impairment or involving acting on these urges. Typically occurs in crowded public places (subways, buses, concerts). The individual rubs their genitals against the victim's buttocks or thighs. Onset: adolescence, almost exclusively male. Typically decreasing with age. Legal consequences: sexual offense charges. Treatment: CBT, relapse prevention, and SSRIs.

Gender Dysphoria

– Psychological distress resulting from incongruence between one's experienced gender and assigned gender at birth. Persists for ≥6 months and causes significant impairment. Onset in childhood (may resolve) or adolescence/adulthood (persistent). Diagnosis: DSM-5 requires marked incongruence between experienced/expressed gender and primary/secondary sex characteristics, with strong desire to be rid of them or to have opposite sex characteristics. Treatment: gender-affirming care including hormone therapy (testosterone/estrogen), puberty blockers for adolescents, and surgical interventions. Access to affirming care improves mental health outcomes.

Generalized Anxiety Disorder

– Characterized by excessive, difficult-to-control anxiety and worry about multiple events or activities occurring more days than not for at least six months. Associated symptoms include restlessness, being easily fatigued, difficulty concentrating, irritability, muscle tension, and sleep disturbance. The anxiety causes significant distress or impairment. It is often comorbid with depression and other anxiety disorders.

Genito-Pelvic Pain/Penetration Disorder

– A sexual dysfunction characterized by persistent or recurrent difficulties with vaginal penetration, marked vulvovaginal or pelvic pain during intercourse, fear or anxiety about penetration, and/or tensing of the pelvic floor muscles, lasting ≥6 months and causing marked distress. Combines the former DSM-IV diagnoses of vaginismus and dyspareunia. Prevalence: 3-15% of women. Etiology: multifactorial (pelvic floor hypertonicity, history of sexual trauma, anxiety, endometriosis, vulvodynia, interstitial cystitis, postpartum/perimenopausal changes, inadequate lubrication). Diagnosis: clinical interview, pelvic exam, and exclusion of organic causes (infection, endometriosis, adhesions). Treatment: multimodal — pelvic floor physical therapy, dilator therapy, CBT/sex therapy, lubricants, topical lidocaine, vaginal estrogen (if atrophic), trigger point injections, and treatment of underlying medical conditions. Graded exposure to penetration is key.

Geriatric Psychiatry

– The subspecialty of psychiatry focused on the prevention, evaluation, diagnosis, and treatment of mental and emotional disorders in older adults. Geriatric patients often present atypically, with depression manifesting as somatic complaints and cognitive decline. Common conditions include late-life depression, dementia, delirium, and late-onset psychosis. Polypharmacy, medical comorbidity, and altered drug metabolism complicate treatment.

Grief

– A natural emotional response to loss, particularly the death of a loved one, but also following other significant losses such as divorce, job loss, or declining health. It encompasses a range of feelings including sadness, anger, guilt, and numbness, often occurring in waves rather than linear stages. While grief is not a mental disorder, prolonged or disabling grief may be diagnosed as prolonged grief disorder and benefit from specialized therapy.

Hallucination

– A perception in the absence of an external stimulus, experienced as real and occurring in objective space. Hallucinations can involve any sensory modality:

auditory (most common, voices commenting or conversing–characteristic of schizophrenia, also in bipolar mania, psychotic depression), visual (formed or unformed–delirium, dementia, Charles Bonnet syndrome in visually impaired with preserved cognition, Lewy body dementia), olfactory (unusual odours–temporal lobe epilepsy, olfactory reference syndrome), gustatory (tastes–seizures, psychosis), tactile (formication–cocaine/amphetamine use, alcohol withdrawal, delusional infestation), and somatic (sensations within the body). Hallucinations are distinct from: illusions (distortions of real external stimuli), hallucinations in sleep (hypnagogic at sleep onset, hypnopompic on waking–normal if not distressing), and pseudohallucinations (experienced as lacking objective reality, located inside the head, often in PTSD). Hallucinations in the context of psychiatric illness must be differentiated from those due to: delirium (acute, fluctuating, medical cause), dementia (chronic, progressive), epilepsy (ictal/postictal, especially temporal lobe), drugs (stimulants, hallucinogens like LSD, psilocybin, ketamine, withdrawal from alcohol/benzodiazepines), sensory deprivation, and migraine (visual aura). Management: treat underlying cause; antipsychotics for psychiatric disorders (olanzapine, risperidone, clozapine for refractory auditory hallucinations); ECT for severe psychotic depression.

Histrionic Personality Disorder

– A pattern of excessive emotionality and attention-seeking. At least five of eight criteria: discomfort when not the center of attention, inappropriate sexually seductive behavior, displays rapidly shifting and shallow emotions, uses physical appearance for attention, speaks in an impressionistic and vague manner, shows self-dramatization and theatricality, is suggestible, and considers relationships more intimate than they actually are.

Hoarding Disorder

– Persistent difficulty discarding or parting with possessions, regardless of their actual value. The individual perceives a need to save items and experiences distress at the thought of discarding them. This results in accumulation of possessions that congest and clutter active living areas. The hoarding is not attributable to another medical condition and causes significant distress or impairment. It is distinct from collecting, which is organized and purposeful.

Hypoactive Sexual Desire Disorder (HSDD)

– A sexual dysfunction characterized by persistently or recurrently deficient or absent sexual thoughts, fantasies, and desire for sexual activity, causing marked distress or interpersonal difficulty. In DSM-5, HSDD in women is merged with female sexual arousal disorder into female sexual interest/arousal disorder. In men, it remains a separate diagnosis. Prevalence: 10-15% of men, 20-30% of women. Causes: psychosocial (relationship conflict, stress, depression, anxiety), hormonal (low testosterone, hyperprolactinemia, menopause), medication-induced (SSRIs, antipsychotics, hormonal contraceptives), and medical (diabetes, thyroid disease, chronic illness). Diagnosis: clinical interview ruling out other psychiatric disorders, medication effects, and medical conditions. Lab work: testosterone (total and free), prolactin, TSH. Treatment: treat underlying cause, modify offending medications, psychotherapy (CBT, sensate focus), and for women: flibanserin (Addyi) or bremelanotide (Vyleesi); for men with low testosterone: testosterone replacement therapy.

Hypomanic Episode

– A distinct period of abnormally elevated, expansive, or irritable mood with increased energy lasting at least four consecutive days. The episode is not severe enough to cause marked impairment in functioning or necessitate hospitalization, and there are no psychotic features. The change in functioning is observable by others. Hypomanic episodes are the defining feature of bipolar II disorder and may transition to full mania or major depression.

Hysteria

– A historical psychiatric term that has been discarded from modern diagnostic classifications (DSM-5, ICD-11). Originally derived from the Greek word for uterus (hystera), reflecting the ancient belief that symptoms originated from a 'wandering uterus'. Previously used to describe a range of symptoms: (1) Motor: paralysis, seizures, pseudoseizures (psychogenic non-epileptic seizures), mutism, aphonia; (2) Sensory: anesthesia, paraesthesia, blindness, deafness; (3) Dissociative: amnesia, fugue, depersonalisation, multiple personality; (4) Somatic: chronic pain, globus hystericus, hyperventilation. The term hysteria was replaced by more specific diagnostic categories: conversion disorder (functional neurological symptom disorder–motor/sensory symptoms without organic cause), somatisation disorder (now somatic symptom disorder, multiple physical complaints), dissociative disorders (dissociative identity disorder, dissociative amnesia). Freud redefined hysteria as resulting from repressed psychological trauma (especially sexual abuse) with unconscious conversion into physical symptoms. Pierre Janet proposed dissociation as the core mechanism. Current understanding integrates psychodynamic, behavioural, cognitive, and neurobiological factors, and does not attribute to the uterus.

Ideas of Reference

– The false belief that unrelated events, objects, or people have a particular and unusual significance specifically for the patient. For example, believing that TV newscasters are sending personal messages or that strangers are conspiring against them. While common in schizotypal personality disorder, they can also occur in schizophrenia and bipolar disorder.

Illness Anxiety Disorder

– Preoccupation with having or acquiring a serious illness, with minimal or no somatic symptoms present. The individual has a high level of anxiety about health, performs excessive health-related behaviors, and experiences maladaptive avoidance. Symptoms persist for at least six months. The individual is not delusional and insight may vary. Previously known as hypochondriasis.

Insomnia Disorder

– Difficulty initiating or maintaining sleep, or early-morning awakening with inability to return to sleep, at least three nights per week for at least three months. The individual experiences clinically significant distress or impairment in functioning. Despite adequate opportunity for sleep, the individual is unable to obtain sufficient sleep. It is not better explained by another sleep disorder, substance use, or medical condition. Cognitive behavioral therapy for insomnia (CBT-I) is first-line treatment. Pharmacological options include benzodiazepine receptor agonists (zolpidem, zaleplon, eszopiclone), melatonin receptor agonists (ramelteon), orexin receptor antagonists (suvorexant, lemborexant), and sedating antidepressants (trazodone, doxepin, mirtazapine). Benzodiazepines are generally avoided due to tolerance and dependence risks. Sleep hygiene measures and stimulus control are essential components of management.

Intermittent Explosive Disorder

– A disorder characterized by recurrent, impulsive, problematic outbursts of verbal or physical aggression that are grossly disproportionate to the provocation. The outbursts are not premeditated and cause marked distress or functional impairment. Age >6 years. Not better explained by another disorder (ADHD, conduct disorder, PTSD, ASD) or substance intoxication. Diagnosis: DSM-5 criteria including aggressive outbursts occurring twice weekly for 3 months (verbal/non-destructive) or 3 outbursts involving damage/injury in the past year. Treatment: CBT (anger management, relaxation), SSRIs (fluoxetine), and mood stabilizers.

Interpersonal Psychotherapy

– A time-limited, evidence-based psychotherapy for depression that focuses on the interpersonal context of mood symptoms. It identifies four problem areas: grief, role disputes, role transitions, and interpersonal deficits. IPT helps patients improve interpersonal functioning and social support. It is an alternative to CBT for depression and has been adapted for eating disorders, postpartum depression, and adolescent depression.

Leukotomy

– A neurosurgical procedure involving the cutting of white matter fibers in the prefrontal cortex, also known as frontal lobotomy or prefrontal leukotomy. Developed by Egas Moniz (Nobel Prize 1949) and Walter Freeman. leukotomy was used for severe psychiatric disorders (schizophrenia, severe depression, OCD) from the 1930s to 1950s, before the advent of effective psychopharmacology. The transorbital lobotomy (Freeman, using an ice pick-like instrument through the orbit) was a simplified outpatient procedure. Use declined dramatically in the 1950s with the introduction of antipsychotics (chlorpromazine) and antidepressants, and due to severe lasting side effects. Side effects of leukotomy: apathy, emotional blunting, inattention, executive dysfunction, cognitive impairment, seizures (5-10%), and death. leukotomy is rarely performed today, but modern stereotactic procedures (anterior capsulotomy, cingulotomy, subcaudate tractotomy) are used in extremely severe, treatment-refractory cases of OCD and depression.

Lithium

– A mood-stabilising medication used primarily for bipolar disorder (mania, hypomania, and prophylaxis of recurrent episodes). See Lithium Toxicity and Bipolar Disorder.

Lithium Toxicity

– Occurs when lithium levels exceed the therapeutic range. Mild toxicity (1.5–2.0 mEq/L): nausea, vomiting, diarrhea, coarse tremor, muscle weakness. Moderate toxicity (2.0–2.5 mEq/L): confusion, ataxia, blurred vision, dysarthria. Severe toxicity (above 2.5 mEq/L): seizures, coma, cardiac arrhythmias, renal failure, and potentially death. Treatment includes stopping lithium, IV fluids, and hemodialysis for severe cases. NSAIDs, ACE inhibitors, and thiazide diuretics increase lithium levels.

Lysergic Acid Diethylamide (LSD)

– A potent hallucinogenic drug (psychedelic) classified as a Schedule 1 controlled substance (high abuse potential, no accepted medical use). LSD acts as a partial agonist at serotonin 5-HT2A receptors in the prefrontal cortex, leading to altered sensory perception, mood, and cognition. Effects: visual hallucinations (geometric patterns, intensified colours, trails), synaesthesia (cross-modal sensory experience, e.g., hearing colours), profound introspection, euphoria/dysphoria, altered sense of time, depersonalization, and mystical experiences. Onset: 30-60 minutes after oral ingestion. Peak: 2-4 hours. Duration: 8-12 hours. Dose: typical 50-200 microgram. Adverse effects: 'bad trip' (anxiety, panic, paranoia, fear of losing control), flashbacks (recurrence of drug effect weeks to months later, Hallucinogen Persisting Perception Disorder–HPPD), exacerbation of pre-existing psychiatric illness (psychosis, bipolar disorder), and acute toxicity (hyperthermia, hypertension, tachycardia, seizures, serotonin syndrome). LSD is not considered physically addictive (does not produce withdrawal or compulsive drug-seeking behaviour), but tolerance develops rapidly (cross-tolerance with other 5-HT2A agonists like psilocybin and mescaline). Medical research: renewed interest in using LSD and other psychedelics (psilocybin, MDMA, ayahuasca) for treatment-resistant depression, anxiety in terminal illness, PTSD, and substance use disorders in controlled clinical settings.

Major Depressive Disorder

– A mood disorder characterized by at least one major depressive episode consisting of five or more symptoms during the same two-week period. At least one symptom must be either depressed mood or loss of interest or pleasure (anhedonia). Symptoms include significant weight change, insomnia or hypersomnia, psychomotor agitation or retardation, fatigue, feelings of worthlessness or excessive guilt, diminished concentration, and recurrent thoughts of death or suicidal ideation. The episode causes clinically significant distress or impairment. MDD is the leading cause of disability worldwide, affecting approximately 7 percent of adults annually. Treatment involves antidepressants (SSRIs, SNRIs, bupropion, mirtazapine), psychotherapy (CBT, interpersonal therapy, behavioural activation), and combination therapy for moderate-to-severe episodes. ECT is reserved for severe, psychotic, or treatment-resistant depression. Prognosis is variable; recurrence is common, and maintenance treatment is recommended.

Male Hypoactive Sexual Desire Disorder

– A sexual dysfunction in men characterized by persistently deficient or absent sexual/erotic thoughts or desire for sexual activity, causing marked distress. Prevalence: 5-15% of men. Distinguish from erectile disorder (men may have ED but intact desire). Causes: low testosterone (primary or secondary hypogonadism), hyperprolactinemia, depression, relationship conflict, medication side effects (SSRIs, antipsychotics, beta-blockers, spironolactone), chronic illness (diabetes, CKD, COPD), and aging. Diagnosis: total and free testosterone, prolactin, TSH, and clinical interview. Treatment: testosterone replacement (if confirmed hypogonadal), treat hyperprolactinemia, switch offending medications, CBT, and sex therapy. Testosterone is not recommended for age-related low T without clear hypogonadism.

Malingering

– The intentional fabrication or exaggeration of physical or psychological symptoms for external incentives (avoiding work, obtaining financial compensation, evading criminal prosecution, obtaining drugs). Not a psychiatric illness (coded as a V-code/Z-code). Suspicion raised by: non-compliance with treatment, antisocial personality disorder, discrepancies between reported symptoms and objective findings, and symptoms that emerge in specific contexts (legal, military, disability evaluation). Diagnosis: collateral information, psychological testing (MMPI-2 validity scales). No psychiatric treatment; management involves confrontation and documentation.

Mania

– A distinct period of abnormally elevated, expansive, or irritable mood with increased goal-directed activity or energy, lasting ≥1 week (or any duration if hospitalization required). Requires ≥3 of: inflated self-esteem, decreased need for sleep, pressured speech, flight of ideas, distractibility, increased goal-directed activity, and excessive involvement in risky activities. Causes significant functional impairment or psychosis. Core feature of bipolar I disorder. Treatment: mood stabilizers (lithium, valproate) and atypical antipsychotics.

Manic Episode

– A distinct period lasting at least seven days (or any duration if hospitalization is required) of abnormally elevated, expansive, or irritable mood with increased energy or activity. At least three symptoms must be present: inflated self-esteem or grandiosity, decreased need for sleep, pressured speech, flight of ideas or racing thoughts, distractibility, increased goal-directed activity or psychomotor agitation, and excessive involvement in pleasurable activities with high potential for painful consequences (e.g., buying sprees, sexual indiscretions, risky investments). The mood disturbance is severe enough to cause marked impairment in social or occupational functioning, requires hospitalisation to prevent harm to self or others, or includes psychotic features. First-line treatment is lithium or valproate, with atypical antipsychotics (olanzapine, quetiapine, risperidone, aripiprazole, and asenapine) as effective alternatives. Benzodiazepines are used as adjunctive therapy for acute agitation. ECT is highly effective for severe or refractory mania.

Melancholic Features

– A specifier for major depressive episodes characterized by a loss of interest or pleasure in all or almost all activities, a depressed mood that is qualitatively distinct from grief, prominent psychomotor retardation or agitation, significant anorexia or weight loss, excessive guilt, and worsening of mood in the morning. It suggests a more severe, biologically driven depression with better response to antidepressants or ECT.

Melatonin Receptor Agonists

– Medications that act on MT1 and MT2 receptors in the suprachiasmatic nucleus to regulate circadian rhythms. Ramelteon is approved for insomnia with sleep onset difficulty. It has no abuse potential and does not cause dependence. It is a useful alternative to benzodiazepines for sleep initiation problems.

Mental Health

– Mental health encompasses emotional, psychologic, and social well-being, affecting how individuals think, feel, behave, and handle stress, relationships, and decision-making. Common mental disorders: anxiety disorders (generalized anxiety, panic, phobias, social anxiety—lifetime prevalence 29%), depressive disorders (major depressive disorder, persistent depressive disorder—lifetime prevalence 21%), bipolar disorder (manic-depressive illness), schizophrenia and other psychotic disorders, and trauma/stressor-related disorders (PTSD, adjustment disorder). The biopsychosocial model recognizes genetic vulnerability (heritability 30–50% for major depression), neurobiologic factors (neurotransmitter dysregulation, HPA-axis abnormalities), psychological factors (maladaptive cognitions, coping deficits), and social determinants (childhood adversity, poverty, discrimination, isolation). Treatment: psychotherapy (CBT, DBT, psychodynamic, interpersonal therapy), pharmacotherapy (antidepressants, mood stabilizers, antipsychotics, anxiolytics), neuromodulation (ECT, TMS, ketamine/esketamine for treatment-resistant depression), and social support interventions. Screening (PHQ-9, GAD-7, AUDIT-C) in primary care improves detection and outcomes.

Mental Status Examination

– A structured assessment of the patient's current psychological functioning performed during the clinical interview. It evaluates appearance, behaviour, speech, mood, affect, thought process, thought content, perceptions, cognition, insight, and judgment. It provides a snapshot of the patient's mental state at the time of evaluation and is documented in the psychiatric note.

Methylphenidate

– See Methylphenidate in Pharmacology.

Monoamine Oxidase Inhibitors

– Antidepressants that inhibit monoamine oxidase, the enzyme responsible for breaking down serotonin, norepinephrine, and dopamine. MAOIs include phenelzine, tranylcypromine, isocarboxazid, and selegiline (transdermal). They are effective for atypical depression and treatment-resistant depression. The major risk is hypertensive crisis from tyramine-rich foods ("cheese reaction"). They are contraindicated with sympathomimetics, meperidine, and other serotonergic agents due to risk of serotonin syndrome. Dietary restrictions require avoidance of aged cheeses, cured meats, fermented foods, and certain alcoholic beverages (tyramine-rich foods). The transdermal selegiline patch bypasses gastrointestinal MAO and does not require dietary restrictions at the lowest dose. MAOIs are reserved for treatment-resistant cases due to their side effect profile and drug interactions.

Mood Stabilizers

– Medications used to treat mood episodes in bipolar disorder and prevent recurrence. The primary mood stabilizers are lithium, valproate, carbamazepine, and lamotrigine. Each has a different mechanism and side effect profile. Lithium is the only agent with demonstrated anti-suicide properties. Valproate is particularly effective for rapid-cycling bipolar disorder. Lamotrigine is more effective for bipolar depression than mania.

Motivational Interviewing

– A client-centered, directive counseling method for enhancing intrinsic motivation to change by exploring and resolving ambivalence. Developed by Miller and Rollnick, it uses the spirit of partnership, acceptance, compassion, and evocation. Core skills include open-ended questions, affirmations, reflections, and summaries. It is particularly effective for substance use disorders and health behavior change.

Narcissistic Personality Disorder

– A pattern of grandiosity, need for admiration, and lack of empathy. At least five of nine criteria: grandiose sense of self-importance, preoccupation with fantasies of unlimited success, belief of being "special," requires excessive admiration, sense of entitlement, interpersonally exploitative, lacks empathy, envious of others or believes others are envious, and shows arrogant, haughty behaviors. May present as grandiose or vulnerable (covert) narcissism.

Narcolepsy

– A chronic sleep disorder characterized by excessive daytime sleepiness and irresistible sleep attacks, with or without cataplexy (sudden bilateral loss of muscle tone triggered by strong emotions). Narcolepsy type 1: with cataplexy, caused by loss of hypocretin/orexin neurons in the hypothalamus. Other features: sleep paralysis (inability to move upon falling asleep or waking), hypnagogic/hypnopompic hallucinations, and disrupted nighttime sleep. Diagnosis: polysomnography followed by MSLT (mean sleep latency <8 minutes, ≥ 2 SOREM periods). Treatment: wake-promoting agents (modafinil, armodafinil, solriamfetol), sodium oxybate for cataplexy/EDS, and SSRIs/SNRIs for cataplexy.

Neuroleptic

– An older term for antipsychotic medications, derived from their ability to produce a state of mental indifference and motor slowing. The term is now largely replaced by "antipsychotic," though it is still used in specific contexts such as neuroleptic malignant syndrome. Typical neuroleptics are first-generation antipsychotics.

Neuroplasticity

– The brain's ability to reorganize its structure, function, and connections in response to experience, learning, and injury. It is relevant to psychiatric treatment because antidepressants, psychotherapy, and ECT are thought to promote neuroplastic changes including neurogenesis in the hippocampus and synaptic remodeling.

Obsessions

– Recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that are experienced as intrusive and cause marked anxiety or distress. Common obsessions include contamination fears, fears of harming oneself or others, need for symmetry, taboo thoughts (sexual, religious, or aggressive), and doubt. The individual recognizes the obsessions as excessive or irrational in many cases (good insight).

Obsessive-Compulsive Disorder

– Characterized by the presence of obsessions, compulsions, or both. Obsessions are recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that cause marked anxiety or distress. The individual attempts to ignore or suppress them or neutralize them with some other thought or action. Compulsions are repetitive behaviors or mental acts that the individual feels driven to perform in response to an obsession or according to rigid rules. They are aimed at preventing or reducing anxiety or preventing a dreaded event, but are not connected in a realistic way with what they are designed to neutralise. Compulsions are time-consuming (taking more than one hour per day) and cause significant distress or impairment. Insight varies: good or fair insight (the individual recognises the beliefs are probably not true), poor insight, or absent insight (delusional beliefs). First-line treatment is CBT with exposure and response prevention (ERP) and SSRIs (fluoxetine, fluvoxamine, sertraline, paroxetine, or escitalopram) at higher doses than used for depression. The combination of CBT and SSRI is superior to either alone. Prognosis is variable; symptoms may be chronic but are usually responsive to treatment.

Obsessive-Compulsive Personality Disorder

– A pattern of preoccupation with orderliness, perfectionism, and mental and interpersonal control, at the expense of flexibility and efficiency. At least four of eight criteria: preoccupied with details, rules, lists, organization, or schedules to the extent that the major point of the activity is lost; shows perfectionism that interferes with task completion; excessively devoted to work; overconscientious and inflexible about morality; unable to discard worn-out objects; reluctant to delegate tasks unless others submit to their way of doing things. Diagnosis is clinical using DSM-5 criteria. Treatment is challenging as patients rarely seek help voluntarily. Psychodynamic therapy and CBT may help address perfectionism and control. Pharmacotherapy targets comorbid depression or anxiety. Prognosis is guarded, as the personality traits are ego-syntonic and resistant to change.

OCD

– See Obsessive-Compulsive Disorder.

Opioid Overdose

– A medical emergency characterized by the classic triad of respiratory depression, miosis (pinpoint pupils), and decreased level of consciousness. It can progress to respiratory arrest, cardiovascular collapse, and death. Treatment involves airway management, naloxone administration (0.4–2 mg IV/IM, may repeat), and monitoring for re-narcotization as naloxone wears off. Fentanyl and its analogs require higher and repeated naloxone doses.

Opioid Use Disorder

– A problematic pattern of opioid use leading to significant impairment or distress. Opioids include heroin, fentanyl, morphine, oxycodone, and hydrocodone. Criteria are the same eleven as for substance use disorders generally. Overdose presents with respiratory depression, miosis, and decreased level of consciousness. Naloxone is the antidote. Treatment includes methadone, buprenorphine, and naltrexone.

Oppositional Defiant Disorder

– A childhood/adolescent disorder characterized by a persistent pattern (≥6 months) of angry/irritable mood, argumentative/defiant behavior, and vindictiveness. Symptoms: often loses temper, is touchy/easily annoyed, angry/resentful, argues with authority figures, actively defies rules, deliberately annoys others, blames others for mistakes, and has been spiteful/vindictive at least twice in the past 6 months. Milder than conduct disorder (no aggression toward people/animals, no destruction of property). Risk factor for development of conduct disorder. Treatment: parent management training, CBT, and school-based interventions.

Orexin Receptor Antagonists

– Medications that block orexin (hypocretin) receptors to promote sleep. Suvorexant and lemborexant are approved for insomnia. They reduce wakefulness by blocking the arousal system. They have lower abuse potential than benzodiazepines and do not cause respiratory depression.

OSFED

– Other Specified Feeding or Eating Disorder, used when the clinical presentation causes significant distress or impairment but does not meet full criteria for any of the specific eating disorders. Examples include atypical anorexia nervosa (all criteria met except weight is within or above normal range), subthreshold bulimia nervosa or binge eating disorder, purging disorder, and night eating syndrome.

Other Mental Health Disorders

– conditions beyond depression and anxiety, such as bipolar disorder (manic-depressive illness cycling between mania/hypomania and depression; lithium is cornerstone therapy), schizophrenia and psychotic disorders (hallucinations, delusions, disorganized thinking; dopamine hypothesis; antipsychotic treatment), obsessive-compulsive disorder (OCD; intrusive thoughts and repetitive rituals; CBT/ERP and SSRIs), post-traumatic stress disorder (PTSD; re-experiencing, avoidance, hyperarousal after trauma; trauma-focused therapy, SSRIs, prazosin for nightmares), eating disorders (anorexia nervosa, bulimia nervosa, binge-eating disorder; requiring medical monitoring for refeeding syndrome), personality disorders (borderline, antisocial, avoidant; dialectical behavior therapy for BPD), and substance use disorders (alcohol, opioids, stimulants; pharmacotherapy—naltrexone, buprenorphine—and psychosocial support). Multidisciplinary treatment: psychotherapy, pharmacotherapy, social support, and coordinated care with medical providers for substance-related and other medical complications.

Other Specified Sexual Dysfunction

– A category for sexual dysfunction symptoms that cause distress but do not meet full criteria for any specific sexual dysfunction. Examples: sexual aversion (extreme avoidance of or aversion to genital sexual contact), inadequate vaginal lubrication without pain, persistent genital arousal disorder (unwanted genital arousal without sexual desire, PGAD), and sexual dysfunction due to medical conditions not otherwise covered. Management is tailored to the specific presentation.

Panic Attack

– An abrupt surge of intense fear or discomfort reaching a peak within minutes. The individual experiences at least four of thirteen symptoms: palpitations, sweating, trembling, shortness of breath, feelings of choking, chest pain, nausea, dizziness, chills or heat, paresthesias, derealization or depersonalization, fear of losing control, and fear of dying. Panic attacks are not limited to panic disorder and occur in many psychiatric and medical conditions.

Panic Disorder

– Recurrent unexpected panic attacks, which are abrupt surges of intense fear or discomfort peaking within minutes. During attacks, individuals experience at least four of thirteen symptoms: palpitations, sweating, trembling, shortness of breath, feelings of choking, chest pain, nausea, dizziness, chills or heat sensations, paresthesias, derealization, fear of losing control, and fear of dying. At least one attack is followed by persistent worry about additional attacks or maladaptive behavioral changes aimed at avoiding future attacks. Panic disorder is a chronic condition with a relapsing-remitting course. The lifetime prevalence is approximately 3-5 percent, with onset typically in late adolescence or early adulthood. The fear of having another panic attack leads to a cycle of escalating anxiety and avoidance. Agoraphobia frequently co-occurs. Diagnosis is clinical using DSM-5 criteria. First-line treatment includes CBT with interoceptive exposure and SSRIs (sertraline, paroxetine, fluoxetine, escitalopram) or SNRIs (venlafaxine). Benzodiazepines (alprazolam, clonazepam) are effective acutely but avoided as first-line due to tolerance and dependence. Prognosis is good with treatment; most patients achieve significant symptom reduction.

Paraphilia

– A condition characterized by intense and persistent sexual interest in atypical objects, situations, or individuals. Examples: exhibitionism, fetishism, frotteurism, pedophilic disorder, sexual masochism, sexual sadism, transvestic disorder, voyeurism, and paraphilia not otherwise specified. Diagnosis requires that the paraphilia causes distress or impairment or involves harm to others. Treatment: CBT, SSRI medications, and antiandrogens in selected cases.

Pedophilic Disorder

– A paraphilic disorder characterized by intense and recurrent sexual attraction to prepubescent children (generally age $\leq$13), lasting $\geq$6 months, and causing distress or impairment or involving acting on these urges. Diagnosis requires the individual is ≥16 years old and ≥5 years older than the child. Assessment: clinical interview, phallometric testing. Treatment: CBT focusing on relapse prevention and cognitive distortions; pharmacotherapy (SSRIs, antiandrogens — medroxyprogesterone acetate, cyproterone acetate, GnRH agonists like leuprolide) to reduce libido. Ethical and legal considerations are paramount.

Perinatal Psychiatry

– The subspecialty focused on psychiatric disorders during pregnancy and the postpartum period. Common conditions include perinatal depression, perinatal anxiety, perinatal OCD, and postpartum psychosis. Medication selection must balance maternal mental health with fetal or neonatal exposure risks. SSRIs are generally considered safe, while lithium and valproate carry teratogenic risks. Breastfeeding considerations influence medication choice.

Persistent Depressive Disorder

– formerly called dysthymia, a chronic depressive mood lasting at least two years in adults (one year in children and adolescents). Patients experience depressed mood for more days than not, along with at least two additional symptoms: poor appetite or overeating, insomnia or hypersomnia, low energy, low self-esteem, poor concentration, or feelings of hopelessness. During a two-year period, the patient has never been symptom-free for more than two months.

Personality Disorder

– A group of mental disorders characterized by enduring maladaptive patterns of thinking, feeling, and behaving that deviate from cultural expectations and cause functional impairment. Cluster A (odd/eccentric): paranoid, schizoid, schizotypal. Cluster B (dramatic/erratic): antisocial, borderline, histrionic, narcissistic. Cluster C (anxious/fearful): avoidant, dependent, obsessive-compulsive. Diagnosis requires onset in adolescence or early adulthood and stability over time. Treatment: psychotherapy (DBT for borderline, CBT, psychodynamic); limited pharmacotherapy for specific symptoms.

Postpartum Depression

– Major depressive or manic episodes occurring during pregnancy or within four weeks of delivery (DSM-5). The term is often used more broadly to cover depressive episodes up to one year postpartum. Risk factors include prior depression, lack of social support, stressful life events, and hormonal fluctuations. Postpartum psychosis is a distinct, more severe entity requiring immediate psychiatric hospitalization.

Postpartum Psychosis

– A psychiatric emergency occurring within the first two weeks postpartum, characterized by rapid onset of psychotic symptoms including confusion, disorientation, hallucinations, and delusions. It occurs in approximately 1–2 per 1000 deliveries and is associated with bipolar disorder. Risk of infanticide and suicide is elevated. Requires immediate psychiatric hospitalization and treatment with mood stabilizers and antipsychotics.

Posttraumatic Stress Disorder (PTSD)

– A psychiatric disorder that develops after exposure to a traumatic event (death, serious injury, sexual violence). Presents with 4 symptom clusters: re-experiencing (intrusive memories, flashbacks, nightmares), avoidance (of trauma reminders), negative alterations in cognition and mood, and hyperarousal (hypervigilance, exaggerated startle, irritability). Duration >1 month with functional impairment. Risk factors: prior trauma, lack of social support, female sex, and peritraumatic dissociation. Diagnosis: DSM-5 criteria. Treatment: trauma-focused CBT (prolonged exposure, cognitive processing therapy), EMDR, and SSRIs (sertraline, paroxetine). Prognosis: chronic without treatment.

Premature (Early) Ejaculation

– A sexual dysfunction characterized by a persistent or recurrent pattern of ejaculation occurring within approximately 1 minute of vaginal penetration and before the individual wishes it, occurring in ≥75% of sexual encounters, lasting ≥6 months, and causing marked distress. Prevalence: 20-30% of men, the most common male sexual dysfunction. Lifelong (primary) vs. acquired (secondary to ED, prostatitis, hyperthyroidism, anxiety, or relationship issues). Diagnosis: clinical DSM-5 criteria (IELT, perceived control, distress). Treatment: first-line — behavioral techniques (stop-start, squeeze technique), topical anesthetics (lidocaine-prilocaine cream). Pharmacotherapy: dapoxetine (short-acting SSRI, on-demand, approved in many countries outside the US), off-label SSRIs (paroxetine 10-40 mg, sertraline 50-200 mg) daily or on-demand, and clomipramine. Treat underlying ED if present.

Premenstrual Dysphoric Disorder

– A severe, debilitating form of premenstrual syndrome characterized by marked mood symptoms (depression, anxiety, irritability, mood swings), physical symptoms (breast tenderness, bloating, fatigue), and functional impairment that occur during the luteal phase and remit within days of menses onset. Symptoms must occur in most menstrual cycles for the past year and cause significant distress. Distinguishing from PMS: greater severity and predominance of affective symptoms. Diagnosis: prospective daily symptom charting for $\geq$2 cycles. Treatment: SSRIs (intermittent luteal phase or continuous), combined oral contraceptives (drospirenone-containing), CBT, and GnRH agonists in refractory cases.

Projection

– A defense mechanism in which unacceptable thoughts, feelings, or impulses are attributed to others. For example, a patient who is angry at the therapist may accuse the therapist of being angry at them. It is common in paranoid thinking, delusional disorders, and personality disorders.

Psychodynamic Therapy

– A form of depth psychology that explores how unconscious processes, early life experiences, and internal conflicts influence current thoughts, feelings, and behaviors. It involves exploration of transference and countertransference, defense mechanisms, and the therapeutic relationship as a vehicle for change. Brief psychodynamic therapy is time-limited and focused on specific conflicts. It is effective for depression, anxiety, personality disorders, and somatic symptom disorders.

Psychogenic Non-Epileptic Seizures

– Seizure-like episodes that are not caused by abnormal electrical activity in the brain. They are a form of conversion disorder with attacks or seizures. Events may include shaking, stiffening, unresponsiveness, and other movements. Diagnosis requires clinical evidence of incompatibility with epilepsy and can be confirmed by video-EEG monitoring showing normal EEG during an event. Treatment includes psychotherapy and education about the diagnosis.

Psychopharmacology

– The study and clinical application of medications used to treat psychiatric disorders. It involves understanding drug mechanisms, pharmacokinetics, drug interactions, and side effect profiles. It is the foundation of biological psychiatry and enables evidence-based medication management for mood, anxiety, psychotic, and cognitive disorders.

PTSD

– See Post-Traumatic Stress Disorder.

Purging Disorder

– Recurrent purging behavior to influence weight or shape in the absence of binge eating. Self-induced vomiting, misuse of laxatives, diuretics, or other medications. It is classified under Other Specified Feeding or Eating Disorder (OSFED). It can lead to electrolyte imbalances, dental erosion, Russell's sign, and esophageal tears.

Reactive Attachment Disorder

– A pattern of inhibited, emotionally withdrawn behavior toward adult caregivers. The child rarely or minimally seeks comfort when distressed and rarely or minimally responds to comfort when distressed. The child shows a pattern of persistently irritable and sad mood that is not limited to interactions with caregivers. It results from a pattern of insufficient care including social neglect, repeated changes of primary caregivers, or rearing in unusual settings.

Restless Legs Syndrome

– An urge to move the legs, usually accompanied by uncomfortable sensations, that begins or worsens during rest, is relieved by movement, and occurs predominantly in the evening or night. The condition causes significant distress or impairment. It is associated with iron deficiency, renal failure, pregnancy, and peripheral neuropathy. Ferritin levels below 50 ng/mL should be supplemented. Dopaminergic agents and alpha-2-delta ligands (gabapentin enacarbil) are used for treatment.

Schizoaffective Disorder

– A disorder characterized by an uninterrupted period of illness during which there is a major mood episode concurrent with criterion A of schizophrenia (delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior, or negative symptoms). The individual must also have delusions or hallucinations for at least two weeks in the absence of a major mood episode. The mood episodes are present for the majority of the total duration of the illness.

Schizoid Personality Disorder

– A pattern of detachment from social relationships with a restricted range of emotional expression. At least four of seven criteria: neither desires nor enjoys close relationships, almost always chooses solitary activities, has little interest in sexual experiences, takes pleasure in few activities, lacks close friends, appears indifferent to praise or criticism, and shows emotional coldness, detachment, or flattened affect.

Schizophrenia

– A severe chronic mental illness characterized by psychotic symptoms (hallucinations, delusions, disorganized thinking) and negative symptoms (avolition, alogia, anhedonia, blunted affect), with cognitive impairment and functional decline. Diagnosis: $\geq$2 core symptoms for $\geq$1 month with continuous signs for $\geq$6 months. Onset typically in late adolescence/early adulthood. Treatment: antipsychotic medications (first-generation: haloperidol; second-generation: risperidone, olanzapine, clozapine for refractory), psychosocial interventions, and family education.

Schizotypal Personality Disorder

– A pattern of social and interpersonal deficits marked by acute discomfort with and reduced capacity for close relationships, cognitive or perceptual distortions, and eccentricities of behavior. Features include ideas of reference, odd beliefs or magical thinking, unusual perceptual experiences, odd thinking and speech, suspiciousness, inappropriate affect, and constricted affect. It shares genetic and phenomenological features with schizophrenia.

Seasonal Affective Disorder

– A specifier of recurrent major depressive episodes that occur at a specific time of year, most commonly winter (with remission in spring/summer). Symptoms: hypersomnia, hyperphagia with carbohydrate craving, weight gain, fatigue, and low energy. Associated with reduced sunlight exposure disrupting circadian rhythms and serotonin regulation. Treatment: light therapy (10,000 lux, 30 minutes upon waking), CBT-SAD, SSRIs/bupropion, and agomelatine. Dawn simulation and vitamin D supplementation may help.

Sedatives

– See Sedatives in Pharmacology.

Selective Mutism

– Consistent failure to speak in specific social situations where speaking is expected despite speaking in other situations. The disturbance interferes with educational or occupational achievement or social communication. Must last for at least one month and is not attributable to lack of knowledge of or comfort with the spoken language. Most commonly seen in children and often co-occurs with social anxiety disorder.

Separation Anxiety Disorder

– Developmentally inappropriate and excessive fear or anxiety concerning separation from attachment figures. The individual may have recurrent excessive distress when anticipating or experiencing separation from home or major attachment figures. Symptoms may include worry about harm befalling attachment figures, worry about experiencing an event causing separation, reluctance to leave home or go out alone, and nightmares involving separation. Must persist for four weeks in children and six months in adults. Diagnosis is clinical using DSM-5 criteria. Treatment includes CBT with graded exposure to separation situations and SSRIs (fluoxetine, sertraline, fluvoxamine) for moderate-to-severe cases. Involving parents in treatment is essential. Prognosis is good with treatment; untreated separation anxiety may persist and predispose to other anxiety disorders in adulthood.

Serotonin-Norepinephrine-Dopamine Reuptake Inhibitors

– A class of antidepressants that inhibit reuptake of serotonin, norepinephrine, and dopamine. The primary agent is bupropion, which also acts as a norepinephrine-dopamine reuptake inhibitor. It is used for depression, smoking cessation, and as augmentation. It lacks sexual side effects and weight gain associated with SSRIs.

Serotonin-Norepinephrine Reuptake Inhibitors

– Antidepressants that block the reuptake of both serotonin and norepinephrine. They include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran. SNRIs are used for major depressive disorder, generalized anxiety disorder, diabetic neuropathy, fibromyalgia, and chronic pain. They are particularly useful in patients with fatigue and poor concentration. Side effects include those of SSRIs plus hypertension (especially at higher doses).

Sexual Masochism Disorder

– A paraphilic disorder characterized by recurrent, intense sexual arousal from the act of being humiliated, beaten, bound, or otherwise made to suffer, lasting ≥6 months and causing distress or impairment or involving acting on these urges. Asphyxiophilia (sexual arousal from oxygen restriction): the most dangerous form, associated with accidental death during solo practice. Hypoxyphilia: arousal from oxygen deprivation. Consensual BDSM activities are not considered pathological. Disorder is diagnosed when the behavior causes clinically significant distress, impairment, or risk of harm. Treatment: CBT and SSRIs.

Sexual Sadism Disorder

– A paraphilic disorder characterized by recurrent, intense sexual arousal from the physical or psychological suffering of another person, lasting ≥6 months and causing distress or impairment or involving acting on these urges with a nonconsenting person. In the context of BDSM with consenting adults, sadistic acts alone do not constitute a disorder. Diagnosis requires that the individual has acted on these urges with a nonconsenting person or the urges cause significant distress or interpersonal difficulty. The most severe form is associated with serial sexual homicide. Treatment: CBT, SSRIs, antiandrogens, and GnRH agonists for severe cases.

Social Anxiety Disorder

– Marked fear or anxiety about one or more social situations in which the individual may be scrutinized by others. The individual fears acting in a way or showing anxiety symptoms that will be negatively evaluated. The social situations almost always provoke fear or anxiety, are avoided or endured with intense anxiety, and are disproportionate to the actual threat. The fear, anxiety, or avoidance is persistent, typically lasting six months or more.

Somatic Symptom Disorder

– Characterized by one or more somatic symptoms that are distressing or result in significant disruption of daily life, accompanied by excessive thoughts, feelings, or behaviors related to the somatic symptoms. The individual experiences persistent high levels of worry about the seriousness of symptoms, an excessive amount of time and energy devoted to health concerns, or disproportionate and persistent concern about the medical implications of symptoms. The problematic symptoms are not better explained by another medical or psychiatric condition. Somatic symptom disorder is common, with prevalence of approximately 5-7 percent in the general population. It is associated with high healthcare utilisation and functional impairment. Diagnosis is clinical using DSM-5 criteria, after appropriate medical evaluation to exclude underlying organic pathology. Treatment includes establishing a consistent therapeutic relationship with a single primary care provider, CBT, graded exercise, and antidepressants (SSRIs, TCAs) for comorbid depression or pain. Prognosis is variable, but most patients improve with appropriate management that avoids unnecessary investigations and procedures.

Specific Learning Disorder

– Persistent difficulties in reading, written expression, or mathematics, despite adequate intelligence and educational opportunity. Diagnosis requires academic skills significantly below expected for age, difficulties for at least six months, and no better explained by intellectual disability, sensory impairment, or neurological condition. Subtypes include dyslexia (reading), dysgraphia (writing), and dyscalculia (mathematics).

Specific Phobia

– Marked fear or anxiety about a specific object or situation that is out of proportion to the actual danger posed. Exposure to the phobic stimulus almost invariably provokes an immediate anxiety response. The object or situation is actively avoided or endured with intense anxiety. The fear, anxiety, or avoidance persists for six months or more. Subtypes include animal, natural environment, blood-injection-injury, situational, and other.

Splitting

– A primitive defense mechanism in which objects or self-representations are viewed as entirely good or entirely bad, with no integration of positive and negative qualities. It is a core feature of borderline personality disorder and contributes to idealization-devaluation patterns in relationships. It reflects impaired object relations.

Stimulant Use Disorder

– A problematic pattern of stimulant use (cocaine, amphetamines, methamphetamine, MDMA) leading to significant impairment. Stimulants increase dopamine and norepinephrine activity. Withdrawal is characterized by fatigue, depression, increased appetite, hypersomnia, psychomotor retardation, and vivid unpleasant dreams. Methamphetamine use is associated with severe dental disease, paranoia, and hallucinations. Treatment is primarily behavioral as no FDA-approved medications exist.

Stress

– A physiological and psychological response to perceived threats or demands that activates the sympathetic nervous system and the hypothalamic-pituitary-adrenal axis. Acute stress can enhance focus and performance in challenging situations, but chronic stress contributes to the development or exacerbation of numerous health conditions, including hypertension, anxiety disorders, depression, gastrointestinal problems, and immunosuppression. Effective stress management techniques include regular physical activity, mindfulness meditation, adequate sleep, social support, and professional counseling when needed.

Substance Abuse Psychiatry

– The subspecialty focused on the evaluation and treatment of substance use disorders and co-occurring psychiatric conditions. Dual diagnosis patients require integrated treatment addressing both disorders simultaneously. Psychopharmacology includes medication-assisted treatment (methadone, buprenorphine, naltrexone for opioid use disorder; acamprosate, disulfiram, naltrexone for alcohol use disorder) alongside psychotherapy including motivational interviewing and CBT.

Substance/Medication-Induced Sexual Dysfunction

– Sexual dysfunction directly caused by substance use or medication effects. Common culprits: SSRIs/SNRIs (decreased desire, delayed ejaculation/anorgasmia, erectile dysfunction — most common cause of drug-induced sexual dysfunction), antipsychotics (hyperprolactinemia causing decreased desire and ED), antihypertensives (beta-blockers, thiazides, alpha-blockers), hormonal contraceptives (decreased desire, vaginal dryness), antiandrogens (spironolactone, finasteride), opioids (decreased desire, ED, delayed ejaculation), and alcohol (acute: erectile difficulty; chronic: decreased desire and testicular atrophy). Management: identify the culprit, dose reduction, drug holiday, switch to alternative (bupropion, mirtazapine for depression; PDE5 inhibitors for ED), and add-on therapy (buspirone for SSRI-induced sexual dysfunction). Bupropion has the lowest rate of sexual side effects among antidepressants.

Substance Use Disorder

– A problematic pattern of substance use leading to clinically significant impairment or distress, manifested by at least two of eleven criteria within a twelve-month period. Criteria include taking the substance in larger amounts or over a longer period than intended, persistent desire or unsuccessful efforts to cut down, great deal of time spent obtaining or recovering from the substance, craving, failure to fulfill obligations, continued use despite social problems, giving up activities, recurrent use in hazardous situations, continued use despite physical or psychological problems, tolerance, and withdrawal. Severity is graded as mild (2-3 criteria), moderate (4-5), or severe (6 or more). Diagnosis is clinical using DSM-5 criteria. Treatment involves motivational interviewing to enhance motivation for change, CBT, contingency management, 12-step facilitation, and medication-assisted treatment where available (methadone, buprenorphine, naltrexone for opioid use disorder; acamprosate, disulfiram, naltrexone for alcohol use disorder). Relapse rates are high, and recovery often requires multiple treatment episodes and long-term support.

Suicidal Ideation

– Thoughts about killing oneself, ranging from passive (wish to be dead, wishes to fall asleep and not wake) to active (thoughts of killing oneself with varying degrees of intent, plan, and means). The C-SSRS grades severity from 1 (wish to be dead) to 5 (active suicidal ideation with specific plan and intent). Assessment of suicidal ideation must include questions about plan, intent, means, and history of attempts.

Suicide

– The intentional act of taking one's own life. Suicide is a major public health problem worldwide, causing 700,000 deaths annually. Risk factors: psychiatric disorders (major depression—most common contributor, bipolar disorder, schizophrenia, borderline personality disorder, PTSD, substance use disorders—especially alcohol), previous suicide attempt (strongest predictor), family history of suicide, access to means (firearms—most common method in the US; hanging, pesticide poisoning—common globally), male sex (3–4:1 ratio of completed suicide vs attempts), older age (peaks in middle-aged and older adults), chronic physical illness, social isolation, unemployment, financial stress, history of childhood abuse/trauma, and LGBTQ+ identity (increased risk in the context of minority stress). Assessment: direct questioning about suicidal thoughts, plans, intent, and means (does not increase risk—contrary to myth). The SAFE-T (Suicide Assessment Five-step Evaluation and Triage) framework. Acute management: ensure safety—remove means, hospitalisation (voluntary or involuntary if imminent risk), suicide watch (1:1 observation), and pharmacological treatment of underlying psychiatric condition. Long-term management: psychotherapy (CBT, dialectical behaviour therapy—DBT for borderline personality disorder), pharmacotherapy (antidepressants—SSRIs, lithium—reduces suicide risk in mood disorders, clozapine—for schizophrenia), and psychosocial interventions (safety planning, crisis hotlines—988 in the US, 111 option 2 in the UK). Postvention: support for bereaved family and friends (suicide bereavement). Prevention: restricting access to means, responsible media reporting, school-based programmes, and gatekeeper training (GPs, teachers, police).

Tardive Dyskinesia

– Involuntary, repetitive, purposeless movements caused by prolonged use of antipsychotic medications, typically developing after months to years of exposure. Movements include choreoathetoid movements of the tongue, face, and extremities (lip smacking, tongue protrusion, grimacing, choreiform movements of limbs). It is caused by dopamine receptor supersensitivity. It may be irreversible. VMAT2 inhibitors (valbenazine, deutetrabenazine) are FDA-approved treatments. Risk is lower with atypical antipsychotics (especially clozapine and quetiapine) compared to typical antipsychotics. Prevention involves using the lowest effective dose, regular screening using the Abnormal Involuntary Movement Scale (AIMS) every 6-12 months, and switching to an atypical antipsychotic with lower D2 affinity if emerging dyskinesia is detected. Patients should be warned about the risk prior to starting antipsychotic treatment.

Therapeutic Alliance

– The collaborative, trusting relationship between therapist and patient that is foundational to effective psychotherapy. It includes agreement on goals, tasks, and the bond between therapist and patient. It is the strongest predictor of psychotherapy outcome across all therapeutic modalities.

Therapeutic Drug Monitoring

– The clinical practice of measuring drug levels in blood to ensure therapeutic concentrations while avoiding toxicity. It is particularly important for medications with narrow therapeutic windows including lithium (0.6–1.2 mEq/L), valproate (50–125 mcg/mL), carbamazepine (4–12 mcg/mL), and clozapine (350–600 ng/mL).

Therapeutic Hypothermia

– Controlled cooling of the body temperature used in psychiatric practice for severe catatonia and neuroleptic malignant syndrome. In NMS, cooling is a key treatment alongside discontinuation of the offending agent and dantrolene administration. Body temperature is reduced to below 38 degrees Celsius using cooling blankets, ice packs, or intravascular cooling catheters.

Thought Blocking

– An abrupt interruption in the train of thought before it is completed, as if the thought was removed or pushed out of the patient's mind. The patient may pause mid-sentence, appear confused, and report that the thought was "taken out" or "blocked." It is a formal thought disorder seen in schizophrenia.

Tourette Disorder

– A neurodevelopmental disorder characterized by multiple motor tics and one or more vocal tics lasting >1 year, with onset before age 18. Motor tics:

eye blinking, head jerking, shoulder shrugging, facial grimacing. Vocal tics: throat clearing, sniffing, grunting, barking, coprolalia (obscene words, occurs in <10%). Premonitory urge precedes tics. Comorbidities: ADHD (60%), OCD (30-50%), anxiety. Course: waxing and waning severity; most improve by late adolescence. Diagnosis: clinical DSM-5 criteria. Treatment: CBIT (habit reversal training), alpha-2 agonists (clonidine, guanfacine), antipsychotics (haloperidol, pimozide) for severe tics.

Tourette Syndrome

– A neurodevelopmental disorder characterized by multiple motor tics and at least one vocal tic occurring over at least one year. Tics are sudden, rapid, recurrent, stereotyped movements or vocalizations. They are often preceded by premonitory urges. Tics typically worsen in adolescence and may improve in adulthood. Comorbidities include ADHD, OCD, and anxiety disorders. Treatments include behavioral therapy and antipsychotics for severe tics.

Tranquillisers

– See Tranquillisers in Pharmacology.

Transference

– A psychoanalytic concept in which feelings, attitudes, and expectations originally directed toward a significant figure from childhood are unconsciously redirected onto the therapist. In psychodynamic therapy, transference is a therapeutic tool for understanding the patient's relational patterns and resolving conflicts.

Transvestic Disorder

– A paraphilic disorder characterized by recurrent, intense sexual arousal from cross-dressing, lasting ≥6 months and causing marked distress or impairment. Almost exclusively in heterosexual males. Transvestic disorder is distinguished from gender dysphoria by the primary motivation: sexual arousal (transvestic) vs. identity incongruence (gender dysphoria). Many cross-dressers do not experience distress and therefore do not meet disorder criteria. Treatment: CBT for distress or guilt, and SSRIs for compulsive behavior. No treatment is indicated for cross-dressing without distress.

Trichotillomania

– Recurrent pulling out of one's hair, resulting in hair loss. The individual has made repeated attempts to decrease or stop hair pulling. The hair pulling causes clinically significant distress or impairment. It is not attributable to another medical condition. The pulling can occur from any body region and may be focused or focused-pulling. It is classified under obsessive-compulsive and related disorders.

Tricyclic Antidepressants

– Older antidepressants that non-selectively inhibit the reuptake of serotonin and norepinephrine. They include amitriptyline, nortriptyline, imipramine, clomipramine, and desipramine. TCAs are effective but have more side effects than SSRIs, including anticholinergic effects (dry mouth, urinary retention, constipation), orthostatic hypotension, weight gain, and cardiac conduction delays. Lethal in overdose due to cardiac arrhythmias. They are used for depression, neuropathic pain, migraine prophylaxis, and off-label for insomnia and anxiety. TCAs have significant anticholinergic, antihistaminergic, and alpha-adrenergic blocking properties. Nortriptyline has the least anticholinergic side effects and is preferred in elderly patients. Due to their narrow therapeutic index and cardiotoxicity in overdose, TCAs are used less frequently than SSRIs but remain important for treatment-resistant depression. Monitoring of ECG and serum drug levels is recommended.

Vagus Nerve Stimulation

– A neuromodulation treatment involving electrical stimulation of the left vagus nerve via an implanted device. It is FDA-approved for treatment-resistant depression and epilepsy. It requires surgical implantation and delivers intermittent stimulation. Improvement may take months to develop. Non-invasive transcutaneous vagus nerve stimulation is being investigated.

Voyeuristic Disorder

– A paraphilic disorder involving recurrent, intense sexual arousal from observing an unsuspecting person who is naked, disrobing, or engaging in sexual activity. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration ≥6 months. Most common in males with onset in adolescence. Treatment: CBT and SSRIs.

Word Salad

– Severe disorganized speech consisting of a jumble of words and phrases that lack meaning or logical connection. It is a hallmark of formal thought disorder in schizophrenia and represents the most severe form of disorganized speech. It reflects profound disruption of language processing.

Chapter 30

Public Health and Preventive Medicine

Adequate intake

– An estimate of the amount of a nutrient needed by healthy people. The Adequate Intake is used when there isn't enough information to set a recommended dietary allowance (RDA).

Antimicrobial Resistance

– Development of resistance to antimicrobial agents. Occurs when microorganisms evolve mechanisms to survive exposure to drugs that previously killed or inhibited them. Driven by overuse and misuse of antibiotics in humans, agriculture, and animal husbandry. Results in treatment failure, prolonged illness, increased healthcare costs, and higher mortality. Major global health threat requiring antimicrobial stewardship, surveillance, and development of new agents.

Asbestosis

– Occupational lung disease caused by inhalation of asbestos fibers. Characterized by diffuse interstitial pulmonary fibrosis. Long latency period of 10 to 20 years. Associated with increased risk of mesothelioma and lung cancer.

Booster Dose

– Additional dose of a vaccine given after the primary series to re-stimulate the immune response and maintain protection. Required for vaccines where immunity wanes over time such as tetanus and pertussis.

Burden of Disease

– Measure of the gap between current health status and an ideal health situation where everyone lives to old age free of disease and disability. Measured in DALYs. Used to set health priorities and allocate resources.

Burden of Illness

– Total impact of a disease on a population measured in terms of mortality, morbidity, disability, and economic costs. Used to set health priorities and allocate research funding.

Byssinosis

– Occupational lung disease caused by inhalation of cotton, flax, or hemp dust. Characterized by chest tightness and wheezing that improves away from work. Also known as brown lung disease.

Carbon Monoxide Poisoning

– Toxic effect of inhaling carbon monoxide an odorless, colorless gas. Binds hemoglobin with high affinity causing tissue hypoxia. Common sources include faulty furnaces, car exhaust, and generators.

Contraindication to Vaccination

– Condition or factor that increases the risk of a serious adverse reaction to a vaccine. Absolute contraindications include severe allergic reaction to a previous dose. Relative contraindications include immunosuppression for live vaccines.

Crossover Trial

– Study design where participants receive both treatments in a random sequence with a washout period between. Each participant serves as their own control. Increases statistical power but requires a washout period to avoid carryover effects.

Disease Prevention

– Actions taken to prevent disease or detect it early before symptoms develop. Includes immunization, screening, and chemoprophylaxis. Distinguished from treatment which occurs after disease has manifested.

Disease Surveillance

– Continuous systematic collection, analysis, interpretation, and dissemination of health data for disease prevention and control. Essential for early detection of outbreaks and monitoring disease trends.

Effect Modification

– Situation where the magnitude of the association between exposure and outcome differs across levels of another variable. Also called interaction. Unlike confounding, effect modification is a real biological phenomenon that should be reported.

Effect Size

– Quantitative measure of the magnitude of an observed effect. Common measures include Cohen's d for mean differences and odds ratios for binary outcomes. Important for assessing clinical significance beyond statistical significance.

Epidemic Curve

– Graphical representation of the number of cases of disease over time during an outbreak. Can be point source, propagated, or intermittent. Provides information about the outbreak pattern, incubation period, and potential source.

Epidemics

– The occurrence of cases of a disease or health-related event in a community or region at a rate clearly in excess of normal expectancy. Epidemics arise from point sources (single exposure, e.g., contaminated food at an event), propagated spread (person-to-person transmission, e.g., influenza, measles), or mixed patterns. Investigation steps: confirm diagnosis, establish case definition, verify cases, plot epidemic curve (cases over time), calculate attack rates, generate hypotheses about source and mode of transmission, implement control measures (isolation, quarantine, vaccination, treatment, sanitation), and communicate findings. A pandemic is an epidemic that has spread across multiple continents. Epidemiology is the scientific discipline that studies the distribution and determinants of health-related states in populations.

Epidemiological study

– An investigation of the links between certain behaviors or risk factors and the occurrence of disease or good health in a population.

Epidemiologic Transition

– Theory describing the shift in patterns of mortality and disease from infectious to chronic noncommunicable diseases as countries develop. Characterized by changing age distributions and life expectancies.

Essential Fatty Acid

– Fatty acids that cannot be synthesized by the body and must be obtained from the diet. Include omega-3 and omega-6 fatty acids. Important for brain development, inflammation regulation, and cardiovascular health.

Folate Deficiency

– Causes megaloblastic anemia and neural tube defects in developing fetuses. Prevention through folic acid supplementation before and during pregnancy. Found in leafy greens, legumes, and fortified grains.

Genetic Epidemiology

– Study of the role of genetic factors in determining health and disease in populations. Includes twin studies, family studies, and genome-wide association studies.

Geriatrician

– A physician who specializes in the care of older patients.

Hazard Ratio

– Ratio of hazard rates between two groups at any point in time. Used in survival analysis to compare the rate of events between treatment and control groups. Similar to relative risk but accounts for time-to-event data.

Health Equity

– Attainment of the highest level of health for all people. Involves addressing social determinants such as income, education, housing, and access to healthcare that produce systematic disparities between population groups. Requires targeted policies and interventions to eliminate avoidable and unfair differences in health outcomes. Distinct from health equality in that equity allocates resources according to need.

Health Inequality

– Differences in health status between different population groups that are avoidable and unfair. Includes inequalities in access to healthcare, healthy food, safe housing, and education.

Herd Immunity

– Protection of a population against an infectious disease when a sufficient proportion is immune either through vaccination or prior infection. The threshold varies by disease. Reduces the risk of infection for susceptible individuals.

Inactivated Vaccine

– Vaccine containing killed pathogens that cannot replicate. Safer than live-attenuated vaccines but require multiple doses and booster shots to maintain protection. Induce humoral immune responses primarily. Examples: influenza (injectable), polio (IPV), hepatitis A, rabies. More stable for storage and transport than live vaccines.

Iodine Deficiency

– Leading cause of preventable intellectual disability worldwide. Causes goiter, hypothyroidism, and cretinism in severe cases. Prevented through iodised salt supplementation programs. Universal salt iodisation is the recommended strategy endorsed by WHO and UNICEF. Pregnant women and young children are most vulnerable to the effects of deficiency.

Iron Deficiency

– Most common nutritional deficiency worldwide. Causes microcytic hypochromic anemia, fatigue, and impaired cognitive development. Risk factors include menstruation, pregnancy, and inadequate dietary intake.

Isolation

– Separation of individuals with a confirmed communicable disease from healthy individuals. Aims to prevent transmission to others. Required for diseases such as tuberculosis, measles, and COVID-19.

Lead Exposure

– Environmental health hazard from lead in paint, water, soil, and dust. Particularly harmful to children causing developmental delays and behavioral problems. Primary prevention through lead abatement and water pipe replacement.

Lead Poisoning

– Occupational and environmental hazard from exposure to lead. Affects the nervous system, kidneys, and hematopoietic system. Particularly dangerous for children causing developmental delays. Measured by blood lead levels.

Lead Time

– Period between detection of disease through screening and the time it would have been diagnosed clinically. Longer lead time does not necessarily mean better prognosis. Used to calculate apparent survival benefit from screening.

Live-Attenuated Vaccine

– Vaccine containing weakened forms of the pathogen that can still replicate but do not cause disease in immunocompetent individuals. Provides strong and long-lasting immunity. Contraindicated in immunocompromised patients.

Low-calorie diet

– A weight-loss plan that limits calories to 800–1,500 a day.

Mercury Exposure

– Environmental health hazard from mercury in fish, industrial emissions, and dental amalgams.

Mesothelioma

– Rare cancer of the mesothelial lining strongly associated with asbestos exposure. Latency period of 20 to 40 years. Poor prognosis with median survival less than one year. Mostly affects the pleura.

Meta-Analysis

– A statistical technique that combines and analyzes the results of multiple independent studies on the same topic to produce a single, more precise estimate of the treatment effect or association. Meta-analysis increases statistical power, improves precision, resolves uncertainty from conflicting studies, and can identify subgroup effects. Key components: systematic literature search, quality assessment of included studies, extraction of effect estimates, calculation of a weighted pooled effect (fixed-effect or random-effects model), assessment of heterogeneity (I^2 statistic, Cochrane Q test), and publication bias assessment (funnel plot, Egger's test). Forest plot: graphical display showing individual study results and the pooled estimate. Limitations: garbage in, garbage out (quality depends on quality of included studies), publication bias (positive results more likely to be published), and heterogeneity (clinical and methodological diversity). Meta-analysis is a key tool in evidence-based medicine and forms the highest level of evidence (systematic reviews/meta-analyses on top of the evidence pyramid).

Multimorbidity

– Co-occurrence of two or more chronic conditions in an individual. Increasingly common with aging populations. Affects quality of life, healthcare utilization, and treatment complexity.

Natural Experiment

– Observational study in which the exposure of interest is not assigned by the researcher but occurs naturally due to external factors such as policy changes, natural disasters, or geographical variation. The researcher leverages this natural variation to estimate causal effects. Strength: allows study of exposures that cannot be randomised for ethical or practical reasons. Limitation: lacks the randomisation of controlled trials, making confounding and bias more difficult to control. Techniques such as difference-in-differences, instrumental variables, and regression discontinuity are used to strengthen causal inference.

Negative Predictive Value

– Proportion of negative test results that are true negatives. Depends on disease prevalence in the tested population. Higher prevalence decreases NPV. Important for ruling out disease.

Neglected Tropical Diseases

– Group of communicable diseases that prevail in tropical and subtropical conditions affecting the poorest populations. Include Chagas disease, dengue, leishmaniasis, and schistosomiasis. Targeted for control and elimination.

Neyman Bias

– Type of selection bias where cases with rapid onset or fatal outcome are missed because they occur and resolve between screening intervals. Also called prevalence-incidence bias. Affects estimates of disease severity from screening studies.

Noise-Induced Hearing Loss

– Permanent sensorineural hearing loss caused by prolonged exposure to loud noise. Most common occupational disease worldwide. Preventable through engineering controls, administrative controls, and hearing protection.

Notifiable Diseases

– Diseases required by law to be reported to public health authorities. Reporting enables tracking of disease trends and outbreak detection. Examples include tuberculosis, measles, and sexually transmitted infections, foodborne illnesses. Reporting enables public health surveillance, outbreak detection, and implementation of control measures. Lists vary by jurisdiction.

Occupational Carcinogen

– Agent in the workplace that increases the risk of cancer. IARC classifies carcinogens into groups from 1 (definitely carcinogenic) to 3 (not classifiable). Includes asbestos, benzene, chromium VI, and formaldehyde.

Occupational Health

– Branch of public health concerned with the health and safety of workers. Includes prevention and management of workplace injuries and illnesses, ergonomic assessment, and promotion of worker wellbeing.

Odds Ratio

– Ratio of the odds of exposure among cases to the odds of exposure among controls. Approximates relative risk when disease is rare. Widely used in case-control studies where direct incidence cannot be calculated.

Opportunity Cost

– Value of the next best alternative foregone when making a choice. In healthcare, represents the benefits that could have been gained from investing resources in a different intervention. Important concept in health economics. When choosing between healthcare interventions, the opportunity cost is the health benefits foregone from not funding the next best alternative. Used in cost-effectiveness analysis and health technology assessment to inform resource allocation decisions.

Outbreak Investigation

– Systematic process to identify the source, mode of transmission, and factors contributing to an epidemic. Aims to implement control measures and prevent future outbreaks. Follows a standard series of steps.

Overdiagnosis

– Detection of disease through screening that would never have caused symptoms or death during the patient's lifetime. Results in unnecessary treatment and associated harms. Particularly problematic in cancer screening.

Overtreatment

– Treatment of conditions detected through screening that would not have caused harm if left undetected.

Pandemic Preparedness

– Planning and capacity building to respond to pandemic threats. Includes surveillance, laboratory capacity, vaccine development, and healthcare system resilience. Guided by the International Health Regulations (IHR 2005). Includes early warning systems, stockpiling of vaccines and antivirals, surge capacity planning, and coordination mechanisms. Lessons from COVID-19, influenza pandemics, and emerging infectious disease threats inform ongoing preparedness efforts.

Patient-Centered Care

– Approach to healthcare that is respectful of and responsive to individual patient preferences, needs, and values. Ensures patient values guide all clinical decisions. Includes shared decision-making, communication, and care coordination. Improves patient satisfaction, treatment adherence, and health outcomes.

Peer Review

– The process by which research submitted for publication or presentation is evaluated by independent experts (peers) in the same field to assess its quality, validity, originality, and significance before acceptance. Types: single-blind (reviewers know authors, but not vice versa), double-blind (neither knows the other), open (both identities are known), and post-publication review. In healthcare, peer review also applies to clinical practice: regular review of doctors' performance by colleagues to maintain and improve standards of care (including case note review, morbidity and mortality meetings, and clinical audit). Peer review is the cornerstone of scientific quality control but is not infallible — it can be subject to bias, delay, and failure to detect major errors or fraud.

Personal Protective Equipment

– Equipment worn to minimize exposure to workplace hazards. Includes respirators, gloves, eye protection, and hearing protection. Last line of defense in the hierarchy of controls.

Primary Prevention

– Prevention of disease before it occurs. Includes health promotion and specific protection measures such as vaccination, health education, and environmental modifications. Aims to reduce the incidence of disease.

Primordial Prevention

– Prevention of the development of risk factors in populations through environmental, economic, social, and behavioural interventions before risk factors emerge. Targets upstream determinants such as poverty, education, and urban planning. Example: policies to reduce air pollution to prevent cardiovascular and respiratory disease. Distinct from primary prevention which targets established risk factors.

Propagated Outbreak

– Outbreak where disease spreads from person to person. Characterized by successive waves of cases corresponding to generations of transmission. The epidemic curve shows multiple peaks.

Proportional Mortality Ratio

– Proportion of deaths due to a specific cause relative to all deaths in a population. Useful for comparing the contribution of different causes of death.

Prospective Cohort Study

– Observational study following a group of individuals over time to determine who develops the outcome of interest. Measures incidence and relative risk. Strong for studying rare exposures. Expensive and time-consuming.

Publication Bias

– Tendency for studies with positive or significant results to be published more readily than those with negative or null results. Can distort meta-analyses and systematic reviews. Addressed through trial registration and comprehensive literature searching.

Quarantine

– Separation and restriction of movement of healthy individuals who may have been exposed to a communicable disease. Aims to prevent disease transmission. Applied during the incubation period to monitor for symptom development. Historically used for plague, cholera, and yellow fever. Legal authority for quarantine varies by jurisdiction. Distinguished from isolation (separation of those already confirmed ill).

Quasi-Experimental Design

– Study design that lacks randomization but includes an intervention. Uses comparison groups that are not randomly assigned. Susceptible to confounding but can be strengthened by techniques such as difference-in-differences analysis.

Radon Exposure

– Natural radioactive gas from the decay of uranium in soil and rock. Second leading cause of lung cancer after smoking. Enters buildings through cracks in foundations. Tested with radon detectors and mitigated by ventilation systems.

Relative Risk

– Ratio of the incidence rate of disease in the exposed population to the incidence rate in the unexposed population. A relative risk greater than 1 indicates increased risk with exposure; less than 1 indicates a protective effect. Used in cohort studies and clinical trials. Important for establishing causal associations. Also called risk ratio.

Relative Risk Reduction

– Proportional reduction in risk between the treatment and control groups. Calculated as the difference in risk divided by the control group risk. Often used in clinical trial reporting.

Risk Communication

– Process of exchanging information about health risks between experts and the public. Includes risk assessment, risk perception, and risk management communication. Important during outbreaks and public health emergencies.

Risk Factor

– Attribute or exposure that is associated with an increased probability of disease. Does not necessarily imply causation. Can be modifiable or nonmodifiable. Important for disease prevention and health promotion.

Scientific Evidence

– Information gathered through systematic observation, experimentation, and reasoning that supports or refutes a hypothesis or theory. In medicine, scientific evidence forms the basis for clinical practice and is ranked by quality (hierarchy of evidence). Levels (from highest to lowest): systematic reviews and meta-analyses, randomized controlled trials (RCTs), cohort studies, case-control studies, cross-sectional studies, case series/case reports, expert opinion, and anecdotal evidence. Evidence-based medicine integrates the best available scientific evidence with clinical expertise and patient values and preferences. The GRADE system rates the quality of evidence as high, moderate, low, or very low based on study design, risk of bias, consistency, directness, and precision.

Sensitivity

– Proportion of individuals with disease who test positive. True positive rate. Important for screening tests where missing cases has serious consequences.

Shared Decision-Making

– Process where healthcare providers and patients collaborate to make clinical decisions. Involves providing evidence-based information about options, eliciting patient preferences, and reaching a mutually agreed plan.

Silicosis

– Occupational lung disease caused by inhalation of crystalline silica dust. Characterized by progressive fibrosis of the lungs. Increased risk of tuberculosis and lung cancer. Common in miners, sandblasters, and stone cutters.

Subunit Vaccine

– Vaccine containing specific components of the pathogen such as proteins or polysaccharides. Cannot cause disease because they do not contain live organisms. Examples include hepatitis B and HPV vaccines.

Syndromic Surveillance

– Monitoring of health-related data that precede diagnosis and signal a potential outbreak. Includes monitoring of symptoms, school absenteeism, and over-the-counter medication sales. Provides earlier warning than traditional surveillance.

Systematic Review

– A structured, comprehensive review of the literature that uses explicit, reproducible methods to identify, select, appraise, and synthesize all available evidence on a specific research question. Key features: pre-specified protocol (PROSPERO registration), comprehensive search strategy (multiple databases: MEDLINE, EMBASE, Cochrane CENTRAL, CINAHL), clear inclusion/exclusion criteria, quality assessment of included studies (risk of bias tools: Cochrane RoB 2 for RCTs, ROBINS-I for non-randomized studies), data extraction, and synthesis (narrative or quantitative via meta-analysis). PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines standardize reporting. Systematic reviews sit at the top of the evidence pyramid and are essential for clinical guideline development and evidence-based practice.

Tertiary Prevention

– Prevention of disease progression and complications after clinical disease has been diagnosed. Includes rehabilitation, disease management, and secondary prevention of complications. Aims to reduce disability and improve quality of life.

Vaccination

– Administration of a vaccine to stimulate the immune system and provide protection against a specific infectious disease. Can be live-attenuated, inactivated, subunit, toxoid, or mRNA-based. Most cost-effective public health intervention.

Vaccine Effectiveness

– Measure of vaccine performance in real-world conditions outside clinical trials. Lower than efficacy due to factors such as vaccine storage, administration, and population heterogeneity. Important for public health decision-making.

Vaccine Efficacy

– Proportionate reduction in disease incidence among vaccinated individuals compared to unvaccinated individuals under ideal conditions. Measured in clinical trials. Different from effectiveness which measures real-world performance.

Vitamin A Deficiency

– Leading cause of preventable childhood blindness worldwide. Causes xerophthalmia, keratomalacia, and increased susceptibility to infections. Common in developing countries with limited access to animal sources and orange fruits and vegetables.

Vitamin D Deficiency

– Causes rickets in children and osteomalacia in adults. Risk factors include limited sun exposure, dark skin, and inadequate dietary intake. Prevented through sunlight exposure, dietary sources, and supplementation.

Zinc Deficiency

– Impairs immune function, growth, and wound healing. Common in developing countries. Risk factors include vegetarian diets, malabsorption, and chronic diseases. Manifests with growth retardation, diarrhea, and recurrent infections.

Chapter 31

Pulmonology

A-a Gradient

– PAO2 minus PaO2, reflecting gas transfer efficiency. Normal approximately 15 mmHg (increases with age: $< 15 + \text{age}/4$). Elevated in V/Q mismatch, shunt, and diffusion impairment. Normal gradient in hypoventilation.

Absent Breath Sounds

– Complete absence of air movement sounds in a lung field, indicating complete airway obstruction, pneumothorax, or large pleural effusion. It is a medical finding requiring urgent evaluation and intervention.

Acute Bronchitis

– Inflammation of the bronchial tubes, most commonly caused by viral infections (rhinovirus, influenza, coronavirus). Presents with cough (productive or nonproductive), sometimes with low-grade fever and malaise. Self-limited condition lasting 1-3 weeks. Chest x-ray typically normal, distinguishing it from pneumonia. Antibiotics not indicated unless pertussis suspected. Treatment: supportive care with rest, hydration, and cough suppressants. Consider asthma or COPD exacerbation if symptoms persist beyond three weeks.

Acute Cough

– Cough lasting < 3 weeks. Most common causes: URI, acute bronchitis, pneumonia. Usually self-limited.

Airway Clearance Techniques

– Physical therapies to mobilize and remove secretions from the airways. Techniques include postural drainage, percussion, vibration, active cycle of breathing, autogenic drainage, positive expiratory pressure, oscillating PEP, and high-frequency chest wall oscillation. They are essential in CF, bronchiectasis, and neuromuscular disease.

Allergen Immunotherapy

– Gradual desensitization to specific allergens through repeated subcutaneous or sublingual exposure. It reduces allergic asthma symptoms and medication needs. Treatment requires 3–5 years for full benefit. It is indicated for allergic asthma with identified specific allergens not controlled by standard therapy.

Allergic Bronchopulmonary Aspergillosis

– A hypersensitivity reaction to Aspergillus fumigatus in asthma or CF patients. Causes central bronchiectasis, elevated total IgE, and eosinophilia. Treatment includes corticosteroids and itraconazole.

Alpha-1 Antitrypsin Deficiency

– An autosomal codominant genetic disorder caused by mutations in the SERPINA1 gene. It predisposes to early-onset panlobular emphysema (lower lobes), liver disease, and panniculitis. PiZZ genotype is most severe. Treatment includes AAT augmentation therapy and smoking avoidance.

Alveolar Gas Equation

– PAO2 = (FiO2 x (Patm - PH2O)) - (PaCO2 / R). Patm = 760 mmHg, PH2O = 47 mmHg, R = 0.8. Allows calculation of A-a gradient.

Anaphylactic Shock

– See Anaphylaxis.

Anoxia

– Complete absence of oxygen supply to tissues, distinguished from hypoxia (reduced oxygen). Anoxia causes rapid cell death, particularly in highly aerobic tissues (brain, heart, kidneys). Causes: airway obstruction (complete—foreign body, laryngeal edema, anaphylaxis), respiratory arrest, severe drowning, carbon monoxide poisoning (carboxyhaemoglobin >50%), and high-altitude exposure. Irreversible brain damage occurs within 4–6 minutes of complete anoxia (cerebral cortex most vulnerable, followed by hippocampus, basal ganglia, and cerebellum). EEG may show isoelectric activity. Hypoxic-ischemic encephalopathy (HIE) is the clinical syndrome following cardiac arrest or perinatal asphyxia. Treatment: immediate restoration of oxygenation (ABC, oxygen, ventilation), therapeutic hypothermia for HIE (reduces neurological injury), and supportive care.

Anti-IgE Therapy

– Omalizumab is a monoclonal antibody that binds free IgE, preventing it from binding to mast cells and basophils. It reduces allergic inflammation and is approved for moderate-to-severe persistent allergic asthma. It reduces exacerbations and the need for oral corticosteroids. It is administered by subcutaneous injection every 2–4 weeks.

Anti-IL4 Receptor Therapy

– Dupilumab is a monoclonal antibody blocking the IL-4 receptor alpha subunit, inhibiting both IL-4 and IL-13 signaling. It is approved for moderate-to-severe asthma with eosinophilic phenotype or oral corticosteroid dependence. It reduces exacerbations and improves lung function.

Anti-IL5 Therapy

– Monoclonal antibodies targeting interleukin-5, a key cytokine in eosinophil production and survival. Mepolizumab, reslizumab, and benralizumab (anti-IL-5 receptor) are approved for severe eosinophilic asthma. They reduce exacerbations and oral corticosteroid requirements in patients with blood eosinophils >= 300.

Apnea-Hypopnea Index

– Number of apneas plus hypopneas per hour of sleep. Apnea = airflow cessation > 10 seconds. Hypopnea = partial obstruction with desaturation or arousal. AHI < 5 normal, 5–14 mild, 15–29 moderate, >= 30 severe.

Apneustic Breathing

– A breathing pattern characterized by a prolonged inspiratory pause, seen in brainstem lesions (particularly the pons). It indicates damage to the apneustic center and is a poor prognostic sign in neurological injury. It may require mechanical ventilation.

Apnoea

– Temporary cessation of breathing, defined as complete cessation of airflow for $\geq$10 seconds. Obstructive apnea: airway collapse despite respiratory effort (common in obstructive sleep apnea). Central apnea: absence of respiratory effort (Cheyne–Stokes respiration in heart failure, high-altitude periodic breathing, opioid-induced central apnea, brainstem stroke). Mixed apnea: features of both. Physiologic apnea: breath-holding, voluntary apnea. Pathological apnea in infants: apnea of prematurity (seen in preterm infants <37 weeks, due to immature respiratory control, treated with caffeine), apparent life-threatening events (ALTEs, now termed brief resolved unexplained events—BRUE). Diagnosis: polysomnography (sleep study). See also Sleep apnea.

Asbestosis

– A chronic, progressive interstitial lung disease caused by inhalation of asbestos fibers, a type of pneumoconiosis with a latency period of 15-40 years from initial exposure. A group of naturally occurring silicate minerals (chrysotile, amosite, crocidolite) once widely used in construction, shipbuilding, and insulation. Chrysotile (serpentine) is the most commonly used; amphibole forms (amosite, crocidolite) are more pathogenic. Inhaled fibers deposit in the terminal bronchioles and alveolar sacs, where they cause inflammation, fibrosis, and progressive scarring of the lung parenchyma.

Aspiration Pneumonia

– Lung infection from aspiration of oropharyngeal or gastric contents. Risk factors include impaired consciousness and dysphagia. Right lower lobe is most commonly affected.

Aspirin-Exacerbated Respiratory Disease

– A triad of asthma, nasal polyps, and aspirin sensitivity due to COX-1 inhibition. It involves overproduction of cysteinyl leukotrienes. Leukotriene receptor antagonists are specifically effective.

Assist-Control Ventilation

– Ventilator delivers a set tidal volume at a set rate. Every breath receives the full set volume. Guarantees minimum rate and minute ventilation. Risk of respiratory alkalosis and auto-PEEP.

Asthma

– A chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and cough. Airflow limitation is often reversible. It involves eosinophilic inflammation, mucus hypersecretion, and smooth muscle hypertrophy.

Asthma-COPD Overlap

– Features of both asthma and COPD in the same patient. Characterized by persistent airflow limitation with features of both diseases: airway inflammation (eosinophilic and neutrophilic), bronchial hyperresponsiveness, and partially reversible obstruction. Treatment: inhaled corticosteroids plus long-acting bronchodilators. Distinguished from pure asthma (usually atopic, reversible) and pure COPD (smoking history, fixed obstruction).

Ataxic Breathing

– An irregular, unpredictable breathing pattern with random variations in rate and depth, seen in medullary lesions. It is the most dangerous respiratory pattern and often precedes respiratory arrest. It requires mechanical ventilation.

Atelectasis

– Complete or partial collapse of lung tissue, resulting in loss of aeration. It may be caused by airway obstruction (resorption atelectasis), pleural disease (compressive atelectasis), surfactant dysfunction (dissolution atelectasis), or scarring (cicatrization atelectasis). It impairs gas exchange and may predispose to infection.

Autogenic Drainage

– A breathing technique using controlled breathing at different lung volumes to mobilize secretions from peripheral to central airways. It involves breathing at low lung volumes (collect), mid lung volumes (loosen), and high lung volumes (evacuate). It requires patient training and cooperation.

Azygos Vein

– A vein that drains the posterior intercostal veins and empties into the superior vena cava. It forms an important collateral pathway when the SVC is obstructed. On chest imaging, enlargement of the azygos vein may indicate SVC obstruction, heart failure, or portal hypertension.

Bagassosis

– An occupational hypersensitivity pneumonitis (extrinsic allergic alveolitis) caused by inhalation of dust from moldy bagasse—the fibrous residue of sugarcane after extraction. Caused by thermophilic actinomycetes (Saccharopolyspora rectivirgula) that colonize stored bagasse. Presents with fever, cough, dyspnea, and malaise 4–8 hours after exposure. Acute episodes resolve with corticosteroid treatment and avoidance; chronic exposure leads to pulmonary fibrosis.

Barotrauma

– Lung injury from excessive pressure during ventilation, causing alveolar rupture (pneumothorax, pneumomediastinum). Risk factors: high pressures, high PEEP. Prevented by lung-protective ventilation.

Berylliosis

– A granulomatous lung disease from beryllium exposure. Chronic berylliosis resembles sarcoidosis. Diagnosis requires beryllium lymphocyte proliferation test.

Biot Breathing

– An irregular breathing pattern with random periods of normal breathing interspersed with apneic episodes, seen in brainstem lesions. It is less organized than Cheyne-Stokes breathing and indicates medullary damage.

Biotrauma

– Inflammatory response to ventilator-induced lung injury with systemic cytokine release. Can lead to multi-organ dysfunction. Minimized by lung-protective ventilation strategies.

BiPAP

– Bilevel Positive Airway Pressure, a form of non-invasive ventilation delivering two levels of pressure: IPAP (inspiratory) for ventilation and EPAP (expiratory) for alveolar recruitment. It is particularly effective for COPD exacerbations with hypercapnic respiratory failure. It reduces work of breathing and improves gas exchange.

Bird Fancier's Lung

– HP from avian proteins in bird droppings or feathers. Common among pigeon breeders. Chronic exposure leads to progressive fibrosis.

Blue Bloater

– A clinical phenotype of chronic bronchitis in COPD. Patients present with chronic cough, sputum production, cyanosis, right heart failure, and airway-centric inflammation. They tend to be younger, have more hypercapnia, and less emphysema on imaging.

Body Plethysmography

– Technique for measuring lung volumes (TLC, RV, FRC) using a sealed chamber. Gold standard for TLC measurement. Useful for diagnosing restriction and hyperinflation.

Breath Sounds

– The sounds produced by airflow through the respiratory tract during breathing, assessed by auscultation with a stethoscope over the chest wall. Normal breath sounds are classified by location and quality: tracheal (loud, harsh, high-pitched heard over the trachea), bronchial (loud, tubular, heard over the manubrium and large airways, with a gap between inspiration and expiration), bronchovesicular (intermediate pitch and intensity, heard over the mainstem bronchi, equal inspiratory and expiratory duration,.

Breath Stacking

– A technique using a manual resuscitation bag or glossopharyngeal breathing to deliver successive breaths without exhaling, stacking tidal volumes to achieve higher lung volumes. It improves cough effectiveness and secretion clearance in patients with weak respiratory muscles.

Bronchial Biopsy

– Tissue sampling of the bronchial wall during bronchoscopy. It is used to diagnose inflammatory conditions (sarcoidosis, eosinophilic bronchitis), infections, and malignancy. Endobronchial biopsy of visible lesions and transbronchial biopsy of parenchymal lesions provide complementary diagnostic information.

Bronchial Stenosis

– Narrowing of a bronchus from trauma, infection, inflammation, or prior intervention. It causes localized wheezing, recurrent infections, and impaired secretion clearance. Diagnosis is by bronchoscopy. Treatment includes balloon dilation, stenting, or surgical resection.

Bronchiectasis

– Permanent, abnormal dilation of bronchi caused by destruction of the bronchial wall from chronic airway inflammation and infection. Causes include CF, immune deficiency, primary ciliary dyskinesia, and post-infectious damage. Features include chronic cough, copious purulent sputum, and recurrent hemoptysis.

Bronchitis

– Inflammation of the bronchial mucosa. Acute bronchitis is typically viral, presents with cough (often productive), and resolves spontaneously. Cough with sputum production for at least 3 months in 2 consecutive years, often due to smoking. Treatment: supportive for acute; bronchodilators, smoking cessation, and pulmonary rehabilitation for chronic.

Bronchoalveolar Lavage

– Diagnostic procedure during bronchoscopy where saline is instilled and aspirated. Analyzed for cell counts, microbiology, and cytology. Useful for Pneumocystis pneumonia, pulmonary alveolar proteinosis, and sarcoidosis diagnosis.

Bronchodilator Reversibility

– Improvement in FEV1 $>$ 12% and $>$ 200 mL after short-acting bronchodilator. Supports asthma diagnosis. Some COPD patients show partial reversibility.

Bronchogenic Cyst

– A congenital cyst derived from abnormal budding of the ventral foregut. It is typically found in the middle mediastinum or near the carina. It is usually asymptomatic but may cause compression of adjacent structures. Treatment is surgical resection if symptomatic.

Bronchophony

– Increased transmission of spoken sounds to the stethoscope over consolidated lung tissue. It is normally heard over the central airways. Excessive bronchophony over peripheral lung fields indicates consolidation. It is part of the constellation of physical examination findings in pneumonia.

Bronchopleural Fistula

– An abnormal communication between the bronchial tree and pleural space. It is a complication of pneumonectomy, lung surgery, or empyema. It causes persistent air leak, subcutaneous emphysema, and respiratory compromise. Treatment includes chest tube drainage, bronchoscopic intervention, and surgical repair.

Bronchopneumonia

– Patchy consolidation centered on bronchioles across multiple lobes. Caused by S. aureus, gram-negative bacteria, and Pseudomonas.

Bronchoscopy

– Endoscopic visualization of the lower airways using flexible or rigid bronchoscope. Indications include diagnosis of lung masses, hemoptysis evaluation, BAL, and foreign body removal. Transbronchial biopsy can sample parenchymal lesions.

Carina

– The ridge at the base of the trachea where it bifurcates into the right and left main bronchi. It is a landmark for endotracheal tube placement and bronchoscopy. The right main bronchus is wider, shorter, and more vertical than the left, making it the preferred path for aspirated foreign bodies.

Central Sleep Apnea

– Apneas from variability in respiratory effort without airway obstruction. Associated with heart failure (Cheyne-Stokes respiration), high altitude, and opioid use.

Centriacinar Emphysema

– A subtype of emphysema in which destruction begins in the respiratory bronchioles and extends peripherally. It predominantly affects the upper lobes and is strongly associated with cigarette smoking. It is the most common form of emphysema and often coexists with chronic bronchitis.

CFTR Modulator Therapy

– Targeted therapies correcting the defective CFTR protein. Elexacaftor-tezacaftor-ivacaftor (Trikafta) is the most effective combination for patients with at least one F508del mutation. These therapies have dramatically changed CF prognosis.

Chest Physiotherapy

– Techniques to mobilize and clear secretions from the lungs, including postural drainage, percussion, vibration, and huffing. It is used in bronchiectasis, CF, and post-operative patients. Positive expiratory pressure devices and oscillating PEP devices are also used.

Chest Tube Insertion

– Placement of a thoracostomy tube in the safe triangle (latissimus dorsi, pectoralis major, nipple level) to drain air, fluid, or blood from the pleural space. Connected to underwater seal drainage.

Chest Wall Deformity

– Abnormalities of the chest wall that can affect lung function. Pectus excavatum (sunken sternum) may compress the heart and lungs. Pectus carinatum (protruding sternum) is usually cosmetic. Scoliosis can cause restrictive lung disease. These may require surgical correction if severe.

Cheyne-Stokes Respiration

– Crescendo-decrescendo breathing pattern with periods of apnea, seen in heart failure and stroke. It is a form of central sleep apnea.

Chlamydophila Pneumonia

– Atypical pneumonia with mild presentation: sore throat, hoarseness, cough. Treatment is macrolides or tetracyclines.

Chronic Bronchitis

– Defined clinically as a productive cough for at least three months in each of two consecutive years, after excluding other causes. It is characterized by hypertrophy of mucus-secreting glands in the bronchi (Reid index > 0.5), goblet cell hyperplasia, and chronic airway inflammation.

Chronic Cough

– Cough lasting > 8 weeks. Top 3 causes in non-smokers with normal CXR: upper airway cough syndrome, asthma, GERD. ACE inhibitor use is another common cause.

Chronic Obstructive Pulmonary Disease

– COPD is a preventable lung disease characterized by persistent respiratory symptoms and airflow limitation due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases (primarily tobacco smoke). Includes emphysema and chronic bronchitis. Diagnosis: spirometry with post-bronchodilator FEV1/FVC <0.70. Management: smoking cessation, bronchodilators, inhaled corticosteroids, pulmonary rehabilitation, oxygen therapy, and lung volume reduction.

Chronic Obstructive Pulmonary Disease (COPD)

– COPD is a progressive lung disease characterized by persistent airflow limitation due to chronic bronchitis and emphysema, most commonly caused by cigarette smoking. Symptoms include dyspnea, chronic cough, sputum production, and frequent respiratory infections. Treatment involves bronchodilators, inhaled corticosteroids, pulmonary rehabilitation, supplemental oxygen, and smoking cessation to reduce exacerbations and improve quality of life.

Chylothorax

– Pleural effusion with chyle from thoracic duct disruption. Triglycerides > 110 mg/dL. Treatment includes dietary modification, octreotide, and surgical ligation if refractory.

Coal Worker's Pneumoconiosis

– Pneumoconiosis from coal dust. Simple CWP shows upper lobe macules. Complicated CWP (progressive massive fibrosis) involves coalescent macules. May be associated with Caplan syndrome.

Coarse Crackles

– Lower-pitched, longer-duration crackles heard throughout inspiration, caused by fluid or secretions in larger airways. They are heard in COPD, bronchiectasis, pneumonia, and pulmonary edema. They are less specific than fine crackles.

Combination Inhalers

– Inhalers containing both an inhaled corticosteroid and a long-acting beta-2 agonist. Examples include fluticasone/salmeterol (Advair), budesonide/formoterol (Symbicort), and fluticasone furoate/vilanterol (Breo Ellipta). They simplify asthma and COPD maintenance therapy and improve adherence.

Community-Acquired Pneumonia

– Pneumonia acquired outside healthcare settings. Common pathogens include S. pneumoniae, H. influenzae, and atypical organisms. Severity assessed by CURB-65.

Compressive Atelectasis

– Atelectasis caused by external compression of the lung by pleural fluid, air, or a mass. The lung volume is reduced by the external force. It is reversible when the compressing force is relieved (e.g., drainage of pleural effusion).

Cor Pulmonale

– Right ventricular hypertrophy and dilation from pulmonary hypertension due to lung disease. Features include peripheral edema, JVD, and hepatomegaly.

Cough

– A complex reflex arc protecting the airway by clearing secretions, irritants, and foreign material from the tracheobronchial tree. The cough reflex involves sensory receptors (rapidly adapting irritant receptors, C-fibers, and mechanoreceptors) in the larynx, trachea, and bronchi, afferent pathways (vagus nerve, superior laryngeal nerve, trigeminal nerve), the cough center in the medulla oblongata, efferent pathways (recurrent laryngeal nerve, phrenic nerve, spinal motor nerves), and effector muscles (diaphragm, abdominal wall, intercostals, and laryngeal muscles) for the coordinated cough response.

Cough Augmentation

– Techniques to improve cough effectiveness in patients with neuromuscular disease or chest wall weakness. Methods include manually assisted cough (abdominal thrust timed with cough), mechanical insufflation-exsufflation (CoughAssist device), and breath stacking with a resuscitation bag. They are essential in Duchenne muscular dystrophy and spinal cord injury.

CPAP

– Continuous Positive Airway Pressure, first-line for OSA. Delivers constant pressure through a mask to prevent airway collapse. Eliminates apneas and improves outcomes. Adherence is the main challenge.

Crazy Paving Pattern

– A HRCT finding characterized by ground-glass opacification with superimposed interlobular septal thickening and intralobular lines. It is seen in pulmonary alveolar proteinosis, lipoid pneumonia, ARDS, pulmonary hemorrhage, and Pneumocystis pneumonia. Despite its name, it is not specific to any single diagnosis.

Cryobiopsy

– A bronchoscopic technique using rapid freezing to obtain larger lung tissue samples than conventional transbronchial biopsy. It provides better histopathological specimens for diagnosing interstitial lung diseases. It requires general anesthesia and bronchoscopic expertise. It is less invasive than surgical lung biopsy.

Cryptogenic Organizing Pneumonia

– An inflammatory lung disease with granulation tissue within alveolar ducts. May be idiopathic or secondary to infection, drugs, or connective tissue disease. Responds well to corticosteroids.

CT Pulmonary Angiography

– Gold standard for PE diagnosis, showing thrombi as filling defects. Sensitivity > 95%, specificity > 97% for central emboli.

CURB-65 Score

– One point each for: Confusion, Urea > 7 mmol/L, RR > 30, BP < 90/60, age >= 65. 0–1: outpatient. 2: consider hospitalization. 3–5: severe.

D-Dimer

– Fibrin degradation product elevated in PE, DVT, sepsis, and trauma. Negative result in low-probability patient excludes PE. Age-adjusted D-dimer improves specificity.

Dead Space Ventilation

– Ventilation of alveoli without perfusion (V/Q = infinity). Anatomic dead space is the conducting airways (150 mL). Physiologic dead space includes alveoli without perfusion. Increased in PE, hypotension, and emphysema. Measured by the Bohr equation.

Decreased Breath Sounds

– Diminished air movement heard on auscultation, indicating decreased ventilation. Causes include pneumothorax, pleural effusion, atelectasis, and mucus plugging. Unilateral decreased breath sounds may indicate tension pneumothorax or large effusion.

Desquamative Interstitial Pneumonia

– A rare ILD characterized by uniform filling of alveoli with macrophages containing brown pigment (hemosiderin-laden). It is strongly associated with smoking. HRCT shows bilateral ground-glass opacities with lower lobe predominance. Smoking cessation and corticosteroids are the main treatments.

Diffusing Capacity (DLCO)

– Ability of lungs to transfer gas from alveoli to blood. Reduced in emphysema, ILD, and anemia. Increased in pulmonary hemorrhage and left-to-right shunts.

Digital Clubbing

– Bulbous enlargement of distal phalanges with loss of nail-bed angle (> 180 degrees). Associated with lung cancer, ILD, bronchiectasis, and cyanotic heart disease.

Directly Observed Therapy

– Supervision of medication ingestion to ensure adherence. Standard of care for TB treatment. Ensures completion of the full treatment course and reduces development of drug resistance.

Dornase Alfa

– A recombinant human DNase that cleaves extracellular DNA in airway secretions, reducing sputum viscosity. It is used in CF to improve lung function and reduce exacerbations. It is administered as a nebulized solution once or twice daily. It has become a standard part of CF airway clearance regimens.

Dry Powder Inhaler

– A breath-actuated inhaler delivering powdered medication. Does not require hand-breath coordination. Requires adequate inspiratory flow. Examples include Advair Diskus and Spiriva HandiHaler.

Dyspnoea

– The subjective sensation of difficult or labored breathing, also known as shortness of breath (SOB). It is a symptom common to many cardiopulmonary and systemic conditions. Causes: respiratory (asthma, COPD, pneumonia, pulmonary embolism, pneumothorax, interstitial lung disease), cardiac (heart failure, myocardial ischemia, arrhythmia, valvular disease), and others (anemia, deconditioning, anxiety, metabolic acidosis, thyrotoxicosis). Assessed using the Borg Scale or modified Medical Research Council (mMRC) scale. Acute dyspnea is a medical emergency requiring rapid assessment of airway, breathing, and circulation; immediate oxygen supplementation and treatment of the underlying cause (bronchodilators, diuretics, anticoagulation, etc.). Chronic dyspnea requires systematic evaluation with pulmonary function testing, echocardiography, and imaging.

Egophony

– An abnormal vocal sound heard during auscultation in which spoken “E” sounds are heard as “A” (the “E-to-A” change). It is a sign of lung consolidation, as in lobar pneumonia. The consolidated lung tissue conducts high-frequency sounds better than air-filled lung.

Electromagnetic Navigation Bronchoscopy

– A guidance system for bronchoscopy that uses electromagnetic fields to navigate to peripheral lung lesions. It allows biopsy of lesions not visible on standard bronchoscopy. It is more accurate than conventional transbronchial biopsy for peripheral lesions. It may reduce the need for CT-guided biopsy.

Emphysema

– A form of COPD characterized by permanent, abnormal enlargement of airspaces distal to the terminal bronchioles, accompanied by destruction of their walls. Centriacinar emphysema affects the respiratory bronchioles and is associated with smoking. Panlobular emphysema involves the entire acinus and is associated with alpha-1 antitrypsin deficiency.

Empyema

– Pus in the pleural space. CT shows peripheral enhancement (split pleura sign). Treatment requires chest tube drainage, intrapleural fibrinolytics, or VATS decortication.

Empyema Necessitans

– Extension of pleural empyema through the chest wall. Presents as a painful, fluctuant mass. Requires drainage and antibiotics.

Endobronchial Foreign Body

– A foreign object lodged in a bronchus, causing partial or complete airway obstruction. It may cause localized wheezing, atelectasis, and post-obstructive pneumonia. It is most common in children and may be initially misdiagnosed as asthma. Bronchoscopy is both diagnostic and therapeutic.

Endobronchial Tumor

– A tumor growing within the bronchial lumen, causing obstruction, atelectasis, and post-obstructive pneumonia. It may be malignant (lung cancer, carcinoid tumor) or benign (hamartoma, papilloma). Symptoms include cough, hemoptysis, and wheezing. Treatment is bronchoscopic removal or debulking plus definitive therapy.

Endobronchial Ultrasound

– Bronchoscopy with ultrasound guidance for transbronchial needle aspiration of mediastinal and hilar lymph nodes. Essential for lung cancer staging. Less invasive than mediastinoscopy.

Endobronchial Valve

– A one-way valve placed bronchoscopically in a target airway to achieve lobar atelectasis in emphysema. It allows air and secretions to exit but prevents air from entering. It is used for bronchoscopic lung volume reduction in selected patients with heterogeneous emphysema and intact fissures.

Endothelin Receptor Antagonists

– Medications that block endothelin-1, a potent vasoconstrictor implicated in pulmonary arterial hypertension. Bosentan (dual ERA), ambrisentan (selective ETA), and macitentan are used for PAH. Side effects include hepatotoxicity (bosentan), anemia, and teratogenicity.

Endotracheal Intubation

– Placement of a tube through the mouth into the trachea for mechanical ventilation. Gold standard for airway management. Confirmation by end-tidal CO2 detection and bilateral breath sounds.

Ethambutol

– First-line TB drug inhibiting arabinosyl transferase. Side effects: optic neuritis (visual acuity and color vision monitoring), hyperuricemia. Dose is weight-based.

Exacerbation of COPD

– An acute worsening of respiratory symptoms beyond normal day-to-day variations, requiring change in medication. It is commonly triggered by respiratory infections (viral, bacterial), air pollution, or non-compliance. Treatment includes increased bronchodilators, systemic corticosteroids, antibiotics if indicated, and oxygen therapy.

Exercise-Induced Bronchoconstriction

– Airway narrowing during or after exercise, presenting with cough, wheeze, and dyspnea. It occurs in up to 90% of asthmatics and is triggered by heat and water loss from the airways. Prevention includes pre-exercise SABA.

Exhaled Nitric Oxide

– A non-invasive measure of airway eosinophilic inflammation. FeNO $>$ 25 ppb in adults suggests eosinophilic airway inflammation and predicts response to inhaled corticosteroids. It is useful for diagnosing eosinophilic asthma, monitoring treatment response, and differentiating asthma from other causes of cough.

Expiratory Grunting

– A forced exhalation against a partially closed glottis, creating an audible sound. It is an end-expiratory maneuver that generates intrinsic PEEP, helping to maintain alveolar recruitment and improve oxygenation. It is commonly seen in children with respiratory distress.

Extensively Drug-Resistant TB

– TB resistant to isoniazid, rifampin, fluoroquinolones, and at least one injectable agent. Very limited treatment options and poor outcomes.

Exudative Pleural Effusion

– From local pleural disease. Causes include pneumonia, malignancy, TB, and PE. Elevated protein, LDH, and cell count.

Farmer's Lung

– The most common HP, caused by thermophilic actinomycetes in moldy hay. Presents acutely 4–8 hours after exposure with fever, cough, and dyspnea.

FEV1/FVC Ratio

– Fraction of air exhaled in first second relative to total. < 0.70 post-bronchodilator indicates obstruction. Normal or elevated in restrictive disease.

Fine Crackles

– High-pitched, brief, discontinuous sounds heard at the end of inspiration, caused by opening of small airways. They are characteristic of ILD, pulmonary fibrosis, and heart failure. They are also called Velcro crackles due to their resemblance to the sound of Velcro being pulled apart.

Flail Chest

– A clinical condition caused by three or more consecutive ribs fractured in two or more places, creating a free-floating segment of chest wall. The segment moves paradoxically inward during inspiration and outward during expiration. It causes severe pain, impaired ventilation, and may require mechanical ventilation. It is often associated with pulmonary contusion.

Flow-Volume Loop

– Graphical representation of airflow during forced breathing. Obstructive: scooped expiratory limb. Restrictive: narrowed loop with preserved shape. Variable extrathoracic obstruction: flattening of inspiratory limb.

Flutter Valve

– A handheld device combining PEP therapy with oscillations during exhalation. The oscillations create shear forces that loosen mucus from airway walls, while the PEP keeps airways open. It is used for airway clearance in bronchiectasis, CF, and COPD.

Forced Oscillation Technique

– Similar to impulse oscillometry, applying small pressure oscillations at various frequencies during tidal breathing to measure respiratory resistance and reactance. It provides information about airway mechanics without requiring forced expiratory maneuvers. It is useful for detecting small airway dysfunction and monitoring treatment response in asthma and COPD.

Forced Vital Capacity

– Total air exhaled after maximal inspiration. Reduced in both obstructive and restrictive disease. In restrictive disease, FVC reduced with preserved ratio.

Foreign Body Aspiration

– Inhalation of a foreign object into the airway, most common in children (ages 1–3) and adults with impaired swallowing. It causes sudden onset choking, coughing, and unilateral wheezing. Right main bronchus is most commonly affected due to its wider, more vertical anatomy. Treatment is bronchoscopic removal.

Haemoptysis

– The expectoration of blood from the lower respiratory tract (UK spelling). Must be differentiated from hematemesis (vomiting blood of upper GI origin) and epistaxis (nasal bleeding). Causes: bronchitis (most common), pneumonia, tuberculosis, bronchiectasis, lung cancer, pulmonary embolism, and coagulopathy. Investigation: chest X-ray, CT thorax, bronchoscopy for persistent or unexplained cases. Management: treat underlying cause; massive hemoptysis (>200 mL in 24 hours) is life-threatening and requires prompt airway protection, lateral decubitus positioning with the bleeding side down, bronchial artery embolisation or surgical resection.

Hamman's Sign

– A crunching or crackling sound heard on auscultation synchronized with the heartbeat, caused by air in the mediastinum (pneumomediastinum). It is a rare but specific finding. It is also described as a mediastinal crunch heard in spontaneous pneumomediastinum.

Hampton's Hump

– Wedge-shaped, pleural-based opacity on CXR representing pulmonary infarction distal to PE.

Hemoptysis

– Expectoration of blood from the lower respiratory tract. Must be differentiated from hematemesis and epistaxis. Causes: bronchitis, pneumonia, TB, bronchiectasis, lung cancer, PE.

Hemothorax

– Blood in pleural space from trauma. Initial chest tube output > 1500 mL or persistent bleeding > 200 mL/hour indicates need for surgery.

Hereditary Hemorrhagic Telangiectasia

– An autosomal dominant vascular disorder (also called Osler-Weber-Rendu syndrome) characterized by telangiectasias on skin and mucous membranes, pulmonary AVMs, hepatic AVMs, and cerebral AVMs. Pulmonary AVMs can cause hypoxemia and paradoxical emboli. Treatment is embolization of pulmonary AVMs.

High-Flow Nasal Cannula

– A device delivering heated and humidified oxygen at flow rates up to 60 L/min through a nasal cannula. It provides a modest amount of PEEP, reduces dead space ventilation, and improves patient comfort compared to non-invasive ventilation. It is used for acute hypoxemic respiratory failure and post-extubation support.

Hospital-Acquired Pneumonia

– Pneumonia developing > 48 hours after hospital admission. Common pathogens include Pseudomonas and MRSA. Empiric treatment covers resistant organisms.

Hypercyanotic Tetralogy

– A complex congenital heart defect with four features: ventricular septal defect, overriding aorta, right ventricular outflow tract obstruction (pulmonary stenosis), and right ventricular hypertrophy. It causes right-to-left shunting and cyanosis. Treatment is surgical repair, typically in infancy.

Hypersensitivity Pneumonitis

– An immune-mediated lung disease from environmental antigen inhalation (molds, bird proteins). Acute, subacute, and chronic forms exist. HRCT shows ground-glass opacities, air trapping, and centrilobular nodules. Chronic HP progresses to fibrosis.

Hypertonic Saline Inhalation

– Nebulized 3–7% sodium chloride solution that draws water into the airway surface liquid by osmosis, improving mucociliary clearance. It is used in CF and bronchiectasis. It may cause bronchospasm and should be preceded by a bronchodilator. It reduces exacerbations in CF.

Hypoxemia

– PaO2 < 80 mmHg on room air. Causes: hypoventilation, V/Q mismatch, shunt, diffusion impairment, low FiO2. A-a gradient helps distinguish causes.

Hypoxia

– Reduced oxygen availability at the tissue level, defined as arterial partial pressure of oxygen (PaO_2) <60 mmHg or arterial oxygen saturation (SpO_2) <90% on room air. Five types: (1) Hypoxic hypoxia (low arterial PO_2)—caused by low inspired oxygen (high altitude), hypoventilation, V/Q mismatch, shunt, or diffusion impairment (pneumonia, COPD, pulmonary edema, ARDS). (2) Anaemic hypoxia (reduced oxygen-carrying capacity)—anemia, carbon monoxide poisoning, methaemoglobinaemia. (3) Stagnant/hypoperfusion hypoxia (reduced blood flow)—heart failure, shock, peripheral vascular disease. (4) Histotoxic hypoxia (impaired tissue utilisation)—cyanide poisoning (blocks cytochrome c oxidase), sepsis. (5) Demand hypoxia (increased O_2 requirement)—exercise, hypermetabolic states (thyrotoxicosis, malignant hyperthermia). Assessment: ABG, pulse oximetry, and calculation of alveolar-arterial (A-a) gradient. Consequences: cellular anaerobic metabolism, lactic acidosis, ATP depletion, and cell death. Treatment: supplemental oxygen (high-flow nasal cannula, non-invasive ventilation, mechanical ventilation), treat underlying cause. The hypoxic ventilatory response is mediated by peripheral chemoreceptors (carotid and aortic bodies).

Idiopathic Pulmonary Fibrosis

– A chronic, progressive fibrosing interstitial pneumonia of unknown cause. Histopathology shows usual interstitial pneumonia (UIP). Features include exertional dyspnea, dry cough, bibasilar crackles, and digital clubbing. Median survival is 3–5 years.

Impulse Oscillometry

– A pulmonary function testing technique that measures respiratory impedance by applying small pressure oscillations during tidal breathing. It assesses large airway resistance (R5), small airway resistance (R5-R20), and reactance. It is more sensitive than spirometry for detecting early small airway disease and is useful in patients who cannot perform spirometry (young children, elderly).

Incentive Spirometry

– A device that encourages deep breathing to prevent atelectasis after surgery. The patient inhales slowly and deeply through the device, maintaining maximal inspiration for several seconds. It promotes alveolar recruitment and prevents mucus plugging. It is standard post-operative care.

Induced Sputum Analysis

– A non-invasive method of assessing airway inflammation by examining sputum cell counts and differentials obtained after hypertonic saline inhalation. Sputum eosinophils $> 3\%$ indicate eosinophilic airway inflammation (asthma, EGPA). Neutrophilic inflammation suggests infection, COPD, or bronchiectasis. It guides anti-inflammatory therapy decisions.

Influenza

– A contagious respiratory illness caused by influenza A or B viruses, characterized by sudden onset of fever, myalgia, headache, malaise, nonproductive cough, sore throat, and nasal congestion. Seasonal epidemics occur annually. Complications: pneumonia (primary viral or secondary bacterial), exacerbation of underlying conditions. Diagnosis: PCR or rapid antigen testing. Treatment: supportive care, neuraminidase inhibitors (oseltamivir). Prevention: annual vaccination.

Inhaled Corticosteroids

– Anti-inflammatory medications delivered directly to the airways, reducing airway inflammation, mucus production, and bronchial hyperresponsiveness. Examples include fluticasone, budesonide, beclomethasone, and mometasone. They are the cornerstone of asthma maintenance therapy. Side effects include oral candidiasis and dysphonia.

Inhaled Nitric Oxide

– A selective pulmonary vasodilator used in acute settings (acute respiratory failure, right heart failure, post-cardiac surgery). It reduces pulmonary vascular resistance and improves oxygenation by dilating well-ventilated pulmonary vessels (improving V/Q matching). It has a short half-life and requires continuous delivery.

Inspiratory Crackles

– Fine, discontinuous sounds heard during inspiration, caused by the sudden opening of collapsed airways. They are a hallmark of interstitial lung disease (bibasilar, Velcro-like crackles) and pulmonary fibrosis. They are also heard in pneumonia, atelectasis, and heart failure.

Interferon-Gamma Release Assay

– Blood tests measuring IFN-gamma release in response to M. tuberculosis antigens (ESAT-6, CFP-10). More specific than PPD and unaffected by BCG. Cannot distinguish latent from active TB.

Interstitial Lung Disease

– A heterogeneous group of disorders with inflammation and fibrosis of lung parenchyma, causing restrictive physiology (reduced FVC and TLC with normal or elevated FEV1/FVC ratio).

Interstitial Pneumonia

– Inflammation centered on the interstitium. Caused by Mycoplasma, Chlamydophila, Legionella, and viruses. Presents with dry cough and diffuse crackles.

Intrapulmonary Shunt

– Perfusion of alveoli without ventilation ($V/Q = 0$). Blood passes through the pulmonary circulation without gas exchange, mixing with oxygenated blood and causing hypoxemia. It is seen in ARDS, pneumonia, and atelectasis. It is refractory to supplemental oxygen.

Ipratropium Bromide

– An inhaled anticholinergic (antimuscarinic) bronchodilator that blocks acetylcholine at airway smooth muscle muscarinic receptors. It reduces bronchospasm and mucus secretion. It is used in COPD and acute asthma exacerbations. It is less effective than beta-2 agonists in asthma but more effective in COPD.

Isoniazid

– First-line TB drug inhibiting mycolic acid synthesis. Side effects: hepatotoxicity, peripheral neuropathy, drug-induced lupus. Most effective for latent TB treatment.

IVC Filter

– A device in the IVC to prevent DVT from reaching the lungs. Indications include recurrent PE despite anticoagulation and absolute contraindication to anticoagulation.

Kartagener Syndrome

– A subset of primary ciliary dyskinesia with situs inversus, chronic sinusitis, and bronchiectasis. Caused by dynein defects impairing both mucociliary clearance and embryonic nodal cilia rotation.

Kartagener Triad

– The classic triad of situs inversus, chronic sinusitis, and bronchiectasis, occurring in approximately 50% of patients with primary ciliary dyskinesia. It is caused by dynein defects impairing both mucociliary clearance and embryonic nodal cilia rotation.

Klebsiella Pneumonia

– Severe, necrotizing pneumonia in alcoholic or debilitated patients. Upper lobe involvement with "bulging fissure sign" and "currant jelly" sputum.

Kussmaul Breathing

– A deep, rapid breathing pattern seen in severe metabolic acidosis (diabetic ketoacidosis, lactic acidosis, uremia). It represents the body's attempt to compensate by blowing off CO2. The increased respiratory rate and depth create the characteristic labored breathing.

Kyphoscoliosis

– An abnormal curvature of the spine in both the coronal (scoliosis) and sagittal (kyphosis) planes. Severe forms cause restrictive lung disease by reducing chest wall compliance and lung volumes. It can lead to chronic respiratory failure and pulmonary hypertension. Treatment includes bracing, surgical correction, and non-invasive ventilation.

Lambert-Eaton Myasthenic Syndrome

– Autoantibodies against presynaptic calcium channels at the neuromuscular junction, associated with SCLC. Proximal weakness, reduced reflexes, facilitation with exercise.

Large Airway Obstruction

– Obstruction of airways with diameter > 2 mm (trachea and main bronchi). It causes significant symptoms including stridor, dyspnea, and impaired secretion clearance. It may be fixed (stenosis, tumor) or variable (tracheomalacia, vocal cord dysfunction). Flow-volume loops show characteristic patterns.

Latent Tuberculosis Infection

– Persistent immune response to M. tuberculosis without active disease. Asymptomatic and noninfectious. 5–10% of immunocompetent individuals develop active TB without treatment. Treatment reduces risk by 60–90%.

Legionella Pneumonia

– From Legionella pneumophila in water systems. Presents with high fever, diarrhea, confusion, and hyponatremia. Diagnosis by urinary antigen test. Treatment is fluoroquinolones or macrolides.

Lfgren Syndrome

– Acute sarcoidosis with bilateral hilar lymphadenopathy, erythema nodosum, and ankle polyarthralgia. Excellent prognosis with spontaneous resolution. Associated with HLA-DR3.

Light's Criteria

– An effusion is exudative if any one is met: pleural protein/serum protein > 0.5, pleural LDH/serum LDH > 0.6, or pleural LDH > 2/3 upper limit of normal serum LDH.

Lipoid Pneumonia

– Lung inflammation caused by accumulation of lipid-laden macrophages in the alveoli. Exogenous lipoid pneumonia results from aspiration of mineral oil, petroleum jelly, or oil-based nose drops. Endogenous lipoid pneumonia occurs distal to bronchial obstruction. HRCT shows ground-glass opacities with low-attenuation areas (fat density).

Lobar Pneumonia

– Consolidation involving an entire lobe. Most commonly caused by S. pneumoniae. Classic progression: congestion, red hepatization, gray hepatization, resolution.

Long-Acting Beta-2 Agonists

– Bronchodilators that provide sustained airway smooth muscle relaxation for 12–24 hours. Examples include salmeterol, formoterol, indacaterol, and olodaterol. They are used for maintenance therapy in asthma and COPD, always in combination with inhaled corticosteroids in asthma.

Low-Dose CT Scan

– CT protocol using reduced radiation for lung cancer screening. Detects small nodules at earlier stages. False positives require follow-up per Fleischner guidelines.

Lower Airway Obstruction

– Narrowing of airways below the carina, causing expiratory wheezing and airflow obstruction on spirometry. Causes include asthma, COPD, bronchiectasis, and foreign body. It is characterized by scooped-out expiratory limb on flow-volume loops and reduced FEV1/FVC ratio.

Lung Abscess

– Localized necrosis with cavitation containing purulent material. Causes include aspiration and anaerobic bacteria. CXR shows thick-walled cavity with air-fluid level.

Lung Disease

– Lung disease encompasses disorders affecting the airways, parenchyma, pleura, pulmonary vasculature, and chest wall. Major categories: obstructive (asthma, COPD—chronic bronchitis and emphysema), restrictive (pulmonary fibrosis, sarcoidosis, asbestosis), infectious (pneumonia, tuberculosis), pulmonary vascular (pulmonary embolism, pulmonary arterial hypertension), and neoplastic (lung cancer, the leading cause of cancer death). Common symptoms: dyspnea, cough (dry or productive), wheezing, hemoptysis, chest pain, and weight loss. Diagnosis: pulmonary function tests (spirometry, DLCO), chest imaging (X-ray, CT), bronchoscopy, and biopsy. Management: smoking cessation (most effective intervention), bronchodilators (beta-agonists, antimuscarinics), inhaled corticosteroids, oxygen therapy, pulmonary rehabilitation, antimicrobials for infection, and lung transplantation for end-stage disease.

Lung Function Tests

– A group of physiological tests that assess the function and capacity of the lungs. See Pulmonary Function Tests.

Lung Protective Ventilation

– A ventilation strategy using low tidal volumes (6 mL/kg ideal body weight), plateau pressure < 30 cmH2O, appropriate PEEP, and permissive hypercapnia to minimize ventilator-induced lung injury. It is the standard of care for ARDS and is used in all mechanically ventilated patients to reduce barotrauma, volutrauma, and biotrauma.

Lung Transplantation

– Surgical replacement of a diseased lung with a healthy lung from a donor. Indications include IPF, COPD, CF, PAH, and bronchiectasis. Options include single lung transplant, double lung transplant, or heart-lung transplant. Post-transplant immunosuppression is required lifelong. Complications include rejection and infection.

Lymphangioleiomyomatosis

– A rare, progressive cystic lung disease caused by proliferation of abnormal smooth muscle-like cells (LAM cells) in the lungs, lymphatics, and kidneys. It occurs almost exclusively in women of childbearing age. HRCT shows thin-walled cysts throughout both lungs. It may be associated with tuberous sclerosis complex. Treatment includes sirolimus (mTOR inhibitor) and lung transplantation.

Lymphocytic Interstitial Pneumonia

– A rare ILD characterized by diffuse polyclonal lymphocytic infiltration of the lung parenchyma. It is strongly associated with Sjgren syndrome and HIV. HRCT shows ground-glass opacities, thin-walled cysts, and lymphadenopathy. Treatment is corticosteroids for symptomatic disease.

Massive Hemoptysis

– > 600 mL in 24 hours or causing hemodynamic compromise. Medical emergency. Treatment: airway protection, bronchial artery embolization (first-line), or surgery.

Mechanical Insufflation-Exsufflation

– A device that delivers positive pressure (insufflation) followed by rapid negative pressure (exsufflation) to simulate a cough. It is used in neuromuscular disease (Duchenne muscular dystrophy, spinal cord injury, ALS) to augment cough effectiveness and clear secretions.

Mediastinoscopy

– Surgical procedure to sample mediastinal lymph nodes through a small incision above the sternum. Gold standard for mediastinal lymph node staging in lung cancer. Being replaced by EBUS in many centers.

Metered-Dose Inhaler

– A pressurized inhaler delivering a metered dose of medication. Requires proper technique for effective delivery. Spacer devices improve drug delivery and reduce oropharyngeal deposition. Most common delivery device for asthma and COPD maintenance.

Methacholine Challenge Test

– Bronchoprovocation test for airway hyperresponsiveness. PC20 < 8 mg/mL is positive. Highly sensitive but less specific for asthma.

Middle Lobe Syndrome

– Chronic or recurrent atelectasis and infection of the right middle lobe due to bronchial obstruction. The long, narrow right middle lobe bronchus is susceptible to compression by lymph nodes, tumors, or mucous plugging. It may be caused by bronchiectasis, malignancy, or granulomatous disease. Treatment depends on the underlying cause.

Miliary TB

– Hematogenous dissemination of M. tuberculosis causing innumerable 1–3 mm granulomas throughout lungs and other organs. CXR shows "millet seed" pattern. More common in immunocompromised patients.

Minute Ventilation

– The total volume of air breathed in one minute (tidal volume x respiratory rate). Normal is 6–8 L/min. Elevated in increased metabolic demand, pain, anxiety, and sepsis. Reduced in neuromuscular disease and sedation.

Mucolytic Therapy

– Medications that reduce mucus viscosity and elasticity, facilitating clearance. N-acetylcysteine (NAC) breaks disulfide bonds in mucus glycoproteins. Dornase alfa (Pulmozyme) cleaves extracellular DNA in CF sputum. Hypertonic saline aerosol draws water into the airway surface liquid. They are used in CF, bronchiectasis, and chronic bronchitis.

Mucus Plugging

– Accumulation of thick, tenacious mucus within airways causing obstruction. It is common in asthma, CF, bronchiectasis, and chronic bronchitis. It impairs gas exchange, promotes infection, and causes atelectasis. Treatment includes airway clearance techniques, mucolytics, and bronchodilators.

Multidrug-Resistant TB

– TB resistant to at least isoniazid and rifampin. Requires second-line drugs for 18–24 months. Worse outcomes and higher mortality than drug-susceptible TB.

Mycoplasma Pneumonia

– Atypical pneumonia most common in ages 5–35. Dry cough, low-grade fever, headache. Patchy infiltrates on CXR disproportionate to symptoms ("walking pneumonia").

Nebulizer

– A device that converts liquid medications into a mist for inhalation. Used for bronchodilators (albuterol, ipratropium) and corticosteroids in acute exacerbations. More effective than MDI in patients with severe obstruction.

Negative Pressure Ventilation

– Ventilation using negative pressure applied around the chest wall to expand the lungs, as opposed to positive pressure applied to the airway. The "iron lung" is a historical example. Modern negative pressure ventilators include the cuirass and wrap. They are rarely used today but may be useful in neuromuscular disease.

Non-Asthmatic Eosinophilic Bronchitis

– Chronic cough with airway eosinophilia (induced sputum eosinophils $> 3\%$) but without bronchial hyperresponsiveness. It is a common cause of chronic cough and does not respond to bronchodilators. It responds well to inhaled corticosteroids. It is distinct from asthma in the absence of variable airflow obstruction.

Non-Invasive Ventilation

– Positive pressure ventilation delivered via a face mask or nasal mask without endotracheal intubation. BiPAP (bilevel positive airway pressure) is most commonly used for acute hypercapnic respiratory failure (COPD exacerbation) and cardiogenic pulmonary edema. It reduces the need for intubation and improves outcomes in selected patients.

Nonspecific Interstitial Pneumonia

– A pattern with uniform, temporally homogeneous inflammation and/or fibrosis. Most common ILD associated with connective tissue disease. HRCT shows bilateral ground-glass opacities with subpleural sparing.

Obesity Hypoventilation Syndrome

– Chronic daytime hypercapnia in obese individuals without other causes. Reduced chest wall compliance and blunted respiratory drive. Often coexists with OSA. Treatment: weight loss and BiPAP.

Obstructive Sleep Apnea

– Repetitive upper airway obstruction during sleep. Presents with snoring, witnessed apneas, daytime sleepiness. AHI quantifies severity: 5–14 mild, 15–29 moderate, >= 30 severe. Treatment: CPAP, oral appliances, weight loss, surgery.

Occupational Asthma

– Asthma from workplace exposures to sensitizers (isocyanates, flour, latex) or irritants. Improves away from work and worsens at work. Serial peak flow monitoring supports diagnosis.

Occupational COPD

– COPD from workplace dust, chemical, or fume exposure independent of smoking. Accounts for 15–20% of COPD cases. High-risk occupations: mining, construction, farming.

Orthopnea

– Dyspnea that occurs when lying flat and is relieved by sitting upright. It is measured by the number of pillows the patient needs to sleep (two-pillow orthopnea). It is a hallmark of left heart failure and is caused by redistribution of fluid from the lower extremities to the lungs in the supine position.

Orthopnoea

– Shortness of breath that occurs when lying flat and is relieved by sitting or standing upright. Orthopnoea is a classic symptom of left-sided heart failure (elevated pulmonary venous pressure worsens pulmonary congestion in the supine position). Other causes: severe COPD, asthma exacerbation, diaphragmatic paralysis, ascites, and obesity. It is quantified by the number of pillows required to sleep comfortably (e.g., two-pillow orthopnoea). Pathophysiology: supine position increases venous return (preload) and reduces lung volumes (cephalad displacement of the diaphragm and abdominal contents), worsening V/Q mismatch and increasing pulmonary capillary pressure. Orthopnoea may develop acutely in pulmonary edema (cardiac asthma) or gradually in worsening chronic heart failure. Related: paroxysmal nocturnal dyspnoea (PND—sudden breathlessness during sleep, relieved by sitting up, also CHF). Management: treat the underlying cause—diuretics, vasodilators, and optimisation of heart failure therapy.

Pancoast Tumor

– Superior sulcus tumor invading brachial plexus, subclavian vessels, and sympathetic chain. Causes Horner syndrome, shoulder/arm pain, and hand muscle atrophy. Treated with chemoradiation followed by surgery.

Panlobular Emphysema

– A subtype involving diffuse destruction of the entire acinus. It predominantly affects the lower lobes and is classically associated with alpha-1 antitrypsin deficiency. The destruction is more uniform compared to centriacinar emphysema.

Parapneumonic Effusion

– Pleural effusion complicating pneumonia. Simple effusions resolve with antibiotics. Complicated effusions (pH < 7.2, glucose < 60 mg/dL) require chest tube drainage.

Paraseptal Emphysema

– A form of emphysema characterized by destruction of alveoli adjacent to the pleura. It typically affects young adults and predisposes to spontaneous pneumothorax due to rupture of subpleural blebs. It may be associated with Marfan syndrome.

Paroxysmal Nocturnal Dyspnea

– Episodes of severe dyspnea that awaken the patient from sleep, typically occurring 1–2 hours after lying down. It is a classic symptom of left heart failure, caused by progressive pulmonary congestion during sleep. It is different from orthopnea in its delayed onset and nocturnal presentation.

Peak Expiratory Flow Rate (PEFR)

– The maximum flow rate achieved during a forced expiratory manoeuvre after maximal inspiration, measured in litres per minute (L/min) using a peak flow meter. PEFR reflects the degree of airflow obstruction and is effort-dependent. Normal values are predicted based on age, sex, and height (e.g., healthy adult male 600 L/min, female 400 L/min). Uses: diagnosis and monitoring of asthma (diurnal variability >20% supports asthma diagnosis), assessment of exacerbation severity (PEFR <50% of predicted indicates severe exacerbation—NICE/BTS guidelines), and guiding asthma action plans (personal best, zone system). PEFR <33% of predicted is life-threatening. Limitations: effort-dependent, poor technique, less reproducible than FEV_1.

PE Anticoagulation

– Initial heparin (LMWH or UFH), followed by warfarin (INR 2–3) for 3–6 months. DOACs are first-line for most patients. Duration depends on provoked vs. unprovoked status.

Percussion Note

– The sound produced by percussing the chest wall.

Normal lung produces a resonant note. Dullness suggests consolidation or fluid. Hyperresonance suggests pneumothorax or emphysema. Stony dullness suggests pleural effusion. Tympany suggests large pneumothorax.

Permissive Hypercapnia

– A ventilator strategy that allows PaCO2 to rise above normal (and pH to fall) in exchange for lung-protective ventilation with lower tidal volumes and airway pressures. It is used in severe ARDS when achieving normal PaCO2 would require harmful ventilation parameters.

Persistent Asthma

– Asthma requiring daily controller therapy. Mild persistent: symptoms > 2 days/week but not daily. Moderate persistent: daily symptoms, nighttime awakenings > 1/week. Severe persistent: symptoms throughout the day, frequent nighttime awakenings, SABA use several times/day, extremely limited activity.

Phosphodiesterase-5 Inhibitors

– Medications that inhibit PDE-5, increasing cGMP and causing smooth muscle relaxation and vasodilation. Sildenafil and tadalafil are used for PAH and erectile dysfunction. In PAH, they reduce pulmonary vascular resistance and improve exercise capacity.

Pink Puffer

– A clinical phenotype of emphysema in COPD. Patients present with severe dyspnea, minimal cough, thin habitus, pursed-lip breathing, accessory muscle use, and relatively normal PaO2 with late hypercapnia. They tend to be older and have more extensive emphysema.

Platypnea

– Dyspnea that occurs in the upright position and is relieved by lying down. It is the opposite of orthopnea. It is rare and may be associated with hepatopulmonary syndrome, intracardiac shunting, or pulmonary arteriovenous malformations.

Pleural Effusion

– Abnormal fluid in the pleural space. Transudates result from altered pressures; exudates from inflammation, infection, or malignancy. Light's criteria differentiate the two.

Pleural Fluid Glucose

– Low pleural fluid glucose (< 60 mg/dL) suggests bacterial empyema, rheumatoid pleurisy, or tuberculosis. It indicates high metabolic activity in the pleural space consuming glucose. Very low glucose is characteristic of empyema and parapneumonic effusions requiring drainage.

Pleural Fluid pH

– Pleural fluid pH < 7.20 indicates a complicated parapneumonic effusion or empyema requiring drainage. Normal pleural fluid pH is approximately 7.60. Low pH reflects high metabolic activity and anaerobic metabolism in the pleural space. It is a key decision point in managing parapneumonic effusions.

Pleural Friction Rub

– A grating, leathery sound heard on auscultation caused by inflamed pleural surfaces rubbing against each other during breathing. It is a sign of pleurisy and may be heard in pneumonia, PE, and pleural effusion. It disappears when fluid accumulates and separates the pleural surfaces.

Pleural Plaque

– Circumscribed fibrous tissue on parietal pleura, hallmark of asbestos exposure. Bilateral, often calcified. Does not indicate asbestosis.

Pleurisy

– Pleural inflammation causing sharp, breathing-related chest pain. Causes: pneumonia, PE, autoimmune disease, malignancy. Treatment addresses the underlying cause.

Pleuritis

– Inflammation of the pleural membranes (visceral and parietal pleura). Presents with pleuritic chest pain: sharp, stabbing pain exacerbated by deep breathing, coughing, or sneezing. Common causes: viral pleurisy (most common), bacterial pneumonia with parapneumonic effusion, pulmonary embolism, tuberculosis, connective tissue diseases (SLE, rheumatoid arthritis), uremia, and malignancy. May be accompanied by pleural friction rub on auscultation. Diagnosis: clinical history, chest X-ray (evaluate for effusion, infiltrates, or pneumothorax), ultrasound, and laboratory evaluation (CBC, CRP, D-dimer to rule out PE). Treatment: addressing the underlying cause; NSAIDs for pain relief; corticosteroids for severe inflammatory pleuritis (e.g., SLE).

Pleurodesis

– Fusion of visceral and parietal pleura to prevent recurrence of effusions or pneumothorax. Chemical pleurodesis uses talc, doxycycline, or bleomycin.

Pneumatocele

– Thin-walled, air-filled cystic spaces in the lung, often in staphylococcal pneumonia. Typically self-resolving.

Pneumoconiosis

– ILDs from mineral dust inhalation (silicosis, asbestosis, coal worker's, berylliosis). They share interstitial fibrosis and risk of infection and malignancy.

Pneumocystis Pneumonia

– Opportunistic pneumonia from P. jirovecii in immunocompromised patients (CD4 < 200). Progressive dyspnea, dry cough, bilateral ground-glass opacities. Elevated LDH. Treatment is TMP-SMX.

Pneumomediastinum

– Air in the mediastinum, often from alveolar rupture with air tracking along bronchovascular sheaths. Causes include asthma exacerbation, trauma, Valsalva maneuver, and esophageal rupture (Boerhaave syndrome). It presents with chest pain, subcutaneous emphysema, and Hamman's crunch (mediastinal crunch synchronous with heartbeat). Usually self-limited.

Pneumonia

– Infection of the lung parenchyma, classified as community-acquired (CAP), hospital-acquired (HAP), or ventilator-associated (VAP). Common pathogens: Streptococcus pneumoniae, Haemophilus influenzae, Mycoplasma pneumoniae, viruses. Presents with cough, fever, dyspnea, and consolidation on imaging. Severity assessment: CURB-65 or PSI. Treatment: empiric antibiotics based on setting and local resistance patterns.

Pneumothorax

– Air in the pleural space causing lung collapse. Primary spontaneous occurs in tall, thin young men. Secondary occurs with underlying lung disease. Tension is a medical emergency.

Positive End-Expiratory Pressure

– Pressure maintained at end-expiration to prevent alveolar collapse. Improves oxygenation by recruiting alveoli and increasing FRC. Balances oxygenation with hemodynamic effects.

Post-Infectious Cough

– A chronic cough persisting 3–8 weeks after a URI, caused by transient bronchial hyperresponsiveness and airway inflammation. It is self-limited and usually resolves without treatment. Inhaled corticosteroids, antitussives, and time are the mainstays of management.

Post-Obstructive Pneumonia

– Pneumonia occurring distal to a bronchial obstruction caused by tumor, foreign body, mucus plug, or stenosis. The obstruction prevents clearance of secretions, creating a favorable environment for infection. It does not resolve with antibiotics alone until the obstruction is relieved.

Post-Operative Atelectasis

– Collapse of alveoli after surgery, the most common post-operative pulmonary complication. Risk factors include general anesthesia, abdominal surgery, obesity, smoking, and pre-existing lung disease. Prevention includes incentive spirometry, early ambulation, and adequate pain control.

Post-Pneumonectomy Syndrome

– A rare complication following pneumonectomy where the remaining lung herniates across the mediastinum, causing compression of airways and vessels. It presents with dyspnea, stridor, and recurrent infections. Treatment is surgical correction with mediastinal repositioning and prosthetic implant placement.

Pott Disease

– TB of the spine involving thoracic or lumbar vertebrae. Causes vertebral destruction, kyphosis (gibbus deformity), and paravertebral abscess. May cause spinal cord compression.

Pressure Support Ventilation

– Spontaneous breathing mode with ventilator-provided pressure support above PEEP for each breath. Reduces work of breathing. Used for weaning from mechanical ventilation.

Primary Ciliary Dyskinesia

– An autosomal recessive disorder of ciliary structure, most commonly caused by dynein arm defects. Leads to impaired mucociliary clearance, chronic sinopulmonary infections, bronchiectasis, and situs inversus (50% of cases).

Primary Tuberculosis

– Initial M. tuberculosis infection causing Ghon focus in lower lobes with hilar lymph node involvement (Ghon complex). Most infections are contained, resulting in latent TB.

Prone Positioning

– Turning supine patients to prone for > 12 hours daily in severe ARDS. Improves oxygenation by recruiting dorsal lung segments. Reduces mortality when PaO2/FiO2 < 150.

Prostacyclin Analogs

– Medications that mimic prostacyclin (PGI2), causing vasodilation and inhibiting platelet aggregation in the pulmonary vasculature. They include epoprostenol (continuous IV), treprostinil (subcutaneous, IV, inhaled, oral), and iloprost (inhaled). They are used for severe PAH.

Pulmonary Alveolar Hemorrhage

– Bleeding into the alveolar spaces from disruption of the alveolar-capillary membrane. Causes include vasculitis (GPA, MPA, Goodpasture), connective tissue disease, medications, and coagulopathy. It presents with hemoptysis, falling hemoglobin, diffuse bilateral infiltrates, and serially bloody returns on BAL.

Pulmonary Alveolar Microlithiasis

– A rare genetic disease caused by SLC34A2 mutations, leading to accumulation of calcium phosphate microliths within the alveoli. HRCT shows diffuse bilateral calcified nodules with a "sandstorm" appearance. It is usually diagnosed in the third to fourth decade. Treatment is supportive; lung transplantation may be needed for progressive disease.

Pulmonary Alveolar Proteinosis

– A rare disease characterized by accumulation of surfactant-like material within the alveoli due to impaired clearance by alveolar macrophages. Most cases are autoimmune (anti-GM-CSF antibodies). Presents with progressive dyspnea and cough. HRCT shows bilateral ground-glass opacities with "crazy paving" pattern. Treatment is whole lung lavage.

Pulmonary Arteriovenous Malformation

– An abnormal direct communication between a pulmonary artery and vein, causing a right-to-left shunt. It may be idiopathic or associated with hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome). It can cause hypoxemia, paradoxical emboli (stroke, brain abscess), and hemoptysis. Treatment is embolization or surgical resection.

Pulmonary Artery Catheter

– A catheter inserted through the right heart into the pulmonary artery to measure hemodynamic parameters (PA pressure, PAWP, cardiac output). Used for diagnosis and management of pulmonary hypertension, heart failure, and critical illness.

Pulmonary Contusion

– Bruising of the lung parenchyma from blunt chest trauma, causing hemorrhage and edema without lung laceration. It impairs gas exchange and can progress to ARDS. It is often associated with rib fractures and flail chest. Treatment is supportive with lung-protective ventilation.

Pulmonary Edema

– Accumulation of fluid in the pulmonary interstitium and alveoli, most commonly due to left-sided heart failure (cardiogenic). Causes increased hydrostatic pressure leads to transudation. Presents with acute dyspnea, orthopnea, crackles, and S3 gallop. Non-cardiogenic causes include ARDS, renal failure, and high altitude. Treatment: diuretics, afterload reduction, oxygen, and positive pressure ventilation if severe.

Pulmonary Embolism

– The obstruction of one or more pulmonary arteries by material lodged in the bloodstream, most commonly a thrombus originating from deep vein thrombosis (DVT) in the lower extremities (90-95 percent of cases). The Virchow triad (venous stasis, hypercoagulability, endothelial injury) underlies DVT/PE pathogenesis. Major risk factors: recent surgery or trauma (especially orthopedic), prolonged immobilization, active cancer, pregnancy and postpartum state, estrogen-containing oral contraceptives and hormone replacement therapy, and inherited thrombophilias.

Pulmonary Fibrosis

– Scarring replacing normal alveolar architecture. Causes: IPF, connective tissue disease, asbestos, radiation, drugs (bleomycin, methotrexate). Features: progressive dyspnea, dry cough, digital clubbing.

Pulmonary Hemosiderosis

– Iron deposition in the lungs from repeated alveolar hemorrhage. It presents with hemoptysis, anemia, and diffuse bilateral infiltrates on imaging. Causes include Goodpasture syndrome, granulomatosis with polyangiitis, and idiopathic pulmonary hemosiderosis. The classic triad is hemoptysis, anemia, and diffuse pulmonary infiltrates.

Pulmonary Infarction

– Lung necrosis from PE when bronchial supply is compromised. Causes pleuritic pain, hemoptysis, and wedge-shaped opacity. Most PE do not cause infarction due to dual blood supply.

Pulmonary Rehabilitation

– Comprehensive intervention including exercise, education, and behavioral modification for chronic respiratory diseases. Improves exercise capacity, reduces dyspnea, and reduces hospitalizations. Recommended for symptomatic COPD.

Pulmonary Sequestration

– A congenital malformation of non-functioning lung tissue that lacks normal communication with the tracheobronchial tree and receives its blood supply from a systemic artery (usually the descending aorta). Intralobar sequestration is within the normal pleural investment; extralobar is outside. It may present with recurrent infections or be incidental.

Pulmonary Vasodilator Therapy

– Medications that reduce pulmonary vascular resistance and pulmonary artery pressure. Used for WHO Group 1 PAH. Includes endothelin receptor antagonists (bosentan), PDE-5 inhibitors (sildenafil), prostacyclin analogs (epoprostenol), and soluble guanylate cyclase stimulators (riociguat).

Pulmonary Wedge Pressure

– The pressure measured by a balloon-tipped pulmonary artery catheter when the balloon is inflated, occluding a branch of the pulmonary artery. It approximates left atrial pressure. Normal is 6–12 mmHg. Elevated in left heart failure (PCWP $>$ 18 mmHg). Low in hypovolemia.

Pyrazinamide

– First-line TB drug effective against intracellular mycobacteria. Important for sterilizing action. Side effects: hepatotoxicity, hyperuricemia (gout).

Reactivation Tuberculosis

– Reactivation of latent infection, typically years later. Characteristically affects apical and posterior upper lobe segments. Features include chronic cough, hemoptysis, night sweats, weight loss, and cavitation.

Rehabilitation

– See Rehabilitation in General.

Relapsing Polychondritis

– A systemic autoimmune disorder causing inflammation and destruction of cartilage throughout the body. Airway involvement can cause tracheal and bronchial stenosis. Other features include auricular chondritis, nasal chondritis, ocular inflammation, and arthritis. Treatment is corticosteroids and immunosuppressants.

Resorption Atelectasis

– Atelectasis caused by complete obstruction of an airway, leading to absorption of trapped air by the pulmonary capillaries. Common causes include mucus plugging, foreign body, tumor, and post-surgical shallow breathing. The affected area appears as increased density on CXR with volume loss.

Respiratory Failure

– Inadequate gas exchange resulting in hypoxemia (PaO_2 <60 mmHg), hypercapnia ($PaCO_2$ >50 mmHg), or both. Type 1 (hypoxemic): V/Q mismatch, shunt, diffusion impairment. Type 2 (hypercapnic): hypoventilation from neuromuscular disease, COPD exacerbation, or opioid overdose. Management: oxygen therapy, non-invasive ventilation, mechanical ventilation, and treatment of the underlying cause.

Respiratory Health

– The proper functioning of the lungs and airways, enabling efficient gas exchange and oxygen delivery to the body. Maintaining respiratory health involves avoiding tobacco smoke, minimizing exposure to air pollutants and occupational hazards, and receiving vaccinations against respiratory infections like influenza and pneumococcal disease. Regular exercise strengthens respiratory muscles and improves lung capacity, while prompt treatment of infections and management of chronic conditions such as asthma or COPD help preserve lung function over a lifetime.

Respiratory Infections

– Illnesses caused by pathogens including viruses, bacteria, and fungi that affect the airways and lungs. Common upper respiratory infections include the common cold, sinusitis, and pharyngitis, while lower respiratory infections such as bronchitis and pneumonia can be more serious, particularly in older adults and immunocompromised individuals. Treatment varies by cause, with viral infections often managed symptomatically and bacterial infections requiring antibiotics; prevention through hand hygiene and vaccination is key.

Respiratory System

– The anatomical and physiological system responsible for gas exchange—delivering oxygen to the blood and removing carbon dioxide. Components: (1) Upper respiratory tract—nose, nasal cavity, paranasal sinuses, pharynx (nasopharynx, oropharynx, laryngopharynx). The nose warms, humidifies, and filters inspired air; the nasal mucosa contains olfactory receptors. (2) Lower respiratory tract—larynx (voice production, airway protection—epiglottis, vocal cords), trachea (C-shaped cartilage rings, 10–12 cm long), bronchi (right main bronchus—wider, shorter, more vertical—common site of aspiration), bronchioles (no cartilage, smooth muscle—asthma), alveolar ducts, and alveoli (300–500 million, total surface area 70–100 m^2), where gas exchange occurs across the thin alveolar-capillary membrane (0.2–0.5 μm). (3) Lungs—right lung (three lobes: upper, middle, lower) and left lung (two lobes: upper, lower—the cardiac notch accommodates the heart). The pleura (visceral and parietal) lines the lungs and chest wall; the pleural cavity contains a thin layer of surfactant-rich fluid for lubrication. (4) Chest wall and muscles of respiration—diaphragm (primary inspiratory muscle, innervated by phrenic nerve C3–C5), intercostal muscles (external—inspiration; internal—expiration), and accessory muscles (sternocleidomastoid, scalenes—used in respiratory distress). Ventilation: the rhythmic contraction of the diaphragm and intercostal muscles creates negative intrapleural pressure, drawing air into the lungs. Control: the respiratory center in the medulla oblongata generates the breathing rhythm, modulated by chemoreceptors (central—CO_2/pH; peripheral—O_2, CO_2, pH in carotid and aortic bodies).

Respiratory Tract Infections

– Infections affecting any part of the respiratory tract, classified by anatomical location into upper respiratory tract infections (URTIs) and lower respiratory tract infections (LRTIs). URTIs: common cold (rhinovirus, coronavirus, adenovirus), pharyngitis (Group A Streptococcus—bacterial, adenovirus, EBV), tonsillitis (Streptococcus pyogenes), sinusitis (Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis, viruses), otitis media (S. pneumoniae, H. influenzae), laryngitis (viruses, vocal strain). LRTIs: acute bronchitis (viruses—influenza, RSV; Bordetella pertussis), bronchiolitis (RSV—infants), pneumonia (community-acquired—S. pneumoniae, H. influenzae, Mycoplasma pneumoniae, Legionella, Chlamydia psittaci, viruses; hospital-acquired—Pseudomonas aeruginosa, MRSA, Enterobacteriaceae; aspiration—anaerobes). Risk factors: extremes of age, smoking, COPD, asthma, immunocompromise, cystic fibrosis, structural lung disease, and institutionalisation. Antibiotic resistance is a growing concern (MRSA, ESBL-producing Enterobacteriaceae, multidrug-resistant S. pneumoniae). Diagnosis: clinical (cough, sputum, fever, dyspnoea, chest pain), chest X-ray (consolidation in pneumonia), sputum culture, blood culture, nasopharyngeal swab (viral PCR), and urinary antigen testing (S. pneumoniae, Legionella). Prevention: hand hygiene, vaccination (influenza, pneumococcal, pertussis, RSV—maternal, palivizumab for high-risk infants), and smoking cessation.

Rhonchi

– Low-pitched, continuous adventitious sounds heard primarily during expiration, caused by secretions in the large airways. They may clear with coughing. They are heard in COPD, bronchitis, and bronchiectasis. They are different from wheezes in their lower pitch and coarser quality.

RIPE Therapy

– Standard 4-drug regimen for active TB: Rifampin, Isoniazid, Pyrazinamide, Ethambutol. Intensive phase 2 months, continuation phase 4 months (total 6 months). Pyridoxine prevents isoniazid-induced neuropathy.

Round Pneumonia

– Pneumonia appearing as a round opacity on CXR rather than typical consolidation. More common in children and elderly. May mimic lung mass.

Siderosis

– Benign pneumoconiosis from iron dust, seen in welders. Causes small opacities on CXR but minimal impairment. Reversible with removal of exposure.

Situs Inversus

– A congenital condition in which the thoracic and abdominal organs are reversed or mirrored from their normal positions. Complete situs inversus (situs inversus totalis) involves all organs. It is usually asymptomatic but may be associated with primary ciliary dyskinesia (Kartagener syndrome), congenital heart disease, and polysplenia or asplenia.

Sleep Apnea (Obstructive Sleep Apnea - OSA)

– A sleep disorder characterized by repeated episodes of partial or complete upper airway obstruction during sleep (obstructive sleep apnea—OSA, most common) or cessation of respiratory drive (central sleep apnea—CSA), causing intermittent hypoxia, hypercapnia, and sleep fragmentation. OSA affects 13% of men and 6% of women (moderate–severe). Risk factors: obesity (BMI ≥30—strongest risk factor, 60–90% of OSA patients), male sex, age >40, menopause, craniofacial abnormalities (retrognathia, macroglossia, tonsillar hypertrophy), smoking, alcohol, and hypothyroidism. Pathophysiology: loss of pharyngeal muscle tone during sleep leads to collapse of the pharyngeal airway (velopharynx, oropharynx, hypopharynx). Clinical: loud snoring, witnessed apnoeas (by bed partner), choking/gasping arousals, unrefreshing sleep, excessive daytime sleepiness (Epworth Sleepiness Scale ≥10), morning headache, nocturia, and cognitive impairment. Consequences: hypertension (dose-dependent, 40% of OSA patients have hypertension), cardiovascular disease (MI, stroke, atrial fibrillation, heart failure), type 2 diabetes, and increased all-cause mortality. Diagnosis: polysomnography (apnea-hypopnoea index—AHI: mild 5–15, moderate 15–30, severe >30 events/hour). Home sleep apnea testing (HSAT) is an alternative for high-risk patients. Treatment: continuous positive airway pressure (CPAP—first-line, splints the airway open, AHI reduction >90%), mandibular advancement devices (MAD—for mild–moderate OSA, <25% of CPAP efficacy), weight loss (5–10% weight loss reduces AHI by 30–50%), positional therapy (for supine-dependent OSA), and surgery (uvulopalatopharyngoplasty, maxillomandibular advancement, hypoglossal nerve stimulation/Inspire—for moderate–severe OSA who fail CPAP). Untreated severe OSA: 5-year mortality 10–15%.

Small Airway Disease

– Disease affecting airways with diameter < 2 mm (bronchioles). It is the earliest site of damage in COPD and is often subclinical until advanced. It is characterized by narrowing, mucus plugging, and inflammation of small airways. PFTs may show air trapping (increased RV/TLC) before FEV1 decline.

Snoring and Sleep Apnea

– Snoring is the sound produced by vibrating soft tissues in the upper airway during sleep, often resulting from partial airway obstruction. Sleep apnea is a more serious condition characterized by repeated episodes of complete or partial airway collapse during sleep, leading to intermittent hypoxia, frequent arousal, and fragmented sleep. Obstructive sleep apnea, the most common form, is associated with obesity, hypertension, and cardiovascular disease; treatment includes continuous positive airway pressure therapy, oral appliances, weight loss, and positional therapy.

Solitary Pulmonary Nodule

– Single, well-circumscribed lesion < 3 cm surrounded by normal lung. Differential includes granuloma, hamartoma, and carcinoma. PET-CT and tissue sampling help determine etiology.

Spirometry

– Most common PFT measuring air volume and flow during forced breathing. Key measurements: FVC, FEV1, FEV1/FVC ratio. Used to diagnose and classify obstructive and restrictive diseases.

Spontaneous Breathing Trial

– A period of unsupported breathing to assess a patient's readiness for extubation. The patient breathes through a T-piece or with minimal pressure support for 30–120 minutes. Successful SBT predicts successful extubation.

Spontaneous Pneumothorax

– Pneumothorax without trauma. Primary in healthy individuals; secondary with underlying lung disease. Treatment depends on size.

Staphylococcal Pneumonia

– Severe pneumonia from S. aureus, often post-influenza. May cause necrotizing pneumonia, lung abscess, and empyema.

Status Asthmaticus

– A severe, life-threatening asthma exacerbation unresponsive to standard bronchodilator therapy. Characterized by severe airflow obstruction, hypoxemia, and hypercapnia. Treatment includes continuous nebulized albuterol, IV magnesium sulfate, systemic corticosteroids, and possibly intubation.

Stridor

– A high-pitched, monophonic, adventitious sound heard primarily during inspiration, caused by narrowing of the extrathoracic or proximal intrathoracic airway. It is a medical emergency when severe. Causes include foreign body aspiration, croup, epiglottitis, and tracheal stenosis.

Subcutaneous Emphysema

– Air in the subcutaneous tissues, often from pneumothorax, pneumomediastinum, or chest trauma. It causes a characteristic crepitus on palpation. It is usually benign and resolves when the underlying air leak is treated. Severe cases may cause skin necrosis and compartment syndrome.

Supplemental Oxygen Therapy

– Administration of oxygen above atmospheric concentration. Indicated for hypoxemia (PaO2 $<$ 60 mmHg or SpO2 $<$ 90%). Delivered via nasal cannula (1–6 L/min), simple mask, non-rebreather mask, or mechanical ventilation. Target SpO2 88–92% in COPD patients to avoid CO2 retention.

SVC Syndrome

– Obstruction of the superior vena cava causing facial swelling, plethora, and distended neck veins. Commonly from lung cancer (SCLC, squamous) and lymphoma. Oncologic emergency.

Tachypnoea

– Abnormally rapid breathing, defined as a respiratory rate >20 breaths per minute in adults (age-dependent: >60 in neonates, >40 in 1–12 months, >30 in 1–5 years, >20 in >5 years). Tachypnoea is a non-specific sign of respiratory or metabolic distress. Causes: (1) Respiratory—hypoxia (pneumonia, asthma/COPD exacerbation, pulmonary embolism, pneumothorax, pulmonary edema, ARDS, interstitial lung disease), airway obstruction, and hypercapnia (COPD, respiratory muscle weakness). (2) Cardiac—heart failure (pulmonary congestion stimulates vagal irritant receptors). (3) Metabolic—metabolic acidosis (diabetic ketoacidosis—Kussmaul breathing, uraemia, lactic acidosis, salicylate poisoning), fever (respiratory rate increases 2–4 breaths/min per 1°C rise). (4) Other—anxiety (panic attack—psychogenic hyperventilation), pain, anemia, hyperthyroidism, pregnancy (progesterone stimulates respiratory center), exercise, and high altitude. (5) Drugs: salicylates, theophylline, β-agonists, and other stimulants. Tachypnoea is often accompanied by shallow breathing (low tidal volume). Assessment: respiratory rate, SpO_2, ABG, chest X-ray, ECG, and focused history. Persistent tachypnoea with hypoxia is a sign of serious illness requiring urgent investigation and treatment.

Tactile Fremitus

– Vibrations transmitted through the lung to the chest wall, felt during palpation. It is increased in consolidation (pneumonia) due to better sound transmission through solid tissue. It is decreased in pleural effusion, pneumothorax, and atelectasis due to impaired transmission.

Talcosis

– Pneumoconiosis from talc dust inhalation or IV injection of crushed oral medications. Causes upper lobe nodular fibrosis. HRCT may show “galaxy sign” of calcified nodules.

Tension Pneumothorax

– Air trapping via a one-way valve causing mediastinal shift, impaired venous return, and cardiovascular collapse. Requires immediate needle decompression (2nd ICS, midclavicular line) followed by chest tube.

Theophylline

– A methylxanthine bronchodilator that inhibits phosphodiesterase and adenosine receptors. It has a narrow therapeutic window (10–20 mcg/mL) and requires therapeutic drug monitoring. Side effects include tachyarrhythmias, seizures, nausea, and diuresis. It is used as add-on therapy for severe asthma or COPD when other treatments are insufficient.

Thermoplasty

– A bronchoscopic procedure that uses radiofrequency energy to ablate smooth muscle in the airways, reducing bronchospasm. It is approved for severe, persistent asthma in adults not controlled with ICS/LABA. It reduces exacerbations and emergency department visits but does not improve lung function.

Thoracentesis

– Removal of pleural fluid via needle insertion through the chest wall. Performed for diagnosis and symptom relief. Ultrasound guidance improves safety.

Thrombolysis for PE

– Systemic thrombolytics (alteplase) for massive PE with hemodynamic instability. Rapid clot dissolution but significant bleeding risk.

Tiotropium Bromide

– A long-acting muscarinic antagonist (LAMA) bronchodilator used for maintenance therapy in COPD. Once-daily dosing provides sustained bronchodilation. It reduces exacerbations and improves lung function. It is often combined with a LABA for additive bronchodilation.

Total Lung Capacity

– Total lung air volume after maximal inspiration. Measured by body plethysmography. Increased in obstructive disease, decreased in restrictive disease. TLC $< 80\%$ confirms restriction.

Tracheal Foreign Body

– A foreign object lodged in the trachea, causing severe respiratory distress, stridor, and complete airway obstruction. It is a medical emergency requiring immediate bronchoscopic removal. In complete obstruction, emergency cricothyrotomy may be needed.

Tracheal Stenosis

– Narrowing of the trachea, often a complication of prolonged endotracheal intubation or tracheostomy. It presents with progressive dyspnea, stridor, and wheezing that may be mistaken for asthma. Diagnosis is by bronchoscopy or CT. Treatment is balloon dilation, stenting, or surgical resection.

Tracheitis

– Inflammation of the tracheal mucosa. Acute bacterial tracheitis: a rare but life-threatening cause of upper airway obstruction in children, typically presenting as a superinfection following viral croup (parainfluenza). Caused by Staphylococcus aureus (most common), Streptococcus pneumoniae, Haemophilus influenzae type b (now rare due to Hib vaccine), and Moraxella catarrhalis. Presents with high fever, stridor (biphasic, worsening), barky cough, hoarseness, respiratory distress, toxic appearance, and copious purulent tracheal secretions. Diagnosis: lateral neck X-ray—tracheal narrowing ('steeple sign' similar to croup, but with irregular tracheal margins and pseudomembrane), bronchoscopy (definitive—shows inflammation, ulceration, and thick purulent secretions/ pseudomembranes). Treatment: hospitalisation, IV antibiotics (anti-staphylococcal—flucloxacillin or cefuroxime, plus metronidazole for anaerobes), humidified oxygen, and intubation if severe respiratory distress (airway protection, suction of thick secretions). Endotracheal intubation may be required for 5–7 days. Complications: airway obstruction, pneumonia, septic shock, and post-intubation stenosis.

Tracheobronchomalacia

– Abnormal softening of the trachea and/or bronchi causing excessive airway collapse during breathing. It may be congenital or acquired. Symptoms include chronic cough, recurrent infections, and expiratory airflow obstruction. Diagnosis is by dynamic CT or bronchoscopy. Treatment includes CPAP, stenting, or surgical reconstruction.

Tracheomalacia

– Softening of the tracheal cartilage causing excessive collapse of the trachea during expiration. It may be congenital or acquired (prolonged intubation, relapsing polychondritis). Symptoms include expiratory stridor and difficulty clearing secretions. Diagnosis is by dynamic CT or bronchoscopy. Treatment may require tracheal stenting or surgery.

Tracheostomy

– Surgical creation of an airway opening in the anterior neck into the trachea. Indications include prolonged ventilation (> 10–14 days), upper airway obstruction, and airway protection.

Transbronchial Biopsy

– Bronchoscopic biopsy of lung parenchyma obtained through the bronchial wall. It is used to diagnose interstitial lung diseases (sarcoidosis, HP), reject lung transplant rejection, and sample peripheral lung lesions. It carries a higher risk of pneumothorax than endobronchial biopsy.

Transudative Pleural Effusion

– From systemic factors altering hydrostatic/oncotic pressures. Most commonly from heart failure. Low protein, low LDH. Treatment addresses the underlying cause.

Trepopnea

– Dyspnea that occurs in one lateral decubitus position but not the other. It is typically worse when lying on the side of the affected lung. It may be caused by unilateral lung disease, pleural effusion, or mediastinal mass.

Upper Airway Obstruction

– Narrowing of the airway above the carina, causing inspiratory stridor, dyspnea, and characteristic changes on flow-volume loops (flattening of the inspiratory limb). Causes include tumors, stenosis, goiter, and vocal cord dysfunction. It is a medical emergency when severe.

Usual Interstitial Pneumonia

– The histopathological pattern in IPF with temporal heterogeneity, fibroblast foci, and honeycombing. HRCT shows basal-predominant, subpleural reticular opacities with traction bronchiectasis.

Ventilation-Perfusion Mismatch

– The most common cause of hypoxemia. Areas of lung are ventilated but not perfused (dead space) or perfused but not ventilated (shunt). It is the mechanism of hypoxemia in COPD, pneumonia, and PE.

Ventilator-Associated Pneumonia

– Pneumonia $>$ 48 hours after intubation. Most common ICU-acquired infection. Diagnosis uses quantitative BAL or protected specimen brush cultures.

Vocal Cord Dysfunction

– Abnormal adduction of the vocal cords during inspiration, causing inspiratory stridor and dyspnea. It may mimic asthma but does not respond to bronchodilators. It is often misdiagnosed as refractory asthma. Diagnosis is by laryngoscopy showing paradoxical vocal cord motion. Treatment is speech therapy and avoiding triggers.

Vocal Resonance

– The transmission of vocal sounds to the chest wall, assessed by having the patient speak while auscultating. It is increased in consolidation and decreased in pleural effusion or pneumothorax. It correlates with bronchophony and whispered pectoriloquy.

Volutrauma

– Lung injury from excessive tidal volume causing alveolar overdistension. Prevented by low tidal volume ventilation (6 mL/kg IBW). Key component of ARDS management.

Wheezes

– High-pitched, continuous, musical sounds heard primarily during expiration, caused by airway narrowing. They are a hallmark of asthma and COPD. Expiratory wheezing indicates intrathoracic airway obstruction; inspiratory wheezing indicates extrathoracic obstruction. A silent chest (absent wheezing) in severe asthma indicates critical obstruction.

Whispered Pectoriloquy

– The ability to hear whispered words clearly through the stethoscope over consolidated lung tissue. It is a sign of pulmonary consolidation. Normally, whispered words are barely audible over the lung fields. It is one of the classic signs of pneumonia on physical examination.

Windpipe

– See Trachea.

Chapter 32

Radiology

Abdominal Radiography

– Abdominal radiography (KUB - kidneys, ureters, bladder) uses supine and upright or decubitus projections to evaluate the gastrointestinal tract, urinary system, and abdominal calcifications. The supine view shows bowel gas pattern, soft tissue outlines, and calcifications. The upright or left lateral decubitus view detects free intraperitoneal air (pneumoperitoneum) under the right hemidiaphragm as a crescentic lucency. Key findings include small bowel obstruction (dilated loops >3 cm, valvulae conniventes visible as circumferential folds, and multiple air-fluid levels on upright imaging), large bowel obstruction (dilated colon >6 cm, haustral folds, and cecal diameter >9 cm predicting perforation), pneumoperitoneum (free air under the diaphragm), and abdominal calcifications (pancreatic, renal, biliary, vascular). Limitations include poor sensitivity for solid organ injury, inability to detect free fluid in small volumes, and limited evaluation of the retroperitoneum.

Bone Densitometry

– Bone densitometry, most commonly performed by dual-energy X-ray absorptiometry (DXA), measures bone mineral density (BMD) to diagnose osteoporosis, assess fracture risk, and monitor treatment response. DXA uses two X-ray energies that are differentially absorbed by bone and soft tissue, enabling calculation of areal BMD (g/cm2) at the lumbar spine (L1-L4), proximal femur (femoral neck, trochanter, total hip), and sometimes the distal one-third radius. Results are reported as T-scores (comparison with young adult mean), Z-scores (comparison with age-matched mean), and the World Health Organization classification: normal (T-score -1.0 or above), osteopenia (T-score -1.0 to -2.5), and osteoporosis (T-score -2.5 or below). DXA is the gold standard for diagnosis of osteoporosis and is used for screening in postmenopausal women over 65, men over 70, and younger patients with risk factors. Limitations include inability to distinguish cortical from trabecular bone, artefact from degenerative changes and compression fractures, and measurements that areal rather than volumetric.

Bone Marrow Imaging

– Bone marrow imaging evaluates the hematopoietic compartments of the skeleton for marrow replacement, infiltration, aplasia, and reconversion using MRI as the primary modality. Normal bone marrow signal depends on the ratio of fat (yellow marrow, bright on T1) to hematopoietic cells (red marrow, intermediate on T1). With aging, conversion from red to yellow marrow progresses centrifugally from the distal extremities toward the axial skeleton. Pathologic marrow replacement (by tumor, fibrosis, or inflammation) results in low T1 signal and variable T2 signal depending on cellularity and water content. STIR (short tau inversion recovery) and fat-saturated T2 sequences suppress fat signal, making pathologic marrow edema more conspicuous. Indications for bone marrow imaging include evaluation of suspected marrow replacement (multiple myeloma, lymphoma, leukemia, metastases), assessment of avascular necrosis (early detection before radiographic changes, with the double-line sign on T2 being pathognomonic), inflammatory arthritis with bone marrow edema, and monitoring of treatment response in haematological malignancies. DWI and ADC mapping help differentiate benign from malignant marrow infiltration.

Bone Scan

– A nuclear medicine imaging study that evaluates the entire skeleton for abnormalities of bone metabolism, using a radiopharmaceutical tracer (technetium-99m methylene diphosphonate, Tc-99m MDP) that is incorporated into the hydroxyapatite crystal matrix of bone at sites of active osteogenesis. The tracer is injected intravenously, followed by imaging 2-4 hours later after skeletal uptake and renal clearance of unbound tracer. Uptake depends on blood flow and osteoblastic activity; increased uptake occurs in fractures, infection, tumors (both primary and metastatic, with osteosarcoma and blastic metastases showing intense uptake), Paget disease of bone (diffuse intense uptake in the affected bone), arthritis, and heterotopic ossification. Decreased uptake (cold defects) occurs in avascular necrosis, photopenic zones of multiple myeloma (lytic lesions without reactive bone formation), and early osteomyelitis before osteoblastic response develops. The bone scan is highly sensitive but not specific for bone pathology; correlation with radiography or cross-sectional imaging is essential for specificity. The three-phase bone scan (angiographic, blood pool, and delayed phases) adds specificity for infection and fracture non-union. SPECT-CT fusion improves anatomic localisation.

Breast MRI

– The most sensitive imaging modality for breast cancer detection, achieving sensitivity exceeding 95 percent, and is indicated for high-risk screening, staging of known breast cancer, and evaluation of treatment response. The examination uses dedicated breast coils with T1 fat-saturated pre- and post-gadolinium sequences and diffusion-weighted imaging. Kinetic analysis characterizes enhancement patterns: initial rapid enhancement with delayed washout (type 3 curve) is highly suspicious (BI-RADS 4C or 5). Indications include high-risk screening (BRCA mutation carriers, lifetime risk >20 percent, history of chest radiation), evaluation of occult primary breast cancer, assessment of extent of disease in newly diagnosed breast cancer (including multifocal, multicentric, and contralateral disease), monitoring of neoadjuvant chemotherapy response, and problem-solving when conventional imaging is inconclusive. Limitations include lower specificity (increased false positives), need for intravenous gadolinium contrast, high cost, and limited availability. Claustrophobia and MRI-incompatible implants are contraindications. BI-RADS MRI lexicon standardises reporting.

Breast Ultrasound

– A problem-solving tool used to evaluate palpable lumps, distinguish cysts from solid masses, guide interventional procedures, and supplement mammography in women with dense breast tissue. High-frequency linear array transducers (12-17 MHz) provide detailed evaluation of superficial breast structures. Simple cysts appear as anechoic, well-circumscribed masses with posterior acoustic enhancement and thin imperceptible walls, requiring no further evaluation (BI-RADS 2). Complicated cysts (indistinct walls, internal echoes, thick walls or septations, solid components) require further evaluation with aspiration, biopsy, or short-interval follow-up depending on the degree of suspicion. Solid masses are characterized by shape, margin, orientation (parallel to skin is usually benign, non-parallel is suspicious), echo pattern, and posterior acoustic features. Suspicious features include spiculated margins, angular margins, marked hypoechogenicity, and calcifications. BI-RADS ultrasound lexicon standardises reporting. color Doppler assesses vascularity, and elastography measures tissue stiffness, with malignant lesions typically being harder than benign ones.

Cardiac CT Angiography

– Cardiac CT angiography (CCTA) uses ECG-gated multidetector CT with rapid intravenous contrast administration to evaluate coronary artery anatomy and detect coronary artery disease. Beta-blockers are administered to reduce heart rate below 65 beats per minute for optimal image quality, and sublingual nitroglycerin dilates coronary arteries to improve visualization. Prospective ECG-triggering (acquiring data during diastole only) reduces radiation dose to 1-3 mSv compared to retrospective ECG-gating (acquiring data throughout the cardiac cycle with higher radiation dose of 10-15 mSv). Coronary artery calcium scoring is performed on non-contrast CT, with Agatston score >400 indicating severe calcified plaque burden and high cardiovascular risk. Coronary artery stenosis is graded as none, mild (<50 percent), moderate (50-69 percent), or severe (70 percent or greater). CCTA has a high negative predictive value for excluding obstructive coronary disease (>99 percent) and is indicated for symptomatic patients with low-to-intermediate pre-test probability. Limitations include artefact from high heart rate, arrhythmia, dense coronary calcification limiting stenosis evaluation, and inability to assess hemodynamic significance.

Cardiac MRI

– The reference standard for quantifying ventricular volumes, mass, and function, using ECG-gated steady-state free precession (SSFP) cine sequences that provide excellent myocardial-blood pool contrast. Standard protocols include cine imaging in short-axis (stack from base to apex), two-chamber, and four-chamber planes for volumetric analysis, T1 and T2 mapping for diffuse fibrosis and edema assessment, late gadolinium enhancement (LGE) for focal fibrosis and scar, and perfusion for myocardial ischemia evaluation. LGE imaging identifies irreversible myocardial injury: ischemic scar appears as subendocardial or transmural hyperenhancement following a coronary artery territory, while non-ischemic patterns include mid-wall (hypertrophic cardiomyopathy), epicardial (myocarditis, sarcoidosis), and global subendocardial (amyloidosis) enhancement. T1 mapping detects diffuse myocardial fibrosis (elevated extracellular volume fraction), and T2 mapping detects myocardial edema (acute myocarditis, stress cardiomyopathy, acute myocardial infarction). CMR is the non-invasive reference standard for assessing biventricular volumes, ejection fraction, and myocardial mass. Indications include assessment of cardiomyopathy (ischemic, dilated, hypertrophic, restrictive, arrhythmogenic right ventricular cardiomyopathy), myocarditis, cardiac masses, pericardial disease, and congenital heart disease. Contraindications include MRI-incompatible devices and severe claustrophobia. Gadolinium-based contrast is contraindicated in advanced renal disease (risk of nephrogenic systemic fibrosis).

Cerebral Angiography

– Digital subtraction cerebral angiography (DSA) remains the gold standard for evaluating intracranial vascular anatomy with superior spatial and temporal resolution compared to CTA or MRA. Femoral or radial artery access is used to catheterize the internal carotid, vertebral, or external carotid arteries, with selective contrast injections and multiple projections demonstrating the circle of Willis, cerebral branches, and venous drainage. Applications include aneurysm evaluation (morphology, neck size, and relationship to parent vessel), arteriovenous malformation evaluation (nidus size, arterial supply, and venous drainage), vasculitis workup, and preoperative tumor embolisation planning. Advances in 3D rotational angiography and cone-beam CT with flat-panel detectors improve the characterisation of complex vascular anatomy. Risks include stroke (1-2 percent from embolic or thrombotic complications at the catheter tip), access site complications (haematoma, pseudoaneurysm, arteriovenous fistula), contrast-induced nephropathy, and radiation exposure to the patient and operator.

Chest Radiography

– The most commonly performed imaging examination, providing a frontal and lateral projection of the thoracic structures including lungs, heart, mediastinum, pleura, chest wall, and upper abdomen. The posteroanterior (PA) view minimizes cardiac magnification by placing the heart closer to the detector, while the anteroposterior (AP) portable view is used for critically ill patients. Lateral view localizes lesions in the anterior/posterior dimension and visualizes the retrosternal space, trachea, and thoracic spine. Key findings include pneumonia (airspace opacities, air bronchograms, silhouettes signs), pulmonary edema (bat-wing perihilar distribution, Kerley B lines, pleural effusions), pneumothorax (visceral pleural line, deep sulcus sign in supine patients), pleural effusion (meniscus sign, blunting of costophrenic angle), cardiomegaly (cardiothoracic ratio >0.5), and mediastinal widening (aortic dissection, mediastinal mass, lymphadenopathy). The lateral view is essential for evaluating the retrosternal airspace, retrocardiac lung, and thoracic spine.

Computed Tomography

– A cross-sectional imaging technique that uses a rotating X-ray source and detector array to acquire projection data from multiple angles, which is then reconstructed into axial, sagittal, and coronal images using filtered back projection or iterative reconstruction algorithms. Modern multidetector CT scanners employ 64 to 320 detector rows, enabling rapid volumetric acquisition with sub-millimeter spatial resolution and isotropic voxel dimensions. CT imaging relies on differential X-ray attenuation by different tissues (bone, soft tissue, fat, air, and contrast material) to generate contrast in the image. The CT number (Hounsfield unit, HU) quantifies tissue attenuation: air -1000 HU, fat -100 to -50 HU, water 0 HU, soft tissue 40-80 HU, acute hemorrhage 60-80 HU, unenhanced blood 50-60 HU, and cortical bone >1000 HU. Intravenous iodinated contrast is used to enhance vascular structures, assess perfusion characteristics of lesions, and detect areas of abnormal enhancement. Oral contrast is used for gastrointestinal imaging, and rectal contrast for colonic studies. CT is the modality of choice for acute trauma, acute abdomen, pulmonary embolism (CTPA), renal colic (CT KUB), and cancer staging. Radiation dose is a concern, particularly in younger patients and those requiring repeated examinations. Dose reduction strategies include iterative reconstruction, tube current modulation, and automatic kV selection. The effective radiation dose for a routine CT chest is approximately 5-7 mSv, and for a CT abdomen/pelvis approximately 8-15 mSv.

Contrast-Enhanced Mammography

– Contrast-enhanced mammography (CEM) combines the anatomic detail of digital mammography with functional information from iodinated contrast administration, detecting tumor neovascularization that increases contrast uptake in malignant lesions. The technique acquires dual-energy images (low energy at 29-33 kVp and high energy at 49-52 kVp) approximately 2 minutes after intravenous iodinated contrast injection. Logarithmic subtraction of the high-energy from low-energy images produces contrast-enhancement map showing areas of abnormal uptake. The main indication is evaluation of breast abnormalities in women with suspected or known breast cancer when MRI is contraindicated or unavailable. CEM has higher sensitivity than digital mammography alone but lower specificity, with false positives from benign enhancing lesions (fibroadenomas, papillomas, and inflammatory changes). BI-RADS contrast-enhanced mammography lexicon standardises reporting. Limitations include need for intravenous access, iodinated contrast (contraindicated in renal impairment and contrast allergy), and radiation exposure (higher than digital mammography but less than mammography plus MRI).

Contrast-Enhanced Ultrasound

– Contrast-enhanced ultrasound (CEUS) uses intravenous microbubble contrast agents (perfluorocarbon or sulfur hexafluoride gas encapsulated in phospholipid shells) that remain entirely intravascular, enabling real-time assessment of tissue perfusion without nephrotoxicity or radiation. Microbubbles oscillate in an ultrasound field, producing harmonic signals that are detected using low mechanical index (MI less than 0.2) imaging to avoid bubble destruction. Applications include focal liver lesion characterisation (e.g., haemangioma showing peripheral nodular enhancement with progressive fill-in, focal nodular hyperplasia showing central scar enhancement, and metastases showing hypoenhancement with rim enhancement), renal mass characterisation, and non-cardiac chest pain evaluation. Limitations include operator dependence, limited penetration in obese patients, inability to evaluate gas-filled structures, and need for intravenous access and contrast injection. Microbubble contrast agents are contraindicated in patients with right-to-left shunts and known hypersensitivity.

Contrast Reactions

– Contrast reactions to iodinated and gadolinium-based contrast agents range from mild self-limited symptoms to severe life-threatening anaphylactoid reactions. Iodinated contrast reactions are classified as mild (flushing, pruritus, urticaria - managed with observation and diphenhydramine), moderate (diffuse urticaria, bronchospasm, laryngeal edema, facial edema - managed with IV diphenhydramine, albuterol nebulizer, and IV fluid bolus), and severe (hypotension, cardiac arrhythmias, pulmonary edema, and anaphylaxis - managed with epinephrine, IV fluids, and advanced life support). Risk factors for contrast reactions include prior contrast reaction, asthma, and atopy. Premedication with corticosteroids and antihistamines is recommended for at-risk patients receiving iodinated contrast. Contrast-induced nephropathy (CIN) is a risk with iodinated contrast in patients with pre-existing renal impairment, with prevention including IV hydration, minimising contrast volume, and using iso-osmolar or low-osmolar contrast agents. Gadolinium-based contrast agents carry a risk of nephrogenic systemic fibrosis in patients with advanced renal disease.

CT Angiography

– CT angiography (CTA) uses multidetector CT with intravenous iodinated contrast to visualize arterial and venous anatomy with high spatial and temporal resolution. Bolus tracking or test bolus techniques time contrast arrival to the vessel of interest. Peripheral CTA evaluates lower extremity aorto-occlusive disease with 3D reconstructions showing stenosis severity, calcification burden, and runoff vessel patency for bypass planning. Aortic CTA detects aneurysms (ascending greater than 5.5 cm, de... descending thoracic greater than 5.5 cm, abdominal greater than 5.0 cm), dissection (intimal flap, true and false lumens), and traumatic aortic injury (pseudoaneurysm, intimal flap at the aortic isthmus). Mesenteric CTA evaluates superior mesenteric artery occlusion or stenosis in acute mesenteric ischemia. Renal CTA evaluates renal artery stenosis (fibromuscular dysplasia, atherosclerosis), and prerenal transplant donor evaluation. Cardiac CTA is discussed separately. Limitations include need for iodinated contrast, radiation exposure, and artefact from calcified plaque limiting stenosis evaluation.

CT Colonography

– A minimally invasive colorectal imaging technique that uses multidetector CT with CO2 insufflation to detect colorectal polyps and cancer, serving as an alternative for patients who cannot undergo or refuse optical colonoscopy. Patient preparation includes cathartic bowel cleansing similar to conventional colonoscopy, along with fecal tagging with oral contrast (barium or iodinated) administered with the prep to label residual stool and fluid. Supine and prone CT acquisitions are obtained, and the images are reviewed using 3D endoluminal fly-through visualisation software. Polyps are classified by size (diminutive <5 mm, small 6-9 mm, and large >10 mm), morphology (sessile, pedunculated, and flat), and location. The CT colonography reporting and data system (C-RADS) standardises findings and management recommendations: no polyp or lesion (C-RADS 1) requires no follow-up for 5 years if the patient is average risk; intermediate polyp 6-9 mm (C-RADS 3) requires CT surveillance or optical colonoscopy; and large polyp >10 mm or mass (C-RADS 4) requires optical colonoscopy. Extracolonic findings are reported separately using the C-RADS extracolonic categories (E1-E4). Sensitivity for polyps >10 mm exceeds 90 percent but is lower for smaller polyps. Limitations include radiation exposure, inability to perform biopsy or polypectomy, and limited sensitivity for flat lesions.

CT Perfusion

– A functional imaging technique that quantifies tissue blood flow by acquiring serial CT images during the first pass of intravenous iodinated contrast, generating parametric maps of cerebral blood volume (CBV), cerebral blood flow (CBF), mean transit time (MTT), and time to maximum (Tmax). The technique requires rapid bolus injection of contrast (40-50 mL at 5-7 mL/sec) followed by acquisition of 40-50 dynamic images over 45-60 seconds. Postprocessing software generates color-coded parametric maps used to identify the ischemic penumbra (tissue at risk but not yet infarcted) in acute stroke. The core infarct is defined by severely reduced CBV (less than 30 percent of normal) and markedly reduced CBF, while the penumbra is defined by prolonged MTT or Tmax with preserved or elevated CBV. The mismatch ratio (penumbra-to-core ratio greater than 1.8) and absolute mismatch volume (greater than 15 mL) are used in selecting patients for endovascular thrombectomy beyond the 6-hour window (DEFUSE-3 and DAWN trial criteria). CT perfusion is also used in oncology to assess tumor angiogenesis, treatment response, and differentiate tumor progression from radiation necrosis. Limitations include radiation exposure, need for good IV access, and susceptibility to motion artefact.

CT Pulmonary Angiography

– The reference standard imaging test for diagnosing pulmonary embolism, using ECG-gated or ungated multidetector CT with rapid intravenous bolus of iodinated contrast timed to opacify the pulmonary arterial tree. The technique requires precise bolus tracking or test bolus timing to achieve peak enhancement (typically 80-150 HU) in the main pulmonary arteries while avoiding excessive contrast in the right heart that produces streak artifact. Acute pulmonary embolism appears as filling defects within the pulmonary arterial lumen on contrast-enhanced CT, with complete occlusion producing the railroad track sign (contrast surrounding a central filling defect in a vessel imaged axially). Chronic PE shows organized, eccentrically located, web-like filling defects and abrupt vessel tapering, along with signs of pulmonary hypertension (enlarged pulmonary artery, right ventricular hypertrophy). The severity of PE is quantified by the clot burden score and by evidence of right ventricular strain (right ventricular-to-left ventricular diameter ratio greater than 1.0 on reconstructed four-chamber views). CTPA is the first-line imaging test for suspected PE in non-pregnant patients with normal renal function, with sensitivity and specificity exceeding 95 percent for proximal PE. Limitations include artefact from breathing, cardiac motion, and suboptimal contrast opacification, as well as the use of iodinated contrast in patients with renal impairment or contrast allergy.

CT Urography

– A comprehensive imaging technique for evaluating the urinary tract, combining non-contrast, nephrographic, and excretory phase acquisitions to assess the renal parenchyma, collecting systems, ureters, and bladder. The protocol begins with a non-contrast phase for stone detection and baseline attenuation measurement, followed by a nephrographic phase at 90-100 seconds post-contrast injection for renal parenchymal evaluation, and an excretory phase at 7-10 minutes when iodinated contrast is excreted into the collecting system. CTU is primarily indicated for evaluation of haematuria (both microscopic and macroscopic), detection of urothelial tumors (transitional cell carcinoma of the renal pelvis, ureter, and bladder), and assessment of congenital urinary tract anomalies. CTU has largely replaced intravenous urography (IVU) for evaluation of the urinary tract. Limitations include radiation exposure (particularly relevant in younger patients with haematuria, where serial studies may be needed), the need for adequate hydration and renal function for contrast excretion, and artefact from bowel motion or inadequate distension of the collecting system.

Digital Radiography

– Digital radiography (DR) has replaced traditional film-screen radiography by using electronic detectors to acquire, store, and display radiographic images with post-processing capabilities. Direct DR uses amorphous selenium photoconductor panels that convert X-rays directly to electrical signals, while indirect DR uses cesium iodide scintillator screens coupled to amorphous silicon photodiode arrays that convert X-rays to light and then to electrical signals. Advantages over film include wider dynamic range, lower radiation dose for equivalent image quality, immediate image availability, and digital image processing (windowing, edge enhancement, and contrast manipulation). Digital radiography systems also facilitate electronic image storage, distribution, and transmission through PACS (Picture Archiving and Communication System). Limitations include reduced spatial resolution compared to film-screen systems (though this is clinically negligible), and detector artefact from dead pixels or calibration drift.

Digital Subtraction Angiography

– The gold standard for vascular imaging, using digital image processing to subtract a pre-contrast mask image from contrast-enhanced images, leaving only the contrast-filled vessels. A high-resolution flat-panel detector acquires images at 15-30 frames per second during intra-arterial injection of iodinated contrast. Pixel-by-pixel logarithmic subtraction removes stationary structures (bone, soft tissue) while preserving moving contrast. Motion artifacts from swallowing, breathing, or patient movement degrade image quality and can simulate pathology such as pseudo-stenosis or dissection. DSA is used for diagnostic evaluation of vascular pathology and as the first step of endovascular interventions including angioplasty, stenting, embolisation, and thrombolysis. It is the gold standard for assessing aneurysm morphology, arteriovenous malformation angioarchitecture, and extent of atherosclerotic disease prior to planned revascularisation. Increasingly, DSA is being replaced by non-invasive CTA and MRA for purely diagnostic indications, but it remains essential for guiding and confirming endovascular therapy.

Doppler Ultrasound

– Doppler ultrasound assesses blood flow velocity and direction using the Doppler shift principle: sound waves reflected by moving red blood cells undergo frequency changes proportional to flow velocity. Color Doppler encodes mean velocity and direction as color overlays on B-mode images (typically red toward, blue away from the transducer), providing qualitative flow assessment. Power Doppler displays the amplitude of the Doppler signal without directional information, offering greater sensitivity for slow and low-volume flow. Spectral Doppler (pulsed wave and continuous wave) measures flow velocity at specific locations or along the beam axis, with the velocity waveform providing information about vascular resistance and stenosis severity: a high-resistance waveform is normal in peripheral arteries (triphasic with reverse flow), while a low-resistance waveform is normal in internal carotid, renal, and hepatic arteries. Spectral Doppler criteria for carotid stenosis include peak systolic velocity thresholds (>125 cm/s for 50-69 percent stenosis, >230 cm/s for 70-99 percent stenosis in the internal carotid artery). Doppler ultrasound is essential for evaluating carotid stenosis, deep vein thrombosis (venous compression ultrasound with absence of flow on color Doppler), organ perfusion (renal artery stenosis, hepatic vascular patency), and fetal well-being (umbilical artery Doppler). Limitations include operator dependence, angle dependence (velocity measurements require a Doppler angle of 60 degrees or less), and inability to visualise through bone or gas.

Echocardiogram

– An ultrasound-based diagnostic imaging modality that uses high-frequency sound waves to create real-time images of the heart, allowing comprehensive assessment of cardiac structure, function, and hemodynamics. The standard transthoracic echocardiogram (TTE) obtains images through the chest wall using multiple acoustic windows (parasternal long and short axis, apical, subcostal, and suprasternal notch views) with various ultrasound modes: two-dimensional (2D) echocardiography provides real-time anatomical imaging of cardiac structures in multiple planes, M-mode (motion mode) provides high-temporal-resolution measurements of chamber dimensions and wall thickness, color Doppler enables qualitative assessment of valvular regurgitation and stenosis, pulsed-wave Doppler provides quantitative flow velocity measurements across valves and in the great vessels, and tissue Doppler imaging assesses myocardial velocity for diastolic function evaluation. TTE is the first-line imaging modality for evaluation of systolic and diastolic function, valvular heart disease (stenosis quantified by pressure gradients and valve area, regurgitation by vena contracta width and effective regurgitant orifice area), pericardial disease (effusion, pericarditis, tamponade), and congenital heart disease. Stress echocardiography uses exercise or pharmacological stress (dobutamine) to detect wall motion abnormalities indicating myocardial ischemia. Transoesophageal echocardiography (TOE) provides superior visualisation of posterior cardiac structures, prosthetic valves, and the left atrial appendage. Limitations include operator dependence, poor acoustic windows in obese patients and those with lung disease, and limited visualisation of the distal ascending aorta and pulmonary arteries.

Fetal MRI

– Fetal MRI provides supplemental imaging when ultrasound findings are inconclusive or require further characterization, using ultrafast sequences (single-shot fast spin echo) to minimize fetal motion artifact without sedation. MRI is typically performed at greater than 20 weeks gestational age when fetal size and myelination provide optimal contrast. The most common indication is evaluation of fetal central nervous system anomalies. CNS findings include ventriculomegaly (atrial width greater than 10 mm), structural CNS abnormalities (agenesis of the corpus callosum, posterior fossa anomalies, holoprosencephaly), and congenital diaphragmatic hernia. Thoracic indications include evaluation of congenital lung lesions (congenital pulmonary airway malformation, pulmonary sequestration, bronchogenic cyst). Abdominal indications include abdominal wall defects (gastroschisis, omphalocele) and renal anomalies. Fetal MRI also plays a role in prenatal counseling and surgical planning for conditions that may require ex-utero intrapartum treatment (EXIT procedure). Safety considerations include the use of 1.5T magnets (avoiding 3T due to increased radiofrequency energy deposition) and avoidance of gadolinium contrast due to fetal risk.

Fluoroscopic Barium Studies

– Fluoroscopic barium studies evaluate the gastrointestinal tract using barium sulfate suspension as positive contrast (radiopaque, appears white) or water-soluble contrast (for suspected perforation) to demonstrate mucosal detail and GI motility. Barium swallow evaluates the esophagus for dysphagia, GERD, strictures, rings (Schatzki ring), webs, achalasia (bird-beak tapering with dilated esophagus), and motility disorders (scleroderma with distal esophageal dysmotility, diffuse esophageal spasm with corkscrew esophagus and pseudodiverticulosis), and post-surgical evaluation (Nissen fundoplication, gastric sleeve). Barium meal evaluates the stomach and duodenum for ulcers, gastric cancer, and gastric outlet obstruction. Small bowel follow-through evaluates the small intestine for Crohn disease (cobblestoning, string sign of terminal ileal stenosis, wall thickening and separation of loops), obstruction, and malabsorption. Barium enema evaluates the colon for diverticulosis, colorectal cancer (apple-core lesion, annular constricting lesion), polyps, and inflammatory bowel disease. Water-soluble contrast (Gastrografin) is used when perforation is suspected, as it is absorbed from the peritoneal cavity and is safer than barium in this setting. Double-contrast studies (using barium plus air) provide superior mucosal detail compared to single-contrast studies.

Fluoroscopy

– Fluoroscopy provides real-time X-ray imaging by using a continuous or pulsed X-ray beam that passes through the patient and is detected by an image intensifier or flat-panel detector, displaying dynamic motion on a monitor. The image intensifier converts X-rays to light via a cesium iodide phosphor, amplifies the light signal using a photocathode and electron optics, and reconverts to a visible image. Modern flat-panel detectors use amorphous silicon photodiodes with cesium iodide or gadolinium oxysulfide scintillator coupled to amorphous silicon photodiodes. Fluoroscopy is used for a wide range of diagnostic and interventional procedures, including barium studies of the gastrointestinal tract, voiding cystourethrography, hysterosalpingography, arthrography, and guidance for interventional procedures such as angiography, stent placement, and percutaneous biopsies. Digital fluoroscopy with pulsed fluoroscopy reduces radiation dose by acquiring images at lower frame rates (e.g., 7.5 or 15 frames per second rather than 30 fps). Last image hold and fluoroscopy loops reduce the need for separate spot images. Radiation dose management is essential, with dose-area product (DAP) monitoring and ALARA principles applied. Operator and patient protection includes lead aprons, thyroid shields, lead glasses, and collimation to reduce scatter radiation.

Gamma Knife

– A non-invasive stereotactic radiosurgery device that delivers precisely focused gamma radiation to intracranial targets. Uses 201 cobalt-60 sources arranged in a hemispherical array, emitting 201 beams that converge at a common focal point (isocentre). The steep dose fall-off outside the target (1-2 mm) spares surrounding tissue. Indications: arteriovenous malformations (AVMs, 70-80% obliteration rate at 3 years), trigeminal neuralgia (70-90% pain relief), acoustic neuroma (vestibular schwannoma), meningioma, pituitary adenoma, cerebral metastases (single or oligometastases), essential tremor (thalamotomy). Fractionation: single-session or hypofractionated (2-5 fractions) for larger targets and metastases adjacent to critical structures. Complications: headache, nausea, alopecia (transient), edema, radiation necrosis (late).

Hepatic Embolization

– Hepatic embolization encompasses image-guided transcatheter techniques that selectively occlude hepatic arterial blood supply to treat tumors, control hemorrhage, and manage vascular malformations. Transarterial chemoembolization (TACE) delivers concentrated chemotherapy (doxorubicin or cisplatin) mixed with lipiodol and followed by gel foam or drug-eluting beads into the feeding hepatic artery branches supplying hepatocellular carcinoma. Drug-eluting bead TACE (DEB-TACE) provides more controlled and sustained drug release. Radioembolisation (selective internal radiation therapy, SIRT) uses yttrium-90 (Y-90) microspheres injected into the hepatic artery to deliver targeted radiation to liver tumors, taking advantage of the dual blood supply (tumors derive most of their blood supply from the hepatic artery while normal liver parenchyma receives most from the portal vein). Bland embolisation uses particles alone without chemotherapy. Indications include unresectable hepatocellular carcinoma, colorectal liver metastases, neuroendocrine tumor liver metastases, and hepatic hemorrhage from trauma or rupture.

The contraindication is main portal vein thrombosis (except segmental). Post-embolisation syndrome (fever, pain, nausea) is common and self-limiting. Complications include hepatic abscess, tumor lysis syndrome, and non-target embolisation (biliary necrosis, cholecystitis, gastroduodenal ulceration).

Hepatic Imaging

– Hepatic imaging encompasses multiple modalities for evaluating focal liver lesions, diffuse liver disease, and vascular anatomy. Ultrasound serves as the initial screening tool, with diffuse fatty infiltration appearing as increased echogenicity relative to the spleen (bright liver), and focal lesions characterized by echogenicity, vascularity on Doppler, and enhancement patterns. CT liver imaging uses multiphasic protocols: non-contrast (calcifications, hemorrhage), hepatic arterial phase at 30-40 seconds post-injection (hypervascular lesions enhance avidly), portal venous phase at 60-80 seconds (best for hypovascular lesions and portal vein evaluation), and delayed phase at 3-5 minutes (washout assessment). MRI liver imaging uses multiphasic gadolinium-enhanced sequences with diffusion-weighted imaging providing additional information about cellularity, and hepatobiliary-specific contrast agents such as gadoxetic acid providing delayed hepatobiliary phase imaging at 10-20 minutes post-injection, which helps distinguish hepatocellular lesions from non-hepatocellular lesions. Focal liver lesions including haemangioma (flash-filling, peripheral nodular discontinuous enhancement with progressive centripetal fill-in), focal nodular hyperplasia (homogeneous hyperenhancement in the arterial phase with isoenhancement in portal venous phase and persistent enhancement on delayed phases, central scar T2 hyperintense), hepatic adenoma (hypervascular with variable washout, heterogeneous due to fat or hemorrhage), hepatocellular carcinoma (arterial hyperenhancement with portal venous or delayed washout, capsule, and mosaic architecture), and metastases (variable enhancement, most commonly hypovascular with rim enhancement).

Hepatobiliary Imaging

– Hepatobiliary imaging encompasses multiple modalities for evaluating the liver, biliary tree, and gallbladder. Hepatobiliary iminodiacetic acid (HIDA) scintigraphy (cholescintigraphy) uses Tc-99m mebrofenin or disofenin, which is taken up by functioning hepatocytes and excreted into the bile ducts, providing functional assessment of the biliary system. Normal studies show hepatic uptake within 5 minutes, visualization of the common bile duct and gallbladder by 30-60 minutes, and intestinal activity within 30-60 minutes. Non-visualisation of the gallbladder beyond 4 hours suggests acute cholecystitis or cystic duct obstruction. Delayed biliary-to-bowel transit suggests biliary obstruction, and retained activity in the liver parenchyma with no excretion suggests hepatocellular dysfunction. Hepatobiliary scintigraphy is also used to evaluate bile leaks after hepatobiliary surgery or trauma, biliary-enteric anastomotic patency after hepaticojejunostomy, and sphincter of Oddi dysfunction. Cholescintigraphy with sincalide (cholecystokinin) stimulation can assess gallbladder ejection fraction, with values below 35 percent suggesting chronic acalculous gallbladder disease. Limitations include poor spatial resolution and radiation exposure.

Hysterosalpingography

– A fluoroscopic examination that evaluates the uterine cavity and fallopian tube patency using iodinated contrast instilled through a catheter placed in the uterine cavity. The procedure is typically performed during the follicular phase (days 6-11) of the menstrual cycle to avoid pregnancy and visualize the endometrial cavity optimally. After cervical cannulation, contrast is slowly injected under fluoroscopic guidance, filling the uterine cavity (normal triangular-shaped cavity), filling of the fallopian tubes (normal linear or serpentine opacification), and peritoneal spillage (free contrast spreading around bowel loops, confirming tubal patency). Tubal obstruction appears as non-opacification of the tube with intrauterine contrast filling only to the level of obstruction (cornual, isthmic, or fimbrial). Hydrosalpinx appears as a dilated, sac-like tubular structure with mucosal folds. Intracavitary filling defects include endometrial polyps (smooth, avoidable, mobile), submucosal fibroids (round, well-defined, distorting contour), intrauterine adhesions (irregular, angular, non-distensible), and endometrial hyperplasia or carcinoma (irregularity). Advantages of HSG include its ability to evaluate both uterine cavity and tubal patency in a single procedure. Limitations include contrast reaction, vasovagal reaction, pain, and risk of infection (prophylactic antibiotics recommended in patients with history of pelvic inflammatory disease). HSG is contraindicated in pregnancy, active pelvic infection, and current menstrual bleeding.

Hysteroscopy Imaging

– Hysteroscopy provides direct visualization of the uterine cavity using a thin endoscope (hysteroscope) inserted through the cervix, with distension media (saline for diagnostic, glycine or sorbitol for operative) creating a clear viewing space. The procedure is performed as outpatient surgery or in-office diagnostic evaluation. Normal findings include a smooth endometrial cavity with tubal ostia in the cornua and a normal endocervical canal. Pathologic findings include endometrial polyps (pedunculated or sessile) and submucosal fibroids (bulging, distorting the cavity), Asherman syndrome (intrauterine adhesions appearing as irregular, scarred bands), endometrial hyperplasia, and endometrial cancer (irregular, friable, vascular mass). Operative hysteroscopy allows biopsy, polypectomy, myomectomy, adhesiolysis, endometrial ablation, foreign body removal, and tubal cannulation under direct vision. Complications include uterine perforation (1-3 percent, higher with cervical stenosis), fluid overload (from absorption of distension media, particularly glycine), hemorrhage, infection, and gas embolism. Contraindications include pregnancy, active pelvic infection, and cervical malignancy.

Image-Guided Abscess Drainage

– Image-guided percutaneous abscess drainage is a minimally invasive alternative to surgical drainage for intra-abdominal, pelvic, thoracic, and soft tissue abscesses, using CT, ultrasound, or fluoroscopy for real-time guidance. The technique involves needle puncture of the abscess cavity under sterile conditions, aspiration of pus for culture and sensitivity, followed by placement of an 8-14 French drainage catheter using the Seldinger technique (guidewire placement through the needle, sequential dilation of the tract, and catheter insertion), securing the catheter to the skin with sutures or adhesive devices, and connecting it to a drainage bag. Catheters are flushed with saline twice daily and removed when drain output is minimal (less than 10 mL per day) and clinical signs of infection have resolved. Pre-procedure imaging evaluation (CT or ultrasound) identifies the safest access route, avoiding bowel, solid organs, and major vessels. The success rate for percutaneous abscess drainage exceeds 80 percent for well-defined, unilocular abscesses with a safe access window. Abscesses with multiple loculations, thick pus, or fistulous communication with the bowel may require surgical drainage. Complications include hemorrhage, bacteraemia, sepsis during catheter manipulation, catheter dislodgement, and bowel perforation. Antibiotic coverage should be guided by culture results, with empiric broad-spectrum antibiotics covering Gram-negative and anaerobic organisms started before the procedure.

Interventional Neuroradiology

– Interventional neuroradiology encompasses minimally invasive endovascular and percutaneous procedures for diagnosis and treatment of neurovascular diseases, performed through femoral or radial artery access under fluoroscopic and angiographic guidance. Endovascular coiling of cerebral aneurysms uses platinum coils deployed through microcatheters positioned within the aneurysm sac, with the goal of complete occlusion (Raymond class 1) to prevent rebleeding. Adjunctive devices include balloon remodeling, stent-assisted coiling, flow diverters (pipeline embolisation device for wide-necked aneurysms of the internal carotid artery), and Woven EndoBridge (WEB) devices for bifurcation aneurysms. Acute stroke intervention includes mechanical thrombectomy with stent retrievers or aspiration catheters for large vessel occlusion. Cerebral vasospasm after subarachnoid hemorrhage is treated with intra-arterial vasodilators (verapamil, milrinone, nicardipine) and balloon angioplasty of proximal spastic segments. Carotid artery stenting is an alternative to carotid endarterectomy for carotid stenosis in high-risk surgical patients. Pre-procedure imaging, careful patient selection, and meticulous technique are essential to minimise periprocedural stroke risk. Complications include stroke (1-3 percent at 30 days for experienced operators), vessel perforation, dissection, groin haematoma, pseudoaneurysm, and contrast-induced nephropathy.

Interventional Radiology

– Interventional radiology (IR) encompasses image-guided minimally invasive procedures that diagnose and treat conditions across virtually every organ system, replacing many open surgical procedures with lower morbidity and faster recovery. Core techniques include percutaneous biopsy (CT or ultrasound-guided), abscess drainage, catheter-directed thrombolysis, embolization, angioplasty and stenting, and thermal ablation. Vascular IR includes for critical limb ischemia, venous disease includes venous access, venous stenting for chronic deep venous occlusion (iliac vein stenting for post-thrombotic syndrome), and inferior vena cava filter placement for recurrent pulmonary embolism despite anticoagulation or contraindication to anticoagulation. Non-vascular IR includes percutaneous biopsy (core needle biopsy for histology, fine needle aspiration for cytology) of lung, thyroid, liver, kidney, prostate, and soft tissue lesions; percutaneous drainage of abscesses, bilomas, urinomas, and seromas; nephrostomy tube insertion for obstructive uropathy; biliary drainage and stenting for obstructive jaundice; percutaneous gastrostomy for nutritional access; and tumor ablation (radiofrequency ablation, microwave ablation, cryoablation, irreversible electroporation, and alcohol ablation) for liver, kidney, lung, bone, and soft tissue tumors. Interventional oncology is a rapidly growing subspecialty integrating IR into the multidisciplinary care of cancer patients. IR procedures are typically performed under moderate sedation or general anesthesia, with real-time image guidance (fluoroscopy, CT, ultrasound, or MRI), sterile technique, and careful monitoring of patient vital signs. Complications include bleeding, infection, contrast reaction, radiation injury, and non-target embolisation.

Lymphangiography

– A fluoroscopic procedure that evaluates the lymphatic system by cannulating lymphatic vessels in the feet and injecting oil-based contrast (lipiodol) that opacifies the inguinal, pelvic, and para-aortic lymph nodes over 24-48 hours. The technique involves subcutaneous injection of blue dye in the web spaces of both feet to identify and isolate lymphatic vessels, followed by cannulation and slow continuous infusion of contrast. Initial images demonstrate the lymph channels and nodes. Delayed images at 24-48 hours show the internal architecture of lymph nodes, including filling defects from metastases or lymphoma. Lymphangiography is now rarely performed, having been largely replaced by CT, MRI (particularly MR lymphangiography), PET-CT, and sentinel lymph node biopsy for most indications. Its remaining indications include evaluation of chylothorax and chylous ascites (identifying lymphatic leaks), pre-operative mapping of lymphatic anatomy, and detection of lymph node metastases in certain cancers (particularly penile and cervical cancer). The procedure is technically challenging, time-consuming, and requires expertise in microsurgical lymph vessel cannulation. Complications include lymphangitis, wound infection, contrast extravasation, oil embolism (lipiodol can cause pulmonary oil embolisation and allergic reactions), and impaired ventilation after extensive pulmonary deposition.

Magnetic Resonance Imaging

– Magnetic resonance imaging (MRI) exploits the magnetic properties of hydrogen protons in water and fat molecules to generate detailed soft tissue contrast without ionizing radiation. In a strong static magnetic field (typically 1.5T or 3T), protons align with the field and precess at the Larmor frequency (63.87 MHz at 1.5T). Radiofrequency pulses at the Larmor frequency tip the net magnetization vector into the transverse plane, and as protons return to equilibrium, they emit detectable signals. Spatial localisation is achieved by applying three orthogonal magnetic field gradients that encode the signal with spatial frequency information (frequency-encoding and phase-encoding gradients), enabling image reconstruction via Fourier transformation by the K-space filling process and the Nyquist-Shannon sampling theorem applied to discrete sampling of spatial frequencies. Key tissue contrast parameters include T1 (spin-lattice or longitudinal relaxation time—the time constant for the net magnetisation vector to return to equilibrium along the longitudinal axis), T2 (spin-spin or transverse relaxation time—the time constant for decay of transverse magnetisation due to irreversible dephasing), T2* (T2 plus dephasing from local magnetic field inhomogeneities), and proton density. T1-weighted sequences (short TR, short TE) provide excellent anatomical detail, with fat appearing bright (short

T1) and fluid appearing dark (long T1, dark CSF). T2-weighted sequences (long TR, long TE) highlight pathology that contains fluid or edema, with fluid appearing bright (long T2, bright CSF). FLAIR (fluid-attenuated inversion recovery) suppresses CSF signal on T2-weighted images, making periventricular and cortical lesions more conspicuous. STIR (short tau inversion recovery) suppresses fat signal, useful for bone marrow edema and optic nerve evaluation. DWI measures the Brownian motion of water molecules, with restricted diffusion (e.g., acute stroke, abscess, cellular tumors) appearing as high signal on DWI and low signal on ADC (apparent diffusion coefficient) maps. DTI (diffusion tensor imaging) maps white matter tracts, and PWI (perfusion-weighted imaging) assesses cerebral blood flow. MRA (magnetic resonance angiography) uses time-of-flight or phase-contrast techniques to visualise blood vessels without contrast. MR spectroscopy measures metabolite concentrations (N-acetylaspartate, choline, creatine, lactate) for tissue characterisation. MRI safety considerations include projectile effects from ferromagnetic objects, radiofrequency heating, peripheral nerve stimulation from gradient switching, hearing damage from acoustic noise, and the risks of gadolinium-based contrast agents (nephrogenic systemic fibrosis in advanced renal disease, gadolinium deposition in brain tissues). Claustrophobia affects up to 10 percent of patients. Implant safety is categorised as MRI-safe, MRI-conditional, or MRI-unsafe.

Mammographic BI-RADS

– BI-RADS (Breast Imaging Reporting and Data System) is the standardized reporting framework developed by the American College of Radiology to reduce variability in breast imaging interpretation and management recommendations. BI-RADS Assessment Categories include Category 0 (incomplete evaluation requiring additional imaging), Category 1 (negative with no abnormal findings), Category 2 (benign findings such as calcified fibroadenomas, fat-containing lesions, or simple cysts), Category 3 (probably benign, less than 2 percent chance of malignancy, requiring 6-month follow-up imaging), Category 4 (suspicious abnormality, biopsy recommended, subdivided into 4A, 4B, and 4C with increasing suspicion), Category 5 (highly suggestive of malignancy, biopsy recommended, greater than or equal to 95 percent chance of malignancy), and Category 6 (known biopsy-proven malignancy). BI-RADS also standardises the description of breast composition (a through d: almost entirely fatty, scattered fibroglandular, heterogeneous dense, and extremely dense, with the latter two masking underlying lesions and reducing mammographic sensitivity), findings (masses: shape, margin, density; calcifications: morphology and distribution; architectural distortion; asymmetries; and associated features), and recommendations (e.g., routine screening, short-interval follow-up, biopsy, or surgical consultation). The BI-RADS system also exists for breast ultrasound and breast MRI with similar standardised categories and lexicon.

Mammography

– A dedicated low-dose X-ray imaging technique for breast cancer screening and diagnosis, using specialized equipment with molybdenum or rhodium targets and filters to produce optimal contrast between glandular and adipose tissues. Standard views include craniocaudal (CC) and mediolateral oblique (MLO) projections, with additional spot compression, magnification, and rolled views for problem-solving. Digital mammography uses flat-panel detectors (direct conversion with amorphous selenium) or indirect conversion (cesium iodide scintillator coupled to amorphous silicon photodiodes), with computer-aided detection (CAD) software highlighting suspicious areas for radiologist review. Digital breast tomosynthesis (DBT) acquires multiple low-dose radiographic projections at different angles during a single compression, reconstructing them into thin-slice (typically 1 mm) images that can be scrolled through, reducing tissue superimposition and improving lesion detection and characterisation. Screening mammography is recommended annually for average-risk women starting at age 40 (American College of Radiology and American Cancer Society guidelines) or biennially from age 50 (US Preventive Services Task Force). Diagnostic mammography is performed for symptomatic women (palpable lump, nipple discharge, skin changes, focal pain) or to evaluate abnormal screening findings. Performance metrics include sensitivity (approximately 75-90 percent overall, lower in dense breasts) and specificity (approximately 85-95 percent). Limitations include false positives with additional imaging and biopsy, false negatives, particularly in dense breasts, and the risk of overdiagnosis of clinically insignificant cancers.

MRI Brain

– MRI of the brain provides superior soft tissue contrast for evaluating neurologic conditions including stroke, neoplasms, demyelination, infection, and congenital anomalies without ionizing radiation. Standard sequences include T1-weighted imaging for anatomy, T2-weighted imaging for pathology, FLAIR for periventricular and cortical lesions, and DWI for acute ischemia. In acute stroke, DWI shows restricted diffusion (high signal on DWI with corresponding low ADC values) within minutes of symptom onset, with FLAIR being negative (DWI-FLAIR mismatch), indicating acute ischemia within the thrombolytic window. MRI with gadolinium demonstrates ring-enhancing lesions in abscesses and brain tumors. MRA evaluates the intracranial and extracranial vasculature for stenosis, occlusion, dissection, aneurysm, and arteriovenous malformations. MR venography (MRV) evaluates dural venous sinus thrombosis. Advanced techniques include functional MRI for eloquent cortex mapping, MR spectroscopy for metabolite analysis, and perfusion-weighted imaging for cerebral blood flow assessment. Contraindications include MRI-incompatible implants, claustrophobia, and inability to lie still for the duration of the examination.

MRI Cardiac Perfusion

– Cardiac MR perfusion imaging evaluates myocardial blood flow by acquiring rapid dynamic images during the first pass of gadolinium-based contrast through the myocardial circulation, detecting areas of reduced perfusion that indicate hemodynamically significant coronary artery disease. The technique uses satated or ungated fast gradient echo sequences (saturation recovery or inversion recovery preparation) with high temporal resolution (1-2 images per heartbeat) during vasodilator stress (adenosine or dipyridamole) or during dobutamine stress. The presence of a subendocardial or transmural perfusion defect (a region of hypoenhanced myocardium relative to remote normal myocardium) indicates reduced blood flow in the territory of a hemodynamically significant coronary stenosis (typically >70 percent diameter stenosis). The extent and severity of perfusion defects are graded semi-quantitatively using the 17-segment American Heart Association model: a normal segment shows homogeneous enhancement, a reversible defect (ischemia) shows hypoenhancement during stress that normalises at rest, and a fixed defect (infarct) shows hypoenhancement on both stress and rest images, with corresponding LGE indicating scar. The ischemic burden quantified by stress perfusion (percentage of left ventricular myocardium affected) guides revascularisation decisions: an ischemic burden of more than 10-15 percent is generally considered a threshold for revascularisation in patients with stable coronary artery disease. Limitations include artefact from dark-rim (Gibbs ringing), motion, and arrhythmia, and the need for gadolinium-based contrast and patient cooperation for breath-holding. Contraindications include adenosine (avoid in asthma, high-grade AV block, and severe hypotension), dobutamine (avoid in arrhythmia and uncontrolled hypertension), and gadolinium (avoid in advanced renal disease).

MRI Liver

– MRI of the liver provides the highest contrast resolution for focal liver lesion characterization, using hepatobiliary-specific gadolinium-based contrast agents that are actively transported into functioning hepatocytes via organic anion transporting polypeptides. Gadoxetic acid (Eovist) is taken up by hepatocytes beginning 10 minutes post-injection, producing hepatobiliary phase images where normal liver parenchyma appears bright, and non-hepatocyte-containing lesions (metastases, intrahepatic cholangiocarcinoma) show as hypointense (black) on the hepatobiliary phase. Haemangiomas show characteristic peripheral nodular discontinuous enhancement with progressive centripetal fill-in on dynamic phases, with a very high T2 signal. Focal nodular hyperplasia shows homogeneous arterial phase hyperenhancement, isoenhancement in portal venous and delayed phases, and isoenhancement or slight hyperenhancement on the hepatobiliary phase (because it contains functioning hepatocytes). Hepatocellular carcinoma in the cirrhotic liver shows arterial hyperenhancement with portal venous or delayed washout, often with a capsule and internal fat (features of LI-RADS 5). Hepatic adenomas are hypervascular with variable washout and may contain macroscopic fat or hemorrhage. Diagnostic accuracy exceeds 90 percent for most focal liver lesion types. The LI-RADS classification system standardises image interpretation and reporting in patients at risk for hepatocellular carcinoma. Limitations include motion artefact from breathing, need for patient cooperation with breath-holding, and the risk of nephrogenic systemic fibrosis in patients with advanced renal disease. Claustrophobia and MRI-incompatible implants are contraindications.

MRI Spine

– MRI of the spine is the gold standard for evaluating spinal cord pathology, disc disease, stenosis, and tumors, providing multiplanar imaging with excellent soft tissue contrast. Standard sequences include sagittal and axial T1-weighted and T2-weighted imaging through the region of interest. T1-weighted sequences show anatomy with CSF appearing dark and fat appearing bright, useful for evaluating bone marrow infiltration (replaced marrow shows low signal on T1). T2-weighted sequences show CSF appears bright, ideal for evaluating disc hydration loss (desiccation), annular fissures, and nerve root compression. Disc herniation is classified by morphology: protrusion (broad-based, focal), extrusion (neck narrower than the herniated portion), and sequestration (free fragment). Spinal stenosis is graded by thecal sac cross-sectional area: mild (100-120 mm2), moderate (80-100 mm2), and severe (<80 mm2). T1-weighted post-gadolinium sequences are essential for evaluating infection (discitis/osteomyelitis shows enhancement of the disc and adjacent vertebral endplates), inflammation (epidural abscess, arachnoiditis), and spinal cord tumors (intramedullary tumors such as ependymomas and astrocytomas enhance, whereas benign syringomyelia does not). STIR sequences are sensitive for bone marrow edema from fracture, infection, or neoplasm. Contraindications include MRI-incompatible implants and severe claustrophobia.

Musculoskeletal MRI

– Musculoskeletal (MSK) MRI provides detailed evaluation of soft tissues, bone marrow, cartilage, ligaments, and tendons, serving as the preferred modality for most non-osseous pathologies. Standard sequences include T1-weighted (anatomy, bone marrow infiltration), T2 fat-saturated or STIR (edema, fluid), proton density (ligament and meniscus assessment), and post-gadolinium (infection, tumor). Knee MRI evaluates meniscal tears (horizontal cleavage, radial, complex, root), cruciate ligament injuries (anterior cruciate ligament tear: discontinuity, abnormal signal, and non-visualisation; posterior cruciate ligament: usually mid-substance tear; medial collateral ligament: edema and discontinuity; lateral collateral ligament: injury often associated with posterolateral corner injury), and meniscal extrusion. Shoulder MRI evaluates rotator cuff tears (supraspinatus, infraspinatus, subscapularis, teres minor), labral tears (SLAP lesions, Bankart lesions), biceps tendon pathology, and glenohumeral instability. Hip MRI evaluates acetabular labral tears, femoroacetabular impingement (CAM and pincer), avascular necrosis of the femoral head (double-line sign on T2), and stress fractures. Wrist and hand MRI evaluates triangular fibrocartilage complex tears, scapholunate and lunotriquetral ligament injuries, and carpal tunnel syndrome findings. Ankle MRI evaluates ligamentous injuries (lateral ankle ligament complex, syndesmosis, deltoid ligament), Achilles tendinopathy and tears, osteochondral lesions of the talar dome, and sinus tarsi syndrome. MSK MRI can also evaluate soft tissue masses (lipoma, sarcoma, desmoid, nerve sheath tumors), with imaging features including signal characteristics on T1, T2, STIR, and post-gadolinium sequences helping to narrow the differential diagnosis.

Musculoskeletal Radiography

– Musculoskeletal radiography remains the initial imaging modality for evaluating bone and joint pathology, providing high spatial resolution for cortical and trabecular bone detail with rapid acquisition and low cost. Standard evaluation includes at least two orthogonal projections (AP and lateral, or oblique views as needed) of the affected region. Fracture evaluation includes assessment of alignment, joint congruence, fracture pattern, and associated soft tissue swelling. The Salter-Harris classification of pediatric fractures types I through V (I: transverse through the growth plate; II: through the growth plate and metaphysis, the most common; III: through the growth plate and epiphysis; IV: through the metaphysis, growth plate, and epiphysis; and V: compression of the growth plate). Osteoarthritis shows joint space narrowing, osteophytes, subchondral sclerosis, and subchondral cysts. Rheumatoid arthritis shows periarticular osteopaenia, marginal erosions, and joint space narrowing. Ankylosing spondylitis shows sacroiliitis, syndesmophytes, and bamboo spine. Gout shows punched-out erosions with overhanging edges and soft tissue tophi. Healing fractures show periosteal reaction and callus formation. Limitations include limited soft tissue contrast compared to MRI and CT, and inability to detect bone marrow edema.

Musculoskeletal Ultrasound

– Musculoskeletal (MSK) ultrasound provides real-time, dynamic evaluation of tendons, ligaments, muscles, nerves, joints, and superficial soft tissues using high-frequency linear transducers. Ultrasound is the preferred initial modality for rotator cuff evaluation, with complete tears appearing as noncompressible hypoechoic gaps in the tendon with retraction, and partial-thickness tears appearing as focal hypoechoic defects at the articular or bursal surface. Achilles tendon assessment demonstrates tendinopathy (thickened, hypoechoic tendon without fibrillar pattern), partial tears (focal hypoechoic or anechoic defects), and complete tears (full-thickness gap with retraction). Plantar fascia assessment shows thickening (greater than 4 mm), hypoechogenicity, and surrounding edema. Dynamic assessment during muscle contraction and joint movement adds diagnostic information not available on static MRI. Advantages include high spatial resolution, real-time dynamic evaluation, color Doppler for hyperaemia, cost-effectiveness, and lack of ionizing radiation. Limitations include operator dependence, small field of view, limited penetration for deep structures, and inability to visualise through bone cortex or air.

Myelography

– A fluoroscopic procedure involving intrathecal injection of iodinated contrast to evaluate the spinal canal, nerve root sleeves, and thecal sac when MRI is contraindicated or insufficient. Lumbar puncture is performed under fluoroscopic or CT guidance, typically at the L3-L4 or L2-L3 interspace, and approximately 8-15 mL of non-ionic contrast (iohexol or iopamidol) is injected into the subarachnoid space. The patient is positioned on a tilting table and moved into Trendelenburg and reverse Trendelenburg positioning, contrast is seen flowing cephalad and caudally within the thecal sac, outlining the spinal cord, nerve root sleeves (which should be symmetrical with sharp, well-defined margins), and the conus medullaris. Extrinsic compression is indicated by extradural defects (from disc herniation, spondylosis, tumor, or epidural abscess), intradural extramedullary defects (from meningioma, schwannoma, or neurofibroma—filling defects within the contrast column displacing the cord or cauda equina), and intramedullary expansion (from tumor or syringomyelia, widening the spinal cord shadow). Complete block shows a column of contrast that ends abruptly at the level of obstruction. Complications include post-dural puncture headache, contrast reaction, arachnoiditis (from the contrast medium, more common with older ionic oil-based contrast agents such as Pantopaque, now replaced by non-ionic water-soluble agents), and intrathecal hemorrhage. Myelography is most commonly used in conjunction with post-myelogram CT (CT myelography), which provides superior delineation of the spinal canal anatomy and nerve roots.

Pediatric Abdominal Imaging

– Pediatric abdominal imaging requires age-appropriate protocols, radiation dose reduction, and knowledge of developmental anatomy and common pediatric conditions. Ultrasound is the initial modality for most pediatric abdominal complaints, avoiding radiation and often not requiring sedation. Intussusception (the most common cause of bowel obstruction in children 6 months to 3 years) shows the target sign (concentric rings of echogenicity) on ultrasound with sensitivity exceeding 98 percent, and is treated with fluoroscopic-guided pneumatic or hydrostatic reduction. Pyloric stenosis (the most common cause of gastric outlet obstruction in infants 2-8 weeks) shows the target sign and antropyloric canal length >16 mm, wall thickness >4 mm, and diameter >14 mm on ultrasound. Appendicitis in children presents with a non-compressible, blind-ending, thick-walled (greater than 2 mm) tubular structure >6 mm in diameter with surrounding increased echogenicity (echogenic mesenteric fat) and hypervascularity. Midgut volvulus (a surgical emergency in neonates and infants) shows inversion of the superior mesenteric artery and vein on ultrasound (the whirlpool sign), requiring urgent upper GI contrast study to confirm malrotation and volvulus. Necrotising enterocolitis in preterm infants shows pneumatosis intestinalis (gas within the bowel wall) on abdominal radiography. CT and MRI are used selectively in children, with radiation dose optimisation and sedation considerations. Contraindications and alternative imaging strategies (ultrasound first, MRI when available) follow the ALARA principle and Image Gently campaign guidelines for radiation dose reduction.

Pediatric Chest Radiograph

– Pediatric chest radiography requires age-specific technique and anatomic knowledge, with the AP view most common in young children and portable radiography standard in the NICU and PICU settings. Thymic tissue is prominent in infants and young children, appearing as a soft tissue density in the anterior mediastinum that may simulate a mediastinal mass (sail sign on the left, thymic wave sign where ribs imprint on the soft thymus). The thymus normally involutes with age and should not be mistaken for pathology on follow-up studies (thymic rebound after chemotherapy). Common pediatric chest radiographic findings include pneumonias (round pneumonias are more common in children than in adults, appearing as rounded airspace opacities without air bronchograms), viral pneumonias (parahilar peribronchial thickening, hyperinflation, and atelectasis), congenital lung malformations (congenital pulmonary airway malformation appearing as a multi-cystic air-filled lesion), and foreign body aspiration (hyperinflation on the affected side on expiratory view due to ball-valve obstruction, with mediastinal shift away from the affected side). In the NICU setting, common findings include hyaline membrane disease/respiratory distress syndrome (reticulogranular ground-glass opacities with air bronchograms), transient tachypnoea of the newborn (fluid in the fissures, perihilar streaking, hyperinflation), meconium aspiration syndrome (diffuse coarse opacities with hyperinflation and patchy atelectasis), and complications of prematurity such as bronchopulmonary dysplasia (chronic lung changes, cystic changes, hyperinflation) and pneumothorax. Congenital heart disease evaluation on radiography assesses cardiomediastinal silhouette, pulmonary vascularity (increased in left-to-right shunts, decreased in right-to-left shunts), and the aortic arch position.

Pediatric Imaging

– Pediatric imaging requires specialized knowledge of developmental anatomy, age-appropriate imaging protocols, and radiation dose optimization in children who are more radiosensitive and have longer remaining lifespans for stochastic effects to manifest. ALARA (As Low As Reasonably Achievable) principles mandate using the lowest radiation dose that achieves diagnostic quality, with size-specific dose estimates replacing tube current-time product as the dose metric. Ultrasound is preferred as the initial imaging modality when feasible, and CT reserved for urgent indications such as trauma. MRI is used for problem-solving without radiation, often requiring sedation or general anesthesia for younger children. Variations in normal pediatric anatomy must be recognised to avoid misinterpretation: the thymus (sail sign), bowel gas patterns in newborns, open fontanelles on head ultrasound (the anterior fontanelle provides an excellent acoustic window for neonatal head ultrasound to evaluate for intraventricular hemorrhage in premature infants, graded I through IV by the Papile classification), and craniosynostosis (premature fusion of the cranial sutures). Common pediatric imaging scenarios include imaging of developmental dysplasia of the hip (Graf classification on ultrasound for infants under 6 months, radiography for older infants), appendicitis (graded compression ultrasound as first-line in children), intussusception (ultrasound diagnosis with the target and doughnut signs), midgut volvulus (upper GI contrast study showing the corkscrew sign of the duodenum), and evaluation of non-accidental injury (skeletal survey for suspected child abuse, looking for classic metaphyseal corner fractures, posterior rib fractures, and healing fractures of different ages). Radiation protection in pediatric imaging includes the use of pediatric-specific protocols, size-based dose modulation, shielding of radiosensitive organs (thyroid, gonads, breast), and avoidance of unnecessary repeat examinations.

Pelvic Ultrasound

– Pelvic ultrasound evaluates the uterus, ovaries, fallopian tubes, and surrounding structures in both gynecologic and non-gynecologic indications, using transabdominal (full bladder, curved array) and transvaginal (endocavitary, high-frequency) approaches. Transvaginal ultrasound provides superior resolution for uterine and ovarian evaluation due to proximity to the transducer. The uterus is assessed for size, position (anteversion vs retroversion), endometrial thickness (premenopausal normal up to 16 mm in postmenopausal women, and normal limits vary by menstrual phase), myometrial assessment (fibroids: well-defined, hypoechoic, causing contour distortion; adenomyosis: diffuse or focal myometrial thickening with a heterogeneous, striated, or cystic appearance, and poor definition of the endometrial-myometrial junction), and adnexal evaluation (ovarian size, follicles, corpus luteum, and masses). Ovarian cysts are common and almost always benign in premenopausal women: simple cysts are anechoic with thin walls and posterior acoustic enhancement. Complex ovarian masses with solid components, thick septations, or papillary projections raise concern for malignancy, prompting surgical referral and further evaluation with MRI and tumor markers. Endometriosis may present as ovarian endometriomas (homogeneous low-level internal echoes, ground-glass appearance, and no internal vascularity—the chocolate cyst). The adnexa should be assessed for hydrosalpinx, tubo-ovarian abscess, and ectopic pregnancy (adnexal mass separate from the ovary, tubal ring sign, and absence of intrauterine pregnancy). Indications include pelvic pain, abnormal uterine bleeding, infertility evaluation, postmenopausal bleeding, and suspected pelvic mass. Transvaginal ultrasound is the preferred approach for detailed uterine and ovarian evaluation. Limitations include operator dependence and limited field of view. In pregnancy, pelvic ultrasound dates the pregnancy, assesses fetal viability and anatomy, and evaluates for complications such as placenta praevia, placental abruption, and ectopic pregnancy.

PET-MRI

– PET-MRI combines the metabolic and molecular information of PET with the superior soft tissue contrast of MRI in a single examination, reducing radiation exposure compared to PET-CT by eliminating the CT component. Hybrid PET-MRI scanners integrate PET detectors within the MRI bore, requiring MRI-compatible PET detector technology (silicon photomultipliers replacing traditional photomultiplier tubes). Simultaneous PET and MRI acquisition enables temporal correlation between metabolic and anatomical information, improving lesion detection and characterisation. Current clinical applications include neurological oncology (brain tumor evaluation, distinguishing tumor progression from radiation necrosis), dementia imaging (amyloid and tau PET-MRI), cardiac sarcoidosis and myocarditis (FDG PET-MRI), prostate cancer (PSMA PET-MRI), and musculoskeletal infection and tumor evaluation. PET-MRI is also used in pediatric oncology to minimise radiation exposure. Challenges include the need for MR-based attenuation correction of PET data (using Dixon-based segmentation or atlas-based methods rather than CT-based methods), longer acquisition times than PET-CT, and the limited availability and high cost of hybrid PET-MRI systems. Despite these challenges, PET-MRI is an increasingly important dual-modality imaging tool, particularly when reducing radiation dose in younger patients and when the superior soft tissue contrast of MRI provides a clear advantage over CT.

Positron Emission Tomography

– A nuclear medicine imaging technique that detects gamma photons produced by the annihilation of positrons emitted from radiopharmaceuticals, providing functional and metabolic information. The most common radiotracer is fluorodeoxyglucose (FDG), a glucose analog labeled with fluorine-18 (half-life 110 minutes), which accumulates in cells with high glucose metabolism such as malignancies, inflammatory cells, and activated brain regions. After intravenous injection, FDG accumulates in tissues with high glucose metabolism and is phosphorylated by hexokinase to FDG-6-phosphate, which is trapped intracellularly (metabolic trapping) and does not undergo further metabolism, enabling imaging of the hexokinase reaction product. Image acquisition occurs approximately 60 minutes after FDG injection, with an increased standardised uptake value (SUV) on delayed images for malignant tissues.

The SUV is a semi-quantitative measure of FDG uptake, normalised to injected dose and patient weight, which helps differentiate benign from malignant lesions. Whole-body PET-CT (non-contrast or contrast-enhanced) is the standard for oncological staging, providing fused anatomical and functional images. The maximum SUV (SUVmax) is the most commonly reported metric for FDG uptake intensity. Benign causes of FDG avidity include infection, inflammation (sarcoidosis, tuberculosis, post-surgical changes, rheumatologic disease), and brown adipose tissue activation (which can be reduced by keeping the patient warm before the scan and by using a low-carbohydrate, high-fat diet the day before the scan). Other PET radiotracers include Ga-68 DOTATATE (somatostatin receptor imaging for neuroendocrine tumors), Ga-68 PSMA (prostate-specific membrane antigen imaging for prostate cancer), F-18 florbetapir and flutemetamol (amyloid PET imaging for Alzheimer disease), F-18 flortaucipir (tau PET imaging), and F-18 DOPA (amino acid transport imaging for brain tumors and hyperinsulinaemic hypoglycaemia). PET-MRI is an alternative to PET-CT that reduces radiation exposure and improves soft tissue contrast, but at higher cost and with longer acquisition times. Limitations of PET include limited spatial resolution (approximately 4-5 mm for clinical PET scanners), false positives from inflammatory processes, false negatives in low-metabolic-rate tumors (e.g., bronchioloalveolar carcinoma, carcinoid, prostate cancer, and mucinous adenocarcinomas), and radiation exposure from both the tracer and the CT component.

Pulmonary Nodule

– A rounded or irregular opacity well delineated from surrounding parenchyma measuring up to 30 mm in diameter, while a mass exceeds 30 mm and is considered malignant until proven otherwise. The Fleischner Society guidelines manage incidentally detected pulmonary nodules based on size, morphology, and patient risk factors. Solid nodules less than 6 mm in low-risk patients require no routine follow-up; nodules 6-8 mm require CT at 6-12 months and consideration of follow-up CT at 3-6 months. Solid nodules 8 mm or larger in high-risk patients require PET-CT, biopsy, or close follow-up at 3 months. Subsolid nodules (ground-glass and part-solid) are managed differently based on size and solid component. Malignant features include spiculated margins, pleural retraction, air bronchograms, and eccentric calcifications. Benign features include diffuse, central, laminated, or popcorn calcifications (granuloma, hamartoma), fat (hamartoma), and stability over two years (benign for solid nodules). The PanCan model (Brock University) and the Lung-RADS classification (for lung cancer screening) provide additional risk stratification. Management follows Fleischner Society guidelines or the British Thoracic Society guidelines, including follow-up CT surveillance, PET-CT, percutaneous biopsy, surgical resection, or surveillance based on the probability of malignancy. Screening with low-dose chest CT reduces lung cancer mortality by 20-25 percent in high-risk populations (age 50-80, 20-pack-year smoking history, current smokers or those who quit within the past 15 years—United States Preventive Services Task Force criteria). Lung cancer screening eligibility and protocols vary by country.

Radiation

– The emission and propagation of energy through space or matter in the form of waves (electromagnetic radiation) or particles (particulate radiation). In medicine, radiation is used diagnostically (X-rays, CT, nuclear medicine) and therapeutically (radiotherapy). Types relevant to medicine: (1) ionizing radiation—has sufficient energy to remove electrons from atoms, causing ionisation and potential biological damage. Includes X-rays, gamma rays, alpha particles, beta particles, neutrons, and protons. X-rays and gamma rays are electromagnetic radiation with wavelengths ranging from 0.01 to 10 nanometres, produced by X-ray tubes (bremsstrahlung and characteristic X-rays) or radioactive decay. (2) Non-ionizing radiation—lower energy, insufficient to ionise atoms but can heat tissue. Includes ultraviolet (UV), visible light, infrared, microwave, and radiofrequency radiation. Medical imaging uses X-rays (plain radiography, fluoroscopy, CT, angiography) and gamma rays (scintigraphy, SPECT, PET). Ultrasound uses high-frequency sound waves (mechanical waves, not radiation—no ionizing effects). MRI uses radiofrequency pulses in a strong magnetic field. Radiation safety: the stochastic (random) risk of cancer increases with cumulative radiation dose (linear no-threshold model—LNT). The ALARA (as low as reasonably achievable) principle guides radiation use. Typical effective doses: chest X-ray 0.02 mSv, mammogram 0.4 mSv, CT head 2 mSv, CT chest 7 mSv, CT abdomen/pelvis 10 mSv (compare to annual background radiation 2.4 mSv). Pregnant patients: fetal radiation risk is highest in the first trimester; alternative imaging (ultrasound, MRI) should be considered.

Radiation enteritis

– An inflammatory condition of the gastrointestinal tract caused by ionizing radiation exposure during cancer treatment, affecting both acute and chronic phases. Acute radiation enteritis occurs during or within weeks of radiation therapy, characterized by mucosal epithelial damage leading to diarrhea, abdominal cramps, nausea, and bleeding, with symptoms correlating with radiation dose (greater than 45 Gy to the pelvis significantly increases risk). Acute changes on imaging include bowel wall thickening, increased enhancement of the mucosa (stratified or layered enhancement pattern), and submucosal edema (target sign). Chronic radiation enteritis, occurring months to years after radiation, is due to progressive obliterative endarteritis and fibrosis leading to ischemia, stricture formation, fistula development, and bowel obstruction. Chronic changes include mural fibrosis (homogeneous wall thickening with reduced enhancement), strictures with proximal bowel dilation, fistulae (particularly enterovesical and enterovaginal fistulae), and adhesions. The portion of bowel most affected correlates with the radiation field: pelvic radiation primarily affects the rectosigmoid colon and distal ileum; abdominal radiation affects the small intestine; and pancreatic or gastric radiation affects the duodenum and stomach. Management of acute radiation enteritis is supportive, including antiemetics, antidiarrhoeals, dietary modification, and parenteral nutrition if symptoms are severe. Management of chronic radiation enteritis depends on the specific complication: strictures may be treated with endoscopic dilation or surgical resection, fistulae may require surgical repair, and bleeding from telangiectasias may be treated with endoscopic argon plasma coagulation.

Radiation Sickness

– See Radiation Sickness in Oncology.

Radioisotope

– See Radioisotope in Oncology.

Renal Ultrasound

– The initial imaging modality for evaluating kidney anatomy, hydronephrosis, renal masses, and diffuse renal parenchymal disease, using curved array transducers for adequate depth penetration. Hydronephrosis is graded as mild (central sinus splitting with mild calyceal dilation), moderate (dilated renal pelvis and calyces with preserved parenchyma), or severe (marked dilation with parenchymal thinning), with causes including ureterolithiasis, ureteropelvic junction obstruction, benign prostatic hyperplasia, and retroperitoneal fibrosis causing extrinsic compression at the level of the ureters, pelvic tumors, and neurogenic bladder. Renal size is measured (normal adult kidney length 9-12 cm, with smaller kidneys suggesting chronic kidney disease and larger kidneys suggestive of polycystic kidney disease, diabetic nephropathy, or HIV nephropathy). Renal masses are characterized as simple cysts (anechoic, well-circumscribed, thin imperceptible walls, posterior acoustic enhancement—Bosniak category I, benign), complicated cysts (Bosniak categories II, IIF, III defined by number and thickness of septations, calcification, and nodularity), or solid masses (which are suspicious for renal cell carcinoma unless proven otherwise). Solid masses require further characterisation with contrast-enhanced CT or MRI for definitive characterisation and staging. color Doppler shows flow in renal vessels and can detect renal artery stenosis (parvus et tardus waveform, tardus-parvus spectral Doppler pattern with prolonged acceleration time and reduced peak systolic velocity), and venous thrombosis (absent flow, enlarged kidney with altered echogenicity). Renal parenchymal disease shows increased cortical echogenicity (grade 1-3), loss of corticomedullary differentiation, and decreased renal length. Bladder evaluation includes wall thickness, presence of masses (transitional cell carcinoma appears as an irregular, papillary, or sessile intraluminal mass), stones, and diverticula (often with a narrow neck). Post-void residual volume is measured after voiding (normal less than 50 mL).

Spinal Cord Compression

– A radiologic emergency requiring urgent imaging and intervention to prevent permanent neurologic deficit. MRI is the reference standard, demonstrating the site, degree, and cause of cord compression with multiplanar capability. Causes include epidural metastatic disease (most common, with breast, lung, prostate, renal, and thyroid primaries), epidural hematoma, epidural abscess-discitis-osteomyelitis, herniated disc (acute cervical or thoracic), primary spinal tumors, and vertebral body collapse from osteoporosis or malignancy (pathological fracture). MRI demonstrates the level of compression (the site of cord deformation), the degree of stenosis (mild, moderate, or severe based on thecal sac cross-sectional area), the presence of cord signal change (T2 hyperintensity indicating cord edema, gliosis, or myelomalacia; T1 hypointensity indicating cavitation or necrosis, both associated with poorer prognosis), and the extent and morphology of the compressive lesion. Management is directed at the cause: surgical decompression (laminectomy, corpectomy, or stabilisation) for acute cord compression, radiotherapy for radiosensitive tumors (lymphoma, myeloma, breast cancer, prostate cancer), and antibiotics with drainage for infection. Corticosteroids (dexamethasone 10 mg IV loading dose followed by 4-6 mg IV every 6 hours) are given immediately to reduce cord edema before definitive treatment in cases of spinal cord compression from metastatic disease. Prognosis depends on the underlying etiology, the severity of the pre-treatment neurological deficit, and the time to treatment. Patients who are ambulatory at presentation have the best prognosis.

Thyroid Ultrasound

– The primary imaging modality for evaluating thyroid nodules, with the Thyroid Imaging Reporting and Data System (TI-RADS) standardizing risk stratification and biopsy recommendations. High-frequency linear transducer (12-15 MHz) provides detailed evaluation of nodule echogenicity, composition, shape, margin, and echogenic foci. TI-RADS categories range from TR1 (benign, no FNA) to TR5 (highly suspicious, FNA recommended at 1 cm or greater). Suspicious features include marked hypoechogenicity, taller-than-wide shape (AP greater than transverse), spiculated or irregular margins, microcalcifications, and extrathyroidal extension. Nodule size thresholds for FNA depend on the TI-RADS category: TR3 nodules $\geq$2.5 cm, TR4 nodules $\geq$1.5 cm, and TR5 nodules $\geq$1.0 cm may be considered for FNA. Additional findings on thyroid ultrasound include thyroiditis (Hashimoto thyroiditis shows a heterogeneous, hypoechoic parenchymal pattern with echogenic septations, sometimes with a micronodular pattern and hypervascularity), Graves disease (diffuse, homogeneous, hypoechoic enlargement with marked hypervascularity—thyroid inferno), and substernal extension of a goitre. Cervical lymph node assessment is essential in thyroid cancer evaluation: abnormal nodes are round (rather than oval), lack a fatty hilum, contain microcalcifications, have cystic components, and show peripheral rather than hilar vascularity.

Ultrasound

– Ultrasound imaging uses high-frequency sound waves (2-18 MHz in clinical practice) transmitted from a piezoelectric transducer into the body, with echoes returning from tissue interfaces to create real-time images. The transducer functions as both transmitter and receiver, with pulse-echo timing determining depth and echo amplitude determining brightness. Ultrasound wave propagation follows the principles of reflection, refraction, and attenuation at tissue boundaries with different acoustic impedances. The amount of echo returning to the transducer depends on the relative difference in acoustic impedance between two tissues (the greater the difference, the stronger the echo, which can be a source of diagnostic information but may also produce artefact). Image resolution is determined by transducer frequency: higher frequencies (10-18 MHz) provide better spatial resolution but less penetration (used for superficial structures such as thyroid, breast, and MSK), while lower frequencies (2-5 MHz) provide deeper penetration but lower resolution (used for abdominal and obstetric imaging). Ultrasound modes include B-mode (brightness mode, the standard 2D gray-scale anatomic imaging mode), M-mode (motion mode for high-temporal-resolution cardiac assessment), color Doppler (flow velocity and direction encoded as color overlays on B-mode images), power Doppler (the amplitude of the Doppler signal, more sensitive for slow flow but without directional information), pulsed-wave spectral

Doppler (velocity at a specific site), and continuous-wave spectral Doppler (velocity along the entire beam path). Ultrasound contrast agents (microbubbles) expand the role of ultrasound to functional evaluation and targeted imaging. Artefacts include acoustic shadowing (from calcifications and stones, blocking transmission of sound, producing a dark region posterior to the calcified structure), posterior acoustic enhancement (from fluid-filled structures such as cysts, increasing through-transmission beyond the cyst and producing a bright region), reverberation (from strong reflectors causing multiple parallel echoes), comet tail artefact (from small calcifications or metallic objects), ring-down artefact (from air bubbles), and mirror image artefact. The advantages of ultrasound include the absence of ionizing radiation, real-time imaging, portability, low cost, and wide availability. Limitations include operator dependence, limited penetration through bone and air, and a small field of view compared to cross-sectional imaging.

Vertebroplasty and Kyphoplasty

– Percutaneous image-guided procedures for painful osteoporotic vertebral compression fractures that have failed conservative management. Vertebroplasty involves direct injection of polymethylmethacrylate (PMMA) bone cement into the fractured vertebral body through a transpedicular approach under fluoroscopic or CT guidance, stabilizing the fracture and providing pain relief in 70-90 percent of patients. Kyphoplasty uses a balloon catheter (inflatable bone tamp) inserted into the vertebral body, inflated to restore vertebral body height and create a cavity, then withdrawn after cement injection under low pressure, which partially reduces the risk of cement extravasation compared to vertebroplasty. Indications include acute or subacute symptomatic osteoporotic vertebral compression fractures (less than 6-8 weeks of pain) refractory to conservative therapy, and painful vertebral metastases. Contraindications include active infection, uncorrectable coagulopathy, and fractures with retropulsion of fragments into the spinal canal. Major complications are rare (less than 2 percent) and include cement extravasation into the spinal canal (epidural space, neural foramen, or disc space), pulmonary cement embolisation (from cement entering the paravertebral veins, migrating to the pulmonary circulation, often asymptomatic but can cause respiratory compromise in severe cases), and infection. Procedural success rates are high, with significant pain relief reported in 70-90 percent of patients within 24-48 hours. Balloon kyphoplasty may provide better restoration of vertebral body height and reduction of kyphotic deformity compared to vertebroplasty, though the clinical significance of this is debated. The evidence base for these procedures has been controversial, with sham-controlled trials (e.g., the INVEST trial) showing no significant benefit over placebo injection, leading to changes in reimbursement and clinical practice guidelines. However, many patients with appropriately selected acute fractures do experience benefit.

VQ Scan

– Ventilation-perfusion (VQ) scintigraphy evaluates pulmonary blood flow and ventilation to diagnose pulmonary embolism, particularly useful when CT pulmonary angiography is contraindicated (renal failure, contrast allergy, pregnancy). The perfusion phase uses Tc-99m macroaggregated albumin (MAA) particles injected intravenously, which lodge in pulmonary capillaries proportional to regional blood flow. The ventilation phase uses inhaled Tc-99m diethylenetriamine pentaacetic acid (DTPA) aerosol or Technegas (ultrafine carbon particles labelled with Tc-99m). Interpretation uses the PIOPED criteria (Prospective Investigation of Pulmonary Embolism Diagnosis), which provide the probability of PE based on the size, number, and location of perfusion defects relative to ventilation abnormalities. A normal VQ scan reliably excludes PE. A high-probability scan (two or more large segmental perfusion defects with normal ventilation) has a greater than 85 percent probability of PE. An intermediate-probability scan requires further testing. VQ scanning is preferred over CTPA in younger patients, patients with renal impairment, and those with contrast allergy, as it avoids iodinated contrast. The radiation dose from VQ scanning is comparable to or slightly lower than CTPA. A chest radiograph is required for correlation with VQ findings (to avoid interpreting radiographic abnormalities as perfusion defects and to exclude an alternative cause for the perfusion abnormality). Limitations include indeterminate results in patients with underlying lung disease causing ventilation abnormalities (COPD, asthma, pneumonia, atelectasis, pleural effusion), and reduced specificity in these patients (intermediate-probability scans are common).

Chapter 33

Rheumatology

Adult-Onset Still Disease

– A rare systemic autoinflammatory disorder of unknown etiology, characterized by quotidian (daily) spiking fevers, evanescent salmon-colored rash, arthritis or arthralgia, sore throat, lymphadenopathy, hepatosplenomegaly, and markedly elevated serum ferritin. The pathogenesis involves excessive production of interleukin-1 (IL-1) and interleukin-18 (IL-18), placing AOSD in the category of autoinflammatory rather than autoimmune disease. Yamaguchi criteria for diagnosis require the presence of five or more features including at least two major criteria (quotidian fever for at least one week, typical rash, and arthralgia or arthritis for at least two weeks) after excluding other conditions. Serum ferritin is markedly elevated, often more than five times the upper limit of normal, with low glycosylated ferritin. Treatment includes high-dose corticosteroids, with steroid-sparing agents such as methotrexate, and biologic therapies targeting IL-1 (anakinra, canakinumab) or IL-6 (tocilizumab) for refractory disease.

ANCA-Associated Vasculitis

– granulomatosis with polyangiitis (GPA, c-ANCA/PR3-ANCA positive), microscopic polyangiitis (MPA, p-ANCA/MPO-ANCA positive), and eosinophilic granulomatosis with polyangiitis (EGPA, p-ANCA/MPO-ANCA positive, asthma, eosinophilia). GPA: upper airway (sinusitis, saddle nose), lower airway (nodules, cavitary lesions), renal (pauci-immune necrotizing GN). MPA: renal-pulmonary syndrome without granulomas. EGPA: asthma, eosinophilia, neuropathy, cardiac, GI, skin. Treatment: glucocorticoids + rituximab or cyclophosphamide for induction, then rituximab or azathioprine/methotrexate for maintenance.

Ankylosing Spondylitis

– A chronic, progressive inflammatory disease primarily affecting the axial skeleton (sacroiliac joints and spine), characterized by inflammation at entheses (where tendons, ligaments, and joint capsules insert into bone), leading to new bone formation, syndesmophyte development, and eventual vertebral fusion (bamboo spine). It is the prototype of the spondyloarthritis family of diseases and is strongly associated with HLA-B27 (present in 80-90 percent of patients, though HLA-B27 is present in 6-8 percent of the general population, meaning most HLA-B27-positive individuals never develop the disease). Clinical features: inflammatory back pain (improved by exercise, worse with rest, morning stiffness >30 minutes), alternating buttock pain, peripheral arthritis (asymmetric, lower limb predominant), enthesitis (Achilles tendinitis, plantar fasciitis), uveitis (acute anterior, recurrent), and in advanced cases, spinal fusion leading to loss of lumbar lordosis, increased thoracic kyphosis, and limited spinal mobility. Diagnosis is based on modified New York criteria requiring sacroiliitis on imaging plus at least one clinical criterion. Treatment includes NSAIDs as first-line, TNF inhibitors (adalimumab, etanercept, infliximab, golimumab, certolizumab), IL-17 inhibitors (secukinumab, ixekizumab), and JAK inhibitors (tofacitinib, upadacitinib) for active disease, along with physiotherapy to maintain spinal mobility.

Antiphospholipid Syndrome

– A systemic autoimmune thrombophilia characterized by arterial and venous thrombosis, pregnancy morbidity, and persistent antiphospholipid antibody positivity. Antiphospholipid antibodies are a heterogeneous group of autoantibodies targeting phospholipid-binding proteins: lupus anticoagulant (LA, detected by prolongation of phospholipid-dependent coagulation assays that fails to correct with mixing study, paradoxically associated with thrombosis), anticardiolipin antibodies (aCL, detected by ELISA), and anti-beta2-glycoprotein I antibodies (anti-beta2GPI). Diagnosis requires at least one clinical criterion (vascular thrombosis or pregnancy morbidity) and at least one laboratory criterion (LA, aCL, or anti-beta2GPI positive on two or more occasions at least 12 weeks apart). Catastrophic APS (CAPS) is a rare, rapidly fatal variant with multiple thromboses in a short period, often triggered by infection, surgery, or anticoagulation withdrawal. Treatment: anticoagulation with warfarin (target INR 2-3 for first thrombosis, 3-4 for recurrent), plus low-dose aspirin in pregnancy; CAPS requires anticoagulation, high-dose glucocorticoids, plasma exchange, and IVIG.

Antisynthetase Syndrome

– Defined by the presence of anti-aminoacyl-tRNA synthetase antibodies (anti-Jo-1 most common, 75%; anti-PL-7, anti-PL-12, anti-EJ, anti-OJ, anti-KS, anti-Zo, anti-Ha, anti-YRS, anti-FRS, anti-TRS, anti-ARS) with clinical triad: interstitial lung disease (most common, often progressive), inflammatory myopathy (DM/PM), and inflammatory arthritis (non-erosive). Also: mechanic's hands (hyperkeratotic fissuring of radial fingers), Raynaud, fever. Treatment: glucocorticoids + rituximab or mycophenolate for ILD and myositis; calcineurin inhibitors (tacrolimus, cyclosporine) for refractory disease.

Arthritis

– Inflammation of one or more joints, characterized by joint pain, swelling, warmth, and stiffness. Arthritis is classified into several major categories: osteoarthritis (degenerative, the most common form, characterized by articular cartilage loss, subchondral sclerosis, and osteophyte formation, typically affecting weight-bearing joints such as knees, hips, and spine as well as hands); rheumatoid arthritis (autoimmune, symmetric polyarthritis affecting small joints of hands and feet with pannus formation causing marginal erosions and periarticular osteopenia); spondyloarthritis (including ankylosing spondylitis, psoriatic arthritis, reactive arthritis, and enteropathic arthritis, typically presenting with inflammatory back pain, asymmetric oligoarthritis, enthesitis, and association with HLA-B27); crystal arthropathies (gout and calcium pyrophosphate deposition disease); septic arthritis (bacterial infection of the joint, a medical emergency requiring urgent drainage and antibiotics); and other arthritides associated with connective tissue diseases, infection, or haemochromatosis.

Autoinflammatory Bone Diseases

– Chronic recurrent multifocal osteomyelitis (CRMO/CNO): see above. Majeed syndrome (LPIN2 mutations): CRMO + congenital dyserythropoietic anemia + Sweet syndrome. DIRA (IL1RN mutations): neonatal pustulosis, periostitis, osteitis. Cherubism (SH3BP2 mutations): bilateral jaw swelling, multilocular radiolucencies, self-limited after puberty. CANDLE syndrome (see above).

Behcet Disease

– A systemic vasculitis of unknown etiology characterized by recurrent oral aphthous ulcers (major criterion), genital ulcers, skin lesions (erythema nodosum, pseudofolliculitis), ocular inflammation (posterior uveitis, retinal vasculitis - leading cause of blindness), neurologic (neuro-Behcet), vascular (arterial aneurysms, venous thrombosis), GI, and arthritic manifestations. Pathergy test positive (skin hyperreactivity). International criteria: recurrent oral ulceration (>=3/year) plus at least two of: genital ulceration, eye lesions, skin lesions, or pathergy test. Treatment: topical steroids for mucocutaneous ulcers, colchicine for mucous membrane involvement, systemic corticosteroids and immunosuppressants (azathioprine, cyclophosphamide, TNF inhibitors) for severe organ-threatening disease including ocular, neurologic, and vascular involvement.

Calcium Pyrophosphate Deposition Disease

– A crystal arthropathy caused by deposition of calcium pyrophosphate dihydrate crystals in articular cartilage (chondrocalcinosis). Presents with acute pseudogout (most common): acute monoarticular inflammation of the knee or wrist, mimicking gout but less painful. Chronic CPPD: degenerative arthritis-like with OA pattern (knee, wrist, MCP joints). Pseudo-rheumatoid: symmetric polyarticular inflammation. Associated with aging, OA, hemochromatosis, hyperparathyroidism, hypomagnesemia, hypothyroidism. Diagnosis: joint aspiration shows weakly positively birefringent rhomboid crystals. Treatment: NSAIDs, colchicine, intra-articular corticosteroids for acute flares; low-dose colchicine for prophylaxis.

CANDLE Syndrome

– Chronic atypical neutrophilic dermatosis with lipodystrophy and elevated temperature (CANDLE, PSMB8 mutations, proteasome dysfunction) causes recurrent fever, annular violaceous plaques, progressive lipodystrophy (loss of subcutaneous fat), joint contractures, developmental delay, basal ganglia calcifications. Treatment: JAK inhibitors (baricitinib, ruxolitinib) improve skin, fever, lipodystrophy.

Childhood Vasculitis

– Kawasaki disease (see medium vessel vasculitis). Henoch-Schönlein purpura (IgA vasculitis): palpable purpura, abdominal pain, arthralgia, renal (IgA nephropathy), self-limited in most; corticosteroids for severe GI/renal. Polyarteritis nodosa in children: similar to adults, hepatitis B association. Takayasu arteritis: adolescent girls. ANCA vasculitis: rare in children.

Cryopyrin-Associated Periodic Syndromes

– Autosomal dominant autoinflammatory disorders (NLRP3 mutations, gain-of-function) causing constitutive inflammasome activation and IL-1beta overproduction. Spectrum: familial cold autoinflammatory syndrome (FCAS, cold-induced urticaria, fever, conjunctivitis), Muckle-Wells syndrome (MWS, progressive sensorineural hearing loss, amyloidosis risk), neonatal-onset multisystem inflammatory disease (NOMID/CINCA, severe CNS involvement, bony overgrowth, chronic meningitis, cognitive impairment). Diagnosis is clinical with genetic confirmation. Treatment: IL-1 inhibitors (anakinra, canakinumab, rilonacept) are dramatically effective, suppressing inflammation and preventing disease progression.

Dactylitis

– Inflammation of an entire digit (finger or toe), producing a diffuse, fusiform swelling often described as sausage-shaped. Most commonly associated with psoriatic arthritis and reactive arthritis (enthesitis-related), but also occurs in sickle cell disease (hand-foot syndrome in young children, caused by dactylitis from vaso-occlusive crises in the small bones of the hands and feet) and tuberculous dactylitis (spina ventosa–bony expansion from granulomatous infection). Diagnosis based on clinical appearance and underlying condition.

Drug-Induced Autoimmunity

– Drug-induced lupus (DIL): procainamide, hydralazine, isoniazid, minocycline, TNF inhibitors, anti-TNF; anti-histone antibodies, positive ANA, negative anti-dsDNA/anti-Sm; symptoms resolve after drug cessation. Drug-induced ANCA vasculitis: propylthiouracil, hydralazine, minocycline, allopurinol; MPO-ANCA positive, pauci-immune GN, pulmonary hemorrhage. Drug-induced subacute cutaneous lupus: hydrochlorothiazide, calcium channel blockers, TNF inhibitors; anti-Ro/SSA positive, annular/papulosquamous photosensitive rash. Diagnosis is clinical with drug history and serology. Management is withdrawal of the offending drug with supportive care; symptoms typically resolve over weeks to months.

Enteropathic Arthritis

– Spondyloarthritis associated with inflammatory bowel disease (Crohn disease, ulcerative colitis), affecting 10-20% of IBD patients. Two patterns: type 1 (pauciarticular, acute, self-limited, parallels IBD activity, large joints), type 2 (polyarticular, chronic, independent of IBD activity, symmetric small joints, resembles RA but RF-negative). Axial involvement (sacroiliitis) in 10-20%, often asymptomatic. Treatment: optimize IBD control (TNF inhibitors, vedolizumab for gut-selective therapy), NSAIDs for symptomatic relief (though they may exacerbate IBD), sulfasalazine, methotrexate, and rarely surgery for refractory arthritis.

Familial Mediterranean Fever

– An autosomal recessive autoinflammatory disorder (MEFV gene mutations, pyrin/inflammasome dysregulation) causing recurrent self-limited febrile attacks with serositis (peritonitis, pleuritis, synovitis), erysipelas-like erythema, lasting 12-72 hours. Predominantly affects Mediterranean populations (Armenians, Turks, Arabs, Sephardic Jews). Complication: AA amyloidosis (renal, hepatic, splenic) from chronic inflammation if untreated. Diagnosis: Tel-Hashomer criteria: major (typical attacks with peritonitis, pleuritis, synovitis, or erysipelas-like erythema) plus minor (incomplete attacks). Treatment: colchicine is highly effective for preventing attacks and AA amyloidosis; IL-1 inhibitors (anakinra, canakinumab) for colchicine-resistant cases.

Fibromyalgia

– A chronic centralized pain syndrome characterized by widespread musculoskeletal pain (>3 months), fatigue, sleep disturbance, cognitive dysfunction (fibro fog), and somatic symptoms, affecting 2-4% of population (female:male 9:1). Pathophysiology: central sensitization, altered pain processing, neurotransmitter dysregulation (serotonin, norepinephrine, dopamine, substance P). 2016 ACR criteria: widespread pain index (WPI) >=7 and symptom severity scale (SSS) >=5 OR WPI 4-6 and SSS >=9, plus generalized pain in at least 4 of 5 regions, symptoms present at similar level for at least 3 months, and absence of another disorder explaining the pain. Management: non-pharmacological (exercise, cognitive behavioural therapy, sleep hygiene, patient education) combined with pharmacotherapy (amitriptyline, duloxetine, milnacipran, pregabalin, gabapentin). Multimodal, individualised treatment is most effective.

Giant Cell Arteritis

– A large-vessel vasculitis of adults >50 (peak 70-80) affecting cranial branches of carotid arteries, with risk of permanent vision loss from anterior ischemic optic neuropathy. Clinical: new-onset headache (temporal), scalp tenderness, jaw claudication, visual disturbances (amaurosis fugax, diplopia), polymyalgia rheumatica symptoms. Diagnosis: temporal artery biopsy (gold standard, skip lesions reduce sensitivity), ultrasound (halo sign), PET-CT (large-vessel involvement (aorta, subclavian). Treatment: high-dose glucocorticoids (prednisolone 40-60 mg/day) promptly to prevent blindness; tocilizumab (IL-6 inhibitor) as steroid-sparing agent; low-dose aspirin for thromboprophylaxis. Methotrexate, azathioprine as alternatives.

Gout

– A crystal deposition disease caused by monosodium urate (MSU) crystal formation in joints and tissues due to chronic hyperuricemia (serum urate >6.8 mg/dL). Pathophysiology: urate overproduction (10%) or underexcretion (90%); crystals activate NLRP3 inflammasome -> IL-1beta release -> neutrophilic inflammation. Clinical: acute monoarticular arthritis (1st MTP podagra 50% first attack), intense pain, erythema, swelling; tophi in chronic disease; urate nephrolithiasis. Diagnosis: synovial fluid aspiration showing negatively birefringent needle-shaped crystals under polarised light microscopy, elevated serum urate. Treatment: acute attack management with NSAIDs, colchicine, or corticosteroids; long-term urate-lowering therapy (allopurinol, febuxostat, probenecid, uricosurics) targeting serum urate <6 mg/dL (5 mg/dL if tophi present). Prophylaxis with NSAIDs or colchicine during initial urate-lowering. Lifestyle modification: avoid purine-rich foods (red meat, shellfish, organ meats), reduce alcohol (especially beer), maintain hydration.

Granulomatosis with Polyangiitis (Wegener's Granulomatosis)

– Granulomatosis with polyangiitis (GPA, formerly Wegener granulomatosis) is a rare, ANCA-associated necrotising granulomatous vasculitis affecting small to medium-sized blood vessels, primarily in the respiratory tract and kidneys. Epidemiology: incidence 8-12 per million, peak age 40-60, slight male predominance. Pathophysiology: pauci-immune vasculitis (no immune complex deposition) with c-ANCA (anti-PR3, 90% in active generalised disease, 50% in localised disease) targeting proteinase 3 (PR3) in neutrophil granules. PR3-ANCA activates neutrophils, causing degranulation, respiratory burst, and endothelial damage. Clinical: upper respiratory tract (chronic sinusitis, epistaxis, nasal crusting, saddle nose deformity from nasal septal perforation, otitis media, subglottic stenosis), lower respiratory tract (cough, dyspnoea, hemoptysis, pulmonary nodules/cavitary lesions on imaging), renal (crescentic pauci-immune glomerulonephritis $\rightarrow$ rapidly progressive glomerulonephritis, haematuria, proteinuria, renal failure), and other (cutaneous vasculitis/purpura, ocular/scleritis/episcleritis, mononeuritis multiplex, arthralgia). Limited GPA: disease confined to the respiratory tract without renal involvement. Diagnosis: clinical, c-ANCA/anti-PR3, tissue biopsy (nasal, lung, or kidney) showing necrotising granulomatous inflammation and vasculitis.

Treatment: induction with high-dose glucocorticoids plus rituximab or cyclophosphamide (efficacy equal); maintenance with rituximab (q6 months) or azathioprine/methotrexate. Methotrexate for non-organ-threatening limited disease. Trimethoprim-sulfamethoxazole for limited upper respiratory tract disease and Pneumocystis jirovecii prophylaxis. Prognosis: without treatment, mortality >90% at 2 years (renal failure or respiratory failure). With treatment, remission in 75-90%, but relapse rate is 30-50% within 5 years.

IgG4-Related Disease

– A fibro-inflammatory condition characterized by tumefactive lesions in multiple organs, dense lymphoplasmacytic infiltrate with >10 IgG4+ plasma cells/HPF and IgG4+/IgG+ ratio >40%, elevated serum IgG4 (>135 mg/dL), obliterative phlebitis, storiform fibrosis. Affects pancreas (type 1 autoimmune pancreatitis), salivary/lacrimal glands (Mikulicz disease), retroperitoneum, orbits, kidneys, lungs, aorta. Mimics malignancy, Sjögren, lymphoma. Treatment: glucocorticoids (first-line), rituximab, azathioprine, mycophenolate, methotrexate; disease-modifying anti-rheumatic drugs are often needed long-term.

Immune-Mediated Necrotizing Myopathy

– A subacute severe proximal myopathy with very high CK (often >10,000), minimal inflammation on biopsy, necrotic fibers. Two major autoantibodies: anti-HMGCR (3-hydroxy-3-methylglutaryl-CoA reductase, statin-associated or spontaneous) and anti-SRP (signal recognition particle). Anti-HMGCR: older, statin exposure, severe; anti-SRP: younger, more cardiac/respiratory involvement, worse prognosis. Treatment: high-dose glucocorticoids + rituximab or IVIG as first-line; additional immunosuppressants (methotrexate, azathioprine, mycophenolate) and physical therapy for rehabilitation.

Inclusion Body Myositis

– The most common acquired myopathy after age 50, characterized by asymmetric distal and proximal weakness (finger flexors, quadriceps), rimmed vacuoles, 15-18 nm tubulofilaments, TDP-43 pathology, endomysial inflammation with CD8+ T cells invading non-necrotic fibers. Refractory to immunotherapy. Clinical: dysphagia (40%), falls from quadriceps weakness. Diagnostic criteria (ENMC 2011): clinical + EMG + biopsy. Treatment: supportive (fall prevention, dysphagia management, speech therapy, and assistive devices for mobility. Clinical trials with arimoclomol, bimagrumab, and follistatin gene therapy are ongoing.

Infection-Related Rheumatic Syndromes

– Post-streptococcal reactive arthritis (PSRA): asymmetric oligoarthritis 1-2 weeks after GAS pharyngitis, HLA-B27 negative, prolonged course, no carditis; distinguish from acute rheumatic fever. Lyme arthritis: Borrelia burgdorferi; monoarticular knee arthritis, positive serology; antibiotic treatment (doxycycline/amoxicillin); refractory Lyme arthritis (post-infectious immune-mediated). Viral arthritis: parvovirus B19 (acute polyarthritis, mimics RA, self-limited), hepatitis B/C (polyarthritis, usually self-limited, cryoglobulin-associated vasculitis), rubella, HIV, and alphaviruses (chikungunya, Ross River). Diagnosis: serology, PCR, exposure history. Treatment: symptomatic (NSAIDs, corticosteroids for severe), treat underlying infection when applicable.

Interferonopathies

– Type I interferonopathies: Aicardi-Goutières syndrome (TREX1, RNASEH2, SAMHD1 mutations, neonatal encephalopathy, calcifications, chilblains, SLE-like); STING-associated vasculopathy with onset in infancy (SAVI, TMEM173 mutations, severe interstitial lung disease, digital ischemia); Proteasome-associated autoinflammatory syndromes (PRAAS, PSMB8 mutations, CANDLE syndrome: lipodystrophy, fever, skin lesions, joint contractures). Treatment: JAK inhibitors (baricitinib, ruxolitinib) target IFN signalling. Baricitinib approved for SAVI and CANDLE. hematopoietic stem cell transplantation in severe Aicardi-Goutières syndrome.

Janus Kinase Inhibitors

– JAK inhibitors (tofacitinib, baricitinib, upadacitinib, filgotinib, abrocitinib) block JAK-STAT signaling downstream of cytokine receptors (IL-6, IFN-gamma, IL-2, IL-7, IL-15, IL-21, GM-CSF, erythropoietin, thrombopoietin). Selectivity: JAK1/JAK3 (tofacitinib), JAK1/JAK2 (baricitinib), JAK1 (upadacitinib, filgotinib, abrocitinib).

Indications: RA, PsA, axial SpA, UC, AD, alopecia areata. Safety: herpes zoster (vaccinate Shingrix before), VTE (risk factors: age, obesity, prior VTE), infections, lipid elevation, transaminitis, neutropenia, anemia, GI perforation. Drug interactions: CYP3A4 (tofacitinib, upadacitinib). Dose adjustment for renal impairment.

Juvenile Dermatomyositis

– Juvenile dermatomyositis (JDM) presents with proximal weakness, heliotrope rash, Gottron papules, calcinosis (20-30%), lipodystrophy, vasculopathy (nailfold capillary changes). Anti-Mi-2 (classic DM, good prognosis), anti-NXP2 (calcinosis, malignancy risk low in children), anti-TIF1-gamma (malignancy risk in adults). Calcinosis is major morbidity. Treatment: glucocorticoids, methotrexate, hydroxychloroquine, IVIG for refractory; rituximab, TNF inhibitors, abatacept for refractory; bisphosphonates for calcinosis. JDM has better prognosis than adult DM; most children achieve remission but calcinosis can cause significant long-term disability.

Juvenile Idiopathic Arthritis

– The most common chronic rheumatic disease of childhood (onset <16 years), encompassing 7 subtypes: systemic JIA (Still disease of childhood, quotidian fever, salmon rash, arthritis, hepatosplenomegaly, serositis, macrophage activation syndrome risk), oligoarticular JIA (4 or fewer joints in first 6 months, high risk of chronic anterior uveitis, ANA positive), polyarticular JIA (5 or more joints, RF-negative or RF-positive), psoriatic JIA (arthritis + psoriasis), enthesitis-related JIA (HLA-B27, male > female, enthesitis, acute uveitis), and undifferentiated JIA. Treatment: NSAIDs, intra-articular corticosteroids, methotrexate, TNF inhibitors, IL-6 inhibitors (tocilizumab), IL-1 inhibitors (anakinra) for systemic JIA. Uveitis screening essential (ANA-positive, oligoarticular).

Juvenile Systemic Sclerosis

– Rare, onset <16 years, more diffuse cutaneous involvement, higher cardiac and renal crisis risk than adults, less pulmonary hypertension. Autoantibodies: anti-Scl-70 (dcSSc), anti-centromere (lcSSc), anti-RNA pol III. Treatment similar to adults but growth considerations; cyclophosphamide for severe ILD; ACE inhibitors for renal crisis; multidisciplinary pediatric rheumatology care.

Lupus

– A chronic autoimmune disease that can affect multiple organ systems. See Systemic Lupus Erythematosus (SLE).

Lupus Erythematosus (Systemic Lupus Erythematosus - SLE)

– See Systemic Lupus Erythematosus (SLE).

Macrophage Activation Syndrome

– A life-threatening complication of rheumatic diseases (especially systemic JIA, AOSD, SLE, Kawasaki) characterized by uncontrolled activation of macrophages and T cells causing hemophagocytosis, cytopenias, hyperferritinemia, hypertriglyceridemia, hypofibrinogenemia, coagulopathy, liver dysfunction, neurologic symptoms. HLH-2004 criteria: fever, splenomegaly, cytopenias (2 lineages), hypertriglyceridemia/hypofibrinogenemia, hemophagocytosis, low NK activity, elevated soluble CD25 (sIL-2R). Hscore or MS score aid diagnosis. Treatment: high-dose corticosteroids, cyclosporine, etoposide (for HLH), anakinra (IL-1 blockade) for MAS in systemic JIA/AOSD. Prompt recognition and aggressive treatment are essential to reduce mortality.

Microscopic Polyangiitis

– A necrotizing vasculitis affecting small blood vessels (capillaries, venules, arterioles) with few or no immune deposits (pauci-immune). Part of the ANCA-associated vasculitis group. Strongly associated with MPO-ANCA (p-ANCA, 80%). Presents with rapidly progressive glomerulonephritis (renal: hematuria, proteinuria, crescents), pulmonary capillaritis (hemoptysis, diffuse alveolar hemorrhage), palpable purpura, mononeuritis multiplex, and other organ involvement. Diagnosis: ANCA serology, renal or lung biopsy showing pauci-immune necrotizing inflammation. Treatment: high-dose corticosteroids + cyclophosphamide or rituximab for induction; azathioprine or rituximab for maintenance. Untreated, rapidly fatal.

Mixed Connective Tissue Disease

– An overlap syndrome with features of systemic lupus erythematosus, systemic sclerosis, and polymyositis/dermatomyositis. Highly associated with anti-U1 RNP antibodies. Presents with Raynaud phenomenon (nearly universal), puffy hands, arthritis, myositis, esophageal dysmotility, and serositis. Renal disease is less common than in SLE. Diagnosis: clinical + anti-U1 RNP positivity without sufficient criteria for other CTDs. Treatment: hydroxychloroquine, corticosteroids, and immunosuppressants (mycophenolate, methotrexate). Prognosis varies with predominant disease manifestation.

Osteitis Pubis

– Inflammation of the pubic symphysis and adjacent bone, causing groin/pelvic pain in athletes (soccer, hockey, running), postpartum women, post-urologic surgery. MRI shows bone marrow edema, sclerosis, fluid at symphysis. Treatment: rest, NSAIDs, physical therapy (core/pelvic floor strengthening), corticosteroid injection; rarely surgical fusion.

Osteoarthritis

– The most common joint disease, characterized by progressive cartilage loss, subchondral bone sclerosis, osteophyte formation, and synovial inflammation, affecting knees, hips, hands (DIP/PIP Heberden/Bouchard nodes), spine. Risk factors: age, obesity, joint injury, malalignment, genetics, female sex. Clinical: activity-related pain, brief morning stiffness (<30 min), crepitus, bony enlargement, functional limitation. Diagnosis: clinical + radiographic (joint space narrowing, osteophytes, subchondral sclerosis, cysts). Management: nonpharmacological (weight loss, exercise, physiotherapy, joint protection), pharmacological (paracetamol, NSAIDs topical/oral, intra-articular corticosteroids, hyaluronic acid injections), and surgical (joint replacement) for end-stage disease.

Osteonecrosis (Avascular Necrosis)

– Osteonecrosis (AVN) is bone death from vascular compromise, affecting femoral head (most common), humeral head, femoral condyles, talus. Risk factors: glucocorticoids (high dose, cumulative), alcohol, trauma, sickle cell disease, SLE, antiphospholipid syndrome, pancreatitis, decompression sickness, radiation, organ transplantation. ARCO classification: stage 0 (normal X-ray, MRI+), I (normal X-ray, MRI+), II (crescent sign), III (subchondral collapse), IV (joint space narrowing, OA). Treatment: core decompression for early stages, osteotomy, and joint replacement for advanced collapse. Bisphosphonates may reduce risk in high-dose glucocorticoid therapy.

Other Types of Arthritis

– Other types of arthritis beyond osteoarthritis and rheumatoid arthritis include psoriatic arthritis (seronegative spondyloarthropathy associated with psoriasis, enthesitis, dactylitis, and nail changes; treated with NSAIDs, DMARDs, TNF inhibitors, IL-17/IL-23 inhibitors), ankylosing spondylitis (axial spondyloarthropathy with inflammatory back pain, sacroiliitis, spinal fusion/bamboo spine; HLA-B27 associated; NSAIDs, TNF inhibitors, IL-17 inhibitors), gout (monosodium urate crystal deposition causing acute monoarticular inflammatory arthritis, tophi, urate nephropathy; acute—colchicine, NSAIDs, corticosteroids; chronic—urate-lowering therapy such as allopurinol, febuxostat), pseudogout (calcium pyrophosphate deposition disease, CPPD), reactive arthritis (post-infectious, urethritis/cervicitis, conjunctivitis), septic arthritis (bacterial joint infection requiring urgent drainage and antibiotics), and juvenile idiopathic arthritis (JIA, childhood arthritis with oligoarticular, polyarticular, or systemic subtypes). Diagnosis requires joint aspiration for crystal analysis and culture, serologies (RF, anti-CCP, ANA, HLA-B27), and imaging. Treatment targets inflammation, prevents joint destruction, and maintains function.

Overlap Myositis

– Patients with features of inflammatory myopathy (DM/PM/IMNM) plus another systemic autoimmune disease (SLE, SSc, MCTD, Sjogren). Anti-PM-Scl (Scl/PM overlap, limited SSc + myositis, good prognosis); anti-Ku (SLE/PM overlap); anti-U1-RNP (MCTD with myositis). Treatment: tailored to predominant organ involvement; often requires combination immunosuppression.

Palindromic Rheumatism

– Characterized by recurrent, self-limited attacks of mono- or oligoarthritis (hours to days), periarticular soft tissue swelling, no radiographic erosions between attacks. 30-50% evolve to RA (anti-CCP positive predicts progression). Treatment: NSAIDs for attacks; hydroxychloroquine reduces frequency; low-dose colchicine; DMARDs if progressing to RA.

Polymyalgia Rheumatica

– An inflammatory syndrome of older adults (age >50, peak 70-80) causing bilateral shoulder and pelvic girdle pain and stiffness lasting >45 minutes, with elevated inflammatory markers (ESR >40, CRP elevated). Strongly associated with giant cell arteritis (15-20% develop GCA; 40-50% of GCA patients have PMR). 2012 ACR/EULAR provisional classification: age >=50, bilateral shoulder pain, abnormal ESR/CRP, morning stiffness >45 min, new hip pain, absence of RF/anti-CCP, absence of other inflammatory arthritis. Treatment: low-dose prednisolone 15-25 mg/day (not exceeding 20 mg/day in most guidelines) with rapid taper; methotrexate as steroid-sparing; tocilizumab for refractory cases. Relapse common during steroid taper.

Polymyositis

– An idiopathic inflammatory myopathy characterized by symmetric, proximal muscle weakness (shoulders, hips) with elevated muscle enzymes (CK, aldolase, LDH). More common in women, peaks at 30-50 years. Diagnosis: EMG (myopathic units with fibrillations), muscle MRI (edema), and muscle biopsy (CD8+ T cell infiltration of MHC-I expressing muscle fibers). Autoantibodies: anti-Jo-1 (antisynthetase syndrome with interstitial lung disease, arthritis, mechanic's hands). Treatment: high-dose corticosteroids, methotrexate, azathioprine, mycophenolate, and IVIG for refractory disease.

Polymyositis and Dermatomyositis

– Idiopathic inflammatory myopathies (IIM) include dermatomyositis (DM), polymyositis (PM), immune-mediated necrotizing myopathy (IMNM), and inclusion body myositis (IBM). DM: heliotrope rash, Gottron papules, Shawl/V-sign, mechanic's hands, calcinosis, ILD, malignancy association (anti-TIF1-gamma, anti-NXP2). PM: proximal weakness, elevated CK, EMG/myopathy on biopsy (CD8+ endomysial inflammation), no rash, rare, often overdiagnosed (many are IMNM or IBM). IMNM: severe proximal weakness, very high CK, necrotic fibers without inflammation, anti-HMGCR or anti-SRP. IBM: age >50, finger flexor and quadriceps weakness, rimmed vacuoles, refractory. Malignancy screening essential in adult DM (anti-TIF1-gamma, anti-NXP2). Treatment: glucocorticoids + steroid-sparing agents (methotrexate, azathioprine, mycophenolate) for DM/PM; IVIG for DM; rituximab for refractory cases; physiotherapy for muscle strengthening.

Psoriatic Arthritis

– An inflammatory arthritis associated with psoriasis (precedes arthritis in 70%, simultaneous 15%, follows 15%), affecting 20-30% of psoriasis patients. CASPAR criteria: inflammatory articular disease plus psoriasis (2 points), nail dystrophy (1), negative RF (1), dactylitis (1), juxta-articular new bone formation (1); score >=3. Patterns: distal interphalangeal predominant, asymmetric oligoarthritis, symmetric polyarthritis (RA-like), spondylitis, arthritis mutilans. Treatment: NSAIDs for mild disease; methotrexate, leflunomide, sulfasalazine as first-line DMARDs; TNF inhibitors, IL-17 inhibitors (secukinumab, ixekizumab), IL-23 inhibitors (guselkumab), and JAK inhibitors (tofacitinib, upadacitinib) for moderate-to-severe disease; physiotherapy and occupational therapy.

Pyoderma Gangrenosum

– A neutrophilic dermatosis causing rapidly expanding, painful ulcers with violaceous undermined borders, pathergy phenomenon. Associated with IBD (50%), hematologic malignancy, rheumatoid arthritis, PAPA syndrome. Peristomal PG in ostomy patients. Treatment: high-potency topical steroids, intralesional steroids, systemic corticosteroids, cyclosporine, TNF inhibitors (infliximab, adalimumab); avoid surgical debridement (worsens via pathergy).

Reactive Arthritis

– A post-infectious spondyloarthritis occurring 1-4 weeks after genitourinary (Chlamydia trachomatis) or gastrointestinal (Salmonella, Shigella, Campylobacter, Yersinia) infection. Classic triad (Reiter syndrome): arthritis, conjunctivitis/uveitis, urethritis/cervicitis (only 1/3 have full triad). HLA-B27 positive in 60-80%. Clinical: asymmetric oligoarthritis (lower extremities), enthesitis (Achilles, plantar fascia), dactylitis, circinate balanitis, keratoderma blennorrhagicum (palmar/plantar pustulosis). Treatment: NSAIDs, corticosteroids (oral or intra-articular), DMARDs (sulfasalazine, methotrexate) for persistent arthritis, TNF inhibitors for refractory cases. Most cases resolve over months but may recur with subsequent infections.

Reiter's Syndrome

– See Reactive Arthritis.

Relapsing Polychondritis

– A rare systemic autoimmune disease characterized by recurrent inflammation and destruction of cartilaginous structures throughout the body, particularly the ears, nose, respiratory tract, and joints. The pathogenesis involves autoimmune targeting of type II collagen (the predominant collagen in hyaline cartilage) and type X collagen (in growth plates). Clinical presentation: auricular chondritis (painful, erythematous, swollen ears with sparing of the earlobe, the classic distinguishing feature), nasal chondritis (saddle nose deformity), laryngotracheal involvement (hoarseness, stridor, life-threatening airway collapse), costochondritis, and non-erosive arthropathy. Ocular inflammation (scleritis, episcleritis, uveitis), audiovestibular involvement (sensorineural hearing loss, vertigo), cardiac involvement (aortic regurgitation from aortitis, pericarditis), and renal and skin involvement may occur. Diagnosis is clinical (McAdam criteria). Treatment: corticosteroids (first-line), dapsone or colchicine for mild disease, and immunosuppressants (methotrexate, azathioprine, cyclophosphamide, mycophenolate, TNF inhibitors) for severe or refractory disease. Airway involvement requires urgent evaluation and possible tracheostomy.

Rheumatoid Arthritis

– A chronic, symmetric, erosive inflammatory polyarthritis primarily affecting small joints of the hands and feet, with a female-to-male ratio of 3:1 and peak onset 40-60 years. Pathogenesis involves genetic susceptibility (HLA-DRB1 shared epitope), environmental triggers (smoking, periodontal disease), and immune dysregulation (citrullination of proteins, anti-CCP antibody production, rheumatoid factor, cytokine-driven synovial inflammation with TNF, IL-6, IL-1). Clinical: symmetric small joint polyarthritis (MCP, PIP, MTP), morning stiffness >1 hour, rheumatoid nodules, extra-articular manifestations (Sjogren, interstitial lung disease, pericarditis, vasculitis, Felty syndrome). Diagnosis: 2010 ACR/EULAR criteria: joint involvement (0-5), serology (RF and anti-CCP, 0-3), acute phase reactants (0-1), symptom duration (0-1). Treatment: treat-to-target strategy with methotrexate as anchor DMARD, plus short-term glucocorticoids; early use of TNF inhibitors, IL-6 inhibitors (tocilizumab), abatacept, JAK inhibitors (tofacitinib, baricitinib, upadacitinib) for inadequate response; rituximab for seropositive disease. Intensive disease management reduces joint damage and disability.

Rheumatology

– The medical specialty concerned with the diagnosis and treatment of diseases affecting the musculoskeletal system, including joints, muscles, bones, tendons, and ligaments, as well as systemic autoimmune and inflammatory conditions. Common rheumatic diseases: inflammatory arthritis (rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, reactive arthritis, gout, pseudogout), osteoarthritis (degenerative joint disease), connective tissue diseases (systemic lupus erythematosus, systemic sclerosis/scleroderma, Sjögren syndrome, mixed connective tissue disease, polymyositis/dermatomyositis), vasculitis (giant cell arteritis, Takayasu arteritis, polyarteritis nodosa, ANCA-associated vasculitis, IgA vasculitis/Henoch–Schönlein purpura), and soft tissue rheumatism (fibromyalgia, tendinopathy, bursitis, adhesive capsulitis/frozen shoulder). Diagnostic tools: clinical history and examination, laboratory tests (rheumatoid factor, anti-CCP, ANA, anti-dsDNA, ANCA, complement, inflammatory markers—ESR, CRP), imaging (X-ray, ultrasound, MRI), and synovial fluid analysis. Treatment: NSAIDs, corticosteroids, conventional synthetic DMARDs (methotrexate, leflunomide, sulfasalazine, hydroxychloroquine), biologic DMARDs (TNF inhibitors—adalimumab, etanercept, infliximab; IL-6 inhibitors—tocilizumab; CTLA4-Ig—abatacept; anti-CD20—rituximab), and targeted synthetic DMARDs (JAK inhibitors—tofacitinib, baricitinib, upadacitinib). Multidisciplinary care: physiotherapy, occupational therapy, podiatry, psychology.

Sarcoidosis

– A multisystem granulomatous disorder of unknown etiology characterized by non-caseating granulomas, predominantly affecting lungs (90%), lymph nodes, skin, eyes, liver, heart, nervous system. Pulmonary stages (Scadding): 0 (normal), I (bilateral hilar lymphadenopathy), II (BHL + parenchymal), III (parenchymal only), IV (fibrosis). Extrapulmonary: Löfgren syndrome (acute, erythema nodosum, bilateral hilar adenopathy, arthritis, fever - good prognosis), Heerfordt syndrome (parotid enlargement, uveitis, fever, facial nerve palsy), cardiac sarcoid (conduction abnormalities, cardiomyopathy, sudden death), neurosarcoid (cranial nerve palsies, meningitis, brain parenchymal lesions), cutaneous (lupus pernio, erythema nodosum). Diagnosis: biopsy of affected tissue showing non-caseating granulomas, exclusion of other granulomatous diseases (TB, fungal). Treatment: corticosteroids first-line; steroid-sparing (methotrexate, azathioprine, mycophenolate, leflunomide); TNF inhibitors (infliximab, adalimumab) for refractory disease.

SAVI Syndrome

– STING-associated vasculopathy with onset in infancy (SAVI, TMEM173 gain-of-function) causes constitutive STING activation -> type I interferon overproduction. Severe interstitial lung disease, digital ischemia/autoamputation, chilblain lupus-like skin lesions, failure to thrive. Treatment: JAK inhibitors (baricitinib, ruxolitinib) reduce IFN signature and improve lung disease.

Sjogren Syndrome

– Primary Sjogren syndrome (pSS) is a chronic autoimmune exocrinopathy characterized by lymphocytic infiltration of lacrimal and salivary glands causing dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia), with female predominance (9:1) and peak onset 40-60. Secondary SS occurs with RA, SLE, SSc. 2016 ACR/EULAR criteria: anti-Ro/SSA positive (3 points) OR focal lymphocytic sialadenitis with focus score >=1 (3 points) OR ocular staining score >=5 (1 point) OR Schirmer test <=5 mm/5 min (1 point). Treatment: symptomatic (artificial tears, saliva substitutes, pilocarpine, cevimeline); systemic immunosuppression (hydroxychloroquine, methotrexate, azathioprine, mycophenolate, rituximab) for extraglandular manifestations (arthritis, ILD, vasculitis, nephritis, neuropathy). Lymphoma risk (3-5 percent, MALT in parotid) requires long-term surveillance.

Sweet Syndrome

– Sweet syndrome (acute febrile neutrophilic dermatosis) presents with fever, neutrophilia, and tender erythematous plaques/nodules with dense dermal neutrophilic infiltrate. Classical (idiopathic), malignancy-associated (AML, solid tumors), drug-induced (G-CSF, granulocyte colony-stimulating factor). Associated with VEXAS syndrome. Treatment: systemic corticosteroids (rapid response); colchicine, dapsone, indomethacin as steroid-sparing; treat underlying malignancy.

Systemic Lupus Erythematosus

– A chronic multisystem autoimmune disease predominantly affecting women (9:1 female:male) of childbearing age, characterized by autoantibody production (ANA 98%, anti-dsDNA 70%, anti-Sm 30%) and immune complex deposition causing tissue injury. 2019 EULAR/ACR classification criteria: ANA positive at >=1:80 plus weighted clinical/immunologic domains (fever, hematologic, neuropsychiatric, mucocutaneous, serosal, musculoskeletal, renal). Clinical manifestations: mucocutaneous (malar rash, discoid rash, photosensitivity, oral ulcers, alopecia), serosal (pleurisy, pericarditis), musculoskeletal (arthritis, myositis), renal (lupus nephritis class I-VI on biopsy), neurologic (seizures, psychosis, mononeuritis multiplex, transverse myelitis), hematologic (hemolytic anemia, leukopenia, lymphopenia, thrombocytopenia, antiphospholipid syndrome), and pulmonary (interstitial lung disease, pulmonary hypertension). Lupus nephritis requires aggressive immunosuppression with mycophenolate or cyclophosphamide plus glucocorticoids. Hydroxychloroquine is recommended for all SLE patients. Belimumab (anti-BLyS), anifrolumab (anti-IFNAR), and calcineurin inhibitors (voclosporin) are newer targeted therapies. Pregnancy in SLE requires careful planning (avoid cyclophosphamide, mycophenolate), hydroxychloroquine continuation, and monitoring for flares and neonatal lupus.

Systemic Sclerosis

– A rare, chronic autoimmune connective tissue disease characterized by progressive fibrosis of the skin and internal organs, vasculopathy (intimal proliferation and luminal narrowing of small vessels), and immune dysregulation, with an incidence of approximately 2 to 20 per million per year and a female-to-male ratio of 4 to 1, peaking between 30 and 50 years of age. The pathogenic triad involves immune activation (T-cell and B-cell infiltration of target tissues, autoantibody production including anti-centromere, anti-Scl-70, anti-RNA polymerase III, and vascular damage). Clinical subsets: limited cutaneous SSc (lcSSc, skin distal to elbows/knees, CREST syndrome: Calcinosis, Raynaud, Esophageal dysmotility, Sclerodactyly, Telangiectasias, anti-centromere, pulmonary hypertension) and diffuse cutaneous SSc (dcSSc, skin proximal to elbows/knees and trunk, anti-Scl-70, anti-RNA pol III, higher risk of ILD and renal crisis). Organ involvement: ILD (NSIP pattern, most common cause of death), pulmonary arterial hypertension (PAH, second leading cause of death), scleroderma renal crisis (SRC, malignant hypertension, AKI, microangiopathic hemolytic anemia), gastrointestinal (GERD, dysphagia, pseudo-obstruction, bacterial overgrowth, fecal incontinence), cardiac (myocardial fibrosis, pericardial effusion, arrhythmias). Treatment: Raynaud (calcium channel blockers, prostacyclin analogues, endothelin receptor antagonists for digital ulcers); ILD (cyclophosphamide, mycophenolate, nintedanib, tocilizumab); PAH (endothelin receptor antagonists, PDE5 inhibitors, prostacyclin); SRC (ACE inhibitors, avoid high-dose glucocorticoids which precipitate SRC); GERD (high-dose proton pump inhibitors). hematopoietic stem cell transplantation for selected patients with rapidly progressive dcSSc.

Vasculitis

– Inflammation of blood vessels causing tissue ischemia and necrosis. Classification by vessel size. Large vessel: giant cell arteritis, Takayasu arteritis. Medium vessel: polyarteritis nodosa (PAN), Kawasaki disease. Small vessel: ANCA-associated vasculitis (microscopic polyangiitis, granulomatosis with polyangiitis, eosinophilic granulomatosis with polyangiitis), IgA vasculitis (Henoch-Schonlein purpura), and anti-GBM disease. Diagnosis: clinical presentation, ANCA, ANA, complement, and tissue biopsy. Treatment: corticosteroids and immunosuppressants (cyclophosphamide, rituximab for ANCA-associated).

VEXAS Syndrome

– VEXAS (Vacuoles, E1 ubiquitin-activating enzyme, X-linked, Autoinflammatory, Somatic) syndrome is an adult-onset autoinflammatory disorder caused by somatic UBA1 mutations in hematopoietic stem cells, affecting men >50. Clinical: recurrent fever, chondritis (relapsing polychondritis-like), vasculitis (Sweet-like, polyarteritis-like), macrocytic anemia, thrombocytopenia, vacuoles in myeloid/erythroid precursors. High mortality from infections, thrombosis, hematologic progression (MDS, AML). Treatment: high-dose corticosteroids for acute inflammation; JAK inhibitors (ruxolitinib, tofacitinib); azacitidine for MDS; supportive care for cytopenias. hematopoietic stem cell transplantation is the only potential cure.

Chapter 34

Surgery

Acute Mesenteric Ischemia

– Sudden interruption of blood flow to the small intestine, with mortality 60-80% if diagnosis delayed. Etiologies: arterial embolism (50%, SMA most common, atrial fibrillation, recent MI, valvular disease), arterial thrombosis (25%, on pre-existing atherosclerosis, often at SMA origin), venous thrombosis (20%, hypercoagulable states, malignancy, OCPs, pregnancy), non-occlusive mesenteric ischemia (NOMI, 5%, low flow state: shock, heart failure, vasopressors). prognosis, with second-look laparotomy to reassess intestinal viability. If extensive infarction, the survival rate is dismal, and palliative care should be considered.

Acute Pancreatitis

– Acute pancreatitis (AP): autodigestion of pancreas by prematurely activated trypsin. Causes: gallstones (40-50%), alcohol (25-35%), hypertriglyceridemia (>1000 mg/dL), ERCP, medications (azathioprine, 6-MP, valproate, estrogens, thiazides), trauma, post-surgical, autoimmune, idiopathic. Severity: Atlanta classification 2012: mild (no organ failure, no local complications), moderately severe (transient organ failure <48h or local complications), severe (persistent organ failure >48h). Revised Atlanta classification includes three severity categories. Local complications include acute peripancreatic fluid collections (APFC), pancreatic pseudocysts (developing >4 weeks after onset), acute necrotic collections (ANC), and walled-off necrosis (WON). Pancreatic and peripancreatic necrosis may become infected in 30-70% of cases after 3-4 weeks, requiring antibiotics and percutaneous or endoscopic drainage. Management: aggressive IV fluids (lactated Ringer's, 250-500 mL/h for first 12-24 hours), analgesia, NPO with NG tube for vomiting, ICU monitoring for severe disease. Antibiotics are not routinely indicated. ERCP with sphincterotomy is indicated for severe gallstone pancreatitis with cholangitis or persistent choledocholithiasis. Necrosectomy (minimally invasive step-up approach via percutaneous drainage, endoscopic transluminal drainage, video-assisted retroperitoneal debridement, or surgical necrosectomy) is reserved for infected necrosis not responding to antibiotics. Complications include multi-organ failure, ARDS, acute kidney injury, splenic vein thrombosis, pseudoaneurysm formation with hemorrhage, and chronic pancreatitis with exocrine and endocrine insufficiency. Prognosis: CT severity index (CTSI), APACHE II, Ranson criteria, and BISAP score predict severity. Mortality of severe acute pancreatitis with persistent organ failure is 30-50%.

Adrenal Tumors

– Adrenal incidentaloma: >1cm on imaging. Workup: hormonal (cortisol after 1mg DST, metanephrines, aldosterone/renin if hypertensive, DHEAS/androgens if virilizing), imaging (CT washout: adenoma >60% absolute washout or >40% relative; MRI chemical shift). Size >4cm or suspicious imaging -> surgery. Pheochromocytoma: catecholamine excess (episodic hypertension, headache, diaphoresis, palpitations); 10% bilateral, 10% malignant, 10% extra-adrenal (paraganglioma), 10% familial (MEN2, VHL, NF1, SDHx). Biochemical confirmation: plasma free metanephrines (elevated). Pre-operative: alpha-blockade (phenoxybenzamine 10 mg TDS for 7-14 days before surgery, starting at 10 mg TDS and titrating up to 20-30 mg TDS to achieve blood pressure control and prevent intraoperative hypertensive crisis, followed by beta-blockade after alpha-blockade to control reflex tachycardia or arrhythmia, such as propranolol 10-

20 mg TDS starting on day 3 of alpha-blockade). Surgery: laparoscopic adrenalectomy (gold standard for pheochromocytoma and other adrenal masses; partial/cortical-sparing adrenalectomy if bilateral disease or genetic syndrome). Cortical-sparing adrenalectomy is preferred for hereditary phaeochromocytoma syndromes (MEN2, VHL) to avoid lifelong steroid dependence. Criteria for surgery: tumors >4 cm or any functioning or suspicious mass in a patient who is a good surgical candidate. Adrenocortical carcinoma (ACC): usually >6cm, irregular, heterogeneous with central necrosis, invasion of adjacent structures, venous tumor thrombus; worst prognosis; treatment is open radical en bloc resection with lymphadenectomy and complete tumor excision achieving R0 resection margins. Mitotane therapy for advanced or metastatic ACC. Adjuvant mitotane therapy after resection reduces recurrence when there is a high risk of recurrence (tumor size >6 cm, Ki-67 >10 percent, positive margins, capsular invasion, or tumor rupture at the time of surgery).

Amputation

– Surgical removal of a limb or part of a limb. Levels: toe, partial foot (transmetatarsal), below-knee (BKA, preserves knee function, 40-60% walk with prosthesis), through-knee, above-knee (AKA), hip disarticulation, and hemipelvectomy. Upper extremity: digit, ray, transradial, transhumeral, and shoulder disarticulation. Indications: peripheral arterial disease with gangrene (most common), trauma (crush, degloving), infection (necrotizing fasciitis, osteomyelitis refractory), malignancy (sarcoma), and severe deformity. Prosthetic fitting: post-operative rigid dressing, early weight-bearing for BKA. Complications: stump pain, phantom limb pain (80-100%), phantom sensation, surgical site infection, and deconditioning.

Anti-Reflux Surgery

– Surgical procedures designed to restore the anti-reflux barrier at the gastroesophageal junction, primarily indicated for gastroesophageal reflux disease (GERD) refractory to medical therapy, patient preference to discontinue lifelong medication, or large hiatal hernias. The most common procedure is laparoscopic Nissen fundoplication, a 360-degree wrap of the gastric fundus around the distal esophagus, restoring the lower esophageal sphincter pressure and preventing transient lower esophageal sphincter relaxation and reducing the volume of refluxate. The wrap is typically 2-3 cm in length and is performed laparoscopically. Pre-operative evaluation includes esophageal manometry to confirm normal peristalsis (essential before a 360-degree wrap), 24-hour pH monitoring to confirm the diagnosis of GERD, upper endoscopy to assess for oesophagitis, Barrett esophagus, and stricture, and a barium swallow to evaluate esophageal anatomy and identify hiatal hernia. Alternatives to Nissen fundoplication include partial wraps (Toupet 270-degree posterior fundoplication for patients with poor esophageal motility, Dor 180-200-degree anterior fundoplication) and the LINX magnetic sphincter augmentation device, which consists of a ring of titanium beads with magnetic cores placed around the gastroesophageal junction. Crural closure of the diaphragmatic hiatus is performed to prevent recurrence of the hiatal hernia. Post-operative complications include dysphagia (common initially, resolves in most cases, may require pneumatic dilation if persistent), gas-bloat syndrome (inability to belch, leading to abdominal distension and discomfort, usually improves over weeks to months), and recurrence of reflux symptoms due to wrap disruption or slippage. Outcomes are excellent, with 85-95 percent patient satisfaction at 10 years.

Appendectomy

– Surgical removal of the appendix, the most common emergency surgical procedure performed worldwide. Appendectomy is indicated for acute appendicitis, the most common surgical emergency with a lifetime risk of 7-8 percent and peak incidence between 10 and 30 years. Laparoscopic appendectomy is the preferred approach, offering lower wound infection rates, faster recovery, less postoperative pain, and better cosmetic outcomes compared to open appendectomy. The procedure involves placement of three to four ports (umbilical camera port, suprapubic and right lower quadrant working ports), identification of the base of the appendix, division of the mesoappendix using cautery or a vessel-sealing device, and ligation of the appendiceal base using endoloops (two proximal and one distal) or a stapler. The appendix is placed in a retrieval bag and removed through the umbilical port. In cases of perforated appendicitis with peritonitis, conversion to open appendectomy may be required for adequate peritoneal lavage. Intravenous antibiotics covering Gram-negative and anaerobic organisms are given pre-operatively. For uncomplicated appendicitis, no post-operative antibiotics are needed beyond 24 hours. For perforated appendicitis, antibiotics are continued for 3-5 days or until the patient is clinically improving. Complicated appendicitis (perforation, abscess, or gangrene) has a higher risk of wound infection and intra-abdominal abscess, which may require percutaneous drainage. The complication rate for laparoscopic appendectomy is approximately 5-10 percent, with wound infection being the most common (1-5 percent). Post-operative recovery is rapid, with most patients discharged within 24 hours and returning to normal activity within 1-2 weeks.

Appendiceal Tumors

– Appendiceal neoplasms: carcinoid (most common, 50%, tip of appendix, <1cm: appendectomy alone; >2cm or positive margins/lymphovascular invasion: right hemicolectomy), mucinous neoplasm (low-grade appendiceal mucinous neoplasm LAMN, high-grade appendiceal mucinous neoplasm HAMN), appendiceal adenocarcinoma (treated like colon cancer: right hemicolectomy), goblet cell carcinoid (adenocarcinoid, amphicrine, right hemicolectomy if >2cm or nodal involvement). Pseudomyxoma peritonei (PMP): disseminated mucinous tumor throughout the peritoneal cavity from ruptured mucinous neoplasm; treatment: cytoreductive surgery (CRS) + hyperthermic intraperitoneal chemotherapy (HIPEC) with mitomycin C. Prognosis: 5-year survival 70% for low-grade, 20-50% for high-grade.

Appendicitis

– Acute appendicitis is the most common surgical emergency, with a lifetime risk of 7-8 percent. Peak incidence 10-30 years. Pathophysiology: obstruction of appendiceal lumen (fecalith, lymphoid hyperplasia, tumor, parasites) -> increased intraluminal pressure -> venous congestion -> arterial compromise -> ischemia, perforation, peritonitis. Clinical: periumbilical pain migrating to RLQ (McBurney point), anorexia, nausea/vomiting, low-grade fever. Signs: RLQ tenderness, rebound tenderness, Rovsing sign (palpation LLQ causes pain in RLQ), psoas sign (pain on right hip extension and thigh flexion), obturator sign (pain on internal rotation of flexed right thigh), Dunphy sign (cough worsens RLQ pain), and Aaron sign (referred pain in epigastrium or precordium when palpating RLQ). Laboratory: leukocytosis (WBC >10,000) with left shift, but WBC may be normal in 10-20% of cases; C-reactive protein (CRP) elevated; procalcitonin may be used. Imaging: graded compression ultrasound (first-line, sensitivity 86%, specificity 93%) shows non-compressible, blind-ending tubular structure >6 mm in diameter, with increased blood flow on color Doppler. CT abdomen with IV contrast is the gold standard for diagnosis (sensitivity 94%, specificity 95%), showing an enlarged appendix >6 mm with wall thickening, periappendiceal fat stranding, and possible abscess or extraluminal air (indicating perforation). Alternative diagnoses to exclude include mesenteric adenitis, terminal ileitis (Crohn disease), Meckel diverticulitis, right-sided diverticulitis, omental infarction, epiploic appendagitis, ureteric colic, pyelonephritis, ovarian cyst torsion or rupture, ectopic pregnancy, pelvic inflammatory disease, and mitelschmerz. The Alvarado score (MANTRELS) and RIPASA score stratify risk. Alvarado score: 1-4 = low risk (observe), 5-6 = moderate (CT), 7-10 = high (surgery). Management: appendectomy (laparoscopic preferred). Antibiotics alone (appendicitis non-operative management) is an option for uncomplicated (non-perforated, no abscess, no faecalith) appendicitis based on the NOTA, APPAC, and CODA trials, but 20-30% recurrence rate within 1 year. Interval appendectomy after non-operative management is no longer routinely recommended. Complications: perforation (incidence 20-30%, higher in elderly and children), abscess formation (percutaneous drainage), wound infection (1-5%), intra-abdominal abscess (1-2%), ileus, and stump leak (rare).

Arterial Line (A-Line)

– A catheter inserted into a peripheral artery (most commonly radial, also femoral, brachial, dorsalis pedis) for continuous blood pressure monitoring and arterial blood gas sampling. Allen test: assesses ulnar collateral flow before radial artery cannulation. Indications: hemodynamic instability (shock, vasopressors), need for frequent ABGs, and advanced hemodynamic monitoring (pulse pressure variation SVV for fluid responsiveness). Transducer: zeroed at the phlebostatic axis (4th intercostal space, midaxillary line), leveled to the patient's heart. Damping: underdamped (overestimates systolic, normal waveform) or overdamped (underestimates systolic, sluggish upstroke). Complications: thrombosis, ischemia (rare if collateral flow is adequate), infection, hematoma, and pseudoaneurysm.

Arterial Ulcer

– A chronic wound caused by arterial insufficiency (peripheral artery disease). Located on distal extremities: toes, heels, lateral malleolus, and sites of pressure. Presents with punched-out, deep, well-demarcated wound with pale or necrotic base, minimal exudate, and cool, hairless, shiny skin. Pain is severe, worse with elevation and night. Associated with claudication and rest pain. Diagnosis: ABI (<0.5 suggests severe ischemia), toe pressure (<30 mmHg), vascular duplex. Treatment: revascularization (endovascular or surgical bypass) is essential for healing. Wound care: debridement, moist healing, infection control. Avoid compression. Amputation if non-salvageable.

Bariatric Surgery

– Weight loss procedures and surgery, collectively known as bariatric surgery, are invasive interventions for severe obesity that produce substantial and sustained weight loss by altering gastrointestinal anatomy and physiology. Common procedures include Roux-en-Y gastric bypass, sleeve gastrectomy, and adjustable gastric banding, which restrict food intake, reduce nutrient absorption, or both. Bariatric surgery is typically reserved for individuals with a body mass index of 40 or greater, or 35 or greater with significant obesity-related complications, and requires lifelong nutritional monitoring and lifestyle adherence.

Benign Prostatic Hyperplasia

– Non-malignant enlargement of the prostate gland due to hyperplasia of stromal and glandular elements, affecting >50% of men by age 60 and >90% by age 85. Presents with lower urinary tract symptoms (LUTS): hesitancy, weak stream, intermittency, straining, incomplete emptying, urgency, frequency, and nocturia. Complications: acute urinary retention, bladder stones, recurrent UTIs, and renal impairment. Diagnosis: digital rectal exam, PSA, AUA symptom index, post-void residual, and uroflowmetry. Treatment: watchful waiting for mild symptoms, alpha-blockers (tamsulosin), 5-alpha-reductase inhibitors (finasteride), combination therapy, and surgical options (TURP, laser prostatectomy, and open prostatectomy for large glands).

Biliary Tract Injuries

– Bile duct injury (BDI) occurs in 0.3-0.5% of laparoscopic cholecystectomies. Strasberg classification: A (cystic duct leak/minor hepatic duct leak), B (occlusion of aberrant right hepatic duct), C (transection of aberrant right hepatic duct without ligation), D (lateral injury to major duct), E (major duct injury: E1-E5 based on level). Recognition: intraoperative (bile leak, visualization) or postoperative (bilious drain output, pain, fever, jaundice). Intraoperative management: if recognized, primary repair over T-tube (if >50% intact wall) or Roux-en-Y hepaticojejunostomy (if transection, ligation, or significant resection of the bile duct). If recognized postoperatively, ERCP with sphincterotomy and stenting (Strasberg A, D) or percutaneous transhepatic cholangiography (PTC) with drainage (Strasberg B, C) may be attempted. Strasberg E injuries require surgical repair via Roux-en-Y hepaticojejunostomy, preferably performed by a hepatobiliary surgeon at a tertiary referral center. The timing of repair is critical: early repair (<72 hours) is acceptable for clean, well-vascularised tissue; otherwise, delayed repair (4-6 weeks) after resolution of inflammation and sepsis is preferred. Pre-operative optimisation includes biliary drainage (ERCP, PTC, or percutaneous transhepatic biliary drainage), control of sepsis with antibiotics (piperacillin-tazobactam or meropenem), and nutritional support (enteral nutrition via nasojejunal tube if possible). Prevention: critical view of safety (CVS) technique during cholecystectomy, low threshold for cholangiography (indocyanine green fluorescence imaging, intraoperative cholangiography, or laparoscopic ultrasound) when anatomy is unclear. Outcomes: Strasberg A and D have excellent outcomes; Strasberg E has a 10-20% stricture rate after repair. Long-term complications include anastomotic stricture (treated with balloon dilation or revision), cholangitis, secondary biliary cirrhosis, and portal hypertension.

Blunt Abdominal Trauma

– Trauma to the abdomen without penetrating injury. Mechanisms: motor vehicle collisions (most common), falls, assaults. Commonly injured organs: spleen (most common, left upper quadrant), liver (right upper quadrant), kidneys (flank), bowel (seat-belt sign, delayed perforation), and pancreas (epigastric). Diagnosis: FAST (Focused Assessment with Sonography in Trauma) for free intraperitoneal fluid; CT scan (gold standard in hemodynamically stable patients). Management: hemodynamically unstable with positive FAST → exploratory laparotomy. Stable: CT for grading; non-operative management of solid organ injuries (spleen/liver) if hemodynamically stable. Bowel and pancreatic injuries often require surgical repair.

BPH

– Abbreviation for Benign Prostatic Hyperplasia.

Breast Surgery

– Breast cancer surgery: breast-conserving therapy (BCT: lumpectomy + radiation) equivalent survival to mastectomy for early stage. Margins: "no ink on tumor" for invasive cancer; 2mm for DCIS. Sentinel lymph node biopsy (SLNB): dual mapping (blue dye + radioisotope); if positive (macrometastasis >2mm): completion ALND (historical) or radiation (ACOSOG Z0011: 1-2 positive SLN + BCT + whole breast RT -> no ALND; AMAROS: ALND vs axillary RT equivalent). Mastectomy types: simple, skin-sparing, nipple-sparing (NSM), and radical modified (Patey: breast + axillary lymph node dissection - ALND). Nipple-areola complex (NAC) is spared if tumor is >2cm from the nipple and no Paget disease of the breast. Oncoplastic techniques integrate plastic surgery principles with oncological resection for improved aesthetic outcomes. Adjuvant therapy: radiation (post-BCT, post-mastectomy if T3/T4 or 4+ positive nodes), endocrine (tamoxifen pre-menopausal, aromatase inhibitor post-menopausal if ER+), HER2-targeted (trastuzumab/pertuzumab if HER2+), chemotherapy (high-risk: triple-negative, HER2+, high-grade, node-positive). BRCA mutation carriers: bilateral mastectomy (risk-reducing) + RRSO (risk-reducing salpingo-oophorectomy) recommended.

Burn Surgery

– Burn assessment: rule of 9s (adult), Lund-Browder (children). Depth: superficial (epidermis), superficial partial (dermis, blanch, blister, painful), deep partial (deeper dermis, white, sluggish capillary refill, less painful), full thickness (all dermis, white/charred, anesthetic). TBSA >20%: IV resuscitation (Parkland: 4 mL/kg/%TBSA LR, half in first 8h, half in next 16h; modified Brooke: 2 mL/kg/%TBSA). Inhalation injury: bronchoscopy, carboxyhemoglobin, cyanide level. Escharotomy: circumferential full-thickness burns of chest (impaired ventilation), extremities (compartment syndrome), neck, or penis. Fasciotomy for compartment syndrome (burn deep to fascia). Tangential excision (within 24-72 hours) with split-thickness skin grafting (STSG: 0.008-0.012 inch), meshed (1:1 to 1:4 for larger surface area) or sheet graft (face, hands, joints). Biological dressings: allograft (cadaver), xenograft (porcine), amniotic membrane, synthetic (Biobrane, Integra, Apligraf, and artificial dermal regeneration template). Scar management: pressure garments, silicone sheeting, laser therapy (pulsed dye laser for hypertrophic scars, fractional CO2 laser for contracture release and scar texture improvement).

CABG (Coronary Artery Bypass Grafting)

– Open heart surgery to bypass stenotic coronary arteries using autologous grafts. On-pump: cardiopulmonary bypass, cardioplegic arrest. Off-pump (OPCAB): beating heart, no bypass machine. Grafts: left internal mammary artery (LIMA to LAD, best patency 95% at 10 years), saphenous vein graft (SVG, patency 60-80% at 10 years), and radial artery (patency between LIMA and SVG). Indications: left main CAD (≥50%), three-vessel CAD, two-vessel CAD with proximal LAD and EF <50%, and failed PCI. Complications: bleeding, AF, stroke, infection (sternal wound), renal failure, and graft failure.

Carotid Endarterectomy

– Surgical removal of atherosclerotic plaque from the carotid bifurcation to prevent stroke. Indications: symptomatic carotid stenosis ≥50% (NASCET criteria: TIA or non-disabling stroke) and asymptomatic stenosis ≥70% (if perioperative risk <3%). Pre-operative: CTA or MRA, antiplatelet therapy. Complications: stroke (2-5%), cranial nerve injury (hypoglossal, vagus, facial marginal mandibular, recurrent laryngeal: 5-10%, most temporary), hematoma, restenosis. Alternative: carotid artery stenting (CAS, for high surgical risk).

Cesarean Section (C-section)

– Surgical delivery of a fetus through incisions in the abdominal wall (laparotomy) and uterus (hysterotomy). Indications: failure to progress, non-reassuring fetal heart tracing, malpresentation (breech, transverse), placenta previa, placental abruption, prior classical C-section, maternal infection (HSV, HIV), and elective (maternal request). Incisions: low transverse (Kerr, most common, heals best for future VBAC), classical (vertical, high risk of rupture in labor). Steps: Pfannenstiel incision, bladder flap, hysterotomy, delivery of the baby, placental removal, closure. Complications: hemorrhage, infection, thromboembolism, bladder/bowel injury, and uterine rupture in future pregnancies (VBAC).

Cholecystectomy

– Surgical removal of the gallbladder. Laparoscopic cholecystectomy (gold standard, >95%): 4 small incisions, short stay. Open cholecystectomy: for severe inflammation, cirrhosis, or failed laparoscopy. Indications: symptomatic cholelithiasis (biliary colic), acute cholecystitis, gallstone pancreatitis, and gallstone ileus. Intraoperative cholangiogram (IOC): assess for CBD stones. Critical view of safety (CVS): before clipping the cystic artery and duct. Complications: bile duct injury (0.3-0.5%), bile leak, bleeding, and retained CBD stones.

Cholecystitis

– Acute cholecystitis is gallbladder inflammation, most commonly from cystic duct obstruction by gallstones (calculous cholecystitis, 90-95%); acalculous cholecystitis (5-10%) in critically ill, trauma, burns, prolonged fasting, TPN, vasculitis. Pathophysiology: stone impaction -> distension -> ischemia -> inflammation -> bacterial translocation (E. coli, Klebsiella, Enterococcus). Clinical: RUQ pain >6 hours, radiating to right scapula, fever, nausea/vomiting. Murphy sign (inspiratory arrest on right subcostal palpation during deep inspiration), present in 60-70% of cases. Laboratory: leukocytosis, elevated CRP, LFTs (elevated ALT/AST in acute cholecystitis, ALP/bilirubin in choledocholithiasis or Mirizzi syndrome). Imaging: RUQ ultrasound (first-line, sensitivity 94%): gallstones, thickened gallbladder wall >3-4 mm (target sign), pericholecystic fluid, sonographic Murphy sign, positive common bile duct >6 mm (or >8 mm post-cholecystectomy). CT abdomen is less sensitive for gallstones but useful for complications (perforation, abscess, emphysematous cholecystitis, gangrenous cholecystitis, Mirizzi syndrome). HIDA scan (cholescintigraphy) shows non-visualisation of the gallbladder (cystic duct obstruction) with 95% sensitivity and specificity. Management: NPO, IV fluids, analgesia, IV antibiotics (ampicillin-sulbactam, piperacillin-tazobactam, or ceftriaxone with metronidazole). Definitive: laparoscopic cholecystectomy (gold standard), ideally within 72 hours of admission (early cholecystectomy reduces morbidity and length of stay compared to delayed surgery at 6-12 weeks). Tokyo Guidelines 2018 (TG18) severity grading: Grade I (mild, no organ dysfunction, mild inflammation), Grade II (moderate, elevated WBC >18,000, palpable tender mass, duration >72 hours, marked inflammation), Grade III (severe, organ dysfunction). Grade I: early laparoscopic cholecystectomy (within 72 hours); Grade II: early laparoscopic cholecystectomy after medical optimisation; Grade III: ICU management, percutaneous cholecystostomy (drainage) if critical, delayed cholecystectomy after resolution. Complications: gallbladder perforation (localised or free), empyema of gallbladder, emphysematous cholecystitis (gas-forming organisms, high mortality, requires urgent cholecystectomy), Mirizzi syndrome (stone impacted in cystic duct or Hartmann pouch compressing common hepatic duct causing obstructive jaundice; may require subtotal cholecystectomy or Roux-en-Y hepaticojejunostomy), post-cholecystectomy syndrome (persistent symptoms after cholecystectomy).

Chronic Pancreatitis

– Chronic pancreatitis (CP): irreversible fibrosis, loss of exocrine/endocrine function. Etiology: alcohol (60-70%), genetic (PRSS1, SPINK1, CFTR, CTRC), obstructive (stricture, tumor), autoimmune (IgG4-related), idiopathic. Clinical: recurrent epigastric pain radiating to back, steatorrhea (exocrine insufficiency), diabetes (endocrine insufficiency), weight loss. Imaging: CT (calcifications, ductal dilation, atrophy), MRCP (ductal changes), EUS (early changes: hyperechoic foci, strands, lobularity), ERCP (therapeutic: stenting, stone extraction, stricture dilation). Cambridge classification (ERCP/MRCP) grade 1-5. Management: pain control (avoid narcotics, consider celiac plexus block for refractory pain), exocrine supplementation (pancreatic enzymes, PERT: lipase 40,000-80,000 units per meal, titrate to symptom relief and normalisation of fecal elastase-1), endocrine (insulin for diabetes), endoscopic (ERCP with sphincterotomy, stone extraction, stricture dilation, stenting), surgical (lateral pancreaticojejunostomy/Puestow procedure for dilated duct >7 mm, Beger/Frey procedure for head-predominant disease, Whipple for mass/head, distal pancreatectomy for body/tail disease, total pancreatectomy with islet autotransplantation for debilitating pain). Complications: pseudocyst (can be internal or external, may require endoscopic drainage, transpapillary drainage, or cystogastrostomy), biliary stricture, duodenal stricture, splenic vein thrombosis leading to left-sided (sinistral) portal hypertension with gastric varices, pancreatic ascites, pancreaticopleural fistula, and pancreatic adenocarcinoma (risk is 2-5% over 20 years, surveillance of the remnant gland with cross-sectional imaging is controversial but may be considered in young patients).

Circumcision

– Surgical removal of the penile foreskin (prepuce). Newborn: most common, Plastibell or Gomco clamp, or Mogen clamp (bloodless). Indications: religious (Jewish Brit Milah), cultural, medical (phimosis, recurrent balanitis, balanitis xerotica obliterans, paraphimosis, and UTI prevention in boys with VUR). Medical benefits: reduced UTI risk (10x decrease in infancy), reduced penile cancer risk, lower HIV/STI transmission. Contraindications: hypospadias, epispadias, chordee (foreskin needed for repair). Complications: bleeding, infection, meatal stenosis, excess/insufficient removal, and urethrocutaneous fistula.

Cleft Lip and Palate

– A congenital anomaly caused by failure of fusion of the facial prominences during embryonic development (4-10 weeks gestation). Cleft lip: failure of maxillary and medial nasal processes to fuse. Cleft palate: failure of palatal shelves to fuse. Can occur together or separately. Incidence: 1 in 700 live births, more common in Asians and Native Americans. Multifactorial: genetic (IRF6, FGFR1) and environmental (maternal smoking, folate deficiency, alcohol). Associated syndromes: Pierre Robin sequence, Van der Woude syndrome, 22q11 deletion. Presents with cosmetic deformity, feeding difficulties, speech abnormalities, and ear infections. Management: multidisciplinary team (ENT, plastic surgery, speech therapy, orthodontics). Surgical repair: cleft lip at 3-6 months, cleft palate at 9-18 months.

Cleft Palate

– A congenital malformation characterized by an opening (cleft) in the palate (roof of the mouth) due to failure of fusion of the palatal shelves during embryonic development (weeks 6–9 of gestation). Incidence: 1 in 1,500–2,500 live births. Cleft palate may occur in isolation or with cleft lip (cleft lip and palate—CLP). Classification: incomplete (soft palate only) vs complete (soft and hard palate), unilateral vs bilateral, and associated with Pierre Robin sequence (micrognathia, glossoptosis, airway obstruction). Risk factors: genetics (multiple genes implicated—IRF6, MSX1, TBX22, FGFR1), maternal smoking, alcohol, diabetes, and folate deficiency. Diagnosis: antenatal ultrasound (around 20-week anomaly scan) or postnatal clinical examination (inspect palate with a tongue depressor). Complications: feeding difficulties (nasal regurgitation, poor weight gain), recurrent otitis media (Eustachian tube dysfunction—hearing loss), speech delay (velopharyngeal insufficiency—hypernasal speech, articulation errors), dental abnormalities (missing or malpositioned teeth), and maxillary hypoplasia. Management: multidisciplinary team (plastic surgeon, ENT, speech therapist, orthodontist, paediatrician, psychologist). Primary palatoplasty (repair of the palate) is performed at 9–12 months (before speech development); techniques: Furlow double-opposing Z-plasty, Bardach two-flap palatoplasty. Secondary surgery for velopharyngeal insufficiency (pharyngeal flap, sphincter pharyngoplasty). Alveolar bone grafting at 6–9 years. Orthognathic surgery for maxillary hypoplasia in adolescence. Speech therapy, audiology, and orthodontic treatment throughout childhood. Long-term outcomes: good speech and feeding outcomes, normal intelligence, and good quality of life with appropriate multidisciplinary care.

Colectomy

– Surgical resection of part or all of the colon. Right hemicolectomy: for cecal/ascending colon cancer. Left hemicolectomy: for descending/sigmoid cancer. Sigmoidectomy: for diverticulitis or sigmoid cancer. Total colectomy: for FAP, UC, or Lynch syndrome. Low anterior resection (LAR): for rectal cancer with anastomosis (coloanal anastomosis). Abdominoperineal resection (APR): for distal rectal cancer, permanent colostomy. Indications: colon cancer, diverticulitis, IBD (UC, Crohn), and ischemic colitis. Complications: anastomotic leak (most feared), hemorrhage, wound infection, and ileus.

Colorectal Surgery

– Colon cancer: right hemicolectomy (cecum to transverse), extended right (to mid-transverse), transverse colectomy, left hemicolectomy (splenic flexure to sigmoid), sigmoid colectomy, anterior resection (low anterior resection - LAR for upper/mid rectal), abdominoperineal resection (APR for low rectal with sphincter involvement), total mesorectal excision (TME) standard for rectal cancer. Lymphadenectomy: minimum 12 nodes. Rectal cancer: neoadjuvant chemoradiation (5-FU/capecitabine + 50.4 Gy) for T3/T4 or node-positive; total neoadjuvant therapy (TNT: induction chemotherapy followed by chemoradiation) for locally advanced; watch-and-wait for complete clinical response (cCR) after TNT. Anal cancer: squamous cell carcinoma; Nigro protocol (mitomycin C + 5-FU + 55-60 Gy radiation); surgery (APR) only for residual/recurrent disease. Diverticulitis: Hinchey classification (I: pericolonic abscess, antibiotics, percutaneous drainage if >4cm; II: pelvic abscess, percutaneous drainage; III: purulent peritonitis, Hartmann or resection with anastomosis + defunctioning stoma; IV: fecal peritonitis, Hartmann procedure). Elective sigmoid resection after 2 episodes of uncomplicated or 1 episode of complicated diverticulitis (current guidelines have shifted towards a more conservative, individualised approach, with surgery considered on a case-by-case basis for recurrent attacks, persistent complications, or immunosuppression).

Craniotomy

– Surgical opening of the skull (bone flap) for access to intracranial contents. Indications: brain tumor resection (glioblastoma, meningioma, mets), hematoma evacuation (SDH, EDH, ICH), aneurysm clipping, AVM resection, and epilepsy surgery (lobectomy). Stereotactic craniotomy: frameless neuronavigation (Stealth, BrainLab) for precise localization. Awake craniotomy: speech/language mapping for tumors in eloquent cortex. Complications: cerebral edema, hemorrhage (epidural, subdural, intraparenchymal), infection (osteomyelitis, meningitis), CSF leak, seizures, and neurologic deficits.

Cryptorchidism

– Failure of one or both testes to descend into the scrotum, the most common congenital genitourinary anomaly in boys (3% at birth, 1% at 1 year). 70% of undescended testes are palpable (inguinal), 30% non-palpable (intra-abdominal, absent, or atrophic). Spontaneous descent by 6 months occurs in about one-third. Complications: impaired fertility (bilateral: 75% reduced, unilateral: 50% reduced), testicular cancer risk (seminoma, 4-5x increased, even after orchiopexy), and psychological effects. Diagnosis: physical exam; ultrasound or laparoscopy for non-palpable. Treatment: orchiopexy (surgical placement in scrotum) between 6-18 months of age. Hormonal therapy (hCG) is less effective. Atrophic or absent testis: orchiectomy. Follow-up: annual exam and testicular self-exam.

Debridement

– The surgical removal of dead, damaged, or infected tissue from a wound to promote healing and prevent infection. Types: sharp/surgical debridement (scalpel, curette, scissors–most rapid and definitive), enzymatic debridement (collagenase, papain-urea), autolytic debridement (moist dressings allow the body's own enzymes to liquefy necrotic tissue), mechanical debridement (wet-to-dry dressings, irrigation, whirlpool), and biologic debridement (sterile maggots–Lucilia sericata larvae consume necrotic tissue). Selection depends on wound type, amount of necrotic tissue, infection status, and patient tolerance.

Diabetic Foot Ulcer

– A chronic wound of the foot in a patient with diabetes, resulting from peripheral neuropathy, peripheral artery disease, and impaired wound healing. Neuropathic ulcers: painless, at pressure points (plantar metatarsal heads, heel), punched-out appearance, surrounded by callus. Ischemic ulcers: painful, at toes or margins of foot, with pale, cool, hairless skin. Risk factors: poor glycemic control, long diabetes duration, prior amputation, neuropathy, PAD, and foot deformity. Diagnosis: monofilament testing, ankle-brachial index, vascular duplex ultrasound, and wound assessment. Treatment: offloading (total contact cast, CAM boot, wheelchair), debridement, infection control (antibiotics, surgical drainage for osteomyelitis), revascularization for ischemic ulcers, and advanced wound care (negative pressure, growth factors, bioengineered skin). Prevention: daily foot inspection, proper footwear, glycemic control, and podiatry referral.

Dilation and Curettage (D&C)

– Surgical dilation of the cervix and curettage (scraping) of the uterine lining. Indications: incomplete or missed abortion (miscarriage management), therapeutic abortion, endometrial sampling (abnormal uterine bleeding, postmenopausal bleeding, endometrial hyperplasia/cancer), and removal of retained products of conception (RPOC). Complications: uterine perforation (1-3%, higher in postmenopausal, retroverted uterus), cervical laceration, hemorrhage, infection, and Asherman syndrome (intrauterine adhesions from overzealous curettage). Now often replaced by hysteroscopy-guided procedures.

Discectomy

– Surgical removal of a herniated intervertebral disc fragment compressing a nerve root. Microscopic discectomy (most common): small incision, microscope, minimal muscle dissection. Indications: radicular pain (sciatica, cervical radiculopathy) refractory to 6-12 weeks of conservative therapy, progressive motor deficit, and cauda equina syndrome (surgical emergency). Success rate: 85-95% for leg pain relief. Also performed via endoscopic, minimally invasive (Tubular/METRx, MIS-TLIF for associated instability), and open (rarely). Complications: dural tear (3-5%), infection, hematoma, recurrent herniation, and nerve root injury.

Drowning

– Respiratory impairment from submersion/immersion in liquid. Drowning can be fatal or non-fatal. Pathophysiology: laryngospasm followed by aspiration of fluid, surfactant washout, pulmonary edema, and hypoxemia. Fresh water: washes out surfactant, causes alveolar collapse. Salt water: draws fluid into alveoli (pulmonary edema). Both cause severe hypoxemia, metabolic acidosis, and multiorgan failure. Risk factors: children 1-4 (bathtubs, pools), alcohol use, lack of supervision, and seizure disorders. Management: immediate rescue, CPR with rescue breaths (most important intervention), ICU care with ventilation, and treatment of aspiration pneumonia and ARDS. Prognosis: associated with submersion time, CPR delay, water temperature, and neurologic status on arrival.

Electrical Injury

– Tissue damage caused by passage of electrical current through the body. Severity depends on voltage, current (alternating vs. direct), pathway, duration, and resistance. High-voltage (>1000V): extensive internal tissue damage (muscle necrosis, compartment syndrome, rhabdomyolysis) with relatively small skin burns (entry/exit wounds). Cardiopulmonary effects: arrhythmias (ventricular fibrillation, asystole), cardiac arrest. Neurologic effects: spinal cord injury, peripheral nerve damage, cognitive deficits. Complications: acute kidney injury (myoglobinuria), compartment syndrome, fractures from falls, and delayed neurologic deficits. Diagnosis: ECG, CK, urine myoglobin, CT/MRI. Treatment: ABCs, cardiac monitoring, IV fluids with alkalinization (bicarbonate), surgical fasciotomy/debridement for compartment syndrome. Tetanus prophylaxis.

Erectile Dysfunction

– The persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance. Affects up to 50% of men aged 40-70. Etiology: organic (vascular disease, diabetes, hypogonadism, neurological disorders, medications, Peyronie disease) and psychogenic (anxiety, depression, relationship issues). Vascular causes: impaired arterial inflow (atherosclerosis) or venous leak. Diagnosis: history, IIEF-5 questionnaire, nocturnal penile tumescence testing, duplex Doppler ultrasound, and hormonal profile. Treatment: PDE5 inhibitors (sildenafil, tadalafil, vardenafil) as first-line; second-line options: intracavernosal injections (alprostadil), vacuum erection devices, and penile prosthesis for refractory cases. Lifestyle modification (weight loss, exercise, smoking cessation) improves outcomes.

Esophageal Surgery

– GERD surgery: Nissen fundoplication (360° wrap, gold standard), partial wraps (Toupet 270°, Dor anterior 180-200°) for dysmotility. LINX device (magnetic sphincter augmentation) alternative. Achalasia: Heller myotomy (laparoscopic/robotic) + partial fundoplication (Dor); POEM (peroral endoscopic myotomy) endoscopic alternative; pneumatic dilation (recurrence 30-50%). Esophageal cancer: adenocarcinoma (distal, Barrett's), squamous (proximal/mid, smoking/alcohol). Staging: EUS, PET-CT, laparoscopy for peritoneal disease. Neoadjuvant chemoradiation (CROSS regimen: carboplatin + paclitaxel + 41.4 Gy) for locally advanced (T2N+ or T3-4 any N). Surgery: Ivor Lewis (right thoracotomy + laparotomy, for mid/distal), McKeown (right thoracotomy + laparotomy + cervical, for proximal), transhiatal (blunt dissection, for distal, higher anastomotic leak rate). Reconstruction: gastric conduit (greater curve based on right gastroepiploic artery, most common), colon interposition (when stomach unavailable). Three-field lymphadenectomy (cervical, mediastinal, abdominal) for squamous cell carcinoma at or above the carina. Anastomotic techniques: stapled (EEA, 21-25 mm, end-to-side) or hand-sewn (two-layer running or interrupted, anastomotic leak rate 5-15%). Wait time of 8-12 weeks after neoadjuvant therapy for surgical resection. Outcomes: 5-year survival for esophageal cancer is 15-25% overall; 40-50% for patients who achieve a pathological complete response (pCR) after neoadjuvant chemoradiation. Salvage oesophagectomy for persistent or recurrent disease after definitive chemoradiation (50.4 Gy + chemotherapy) has higher morbidity (30-50% complication rate, especially anastomotic leak and pulmonary complications).

Excision

– The surgical removal of a piece of tissue, an organ, or a lesion. Excisional biopsy removes the entire lesion with a margin of normal tissue, used for definitive diagnosis and treatment of small tumors (suspected melanoma, breast lump, lymph node). Wide local excision removes the lesion with an additional margin of healthy tissue for oncologic clearance. In biochemistry, excision repair: the removal of damaged DNA segments. Surgical principles: hemostasis, closure method (primary, secondary, or tertiary intention), and specimen orientation for pathology. Margin status (clear, close, or involved) is the most important prognostic factor for local recurrence in cancer surgery.

Fasciotomy

– Surgical incision through the fascia to relieve compartment syndrome. Indications: acute compartment syndrome (trauma, reperfusion, burns, crush injury). Classic signs: pain out of proportion, pain with passive stretch, paresthesias, pallor, pulselessness (late). Diagnosis: clinical, compartment pressure >30 mmHg OR delta pressure (diastolic BP minus compartment pressure) <30 mmHg. Leg: 4 compartments (anterior, lateral, superficial posterior, deep posterior) via two incisions. Forearm: volar (Henry approach) and dorsal. Foot, thigh, hand, abdomen (for abdominal compartment syndrome) also possible. Fasciotomy wounds are left open for 48-72 hours, then delayed primary closure or skin grafting.

Femoral Hernia

– A hernia of abdominal contents through the femoral ring, into the femoral canal, medial to the femoral vessels and below the inguinal ligament. More common in women (3:1) and more prone to strangulation (40%) than inguinal hernias due to narrow, rigid femoral ring. Presents as a small, non-reducible mass below and lateral to the pubic tubercle (distinguishing from inguinal hernia). Diagnosis: clinical; ultrasound/CT for equivocal cases. Treatment: surgical repair (McVay, mesh plug, or laparoscopic TEP/TAPP) due to high strangulation risk. Emergency surgery for strangulated or incarcerated femoral hernia.

Frostbite

– Tissue injury caused by freezing of skin and underlying tissues. Severity grading: first degree (superficial, numbness, erythema), second degree (blisters, clear fluid, pain), third degree (hemorrhagic blisters, necrosis, full-thickness skin loss), fourth degree (deep tissue necrosis, mummification, autoamputation). Common sites: fingers, toes, nose, ears, cheeks. Predisposing factors: cold exposure, wind chill, vasoconstriction, immobility, alcohol, smoking, diabetes. Pathophysiology: ice crystal formation, endothelial injury, thrombosis, and reperfusion injury upon rewarming. Diagnosis: clinical. Treatment: rapid rewarming in water 37-39°C (100-104°F) for 15-30 minutes; ibuprofen, tetanus prophylaxis, wound care, and delayed amputation (march of demarcation). Thrombolytic therapy (tPA) within 24 hours for severe cases to reduce amputation rate.

Gallbladder Cancer

– Aggressive, often diagnosed incidentally after cholecystectomy for cholelithiasis (1-2% of cholecystectomies). Risk factors: gallstones (chronic inflammation), porcelain gallbladder (calcified wall, 10-25% cancer risk), anomalous pancreaticobiliary junction, gallbladder polyps >1cm, typhoid carrier state. T staging: T1a (lamina propria), T1b (muscularis), T2 (perimuscular connective tissue), T3 (perforates serosa or invades liver/adjacent organ), T4 (main portal vein/hepatic artery, extensive). Lymphatic spread: cystic, pericholedochal, hilar, peripancreatic, periduodenal, celiac, retroperitoneal nodes. Surgical management: T1a: simple cholecystectomy alone is sufficient (no lymphadenectomy required, 5-year survival >95%); T1b: extended cholecystectomy (cholecystectomy + en bloc wedge resection of liver segments IVb and V + portal lymphadenectomy) is the minimum recommended operation (5-year survival 80-85%); T2: extended cholecystectomy + portal lymphadenectomy (5-year survival 50-60%); T3: extended cholecystectomy + portal lymphadenectomy ± bile duct resection/right hepatectomy (5-year survival 20-30%); T4: unresectable (palliative: biliary stenting, chemotherapy). Incidental gallbladder cancer discovered after simple cholecystectomy: refer to a hepatobiliary center for completion radical cholecystectomy (wedge resection of gallbladder bed, portal lymphadenectomy, ± bile duct excision and reconstruction with Roux-en-Y hepaticojejunostomy) within 4 weeks of the initial cholecystectomy. Chemotherapy: gemcitabine + cisplatin for advanced disease (ABC-02 and BILCAP trials). Prognosis: overall 5-year survival 5-15%, worse if lymph node involvement (5-year survival <10% with nodal metastases). Adjuvant chemotherapy with capecitabine (BILCAP) or gemcitabine-based regimens may improve survival after curative resection.

Gastric Surgery

– Gastric cancer: D2 lymphadenectomy standard (stations 1-12a, 12b). Gastrectomy types: total, distal subtotal (Billroth I gastroduodenostomy, Billroth II gastrojejunostomy, Roux-en-Y), proximal, pylorus-preserving. Reconstruction: Roux-en-Y preferred (prevents bile reflux). Early gastric cancer (T1a): ESD (endoscopic submucosal dissection) if no LVI, well-differentiated, <2cm (expanded criteria). Neoadjuvant: FLOT (5-FU, leucovorin, oxaliplatin, docetaxel) superior to surgery alone. Palliative: systemic chemotherapy, palliative resection for bleeding/obstruction/perforation, stenting for GOO (gastric outlet obstruction). GIST: imatinib (neoadjuvant for downstaging, adjuvant if high risk: >10cm, mitotic >10/HPF, rupture), surgery (R0, avoid rupture). MALT lymphoma: H. pylori eradication first; only treat with surgery or chemoradiation if not responsive. Perforated peptic ulcer: omental patch repair (Graham patch: interrupted 2-0 silk sutures through the full thickness of each side of the perforation, tied over a pedicle of omentum, which is then secured in place with a second set of sutures), laparoscopic (preferred if hemodynamically stable, no contamination, surgeon experience), open if peritonitis. Intraoperative decision-making: if the perforation is >2cm, there is a suspicion of malignancy, or the edges are oedematous and friable, a more extensive resection may be required (distal gastrectomy with Billroth II or Roux-en-Y reconstruction). Post-operative: PPI infusion (omeprazole 80 mg bolus + 8 mg/hour), H. pylori testing and eradication (triple therapy: PPI + amoxicillin + clarithromycin or metronidazole, 14-day course), return to oral intake after resolution of ileus.

Grafting

– The surgical transfer of tissue from one site to another, or from one individual to another. Types: (1) Autograft: same individual (gold standard, e.g., skin graft for burns, bone graft for non-union, saphenous vein graft for CABG). (2) Allograft: same species, different individual (cadaveric bone, cornea, heart valve, kidney). (3) Xenograft: different species (porcine heart valve, porcine/bovine skin for burns). (4) Isograft: identical twin (syngeneic). (5) Synthetic graft: prosthetic material (Dacron vascular graft, Gore-Tex, metal implants, polyethylene for joint replacement). Skin grafts: split-thickness (epidermis + part of dermis) or full-thickness (epidermis + full dermis). Flaps: tissue transferred with its blood supply intact (pedicled or free). Complications: graft rejection (allografts, xenografts), infection, haematoma, seroma, graft failure (ischemia), contracture. Immunosuppression (tacrolimus, mycophenolate, corticosteroids) is required for allograft organ transplantation.

Heimlich Manoeuvre

– An emergency procedure for relieving upper airway obstruction caused by a foreign body (choking). The maneuver is performed on a conscious, choking adult who cannot speak, cough effectively, or breathe. Technique: the rescuer stands behind the victim, wraps their arms around the victim's waist, makes a fist with one hand (thumb side against the victims abdomen, midline above the navel and well below the xiphoid process), grasps the fist with the other hand, and performs upward, inward thrusts (abdominal thrusts). For obese or pregnant victims, chest thrusts (similar position but on the lower sternum) are used instead. The maneuver abruptly increases intra-abdominal and intrathoracic pressure, forcing air out of the lungs and creating a cough-like force to expel the obstructing object. For a choking infant (under 1 year): hold the infant face-down along the rescuer's forearm, with the head lower than the trunk, and deliver 5 back blows between the shoulder blades, followed by 5 chest thrusts (face-up, two fingers on lower sternum). The Heimlich maneuver is contraindicated for: infants under 1 year (use back blows and chest thrusts), unconscious victims (start CPR), and cases where the obstruction is not complete (partial obstruction with good air exchange, encourage coughing). After successful expulsion, the patient should have a medical evaluation to exclude complications (esophageal rupture, gastric rupture, pneumothorax, abdominal organ injury).

Hemicolectomy

– See Colectomy.

Hemorrhoidectomy

– Surgical excision of symptomatic hemorrhoids. Indications: Grade III-IV internal hemorrhoids (prolapsed, irreducible), external hemorrhoids with uncontrolled bleeding or recurrent thrombosis, mixed hemorrhoids. Techniques: closed (Ferguson, primary closure), open (Milligan-Morgan, leave wounds open), and stapled hemorrhoidopexy (PPH, for circumferential prolapse, lower pain but higher recurrence). Laser, bipolar, and harmonic scalpel techniques also available. Postoperative pain is significant (particularly with open technique). Recovery: 2-4 weeks. Complications: pain, bleeding, urinary retention, anal stenosis, and incontinence.

Hepatic Surgery

– Liver resection: Couinaud segments. Indications: colorectal liver metastases (CRLM, up to 70% 5-year survival with R0), HCC (Child-Pugh A, portal pressure <10 mmHg, future liver remnant >20% normal, >40% cirrhotic), cholangiocarcinoma, benign tumors (adenoma >5cm, symptomatic hemangioma, FNH diagnostic uncertainty). Preoperative: PVE (portal vein embolization) for FLR <20%/40%; ALPPS (associating liver partition and portal vein ligation) for rapid hypertrophy (2 weeks). Techniques: open, laparoscopic, robotic. Pringle manoeuvre (clamp hepatoduodenal ligament, 15-20 min on, 5 min off, ischemic preconditioning may reduce ischemia-reperfusion injury). Low central venous pressure (CVP <5 mmHg) anesthesia. Parenchymal transection: crush-clamp (Kelly clamp), CUSA (cavitron ultrasonic surgical aspirator), LigaSure, Harmonic scalpel. Hepatic vein management: extrahepatic control (right, middle, left suprahepatic veins) or intrahepatic approach (anterior approach with hanging manoeuvre). Complications: bile leak (3-10%), PHLF (post-hepatectomy liver failure, defined by ISGLS peak bilirubin >50 mumol/L + INR >1.5 on postoperative day 5; management: medical support, albumin, diuretics, TIPS, or liver transplantation in severe cases), hemorrhage, abscess, ascites. Non-resectional techniques: RFA (radiofrequency ablation), MWA (microwave ablation), IRE (irreversible electroporation), SBRT (stereotactic body radiotherapy), TACE (transarterial chemoembolization), and TARE (transarterial radioembolization with yttrium-90), each indicated for specific tumor types and locations where surgical resection is not feasible. Liver transplantation: Milan criteria for HCC (single $\leq$5 cm or up to 3 nodules each $\leq$3 cm, no extrahepatic disease, no major vascular invasion) with 5-year survival >70%.

Hernia

– The protrusion of an organ or tissue through an abnormal opening in the wall of the cavity that normally contains it. Most hernias occur in the abdominal wall. Types: (1) Inguinal hernia (most common): indirect (through the deep inguinal ring, lateral to inferior epigastric vessels, congenital or acquired, more common in younger men) vs direct (through Hesselbach triangle, medial to inferior epigastric vessels, acquired, older men). (2) Femoral hernia: through the femoral ring below the inguinal ligament, more common in women, higher risk of strangulation (40%) vs inguinal (10%). (3) Umbilical hernia: through the umbilical ring, common in infants (usually resolves spontaneously by age 2-4) and adults (obesity, pregnancy, ascites). (4) Incisional (ventral) hernia: through a surgical scar, risk factors: wound infection, obesity, smoking, COPD. (5) Hiatus hernia: see separate entry. (6) Epigastric hernia: through the linea alba. (7) Spigelian hernia: through the linea semilunaris. (8) Obturator hernia: through the obturator foramen (rare, elderly thin women). (9) Lumbar hernia: through the lumbar triangle. (10) Diaphragmatic hernia: through the diaphragm. Diagnosis: history (lump or bulge, worse on standing/straining, cough impulse), examination (inspection: visible bulge; palpation: cough impulse; reducibility). Complications: irreducibility (cannot be reduced), obstruction (bowel obstruction), strangulation (vascular compromise–ischemia, gangrene, surgical emergency). Treatment: surgical repair (open or laparoscopic; tension-free mesh repair is standard–Lichtenstein for inguinal, laparoscopic transabdominal preperitoneal TAPP or totally extraperitoneal TEP). Watchful waiting is an option for minimally symptomatic inguinal hernias in men.

Hernia Repair (Herniorrhaphy)

– Surgical repair of a hernia (inguinal, femoral, umbilical, incisional, hiatal). Inguinal hernia repair: open (Lichtenstein tension-free mesh, Shouldice, Bassini) or laparoscopic (TEP: totally extraperitoneal, TAPP: transabdominal preperitoneal). Mesh reinforcement reduces recurrence (1-3% for mesh vs. 5-15% for primary tissue repair). Indications: symptomatic hernia, incarceration/strangulation risk, and groin pain. Emergency repair for strangulated hernia (bowel resection may be needed). Complications: recurrence, chronic groin pain (5-10%, especially open), mesh infection, seroma, hematoma, and cord injury (vas deferens, testicular vessels).

Hiatus Hernia

– The protrusion of part of the stomach through the esophageal hiatus of the diaphragm into the chest cavity. Types: (1) Sliding hernia (type I, 95%): the gastro-esophageal junction (GOJ) and a portion of the gastric cardia slide above the diaphragm through the hiatus. The GOJ is displaced cephalad, and the angle of His is lost, predisposing to GORD. Associated with: obesity, pregnancy, increased intra-abdominal pressure. (2) Para-esophageal hernia (type II): the GOJ remains in its normal position below the diaphragm, but a portion of the gastric fundus herniates alongside the esophagus. (3) Mixed (type III): both GOJ and fundus herniate. (4) Type IV: herniation of other abdominal organs (colon, omentum, spleen) through the hiatus. Para-esophageal hernias (types II-IV) can incarcerate or strangulate. Symptoms: many sliding hernias are asymptomatic; associated with GORD (heartburn, regurgitation, dysphagia, chest pain, chronic cough). Large para-esophageal hernias may cause: postprandial fullness, chest pain, dyspnoea, dysphagia, vomiting, and rarely acute gastric volvulus (Borchardt triad: severe epigastric pain, retching without vomiting, inability to pass a nasogastric tube–surgical emergency). Diagnosis: barium swallow (GOJ above diaphragm), upper endoscopy (GOJ displacement, Los Angeles classification of oesophagitis), high-resolution manometry. Management: asymptomatic sliding hernias require no treatment; symptomatic GORD: PPI therapy, lifestyle measures; para-esophageal hernias with symptoms or risk of volvulus: laparoscopic Nissen fundoplication with hernia reduction and crural repair (hiatoplasty).

Hypospadias

– A congenital anomaly in which the urethral meatus opens on the ventral (underside) of the penis, proximal to the glans. Incidence: 1 in 250 male births. Classification: distal (glanular, coronal — 70%), midshaft (10%), proximal (penoscrotal, scrotal, perineal — 20%). Associated with chordee (ventral penile curvature), undescended testis, and inguinal hernia. Most cases are idiopathic; genetic and endocrine factors are implicated. Diagnosis: clinical exam — note the position of the meatus and presence of chordee. Do not circumcise before surgical repair (foreskin may be needed for urethroplasty). Surgical repair (urethroplasty) at 6-18 months: single-stage (TIP / Snodgrass repair for distal) or two-stage (for proximal). Complications: fistula, stricture, and residual chordee.

Hysterectomy

– Surgical removal of the uterus. Total hysterectomy: removal of uterus and cervix. Subtotal (supracervical): removes uterus, leaves cervix. Radical hysterectomy: uterus, cervix, parametrium, upper vagina, and pelvic lymph nodes (for cervical cancer). Routes: abdominal (open, most common for large uteri/malignancy), vaginal (for prolapse, small uterus), laparoscopic (TLH, less pain, shorter stay), and robotic (da Vinci, improved dexterity). Indications: uterine fibroids (most common), endometriosis, adenomyosis, uterine prolapse, abnormal uterine bleeding, endometrial cancer, cervical cancer, and ovarian cancer. Complications: bleeding, infection, ureter/bladder/bowel injury, and vaginal cuff dehiscence.

Incisional Hernia

– A hernia through a previous surgical incision scar, the most common long-term complication of abdominal surgery (10-20% of laparotomies). Risk factors: wound infection, obesity, smoking, malnutrition, COPD, and multiple previous surgeries. Present as a bulge at or near the incision site, worse with standing or straining. Can cause pain, obstruction, and strangulation. Diagnosis: clinical; CT for evaluation of size, number of defects, and contents. Treatment: surgical repair with mesh (open sublay, onlay, or laparoscopic IPOM) to reduce recurrence. Component separation for large/complex hernias.

Inguinal Hernia

– Protrusion of abdominal contents through the inguinal canal. Indirect (lateral to inferior epigastric vessels, through deep inguinal ring): congenital, patent processus vaginalis; most common overall (60-70%), male predominance. Direct (medial to inferior epigastric vessels, through Hesselbach triangle): acquired, weakness of transversalis fascia; older men. Pantaloon: both indirect and direct. Femoral hernia (below inguinal ligament, through femoral canal): more common in women, right-sided predominance. Clinical: groin bulge (medial to pubic tubercle, below inguinal ligament, irreducible, may cause obstruction or strangulation). Imaging: ultrasound (sensitivity 95%, Valsalva enhancement increases sensitivity) or MRI if diagnosis uncertain. Management: elective repair (symptomatic; open or laparoscopic: TEP - totally extraperitoneal, TAPP - transabdominal preperitoneal). Mesh (polypropylene, lightweight macroporous) preferred over tissue repair (Bassini, Shouldice, McVay, Desarda, Lichtenstein) due to <5% recurrence. Lichtenstein: open tension-free mesh onlay repair, gold standard for open inguinal hernia repair. Laparoscopic: lower recurrence for bilateral and recurrent hernias, faster recovery, less chronic pain (in experienced hands, 1-2% for TEP). Emergency surgery for obstruction/strangulation: McBurney or Lotheissen approach (low midline incision for control of incarcerated bowel), viable bowel → herniorrhaphy, nonviable bowel → resection + herniorrhaphy. Femoral hernia repair: McEvedy approach (retroperitoneal, high approach, through a lower midline or transverse lower abdominal incision) or laparoscopic TEP/TAPP; locking sutures or mesh plug (Bard plug, PerFix plug) or open suture repair (Moschcowitz type, a suture repair of the femoral canal). Complications: recurrence (1-10%, higher with tissue repairs and laparoscopic learning curve), chronic groin pain (10-20%, higher with open, meshoma, nerve entrapment, triple neurectomy for refractory cases), seroma, haematoma, infection (mesh infection <1%, may require mesh removal), testicular atrophy or ischemic orchitis (1-2% during open repair for recurrent hernias or large scrotal hernias), and urinary retention (5-10%, transient) in older male patients after general or spinal anesthesia.

Jejunostomy

– See Feeding Jejunostomy.

Laceration

– A wound caused by tearing of body tissue, typically caused by trauma from a sharp object (knife, glass, metal) or blunt force that splits the skin. Lacerations involve the skin and may extend into subcutaneous tissue, muscle, tendon, nerve, or blood vessels. Assessment: location, length, depth, contamination, foreign bodies, neurovascular status (sensation, motor function, capillary refill, distal pulses), and involvement of deeper structures (tendon function, bone exposure). Management: (1) hemostasis: direct pressure, elevation. (2) Wound cleaning: irrigation with normal saline (high-pressure, 250-500 mL, reduces bacterial load). (3) anesthesia: local infiltration with lidocaine 1-2% (with or without adrenaline, max 4.5 mg/kg plain, 7 mg/kg with adrenaline) or topical lidocaine-prilocaine cream for wound margins. (4) Exploration: assess for foreign bodies, tendon/nerve/vascular injury. (5) Debridement: excision of devitalised tissue, ragged wound edges (freshening). (6) Closure: primary closure (sutures–simple interrupted, continuous/mattress, subcuticular; tissue adhesive; adhesive strips; staples) for clean, non-infected wounds; delayed primary closure (3-5 days) for contaminated wounds; healing by secondary intention for infected or high-risk wounds. Tetanus prophylaxis: clean wound–tetanus toxoid if last dose >10 years; contaminated wound–tetanus toxoid + immunoglobulin if last dose >5 years or unknown. Antibiotics: not routinely indicated for simple lacerations; consider for contaminated wounds, human/animal bites, immunocompromised patients. Sutures removal timing: face 3-5 days, scalp 7 days, trunk/arms 7-10 days, legs 10-14 days. Complications: wound infection (2-5%), dehiscence, hypertrophic scarring/keloid, retained foreign body, nerve damage.

Laminectomy

– Surgical removal of the lamina (posterior vertebral arch) to decompress the spinal canal. Indications: spinal stenosis (lumbar and cervical), herniated disc (with central stenosis), and spinal tumor. Decompression of the entire spinal canal. Often combined with discectomy (lumbar) or foraminotomy (nerve root decompression). May require fusion if instability is present (laminectomy with fusion, PLF). Complications: CSF leak (dural tear), spinal instability (post-laminectomy kyphosis), epidural hematoma, infection, and neurologic injury.

Large Bowel Obstruction

– Less common than SBO. Causes: colorectal cancer (60%), diverticular stricture (20%), volvulus (sigmoid 60%, cecal 40%), fecal impaction, Ogilvie syndrome (acute colonic pseudo-obstruction), ischemic stricture, endometriosis, radiation stricture. Clinical: abdominal distension (prominent), constipation/obstipation, less vomiting than SBO, pain may be less severe. Imaging: abdominal X-ray (dilated colon >6cm, cecum >9cm predicts perforation), CT with contrast (transition point, cause, closed loop obstruction, pneumatosis intestinalis indicating ischemia, portal venous gas, pneumoperitoneum - all signs of ischemia or perforation). Water-soluble contrast enema can differentiate pseudo-obstruction (Ogilvie) from mechanical obstruction (transition point, cut-off sign). Management: NPO, NGT, IV fluids, correct electrolyte abnormalities. Specific management depends on the cause. Sigmoid volvulus: endoscopic detorsion (rigid sigmoidoscopy with placement of a flatus tube, success rate 80-90%) + elective sigmoid colectomy (resection with primary anastomosis) during the same admission (recurrence 40-60% without surgery). Cecal volvulus: colonoscopic detorsion (less successful, 30-40%) + right hemicolectomy. Colorectal cancer: segmental colectomy with primary anastomosis (elective); emergency: resection with stoma (Hartmann) or resection with on-table colonic lavage and primary anastomosis (controversial, increased leak risk). Stenting (self-expanding metallic stent, SEMS) as bridge to surgery (if no perforation, no peritonitis, and the tumor is accessible, allows for medical optimisation and elective resection with primary anastomosis avoiding a stoma in 80-90% of cases) or as definitive palliation (for metastatic disease, high surgical risk). Ogilvie syndrome (acute colonic pseudo-obstruction): conservative (NPO, NGT, correction of electrolytes, mobilisation), neostigmine (2 mg IV, continuous cardiac monitoring for bradycardia and bronchospasm, only in the absence of mechanical obstruction confirmed by CT or water-soluble contrast enema; response within 30 minutes in 60-90% of patients), colonoscopic decompression (suction of air, aspiration of liquid stool) for neostigmine failure, cecostomy (percutaneous or surgical) for recurrence. Contraindications to neostigmine: mechanical obstruction, bradycardia, hypotension, bronchial asthma, peptic ulcer disease, pregnancy. Surgical intervention (laparotomy with colectomy, subtotal colectomy with terminal ileostomy and Hartmann closure of the distal stump) for perforation, ischemia, or failure of all conservative measures.

Lobectomy

– Surgical removal of a lobe of an organ. Most commonly lung lobectomy (removal of one lobe of the lung, performed for lung cancer, bronchiectasis, abscess, or benign tumor) or temporal lobectomy (removal of the anterior temporal lobe for drug-resistant temporal lobe epilepsy). Lung lobectomy: the standard surgical treatment for early-stage (stage I-II) non-small cell lung cancer (NSCLC). Approaches: open thoracotomy (posterolateral) or video-assisted thoracoscopic surgery (VATS, minimally invasive, less post-operative pain, shorter hospital stay, equivalent oncologic outcomes). The remaining lung lobes expand to fill the thoracic cavity. Pulmonary function loss corresponds to the number of segments removed (each lobe contains 2-5 bronchopulmonary segments). Compensatory hyperinflation of remaining lung occurs over weeks. Temporal lobectomy: anterior temporal lobectomy (ATL) for epilepsy involves resection of 3-4 cm of the anterior temporal neocortex plus the amygdala and anterior 2-3 cm of the hippocampus. Seizure-free outcome: 60-70% in well-selected patients with mesial temporal sclerosis. Complications: visual field deficit (superior quadrantanopia, due to optic radiation Meyer loop injury, 50-80%), language deficits (dominant hemisphere), memory impairment. Other lobectomies: hepatic lobectomy (liver resection, for cancer or living donor), thyroid lobectomy (removal of one thyroid lobe, for thyroid nodule/cancer), and cerebral lobectomy (removal of a whole brain lobe, for severe trauma or tumor).

Mastectomy

– Surgical removal of the breast. Total (simple) mastectomy: removal of breast tissue, areola, and nipple. Modified radical mastectomy (MRM): total mastectomy + axillary lymph node dissection (levels I-III), for invasive breast cancer. Skin-sparing mastectomy: preserves breast skin envelope for immediate reconstruction (implant or autologous). Nipple-sparing mastectomy: preserves nipple-areolar complex (for small, peripherally located tumors). Indications: breast cancer (locally advanced, multicentric, inflammatory), and risk-reducing in BRCA carriers (prophylactic bilateral mastectomy). Sentinel lymph node biopsy (SLNB) replaced ALND for node-negative patients. Reconstruction: implant-based or autologous (TRAM, DIEP, latissimus dorsi flap).

Necrotizing Fasciitis

– A life-threatening soft tissue infection characterized by rapid necrosis of the subcutaneous fascia and subcutaneous tissue, with relative sparing of the overlying skin in early stages. It is a surgical emergency requiring immediate wide surgical debridement and broad-spectrum antibiotics. Type I (polymicrobial, 60-75 percent of cases) occurs in diabetic, immunocompromised, or post-surgical patients and involves aerobic and anaerobic bacteria (E. coli, Klebsiella, Bacteroides, Peptostreptococcus, Clostridium species). Type II (monomicrobial, 15-30 percent) is caused by Group A Streptococcus (Streptococcus pyogenes) with or without Staphylococcus aureus, often in healthy individuals, and may present with toxic shock syndrome. Type III (gram-negative, marine-related: Vibrio vulnificus, Aeromonas hydrophila) occurs after exposure to seawater or freshwater, especially in patients with chronic liver disease. Type IV (fungal: Candida, Mucor, Aspergillus) occurs in severely immunocompromised patients and neutropenic patients. LRINEC score (Laboratory Risk Indicator for Necrotizing Fasciitis) stratifies risk: CRP >150 mg/L + WBC $>15{,}000$ + Hb <11 g/dL + Na <135 mmol/L + Cr >1.6 mg/dL + glucose >10 mmol/L, with a score ≥ 6 raising suspicion (positive predictive value 92%) and ≥ 8 strongly predictive (positive predictive value 93-99%). Management: aggressive IV fluid resuscitation, broad-spectrum antibiotics (piperacillin-tazobactam 4.5 g IV q6h or meropenem 1 g IV q8h + clindamycin 600-900 mg IV q8h to suppress toxin production, particularly for streptococcal toxic shock syndrome), urgent surgical debridement (fasciotomy with excision of all devitalised tissue, to the level of healthy bleeding muscle, re-exploration within 24-48 hours is mandatory). IVIG is used as adjunctive therapy for streptococcal toxic shock syndrome (2 g/kg loading dose, then 0.4 g/kg/day for 3-5 days). Hyperbaric oxygen (HBO) therapy is controversial but may be beneficial as an adjunct. Post-operative care includes repeat debridement such that the wound is thoroughly debrided within 24-72 hours, negative pressure wound therapy (NPWT) to promote granulation, and delayed primary closure or split-thickness skin grafting (STSG) once the wound is clean. Mortality: 20-40%, >50% if septic shock at presentation. Amputation rate is 15-25%. Poor prognostic factors include advanced age (>60), diabetes mellitus, delay in surgery (>24 hours from admission to first debridement), involvement of the trunk, and streptococcal toxic shock syndrome. Renal failure, immune compromise, LRINEC score ≥8 and the presence of bullae or skin necrosis are also poor prognostic signs.

Nephrectomy

– Surgical removal of a kidney. Radical nephrectomy: kidney + Gerota fascia + adrenal gland (for renal cell carcinoma). Partial nephrectomy (nephron-sparing): for small renal tumors (<7 cm, renal function preservation). Simple nephrectomy: for benign disease (non-functioning kidney, chronic infection, massive hydronephrosis). Laparoscopic/robotic approach is standard (less pain, shorter stay). Indications: renal cell carcinoma (most common), trauma, and living donor nephrectomy. Complications: bleeding, infection, pneumothorax, splenic or pancreatic injury (left), and vascular injury.

Nephrostomy Tube

– A percutaneously placed catheter in the renal pelvis to drain urine, bypassing a ureteral obstruction. Indications: ureteral obstruction (calculus, tumor, stricture) causing hydronephrosis, pyonephrosis (infected, obstructed kidney — emergency drainage), and as a temporary measure before definitive surgery. Placed by interventional radiology under ultrasound/CT guidance. Sizes: 8-14 Fr. Output: normal 1-2 L/day. Tube is secured to the skin and connected to a drainage bag. Changed q6-12 weeks. Complications: bleeding (hematuria, perirenal hematoma), infection, tube dislodgement/obstruction, and adjacent organ injury (colon, pleura).

Open-Heart Surgery

– Any surgical procedure involving opening the chest wall (sternotomy or thoracotomy) and operating on the heart, usually with cardiopulmonary bypass (the heart–lung machine). Procedures: coronary artery bypass grafting (CABG—most common), valve repair/replacement (aortic, mitral), septal defect repair (ASD, VSD), heart transplantation, and ventricular assist device implantation. Cardiopulmonary bypass drains venous blood, oxygenates it, and returns it to the arterial system, allowing the surgeon to operate on a still, bloodless heart. Cardioplegia solution (high potassium) stops the heart in diastole, reducing myocardial oxygen demand during ischemia. Risks: bleeding, infection (mediastinitis, sternal wound), stroke (aortic manipulation—embolic), acute kidney injury, post-pericardiotomy syndrome, and cognitive dysfunction ('pump head'—typically transient). Minimally invasive approaches (mini-thoracotomy, robotic-assisted) reduce recovery time and blood loss. Off-pump CABG avoids cardiopulmonary bypass but has similar outcomes in selected patients.

Organ Transplants

– The surgical transfer of an organ from a donor to a recipient with end-stage organ failure. Solid organ transplants: kidney (most common, >90,000 per year worldwide), liver, heart, lung, pancreas, and intestine. Donor types: deceased donor (brainstem death or donation after circulatory death—DCD), living donor (kidney, partial liver, partial lung). Immunosuppression: induction (anti-thymocyte globulin, basiliximab) and maintenance (calcineurin inhibitors—tacrolimus, cyclosporine; antiproliferatives—mycophenolate mofetil; corticosteroids). Rejection: hyperacute (minutes—preformed antibodies, ABO incompatibility), acute (days to months—T-cell mediated, treatable with high-dose steroids), and chronic (months to years—multifactorial, progressive graft dysfunction). Organ allocation is managed by national bodies (NHSBT in the UK, UNOS in the US). Waiting list mortality is driven by organ shortage. Xenotransplantation (pig organs—genetically modified to reduce hyperacute rejection) is emerging as a potential solution.

Pancreatic Neuroendocrine Tumors

– Pancreatic neuroendocrine tumors (pNETs): functional (insulinoma, gastrinoma, glucagonoma, VIPoma, somatostatinoma) or non-functional (60-70%). WHO 2017 grading: NET G1 (Ki-67 <3%, mitotic <2/10 HPF), NET G2 (Ki-67 3-20%, mitotic 2-20/10 HPF), NEC G3 (Ki-67 >20%, mitotic >20/10 HPF). Functional: insulinoma (hypoglycemia, Whipple triad, 90% benign, enucleation/distal pancreatectomy), gastrinoma (Zollinger-Ellison syndrome, ulcers, diarrhea, MEN1 association, duodenal > pancreatic, surgery if localized and resectable, PPI (gastrinoma: high-dose PPI, 80 mg omeprazole daily, can be increased to 120 mg daily for symptom control), octreotide for symptom control), glucagonoma (necrolytic migratory erythema, diabetes, weight loss, DVT, distal pancreatectomy + octreotide), VIPoma (WDHA syndrome: watery diarrhea, hypokalaemia, achlorhydria, VIP levels >75 pmol/L, octreotide, surgery), somatostatinoma (diabetes, gallstones, steatorrhea, surgery). Non-functional: 60-70%, large >2cm at diagnosis, no hormonal syndrome, surgery (resection or enucleation if <2cm and favourable location). pNETs in MEN1: multiple, less aggressive, surgery for >2cm or functioning tumors in the pancreatic head; enucleation preferred for superficial tumors while deeper lesions may require pancreaticoduodenectomy; observation for small (<2cm) non-functioning pNETs with surveillance every 6-12 months (in MEN1 and sporadic cases). Syndromic associations: MEN1 (multiple pNETs, parathyroid adenomas, pituitary adenomas, carcinoid tumors), VHL (serous cystadenomas, pNETs, renal cell carcinoma, phaeochromocytoma, retinal angiomas, cerebellar haemangioblastomas), tuberous sclerosis, NF1. Prognosis: 5-year survival 90% for well-differentiated localised NET, 30-50% for metastatic (grade-dependent; Ki-67 is the most important independent prognostic factor). WHO classification 2019: well-differentiated neuroendocrine tumor (NET G1, G2, G3) and poorly differentiated neuroendocrine carcinoma (NEC G3, small cell or large cell type), with significant difference in prognosis and management (NET G3 is treated as an indolent well-differentiated NET rather than aggressive NEC G3).

Pancreatic Surgery

– Pancreaticoduodenectomy (Whipple): pancreatic head, duodenum, gallbladder, distal stomach (pylorus-preserving standard), common bile duct; reconstruction: pancreaticojejunostomy (duct-to-mucosa or invagination), hepaticojejunostomy, gastrojejunostomy/duodenojejunostomy. Distal pancreatectomy: body/tail; spleen-preserving (Kimura, Warshaw) or splenectomy. Central pancreatectomy: neck tumors, preserve head and tail. Enucleation: small pNETs/insulinomas away from duct. Complications: POPF (pancreatic fistula; ISGPS definition: amylase >3x upper limit of normal in drain fluid on or after post-operative day 3; biochemical leak (grade A, no change in management) vs grade B (requires change in management, e.g., prolonged drainage, total parenteral nutrition, somatostatin analogues, antibiotics, percutaneous drainage) vs grade C (major change in management, ICU stay, reoperation, or death); management: NPO, TPN or enteral nutrition via nasojejunal tube if possible, octreotide (somatostatin 250-500 mcg subcutaneously TDS after gastrectomy or after Whipple procedure to reduce exocrine pancreatic secretion, though evidence is mixed and routine use is not recommended), ERCP with pancreatic duct stenting for persistent high-output fistula; percutaneous drainage of fluid collections; reoperation for uncontrolled fistula or bleeding), DGE (delayed gastric emptying, ISGPS definition: NGT >500 mL/day lasting beyond post-operative day 3; management: prokinetics, NGT decompression, enteral nutrition via surgical jejunostomy tube placed at the time of pancreaticoduodenectomy or by nasojejunal tube; most cases resolve spontaneously within 2-4 weeks; persistent DGE may require endoscopic dilation or stenting if there is mechanical obstruction at the gastrojejunostomy), post-pancreatectomy hemorrhage (PPH; early <24 hours, late >24 hours; ISGPS classification: grade A no intervention, grade B intervention without ICU, grade C ICU ± reoperation ± embolisation; management: resuscitation, endoscopy or angiography with embolisation, reoperation), and pancreatic ductal adenocarcinoma (PDAC) recurrence (locoregional, liver, peritoneal, distant). Adjuvant chemotherapy: FOLFIRINOX (modified) for 6 months after resection (PRODIGE 24 trial - median survival 54.4 months vs 35.0 months for gemcitabine) or gemcitabine + capecitabine (ESPAC-

4 trial). Median survival for resected PDAC is 24-30 months; 5-year survival 15-25% (improving with modern multi-agent chemotherapy). Borderline resectable PDAC: neoadjuvant chemotherapy (FOLFIRINOX or gemcitabine + nab-paclitaxel) for 2-6 months, then restaging CT and CA 19-9; proceed to surgery if no progression and venous involvement is reconstructable. Arterial involvement (celiac axis, superior mesenteric artery, common hepatic artery) generally precludes resection unless a modified approach (e.g., Appleby procedure for celiac axis involvement) is considered in selected cases.

Paraphimosis

– A urological emergency in which the retracted foreskin cannot be returned to its normal position over the glans penis, causing constriction at the coronal sulcus. The trapped foreskin acts as a tourniquet, causing venous congestion, edema, pain, and eventually arterial compromise and ischemia of the glans. Causes: iatrogenic (failure to reduce the foreskin after catheterisation, cystoscopy, or penile examination—most common), poor hygiene, and balanitis. Predisposing factors: phimosis (narrow, non-retractile foreskin), lichen sclerosus (balanitis xerotica obliterans). Presentation: a painful, swollen glans with a constricting band of retracted foreskin behind the corona. Treatment: (1) Gentle manual reduction—firm, sustained compression of the oedematous glans for 5–10 minutes (ice pack or granulated sugar to reduce edema), then push the glans posteriorly while pulling the foreskin forward. (2) If manual reduction fails—dorsal slit (incision of the constricting band under local anesthesia) or emergency circumcision. (3) Definitive: elective circumcision after resolution of acute inflammation. Complications of paraphimosis: ulceration, necrosis of the glans, and urethral injury.

Parathyroid Surgery

– Primary hyperparathyroidism (PHPT): single adenoma (85%), hyperplasia (15%), double adenoma (2%), carcinoma (<1%). Criteria for surgery (NIH): age <50, calcium >1 above upper limit, creatinine clearance <60, BMD T-score <-2.5, vertebral fracture, nephrolithiasis, nephrocalcinosis. Localization: sestamibi scan (sensitivity 80-90%), US (sensitivity 70-80%), 4D-CT, MRI, PET-Choline. Minimally invasive parathyroidectomy (MIP) for single adenoma with concordant imaging; bilateral exploration for negative or discordant imaging. Intraoperative PTH monitoring (IOPTH): drop of >50% from baseline 10 minutes after gland excision confirms cure with 95% positive predictive value. Four-gland exploration: identify 4 glands, resect enlarged, biopsy normal. Auto-transplantation: implant parathyroid tissue (typically 10-15 fragments of 1-2mm each) into brachioradialis or sternocleidomastoid muscle if total parathyroidectomy is necessary (reoperative surgery, secondary/tertiary hyperparathyroidism, multiple gland hyperplasia, MEN1, familial hypocalciuric hypercalcaemia). Reoperative parathyroid surgery: persistent (never normocalcaemic) or recurrent (normocalcaemic for >6 months) PHPT; requires pre-operative imaging (4D-CT, sestamibi with SPECT/CT, MRI, PET-Choline) and intraoperative nerve monitoring (recurrent laryngeal nerve is at risk of injury in reoperative neck surgery). Complications: recurrent laryngeal nerve injury (<1% in primary surgery, 3-5% in reoperative), hypoparathyroidism (transient 10-20%, permanent <2%), haematoma (1-2%), infection, and hungry bone syndrome (severe, prolonged hypocalcaemia after resection of severe PHPT with high pre-operative calcium and ALP; management: IV calcium gluconate infusion, oral calcium carbonate 3-6 g/day, calcitriol 0.5-2 mcg/day, magnesium replacement if indicated; worst in patients with pre-operative calcium >3.0 mmol/L and markedly elevated PTH and alkaline phosphatase). Secondary hyperparathyroidism (renal failure): medical management (phosphate binders, vitamin D analogues, cinacalcet); subtotal parathyroidectomy (3.5 gland) or total parathyroidectomy with auto-transplantation for refractory symptomatic hyperparathyroidism with persistent hypercalcaemia despite maximal medical therapy, or for tertiary hyperparathyroidism after kidney transplantation with persistent hypercalcaemia >6 months post-transplant. Tertiary hyperparathyroidism: autonomous PTH secretion despite normal renal function after renal transplantation; parathyroidectomy indicated if hypercalcaemia persists beyond 6 months from transplantation.

Penetrating Abdominal Trauma

– Trauma to the abdomen caused by a missile (gunshot wound) or sharp object (stab wound). GSWs: high energy transfer, produce blast cavity, high risk of intra-abdominal injury (90%), mandatory exploration. Stab wounds: lower energy, 30-50% do not penetrate peritoneum; local wound exploration to determine peritoneal violation. Zones: anterior abdomen (xiphoid to pubis, costal margins), flank (midaxillary to posterior axillary line), and back (posterior to posterior axillary line). Diagnosis: CT with triple contrast (IV, oral, rectal) in stable patients; diagnostic laparoscopy for equivocal cases. Management: hemodynamic instability or peritonitis → laparotomy. Stable: serial exams, CT, and selective non-operative management.

Phimosis

– The inability to retract the prepuce (foreskin) over the glans penis. Physiologic phimosis: normal in prepubertal boys, resolves with age. Pathologic phimosis: scarring of the foreskin due to balanitis xerotica obliterans (lichen sclerosus), recurrent infection, or trauma. Presents with ballooning of the foreskin during urination, recurrent balanitis, and urinary tract infections. Complications: paraphimosis (retracted foreskin cannot be reduced, causing glandular ischemia), and in severe cases, urinary obstruction. Treatment: topical corticosteroids (betamethasone) for mild cases; preputioplasty or circumcision for refractory pathologic phimosis. Physiologic phimosis in children does not require treatment.

PICC Line (Peripherally Inserted Central Catheter)

– A long, flexible catheter inserted into the basilic or cephalic vein and advanced so the tip sits in the superior vena cava. Inserted at the bedside, often by a specialized PICC nurse, with ultrasound and ECG or X-ray confirmation. Indications: long-term IV antibiotics (>2 weeks), TPN, chemotherapy, and frequent blood draws. Sizes: 4-6 Fr single, double, or triple lumen. Duration: weeks to months. Advantages over tunneled central lines: lower risk of pneumothorax, insertion by non-physicians. Complications: infection (line-related bloodstream infection), thrombosis, phlebitis, malposition, and catheter fracture.

Postoperative Complications

– Clavien-Dindo classification: Grade I (any deviation, no treatment), Grade II (pharmacologic), Grade III (surgical/endoscopic/radiologic intervention: IIIa no GA, IIIb GA), Grade IV (ICU: IVa single organ, IVb multi-organ), Grade V (death). Common: ileus (delayed GI motility, NPO, NGT, ambulation, gum chewing, neostigmine for Ogilvie), anastomotic leak (clinical: pain, fever, leukocytosis; CT: extraluminal contrast, fluid; management: antibiotics, percutaneous drainage, diversion, reoperation), wound infection (SSI, superficial/deep/organ space; management: antibiotics, drainage, wound care, negative pressure wound therapy or vacuum-assisted closure), surgical site infection prevention bundle, pulmonary complications (pneumonia, atelectasis: IS/CPT, O2, incentive spirometry 10 breaths per hour, early mobilisation), DVT/PE (PESI score, Caprini risk assessment, pharmacologic prophylaxis: enoxaparin 40 mg SC daily or 30 mg BID or unfractionated heparin 5,000 units TID; mechanical prophylaxis: SCD, TED stockings; systemic anticoagulation if PE confirmed). ERAS (Enhanced Recovery After Surgery) pathways: multidisciplinary approach to reduce surgical stress and accelerate recovery; components include pre-operative counseling, avoidance of prolonged fasting (carbohydrate loading 2 hours before surgery), no bowel preparation, epidural analgesia, goal-directed fluid therapy (avoiding excessive crystalloid administration), avoidance of NG tubes and drains, early mobilisation (day of surgery for ambulatory patients, or as soon as the patient can sit in a chair), early oral intake (clear liquids within 4-6 hours, progression to full fluids and then solids as tolerated), and multimodal non-opioid analgesia (paracetamol, NSAIDs, gabapentinoids, lidocaine patches, regional anesthesia blocks); ERAS compliance >70% associated with reduced length of stay (LOS) and complication rates.

Postoperative Ileus

– Temporary impairment of gastrointestinal motility after surgery, most common after abdominal surgery. Pathophysiology: autonomic dysfunction (sympathetic overactivity), inflammatory mediators, opioids, and electrolyte abnormalities. Presents with abdominal distension, nausea/vomiting, inability to tolerate oral intake, and absence of flatus/stool. Normal postoperative GI recovery: small bowel (24h), stomach (48h), colon (72-120h). Distinguish from mechanical obstruction (pain is crampy, bowel sounds high-pitched). Management: NPO, IV fluids, electrolyte correction, early ambulation, minimize opioids (use multimodal analgesia), gum chewing, and alvimopan (mu-opioid receptor antagonist). Neostigmine for Ogilvie syndrome (colonic pseudo-obstruction).

Prostatectomy

– Surgical removal of the prostate. Radical prostatectomy: for localized prostate cancer (removal of prostate, seminal vesicles, vas deferens, pelvic lymph node dissection). Approaches: open retropubic (RRP), laparoscopic, and robotic-assisted laparoscopic (RALP, most common in the US). Nerve-sparing technique: preserves neurovascular bundles for erectile function and urinary continence. Indications: localized prostate cancer (Gleason ≥ 7, PSA >10, or higher stage). Transurethral resection of the prostate (TURP): for BPH (removes central prostate tissue via resectoscope through the urethra), not a cancer operation.

Ranula

– A mucocele (mucus extravasation cyst) of the sublingual salivary gland, presenting as a blue-translucent, fluctuant swelling in the floor of the mouth, typically lateral to the lingual frenulum. A ranula results from trauma or obstruction of a sublingual gland duct, causing mucus accumulation in the sublingual space. If the ranula extends through the mylohyoid muscle into the submental or submandibular space, it is termed a 'plunging' or 'cervical' ranula (presents as a neck mass). Ranulas are benign and painless unless secondarily infected. Differential diagnosis: dermoid cyst, thyroglossal duct cyst, lymphatic malformation, and submandibular gland swelling. Treatment: excision of the ranula and the ipsilateral sublingual gland (lowest recurrence rate—$<5\%$), marsupialisation (incision and drainage, suturing the cyst wall to the oral mucosa—higher recurrence rate), or laser ablation. Plunging ranulas require transcervical excision of the sublingual gland. Needle aspiration is not curative. Simple oral ranulas can be managed conservatively if small and asymptomatic.

Renal Transplant (Renal Transplantation)

– See Kidney Transplantation.

Skin Graft

– Transfer of skin from a donor site to a recipient site to cover a wound. Split-thickness skin graft (STSG): epidermis + partial dermis (0.008-0.018 inches), harvested with dermatome, donor site heals spontaneously (2-3 weeks). Full-thickness skin graft (FTSG): epidermis + full dermis, donor site requires closure, better cosmetic outcome. Types: autograft (same person), allograft (cadaveric), xenograft (animal), and synthetic (Integra, Biobrane). Uses: burn wounds, traumatic skin loss, chronic ulcers, and post-oncologic reconstruction. Graft take: requires vascularized wound bed, immobilization, and prevention of infection/shear.

Small Bowel Obstruction

– Mechanical blockage of the intestinal lumen. Causes: adhesions (75%), hernias (15%), malignancy, Crohn disease strictures, volvulus, intussusception, gallstone ileus, and bezoars. Clinical: crampy abdominal pain, nausea/vomiting (early in proximal SBO), obstipation, and abdominal distension. Signs include high-pitched bowel sounds, localised tenderness (suggesting ischemia), and peritonitis. Laboratory: leukocytosis, elevated lactate (indicator of strangulation), elevated BUN/creatinine (dehydration), and metabolic alkalosis. Imaging: abdominal X-ray shows dilated small bowel loops >3 cm with air-fluid levels. CT abdomen with IV contrast is the gold standard, demonstrating a transition point with proximal dilation and distal collapse, closed-loop obstruction (U-shaped configuration with convergent mesenteric vessels, high risk of ischemia), pneumatosis intestinalis, and signs of perforation; sensitivity $>90\%$ for high-grade SBO. Management: NPO, nasogastric decompression, IV fluids, and CT with contrast. Non-operative observation is appropriate for adhesive SBO without strangulation features (60-80% resolve over 2-5 days). A water-soluble contrast challenge (Gastrografin) may be both diagnostic (predicts resolution if contrast reaches colon within 4-24 hours) and therapeutic. Operative indications: complete obstruction, peritonitis, strangulation, closed-loop obstruction, failure of non-operative management >48-72 hours, or suspected ischemia. Surgery is via laparoscopy (adhesive SBO) or laparotomy (complicated cases), with adhesiolysis, bowel resection for non-viable segments, and primary anastomosis. Hartmann procedure is reserved for peritonitis or hemodynamic instability. Mortality: 3-5% for simple obstruction, 25-30% for strangulated obstruction.

Smoke Inhalation Injury

– Pulmonary injury caused by inhalation of toxic products of combustion. Three components: thermal injury (supraglottic: heat causes edema of the upper airway), chemical irritation (lower airway: aldehydes, ammonia, chlorine, hydrogen chloride cause bronchospasm, mucosal edema, and sloughing), and systemic toxicity (carbon monoxide: carboxyhemoglobin >10% causes tissue hypoxia; cyanide: impairs cellular respiration). Presents with facial burns, singed nasal hairs, carbonaceous sputum, hoarseness, stridor, and dyspnea. Diagnosis: bronchoscopy (gold standard), carboxyhemoglobin level. Treatment: 100% oxygen (shortens CO half-life to 60 min); intubation for upper airway edema; mechanical ventilation with lung-protective strategy; bronchodilators; aggressive pulmonary toilet. Hyperbaric oxygen for severe CO poisoning (neuropsychiatric). Hydroxocobalamin for cyanide toxicity.

Spinal Fusion

– Surgical union of two or more vertebrae using bone graft or bone graft substitutes with instrumentation (rods, screws, cages). Approaches: posterior (posterolateral fusion, transforaminal lumbar interbody fusion TLIF), anterior (anterior lumbar interbody fusion ALIF), and lateral (extreme lateral interbody fusion XLIF). Indications: degenerative disc disease, spondylolisthesis, spinal instability, scoliosis, and trauma. Bone graft sources: autograft (iliac crest, gold standard), allograft, BMP (bone morphogenetic protein), and synthetic. Complications: pseudoarthrosis (non-union), hardware failure, adjacent segment disease, infection, and nerve root injury.

Splenectomy

– Indications: trauma (grade IV/V, hemodynamic instability), ITP (refractory), hereditary spherocytosis/elliptocytosis, thalassemia major, sickle cell (sequestration crisis), TTP (refractory), lymphoma staging (historical), hypersplenism, splenic cysts/tumors, Wandering spleen. Laparoscopic preferred. Vaccinations: pneumococcal (PCV20 or PCV15+PPSV23), meningococcal (MenACWY + MenB), Hib - ideally 2 weeks pre-op; post-op if emergency. Antibiotic prophylaxis: penicillin V daily (children <5 or <2 years post-splenectomy or with continued immunosuppression) or amoxicillin daily (adults, alternatively: penicillin V 250 mg BID or amoxicillin 500 mg daily) for at least 2 years after splenectomy or for life if ongoing immunosuppression; immunocompromised patients require lifelong prophylaxis. Fever: stationary IV ceftriaxone plus azithromycin; seek immediate medical attention (splenectomy alert card/wallet card, medical bracelet, fever algorithm for carers). Overwhelming post-splenectomy infection (OPSI): encapsulated bacteria (S. pneumoniae 50-60%, H. influenzae type b 10-15%, N. meningitidis 10-15%, E. coli, Group B Streptococcus, Capnocytophaga canimorsus from dog bites); rapidly progressive (fulminant) sepsis (mortality 50-70%); presentation: fever, rigors, hypotension, DIC, multi-organ failure within hours; management: IV fluids, broad-spectrum IV antibiotics (ceftriaxone 2 g IV 12-hourly or meropenem 1 g IV 8-hourly) immediately, ICU, vasopressors. Platelet count: rises after splenectomy (thrombocytosis usually transient, peaks at 1-3 weeks, resolves by 2-3 months; platelet count >450,000-1,000,000); aspirin 81 mg daily if platelet count >1,000,000 (to reduce risk of thrombosis). Complications: portal vein thrombosis (3-10%, increased if splenomegaly, myeloproliferative disorder, or hemolytic anemia; duplex ultrasound at day 7 post-operatively in high-risk patients, or CT with portal venous phase imaging if symptomatic; treatment: therapeutic anticoagulation with LMWH or warfarin for 3-6 months or lifelong if there is thrombophilia or persistent splanchnic vein thrombus). Long-term: re-immunise with PPSV23 every 5 years (or per guidelines; Centers for Disease Control and Prevention recommends a single PPSV23 booster 5 years after the initial dose in asplenic adults, and a second booster at age 65 or above if at least 5 years since last). The risk of OPSI is 0.23-4.4% per year, with a lifetime risk of 5-10%.

Stoma Complications

– Complications related to intestinal or urinary stomas. Early (<30 days): ischemia/necrosis (assess with test tube/endoscope, if → mucosa or at fascia → revision), mucocutaneous separation, retraction, parastomal abscess, high output (>1500 mL/day, may cause dehydration/electrolyte depletion → antidiarrheals, loperamide, codeine, octreotide, dietary modification). Late: parastomal hernia (most common late complication, abdominal wall defect → support belt, surgical mesh repair), prolapse (full-thickness protrusion → reduction, repair), stenosis (fibrous ring at skin level → dilation, revision), and skin irritation (leakage → proper appliance fit, skin barriers, powder, paste). Patient education and stoma nurse consultation essential.

Surgical Site Infection

– SSI classification (CDC): superficial incisional (skin/subcutaneous, <30 days), deep incisional (fascia/muscle, <30/90 days with implant), organ/space (any part opened/manipulated, <30/90 days with implant). Risk factors: patient (diabetes, obesity, smoking, immunosuppression, malnutrition, age), operative (contamination class, duration, technique, hair removal, hypothermia, hyperglycemia), postoperative (hyperglycemia, hypothermia, hematoma). Prevention: preoperative shower (CHG), appropriate antibiotic (cefazolin or cefoxitin, or ceftriaxone + metronidazole for colorectal surgery; vancomycin or clindamycin for penicillin-allergic patients) within 60 minutes before incision (120 minutes for vancomycin and fluoroquinolones due to longer infusion time), hair removal with clippers (not razors), normothermia (active warming with forced-air warming blankets), oxygenation (FiO_2 80% during and 6 hours after surgery), maintain euglycaemia (glucose <10 mmol/L, tighter range of 6-10 mmol/L, with insulin sliding scale if needed), wound protector, glove change before closure, and post-operative normothermia, euglycaemia, and oxygen. Antibiotic re-dosing for prolonged surgery (>2 half-lives, approximately 2-4 hours for cefazolin, 4-6 hours for metronidazole, 6-8 hours for ciprofloxacin), appropriate weight-based dosing (e.g., vancomycin 15 mg/kg obese, 30 mg/kg for cefazolin in morbid obesity). Surgical technique: gentle tissue handling, meticulous hemostasis, avoidance of dead space, tension-free closure, use of knotless barbed sutures (e.g., Stratafix, V-Loc) for fascial closure and skin closure techniques: subcuticular absorbable monofilament sutures (skin closure in layers reduces risk of wound infection) or metallic staples (which have the advantage of speed but are not recommended for closure of laparoscopic port sites >5 mm). When closing the fascia, use a continuous slowly absorbable monofilament suture such as PDS (polydioxanone) 1 or 2 for laparotomy incisions, a suture length to wound length ratio of at least 4:1 (the "small bites" technique uses 5 mm bites 5 mm apart, reducing incisional hernia). Prophylactic negative pressure wound therapy (ciNPT, Prevena or PICO system) for high-risk wounds (contaminated, dirty, emergency laparotomy, obese, diabetic, stoma, or immunocompromised patients). Management of established SSI: wound culture, open wound, drainage, debridement, packing with saline-soaked gauze or alginate dressings, VAC or negative pressure wound therapy (NPWT) for large, deep, or chronic wounds. Antibiotics only if cellulitis, systemic signs, or organ/space infection. SSI rates: clean <2%, clean-contaminated <5%, contaminated <15%, dirty >25%.

Suture

– A thread or strand of material used to approximate tissues (bring wound edges together) and hold them in place until healing occurs. Sutures are classified by: (1) Absorbability—absorbable (catgut, polyglactin/Vicryl, poliglecaprone/Monocryl, polydioxanone/PDS—degrade by hydrolysis over weeks to months, used for deep tissues, internal wounds) and non-absorbable (nylon/Ethilon, polypropylene/Prolene, silk, stainless steel—remain permanently, used for skin closure, vascular anastomoses, tendons). (2) Structure—monofilament (single strand—less tissue trauma, lower infection risk, but more difficult to handle, greater knot slippage) and multifilament (braided—easier handling, better knot security, higher infection risk—capillary action harbours bacteria). (3) Needle type—cutting (for skin, tough tissues), reverse cutting, round-bodied (for soft tissues—bowel, vessels, fascia), and tapercut. Suture size: from 10-0 (ophthalmic microsurgery) to 0–2 (fascia, orthopedics). Suture techniques: simple interrupted (most common, secure, allows individual knot removal), continuous/running (faster, better wound edge eversion, but if one stitch breaks, the entire suture line may fail), horizontal mattress (everts wound edges, tension reduction), vertical mattress (everts edges, good for deep closure), subcuticular (intradermal—best cosmetic result, eliminates suture marks), and purse-string. Alternatives to sutures: surgical staplers (fast, consistent—bowel anastomoses, skin closure), tissue adhesives (cyanoacrylate/Dermabond—for superficial wounds, topical use), adhesive strips (Steri-Strips—for superficial, low-tension wounds), and knotless barbed sutures (Knotless Tissue Control—used in laparoscopic and robotic surgery).

TACO

– Transfusion-Associated Circulatory Overload, a complication of blood transfusion due to volume overload causing pulmonary edema. Risk factors: pre-existing cardiac or renal disease, extremes of age, large transfusion volume, rapid infusion rate. Presents with dyspnea, orthopnea, hypertension, JVD, pulmonary edema, and BNP elevation within 6-12 hours of transfusion. Distinguish from TRALI: TRALI has hypotension, fever, and no fluid overload signs. Management: diuresis (furosemide), oxygen, and slow or split transfusion units in high-risk patients. Prevention: slow transfusion rate, pre-transfusion furosemide.

Testicular Torsion

– A surgical emergency caused by twisting of the spermatic cord, leading to occlusion of the testicular artery and venous outflow, resulting in testicular ischemia and infarction if not treated promptly. Peak incidence: neonatal period and adolescence (12–18 years). Anatomy: the bell-clapper deformity (high insertion of the tunica vaginalis, allowing the testis to rotate freely within the scrotum) is a bilateral predisposing factor present in 12% of males. Torsion is typically intravaginal (twisting of the testis within the tunica vaginalis). Clinical presentation: sudden onset of severe, unilateral scrotal pain (often waking the patient from sleep), nausea and vomiting, lower abdominal pain, and the affected testis is swollen, tender, and elevated (high-riding testis) with a horizontal lie. The cremasteric reflex is absent on the affected side. Prehn sign (elevation relieves pain in epididymitis) is unreliable. Differential diagnosis: epididymo-orchitis (gradual onset, dysuria, fever, intact cremasteric reflex), torsion of the appendix testis (tender nodule at the superior pole, 'blue dot sign'), and strangulated inguinal hernia. Diagnosis: Doppler ultrasound (reduced or absent testicular blood flow) is the gold standard imaging; sensitivity >95%. Management: manual detorsion (rotate the testis outward—'open the book'—medial to lateral; success confirmed by pain relief and return of Doppler flow) should be attempted immediately, followed by emergency surgical exploration (scrotal exploration, detorsion, and orchiopexy: fixation of both testes to the scrotal wall with non-absorbable sutures). If the testis is non-viable (black, no bleeding from a small incision): orchiectomy. Salvage rates: >90% if surgery within 6 hours, 50% at 12 hours, <10% after 24 hours. Bilateral orchiopexy is mandatory due to the high rate of bilateral anatomical predisposition.

Thoracotomy

– A surgical incision into the chest wall to access the thoracic cavity (pleural space, lungs, heart, great vessels, esophagus, diaphragm, mediastinum). Types: (1) Posterolateral thoracotomy (most common)—incision below the tip of the scapula, dividing the latissimus dorsi and serratus anterior, entering the chest through the 4th, 5th, or 6th intercostal space; provides excellent access for pulmonary lobectomy, pneumonectomy, oesophagectomy, and thoracic aortic surgery. (2) Anterolateral thoracotomy—emergency access for trauma, pericardial window, or lung biopsy. (3) Median sternotomy—midline division of the sternum (sternotomy), used for cardiac surgery (CABG, valve surgery, thymectomy, anterior mediastinal mass). (4) Clamshell incision (bilateral anterolateral thoracotomy with transverse sternotomy)—for bilateral lung transplantation, massive trauma. (5) Axillary thoracotomy—for first rib resection, sympathectomy. Complications: post-thoracotomy pain (significant, long-lasting—use of epidural analgesia, paravertebral block, intercostal nerve blocks), wound infection, hemorrhage, persistent air leak, pneumonia, and atelectasis. Minimally invasive alternatives: video-assisted thoracoscopic surgery (VATS)—reduces pain, hospital stay, and recovery time.

Thyroid Surgery

– lobectomy, subtotal thyroidectomy, and total thyroidectomy. Indications: Bethesda V/VI nodules on cytology, thyroid cancer (papillary >1.5 cm, follicular, medullary, anaplastic), Graves disease (compressive symptoms, failed medical therapy, severe ophthalmopathy), and multinodular goiter with compressive symptoms or substernal extension. Lobectomy is appropriate for Bethesda III/IV nodules and low-risk papillary microcarcinoma (<1.5 cm, solitary, no nodal involvement). Total thyroidectomy is standard for bilateral disease, high-risk features, medullary carcinoma, and anaplastic carcinoma (palliative for airway obstruction). Central compartment dissection (level VI) is performed for clinically involved nodes; prophylactic dissection is controversial due to increased risk of nerve injury and hypoparathyroidism. Lateral neck dissection (levels II-V) is reserved for confirmed cervical metastases. Intraoperative nerve monitoring facilitates identification of the recurrent laryngeal nerve. The external branch of the superior laryngeal nerve is at risk during ligation of the superior thyroid pole. The parathyroid glands are at risk of devascularization or accidental excision. Complications: recurrent laryngeal nerve injury (2-5% transient, <1% permanent) presents with hoarseness and aspiration; bilateral injury causes stridor requiring tracheostomy. Hypoparathyroidism (transient 10-30%, permanent 1-2%) requires calcium and calcitriol supplementation, guided by post-operative PTH monitoring. Haematoma (0.5-2%) requires emergency evacuation to prevent airway compromise. Wound infection and hypertrophic scarring are less common. TSH suppression therapy reduces recurrence risk in thyroid cancer. Radioactive iodine ablation is given for high-risk differentiated cancer after total thyroidectomy.

TIPS (Transjugular Intrahepatic Portosystemic Shunt)

– A radiologically placed shunt connecting the portal vein to a hepatic vein within the liver, creating a portosystemic shunt to reduce portal pressure. Indications: variceal bleeding (refractory or recurrent), refractory ascites, and hepatic hydrothorax. Procedure: transjugular access, hepatic vein cannulation, needle puncture through liver parenchyma into the portal vein, balloon dilation, and covered stent placement. Portal pressure gradient goal: <12 mmHg. Contraindications: severe pulmonary hypertension, right heart failure, severe hepatic encephalopathy, and severe liver failure (MELD >18-20). Complications: encephalopathy (30-40%), shunt stenosis/occlusion, bleeding, infection, and radiation exposure.

Total Hip Arthroplasty (THA)

– Replacement of the hip joint with prosthetic components. Acetabular component: metal shell + polyethylene liner. Femoral component: metal stem + femoral head (cobalt-chrome or ceramic). Fixation: cemented (PMMA, for elderly, osteoporotic bone), cementless (porous-coated, for younger patients, biologic fixation), and hybrid. Approaches: posterior (most common, dislocation risk 2-5%), direct anterior (DAA, muscle-sparing, less pain), anterolateral, and lateral (Hardinge). Indications: end-stage osteoarthritis, avascular necrosis, rheumatoid arthritis, and hip fracture (femoral neck). Complications: dislocation, infection (1-2%), VTE, periprosthetic fracture, leg length discrepancy, and aseptic loosening (long-term).

Total Knee Arthroplasty (TKA)

– Replacement of the knee joint with prosthetic components. Femoral component (cobalt-chrome), tibial component (titanium tray + polyethylene insert), and patellar component (polyethylene, optional). Fixation: cement (PMMA) most common in older adults; cementless in younger. Alignment: mechanical (neutral, most common), kinematic, and anatomic. Indications: end-stage osteoarthritis (most common), inflammatory arthritis, and post-traumatic arthritis. Complications: infection (1-2%), stiffness (manipulation under anesthesia), instability, patellar complications (maltracking, fracture), periprosthetic fracture, VTE, and aseptic loosening.

Tracheostomy

– A surgical opening (stoma) in the trachea through the neck. Percutaneous dilational tracheostomy (PDT, ICU bedside, Seldinger technique, most common) vs. open surgical tracheostomy (OR, for difficult anatomy, large goiter, obese neck). Indications: prolonged mechanical ventilation (expected >14-21 days, reduces VAP, sedation, and ICU length of stay), upper airway obstruction, and severe secretion management. Tracheostomy tube: cuffed (for ventilation) or uncuffed/fenestrated (for spontaneous breathing and speech). Complications: bleeding, infection, pneumothorax, subcutaneous emphysema, tracheal stenosis, tracheomalacia, and stomal infection.

Tracheotomy

– See Tracheostomy.

TRALI

– Transfusion-Related Acute Lung Injury, a life-threatening complication of blood transfusion characterized by acute hypoxemia and bilateral pulmonary edema within 6 hours of transfusion, without evidence of volume overload. Pathophysiology: donor antibodies (anti-HLA, anti-HNA) in plasma-rich products (FFP, platelets) activate recipient neutrophils, causing endothelial injury and capillary leak in the lungs. Presents with dyspnea, hypoxemia, fever, and bilateral infiltrates on CXR (mimics ARDS). Management: supportive care (oxygen, mechanical ventilation), similar to ARDS. Prevention: use male-only plasma to reduce antibody risk. Mortality 5-10%.

Transfusion Reaction

– An adverse reaction to transfusion of blood products. Acute hemolytic transfusion reaction: ABO incompatibility, immediate fever/chills, hypotension, back pain, DIC, renal failure; management: STOP transfusion, maintain BP/urine output, supportive care. Febrile non-hemolytic transfusion reaction: most common, fever/chills without hemolysis. Allergic: urticaria; treat with antihistamines. Anaphylactic: severe allergic, IgA deficiency; treat with epinephrine. Transfusion-related acute lung injury (TRALI): acute hypoxemia, bilateral pulmonary edema within 6 hours; supportive care. Transfusion-associated circulatory overload (TACO): pulmonary edema from volume overload; diuretics. Bacterial contamination: sepsis from contaminated unit.

Transplant Rejection

– Immune-mediated destruction of a transplanted organ or tissue, classified by timing and mechanism. Hyperacute: preformed donor-specific antibodies, minutes to hours post-transplant, immediate graft thrombosis, irreversible; prevention by cross-matching. Acute: T-cell mediated (most common, days to weeks/months), responds to immunosuppression; presents with graft dysfunction and biopsy findings. Chronic: antibody-mediated, progressive, irreversible fibrosis and graft dysfunction over months to years. Diagnosis: graft biopsy (Banff classification). Treatment: acute rejection — pulse corticosteroids, thymoglobulin, IVIG, plasmapheresis; chronic — optimize immunosuppression, manage risk factors.

Transplant Surgery

– Transplant surgery involves replacing a failing organ with a healthy donor organ. Kidney transplant: living donors (laparoscopic nephrectomy, left kidney preferred) or deceased donors (standard criteria, expanded criteria, DCD). Induction immunosuppression includes basiliximab (standard risk) or thymoglobulin/alemtuzumab (high immunologic risk). Maintenance typically uses tacrolimus, mycophenolate, and prednisone; belatacept offers CNI-free options for EBV-positive patients. Rejection: hyperacute (preformed antibodies, immediate graft loss), acute cellular (T-cell mediated, responds to steroid pulse or ATG), acute antibody-mediated (C4d+ on biopsy with donor-specific antibodies, treated with IVIG, plasmapheresis, rituximab), and chronic allograft nephropathy (interstitial fibrosis and tubular atrophy). Liver transplant allocation uses the MELD-Na score. Donor hepatectomy may be whole or split/living donor. Recipient hepatectomy uses bicaval or piggyback technique with portal vein, hepatic artery, and biliary reconstruction (duct-to-duct or Roux-en-Y hepaticojejunostomy). Complications: primary non-function, hepatic artery thrombosis (3-5%, requires retransplantation unless collateralized), biliary leak/stricture, portal vein thrombosis, and rejection. PTLD is an EBV-driven lymphoproliferative complication managed by immunosuppression reduction and rituximab. Pancreas transplant (SPK, PAK, PTA) is performed for type 1 diabetes with ESRD; enteric drainage is preferred for exocrine secretion. Long-term management: immunosuppression optimization, infection prophylaxis (CMV, EBV, fungal), and surveillance for graft dysfunction, PTLD, and cardiovascular disease. One-year graft survival exceeds 90% for kidney and 85% for liver; 5-year survival >70% for both.

Trauma Surgery

– Trauma surgery encompasses the acute care of injured patients, guided by ATLS (ABCDE) principles: Airway maintenance with cervical spine protection, Breathing and ventilation, Circulation with hemorrhage control, Disability/neurologic assessment, and Exposure/environmental control. Hemorrhage control is the priority, achieved via direct pressure, tourniquets for extremity injuries, pelvic binders for pelvic fractures, and REBOA for exsanguinating torso hemorrhage. Damage control surgery is indicated for patients with the lethal triad of hypothermia, acidosis, and coagulopathy. Phase 1 involves abbreviated laparotomy with bleeding control (packing, shunting) and contamination control. Phase 2 is ICU resuscitation to correct hypothermia (active rewarming), coagulopathy (massive transfusion at 1:1:1 ratio of pRBCs, FFP, platelets; tranexamic acid 1 g IV within 3 hours of injury), and acidosis. Phase 3 is return to the OR for definitive repair after physiologic normalization. Damage control resuscitation incorporates permissive hypotension (SBP 80-90 mmHg) in penetrating torso trauma without head injury. Definitive surgical management depends on injury pattern and may include splenectomy, liver packing and resection, bowel resection, vascular repair, and fracture fixation. Resuscitative thoracotomy is indicated for penetrating chest trauma with witnessed arrest and short transport time (best survival for cardiac tamponade). Tension pneumothorax requires immediate needle decompression (14G, 2nd ICS midclavicular line) followed by tube thoracostomy. Massive haemothorax (>1500 mL initial output) requires expeditious thoracotomy. Abdominal compartment syndrome (bladder pressure >20 mmHg with organ dysfunction) may necessitate decompressive laparotomy with temporary closure. FAST examination and CT imaging guide surgical decision-making. Enteral nutrition is initiated within 24-48 hours. Outcome depends on injury severity, promptness of intervention, and successful resuscitation.

Traumatic Brain Injury

– Disruption of brain function caused by an external mechanical force. Severity: mild (GCS 13-15, concussion), moderate (GCS 9-12), severe (GCS 3-8). Primary injury: direct damage at moment of impact (contusion, diffuse axonal injury, intracranial hemorrhage). Secondary injury: evolves over hours to days (cerebral edema, increased ICP, hypoxia, hypotension, ischemia, seizures). Intracranial hemorrhage: epidural (arterial, lenticular), subdural (venous, crescentic), intracerebral, subarachnoid. Management: ABCs, prevent secondary injury (maintain SBP >90, SpO2 >94%, PaCO2 35-45), ICP monitoring (goal <22), CPP >60 mmHg, head of bed 30°, hyperosmolar therapy (mannitol, hypertonic saline), seizure prophylaxis, and surgical evacuation of mass lesions.

Trephining (Trepanation)

– A surgical procedure in which a hole is drilled or scraped into the skull (cranium), exposing the dura mater. Historically, trephining dates back to Neolithic times (10,000 BCE) and was performed for ritual, therapeutic (releasing 'evil spirits'), and post-traumatic indications. Modern medical trephination: (1) Burr hole—small craniotomy using a handheld or electric drill (burr) for evacuation of intracranial haematomas (epidural, subdural—including twist-drill craniostomy for chronic subdural haematoma), placement of external ventricular drain (EVD) for intracranial pressure monitoring or CSF drainage, and brain biopsy. (2) Decompressive craniectomy—removal of a large skull flap to treat refractory intracranial hypertension (traumatic brain injury, malignant middle cerebral artery infarction, cerebral venous sinus thrombosis). (3) Craniotomy—larger opening for definitive neurosurgery (tumor resection, aneurysm clipping, epilepsy surgery). (4) Stereotactic trephination—frame-based or frameless stereotactic navigation for biopsy, deep brain stimulation electrode placement, or tumor ablation. Instruments: Hudson brace, perforator and burr, craniotome, and high-speed drills. Complications: bleeding (epidural, subdural, intracerebral), infection (osteomyelitis, meningitis, brain abscess), CSF leak, pneumocephalus, seizures, and brain herniation.

Tubal Ligation

– Surgical sterilization by occluding the fallopian tubes, preventing the ovum from reaching the uterus. Postpartum tubal ligation: after vaginal or cesarean delivery (Pomeroy technique, Parkland). Interval: laparoscopic (bands, clips, bipolar cautery) or mini-laparotomy (Hulka or Filshie clip). Reversal possible with tubal reanastomosis (pregnancy rate 50-80%). Female sterilization: 99% effective, permanent. Complications: ectopic pregnancy if procedure fails (higher after bipolar cautery than clips). Essure (hysteroscopic coils) removed from the market.

Umbilical Hernia

– A hernia through the umbilical ring, common in infants (most umbilical hernias close spontaneously by age 4-5) and adults (risk factors: obesity, pregnancy, ascites, cirrhosis). Presents as a reducible bulge at the umbilicus, worse with increased intra-abdominal pressure. In adults: risk of incarceration and strangulation is higher. Diagnosis: clinical. Treatment: in children: observe, surgically repair if persists >age 4-5 or >2 cm. In adults: surgical repair (primary suture repair, mesh onlay, or laparoscopic IPOM) due to risk of complications.

Ureteral Stent (Double-J Stent)

– A tube placed in the ureter from the renal pelvis to the bladder to maintain patency, with pigtail coils at both ends (hence double-J). Indications: ureteral obstruction (stones, stricture, malignancy), after ureteroscopy (prevent post-op obstruction), and as a pre-treatment for complex ureteral stones. Placed cystoscopically or antegrade through a nephrostomy. Sizes: 6-8 Fr, lengths 22-30 cm. Left in for weeks to months (though should not exceed 3 months due to encrustation). Complications: stent-related discomfort (urgency, frequency, hematuria, flank pain — very common), infection, migration, encrustation (forgotten stent), and stone formation.

Urethral Stricture

– Narrowing of the urethra due to fibrosis of the corpus spongiosum, most commonly caused by trauma (pelvic fracture, straddle injury, iatrogenic from catheterisation or urethral instrumentation), infection (gonococcal urethritis leading to stricture years later, now less common in the antibiotic era), and lichen sclerosus (Balanitis Xerotica Obliterans, BXO). Presents with progressive urinary obstruction: weak stream, hesitancy, straining, incomplete bladder emptying, frequency, and recurrent UTIs. Complete obstruction causes acute urinary retention. Diagnosis: retrograde urethrogram demonstrates the location and length of the stricture. Urethroscopy confirms and may allow simultaneous treatment. Treatment depends on location, length, and severity: urethral dilation (temporary, high recurrence rate), internal urethrotomy (optical incision of the stricture, 50% recurrence at 2 years for bulbar strictures <2 cm), and urethroplasty (open surgical reconstruction, the gold standard for recurrent or long strictures, success rates >90% for primary bulbar repairs). Complications include recurrence, erectile dysfunction, and incontinence.

Urology

– Surgical specialty focused on the diagnosis and treatment of diseases of the male and female urinary tract (kidneys, ureters, bladder, urethra) and the male reproductive system (prostate, penis, testes, seminal vesicles). Subspecialties: urologic oncology (prostate, bladder, kidney, testicular cancer), endourology (minimally invasive stone surgery, ureteroscopy, PCNL), female urology (incontinence, pelvic organ prolapse), neurourology (voiding dysfunction, neurogenic bladder), andrology (male infertility, erectile dysfunction), and pediatric urology (congenital anomalies, hypospadias, undescended testes). Common procedures: transurethral resection of the prostate (TURP), cystoscopy, vasectomy, nephrectomy, and radical prostatectomy.

Varicocele

– Abnormal dilatation of the pampiniform plexus of veins in the spermatic cord, caused by incompetent or absent valves in the internal spermatic (testicular) vein. Left-sided varicoceles are far more common (85-90%) because the left testicular vein inserts into the left renal vein at a right angle, whereas the right testicular vein inserts obliquely into the inferior vena cava. Acute-onset right-sided varicocele raises suspicion for retroperitoneal pathology (renal cell carcinoma with tumor thrombus extending into the IVC, retroperitoneal mass). Presents as a scrotal swelling described as a bag of worms on palpation, typically painless or causing a dull ache/heavy sensation that worsens with standing or Valsalva and improves with lying supine. Grading: Grade I (palpable only with Valsalva), Grade II (palpable without Valsalva), Grade III (visible through the scrotal skin). Varicocele is the most common correctable cause of male infertility: elevated testicular temperature from venous congestion impairs spermatogenesis (reduced sperm count, motility, and increased abnormal morphology). Diagnosis: clinical examination and scrotal ultrasound with

Doppler. Treatment: indicated for symptomatic varicocele, infertility with abnormal semen analysis, testicular hypotrophy in adolescents, or patient preference. Options: microscopic varicocelectomy (gold standard, highest success, lowest recurrence), laparoscopic varicocelectomy, or percutaneous embolization. Subclinical varicoceles (detected only on ultrasound) do not require treatment.

Vascular Access

– Vascular access for hemodialysis includes arteriovenous fistulas (AVF), arteriovenous grafts (AVG), and central venous catheters (CVC). The fistula first initiative prioritizes AVF as the gold standard due to best long-term patency and lowest complication rates. Common configurations: radiocephalic (wrist), brachiocephalic (elbow), and brachiobasilic transposition (upper arm). Preoperative vein mapping by ultrasound is essential for planning. AVGs (PTFE) mature faster (2-4 weeks vs. 2-4 months for AVF) and are useful when native veins are inadequate, but have higher rates of thrombosis and infection. Tunneled CVCs allow immediate use and serve as a bridge to maturation or for patients with exhausted peripheral options, but carry the highest infection and central venous stenosis risk. Non-tunneled CVCs are for short-term emergent use. Maturation criteria: vein diameter >6 mm, depth <6 mm, flow >600 mL/min. Indications: end-stage renal disease requiring hemodialysis. Complications: stenosis (angioplasty, stenting), thrombosis (thrombectomy, catheter-directed thrombolysis), steal syndrome (hand ischemia managed by DRIL or PAI procedures), aneurysm and pseudoaneurysm (resection, ligation), infection (graft excision, antibiotics), and high-output cardiac failure (revision, banding, or ligation). CVC complications: pneumothorax (reduced by ultrasound guidance), arterial puncture, catheter-related bloodstream infection (removal, tip culture, antibiotics), central venous stenosis, and air embolism. Patency: AVF > AVG > CVC.

Vasectomy

– Male surgical sterilization: cutting and sealing the vas deferens. No-scalpel vasectomy: puncture technique, less bleeding, faster recovery. Conventional: small incisions. Fascial interposition: prevents recanalization. Post-procedure: semen analysis at 12 weeks or 20 ejaculates (azoospermia confirms sterility). Failure rate: <1% (from recanalization). Reversal: vasovasostomy (microsurgical, success 70-95% depending on time since vasectomy). Complications: hematoma, infection, chronic scrotal pain (congestive epididymitis, post-vasectomy pain syndrome, 1-2%), and sperm granuloma.

Venous Stasis Ulcer

– A chronic wound caused by chronic venous insufficiency, most common type of leg ulcer (70-80%). Located on the medial gaiter area (medial malleolus). Presents with shallow, irregular ulcer with granulation tissue, surrounded by hemosiderin deposition (brownish discoloration), lipodermatosclerosis, edema, and varicose veins. Often painful and associated with venous insufficiency (CEAP classification). Diagnosis: clinical; venous duplex ultrasound to assess reflux. Treatment: compression therapy (multi-layer bandaging, graded compression stockings 30-40 mmHg) is the mainstay. Adjunctive: leg elevation, wound debridement, moist wound healing, and pentoxifylline. Refractory: venous ablation or surgery.

Venous Ulcer

– See Venous Stasis Ulcer.

Wound Dehiscence

– Separation of the layers of a surgical wound after closure, most commonly occurring within 3-14 days postoperatively. Risk factors: wound infection, poor nutrition (low albumin), immunosuppression, obesity, diabetes, smoking, increased intra-abdominal pressure (coughing, straining, ileus), and poor surgical technique (tight sutures, extensive electrocautery). Presents with sudden serosanguinous drainage from the wound, often with a pop sensation. Superficial dehiscence: skin and subcutaneous layers open. Deep (fascial) dehiscence: separation of the fascia, often with evisceration (viscera protruding through the wound) → surgical emergency. Management: superficial → wound care and NPWT. Deep → emergent return to OR for closure with retention sutures.

Xenograft (Heterograft)

– A surgical graft taken from a donor of a different species (xenos = foreign, Greek). Xenografts are used as temporary biological dressings or as tissue replacements. The most common xenograft source is porcine (pig) skin, used as a temporary wound covering for extensive burns (partial-thickness and deep partial-thickness burns) and large traumatic skin loss. Porcine xenografts adhere to the wound bed, reduce fluid and protein loss, decrease pain, and provide a barrier against bacterial contamination while the patient's own skin regenerates or until autografting is possible. Other xenograft sources include bovine (cow) pericardium (used in cardiac surgery for valve reconstruction, pericardial patch repair, and as a bioprosthetic heart valve material — Carpentier-Edwards, Hancock valves), and bovine bone grafts (used in orthopaedic and dental surgery for bone void filling, spinal fusion, and maxillofacial reconstruction). Xenografts trigger a strong immune response (hyperacute rejection mediated by pre-existing xenoreactive antibodies against Gal-alpha-1,3-Gal epitopes on porcine cells) and are therefore rejected within 7-14 days if used as viable tissue, limiting their use to temporary coverage. Genetically modified pigs (knockout of Gal epitopes) have been developed to reduce immunogenicity and enable longer-term xenograft survival. The first pig-to-human heart transplant (xenotransplantation) was performed in 2022 (University of Maryland, 10 gene-modified pig heart, recipient survived 2 months). Compared to allografts (human donor grafts) and autografts (patient's own tissue), xenografts have the advantages of unlimited availability and no donor site morbidity, but are inferior in durability and immunocompatibility for permanent replacement.

Chapter 35

General

Acid-Base Disorder

– Disturbance of the body's acid-base balance, classified by primary disturbance and compensation. Respiratory acidosis: hypoventilation, high PaCO2, low pH. Respiratory alkalosis: hyperventilation, low PaCO2, high pH. Metabolic acidosis: low HCO3, low pH; with high AG (DKA, lactic acidosis, renal failure, toxins) or normal AG (diarrhea, RTA). Metabolic alkalosis: high HCO3, high pH; from vomiting, diuretics, or hyperaldosteronism. Mixed disorders: two or more primary disturbances simultaneously. Diagnosis: ABG + electrolytes, using Winters formula for appropriate compensation. Management: treat the underlying cause.

Anal Intercourse

– The insertion of the penis into the anus and rectum for sexual pleasure. The anal canal is rich in nerve endings, making it a sensitive erogenous zone. The rectum lacks natural lubrication, so artificial lubricants are essential to prevent mucosal trauma. Risks: STI transmission is higher than with vaginal intercourse (especially HIV, HPV, hepatitis A/B, gonorrhea, chlamydia, and syphilis) due to the fragile rectal mucosa and increased HIV target cells (CD4+ T-cells) in the rectal tissue. Condom use significantly reduces STI risk. Other risks: anal fissures, fecal incontinence (rare with healthy sphincter), and foreign body retention. Anal intercourse does not cause hemorrhoids.

Anorexia

– Loss of appetite, distinct from anorexia nervosa (the eating disorder). Causes include: infections (most common), malignancies (especially GI, pancreatic, lung), chronic diseases (CKD, COPD, heart failure, liver disease), medications (chemotherapy, opioids, antibiotics), psychological disorders (depression, anxiety), and endocrine disorders (Addison's disease, hypothyroidism). Pathophysiology involves cytokines (TNF-α, IL-6), leptin dysregulation, and altered hypothalamic appetite centers. Consequences: malnutrition, weight loss, sarcopenia, and immune dysfunction. Management: treat underlying cause, nutritional support (oral supplements, enteral feeding if severe), appetite stimulants (megestrol acetate, corticosteroids, dronabinol), and dietary modifications (small frequent meals, preferred foods).

Anuria

– Absence of urine output (<100 mL/day). Classification by cause: prerenal (severe hypovolaemia, shock, renal artery occlusion), renal (acute tubular necrosis, acute cortical necrosis, rapidly progressive glomerulonephritis, bilateral renal artery occlusion), and postrenal (bilateral ureteric obstruction, bladder outlet obstruction, urethral obstruction). Anuria is a medical emergency requiring urgent investigation: renal ultrasound to exclude obstruction, bladder catheterisation, urinalysis, serum creatinine, and renal biopsy if indicated. Management: treat underlying cause (fluid resuscitation for prerenal, relief of obstruction for postrenal, dialysis for severe renal failure). Complications: hyperkalaemia, volume overload, uraemia, and metabolic acidosis.

Beriberi

– A nutritional deficiency disease caused by inadequate thiamine (vitamin B1), occurring in two forms. Wet beriberi presents with high-output heart failure, peripheral vasodilation, edema, and tachycardia due to impaired cardiac energy metabolism. Dry beriberi presents with peripheral neuropathy (symmetric sensory and motor deficits, paresthesias, muscle wasting). Risk factors: polished rice as dietary staple, alcoholism, malabsorption, and pregnancy. Treatment: thiamine supplementation (parenteral initially for severe cases). Wernicke-Korsakoff syndrome is the neuropsychiatric manifestation of thiamine deficiency in alcoholism.

Bites and Stings

– Injuries caused by the biting or stinging apparatus of arthropods (insects, spiders, ticks), marine animals, snakes, and mammals. Insect stings (bees, wasps, hornets, ants): local reaction (pain, erythema, edema, pruritus at sting site, self-limited over hours to days); large local reaction (extensive swelling spreading beyond the sting site, diameter >10 cm, peaking at 24-48 hours, resolves in 5-10 days); systemic allergic reaction/anaphylaxis (urticaria, angioedema, bronchospasm, hypotension, cardiovascular collapse — medical emergency requiring intramuscular epinephrine). Management: remove stinger (scrape, do not squeeze), cold compress, antihistamines (cetirizine, loratadine), and NSAIDs for pain. Anaphylaxis: IM epinephrine (0.3-0.5 mg, anterolateral thigh, repeat q5-15min as needed), IV fluids, supplemental oxygen, and hospital observation. Venom immunotherapy for patients with prior systemic reactions. Tick bites: Lyme disease (Borrelia burgdorferi, erythema migrans rash), Rocky Mountain spotted fever, and tick-borne encephalitis. Remove tick with fine-tipped tweezers, grasp as close to skin as possible, pull straight upward. No folklore remedies (matches, petroleum jelly, nail polish). Snake bites: pit vipers (rattlesnakes, copperheads, cottonmouths) cause local tissue destruction, coagulopathy, and systemic effects; treatment is antivenom (Fab fragments). Coral snakes cause neurotoxicity (descending paralysis, respiratory failure). Management: immobilisation, splinting, pressure immobilisation bandage for elapid bites (not for pit viper bites, which increases local necrosis). Mammal bites (dogs, cats, humans): high risk of infection (Pasteurella multocida in cat bites, Capnocytophaga canimorsus in dog bites, Eikenella corrodens in human bites). Management: wound irrigation, debridement, rabies prophylaxis if indicated, tetanus vaccination update, and prophylactic antibiotics (amoxicillin-clavulanate) for high-risk bites (cat, hand, deep, delayed presentation, immunosuppressed). Marine stings (jellyfish, Portuguese man-of-war): nematocyst discharge causes painful, linear, urticarial eruptions. Management: rinse with vinegar (acetic acid) to inactivate nematocysts, remove tentacles with gloved hands, hot water immersion (45°C for 20 minutes) for pain, and antivenom for box jellyfish (Chironex fleckeri) stings.

Body Mass Index (BMI)

– A measurement that estimates body fat based on weight and height, calculated as weight in kilograms divided by height in meters squared (kg/m²). WHO classification: underweight <18.5, normal 18.5-24.9, overweight 25-29.9, obese class I 30-34.9, class II 35-39.9, class III (morbid obesity) ≥40. BMI is a screening tool, not a diagnostic test for body fatness, and does not account for muscle mass, bone density, body composition, or fat distribution. Limitations: overestimates body fat in athletes (high muscle mass), underestimates in older adults (low muscle mass), and does not reflect visceral adipose tissue (the most metabolically harmful fat). Waist circumference is a better predictor of cardiovascular risk than BMI alone. Clinically, BMI is used to stratify risk for obesity-related conditions (diabetes, hypertension, CVD, sleep apnea, certain cancers) and guide treatment decisions (bariatric surgery eligibility typically requires BMI ≥40 or ≥35 with comorbidities).

Body Temperature

– See Temperature (Body Temperature).

Cachexia

– A complex metabolic syndrome characterized by weight loss, muscle wasting, and systemic inflammation, occurring in chronic diseases such as cancer (most common), heart failure, COPD, chronic kidney disease, and AIDS. Diagnostic criteria: >5% weight loss over 6 months in the absence of simple starvation, or >2% weight loss with BMI <20 or sarcopenia. Pathophysiology: pro-inflammatory cytokines (TNF-alpha, IL-6), tumor-derived factors, hypermetabolism, and catabolic state. Unlike starvation, cachexia cannot be fully reversed by nutritional supplementation alone. Management: treat underlying disease, multimodal approach (nutritional support, exercise, anti-inflammatory agents, progestins like megestrol acetate, corticosteroids for short-term appetite stimulation). Prognosis: poor; cachexia is an independent predictor of mortality.

Confidentiality

– The ethical and legal duty of healthcare professionals to protect patient information from unauthorized disclosure. Founded on the principle of respect for patient autonomy and trust, confidentiality is a core requirement of medical ethics (GMC guidelines) and law (GDPR, Data Protection Act, Caldicott principles). Information should only be shared with the patient's explicit consent, or when required by law (public health reporting, court order), or when necessary to protect the patient or others from serious harm (safeguarding, GMC duty of candor). Breach of confidentiality can result in disciplinary action, professional sanctions, and legal liability. Exceptions: serious communicable disease reporting (HIV, TB), child protection concerns, fitness to drive notification, and terrorism prevention. The duty of confidentiality continues after death (access to records of the deceased requires executor or next-of-kin consent).

Debility

– A state of generalized weakness, loss of strength, and reduced functional capacity, often associated with chronic illness, malnutrition, prolonged bed rest, or advanced age. Debility is a nonspecific term that may reflect underlying conditions such as sarcopenia, cachexia, deconditioning, or frailty. Management focuses on addressing the underlying cause, nutritional support, physical therapy, and gradual mobilization to prevent further functional decline.

Deficiency Diseases

– Diseases caused by a lack of specific essential nutrients in the diet, including vitamins, minerals, or macronutrients. Examples: scurvy (vitamin C deficiency–gingival hypertrophy, perifollicular hemorrhages, impaired wound healing), beriberi (thiamine deficiency–wet: heart failure, dry: neuropathy), pellagra (niacin deficiency–dermatitis, diarrhea, dementia), rickets (vitamin D deficiency in children–bowing deformities, growth retardation), osteomalacia (vitamin D deficiency in adults–bone pain, fractures), kwashiorkor (protein deficiency–edema, hepatomegaly), marasmus (calorie deficiency–wasting, growth failure), iron deficiency anemia, and vitamin B12 deficiency. Treatment: replacement of the deficient nutrient and treatment of the underlying cause.

Detoxification

– The process of removing toxic substances from the body or rendering them harmless. In medical contexts, medically supervised withdrawal from addictive substances (alcohol, opioids, benzodiazepines), involving gradual dose reduction or substitution with longer-acting agents to manage withdrawal symptoms safely. Physiologic detoxification is performed primarily by the liver (phase I: cytochrome P450 oxidation, reduction, hydrolysis; phase II: conjugation with glucuronide, sulfate, glutathione, or glycine to increase water solubility for renal or biliary excretion). Detoxification is distinct from addiction treatment and is only the first step in substance use disorder management.

Douche

– A stream of water or solution directed into a body cavity, most commonly the vagina, for cleansing, therapeutic, or contraceptive purposes. Medical use: therapeutic douches with antiseptic solutions (povidone-iodine, vinegar, chlorhexidine) for treating specific vaginal infections (trichomoniasis, bacterial vaginosis) as adjunctive therapy. Routine vaginal douching is not recommended and is associated with adverse outcomes: disruption of normal vaginal flora (increased risk of bacterial vaginosis, candida overgrowth), pelvic inflammatory disease, ectopic pregnancy, and increased HIV transmission risk.

Dropsy (Edema)

– An historic term for edema, referring to the abnormal accumulation of serous fluid in the tissues or body cavities. Types: peripheral edema (dependent: feet, ankles, legs, sacrum), pulmonary edema (fluid in lung interstitium and alveoli–causes include left heart failure, ARDS, high altitude), ascites (fluid in peritoneal cavity–cirrhosis, heart failure, malignancy), anasarca (generalized severe edema). Pathophysiology: increased hydrostatic pressure (heart failure, venous obstruction), decreased plasma oncotic pressure (nephrotic syndrome, cirrhosis, malnutrition), increased capillary permeability (inflammation, burns, sepsis), and lymphatic obstruction (lymphedema). Treatment: address underlying cause, sodium restriction, diuretics, and mechanical compression.

Ejaculation

– The expulsion of seminal fluid from the male urethra during sexual climax, a reflex coordinated by the sympathetic nervous system (T10–L2) and involving contraction of the epididymis, vas deferens, seminal vesicles, prostate, and bulbospongiosus muscle. The process has two phases: emission (deposition of sperm and seminal fluid into the posterior urethra) and expulsion (rhythmic contraction of the pelvic floor muscles propelling semen through the urethra). Ejaculatory disorders include premature ejaculation (most common), delayed ejaculation, retrograde ejaculation (semen flows backward into the bladder), anejaculation (no ejaculation), and painful ejaculation. Causes: neurologic, pharmacologic (SSRIs, alpha-blockers), psychologic, and structural.

Electrolyte Imbalance

– Abnormal serum concentration of electrolytes (sodium, potassium, calcium, magnesium, phosphate, chloride). Hyponatremia (<135 mEq/L): most common electrolyte disorder; causes include SIADH, heart failure, hypovolemia, and polydipsia. Hypernatremia (>145 mEq/L): free water deficit from poor intake, diabetes insipidus, or diuretics. Hypokalemia (<3.5 mEq/L): diuretics, vomiting, diarrhea, hyperaldosteronism. Hyperkalemia (>5.0 mEq/L): renal failure, ACEi, K-sparing diuretics. Hypocalcemia, hypercalcemia, hypomagnesemia, and hypermagnesemia each have specific causes and clinical presentations. Management: correct the underlying cause and replace/remove electrolyte as needed, with careful monitoring to avoid rapid shifts.

Emission

– The discharge or release of a substance, usually a fluid.

Erectile Dysfunction (ED)

– The persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance, affecting approximately 50% of men aged 40–70. Etiology is predominantly organic (vascular, neurogenic, hormonal, or medication-induced) but often has psychogenic components. ED shares risk factors with cardiovascular disease—hypertension, diabetes, hyperlipidemia, smoking, obesity—and is considered an early marker of endothelial dysfunction. Diagnosis includes history (validated questionnaires like IIEF-5), morning erections (present in psychogenic, absent in organic), and laboratory testing (testosterone, glucose, lipids). First-line treatment: phosphodiesterase-5 inhibitors (sildenafil, tadalafil, vardenafil, avanafil). Second-line: vacuum erection devices, intracavernosal injections (alprostadil, papaverine), and intraurethral suppositories. Surgical: penile prosthesis (inflatable or malleable) for refractory cases. Lifestyle modification (exercise, weight loss, smoking cessation) improves outcomes.

Euthanasia

– The deliberate act of ending a person's life to relieve suffering, typically performed by a physician. Active euthanasia involves directly administering a lethal substance (e.g., barbiturate overdose followed by neuromuscular blockade) at the patient's explicit request. Voluntary euthanasia is performed with the patient's informed consent. Physician-assisted suicide (PAS) involves the physician prescribing a lethal dose of medication that the patient self-administers. Euthanasia is legal in a growing number of countries and jurisdictions under strict regulatory conditions (Belgium, Netherlands, Luxembourg, Canada, Colombia, Spain, New Zealand, and several Australian states, as well as PAS in Switzerland, Germany, Austria, and several US states). It is distinct from palliative sedation (using medication to relieve refractory suffering at the end of life, not intended to hasten death) and withdrawal of life-sustaining treatment (allowing natural death).

Frailty

– A clinical syndrome in older adults characterized by decreased physiologic reserve and increased vulnerability to stressors, resulting in increased risk of falls, hospitalization, disability, and death. Fried frailty phenotype: $\geq$3 of 5 criteria present — unintentional weight loss (>10 lbs in past year), exhaustion (self-reported), weakness (grip strength), slow gait speed, and low physical activity. Frailty is distinct from disability and comorbidity. Pre-frail: 1-2 criteria. Management: multi-domain intervention (exercise including resistance training, nutritional optimization with protein supplementation, medication review, and management of comorbidities). Frailty is potentially reversible with appropriate intervention.

Galvanism

– The application of a direct electric current for therapeutic or diagnostic purposes. Named after Luigi Galvani (1737-1798). Historical uses: galvanic stimulation of muscles, iontophoresis (introduction of ionised drugs through skin), electroconvulsive therapy (modified). Modern context: galvanic vestibular stimulation (research tool), dental galvanism (electrochemical reaction between dissimilar dental metals causing oral discomfort or metallic taste).

Gargle

– The act of rinsing the mouth and throat with a liquid, holding it at the back of the throat while exhaling to create a bubbling action. Used for relief of sore throat or as a delivery method for antiseptics, anaesthetics, and anti-inflammatory agents. Common gargle solutions: warm saline, chlorhexidine (antibacterial), benzydamine (NSAID), lidocaine (local anesthetic), hydrogen peroxide (dilute). Not effective for treating bacterial pharyngitis. Gargling does not reach the larynx or lower airways.

General Practitioner (GP)

– A physician providing comprehensive primary medical care to individuals and families in the community. GPs practice general practice/family medicine, managing undifferentiated presentations, chronic disease, prevention, and coordinating specialist referrals. Training: 4-6 years medical school + 2 years foundation training + 3 years GP specialty training. Gatekeeper role: GPs triage and refer to specialist care, reducing healthcare costs. The role includes continuity of care, home visits, palliative care, and holistic biopsychosocial management. UK: approximately 1 GP per 1,800 patients.

Geriatrics (see Medicine of Ageing)

– The medical specialty concerned with the health and care of older adults. See Medicine of Ageing.

Germ

– A colloquial term for a microorganism that causes disease, typically bacteria, viruses, fungi, and protozoa. Not a scientific classification. The germ theory of disease, established by Pasteur, Koch, and Lister in the 19th century, states that microorganisms are the cause of many infectious diseases. Koch's postulates established criteria for proving causation. Modern understanding includes: pathogenicity (ability to cause disease), virulence (severity of disease), transmission (direct contact, airborne, vector-borne, fomite, food/waterborne).

Grippe (see Influenza)

– An older term for influenza. See Influenza.

Haem- / Haema- / Haemato- / Haemo- (Prefixes relating to blood)

– Prefixes derived from the Greek word haima (blood), used in medical terminology to denote blood-related conditions and procedures. Examples: hemoglobin (oxygen-carrying protein), hemorrhage (bleeding), hemostasis (stopping bleeding), hematology (study of blood), haematuria (blood in urine), hemoptysis (coughing blood), haemangioma (blood vessel tumor), haematocrit (percentage of blood volume occupied by red cells), hemophilia (bleeding disorder), haemarthrosis (blood in a joint), haemodialysis (blood filtration), haemolysis (breakdown of red cells), hemolytic anemia (anemia from red cell destruction), haemopoietic (blood-forming). In American English, the prefix is typically spelled hem-, hema-, hemato-, hemo-.

Hiccough (Hiccup)

– An involuntary, repetitive, spasmodic contraction of the diaphragm followed by sudden closure of the glottis, producing the characteristic 'hic' sound. Hiccups are caused by irritation or stimulation of the phrenic nerve (C3-C5, innervates the diaphragm) or the vagus nerve (CN X). Acute (<48 hours): most are benign and self-limiting, triggered by: carbonated beverages, eating too quickly, spicy foods, alcohol, sudden temperature change (hot then cold), excitement, stress. Persistent (>48 hours) or intractable (>1 month): underlying pathology should be sought. Causes: (1) Gastrointestinal: GORD, hiatus hernia, gastric distension, gastritis, hepatitis, pancreatitis, bowel obstruction. (2) Central nervous system: stroke, multiple sclerosis, brainstem tumor, encephalitis, meningitis. (3) Thoracic: pneumonia, pleurisy, pericarditis, myocardial infarction, aortic aneurysm, mediastinal mass, lung cancer. (4) Metabolic: uraemia, hyponatraemia, diabetes, alcohol use disorder. (5) Drugs: corticosteroids, benzodiazepines, barbiturates, methyldopa, chemotherapy (platinum-based). (6) Psychogenic: anxiety, conversion disorder, malingering. Management: acute: breath-holding, drinking cold water, Valsalva manoeuvre, pulling knees to chest, eating a spoonful of sugar, or using a paper bag (rebreathing). Persistent/intractable: chlorpromazine (dopamine antagonist, 25-50 mg IM/IV or 25 mg PO TID, first-line, FDA approved), metoclopramide, baclofen, gabapentin, nifedipine, omeprazole. In severe cases: phrenic nerve block or surgical phrenic nerve crush (rare).

Histology

– The microscopic study of the structure of tissues and cells. Histology involves preparing thin tissue sections (3-10 micrometres) by fixation (formalin, paraffin embedding), sectioning (microtome), staining (haematoxylin and eosin, H&E), and examination under a light or electron microscope. Basic tissue types: (1) Epithelium: lining of surfaces and cavities, glandular tissue; classified by cell shape (squamous, cuboidal, columnar) and layering (simple, stratified, pseudostratified). (2) Connective tissue: supports and connects other tissues; includes bone, cartilage, adipose, blood, and loose/dense connective tissue; consists of cells (fibroblasts, osteocytes, chondrocytes, adipocytes, macrophages, mast cells) embedded in extracellular matrix (collagen, elastin, proteoglycans). (3) Muscle tissue: skeletal (striated, voluntary), cardiac (striated, involuntary, intercalated discs), smooth (non-striated, involuntary). (4) Nervous tissue: neurons (conduct electrical impulses) and glial cells (support, myelinate, nourish, defend). Special stains: Masson trichrome (collagen), periodic acid-Schiff (PAS, glycogen/mucin), silver stains (reticulin, nerve fibers), immunohistochemistry (antigen detection using labelled antibodies). Histopathology is the histological diagnosis of disease (cancer grading/staging, identification of inflammatory or infectious processes).

Hospice

– A philosophy and model of care that focuses on providing comfort, dignity, and quality of life for patients with terminal illness and their families when curative treatment is no longer the goal. Hospice care is multidisciplinary: medical (pain and symptom management, palliative medicine specialist), nursing (24/7 availability, symptom monitoring), social work (emotional support, practical assistance, advance care planning, referral to financial/legal services), chaplaincy/spiritual care, bereavement support (for family for up to 13 months after death), physiotherapy, occupational therapy, dietitian, and trained volunteers. Settings: inpatient hospice units (free-standing, NHS-funded or charitable), home hospice care (most preferred, enabling patients to die at home if desired), hospital palliative care teams, and day hospice. Admission criteria: life expectancy of 6 months or less (though many have longer trajectories), patient and family agree to focus on comfort rather than life-prolonging treatment. The modern hospice movement began with St. Christopher's Hospice in London (1967, founded by Dame Cicely Saunders), emphasising total pain (physical, psychological, social, spiritual). Over 200 hospices operate in the UK (mostly charitable) and over 5,000 in the US.

Hyperpyrexia

– Extremely elevated body temperature >41.5°C (106.7°F), a medical emergency. Causes: heat stroke (environmental, classic or exertional, with CNS dysfunction), malignant hyperthermia (anesthesia trigger: succinylcholine, volatile anesthetics, treat with dantrolene), neuroleptic malignant syndrome (antipsychotics, rigidity, autonomic instability, elevated CK, treat with dantrolene and bromocriptine), serotonin syndrome (SSRIs/MAOIs, clonus, hyperreflexia, agitation, treat with cyproheptadine and supportive care), CNS infection/injury, and thyrotoxicosis. Management: ABCs, immediate cooling (evaporative, ice packs, cold IV fluids, cooling blanket), dantrolene, and treatment of underlying cause. Remove from heat source.

Hypertension (High Blood Pressure)

– See Hypertension.

Hyperventilation

– Breathing in excess of metabolic requirements, resulting in decreased partial pressure of arterial carbon dioxide (PaCO2) and respiratory alkalosis. Acute hyperventilation causes: perioral and digital paraesthesiae (numbness, tingling of lips, hands, feet), carpopedal spasm (tetany due to decreased ionised calcium from alkalosis–Chvostek and Trousseau signs), lightheadedness, dizziness, chest tightness, palpitations, anxiety, panic, and rarely loss of consciousness. Causes: anxiety and panic disorder (most common, acute or chronic hyperventilation syndrome), pain, fever, sepsis, pulmonary embolism (tachypnoea, hypoxia, hyperventilation), head injury (lesions affecting the respiratory center), metabolic acidosis (Kussmaul breathing as compensatory hyperventilation, e.g., DKA), drugs (salicylate overdose), pregnancy (progesterone stimulates the respiratory center), and neurological disorders (stroke, brainstem lesions). Diagnosis: clinical (typical symptoms, normal oxygen saturation, history of anxiety/panic), arterial blood gas (decreased PaCO2 <35 mmHg, elevated pH >7.45, normal PO2). Hyperventilation can be distinguished from pathological tachypnoea by the absence of hypoxia and underlying cause. Management: (1) Acute: reassurance, rebreathing into a paper bag (controversial and risk of hypoxia safer to use slow breathing technique), breathing retraining (5-second inhale, 5-second hold, 5-second exhale), treatment of underlying cause. (2) Chronic: cognitive behavioural therapy (CBT), breathing exercises, management of anxiety/panic disorder (SSRI, buspirone, relaxation techniques).

Hypnosis

– A state of focused attention, reduced peripheral awareness, and heightened suggestibility, resembling a trance-like state but distinct from sleep. Hypnosis is induced by a procedure (hypnotic induction) involving focused concentration, relaxation, and guided imagery. Characteristics: absorption (deep involvement in the experience, narrowed attention), dissociation (automaticity, detachment from immediate environment), suggestibility (heightened responsiveness to suggestions, both verbal and nonverbal). The subject remains aware, oriented, and able to resist suggestions. Clinical applications: (1) Pain management (acute and chronic pain, labour, dental procedures, cancer-related pain), (2) Anxiety and stress reduction (medical procedures, phobias, performance anxiety), (3) Irritable bowel syndrome (good evidence, NICE recommended), (4) Smoking cessation and habit modification (moderate evidence), (5) Post-traumatic stress disorder (as adjunctive therapy), (6) Nausea and vomiting (chemotherapy-related). Hypnotherapy is a regulated profession in many countries; clinical hypnosis should only be practiced by appropriately trained healthcare professionals. Contraindications: severe mental illness with psychotic features (delusions, hallucinations–hypnosis may worsen), dissociative identity disorder, epilepsy (rarely, may trigger seizure). Theories of mechanism: involvement of prefrontal cortex deactivation (reduced executive function and reality testing) and enhanced connectivity between the prefrontal cortex and insula (increased body awareness and interoception).

Hysterectomy

– Surgical removal of the uterus. Hysterectomy is the most common major gynaecological procedure. Types: (1) Total hysterectomy (removal of uterus and cervix–most common); (2) Subtotal/supracervical hysterectomy (removal of uterus, cervix retained–lower risk of ureteric damage and vault prolapse, continued risk of cervical cancer); (3) Radical hysterectomy (uterus, cervix, upper vagina, parametrium, pelvic lymph nodes–for cervical cancer, Wertheim procedure). Surgical approaches: abdominal (open, via Pfannenstiel or midline incision–most common in UK, preferred for large uteri, extensive pathology, or when concomitant surgery needed), vaginal (through the vaginal vault–less invasive, faster recovery, but smaller operating field), laparoscopic (total laparoscopic hysterectomy, TLH), and robot-assisted laparoscopic (da Vinci system). Indications: uterine fibroids (most common indication, 40%), endometriosis (20%), uterine prolapse (15%), abnormal uterine bleeding refractory to medical management (10%), endometrial cancer, cervical cancer, ovarian cancer, chronic pelvic pain, adenomyosis. Bilateral salpingo-oophorectomy (removal of ovaries and fallopian tubes) may be performed concurrently (especially if patient is postmenopausal or at high risk of ovarian cancer, e.g., BRCA mutation carriers). Complications: hemorrhage (1-3%), infection (wound, pelvic, urinary tract), ureteric injury (0.5-1%), bladder injury (1-2%), bowel injury (rare, 0.1%), DVT/PE, vault haematoma, and long-term: pelvic organ prolapse, urinary incontinence (increased in 20-40% after abdominal hysterectomy, decreased after vaginal), loss of fertility, early menopause if oophorectomy performed, changes in sexual function (may improve or remain unchanged).

ICU (Intensive Care Unit)

– A specialized hospital unit providing continuous monitoring and advanced organ support for critically ill patients. ICU admission criteria include: airway compromise, respiratory failure (PaO_2/FiO_2 <300), hemodynamic instability (MAP <65 despite fluids), severe metabolic disturbance, and multi-organ dysfunction. Staffing comprises a multi-disciplinary team: intensivists, ICU nurses, physiotherapists, pharmacists, and dietitians. Core interventions: mechanical ventilation, vasopressors/inotropes, renal replacement therapy, central venous and arterial line monitoring, and nutritional support. Prognostic scoring systems (APACHE II, SOFA, SAPS II) quantify mortality risk and guide clinical decision-making. Common ICU complications: ventilator-associated pneumonia, catheter-related bloodstream infection, pressure injuries, and ICU-acquired weakness.

Improving Sleep

– Adopting habits and environmental modifications to enhance the quality and duration of restorative sleep. Core strategies include maintaining a consistent sleep schedule, creating a dark and cool sleep environment, reducing screen exposure before bed, and avoiding caffeine and alcohol in the evening. Cognitive behavioral therapy for insomnia (CBT-I) is the first-line treatment for chronic sleep difficulties.

Junk Food (see Nutrition)

– See Nutrition.

Lesion

– A general term for any abnormal change in tissue structure due to disease or injury. Lesions can occur in any organ or tissue and can be classified by: (1) morphology: macule, papule, plaque, nodule, vesicle, bulla, pustule, ulcer, erosion, fissure, tumor; (2) location: skin, organ-specific, or systemic; (3) etiology: traumatic, infectious, inflammatory, neoplastic (benign or malignant), degenerative, congenital, vascular, metabolic; (4) imaging characteristics: radiolucent, radiopaque, contrast-enhancing, cystic, solid. A lesion is a descriptive finding, not a diagnosis; further characterisation (biopsy, imaging) is needed to determine the specific pathology.

Leucocyte (White Blood Cell)

– See White Blood Cell / Leucocyte.

Libido and Sexual Performance

– Libido (sexual desire) is modulated by a complex interplay of hormonal (testosterone, estrogen, prolactin), neurologic, psychologic, and relational factors. Low libido in men is commonly associated with hypogonadism, depression, medication side effects (SSRIs, antihypertensives), and chronic illness; in women, contributing factors include hormonal changes (menopause, postpartum), relationship distress, stress, and body image concerns. Sexual performance encompasses erectile function, ejaculatory control, and orgasmic capacity. Evaluation includes validated questionnaires, laboratory testing (testosterone, prolactin, thyroid function, glucose, lipids), and psychologic assessment. Treatment targets the underlying cause: testosterone replacement for hypogonadism, PDE5 inhibitors for ED, SSRIs or topical anesthetics for premature ejaculation, and psychosexual therapy or couples counseling for relationship or psychogenic contributors.

Living Will (Advance Statements)

– A legally binding written document that specifies a person's preferences for medical treatment if they become unable to make decisions for themselves in the future (loss of capacity). Living wills are a type of advance directive (or advance decision/advance statement in the UK). They allow individuals to refuse specific treatments (e.g., cardiopulmonary resuscitation, mechanical ventilation, artificial nutrition and hydration, antibiotics) in specified future scenarios (e.g., terminal illness, permanent unconsciousness/vegetative state, severe dementia). Legal requirements: the document must be signed and witnessed, signed in the presence of a witness, and it must clearly state the treatments being refused and the circumstances in which the refusal applies. In the UK: the Mental Capacity Act 2005 provides the legal framework for advance decisions. They are legally binding for refusing life-sustaining treatment if they are valid and applicable to the current circumstances. A valid advance decision can be overridden if the person has capacity and changes their mind when the situation arises. Related concepts: (1) Lasting Power of Attorney (LPA): allowed in the UK from 2007 for health and welfare decisions as well as decisions about property and financial affairs (in England and Wales), giving a named person legal authority to make decisions on behalf of the incapacitated patient. (2) Do Not Attempt Resuscitation (DNAR/DNACPR): a medical order signed by a senior clinician indicating that CPR should not be attempted in the event of cardiac arrest. (3) Physician Orders for Life-Sustaining Treatment (POLST): a medical order form used in the US that translates patient preferences into actionable medical orders. Advance care planning (ACP) discussions are encouraged to clarify patient values, goals, and preferences for future medical care.

Lockjaw (Tetanus)

– See Tetanus.

Masturbation

– Self-stimulation of the genitalia for sexual pleasure, a normal and common human behavior across all ages and genders. Prevalence: >90% of males and >60% of females report having masturbated at some point in their lives. It is a healthy component of psychosexual development, stress relief, and sexual self-exploration. There are no adverse medical consequences. Excessive or compulsive masturbation may rarely be associated with underlying psychiatric conditions (e.g., obsessive-compulsive disorder, hypersexuality). Cultural and religious attitudes vary widely. Medical benefits: may reduce stress, improve sleep, relieve sexual tension, and (in males) may reduce the risk of prostate cancer by facilitating the clearance of prostatic secretions.

Medical Devices

– Instruments, apparatuses, implants, or software used for diagnosis, monitoring, treatment, or prevention of disease, ranging from simple tongue depressors to complex implantable defibrillators and robotic surgical systems. Classification by risk: Class I (low risk, e.g., bandages, stethoscopes), Class II (moderate risk, e.g., infusion pumps, glucose monitors, MRI machines), and Class III (high risk, e.g., pacemakers, prosthetic heart valves, breast implants). Key categories: cardiovascular (stents, defibrillators, pacemakers), orthopedic (joint replacements, plates/screws), neurologic (deep brain stimulators, cochlear implants), diagnostic imaging (CT, MRI, ultrasound), and wearable/remote monitoring devices (continuous glucose monitors, smartwatch ECG). Regulatory oversight (FDA, CE marking) ensures safety and efficacy through premarket approval or clearance (510(k)). Device-related complications include infection, malfunction, thrombosis, and foreign body reaction.

Medical Tests

– Diagnostic procedures used to screen for, confirm, or monitor disease through analysis of biologic specimens, imaging, physiologic measurements, or genetic material. Categories: laboratory tests (complete blood count, comprehensive metabolic panel, cardiac biomarkers, thyroid function, HbA1c, cultures, PCR), imaging studies (X-ray, CT, MRI, ultrasound, PET, nuclear scintigraphy), physiologic testing (ECG, Holter monitor, pulmonary function tests, echocardiography), endoscopic procedures (colonoscopy, esophagogastroduodenoscopy, bronchoscopy, cystoscopy), and genetic/molecular testing (karyotype, microarray, NGS panels). Interpretation requires understanding of sensitivity, specificity, positive/negative predictive values, and reference ranges adjusted for age, sex, and comorbidities. Appropriate use reduces overdiagnosis and incidentalomas; shared decision-making about testing is recommended.

Mediterranean Diet

– The Mediterranean diet is a plant-forward eating pattern inspired by traditional cuisines of Greece, Italy, and Spain, characterized by high consumption of olive oil (primary fat), vegetables, fruits, legumes, whole grains, nuts, and seeds; moderate intake of fish and seafood, poultry, dairy (yogurt, cheese), and red wine; and limited red meat, processed meats, and sweets. Rich in monounsaturated fats, polyphenols, omega-3 fatty acids, and fiber, it exerts anti-inflammatory and antioxidant effects. Robust evidence demonstrates reduction in cardiovascular events (PREDIMED trial—30% relative risk reduction), improved glycemic control, lower LDL cholesterol, reduced blood pressure, decreased cancer incidence, and slower cognitive decline. It is the recommended dietary pattern for cardiovascular disease prevention by AHA/ACC guidelines and is associated with longevity in observational studies.

Mind-Body Exercise

– Mind–body exercise integrates physical movement with intentional focus on breath, body awareness, and mental concentration, including yoga, Tai Chi, Qigong, and Pilates. Yoga combines postures (asanas), breathing techniques (pranayama), and meditation; systematic reviews demonstrate benefit in reducing low back pain, improving flexibility, strength, and balance, and decreasing anxiety and depressive symptoms. Tai Chi—a slow, flowing Chinese martial art emphasizing weight shifting, coordination, and breath synchronicity—is effective for fall prevention in older adults (reducing fall risk by 30%), improving balance in Parkinson's disease, and reducing pain in knee osteoarthritis. Mechanisms include modulation of the autonomic nervous system (increased parasympathetic activity), hypothalamic–pituitary–adrenal axis regulation, and enhanced interoceptive awareness. Mind–body exercise is low-impact, accessible, and can be adapted for various fitness levels and health conditions.

MIND Diet

– The MIND diet (Mediterranean-DASH Intervention for Neurodegenerative Delay) is a hybrid eating pattern combining elements of the Mediterranean and DASH diets, specifically designed to preserve cognitive function and reduce dementia risk. Emphasis is on green leafy vegetables (≥6 servings/week), berries (≥2 servings/week, specifically blueberries and strawberries for their flavonoid content), nuts, whole grains, fish, poultry, olive oil, and wine (≤1 glass/day); red meat, butter/margarine, cheese, pastries/sweets, and fried foods are limited. The MIND diet trial (Morris et al.) showed that higher adherence was associated with significantly slower cognitive decline equivalent to 7.5 years of younger age; observational data suggest even moderate adherence reduces Alzheimer disease risk by 35%. Protective mechanisms: antioxidant (vitamin E, flavonoids), anti-inflammatory (polyunsaturated fats, polyphenols), and vascular (blood pressure reduction, improved cerebral blood flow) effects.

Mindfulness and Meditation

– Mindfulness is the practice of nonjudgmental, present-moment awareness cultivated through meditation techniques, including focused attention (breath, body scan), open monitoring (observing thoughts/emotions without attachment), and loving-kindness meditation. Neurobiologic correlates include decreased default mode network activity (reduced rumination), increased prefrontal cortex activation (enhanced executive control), and reduced amygdala reactivity (stress regulation). Meta-analyses demonstrate moderate-to-strong effects in reducing anxiety, depression, stress, and pain severity; mindfulness-based stress reduction (MBSR) and mindfulness-based cognitive therapy (MBCT) are evidence-based programs with efficacy equivalent to antidepressants for preventing depression relapse. Typical training: 8-week program with 2–3 hours/week group sessions plus 45 minutes daily home practice. Benefits accumulate with consistent practice; smartphone-based apps (Headspace, Calm) improve accessibility but produce smaller effects than instructor-led programs.

Nutrition

– The science of food intake, digestion, absorption, metabolism, and utilization of nutrients essential for health, growth, and disease prevention. Macronutrients: carbohydrates (primary energy source, 45–65% of total calories, emphasizing whole grains and fiber ≥25–38 g/day), proteins (0.8 g/kg body weight/day, higher for athletes, illness, and older adults), and fats (20–35% of calories, prioritizing unsaturated—olive oil, avocado, nuts, fatty fish—limiting saturated <10% and trans fats). Micronutrients: vitamins (A, D, E, K, C, B-complex) and minerals (calcium, iron, zinc, magnesium, potassium, iodine) with specific RDIs; deficiencies occur in restrictive diets, malabsorption, pregnancy, and chronic disease. Dietary patterns (Mediterranean, DASH, MIND, plant-based) reduce cardiovascular disease, diabetes, cancer, and all-cause mortality, whereas high intake of ultra-processed foods, added sugars, sodium, and red/processed meats increases chronic disease risk. Screening (MNA-SF for malnutrition risk) and counseling are essential components of primary care and chronic disease management, ideally delivered by registered dietitians.

Oral Sex (Fellatio, Cunnilingus)

– Oral stimulation of the genitals for sexual pleasure. Fellatio: oral stimulation of the penis. Cunnilingus: oral stimulation of the vulva and clitoris. Analingus (rimming): oral stimulation of the anus. Oral sex carries lower STI risk than unprotected vaginal or anal intercourse, but STIs can still be transmitted (HSV-1/2, gonorrhea (pharyngeal), chlamydia (pharyngeal), syphilis, HPV (oropharyngeal cancer risk with HPV-16), and HIV (lower risk than anal/vaginal). Pharyngeal gonorrhea and chlamydia are often asymptomatic but can be transmitted to partners. Dental dams and condoms reduce risk. Oral sex is a common cause of primary HSV-1 infection (cold sores transmitted to the genitals or vice versa).

Osteopathy

– A system of manual medicine and healthcare founded by Andrew Taylor Still (1828–1917) based on the principle that the body's structure (musculoskeletal system) and function are interrelated, and that the body has inherent self-regulatory and healing mechanisms. Osteopathic practitioners (osteopaths in the UK; Doctors of Osteopathic Medicine, DOs, in the US) use hands-on techniques including soft tissue massage, joint articulation, myofascial release, high-velocity low-amplitude thrust, muscle energy technique, and cranial osteopathy to diagnose and treat musculoskeletal pain, dysfunction, and associated conditions. In the UK, osteopathy is regulated by the General Osteopathic Council (GOsC) and is a statutory regulated profession; osteopaths complete a 4-year integrated Master's degree (MOst). Osteopathy is distinct from orthopaedic surgery but often works alongside it in managing back pain, neck pain, and sports injuries. Evidence supports osteopathic manipulative treatment for chronic low back pain, although the evidence base for cranial osteopathy is limited.

Physical Therapy

– A healthcare profession that uses evidence-based interventions to restore, maintain, and optimize movement, function, and quality of life across the lifespan after injury, surgery, or in chronic conditions. Components: therapeutic exercise (strengthening, stretching, range of motion, aerobic conditioning, balance training), manual therapy (joint mobilization/manipulation, soft tissue massage, myofascial release, stretching), modalities (therapeutic ultrasound, electrical stimulation—TENS, NMES, iontophoresis, phonophoresis, heat/ice, traction), functional training (gait training, transfer training, ergonomic assessment), and patient education. PT is first-line treatment for many musculoskeletal conditions (low back pain, knee osteoarthritis, rotator cuff tendinopathy, patellofemoral pain syndrome), postoperative rehabilitation (joint replacement, ACL reconstruction, rotator cuff repair), neurologic conditions (stroke, Parkinson's, spinal cord injury, multiple sclerosis), vestibular disorders, and fall prevention. Evidence supports early PT for acute low back pain reduces chronicity; supervised exercise is superior to home exercise alone for most conditions. Physical therapists are licensed doctoral-level practitioners (DPT) with direct access in all US states.

Pressure Injury

– Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences. Stages: Stage 1 (non-blanchable erythema, intact skin), Stage 2 (partial-thickness skin loss with exposed dermis, shallow open ulcer), Stage 3 (full-thickness skin loss, subcutaneous fat visible), Stage 4 (full-thickness tissue loss with exposed bone, muscle, or tendon), Unstageable (full-thickness tissue loss with eschar or slough obscuring the wound base), and Deep Tissue Injury (persistent non-blanchable deep red/maroon discoloration). Common sites: sacrum, heels, ischial tuberosities, trochanters, and occiput. Risk factors: immobility, malnutrition, incontinence, sensory loss, and decreased perfusion. Prevention: repositioning q2h, pressure-relieving surfaces (air mattress), moisture management, and nutritional support. Treatment: wound debridement, infection control, moist wound healing, NPWT, and surgical closure (flap) for advanced stages.

Pyrexia

– Body temperature >38°C (100.4°F), also known as fever. A regulated rise in body temperature mediated by pyrogens (exogenous: microbial products, endotoxins; endogenous: IL-1, IL-6, TNF, prostaglandin E_2) acting on the hypothalamic thermoregulatory center. Pyrexia is a protective host response (enhances immune function, inhibits pathogen growth), not a disease itself. Fever of unknown origin (FUO): temperature >38.3°C for >3 weeks without diagnosis despite 3 outpatient visits or 3 inpatient days. Patterns: continuous (typhoid, UTI), intermittent (malaria, sepsis, abscess), relapsing (borreliosis, lymphoma), and Pel-Ebstein (Hodgkin lymphoma). Management: treat underlying cause; antipyretics (paracetamol, NSAIDs) for patient comfort. Fever itself does not require treatment unless >40°C or in vulnerable patients (young children, pregnancy, cardiac disease).

Rehabilitation

– A multidisciplinary process aimed at restoring and optimising physical, cognitive, social, and psychological function in individuals with disabling conditions, enabling them to achieve the highest possible level of independence and quality of life. Rehabilitation addresses impairments (loss of body function—weakness, pain, reduced range of motion), activity limitations (difficulty performing tasks—walking, dressing, bathing), and participation restrictions (social, vocational, recreational). Settings: inpatient rehabilitation units (stroke, traumatic brain injury, spinal cord injury, amputation, major trauma, post-surgical), outpatient rehabilitation (musculoskeletal injuries, chronic conditions), and community-based rehabilitation. Core rehabilitation disciplines: physiotherapy (mobility, strength, balance, gait training), occupational therapy (activities of daily living, upper limb function, cognitive rehabilitation, home modifications), speech and language therapy (communication, swallowing, voice), orthotics and prosthetics, psychological support (adjustment to disability, mood disorders, pain management), social work (discharge planning, benefits, community resources), and rehabilitation nursing. Specific rehabilitation programs: cardiac rehabilitation (exercise, education, risk factor modification—reduces mortality by 20–25% after MI), pulmonary rehabilitation (COPD—improves exercise capacity, quality of life, reduces hospitalisations), stroke rehabilitation (early mobilisation, task-specific training, constraint-induced movement therapy), and neurological rehabilitation (multiple sclerosis, Parkinson disease, spinal cord injury). The WHO International Classification of Functioning, Disability and Health (ICF) provides a biopsychosocial framework for rehabilitation assessment and goal setting.

Remission

– A period during which the signs and symptoms of a chronic disease are reduced or absent. In cancer, remission is classified as: complete remission (CR)—disappearance of all detectable tumor (clinical, radiographic, and laboratory evidence of disease) for at least 1 month; partial remission (PR)—≥50% reduction in tumor size and/or extent of disease; and stable disease (SD)—no significant change. Remission is not synonymous with cure, as microscopic residual disease may persist and lead to relapse. In autoimmune diseases (rheumatoid arthritis, systemic lupus erythematosus, inflammatory bowel disease), remission is defined by clinical (absence of symptoms), biochemical (normal inflammatory markers), and/or endoscopic (mucosal healing in IBD) criteria. In psychiatric disorders (depression—remission is defined as a Hamilton Depression Rating Scale score ≤7; schizophrenia—remission is defined by sustained improvement in core symptoms for ≥6 months). The goal of treatment is to achieve and maintain remission. Monitoring during remission includes regular follow-up, surveillance imaging, and laboratory tests to detect early relapse.

Sarcopenia

– Progressive, generalized loss of skeletal muscle mass, strength, and function associated with aging. Diagnostic criteria (EWGSOP2): low muscle strength (grip strength <27 kg men, <16 kg women) plus low muscle quantity/quality (DEXA, CT, or bioimpedance). Affects 10-25% of adults >65, rising to >50% in those >80. Etiology: multifactorial (decreased protein synthesis, anabolic resistance, inflammation, mitochondrial dysfunction, physical inactivity, and malnutrition). Consequences: falls, fractures, frailty, decreased mobility, loss of independence, and increased mortality. Management: resistance exercise (strongest evidence), adequate protein intake (1.2-1.5 g/kg/day), vitamin D supplementation, and treatment of underlying conditions.

Scurvy

– A disease caused by severe vitamin C (ascorbic acid) deficiency. Vitamin C is a cofactor for proline and lysine hydroxylation in collagen synthesis, and a potent antioxidant. Scurvy is rare in developed countries but occurs in people with severely restricted diets (elderly, alcoholism, eating disorders, autism with food selectivity, malabsorption, homelessness). Symptoms develop 8–12 weeks after vitamin C depletion: fatigue, malaise, lethargy, joint and muscle pain, follicular hyperkeratosis (perifollicular haemorrhages), ecchymoses, gingival swelling and bleeding (hypertrophic, spongy gums), impaired wound healing, corkscrew hairs, and anemia. Advanced disease: haemarthrosis, subperiosteal hemorrhage (painful, especially in children), jaundice, edema, and sudden death. Diagnosis: clinical (dietary history, physical exam) and low serum ascorbic acid (<11 μmol/L). Treatment: vitamin C 100 mg PO TID for 1 week, then 100 mg daily (higher doses IV for severe disease). Symptoms improve within 24–48 hours; gingival bleeding resolves in 2–3 days; complete recovery in weeks. Prevention: dietary vitamin C (citrus fruits, berries, kiwi, tomatoes, broccoli, peppers).

Semen

– The fluid ejaculated from the male reproductive tract, containing sperm (produced in the testes) suspended in seminal plasma (produced by the seminal vesicles, prostate, and bulbourethral glands). Semen provides nutrients (fructose, amino acids), protective agents (zinc, antioxidants), and a medium for sperm transport. Normal semen parameters (WHO 2021): volume $\geq$1.4 mL, sperm concentration $\geq$16 million/mL, total sperm count $\geq$39 million, progressive motility $\geq$30%, normal morphology $\geq$4%.

Sexual Intercourse (Coitus)

– The insertion of a penis into a partner's body for sexual pleasure, reproduction, or both. Most commonly vaginal intercourse (penile-vaginal penetration) but may also include anal intercourse. Sexual intercourse is a key component of human pair bonding and reproduction. From a medical perspective, it is associated with risks including sexually transmitted infections (STIs), unintended pregnancy, and physical injury (e.g., vaginal or penile trauma). Safe sex practices (condoms, contraception, PrEP, vaccination) reduce these risks. Sexual dysfunction can affect any stage of the sexual response cycle (desire, arousal, orgasm).

SIRS

– Systemic Inflammatory Response Syndrome, a clinical response to a nonspecific insult (infection, trauma, pancreatitis, burns, autoimmune). Diagnosis requires $\geq$2 of: temperature >38°C or <36°C, heart rate >90 bpm, respiratory rate >20 breaths/min or PaCO2 <32 mmHg, WBC >12,000 or <4,000 or >10% bands. SIRS is non-specific but was historically used as a screening tool for sepsis (qSOFA and SOFA scores are now preferred). Sepsis: SIRS + confirmed infection. Severe sepsis: sepsis + organ dysfunction. Septic shock: sepsis + hypotension despite adequate fluid resuscitation. Management: source control, IV fluids, vasopressors, and antibiotics.

Sleep Hygiene

– A set of behavioral and environmental practices designed to promote consistent, uninterrupted, and restorative sleep. Key elements include maintaining a regular sleep schedule, creating a cool, dark, and quiet sleep environment, avoiding caffeine and electronic screens before bedtime, and engaging in relaxing pre-sleep routines. Poor sleep hygiene is a common contributor to insomnia, and adopting these habits is often the first-line recommendation for improving sleep quality without medication.

Sperm

– The male gamete (reproductive cell), produced in the seminiferous tubules of the testis through spermatogenesis. Sperm consists of: head (contains the haploid nucleus and acrosome containing enzymes for egg penetration), midpiece (contains mitochondria for energy), and tail (flagellum for motility). Approximately 1,500 sperm are produced per second. Sperm travel from the testis through the epididymis (maturation), vas deferens, seminal vesicles, and urethra during ejaculation.

Stem Cells

– Undifferentiated cells capable of self-renewal (producing identical daughter cells) and differentiation into specialized cell types. Classification by potency: totipotent (zygote—can form all cell types including extraembryonic tissues), pluripotent (embryonic stem cells—can form all three germ layers: ectoderm, endoderm, mesoderm), multipotent (adult/somatic stem cells—limited to a specific lineage: hematopoietic stem cells → all blood cells; mesenchymal stem cells → bone, cartilage, fat; neural stem cells → neurons, glia), and oligopotent/unipotent. Sources: embryonic stem cells (derived from the inner cell mass of blastocysts—ethical concerns), induced pluripotent stem cells (iPSCs—somatic cells reprogrammed by Yamanaka factors: OCT4, SOX2, KLF4, c-MYC), adult stem cells (bone marrow, adipose tissue, umbilical cord blood), and perinatal stem cells (amniotic fluid, placenta). Clinical applications: hematopoietic stem cell transplantation (bone marrow transplant for leukemia, lymphoma, multiple myeloma, aplastic anemia), mesenchymal stem cells (orthopedics—cartilage repair, graft-versus-host disease, Crohn disease—clinical trials), limbal stem cell transplantation (corneal regeneration), and iPSC-derived cell therapies (retinal pigment epithelium for macular degeneration—clinical trials). Challenges: teratoma formation (pluripotent cells), immune rejection, ethical controversies (embryonic stem cells), and regulatory hurdles.

Temperature (Body Temperature)

– The balance between heat production (metabolism, muscle activity) and heat loss (radiation, conduction, convection, evaporation). Normal core body temperature: 36.5–37.5°C (97.7–99.5°F), regulated by the hypothalamus (preoptic anterior hypothalamus—thermoregulatory center). Temperature is measured orally (sublingual—0.3–0.5°C below core), axillary (0.5–1.0°C below core), tympanic (close to core, infrared), rectal (closest to core), or via esophageal/pulmonary artery catheter (gold standard—core). Circadian variation: lowest at 4–6 AM, highest at 4–6 PM. Effects of exercise, digestion, menstrual cycle (0.5°C rise post-ovulation). Hyperthermia: elevated temperature due to failed thermoregulation (heat stroke, malignant hyperthermia, neuroleptic malignant syndrome)—does not respond to antipyretics. Fever (pyrexia): regulated temperature elevation driven by cytokines (IL-1, IL-6, TNF, prostaglandin E_2) in response to infection, inflammation, or tissue damage—responds to antipyretics. Hypothermia: core temperature <35°C. Fever of unknown origin (FUO): temperature >38.3°C (101°F) for ≥ 3 weeks with no diagnosis after 1 week of inpatient investigation.

Testicular Disorders

– conditions affecting the testes, including testicular torsion (spermatic cord twisting causing ischemic pain and swelling, a surgical emergency requiring detorsion within 6 hours for testicular salvage), epididymitis (inflammation of the epididymis, often from bacterial infection—STI in young men, urinary pathogens in older men—treated with antibiotics and supportive care), orchitis (testicular inflammation, often viral from mumps), hydrocele (fluid collection in the tunica vaginalis, benign), varicocele (dilated pampiniform plexus veins, the most common correctable cause of male infertility), and testicular cancer (most common solid malignancy in young men 15–35, typically presenting as a painless testicular mass, diagnosed with scrotal ultrasound and tumor markers—AFP, hCG, LDH—and treated with radical inguinal orchiectomy with possible chemotherapy or radiation). Evaluation includes scrotal ultrasound, urinalysis, STI screening, and tumor markers.

Unconsciousness

– See Coma.

Uranism

– An archaic and now obsolete term for male homosexuality, derived from the Greek goddess Aphrodite Urania (heavenly Aphrodite), used historically in 19th-century sexology literature (Karl Heinrich Ulrichs, Magnus Hirschfeld). The term is no longer used in modern medical or psychiatric practice; the preferred terminology is homosexuality or same-sex attraction.

Vaginal Intercourse

– The insertion of the penis into the vagina for sexual pleasure or reproduction. The most common form of sexual intercourse. Vaginal intercourse can lead to pregnancy if sperm are deposited in the vagina. From a medical perspective, it carries risks of STIs (HIV, gonorrhea, chlamydia, HPV, herpes, syphilis), unintended pregnancy, and rarely trauma (vaginal lacerations, especially in postmenopausal women or with inadequate lubrication). Contraception (condoms, hormonal methods, IUDs) and safe sex practices reduce these risks. Normal physiologic changes during vaginal intercourse include vaginal lubrication (Bartholin glands), vasocongestion, and orgasm.

Valsalva's Manoeuvre

– Forced expiratory effort against a closed airway (glottis closed or mouth and nose pinched shut), typically performed by bearing down as if straining during a bowel movement. The manoeuvre has four phases: Phase I (onset of straining): increased intrathoracic pressure compresses the aorta, causing a transient rise in blood pressure. Phase II (sustained strain): decreased venous return to the heart reduces cardiac output and blood pressure, with compensatory reflex tachycardia and peripheral vasoconstriction (baroreceptor-mediated). Phase III (release): sudden drop in intrathoracic pressure causes a transient fall in blood pressure. Phase IV (overshoot): cardiac output returns but peripheral vessels remain constricted, causing blood pressure to overshoot above baseline, with reflex bradycardia. Clinical uses: assessing heart murmurs (dynamic auscultation — most murmurs decrease with Valsalva, except hypertrophic obstructive cardiomyopathy [increases] and mitral valve prolapse [moves earlier]), terminating supraventricular tachycardia (via vagal stimulation increasing vagal tone to the AV node), autonomic function testing (heart rate and blood pressure responses), detecting abdominal hernias, and diagnosing pelvic organ prolapse. Contraindications: severe aortic stenosis, recent MI, glaucoma, and retinopathy.

Wasp Stings

– See Bites and Stings.

Waterbed

– A bed with a mattress filled with water, historically used in medical settings for the prevention and management of pressure ulcers (decubitus ulcers) in immobile or bedridden patients. The water-filled mattress distributes body weight evenly, reducing interface pressure over bony prominences (sacrum, heels, hips) compared to standard hospital mattresses, thereby reducing the risk of pressure ulcer formation. Waterbeds are also used in neonatal intensive care for preterm infants: the warm, fluid environment mimics the intrauterine environment, providing gentle containment and vestibular stimulation. In home care, waterbeds are occasionally recommended for patients with chronic pain (fibromyalgia, back pain) or widespread burns, though their use has declined with the availability of advanced pressure-relieving surfaces (low-air-loss and alternating-pressure mattresses). Waterbeds are heavy (difficult to move or clean), prone to leaking, and may cause claustrophobia or motion sickness in some patients.

Water Intoxication

– A potentially life-threatening condition of hyponatremia (low serum sodium) caused by excessive water intake exceeding the kidneys' maximal free water excretion capacity (0.7-1 L/hour in healthy adults). Water intoxication dilutes extracellular sodium, causing an osmotic shift of water into cells, particularly dangerous in the brain (cerebral edema). Causes: primary polydipsia (psychogenic polydipsia in psychiatric patients, especially schizophrenia), iatrogenic (excessive hypotonic IV fluids, postoperative, use of hypotonic irrigating solutions during TURP or hysteroscopy → TURP syndrome), infant water intoxication (overdiluted formula, forced water ingestion), exercise-associated hyponatremia (marathon runners, endurance athletes who overhydrate with plain water), and MDMA/ecstasy use (increased thirst + SIADH-like effect). Symptoms: mild (nausea, headache, confusion, lethargy), moderate (vomiting, disorientation, muscle cramps), severe (seizures, coma, brainstem herniation, respiratory arrest). Diagnosis: low serum sodium (<135 mEq/L), low serum osmolality (<275 mOsm/kg), and high urine osmolality. Treatment: mild-moderate — water restriction, observation. Severe: hypertonic saline (3% NaCl, 1-2 mL/kg IV bolus, repeat q10min until symptoms resolve or sodium reaches 120-125 mEq/L). Goal is to raise sodium by 4-6 mEq/L to reverse cerebral edema, avoiding overly rapid correction (>8-10 mEq/L in 24 hours) which risks osmotic demyelination syndrome (central pontine myelinolysis).

Weight Loss

– The intentional reduction of body weight, typically through a combination of dietary modification, increased physical activity, and behavioral changes. Achieving sustainable weight loss requires creating a consistent calorie deficit while preserving lean muscle mass through adequate protein intake and strength training. Even modest weight loss of 5 to 10 percent of body weight can produce significant health benefits, including improved blood glucose control, reduced blood pressure, and decreased risk of cardiovascular disease.

Weight Loss and Management

– Weight loss and management encompasses the ongoing process of achieving and maintaining a healthy body weight through sustainable lifestyle modifications. Effective weight management integrates a balanced, calorie-controlled diet, regular physical activity, behavioral strategies such as self-monitoring and goal setting, and adequate sleep and stress management. Long-term success requires a shift from short-term dieting to permanent lifestyle changes, as weight regain is common without sustained behavioral support and environmental modifications.

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